



**Swiss Centre  
for Life Cycle  
Inventories**

A joint initiative  
of the ETH domain and  
Swiss Federal Offices



# Life Cycle Inventories of Metals

Data v2.1 (2009)



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ecoinvent v2.1 report No. 10

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## Project "ecoinvent data v2.1"

|                                            |                                                                                                                                                                                                                       |
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## Life Cycle Inventories of Metals

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## Foreword to Version 2.1

This report is a revised and extended version of the report from Althaus et al. (2004) on the datasets of the version 1.1. It comprises

- corrections of reported errors,
- extension of inventories by new metals and the related updating of established inventories and
- some restructuring in the report structure.

The changes to version 1.2 and 1.3 as documented in Frischknecht et al. (2006) are integrated entirely where it concerned the datasets contained in this report.

The extension of the metals database consists of primary Cadmium, primary Indium, Gold and Silver (both primary and secondary as well as supply mix), Lead (secondary and supply mix), Tantalum (capacitor powder) and the metals Gallium, Arsenic and Tellurium as semiconductors.

Since most of these metals are by-products from the Zinc / Lead cycle, a new Part has been created that hosts the respective content from the original Part VIII Other Metals in Althaus et al. (2004) which in turn has been suspended. Its remaining contents have been reallocated: The copper alloys brass, bronze and brazing solders to Part II Copper; Magnesium and Tin to new individual parts.

The original inventories as they were established in version 1.1 through 1.3 are not altered. No general changes were made to actualise dated information such as supply and use patterns. Further the discussion of selected LCI / LCIA results has not been updated with the v2.0 calculation results. The respective chapters have been deleted.

The Overview on the following page provides a schematic representation of the changes of this report as compared to the report on data v1.1 in Althaus et al. (2004).

Compared to version 2.0, the version 2.1 contains a new part on antimony and several corrections and updates.

## References

- |                            |                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Althaus et al. (2004)      | Althaus H.-J., Blaser S., Classen M. and Jungbluth N. (2004) Life Cycle Inventories of Metals. Final report ecoinvent 2000 No. 10. EMPA Dübendorf, Swiss Centre for Life Cycle Inventories, Dübendorf, CH, Online-Version under: <a href="http://www.ecoinvent.ch">www.ecoinvent.ch</a> .                                                                       |
| Frischknecht et al. (2006) | Frischknecht R., Althaus H.-J., Bauer C., Capello C., Doka G., Dones R., Faist Emmenegger M., Hirschier R., Jungbluth N., Kellenberger D., Margni M., Nemecek T. and Spielmann M. (2006) Documentation of changes implemented in ecoinvent Data v1.2 and v1.3. ecoinvent report No. 16. EMPA Dübendorf, Swiss Centre for Life Cycle Inventories, Dübendorf, CH. |



## Overview

Following diagram indicates the relation between the contents of the preceding report on v1.1 data and this report on v2.1 data. The report on version v2.0 was structured as v2.1 without part XVI.

| Data v1.1 (2004) |                       | Data v2.1 (2009)                        |                                                                                                                                                                                                                                                                                                     |
|------------------|-----------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Part I           | Aluminium             | Part I                                  | Aluminium                                                                                                                                                                                                                                                                                           |
| Part II          | Iron and Steel        | Part II                                 | Iron and Steel                                                                                                                                                                                                                                                                                      |
| Part III         | Copper                | Part III <b>(updated)</b>               | Copper                                                                                                                                                                                                                                                                                              |
| Part IV          | Nickel                | Part IV                                 | Nickel                                                                                                                                                                                                                                                                                              |
| Part V           | Platinum Group Metals | Part V                                  | Platinum Group Metals                                                                                                                                                                                                                                                                               |
| Part VI          | Chromium              | Part VI                                 | Chromium                                                                                                                                                                                                                                                                                            |
| Part VII         | Manganese             | Part VII                                | Manganese                                                                                                                                                                                                                                                                                           |
| Part VIII        | Other metals          | <div> <b>content transferred</b> </div> | Part VIII <b>(updated)</b><br>Lead, Zinc, Cadmium and Indium<br><br>Part IX <b>(new)</b><br>Gold and Silver<br><br>Part X <b>(new)</b><br>Semiconductors: Tellurium, Gallium, Cadmium Compounds<br><br>Part XI <b>(new)</b><br>Tantalum powder<br><br>Part XII<br>Tin<br><br>Part XIII<br>Magnesium |
| Part IX          | Auxiliary Processes   |                                         | Part XIV<br>Auxiliary Processes                                                                                                                                                                                                                                                                     |
| Part X           | Metal processing      |                                         | Part XV<br>Metal processing                                                                                                                                                                                                                                                                         |
|                  |                       |                                         | Part XVI<br>Antimony                                                                                                                                                                                                                                                                                |
|                  |                       |                                         |                                                                                                                                                                                                                                                                                                     |
|                  |                       |                                         |                                                                                                                                                                                                                                                                                                     |

## Part I

# Aluminium

Data v2.1 (2009)

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Classen M., Althaus H.-J., Blaser S., Tuchscheid M., Jungbluth N., Doka G., Faist Emmenegger M. and Scharnhorst W. (2009) Life Cycle Inventories of Metals. Final report ecoinvent data v2.1, No 10. EMPA Dübendorf, Swiss Centre for Life Cycle Inventories, Dübendorf, CH, Online-Version under: [www.ecoinvent.ch](http://www.ecoinvent.ch).

Final report of updated Swiss National Life Cycle Inventory Database "ecoinvent", version 2.1

## Acknowledgement

We thank Bernard de Gélas and Christian Leroy of the European Aluminium Association (EAA) for providing the unit process data of their study and for their help with the interpretation of the data. We also thank Eirik Nordheim of the EAA and Gerald Reblizer of Alcan Technology & Management for their valuable comments to the report and the data.

## Comments on this version 2.1

This part corresponds to the original report in Althaus et al. (2004) that has been updated with the changes reported in Frischknecht et al. (2006), where the factor for resource consumption in the bauxite mining was corrected. Beside this change the CO emission in the dataset “aluminium, primary, liquid, at plant” (RER) which was inventoried as “biogenic” in earlier versions was corrected to “fossil”.

Chapter “6 Cumulative Results and Interpretation” of the original report has not been adapted and was deleted for this report.

Note that the flow names in the tables displaying unit processes have not been updated to reflect the name changes with version 2.0. This namely affects the designation of transportation service flows.

## Summary

The aluminium production is modelled for the European situation, primarily based on data from the European Aluminium Association (EAA (2000) and personal communications. The mining represents a world-wide situation primarily based on Mori & Adelhardt (1998).

Some of the datasets described in this report make only sense in the context of the aluminium production and are therefore not intended for other uses. Intermediate products like aluminium hydroxide and aluminium oxide however can be used to represent these chemicals in other applications.

The datasets to be chosen to represent aluminium in a LCA study are either the primary aluminium (chapter 5.3), one of the secondary aluminium datasets (chapter 5.4) or one of the production mixes (chapter 5.5). The decision whether to use primary, secondary or a mix of the two depends on the goal and scope of the LCA to be conducted.



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# 1 Introduction

This part deals with primary and secondary aluminium production. Because bauxite, aluminium hydrate and aluminium oxide are mainly consumed in the aluminium production, these chemicals are also included in this report.

A considerable part of the aluminium consumed today is secondary material. Thus also the collection of aluminium scrap and the recycling process are included.

A simplified overview of the processes is given in Fig. 1.1.

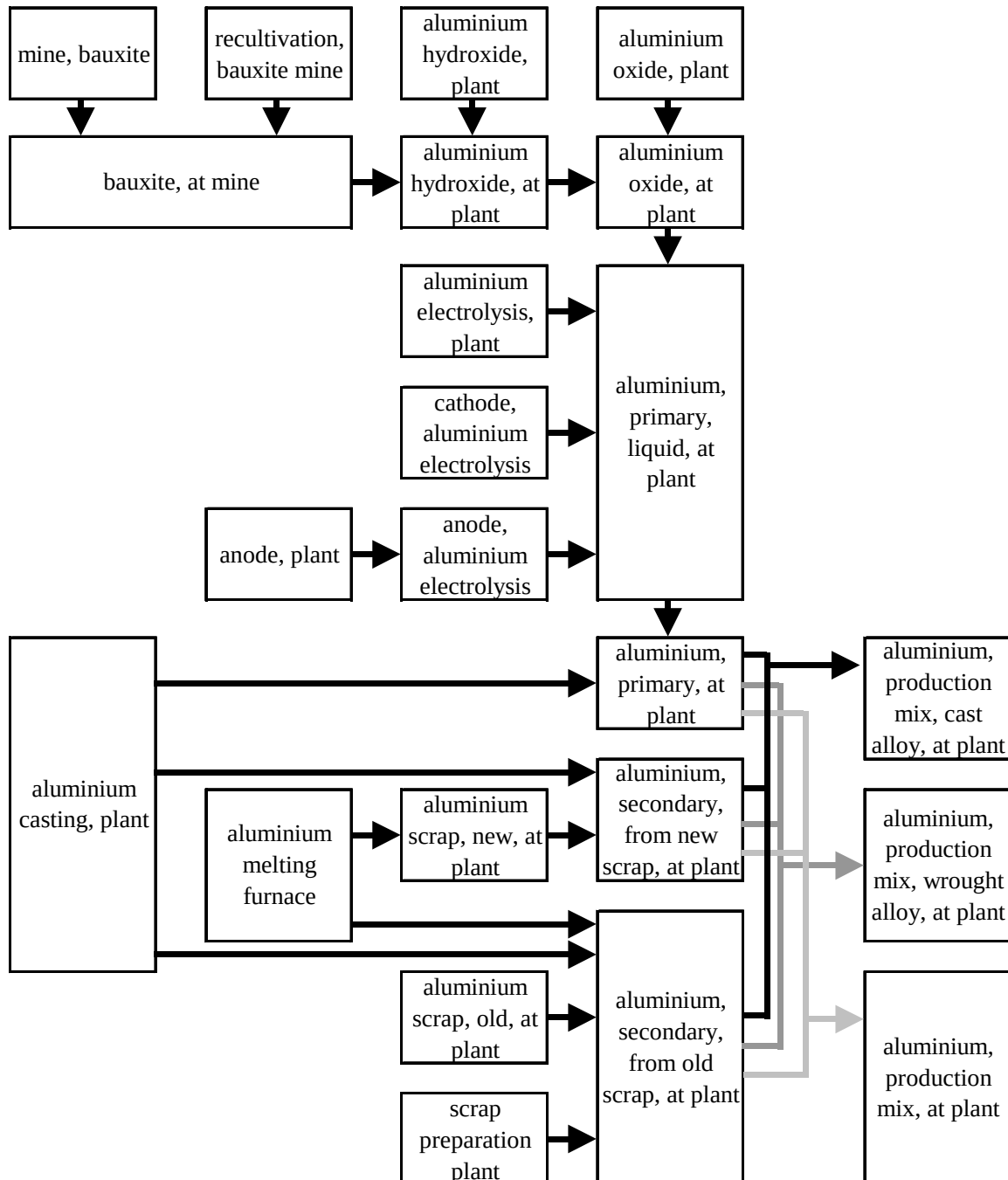


Fig. 1.1 Simplified overview of the processes for the aluminium production



## 2 Reserves and Resources of Materials

Aluminium is the most important light metal in the outer earth crust. It never occurs in nature in its metal form but only in compounds with other elements. Bauxite, the main resource for the aluminium production, is used to 85% for this purpose (Mori & Adelhardt (1998)).

Bauxite, the mineral from which aluminium oxide is gained, is mined mainly in the regions north and south of the tropical rainforests and most of the aluminium oxide is produced close to the mining sites (Tab. 2.1). The yearly bauxite mine production was 71 Mt/a in 1983 and the estimated reserves are 22.5 Gt (Hudson et al. (1997)). The yearly bauxite consumption rose to 140 Mt/a in 2002<sup>1</sup>.

**Tab. 2.1 World wide bauxite and aluminium oxide production (Mori & Adelhardt (1998), p. 20)**

|                                       | Aus-<br>tralia | Guinea | Ja-<br>maica | Brasil | China | Vene-<br>zuela | India | Russia | USA | Europe |
|---------------------------------------|----------------|--------|--------------|--------|-------|----------------|-------|--------|-----|--------|
| % of world bauxite production         | 38             | 13     | 10           | 9      | 6     | 5              | 5     | 3      |     |        |
| % of world aluminium oxide production | 30             |        | 7            |        | 5     |                |       | 6      | 30  | 10     |

98% of the industrial production of aluminium is done by electrolysis of aluminium oxide (Hall-Heroult process). This process requires a big amount of electricity. Thus most of the aluminium smelters are located in areas with cheap electricity be it from coal, nuclear or from hydro power plants. The electricity used to produce aluminium for the European market is produced according to Tab. 2.2 (EAA (2000)).

**Tab. 2.2 Electricity distributed by energy source for the production of primary aluminium for the European market**

| Energy re-<br>source | brown coal | hard coal | natural<br>gas | crude oil | nuclear<br>electricity | hydro<br>electricity |
|----------------------|------------|-----------|----------------|-----------|------------------------|----------------------|
| Share [%]            | 5.1        | 20        | 4.6            | 3.1       | 14.6                   | 52.6                 |

Because of the high energy input needed in primary aluminium production, secondary aluminium is an important resource. The overall share of secondary aluminium on the world aluminium market is estimated to be 32%<sup>2</sup> by industrial experts<sup>3</sup>.

<sup>1</sup> Eirik Nordheim, European Aluminium Association. Personal communication, 15.04.2004

<sup>2</sup> This figure is so low because the aluminium market is growing. The share of aluminium that is recycled after use is very variable depending on the application (e.g. in the building sector in Europe between 92% and 98%).

<sup>3</sup> Dr. G. Rombach, Project manager Recycling, Hydro Aluminium Deutschland GmbH. Personal communication, 1.10.2002 <http://www.aluminium.org/statistics/overview.asp>, accessed 18.12.2002





## 3 Characterisation of the materials

### 3.1 Bauxite

The term bauxite [CAS Nr. 1318-16-7] is used for sedimentary rocks that contain economically recoverable quantities of the aluminium minerals gibbsite, boehmite, and diaspore. The average crude ore that is mined contains 47.9% bauxite. The world average aluminium oxide content of bauxite is 44.6%, the European average content is 53%. Other important composites of bauxite are  $\text{SiO}_2$ ,  $\text{Fe}_2\text{O}_3$  and  $\text{TiO}_2$ .

### 3.2 Aluminium hydroxide

There are various modifications of aluminium hydroxides. The best defined crystalline forms are the three trihydroxides,  $\text{Al}(\text{OH})_3$  gibbsite, bayerite and nordstrandite.

Aluminium hydroxide obtained from the Bayer process is a 99.5% pure white powder. It is either calcinated to aluminium oxide ore sold as wet or dry aluminium hydrate. The specifications of the products of the "Aluminium Oxid Stade GmbH" (AOS) are given in Tab. 3.2.

Tab. 3.1 Product Specification for aluminium hydroxide<sup>4</sup>

| Chemical and physical analysis   |   | Typical Value | Min. Val. | Max. Val. | Quality     |
|----------------------------------|---|---------------|-----------|-----------|-------------|
| Moisture (20°-105° C)            | % | 7.0           | 5         | 9         | Wet         |
| $\text{Al}_2\text{O}_3$          | % | 60.5          | 59        | 62        | Wet         |
| Moisture (20°-105° C)            | % | 0.02          | 0.00      | 0.1       | Dry         |
| $\text{Al}_2\text{O}_3$          | % | 65            | 64.5      | 65.5      | Dry         |
| $\text{SiO}_2$                   | % | 0.004         | 0.003     | 0.007     | Wet and dry |
| $\text{Fe}_2\text{O}_3$          | % | 0.009         | 0.005     | 0.016     | Wet and dry |
| $\text{TiO}_2$                   | % | < 0.001       | 0.000     | 0.003     | Wet and dry |
| $\text{CaO}$                     | % | 0.004         | 0.000     | 0.007     | Wet and dry |
| $\text{Na}_2\text{O}$            | % | 0.22          | 0.18      | 0.26      | Wet and dry |
| $\text{V}_2\text{O}_5$           | % | < 0.0007      | 0.000     | 0.001     | Wet and dry |
| $\text{P}_2\text{O}_5$           | % | < 0.0004      | 0.000     | 0.001     | Wet and dry |
| $\text{Ga}_2\text{O}_3$          | % | 0.007         | 0.002     | 0.013     | Wet and dry |
| Ignition loss (105° - 1200° C)   | % | 34.6          |           |           | Wet and dry |
| Particle size < 45 $\mu\text{m}$ | % | 4             | 1         | 12        | Wet and dry |
| Particle size < 90 $\mu\text{m}$ | % | 45            | 30        | 60        | Wet and dry |

### 3.3 Aluminium Oxide

Hexagonally closest packed aluminium oxide,  $\alpha\text{-Al}_2\text{O}_3$ , is the only stable oxide in the  $\text{Al}_2\text{O}_3\text{-H}_2\text{O}$ -system. The mineral is also called corundum. Corundum is common in igneous and metamorphic rocks. Red and blue varieties of gem quality are called ruby and sapphire, respectively.

The specifications of the  $\text{Al}_2\text{O}_3$  of the "Aluminium Oxid Stade GmbH" (AOS) are given in Tab. 3.2.

<sup>4</sup> [http://www.aos-stade.de/html/home\\_gb.html](http://www.aos-stade.de/html/home_gb.html), accessed 16.12.2002

Tab. 3.2 Product Specification for aluminium oxide<sup>5</sup>

| Chemical and physical analysis |                   | Typical Value | Min. Val. | Max. Val. |
|--------------------------------|-------------------|---------------|-----------|-----------|
| Moisture (20°-300° C)          | %                 | 0.10          | 0.00      | 0.40      |
| Bulk density (lose)            | kg/m <sup>3</sup> | 970           | 950       | 1000      |
| Al <sub>2</sub> O <sub>3</sub> | %                 | 99.0          | 98.5      | 99.5      |
| SiO <sub>2</sub>               | %                 | 0.006         | 0.004     | 0.010     |
| Fe <sub>2</sub> O <sub>3</sub> | %                 | 0.014         | 0.008     | 0.025     |
| TiO <sub>2</sub>               | %                 | 0.001         | 0.000     | 0.004     |
| CaO                            | %                 | 0.006         | 0.000     | 0.010     |
| Na <sub>2</sub> O              | %                 | 0.34          | 0.28      | 0.39      |
| V <sub>2</sub> O <sub>5</sub>  | %                 | 0.001         | 0.000     | 0.002     |
| P <sub>2</sub> O <sub>5</sub>  | %                 | < 0.0006      | 0.000     | 0.001     |
| Ga <sub>2</sub> O <sub>3</sub> | %                 | 0.011         | 0.003     | 0.020     |
| Ignition loss (300° - 1200° C) | %                 | 0.53          | 0.45      | 0.60      |
| Angle of repose degree         | °                 | 34            | 32        | 36        |
| BET surface area               | m <sup>2</sup> /g | 50            | 45        | 60        |
| Particle size < 45 µm          | %                 | 8             | 5         | 15        |
| Particle size < 90 µm          | %                 | 55            | 45        | 65        |

### 3.4 Aluminium

Aluminium is a light grey coloured light metal. It has high specific strength, low specific weight, high electrical and thermal conductivity, good ductility or high hardness (depending on alloy) and due to a relatively stable and dense oxide layer on its surface, it exhibits good resistance against corrosion. Some of the most important chemical and physical properties of aluminium are given in Tab. 3.3.

Tab. 3.3 Physical properties of Aluminium

| Property                                             | Value                            | Unit              |
|------------------------------------------------------|----------------------------------|-------------------|
| Atomic number                                        | 13                               | -                 |
| Atomic weight                                        | 27                               | g/mol             |
| Density                                              | 2700                             | kg/m <sup>3</sup> |
| Melting point                                        | 660.4                            | °C                |
| Boiling point                                        | 2467                             | °C                |
| Thermal conductivity (at 20°C)                       | 239                              | W/mK              |
| Modulus of elasticity (99.99% Al)                    | $7.1 \cdot 10^{10}$              | N/m <sup>2</sup>  |
| Tensile strength (99.99% Al, soft / strain-hardened) | $5 \cdot 10^7$ / $13 \cdot 10^7$ | N/m <sup>2</sup>  |
| Tensile strength (Dural)                             | $40 \cdot 10^7$                  | N/m <sup>2</sup>  |
| Specific electrical resistance                       | $2.82 \cdot 10^{-8}$             | Ω m               |

Most technically used aluminium is alloyed. Important alloy elements for aluminium are silicon (Si), zinc (Zn), copper (Cu) and magnesium (Mg). Basically, two types of aluminium alloys are commonly distinguished: wrought alloys and casting alloys. As a rule of thumb, casting alloys tend to have higher contents of non-aluminium metals, especially of iron, silicon and/or zinc, whereas wrought alloys tend to have higher magnesium or sometimes manganese contents.

<sup>5</sup> [http://www.aos-stade.de/html/home\\_gb.html](http://www.aos-stade.de/html/home_gb.html), accessed 16.12.2002

## 4 Use / Application of Material

### 4.1 Bauxite, aluminium hydroxide and aluminium oxide

About 75% of the bauxite and 90% of the aluminium hydroxide and aluminium oxide is used for aluminium production. Other uses are listed in Tab. 4.1.

Tab. 4.1 Uses of aluminium precursors other than aluminium production

| bauxite          | aluminium hydroxide                             | aluminium oxide                                     |
|------------------|-------------------------------------------------|-----------------------------------------------------|
| Fireproof bricks | Filler for paints                               | Pigment in paints                                   |
| Abrasives        | Filler for paper                                | Filler                                              |
| Catalysts        | Filler for plastics (fire retardant)            | Ceramics                                            |
| Adsorbents       | Raw material for aluminium compounds            | Cosmetics and pharmaceuticals (e.g. sun protection) |
|                  | Cosmetics and pharmaceuticals (e.g. toothpaste) |                                                     |

### 4.2 Aluminium

In 1995, 20.4 Million t of primary aluminium has been used world wide (Mori & Adelhardt (1998)). The amount has increased steadily in the last decades. The biggest consumer of aluminium is the transport industry. Also important are the building and packaging sectors. (See Fig. 4.1 and Fig. 4.2)

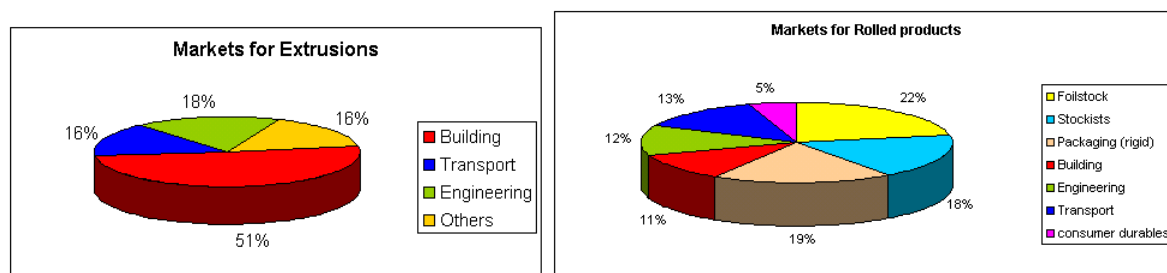


Fig. 4.1 Markets for aluminium products in Europe (Estimation EAA)<sup>6</sup>

Secondary aluminium is mainly used in cast alloys because they are less sensitive to high content of silicon and other metals in the scrap and thus can be produced more economically from scrap.

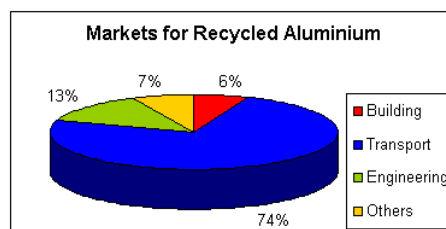


Fig. 4.2 Market for recycled aluminium in Europe in 2000 (estimation EAA)<sup>6</sup>

<sup>6</sup> <http://www.aluminium.org/statistics/overview.asp>, accessed 18.12.2002



## 5 System Characterisation

### 5.1 Bauxite mining

This part is mainly based on Mori & Adelhardt (1998).

#### 5.1.1 Production process and infrastructure

More than 95% of the bauxite today is gained from open mines. The thickness of the recoverable ore deposit usually is between 3.5 and 7 meters. The cap of the deposit, which in average is less than one meter thick, is removed as overburden and deposited nearby to be used for the recultivation when the mine has been depleted. The ore is won in stripes using hydraulic excavators and / or belt loaders. Drilling and blasting is only needed in some mines with exceptionally hard bauxite. The bauxite is transported to the mobile grinder and possibly to a dryer from where it is shipped to the aluminium oxide extraction plant.

According to the International Aluminium Institute, about 80% of the bauxite mines are recultivated after their depletion and 97% of the plants have plans for rehabilitating the area currently mined.

#### 5.1.2 Emissions

The only relevant direct emissions from bauxite mining are the dust emissions to air.

#### 5.1.3 Waste

There is no relevant waste produced in the mining process. The overburden is used in recultivation of the mine and not treated as waste although due to finer fractions and possibly increased weathering higher amounts of components of the backfilled material might get into the ground water compared to the situation before the mining activity.

#### 5.1.4 Bauxite mining in ecoinvent

The background information to the processes are given in Tab. 5.1, flows and their sources are shown in:

- |                               |          |
|-------------------------------|----------|
| - bauxite, at mine            | Fig. 5.1 |
| - mine, bauxite               | Fig. 5.2 |
| - recultivation, bauxite mine | Fig. 5.3 |

For a lack of information on this topic, the land use, the size and type of buildings and the weight of the stationary machines used are estimated. The estimation is based on information on other mines (gypsum, limestone).

#### Data quality

Data for bauxite at mine are of good overall quality. The infrastructure is roughly estimated but compared to the actual mining activities the influence is negligible. The values for the mining processes stem from a recent and comprehensive source (Mori & Adelhardt (1998)).

Tab. 5.1 Meta information for the bauxite mining processes

| Name                                  | bauxite, at mine                                                                                                                                                                                                  | mine, bauxite                                                                                                                                                                                                          | recultivation, bauxite mine                                                                                                  |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Location                              | GLO <sup>7</sup>                                                                                                                                                                                                  | GLO                                                                                                                                                                                                                    | GLO                                                                                                                          |
| Infrastructure Process                | 0                                                                                                                                                                                                                 | 1                                                                                                                                                                                                                      | 1                                                                                                                            |
| Unit                                  | kg                                                                                                                                                                                                                | unit                                                                                                                                                                                                                   | m2                                                                                                                           |
| Data Set Version                      | 2.0                                                                                                                                                                                                               | 2.0                                                                                                                                                                                                                    | 1.0                                                                                                                          |
| Included Processes                    | Includes the mining with its land transformation and occupation and the first mechanical grinding and separation of the ore from the gangue material. Also included is the drying of the bauxite to 11% moisture. | Includes the infrastructure and their disposal (buildings with their land use and machines) needed to provide the dried bauxite. Does not include the land use for the mine itself nor that for the roads in the mine. | Includes Land transformation and the machines needed. No refill material other than the overburdens from mining is included. |
| Amount                                | 1                                                                                                                                                                                                                 | 1                                                                                                                                                                                                                      | 1                                                                                                                            |
| Local Name                            | Bauxiterz, ab Abbau                                                                                                                                                                                               | Mine, Bauxit                                                                                                                                                                                                           | Rekultivierung, Bauxitmine                                                                                                   |
| Synonyms                              | aluminium ore//Aluminiumerz                                                                                                                                                                                       |                                                                                                                                                                                                                        |                                                                                                                              |
| General Comment to reference function | Refers to the dry bauxite mass (11% moisture). The Al <sub>2</sub> O <sub>3</sub> content of the bauxite is 53% (w/w)                                                                                             | It is assumed that the mine is in use for 20 years and has a yearly output of 1E6 t/a                                                                                                                                  | Refers to the recultivation of a mine with bauxite content of about 50% (w/w) in crude ore                                   |
| Start Date                            | 1994                                                                                                                                                                                                              | 1998                                                                                                                                                                                                                   | 1998                                                                                                                         |
| End Date                              | 2002                                                                                                                                                                                                              | 2002                                                                                                                                                                                                                   | 2002                                                                                                                         |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                 | 1                                                                                                                                                                                                                      | 1                                                                                                                            |
| Geography text                        | Some European datasets are used for global processes.                                                                                                                                                             | European data are used for buildings and machines                                                                                                                                                                      |                                                                                                                              |
| Technology text                       | Average bauxite mining technology world-wide (open mining)                                                                                                                                                        | Average technology for the aluminium consumed in Europe                                                                                                                                                                | Average bauxite mine recultivation world-wide (open mining)                                                                  |
| Representativeness [%]                | 79                                                                                                                                                                                                                |                                                                                                                                                                                                                        |                                                                                                                              |
| Production Volume                     | 111E6 t dry bauxite (=137E6 t crude bauxite) in 1994                                                                                                                                                              | unknown                                                                                                                                                                                                                | unknown                                                                                                                      |
| Sampling Procedure                    | Literature / Survey                                                                                                                                                                                               | estimation                                                                                                                                                                                                             | Literature / Survey                                                                                                          |
| Extrapolations                        | see Geography                                                                                                                                                                                                     | none                                                                                                                                                                                                                   | none                                                                                                                         |
| Uncertainty Adjustments               | Uncertainty is adjusted for geographical mismatch (using the pedigree matrix).                                                                                                                                    | none                                                                                                                                                                                                                   | none                                                                                                                         |

<sup>7</sup> The country code should actually be RER, since the subsequent processes all are for RER. This will be changed in v2.2

| General Flow information                     |              |                         |                                                                                                                                                                                      | Representation in ecoinvent |                   |                |          |                                                        |            |                  | Uncertainty                                                      |      |           |                  |
|----------------------------------------------|--------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-------------------|----------------|----------|--------------------------------------------------------|------------|------------------|------------------------------------------------------------------|------|-----------|------------------|
| Input                                        | Process Name | Output                  | Remarks                                                                                                                                                                              | Category                    | Sub category      | Infrastructure | Location | Module name in ecoinvent                               | Mean value | Unit             | Source mean value                                                | Type | StDev 95% | General Comment  |
| Bauxite (resource)                           | m            | bauxite, at mine: input | Bauxite = 47.9 % in crude ore, Al <sub>2</sub> O <sub>3</sub> = 53 % in bauxite, Al = 52.9 % in Al <sub>2</sub> O <sub>3</sub><br>--> 0.281 kg Al = 1 kg Bauxite = 2.09 kg crude ore | resource                    | in ground         |                |          | Aluminium, 24% in bauxite, 11% in crude ore, in ground | 2.81E-01   | kg               | Mori & Adelhardt 1998 (p72)                                      | 1    | 1.06      | (1,2,2,1,1,1,12) |
| Land use (mine) [m <sup>2</sup> a]           | m            |                         | assumption that the massive size of 3.5 to 7 m is depleted in 2 year                                                                                                                 | resource                    | land              |                |          | Occupation, mineral extraction site                    | 3.35E-04   | m <sup>2</sup> a | Mori & Adelhardt 1998 (p. 43)                                    | 1    | 1.50      | (1,2,2,1,1,1,7)  |
| Land transformation (mine) [m <sup>2</sup> ] | m            |                         |                                                                                                                                                                                      | resource                    | land              |                |          | Transformation, to mineral extraction site             | 1.67E-04   | m <sup>2</sup>   | Mori & Adelhardt 1998 (p. 43)                                    | 1    | 2.00      | (1,2,2,1,1,1,8)  |
| Land transformation (mine) [m <sup>2</sup> ] | m            |                         |                                                                                                                                                                                      | resource                    | land              |                |          | Transformation, from arable                            | 1.07E-04   | m <sup>2</sup>   | Mori & Adelhardt 1998 (p. 43)                                    | 1    | 2.00      | (1,2,2,1,1,1,8)  |
| Land transformation (mine) [m <sup>2</sup> ] | m            |                         |                                                                                                                                                                                      | resource                    | land              |                |          | Transformation, from arable, non-irrigated, fallow     | 3.41E-05   | m <sup>2</sup>   | Mori & Adelhardt 1998 (p. 43)                                    | 1    | 2.00      | (1,2,2,1,1,1,8)  |
| Land transformation (mine) [m <sup>2</sup> ] | m            |                         | torpical rainforest                                                                                                                                                                  | resource                    | land              |                |          | Transformation, from forest, extensive                 | 2.67E-05   | m <sup>2</sup>   | Mori & Adelhardt 1998 (p. 43)                                    | 1    | 2.00      | (1,2,2,1,1,1,8)  |
| Recultivation mine                           | m            |                         | 80% of mines are recultivated                                                                                                                                                        | metals                      | extraction        | No             | GLO      | recultivation, bauxite mine                            | 1.34E-04   | m <sup>2</sup>   | Mori & Adelhardt 1998 (p. 43)                                    | 1    | 2.00      | (1,2,2,1,1,1,8)  |
| Water                                        | m            |                         | Other source (EAA 2001) gives 2.88E-5 m <sup>3</sup>                                                                                                                                 | resource                    | in water          |                |          | Water, unspecified natural origin                      | 5.78E-04   | m <sup>3</sup>   | Mori & Adelhardt 1998                                            | 1    | 1.06      | (1,2,2,1,1,1,12) |
| diesel, burned in building machine [MJ]      | m            |                         | Input includes infrastructure and emissions of building machines according to their fuel consumption                                                                                 | construction processes      | machinery         | No             | GLO      | diesel, burned in building machine                     | 4.39E-02   | MJ               | Mori & Adelhardt 1998                                            | 1    | 2.00      | (1,2,2,1,1,1,5)  |
| Electrical power                             | m            |                         |                                                                                                                                                                                      | electricity                 | production mix    | No             | UCTE     | electricity, medium voltage, production UCTE, at grid  | 2.80E-03   | kWh              | Mori & Adelhardt 1998                                            | 1    | 1.06      | (1,2,2,1,1,1,2)  |
| Blasting [kg explosive]                      | m            |                         | Blasting includes Production of explosive and emissions of blasting                                                                                                                  | construction processes      | civil engineering | No             | RER      | blasting                                               | 1.56E-04   | kg               | Mori & Adelhardt 1998                                            | 1    | 1.06      | (1,2,2,1,1,1,1)  |
| Infrastructure (mine)                        | m            |                         | Input includes infrastructure needed at mine (buildings, machines,...) (exclusive building machines)                                                                                 | metals                      | extraction        | Yes            | GLO      | mine, bauxite                                          | 8.33E-13   | unit             | calculated (Mine with output of 1E6 t/year, in use for 20 years) | 1    | 3.00      | (1,2,2,1,1,1,9)  |

Fig. 5.1 Flows for "bauxite, at mine" and their representation in the ecoinvent database: The percentages given in the resource name are valid for the global situation. The should, however, be valid for the European situation and thus be 28.1% in Bauxite, 13.4% in crude ore.



| General Flow information    |              |                  |                                                                             | Representation in ecoinvent |                        |                |          |                                     |            |      | Uncertainty                                                                                 |      |          |                  |  |
|-----------------------------|--------------|------------------|-----------------------------------------------------------------------------|-----------------------------|------------------------|----------------|----------|-------------------------------------|------------|------|---------------------------------------------------------------------------------------------|------|----------|------------------|--|
| Input                       | Process Name | Output           | Remarks                                                                     | Category                    | Sub category           | Infrastructure | Location | Modul name in ecoinvent             | Mean value | Unit | Source mean value                                                                           | Type | StDv 95% | General Comment  |  |
| bauxite, at mine:<br>Output | m            | waste heat       | from electricity use                                                        | air                         | low population density |                |          | Heat, waste                         | 1.01E-02   | MJ   | calculated                                                                                  | 1    | 1.08     | (1,1,2,1,1,3,13) |  |
|                             | m            | Dust to air [kg] | 50% of "Dust" is > 10 um                                                    | air                         | low population density |                |          | Particulates, > 10 um               | 1.60E-03   | kg   | total amount: EAA 2001 (pers. communication), fraction: CEPMEIP Database - Emission Factors | 1    | 1.51     | (1,3,2,3,1,3,25) |  |
|                             | m            | Dust to air [kg] | 45 % of "Dust" is between 2.5 and 10 um                                     | air                         | low population density |                |          | Particulates, > 2.5 um, and < 10 um | 1.44E-03   | kg   |                                                                                             | 1    | 2.01     | (1,3,2,3,1,3,26) |  |
|                             | m            | Dust to air [kg] | 5 % of "Dust" is below 2.5 um                                               | air                         | low population density |                |          | Particulates, < 2.5 um              | 1.60E-04   | kg   |                                                                                             | 1    | 3.01     | (1,3,2,3,1,3,27) |  |
|                             | m            | Waste [kg]       | The waste is used in the recultivation of the mine and thus not inventoried |                             |                        |                |          |                                     | -4.74E-01  | kg   | mass balance                                                                                |      |          |                  |  |
|                             | m            | Bauxite, at mine |                                                                             | metals                      | extraction             | No             | GLO      | bauxite, at mine                    | 1.00E+00   | kg   |                                                                                             |      |          |                  |  |

Fig. 5.1 Flows for "bauxite, at mine" and their representation in the ecoinvent database (cont.)

| General Flow information            |    |               |               | Representation in ecoinvent |                                                          |                        |              |                |          |                                                  |               | Uncertainty |                                               |      |          |                                        |
|-------------------------------------|----|---------------|---------------|-----------------------------|----------------------------------------------------------|------------------------|--------------|----------------|----------|--------------------------------------------------|---------------|-------------|-----------------------------------------------|------|----------|----------------------------------------|
| Input                               |    | Process Name  |               | Output                      | Remarks                                                  | Category               | Sub category | Infrastructure | Location | Module name in ecoinvent                         | Mean value    | Unit        | Source mean value                             | Type | StdV 95% | General Comment                        |
| Land use for roads in mine area     | ML | mine, bauxite |               |                             | Is inventoried in Land use for mine                      |                        |              |                |          |                                                  |               |             |                                               |      |          |                                        |
| Land use for buildings              | ML |               |               |                             | Assumed that mine is in use for 20 years                 | resource               | land         |                |          | Occupation, industrial area, built up            | 3.00E+05      | m2a         | estimation, based on knowledge of other mines | 1    | 2.23     | (5,n.A.,n.A.,n.A.,n.A.,8); rough guess |
| Land use for buildings              | ML |               |               |                             | rough assumption                                         | resource               | land         |                |          | Transformation, to industrial area, built up     | 1.50E+04      | m2          | estimation, based on knowledge of other mines | 1    | 2.23     | (5,n.A.,n.A.,n.A.,n.A.,8); rough guess |
| Land use for buildings              | ML |               |               |                             | 55 %from agriculture / silviculture                      | resource               | land         |                |          | Transformation, from arable                      | 8.25E+03      | m2          | Mori & Adelhardt 1998                         | 1    | 2.23     | (5,n.A.,n.A.,n.A.,n.A.,8); rough guess |
| Land use for buildings              | ML |               |               |                             | 27%from extensive agriculture (steppe, savannah, desert) | resource               | land         |                |          | Transformation, from arable                      | 4.05E+03      | m2          | Mori & Adelhardt 1998                         | 1    | 2.23     | (5,n.A.,n.A.,n.A.,n.A.,8); rough guess |
| Land use for buildings              | ML |               |               |                             | 18%from tropical rainforest                              | resource               | land         |                |          | Transformation, from forest, extensive           | 2.70E+03      | m2          | Mori & Adelhardt 1998                         | 1    | 2.23     | (5,n.A.,n.A.,n.A.,n.A.,8); rough guess |
| multi storage buildings             | ML |               |               |                             |                                                          | construction processes | buildings    | Yes            | RER      | building, multi-storey                           | 1.50E+03      | m3          | estimation, based on knowledge of other mines | 1    | 3.23     | (5,n.A.,n.A.,n.A.,n.A.,9); rough guess |
| halls for dryer etc.                | ML |               |               |                             |                                                          | construction processes | buildings    | Yes            | CH       | building, hall                                   | 1.00E+04      | m2          | estimation, based on knowledge of other mines | 1    | 3.23     | (5,n.A.,n.A.,n.A.,n.A.,9); rough guess |
| machines for crushing, washing etc. | ML |               |               |                             | 13.13209803                                              | construction processes | machinery    | Yes            | RER      | Industrial machine, heavy, unspecified, at plant | 1.00E+04      | kg          | estimation, based on knowledge of other mines | 1    | 3.23     | (5,n.A.,n.A.,n.A.,n.A.,9); rough guess |
|                                     | ML |               | Mine, bauxite |                             |                                                          |                        | metals       | extraction     | Yes      | GLO                                              | mine, bauxite | 1.00E+00    | unit                                          |      |          |                                        |

Fig. 5.2 Flows for "mine, bauxite" and their representation in the ecoinvent database

| General Flow information                |   |                             |  | Representation in ecoinvent |                                                                             |          |              |                |          |                                                  |                             | Uncertainty |                              |      |          |                 |
|-----------------------------------------|---|-----------------------------|--|-----------------------------|-----------------------------------------------------------------------------|----------|--------------|----------------|----------|--------------------------------------------------|-----------------------------|-------------|------------------------------|------|----------|-----------------|
| Input                                   |   | Process Name                |  | Output                      | Remarks                                                                     | Category | Sub category | Infrastructure | Location | Module name in ecoinvent                         | Mean value                  | Unit        | Source mean value            | Type | StdV 95% | General Comment |
| diesel, burned in building machine [MJ] | m | recultivation, bauxite mine |  |                             | half of the demand for depletion as half of the amount of material is moved |          |              | No             | GLO      | diesel, burned in building machine               | # BEZUG!                    | MJ          | estimated                    | 1    | 2.10     | (4,1,2,1,3,1,5) |
| Land transformation (from mine) [m2]    | m |                             |  |                             | 100%from mine                                                               | resource | land         |                |          | Transformation, from mineral extraction site     | 1.00E+00                    | m2          | calculated                   | 1    | 2.00     | (1,2,2,1,1,1,8) |
| Land transformation (from mine) [m2]    | m |                             |  |                             | 55%to farm land                                                             | resource | land         |                |          | Transformation, to arable                        | 5.50E-01                    | m2          | Mori & Adelhardt 1998 (p.41) | 1    | 2.00     | (1,2,2,1,1,1,8) |
| Land transformation (from mine) [m2]    | m |                             |  |                             | 22%to extensive farm land                                                   | resource | land         |                |          | Transformation, to arable, non-irrigated, fallow | 2.20E-01                    | m2          | Mori & Adelhardt 1998 (p.41) | 1    | 2.00     | (1,2,2,1,1,1,8) |
| Land transformation (from mine) [m2]    | m |                             |  |                             | 22%to forest                                                                | resource | land         |                |          | Transformation, to forest                        | 2.20E-01                    | m2          | Mori & Adelhardt 1998 (p.41) | 1    | 2.00     | (1,2,2,1,1,1,8) |
| Land transformation (from mine) [m2]    | m |                             |  |                             | 1%to industrial area                                                        | resource | land         |                |          | Transformation, to industrial area               | 1.00E-02                    | m2          | Mori & Adelhardt 1998 (p.41) | 1    | 2.00     | (1,2,2,1,1,1,8) |
|                                         | m |                             |  | Recultivation mine          |                                                                             |          | metals       | extraction     | Yes      | GLO                                              | recultivation, bauxite mine | 1.00E+00    | m2                           |      |          |                 |

Fig. 5.3 Flows for "recultivation bauxite mine" and their representation in the ecoinvent database

## 5.2 Aluminium hydroxide and aluminium oxide production

This part is mainly based on information from Aluminium Oxid Stade GmbH<sup>8</sup> and Queensland Alumina Limited<sup>9</sup> and on (Mori & Adelhardt (1998)) and (Hudson et al. (1997)).

### 5.2.1 Production process and infrastructure

#### Overview

98% of the world wide production of aluminium hydroxide and aluminium oxide is based on the Bayer-Process. Austrian Karl-Josef-Bayer received patents in 1887 and 1892 for his process to extract aluminium hydroxide from bauxite using caustic soda.

The aluminium components of bauxite are dissolved by addition of caustic soda. By means of sedimentation and filtration the residue of the digestion is separated. Out of the clear sodium aluminate solution the aluminium hydroxide is precipitated by addition of seeding crystals. After filtration, the aluminium hydroxide is calcined to aluminium oxide at approx. 1000° C. The calcination in the circulating fluidised bed calciner is to remove the water of crystallisation from the hydroxide.

The operation of the process is outlined in Fig. 5.4.

The infrastructure is estimated based on an average sized aluminium oxide refinery (the Aluminium Oxid Stade GmbH (AOS)) in Germany. The plant is spread over an area of 55 hectares. The production was started up at the end of 1973. The nominal production capacity is on a current level of approx. 800'000 metric tons of alumina per year. Fig. 5.5 gives an overview of the refinery.

#### Delivery of raw materials

Most of the refineries are located close to a bauxite mine and / or they have their own port. AOS has a port with outside and inside berth with a usable length of the pier of about 200 m. The unloading is performed by grab with an unloading rate of approx. 600 t/h. The bauxite is transported to the bauxite storage area or the storage building in an enclosed conveyor system.

#### Bauxite storage

AOS open storage capacity is more than 1 mil metric tons. Additionally there is a storage building with approx. 100'000 t capacity.

#### Grinding

The bauxite is ground to a size of maximum 20 mm under addition of recycled digestion liquor (caustic soda solution). Lime can be introduced for phosphate control and mud conditioning. After grinding, the bauxite-liquor mixture (slurry) is pumped into storage tanks.

#### Desilication

The silica component of the bauxite is chemically attacked by caustic soda, causing alumina and soda losses by combining to form solid desilication products. To desilicate the slurry prior to digestion, it is heated and held at atmospheric pressure in pre-treatment tanks, reducing the build-up of scale in tanks and pipes. Most desilication products pass out with the mud waste as sodium aluminium silicate compounds.

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<sup>8</sup> <http://www.aos-stade.de>, accessed 16.12.2002

<sup>9</sup> <http://www.qal.com.au>, accessed 16.12.02



The bauxite slurry is pumped through the approximately 4'000 m long heat exchanger system of four

Thus about 97% of the aluminium components are extracted from the bauxite. After digestion about 30% of the bauxite mass remains in suspension as thin red mud slurry of silicates, and oxides of iron and titanium. The mud-laden liquor leaving the digestion vessel is flash-cooled to atmospheric boiling point by flowing through a series of flash vessels which operate at successively lower pressures. The energy released by flash evaporation is utilised to pre-heat the bauxite slurry. The condensate generated, which comprises approximately 75 % of the total condensate generated in the plant, is used for washing processes.

### **Dilution**

The bauxite slurry, concentrated by the evaporation of water during digestion, is diluted with wash water from subsequent operations. At AOS 2 dilution tanks with each 1'000 m<sup>3</sup> and a height of 15 m are used. Measured addition of digestion liquor adjusts the desired molar ratio to reduce recovery losses.

### **Red Mud Separation**

The coarse fraction of the undissolved bauxite components (sand) is separated by a hydroseparator and a rake classifier. Next, the slurry flows to the settlers where the remaining undissolved components of the bauxite (bauxite residue (BR) or red mud) are separated by sedimentation. Flocculants might be added to the settler feed stream to improve the rate of mud settling and achieve good clarity in the overflow liquor. The underflow of the settlers is pumped to the first of two washers for pre-washing of the BR. The filtrate is pumped to the washers in a countercurrent stream for dilution. Slaked lime is added to the dilute caustic liquor in the washing process to remove carbonate (Na<sub>2</sub>CO<sub>3</sub>) which forms by reaction with compounds in bauxite and also from the atmosphere and which reduces the effectiveness of liquor to dissolve alumina. Lime regenerates caustic soda, allowing the insoluble calcium carbonate to be removed with the waste mud.



The settled residue from the second washer is pumped to BR filtration, separated by vacuum drum filters and washed with condensate. The filtrate is used in the washer while the filter cake is pumped to the BR disposal area.

### **Liquor Cleaning**

The rest of the undissolved contaminants, such as iron oxide, are substantially filtered out by Kelly filters (pressure filter) using polypropylene filter cloth with aluminium hydroxide as a filter aid. As the contaminants are removed, the aluminium hydroxide is recovered as the filter cake.

### **Liquor Cooling**

The clear supersaturated alumina solution from the top stream of the settlers is cooled down to the temperature required by the precipitation process through multi-stage flash evaporation. Waste heat is used to pre-heat the spent liquor from precipitation.

### **Precipitation**

The cleaned aluminate liquor flows to the agitated precipitation tanks. In these, the super saturated liquor of aluminium hydroxide is seeded (aluminium hydroxide) in two steps and aluminium hydroxide is crystallised over 40-50 hours.

### **Hydroxide Filtration**

The fine fraction of the hydroxide is filtered on disc filters and is returned as seed to the precipitation process. The filtrate, after measured addition of caustic soda, flows back as digestion liquor primarily to the tube digester and grinding facilities. After double filtration and washing, the coarse fraction of the hydrate is calcined, dried, or transported to the hydroxide storage facility.

## **Calcination**

The feed material is calcined to remove both free moisture and chemically-combined water. By calcination, aluminium hydroxide is transformed into aluminium oxide at temperatures around 1'000° C. Calciners can use rotary kilns or the fluid bed process. Fluid bed facilities have proven themselves very reliable, are more energy efficient and don't contaminate the product like the rotary kilns.

Rotary or satellite coolers are used to cool the calcined alumina from the rotary kilns, and to pre-heat secondary combustion air for the kilns. Fluidised-bed coolers further reduce alumina temperature to less than 90°C before it is discharged on to conveyor belts which carry it to storage buildings or silos where it is stockpiled for shipment.

### **5.2.2 Emissions**

#### **Air**

There are no relevant direct air emissions from these processes beside dust from bauxite handling and water from the calcination.

At some sites the area of bauxite unloading and the conveyor belt system for transportation of input materials and products, are enclosed and feature efficient dust removing facilities.

#### **Water**

Process water is recycled. Waste water is created only by the operation of the bauxite residue disposal. Rain water has to be drained from the disposal site. This water is treated before being emitted to nature. After the treatment some NaOH, suspended particles and oxidisable compounds remain in the water.

### **5.2.3 Waste**

Different waste fractions are resulting from the production of alumina. The main part is the red mud or bauxite residue, consisting to about 40 % of water and the other part of mineral matters, being undissolved in the caustic soda - this is principally inert iron and titanium oxides and silica compounds - deriving from the bauxite input. This waste is disposed in special landfills (ponds). Red mud ponds are controlled landfills with base lining and leachate collection. The red mud disposal is therefore inventoried with the model for the residual landfill (Doka (2003), Part III). The red mud is not solidified with cement (like e.g. landfilled incineration residues are). Still the calcium content of red mud is high enough (2.2%) to estimate a carbonate phase of longer than 60'000 years (as in the residual landfill model). No adaptation of transfer coefficients in the model is necessary.

A smaller part is sand waste. For this waste the same composition as for the red mud is assumed.

A generic red mud composition is derived from literature data for red mud from Australia, Jamaica, India, Hungary, Yugoslavia, Turkey, and Suriname (Altundogan et al. (2002), Anonymous (2001), McLaughlin et al. (1998), NPI (1999)). It is given in Tab. 5.2.

Tab. 5.2 Composition of red mud in this project

| Element | Amount [% w/w] | Element | Amount [% w/w] | Element | Amount [% w/w] | Element | Amount [% w/w] |
|---------|----------------|---------|----------------|---------|----------------|---------|----------------|
| O       | 37.97%         | Cd      | 0.0005%        | Pb      | 0.0014%        | Si      | 10.4474%       |
| C       | 0.7607%        | Co      | 0.0022%        | Sb      | 0.0000%        | Fe      | 29.6417%       |
| S       | 0.0800%        | Cr      | 0.0751%        | Se      | 0.0005%        | Ca      | 2.2446%        |
| P       | 0.1475%        | Cu      | 0.0037%        | V       | 0.0535%        | Al      | 8.4217%        |
| B       | 0.0010%        | Hg      | 0.0000%        | Zn      | 0.0037%        | Na      | 5.9688%        |
| F       | 0.1130%        | Mn      | 0.0363%        | Be      | 0.0001%        | Na      | 5.9688%        |
| As      | 0.0029%        | Ni      | 0.0005%        | Ti      | 4.0203%        | Na      | 5.9688%        |

### 5.2.4 Aluminium hydroxide and aluminium oxide production in ecoinvent

The available information about the infrastructure needed for aluminium oxide production is non-quantitative. Tab. 5.3 and Tab. 5.4 show the details of the assumptions made to estimate the infrastructure expenditures.

Tab. 5.3 Estimated infrastructure for aluminium hydroxide plant

| aluminium hydroxide plant                                                       | Demand        | Unit | Life time | Life time demand |
|---------------------------------------------------------------------------------|---------------|------|-----------|------------------|
| Yearly production capacity                                                      | 1'220'000'000 | kg   | 50        | 1'220'000'000    |
| Area needed without calcination and aluminium oxide storage (95% of total) [m2] | 522'500       | m2   | 50        | 522'500          |
| Administration building (50x50x10 m3)                                           | 25'000        | m3   | 50        | 25'000           |
| Concrete for piers [m3]                                                         | 1'200         | m3   | 50        | 1'200            |
| Grab for unloading [kg]                                                         | 100'000       | kg   | 25        | 200'000          |
| conveyor to bauxite storage [m]                                                 | 300           | m    | 25        | 600              |
| Storage building (hall for 100'000 t = 50'000 m3; average height 10m) [m2]      | 5'000         | m2   | 50        | 5'000            |
| Taking up device [kg]                                                           | 10'000        | kg   | 25        | 20'000           |
| conveyor bauxite storage to grinder [m]                                         | 20            | m    | 25        | 40               |
| Rod mills for grinding [kg]                                                     | 10'000        | kg   | 25        | 20'000           |
| Tube digester (4000 m length) [kg]                                              | 28'000        | kg   | 25        | 56'000           |
| High pressure diaphragm pump [kg]                                               | 10'000        | kg   | 25        | 20'000           |
| Dilution tank (2 tanks to 1000 m3 --> 4.6m radius and 15m height) [kg]          | 39'200        | kg   | 25        | 78'400           |
| Red mud separation (3 settlers)                                                 | 1'414         | m3   | 50        | 1'414            |
| Red mud separation (2 countercurrent washers)                                   | 705           | m3   | 50        | 705              |
| Red mud separation (12 vacuum drum filters)                                     | 63'335        | kg   | 25        | 126'669          |
| Liquor cleaning                                                                 | 40'000        | kg   | 25        | 80'000           |
| Precipitation tanks (30 tanks, 3000 m3 each, 15 m high --> radius 8m)           | 606'956       | kg   | 25        | 1'213'911        |
| Hydroxide filters (11 Filters, radius 4 m, height 10 m)                         | 135'465       | kg   | 25        | 270'931          |
| Hydroxide silo (500 t = 500 m3; height 10m radius 4m)                           | 12'315        | kg   | 25        | 24'630           |
| Hydroxide storage hall (6000 t = 6000 m3, average height 10 m)                  | 600           | m2   | 50        | 600              |



Tab. 5.4 Estimated infrastructure for aluminium oxide plant

| aluminium oxide plant                                                      | Demand      | Unit | Life time | Life time demand |
|----------------------------------------------------------------------------|-------------|------|-----------|------------------|
| Yearly production capacity                                                 | 800'000'000 | kg   | 50        | 800'000'000      |
| Area needed for calcination and aluminium oxide storage (5% of total) [m2] | 27'500      | m2   | 50        | 27'500           |
| 3 fluid bed calciners                                                      | 20'000      | kg   | 25        | 40'000           |
| Concrete silo (65000 m3; height 25 m, radius 30m)                          | 1'555       | m3   | 50        | 1'555            |
| Steel silo (2000 m3, height 15 m, radius 6.5m)                             | 30'733      | kg   | 25        | 61'465           |
| conveyor belt                                                              | 100         | m    | 25        | 200              |

The other background information to the processes is given in Tab. 5.5, flows and their representation in the database is shown in:

- aluminium hydroxide, at plant Fig. 5.6
- aluminium hydroxide, plant Fig. 5.7
- aluminium oxide, at plant Fig. 5.8
- aluminium oxide, plant Fig. 5.9

### Data quality

Data for the steps in the aluminium oxide production are of good overall quality. The infrastructure is roughly estimated but compared to the other activities the influence is negligible. The values for the processes stem from a recent and comprehensive source (EAA (2000)).

Tab. 5.5 Meta information for the aluminium hydroxide and aluminium oxide processes

| Name                                  | aluminium hydroxide, at plant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | aluminium hydroxide, plant                                                                                                                                                                                  |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location                              | RER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | RER                                                                                                                                                                                                         |
| Infrastructure Process                | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1                                                                                                                                                                                                           |
| Unit                                  | kg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | unit                                                                                                                                                                                                        |
| Data Set Version                      | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2.0                                                                                                                                                                                                         |
| Included Processes                    | Includes bauxite and ancillary materials with their transports to the calcination plant, the extraction of $\text{Al}(\text{OH})_3$ (grinding, digestion and precipitation) and the disposal of the wastes.                                                                                                                                                                                                                                                                                                                                                                       | Part of the $\text{Al}_2\text{O}_3$ plant needed for $\text{Al}(\text{OH})_3$ fabrication including port, storage and process infrastructure. Also includes Building for administration. Includes disposal. |
| Amount                                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1                                                                                                                                                                                                           |
| Local Name                            | Aluminiumhydroxid, ab Werk                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Aluminiumhydroxid, Fabrik                                                                                                                                                                                   |
| Synonyms                              | Bay-erit//Hydragillit//Gibbsit//Tonerdehydrat//bayerite//hydragillite//gibbsite//diaspore// $\text{Al}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$ //Aluminiumtrihydrate//aluminium hydroxide                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                             |
| General Comment to reference function | $\text{Al}(\text{OH})_3$ extraction and calcination to $\text{Al}_2\text{O}_3$ is done in the same plant in line. Only cumulated data for the whole process are available. In- and outputs of $\text{Al}(\text{OH})_3$ are calculated according to the stoichiometry ( $1.530 \text{ kg } \text{Al}(\text{OH})_3 \rightarrow 1.000 \text{ kg } \text{Al}_2\text{O}_3 + 0.530 \text{ kg } \text{H}_2\text{O}$ ). Weak data for the energy consumption fractions of the two process parts were available. It is however not known if they relate to wet or dry aluminium hydroxide. | Medium sized plant. Output: 1'220'000 t/a, life time: 50 years                                                                                                                                              |
| Start Date                            | 1995                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2002                                                                                                                                                                                                        |
| End Date                              | 2002                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2002                                                                                                                                                                                                        |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1                                                                                                                                                                                                           |
| Geography text                        | Some Swiss datasets are used for European processes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Some Swiss datasets are used for European processes.                                                                                                                                                        |
| Technology text                       | Average technology for the aluminium consumed in Europe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Technology of one medium sized plant in Germany                                                                                                                                                             |
| Representativeness [%]                | 98                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                             |
| Production Volume                     | unknown                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | unknown                                                                                                                                                                                                     |
| Sampling Procedure                    | Literature / Survey                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Internet search                                                                                                                                                                                             |
| Extrapolations                        | see Geography                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | see Geography                                                                                                                                                                                               |
| Uncertainty Adjustments               | Uncertainty is adjusted for geographical mismatch (using the pedigree matrix).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Uncertainty is adjusted for geographical mismatch (using the pedigree matrix).                                                                                                                              |

Tab. 5.5 Meta information for the aluminium hydroxide and aluminium oxide processes (cont.)

| Name                                  | aluminium oxide, at plant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | aluminium oxide, plant                                                                                                                                               |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location                              | RER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | RER                                                                                                                                                                  |
| Infrastructure Process                | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1                                                                                                                                                                    |
| Unit                                  | kg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | unit                                                                                                                                                                 |
| Data Set Version                      | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2.0                                                                                                                                                                  |
| Included Processes                    | Includes calcination of $\text{Al}(\text{OH})_3$ to $\text{Al}_2\text{O}_3$ and its infrastructure. No transports are included because $\text{Al}_2\text{O}_3$ is produced in the same plant as $\text{Al}(\text{OH})_3$ .                                                                                                                                                                                                                                                                                                                                                        | Part of the $\text{Al}_2\text{O}_3$ plant needed only for $\text{Al}_2\text{O}_3$ fabrication (= calcination and $\text{Al}_2\text{O}_3$ storage) including disposal |
| Amount                                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1                                                                                                                                                                    |
| Local Name                            | Aluminiumoxid, ab Werk                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Aluminiumoxid, Fabrik                                                                                                                                                |
| Synonyms                              | Tonerde//Korund//Corundum//alumina                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                      |
| General Comment to reference function | $\text{Al}(\text{OH})_3$ extraction and calcination to $\text{Al}_2\text{O}_3$ is done in the same plant in line. Only cumulated data for the whole process are available. In- and outputs of $\text{Al}(\text{OH})_3$ are calculated according to the stoichiometry ( $1.530 \text{ kg } \text{Al}(\text{OH})_3 \rightarrow 1.000 \text{ kg } \text{Al}_2\text{O}_3 + 0.530 \text{ kg } \text{H}_2\text{O}$ ). Weak data for the energy consumption fractions of the two process parts were available. It is however not known if they relate to wet or dry aluminium hydroxide. | Medium sized plant. Output: 800'000 t/a, life time: 50 years                                                                                                         |
| Start Date                            | 1995                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2002                                                                                                                                                                 |
| End Date                              | 2002                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2002                                                                                                                                                                 |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1                                                                                                                                                                    |
| Geography text                        | Some Swiss datasets are used for European processes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Some Swiss datasets are used for European processes.                                                                                                                 |
| Technology text                       | Average technology for the aluminium consumed in Europe                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Technology of one medium sized plant in Germany                                                                                                                      |
| Representativeness [%]                | 98                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                      |
| Production Volume                     | unknown                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | unknown                                                                                                                                                              |
| Sampling Procedure                    | Literature / Survey                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Internet search                                                                                                                                                      |
| Extrapolations                        | see Geography                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | see Geography                                                                                                                                                        |
| Uncertainty Adjustments               | Uncertainty is adjusted for geographical mismatch (using the pedigree matrix).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Uncertainty is adjusted for geographical mismatch (using the pedigree matrix).                                                                                       |

| General Flow information                                      |                                              |        |                                                   | Representation in ecoinvent |              |                |          |                                                                     |            |      |                   | Uncertainty |          |                         |  |
|---------------------------------------------------------------|----------------------------------------------|--------|---------------------------------------------------|-----------------------------|--------------|----------------|----------|---------------------------------------------------------------------|------------|------|-------------------|-------------|----------|-------------------------|--|
| Input                                                         | Process Name                                 | Output | Remarks                                           | Category                    | Sub category | Infrastructure | Location | Module name in ecoinvent                                            | Mean value | Unit | Source mean value | Type        | StdV 95% | General Comment         |  |
| Bauxite [kg]                                                  | aluminium hydroxide, at plant; Inputs part 1 |        |                                                   | metals                      | extraction   | No             | GLO      | bauxite, at mine                                                    | 1.40E+00   | kg   | EAA 2000 (p. 49)  | 1           | 1.08     | (1,1,2,1,1,3,3)         |  |
| Transport of Bauxite from mine to plant [tkm]                 |                                              |        | Transport by ship (ocean)                         | transport systems           | ship         | No             | OCE      | transport, transoceanic freight ship                                | 7.11E+00   | tkm  | EAA 2000 (p. 50)  | 1           | 2.00     | (1,1,2,1,1,3,5)         |  |
| Transport of Bauxite from mine to plant [tkm]                 |                                              |        | Transport by ship (costal)                        | transport systems           | ship         | No             | RER      | transport, barge                                                    | 2.00E-03   | tkm  | EAA 2000 (p. 50)  | 1           | 2.00     | (1,1,2,1,1,3,5)         |  |
| Transport of Bauxite from mine to plant [tkm]                 |                                              |        | Transport by Road                                 | transport systems           | road         | No             | RER      | transport, lorry 32t                                                | 3.34E-01   | tkm  | EAA 2000 (p. 50)  | 1           | 2.00     | (1,1,2,1,1,3,5)         |  |
| Transport of Bauxite from mine to plant [tkm]                 |                                              |        | Transport by Rail                                 | transport systems           | train        | No             | RER      | transport, freight, rail                                            | 1.10E-02   | tkm  | EAA 2000 (p. 50)  | 1           | 2.00     | (1,1,2,1,1,3,5)         |  |
| Grinding of bauxite [kg / kg Al <sub>2</sub> O <sub>3</sub> ] |                                              |        | Energy is included in general power consumption   |                             |              |                |          |                                                                     |            |      |                   |             |          |                         |  |
| sodium hydroxide, 50% in H <sub>2</sub> O [kg]                |                                              |        | only weight of NaOH in solution (without solvent) | chemicals                   | inorganics   | No             | RER      | sodium hydroxide, 50% in H <sub>2</sub> O, production mix, at plant | 3.93E-02   | kg   | EAA 2000 (p. 49)  | 1           | 1.08     | (1,1,2,1,1,3,4)         |  |
| Transport of NaOH from plant to plant [tkm]                   |                                              |        | Rail; Standard distance from Ecoinvent 2000       | transport systems           | train        | No             | RER      | transport, freight, rail                                            | 4.71E-02   | tkm  | standard distance | 1           | 2.09     | (4,5,nA.,nA.,nA.,nA.,5) |  |
| Transport of NaOH from plant to plant [tkm]                   |                                              |        | Road; Standard distance from Ecoinvent 2000       | transport systems           | road         | No             | RER      | transport, lorry 32t                                                | 7.85E-03   | tkm  | standard distance | 1           | 2.09     | (4,5,nA.,nA.,nA.,nA.,5) |  |
| Heating of slurry and solution [MJ]                           |                                              |        | Heat included in general heat consumption         |                             |              |                |          |                                                                     |            |      | EAA 2000          |             |          |                         |  |
| CaO for desilication [kg]                                     |                                              |        |                                                   | construction materials      | additives    | No             | CH       | quicklime, milled, loose, at plant                                  | 3.01E-02   | kg   | EAA 2000 (p. 49)  | 1           | 1.08     | (1,1,2,1,1,3,4)         |  |

Fig. 5.6 Flows for "aluminium hydroxide, at plant" and their representation in the ecoinvent database

| General Flow information                                                               |   |                                              |  | Representation in ecoinvent |                                                                                                 |                   |                 |                |          |                                                       |            | Uncertainty |                               |      |          |                                             |
|----------------------------------------------------------------------------------------|---|----------------------------------------------|--|-----------------------------|-------------------------------------------------------------------------------------------------|-------------------|-----------------|----------------|----------|-------------------------------------------------------|------------|-------------|-------------------------------|------|----------|---------------------------------------------|
| Input                                                                                  |   | Process Name                                 |  | Output                      | Remarks                                                                                         | Category          | Sub category    | Infrastructure | Location | Module name in ecoinvent                              | Mean value | Unit        | Source mean value             | Type | StdV 95% | General Comment                             |
| Transport of CaO from plant to plant [tkm]                                             | m | aluminium hydroxide, at plant; Inputs part 2 |  |                             | Rail; Standard distance (for cement) from Ecoinvent 2000                                        | transport systems | train           | No             | RER      | transport, freight, rail                              | 3.01E-03   | tkm         | standard distance             | 1    | 2.09     | (4,5,n.A.,n.A.,n.A.,n.A.,5)                 |
| Transport of CaO from plant to plant [tkm]                                             | m |                                              |  |                             | Rail; Standard distance (for cement) from Ecoinvent 2000                                        | transport systems | road            | No             | RER      | transport, lorry 32t                                  | 3.01E-03   | tkm         | standard distance             | 1    | 2.09     | (4,5,n.A.,n.A.,n.A.,n.A.,5)                 |
| Separation of Al(OH) <sub>3</sub> -Solution from residue solids                        | m |                                              |  |                             | Not inventoried, Energy included in general power / heat consumption                            |                   |                 |                |          |                                                       |            |             |                               |      |          |                                             |
| Water-cooling for persipitation of Al(OH) <sub>3</sub>                                 | m |                                              |  |                             | Spec. Weight of water: 998 kg/m3                                                                | resource          | in water        |                |          | Water, cooling, unspecified natural origin            | 2.42E-03   | m3          | EAA 2001(pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,4)                             |
| Electrical power used in plant (process, lighting, office work, room cooling...) [kWh] | m |                                              |  |                             | Includes all power consumed (93 % of total alumina production)                                  | electricity       | production mix  | No             | UCTE     | electricity, medium voltage, production UCTE, at grid | 1.40E-01   | kWh         | EAA 2001(pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,2)                             |
| Water for washing of persipitation & residue [kg]                                      | m |                                              |  |                             | Surplus water is evaporated, remaining solution is reused in process: Steam is not inventoried. | resource          | in water        |                |          | Water, unspecified natural origin                     | 1.05E-05   | m3          | EAA 2001(pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,4)                             |
| Heat energy for evaporation of wash water [MJ]                                         | m |                                              |  |                             | Included in general heat consumption                                                            |                   |                 |                |          |                                                       |            |             |                               |      |          |                                             |
| General heat from coal                                                                 | m |                                              |  |                             | Includes all heat from coal consumed (100 %of total alumina production)                         | hard coal         | heating systems | No             | RER      | heat, at hard coal industrial furnace 1-10MW          | 1.63E-01   | MJ          | EAA 2001(pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,1)                             |
| General heat from natural gas                                                          | m |                                              |  |                             | Includes all heat from natural gas consumed (36 %of total alumina production)                   | natural gas       | heating systems | No             | RER      | heat, natural gas, at industrial furnace >100kW       | 2.85E-01   | MJ          | EAA 2001(pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,1)                             |
| General heat from oil                                                                  | m |                                              |  |                             | Includes all heat from oil consumed (65 %of total alumina production)                           | oil               | heating systems | No             | CH       | heat, light fuel oil, at industrial furnace 1MW       | 3.59E+00   | MJ          | EAA 2001(pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,1)                             |
| Infrastructure (plant) [piece]                                                         | m |                                              |  |                             | Includes land use                                                                               | metals            | extraction      | Yes            | RER      | aluminium hydroxide, plant                            | 1.64E-11   | unit        | calculated                    | 1    | 3.23     | (5,n.A.,n.A.,n.A.,n.A.,n.A.,9); rough guess |

Fig. 5.6 Flows for "aluminium hydroxide, at plant" and their representation in the ecoinvent database (cont.)

| General Flow information               |              |                                             |                                                                             | Representation in ecoinvent |                            |                |          |                                                                                  |            |      | Uncertainty                    |      |          |                  |
|----------------------------------------|--------------|---------------------------------------------|-----------------------------------------------------------------------------|-----------------------------|----------------------------|----------------|----------|----------------------------------------------------------------------------------|------------|------|--------------------------------|------|----------|------------------|
| Input                                  | Process Name | Output                                      | Remarks                                                                     | Category                    | Sub category               | Infrastructure | Location | Module name in ecoinvent                                                         | Mean value | Unit | Source mean value              | Type | StdV 95% | General Comment  |
| aluminium hydroxide, at plant; Outputs |              | ML waste heat                               | from electricity use                                                        | air                         | unspecified                |                |          | Heat, waste                                                                      | 5.03E-01   | MJ   | calculated                     | 1    | 1.08     | (1,1,2,1,1,3,13) |
|                                        |              | ML Dust from bauxite handling               | is included in bauxite at mine --> not inventoried                          |                             |                            |                |          |                                                                                  |            |      |                                |      |          |                  |
|                                        |              | ML NaOH in water                            | Na calculated from NaOH emission based on stoichiometry                     | water                       | unspecified                |                |          | Sodium, ion                                                                      | 1.28E-03   | kg   | EAA 2001 (pers. communication) | 1    | 1.51     | (1,1,2,1,1,3,33) |
|                                        |              | ML suspended particles in water             |                                                                             | water                       | unspecified                |                |          | Suspended solids, unspecified                                                    | 1.70E-04   | kg   | EAA 2001 (pers. communication) | 1    | 1.51     | (1,1,2,1,1,3,33) |
|                                        |              | ML COD in water                             |                                                                             | water                       | unspecified                |                |          | COD, Chemical Oxygen Demand                                                      | 4.59E-05   | kg   | EAA 2001 (pers. communication) | 1    | 1.51     | (1,1,2,1,1,3,32) |
|                                        |              | ML BOD                                      | calculated from COD, assuming C <sub>6</sub> H <sub>12</sub> O <sub>6</sub> | water                       | unspecified                |                |          | BOD <sub>5</sub> , Biological Oxygen Demand                                      | 4.59E-05   | kg   | calculated                     | 1    | 1.51     | (1,1,2,1,1,3,32) |
|                                        |              | ML TOC                                      |                                                                             | water                       | unspecified                |                |          | TOC, Total Organic Carbon                                                        | 1.80E-05   | kg   | calculated                     | 1    | 1.51     | (1,1,2,1,1,3,32) |
|                                        |              | ML DOC                                      |                                                                             | water                       | unspecified                |                |          | DOC, Dissolved Organic Carbon                                                    | 1.80E-05   | kg   | calculated                     | 1    | 1.51     | (1,1,2,1,1,3,32) |
|                                        |              | ML Sand to Disposal [kg]                    | composition assumed as red mud                                              | waste management            | residual material landfill | No             | CH       | disposal, redmud from bauxite digestion, 0% water, to residual material landfill | 3.33E-02   | kg   | EAA 2001 (pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,6)  |
|                                        |              | ML solid waste unspecified to Disposal [kg] |                                                                             | waste management            | inert material landfill    | No             | CH       | disposal, inert waste, 5% water, to inert material landfill                      | 2.01E-02   | kg   | EAA 2001 (pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,6)  |
|                                        |              | ML Bauxite residue to Disposal [kg]         | Red mud                                                                     | waste management            | residual material landfill | No             | CH       | disposal, redmud from bauxite digestion, 0% water, to residual material landfill | 4.37E-01   | kg   | EAA 2001 (pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,6)  |
|                                        |              | ML Heat, waste (recovered) [MJ]             | recovered for process --> not inventoried                                   |                             |                            |                |          |                                                                                  |            |      |                                |      |          |                  |
|                                        |              | ML Aluminium hydroxide, at plant [kg]       |                                                                             | chemicals                   | inorganics                 | No             | RER      | aluminium hydroxide, at plant                                                    | 1.00E+00   | kg   |                                |      |          |                  |

Fig. 5.6 Flows for "aluminium hydroxide, at plant" and their representation in the ecoinvent database (cont.)

| General Flow information                                       |   |                                    |  | Representation in ecoinvent                 |                        |           |              |                |                                                |                              |            | Uncertainty |                   |      |                 |                 |
|----------------------------------------------------------------|---|------------------------------------|--|---------------------------------------------|------------------------|-----------|--------------|----------------|------------------------------------------------|------------------------------|------------|-------------|-------------------|------|-----------------|-----------------|
| Input                                                          |   | Process Name                       |  | Output                                      | Remarks                | Category  | Sub category | Infrastructure | Location                                       | Module name in ecoinvent     | Mean value | Unit        | Source mean value | Type | StdV 95%        | General Comment |
| Total land use (transformation)                                | m | aluminium hydroxide, plant; part 1 |  |                                             |                        | resource  | land         |                |                                                | Transformation, from unknown | 5.23E+05   | m2          | estimated         | 1    | 2.03            | (1,4,1,1,1,4,8) |
| Land occupation for buildings and open bauxite storage area    | m |                                    |  | assumption: 80% of total area is built up   | resource               | land      |              |                | Occupation, industrial area, built up          | 2.09E+07                     | m2a        | estimated   | 1                 | 1.54 | (2,4,1,1,1,4,7) |                 |
| Land transformation to buildings and open bauxite storage area | m |                                    |  |                                             | resource               | land      |              |                | Transformation, to industrial area, built up   | 4.18E+05                     | m2         | estimated   | 1                 | 2.03 | (2,4,1,1,1,4,8) |                 |
| Land occupation for vegetation area                            | m |                                    |  | assumption: 20% of total area is vegetation | resource               | land      |              |                | Occupation, industrial area, vegetation        | 5.23E+06                     | m2a        | estimated   | 1                 | 1.54 | (2,4,1,1,1,4,7) |                 |
| Land transformation to vegetation area                         | m |                                    |  |                                             | resource               | land      |              |                | Transformation, to industrial area, vegetation | 1.05E+05                     | m2         | estimated   | 1                 | 2.03 | (2,4,1,1,1,4,8) |                 |
| Building, multi story                                          | m |                                    |  |                                             | construction processes | buildings | Yes          | RER            | building, multi-storey                         | 2.50E+04                     | m3         | estimated   | 1                 | 3.04 | (3,4,1,1,1,4,9) |                 |
| Building, hall                                                 | m |                                    |  |                                             | construction processes | buildings | Yes          | CH             | building, hall                                 | 5.60E+03                     | m2         | estimated   | 1                 | 3.04 | (3,4,1,1,1,4,9) |                 |
| concrete in reinforced concrete                                | m |                                    |  |                                             | construction materials | concrete  | No           | CH             | concrete, normal, at plant                     | 3.02E+03                     | m3         | estimated   | 1                 | 3.04 | (3,4,1,1,1,4,9) |                 |
| steel in reinforced concrete                                   | m |                                    |  |                                             |                        | metals    | extraction   | No             | RER                                            | reinforcing steel, at plant  | 2.09E+06   | kg          | estimated         | 1    | 3.04            | (3,4,1,1,1,4,9) |

Fig. 5.7 Flows for "aluminium hydroxide, plant" and their representation in the ecoinvent database

| General Flow information                |   |                                    |   | Representation in ecoinvent     |                                                                                                                |                        |                     |                |          |                                                                 |            | Uncertainty |                   |      |          |                 |
|-----------------------------------------|---|------------------------------------|---|---------------------------------|----------------------------------------------------------------------------------------------------------------|------------------------|---------------------|----------------|----------|-----------------------------------------------------------------|------------|-------------|-------------------|------|----------|-----------------|
| Input                                   |   | Process Name                       |   | Output                          | Remarks                                                                                                        | Category               | Sub category        | Infrastructure | Location | Module name in ecoinvent                                        | Mean value | Unit        | Source mean value | Type | StdV 95% | General Comment |
| heavy machine, unspecified              | m | aluminium hydroxide, plant; part 2 | m |                                 |                                                                                                                | construction processes | machinery           | Yes            | RER      | Industrial machine, heavy, unspecified, at plant                | 2.20E+05   | kg          | estimated         | 1    | 3.04     | (3,4,1,1,14,9)  |
| Poor concrete for additional foundation |   |                                    |   |                                 |                                                                                                                | construction materials | concrete            | No             | CH       | concrete, sole plate and foundation, at plant                   | 1.00E+02   | m3          | estimated         | 1    | 3.04     | (3,4,1,1,14,9)  |
| Conveyor belt                           |   |                                    |   |                                 |                                                                                                                | construction processes | machinery           | Yes            | RER      | conveyor belt, at plant                                         | 6.40E+02   | m           | estimated         | 1    | 3.04     | (3,4,1,1,14,9)  |
| Chromium steel (stainless)              |   |                                    |   |                                 |                                                                                                                | metals                 | extraction          | No             | RER      | chromium steel 18/8, at plant                                   | 1.87E+06   | kg          | estimated         | 1    | 3.04     | (3,4,1,1,14,9)  |
| Steel low alloyed                       |   |                                    |   |                                 |                                                                                                                | metals                 | extraction          | No             | RER      | steel, low-alloyed, at plant                                    | 2.46E+04   | kg          | estimated         | 1    | 3.04     | (3,4,1,1,14,9)  |
|                                         | m |                                    |   | reinforced concrete to disposal |                                                                                                                | waste management       | building demolition | No             | CH       | disposal, building, reinforced concrete, to final disposal      | 8.73E+06   | kg          | estimated         | 1    | 3.04     | (3,4,1,1,14,9)  |
|                                         | m |                                    |   | foundation concrete to disposal |                                                                                                                | waste management       | building demolition | No             | CH       | disposal, building, concrete, not reinforced, to final disposal | 2.20E+05   | kg          | estimated         | 1    | 3.04     | (3,4,1,1,14,9)  |
|                                         | m |                                    |   | aluminium hydroxide plant       | Parts of the aluminium oxyde plant needed for the hydroxide production. Output 1220000000 kg/a, life time 50 a | metals                 | extraction          | Yes            | RER      | aluminium hydroxide, plant                                      | 1.00E+00   | unit        |                   |      |          |                 |

Fig. 5.7 Flows for "aluminium hydroxide, plant" and their representation in the ecoinvent database (cont.)



| General Flow information                                                               |                           |                                |                                                                                | Representation in ecoinvent |                 |                |          |                                                       |            |      | Uncertainty                    |      |          |                                             |
|----------------------------------------------------------------------------------------|---------------------------|--------------------------------|--------------------------------------------------------------------------------|-----------------------------|-----------------|----------------|----------|-------------------------------------------------------|------------|------|--------------------------------|------|----------|---------------------------------------------|
| Input                                                                                  | Process Name              | Output                         | Remarks                                                                        | Category                    | Sub category    | Infrastructure | Location | Module name in ecoinvent                              | Mean value | Unit | Source mean value              | Type | StdV 95% | General Comment                             |
| Aluminiumhydroxide (Al(OH) <sub>3</sub> )                                              | aluminium oxide, at plant |                                |                                                                                | chemicals                   | inorganics      | No             | RER      | aluminium hydroxide, at plant                         | 1.53E+00   | kg   | calculated                     | 1    | 1.08     | (1,1,2,1,1,3,3)                             |
| General heat from natural gas                                                          |                           |                                | Includes all heat from natural gas consumed (64 % of total alumina production) | natural gas                 | heating systems | No             | RER      | heat, natural gas, at industrial furnace >100 kW      | 5.07E-01   | MJ   | EAA 2001 (pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,1)                             |
| General heat from oil                                                                  |                           |                                | Includes all heat from oil consumed (35 % of total alumina production)         | oil                         | heating systems | No             | CH       | heat, light fuel oil, at industrial furnace 1M W      | 1.94E+00   | MJ   | EAA 2001 (pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,1)                             |
| Electrical power used in plant (process, lighting, office work, room cooling...) [kWh] |                           |                                | Includes all power consumed (7 % of total alumina production)                  | electricity                 | production mix  | No             | UCTE     | electricity, medium voltage, production UCTE, at grid | 1.05E-02   | kWh  | EAA 2001 (pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,2)                             |
| Infrastructure (plant) [piece]                                                         |                           |                                | Includes land use                                                              | metals                      | extraction      | Yes            | RER      | aluminium oxide, plant                                | 2.50E-11   | unit | estimated                      | 1    | 3.23     | (5,n.A.,n.A.,n.A.,n.A.,n.A.,9); rough guess |
|                                                                                        |                           |                                |                                                                                |                             |                 |                |          |                                                       |            |      |                                |      |          |                                             |
|                                                                                        |                           | waste heat                     | from electricity use                                                           | air                         | unspecified     |                |          | Heat, waste                                           | 3.79E-02   | MJ   | calculated                     | 1    | 1.08     | (1,1,2,1,1,3,13)                            |
|                                                                                        |                           | Steam / Vapour                 | Steam (0.530 kg / kg Al <sub>2</sub> O <sub>3</sub> ) not inventoried          |                             |                 |                |          |                                                       |            |      |                                |      |          |                                             |
|                                                                                        |                           | Aluminium oxide, at plant [kg] |                                                                                | chemicals                   | inorganics      | No             | RER      | aluminium oxide, at plant                             | 1.00E+00   | kg   |                                |      |          |                                             |

Fig. 5.8 Flows for "aluminium oxide, at plant" and their representation in the ecoinvent database

| General Flow information                                       |                        |                                 |                                                                                                            | Representation in ecoinvent |                     |                |          |                                                            |            |      |                   | Uncertainty |          |                 |  |
|----------------------------------------------------------------|------------------------|---------------------------------|------------------------------------------------------------------------------------------------------------|-----------------------------|---------------------|----------------|----------|------------------------------------------------------------|------------|------|-------------------|-------------|----------|-----------------|--|
| Input                                                          | Process Name           | Output                          | Remarks                                                                                                    | Category                    | Sub category        | Infrastructure | Location | Module name in ecoinvent                                   | Mean value | Unit | Source mean value | Type        | StdV 95% | General Comment |  |
| Total land use (transformation)                                | aluminium oxide, plant |                                 |                                                                                                            | resource                    | land                |                |          | Transformation, from unknown                               | 2.75E+04   | m2   | estimated         | 1           | 2.03     | (1,4,1,1,1,4,8) |  |
| Land occupation for buildings and open bauxite storage area    |                        |                                 | assumption: 80% of total area is built up                                                                  | resource                    | land                |                |          | Occupation, industrial area, built up                      | 1.10E+06   | m2a  | estimated         | 1           | 1.54     | (2,4,1,1,1,4,7) |  |
| Land transformation to buildings and open bauxite storage area |                        |                                 |                                                                                                            | resource                    | land                |                |          | Transformation, to industrial area, built up               | 2.20E+04   | m2   | estimated         | 1           | 2.03     | (2,4,1,1,1,4,8) |  |
| Land occupation for vegetation area                            |                        |                                 | assumption: 20% of total area is vegetation                                                                | resource                    | land                |                |          | Occupation, industrial area, vegetation                    | 2.75E+05   | m2a  | estimated         | 1           | 1.54     | (2,4,1,1,1,4,7) |  |
| Land transformation to vegetation area                         |                        |                                 |                                                                                                            | resource                    | land                |                |          | Transformation, to industrial area, vegetation             | 5.50E+03   | m2   | estimated         | 1           | 2.03     | (2,4,1,1,1,4,8) |  |
| concrete in reinforced concrete                                |                        |                                 |                                                                                                            | construction materials      | concrete            | No             | CH       | concrete, normal, at plant                                 | 1.42E+03   | m3   | estimated         | 1           | 3.04     | (3,4,1,1,1,4,9) |  |
| steel in reinforced concrete                                   |                        |                                 |                                                                                                            | metals                      | extraction          | No             | RER      | reinforcing steel, at plant                                | 9.80E+05   | kg   | estimated         | 1           | 3.04     | (3,4,1,1,1,4,9) |  |
| Conveyor belt                                                  |                        |                                 |                                                                                                            | construction processes      | machinery           | Yes            | RER      | conveyor belt, at plant                                    | 2.00E+02   | m    | estimated         | 1           | 3.04     | (3,4,1,1,1,4,9) |  |
| Chromium steel (stainless)                                     |                        |                                 |                                                                                                            | metals                      | extraction          | No             | RER      | chromium steel 18/8, at plant                              | 4.00E+04   | kg   | estimated         | 1           | 3.04     | (3,4,1,1,1,4,9) |  |
| Steel low alloyed                                              |                        |                                 |                                                                                                            | metals                      | extraction          | No             | RER      | steel, low-alloyed, at plant                               | 6.15E+04   | kg   | estimated         | 1           | 3.04     | (3,4,1,1,1,4,9) |  |
|                                                                |                        | reinforced concrete to disposal |                                                                                                            | waste management            | building demolition | No             | CH       | disposal, building, reinforced concrete, to final disposal | 4.09E+06   | kg   | estimated         | 1           | 3.04     | (3,4,1,1,1,4,9) |  |
|                                                                |                        | aluminium oxide plant           | Only parts of the plant that are not needed for aluminium hydroxide. Output 800000000 kg/a, life time 50 a | metals                      | extraction          | Yes            | RER      | aluminium oxide, plant                                     | 1.00E+00   | unit |                   |             |          |                 |  |

Fig. 5.9 Flows for "aluminium oxide, plant" and their representation in the ecoinvent database

## 5.3 Primary aluminium

This part is mainly based on (Hudson et al. (1997)) and (Mori & Adelhardt (1998)).

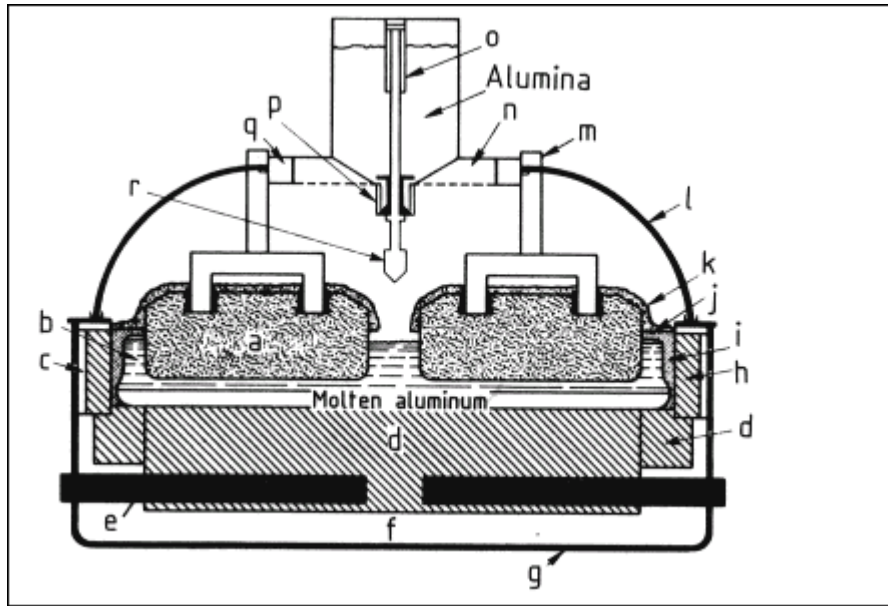
### 5.3.1 Production process and infrastructure

All commercial production of aluminium today is done in Hall-Héroult cells. Two major types of cells are currently in use: those employing prebaked carbon anodes and those employing self-baking Söderberg anodes. The Hall-Héroult cell with prebaked anodes is shown in Fig. 5.10. Essentially pure alumina is fed into the cryolite base electrolyte. Electric current deposits aluminium into a pool of molten aluminium held under the electrolyte in the carbon lined cavity of the cell. Oxygen from the alumina deposits electrolytically onto the carbon anode dipping into the electrolyte and reacts with (burns) the anode. Cells typically range from 9 to 12 m long, 3 to 4 m wide, and 1 to 1.2 m high. Thermal insulation surrounds the carbon lining of the cell to control heat losses. Although carbon is the material known to withstand best the combined corrosive action of molten fluorides and molten aluminium, even carbon would have a very limited life in contact with the electrolyte at the sides of the cell were it is not protected by a layer of frozen electrolyte. The thermal insulation is adjusted carefully to maintain a protective coating on the walls but not on the bottom, which must remain substantially bare for electrical contact. Steel collector bars in the carbon cathode conduct electric current from the cell. These bars are inserted into holes that have been sized carefully so that thermal expansion forms a tight electrical contact, or cemented in place with carbonaceous cement containing metal particles, or bonded in place with cast iron.

The electrical resistivity of prebaked anodes ranges from 0.005 to 0.006  $\Omega \cdot \text{cm}$ . Current density at the anode face is 0.6 – 1.3  $\text{A}/\text{cm}^2$ .

The Söderberg anode uses a premixed "paste" of petroleum coke and coal-tar pitch. This mixture is added at the top of a rectangular steel casing that is typically 6 – 8 m long, 2 m wide, and 1 m high. Heat from the electrolyte and heat from the electric current passing through the anode bakes the carbonaceous mix as it progresses through the casing.

The baked portion extends past the casing and into the molten electrolyte. Baked mix replaces anode being consumed at the bottom surface. Electric current enters the anode through either vertical or sloping steel spikes. These spikes are pulled and reset to a higher level as they approach the lower surface. Söderberg anodes have an electrical resistivity about 30 % higher than that of prebaked anodes. Because of the resulting lower power efficiency and the greater difficulty in collecting and disposing of baking fumes, Söderberg anodes are being replaced by prebaked anodes, even though the former save the capital, labour, and energy required to manufacture the latter.



**Fig. 5.10 Hall-Héroult cell with prebaked anode:** a) carbon anode b) electrolyte, c) insulation, d) carbon lining, e) current collector bar, f) thermal insulation, g) steel shell, h) carbon block, i) ledge, j) crust, k) aluminium oxide cover, l) removable covers, m) anode rods, n) fume collection, o) air cylinder, p) feeder, q) current supply, r) crust breaker

Metal tapped from the Hall-Héroult cell may be treated in one of several ways. First, it may be transported in the molten state to a holding furnace, where, along with solid scrap, it is alloyed, fluxed, filtered, and cast into a form suitable for fabrication by rolling, forging, etc., or for remelting by a foundry. It may be transported further (over distances up to 200 km) on trucks, equipped with well-insulated containers to prevent freezing, to a foundry where it is alloyed and used to make shaped castings. Both these methods of utilizing the molten metal are favoured because energy required for remelting is conserved.

Alternatively, where ingot casting facilities or foundries are not nearby, metal is cast using cast iron moulds into "primary" ingots for remelting. Generally, these unalloyed remelt ingots come as stackable 22-, 340-, or 680-kg ingots. Casting machines are often fully automated, with production rates of up to 20000 kg/h.

Infrastructure is estimated based on information of an average sized electrolysis plant with cast house in Norway. Production capacity is about 130'000 t/year. About 3'000 tons is secondary material (for alloying) that can be neglected for our purpose. The aluminium is cast in 7 meter long billets with a diameter that varies between 178 and 314 mm. Later the billets are sawed, packed and driven to the quay for shipment. The most important data about the plant are given in Tab. 5.6.

**Tab. 5.6 Infrastructure data for a medium sized electrolysis plant with cast house (Soral)<sup>10</sup>**

|                      |                             |
|----------------------|-----------------------------|
| Production Capacity  | 130.000 tons                |
| Area:                | 976,000 m <sup>2</sup>      |
| Length -Pot-rooms    | 857 m                       |
| Breadth- Pot-rooms   | 23 m                        |
| Electrolysis pots    | 320                         |
| Cast -house          | 12.200 m <sup>2</sup>       |
| Height- Silos        | 55 m and 62 m               |
| Capacity-Silos       | 30.000 tons and 40.000 tons |
| Length-Conveyor Belt | 900 m                       |
| Length-Quay          | 240 m                       |
| Capacity-Quay        | 70.000 tdw                  |
| Storage house        | 9500 m <sup>2</sup>         |

### 5.3.2 Emissions

#### Air

Relevant direct air emissions from the anode production (baking) and from the electrolysis itself are CO, CO<sub>2</sub>, fluorides, dust, polycyclic aromatic hydrocarbons, Nitrogen oxides and sulphur oxides. The casting is responsible for some direct fluoride emissions. The other processes don't emit to air directly.

#### Water

There are no relevant direct water emissions from these processes

### 5.3.3 Waste

Several different waste streams are generated. Some of them are recycled or reused and thus not inventoried. The others are deposited in landfills. (See Fig. 5.11, Fig. 5.12, Fig. 5.13 and Fig. 5.14)

#### Anode production waste

In the anode production carbon and tar wastes are produced. The compositions are unknown. 100% carbon is assumed for tar. A certain amount of waste might be recycled. However, since no data on the share to recycling is available, disposal in a residual material landfill is assumed.

The "carbon waste" refers to the waste of the anode butt recycling. Thus it is inventoried the same way as the carbon fraction of spent pot lining.

#### Cathode production waste

Because the cathode production includes the disposal of the old cathode, the waste of this process is basically the spent pot lining. It is separated into a carbon fraction (1<sup>st</sup> cut) and refractory (brick) fraction (2<sup>nd</sup> cut). A certain amount of these wastes might be recycled. However, since no data on the share to recycling is available, both wastes are assumed to be landfilled in a residual material landfill. The ranges for their chemical composition that were presented by R. P. Pawlek at the 10<sup>th</sup> international Al symposium in 1999<sup>11</sup> and the values used in this project are given in Tab. 5.7.

#### Electrolysis waste

The carbon waste of the electrolysis is basically the waste fraction from used anodes that can't be recycled. It is inventoried as the carbon fraction of SPL.

<sup>10</sup> <http://www.soral.no/en/produksjon.html>, accessed 16.12.2002

<sup>11</sup> <http://www.aluminium.sk/prednasky/32.htm>, accessed 19.12.2002

### Aluminium casting waste

The fibre and refractory wastes of the casting process are inventoried as inert material to landfill. The dross composition given in

Tab. 5.8 stems from the environment agency of the UK<sup>12</sup>. Disposal in a residual material landfill is assumed.

Data for the filter dust composition are given in Tab. 5.9. Disposal in a residual material landfill is assumed.

**Tab. 5.7 Composition of spent pot liner**

| Element or compound<br>(Metals mostly as oxides) | unit | Carbon               |              | Refractory           |              |
|--------------------------------------------------|------|----------------------|--------------|----------------------|--------------|
|                                                  |      | Pawlek <sup>11</sup> | This project | Pawlek <sup>11</sup> | This project |
| SiO <sub>2</sub>                                 | %    | 0 – 6                | 2.96         | 10 – 50              | 36.94        |
| Al <sub>2</sub> O <sub>3</sub>                   | %    | 0 – 10               | 4.94         | 10 – 50              | 36.94        |
| metal Al                                         | %    | 0 – 5                | 2.47         | 0                    | 0.00         |
| Fe <sub>2</sub> O <sub>3</sub>                   | %    |                      | 0.00         | 0 – 3                | 1.80         |
| CaO                                              | %    | 1 – 6                | 3.46         | 1 – 8                | 5.49         |
| K <sub>2</sub> O                                 | %    | <0.5                 | 0.49         | <0.5                 | 0.30         |
| MnO                                              | %    | 0 – 0.02             | 0.01         | 0 – 0.02             | 0.01         |
| TiO <sub>2</sub>                                 | %    |                      | 0.00         | 0 – 0.1              | 0.06         |
| P <sub>2</sub> O <sub>5</sub>                    | g/t  | 0 – 650              | 0.03         | 0 – 300              | 0.02         |
| Na                                               | %    | 8 – 12               | 9.88         | 6 – 10               | 9.79         |
| F                                                | %    | 6 – 10               | 7.90         | 4 – 10               | 8.59         |
| Cl                                               | %    | <0.06                | 0.06         | <0.06                | 0.04         |
| CN free                                          | g/t  | 0 – 2,000            | 0.10         | 0 – 500              | 0.00         |
| CN total                                         | g/t  | 0 – 5,000            | 2.47         | 0 – 1,000            | 0.06         |
| Carbon                                           | %    | -                    | 66.46        | 0                    | 0.00         |

**Tab. 5.8 Composition of dross<sup>12</sup>**

| Compound          | Unit    | Literature <sup>12</sup> | This project |
|-------------------|---------|--------------------------|--------------|
| Aluminium         | % (w/w) | 8-50                     | 30           |
| Aluminium oxide   | % (w/w) | 30-90                    | 61           |
| Aluminium nitride | % (w/w) | 0-10                     | 6            |
| Aluminium carbide | % (w/w) | 0-5                      | 3            |

<sup>12</sup> [http://www.environment-agency.gov.uk/commondata/105385/swen042.pdf?lang=\\_e](http://www.environment-agency.gov.uk/commondata/105385/swen042.pdf?lang=_e), accessed 19.12.2002

Tab. 5.9 Composition of filter dust<sup>13</sup>

| Element | Amount [% w/w] | Element | Amount [% w/w] |
|---------|----------------|---------|----------------|
| Al      | 33.820         | Cu      | 0.057          |
| Na      | 6.308          | Ni      | 0.051          |
| F       | 9.051          | Cr      | 0.005          |
| C       | 11.586         | Pb      | 0.039          |
| Fe      | 0.545          | Zn      | 0.003          |
| Si      | 0.046          | V       | 0.048          |
| Ti      | 0.039          | P       | 0.058          |
| Mg      | 0.090          | O       | 38.224         |
| Ga      | 0.030          |         |                |

### 5.3.4 Primary aluminium in ecoinvent

The available information about the infrastructure needed for aluminium production is non-quantitative. Tab. 5.11 and Tab. 5.12 show the details of the assumptions made to estimate the infrastructure expenditures.

Tab. 5.10 Estimated infrastructure for anode plant

| anode, plant                                       | Demand     | Unit           | Life time | Life time demand |
|----------------------------------------------------|------------|----------------|-----------|------------------|
| Yearly production capacity                         | 80'000'000 | kg             | 50        | 80'000'000       |
| Area of plant (assumption based on casting plant)  | 300'308    | m <sup>2</sup> | 50        | 300'308          |
| Administration building (50x20x10 m <sup>3</sup> ) | 10'000     | m <sup>3</sup> | 50        | 10'000           |
| Production hall                                    | 7'508      | m <sup>2</sup> | 50        | 7'508            |
| Production machines                                | 615'385    | kg             | 25        | 1'230'769        |
| Storage hall                                       | 5'846      | m <sup>2</sup> | 50        | 5'846            |

Tab. 5.11 Estimated infrastructure for aluminium electrolysis plant

| aluminium electrolysis, plant                         | Demand      | Unit           | Life time | Life time demand |
|-------------------------------------------------------|-------------|----------------|-----------|------------------|
| Yearly production capacity                            | 130'000'000 | kg             | 50        | 130'000'000      |
| Area of plant (50% of total, rest casting plant)      | 488'000     | m <sup>2</sup> | 50        | 488'000          |
| Administration building (50x50x10 m <sup>3</sup> )    | 25'000      | m <sup>3</sup> | 50        | 25'000           |
| Concrete for piers [m <sup>3</sup> ]                  | 1'200       | m <sup>3</sup> | 50        | 1'200            |
| Grab for unloading [kg]                               | 100'000     | kg             | 25        | 200'000          |
| conveyor to storage [m]                               | 900         | m              | 25        | 1'800            |
| Silo (30000 m <sup>3</sup> , height 55 m, radius 13m) | 194'402     | kg             | 25        | 388'804          |
| Silo (40000 m <sup>3</sup> , height 62 m, radius 14m) | 233'986     | kg             | 25        | 467'972          |
| Area of pot rooms (halls)                             | 19'711      | m <sup>2</sup> | 50        | 19'711           |
| Steel shell for 320 pots (10 by 3.5 by 1.1 m)         | 1'116'640   | kg             | 25        | 2'233'280        |

<sup>13</sup> [http://sperber.lrh.rwth-aachen.de/sfb/subpro/sub\\_pro2/entsorg/filt1\\_de.htm](http://sperber.lrh.rwth-aachen.de/sfb/subpro/sub_pro2/entsorg/filt1_de.htm), accessed 19.12.2002

Tab. 5.12 Estimated infrastructure for aluminium casting plant

| aluminium casting, plant                              | Demand      | Unit | Life time | Life time demand |
|-------------------------------------------------------|-------------|------|-----------|------------------|
| Yearly production capacity                            | 130'000'000 | kg   | 50        | 130'000'000      |
| Area of plant (50% of total, rest electrolysis plant) | 488'000     | m2   | 50        | 488'000          |
| Administration building (20x10x10 m3)                 | 2'000       | m3   | 50        | 2'000            |
| Cast house                                            | 12'200      | m2   | 50        | 12'200           |
| Casting machines                                      | 1'000'000   | kg   | 25        | 2'000'000        |
| Storage house                                         | 9'500       | m2   | 50        | 9'500            |

The other background information to the processes is given in Tab. 5.13, flows and their representation in the database is shown in:

- anode, aluminium electrolysis Fig. 5.11
- anode, plant Fig. 5.12
- cathode, aluminium electrolysis Fig. 5.13
- aluminium, primary, liquid, at plant Fig. 5.14
- aluminium electrolysis, plant Fig. 5.15
- aluminium, primary, at plant Fig. 5.16
- aluminium casting, plant Fig. 5.17

### Data quality

Data for the steps in the primary aluminium production are of good overall quality. The infrastructure is roughly estimated but compared to the other activities the influence is negligible. The values for the processes stem from a recent and comprehensive source (EAA (2000)).



Tab. 5.13 Meta information for the aluminium electrolysis and casting processes

| Name                                  | anode, aluminium electrolysis                                                                                    | anode, plant                                                                                                | cathode, aluminium electrolysis                                                                                                                                                                                                                                                                                                              |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location                              | RER                                                                                                              | RER                                                                                                         | RER                                                                                                                                                                                                                                                                                                                                          |
| Infrastructure Process                | 0                                                                                                                | 1                                                                                                           | 0                                                                                                                                                                                                                                                                                                                                            |
| Unit                                  | kg                                                                                                               | unit                                                                                                        | kg                                                                                                                                                                                                                                                                                                                                           |
| Data Set Version                      | 2.0                                                                                                              | 2.0                                                                                                         | 2.0                                                                                                                                                                                                                                                                                                                                          |
| Included Processes                    | Includes anode production (with its plant), transports of materials to the plant and the disposal of the wastes. | Production hall, machinery and administrative buildings (including disposal)                                | Includes cathode production and removal, transports of materials to the plant and the disposal of the wastes. Because the cathode is produced in the electrolysis plant, no cathode production plant is balanced. No data have been found on air emissions from baking. It is assumed that these emissions are included in the electrolysis. |
| Amount                                | 1                                                                                                                | 1                                                                                                           | 1                                                                                                                                                                                                                                                                                                                                            |
| Local Name                            | Anode, Aluminium Elektrolyse                                                                                     | Anode, Fabrik                                                                                               | Kathode, Aluminium Elektrolyse                                                                                                                                                                                                                                                                                                               |
| Synonyms                              |                                                                                                                  |                                                                                                             |                                                                                                                                                                                                                                                                                                                                              |
| General Comment to reference function |                                                                                                                  | Medium sized plant. Output: 80'000 t/a, life time: 50 years. Assumption is based on data for casting plant. |                                                                                                                                                                                                                                                                                                                                              |
| Start Date                            | 1995                                                                                                             | 2002                                                                                                        | 1995                                                                                                                                                                                                                                                                                                                                         |
| End Date                              | 2002                                                                                                             | 2002                                                                                                        | 2002                                                                                                                                                                                                                                                                                                                                         |
| Data Valid For Entire Period          | 1                                                                                                                | 1                                                                                                           | 1                                                                                                                                                                                                                                                                                                                                            |
| Geography text                        | Some Swiss datasets are used for European processes.                                                             | Some Swiss datasets are used for European processes.                                                        | Some Swiss datasets are used for European processes.                                                                                                                                                                                                                                                                                         |
| Technology text                       | Average technology for the aluminium consumed in Europe (85% prebaked, 15% Söderberg)                            | Assumed technology                                                                                          | Average technology for the aluminium consumed in Europe (85% prebaked, 15% Söderberg)                                                                                                                                                                                                                                                        |
| Representativeness [%]                | 98                                                                                                               |                                                                                                             | 98                                                                                                                                                                                                                                                                                                                                           |
| Production Volume                     | unknown                                                                                                          | unknown                                                                                                     | unknown                                                                                                                                                                                                                                                                                                                                      |
| Sampling Procedure                    | Literature / Survey                                                                                              | Internet search                                                                                             | Literature / Survey                                                                                                                                                                                                                                                                                                                          |
| Extrapolations                        | see Geography                                                                                                    | see Geography                                                                                               | see Geography                                                                                                                                                                                                                                                                                                                                |
| Uncertainty Adjustments               | Uncertainty is adjusted for geographical mismatch (using the pedigree matrix).                                   | Uncertainty is adjusted for geographical mismatch (using the pedigree matrix).                              | Uncertainty is adjusted for geographical mismatch (using the pedigree matrix).                                                                                                                                                                                                                                                               |

Tab. 5.13 Meta information for the aluminium electrolysis and casting processes (cont.)

| Name                                  | aluminium, primary, liquid, at plant                                                                                        | aluminium electrolysis, plant                                                                                                                                     | aluminium, primary, at plant                                                                                                    | aluminium casting, plant                                                                                                                                               |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location                              | RER                                                                                                                         | RER                                                                                                                                                               | RER                                                                                                                             | RER                                                                                                                                                                    |
| Infrastructure Process                | 0                                                                                                                           | 1                                                                                                                                                                 | 0                                                                                                                               | 1                                                                                                                                                                      |
| Unit                                  | kg                                                                                                                          | unit                                                                                                                                                              | kg                                                                                                                              | unit                                                                                                                                                                   |
| Data Set Version                      | 2.0                                                                                                                         | 2.0                                                                                                                                                               | 2.0                                                                                                                             | 2.0                                                                                                                                                                    |
| Included Processes                    | Includes liquid aluminium production (with its plant), transports of materials to the plant and the disposal of the wastes. | Port, production hall, machinery and administrative buildings (including disposal)                                                                                | Includes cast aluminium ingot production (with its plant), transports of materials to the plant and the disposal of the wastes. | Production hall, machinery and administrative buildings (including disposal)                                                                                           |
| Amount                                | 1                                                                                                                           | 1                                                                                                                                                                 | 1                                                                                                                               | 1                                                                                                                                                                      |
| Local Name                            | Aluminium, primär, flüssig, ab Werk                                                                                         | Aluminiumelektrolyse, Werk                                                                                                                                        | Aluminium, primär, ab Werk                                                                                                      | Aluminium giessen, Werk                                                                                                                                                |
| Synonyms                              |                                                                                                                             |                                                                                                                                                                   |                                                                                                                                 |                                                                                                                                                                        |
| General Comment to reference function |                                                                                                                             | Medium sized plant. Output: 130'000 t/a, life time: 50 years. Site is shared with casting plant. Total land use is allocated 50% to electrolysis, 50% to casting. |                                                                                                                                 | Medium sized plant. Output: 130'000 t/a, life time: 50 years. Site is shared with electrolysis plant. Total land use is allocated 50% to electrolysis, 50% to casting. |
| Start Date                            | 1995                                                                                                                        | 2002                                                                                                                                                              | 1995                                                                                                                            | 2002                                                                                                                                                                   |
| End Date                              | 2002                                                                                                                        | 2002                                                                                                                                                              | 2002                                                                                                                            | 2002                                                                                                                                                                   |
| Data Valid For Entire Period          | 1                                                                                                                           | 1                                                                                                                                                                 | 1                                                                                                                               | 1                                                                                                                                                                      |
| Geography text                        | Some Swiss datasets are used for European processes.                                                                        | Some Swiss datasets are used for European processes.                                                                                                              | Some Swiss datasets are used for European processes.                                                                            | Some Swiss datasets are used for European processes.                                                                                                                   |
| Technology text                       | Average technology for the aluminium consumed in Europe                                                                     | Technology of one medium sized plant in Norway                                                                                                                    | Average technology for the aluminium consumed in Europe                                                                         | Technology of one medium sized plant in Norway                                                                                                                         |
| Representativeness [%]                | 98                                                                                                                          |                                                                                                                                                                   | 98                                                                                                                              |                                                                                                                                                                        |
| Production Volume                     | unknown                                                                                                                     | unknown                                                                                                                                                           | unknown                                                                                                                         | unknown                                                                                                                                                                |
| Sampling Procedure                    | Literature / Survey                                                                                                         | Internet search                                                                                                                                                   | Literature / Survey                                                                                                             | Internet search                                                                                                                                                        |
| Extrapolations                        | see Geography                                                                                                               | see Geography                                                                                                                                                     | see Geography                                                                                                                   | see Geography                                                                                                                                                          |
| Uncertainty Adjustments               | Uncertainty is adjusted for geographical mismatch (using the pedigree matrix).                                              | Uncertainty is adjusted for geographical mismatch (using the pedigree matrix).                                                                                    | Uncertainty is adjusted for geographical mismatch (using the pedigree matrix).                                                  | Uncertainty is adjusted for geographical mismatch (using the pedigree matrix).                                                                                         |

| General Flow information                      |   |                                       |  | Representation in ecoinvent                      |                        |            |              |                |                                        |                             |            | Uncertainty                                                |                                                            |      |                             |                 |
|-----------------------------------------------|---|---------------------------------------|--|--------------------------------------------------|------------------------|------------|--------------|----------------|----------------------------------------|-----------------------------|------------|------------------------------------------------------------|------------------------------------------------------------|------|-----------------------------|-----------------|
| Input                                         |   | Process Name                          |  | Output                                           | Remarks                | Category   | Sub category | Infrastructure | Location                               | Module name in ecoinvent    | Mean value | Unit                                                       | Source mean value                                          | Type | StdV 95%                    | General Comment |
| Petrol coke [kg]                              | ⌚ | anode, aluminium electrolysis; part 1 |  |                                                  |                        | oil        | fuels        | No             | RER                                    | petroleum coke, at refinery | 6.49E-01   | kg                                                         | EAA 2000 (p. 49) / personal communication B. de Gelas, EAA | 1    | 1.08                        | (1,1,2,1,1,3,4) |
| Transport of petrol coke to anode plant [tkm] | ⌚ |                                       |  | Transport by road                                | transport systems      | road       | No           | RER            | transport, lorry 32t                   | 6.49E-02                    | tkm        | standard distance                                          | 1                                                          | 2.09 | (4,5,n.A.,n.A.,n.A.,n.A.,5) |                 |
| Transport of petrol coke to anode plant [tkm] | ⌚ |                                       |  | Transport by rail                                | transport systems      | train      | No           | RER            | transport, freight, rail               | 1.30E-01                    | tkm        | standard distance                                          | 1                                                          | 2.09 | (4,5,n.A.,n.A.,n.A.,n.A.,5) |                 |
| Pitch [kg]                                    | ⌚ |                                       |  | inventoried as bitumen                           | oil                    | fuels      | No           | RER            | bitumen, at refinery                   | 1.64E-01                    | kg         | EAA 2000 (p. 49) / personal communication B. de Gelas, EAA | 1                                                          | 1.08 | (1,1,2,1,1,3,4)             |                 |
| Transport of pitch to anode plant [tkm]       | ⌚ |                                       |  | Transport by road                                | transport systems      | road       | No           | RER            | transport, lorry 32t                   | 1.64E-02                    | tkm        | standard distance                                          | 1                                                          | 2.09 | (4,5,n.A.,n.A.,n.A.,n.A.,5) |                 |
| Transport of pitch to anode plant [tkm]       | ⌚ |                                       |  | Transport by rail                                | transport systems      | train      | No           | RER            | transport, freight, rail               | 3.28E-02                    | tkm        | standard distance                                          | 1                                                          | 2.09 | (4,5,n.A.,n.A.,n.A.,n.A.,5) |                 |
| anode butts                                   | ⌚ |                                       |  | recycling of used butts --> not inventoried      |                        |            |              |                |                                        | 1.88E-01                    |            | personal communication B. de Gelas, EAA                    |                                                            |      |                             |                 |
| cast iron for stub [kg]                       | ⌚ |                                       |  | Reportet under "Electrolysis" but used in Anodes | metals                 | extraction | No           | RER            | cast iron, at plant                    | 7.81E-03                    | kg         | EAA 2001(pers. communication)                              | 1                                                          | 1.08 | (1,1,2,1,1,3,4)             |                 |
| Transport of iron to anode plant [tkm]        | ⌚ |                                       |  | Transport by road                                | transport systems      | road       | No           | RER            | transport, lorry 32t                   | 7.81E-04                    | tkm        | standard distance                                          | 1                                                          | 2.09 | (4,5,n.A.,n.A.,n.A.,n.A.,5) |                 |
| Transport of iron to anode plant [tkm]        | ⌚ |                                       |  | Transport by rail                                | transport systems      | train      | No           | RER            | transport, freight, rail               | 1.56E-03                    | tkm        | standard distance                                          | 1                                                          | 2.09 | (4,5,n.A.,n.A.,n.A.,n.A.,5) |                 |
| Refractory materials                          | ⌚ |                                       |  |                                                  | construction materials | bricks     | No           | DE             | refractory, fireclay, packed, at plant | 1.01E-02                    | kg         | EAA 2001(pers. communication)                              | 1                                                          | 1.08 | (1,1,2,1,1,3,4)             |                 |
| Transport of Refractory materials             | ⌚ |                                       |  | Transport by road                                | transport systems      | road       | No           | RER            | transport, lorry 32t                   | 1.01E-03                    | tkm        | standard distance                                          | 1                                                          | 2.09 | (4,5,n.A.,n.A.,n.A.,n.A.,5) |                 |
| Transport of Refractory materials             | ⌚ |                                       |  | Transport by rail                                | transport systems      | train      | No           | RER            | transport, freight, rail               | 2.02E-03                    | tkm        | standard distance                                          | 1                                                          | 2.09 | (4,5,n.A.,n.A.,n.A.,n.A.,5) |                 |

Fig. 5.11 Flows for "anode, aluminium electrolysis" and their representation in the ecoinvent database

| General Flow information                       |                                       |                             |                                             | Representation in ecoinvent |                 |                |          |                                                       |            |      | Uncertainty                                                    |      |          |                                             |
|------------------------------------------------|---------------------------------------|-----------------------------|---------------------------------------------|-----------------------------|-----------------|----------------|----------|-------------------------------------------------------|------------|------|----------------------------------------------------------------|------|----------|---------------------------------------------|
| Input                                          | Process Name                          | Output                      | Remarks                                     | Category                    | Sub category    | Infrastructure | Location | Module name in ecoinvent                              | Mean value | Unit | Source mean value                                              | Type | StdV 95% | General Comment                             |
| General heat from natural gas                  | anode, aluminium electrolysis; part 2 |                             | Calcination of petrol coke, baking of anode | natural gas                 | heating systems | No             | RER      | heat, natural gas, at industrial furnace >100 kW      | 1.90E+00   | MJ   | EAA 2001 (pers. communication)                                 | 1    | 1.08     | (1,1,2,1,1,3,1)                             |
| General heat from oil                          |                                       |                             | Calcination of petrol coke, baking of anode | oil                         | heating systems | No             | CH       | heat, light fuel oil, at industrial furnace 1M W      | 9.35E-01   | MJ   | EAA 2001 (pers. communication)                                 | 1    | 1.08     | (1,1,2,1,1,3,1)                             |
| Cooling water [m3]                             |                                       |                             |                                             | resource                    | in water        |                |          | Water, cooling, unspecified natural origin            | 4.31E-03   | m3   | EAA 2001 (pers. communication)                                 | 1    | 1.08     | (1,1,2,1,1,3,4)                             |
| Electrical power used in plant (process) [kWh] |                                       |                             | generic data from Ecoinvent 2000            | electricity                 | production mix  | No             | UCTE     | electricity, medium voltage, production UCTE, at grid | 1.31E-01   | kWh  | EAA 2001 (pers. communication)                                 | 1    | 1.08     | (1,1,2,1,1,3,2)                             |
| Infrastructure (plant) [piece]                 |                                       |                             |                                             | metals                      | extraction      | Yes            | RER      | anode, plant                                          | 2.50E-10   | unit | estimated                                                      | 1    | 3.23     | (5,n.A.,n.A.,n.A.,n.A.,n.A.,9); rough guess |
|                                                |                                       | waste heat                  | from electricity use                        | air                         | unspecified     |                |          | Heat, waste                                           | 4.72E-01   | MJ   | calculated                                                     | 1    | 1.08     | (1,1,2,1,1,3,13)                            |
|                                                |                                       | BaP (Benzo-a-Pyrene) to air |                                             | air                         | unspecified     |                |          | Benzo(a)pyrene                                        | 3.18E-06   | kg   | EAA 2001 (pers. communication)                                 | 1    | 3.00     | (1,1,2,1,1,3,21)                            |
|                                                |                                       | CO to air                   |                                             | air                         | unspecified     |                |          | Carbon monoxide, fossil                               | 1.04E-03   | kg   | EAA 2001 (pers. communication)                                 | 1    | 5.01     | (1,1,2,1,1,3,17)                            |
|                                                |                                       | CO2 to air                  |                                             | air                         | unspecified     |                |          | Carbon dioxide, fossil                                | 2.53E-01   | kg   | EAA 2001 (pers. communication)                                 | 1    | 1.08     | (1,1,2,1,1,3,24)                            |
|                                                |                                       | Dust to air                 | 4.2% > 10 um                                | air                         | unspecified     |                |          | Particulates, > 10 um                                 | 1.30E-04   | kg   | total EAA 2001 (pers. communication), fraction www.iiasa.ac.at | 1    | 1.51     | (1,1,2,1,1,3,25)                            |
|                                                |                                       | Dust to air                 | 10 um > 30% > 2.5 um                        | air                         | unspecified     |                |          | Particulates, > 2.5 um, and < 10 um                   | 9.30E-05   | kg   | total EAA 2001 (pers. communication), fraction www.iiasa.ac.at | 1    | 2.00     | (1,1,2,1,1,3,26)                            |
|                                                |                                       | Dust to air                 | 28% < 2.5 um                                | air                         | unspecified     |                |          | Particulates, < 2.5 um                                | 8.68E-05   | kg   | total EAA 2001 (pers. communication), fraction www.iiasa.ac.at | 1    | 3.00     | (1,1,2,1,1,3,27)                            |

Fig. 5.11 Flows for "anode, aluminium electrolysis" and their representation in the ecoinvent database (cont.)

| General Flow information              |                                       |                                     |                  | Representation in ecoinvent |              |                |          |                                                                                    |            |      | Uncertainty                    |      |          |                  |
|---------------------------------------|---------------------------------------|-------------------------------------|------------------|-----------------------------|--------------|----------------|----------|------------------------------------------------------------------------------------|------------|------|--------------------------------|------|----------|------------------|
| Input                                 | Process Name                          | Output                              | Remarks          | Category                    | Sub category | Infrastructure | Location | Module name in ecoinvent                                                           | Mean value | Unit | Source mean value              | Type | StdV 95% | General Comment  |
| anode, aluminium electrolysis; part 3 | fluoride (gaseous) to air             | inventoried as HF                   | air              | unspecified                 |              |                |          | Hydrogen fluoride                                                                  | 9.00E-05   | kg   | EAA 2001 (pers. communication) | 1    | 1.51     | (1,1,2,1,1,3,31) |
|                                       | fluoride (non gaseous) to air         | inventoried as particulate emission | air              | unspecified                 |              |                |          | Particulates, > 2.5 um, and < 10 um                                                | 5.00E-05   | kg   | EAA 2001 (pers. communication) | 1    | 2.00     | (1,1,2,1,1,3,26) |
|                                       | polycyclic aromatic HC to air         | without BaP                         | air              | unspecified                 |              |                |          | PAH, polycyclic aromatic hydrocarbons                                              | 9.48E-05   | kg   | EAA 2001 (pers. communication) | 1    | 3.00     | (1,1,2,1,1,3,21) |
|                                       | NOx to air                            |                                     | air              | unspecified                 |              |                |          | Nitrogen oxides                                                                    | 1.90E-04   | kg   | EAA 2001 (pers. communication) | 1    | 1.51     | (1,1,2,1,1,3,31) |
|                                       | SO2 to air                            |                                     | air              | unspecified                 |              |                |          | Sulfur dioxide                                                                     | 8.40E-04   | kg   | EAA 2001 (pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,15) |
|                                       | carbon material for recycling         | not inventoried                     |                  |                             |              |                |          |                                                                                    | 2.65E-02   | kg   | EAA 2001 (pers. communication) |      |          |                  |
|                                       | carbon waste                          | to landfill, non-hazardous          | waste management | residual material landfill  | No           | CH             |          | disposal, refractory SPL, Al electrolysis, 0% water, to residual material landfill | 3.90E-03   | kg   | EAA 2001 (pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,6)  |
|                                       | refractory waste                      | to landfill, non-hazardous          | waste management | inert material landfill     | No           | CH             |          | disposal, inert waste, 5% water, to inert material landfill                        | 1.05E-02   | kg   | EAA 2001 (pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,6)  |
|                                       | tar waste                             | to landfill, non-hazardous          | waste management | sanitary landfill           | No           | CH             |          | disposal, asphalt, 0.1% water, to sanitary landfill                                | 7.00E-04   | kg   | EAA 2001 (pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,6)  |
|                                       | Water / steam                         | not inventoried                     |                  |                             |              |                |          |                                                                                    |            |      |                                |      |          |                  |
|                                       | Anode for Aluminium electrolysis [kg] |                                     | metals           | extraction                  | No           | RER            |          | anode, aluminium electrolysis                                                      | 1.00E+00   | kg   |                                |      |          |                  |

Fig. 5.11 Flows for "anode, aluminium electrolysis" and their representation in the ecoinvent database (cont.)

| General Flow information                                       |              |                                        |                                            | Representation in ecoinvent |                     |                        |              |                                                                 |               |      |                      | Uncertainty |             |                                |
|----------------------------------------------------------------|--------------|----------------------------------------|--------------------------------------------|-----------------------------|---------------------|------------------------|--------------|-----------------------------------------------------------------|---------------|------|----------------------|-------------|-------------|--------------------------------|
| Input                                                          | Process Name | Output                                 | Remarks                                    | Cate<br>gory                | Sub<br>category     | Infra<br>struc<br>ture | Loca<br>tion | Modul name in<br>ecoinvent                                      | Mean<br>value | Unit | Source mean<br>value | Typ<br>e    | StDv<br>95% | General<br>Comment             |
| Total land use (transformation)                                | anode, plant |                                        |                                            | resource                    | land                |                        |              | Transformation, from unknown                                    | 3.00E+05      | m2   | estimated            | 1           | 2.23        | (5,n.A.,n.A.,n.A.,n.A.,n.A.,8) |
| Land occupation for buildings and open bauxite storage area    |              |                                        | assumption: 80%of total area is built up   | resource                    | land                |                        |              | Occupation, industrial area, built up                           | 1.20E+07      | m2a  | estimated            | 1           | 1.77        | (5,n.A.,n.A.,n.A.,n.A.,n.A.,7) |
| Land transformation to buildings and open bauxite storage area |              |                                        |                                            | resource                    | land                |                        |              | Transformation, to industrial area, built up                    | 2.40E+05      | m2   | estimated            | 1           | 2.23        | (5,n.A.,n.A.,n.A.,n.A.,n.A.,8) |
| Land occupation for vegetation area                            |              |                                        | assumption: 20%of total area is vegetation | resource                    | land                |                        |              | Occupation, industrial area, vegetation                         | 3.00E+06      | m2a  | estimated            | 1           | 1.77        | (5,n.A.,n.A.,n.A.,n.A.,n.A.,7) |
| Land transformation to vegetation area                         |              |                                        |                                            | resource                    | land                |                        |              | Transformation, to industrial area, vegetation                  | 6.01E+04      | m2   | estimated            | 1           | 2.23        | (5,n.A.,n.A.,n.A.,n.A.,n.A.,8) |
| Building, multi story                                          |              |                                        |                                            | construction processes      | buildings           | Yes                    | RER          | building, multi-storey                                          | 1.00E+04      | m3   | estimated            | 1           | 3.23        | (5,n.A.,n.A.,n.A.,n.A.,n.A.,9) |
| Building, hall                                                 |              |                                        |                                            | construction processes      | buildings           | Yes                    | CH           | building, hall                                                  | 1.34E+04      | m2   | estimated            | 1           | 3.23        | (5,n.A.,n.A.,n.A.,n.A.,n.A.,9) |
| heavy machine, unspecific                                      |              |                                        |                                            | construction processes      | machinery           | Yes                    | RER          | Industrial machine, heavy, unspecified , at plant               | 1.23E+06      | kg   | estimated            | 1           | 3.23        | (5,n.A.,n.A.,n.A.,n.A.,n.A.,9) |
| Poor concrete for additional foundation                        |              |                                        |                                            | construction materials      | concrete            | No                     | CH           | concrete, sole plate and foundation, at plant                   | 5.59E+02      | m3   | estimated            | 1           | 3.23        | (5,n.A.,n.A.,n.A.,n.A.,n.A.,9) |
|                                                                |              | foundation concrete to disposal        |                                            | waste management            | building demolition | No                     | CH           | disposal, building, concrete, not reinforced, to final disposal | 1.23E+06      | kg   | estimated            | 1           | 3.23        | (5,n.A.,n.A.,n.A.,n.A.,n.A.,9) |
|                                                                | anode plant  | Output: 80000000 kg/a, life time: 50 a | metals                                     | extraction                  | Yes                 | RER                    | anode, plant | 1.00E+00                                                        | unit          |      |                      |             |             |                                |

Fig. 5.12 Flows for "anode, plant" and their representation in the ecoinvent database

| General Flow information                                      |   |                                                |  |                        | Representation in ecoinvent      |            |              |                |                                        |                          |            | Uncertainty                   |                   |      |                             |                 |
|---------------------------------------------------------------|---|------------------------------------------------|--|------------------------|----------------------------------|------------|--------------|----------------|----------------------------------------|--------------------------|------------|-------------------------------|-------------------|------|-----------------------------|-----------------|
| Input                                                         |   | Process Name                                   |  | Output                 | Remarks                          | Category   | Sub category | Infrastructure | Location                               | Module name in ecoinvent | Mean value | Unit                          | Source mean value | Type | StdV 95%                    | General Comment |
| Carbon blocks [kg]                                            | ⌚ | cathode, aluminium electrolysis; Inputs part 1 |  |                        | generic data from Ecoinvent 2000 | hard coal  | fuels        | No             | RER                                    | hard coal coke, at plant | 1.29E+01   | MJ                            | EAA 2000          | 1    | 1.08                        | (1,1,2,1,1,3,4) |
| Transport of carbon blocks to electrolysis plant [tkm]        | ⌚ |                                                |  | Transport by road      | transport systems                | road       | No           | RER            | transport, lorry 32t                   | 1.29E+00                 | tkm        | standard distance             | 1                 | 2.09 | (4,5,n.A.,n.A.,n.A.,n.A.,5) |                 |
| Transport of carbon blocks to electrolysis plant [tkm]        | ⌚ |                                                |  | Transport by rail      | transport systems                | train      | No           | RER            | transport, freight, rail               | 2.58E+00                 | tkm        | standard distance             | 1                 | 2.09 | (4,5,n.A.,n.A.,n.A.,n.A.,5) |                 |
| cast iron for electrical contacts [kg]                        | ⌚ |                                                |  |                        | metals                           | extraction | No           | RER            | cast iron, at plant                    | 4.44E-02                 | kg         | EAA 2001(pers. communication) | 1                 | 1.08 | (1,1,2,1,1,3,4)             |                 |
| Transport of iron to electrolysis plant [tkm]                 | ⌚ |                                                |  | Transport by road      | transport systems                | road       | No           | RER            | transport, lorry 32t                   | 4.44E-03                 | tkm        | standard distance             | 1                 | 2.09 | (4,5,n.A.,n.A.,n.A.,n.A.,5) |                 |
| Transport of iron to electrolysis plant [tkm]                 | ⌚ |                                                |  | Transport by rail      | transport systems                | train      | No           | RER            | transport, freight, rail               | 8.89E-03                 | tkm        | standard distance             | 1                 | 2.09 | (4,5,n.A.,n.A.,n.A.,n.A.,5) |                 |
| Refractory materials [kg]                                     | ⌚ |                                                |  |                        | construction materials           | bricks     | No           | DE             | refractory, fireclay, packed, at plant | 4.78E-01                 | kg         | EAA 2001(pers. communication) | 1                 | 1.08 | (1,1,2,1,1,3,4)             |                 |
| Transport of refractory materials to electrolysis plant [tkm] | ⌚ |                                                |  | Transport by road      | transport systems                | road       | No           | RER            | transport, lorry 32t                   | 4.78E-02                 | tkm        | standard distance             | 1                 | 2.09 | (4,5,n.A.,n.A.,n.A.,n.A.,5) |                 |
| Transport of refractory materials to electrolysis plant [tkm] | ⌚ |                                                |  | Transport by rail      | transport systems                | train      | No           | RER            | transport, freight, rail               | 9.57E-02                 | tkm        | standard distance             | 1                 | 2.09 | (4,5,n.A.,n.A.,n.A.,n.A.,5) |                 |
| Collar/ramming paste [kg]                                     | ⌚ |                                                |  | inventoried as bitumen | oil                              | fuels      | No           | RER            | bitumen, at refinery                   | 1.94E-01                 | kg         | EAA 2001(pers. communication) | 1                 | 1.08 | (1,1,2,1,1,3,4)             |                 |
| Transport of paste to electrolysis plant [tkm]                | ⌚ |                                                |  | Transport by road      | transport systems                | road       | No           | RER            | transport, lorry 32t                   | 1.94E-02                 | tkm        | standard distance             | 1                 | 2.09 | (4,5,n.A.,n.A.,n.A.,n.A.,5) |                 |
| Transport of paste to electrolysis plant [tkm]                | ⌚ |                                                |  | Transport by rail      | transport systems                | train      | No           | RER            | transport, freight, rail               | 3.89E-02                 | tkm        | standard distance             | 1                 | 2.09 | (4,5,n.A.,n.A.,n.A.,n.A.,5) |                 |

Fig. 5.13 Flows for "cathode, aluminium electrolysis" and their representation in the ecoinvent database

| General Flow information                                                |                                                |        |                                  | Representation in ecoinvent |                   |                |          |                                                       |            |      | Uncertainty                                                |      |          |                         |
|-------------------------------------------------------------------------|------------------------------------------------|--------|----------------------------------|-----------------------------|-------------------|----------------|----------|-------------------------------------------------------|------------|------|------------------------------------------------------------|------|----------|-------------------------|
| Input                                                                   | Process Name                                   | Output | Remarks                          | Category                    | Sub category      | Infrastructure | Location | Module name in ecoinvent                              | Mean value | Unit | Source mean value                                          | Type | Stdv 95% | General Comment         |
| Aluminium oxide [kg]                                                    | cathode, aluminium electrolysis; Inputs part 2 |        |                                  | chemicals                   | inorganics        | No             | RER      | aluminium oxide, at plant                             | 2.22E-02   | kg   | EAA 2001(pers. communication)                              | 1    | 1.08     | (1,1,2,1,1,3,4)         |
| Transport of Al <sub>2</sub> O <sub>3</sub> to electrolysis plant [tkm] |                                                |        | Transport by road                | transport systems           | road              | No             | RER      | transport, lorry 32t                                  | 1.50E-02   | tkm  | EAA 2000 (p. 50) / personal communication B. de Gelas, EAA | 1    | 2.00     | (1,1,2,1,1,3,5)         |
| Transport of Al <sub>2</sub> O <sub>3</sub> to electrolysis plant [tkm] |                                                |        | Transport by rail                | transport systems           | train             | No             | RER      | transport, freight, rail                              | 4.20E-02   | tkm  | EAA 2000 (p. 50) / personal communication B. de Gelas, EAA | 1    | 2.00     | (1,1,2,1,1,3,5)         |
| Transport of Al <sub>2</sub> O <sub>3</sub> to electrolysis plant [tkm] |                                                |        | Transport by ship (ocean)        | transport systems           | ship              | No             | OCE      | transport, transoceanic freight ship                  | 3.74E+00   | tkm  | EAA 2000 (p. 50) / personal communication B. de Gelas, EAA | 1    | 2.00     | (1,1,2,1,1,3,5)         |
| Transport of Al <sub>2</sub> O <sub>3</sub> to electrolysis plant [tkm] |                                                |        | Transport by ship (costal)       | transport systems           | ship              | No             | RER      | transport, barge                                      | 2.04E-01   | tkm  | EAA 2000 (p. 50) / personal communication B. de Gelas, EAA | 1    | 2.00     | (1,1,2,1,1,3,5)         |
| Blasting [kg]                                                           |                                                |        |                                  | construction processes      | civil engineering | No             | RER      | blasting                                              | 2.22E-02   | kg   | EAA 2000                                                   | 1    | 1.08     | (1,1,2,1,1,3,1)         |
| Transport of blasting materials [tkm]                                   |                                                |        | Transport by road                | transport systems           | road              | No             | RER      | transport, lorry 32t                                  | 2.22E-03   | tkm  | standard distance                                          | 1    | 2.09     | (4,5,nA.,nA.,nA.,nA.,5) |
| Transport of blasting materials [tkm]                                   |                                                |        | Transport by rail                | transport systems           | train             | No             | RER      | transport, freight, rail                              | 4.44E-03   | tkm  | standard distance                                          | 1    | 2.09     | (4,5,nA.,nA.,nA.,nA.,5) |
| Steel bars (electrical contacts) [kg]                                   |                                                |        |                                  | metals                      | extraction        | No             | RER      | reinforcing steel, at plant                           | 2.83E-01   | kg   | EAA 2000                                                   | 1    | 1.08     | (1,1,2,1,1,3,4)         |
| Transport of steel bars [tkm]                                           |                                                |        | Transport by road                | transport systems           | road              | No             | RER      | transport, lorry 32t                                  | 2.83E-02   | tkm  | standard distance                                          | 1    | 2.09     | (4,5,nA.,nA.,nA.,nA.,5) |
| Transport of steel bars [tkm]                                           |                                                |        | Transport by rail                | transport systems           | train             | No             | RER      | transport, freight, rail                              | 5.67E-02   | tkm  | standard distance                                          | 1    | 2.09     | (4,5,nA.,nA.,nA.,nA.,5) |
| General heat from natural gas                                           |                                                |        |                                  | natural gas                 | heating systems   | No             | RER      | heat, natural gas, at industrial furnace >100kW       | 1.45E+00   | MJ   | EAA 2000                                                   | 1    | 1.08     | (1,1,2,1,1,3,1)         |
| General heat from oil                                                   |                                                |        |                                  | oil                         | heating systems   | No             | CH       | heat, light fuel oil, at industrial furnace 1MW       | 2.78E-01   | MJ   | EAA 2000                                                   | 1    | 1.08     | (1,1,2,1,1,3,1)         |
| Electrical power used in plant (process) [kWh]                          |                                                |        | generic data from Ecoinvent 2000 | electricity                 | production mix    | No             | UCTE     | electricity, medium voltage, production UCTE, at grid | 2.22E-01   | kWh  | EAA 2000                                                   | 1    | 1.08     | (1,1,2,1,1,3,2)         |
| Infrastructure (plant) [piece]                                          |                                                |        | Included in Electrolysis plant   |                             |                   |                |          |                                                       |            |      |                                                            |      |          |                         |

Fig. 5.13 Flows for "cathode, aluminium electrolysis" and their representation in the ecoinvent database (cont.)



| General Flow information                    |              |                                                                                         |                                   | Representation in ecoinvent |                            |                |          |                                                                                    |            |      | Uncertainty                    |      |          |                  |
|---------------------------------------------|--------------|-----------------------------------------------------------------------------------------|-----------------------------------|-----------------------------|----------------------------|----------------|----------|------------------------------------------------------------------------------------|------------|------|--------------------------------|------|----------|------------------|
| Input                                       | Process Name | Output                                                                                  | Remarks                           | Category                    | Sub category               | Infrastructure | Location | Modul name in ecoinvent                                                            | Mean value | Unit | Source mean value              | Type | StdV 95% | General Comment  |
| cathode, aluminium electrolysis;<br>Outputs | m            | waste heat                                                                              | from electricity use              | air                         | unspecified                |                |          | Heat, waste                                                                        | 8.01E-01   | MJ   | calculated                     | 1    | 1.08     | (1,1,2,1,1,3,13) |
|                                             | m            | Direct emissions of cathode production (CO <sub>2</sub> , CO, SO <sub>x</sub> , ...) ?? | inventoried with electrolysis     |                             |                            |                |          |                                                                                    |            |      | EAA 2001 (pers. communication) |      |          |                  |
|                                             | m            | Al <sub>2</sub> O <sub>3</sub> residue [kg]                                             | reuse --> not inventoried         |                             |                            |                |          |                                                                                    | 2.21E-02   | kg   | EAA 2001 (pers. communication) |      |          |                  |
|                                             | m            | Solid waste unspecified [kg]                                                            | to landfill (controlled)          | waste management            | residual material landfill | No             | CH       | disposal, refractory SPL, Al electrolysis, 0% water, to residual material landfill | 5.52E-03   | kg   | EAA 2001 (pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,6)  |
|                                             | m            | Carbon waste [kg]                                                                       | to landfill, non-hazardous        | waste management            | residual material landfill | No             | CH       | disposal, refractory SPL, Al electrolysis, 0% water, to residual material landfill | 4.25E-01   | kg   | EAA 2001 (pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,6)  |
|                                             | m            | Carbon waste [kg]                                                                       | reuse as fuel --> not inventoried |                             |                            |                |          |                                                                                    | 3.65E-01   | kg   | EAA 2001 (pers. communication) |      |          |                  |
|                                             | m            | Steel scrap [kg]                                                                        | recycling --> not blanced         |                             |                            |                |          |                                                                                    | 2.87E-01   | kg   | EAA 2001 (pers. communication) |      |          |                  |
|                                             | m            | refractory materials [kg]                                                               | to landfill, non-hazardous        | waste management            | residual material landfill | No             | CH       | disposal, refractory SPL, Al electrolysis, 0% water, to residual material landfill | 8.40E-01   | kg   | EAA 2001 (pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,6)  |
|                                             | m            | refractory materials [kg]                                                               | reuse --> not inventoried         |                             |                            |                |          |                                                                                    | 9.39E-02   | kg   | EAA 2001 (pers. communication) |      |          |                  |
|                                             | m            | Heat, waste [MJ]                                                                        | recovered --> not inventoried     |                             |                            |                |          |                                                                                    |            |      |                                |      |          |                  |
|                                             | m            | Cathode for Aluminium electrolysis [kg]                                                 |                                   | metals                      | extraction                 | No             | RER      | cathode, aluminium electrolysis                                                    | 1.00E+00   | kg   |                                |      |          |                  |

Fig. 5.13 Flows for "cathode, aluminium electrolysis" and their representation in the ecoinvent database (cont.)

| General Flow information                                              |                                              |        |                                  | Representation in ecoinvent |              |                |          |                                      |            |      |                                                            | Uncertainty |          |                 |  |
|-----------------------------------------------------------------------|----------------------------------------------|--------|----------------------------------|-----------------------------|--------------|----------------|----------|--------------------------------------|------------|------|------------------------------------------------------------|-------------|----------|-----------------|--|
| Input                                                                 | Process Name                                 | Output | Remarks                          | Category                    | Sub category | Infrastructure | Location | Module name in ecoinvent             | Mean value | Unit | Source mean value                                          | Type        | Stdv 95% | General Comment |  |
| Aluminium oxide, at plant [kg]                                        | aluminium, primary, liquid, at plant; part 1 |        |                                  | chemicals                   | inorganics   | No             | RER      | aluminium oxide, at plant            | 1.92E+00   | kg   | EAA 2000 (p. 49)                                           | 1           | 1.08     | (1,1,2,1,1,3,3) |  |
| Transport of Al <sub>2</sub> O <sub>3</sub> from plant to plant [tkm] |                                              |        | Transport by ship (ocean)        | transport systems           | ship         | No             | OCE      | transport, transoceanic freight ship | 3.74E+00   | tkm  | EAA 2000 (p. 50) / personal communication B. de Gelas, EAA | 1           | 2.00     | (1,1,2,1,1,3,5) |  |
| Transport of Al <sub>2</sub> O <sub>3</sub> from plant to plant [tkm] |                                              |        | Transport by ship (coastal)      | transport systems           | ship         | No             | RER      | transport, barge                     | 2.04E-01   | tkm  | EAA 2000 (p. 50) / personal communication B. de Gelas, EAA | 1           | 2.00     | (1,1,2,1,1,3,5) |  |
| Transport of Al <sub>2</sub> O <sub>3</sub> from plant to plant [tkm] |                                              |        | Transport by Road                | transport systems           | road         | No             | RER      | transport, lorry 32t                 | 1.50E-02   | tkm  | EAA 2000 (p. 50) / personal communication B. de Gelas, EAA | 1           | 2.00     | (1,1,2,1,1,3,5) |  |
| Transport of Al <sub>2</sub> O <sub>3</sub> from plant to plant [tkm] |                                              |        | Transport by Rail                | transport systems           | train        | No             | RER      | transport, freight, rail             | 4.20E-02   | tkm  | EAA 2000 (p. 50) / personal communication B. de Gelas, EAA | 1           | 2.00     | (1,1,2,1,1,3,5) |  |
| Anode for Aluminium electrolysis [kg]                                 |                                              |        |                                  | metals                      | extraction   | No             | RER      | anode, aluminium electrolysis        | 4.48E-01   | kg   | EAA 2000 (p. 49)                                           | 1           | 1.08     | (1,1,2,1,1,3,4) |  |
| Transport of anode to electrolysis plant [tkm]                        |                                              |        | Transport by ship (ocean)        | transport systems           | ship         | No             | OCE      | transport, transoceanic freight ship | 6.40E-02   | tkm  | EAA 2001(pers. communication)                              | 1           | 2.00     | (1,1,2,1,1,3,5) |  |
| Transport of anode to electrolysis plant [tkm]                        |                                              |        | Transport by ship (coastal)      | transport systems           | ship         | No             | RER      | transport, barge                     | 2.64E-02   | tkm  | EAA 2001(pers. communication)                              | 1           | 2.00     | (1,1,2,1,1,3,5) |  |
| Transport of anode to electrolysis plant [tkm]                        |                                              |        | Transport by Road                | transport systems           | road         | No             | RER      | transport, lorry 32t                 | 1.61E-02   | tkm  | EAA 2001(pers. communication)                              | 1           | 2.00     | (1,1,2,1,1,3,5) |  |
| Transport of anode to electrolysis plant [tkm]                        |                                              |        | Transport by Rail                | transport systems           | train        | No             | RER      | transport, freight, rail             | 1.61E-02   | tkm  | EAA 2001(pers. communication)                              | 1           | 2.00     | (1,1,2,1,1,3,5) |  |
| Cathode for Aluminium electrolysis [kg]                               |                                              |        |                                  | metals                      | extraction   | No             | RER      | cathode, aluminium electrolysis      | 1.81E-02   | kg   | EAA 2000                                                   | 1           | 1.08     | (1,1,2,1,1,3,4) |  |
| Cryolite [kg]                                                         |                                              |        | generic data from Ecoinvent 2000 | chemicals                   | inorganics   | No             | RER      | cryolite, at plant                   | 1.60E-03   | kg   | EAA 2000                                                   | 1           | 1.08     | (1,1,2,1,1,3,4) |  |

Fig. 5.14 Flows for "aluminium, primary, liquid, at plant" and their representation in the ecoinvent database

| General Flow information                                  |                                              |                               |                               | Representation in ecoinvent |                 |                |          |                                                          |            |      |                                | Uncertainty |          |                                             |  |
|-----------------------------------------------------------|----------------------------------------------|-------------------------------|-------------------------------|-----------------------------|-----------------|----------------|----------|----------------------------------------------------------|------------|------|--------------------------------|-------------|----------|---------------------------------------------|--|
| Input                                                     | Process Name                                 | Output                        | Remarks                       | Category                    | Sub category    | Infrastructure | Location | Module name in ecoinvent                                 | Mean value | Unit | Source mean value              | Type        | StdV 95% | General Comment                             |  |
| Transport of cryolite to electrolysis plant [tkm]         | aluminium, primary, liquid, at plant; part 2 |                               | Transport by Road             | transport systems           | road            | No             | RER      | transport, lorry >16t, fleet average                     | 1.60E-04   | tkm  | standard distance              | 1           | 2.09     | (4,5,n.A.,n.A.,n.A.,n.A.,5)                 |  |
| Transport of cryolite to electrolysis plant [tkm]         |                                              |                               | Transport by Rail             | transport systems           | train           | No             | RER      | transport, freight, rail                                 | 3.20E-04   | tkm  | standard distance              | 1           | 2.09     | (4,5,n.A.,n.A.,n.A.,n.A.,5)                 |  |
| Aluminium fluoride [kg]                                   |                                              |                               |                               | metals                      | extraction      | No             | RER      | aluminium fluoride, at plant                             | 1.87E-02   | kg   | EAA 2000                       | 1           | 1.08     | (1,1,2,1,1,3,4)                             |  |
| Transport of AlF <sub>3</sub> to electrolysis plant [tkm] |                                              |                               | Transport by Road             | transport systems           | road            | No             | RER      | transport, lorry >16t, fleet average                     | 1.87E-03   | tkm  | standard distance              | 1           | 2.09     | (4,5,n.A.,n.A.,n.A.,n.A.,5)                 |  |
| Transport of AlF <sub>3</sub> to electrolysis plant [tkm] |                                              |                               | Transport by Rail             | transport systems           | train           | No             | RER      | transport, freight, rail                                 | 3.74E-03   | tkm  | standard distance              | 1           | 2.09     | (4,5,n.A.,n.A.,n.A.,n.A.,5)                 |  |
| General heat from natural gas                             |                                              |                               |                               | natural gas                 | heating systems | No             | RER      | heat, natural gas, at industrial furnace >100kW          | 8.40E-02   | MJ   | EAA 2000                       | 1           | 1.08     | (1,1,2,1,1,3,1)                             |  |
| General heat from oil                                     |                                              |                               |                               | oil                         | heating systems | No             | CH       | heat, light fuel oil, at industrial furnace 1MW          | 8.90E-02   | MJ   | EAA 2000                       | 1           | 1.08     | (1,1,2,1,1,3,1)                             |  |
| Electrical power used in plant (process) [kWh]            |                                              |                               | Aluminium industry mix        | electricity                 | supply mix      | No             | GLO      | electricity, medium voltage, aluminium industry, at grid | 1.56E+01   | kWh  | EAA 2001 (pers. communication) | 1           | 1.08     | (1,1,2,1,1,3,2)                             |  |
| Infrastructure (plant) [piece]                            |                                              |                               |                               | metals                      | extraction      | Yes            | RER      | aluminium electrolysis, plant                            | 1.54E-10   | unit | estimated                      | 1           | 3.23     | (5,n.A.,n.A.,n.A.,n.A.,n.A.,9); rough guess |  |
|                                                           |                                              | waste heat                    | from electricity use          | air                         | unspecified     |                |          | Heat, waste                                              | 5.60E+01   | MJ   | calculated                     | 1           | 1.08     | (1,1,2,1,1,3,13)                            |  |
|                                                           |                                              | BaP (Benzo-a-Pyrene) to air   |                               | air                         | unspecified     |                |          | Benzo(a)pyrene                                           | 1.30E-06   | kg   | EAA 2001 (pers. communication) | 1           | 3.00     | (1,1,2,1,1,3,21)                            |  |
|                                                           |                                              | CF <sub>4</sub>               | 90% of polyfluorinated carbon | air                         | unspecified     |                |          | Methane, tetrafluoro-, R-14                              | 2.52E-04   | kg   | EAA 2001 (pers. communication) | 1           | 1.51     | (1,1,2,1,1,3,31)                            |  |
|                                                           |                                              | C <sub>2</sub> F <sub>6</sub> | 10% of polyfluorinated carbon | air                         | unspecified     |                |          | Ethane, hexafluoro-, HFC-116                             | 2.80E-05   | kg   | EAA 2001 (pers. communication) | 1           | 1.51     | (1,1,2,1,1,3,31)                            |  |
|                                                           |                                              | CO to air                     |                               | air                         | unspecified     |                |          | Carbon monoxide, fossil                                  | 9.17E-02   | kg   | EAA 2001 (pers. communication) | 1           | 5.01     | (1,1,2,1,1,3,17)                            |  |

Fig. 5.14 Flows for "aluminium, primary, liquid, at plant" and their representation in the ecoinvent database (cont.)

| General Flow information                     |              |                               |                                                                             | Representation in ecoinvent |                            |                |          |                                                                                    |            |      | Uncertainty                    |      |          |                  |
|----------------------------------------------|--------------|-------------------------------|-----------------------------------------------------------------------------|-----------------------------|----------------------------|----------------|----------|------------------------------------------------------------------------------------|------------|------|--------------------------------|------|----------|------------------|
| Input                                        | Process Name | Output                        | Remarks                                                                     | Category                    | Sub category               | Infrastructure | Location | Module name in ecoinvent                                                           | Mean value | Unit | Source mean value              | Type | StdV 95% | General Comment  |
| aluminium, primary, liquid, at plant; part 3 | ML           | CO2 to air                    |                                                                             | air                         | unspecified                |                |          | Carbon dioxide, fossil                                                             | 1.50E+00   | kg   | EAA 2001 (pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,24) |
|                                              | ML           | Dust to air                   | particules that are not retained by the cyclone --> Particule size < 2.5 um | air                         | unspecified                |                |          | Particulates, < 2.5 um                                                             | 2.61E-03   | kg   | EAA 2001 (pers. communication) | 1    | 3.00     | (1,1,2,1,1,3,20) |
|                                              | ML           | fluoride (gaseous) to air     | inventoried as HF                                                           | air                         | unspecified                |                |          | Hydrogen fluoride                                                                  | 5.39E-04   | kg   | EAA 2001 (pers. communication) | 1    | 1.51     | (1,1,2,1,1,3,31) |
|                                              | ML           | fluoride (non gaseous) to air | inventoried as particulate emission                                         | air                         | unspecified                |                |          | Particulates, > 2.5 um, and < 10um                                                 | 6.09E-04   | kg   | EAA 2001 (pers. communication) | 1    | 2.00     | (1,1,2,1,1,3,26) |
|                                              | ML           | polycyclic aromatic HC to air | without BaP                                                                 | air                         | unspecified                |                |          | PAH, polycyclic aromatic hydrocarbons                                              | 4.57E-05   | kg   | EAA 2001 (pers. communication) | 1    | 3.00     | (1,1,2,1,1,3,21) |
|                                              | ML           | NOx to air                    |                                                                             | air                         | unspecified                |                |          | Nitrogen oxides                                                                    | 6.39E-05   | kg   | EAA 2001 (pers. communication) | 1    | 1.41     | (1,1,2,1,1,3,29) |
|                                              | ML           | SO2 to air                    |                                                                             | air                         | unspecified                |                |          | Sulfur dioxide                                                                     | 8.83E-03   | kg   | EAA 2001 (pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,15) |
|                                              | ML           | carbon material for recycling | not inventoried                                                             |                             |                            |                |          |                                                                                    | 3.00E-03   | kg   | EAA 2001 (pers. communication) |      |          |                  |
|                                              | ML           | carbon waste                  | to landfill (controlled)                                                    | waste management            | residual material landfill | No             | CH       | disposal, refractory SPL, Al electrolysis, 0% water, to residual material landfill | 1.90E-03   | kg   | EAA 2001 (pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,6)  |
|                                              | ML           | cast iron                     | recycled --> not inventoried                                                |                             |                            |                |          |                                                                                    | 1.80E-03   | kg   | EAA 2001 (pers. communication) |      |          |                  |
|                                              | ML           | crushed bath sold             | recycled --> not inventoried                                                |                             |                            |                |          |                                                                                    | 4.10E-03   | kg   | EAA 2001 (pers. communication) |      |          |                  |
|                                              | ML           | dust from Cyclones/ESP        | to landfill (controlled)                                                    | waste management            | residual material landfill | No             | CH       | disposal, filter dust Al electrolysis, 0% water, to residual material landfill     | 2.00E-03   | kg   | EAA 2001 (pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,6)  |
|                                              | ML           | Solid waste unspecified       | to landfill, non-hazardous                                                  | waste management            | inert material landfill    | No             | CH       | disposal, inert waste, 5% water, to inert material landfill                        | 5.00E-03   | kg   | EAA 2001 (pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,6)  |
|                                              | ML           | soot                          | to landfill (controlled)                                                    | waste management            | sanitary landfill          | No             | CH       | disposal, bitumen, 1.4% water, to sanitary landfill                                | 1.20E-03   | kg   | EAA 2001 (pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,6)  |
|                                              | ML           | Aluminium, liquid [kg]        |                                                                             | metals                      | extraction                 | No             | RER      | aluminium, primary, liquid, at plant                                               | 1.00E+00   | kg   |                                |      |          |                  |

Fig. 5.14 Flows for "aluminium, primary, liquid, at plant" and their representation in the ecoinvent database (cont.)

## 5. System Characterisation

| General Flow information                                       |                               |                                 |                                             | Representation in ecoinvent |                     |                |          |                                                                 |            |      |                   | Uncertainty |          |                 |  |
|----------------------------------------------------------------|-------------------------------|---------------------------------|---------------------------------------------|-----------------------------|---------------------|----------------|----------|-----------------------------------------------------------------|------------|------|-------------------|-------------|----------|-----------------|--|
| Input                                                          | Process Name                  | Output                          | Remarks                                     | Category                    | Sub category        | Infrastructure | Location | Module name in ecoinvent                                        | Mean value | Unit | Source mean value | Type        | StdV 95% | General Comment |  |
| Total land use (transformation)                                | aluminium electrolysis, plant |                                 |                                             | resource                    | land                |                |          | Transformation, from unknown                                    | 4.88E+05   | m2   | estimated         | 1           | 2.03     | (2,4,1,1,14,8)  |  |
| Land occupation for buildings and open bauxite storage area    |                               |                                 | assumption: 80% of total area is built up   | resource                    | land                |                |          | Occupation, industrial area, built up                           | 1.95E+07   | m2a  | estimated         | 1           | 1.54     | (2,4,1,1,14,7)  |  |
| Land transformation to buildings and open bauxite storage area |                               |                                 |                                             | resource                    | land                |                |          | Transformation, to industrial area, built up                    | 3.90E+05   | m2   | estimated         | 1           | 2.03     | (2,4,1,1,14,8)  |  |
| Land occupation for vegetation area                            |                               |                                 | assumption: 20% of total area is vegetation | resource                    | land                |                |          | Occupation, industrial area, vegetation                         | 4.88E+06   | m2a  | estimated         | 1           | 1.54     | (2,4,1,1,14,7)  |  |
| Land transformation to vegetation area                         |                               |                                 |                                             | resource                    | land                |                |          | Transformation, to industrial area, vegetation                  | 9.76E+04   | m2   | estimated         | 1           | 2.03     | (2,4,1,1,14,8)  |  |
| Building, multi story                                          |                               |                                 |                                             | construction processes      | buildings           | Yes            | RER      | building, multi-storey                                          | 2.50E+04   | m3   | estimated         | 1           | 3.04     | (3,4,1,1,14,9)  |  |
| Building, hall                                                 |                               |                                 |                                             | construction processes      | buildings           | Yes            | CH       | building, hall                                                  | 1.97E+04   | m2   | estimated         | 1           | 3.04     | (3,4,1,1,14,9)  |  |
| concrete in reinforced concrete                                |                               |                                 |                                             | construction materials      | concrete            | No             | CH       | concrete, normal, at plant                                      | 1.09E+03   | m3   | estimated         | 1           | 3.04     | (3,4,1,1,14,9)  |  |
| steel in reinforced concrete                                   |                               |                                 |                                             | metals                      | extraction          | No             | RER      | reinforcing steel, at plant                                     | 7.56E+05   | kg   | estimated         | 1           | 3.04     | (3,4,1,1,14,9)  |  |
| heavy machine, unspecific                                      |                               |                                 |                                             | construction processes      | machinery           | Yes            | RER      | Industrial machine, heavy, unspecified, at plant                | 2.00E+05   | kg   | estimated         | 1           | 3.04     | (3,4,1,1,14,9)  |  |
| Poor concrete for additional foundation                        |                               |                                 |                                             | construction materials      | concrete            | No             | CH       | concrete, sole plate and foundation, at plant                   | 9.09E+01   | m3   | estimated         | 1           | 3.04     | (3,4,1,1,14,9)  |  |
| Conveyor belt                                                  |                               |                                 |                                             | construction processes      | machinery           | Yes            | RER      | conveyor belt, at plant                                         | 1.80E+03   | m    | estimated         | 1           | 3.04     | (3,4,1,1,14,9)  |  |
| Steel low alloyed                                              |                               |                                 |                                             | metals                      | extraction          | No             | RER      | steel, low-alloyed, at plant                                    | 8.57E+05   | kg   | estimated         | 1           | 3.04     | (3,4,1,1,14,9)  |  |
|                                                                |                               | reinforced concrete to disposal |                                             | waste management            | building demolition | No             | CH       | disposal, building, reinforced concrete, to final disposal      | 3.16E+06   | kg   | estimated         | 1           | 3.04     | (3,4,1,1,14,9)  |  |
|                                                                |                               | foundation concrete to disposal |                                             | waste management            | building demolition | No             | CH       | disposal, building, concrete, not reinforced, to final disposal | 2.00E+05   | kg   | estimated         | 1           | 3.04     | (3,4,1,1,14,9)  |  |
|                                                                |                               | aluminium electrolysis plant    | Output: 130000000 kg/a, life time: 50 a     | metals                      | extraction          | Yes            | RER      | aluminium electrolysis, plant                                   | 1.00E+00   | unit |                   |             |          |                 |  |

Fig. 5.15 Flows for "aluminium electrolysis, plant" and their representation in the ecoinvent database

| General Flow information                                                               |                                      |        |                                  | Representation in ecoinvent |                              |                |          |                                                          |            |      | Uncertainty                   |      |          |                                             |
|----------------------------------------------------------------------------------------|--------------------------------------|--------|----------------------------------|-----------------------------|------------------------------|----------------|----------|----------------------------------------------------------|------------|------|-------------------------------|------|----------|---------------------------------------------|
| Input                                                                                  | Process Name                         | Output | Remarks                          | Category                    | Sub category                 | Infrastructure | Location | Module name in ecoinvent                                 | Mean value | Unit | Source mean value             | Type | StdV 95% | General Comment                             |
| Aluminium, liquid [kg]                                                                 | aluminium, primary, at plant; Inputs |        |                                  | metals                      | extraction                   | No             | RER      | aluminium, primary, liquid, at plant                     | 1.00E+00   | kg   | EAA 2001(pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,3)                             |
| Cooling water [m3]                                                                     |                                      |        | Spec. Weight of water: 998 kg/m3 | resource                    | in water                     |                |          | Water, cooling, unspecified natural origin               | 4.71E-03   | m3   | EAA 2001(pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,4)                             |
| Refractory materials [kg]                                                              |                                      |        |                                  | construction materials      | bricks                       | No             | DE       | refractory, fireclay, packed, at plant                   | 7.00E-04   | kg   | EAA 2001(pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,4)                             |
| Alloying elements [kg]                                                                 |                                      |        |                                  | metals                      | extraction                   | No             | NO       | M G-silicon, at plant                                    | 1.08E-02   | kg   | EAA 2001(pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,4)                             |
| Nitrogen gas [kg]                                                                      |                                      |        |                                  | chemicals                   | inorganics                   | No             | RER      | nitrogen, liquid, at plant                               | 6.00E-04   | kg   | EAA 2001(pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,4)                             |
| Argon gas [kg]                                                                         |                                      |        |                                  | chemicals                   | inorganics                   | No             | RER      | argon, liquid, at plant                                  | 1.50E-03   | kg   | EAA 2001(pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,4)                             |
| Chlorine gas [kg]                                                                      |                                      |        |                                  | chemicals                   | inorganics                   | No             | RER      | chlorine, liquid, production mix, at plant               | 1.00E-04   | kg   | EAA 2001(pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,4)                             |
| Fibre material [kg]                                                                    |                                      |        |                                  | insulation materials        | production                   | No             | CH       | rock wool, at plant                                      | 1.10E-04   | kg   | EAA 2001(pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,4)                             |
| Fluxing salts [kg]                                                                     |                                      |        |                                  | chemicals                   | inorganics                   | No             | RER      | cryolite, at plant                                       | 4.00E-04   | kg   | EAA 2001(pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,4)                             |
| Mineral / vegetal oils [kg]                                                            |                                      |        |                                  | chemicals                   | organics                     | No             | MY       | crude palm oil, at plant                                 | 8.00E-05   | kg   | EAA 2001(pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,4)                             |
| Packaging materials [kg]                                                               |                                      |        |                                  | paper & cardboard           | cardboard & corrugated board | No             | RER      | corrugated board, mixed fibre, single wall, at plant     | 1.80E-03   | kg   | EAA 2001(pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,4)                             |
| Transport of materials [tkm]                                                           |                                      |        |                                  | transport systems           | road                         | No             | RER      | transport, lorry 32t                                     | 1.61E-03   | tkm  | standard distance             | 1    | 2.09     | (4,5,n.A.,n.A.,n.A.,n.A.,5)                 |
| Transport of materials [tkm]                                                           |                                      |        |                                  | transport systems           | train                        | No             | RER      | transport, freight, rail                                 | 3.22E-03   | tkm  | standard distance             | 1    | 2.09     | (4,5,n.A.,n.A.,n.A.,n.A.,5)                 |
| Electrical power used in plant (process, lighting, office work, room cooling...) [kWh] |                                      |        |                                  | electricity                 | supply mix                   | No             | GLO      | electricity, medium voltage, aluminium industry, at grid | 1.60E-02   | kWh  | EAA 2001(pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,2)                             |
| General heat from oil                                                                  |                                      |        |                                  | oil                         | heating systems              | No             | CH       | heat, light fuel oil, at industrial furnace 1MW          | 4.41E-01   | MJ   | EAA 2001(pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,1)                             |
| General heat from natural gas                                                          |                                      |        |                                  | natural gas                 | heating systems              | No             | RER      | heat, natural gas, at industrial furnace >100kW          | 6.41E-01   | MJ   | EAA 2001(pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,1)                             |
| Infrastructure (plant) [unit]                                                          |                                      |        |                                  | metals                      | extraction                   | Yes            | RER      | aluminium casting, plant                                 | 1.54E-10   | unit | estimated                     | 1    | 3.23     | (5,n.A.,n.A.,n.A.,n.A.,n.A.,9); rough guess |

Fig. 5.16 Flows for "aluminium, primary, at plant" and their representation in the ecoinvent database

| General Flow information                 |   |                                   |  |                              | Representation in ecoinvent |                  |                            |                |          |                                                                               |            | Uncertainty |                               |      |          |                  |
|------------------------------------------|---|-----------------------------------|--|------------------------------|-----------------------------|------------------|----------------------------|----------------|----------|-------------------------------------------------------------------------------|------------|-------------|-------------------------------|------|----------|------------------|
| Input                                    |   | Process Name                      |  | Output                       | Remarks                     | Category         | Sub category               | Infrastructure | Location | Module name in ecoinvent                                                      | Mean value | Unit        | Source mean value             | Type | StdV 95% | General Comment  |
| aluminium, primary, at plant;<br>Outputs | m | waste heat                        |  | from electricity use         |                             | air              | unspecified                |                |          | Heat, waste                                                                   | 5.76E-02   | MJ          | calculated                    | 1    | 1.08     | (1,1,2,1,1,3,13) |
|                                          | m | fluoride (gaseous) to air         |  | inventoried as HF            |                             | air              | unspecified                |                |          | Hydrogen fluoride                                                             | 3.00E-06   | kg          | EAA 2001(pers. communication) | 1    | 1.51     | (1,1,2,1,1,3,31) |
|                                          | m | fluoride (non gaseous) to air     |  | inventoried as particulates  |                             | air              | unspecified                |                |          | Particulates, > 2.5 um, and < 10 um                                           | 7.00E-06   | kg          | EAA 2001(pers. communication) | 1    | 2.00     | (1,1,2,1,1,3,26) |
|                                          | m | Dross (fines)                     |  | to landfill (controlled)     |                             | waste management | residual material landfill | No             | CH       | disposal, dross from Al electrolysis, 0% water, to residual material landfill | 1.10E-04   | kg          | EAA 2001(pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,6)  |
|                                          | m | waste (fiber material)            |  | to landfill, non-hazardous   |                             | waste management | inert material landfill    | No             | CH       | disposal, inert waste, 5%water, to inert material landfill                    | 9.00E-05   | kg          | EAA 2001(pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,6)  |
|                                          | m | refractory waste                  |  | to landfill, non-hazardous   |                             | waste management | inert material landfill    | No             | CH       | disposal, inert waste, 5%water, to inert material landfill                    | 9.00E-04   | kg          | EAA 2001(pers. communication) | 1    | 1.08     | (1,1,2,1,1,3,6)  |
|                                          | m | skimmings and dross for recycling |  | recycled --> not inventoried |                             |                  |                            |                |          |                                                                               | 1.86E-02   | kg          | EAA 2001(pers. communication) |      |          |                  |
|                                          | m | swarf / turnings                  |  | recycled --> not inventoried |                             |                  |                            |                |          |                                                                               | 8.40E-04   | kg          | EAA 2001(pers. communication) |      |          |                  |
|                                          | m | aluminium, 0% recycling, at plant |  |                              |                             | metals           | extraction                 | No             | RER      | aluminium, primary, at plant                                                  | 1.00E+00   | kg          |                               |      |          |                  |

Fig. 5.16 Flows for "aluminium, primary, at plant" and their representation in the ecoinvent database (cont.)

| General Flow information                                       |                          |                                 |                                             | Representation in ecoinvent |                     |                |          |                                                                 |            |      |                   | Uncertainty |          |                 |  |
|----------------------------------------------------------------|--------------------------|---------------------------------|---------------------------------------------|-----------------------------|---------------------|----------------|----------|-----------------------------------------------------------------|------------|------|-------------------|-------------|----------|-----------------|--|
| Input                                                          | Process Name             | Output                          | Remarks                                     | Category                    | Sub category        | Infrastructure | Location | Module name in ecoinvent                                        | Mean value | Unit | Source mean value | Type        | StdV 95% | General Comment |  |
| Total land use (transformation)                                | aluminium casting, plant |                                 |                                             | resource                    | land                |                |          | Transformation, from unknown                                    | 4.88E+05   | m2   | estimated         | 1           | 2.03     | (2,4,1,1,1,4,8) |  |
| Land occupation for buildings and open bauxite storage area    |                          |                                 | assumption: 80% of total area is built up   | resource                    | land                |                |          | Occupation, industrial area, built up                           | 1.95E+07   | m2a  | estimated         | 1           | 1.54     | (2,4,1,1,1,4,7) |  |
| Land transformation to buildings and open bauxite storage area |                          |                                 |                                             | resource                    | land                |                |          | Transformation, to industrial area, built up                    | 3.90E+05   | m2   | estimated         | 1           | 2.03     | (2,4,1,1,1,4,8) |  |
| Land occupation for vegetation area                            |                          |                                 | assumption: 20% of total area is vegetation | resource                    | land                |                |          | Occupation, industrial area, vegetation                         | 4.88E+06   | m2a  | estimated         | 1           | 1.54     | (2,4,1,1,1,4,7) |  |
| Land transformation to vegetation area                         |                          |                                 |                                             | resource                    | land                |                |          | Transformation, to industrial area, vegetation                  | 9.76E+04   | m2   | estimated         | 1           | 2.03     | (2,4,1,1,1,4,8) |  |
| Building, multi-story                                          |                          |                                 |                                             | construction processes      | buildings           | Yes            | RER      | building, multi-storey                                          | 2.00E+03   | m3   | estimated         | 1           | 3.04     | (3,4,1,1,1,4,9) |  |
| Building, hall                                                 |                          |                                 |                                             | construction processes      | buildings           | Yes            | CH       | building, hall                                                  | 2.17E+04   | m2   | estimated         | 1           | 3.04     | (3,4,1,1,1,4,9) |  |
| heavy machine, unspecified                                     |                          |                                 |                                             | construction processes      | machinery           | Yes            | RER      | Industrial machine, heavy, unspecified, at plant                | 2.00E+06   | kg   | estimated         | 1           | 3.04     | (3,4,1,1,1,4,9) |  |
| Poor concrete for additional foundation                        |                          |                                 |                                             | construction materials      | concrete            | No             | CH       | concrete, sole plate and foundation, at plant                   | 9.09E+02   | m3   | estimated         | 1           | 3.04     | (3,4,1,1,1,4,9) |  |
|                                                                |                          | foundation concrete to disposal |                                             | waste management            | building demolition | No             | CH       | disposal, building, concrete, not reinforced, to final disposal | 2.00E+06   | kg   | estimated         | 1           | 3.04     | (3,4,1,1,1,4,9) |  |
|                                                                |                          | aluminium casting plant         | Output: 130000000 kg/a, life time: 50 a     | metals                      | extraction          | Yes            | RER      | aluminium casting, plant                                        | 1.00E+00   | unit |                   |             |          |                 |  |

Fig. 5.17 Flows for "aluminium casting, plant" and their representation in the ecoinvent database



## 5.4 Secondary aluminium

This part is mainly based on EAA (2000).

### 5.4.1 Production process and infrastructure

A wide variety of aluminium scrap is processed in the secondary aluminium industry. A basic differentiation is made between "new" and "old" scrap.

New scrap is surplus material that arises during production and fabrication of aluminium products up to the point where they are sold to the final consumer. It is usually of known quality and alloy and often uncoated. It can then be remelted with little preparation. Some new scrap is coated with paints or plastics. This scrap has to be de-coated in an oven or a mesh conveyer where the coatings are volatilised or burned.

Old scrap or post-consumer scrap is the aluminium recovered after the product has been used by the final consumer. It comes into the secondary aluminium industry via metal merchants and waste management companies which recover aluminium from e.g. vehicles, household goods etc. This is usually done by shredding, magnetic separation and sink-and -float installations or eddy current installations successively. Old scrap is usually recycled together with two-thirds of new scrap from aluminium processing.

Different melting processes are used depending on composition of the scrap, process availability and economics amongst other variables. Molten metal fluxing and filtration technologies are used for quality control of the resulting alloys.

### 5.4.2 Emissions

#### Air

If new scrap is clean and uncoated no emissions arise from cleaning or de-coating. Melting and alloying of new scrap emits HCl and chlorides. Old scrap preparation leads to dust and also HCl and chloride emission. Melting and alloying of old scrap leads to dust and different inorganic emissions.

The size fractions of the dust emissions are derived by the medium level emissions in cepmeip emission factor database<sup>14</sup>.

#### Water

There are no reports of direct water emissions from these processes. The preparation of old scrap however is expected to emit certain substances to water.

### 5.4.3 Waste

#### New scrap production waste

In the preparation of uncoated new scrap no waste is generated.

#### Aluminium from new scrap production waste

The secondary aluminium production from new scrap generates only some refractory material waste which is inert and deposited in landfills.

#### Old scrap production waste

The preparation of old scrap generates different waste fractions:

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<sup>14</sup> [http://www.air.sk/tno/cepmeip/em\\_factors\\_results.php?](http://www.air.sk/tno/cepmeip/em_factors_results.php?), accessed 08.01.03

- Waste oil is assumed to be deposited as hazardous waste in an underground deposit.
- No composition data could be found for ball mill dust, waste filter materials and waste sediments. All the fractions are assumed to consist mainly of organic coatings of the scrap and to a lower extent of aluminium oxide. Deposition as hazardous waste in an underground deposit is assumed.
- Dirt is assumed to be an inert waste deposited in inert material landfills.
- Filter dust composition is assumed to be the same as in the electrolysis process (see Tab. 5.9). Disposal in a residual material landfill is assumed.

#### Aluminium from old scrap processing waste

The secondary aluminium production from old scrap also generates "ball mill dust" and "filter dust". These wastes are treated as the respective wastes from the old scrap production.

#### 5.4.4 Secondary Aluminium in ecoinvent

The available information about the infrastructure needed for aluminium production is non-quantitative. Tab. 5.14 and Tab. 5.15 show the details of the assumptions made to estimate the infrastructure expenditures.

Tab. 5.14 Estimated infrastructure for aluminium melting furnace

| aluminium melting furnace                         |            | Unit | Life time [a] | Life time demand |
|---------------------------------------------------|------------|------|---------------|------------------|
| Yearly production capacity                        | 10'000'000 | kg   | 50            | 10'000'000       |
| Area of plant (assumption based on casting plant) | 37'538     | m2   | 50            | 37'538           |
| Production hall                                   | 938        | m2   | 50            | 938              |
| Furnace                                           | 76'923     | kg   | 25            | 153'846          |

Tab. 5.15 Estimated infrastructure for scrap preparation plant

| scrap preparation plant                           | Demand     | Unit | Life time [a] | Life time demand |
|---------------------------------------------------|------------|------|---------------|------------------|
| Yearly production capacity                        | 10'000'000 | kg   | 50            | 10'000'000       |
| Area of plant (assumption based on casting plant) | 37'538     | m2   | 50            | 37'538           |
| Administration building (50x50x10 m3)             | 25'000     | m3   | 50            | 25'000           |
| Production hall                                   | 938        | m2   | 50            | 938              |
| conveyor belt [m]                                 | 100        | m    | 25            | 200              |
| Production machines                               | 76'923     | kg   | 25            | 153'846          |
| Floating tanks 20 x 5 x 1 m                       | 21         | m3   | 50            | 21               |
| Storage hall                                      | 731        | m2   | 50            | 731              |

The other background information to the processes are given in, flows are shown in Tab. 5.16:

- aluminium scrap, new, at plant Fig. 5.18
- aluminium, secondary, from new scrap, at plant Fig. 5.19
- aluminium melting furnace Fig. 5.20

- aluminium scrap, old, at plant Fig. 5.21
- aluminium, secondary, from old scrap, at plant Fig. 5.22
- scrap preparation plant Fig. 5.23

Tab. 5.16 Meta information for the secondary aluminium production processes

| Name                                  | aluminium scrap, new, at plant                                                    | aluminium, secondary, from new scrap, at plant                                 | aluminium melting furnace                                                                                                                 |
|---------------------------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Location                              | RER                                                                               | RER                                                                            | RER                                                                                                                                       |
| Infrastructure Process                | 0                                                                                 | 0                                                                              | 1                                                                                                                                         |
| Unit                                  | kg                                                                                | kg                                                                             | unit                                                                                                                                      |
| Data Set Version                      | 2.0                                                                               | 2.0                                                                            | 2.0                                                                                                                                       |
| Included Processes                    | Includes only transports of new scrap to secondary aluminium production.          | Melting, alloying and casting of new scrap to secondary aluminium billets.     | Production hall and machinery (including disposal). Does not include administrative buildings because they are included in casting plant. |
| Amount                                | 1                                                                                 | 1                                                                              | 1                                                                                                                                         |
| Local Name                            | Aluminiumschrott, neu, ab Werk                                                    | Aluminium, sekundär, aus neuem Schrott, ab Werk                                | Aluminium Schmelzofen                                                                                                                     |
| Synonyms                              |                                                                                   |                                                                                |                                                                                                                                           |
| General Comment to reference function | It is assumed that the new scrap is not coated and can be used without treatment. |                                                                                | Medium sized plant. Output: 10'000 t/a, life time: 50 years. Assumption is based on data for casting plant.                               |
| Start Date                            | 1995                                                                              | 1995                                                                           | 2002                                                                                                                                      |
| End Date                              | 2002                                                                              | 2002                                                                           | 2002                                                                                                                                      |
| Data Valid For Entire Period          | 1                                                                                 | 1                                                                              | 1                                                                                                                                         |
| Geography text                        | Some Swiss datasets are used for European processes.                              | Some Swiss datasets are used for European processes.                           | Some Swiss datasets are used for European processes.                                                                                      |
| Technology text                       | Average technology for the aluminium recycled in Europe                           | Average technology for the aluminium recycled in Europe                        | Assumed technology                                                                                                                        |
| Representativeness [%]                |                                                                                   |                                                                                |                                                                                                                                           |
| Production Volume                     | unknown                                                                           | unknown                                                                        | unknown                                                                                                                                   |
| Sampling Procedure                    | Literature / Survey                                                               | Literature / Survey                                                            | Internet search                                                                                                                           |
| Extrapolations                        | see Geography                                                                     | see Geography                                                                  | see Geography                                                                                                                             |
| Uncertainty Adjustments               | Uncertainty is adjusted for geographical mismatch (using the pedigree matrix).    | Uncertainty is adjusted for geographical mismatch (using the pedigree matrix). | Uncertainty is adjusted for geographical mismatch (using the pedigree matrix).                                                            |

Tab. 5.16 Meta information for the secondary aluminium production processes (cont.)

| Name                                  | aluminium scrap, old, at plant                                                             | scrap preparation plant                                                                                     | aluminium, secondary, from old scrap, at plant                                 |
|---------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Location                              | RER                                                                                        | RER                                                                                                         | RER                                                                            |
| Infrastructure Process                | 0                                                                                          | 1                                                                                                           | 0                                                                              |
| Unit                                  | kg                                                                                         | unit                                                                                                        | kg                                                                             |
| Data Set Version                      | 2.0                                                                                        | 2.0                                                                                                         | 2.0                                                                            |
| Included Processes                    | Collecting, sorting and preparing (cleaning, pressing...) of post consumer aluminium scrap | Production hall, machinery and administrative buildings (including disposal)                                | Melting, alloying and casting of old scrap to secondary aluminium billets.     |
| Amount                                | 1                                                                                          | 1                                                                                                           | 1                                                                              |
| Local Name                            | Aluminiumschrott, alt, ab Werk                                                             | Sortieranlage Schrott                                                                                       | Aluminium, sekundär, aus altem Schrott, ab Werk                                |
| Synonyms                              | post consumer aluminium                                                                    |                                                                                                             |                                                                                |
| General Comment to reference function |                                                                                            | Medium sized plant. Output: 10'000 t/a, life time: 50 years. Assumption is based on data for casting plant. |                                                                                |
| Start Date                            | 2002                                                                                       | 2002                                                                                                        | 1995                                                                           |
| End Date                              | 2002                                                                                       | 2002                                                                                                        | 2002                                                                           |
| Data Valid For Entire Period          | 1                                                                                          | 1                                                                                                           | 1                                                                              |
| Geography text                        | Some Swiss datasets are used for European processes.                                       | Some Swiss datasets are used for European processes.                                                        | Some Swiss datasets are used for European processes.                           |
| Technology text                       | Average technology for the aluminium recycled in Europe                                    | Assumed technology                                                                                          | Average technology for the aluminium recycled in Europe                        |
| Representativeness [%]                |                                                                                            |                                                                                                             |                                                                                |
| Production Volume                     | unknown                                                                                    | unknown                                                                                                     | unknown                                                                        |
| Sampling Procedure                    | Literature / Survey                                                                        | Internet search                                                                                             | Literature / Survey                                                            |
| Extrapolations                        | see Geography                                                                              | see Geography                                                                                               | see Geography                                                                  |
| Uncertainty Adjustments               | Uncertainty is adjusted for geographical mismatch (using the pedigree matrix).             | Uncertainty is adjusted for geographical mismatch (using the pedigree matrix).                              | Uncertainty is adjusted for geographical mismatch (using the pedigree matrix). |

### Data quality

Data for the steps in the secondary aluminium production are of satisfactory overall quality. The values for the smelting processes stem from a recent and comprehensive source. The proxies chosen to represent the alloying elements in ecoinvent however are decreasing the overall quality. The collection of the scrap and the infrastructure are roughly estimated. The infrastructure is of minor importance compared to the other activities but the transports (scrap collection) could have a noticeable impact.

| General Flow information     |                                |                      |                                    | Representation in ecoinvent |              |                |          |                                |            |      | Uncertainty       |      |          |                         |
|------------------------------|--------------------------------|----------------------|------------------------------------|-----------------------------|--------------|----------------|----------|--------------------------------|------------|------|-------------------|------|----------|-------------------------|
| Input                        | Process Name                   | Output               | Remarks                            | Category                    | Sub category | Infrastructure | Location | Modul name in ecoinvent        | Mean value | Unit | Source mean value | Type | StdV 95% | General Comment         |
| New scrap from production    | aluminium scrap, new, at plant |                      | from recycling --> not inventoried |                             |              |                |          |                                | 1.00E+00   | kg   | calculated        |      |          |                         |
| Transport of inputs to plant |                                |                      | road: 100 km                       | transport systems           | road         | No             | RER      | transport, lorry 32t           | 1.00E-01   | tkm  | standard distance | 1    | 2.09     | (4,5,nA.,nA.,nA.,nA.,5) |
| Transport of inputs to plant |                                |                      | rail: 200 km                       | transport systems           | train        | No             | RER      | transport, freight, rail       | 2.00E-01   | tkm  | standard distance | 1    | 2.09     | (4,5,nA.,nA.,nA.,nA.,5) |
|                              |                                | aluminium scrap, new |                                    | metals                      | extraction   | No             | RER      | aluminium scrap, new, at plant | 1.00E+00   | kg   |                   |      |          |                         |

Fig. 5.18 Flows for "aluminium scrap, new, at plant" and their representation in the ecoinvent database

| General Flow information           |                                                        |        |                  | Representation in ecoinvent |                 |                |          |                                                       |            |      | Uncertainty       |      |          |                 |
|------------------------------------|--------------------------------------------------------|--------|------------------|-----------------------------|-----------------|----------------|----------|-------------------------------------------------------|------------|------|-------------------|------|----------|-----------------|
| Input                              | Process Name                                           | Output | Remarks          | Category                    | Sub category    | Infrastructure | Location | Modul name in ecoinvent                               | Mean value | Unit | Source mean value | Type | StdV 95% | General Comment |
| New aluminium scrap                | aluminium, secondary, from new scrap, at plant; part 1 |        |                  | metals                      | extraction      | No             | RER      | aluminium scrap, new, at plant                        | 1.01E+00   | kg   | EAA 2000          | 1    | 1.08     | (1,1,2,1,1,3,3) |
| Alloying elements (pure form)      |                                                        |        | Silicon as proxy | metals                      | extraction      | No             | NO       | M G-silicon, at plant                                 | 1.14E-02   | kg   | EAA 2000          | 1    | 1.22     | (1,1,2,1,3,3,4) |
| Alloying elements (master alloys)  |                                                        |        | Copper as proxy  | metals                      | extraction      | No             | RER      | copper, at regional storage                           | 1.04E-02   | kg   | EAA 2000          | 1    | 1.22     | (1,1,2,1,3,3,4) |
| electricity                        |                                                        |        |                  | electricity                 | production mix  | No             | UCTE     | electricity, medium voltage, production UCTE, at grid | 1.74E-01   | kWh  | EAA 2000          | 1    | 1.08     | (1,1,2,1,1,3,2) |
| Thermal energy from gas            |                                                        |        |                  | natural gas                 | heating systems | No             | RER      | heat, natural gas, at boiler modulating >100kW        | 3.04E+00   | MJ   | EAA 2000          | 1    | 1.08     | (1,1,2,1,1,3,1) |
| Thermal energy from heavy fuel oil |                                                        |        |                  | oil                         | heating systems | No             | RER      | heat, heavy fuel oil, at industrial furnace 1M W      | 7.00E-02   | MJ   | EAA 2000          | 1    | 1.08     | (1,1,2,1,1,3,1) |
| Thermal energy from light fuel oil |                                                        |        |                  | oil                         | heating systems | No             | CH       | heat, light fuel oil, at industrial furnace 1M W      | 3.32E-03   | MJ   | EAA 2000          | 1    | 1.08     | (1,1,2,1,1,3,1) |

Fig. 5.19 Flows for "aluminium, secondary, from new scrap, at plant" and their representation in the ecoinvent database

| General Flow information |                                                        |                                    |                                  | Representation in ecoinvent |                         |                |          |                                                            |            |      | Uncertainty       |      |          |                                             |
|--------------------------|--------------------------------------------------------|------------------------------------|----------------------------------|-----------------------------|-------------------------|----------------|----------|------------------------------------------------------------|------------|------|-------------------|------|----------|---------------------------------------------|
| Input                    | Process Name                                           | Output                             | Remarks                          | Category                    | Sub category            | Infrastructure | Location | Modul name in ecoinvent                                    | Mean value | Unit | Source mean value | Type | StDv 95% | General Comment                             |
| Cooling water            | aluminium, secondary, from new scrap, at plant; part 2 |                                    |                                  | resource                    | in water                |                |          | Water, cooling, unspecified natural origin                 | 7.80E-02   | m3   | EAA 2000          | 1    | 1.08     | (1,1,2,1,1,3,4)                             |
| argon                    |                                                        |                                    | 1.664 kg/m3                      | chemicals                   | inorganics              | No             | RER      | argon, liquid, at plant                                    | 2.50E-03   | kg   | EAA 2000          | 1    | 1.08     | (1,1,2,1,1,3,4)                             |
| nitrogen                 |                                                        |                                    | 1.251 kg/m3                      | chemicals                   | inorganics              | No             | RER      | nitrogen, liquid, at plant                                 | 1.25E-03   | kg   | EAA 2000          | 1    | 1.08     | (1,1,2,1,1,3,4)                             |
| chlorine                 |                                                        |                                    | 3.214 kg/m3                      | chemicals                   | inorganics              | No             | RER      | chlorine, liquid, production mix, at plant                 | 3.86E-05   | kg   | EAA 2000          | 1    | 1.08     | (1,1,2,1,1,3,4)                             |
| salt                     |                                                        |                                    |                                  | chemicals                   | inorganics              | No             | RER      | sodium chloride, powder, at plant                          | 8.90E-04   | kg   | EAA 2000          | 1    | 1.08     | (1,1,2,1,1,3,4)                             |
| refractories             |                                                        |                                    |                                  | construction materials      | bricks                  | No             | DE       | refractory, fireclay, packed, at plant                     | 1.40E-03   | kg   | EAA 2000          | 1    | 1.08     | (1,1,2,1,1,3,4)                             |
| Infrastructure melting   |                                                        |                                    |                                  | metals                      | extraction              | Yes            | RER      | aluminium melting furnace                                  | 2.00E-09   | unit | estimated         | 1    | 3.23     | (5,n.A.,n.A.,n.A.,n.A.,n.A.,9); rough guess |
| Infrastructure casting   |                                                        |                                    |                                  | metals                      | extraction              | Yes            | RER      | aluminium casting, plant                                   | 1.54E-10   | unit | estimated         | 1    | 3.23     | (5,n.A.,n.A.,n.A.,n.A.,n.A.,9); rough guess |
|                          |                                                        | HCl                                |                                  | air                         | unspecified             |                |          | Hydrogen chloride                                          | 3.30E-05   | kg   | EAA 2000          | 1    | 1.51     | (1,1,2,1,1,3,31)                            |
|                          |                                                        | chlorides                          |                                  | air                         | unspecified             |                |          | Halogenated hydrocarbons, chlorinated                      | 3.90E-06   | kg   | EAA 2000          | 1    | 1.51     | (1,1,2,1,1,3,31)                            |
|                          |                                                        | non hazardous waste                |                                  | waste management            | inert material landfill | No             | CH       | disposal, inert waste, 5%water, to inert material landfill | 1.50E-03   | kg   | EAA 2000          | 1    | 1.08     | (1,1,2,1,1,3,6)                             |
|                          |                                                        | skimmings                          | to recycling --> not inventoried |                             |                         |                |          |                                                            | 3.27E-02   | kg   | EAA 2000          |      |          |                                             |
|                          |                                                        | secondary aluminium from new scrap |                                  | metals                      | extraction              | No             | RER      | aluminium, secondary, from new scrap, at plant             | 1.00E+00   | kg   |                   |      |          |                                             |

Fig. 5.19 Flows for "aluminium, secondary, from new scrap, at plant" and their representation in the ecoinvent database (cont.)

| General Flow information                                       |                           |                                        |         | Representation in ecoinvent                |                        |                        |                           |                                                                 |                                                  |          | Uncertainty          |           |             |                                |                                |
|----------------------------------------------------------------|---------------------------|----------------------------------------|---------|--------------------------------------------|------------------------|------------------------|---------------------------|-----------------------------------------------------------------|--------------------------------------------------|----------|----------------------|-----------|-------------|--------------------------------|--------------------------------|
| Input                                                          | Process Name              | Output                                 | Remarks | Cate<br>gory                               | Sub<br>category        | Infra<br>struc<br>ture | Loca<br>tion              | Modul name in<br>ecoinvent                                      | Mean<br>value                                    | Unit     | Source mean<br>value | Typ<br>e  | StdV<br>95% | General<br>Comment             |                                |
| Total land use (transformation)                                | aluminium melting furnace |                                        |         | resource                                   | land                   |                        |                           | Transformation, from unknown                                    | 3.75E+04                                         | m2       | estimated            | 1         | 2.23        | (5,n.A.,n.A.,n.A.,n.A.,n.A.,8) |                                |
| Land occupation for buildings and open bauxite storage area    |                           |                                        |         | assumption: 80%of total area is built up   | resource               | land                   |                           |                                                                 | Occupation, industrial area, built up            | 1.50E+06 | m2a                  | estimated | 1           | 1.77                           | (5,n.A.,n.A.,n.A.,n.A.,n.A.,7) |
| Land transformation to buildings and open bauxite storage area |                           |                                        |         |                                            | resource               | land                   |                           |                                                                 | Transformation, to industrial area, built up     | 3.00E+04 | m2                   | estimated | 1           | 2.23                           | (5,n.A.,n.A.,n.A.,n.A.,n.A.,8) |
| Land occupation for vegetation area                            |                           |                                        |         | assumption: 20%of total area is vegetation | resource               | land                   |                           |                                                                 | Occupation, industrial area, vegetation          | 3.75E+05 | m2a                  | estimated | 1           | 1.77                           | (5,n.A.,n.A.,n.A.,n.A.,n.A.,7) |
| Land transformation to vegetation area                         |                           |                                        |         |                                            | resource               | land                   |                           |                                                                 | Transformation, to industrial area, vegetation   | 7.51E+03 | m2                   | estimated | 1           | 2.23                           | (5,n.A.,n.A.,n.A.,n.A.,n.A.,8) |
| Building, hall                                                 |                           |                                        |         |                                            | construction processes | buildings              | Yes                       | CH                                                              | building, hall                                   | 9.38E+02 | m2                   | estimated | 1           | 3.23                           | (5,n.A.,n.A.,n.A.,n.A.,n.A.,9) |
| heavy machine, unspecific                                      |                           |                                        |         |                                            | construction processes | machinery              | Yes                       | RER                                                             | Industrial machine, heavy, unspecified, at plant | 1.54E+05 | kg                   | estimated | 1           | 3.23                           | (5,n.A.,n.A.,n.A.,n.A.,n.A.,9) |
| Poor concrete for additional foundation                        |                           |                                        |         |                                            | construction materials | concrete               | No                        | CH                                                              | concrete, sole plate and foundation, at plant    | 6.99E+01 | m3                   | estimated | 1           | 3.23                           | (5,n.A.,n.A.,n.A.,n.A.,n.A.,9) |
|                                                                |                           | foundation concrete to disposal        |         | waste management                           | building demolition    | No                     | CH                        | disposal, building, concrete, not reinforced, to final disposal | 1.54E+05                                         | kg       | estimated            | 1         | 3.23        | (5,n.A.,n.A.,n.A.,n.A.,n.A.,9) |                                |
|                                                                | aluminium melting furnace | Output: 10000000 kg/a, life time: 50 a | metals  | extraction                                 | Yes                    | RER                    | aluminium melting furnace | 1.00E+00                                                        | unit                                             |          |                      |           |             |                                |                                |

Fig. 5.20 Flows for "aluminium melting furnace" and their representation in the ecoinvent database

| General Flow information                   |                                        |                     |                                                             | Representation in ecoinvent |                 |                |          |                                                       |            |      | Uncertainty         |      |          |                                        |
|--------------------------------------------|----------------------------------------|---------------------|-------------------------------------------------------------|-----------------------------|-----------------|----------------|----------|-------------------------------------------------------|------------|------|---------------------|------|----------|----------------------------------------|
| Input                                      | Process Name                           | Output              | Remarks                                                     | Category                    | Sub category    | Infrastructure | Location | Module name in ecoinvent                              | Mean value | Unit | Source mean value   | Type | StdV 95% | General Comment                        |
| old scrap from consumers (including water) | aluminium scrap, old, at plant; part 1 |                     | from recycling --> not inventoried                          |                             |                 |                |          |                                                       | 1.23E+00   | kg   | EAA 2000            |      |          |                                        |
| Transport of inputs to plant               |                                        |                     | road: everything 100 km                                     | transport systems           | road            | No             | RER      | transport, lorry 32t                                  | 1.23E-01   | tkm  | standard distance   | 1    | 2.09     | (4,5,nA.,nA.,nA.,nA.,5)                |
| Transport of inputs to plant               |                                        |                     | rail: metals / lime: 200 km, rest 600 km                    | transport systems           | train           | No             | RER      | transport, freight, rail                              | 2.46E-01   | tkm  | standard distance   | 1    | 2.09     | (4,5,nA.,nA.,nA.,nA.,5)                |
| electrical                                 |                                        |                     |                                                             | electricity                 | production mix  | No             | UCTE     | electricity, medium voltage, production UCTE, at grid | 6.40E-02   | kWh  | EAA 2000            | 1    | 1.08     | (1,1,2,1,1,3,2)                        |
| thermal from heavy fuel oil                |                                        |                     |                                                             | oil                         | heating systems | No             | RER      | heat, heavy fuel oil, at industrial furnace 1M W      | 1.66E-01   | MJ   | EAA 2000            | 1    | 1.08     | (1,1,2,1,1,3,1)                        |
| thermal from gas                           |                                        |                     |                                                             | natural gas                 | heating systems | No             | RER      | heat, natural gas, at boiler modulating >100kW        | 1.02E+00   | MJ   | EAA 2000            | 1    | 1.08     | (1,1,2,1,1,3,1)                        |
| detergent                                  |                                        |                     |                                                             | chemicals                   | organics        | No             | GLO      | chemicals organic, at plant                           | 2.60E-04   | kg   | EAA 2000            | 1    | 1.22     | (1,1,2,1,3,3,4)                        |
| hydraulic oil                              |                                        |                     |                                                             | chemicals                   | organics        | No             | RER      | lubricating oil, at plant                             | 7.30E-06   | kg   | EAA 2000            | 1    | 1.22     | (1,1,2,1,3,3,4)                        |
| light oil                                  |                                        |                     |                                                             | oil                         | fuels           | No             | RER      | light fuel oil, at regional storage                   | 8.30E-05   | kg   | EAA 2000            | 1    | 1.08     | (1,1,2,1,1,3,4)                        |
| alloying additives                         |                                        |                     | Copper as proxy                                             | metals                      | extraction      | No             | RER      | copper, at regional storage                           | 9.60E-04   | kg   | EAA 2000            | 1    | 1.22     | (1,1,2,1,3,3,4)                        |
| ferrosilicon                               |                                        |                     | Silicon as proxy                                            | metals                      | extraction      | No             | NO       | M G-silicon, at plant                                 | 5.80E-04   | kg   | EAA 2000            | 1    | 1.22     | (1,1,2,1,3,3,4)                        |
| lime                                       |                                        |                     |                                                             | construction materials      | binder          | No             | CH       | lime, hydrated, packed, at plant                      | 2.80E-04   | kg   | EAA 2000            | 1    | 1.08     | (1,1,2,1,1,3,4)                        |
| Infrastructure (plant) [piece]             |                                        |                     | Output: 10000000 kg/a, life time: 50 a                      | metals                      | extraction      | Yes            | RER      | scrap preparation plant                               | 2.00E-09   | unit | estimated           | 1    | 3.23     | (5,nA.,nA.,nA.,nA.,nA.,9); rough guess |
|                                            |                                        | chloride-gas in air |                                                             | air                         | unspecified     |                |          | Halogenated hydrocarbons, chlorinated                 | 4.60E-07   | kg   | EAA 2000            | 1    | 1.51     | (1,1,2,1,1,3,31)                       |
|                                            |                                        | dust                | total dust: 0.000029, 32%<2.5 um, 48%>2.5<10 um, 20%> 10 um | air                         | unspecified     |                |          | Particulates, > 10 um                                 | 9.28E-06   | kg   | EAA 2000 / cep meip | 1    | 1.51     | (1,1,2,1,1,3,25)                       |
|                                            |                                        | dust                |                                                             | air                         | unspecified     |                |          | Particulates, > 2.5 um, and < 10 um                   | 1.39E-05   | kg   | EAA 2000 / cep meip | 1    | 2.00     | (1,1,2,1,1,3,26)                       |
|                                            |                                        | dust                |                                                             | air                         | unspecified     |                |          | Particulates, < 2.5 um                                | 5.80E-06   | kg   | EAA 2000 / cep meip | 1    | 3.00     | (1,1,2,1,1,3,27)                       |

Fig. 5.21 Flows for "aluminium scrap, old, at plant" and their representation in the ecoinvent database



| General Flow information               |              |                           |                                                          | Representation in ecoinvent |                            |                |          |                                                                               |            |      | Uncertainty       |      |          |                   |
|----------------------------------------|--------------|---------------------------|----------------------------------------------------------|-----------------------------|----------------------------|----------------|----------|-------------------------------------------------------------------------------|------------|------|-------------------|------|----------|-------------------|
| Input                                  | Process Name | Output                    | Remarks                                                  | Category                    | Sub category               | Infrastructure | Location | Module name in ecoinvent                                                      | Mean value | Unit | Source mean value | Type | StdV 95% | General Comment   |
| aluminium scrap, old, at plant; part 2 | m            | HCl(hydrogen chloride)    |                                                          | air                         | unspecified                |                |          | Hydrogen chloride                                                             | 1.10E-05   | kg   | EAA 2000          | 1    | 1.51     | (1,1,2,1,1,3,3,1) |
|                                        | m            | oil (waste)               |                                                          | waste management            | underground deposit        | No             | DE       | disposal, hazardous waste, 0%water, to underground deposit                    | 2.60E-03   | kg   | EAA 2000          | 1    | 1.08     | (1,1,2,1,1,3,6)   |
|                                        | m            | ball mill dust (waste)    | composition unknown, hazardous waste deposition assumed  | waste management            | underground deposit        | No             | DE       | disposal, hazardous waste, 0%water, to underground deposit                    | 4.81E-02   | kg   | EAA 2000          | 1    | 1.08     | (1,1,2,1,1,3,6)   |
|                                        | m            | by-product: Al-Mg         | to other use --> not inventoried                         |                             |                            |                |          |                                                                               | 8.30E-04   | kg   | EAA 2000          |      |          |                   |
|                                        | m            | by-product: iron scrap    | to other use --> not inventoried                         |                             |                            |                |          |                                                                               | 9.52E-03   | kg   | EAA 2000          |      |          |                   |
|                                        | m            | dirt                      |                                                          | waste management            | inert material landfill    | No             | CH       | disposal, inert waste, 5%water, to inert material landfill                    | 1.87E-03   | kg   | EAA 2000          | 1    | 1.08     | (1,1,2,1,1,3,6)   |
|                                        | m            | filter dust               | composition and treatment as filter dust of electrolysis | waste management            | residual material landfill | No             | CH       | disposal, filter dust Al electrolysis, 0%water, to residual material landfill | 3.28E-03   | kg   | EAA 2000          | 1    | 1.08     | (1,1,2,1,1,3,6)   |
|                                        | m            | refractory waste          |                                                          | waste management            | inert material landfill    | No             | CH       | disposal, inert waste, 5%water, to inert material landfill                    | 8.70E-05   | kg   | EAA 2000          | 1    | 1.08     | (1,1,2,1,1,3,6)   |
|                                        | m            | rubber                    |                                                          | waste management            | municipal incineration     | No             | CH       | disposal, rubber, unspecified, 0%water, to municipal incineration             | 2.36E-02   | kg   | EAA 2000          | 1    | 1.08     | (1,1,2,1,1,3,6)   |
|                                        | m            | solid waste - unspecified |                                                          | waste management            | municipal incineration     | No             | CH       | disposal, municipal solid waste, 22.9% water, to municipal incineration       | 3.23E-03   | kg   | EAA 2000          | 1    | 1.08     | (1,1,2,1,1,3,6)   |
|                                        | m            | stones                    |                                                          | waste management            | inert material landfill    | No             | CH       | disposal, inert waste, 5%water, to inert material landfill                    | 4.16E-03   | kg   | EAA 2000          | 1    | 1.08     | (1,1,2,1,1,3,6)   |
|                                        | m            | waste filter material     | composition unknown, hazardous waste deposition assumed  | waste management            | underground deposit        | No             | DE       | disposal, hazardous waste, 0%water, to underground deposit                    | 4.40E-06   | kg   | EAA 2000          | 1    | 1.08     | (1,1,2,1,1,3,6)   |
|                                        | m            | waste sediment            | composition unknown, hazardous waste deposition assumed  | waste management            | underground deposit        | No             | DE       | disposal, hazardous waste, 0%water, to underground deposit                    | 3.33E-03   | kg   | EAA 2000          | 1    | 1.08     | (1,1,2,1,1,3,6)   |
|                                        | m            | Non ferrous metal scrap   | to other use --> not inventoried                         |                             |                            |                |          |                                                                               | 4.04E-02   | kg   | EAA 2000          |      |          |                   |
|                                        | m            | aluminium scrap, old      |                                                          | metals                      | extraction                 | No             | RER      | aluminium scrap, old, at plant                                                | 1.00E+00   | kg   |                   |      |          |                   |

Fig. 5.21 Flows for "aluminium scrap, old, at plant" and their representation in the ecoinvent database (cont.)

# 5. System Characterisation

| General Flow information     |                                                        |        |                                                                     | Representation in ecoinvent |              |                |                                                        |                                       |            |          |                   | Uncertainty |                 |                                             |  |
|------------------------------|--------------------------------------------------------|--------|---------------------------------------------------------------------|-----------------------------|--------------|----------------|--------------------------------------------------------|---------------------------------------|------------|----------|-------------------|-------------|-----------------|---------------------------------------------|--|
| Input                        | Process Name                                           | Output | Remarks                                                             | Category                    | Sub category | Infrastructure | Location                                               | Module name in ecoinvent              | Mean value | Unit     | Source mean value | Type        | StdV 95%        | General Comment                             |  |
| Prepared old scrap           | aluminium, secondary, from old scrap, at plant; Inputs |        | aluminium prepared scrap (net from 1110 kg - 76.65 kg al metallics) | metals                      | extraction   | No             | RER                                                    | aluminium scrap, old, at plant        | 1.03E+00   | kg       | EAA 2000          | 1           | 108             | (1,1,2,1,1,3,3)                             |  |
| electrical                   |                                                        |        | electricity                                                         | production mix              | No           | UCTE           | electricity, medium voltage, production UCTE, at grid  | 2.88E-01                              | kWh        | EAA 2000 | 1                 | 108         | (1,1,2,1,1,3,2) |                                             |  |
| thermal from heavy oil       |                                                        |        | oil                                                                 | heating systems             | No           | RER            | heat, heavy fuel oil, at industrial furnace 1M W       | 5.13E-01                              | MJ         | EAA 2000 | 1                 | 108         | (1,1,2,1,1,3,1) |                                             |  |
| thermal from gas             |                                                        |        | natural gas                                                         | heating systems             | No           | RER            | heat, natural gas, at boiler modulating >100kW         | 8.27E+00                              | MJ         | EAA 2000 | 1                 | 108         | (1,1,2,1,1,3,1) |                                             |  |
| chlorine                     |                                                        |        | chemicals                                                           | inorganics                  | No           | RER            | chlorine, liquid, production mix, at plant             | 1.60E-03                              | kg         | EAA 2000 | 1                 | 108         | (1,1,2,1,1,3,4) |                                             |  |
| nitrogen                     |                                                        |        | chemicals                                                           | inorganics                  | No           | RER            | nitrogen, liquid, at plant                             | 1.80E-03                              | kg         | EAA 2000 | 1                 | 108         | (1,1,2,1,1,3,4) |                                             |  |
| hydrochloric acid            |                                                        |        | chemicals                                                           | inorganics                  | No           | RER            | hydrochloric acid, 30% in H2O, at plant                | 2.00E-04                              | kg         | EAA 2000 | 1                 | 108         | (1,1,2,1,1,3,4) |                                             |  |
| light oil                    |                                                        |        | oil                                                                 | fuels                       | No           | RER            | light fuel oil, at regional storage                    | 2.40E-06                              | kg         | EAA 2000 | 1                 | 108         | (1,1,2,1,1,3,4) |                                             |  |
| sodium hydroxide             |                                                        |        | chemicals                                                           | inorganics                  | No           | RER            | sodium hydroxide, 50% in H2O, production mix, at plant | 1.59E-03                              | kg         | EAA 2000 | 1                 | 108         | (1,1,2,1,1,3,4) |                                             |  |
| sulphuric acid               |                                                        |        | chemicals                                                           | inorganics                  | No           | RER            | sulphuric acid, liquid, at plant                       | 7.96E-03                              | kg         | EAA 2000 | 1                 | 108         | (1,1,2,1,1,3,4) |                                             |  |
| water (consumed)             |                                                        |        | resource                                                            | in water                    |              |                | Water, unspecified natural origin                      | 7.96E-03                              | m3         | EAA 2000 | 1                 | 108         | (1,1,2,1,1,3,4) |                                             |  |
| Silicon as alloying additive |                                                        |        | metals                                                              | extraction                  | No           | NO             | M G-silicon, at plant                                  | 1.23E-02                              | kg         | EAA 2000 | 1                 | 108         | (1,1,2,1,1,3,4) |                                             |  |
| Copper as alloying additive  |                                                        |        | metals                                                              | extraction                  | No           | RER            | copper, at regional storage                            | 7.50E-04                              | kg         | EAA 2000 | 1                 | 108         | (1,1,2,1,1,3,4) |                                             |  |
| other alloying additives     |                                                        |        | Zinc as proxy                                                       | metals                      | extraction   | No             | RER                                                    | zinc for coating, at regional storage | 6.50E-02   | kg       | EAA 2000          | 1           | 122             | (1,1,2,1,3,3,4)                             |  |
| lime                         |                                                        |        |                                                                     | construction materials      | binder       | No             | CH                                                     | lime, hydrated, packed, at plant      | 7.41E-03   | kg       | EAA 2000          | 1           | 108             | (1,1,2,1,1,3,4)                             |  |
| salt                         |                                                        |        |                                                                     | chemicals                   | inorganics   | No             | RER                                                    | sodium chloride, powder, at plant     | 1.37E-02   | kg       | EAA 2000          | 1           | 108             | (1,1,2,1,1,3,4)                             |  |
| Transport of inputs to plant |                                                        |        | road: metals, lime & chemicals 100 km, gases 50 km                  | transport systems           | road         | No             | RER                                                    | transport, lorry 32t                  | 1.15E-01   | tkm      | standard distance | 1           | 2.09            | (4,5,n.A.,n.A.,n.A.,n.A.,5)                 |  |
| Transport of inputs to plant |                                                        |        | rail: metals & lime 200 km, chemicals 600 km, gases 100 km          | transport systems           | train        | No             | RER                                                    | transport, freight, rail              | 2.39E-01   | tkm      | standard distance | 1           | 2.09            | (4,5,n.A.,n.A.,n.A.,n.A.,5)                 |  |
| Infrastructure melting       |                                                        |        |                                                                     | metals                      | extraction   | Yes            | RER                                                    | aluminium melting furnace             | 2.00E-09   | unit     | standard distance | 1           | 3.23            | (5,n.A.,n.A.,n.A.,n.A.,n.A.,9); rough guess |  |
| Infrastructure casting       |                                                        |        |                                                                     | metals                      | extraction   | Yes            | RER                                                    | aluminium casting, plant              | 1.54E-10   | unit     | estimated         | 1           | 3.23            | (5,n.A.,n.A.,n.A.,n.A.,n.A.,9); rough guess |  |

Fig. 5.22 Flows for "aluminium, secondary, from old scrap, at plant" and their representation in the ecoinvent database

| General Flow information                        |              |                                      |                                                             | Representation in ecoinvent |                            |                |          |                                                                               |            |      | Uncertainty        |      |          |                   |
|-------------------------------------------------|--------------|--------------------------------------|-------------------------------------------------------------|-----------------------------|----------------------------|----------------|----------|-------------------------------------------------------------------------------|------------|------|--------------------|------|----------|-------------------|
| Input                                           | Process Name | Output                               | Remarks                                                     | Category                    | Sub category               | Infrastructure | Location | Module name in ecoinvent                                                      | Mean value | Unit | Source mean value  | Type | StdV 95% | General Comment   |
| aluminium, secondary, from old scrap, at plant; | Outputs      | ammonia (NH <sub>3</sub> )           | emissions to air                                            | air                         | unspecified                |                |          | Ammonia                                                                       | 2.00E-05   | kg   | EAA 2000           | 1    | 1.51     | (1,1,2,1,1,3,3,1) |
|                                                 |              | chloride-gas in air                  |                                                             | air                         | unspecified                |                |          | Halogenated hydrocarbons, chlorinated                                         | 5.00E-05   | kg   | EAA 2000           | 1    | 1.51     | (1,1,2,1,1,3,3,1) |
|                                                 |              | chlorine-gas in air                  |                                                             | air                         | unspecified                |                |          | Chlorine                                                                      | 4.90E-07   | kg   | EAA 2000           | 1    | 1.51     | (1,1,2,1,1,3,3,1) |
|                                                 |              | dust                                 | total dust: 0.000024, 32%<2.5 um, 48%>2.5<10 um, 20%> 10 um | air                         | unspecified                |                |          | Particulates, > 10 um                                                         | 7.68E-07   | kg   | EAA 2000 / cepmeip | 1    | 1.51     | (1,1,2,1,1,3,2,5) |
|                                                 |              | dust                                 |                                                             | air                         | unspecified                |                |          | Particulates, > 2.5 um, and < 10um                                            | 1.15E-06   | kg   | EAA 2000 / cepmeip | 1    | 2.00     | (1,1,2,1,1,3,2,6) |
|                                                 |              | dust                                 |                                                             | air                         | unspecified                |                |          | Particulates, < 2.5 um                                                        | 4.80E-07   | kg   | EAA 2000 / cepmeip | 1    | 3.00     | (1,1,2,1,1,3,2,7) |
|                                                 |              | H <sub>2</sub> S (hydrogen sulphide) |                                                             | air                         | unspecified                |                |          | Hydrogen sulfide                                                              | 2.80E-06   | kg   | EAA 2000           | 1    | 1.51     | (1,1,2,1,1,3,3,1) |
|                                                 |              | HCl(hydrogen chloride)               |                                                             | air                         | unspecified                |                |          | Hydrogen chloride                                                             | 4.10E-06   | kg   | EAA 2000           | 1    | 1.51     | (1,1,2,1,1,3,3,1) |
|                                                 |              | HF (hydrogen fluoride)               |                                                             | air                         | unspecified                |                |          | Hydrogen fluoride                                                             | 3.50E-06   | kg   | EAA 2000           | 1    | 1.51     | (1,1,2,1,1,3,3,1) |
|                                                 |              | nitrogen                             | not inventoried                                             |                             |                            |                |          |                                                                               | 2.45E-03   | kg   | EAA 2000           |      |          |                   |
|                                                 |              | phosphine                            | Phosphorus as proxy                                         | air                         | unspecified                |                |          | Phosphorus                                                                    | 5.20E-07   | kg   | EAA 2000           | 1    | 1.08     | (1,1,2,1,1,3,6)   |
|                                                 |              | ball mill dust                       | composition unknown, hazardous waste deposition assumed     | waste management            | underground deposit        | No             | DE       | disposal, hazardous waste, 0%water, to underground deposit                    | 1.45E-02   | kg   | EAA 2000           | 1    | 1.08     | (1,1,2,1,1,3,6)   |
|                                                 |              | by-product: iron scrap               | not inventoried                                             |                             |                            |                |          |                                                                               | 2.18E-03   | kg   | EAA 2000           |      |          |                   |
|                                                 |              | filter dust                          |                                                             | waste management            | residual material landfill | No             | CH       | disposal, filter dust Al electrolysis, 0%water, to residual material landfill | 9.63E-03   | kg   | EAA 2000           | 1    | 1.08     | (1,1,2,1,1,3,6)   |
|                                                 |              | refractory waste                     |                                                             | waste management            | inert material landfill    | No             | CH       | disposal, inert waste, 5%water, to inert material landfill                    | 2.02E-03   | kg   | EAA 2000           | 1    | 1.08     | (1,1,2,1,1,3,6)   |
|                                                 |              | solid waste - unspecified            |                                                             | waste management            | municipal incineration     | No             | CH       | disposal, municipal solid waste, 22.9% water, to municipal incineration       | 8.00E-05   | kg   | EAA 2000           | 1    | 1.08     | (1,1,2,1,1,3,6)   |
|                                                 |              | Aluminium oxide                      | not inventoried                                             |                             |                            |                |          |                                                                               | 1.19E-01   | kg   | EAA 2000           |      |          |                   |
|                                                 |              | secondary aluminium from old scrap   |                                                             | metals                      | extraction                 | No             | RER      | aluminium, secondary, from old scrap, at plant                                | 1.00E+00   | kg   |                    |      |          |                   |

Fig. 5.22 Flows for "aluminium, secondary, from old scrap, at plant" and their representation in the ecoinvent database (cont.)

| General Flow information                                       |    |                         |                                 | Representation in ecoinvent                |                        |                     |                |          |                                                                 |                             |          | Uncertainty       |           |          |                           |                           |
|----------------------------------------------------------------|----|-------------------------|---------------------------------|--------------------------------------------|------------------------|---------------------|----------------|----------|-----------------------------------------------------------------|-----------------------------|----------|-------------------|-----------|----------|---------------------------|---------------------------|
| Input                                                          |    | Process Name            | Output                          | Remarks                                    | Category               | Sub category        | Infrastructure | Location | Module name in ecoinvent                                        | Mean value                  | Unit     | Source mean value | Type      | StdV 95% | General Comment           |                           |
| Total land use (transformation)                                | ML | scrap preparation plant |                                 |                                            | resource               | land                |                |          | Transformation, from unknown                                    | 3.75E+04                    | m2       | estimated         | 1         | 2.23     | (5,nA.,nA.,nA.,nA.,nA.,8) |                           |
| Land occupation for buildings and open bauxite storage area    | ML |                         |                                 | assumption: 80%of total area is built up   | resource               | land                |                |          | Occupation, industrial area, built up                           | 1.50E+06                    | m2a      | estimated         | 1         | 1.77     | (5,nA.,nA.,nA.,nA.,nA.,7) |                           |
| Land transformation to buildings and open bauxite storage area | ML |                         |                                 |                                            | resource               | land                |                |          | Transformation, to industrial area, built up                    | 3.00E+04                    | m2       | estimated         | 1         | 2.23     | (5,nA.,nA.,nA.,nA.,nA.,8) |                           |
| Land occupation for vegetation area                            | ML |                         |                                 | assumption: 20%of total area is vegetation | resource               | land                |                |          | Occupation, industrial area, vegetation                         | 3.75E+05                    | m2a      | estimated         | 1         | 1.77     | (5,nA.,nA.,nA.,nA.,nA.,7) |                           |
| Land transformation to vegetation area                         | ML |                         |                                 |                                            | resource               | land                |                |          | Transformation, to industrial area, vegetation                  | 7.51E+03                    | m2       | estimated         | 1         | 2.23     | (5,nA.,nA.,nA.,nA.,nA.,8) |                           |
| Building, multi story                                          | ML |                         |                                 |                                            | construction processes | buildings           | Yes            | RER      | building, multi-storey                                          | 2.50E+04                    | m3       | estimated         | 1         | 3.23     | (5,nA.,nA.,nA.,nA.,nA.,9) |                           |
| Building, hall                                                 | ML |                         |                                 |                                            | construction processes | buildings           | Yes            | CH       | building, hall                                                  | 1.67E+03                    | m2       | estimated         | 1         | 3.23     | (5,nA.,nA.,nA.,nA.,nA.,9) |                           |
| concrete in reinforced concrete                                | ML |                         |                                 |                                            | construction materials | concrete            | No             | CH       | concrete, normal, at plant                                      | 1.91E+01                    | m3       | estimated         | 1         | 3.23     | (5,nA.,nA.,nA.,nA.,nA.,9) |                           |
| steel in reinforced concrete                                   | ML |                         |                                 |                                            |                        | metals              | extraction     | No       | RER                                                             | reinforcing steel, at plant | 1.32E+04 | kg                | estimated | 1        | 3.23                      | (5,nA.,nA.,nA.,nA.,nA.,9) |
| heavy machine, unspecific                                      | ML |                         |                                 |                                            | construction processes | machinery           | Yes            | RER      | Industrial machine, heavy, unspecified, at plant                | 1.54E+05                    | kg       | estimated         | 1         | 3.23     | (5,nA.,nA.,nA.,nA.,nA.,9) |                           |
| Poor concrete for additional foundation                        | ML |                         |                                 |                                            | construction materials | concrete            | No             | CH       | concrete, sole plate and foundation, at plant                   | 6.99E+01                    | m3       | estimated         | 1         | 3.23     | (5,nA.,nA.,nA.,nA.,nA.,9) |                           |
| Conveyor belt                                                  | ML |                         |                                 |                                            | construction processes | machinery           | Yes            | RER      | conveyor belt, at plant                                         | 2.00E+02                    | m        | estimated         | 1         | 3.23     | (5,nA.,nA.,nA.,nA.,nA.,9) |                           |
|                                                                | ML |                         | reinforced concrete to disposal |                                            | waste management       | building demolition | No             | CH       | disposal, building, reinforced concrete, to final disposal      | 5.53E+04                    | kg       | estimated         | 1         | 3.23     | (5,nA.,nA.,nA.,nA.,nA.,9) |                           |
|                                                                | ML |                         | foundation concrete to disposal |                                            | waste management       | building demolition | No             | CH       | disposal, building, concrete, not reinforced, to final disposal | 1.54E+05                    | kg       | estimated         | 1         | 3.23     | (5,nA.,nA.,nA.,nA.,nA.,9) |                           |
|                                                                | ML |                         | scrap preparation plant         | Output: 10000000 kg/a, life time: 50 a     | metals                 | extraction          | Yes            | RER      | scrap preparation plant                                         | 1.00E+00                    | unit     |                   |           |          |                           |                           |

Fig. 5.23 Flows for "scrap preparation plant" and their representation in the ecoinvent database

## 5.5 Aluminium production mix

The share of primary and secondary metal on the world wide production of different aluminium alloys is given in Tab. 5.17. These mixes are proposed to be used in attributional LCA, if the scope of the study does not imply a special share of secondary material for a certain product. They are used as default within the ecoinvent database. In consequential LCA the value corrected substitution as proposed in Werner (2003); Werner & Richter (2000) might be better suited for modelling the life cycles of aluminium products.

Tab. 5.17 Production mix of "average" aluminium alloys<sup>15</sup>

| Alloy         | primary aluminium | secondary aluminium<br>from new scrap | secondary aluminium<br>from old scrap | Share on production |
|---------------|-------------------|---------------------------------------|---------------------------------------|---------------------|
| Cast alloy    | 20 %              | 47 %                                  | 33 %                                  | 31 %                |
| Wrought alloy | 90 %              | 10 %                                  | 0 %                                   | 69 %                |
| Unspecific    | 68 %              | 22 %                                  | 10 %                                  | 100 %               |

### 5.5.1 Aluminium production mixes in ecoinvent

The background information to the processes are given in, flows are shown in Tab. 5.18:

- aluminium, production mix, cast alloy, at plant Fig. 5.24
- aluminium, production mix, wrought alloy, at plant Fig. 5.25
- aluminium, production mix, at plant Fig. 5.26

#### Data quality

The shares are estimates. Nevertheless they are believed to give a good representation of the overall situation.

<sup>15</sup> Dr. G. Rombach, Project manager Recycling, Hydro Aluminium Deutschland GmbH. Personal communication, 1.10.2002

Tab. 5.18 Meta information for the aluminium production mixes

| Name                                  | aluminium, production mix, cast alloy, at plant                                           | aluminium, production mix, wrought alloy, at plant                                        | aluminium, production mix, at plant                                                       |
|---------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Location                              | RER                                                                                       | RER                                                                                       | RER                                                                                       |
| Infrastructure Process                | 0                                                                                         | 0                                                                                         | 0                                                                                         |
| Unit                                  | kg                                                                                        | kg                                                                                        | kg                                                                                        |
| Data Set Version                      | 2.0                                                                                       | 2.0                                                                                       | 2.0                                                                                       |
| Included Processes                    | Mix of primary and secondary aluminium according to their share on world-wide production. | Mix of primary and secondary aluminium according to their share on world-wide production. | Mix of primary and secondary aluminium according to their share on world-wide production. |
| Amount                                | 1                                                                                         | 1                                                                                         | 1                                                                                         |
| Local Name                            | Aluminium, Produktionsmix, Gusslegierung, ab Werk                                         | Aluminium, Produktionsmix, Knetlegierung, ab Werk                                         | Aluminium, Produktionsmix, ab Werk                                                        |
| Synonyms                              |                                                                                           |                                                                                           |                                                                                           |
| General Comment to reference function |                                                                                           |                                                                                           |                                                                                           |
| Start Date                            | 2002                                                                                      | 2002                                                                                      | 2002                                                                                      |
| End Date                              | 2002                                                                                      | 2002                                                                                      | 2002                                                                                      |
| Data Valid For Entire Period          | 1                                                                                         | 1                                                                                         | 1                                                                                         |
| Geography text                        | World-wide production mix as proxy for European consumption                               | World-wide production mix as proxy for European consumption                               | World-wide production mix as proxy for European consumption                               |
| Technology text                       | Mix                                                                                       | Mix                                                                                       | Mix                                                                                       |
| Representativeness [%]                |                                                                                           |                                                                                           |                                                                                           |
| Production Volume                     | unknown                                                                                   | unknown                                                                                   | unknown                                                                                   |
| Sampling Procedure                    | Interview with expert                                                                     | Interview with expert                                                                     | Interview with expert                                                                     |
| Extrapolations                        | None                                                                                      | None                                                                                      | None                                                                                      |
| Uncertainty Adjustments               | None                                                                                      | None                                                                                      | None                                                                                      |

| General Flow information           |                                                 |                               |         | Representation in ecoinvent |              |                |          |                                                 |            |      | Uncertainty                                                                                                    |      |          |                 |
|------------------------------------|-------------------------------------------------|-------------------------------|---------|-----------------------------|--------------|----------------|----------|-------------------------------------------------|------------|------|----------------------------------------------------------------------------------------------------------------|------|----------|-----------------|
| Input                              | Process Name                                    | Output                        | Remarks | Category                    | Sub category | Infrastructure | Location | Module name in ecoinvent                        | Mean value | Unit | Source mean value                                                                                              | Type | StdV 95% | General Comment |
| primary aluminium                  | aluminium, production mix, cast alloy, at plant | aluminium mix for cast alloys |         | metals                      | extraction   | No             | RER      | aluminium, primary, at plant                    | 2.00E-01   | kg   | Dr. G. Rombach, Project manager Recycling, Hydro Aluminium Deutschland GmbH. Personal communication, 1.10.2002 | 1    | 124      | (3,1,1,1,5,4)   |
| secondary aluminium from old scrap |                                                 |                               |         | metals                      | extraction   | No             | RER      | aluminium, secondary, from old scrap, at plant  | 3.30E-01   | kg   |                                                                                                                | 1    | 124      | (3,1,1,1,5,4)   |
| secondary aluminium from new scrap |                                                 |                               |         | metals                      | extraction   | No             | RER      | aluminium, secondary, from new scrap, at plant  | 4.70E-01   | kg   |                                                                                                                | 1    | 124      | (3,1,1,1,5,4)   |
|                                    |                                                 |                               |         | metals                      | extraction   | No             | RER      | aluminium, production mix, cast alloy, at plant | 1.00E+00   | kg   |                                                                                                                |      |          |                 |

Fig. 5.24 Flows for "aluminium, production mix, cast alloy, at plant" and their representation in the ecoinvent database

| General Flow information           |                                                    |                                 |         | Representation in ecoinvent |              |                |          |                                                    |            |      | Uncertainty                                                                                                    |      |          |                 |
|------------------------------------|----------------------------------------------------|---------------------------------|---------|-----------------------------|--------------|----------------|----------|----------------------------------------------------|------------|------|----------------------------------------------------------------------------------------------------------------|------|----------|-----------------|
| Input                              | Process Name                                       | Output                          | Remarks | Category                    | Sub category | Infrastructure | Location | Module name in ecoinvent                           | Mean value | Unit | Source mean value                                                                                              | Type | StdV 95% | General Comment |
| primary aluminium                  | aluminium, production mix, wrought alloy, at plant | aluminium mix for wrought alloy |         | metals                      | extraction   | No             | RER      | aluminium, primary, at plant                       | 9.00E-01   | kg   | Dr. G. Rombach, Project manager Recycling, Hydro Aluminium Deutschland GmbH. Personal communication, 1.10.2002 | 1    | 124      | (3,1,1,1,5,4)   |
| secondary aluminium from new scrap |                                                    |                                 |         | metals                      | extraction   | No             | RER      | aluminium, secondary, from new scrap, at plant     | 1.00E-01   | kg   |                                                                                                                | 1    | 124      | (3,1,1,1,5,4)   |
|                                    |                                                    |                                 |         | metals                      | extraction   | No             | RER      | aluminium, production mix, wrought alloy, at plant | 1.00E+00   | kg   |                                                                                                                |      |          |                 |

Fig. 5.25 Flows for "aluminium, production mix, wrought alloy, at plant" and their representation in the ecoinvent database

| General Flow information           |   |                                     |               | Representation in ecoinvent |         |            |              |                |                                                |                              |            | Uncertainty |                                                                                                                |      |               |                 |
|------------------------------------|---|-------------------------------------|---------------|-----------------------------|---------|------------|--------------|----------------|------------------------------------------------|------------------------------|------------|-------------|----------------------------------------------------------------------------------------------------------------|------|---------------|-----------------|
| Input                              |   | Process Name                        |               | Output                      | Remarks | Category   | Sub category | Infrastructure | Location                                       | Module name in ecoinvent     | Mean value | Unit        | Source mean value                                                                                              | Type | StdV 95%      | General Comment |
| primary aluminium                  | m | aluminium, production mix, at plant |               |                             |         | metals     | extraction   | No             | RER                                            | aluminium, primary, at plant | 6.80E-01   | kg          | Dr. G. Rombach, Project manager Recycling, Hydro Aluminium Deutschland GmbH. Personal communication, 1.10.2002 | 1    | 1.24          | (3,1,1,1,5,4)   |
| secondary aluminium from old scrap | m |                                     |               |                             | metals  | extraction | No           | RER            | aluminium, secondary, from old scrap, at plant | 1.04E-01                     | kg         |             | 1                                                                                                              | 1.24 | (3,1,1,1,5,4) |                 |
| secondary aluminium from new scrap | m |                                     |               |                             | metals  | extraction | No           | RER            | aluminium, secondary, from new scrap, at plant | 2.16E-01                     | kg         |             | 1                                                                                                              | 1.24 | (3,1,1,1,5,4) |                 |
|                                    | m |                                     | aluminium mix |                             | metals  | extraction | No           | RER            | aluminium, production mix, at plant            | 1.00E+00                     | kg         |             |                                                                                                                |      |               |                 |

Fig. 5.26 Flows for "aluminium, production mix, at plant" and their representation in the ecoinvent database





## 6 Literature

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## Part II

# Iron and Steel

Data v2.1 (2009)

Author: Hans-Jörg Althaus, EMPA, Dübendorf

Review Roland Hirschler, EMPA, St. Gallen

Update Hans-Jörg Althaus, Mischa Classen, EMPA, Dübendorf

Classen M., Althaus H.-J., Blaser S., Tuchscheid M., Jungbluth N., Doka G., Faist Emmenegger M. and Scharnhorst W. (2009) Life Cycle Inventories of Metals. Final report ecoinvent data v2.1, No 10. EMPA Dübendorf, Swiss Centre for Life Cycle Inventories, Dübendorf, CH, Online-Version under: [www.ecoinvent.ch](http://www.ecoinvent.ch).

Final report of updated Swiss National Life Cycle Inventory Database "ecoinvent", version 2.1

## Comments on version 2.1

This part corresponds to the original report in Althaus et al. (2004) that has been updated with the changes reported in Frischknecht et al. (2006). The changes concerned emission factors for dioxine (factor 1000 reduced in sinter, pellet and pig iron), PAH (factor 1000 lower in pellet) and CO<sub>2</sub> (factor 1000 in pig iron), as electricity demand (factor 3.6 lower in cast iron, electric steel, both chromium and un- / low alloyed) and coke consumption for pig iron (factor 28.6 higher).

In version 2.0, changes have been made to the following datasets:

- iron ore, 65% Fe, at beneficiation / GLO [kg]:  
all water emissions decreased approx. a factor of 100
- sinter, iron, at plant / GLO [kg]:  
PAH and PCB emissions decreased by a factor of 1000. VOC emissions increased slightly due to decreased PAH and PCB.
- pig iron, at plant / GLO [kg]:  
additional demand of 0.12 MJ natural gas,  
higher waste heat generation (accordingly),  
slightly higher transports due to increased coke consumption,  
higher amount of waste water to treatment (factor 1000)  
and higher CO<sub>2</sub> emission (approx. factor 2; newly calculated from C-balance)
- steel, electric, un- and low-alloyed, at plant / RER [kg]  
transports decreased by a factor of approx. 8
- steel, electric, chromium steel 18/8, at plant/ RER [kg]  
transports decreased by a factor of approx. 8
- cast iron, at plant/ RER [kg]  
transports decreased by a factor of approx. 8

Chapter “6 Cumulative Results and Interpretation” of the original report has not been adapted and was deleted for this report. The flow names in the tables displaying unit process data have been updated with the actual names of version 2.0. This affects the designation of namely the transportation service flows that have been changed respective to the preceding version.

## Comments on this version 2.1

This version corresponds to the version 2.0 with a changed composition of low alloyed steel.

## Summary

The steel production is modelled for the European situation, primarily based on data from the description of the current situation in the “Best Available Techniques Reference Document on the Production of Iron and Steel” IPPC (2000) and on process descriptions.

The mining, beneficiation and partly the sinter and pellet production represent a world-wide situation primarily based on Roth et al. (1999).

Some of the datasets described in this report make only sense in the context of the steel production and are therefore not intended for other uses.

The datasets to be chosen to represent iron or steel in a LCA study are either one of the primary (converter) steel datasets (chapter 5.6), one of the secondary (electric) steel datasets (chapter 5.7), the cast iron (chapter 5.8) or one of the production mixes (chapter 5.9). The decision whether to use primary, secondary or a mix of the two depends on the goal and scope of the LCA to be conducted.



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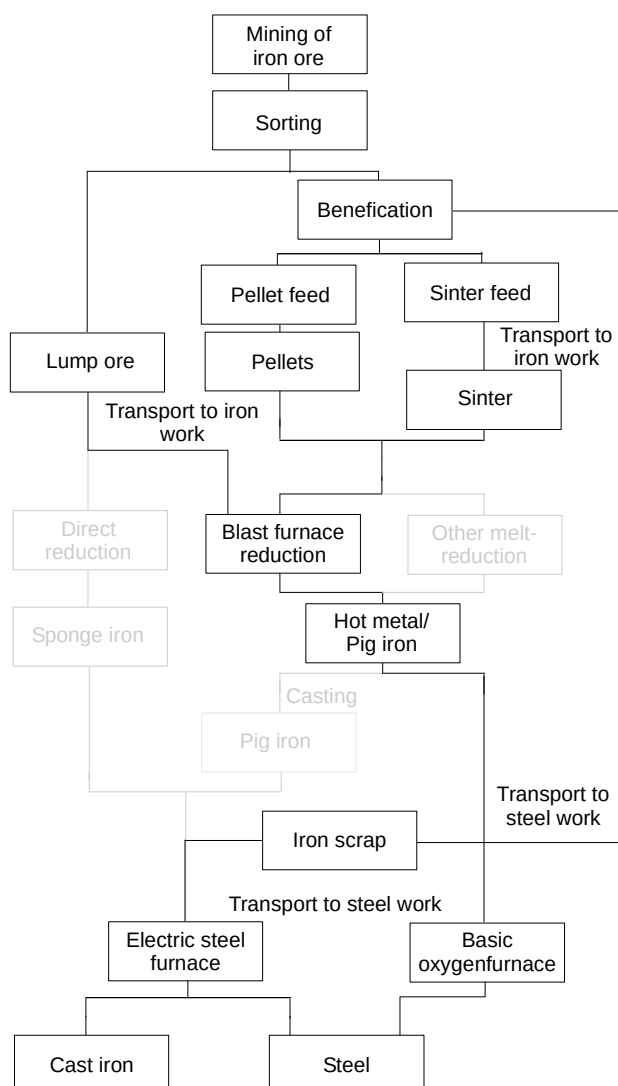


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# 1 Introduction

This part deals with primary and secondary iron and steel production. A rough overview of the cradle to gate processes of the cast iron and steel production are given in Fig. 1.1.



**Fig. 1.1** Overview of cast iron and steel production (The grey materials and processes are of minor relevance and not considered in the ecoinvent data)

## 2 Reserves and Resources of Materials

Iron is the 4<sup>th</sup> most common element in the outer earth crust. It occurs in nature in compounds with other elements. The economically most important iron minerals are hematite and magnetite.

Iron ore is mined mainly in China, Brazil and in Australia. (Tab. 2.1). The yearly iron ore production was about  $1 \cdot 10^9$  t/a in 1996 and the estimated reserves are about  $150 \cdot 10^9$  t. (Roth et al. (1999))

**Tab. 2.1 World wide iron ore production in 1996 (Roth et al. (1999), appendix 2 & 4)**

|                                     | <b>Western Europe</b> | <b>Eastern Europe</b> | <b>Asia</b> | <b>Africa</b> | <b>Northern America</b> | <b>South America</b> | <b>Australia</b> | <b>World</b> |
|-------------------------------------|-----------------------|-----------------------|-------------|---------------|-------------------------|----------------------|------------------|--------------|
| Iron ore [% of world production]    | 3%                    | 12%                   | 34%         | 5%            | 9%                      | 22%                  | 15%              | 100%         |
| Iron in ore [% of world production] | 4%                    | 12%                   | 25%         | 5%            | 11%                     | 26%                  | 17%              | 100%         |
| Iron concentration in ore [%]       | 57%                   | 54%                   | 39%         | 61%           | 64%                     | 65%                  | 63%              | 54%          |

Most of the iron ore is reduced to iron in blast furnaces. About 0.5 kg of coke and smaller amounts of other fossil energy carriers and electricity are needed for the reduction.

Iron and steel scrap are also important resources because of the lower primary energy consumption and lower CO<sub>2</sub> emissions, compared to the blast furnace – converter production route. The share of secondary iron on the European market is 34% according to Frischknecht et al. (1996) and the world average of steel scrap used in steel production is about 40% Steffen (1997).

## 3 Characterisation of the materials

### 3.1 Iron ore

Iron ores are mixtures of up to 400 different minerals. The most important ones are the iron oxides and sulphides. Economically interesting are those ores with high iron content. Two third of the ore mined today come from banded iron formations (BIF). The average iron content of BIF's is between 15% and 45% but the more interesting ores contain up to 68% iron. Although the world average iron content in the ores mined in the years 1975 to 1996 is 56% (calculated from Roth et al. (1999), appendices 1 - 4), the iron content of raw ores used for the input- output tables in Roth et al. (1999) is only 46%. The authors don't mention this discrepancy. It is likely that different methods of calculation are responsible for it. In the ecoinvent inventories we use an iron content of 46% to be compatible with the flow data given in Roth et al. (1999).

An enrichment of this ore to an iron content of 65% in world average is done at the mine. 13% of the enriched ore are big lumps that can be fed to the blast furnace directly. The rest is more or less fine grade ore that has to be agglomerated to sinter or pellets for use in the blast furnace.

### 3.2 Sinter and Pellets

As a rule, blast-furnace burden contains lump ore, sinter, and pellets. Pellets are made from both concentrate and iron-rich fine ore. If the beneficiation process yields very fine-grained concentrates (e.g., <0.1 mm), these are usually converted to pellets at the mine and transported in this form. Sinter is generally produced at the ironworks. The varying loads imposed on material during transport to the blast furnace mean that pellets must satisfy stringent mechanical strength requirements. Pellets are also nearly always made from one well-defined ore or concentrate, whereas sinter is produced from pre-designed mixtures of ores and additives.

Sinter and pellets differ in the way the particles are bound into the agglomerate:

In pellets, the ore particles are connected chiefly by bridges of ore. The formation of iron oxide bridges is supported by additional slag bridges; it must be remembered that the quantity of slag in many pellet qualities is only 3 – 4 % (w/w) (corresponding to 6 – 7 % (v/v)).

Sinter consists of a calcium ferrite matrix, with siliceous and aluminous constituents dissolved and primary hematite embedded in it. Whereas pellets are thoroughly oxidised (a maximum of 0.5 % (w/w) FeO), sinter contains 4 – 6 % (w/w) FeO, which is linked to silica and aluminium oxide in the matrix. (Meiler (1997))

### 3.3 Hot metal / Pig iron

Pig iron is the iron produced in the blast furnace. It is an alloy of 94% Fe and more than 2% C (world average 4.5%). Liquid pig iron is also referred to as "hot metal".

### 3.4 Sponge iron

Sponge iron is produced by direct reduction of iron ore in the solid state. This process is only possible with special ores with high iron content and low content on acid slag components. About 6% of the total world pig iron and sponge iron production is sponge iron.

### 3.5 Cast iron

The term cast iron is used for unalloyed, low-alloy, and highly alloyed iron – carbon cast material with carbon content ca. 2.8 – 4 %. The carbon is for the most part not chemically bonded to the iron, but is

present in elemental form. The properties of cast iron can be modified within wide limits by adding elements such as nickel, chromium, manganese, copper, and silicon, which change the metallic structure. Corrosion-resistant and high-temperature grades are used, e.g., in the chemical industry, in furnaces, and in the automobile industry. The cryogenics industry uses low-temperature grades that do not become brittle at temperatures far below 0 °C. Other important characteristics include its property of damping vibration, its good resistance to corrosion by weathering, and its good machining properties. (Kucharcik (1997))

## 3.6 Steel

Steels are forgeable iron alloys with a carbon content of less than 2.1% (w/w). More than 90% of the world iron production is used for steel making and more than 3500 different grades of steel are available today<sup>1</sup>.

The following iron-bearing materials are used for crude steel production:

- Iron melt from blast furnaces (hot metal), mainly used for steel production by the basic oxygen furnace (BOF) process.
- Scrap, sponge iron, and solidified blast-furnace iron (pig iron), mainly used for melting steel in electric furnaces.

According to the production process, steels can be classified as converter (BOF) or electric steel.

### 3.6.1 Unalloyed steel

Unalloyed steels or carbon steels are those in which the level of an individual element in no case reaches or exceeds the limiting concentration given in Tab. 3.1 (Kalla & Kalwa (1997)). Typical unalloyed steels are the reinforcement steels used in the building industry.

**Tab. 3.1 Limiting concentration of elements in unalloyed steel according to EN 10 020 Kalla & Kalwa (1997)**

| Element            | Al              | B      | Bi   | Co   | Cr <sup>1</sup> | Cu <sup>1</sup> | La             | Mn                | Mo <sup>1</sup> | Nb <sup>2</sup>     |
|--------------------|-----------------|--------|------|------|-----------------|-----------------|----------------|-------------------|-----------------|---------------------|
| max. concentration | 0.10            | 0.0008 | 0.10 | 0.10 | 0.30            | 0.40            | 0.05           | 1.65 <sup>3</sup> | 0.08            | 0.06                |
| Element            | Ni <sup>1</sup> | Pb     | Se   | Si   | Te              | Ti <sup>2</sup> | V <sup>2</sup> | W                 | Zr <sup>2</sup> | Others <sup>4</sup> |
| max. concentration | 0.30            | 0.40   | 0.10 | 0.50 | 0.10            | 0.05            | 0.10           | 0.10              | 0.05            | 0.05                |

<sup>1</sup> If two, three or four of these elements are present in concentrations less than the maximum permitted, their total concentration must not exceed 70% of the sum of the maximum

<sup>2</sup> The same rule applies to these elements

<sup>3</sup> If the manganese content is quoted as minimum, this value applies

<sup>4</sup> Except C, N, O and S

### 3.6.2 Low-alloyed steel

Low-alloy steel contains more than the minimum given in Tab. 3.1 of at least one of the elements and less than 5% alloying elements in total.

<sup>1</sup> <http://www.worldsteel.org/understanding/18>; accessed 15.01.2003

For this project a composition of the average low-alloyed steel is derived from literature and from the production volumes of the different alloying elements. Tab. 3.2 shows the composition and the sources of the individual numbers.

**Tab. 3.2 Composition of low-alloyed steel**

| Element | concentration | Source                                                                                                                                                                                                                                                                                                                 |
|---------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cr      | 2.15%         | Papp 1994 <sup>2</sup> , Tab 19                                                                                                                                                                                                                                                                                        |
| Ni      | 1%            | Estimation                                                                                                                                                                                                                                                                                                             |
| Mn      | 1.1%          | Jones 1994 <sup>3</sup>                                                                                                                                                                                                                                                                                                |
| Mo      | 0.034%        | Calculated from Mo content in recycled steel (26700 t Mo in 78 Mio t steel according to Sibley 2004 <sup>4</sup> )                                                                                                                                                                                                     |
| Nb      | 0.066%        | Estimated for this study. Mean value between 0.004%, calculated from Nb content in recycled steel (2900 t Nb in 78 Mio t steel according to Sibley 2004) and 0.129%, calculated from Nb use in steel making (80% of 10000 t used in 6.19 mio t steel according to USGS Mineral commodity summaries 2008 <sup>5</sup> ) |
| W       | 0.084%        | Calculated from W use in steel making (5200 t used in 6.19 mio t steel according to USGS Mineral commodity summaries 2008)                                                                                                                                                                                             |
| V       | 0.059%        | Calculated from V use in steel making (91% of 4030 t used in 6.19 mio t steel according to USGS Mineral commodity summaries 2008)                                                                                                                                                                                      |
| Cu      | 0%            | Assumption. Cu is not wanted in steel                                                                                                                                                                                                                                                                                  |

### 3.6.3 Stainless steel

Stainless steel is the generic name for a number of different steels used primarily for their resistance to corrosion. The one key element they all share is a minimum of 10.5% (w/w) of chromium. Although other elements, particularly nickel and molybdenum, are added to improve corrosion resistance, chromium is always the deciding factor. The vast majority of steel produced in the world is carbon and alloy steel, with the more expensive stainless steels representing a small, but valuable niche market<sup>6</sup>. Of all the stainless steels the grade 304 steel (18/8 stainless, 18% chromium 8% nickel) is most commonly used. Typical applications for this grade are cutlery, food and chemical production infrastructure and containers and more and more architectural applications.

In this project we balance only 18/8 stainless steel.

<sup>2</sup> Chromium Life Cycle Study. <http://pubs.usgs.gov/usbmic/ic-9411/ic-9411.pdf>; accessed 17.12.2008

<sup>3</sup> Manganese Materials Flow Patterns. <http://pubs.usgs.gov/usbmic/ic-9399/9399.pdf>; accessed 17.12.2008

<sup>4</sup> Flow Studies for Recycling Metal Commodities in the United States. [http://pubs.usgs.gov/circ/2004/1196am/c1196am\\_v2.pdf](http://pubs.usgs.gov/circ/2004/1196am/c1196am_v2.pdf); Accessed 17.12.2008

<sup>5</sup> <http://minerals.usgs.gov/minerals/pubs/mcs/2008/mcs2008.pdf> ; accessed 17.12.2008

<sup>6</sup> [http://www.worldstainless.org/what\\_basic.php](http://www.worldstainless.org/what_basic.php); accessed 15.01.2003

## **4 Use / Application of Material**

### **4.1 Iron ore**

Most of the iron ore is used for iron and steel production.

### **4.2 Pig iron / hot metal**

Most of the pig iron is used for steel production before it is solidified. A smaller part is cast to ingots that are used for cast iron production later.

### **4.3 Cast iron**

About 40 % of cast iron is used in the automobile industry. Other applications include mechanical engineering, the building industry (e.g. radiators, boilers, sanitary ware and pipes), chemical plant, ship-building, and mining. The former demand by steelworks for cast iron ingot moulds has been greatly diminished by the introduction of the continuous casting process. (Kucharcik (1997))

### **4.4 Steel**

Steel is one of the most important materials in the industrialised world. It is widely used in all industries and in many products. The yearly world steel production has risen exponentially between 1940 and 1970 to about 700 million tonnes. Since then it has shown a slight increase to about 850 million tonnes in the year 2000.

#### **4.4.1 Unalloyed and low-alloyed steel**

Unalloyed steel is mainly used as reinforcing steel in the building industry. There are many other applications in industrial machines, vehicles etc. for unalloyed and low-alloyed steel.

#### **4.4.2 Stainless steel**

Stainless steel is often used in the chemical and food industry. Again there are many other applications for stainless steel.

## 5 System Characterisation

### 5.1 General overview

A rough overview of the cradle to gate processes of the cast iron and steel production are given in Fig. 5.1.

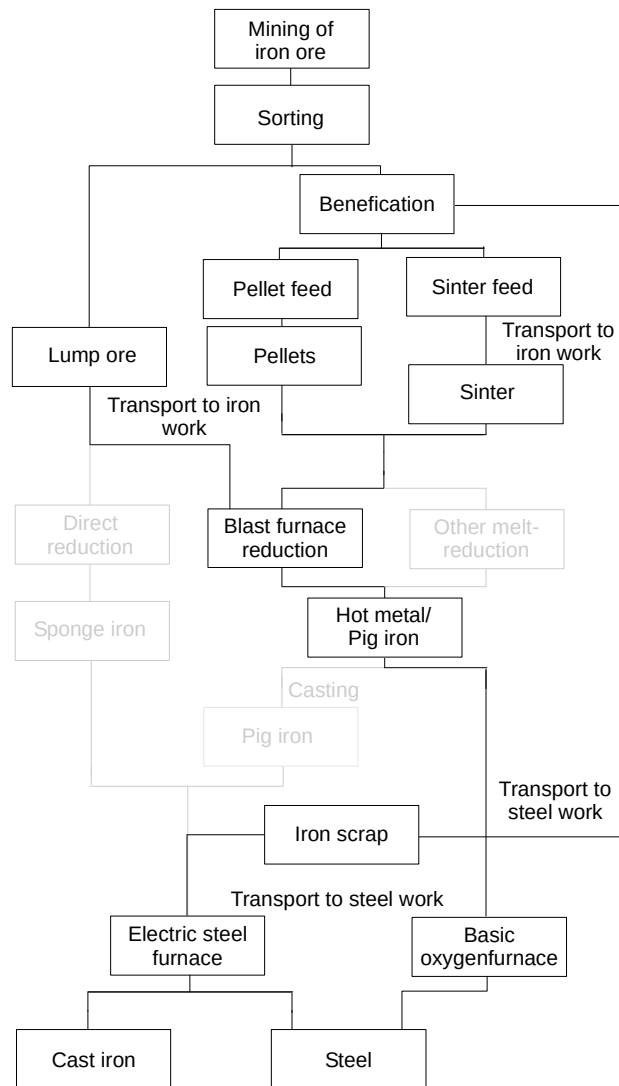


Fig. 5.1 Overview of cast iron and steel production (The grey materials and processes are of minor relevance and not considered in the ecoinvent data)

### 5.2 Iron ore mining and beneficiation

This part is mainly based on (Roth et al. (1999)) and (Meiler (1997)).

#### 5.2.1 Production process and infrastructure

Most of the iron ore today is gained from open mines. The thickness of the recoverable ore deposit usually is in the range of 100 meters. The cap of the deposit is removed as overburden and deposited nearby to be used for the recultivation when the mine has been depleted. The ore is won in stripes us-



ing hydraulic excavators and / or belt loaders. Drilling and blasting is needed in some mines with hard ore.

The only processing steps used for ores mined from rich deposits (rich ores) are crushing and screening. Such ores come mainly from the weathering zones of itabirite deposits. At some mines, washing is combined with the screening of fines ( $< 6 - 8$  mm) for sintering; very fine material containing kaolin is thus separated, and the aluminium oxide content is reduced. Low-grade ore is concentrated to lower transport costs and bring about a chemical composition suitable for the subsequent reduction step (lower silica content). The concentration of iron ore is generally based on physical differences between the iron-bearing and gangue minerals (spec. weight, magnetic properties).

Harmful components present in the ore must be removed. Such components include phosphorus, arsenic, chromium, copper, vanadium, alkali, sulphur and titanium; these either make metallurgical processing more difficult or degrade the iron and steel quality. A small quantity of titanium ( $< 0.5$  %) is, however, added to the burden at some ironworks, because it has been found to increase the life of the blast-furnace brickwork.

### **5.2.2 Emissions**

#### **Air**

The only relevant direct emissions to air from iron ore mining and beneficiation are the dust emissions.

#### **Water**

Heavy metal and hydrocarbon emissions to water occur in the beneficiation process. They are estimated according to the maximum values given in Anonymous (1998)

### **5.2.3 Waste**

There is no relevant waste produced in the mining and beneficiation process. The overburden is used in recultivation of the mine and not treated as waste.

### **5.2.4 Iron ore mining and beneficiation in ecoinvent**

The background information to the processes are given in Tab. 5.1, flows are shown in:

- |                                      |          |
|--------------------------------------|----------|
| - iron ore, 46% Fe, at mine          | Fig. 5.2 |
| - mine, iron                         | Fig. 5.3 |
| - recultivation, iron mine           | Fig. 5.4 |
| - iron ore, 65% Fe, at beneficiation | Fig. 5.5 |

For a lack of information on this topic, the land use, the size and type of buildings and the weight of the stationary machines used in the iron mines are estimated. The estimation is based on information on other mines (gypsum, limestone).

**Tab. 5.1 Meta information for the iron ore mining and beneficiation processes**

|                                       |                                                                                         |                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                         |                                                                                                    |
|---------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Name                                  | iron ore, 46% Fe, at mine                                                               | recultivation, iron mine                                                                                                                                                                           | iron ore, 65% Fe, at beneficiation                                                                                                                                                                                                                                                                                      | mine, iron                                                                                         |
| Location                              | GLO                                                                                     | GLO                                                                                                                                                                                                | GLO                                                                                                                                                                                                                                                                                                                     | GLO                                                                                                |
| Infrastructure Process                | 0                                                                                       | 1                                                                                                                                                                                                  | 0                                                                                                                                                                                                                                                                                                                       | 1                                                                                                  |
| Unit                                  | kg                                                                                      | m2                                                                                                                                                                                                 | kg                                                                                                                                                                                                                                                                                                                      | unit                                                                                               |
| Data Set Version                      | 2.0                                                                                     | 2.0                                                                                                                                                                                                | 2.0                                                                                                                                                                                                                                                                                                                     | 2.0                                                                                                |
| Included Processes                    | Mining with its direct land use and transport of the crude ore to the enrichment plant. | Recultivation and the land transformation.                                                                                                                                                         | Milling and sorting of crude ore.                                                                                                                                                                                                                                                                                       | Infrastructure of the mine (without land use of the open mine itself) and of the enrichment plant. |
| Amount                                | 1                                                                                       | 1                                                                                                                                                                                                  | 1                                                                                                                                                                                                                                                                                                                       | 1                                                                                                  |
| Local Name                            | Eisenerz, 46% Fe, ab Abbau                                                              | Rekultivierung, Eisenmine                                                                                                                                                                          | Eisenerz, 65% Fe, ab Anreicherung                                                                                                                                                                                                                                                                                       | Mine, Eisen                                                                                        |
| Synonyms                              |                                                                                         |                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                         |                                                                                                    |
| General Comment to reference function |                                                                                         | It is assumed, that the crude ore residues are used for recultivation. The same amount of energy used in building machines that is used per kg crude ore is used per kg residue for recultivation. | This ore has different grain sizes. It can be lump ore that can be used directly in the blast furnace or it can be ore of smaller grain size that is used for sinter and pellet production. Water emissions are estimated according to the maximal levels for financial support of mining activities by the world bank. | These data are a rough assumption. Yearly output is 40000000000 kg, life Time 30 years             |
| Start Date                            | 1999                                                                                    | 2000                                                                                                                                                                                               | 1998                                                                                                                                                                                                                                                                                                                    | 2000                                                                                               |
| End Date                              | 2000                                                                                    | 2000                                                                                                                                                                                               | 2000                                                                                                                                                                                                                                                                                                                    | 2000                                                                                               |
| Data Valid For Entire Period          | 1                                                                                       | 1                                                                                                                                                                                                  | 1                                                                                                                                                                                                                                                                                                                       | 1                                                                                                  |
| Other Period Text                     |                                                                                         |                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                         |                                                                                                    |
| Geography text                        | Data relate to the global average                                                       |                                                                                                                                                                                                    | Data relate to the global average                                                                                                                                                                                                                                                                                       |                                                                                                    |
| Technology text                       | Data relate to open mining                                                              | Data relate to the recultivation of open mines                                                                                                                                                     | Milling and mechanical sorting. Average iron yield is 84%.                                                                                                                                                                                                                                                              | Buildings and stationary machines.                                                                 |
| Representativeness [%]                | 65                                                                                      |                                                                                                                                                                                                    | 65                                                                                                                                                                                                                                                                                                                      |                                                                                                    |
| Production Volume                     | 1.02E12 kg/a                                                                            | unknown                                                                                                                                                                                            | 6.14E11 kg/a                                                                                                                                                                                                                                                                                                            | unknown                                                                                            |
| Sampling Procedure                    | based on literature                                                                     | estimation                                                                                                                                                                                         | based on literature                                                                                                                                                                                                                                                                                                     | estimation                                                                                         |
| Extrapolations                        | none                                                                                    | none                                                                                                                                                                                               | none                                                                                                                                                                                                                                                                                                                    | none                                                                                               |
| Uncertainty Adjustments               | none                                                                                    | none                                                                                                                                                                                               | none                                                                                                                                                                                                                                                                                                                    | none                                                                                               |

| General Flow information                |                           |                    |                                                                                                                                                  | Representation in ecoinvent |                        |                |          |                                                       |            |      | Uncertainty                                         |      |          |                   |
|-----------------------------------------|---------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|------------------------|----------------|----------|-------------------------------------------------------|------------|------|-----------------------------------------------------|------|----------|-------------------|
| Input                                   | Process Name              | Output             | Remarks                                                                                                                                          | Category                    | Sub category           | Infrastructure | Location | Module name in ecoinvent                              | Mean value | Unit | Source mean value                                   | Type | StdV 95% | General Comment   |
| iron in ore (resource)                  | iron ore, 46% Fe, at mine |                    | crude iron ore = 35.1 % in mined material, Fe = 46.19 % in iron ore --> 0.462 kg Fe in ore corresponds to 1 kg ore and to 2.85 kg mined material | resource                    | in ground              |                |          | Iron, 46% in ore, 25% in crude ore, in ground         | 4.62E-01   | kg   | Roth et al. 1999                                    | 1    | 1.05     | (1,2,1,1,1,1,12)  |
| land use (mine) [m2a]                   |                           |                    | assumption that mine is depleted in 30 years.                                                                                                    | resource                    | land                   |                |          | Occupation, mineral extraction site                   | 1.20E-04   | m2a  | calculated from data for 1 mine in Roth et al. 1999 | 1    | 1.54     | (2,4,1,1,1,4,7)   |
| land transformation from unknown        |                           |                    | mine of 4.8 km2 with an annual production of 40 Mio. t/a.                                                                                        | resource                    | land                   |                |          | Transformation, from unknown                          | 2.13E-06   | m2   | calculated from data for 1 mine in Roth et al. 1999 | 1    | 2.03     | (2,4,1,1,1,4,8)   |
| land transformation to mine             |                           |                    | mine of 4.8 km2 with an annual production of 40 Mio. t/a.                                                                                        | resource                    | land                   |                |          | Transformation, to mineral extraction site            | 2.13E-06   | m2   | calculated from data for 1 mine in Roth et al. 1999 | 1    | 2.03     | (2,4,1,1,1,4,8)   |
| reclamation mine                        |                           |                    | Assumption that 80% of mine area is reclaimed                                                                                                    | metals                      | extraction             | No             | GLO      | reclamation, iron mine                                | 1.70E-06   | m2   | estimated / calculated                              | 1    | 3.05     | (4,nA,1,1,1,nA,9) |
| water                                   |                           |                    | recycled water from mine --> not inventoried                                                                                                     |                             |                        |                |          |                                                       | 1.15E-05   | m3   | Roth et al. 1999                                    |      |          |                   |
| diesel, burned in building machine [MJ] |                           |                    |                                                                                                                                                  | construction processes      | machinery              | No             | GLO      | diesel, burned in building machine                    | 2.55E-02   | MJ   | Roth et al. 1999                                    | 1    | 2.00     | (1,2,1,1,1,1,5)   |
| electricity                             |                           |                    |                                                                                                                                                  | electricity                 | production mix         | No             | UCTE     | electricity, medium voltage, production UCTE, at grid | 1.42E-03   | kWh  | Roth et al. 1999                                    | 1    | 1.06     | (1,2,1,3,1,1,2)   |
| blasting [kg explosive]                 |                           |                    |                                                                                                                                                  | construction processes      | civil engineering      | No             | RER      | blasting                                              | 2.71E-04   | kg   | Roth et al. 1999                                    | 1    | 1.06     | (1,2,1,3,1,1,1)   |
| infrastructure (mine) [unit]            |                           |                    | includes infrastructure needed at mine (buildings, access roads,...) (exclusive building machines)                                               | metals                      | extraction             | Yes            | GLO      | mine, iron                                            | 8.33E-13   | unit | estimation, based on knowledge of other mines       | 1    | 3.05     | (4,nA,1,1,1,nA,9) |
|                                         |                           | waste heat         | from electricity use                                                                                                                             | air                         | low population density |                |          | Heat, waste                                           | 5.13E-03   | MJ   | calculated                                          | 1    | 1.06     | (1,2,1,3,1,1,13)  |
|                                         |                           | dust, coarse       | 50% of "dust" is > 10 um                                                                                                                         | air                         | low population density |                |          | Particulates, > 10 um                                 | 1.44E-03   | kg   | estimated according to bauxite mining               | 1    | 1.63     | (2,5,1,3,3,3,25)  |
|                                         |                           | dust, medium       | 45 % of "dust" is between 2.5 and 10 um                                                                                                          | air                         | low population density |                |          | Particulates, > 2.5 um, and < 10um                    | 1.30E-03   | kg   |                                                     | 1    | 2.10     | (2,5,1,3,3,3,26)  |
|                                         |                           | dust, fine         | 5 % of "dust" is below 2.5 um                                                                                                                    | air                         | low population density |                |          | Particulates, < 2.5 um                                | 1.44E-04   | kg   |                                                     | 1    | 3.10     | (2,5,1,3,3,3,27)  |
|                                         |                           | water              | from mine, partly recycled --> not inventoried                                                                                                   |                             |                        |                |          |                                                       | 1.07E-04   | m3   | Roth et al. 1999                                    |      |          |                   |
|                                         |                           | waste              | the waste is used in the reclamation of the mine and thus not inventoried                                                                        |                             |                        |                |          |                                                       | 8.82E-04   | kg   | Roth et al. 1999                                    |      |          |                   |
|                                         |                           | unrefined iron ore | 46.19% Fe                                                                                                                                        | metals                      | extraction             | No             | GLO      | iron ore, 46% Fe, at mine                             | 1.00E+00   | kg   |                                                     |      |          |                   |

Fig. 5.2 Flows for "iron ore, 46% Fe, at mine" and its representation in the ecoinvent database

| General Flow information                |                          |                    |                                                                        | Representation in ecoinvent |              |                |          |                                              |             |      | Uncertainty       |      |          |                                   |
|-----------------------------------------|--------------------------|--------------------|------------------------------------------------------------------------|-----------------------------|--------------|----------------|----------|----------------------------------------------|-------------|------|-------------------|------|----------|-----------------------------------|
| Input                                   | Process Name             | Output             | Remarks                                                                | Category                    | Sub category | Infrastructure | Location | Module name in ecoinvent                     | Mean value  | Unit | Source mean value | Type | StdV 95% | General Comment                   |
| diesel, burned in building machine [MJ] | recultivation, iron mine |                    | the same amount per kg refilled material as per kg crude ore in mining |                             |              | No             | GLO      | diesel, burned in building machine           | 5.97E+03 MJ |      | estimated         | 1    | 5.00     | (2,4,1,1,1,4,5); rough estimation |
| refill material                         |                          |                    | waste from mining and enrichment --> not inventoried                   |                             |              |                |          |                                              | 2.34E+05 kg |      |                   |      |          |                                   |
| land transformation (from mine) [m2]    |                          |                    |                                                                        | resource                    | land         |                |          | Transformation, from mineral extraction site | 1.00E+00 m2 |      | calculated        | 1    | 5.00     | (2,4,1,1,1,4,8); rough estimation |
| land transformation (to unknown) [m2]   |                          |                    | land occupation is included in the mining activity                     | resource                    | land         |                |          | Transformation, to unknown                   | 1.00E+00 m2 |      | calculated        | 1    | 5.00     | (2,4,1,1,1,4,8); rough estimation |
|                                         |                          | recultivation mine |                                                                        | metals                      | extraction   | No             | GLO      | recultivation, iron mine                     | 1.00E+00 m2 |      |                   |      |          |                                   |

Fig. 5.3 Flows for "mine, iron" and its representation in the ecoinvent database

| General Flow information                |                          |                    |                                                                        | Representation in ecoinvent |              |                |          |                                              |             |      |                   | Uncertainty |          |                                   |  |
|-----------------------------------------|--------------------------|--------------------|------------------------------------------------------------------------|-----------------------------|--------------|----------------|----------|----------------------------------------------|-------------|------|-------------------|-------------|----------|-----------------------------------|--|
| Input                                   | Process Name             | Output             | Remarks                                                                | Category                    | Sub category | Infrastructure | Location | Module name in ecoinvent                     | Mean value  | Unit | Source mean value | Type        | StdV 95% | General Comment                   |  |
| diesel, burned in building machine [MJ] | recultivation, iron mine |                    | the same amount per kg refilled material as per kg crude ore in mining |                             |              | No             | GLO      | diesel, burned in building machine           | 5.97E+03 MJ |      | estimated         | 1           | 5.00     | (2,4,1,1,1,4,5); rough estimation |  |
| refill material                         |                          |                    | waste from mining and enrichment --> not inventoried                   |                             |              |                |          |                                              | 2.34E+05 kg |      |                   |             |          |                                   |  |
| land transformation (from mine) [m2]    |                          |                    |                                                                        | resource                    | land         |                |          | Transformation, from mineral extraction site | 1.00E+00 m2 |      | calculated        | 1           | 5.00     | (2,4,1,1,1,4,8); rough estimation |  |
| land transformation (to unknown) [m2]   |                          |                    | land occupation is included in the mining activity                     | resource                    | land         |                |          | Transformation, to unknown                   | 1.00E+00 m2 |      | calculated        | 1           | 5.00     | (2,4,1,1,1,4,8); rough estimation |  |
|                                         |                          | recultivation mine |                                                                        | metals                      | extraction   | No             | GLO      | recultivation, iron mine                     | 1.00E+00 m2 |      |                   |             |          |                                   |  |

Fig. 5.4 Flows for "recultivation iron mine" and its representation in the ecoinvent database

| General Flow information          |                                            |                |                                                                                                                                   | Representation in ecoinvent |                        |                |          |                                                       |              |      | Uncertainty                                        |      |          |                  |
|-----------------------------------|--------------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------|------------------------|----------------|----------|-------------------------------------------------------|--------------|------|----------------------------------------------------|------|----------|------------------|
| Input                             | Process Name                               | Output         | Remarks                                                                                                                           | Category                    | Sub category           | Infrastructure | Location | Module name in ecoinvent                              | Mean value   | Unit | Source mean value                                  | Type | StdV 95% | General Comment  |
| unrefined iron ore                | iron ore, 65% Fe, at beneficiation; part 1 |                | raw ore, 46.19% Fe                                                                                                                | metals                      | extraction             | No             | GLO      | iron ore, 46% Fe, at mine                             | 1.66E+00 kg  |      | Roth et al. 1999                                   | 1    | 1.05     | (1,2,1,1,1,1,3)  |
| water                             |                                            |                | only fresh water. 1.329E-4 m3 recycled water from mine is not inventoried.                                                        | resource                    | in water               |                |          | Water, unspecified natural origin                     | 1.52E-03 m3  |      | Roth et al. 1999                                   | 1    | 1.05     | (1,2,1,1,1,1,12) |
| electricity                       |                                            |                |                                                                                                                                   | electricity                 | production mix         | No             | UCTE     | electricity, medium voltage, production UCTE, at grid | 1.87E-02 kWh |      | Roth et al. 1999                                   | 1    | 1.06     | (1,2,1,3,1,1,2)  |
| infrastructure (refinery) [unite] |                                            |                | included in infrastructure of mine --> not inventoried here                                                                       |                             |                        |                |          |                                                       |              |      |                                                    |      |          |                  |
|                                   |                                            | waste heat     | from electricity use                                                                                                              | air                         | low population density |                |          | Heat, waste                                           | 6.74E-02 MJ  |      | calculated                                         | 1    | 1.06     | (1,2,1,3,1,1,13) |
|                                   |                                            | dust           | included in mining --> not inventoried here                                                                                       |                             |                        |                |          |                                                       |              |      |                                                    |      |          |                  |
|                                   |                                            | water          | water itself not inventoried                                                                                                      |                             |                        |                |          |                                                       | 1.16E-03 m3  |      | Roth et al. 1999                                   |      |          |                  |
|                                   |                                            | oil and grease | water emissions based on maximum levels to be achieved for world bank support of a project. Half of maximum assumed as mean value | water                       | unspecified            |                |          | Oils, unspecified                                     | 5.79E-06 kg  |      | estimated according to world bank (Anonymous 1998) | 1    | 3.18     | (4,5,1,3,3,5,34) |
|                                   |                                            | cyanide        |                                                                                                                                   | water                       | unspecified            |                |          | Cyanide                                               | 5.79E-07 kg  |      |                                                    | 1    | 1.73     | (4,5,1,3,3,5,33) |
|                                   |                                            | COD            |                                                                                                                                   | water                       | unspecified            |                |          | COD, Chemical Oxygen Demand                           | 8.69E-05 kg  |      |                                                    | 1    | 1.73     | (4,5,1,3,3,5,32) |

Fig. 5.5 Flows for "iron ore, 65% Fe, at beneficiation" and its representation in the ecoinvent database

| General Flow information                   |              |                  |                                                                                  | Representation in ecoinvent |              |                |          |                                    |             |                  | Uncertainty       |      |          |                  |
|--------------------------------------------|--------------|------------------|----------------------------------------------------------------------------------|-----------------------------|--------------|----------------|----------|------------------------------------|-------------|------------------|-------------------|------|----------|------------------|
| Input                                      | Process Name | Output           | Remarks                                                                          | Category                    | Sub category | Infrastructure | Location | Module name in ecoinvent           | Mean value  | Unit             | Source mean value | Type | StdV 95% | General Comment  |
| iron ore, 65% Fe, at beneficiation; part 2 | →            | As               |                                                                                  | water                       | unspecified  |                |          | Arsenic, ion                       | 5.79E-08 kg |                  |                   | 1    | 1.73     | (4,5,1,3,3,5,33) |
|                                            |              | Cd               |                                                                                  | water                       | unspecified  |                |          | Cadmium, ion                       | 5.79E-08 kg |                  |                   | 1    | 5.21     | (4,5,1,3,3,5,35) |
|                                            |              | Cr (VI)          |                                                                                  | water                       | unspecified  |                |          | Chromium VI                        | 5.79E-08 kg |                  |                   | 1    | 5.21     | (4,5,1,3,3,5,35) |
|                                            |              | Cu               |                                                                                  | water                       | unspecified  |                |          | Copper, ion                        | 2.90E-07 kg |                  |                   | 1    | 5.21     | (4,5,1,3,3,5,35) |
|                                            |              | Fe               |                                                                                  | water                       | unspecified  |                |          | Iron, ion                          | 2.03E-06 kg |                  |                   | 1    | 5.21     | (4,5,1,3,3,5,35) |
|                                            |              | Pb               |                                                                                  | water                       | unspecified  |                |          | Lead                               | 1.16E-07 kg |                  |                   | 1    | 5.21     | (4,5,1,3,3,5,35) |
|                                            |              | Hg               |                                                                                  | water                       | unspecified  |                |          | Mercury                            | 5.79E-09 kg |                  |                   | 1    | 5.21     | (4,5,1,3,3,5,35) |
|                                            |              | Ni               |                                                                                  | water                       | unspecified  |                |          | Nickel, ion                        | 2.90E-07 kg |                  |                   | 1    | 5.21     | (4,5,1,3,3,5,35) |
|                                            |              | Zn               |                                                                                  | water                       | unspecified  |                |          | Zinc, ion                          | 1.16E-06 kg |                  |                   | 1    | 5.21     | (4,5,1,3,3,5,35) |
|                                            |              | BOD              | calculated from COD, assuming C6H12O6                                            | water                       | unspecified  |                |          | BOD5, Biological Oxygen Demand     | 8.69E-05 kg | calculated       |                   | 1    | 1.73     | (4,5,1,3,3,5,32) |
|                                            |              | TOC              |                                                                                  | water                       | unspecified  |                |          | TOC, Total Organic Carbon          | 3.14E-06 kg | calculated       |                   | 1    | 1.73     | (4,5,1,3,3,5,32) |
|                                            |              | DOC              |                                                                                  | water                       | unspecified  |                |          | DOC, Dissolved Organic Carbon      | 3.14E-06 kg | calculated       |                   | 1    | 1.73     | (4,5,1,3,3,5,32) |
|                                            |              | waste [kg]       | the waste is used in the recultivation of the mine and thus not inventoried      |                             |              |                |          |                                    | 6.58E-01 kg | Roth et al. 1999 |                   |      |          |                  |
|                                            |              | refined iron ore | 64.19 % Fe content. 13% as lump ore, 70% to sinter and 27% to pellet fabrication | metals                      | extraction   | No             | GLO      | iron ore, 65% Fe, at beneficiation | 1.00E+00 kg |                  |                   |      |          |                  |

Fig. 5.5 Flows for "iron ore, 65% Fe, at beneficiation" and its representation in the ecoinvent database (cont.)

### **Data quality**

Data uncertainty is derived using the Pedigree matrix as described in (Frischknecht et al. (2003)) except for the recultivation of the mine and for the mine infrastructure where this method would result in values far too low.

Data for iron ore mining are of good overall quality although the infrastructure is roughly estimated. Water emission data for beneficiation are based on threshold values and thus exhibit a greater uncertainty.

## **5.3 Agglomeration: sinter and pellet production**

This part is mainly based on information from (Meiler (1997)).

### **5.3.1 Production process and infrastructure**

Very fine material is undesirable in the reduction equipment. Modern high-performance blast furnaces require physical and metallurgical preparation of the burden. Agglomeration on one hand is used as a process of size enlargement aimed at improving the permeability of the blast-furnace burden. On the other hand slagging materials are introduced via the agglomerates.

As a rule, blast-furnace burden contains lump ore, sinter, and pellets. Pellets are made from both concentrate and iron-rich fine ore. If the beneficiation process yields very fine-grained concentrates (e.g., <0.1 mm), these are usually converted to pellets at the mine and transported in this form. Sinter is generally produced at the ironworks. The varying loads imposed on material during transport to the blast furnace mean that pellets must satisfy stringent mechanical strength requirements. Pellets are also nearly always made from one well-defined ore or concentrate, whereas sinter is produced from pre-designed mixtures of ores and additives.

Sinter and pellets differ in the way the particles are bound into the agglomerate. Pellets are burned at a temperature below the melting points of the constituents. Their strength gain is adequate only when the reaction temperature is controlled to  $\pm 10$  °C over long time periods. For this reason, hot gas is used to supply heat; it can be generated by the combustion of fuel gas, oil, or even coal. In sintering, on the other hand, the constituents of the mixture are melted. This happens at temperatures so high that the reaction itself goes to completion within minutes, but in a thin layer that continuously advances through the material bed concurrently to the gas.

In pelletising spherical green pellets (green balls) of appropriate size (ca. 9 – 16 mm in diameter) are first formed by a pelletising disk or drum. Then the pellets are thermally hardened of at 1200 – 1320 °C.

Iron-ore sinters are virtually all produced in travelling-grate installations. Along with ore and concentrate, the mixture for sintering also contains recycle material from the ironworks, provided no build-up of undesirable or detrimental substances results.

Coke is the dominant sinter plant energy input (about 85%), with electricity and gas (COG and/or BF gas and/or natural gas) supplying the remainder in equal shares (IPPC (2000)).

### **5.3.2 Emissions**

#### **Air**

Sintering and pelletising leads to the emission of dust, various heavy metals and hydrocarbons, hydrogen halogenides, NO<sub>x</sub>, SO<sub>x</sub> and CO<sub>2</sub> from the burning of the fuel gas and some of the coke. These pro-



cesses are also a major source of dioxide emissions. The dioxin emission value for the sinter production stems from (Anonymous (2002)).

Dust grain size analysis before treatment shows two distinctive peaks. The first and higher peak is at about 1  $\mu\text{m}$  while the second one is at 100  $\mu\text{m}$  (IPPC (2000)). The abatement used in sinter plants has a high efficiency in removing the larger particles while the smaller alkali and lead chlorides are hard to remove in the electrostatic precipitators due to their high resistivity. Thus a particle size of < 2.5  $\mu\text{m}$  is assumed for the dust emission after abatement.

## **Water**

The water used in the agglomeration processes evaporates. Thus no relevant water emissions occur.

### **5.3.3 Waste**

Normally all solid wastes originating from the agglomeration plant (dusts from de-dusting devices and sieving fractions) are recycled to the strand. However, in sinter plants there can be two exceptions: The first concerns sludge from wet waste gas treatment systems, which is put to landfill. At present, there are only two plants in Europe operating such a system. The second exception is filter dust from the last field of electrostatic precipitators (ESP). Most European sinter plants are operated with fully closed dust cycles. However, some plants exclude fine dust from the last field of the ESP. This dust mainly consists of alkali and metal chlorides. This so-called partly open filter dust cycle is carried out in order to improve the operation of ESP or (in one case) of the bag filter, or to reduce alkali and metal chloride emissions. (IPPC (2000))

In this project we neglect the wastes to landfill because:

- their amount is not known (but small)
- their composition is not known
- their impact is well below the uncertainty of the impact of the air emissions

### **5.3.4 Transports**

Sinter usually is produced near the blast furnace while pellets are produced near the beneficiation plant. This is because pellets are made of fraction of the ore so fine that it would ask for elaborate technologies for transportation (especially loading and unloading).

Masses for transportation of the ore to the sinter plant and of the lump ore and pellets to the blast furnace are estimated according to the supply and demand situation in the different regions of the world. It is assumed, that as much ore as possible is used within these regions. Thus Western Europe and Asia are net importers while Eastern Europe / GUS, South America, Australia and Africa are net exporters. North America produces almost exactly as much iron ore as it consumes. An export mix is generated according to the export capacity of the net exporters and it is assumed that the imports of the net importers correspond to this mix. Distances within the regions are neglected and those between the regions and the means of transpiration are assumed. Data are given in Tab. 5.2.

Tab. 5.2 Transport masses and distances for iron ore and pellets

| Region           | Iron ore production 1995 |            | Pig iron production 1995 |            | Ore needed for pig iron production 1995 |            | Import demand total | Export composition |
|------------------|--------------------------|------------|--------------------------|------------|-----------------------------------------|------------|---------------------|--------------------|
|                  | Mio t                    | % of world | Mio t                    | % of world | Mio t                                   | % of world | Mio t               | %                  |
| West Europe      | 34                       | 3.4        | 125                      | 23.8       | 211                                     | 23.8       | 177                 |                    |
| East Europe /GUS | 131                      | 13.1       | 60                       | 11.3       | 100                                     | 11.3       | -31                 | 9.1                |
| Asia             | 352                      | 35.1       | 235                      | 44.5       | 394                                     | 44.5       | 43                  |                    |
| Africa           | 50                       | 4.9        | 9                        | 1.6        | 14                                      | 1.6        | -35                 | 10.4               |
| North America    | 98                       | 9.7        | 59                       | 11.3       | 100                                     | 11.3       | 2                   |                    |
| South America    | 193                      | 19.3       | 32                       | 6.1        | 54                                      | 6.1        | -139                | 41.2               |
| Australia        | 145                      | 14.5       | 7                        | 1.4        | 13                                      | 1.4        | -133                | 39.3               |

| Import      | Import [Mio t]               | Distance Train [km] | Distance ship (ocean) [km] | Distance ship (coastal) [km] |
|-------------|------------------------------|---------------------|----------------------------|------------------------------|
| <b>to</b>   | <b>from East Europe /GUS</b> |                     |                            |                              |
| West Europe | 16                           | 2500                | 0                          | 0                            |
| Asia        | 4                            | 1000                | 14366.761                  | 0                            |
| <b>to</b>   | <b>from Africa</b>           |                     |                            |                              |
| West Europe | 18                           | 1000                | 6009.615                   | 173.772                      |
| Asia        | 4                            | 1000                | 17444.778                  | 0                            |
| <b>to</b>   | <b>from South America</b>    |                     |                            |                              |
| West Europe | 73                           | 1000                | 8495.52                    | 173.772                      |
| Asia        | 18                           | 1000                | 18344.209                  | 0                            |
| <b>to</b>   | <b>from Australia</b>        |                     |                            |                              |
| West Europe | 70                           | 1000                | 15526.85                   | 173.772                      |
| Asia        | 17                           | 1000                | 7108.562                   | 0                            |

### 5.3.5 Sinter and pellet production in ecoinvent

The background information to the processes are given in Tab. 5.3, flows are shown in:

- sinter, iron, at plant Fig. 5.6

- pellets, iron, at plant Fig. 5.7

Where (IPPC (2000)) is given as the source, arithmetic means of the low and high values are used in this project.

No data on the infrastructure are available. Thus the aluminium oxide plant (calcination plant) was chosen to represent the sinter and the pelletising plant.

**Tab. 5.3 Meta information for the iron ore agglomeration processes**

|                                       |                                                                                                                                        |                                                                                                                        |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Name                                  | sinter, iron, at plant                                                                                                                 | pellets, iron, at plant                                                                                                |
| Location                              | GLO                                                                                                                                    | GLO                                                                                                                    |
| Infrastructure Process                | 0                                                                                                                                      | 0                                                                                                                      |
| Unit                                  | kg                                                                                                                                     | kg                                                                                                                     |
| Data Set Version                      | 2.0                                                                                                                                    | 2.0                                                                                                                    |
| Included Processes                    | Blending, mixing and sintering.<br>Emissions are abated                                                                                | Blending, mixing and sintering.<br>Emissions are abated                                                                |
| Amount                                | 1                                                                                                                                      | 1                                                                                                                      |
| Local Name                            | Sinter, Eisen, ab Werk                                                                                                                 | Pellets, Eisen, ab Werk                                                                                                |
| Synonyms                              |                                                                                                                                        |                                                                                                                        |
| General Comment to reference function | Inputs and air emissions from different sources. Transports of iron ore calculated according to information about exports and imports. | Inputs and air emissions from different sources. No transport of iron ore because pellets are fabricated at mine       |
| Start Date                            | 1999                                                                                                                                   | 1999                                                                                                                   |
| End Date                              | 2002                                                                                                                                   | 2002                                                                                                                   |
| Data Valid For Entire Period          | 1                                                                                                                                      | 1                                                                                                                      |
| Other Period Text                     |                                                                                                                                        |                                                                                                                        |
| Geography text                        | Inputs relate to the global average.<br>Emissions relate to European plants                                                            | Inputs relate to the global average.<br>Emissions relate to European plants                                            |
| Technology text                       | Down draft sintering on continuous travelling grates                                                                                   | Strate grate and grate kiln process                                                                                    |
| Representativeness [%]                | 65                                                                                                                                     | 65                                                                                                                     |
| Production Volume                     | 3.83E11 kg/a                                                                                                                           | 1.54E11 kg/a                                                                                                           |
| Sampling Procedure                    | Main in- and outputs based on world average, emissions based on some plants in the EU. Everything based on literature.                 | Main in- and outputs based on world average, emissions based on some plants in the EU. Everything based on literature. |
| Extrapolations                        | see Geography                                                                                                                          | see Geography                                                                                                          |
| Uncertainty Adjustments               | none                                                                                                                                   | none                                                                                                                   |

### Data quality

Data uncertainty is derived using the Pedigree matrix as described in (Frischknecht et al. (2003)) except for the infrastructure where this method would result in values far too low.

Data of input and main output materials for sintering and pelletising are of good overall quality.

Emission data for pelletising stem from all 5 pelletising plants in Europe. These 5 plants produced about 15 Mt of the 35 Mt pellets used in Europe in 1995. Thus data for European consumption are of rather high uncertainty.

Emission data for sintering stem from 5 of the 47 sinter plants in Europe. Thus data for European consumption are of rather high uncertainty.

A rough proxy is used for the infrastructure.

| General Flow information       |                                |            |                                                                                                                                                            | Representation in ecoinvent |                |                |          |                                                       |               |      | Uncertainty                |      |          |                                       |
|--------------------------------|--------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------|----------------|----------|-------------------------------------------------------|---------------|------|----------------------------|------|----------|---------------------------------------|
| Input                          | Process Name                   | Output     | Remarks                                                                                                                                                    | Category                    | Sub category   | Infrastructure | Location | Module name in ecoinvent                              | Mean value    | Unit | Source mean value          | Type | StdV 95% | General Comment                       |
| refined iron ore [kg]          | sinter, iron, at plant; part 1 |            |                                                                                                                                                            | metals                      | extraction     | No             | GLO      | iron ore, 65% Fe, at beneficiation                    | 1.05E+00 kg   |      | Roth et al. 1999           | 1    | 1.05     | (1,2,1,1,1,1,3)                       |
| transport ship (ocean)         |                                |            | refined ore from mine to plant by ship (ocean)                                                                                                             | transport systems           | ship           | No             | OCE      | transport, transoceanic freight ship                  | 2.84E+00 tkm  |      | estimated                  | 1    | 2.01     | (2,nA,1,1,1,3,5)                      |
| transport ship (costal)        |                                |            | refined ore from mine to plant by ship (costal)                                                                                                            | transport systems           | ship           | No             | RER      | transport, barge                                      | 3.15E-02 tkm  |      | estimated                  | 1    | 2.01     | (2,nA,1,3,1,3,5)                      |
| transport rail                 |                                |            | refined ore from mine to plant by rail                                                                                                                     | transport systems           | train          | No             | RER      | transport, freight, rail                              | 2.89E-01 tkm  |      | estimated                  | 1    | 2.01     | (2,nA,1,3,1,3,5)                      |
| lime                           |                                |            |                                                                                                                                                            | construction materials      | additives      | No             | CH       | quicklime, in pieces, loose, at plant                 | 5.00E-02 kg   |      | Roth et al. 1999           | 1    | 1.06     | (1,2,1,3,1,1,3)                       |
| transport rail                 |                                |            | limestone & coke: 200 km                                                                                                                                   | transport systems           | train          | No             | RER      | transport, freight, rail                              | 2.00E-02 tkm  |      | estimated                  | 1    | 2.09     | (4,5,nA,nA,nA,nA,5)                   |
| transport road                 |                                |            | 20 km                                                                                                                                                      | transport systems           | road           | No             | RER      | transport, lorry >16t, fleet average                  | 2.00E-03 tkm  |      | estimated                  | 1    | 2.09     | (4,5,nA,nA,nA,nA,5)                   |
| fuel gas for heating           |                                |            | natural gas, COG and BFGas are used, only consumption of natural gas is inventoried with the assumption that half of the energy is supplied by natural gas | natural gas                 | fuels          | No             | RER      | natural gas, high pressure, at consumer               | 3.63E-02 MJ   |      | Roth et al. 1999           | 1    | 1.06     | (1,2,1,3,1,1,1)                       |
| compressed air                 |                                |            | not inventoried                                                                                                                                            |                             |                |                |          |                                                       | 3.00E-03 Nm3  |      | Roth et al. 1999           |      |          |                                       |
| suction air                    |                                |            | not inventoried                                                                                                                                            |                             |                |                |          |                                                       | 2.00E+00 Nm3  |      | Roth et al. 1999           |      |          |                                       |
| coke chippings                 |                                |            | 28.6 MJ/kg                                                                                                                                                 | hard coal                   | fuels          | No             | RER      | hard coal coke, at plant                              | 1.43E+00 MJ   |      | Roth et al. 1999           | 1    | 1.06     | (1,2,1,3,1,1,3)                       |
| water                          |                                |            |                                                                                                                                                            | resource                    | in water       |                |          | Water, unspecified natural origin                     | 5.00E-04 m3   |      | Roth et al. 1999           | 1    | 1.05     | (1,2,1,1,1,1,12)                      |
| electricity                    |                                |            | includes power consumed for air compression and suction and for exhaust gaz cleaning                                                                       | electricity                 | production mix | No             | UCTE     | electricity, medium voltage, production UCTE, at grid | 1.00E-02 kWh  |      | Roth et al. 1999           | 1    | 1.06     | (1,2,1,3,1,1,2)                       |
| infrastructure (plant) [piece] |                                |            | aluminium oxide plant used as proxy                                                                                                                        | metals                      | extraction     | Yes            | RER      | aluminium oxide, plant                                | 2.50E-11 unit |      | estimated                  | 1    | 5.00     | (5,5,nA,nA,nA,nA,9); rough estimation |
|                                |                                | waste heat | from electricity fuel gas and coke use (assumption all coke is burned)                                                                                     | air                         | unspecified    |                |          | Heat, waste                                           | 1.54E+00 MJ   |      | calculated                 | 1    | 1.06     | (1,2,1,3,1,1,13)                      |
|                                |                                | dust, fine | all "dust" is below 2.5 um                                                                                                                                 | air                         | unspecified    |                |          | Particulates, < 2.5 um                                | 2.06E-04 kg   |      | calculated after IPPC 2000 | 1    | 3.01     | (2,3,2,3,1,3,27)                      |
|                                |                                | Cd         |                                                                                                                                                            | air                         | unspecified    |                |          | Cadmium                                               | 1.93E-08 kg   |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22)                      |
|                                |                                | Cr         |                                                                                                                                                            | air                         | unspecified    |                |          | Chromium                                              | 2.52E-08 kg   |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22)                      |
|                                |                                | Cu         |                                                                                                                                                            | air                         | unspecified    |                |          | Copper                                                | 7.66E-08 kg   |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22)                      |
|                                |                                | Hg         |                                                                                                                                                            | air                         | unspecified    |                |          | Mercury                                               | 7.57E-08 kg   |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22)                      |

Fig. 5.6 Flows for "sinter, iron, at plant" and its representation in the ecoinvent database

5. System Characterisation

| General Flow information       |                    |                                                                                                                                                      |         | Representation in ecoinvent |              |                |          |                                                          |             |      | Uncertainty                |      |          |                  |
|--------------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-----------------------------|--------------|----------------|----------|----------------------------------------------------------|-------------|------|----------------------------|------|----------|------------------|
| Input                          | Process Name       | Output                                                                                                                                               | Remarks | Category                    | Sub category | Infrastructure | Location | Module name in ecoinvent                                 | Mean value  | Unit | Source mean value          | Type | StdV 95% | General Comment  |
| sinter, iron, at plant; part 2 | → Mn               |                                                                                                                                                      |         | air                         | unspecified  |                |          | Manganese                                                | 1.93E-07 kg |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22) |
|                                | → Ni               |                                                                                                                                                      |         | air                         | unspecified  |                |          | Nickel                                                   | 1.93E-08 kg |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22) |
|                                | → Pb               |                                                                                                                                                      |         | air                         | unspecified  |                |          | Lead                                                     | 3.23E-06 kg |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22) |
|                                | → Ti               |                                                                                                                                                      |         | air                         | unspecified  |                |          | Titanium                                                 | 1.60E-08 kg |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22) |
|                                | → V                |                                                                                                                                                      |         | air                         | unspecified  |                |          | Vanadium                                                 | 1.15E-08 kg |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22) |
|                                | → Zn               |                                                                                                                                                      |         | air                         | unspecified  |                |          | Zinc                                                     | 8.26E-07 kg |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22) |
|                                | → HCl              |                                                                                                                                                      |         | air                         | unspecified  |                |          | Hydrogen chloride                                        | 3.76E-05 kg |      | calculated after IPPC 2000 | 1    | 1.52     | (2,3,2,3,1,3,31) |
|                                | → HF               |                                                                                                                                                      |         | air                         | unspecified  |                |          | Hydrogen fluoride                                        | 2.25E-06 kg |      | calculated after IPPC 2000 | 1    | 1.52     | (2,3,2,3,1,3,31) |
|                                | → NOx              |                                                                                                                                                      |         | air                         | unspecified  |                |          | Nitrogen oxides                                          | 5.27E-04 kg |      | calculated after IPPC 2000 | 1    | 1.52     | (2,3,2,3,1,3,16) |
|                                | → SO2              |                                                                                                                                                      |         | air                         | unspecified  |                |          | Sulfur dioxide                                           | 1.26E-03 kg |      | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,15) |
|                                | → CO               |                                                                                                                                                      |         | air                         | unspecified  |                |          | Carbon monoxide, fossil                                  | 2.57E-02 kg |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,17) |
|                                | → CO2              |                                                                                                                                                      |         | air                         | unspecified  |                |          | Carbon dioxide, fossil                                   | 2.04E-01 kg |      | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,14) |
|                                | → VOC              | VOC without PAH and PCB                                                                                                                              |         | air                         | unspecified  |                |          | Hydrocarbons, aliphatic, alkanes, unspecified            | 1.38E-04 kg |      | calculated after IPPC 2000 | 1    | 1.52     | (2,3,2,3,1,3,16) |
|                                | → PAH              |                                                                                                                                                      |         | air                         | unspecified  |                |          | PAH, polycyclic aromatic hydrocarbons                    | 4.72E-10 kg |      | calculated after IPPC 2000 | 1    | 3.01     | (2,3,2,3,1,3,21) |
|                                | → PCB              |                                                                                                                                                      |         | air                         | unspecified  |                |          | Polychlorinated biphenyls                                | 6.42E-12 kg |      | calculated after IPPC 2000 | 1    | 3.01     | (2,3,2,3,1,3,21) |
|                                | → dioxin (I-TEQ)   | Quass gives a range of 1 to 10 ng / m3 flue gas (→2.1 to 21 ng/kg sinter). The European Dioxine Inventory uses 7 ng/kg sinter as average for Europe) |         | air                         | unspecified  |                |          | Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin | 7.00E-12 kg |      | Anonymus 2002              | 1    | 3.01     | (1,3,2,3,1,2,21) |
|                                | → dust to landfill | totally recycled in the vast majority of plants. The small part that is not recycled in few plants is neglected                                      |         |                             |              |                |          |                                                          |             |      |                            |      |          |                  |
|                                | → sludge           | applies only for two plants that have a wet scrubber → neglected.                                                                                    |         |                             |              |                |          |                                                          |             |      |                            |      |          |                  |
|                                | → sinter           |                                                                                                                                                      |         | metals                      | extraction   | No             | GLO      | sinter, iron, at plant                                   | 1.00E+00 kg |      |                            |      |          |                  |

Fig. 5.6 Flows for "sinter, iron, at plant" and its representation in the ecoinvent database (cont.)

# 5. System Characterisation

| General Flow information       |                                 |            |                                                                                                   | Representation in ecoinvent |                 |                |          |                                                       |               |      | Uncertainty                |      |           |                                       |
|--------------------------------|---------------------------------|------------|---------------------------------------------------------------------------------------------------|-----------------------------|-----------------|----------------|----------|-------------------------------------------------------|---------------|------|----------------------------|------|-----------|---------------------------------------|
| Input                          | Process Name                    | Output     | Remarks                                                                                           | Category                    | Sub category    | Infrastructure | Location | Module name in ecoinvent                              | Mean value    | Unit | Source mean value          | Type | StDev 95% | General Comment                       |
| refined iron ore               | pellets, iron, at plant; part 1 |            |                                                                                                   | metals                      | extraction      | No             | GLO      | iron ore, 65% Fe, at beneficiation                    | 1.05E+00 kg   |      | Roth et al. 1999           | 1    | 1.05      | (1,2,1,1,1,1,3)                       |
| bentonite                      |                                 |            |                                                                                                   | construction materials      | additives       | No             | DE       | bentonite, at processing                              | 4.00E-02 kg   |      | Roth et al. 1999           | 1    | 1.06      | (1,2,1,3,1,1,4)                       |
| transport rail                 |                                 |            | bentonite: 600 km; coal: 200km                                                                    | transport systems           | train           | No             | RER      | transport, freight, rail                              | 2.41E-02 tkm  |      | estimated                  | 1    | 2.09      | (4,5,nA,nA,nA,5)                      |
| transport road                 |                                 |            | bentonite: 100 km; coal: 20km                                                                     | transport systems           | road            | No             | CH       | transport, lorry >28t, fleet average                  | 4.01E-03 tkm  |      | estimated                  | 1    | 2.09      | (4,5,nA,nA,nA,nA,5)                   |
| coal dust                      |                                 |            | 28.6 MJ/kg                                                                                        | hard coal                   | fuels           | No             | UCTE     | hard coal mix, at regional storage                    | 1.30E-02 kg   |      | Roth et al. 1999           | 1    | 1.06      | (1,2,1,3,1,1,11)                      |
| water                          |                                 |            |                                                                                                   | resource                    | in water        |                |          | Water, unspecified natural origin                     | 9.00E-05 m3   |      | Roth et al. 1999           | 1    | 1.05      | (1,2,1,1,1,1,12)                      |
| fuel gas for heating           |                                 |            | generic heat module is inventoried (because no detailed information about emissions is available) | natural gas                 | heating systems | No             | RER      | natural gas, burned in industrial furnace >100kW      | 6.90E-02 MJ   |      | Roth et al. 1999           | 1    | 1.06      | (1,2,1,3,1,1,11)                      |
| electricity                    |                                 |            |                                                                                                   | electricity                 | production mix  | No             | UCTE     | electricity, medium voltage, production UCTE, at grid | 2.50E-02 kWh  |      | Roth et al. 1999           | 1    | 1.06      | (1,2,1,3,1,1,2)                       |
| infrastructure (plant) [piece] |                                 |            | aluminium oxide plant used as proxy                                                               | metals                      | extraction      | Yes            | RER      | aluminium oxide, plant                                | 2.50E-11 unit |      | estimated                  | 1    | 5.00      | (5,5,nA,nA,nA,nA,9); rough estimation |
|                                |                                 | waste heat | from electricity fuel gas and coal use (assumption all coal is burned)                            | air                         | unspecified     |                |          | Heat, waste                                           | 5.31E-01 MJ   |      | calculated                 | 1    | 1.06      | (1,2,1,3,1,1,13)                      |
|                                |                                 | dust, fine | all "dust" is below 2.5 um                                                                        | air                         | unspecified     |                |          | Particulates, < 2.5 um                                | 7.50E-05 kg   |      | calculated after IPPC 2000 | 1    | 3.01      | (2,3,2,3,1,3,27)                      |
|                                |                                 | Cd         |                                                                                                   | air                         | unspecified     |                |          | Cadmium                                               | 2.10E-10 kg   |      | calculated after IPPC 2000 | 1    | 5.01      | (2,3,2,3,1,3,22)                      |
|                                |                                 | Cr         |                                                                                                   | air                         | unspecified     |                |          | Chromium                                              | 2.70E-09 kg   |      | calculated after IPPC 2000 | 1    | 5.01      | (2,3,2,3,1,3,22)                      |
|                                |                                 | Cu         |                                                                                                   | air                         | unspecified     |                |          | Copper                                                | 4.60E-09 kg   |      | calculated after IPPC 2000 | 1    | 5.01      | (2,3,2,3,1,3,22)                      |
|                                |                                 | Hg         |                                                                                                   | air                         | unspecified     |                |          | Mercury                                               | 2.50E-10 kg   |      | calculated after IPPC 2000 | 1    | 5.01      | (2,3,2,3,1,3,22)                      |
|                                |                                 | Mn         |                                                                                                   | air                         | unspecified     |                |          | Manganese                                             | 2.30E-08 kg   |      | calculated after IPPC 2000 | 1    | 5.01      | (2,3,2,3,1,3,22)                      |

Fig. 5.7 Flows for "pellets, iron, at plant" and its representation in the ecoinvent database

| General Flow information        |              |                              |                 | Representation in ecoinvent |              |                |          |                                                          |            |      | Uncertainty                |      |          |                  |
|---------------------------------|--------------|------------------------------|-----------------|-----------------------------|--------------|----------------|----------|----------------------------------------------------------|------------|------|----------------------------|------|----------|------------------|
| Input                           | Process Name | Output                       | Remarks         | Category                    | Sub category | Infrastructure | Location | Module name in ecoinvent                                 | Mean value | Unit | Source mean value          | Type | StdV 95% | General Comment  |
| pellets, iron, at plant; part 2 | →            | Ni                           |                 | air                         | unspecified  |                |          | Nickel                                                   | 1.50E-08   | kg   | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22) |
|                                 | →            | Pb                           |                 | air                         | unspecified  |                |          | Lead                                                     | 6.65E-08   | kg   | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22) |
|                                 | →            | V                            |                 | air                         | unspecified  |                |          | Vanadium                                                 | 8.55E-08   | kg   | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22) |
|                                 | →            | Zn                           |                 | air                         | unspecified  |                |          | Zinc                                                     | 5.62E-08   | kg   | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22) |
|                                 | →            | HCl                          |                 | air                         | unspecified  |                |          | Hydrogen chloride                                        | 2.50E-05   | kg   | calculated after IPPC 2000 | 1    | 1.52     | (2,3,2,3,1,3,31) |
|                                 | →            | HF                           |                 | air                         | unspecified  |                |          | Hydrogen fluoride                                        | 1.99E-05   | kg   | calculated after IPPC 2000 | 1    | 1.52     | (2,3,2,3,1,3,31) |
|                                 | →            | NOx                          |                 | air                         | unspecified  |                |          | Nitrogen oxides                                          | 3.15E-04   | kg   | calculated after IPPC 2000 | 1    | 1.52     | (2,3,2,3,1,3,16) |
|                                 | →            | SO2                          |                 | air                         | unspecified  |                |          | Sulfur dioxide                                           | 1.34E-04   | kg   | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,15) |
|                                 | →            | CO                           |                 | air                         | unspecified  |                |          | Carbon monoxide, fossil                                  | 2.10E-04   | kg   | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,17) |
|                                 | →            | CO2                          |                 | air                         | unspecified  |                |          | Carbon dioxide, fossil                                   | 2.37E-02   | kg   | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,14) |
|                                 | →            | VOC                          | VOC without PAH | air                         | unspecified  |                |          | Hydrocarbons, aliphatic, alkanes, unspecified            | 2.25E-05   | kg   | calculated after IPPC 2000 | 1    | 1.52     | (2,3,2,3,1,3,16) |
|                                 | →            | PAH                          |                 | air                         | unspecified  |                |          | PAH, polycyclic aromatic hydrocarbons                    | 1.90E-10   | kg   | calculated after IPPC 2000 | 1    | 3.01     | (2,3,2,3,1,3,21) |
|                                 | →            | dioxin (I-TEQ)               |                 | air                         | unspecified  |                |          | Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin | 5.70E-15   | kg   | calculated after IPPC 2000 | 1    | 3.01     | (2,3,2,3,1,3,21) |
|                                 | →            | pellets, iron, at plant [kg] |                 | metals                      | extraction   | No             | GLO      | pellets, iron, at plant                                  | 1.00E+00   | kg   |                            |      |          |                  |

Fig. 5.7 Flows for "pellets, iron, at plant" and its representation in the ecoinvent database (cont.)

## 5.4 Blast furnace process: pig iron / hot metal production

This part is mainly based on Roth et al. (1999) and IPPC (2000).

### 5.4.1 Production process and infrastructure

The blast furnace remains by far the most important process for the production of pig iron. The technique is likely to continue to dominate pig iron production for at least the next 20 years.

A blast furnace is a closed system into which iron bearing materials (iron ore lump, sinter and/or pellets), additives (slag formers such as limestone) and reducing agents (coke) are continuously fed from the top of the furnace shaft through a charging system that prevents escape of blast furnace gas (BF gas). Fig. 5.8 shows a simplified layout of a blast furnace consisting of the furnace itself, the cast house, the hot stoves and two-stage treatment of BF gas.

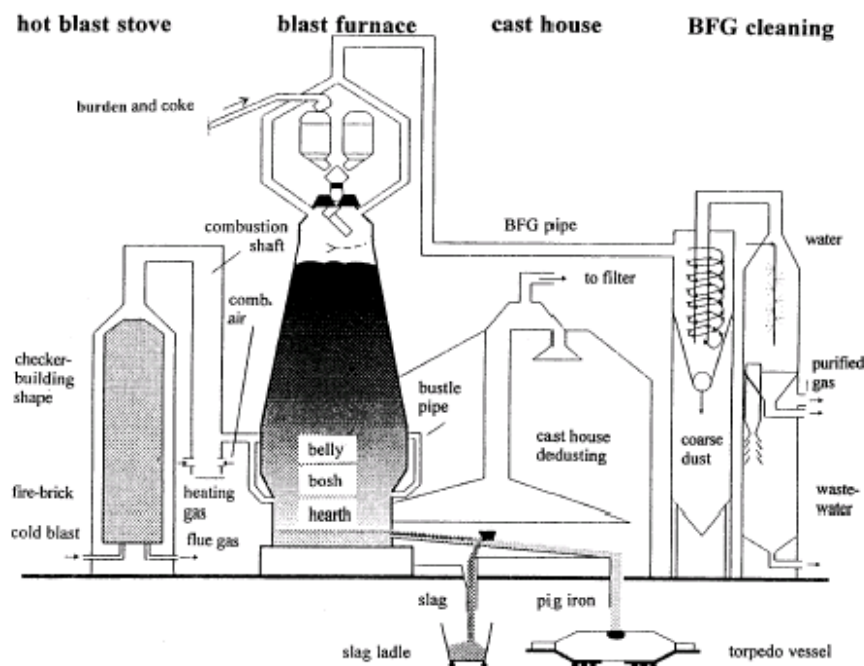


Fig. 5.8 Simplified scheme of a blast furnace (IPPC (2000))

No data have been found on the infrastructure needed for the process. Assumptions made in this project are based on different pictures and information on the aluminium production infrastructure (see Tab. 5.4).

The infrastructure is modelled to be used for 50 years. The replacement of the machines with a life-time of only 25 years is taken into account by doubling the amount. Thus the values given in Tab. 5.5 are inventoried. For the waste disposal it is assumed that the conveyer belts and the machines are completely recycled. The disposal of the buildings is already inventoried in the building modules. Thus only the concrete and the refractory materials are disposed.



Tab. 5.4 Assumptions for infrastructure of blast furnaces

| blast furnace                                         | amount        | unit           | lifetime [a] | Material / Module          |
|-------------------------------------------------------|---------------|----------------|--------------|----------------------------|
| yearly production capacity                            | 1'500'000'000 | kg             | 50           | Yearly production capacity |
| area of plant                                         | 976'000       | m <sup>2</sup> | 50           | Transformed area           |
| administration building (50x50x10 m <sup>3</sup> )    | 25'000        | m <sup>3</sup> | 50           | Building, multi story      |
| concrete for piers [m <sup>3</sup> ]                  | 1'200         | m <sup>3</sup> | 50           | reinforced concrete        |
| grab for unloading [kg]                               | 100'000       | kg             | 25           | heavy machine, unspecific  |
| conveyor to storage [m]                               | 900           | m              | 25           | conveyor belt              |
| silo (40000 m <sup>3</sup> , height 62 m, radius 14m) | 233'986       | kg             | 25           | Steel low alloyed          |
| furnace & cowper, concrete                            | 30            | m <sup>3</sup> | 25           | reinforced concrete        |
| furnace & cowper, refractory materials                | 100'000       | kg             | 25           | firebrick                  |
| cast house                                            | 10'000        | m <sup>2</sup> | 50           | Building, hall             |
| casting machines                                      | 10'000'000    | kg             | 25           | heavy machine, unspecific  |
| storage house                                         | 9'500         | m <sup>2</sup> | 50           | Building, hall             |

Tab. 5.5 Infrastructure of the blast furnace

|                                              | blast furnace | Unit           | Remarks                 |
|----------------------------------------------|---------------|----------------|-------------------------|
| Yearly production capacity                   | 1'500'000'000 | kg             |                         |
| Transformed area                             | 976'000       | m <sup>2</sup> |                         |
| Building, multi story                        | 25'000        | m <sup>3</sup> |                         |
| Building, hall                               | 19'500        | m <sup>2</sup> |                         |
| reinforced concrete                          | 1'260         | m <sup>3</sup> |                         |
| concrete in reinforced concrete (91 % (v/v)) | 1'147         | m <sup>3</sup> |                         |
| steel in reinforced concrete (9 % (v/v))     | 793'800       | kg             |                         |
| heavy machine, unspecific                    | 20'200'000    | kg             |                         |
| Poor concrete for additional foundation      | 9'182         | m <sup>3</sup> | same weight as machines |
| conveyor belt                                | 1'800         | m              |                         |
| Steel low alloyed                            | 467'972       | kg             |                         |
| firebrick                                    | 200'000       | kg             |                         |

By-products of the pig iron production in blast furnaces are:

- Slag that is used in road building, concrete aggregate, thermal insulation (mineral wool) and as a cement replacement. Total utilisation of blast furnace slag is a target which has already been met in many cases.
- BF gas that contains about 20-28% CO, 1–5% H<sub>2</sub>, inert compounds (50-55% N<sub>2</sub>, 17-25% CO<sub>2</sub>), some sulphur and cyanide compounds and large amounts of dust from the burden. After treatment and enrichment, BF gas can be used in the hot blast stove or e.g. in the sinter production.
- Electricity recovered of the top gas pressure energy. Many plants with adequate top gas pressure (> 1.5 bar gauge) and sufficient pressure drop are suitable for installation of expansion turbines.

## 5.4.2 Emissions

### Air

Relevant direct air emissions from the pig iron production are CO, CO<sub>2</sub>, nitrogen oxides, sulphur oxides and heavy metals. The casting is responsible for most of the direct dust emissions.

CO<sub>2</sub> emission given in IPPC (2000) are considerably lower than the calculated C-balance suggests (Tab. 5.6). The calculated value from the C-balance (0.849 kg/kg) is used in ecoinvent v2.0 Data.

**Tab. 5.6 C-balance of blast furnace process**

|        | Flow            | comment                                                                        | Amount   | Unit            | kg C/ unit flow | kg C/ kg pig iron |
|--------|-----------------|--------------------------------------------------------------------------------|----------|-----------------|-----------------|-------------------|
| Input  | coke            | 0.34 kg/kg @ 28.6 MJ/kg                                                        | 9.72E+00 | MJ              | 0.031           | 0.306             |
|        | coal            |                                                                                | 1.50E-01 | kg              | 0.900           | 0.135             |
|        | BF gas          | energy content = 3 MJ/Nm <sup>3</sup> ;<br>C content = 0.241kg/Nm <sup>3</sup> | 9.00E-01 | Nm <sup>3</sup> | 0.241           | 0.217             |
|        | coke oven gas   | energy content = 3 MJ/Nm <sup>3</sup> ;<br>C content = 0.241kg/Nm <sup>3</sup> | 1.20E-02 | Nm <sup>3</sup> | 0.241           | 0.003             |
|        | natural gas     | 0.003 Nm <sup>3</sup> @ 40 MJ/Nm <sup>3</sup>                                  | 1.20E-01 | MJ              | 0.020           | 0.002             |
| Output | CO              |                                                                                | 1.34E-03 | kg              | 0.429           | 0.001             |
|        | CO <sub>2</sub> |                                                                                | 8.49E-01 | kg              | 0.273           | 0.232             |
|        | pig iron        |                                                                                | 1.00E+00 | kg              | 0.045           | 0.045             |
|        | BF gas          | energy content = 3 MJ/Nm <sup>3</sup> ;<br>C content = 0.241kg/Nm <sup>3</sup> | 1.60E+00 | Nm <sup>3</sup> | 0.241           | 0.386             |

## Water

There are no relevant direct water emissions from these processes

## 5.4.3 Waste

### Solid waste

Several different waste streams are generated. Some of them are recycled or used for different purposes and thus not inventoried. Others can not be used due to a high contamination and are deposited in landfills.

The sludge from BF gas treatment can't be reused because the zinc and lead compounds are mainly separated in the scrubber thus leading to a high contamination of the sludge. Sludge composition data are average data from IPPC (2000). The difference to 100 % is assumed to be water. (see Tab. 5.7).

There is no representative information available on the fate of the rubble waste/by-product. In some steelworks it is partly recycled to the BOF or reused for the production of new refractory material. In other cases it is put to landfill.

**Tab. 5.7 Typical composition in [% (w/w)] of sludge from BF gas treatment (IPPC (2000), Tab. 7.6)**

|                                | (IPPC (2000), Tab. 7.6) |      | This project |
|--------------------------------|-------------------------|------|--------------|
|                                | Min                     | Max  |              |
| C                              | 15                      | 47   | 31           |
| Fe                             | 7                       | 35   | 21           |
| Pb                             | 0.8                     | 2    | 1.4          |
| Zn                             | 0.1                     | 10   | 5.05         |
| Mn                             | 0.12                    | 0.14 | 0.13         |
| Al <sub>2</sub> O <sub>3</sub> | 0.8                     | 4.6  | 2.7          |
| S                              | 2.4                     | 2.5  | 2.45         |
| SiO <sub>2</sub>               | 3                       | 9    | 6            |
| P <sub>2</sub> O <sub>5</sub>  | 0.1                     | 0.44 | 0.27         |
| CaO                            | 3.5                     | 18   | 10.75        |
| MgO                            | 3.5                     | 17   | 10.25        |
| Na <sub>2</sub> O              | 0.15                    | 0.24 | 0.195        |
| K <sub>2</sub> O               | 0.08                    | 0.36 | 0.22         |
| water                          |                         |      | 8.585        |

### Waste water

Overflow of water from slag granulation primarily depends on water availability and is in the range 0.125 – 10 m<sup>3</sup>/t pig iron produced. Information on chemical composition from one German plant is given in Tab. 5.8 (IPPC (2000), Tab 7.8).

**Tab. 5.8 Waste water composition from pig iron production (IPPC (2000), Tab 7.8)**

| Parameter | Unit                | Number of measurements | Average concentration | Median concentration | Emission factor | Unit                |
|-----------|---------------------|------------------------|-----------------------|----------------------|-----------------|---------------------|
| Flow      | [m <sup>3</sup> /d] |                        | 16000                 |                      | 1.96            | [m <sup>3</sup> /t] |
| Pb        | [µg/l]              | 19                     | 3.45                  | 2.0                  | 3.92            | [mg/t pig iron]     |
| Cr        | [µg/l]              | 19                     | 3.32                  | 2.0                  | 3.92            | [mg/t pig iron]     |
| Cu        | [µg/l]              | 19                     | 5.99                  | 4.20                 | 8.232           | [mg/t pig iron]     |
| Zn        | [µg/l]              | 19                     | 37.63                 | 20.00                | 39.2            | [mg/t pig iron]     |
| Ni        | [µg/l]              | 19                     | 4.91                  | 3.00                 | 5.88            | [mg/t pig iron]     |
| TOC       | [mg/l]              | 19                     | 4.40                  | 4.10                 | 8.036           | [g/t pig iron]      |
| COD       | [mg/l]              | 19                     | 20.62                 | 20.30                | 39.788          | [g/t pig iron]      |

Emission factor is based on the median

### 5.4.4 Pig iron / hot metal in ecoinvent

The background information to the processes is given in Tab. 5.9, flows are shown in:

- pig iron, at plant Fig. 5.9
- blast furnace Fig. 5.10

Where (IPPC (2000)) is given as the source, arithmetic means of the low and high values are used in this project.

Transport distances for iron ore and pellets are shown in Tab. 5.2.

## Data quality

Data uncertainty is derived using the Pedigree matrix as described in Frischknecht et al. (2003) except for the infrastructure where this method would result in values far too low.

Data of input and main output materials for pig iron production are of good overall quality.

Emission data for pig iron production stem from 4 of 81 blast furnaces in Europe. Thus data for European consumption are of rather high uncertainty.

Data for the infrastructure is roughly estimated.

**Tab. 5.9 Meta information for the blast furnace processes**

|                                       |                                                                                                                                        |                                                                                                                               |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Name                                  | pig iron, at plant                                                                                                                     | blast furnace                                                                                                                 |
| Location                              | GLO                                                                                                                                    | RER                                                                                                                           |
| Infrastructure Process                | 0                                                                                                                                      | 1                                                                                                                             |
| Unit                                  | kg                                                                                                                                     | unit                                                                                                                          |
| Data Set Version                      | 2.0                                                                                                                                    | 2.0                                                                                                                           |
| Included Processes                    | Blast furnace process. Emissions are abated                                                                                            | Infrastructure for input material unloading and storage, blast furnace processes and casting and for administrative buildings |
| Amount                                | 1                                                                                                                                      | 1                                                                                                                             |
| Local Name                            | Roheisen, ab Werk                                                                                                                      | Hochofen                                                                                                                      |
| Synonyms                              | hot metal                                                                                                                              |                                                                                                                               |
| General Comment to reference function | Inputs and air emissions from different sources. Transports of iron ore calculated according to information about exports and imports. | Assumptions for medium sized plant. Output: 1'500'000 t/a, life time: 50 years. Site is shared with BOF converter.            |
| Start Date                            | 1999                                                                                                                                   | 2002                                                                                                                          |
| End Date                              | 2002                                                                                                                                   | 2002                                                                                                                          |
| Data Valid For Entire Period          | 1                                                                                                                                      | 1                                                                                                                             |
| Other Period Text                     |                                                                                                                                        |                                                                                                                               |
| Geography text                        | Inputs relate to the global average. Emissions relate to European plants                                                               | Some Swiss datasets are used for European processes.                                                                          |
| Technology text                       | Blast furnace process                                                                                                                  | Assumed technology of medium sized plant.                                                                                     |
| Representativeness [%]                | 90                                                                                                                                     |                                                                                                                               |
| Production Volume                     | 5.30E11 kg/a                                                                                                                           | unknown                                                                                                                       |
| Sampling Procedure                    | Main in- and outputs based on world average, emissions based on some plants in the EU. Everything based on literature.                 | Internet search                                                                                                               |
| Extrapolations                        | see Geography                                                                                                                          | see Geography                                                                                                                 |
| Uncertainty Adjustments               | none                                                                                                                                   | none                                                                                                                          |

| General Flow information       |                            |              |                                                                                                                                                                    | Representation in ecoinvent |              |                 |           |                                         |               |      | Uncertainty                |       |          |                      |
|--------------------------------|----------------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|--------------|-----------------|-----------|-----------------------------------------|---------------|------|----------------------------|-------|----------|----------------------|
| Input                          | Process Name               | Output       | Remarks                                                                                                                                                            | Cate gory                   | Sub category | Infra structure | Loca tion | Modul name in ecoinvent                 | Mean value    | Unit | Source mean value          | Typ e | StDv 95% | General Comment      |
| sinter                         | pig iron, at plant; part 1 |              |                                                                                                                                                                    | metals                      | extraction   | No              | GLO       | sinter, iron, at plant                  | 1.05E+00 kg   |      | Roth et al. 1999, Tab 24   | 1     | 1.05     | (1,2,1,1,1,1,3)      |
| pellets                        |                            |              |                                                                                                                                                                    | metals                      | extraction   | No              | GLO       | pellets, iron, at plant                 | 4.00E-01 kg   |      | Roth et al. 1999, Tab 24   | 1     | 1.05     | (1,2,1,1,1,1,3)      |
| lump ore                       |                            |              |                                                                                                                                                                    | metals                      | extraction   | No              | GLO       | iron ore, 65% Fe, at beneficiation      | 1.50E-01 kg   |      | Roth et al. 1999, Tab 24   | 1     | 1.05     | (1,2,1,1,1,1,3)      |
| transport ship (ocean)         |                            |              | pellets and lump ore from mine to plant by ship (ocean)                                                                                                            | transport systems           | ship         | No              | OCE       | transport, transoceanic freight ship    | 1.49E+00 tkm  |      | estimated                  | 1     | 2.01     | (2,nA,1,1,1,3,5)     |
| transport ship (costal)        |                            |              | pellets and lump ore from mine to plant by ship (costal)                                                                                                           | transport systems           | ship         | No              | RER       | transport, barge                        | 1.65E-02 tkm  |      | estimated                  | 1     | 2.01     | (2,nA,1,3,1,3,5)     |
| transport rail                 |                            |              | pellets and lump ore from mine to plant by rail                                                                                                                    | transport systems           | train        | No              | RER       | transport, freight, rail                | 1.51E-01 tkm  |      | estimated                  | 1     | 2.01     | (2,nA,1,3,1,3,5)     |
| additives                      |                            |              | limestone, according to IPPC 2000                                                                                                                                  | construction materials      | additives    | No              | CH        | limestone, at mine                      | 1.00E-02 kg   |      | Roth et al. 1999, Tab 24   | 1     | 1.06     | (1,2,1,3,1,1,4)      |
| refractory materials           |                            |              |                                                                                                                                                                    | construction materials      | bricks       | No              | DE        | refractory, fireclay, packed, at plant  | 2.00E-03 kg   |      | Roth et al. 1999, Tab 24   | 1     | 1.06     | (1,2,1,3,1,1,4)      |
| slag                           |                            |              | almost total utilisation --> not inventoried                                                                                                                       |                             |              |                 |           |                                         | 2.70E-01 kg   |      | Roth et al. 1999, Tab 24   |       |          |                      |
| water                          |                            |              |                                                                                                                                                                    | resource                    | in water     |                 |           | Water, unspecified natural origin       | 6.00E-03 m3   |      | Roth et al. 1999, Tab 24   | 1     | 1.05     | (1,2,1,1,1,1,12)     |
| coke                           |                            |              | 0.34 kg/kg @ 28.6 MJ/kg                                                                                                                                            | hard coal                   | fuels        | No              | RER       | hard coal coke, at plant                | 9.72E+00 MJ   |      | Roth et al. 1999, Tab 24   | 1     | 1.06     | (1,2,1,3,1,1,11)     |
| coal                           |                            |              |                                                                                                                                                                    | hard coal                   | fuels        | No              | UCTE      | hard coal mix, at regional storage      | 1.50E-01 kg   |      | Roth et al. 1999, Tab 24   | 1     | 1.06     | (1,2,1,3,1,1,11)     |
| electricity total              |                            |              | not inventoried because the same amount is generated (cf. outputs)                                                                                                 |                             |              |                 |           |                                         | 1.00E-01 kWh  |      | Roth et al. 1999, Tab 24   | 1     | 1.06     | (1,2,1,3,1,1,11)     |
| steam                          |                            |              | not inventoried as internally generated                                                                                                                            |                             |              |                 |           |                                         | 3.00E-02 kg   |      | Roth et al. 1999, Tab 24   |       |          |                      |
| BF gas                         |                            |              | Output higher than input                                                                                                                                           |                             |              |                 |           |                                         | 9.00E-01 Nm3  |      | Roth et al. 1999, Tab 24   |       |          |                      |
| coke oven gas                  |                            |              | cut off from coke oven                                                                                                                                             |                             |              |                 |           |                                         | 1.20E-02 Nm3  |      | Roth et al. 1999, Tab 24   |       |          |                      |
| natural gas                    |                            |              | 0.003 Nm3 @ 40 MJ/Nm3                                                                                                                                              | natural gas                 | fuels        | No              | RER       | natural gas, high pressure, at consumer | 1.20E-01 MJ   |      | Roth et al. 1999, Tab 24   | 1     | 1.06     | (1,2,1,3,1,1,11)     |
| transport road                 |                            |              | limestone, refractories, coal & coke: 20 km;                                                                                                                       | transport systems           | road         | No              | RER       | transport, lorry >16t, fleet average    | 1.00E-02 tkm  |      | estimated                  | 1     | 2.09     | (4,5,nA,nA,nA,nA,5)  |
| transport rail                 |                            |              | limestone, refractories, coal & coke: 200 km;                                                                                                                      | transport systems           | train        | No              | RER       | transport, freight, rail                | 1.00E-01 tkm  |      | estimated                  | 1     | 2.09     | (4,5,nA,nA,nA,nA,5)  |
| infrastructure (blast furnace) |                            |              |                                                                                                                                                                    | metals                      | extraction   | Yes             | RER       | blast furnace                           | 1.33E-11 unit |      | estimated                  | 1     | 3.23     | (5,nA,nA,nA,nA,nA,9) |
|                                |                            | waste heat   | from burning of coke, coal and natural gas. Heat from burning of BFGas and from electricity use is not included because net energy consumption is zero or negative | air                         | unspecified  |                 |           | Heat, waste                             | 1.43E+01 MJ   |      | calculated                 | 1     | 1.06     | (1,2,1,3,1,1,13)     |
|                                |                            | dust, coarse | 5% > 10 um, 5% between 2.5 and 10 um and 90% < 2.5 um (CEPMEIP database)                                                                                           | air                         | unspecified  |                 |           | Particulates, > 10 um                   | 1.60E-06 kg   |      | calculated after IPPC 2000 | 1     | 1.52     | (2,3,2,3,1,3,25)     |
|                                |                            | dust, medium |                                                                                                                                                                    | air                         | unspecified  |                 |           | Particulates, > 2.5 um, and < 10um      | 1.60E-06 kg   |      | calculated after IPPC 2000 | 1     | 2.01     | (2,3,2,3,1,3,26)     |
|                                |                            | dust, fine   |                                                                                                                                                                    | air                         | unspecified  |                 |           | Particulates, < 2.5 um                  | 2.87E-05 kg   |      | calculated after IPPC 2000 | 1     | 3.01     | (2,3,2,3,1,3,27)     |

Fig. 5.9 Flows for "pig iron, at plant" and its representation in the ecoinvent database

5. System Characterisation

| General Flow information   |                                   |                                                                                                                                                                                                                                                                                                      |         | Representation in ecoinvent |                            |                |          |                                                                                  |              |      | Uncertainty                |      |          |                  |
|----------------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-----------------------------|----------------------------|----------------|----------|----------------------------------------------------------------------------------|--------------|------|----------------------------|------|----------|------------------|
| Input                      | Process Name                      | Output                                                                                                                                                                                                                                                                                               | Remarks | Category                    | Sub category               | Infrastructure | Location | Module name in ecoinvent                                                         | Mean value   | Unit | Source mean value          | Type | StdV 95% | General Comment  |
| pig iron, at plant; part 2 | → Mn                              |                                                                                                                                                                                                                                                                                                      |         | air                         | unspecified                |                |          | Manganese                                                                        | 7.45E-08 kg  |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22) |
|                            | → Ni                              |                                                                                                                                                                                                                                                                                                      |         | air                         | unspecified                |                |          | Nickel                                                                           | 1.60E-08 kg  |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22) |
|                            | → Pb                              |                                                                                                                                                                                                                                                                                                      |         | air                         | unspecified                |                |          | Lead                                                                             | 6.91E-08 kg  |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22) |
|                            | → SOx                             |                                                                                                                                                                                                                                                                                                      |         | air                         | unspecified                |                |          | Sulfur dioxide                                                                   | 1.33E-04 kg  |      | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,15) |
|                            | → NOx                             |                                                                                                                                                                                                                                                                                                      |         | air                         | unspecified                |                |          | Nitrogen oxides                                                                  | 7.98E-05 kg  |      | calculated after IPPC 2000 | 1    | 1.52     | (2,3,2,3,1,3,16) |
|                            | → H2S                             |                                                                                                                                                                                                                                                                                                      |         | air                         | unspecified                |                |          | Hydrogen sulfide                                                                 | 1.07E-05 kg  |      | calculated after IPPC 2000 | 1    | 1.52     | (2,3,2,3,1,3,31) |
|                            | → CO                              |                                                                                                                                                                                                                                                                                                      |         | air                         | unspecified                |                |          | Carbon monoxide, fossil                                                          | 1.34E-03 kg  |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,17) |
|                            | → CO2                             | Average given in IPPC 2000 = 0.415 kg/kg. Value used calculated from C-balance                                                                                                                                                                                                                       |         | air                         | unspecified                |                |          | Carbon dioxide, fossil                                                           | 8.49E-01 kg  |      | calculated from C-balance  | 1    | 1.11     | (2,3,2,3,1,3,14) |
|                            | → dioxin (I-TEQ)                  |                                                                                                                                                                                                                                                                                                      |         | air                         | unspecified                |                |          | Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin                         | 2.66E-15 kg  |      | calculated after IPPC 2000 | 1    | 3.01     | (2,3,2,3,1,3,21) |
|                            | → slags                           | used in road construction --> not inventoried                                                                                                                                                                                                                                                        |         |                             |                            |                |          |                                                                                  | 2.61E-01 kg  |      | calculated after IPPC 2000 |      |          |                  |
|                            | → top gas dust                    | used in sinter plant --> not inventoried                                                                                                                                                                                                                                                             |         |                             |                            |                |          |                                                                                  | 1.17E-02 kg  |      | calculated after IPPC 2000 |      |          |                  |
|                            | → BF gas sludge                   |                                                                                                                                                                                                                                                                                                      |         | waste management            | residual material landfill | No             | CH       | disposal, sludge, pig iron production, 8.6% water, to residual material landfill | 4.26E-03 kg  |      | calculated after IPPC 2000 | 1    | 3.01     | (2,3,2,3,1,3,21) |
|                            | → dust from cast house de-dusting | used in sinter plant --> not inventoried                                                                                                                                                                                                                                                             |         |                             |                            |                |          |                                                                                  | 1.06E-03 kg  |      | calculated after IPPC 2000 |      |          |                  |
|                            | → rubble (refractory material)    | although partly recycled total amount is assumed to be disposed in inert material landfill.                                                                                                                                                                                                          |         | waste management            | inert material landfill    | No             | CH       | disposal, inert waste, 5% water, to inert material landfill                      | 2.07E-02 kg  |      | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,6)  |
|                            | → waste water                     |                                                                                                                                                                                                                                                                                                      |         | waste management            | wastewater treatment       | No             | CH       | treatment, pig iron production effluent, to wastewater treatment, class 3        | 1.81E-03 m3  |      | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,6)  |
|                            | → pig iron                        | 94 % Fe                                                                                                                                                                                                                                                                                              |         | metals                      | extraction                 | No             | GLO      | pig iron, at plant                                                               | 1.00E+00 kg  |      |                            |      |          |                  |
|                            | → BF gas                          | net output 0.7 Nm3 (input=900, output=1600 Nm3); energy content = 3 MJ/Nm3; C content = 0.241kg/Nm3 only if of top gas turbine is present. It is assumed that this is the case for 50% of the blast furnaces. Thus the electricity consumption of the average process is covered by this production. |         |                             |                            |                |          |                                                                                  | 1.60E+00 Nm3 |      | Roth et al. 1999           |      |          |                  |
|                            | → electricity                     |                                                                                                                                                                                                                                                                                                      |         |                             |                            |                |          |                                                                                  | 2.08E-01 kWh |      | IPPC 2000                  |      |          |                  |

Fig. 5.9 Flows for "pig iron, at plant" and its representation in the ecoinvent database (cont.)

| General Flow information                                       |                       |        |                                             | Representation in ecoinvent |              |                |          |                                                |            |      |                   | Uncertainty |          |                                        |  |
|----------------------------------------------------------------|-----------------------|--------|---------------------------------------------|-----------------------------|--------------|----------------|----------|------------------------------------------------|------------|------|-------------------|-------------|----------|----------------------------------------|--|
| Input                                                          | Process Name          | Output | Remarks                                     | Category                    | Sub category | Infrastructure | Location | Module name in ecoinvent                       | Mean value | Unit | Source mean value | Type        | StdV 95% | General Comment                        |  |
| Total land use (transformation)                                | blast furnace; part 1 |        |                                             | resource                    | land         |                |          | Transformation, from unknown                   | 9.76E+05   | m2   | estimated         | 1           | 5.00     | (5,nA,nA,nA,nA,nA,8); rough estimation |  |
| Land occupation for buildings and open storage area            |                       |        |                                             | resource                    | land         |                |          | Occupation, industrial area, built up          | 3.90E+07   | m2a  | estimated         | 1           | 5.00     | (5,nA,nA,nA,nA,nA,7); rough estimation |  |
| Land transformation to buildings and open bauxite storage area |                       |        | assumption: 80% of total area is built up   | resource                    | land         |                |          | Transformation, to industrial area, built up   | 7.81E+05   | m2   | estimated         | 1           | 5.00     | (5,nA,nA,nA,nA,nA,8); rough estimation |  |
| Land occupation for vegetation area                            |                       |        |                                             | resource                    | land         |                |          | Occupation, industrial area, vegetation        | 9.76E+06   | m2a  | estimated         | 1           | 5.00     | (5,nA,nA,nA,nA,nA,7); rough estimation |  |
| Land transformation to vegetation area                         |                       |        | assumption: 20% of total area is vegetation | resource                    | land         |                |          | Transformation, to industrial area, vegetation | 1.95E+05   | m2   | estimated         | 1           | 5.00     | (5,nA,nA,nA,nA,nA,8); rough estimation |  |
| Building, multi story                                          |                       |        |                                             | construction processes      | buildings    | Yes            | RER      | building, multi-storey                         | 2.50E+04   | m3   | estimated         | 1           | 5.00     | (5,nA,nA,nA,nA,nA,9); rough estimation |  |
| Building, hall                                                 |                       |        |                                             | construction processes      | buildings    | Yes            | CH       | building, hall                                 | 1.95E+04   | m2   | estimated         | 1           | 5.00     | (5,nA,nA,nA,nA,nA,9); rough estimation |  |
| concrete in reinforced concrete                                |                       |        |                                             | construction materials      | concrete     | No             | CH       | concrete, normal, at plant                     | 1.15E+03   | m3   | estimated         | 1           | 5.00     | (5,nA,nA,nA,nA,nA,9); rough estimation |  |
| steel in reinforced concrete                                   |                       |        |                                             | metals                      | extraction   | No             | RER      | reinforcing steel, at plant                    | 7.94E+05   | kg   | estimated         | 1           | 5.00     | (5,nA,nA,nA,nA,nA,9); rough estimation |  |

Fig. 5.10 Flows for "blast furnace" and its representation in the ecoinvent database

| General Flow information                |                       |                                 |                                          | Representation in ecoinvent |                     |                |          |                                                                 |               |           | Uncertainty       |      |          |                                        |
|-----------------------------------------|-----------------------|---------------------------------|------------------------------------------|-----------------------------|---------------------|----------------|----------|-----------------------------------------------------------------|---------------|-----------|-------------------|------|----------|----------------------------------------|
| Input                                   | Process Name          | Output                          | Remarks                                  | Category                    | Sub category        | Infrastructure | Location | Module name in ecoinvent                                        | Mean value    | Unit      | Source mean value | Type | StdV 95% | General Comment                        |
| heavy machine, unspecified              | blast furnace; part 2 |                                 |                                          | construction processes      | machinery           | Yes            | RER      | industrial machine, heavy, unspecified, at plant                | 2.02E+07 kg   | estimated |                   | 1    | 5.00     | (5,nA,nA,nA,nA,nA,9); rough estimation |
| Poor concrete for additional foundation |                       |                                 |                                          | construction materials      | concrete            | No             | CH       | concrete, sole plate and foundation, at plant                   | 9.18E+03 m3   | estimated |                   | 1    | 5.00     | (5,nA,nA,nA,nA,nA,9); rough estimation |
| Conveyor belt                           |                       |                                 |                                          | construction processes      | machinery           | Yes            | RER      | conveyor belt, at plant                                         | 1.80E+03 m    | estimated |                   | 1    | 5.00     | (5,nA,nA,nA,nA,nA,9); rough estimation |
| Steel low alloyed                       |                       |                                 |                                          | metals                      | extraction          | No             | RER      | steel, low-alloyed, at plant                                    | 4.68E+05 kg   | estimated |                   | 1    | 5.00     | (5,nA,nA,nA,nA,nA,9); rough estimation |
| firebrick                               |                       |                                 |                                          | construction materials      | bricks              | No             | DE       | refractory, basic, packed, at plant                             | 2.00E+05 kg   | estimated |                   | 1    | 5.00     | (5,nA,nA,nA,nA,nA,9); rough estimation |
|                                         |                       | reinforced concrete to disposal |                                          | waste management            | building demolition | No             | CH       | disposal, building, reinforced concrete, to final disposal      | 3.32E+06 kg   | estimated |                   | 1    | 5.00     | (5,nA,nA,nA,nA,nA,9); rough estimation |
|                                         |                       | foundation concrete to disposal |                                          | waste management            | building demolition | No             | CH       | disposal, building, concrete, not reinforced, to final disposal | 2.02E+07 kg   | estimated |                   | 1    | 5.00     | (5,nA,nA,nA,nA,nA,9); rough estimation |
|                                         |                       | firebricks to disposal          |                                          | waste management            | building demolition | No             | CH       | disposal, building, brick, to final disposal                    | 2.00E+05 kg   | estimated |                   | 1    | 5.00     | (5,nA,nA,nA,nA,nA,9); rough estimation |
|                                         |                       | blast furnace                   | Output: 1500000000 kg/a, life time: 50 a | metals                      | extraction          | Yes            | RER      | blast furnace                                                   | 1.00E+00 unit |           |                   |      |          |                                        |

Fig. 5.10 Flows for "blast furnace" and its representation in the ecoinvent database (cont.)



## 5.5 Making available of iron scrap

About 35% of the cast iron and steel production is based on scrap as iron source. While in the BOF converter scrap is only used to control the temperature of the melt, it is the only (or at least main) iron bearing input in the electric arc furnace.

A wide variety of iron scrap is processed in the secondary iron and steel industry. A basic differentiation is made between "new" and "old" scrap.

New scrap is surplus material that arises during production and fabrication of iron and steel products up to the point where they are sold to the final consumer. It is usually of known quality and alloy and often uncoated. Some new scrap is coated with paints or plastics.

Old scrap or post-consumer scrap is the iron and steel recovered after the product has been used by the final consumer. It comes into the secondary iron and steel industry via metal merchants and waste management companies which recover aluminium from e.g. vehicles, household goods etc. This is usually done by shredding, magnetic separation and "sink-and-float" installations or eddy current installations successively.

Because of the high process temperature and the addition of slag builder old and new scrap can be remelted with little preparation (no de-coating). No information on the share of the two types on the overall scrap market and on specific treatment or emissions from remelting of one or the other was found.

### 5.5.1 Iron scrap in ecoinvent

The background information to the processes are given in Tab. 5.10, flows are shown in Fig. 5.11.

#### Data quality

Data are based on rough assumptions and of poor quality.

Data uncertainty is not derived using the Pedigree matrix as described in (Frischknecht et al. (2003)) because the resulting values would seem to low.

**Tab. 5.10 Meta information for the iron scrap collection and sorting processes**

|                                       |                                                                                                |
|---------------------------------------|------------------------------------------------------------------------------------------------|
| Name                                  | iron scrap, at plant                                                                           |
| Location                              | RER                                                                                            |
| Infrastructure Process                | 0                                                                                              |
| Unit                                  | kg                                                                                             |
| Data Set Version                      | 2.0                                                                                            |
| Included Processes                    | Collecting of new and old iron scrap, transport to scrap-yard, sorting and pressing to blocks. |
| Amount                                | 1                                                                                              |
| Local Name                            | Eisenschrott, ab Werk                                                                          |
| Synonyms                              |                                                                                                |
| General Comment to reference function | Data based on assumptions.                                                                     |
| Start Date                            | 2002                                                                                           |
| End Date                              | 2002                                                                                           |
| Data Valid For Entire Period          | 1                                                                                              |
| Other Period Text                     |                                                                                                |
| Geography text                        | Data relate to plants in the EU                                                                |
| Technology text                       | Assumed technology of medium sized plant.                                                      |
| Representativeness [%]                |                                                                                                |
| Production Volume                     | unknown                                                                                        |
| Sampling Procedure                    | Internet search                                                                                |
| Extrapolations                        | none                                                                                           |
| Uncertainty Adjustments               | none                                                                                           |

| General Flow information                        |                      |            |                                                                         | Representation in ecoinvent |                |                |          |                                                       |            |      |                   | Uncertainty |          |                  |  |
|-------------------------------------------------|----------------------|------------|-------------------------------------------------------------------------|-----------------------------|----------------|----------------|----------|-------------------------------------------------------|------------|------|-------------------|-------------|----------|------------------|--|
| Input                                           | Process Name         | Output     | Remarks                                                                 | Category                    | Sub category   | Infrastructure | Location | Module name in ecoinvent                              | Mean value | Unit | Source mean value | Type        | StdV 95% | General Comment  |  |
| scrap from production and end-user              | iron scrap, at plant |            | from recycling --> not inventoried                                      |                             |                |                |          |                                                       | 1.00E+00   | kg   | calculated        |             |          |                  |  |
| transport road                                  |                      |            | road: 100 km                                                            | transport systems           | road           | No             | RER      | transport, lorry >16t, fleet average                  | 1.00E-01   | tkm  | estimated         | 1           | 5.00     | rough estimation |  |
| transport rail                                  |                      |            | rail: 200 km                                                            | transport systems           | train          | No             | RER      | transport, freight, rail                              | 2.00E-01   | tkm  | estimated         | 1           | 5.00     | rough estimation |  |
| diesel powered equipment for sorting & pressing |                      |            |                                                                         | construction processes      | machinery      | No             | GLO      | diesel, burned in building machine                    | 1.00E-01   | MJ   | estimated         | 1           | 5.00     | rough estimation |  |
| electricity use for sorting and pressing        |                      |            |                                                                         | electricity                 | production mix | No             | UCTE     | electricity, medium voltage, production UCTE, at grid | 1.00E-02   | kWh  | estimated         | 1           | 5.00     | rough estimation |  |
| Infrastructure                                  |                      |            | aluminium scrap plant as proxy (output: 20000000 kg/a, life time: 50 a) | metals                      | extraction     | Yes            | RER      | scrap preparation plant                               | 1.00E-09   | unit | estimated         | 1           | 5.00     | rough estimation |  |
|                                                 |                      | waste heat | from electricity                                                        | air                         | unspecified    |                |          | Heat, waste                                           | 3.60E-02   | MJ   | calculated        | 1           | 5.00     | rough estimation |  |
|                                                 |                      | iron scrap |                                                                         | metals                      | extraction     | No             | RER      | iron scrap, at plant                                  | 1.00E+00   | kg   |                   |             |          |                  |  |

Fig. 5.11 Flows for "iron scrap" and its representation in the ecoinvent database

## 5.6 Basic oxygen furnace (BOF) steel making

This part is mainly based on (IPPC (2000)).

The objective in oxygen steel making is to burn (i.e., oxidise) the undesirable impurities contained in the metallic feedstock. The main elements thus converted into oxides are carbon, silicon, manganese, phosphorus, and sulphur. The purpose of this oxidation process, therefore, is:

- to reduce the carbon content to a specified level (from approximately 4% to less than 1 %, but often lower)
- to adjust the contents of desirable foreign elements
- to remove undesirable impurities to the greatest possible extent

### 5.6.1 Production process and infrastructure

The production of steel by the BOF process is a discontinuous process which involves the following steps:

- transfer and storage of hot metal
- pre-treatment of hot metal (desulphurisation)
- oxidation in the BOF (decarburisation and oxidation of impurities)
- secondary metallurgical treatment
- casting (continuous or/and ingot)

The individual steps and their associated emissions are summarised in Fig. 5.12. More information on the individual steps is given in (IPPC (2000)).

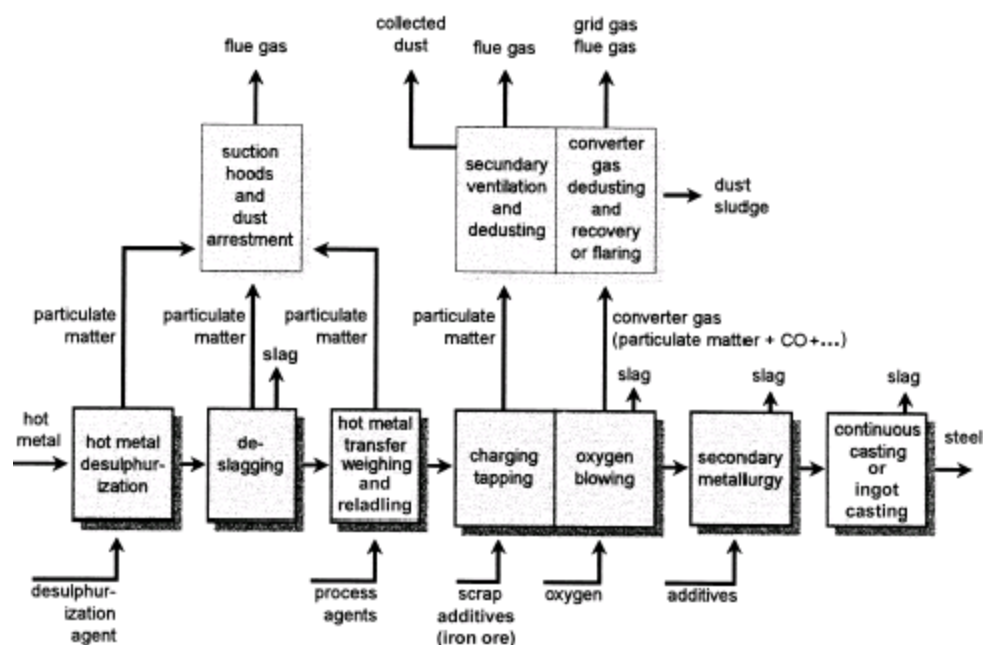


Fig. 5.12 Individual steps in BOF steel making (IPPC (2000))

Desulphurisation can be done in the torpedo ladles used for transporting the hot metal from the blast furnace to the BOF. The most widespread hot metal desulphurisation method used in Europe today is

that based on calcium carbide, which has superseded the previous soda process for waste disposal and air quality management reasons.

There are several types of reactors used for the basic oxygen steel making process. The most commonly used type is the LD converter (Linz-Donawitz) applied for pig iron with low phosphorous content; in the case of high phosphorous content a modified process is used (LD/AC process = Linz-Donawitz/Arbed-CRM-process). The converter is a pear-shaped, refractory-lined reactor into which a water-cooled oxygen lance is lowered. Through this lance pure oxygen (>99%) from an air separation plant is blown onto the liquid pig iron. The amount of oxygen consumed depends on the composition of the hot metal (i.e. C, Si, P content). Thus undesired impurities are oxidised with subsequent removal with the off-gas or slag in the BOF furnace. The energy required to raise the temperature and melt the input materials is supplied by the exothermic oxidation reactions, so that no additional heat input is required, on the one hand, and scrap or ore have to be added to balance heat on the other hand. Usually, approximately 10-20% of the converter charge is scrap, but values up to 40% are sometimes used. The amount of scrap charged depends on the pre-treatment given to the pig iron and the required liquid steel tapping temperature. Variations in the market value of scrap and the required steel specifications also have an influence.

The oxidising process in the converter is usually followed by post-treatment comprising a number of metallurgical operations referred to as "secondary metallurgy". The main objectives of secondary metallurgy are:

- mixing and homogenising
- adjustment of chemical compositions to close analysis tolerances
- temperature adjustment in time for the downstream casting process
- deoxidisation
- removal of undesirable gases such as hydrogen and nitrogen
- improvement of the oxidic purity by separating non-metallic inclusions.

An overview of secondary metallurgy operations is given in (IPPC (2000)).

Once the final steel quality has been achieved the steel is conveyed in a casting ladle to the casting machines. Today, the method of choice (>95% in Europe) is continuous casting, whereby the steel is cast in a continuous strand.

No data have been found on the infrastructure needed for the process. Assumptions made in this project are based on different pictures and information on the aluminium production infrastructure (see Tab. 5.11).

**Tab. 5.11 Assumptions for infrastructure of BOF converters**

| <b>blast oxygen furnace converter</b>       | <b>amount</b> | <b>unit</b>    | <b>lifetime [a]</b> | <b>Material / Module</b>   |
|---------------------------------------------|---------------|----------------|---------------------|----------------------------|
| yearly production capacity (5000h; 300 t/h) | 1'500'000'000 | kg             | 50                  | Yearly production capacity |
| area of plant (500 x 100 m)                 | 500'000       | m <sup>2</sup> | 50                  | Transformed area           |
| administration building (20x10x10 m3)       | 2'000         | m <sup>3</sup> | 50                  | Building, multi story      |
| production hall (70% of total area)         | 400'000       | m <sup>2</sup> | 50                  | Building, hall             |
| torpedo ladle (without lining)              | 30'000        | kg             | 10                  | heavy machine, unspecific  |
| furnace & ladle                             | 1'000'000     | kg             | 25                  | heavy machine, unspecific  |
| lining for furnace & ladle                  | 100'000       | kg             | 25                  | firebrick                  |
| casting machines                            | 10'000'000    | kg             | 25                  | heavy machine, unspecific  |

The infrastructure is modelled to be used for 50 years. The replacement of the machines with a life-time of only 25 years is taken into account by doubling the amount, the amount of torpedo ladles is inventoried five times. The infrastructure is disposed after its useful life in inert material landfills. Thus the values given in Tab. 5.12 are inventoried. For the waste disposal it is assumed that the conveyer belts and the machines are completely recycled. The disposal of the buildings is already inventoried in the building modules. Thus only the concrete and the refractory materials are disposed.

**Tab. 5.12 Infrastructure of BOF converters**

|                                         | <b>blast oxygen furnace converter</b> | <b>Unit</b>    | <b>Remarks</b>          |
|-----------------------------------------|---------------------------------------|----------------|-------------------------|
| Yearly production capacity              | 1'500'000'000                         | kg             |                         |
| Transformed area                        | 500'000                               | m <sup>2</sup> |                         |
| Building, multi story                   | 2'000                                 | m <sup>3</sup> |                         |
| Building, hall                          | 400'000                               | m <sup>2</sup> |                         |
| heavy machine, unspecific               | 22'150'000                            | kg             |                         |
| Poor concrete for additional foundation | 10'068                                | m3             | same weight as machines |
| firebrick                               | 200'000                               | kg             |                         |

## 5.6.2 Emissions

### Air

The gases produced during oxygen blowing (converter gas) contain large amounts of carbon monoxide. In many steel making plants, measures have been taken to recover the converter gas and use it as an energy source. Both "open combustion" and "suppressed combustion" systems are in use. Open combustion systems introduce air into the converter flue gas duct, thus combusting the carbon monoxide. The heat generated is later recovered in a waste heat boiler. In suppressed combustion, a skirt is lowered over the converter mouth during oxygen blowing. Thus, ambient oxygen can not enter the flue gas duct and the combustion of carbon monoxide is prevented. The CO-rich flue gas can be collected, cleaned and stored for subsequent use as fuel. A main advantage of suppressed combustion is the smaller flue gas flow since no combustion occurs and no additional air-nitrogen is introduced. This results in higher productivity since oxygen blowing speed can be increased.

The oxygen steel making process also generates considerable quantities of particulate matter, during charging of scrap and hot metal, blowing and during tapping of slag and liquid steel. All steel making shops in the EU have taken measures to reduce particulate matter emissions. The overall emissions given are in the lowest range in (Berdowski et al. (1995)) thus all particles are assumed to be below 2.5 µm.

### Water

There are no reports of direct water emissions from these processes.

## 5.6.3 Waste

During the steel making process, slag is formed. Slag control is intended to effectively reduce the amount of undesirable substances contained in the hot metal and to generate slag of high quality that will be suitable for subsequent processing and usage. Usually, the slag is cooled and crushed, after which metallic iron is recovered by magnetic separation. The technical properties of the slag make it suitable for many kinds of application in civil and hydraulic engineering. Because of its structure, LD

slag has high abrasion resistance and is therefore often used for road construction. It is also put to other uses or disposed of in landfills.

Desulphurisation slag is a heterogeneous slag which is only partially melted. The composition of desulphurisation slag strongly depends on the used desulphurisation agents. The typical composition of such a slag are given in Tab. 5.13.

**Tab. 5.13 Chemical composition of slag from pig iron desulphurisation [% (w/w)] (IPPC (2000), Tab.8.3)**

| CaO  | SiO <sub>2</sub> | Al <sub>2</sub> O <sub>3</sub> | MgO | Total Fe | Metallic Fe | MnO   | P <sub>2</sub> O <sub>5</sub> | Cr <sub>2</sub> O <sub>3</sub> | Free CaO | S   | CaO/SiO <sub>2</sub> |
|------|------------------|--------------------------------|-----|----------|-------------|-------|-------------------------------|--------------------------------|----------|-----|----------------------|
| 27.0 | 18               | 8                              | 10  | 20       | 15          | ≤ 0.5 | ≤ 0.2                         | ≤ 0.1                          | ≤ 5      | ≤ 4 | 1.5                  |

In this project ≤ x is interpreted as x/2

The relatively high sulphur content and unsatisfactory mechanical properties do not make desulphurisation slag ideal for reuse. Partly they are used for landfill construction or for noise protection barriers. In Europe 41% is landfilled.

The slag from basic oxygen steel making makes up the largest share of residues from BOF steel making. The chemical composition of this slag depends on the processes employed (see Tab. 5.14).

BOF slag can be reused by returning it to the iron-making process. It can also be used for fertiliser manufacture or for road construction. Lime phosphate slag (from LD/AC or OBM processes) is used entirely for fertiliser production. BOF slag is normally used in civil and hydraulic engineering, road construction and cement industry. Thereby the relatively high free CaO content has to be considered. Nevertheless, in the EU a considerable percentage (26%) of BOF slag is still put to landfill. For the composition of BOF slag used in this project see Tab. 5.14. Katz & Salem (1994) give a chromium content for steel mill slag of 100-200 ppm and 9'000-48'000 ppm. This indicates a rather wide range in which the value used in this project is included.

**Tab. 5.14 Chemical composition of BOF slag [% (w/w)] (IPPC (2000), Tab.8.8)**

| Process                        | LD/AC | LD   | AOD   | This project |
|--------------------------------|-------|------|-------|--------------|
| CaO                            | 50.0  | 50.0 | 53.0  | 51           |
| SiO <sub>2</sub>               | 9.0   | 15.0 | 28.0  | 16           |
| Al <sub>2</sub> O <sub>3</sub> | ≤ 2   | ≤ 2  | 3.0   | 1.6          |
| MgO                            | ≤ 3   | ≤ 3  | 5.0   | 2.6          |
| Total Fe                       | 12.0  | 16.0 | ≤ 2   | 8.9          |
| Metallic Fe                    | ≤ 1   | ≤ 1  | ≤ 1   | 0.5          |
| MnO                            | 2.0   | ≤ 4  | ≤ 1   | 1.5          |
| P <sub>2</sub> O <sub>5</sub>  | 15.0  | ≤ 2  | ≤ 0.5 | 8.7          |
| Cr <sub>2</sub> O <sub>3</sub> | ≤ 1   | ≤ 1  | ≤ 2   | 0.66         |
| Free CaO                       | ≤ 7   | ≤ 10 | ≤ 5   | 3.3          |
| S                              | -     | -    | ≤ 1   | 0.16         |
| CaO/SiO <sub>2</sub>           | 4     | 2.5  | 1.8   | 3.1          |

Legend: LD/AC = Linz-Donawitz/Arbed-CRM process; LD = Linz-Donawitz process;  
AOD = Argon-Oxygen-Decarburisation

This project = 0.57 \* LD/AC + 0.11 \* LD + 0.32 \* AOD based on capacity in Europe (IPPC (2000), Tab.8.1) with ≤ x interpreted as x/2

Coarse dust waste is usually reused while fine dust, due to high Pb and Zn contamination is partly landfilled. About 60% of the total dust waste is fine dust of which 42% is landfilled (IPPC (2000), Tab.8.6, Fig. 8.15). Thus it is assumed that 25% of the total dust is landfilled. The composition of this dust is assumed according to the information for fine dust of one plant given in Tab. 5.15. The composition of this dust seems rather incomplete. Especially the lack of Cr and Ni in the dust also for stainless steel probably is due to these elements not being measured. Considering the high environmental relevance of these elements in landfills, this data gap would lead to considerable underestimation of environmental effects. Data for the electric arc furnace (EAF) dusts are much more comprehensible (cf. Tab. 5.21). For this reason, the disposal modules of EAF dusts are inventoried as proxy for the disposal of the BOF dust.

**Tab. 5.15 Chemical composition of the BOF dust fraction that is landfilled [% (w/w)]. Data not used in this project (cf text) (IPPC (2000), Tab.8.9)**

| Parameter   | Coarse dust | Fine dust   |
|-------------|-------------|-------------|
| Total Fe    | 30 – 85     | 54 – 70     |
| Metallic Fe | 72          | 20          |
| CaO         | 8 – 21      | 3 – 11      |
| Zn          | 0.01 – 0.4  | 1.4 – 3.2   |
| Pb          | 0.01 – 0.04 | 0.2 – 1.0   |
| S           | 0.02 – 0.06 | 0.07 – 0.12 |
| C           | 1.4         | 0.7         |

#### 5.6.4 Converter steel in ecoinvent

The background information to the processes is given in Tab. 5.17, flows are shown in:

- steel, converter, unalloyed, at plant Fig. 5.13
- steel, converter, low-alloyed, at plant Fig. 5.14
- steel, converter, chromium steel 18/8, at plant Fig. 5.15
- blast oxygen furnace converter Fig. 5.16

If not stated otherwise, arithmetic means of the low and high values given in (IPPC (2000)) are used in this project.

Because the different slag fractions of the BOF process that are not recycled or reused can be assumed to be landfilled together, these wastes are treated as one having an average composition according to Tab. 5.16.



**Tab. 5.16 Chemical composition of the average BOF slag to landfill [% (w/w)]**

|                                | desulphurisation slag | BOF slag | BOF slag, average |
|--------------------------------|-----------------------|----------|-------------------|
| CaO                            | 27                    | 51       | 48.3              |
| SiO <sub>2</sub>               | 18                    | 16       | 16.5              |
| Al <sub>2</sub> O <sub>3</sub> | 8                     | 1.6      | 2.51              |
| MgO                            | 10                    | 2.6      | 3.66              |
| Total Fe                       | 20                    | 8.9      | 10.5              |
| Metallic Fe                    | 15                    | 0.5      | 2.51              |
| MnO                            | 0.25                  | 1.5      | 1.35              |
| P <sub>2</sub> O <sub>5</sub>  | 0.1                   | 8.7      | 7.62              |
| Cr <sub>2</sub> O <sub>3</sub> | 0.05                  | 0.66     | 0.584             |
| Free CaO                       | 0.25                  | 3.3      | 2.92              |
| S                              | 2                     | 0.16     | 0.417             |
| CaO/SiO <sub>2</sub>           | 1.5                   | 3.1      | 2.92              |

BOF waste, average = 0.14 \* desulphurisation slag + 0.86 \* BOF slag (according to amount to landfill),  
linear extrapolation to 100% (from 102%)

For the elemental composition, CaOSiO<sub>2</sub> is interpreted as 50% CaO and 50% SiO<sub>2</sub>

### Data quality

Data uncertainty is derived using the Pedigree matrix as described in (Frischknecht et al. (2003)) except for the infrastructure where this method would result in values far too low.

Data of input and main output materials as well as emission data for BOF steel production stem from 4 of 95 basic oxygen furnaces in Europe. Thus data for European consumption are of rather high uncertainty.

Data for the infrastructure is roughly estimated.

Tab. 5.17 Meta information for the BOF steel production processes

|                                       |                                                                                                   |                                                                                                   |                                                                                                   |                                                                                                                                         |
|---------------------------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Name                                  | steel, converter, unalloyed, at plant                                                             | steel, converter, low-alloyed, at plant                                                           | steel, converter, chromium steel 18/8, at plant                                                   | blast oxygen furnace converter                                                                                                          |
| Location                              | RER                                                                                               | RER                                                                                               | RER                                                                                               | RER                                                                                                                                     |
| Infrastructure Process                | 0                                                                                                 | 0                                                                                                 | 0                                                                                                 | 1                                                                                                                                       |
| Unit                                  | kg                                                                                                | kg                                                                                                | kg                                                                                                | unit                                                                                                                                    |
| Data Set Version                      | 2.0                                                                                               | 2.0                                                                                               | 2.0                                                                                               | 2.0                                                                                                                                     |
| Included Processes                    | Transports of hot metal and other input materials to converter, steel making process and casting. | Transports of hot metal and other input materials to converter, steel making process and casting. | Transports of hot metal and other input materials to converter, steel making process and casting. | Infrastructure for input material unloading, for BOF and secondary metallurgical processes and casting and for administrative buildings |
| Amount                                | 1                                                                                                 | 1                                                                                                 | 1                                                                                                 | 1                                                                                                                                       |
| Local Name                            | Blasstahl, unlegiert, ab Werk                                                                     | Blasstahl, niedriglegiert, ab Werk                                                                | Blasstahl, Chromstahl 18/8, ab Werk                                                               | Sauerstoff-blaskonverter, Stahl                                                                                                         |
| Synonyms                              |                                                                                                   |                                                                                                   |                                                                                                   |                                                                                                                                         |
| General Comment to reference function | This process produces primary steel. Scrap is only used for cooling the liquid steel.             | This process produces primary steel. Scrap is only used for cooling the liquid steel.             | This process produces primary steel. Scrap is only used for cooling the liquid steel.             | Assumptions for medium sized plant. Output: 1'500'000 t/a, life time: 50 years. Site is shared with blast furnace.                      |
| Start Date                            | 2001                                                                                              | 2001                                                                                              | 2001                                                                                              | 2002                                                                                                                                    |
| End Date                              | 2001                                                                                              | 2001                                                                                              | 2001                                                                                              | 2002                                                                                                                                    |
| Data Valid For Entire Period          | 1                                                                                                 | 1                                                                                                 | 1                                                                                                 | 1                                                                                                                                       |
| Other Period Text                     |                                                                                                   |                                                                                                   |                                                                                                   |                                                                                                                                         |
| Geography text                        | Data relate to plants in the EU                                                                   | Data relate to plants in the EU                                                                   | Data relate to plants in the EU                                                                   | Some Swiss datasets are used for European processes.                                                                                    |
| Technology text                       | EU technology mix (mainly LD converter)                                                           | EU technology mix (mainly LD converter)                                                           | EU technology mix (mainly LD converter)                                                           | Assumed technology of medium sized plant.                                                                                               |
| Representativeness [%]                |                                                                                                   |                                                                                                   |                                                                                                   |                                                                                                                                         |
| Production Volume                     | unknown                                                                                           | unknown                                                                                           | unknown                                                                                           | unknown                                                                                                                                 |
| Sampling Procedure                    | based on literature                                                                               | based on literature                                                                               | based on literature                                                                               | Internet search                                                                                                                         |
| Extrapolations                        | none                                                                                              | none                                                                                              | none                                                                                              | see Geography                                                                                                                           |
| Uncertainty Adjustments               | none                                                                                              | none                                                                                              | none                                                                                              | none                                                                                                                                    |

| General Flow information |                                               |             |                                                                             | Representation in ecoinvent |                |                |          |                                                                                               |              |      |                            | Uncertainty |          |                  |  |
|--------------------------|-----------------------------------------------|-------------|-----------------------------------------------------------------------------|-----------------------------|----------------|----------------|----------|-----------------------------------------------------------------------------------------------|--------------|------|----------------------------|-------------|----------|------------------|--|
| Input                    | Process Name                                  | Output      | Remarks                                                                     | Category                    | Sub category   | Infrastructure | Location | Module name in ecoinvent                                                                      | Mean value   | Unit | Source mean value          | Type        | StdV 95% | General Comment  |  |
| pig iron / hot metal     | steel, converter, unalloyed, at plant; part 1 |             |                                                                             |                             |                |                |          |                                                                                               |              |      |                            |             |          |                  |  |
| iron scrap               |                                               |             |                                                                             | metals                      | extraction     | No             | GLO      | pig iron, at plant                                                                            | 9.00E-01 kg  |      | calculated after IPPC 2000 | 1           | 1.11     | (2,3,2,3,1,3,3)  |  |
| iron ore                 |                                               |             |                                                                             | metals                      | extraction     | No             | RER      | iron scrap, at plant                                                                          | 2.13E-01 kg  |      | calculated after IPPC 2000 | 1           | 1.11     | (2,3,2,3,1,3,3)  |  |
| other Fe material        |                                               |             |                                                                             | metals                      | extraction     | No             | GLO      | iron ore, 65% Fe, at beneficiation                                                            | 1.35E-02 kg  |      | calculated after IPPC 2000 | 1           | 1.11     | (2,3,2,3,1,3,3)  |  |
| coke                     |                                               |             |                                                                             | metals                      | extraction     | No             | GLO      | iron ore, 65% Fe, at beneficiation                                                            | 8.50E-03 kg  |      | calculated after IPPC 2000 | 1           | 1.11     | (2,3,2,3,1,3,3)  |  |
| lime                     |                                               | 28.6 MJ/kg  |                                                                             | hard coal                   | fuels          | No             | RER      | hard coal coke, at plant                                                                      | 2.50E-04 MJ  |      | calculated after IPPC 2000 | 1           | 1.11     | (2,3,2,3,1,3,11) |  |
| dolomite                 |                                               |             |                                                                             | construction materials      | additives      | No             | CH       | quicklime, in pieces, loose, at plant                                                         | 4.25E-02 kg  |      | calculated after IPPC 2000 | 1           | 1.11     | (2,3,2,3,1,3,4)  |  |
| alloys                   |                                               |             |                                                                             | construction materials      | others         | No             | RER      | dolomite, at plant                                                                            | 2.75E-03 kg  |      | calculated after IPPC 2000 | 1           | 1.11     | (2,3,2,3,1,3,4)  |  |
| oxygen (99%)             |                                               |             |                                                                             | metals                      | extraction     | No             | GLO      | ferronickel, 25% Ni, at plant                                                                 | 6.00E-03 kg  |      | calculated after IPPC 2000 | 1           | 1.11     | (2,3,2,3,1,3,3)  |  |
| Energy from natural gas  |                                               | 1.429 kg/m3 |                                                                             | chemicals                   | inorganics     | No             | RER      | oxygen, liquid, at plant                                                                      | 7.15E-02 kg  |      | calculated after IPPC 2000 | 1           | 1.11     | (2,3,2,3,1,3,4)  |  |
| electricity              |                                               |             |                                                                             | natural gas                 | fuels          | No             | RER      | natural gas, high pressure, at consumer electricity, medium voltage, production UCTE, at grid | 3.75E-02 MJ  |      | calculated after IPPC 2000 | 1           | 1.11     | (2,3,2,3,1,3,11) |  |
| steam                    |                                               |             |                                                                             | electricity                 | production mix | No             | UCTE     |                                                                                               | 2.19E-02 kWh |      | calculated after IPPC 2000 | 1           | 1.11     | (2,3,2,3,1,3,2)  |  |
| compressed air           |                                               |             | net steam production --> consumption not inventoried                        |                             |                |                |          |                                                                                               | 8.50E-02 MJ  |      | calculated after IPPC 2000 |             |          |                  |  |
| water                    |                                               |             | not inventoried. Energy for compressing included in electricity consumption |                             |                |                |          |                                                                                               | 1.10E-02 m3  |      | calculated after IPPC 2000 |             |          |                  |  |
| transport ship (ocean)   |                                               |             |                                                                             | resource                    | in water       |                |          | Water, unspecified natural origin                                                             | 2.70E-03 m3  |      | calculated after IPPC 2000 | 1           | 1.11     | (2,3,2,3,1,3,12) |  |
| transport ship (costal)  |                                               |             | iron ore from mine to plant by ship (costal)                                | transport systems           | ship           | No             | OCE      | transport, transoceanic freight ship                                                          | 5.94E-02 tkm |      | calculated (Tab. 4.2)      | 1           | 2.01     | (2,nA,1,1,1,3,5) |  |
|                          |                                               |             | iron ore from mine to plant by ship (ocean)                                 | transport systems           | ship           | No             | RER      | transport, barge                                                                              | 6.60E-04 tkm |      | calculated (Tab. 4.2)      | 1           | 2.01     | (2,nA,1,3,1,3,5) |  |

Fig. 5.13 Flows for "steel, converter, unalloyed, at plant" and its representation in the ecoinvent database

| General Flow information |                                               |                |                                                                                                                                              | Representation in ecoinvent |              |                |          |                                                          |            |      |                            | Uncertainty |          |                      |  |
|--------------------------|-----------------------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|--------------|----------------|----------|----------------------------------------------------------|------------|------|----------------------------|-------------|----------|----------------------|--|
| Input                    | Process Name                                  | Output         | Remarks                                                                                                                                      | Category                    | Sub category | Infrastructure | Location | Module name in ecoinvent                                 | Mean value | Unit | Source mean value          | Type        | StdV 95% | General Comment      |  |
| transport road           | steel, converter, unalloyed, at plant; part 2 |                | iron scrap: 100 km; coke, lime, dolomite, alloys: 20 km; iron ore from mine to plant by train                                                | transport systems           | road         | No             | RER      | transport, lorry >16t, fleet average                     | 2.23E-02   | tkm  | estimated                  | 1           | 2.09     | (4,5,nA,nA,nA,nA,5)  |  |
| transport rail           |                                               |                | iron scrap: 600 km; coke, lime, dolomite, alloys: 200 km; iron ore from mine to plant by train                                               | transport systems           | train        | No             | RER      | transport, freight, rail                                 | 1.44E-01   | tkm  | estimated                  | 1           | 2.09     | (4,5,nA,nA,nA,nA,5)  |  |
| infrastructure           |                                               |                | for transport of hot metal, BOF process, secondary metallurgy and casting                                                                    | metals                      | extraction   | Yes            | RER      | blast oxygen furnace converter                           | 1.33E-11   | unit | estimated                  | 1           | 3.23     | (5,nA,nA,nA,nA,nA,9) |  |
|                          |                                               | waste heat     | from coke, natural gas and electricity. Waste heat from oxidation of C in pig iron to CO and CO2 already included in waste heat of pig iron. | air                         | unspecified  |                |          | Heat, waste                                              | 1.17E-01   | MJ   | calculated                 | 1           | 1.08     | (1,3,2,3,1,2,13)     |  |
|                          |                                               | dust, fine     | all dust is fine according to CEPMEIP database                                                                                               | air                         | unspecified  |                |          | Particulates, < 2.5 um                                   | 4.75E-05   | kg   | calculated after IPPC 2000 | 1           | 3.01     | (2,3,2,3,1,3,27)     |  |
|                          |                                               | Cr             |                                                                                                                                              | air                         | unspecified  |                |          | Chromium                                                 | 1.85E-07   | kg   | calculated after IPPC 2000 | 1           | 5.01     | (2,3,2,3,1,3,22)     |  |
|                          |                                               | Cu             |                                                                                                                                              | air                         | unspecified  |                |          | Copper                                                   | 2.50E-08   | kg   | calculated after IPPC 2000 | 1           | 5.01     | (2,3,2,3,1,3,22)     |  |
|                          |                                               | Pb             |                                                                                                                                              | air                         | unspecified  |                |          | Lead                                                     | 5.15E-07   | kg   | calculated after IPPC 2000 | 1           | 5.01     | (2,3,2,3,1,3,22)     |  |
|                          |                                               | Mn             |                                                                                                                                              | air                         | unspecified  |                |          | Manganese                                                | 6.05E-07   | kg   | calculated after IPPC 2000 | 1           | 5.01     | (2,3,2,3,1,3,22)     |  |
|                          |                                               | NOx            |                                                                                                                                              | air                         | unspecified  |                |          | Nitrogen oxides                                          | 1.25E-05   | kg   | calculated after IPPC 2000 | 1           | 1.52     | (2,3,2,3,1,3,16)     |  |
|                          |                                               | CO             |                                                                                                                                              | air                         | unspecified  |                |          | Carbon monoxide, fossil                                  | 4.73E-03   | kg   | calculated after IPPC 2000 | 1           | 5.01     | (2,3,2,3,1,3,17)     |  |
|                          |                                               | CO2            |                                                                                                                                              | air                         | unspecified  |                |          | Carbon dioxide, fossil                                   | 7.56E-02   | kg   | calculated after IPPC 2000 | 1           | 1.11     | (2,3,2,3,1,3,14)     |  |
|                          |                                               | PAH            |                                                                                                                                              | air                         | unspecified  |                |          | PAH, polycyclic aromatic hydrocarbons                    | 1.20E-10   | kg   | calculated after IPPC 2000 | 1           | 3.01     | (2,3,2,3,1,3,21)     |  |
|                          |                                               | dioxin (I-TEQ) |                                                                                                                                              | air                         | unspecified  |                |          | Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin | 3.05E-14   | kg   | calculated after IPPC 2000 | 1           | 3.01     | (2,3,2,3,1,3,21)     |  |

Fig. 5.13 Flows for "steel, converter, unalloyed, at plant" and its representation in the ecoinvent database (cont.)

5. System Characterisation

| General Flow information                      |   |              |  | Representation in ecoinvent    |                                                                                                                 |                  |                            |                  |           |                                                                                 |            | Uncertainty |                            |       |          |                 |
|-----------------------------------------------|---|--------------|--|--------------------------------|-----------------------------------------------------------------------------------------------------------------|------------------|----------------------------|------------------|-----------|---------------------------------------------------------------------------------|------------|-------------|----------------------------|-------|----------|-----------------|
| Input                                         |   | Process Name |  | Output                         | Remarks                                                                                                         | Cate gory        | Sub category               | Infra struc ture | Loca tion | Modul name in ecoinvent                                                         | Mean value | Unit        | Source mean value          | Typ e | StDv 95% | General Comment |
| steel, converter, unalloyed, at plant; part 3 | → |              |  | desulphurisation slag          | amount inventoried corresponds to the 41% of the slag that is landfilled. The rest is recycled or reused        | waste management | residual material landfill | No               | CH        | disposal, basic oxygen furnace wastes, 0% water, to residual material landfill  | 4.39E-03   | kg          | calculated after IPPC 2000 | 1     | 1.11     | (2,3,2,3,1,3,6) |
|                                               | → |              |  | BOF slag                       | amount inventoried corresponds to the 26% of the slag that is landfilled. The rest is recycled or reused        | waste management | residual material landfill | No               | CH        | disposal, basic oxygen furnace wastes, 0% water, to residual material landfill  | 2.54E-02   | kg          | calculated after IPPC 2000 | 1     | 1.11     | (2,3,2,3,1,3,6) |
|                                               | → |              |  | slag from secondary metallurgy | assumed to be composed and used / disposed as BOF slag --> amount inventoried represents only 26% of total slag | waste management | residual material landfill | No               | CH        | disposal, basic oxygen furnace wastes, 0% water, to residual material landfill  | 2.34E-03   | kg          | calculated after IPPC 2000 | 1     | 1.11     | (2,3,2,3,1,3,6) |
|                                               | → |              |  | spittings                      | recycled --> not inventoried                                                                                    |                  |                            |                  |           |                                                                                 | 4.50E-03   | kg          | calculated after IPPC 2000 |       |          |                 |
|                                               | → |              |  | dust                           | amount inventoried corresponds to the 25% of the total dust that is landfilled. The rest is recycled or reused  | waste management | residual material landfill | No               | CH        | disposal, dust, unalloyed EAF steel, 15.4% water, to residual material landfill | 1.06E-03   | kg          | calculated after IPPC 2000 | 1     | 1.11     | (2,3,2,3,1,3,6) |
|                                               | → |              |  | slag from casting              | recycled --> not inventoried                                                                                    |                  |                            |                  |           |                                                                                 | 4.50E-03   | kg          | calculated after IPPC 2000 |       |          |                 |
|                                               | → |              |  | mill scale                     | recycled --> not inventoried                                                                                    |                  |                            |                  |           |                                                                                 | 3.60E-03   | kg          | calculated after IPPC 2000 |       |          |                 |
|                                               | → |              |  | rubble (refractory material)   | although partly recycled total amount is assumed to be disposed in inert material landfill.                     | waste management | inert material landfill    | No               | CH        | disposal, inert waste, 5% water, to inert material landfill                     | 2.90E-03   | kg          | calculated after IPPC 2000 | 1     | 1.11     | (2,3,2,3,1,3,6) |
|                                               | → |              |  | waste water                    | amount and composition not known                                                                                |                  |                            |                  |           |                                                                                 |            |             | calculated after IPPC 2000 |       |          |                 |
|                                               | → |              |  | BOFgas                         | internally used --> not inventoried                                                                             |                  |                            |                  |           |                                                                                 | 4.20E-01   | MJ          | calculated after IPPC 2000 |       |          |                 |
|                                               | → |              |  | steam                          | internally used --> not inventoried                                                                             |                  |                            |                  |           |                                                                                 | 1.35E-01   | MJ          | calculated after IPPC 2000 |       |          |                 |
|                                               | → |              |  | unalloyed BOF steel slab       |                                                                                                                 | metals           | extraction                 | No               | RER       | steel, converter, unalloyed, at plant                                           | 1.00E+00   | kg          |                            |       |          |                 |

Fig. 5.13 Flows for "steel, converter, unalloyed, at plant" and its representation in the ecoinvent database (cont.)

5. System Characterisation

| General Flow information |                                                 |        |                                                                             | Representation in ecoinvent |                |                |          |                                                                                               |            |      | Uncertainty                |      |          |                     |
|--------------------------|-------------------------------------------------|--------|-----------------------------------------------------------------------------|-----------------------------|----------------|----------------|----------|-----------------------------------------------------------------------------------------------|------------|------|----------------------------|------|----------|---------------------|
| Input                    | Process Name                                    | Output | Remarks                                                                     | Category                    | Sub category   | Infrastructure | Location | Module name in ecoinvent                                                                      | Mean value | Unit | Source mean value          | Type | StdV 95% | General Comment     |
| pig iron / hot metal     | steel, converter, low-alloyed, at plant; part 1 |        |                                                                             | metals                      | extraction     | No             | GLO      | pig iron, at plant                                                                            | 9.00E-01   | kg   | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,3)     |
| iron scrap               |                                                 |        |                                                                             | metals                      | extraction     | No             | RER      | iron scrap, at plant                                                                          | 1.25E-01   | kg   | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,3)     |
| ferronickel              |                                                 |        | 1% Ni in alloy                                                              | metals                      | extraction     | No             | GLO      | ferronickel, 25% Ni, at plant                                                                 | 4.00E-02   | kg   | calculated                 | 1    | 1.07     | (2,nA,nA,nA,1,nA,3) |
| ferrochromium            |                                                 |        | 2.15% Cr in alloy                                                           | metals                      | extraction     | No             | GLO      | ferrochromium, high-carbon, 68% Cr, at plant                                                  | 3.16E-02   | kg   | calculated                 | 1    | 1.07     | (2,nA,nA,nA,1,nA,3) |
| ferromanganese           |                                                 |        | 1.1% Mn in alloy                                                            | metals                      | extraction     | No             | RER      | ferromanganese, high-coal, 74.5% Mn, at regional storage                                      | 1.53E-02   | kg   | calculated                 | 1    | 1.07     | (2,nA,nA,nA,1,nA,3) |
| molybdenite              |                                                 |        | 0.034% Mo in alloy (molybdenite contains 57% Mo)                            | metals                      | extraction     | No             | GLO      | molybdenite, at plant                                                                         | 5.96E-04   | kg   | calculated                 | 1    | 1.07     | (2,nA,nA,nA,1,nA,3) |
| Niobium                  |                                                 |        | 0.066% Nb in alloy (ferronickel as proxy)                                   | metals                      | extraction     | No             | GLO      | ferronickel, 25% Ni, at plant                                                                 | 2.64E-03   | kg   | calculated                 | 1    | 1.07     | (2,nA,nA,nA,1,nA,3) |
| Tungsten                 |                                                 |        | 0.084% W in alloy (ferrochromium used as proxy)                             | metals                      | extraction     | No             | GLO      | ferrochromium, high-carbon, 68% Cr, at plant                                                  | 1.24E-03   | kg   | calculated                 | 1    | 1.07     | (2,nA,nA,nA,1,nA,3) |
| Vanadium                 |                                                 |        | 0.059% V in alloy (ferronickel as proxy)                                    | metals                      | extraction     | No             | GLO      | ferronickel, 25% Ni, at plant                                                                 | 2.36E-03   | kg   | calculated                 | 1    | 1.07     | (2,nA,nA,nA,1,nA,3) |
| iron ore                 |                                                 |        |                                                                             | metals                      | extraction     | No             | GLO      | iron ore, 65% Fe, at beneficiation                                                            | 1.35E-02   | kg   | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,3)     |
| other Fe material        |                                                 |        | as iron ore                                                                 | metals                      | extraction     | No             | GLO      | iron ore, 65% Fe, at beneficiation                                                            | 8.50E-03   | kg   | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,3)     |
| coke                     |                                                 |        | 28.6 MJ/kg                                                                  | hard coal                   | fuels          | No             | RER      | hard coal coke, at plant                                                                      | 2.50E-04   | MJ   | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,11)    |
| lime                     |                                                 |        |                                                                             | construction materials      | additives      | No             | CH       | quicklime, in pieces, loose, at plant                                                         | 4.25E-02   | kg   | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,4)     |
| dolomite                 |                                                 |        |                                                                             | construction materials      | others         | No             | RER      | dolomite, at plant                                                                            | 2.75E-03   | kg   | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,4)     |
| oxygen (99%)             |                                                 |        | 1.429 kg/m3                                                                 | chemicals                   | inorganics     | No             | RER      | oxygen, liquid, at plant                                                                      | 7.15E-02   | kg   | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,4)     |
| Energy from natural gas  |                                                 |        |                                                                             | natural gas                 | fuels          | No             | RER      | natural gas, high pressure, at consumer electricity, medium voltage, production UCTE, at grid | 3.75E-02   | MJ   | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,11)    |
| electricity              |                                                 |        |                                                                             | electricity                 | production mix | No             | UCTE     |                                                                                               | 2.19E-02   | kWh  | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,2)     |
| steam                    |                                                 |        | net steam production ---> consumption not inventoried                       |                             |                |                |          |                                                                                               | 8.50E-02   | MJ   | calculated after IPPC 2000 |      |          |                     |
| compressed air           |                                                 |        | not inventoried. Energy for compressing included in electricity consumption |                             |                |                |          |                                                                                               | 1.10E-02   | m3   | calculated after IPPC 2000 |      |          |                     |
| water                    |                                                 |        |                                                                             | resource                    | in water       |                |          | Water, unspecified natural origin                                                             | 2.70E-03   | m3   | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,12)    |

Fig. 5.14 Flows for "steel, converter, low-alloyed, at plant" and its representation in the ecoinvent database

| General Flow information |                                                 |                |                                                                                                                                              | Representation in ecoinvent |              |                |          |                                                          |               |      |                            | Uncertainty |          |                      |
|--------------------------|-------------------------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|--------------|----------------|----------|----------------------------------------------------------|---------------|------|----------------------------|-------------|----------|----------------------|
| Input                    | Process Name                                    | Output         | Remarks                                                                                                                                      | Category                    | Sub category | Infrastructure | Location | Module name in ecoinvent                                 | Mean value    | Unit | Source mean value          | Type        | StdV 95% | General Comment      |
| transport ship (ocean)   | steel, converter, low-alloyed, at plant; part 2 |                | iron ore from mine to plant by ship (costal)                                                                                                 | transport systems           | ship         | No             | OCE      | transport, transoceanic freight ship                     | 5.94E-02 tkm  |      | calculated (Tab. 4.2)      | 1           | 2.01     | (2,nA,1,1,1,3,5)     |
| transport ship (costal)  |                                                 |                | iron ore from mine to plant by ship (ocean)                                                                                                  | transport systems           | ship         | No             | RER      | transport, barge                                         | 6.60E-04 tkm  |      | calculated (Tab. 4.2)      | 1           | 2.01     | (2,nA,1,3,1,3,5)     |
| transport road           |                                                 |                | iron scrap: 100 km; coke, lime, dolomite, alloys: 20 km; iron ore from mine to plant by train                                                | transport systems           | road         | No             | RER      | transport, lorry >16t, fleet average                     | 1.53E-02 tkm  |      | estimated                  | 1           | 2.09     | (4,5,nA,nA,nA,nA,5)  |
| transport rail           |                                                 |                | iron scrap: 600 km; coke, lime, dolomite, alloys: 200 km; iron ore from mine to plant by train                                               | transport systems           | train        | No             | RER      | transport, freight, rail                                 | 1.08E-01 tkm  |      | estimated                  | 1           | 2.09     | (4,5,nA,nA,nA,nA,5)  |
| infrastructure           |                                                 |                | for transport of hot metal, BOF process, secondary metallurgy and casting                                                                    | metals                      | extraction   | Yes            | RER      | blast oxygen furnace converter                           | 1.33E-11 unit |      | estimated                  | 1           | 3.23     | (5,nA,nA,nA,nA,nA,9) |
|                          |                                                 | waste heat     | from coke, natural gas and electricity. Waste heat from oxidation of C in pig iron to CO and CO2 already included in waste heat of pig iron. | air                         | unspecified  |                |          | Heat, waste                                              | 1.17E-01 MJ   |      | calculated                 | 1           | 1.08     | (1,3,2,3,1,2,13)     |
|                          |                                                 | dust, fine     | all dust is fine according to CEPMEIP database                                                                                               | air                         | unspecified  |                |          | Particulates, < 2.5 um                                   | 4.75E-05 kg   |      | calculated after IPPC 2000 | 1           | 3.01     | (2,3,2,3,1,3,27)     |
|                          |                                                 | Cr             |                                                                                                                                              | air                         | unspecified  |                |          | Chromium                                                 | 1.85E-07 kg   |      | calculated after IPPC 2000 | 1           | 5.01     | (2,3,2,3,1,3,22)     |
|                          |                                                 | Cu             |                                                                                                                                              | air                         | unspecified  |                |          | Copper                                                   | 2.50E-08 kg   |      | calculated after IPPC 2000 | 1           | 5.01     | (2,3,2,3,1,3,22)     |
|                          |                                                 | Pb             |                                                                                                                                              | air                         | unspecified  |                |          | Lead                                                     | 5.15E-07 kg   |      | calculated after IPPC 2000 | 1           | 5.01     | (2,3,2,3,1,3,22)     |
|                          |                                                 | Mn             |                                                                                                                                              | air                         | unspecified  |                |          | Manganese                                                | 6.05E-07 kg   |      | calculated after IPPC 2000 | 1           | 5.01     | (2,3,2,3,1,3,22)     |
|                          |                                                 | NOx            |                                                                                                                                              | air                         | unspecified  |                |          | Nitrogen oxides                                          | 1.25E-05 kg   |      | calculated after IPPC 2000 | 1           | 1.52     | (2,3,2,3,1,3,16)     |
|                          |                                                 | CO             |                                                                                                                                              | air                         | unspecified  |                |          | Carbon monoxide, fossil                                  | 4.73E-03 kg   |      | calculated after IPPC 2000 | 1           | 5.01     | (2,3,2,3,1,3,17)     |
|                          |                                                 | CO2            |                                                                                                                                              | air                         | unspecified  |                |          | Carbon dioxide, fossil                                   | 7.56E-02 kg   |      | calculated after IPPC 2000 | 1           | 1.11     | (2,3,2,3,1,3,14)     |
|                          |                                                 | PAH            |                                                                                                                                              | air                         | unspecified  |                |          | PAH, polycyclic aromatic hydrocarbons                    | 1.20E-10 kg   |      | calculated after IPPC 2000 | 1           | 3.01     | (2,3,2,3,1,3,21)     |
|                          |                                                 | dioxin (I-TEQ) |                                                                                                                                              | air                         | unspecified  |                |          | Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin | 3.05E-14 kg   |      | calculated after IPPC 2000 | 1           | 3.01     | (2,3,2,3,1,3,21)     |

Fig. 5.14 Flows for "steel, converter, low-alloyed, at plant" and its representation in the ecoinvent database (cont.)

5. System Characterisation

| General Flow information                        |   |              |  | Representation in ecoinvent    |                                                                                                                 |                  |                            |                  |           |                                                                                 |            | Uncertainty |                            |       |          |                 |
|-------------------------------------------------|---|--------------|--|--------------------------------|-----------------------------------------------------------------------------------------------------------------|------------------|----------------------------|------------------|-----------|---------------------------------------------------------------------------------|------------|-------------|----------------------------|-------|----------|-----------------|
| Input                                           |   | Process Name |  | Output                         | Remarks                                                                                                         | Cate gory        | Sub category               | Infra struc ture | Loca tion | Modul name in ecoinvent                                                         | Mean value | Unit        | Source mean value          | Typ e | StDv 95% | General Comment |
| steel, converter, low-alloyed, at plant; part 3 | → |              |  | desulphurisation slag          | amount inventoried corresponds to the 41% of the slag that is landfilled. The rest is recycled or reused        | waste management | residual material landfill | No               | CH        | disposal, basic oxygen furnace wastes, 0% water, to residual material landfill  | 4.39E-03   | kg          | calculated after IPPC 2000 | 1     | 1.11     | (2,3,2,3,1,3,6) |
|                                                 | → |              |  | BOF slag                       | amount inventoried corresponds to the 26% of the slag that is landfilled. The rest is recycled or reused        | waste management | residual material landfill | No               | CH        | disposal, basic oxygen furnace wastes, 0% water, to residual material landfill  | 2.54E-02   | kg          | calculated after IPPC 2000 | 1     | 1.11     | (2,3,2,3,1,3,6) |
|                                                 | → |              |  | slag from secondary metallurgy | assumed to be composed and used / disposed as BOF slag --> amount inventoried represents only 26% of total slag | waste management | residual material landfill | No               | CH        | disposal, basic oxygen furnace wastes, 0% water, to residual material landfill  | 2.34E-03   | kg          | calculated after IPPC 2000 | 1     | 1.11     | (2,3,2,3,1,3,6) |
|                                                 | → |              |  | spittings                      | recycled --> not inventoried                                                                                    |                  |                            |                  |           |                                                                                 | 4.50E-03   | kg          | calculated after IPPC 2000 |       |          |                 |
|                                                 | → |              |  | dust                           | amount inventoried corresponds to the 25% of the total dust that is landfilled. The rest is recycled or reused  | waste management | residual material landfill | No               | CH        | disposal, dust, unalloyed EAF steel, 15.4% water, to residual material landfill | 1.06E-03   | kg          | calculated after IPPC 2000 | 1     | 1.11     | (2,3,2,3,1,3,6) |
|                                                 | → |              |  | slag from casting              | recycled --> not inventoried                                                                                    |                  |                            |                  |           |                                                                                 | 4.50E-03   | kg          | calculated after IPPC 2000 |       |          |                 |
|                                                 | → |              |  | mill scale                     | recycled --> not inventoried                                                                                    |                  |                            |                  |           |                                                                                 | 3.60E-03   | kg          | calculated after IPPC 2000 |       |          |                 |
|                                                 | → |              |  | rubble (refractory material)   | although partly recycled total amount is assumed to be disposed in inert material landfill.                     | waste management | inert material landfill    | No               | CH        | disposal, inert waste, 5% water, to inert material landfill                     | 2.90E-03   | kg          | calculated after IPPC 2000 | 1     | 1.11     | (2,3,2,3,1,3,6) |
|                                                 | → |              |  | waste water                    | amount and composition not known                                                                                |                  |                            |                  |           |                                                                                 |            |             | calculated after IPPC 2000 |       |          |                 |
|                                                 | → |              |  | BOFgas                         | internally used --> not inventoried                                                                             |                  |                            |                  |           |                                                                                 | 4.20E-01   | MJ          | calculated after IPPC 2000 |       |          |                 |
|                                                 | → |              |  | steam                          | internally used --> not inventoried                                                                             |                  |                            |                  |           |                                                                                 | 1.35E-01   | MJ          | calculated after IPPC 2000 |       |          |                 |
|                                                 | → |              |  | low-alloyed BOF steel slab     |                                                                                                                 | metals           | extraction                 | No               | RER       | steel, converter, low-alloyed, at plant                                         | 1.00E+00   | kg          |                            |       |          |                 |

Fig. 5.14 Flows for "steel, converter, low-alloyed, at plant" and its representation in the ecoinvent database (cont.)



| General Flow information |                                                         |        |                                                                             | Representation in ecoinvent |                |                |          |                                                       |              |      | Uncertainty                |      |          |                     |
|--------------------------|---------------------------------------------------------|--------|-----------------------------------------------------------------------------|-----------------------------|----------------|----------------|----------|-------------------------------------------------------|--------------|------|----------------------------|------|----------|---------------------|
| Input                    | Process Name                                            | Output | Remarks                                                                     | Category                    | Sub category   | Infrastructure | Location | Module name in ecoinvent                              | Mean value   | Unit | Source mean value          | Type | StdV 95% | General Comment     |
| pig iron / hot metal     | steel, converter, chromium steel 18/8, at plant; part 1 |        |                                                                             | metals                      | extraction     | No             | GLO      | pig iron, at plant                                    | 5.28E-01 kg  |      | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,3)     |
| ferronickel              |                                                         |        | 8% Ni in alloy                                                              | metals                      | extraction     | No             | GLO      | ferronickel, 25% Ni, at plant                         | 3.20E-01 kg  |      | calculated                 | 1    | 1.07     | (2,nA,nA,nA,1,nA,3) |
| ferrochromium            |                                                         |        | 18% Cr in alloy                                                             | metals                      | extraction     | No             | GLO      | ferrochromium, high-carbon, 68% Cr, at plant          | 2.65E-01 kg  |      | calculated                 | 1    | 1.07     | (2,nA,nA,nA,1,nA,3) |
| iron ore                 |                                                         |        |                                                                             | metals                      | extraction     | No             | GLO      | iron ore, 65% Fe, at beneficiation                    | 1.35E-02 kg  |      | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,3)     |
| other Fe material        |                                                         |        | as iron ore                                                                 | metals                      | extraction     | No             | GLO      | iron ore, 65% Fe, at beneficiation                    | 8.50E-03 kg  |      | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,3)     |
| coke                     |                                                         |        | 28.6 MJ/kg                                                                  | hard coal                   | fuels          | No             | RER      | hard coal coke, at plant                              | 2.50E-04 MJ  |      | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,11)    |
| lime                     |                                                         |        |                                                                             | construction materials      | additives      | No             | CH       | quicklime, in pieces, loose, at plant                 | 4.25E-02 kg  |      | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,4)     |
| dolomite                 |                                                         |        |                                                                             | construction materials      | others         | No             | RER      | dolomite, at plant                                    | 2.75E-03 kg  |      | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,4)     |
| oxygen (99%)             |                                                         |        | 1.429 kg/m3                                                                 | chemicals                   | inorganics     | No             | RER      | oxygen, liquid, at plant                              | 7.15E-02 kg  |      | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,4)     |
| energy from natural gas  |                                                         |        |                                                                             | natural gas                 | fuels          | No             | RER      | natural gas, high pressure, at consumer               | 3.75E-02 MJ  |      | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,11)    |
| electricity              |                                                         |        |                                                                             | electricity                 | production mix | No             | UCTE     | electricity, medium voltage, production UCTE, at grid | 2.19E-02 kWh |      | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,2)     |
| steam                    |                                                         |        | net steam production --> consumption not inventoried                        |                             |                |                |          |                                                       | 8.50E-02 MJ  |      | calculated after IPPC 2000 |      |          |                     |
| compressed air           |                                                         |        | not inventoried. Energy for compressing included in electricity consumption |                             |                |                |          |                                                       | 1.10E-02 m3  |      | calculated after IPPC 2000 |      |          |                     |
| water                    |                                                         |        |                                                                             | resource                    | in water       |                |          | Water, unspecified natural origin                     | 2.70E-03 m3  |      | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,12)    |

Fig. 5.15 Flows for "steel, converter, chromium steel 18/8, at plant" and its representation in the ecoinvent database

| General Flow information |                                                         |                |                                                                                                                                              | Representation in ecoinvent |              |                |          |                                                          |               |      | Uncertainty                |      |          |                      |
|--------------------------|---------------------------------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|--------------|----------------|----------|----------------------------------------------------------|---------------|------|----------------------------|------|----------|----------------------|
| Input                    | Process Name                                            | Output         | Remarks                                                                                                                                      | Category                    | Sub category | Infrastructure | Location | Module name in ecoinvent                                 | Mean value    | Unit | Source mean value          | Type | StdV 95% | General Comment      |
| transport ship (ocean)   | steel, converter, chromium steel 18/8, at plant; part 2 |                | iron ore from mine to plant by ship (costal)                                                                                                 | transport systems           | ship         | No             | OCE      | transport, transoceanic freight ship                     | 5.94E-02 tkm  |      | calculated (Tab. 4.2)      | 1    | 2.01     | (2,nA,1,1,1,3,5)     |
| transport ship (costal)  |                                                         |                | iron ore from mine to plant by ship (ocean)                                                                                                  | transport systems           | ship         | No             | RER      | transport, barge                                         | 6.60E-04 tkm  |      | calculated (Tab. 4.2)      | 1    | 2.01     | (2,nA,1,3,1,3,5)     |
| transport road           |                                                         |                | coke, lime, dolomite, alloys: 20 km; iron ore from mine to plant by train                                                                    | transport systems           | road         | No             | RER      | transport, lorry >16t, fleet average                     | 1.26E-02 tkm  |      | estimated                  | 1    | 2.09     | (4,5,nA,nA,nA,nA,5)  |
| transport rail           |                                                         |                | coke, lime, dolomite, alloys: 200 km; iron ore from mine to plant by train                                                                   | transport systems           | train        | No             | RER      | transport, freight, rail                                 | 1.32E-01 tkm  |      | estimated                  | 1    | 2.09     | (4,5,nA,nA,nA,nA,5)  |
| infrastructure           |                                                         |                | for transport of hot metal, BOF process, secondary metallurgy and casting                                                                    | metals                      | extraction   | Yes            | RER      | blast oxygen furnace converter                           | 1.33E-11 unit |      | estimated                  | 1    | 3.23     | (5,nA,nA,nA,nA,nA,9) |
|                          |                                                         | waste heat     | from coke, natural gas and electricity. Waste heat from oxidation of C in pig iron to CO and CO2 already included in waste heat of pig iron. | air                         | unspecified  |                |          | Heat, waste                                              | 1.17E-01 MJ   |      | calculated                 | 1    | 1.08     | (1,3,2,3,1,2,13)     |
|                          |                                                         | dust, fine     | all dust is fine according to CEPMEIP database                                                                                               | air                         | unspecified  |                |          | Particulates, < 2.5 um                                   | 4.75E-05 kg   |      | calculated after IPPC 2000 | 1    | 3.01     | (2,3,2,3,1,3,27)     |
|                          |                                                         | Cr             |                                                                                                                                              | air                         | unspecified  |                |          | Chromium                                                 | 1.85E-07 kg   |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22)     |
|                          |                                                         | Cu             |                                                                                                                                              | air                         | unspecified  |                |          | Copper                                                   | 2.50E-08 kg   |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22)     |
|                          |                                                         | Pb             |                                                                                                                                              | air                         | unspecified  |                |          | Lead                                                     | 5.15E-07 kg   |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22)     |
|                          |                                                         | Mn             |                                                                                                                                              | air                         | unspecified  |                |          | Manganese                                                | 6.05E-07 kg   |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22)     |
|                          |                                                         | NOx            |                                                                                                                                              | air                         | unspecified  |                |          | Nitrogen oxides                                          | 1.25E-05 kg   |      | calculated after IPPC 2000 | 1    | 1.52     | (2,3,2,3,1,3,16)     |
|                          |                                                         | CO             |                                                                                                                                              | air                         | unspecified  |                |          | Carbon monoxide, fossil                                  | 4.73E-03 kg   |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,17)     |
|                          |                                                         | CO2            |                                                                                                                                              | air                         | unspecified  |                |          | Carbon dioxide, fossil                                   | 7.56E-02 kg   |      | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,14)     |
|                          |                                                         | PAH            |                                                                                                                                              | air                         | unspecified  |                |          | PAH, polycyclic aromatic hydrocarbons                    | 1.20E-10 kg   |      | calculated after IPPC 2000 | 1    | 3.01     | (2,3,2,3,1,3,21)     |
|                          |                                                         | dioxin (I-TEQ) |                                                                                                                                              | air                         | unspecified  |                |          | Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin | 3.05E-14 kg   |      | calculated after IPPC 2000 | 1    | 3.01     | (2,3,2,3,1,3,21)     |

Fig. 5.15 Flows for "steel, converter, chromium steel 18/8, at plant" and its representation in the ecoinvent database (cont.)

| General Flow information                                |   |              |  | Representation in ecoinvent    |                                                                                                                 |                  |                            |                  |           |                                                                                |            | Uncertainty |                            |       |          |                 |
|---------------------------------------------------------|---|--------------|--|--------------------------------|-----------------------------------------------------------------------------------------------------------------|------------------|----------------------------|------------------|-----------|--------------------------------------------------------------------------------|------------|-------------|----------------------------|-------|----------|-----------------|
| Input                                                   |   | Process Name |  | Output                         | Remarks                                                                                                         | Cate gory        | Sub category               | Infra struc ture | Loca tion | Modul name in ecoinvent                                                        | Mean value | Unit        | Source mean value          | Typ e | StDv 95% | General Comment |
| steel, converter, chromium steel 18/8, at plant; part 3 | → |              |  | desulphurisation slag          | amount inventoried corresponds to the 41% of the slag that is landfilled. The rest is recycled or reused        | waste management | residual material landfill | No               | CH        | disposal, basic oxygen furnace wastes, 0% water, to residual material landfill | 4.39E-03   | kg          | calculated after IPPC 2000 | 1     | 1.11     | (2,3,2,3,1,3,6) |
|                                                         | → |              |  | BOF slag                       | amount inventoried corresponds to the 26% of the slag that is landfilled. The rest is recycled or reused        | waste management | residual material landfill | No               | CH        | disposal, basic oxygen furnace wastes, 0% water, to residual material landfill | 2.54E-02   | kg          | calculated after IPPC 2000 | 1     | 1.11     | (2,3,2,3,1,3,6) |
|                                                         | → |              |  | slag from secondary metallurgy | assumed to be composed and used / disposed as BOF slag --> amount inventoried represents only 26% of total slag | waste management | residual material landfill | No               | CH        | disposal, basic oxygen furnace wastes, 0% water, to residual material landfill | 2.34E-03   | kg          | calculated after IPPC 2000 | 1     | 1.11     | (2,3,2,3,1,3,6) |
|                                                         | → |              |  | spittings                      | recycled --> not inventoried                                                                                    |                  |                            |                  |           |                                                                                | 4.50E-03   | kg          | calculated after IPPC 2000 |       |          |                 |
|                                                         | → |              |  | dust                           | amount inventoried corresponds to the 25% of the total dust that is landfilled. The rest is recycled or reused  | waste management | residual material landfill | No               | CH        | disposal, dust, alloyed EAF steel, 15.4% water, to residual material landfill  | 1.06E-03   | kg          | calculated after IPPC 2000 | 1     | 1.11     | (2,3,2,3,1,3,6) |
|                                                         | → |              |  | slag from casting              | recycled --> not inventoried                                                                                    |                  |                            |                  |           |                                                                                | 4.50E-03   | kg          | calculated after IPPC 2000 |       |          |                 |
|                                                         | → |              |  | mill scale                     | recycled --> not inventoried                                                                                    |                  |                            |                  |           |                                                                                | 3.60E-03   | kg          | calculated after IPPC 2000 |       |          |                 |
|                                                         | → |              |  | rubble (refractory material)   | although partly recycled total amount is assumed to be disposed in inert material landfill.                     | waste management | inert material landfill    | No               | CH        | disposal, inert waste, 5% water, to inert material landfill                    | 2.90E-03   | kg          | calculated after IPPC 2000 | 1     | 1.11     | (2,3,2,3,1,3,6) |
|                                                         | → |              |  | waste water                    | amount and composition not known                                                                                |                  |                            |                  |           |                                                                                |            |             | calculated after IPPC 2000 |       |          |                 |
|                                                         | → |              |  | BOFgas                         | internally used --> not inventoried                                                                             |                  |                            |                  |           |                                                                                | 4.20E-01   | MJ          | calculated after IPPC 2000 |       |          |                 |
|                                                         | → |              |  | steam                          | internally used --> not inventoried                                                                             |                  |                            |                  |           |                                                                                | 1.35E-01   | MJ          | calculated after IPPC 2000 |       |          |                 |
|                                                         | → |              |  | BOF chromium steel slab        |                                                                                                                 | metals           | extraction                 | No               | RER       | steel, converter, chromium steel 18/8, at plant                                | 1.00E+00   | kg          |                            |       |          |                 |

Fig. 5.15 Flows for "steel, converter, chromium steel 18/8, at plant" and its representation in the ecoinvent database (cont.)

| General Flow information                                       |                                |                                 |                                             | Representation in ecoinvent |                     |                |          |                                                                 |               |      |                   | Uncertainty |          |                                        |
|----------------------------------------------------------------|--------------------------------|---------------------------------|---------------------------------------------|-----------------------------|---------------------|----------------|----------|-----------------------------------------------------------------|---------------|------|-------------------|-------------|----------|----------------------------------------|
| Input                                                          | Process Name                   | Output                          | Remarks                                     | Category                    | Sub category        | Infrastructure | Location | Module name in ecoinvent                                        | Mean value    | Unit | Source mean value | Type        | StdV 95% | General Comment                        |
| total land use (transformation)                                | blast oxygen furnace converter |                                 |                                             | resource                    | land                |                |          | Transformation, from unknown                                    | 5.00E+05 m2   |      | estimated         | 1           | 5.00     | (5,nA,nA,nA,nA,nA,8); rough estimation |
| land occupation for buildings and open storage area            |                                |                                 |                                             | resource                    | land                |                |          | Occupation, industrial area, built up                           | 2.00E+07 m2a  |      | estimated         | 1           | 5.00     | (5,nA,nA,nA,nA,nA,7); rough estimation |
| land transformation to buildings and open bauxite storage area |                                |                                 | assumption: 80% of total area is built up   | resource                    | land                |                |          | Transformation, to industrial area, built up                    | 4.00E+05 m2   |      | estimated         | 1           | 5.00     | (5,nA,nA,nA,nA,nA,8); rough estimation |
| land occupation for vegetation area                            |                                |                                 |                                             | resource                    | land                |                |          | Occupation, industrial area, vegetation                         | 5.00E+06 m2a  |      | estimated         | 1           | 5.00     | (5,nA,nA,nA,nA,nA,7); rough estimation |
| land transformation to vegetation area                         |                                |                                 | assumption: 20% of total area is vegetation | resource                    | land                |                |          | Transformation, to industrial area, vegetation                  | 1.00E+05 m2   |      | estimated         | 1           | 5.00     | (5,nA,nA,nA,nA,nA,8); rough estimation |
| building, multi story                                          |                                |                                 |                                             | construction processes      | buildings           | Yes            | RER      | building, multi-storey                                          | 2.00E+03 m3   |      | estimated         | 1           | 5.00     | (5,nA,nA,nA,nA,nA,9); rough estimation |
| building, hall                                                 |                                |                                 |                                             | construction processes      | buildings           | Yes            | CH       | building, hall                                                  | 4.00E+05 m2   |      | estimated         | 1           | 5.00     | (5,nA,nA,nA,nA,nA,9); rough estimation |
| heavy machine, unspecified                                     |                                |                                 |                                             | construction processes      | machinery           | Yes            | RER      | industrial machine, heavy, unspecified, at plant                | 2.22E+07 kg   |      | estimated         | 1           | 5.00     | (5,nA,nA,nA,nA,nA,9); rough estimation |
| poor concrete for additional foundation                        |                                |                                 |                                             | construction materials      | concrete            | No             | CH       | concrete, sole plate and foundation, at plant                   | 1.01E+04 m3   |      | estimated         | 1           | 5.00     | (5,nA,nA,nA,nA,nA,9); rough estimation |
| firebrick                                                      |                                |                                 |                                             | construction materials      | bricks              | No             | DE       | refractory, basic, packed, at plant                             | 2.00E+05 kg   |      | estimated         | 1           | 5.00     | (5,nA,nA,nA,nA,nA,9); rough estimation |
|                                                                |                                | foundation concrete to disposal |                                             | waste management            | building demolition | No             | CH       | disposal, building, concrete, not reinforced, to final disposal | 2.22E+07 kg   |      | estimated         | 1           | 5.00     | (5,nA,nA,nA,nA,nA,9); rough estimation |
|                                                                |                                | firebricks to disposal          |                                             | waste management            | building demolition | No             | CH       | disposal, building, brick, to final disposal                    | 2.00E+05 kg   |      | estimated         | 1           | 5.00     | (5,nA,nA,nA,nA,nA,9); rough estimation |
|                                                                |                                | BOF converter                   | Output: 1500000000 kg/a, life time: 50 a    | metals                      | extraction          | Yes            | RER      | blast oxygen furnace converter                                  | 1.00E+00 unit |      |                   |             |          |                                        |

Fig. 5.16 Flows for "blast oxygen furnace converter" and its representation in the ecoinvent database

## 5.7 Electric arc furnace (EAF) steel making

This part is mainly based on (IPPC (2000)).

The direct smelting of iron-containing materials, such as scrap is usually performed in electric arc furnaces (EAF) which play an important and increasing role in modern steel works concepts. Today the percentage of electric arc furnace steel of the overall steel production in the EU is 35.3 %.

The major feed stock for the EAF is ferrous scrap, which may comprise of scrap from inside the steel-works, cut-offs from steel product manufacturers (e.g. vehicle builders) and capital or post-consumer scrap (e.g. end of life products). Direct-reduced iron (DRI) is also increasingly being used as a feed-stock due both to its low gangue content and variable scrap prices. As in the BOF, a slag is formed from lime to collect undesirable components in the steel.

### 5.7.1 Production process and infrastructure

The production of steel by the EAF process is a discontinuous process which involves the following steps:

- raw material handling and storage
- furnace charging with/without scrap preheating
- EAF scrap melting
- steel and slag tapping
- ladle furnace treatments for quality adjustment
- slag handling
- continuous casting

For high alloyed and special steels (15% of European steel production), the operation sequence is more complex and tailor-made for the end-products. In addition to the mentioned operations for carbon steels various ladle treatments (secondary metallurgy) are carried out like

- desulphurisation
- degassing for the elimination of dissolved gases like nitrogen and hydrogen
- decarburisation (AOD=Argon-Oxygen-Decarburisation or VOD=Vacuum-Oxygen-Decarburisation)

The actual melting is done by lowering graphite electrodes to the scrap until they strike an arc that melts the scrap. More information on the individual steps is given in (IPPC (2000)).

No data have been found on the infrastructure needed for the process. Assumptions made in this project are based on different pictures and information on the aluminium production infrastructure (see Tab. 5.18).

**Tab. 5.18 Assumptions for infrastructure of EAF converters**

| electric arc furnace converter                     | amount      | unit           | lifetime [a] | Material / Module          |
|----------------------------------------------------|-------------|----------------|--------------|----------------------------|
| yearly production capacity (5000h; 100 t/h)        | 500'000'000 | kg             | 50           | Yearly production capacity |
| area of plant (500 x 100 m)                        | 500'000     | m <sup>2</sup> | 50           | Transformed area           |
| administration building (50x50x10 m <sup>3</sup> ) | 2'000       | m <sup>3</sup> | 50           | Building, multi story      |
| production hall (50% of total area)                | 250'000     | m <sup>2</sup> | 50           | Building, hall             |
| furnace & ladle                                    | 500'000     | kg             | 25           | heavy machine, unspecific  |
| lining for furnace & ladle                         | 50'000      | kg             | 25           | firebrick                  |
| casting machines                                   | 5'000'000   | kg             | 25           | heavy machine, unspecific  |

The infrastructure is modelled to be used for 50 years. The replacement of the machines with a lifetime of only 25 years is taken into account by doubling the amount. The infrastructure is disposed after its useful life in inert material landfills. Thus the values given in Tab. 5.21 are inventoried. For the waste disposal it is assumed that the conveyer belts and the machines are completely recycled. The disposal of the buildings is already inventoried in the building modules. Thus only the concrete and the refractory materials are disposed.

**Tab. 5.19 Infrastructure of EAF converters**

|                                         | electric arc furnace converter | Unit           | Remarks                 |
|-----------------------------------------|--------------------------------|----------------|-------------------------|
| Yearly production capacity              | 500'000'000                    | kg             |                         |
| Transformed area                        | 500'000                        | m <sup>2</sup> |                         |
| Building, multi story                   | 2'000                          | m <sup>3</sup> |                         |
| Building, hall                          | 250'000                        | m <sup>2</sup> |                         |
| heavy machine, unspecific               | 11'000'000                     | kg             |                         |
| Poor concrete for additional foundation | 5'000                          | m <sup>3</sup> | same weight as machines |
| firebrick                               | 100'000                        | kg             |                         |

## 5.7.2 Emissions

### Air

Relevant emissions of heavy metals varying largely with the scrap quality occur in electric steel making. Also varying with the scrap quality are chlorinated hydrocarbon emissions such as PCB and dioxins. Other VOC emissions can be remarkably high and correspond with the use of coal.

The overall particle emissions given are in the medium range in (Berdowski et al. (1995)) thus 42.5% of particles are assumed to be smaller than 2.5 µm, 42.5% of particles are assumed to be between 2.5 µm and 10 µm and 15% of particles are assumed to be larger than 10 µm.

### Water

Drainage water from unpaved scrap-yards can be contaminated, especially in case of oil/emulsion containing scrap like turnings. There is no information available on quantities and pollution of drainage water. Usually it is at least treated in an oil separator prior to be discharged.

### Soil

Soil contamination may arise from contaminated scrap in scrap-yards. No information on quantities and pollutants is available.

### 5.7.3 Waste

The main waste generated in EAF steel making is slag. Their composition slightly depends on the alloy and on the sub-process they are generated in. The differences are not significant and therefore neglected in this study. One average composition is established (see Tab. 5.20). The recycling fraction also depends on the alloy. While slag from carbon steel and low alloyed steel production are landfilled to 69% and 59% respectively, only 53% of the slag from high alloyed steel production is landfilled.

Tab. 5.20 Chemical composition of slag from EAF steel making [% (w/w)]

| CaO               | SiO <sub>2</sub> | Al <sub>2</sub> O <sub>3</sub> | MgO   | Total Fe | TiO <sub>2</sub> | MnO | P <sub>2</sub> O <sub>5</sub> | Cr <sub>2</sub> O <sub>3</sub> | Free CaO | S     | C    |
|-------------------|------------------|--------------------------------|-------|----------|------------------|-----|-------------------------------|--------------------------------|----------|-------|------|
| 36                | 14               | 5.6                            | 8.7   | 22       | 0.31             | 8.2 | 0.31                          | 1.5                            | 2.1      | 0.021 | 0.34 |
| Na <sub>2</sub> O | K <sub>2</sub> O | V <sub>2</sub> O <sub>5</sub>  | ZnO   | CuO      | NiO              |     |                               |                                |          |       |      |
| 0.47              | 0.11             | 0.18                           | 0.021 | 0.031    | 0.21             |     |                               |                                |          |       |      |

Arithmetic mean of values in (IPPC (2000), Tab. 9.4), linear adjusted to 100% (from 97.5%). In this project  $\leq x$  is interpreted as  $x/2$

As already mentioned the treatment of off gases (mostly primary off gases together with secondary off gases) is very often performed in bag filters. The composition of dusts from production of carbon, low alloyed and high alloyed steel can be seen from Tab. 5.21.

Wastewater also results from off gas treatment in a wet scrubber and from direct cooling from continuous casting. For the former wastewater no information on applied treatment techniques and discharged quantities and its pollution is available. Latter wastewater contains mill scale (1–3 g/l) and oil/grease. Usually this wastewater is treated together with other streams from the rolling mills.

Tab. 5.21 Chemical composition of dust from EAF steel making [% (w/w)]

| Component                      | Dust from carbon/low-alloyed steel production | Dust from high-alloyed / stainless steel production | Dust from carbon/low-alloyed EAF steel in this project | Dust from high-alloyed EAF steel in this project |
|--------------------------------|-----------------------------------------------|-----------------------------------------------------|--------------------------------------------------------|--------------------------------------------------|
|                                | (IPPC (2000), Tab. 9.6)                       |                                                     | [% (w/w)]                                              | [% (w/w)]                                        |
| Fe tot                         | 25 – 50                                       | 30 – 40                                             | 3.59E+01                                               | 3.25E+01                                         |
| SiO <sub>2</sub>               | 1.5 – 5                                       | 7 – 10                                              | 3.11E+00                                               | 7.88E+00                                         |
| CaO                            | 4 – 15                                        | 5 – 17                                              | 9.10E+00                                               | 1.02E+01                                         |
| Al <sub>2</sub> O <sub>3</sub> | 0.3 – 0.7                                     | 1 – 4                                               | 4.79E-01                                               | 2.32E+00                                         |
| MgO                            | 1 – 5                                         | 2 – 5                                               | 2.87E+00                                               | 3.25E+00                                         |
| P <sub>2</sub> O <sub>5</sub>  | 0.2 – 0.6                                     | 0.01 – 0.1                                          | 3.83E-01                                               | 5.10E-02                                         |
| MnO                            | 2.5 – 5.5                                     | 3 – 6                                               | 3.83E+00                                               | 4.17E+00                                         |
| Cr <sub>2</sub> O <sub>3</sub> | 0.2 – 1                                       | 10 – 20                                             | 5.75E-01                                               | 1.39E+01                                         |
| Na <sub>2</sub> O              | 1.5 – 1.9                                     | n/a                                                 | 1.63E+00                                               | 1.63E+00                                         |
| K <sub>2</sub> O               | 1.2 – 1.5                                     | n/a                                                 | 1.29E+00                                               | 1.29E+00                                         |
| Zn                             | 10 – 35                                       | 2 – 10                                              | 2.16E+01                                               | 5.56E+00                                         |
| Pb                             | 0.8 – 6                                       | 0.5 – 2                                             | 3.26E+00                                               | 1.16E+00                                         |
| Cd                             | 0.02 – 0.1                                    | 0.01 – 0.08                                         | 5.75E-02                                               | 4.17E-02                                         |
| Cu                             | 0.15 – 0.4                                    | 0.01 – 0.3                                          | 2.64E-01                                               | 1.44E-01                                         |
| Ni                             | 0.02 – 0.04                                   | 2 – 4                                               | 2.87E-02                                               | 2.78E+00                                         |
| V                              | 0.02 – 0.05                                   | 0.1 – 0.3                                           | 3.35E-02                                               | 1.85E-01                                         |
| Co                             | 0.001 – 0.002                                 | n/a                                                 | 1.44E-03                                               | 1.44E-03                                         |
| As                             | 0.003 – 0.08                                  | n/a                                                 | 3.98E-02                                               | 3.98E-02                                         |
| Hg                             | 0.0001 – 0.001                                | n/a                                                 | 5.27E-04                                               | 5.27E-04                                         |
| Cl                             | 1.5 – 4                                       | n/a                                                 | 2.64E+00                                               | 2.64E+00                                         |
| F                              | 0.02 – 0.9                                    | 0.01 – 0.05                                         | 4.41E-01                                               | 2.78E-02                                         |
| S                              | 0.5 – 1                                       | 0.1 – 0.3                                           | 7.19E-01                                               | 1.85E-01                                         |
| C                              | 0.5 – 2                                       | 0.5 – 1                                             | 1.20E+00                                               | 6.96E-01                                         |
| Basicity                       | 2.0 – 6.5                                     | n/a                                                 | 4.25E+00                                               | 4.25E+00                                         |
| Moisture                       | 6 – 16                                        | n/a                                                 | 1.05E+01                                               | 1.05E+01                                         |
|                                | (Energetics (2000), Tab. 5-5)                 |                                                     |                                                        |                                                  |
| Fe                             | 28.5                                          | 31.7                                                |                                                        |                                                  |
| Zn                             | 19                                            | 1                                                   |                                                        |                                                  |
| Cd                             | < 0.1                                         | 0.16                                                |                                                        |                                                  |
| Cr                             | 0.39                                          | 10.2                                                |                                                        |                                                  |
| CaO / MgO                      | 10.7                                          | 3.1                                                 |                                                        |                                                  |

This project = arithmetic mean values, linear adjusted to 100% (from 104% and 108% for carbon/low-alloyed steel and high-alloyed steel respectively)

#### 5.7.4 Electric steel in ecoinvent

The background information to the processes are given in Tab. 5.22, flows are shown in:

- steel, electric, un- and low-alloyed, at plant Fig. 5.17
- steel, electric, chromium steel 18/8, at plant Fig. 5.18
- electric arc furnace converter Fig. 5.19

If not stated otherwise, arithmetic means of the low and high values given in (IPPC (2000)) are used in this project.



Tab. 5.22 Meta information for the EAF steel production processes

|                                       |                                                                                                                                                             |                                                                                                                                                             |                                                                                                                         |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Name                                  | steel, electric, un- and low-alloyed, at plant                                                                                                              | steel, electric, chromium steel 18/8, at plant                                                                                                              | electric arc furnace converter                                                                                          |
| Location                              | RER                                                                                                                                                         | RER                                                                                                                                                         | RER                                                                                                                     |
| Infrastructure Process                | 0                                                                                                                                                           | 0                                                                                                                                                           | 1                                                                                                                       |
| Unit                                  | kg                                                                                                                                                          | kg                                                                                                                                                          | unit                                                                                                                    |
| Data Set Version                      | 2.0                                                                                                                                                         | 2.0                                                                                                                                                         | 2.0                                                                                                                     |
| Included Processes                    | Transports of scrap metal and other input materials to electric arc furnace, steel making process and casting.                                              | Transports of scrap metal and other input materials to electric arc furnace, steel making process and casting.                                              | Infrastructure for input material unloading and storage, for EAF processes and casting and for administrative buildings |
| Amount                                | 1                                                                                                                                                           | 1                                                                                                                                                           | 1                                                                                                                       |
| Local Name                            | Elektrostahl, un- und niedriglegiert, ab Werk                                                                                                               | Elektrostahl, Chromstahl 18/8, ab Werk                                                                                                                      | Elektrolichtbogenofen, Stahl                                                                                            |
| Synonyms                              |                                                                                                                                                             |                                                                                                                                                             |                                                                                                                         |
| General Comment to reference function | This process produces secondary steel. Only scrap is used as iron bearing input.                                                                            | This process produces secondary steel. Only scrap is used as iron bearing input.                                                                            | Assumptions for medium sized plant. Output: 500'000 t/a, life time: 50 years.                                           |
| Start Date                            | 2001                                                                                                                                                        | 2001                                                                                                                                                        | 2002                                                                                                                    |
| End Date                              | 2001                                                                                                                                                        | 2001                                                                                                                                                        | 2002                                                                                                                    |
| Data Valid For Entire Period          | 1                                                                                                                                                           | 1                                                                                                                                                           | 1                                                                                                                       |
| Other Period Text                     |                                                                                                                                                             |                                                                                                                                                             |                                                                                                                         |
| Geography text                        | Data relate to plants in the EU                                                                                                                             | Data relate to plants in the EU                                                                                                                             | Some Swiss datasets are used for European processes.                                                                    |
| Technology text                       | EU technology mix (mainly furnace with 4th hole, partly with additional evacuation of building atmosphere). An average for different alloys is represented. | EU technology mix (mainly furnace with 4th hole, partly with additional evacuation of building atmosphere). An average for different alloys is represented. | Assumed technology of medium sized plant.                                                                               |
| Representativeness [%]                |                                                                                                                                                             |                                                                                                                                                             |                                                                                                                         |
| Production Volume                     | unknown                                                                                                                                                     | unknown                                                                                                                                                     | unknown                                                                                                                 |
| Sampling Procedure                    | based on literature                                                                                                                                         | based on literature                                                                                                                                         | Internet search                                                                                                         |
| Extrapolations                        | none                                                                                                                                                        | none                                                                                                                                                        | see Geography                                                                                                           |
| Uncertainty Adjustments               | none                                                                                                                                                        | none                                                                                                                                                        | none                                                                                                                    |

### Data quality

Data uncertainty is derived using the Pedigree matrix as described in (Frischknecht et al. (2003)) except for the infrastructure where this method would result in values far too low.

Data of input and main output materials as well as emission data for EAF steel production stem from an unknown number of 203 electric arc furnaces in Europe. Thus data for European consumption are of rather high uncertainty.

Data for the infrastructure is roughly estimated.

| General Flow information      |                                                        |        |                                                            | Representation in ecoinvent |                |                |          |                                                       |               |      | Uncertainty                   |      |          |                   |
|-------------------------------|--------------------------------------------------------|--------|------------------------------------------------------------|-----------------------------|----------------|----------------|----------|-------------------------------------------------------|---------------|------|-------------------------------|------|----------|-------------------|
| Input                         | Process Name                                           | Output | Remarks                                                    | Category                    | Sub category   | Infrastructure | Location | Module name in ecoinvent                              | Mean value    | Unit | Source mean value             | Type | StdV 95% | General Comment   |
| iron scrap                    | steel, electric, un- and low-alloyed, at plant; part 1 |        |                                                            | metals                      | extraction     | No             | RER      | iron scrap, at plant                                  | 1.11E+00 kg   |      | calculated after IPPC 2000    | 1    | 1.11     | (2,3,2,3,1,3,3)   |
| lime                          |                                                        |        |                                                            | construction materials      | additives      | No             | CH       | quicklime, in pieces, loose, at plant                 | 5.50E-02 kg   |      | calculated after IPPC 2000    | 1    | 1.11     | (2,3,2,3,1,3,4)   |
| coal                          |                                                        |        |                                                            | hard coal                   | fuels          | No             | UCTE     | hard coal mix, at regional storage                    | 1.40E-02 kg   |      | calculated after IPPC 2000    | 1    | 1.11     | (2,3,2,3,1,3,11)  |
| graphite electrode            |                                                        |        |                                                            | metals                      | extraction     | No             | RER      | anode, aluminium electrolysis                         | 3.00E-03 kg   |      | calculated after IPPC 2000    | 1    | 1.11     | (2,3,2,3,1,3,4)   |
| refractory materials (lining) |                                                        |        |                                                            | construction materials      | bricks         | No             | DE       | refractory, basic, packed, at plant                   | 1.35E-02 kg   |      | calculated after IPPC 2000    | 1    | 1.11     | (2,3,2,3,1,3,4)   |
| oxygen (99%)                  |                                                        |        | density 1.429 kg/m3                                        | chemicals                   | inorganics     | No             | RER      | oxygen, liquid, at plant                              | 5.07E-02 kg   |      | calculated after IPPC 2000    | 1    | 1.11     | (2,3,2,3,1,3,4)   |
| other energy                  |                                                        |        | all other energy assumed as natural gas                    | natural gas                 | fuels          | No             | RER      | natural gas, high pressure, at consumer               | 9.75E-01 MJ   |      | calculated after IPPC 2000    | 1    | 1.11     | (2,3,2,3,1,3,11)  |
| electricity                   |                                                        |        |                                                            | electricity                 | production mix | No             | UCTE     | electricity, medium voltage, production UCTE, at grid | 4.24E-01 kWh  |      | calculated after IPPC 2000    | 1    | 1.11     | (2,3,2,3,1,3,2)   |
| transport road                |                                                        |        | scrap, lime, refractories, coal & electrode: 100 km;       | transport systems           | road           | No             | RER      | transport, lorry >16t, fleet average                  | 1.19E-01 tkm  |      | estimated (standard distance) | 1    | 2.09     | (4,5,nA,nA,nA,5)  |
| transport rail                |                                                        |        | scrap, lime, refractories & coal: 200 km; electrode: 600km | transport systems           | train          | No             | RER      | transport, freight, rail                              | 1.21E-01 tkm  |      | estimated (standard distance) | 1    | 2.09     | (4,5,nA,nA,nA,5)  |
| water                         |                                                        |        | closed cycle --> not inventoried                           |                             |                |                |          |                                                       | 0.00E+00      |      | IPPC 2000                     |      |          |                   |
| infrastructure                |                                                        |        | for EAF process, secondary metallurgy and casting          | metals                      | extraction     | Yes            | RER      | electric arc furnace converter                        | 4.00E-11 unit |      | estimated                     | 1    | 3.23     | (5,nA,nA,nA,nA,9) |

Fig. 5.17 Flows for "steel, electric, and- and low-alloyed, at plant" and its representation in the ecoinvent database

| General Flow information                               |              |              |                                                               | Representation in ecoinvent |              |                |          |                                    |             |      | Uncertainty                |      |          |                  |
|--------------------------------------------------------|--------------|--------------|---------------------------------------------------------------|-----------------------------|--------------|----------------|----------|------------------------------------|-------------|------|----------------------------|------|----------|------------------|
| Input                                                  | Process Name | Output       | Remarks                                                       | Category                    | Sub category | Infrastructure | Location | Module name in ecoinvent           | Mean value  | Unit | Source mean value          | Type | StdV 95% | General Comment  |
| steel, electric, un- and low-alloyed, at plant; part 2 | →            | waste heat   | from coal, electrode (29 MJ/kg), other energy and electricity | air                         | unspecified  |                |          | Heat, waste                        | 3.01E+00 MJ |      | calculated                 | 1    | 1.08     | (1,3,2,3,1,2,13) |
|                                                        | →            | dust, coarse | 15% > 10 um                                                   | air                         | unspecified  |                |          | Particulates, > 10 um              | 5.86E-05 kg |      | calculated after IPPC 2000 | 1    | 1.52     | (2,3,2,3,1,3,25) |
|                                                        | →            | dust, medium | 42,5% between 2.5 um and 10 um                                | air                         | unspecified  |                |          | Particulates, > 2.5 um, and < 10um | 1.66E-04 kg |      | calculated after IPPC 2000 | 1    | 2.01     | (2,3,2,3,1,3,26) |
|                                                        | →            | dust, fine   | 42.5% < 2.5 um                                                | air                         | unspecified  |                |          | Particulates, < 2.5 um             | 1.66E-04 kg |      | calculated after IPPC 2000 | 1    | 3.01     | (2,3,2,3,1,3,27) |
|                                                        | →            | Hg           |                                                               | air                         | unspecified  |                |          | Mercury                            | 2.24E-06 kg |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22) |
|                                                        | →            | Pb           |                                                               | air                         | unspecified  |                |          | Lead                               | 1.81E-06 kg |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22) |
|                                                        | →            | Cr           |                                                               | air                         | unspecified  |                |          | Chromium                           | 1.25E-06 kg |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22) |
|                                                        | →            | Ni           |                                                               | air                         | unspecified  |                |          | Nickel                             | 7.01E-07 kg |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22) |
|                                                        | →            | Zn           |                                                               | air                         | unspecified  |                |          | Zinc                               | 2.29E-05 kg |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22) |
|                                                        | →            | Cd           |                                                               | air                         | unspecified  |                |          | Cadmium                            | 3.65E-08 kg |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22) |
|                                                        | →            | Cu           |                                                               | air                         | unspecified  |                |          | Copper                             | 2.31E-07 kg |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22) |
|                                                        | →            | HF           |                                                               | air                         | unspecified  |                |          | Hydrogen fluoride                  | 2.35E-06 kg |      | calculated after IPPC 2000 | 1    | 1.52     | (2,3,2,3,1,3,31) |
|                                                        | →            | HCl          |                                                               | air                         | unspecified  |                |          | Hydrogen chloride                  | 5.20E-06 kg |      | calculated after IPPC 2000 | 1    | 1.52     | (2,3,2,3,1,3,31) |
|                                                        | →            | SO2          |                                                               | air                         | unspecified  |                |          | Sulfur dioxide                     | 7.70E-05 kg |      | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,15) |
|                                                        | →            | NOx          |                                                               | air                         | unspecified  |                |          | Nitrogen oxides                    | 1.80E-04 kg |      | calculated after IPPC 2000 | 1    | 1.52     | (2,3,2,3,1,3,16) |
|                                                        | →            | CO           |                                                               | air                         | unspecified  |                |          | Carbon monoxide, fossil            | 2.32E-03 kg |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,17) |

Fig. 5.17 Flows for "steel, electric, and- and low-alloyed, at plant" and its representation in the ecoinvent database (cont.)

| General Flow information                               |              |                                |                                                                                                                | Representation in ecoinvent |                            |                |          |                                                                                  |             |      | Uncertainty                |      |          |                  |
|--------------------------------------------------------|--------------|--------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------|----------------------------|----------------|----------|----------------------------------------------------------------------------------|-------------|------|----------------------------|------|----------|------------------|
| Input                                                  | Process Name | Output                         | Remarks                                                                                                        | Category                    | Sub category               | Infrastructure | Location | Module name in ecoinvent                                                         | Mean value  | Unit | Source mean value          | Type | StdV 95% | General Comment  |
| steel, electric, un- and low-alloyed, at plant; part 3 | →            | TOC (total organic carbon)     | as aromatic hydrocarbons (92% C); individually mentioned sources of organic carbon are subtracted              | air                         | unspecified                |                |          | Hydrocarbons, aromatic                                                           | 7.70E-05 kg |      | calculated after IPPC 2000 | 1    | 1.52     | (2,3,2,3,1,3,18) |
|                                                        | →            | benzene                        | (92% C)                                                                                                        | air                         | unspecified                |                |          | Benzene                                                                          | 2.29E-06 kg |      | calculated after IPPC 2000 | 1    | 2.01     | (2,3,2,3,1,3,23) |
|                                                        | →            | hexachlorobenzene              | (25 %C)                                                                                                        | air                         | unspecified                |                |          | Benzene, hexachloro-                                                             | 2.00E-08 kg |      | calculated after IPPC 2000 | 1    | 2.01     | (2,3,2,3,1,3,23) |
|                                                        | →            | PAH                            | (92%C); benzene and hexachlorobenzene are not included                                                         | air                         | unspecified                |                |          | PAH, polycyclic aromatic hydrocarbons                                            | 3.73E-08 kg |      | calculated after IPPC 2000 | 1    | 3.01     | (2,3,2,3,1,3,21) |
|                                                        | →            | PCB                            | (50%C)                                                                                                         | air                         | unspecified                |                |          | Polychlorinated biphenyls                                                        | 2.33E-08 kg |      | calculated after IPPC 2000 | 1    | 3.01     | (2,3,2,3,1,3,21) |
|                                                        | →            | dioxin (I-TEQ)                 | (45% C)                                                                                                        | air                         | unspecified                |                |          | Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin                         | 4.54E-12 kg |      | calculated after IPPC 2000 | 1    | 3.01     | (2,3,2,3,1,3,21) |
|                                                        | →            | slag from furnace              | amount inventoried corresponds to the 64% of the slag that is landfilled. The rest is recycled or reused       | waste management            | residual material landfill | No             | CH       | disposal, slag, unalloyed electr. steel, 0% water, to residual material landfill | 8.00E-02 kg |      | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,6)  |
|                                                        | →            | slag from ladle                | amount inventoried corresponds to the 64% of the slag that is landfilled. The rest is recycled or reused       | waste management            | residual material landfill | No             | CH       | disposal, slag, unalloyed electr. steel, 0% water, to residual material landfill | 1.28E-02 kg |      | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,6)  |
|                                                        | →            | dust                           | amount inventoried corresponds to the 64% of the total dust that is landfilled. The rest is recycled or reused | waste management            | residual material landfill | No             | CH       | disposal, dust, unalloyed EAF steel, 15.4% water, to residual material landfill  | 9.60E-03 kg |      | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,6)  |
|                                                        | →            | refractory material            | although partly recycled total amount is assumed to be disposed in inert material landfill.                    | waste management            | inert material landfill    | No             | CH       | disposal, inert waste, 5% water, to inert material landfill                      | 5.00E-03 kg |      | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,6)  |
|                                                        | →            | un-/low-alloyed EAF steel slab |                                                                                                                | metals                      | extraction                 | No             | RER      | steel, electric, un- and low-alloyed, at plant                                   | 1.00E+00 kg |      |                            |      |          |                  |

Fig. 5.17 Flows for "steel, electric, and- and low-alloyed, at plant" and its representation in the ecoinvent database (cont.)

| General Flow information      |   |                                                        |  |              | Representation in ecoinvent                                        |                        |                |                  |                                              |                                                       |               | Uncertainty                |                               |       |          |                     |                      |
|-------------------------------|---|--------------------------------------------------------|--|--------------|--------------------------------------------------------------------|------------------------|----------------|------------------|----------------------------------------------|-------------------------------------------------------|---------------|----------------------------|-------------------------------|-------|----------|---------------------|----------------------|
| Input                         |   | Process Name                                           |  | Output       | Remarks                                                            | Cate gory              | Sub category   | Infra struc ture | Loca tion                                    | Modul name in ecoinvent                               | Mean value    | Unit                       | Source mean value             | Typ e | StDv 95% | General Comment     |                      |
|                               |   |                                                        |  |              |                                                                    |                        |                |                  |                                              |                                                       |               |                            |                               |       |          |                     |                      |
| iron scrap                    | → | steel, electric, chromium steel 18/8, at plant; part 1 |  |              |                                                                    | metals                 | extraction     | No               | RER                                          | iron scrap, at plant                                  | 5.20E-01 kg   |                            | calculated after IPPC 2000    | 1     | 1.11     | (2,3,2,3,1,3,3)     |                      |
| nickel                        | → |                                                        |  | 8% in alloy  | metals                                                             | extraction             | No             | GLO              | ferronickel, 25% Ni, at plant                | 3.20E-01 kg                                           |               | calculated                 |                               | 1     | 1.07     | (2,nA,nA,nA,1,nA,3) |                      |
| chromium                      | → |                                                        |  | 18% in alloy | metals                                                             | extraction             | No             | GLO              | ferrochromium, high-carbon, 68% Cr, at plant | 2.65E-01 kg                                           |               | calculated                 |                               | 1     | 1.07     | (2,nA,nA,nA,1,nA,3) |                      |
| lime                          | → |                                                        |  |              | construction materials                                             | additives              | No             | CH               | quicklime, in pieces, loose, at plant        | 5.50E-02 kg                                           |               | calculated after IPPC 2000 |                               | 1     | 1.11     | (2,3,2,3,1,3,4)     |                      |
| coal                          | → |                                                        |  |              | hard coal                                                          | fuels                  | No             | UCTE             | hard coal mix, at regional storage           | 1.40E-02 kg                                           |               | calculated after IPPC 2000 |                               | 1     | 1.11     | (2,3,2,3,1,3,11)    |                      |
| graphite electrode            | → |                                                        |  |              |                                                                    | metals                 | extraction     | No               | RER                                          | anode, aluminium electrolysis                         | 3.00E-03 kg   |                            | calculated after IPPC 2000    |       | 1        | 1.11                | (2,3,2,3,1,3,4)      |
| refractory materials (lining) | → |                                                        |  |              |                                                                    | construction materials | bricks         | No               | DE                                           | refractory, basic, packed, at plant                   | 1.35E-02 kg   |                            | calculated after IPPC 2000    |       | 1        | 1.11                | (2,3,2,3,1,3,4)      |
| oxygen (99%)                  | → |                                                        |  |              | density 1.429 kg/m3                                                | chemicals              | inorganics     | No               | RER                                          | oxygen, liquid, at plant                              | 5.07E-02 kg   |                            | calculated after IPPC 2000    |       | 1        | 1.11                | (2,3,2,3,1,3,4)      |
| other energy                  | → |                                                        |  |              | all other energy assumed as natural gas                            | natural gas            | fuels          | No               | RER                                          | natural gas, high pressure, at consumer               | 9.75E-01 MJ   |                            | calculated after IPPC 2000    |       | 1        | 1.11                | (2,3,2,3,1,3,11)     |
| electricity                   | → |                                                        |  |              |                                                                    | electricity            | production mix | No               | UCTE                                         | electricity, medium voltage, production UCTE, at grid | 4.24E-01 kWh  |                            | calculated after IPPC 2000    |       | 1        | 1.11                | (2,3,2,3,1,3,2)      |
| transport road                | → |                                                        |  |              | scrap, alloys, lime, refractories, coal & electrode: 100 km;       | transport systems      | road           | No               | RER                                          | transport, lorry >16t, fleet average                  | 1.19E-01 tkm  |                            | estimated (standard distance) |       | 1        | 2.09                | (4,5,nA,nA,nA,nA,5)  |
| transport rail                | → |                                                        |  |              | scrap, alloys, lime, refractories & coal: 200 km; electrode: 600km | transport systems      | train          | No               | RER                                          | transport, freight, rail                              | 1.21E-01 tkm  |                            | estimated (standard distance) |       | 1        | 2.09                | (4,5,nA,nA,nA,nA,5)  |
| water                         | → |                                                        |  |              | closed cycle --> not inventoried                                   |                        |                |                  |                                              |                                                       | 0.00E+00      |                            | IPPC 2000                     |       |          |                     |                      |
| infrastructure                | → |                                                        |  |              | for EAF process, secondary metallurgy and casting                  | metals                 | extraction     | Yes              | RER                                          | electric arc furnace converter                        | 4.00E-11 unit |                            | estimated                     |       | 1        | 3.23                | (5,nA,nA,nA,nA,nA,9) |

Fig. 5.18 Flows for "steel, electric, chromium steel 18/8, at plant" and its representation in the ecoinvent database

| General Flow information                               |              |              |                                                               | Representation in ecoinvent |              |                |          |                                    |             |      | Uncertainty                |      |          |                  |
|--------------------------------------------------------|--------------|--------------|---------------------------------------------------------------|-----------------------------|--------------|----------------|----------|------------------------------------|-------------|------|----------------------------|------|----------|------------------|
| Input                                                  | Process Name | Output       | Remarks                                                       | Category                    | Sub category | Infrastructure | Location | Module name in ecoinvent           | Mean value  | Unit | Source mean value          | Type | StdV 95% | General Comment  |
| steel, electric, chromium steel 18/8, at plant; part 2 | →            | waste heat   | from coal, electrode (29 MJ/kg), other energy and electricity | air                         | unspecified  |                |          | Heat, waste                        | 3.01E+00 MJ |      | calculated                 | 1    | 1.08     | (1,3,2,3,1,2,13) |
|                                                        | →            | dust, coarse | 15% > 10 um                                                   | air                         | unspecified  |                |          | Particulates, > 10 um              | 5.86E-05 kg |      | calculated after IPPC 2000 | 1    | 1.52     | (2,3,2,3,1,3,25) |
|                                                        | →            | dust, medium | 42,5% between 2.5 um and 10 um                                | air                         | unspecified  |                |          | Particulates, > 2.5 um, and < 10um | 1.66E-04 kg |      | calculated after IPPC 2000 | 1    | 2.01     | (2,3,2,3,1,3,26) |
|                                                        | →            | dust, fine   | 42.5% < 2.5 um                                                | air                         | unspecified  |                |          | Particulates, < 2.5 um             | 1.66E-04 kg |      | calculated after IPPC 2000 | 1    | 3.01     | (2,3,2,3,1,3,27) |
|                                                        | →            | Hg           |                                                               | air                         | unspecified  |                |          | Mercury                            | 2.24E-06 kg |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22) |
|                                                        | →            | Pb           |                                                               | air                         | unspecified  |                |          | Lead                               | 1.81E-06 kg |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22) |
|                                                        | →            | Cr           |                                                               | air                         | unspecified  |                |          | Chromium                           | 1.25E-06 kg |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22) |
|                                                        | →            | Ni           |                                                               | air                         | unspecified  |                |          | Nickel                             | 7.01E-07 kg |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22) |
|                                                        | →            | Zn           |                                                               | air                         | unspecified  |                |          | Zinc                               | 2.29E-05 kg |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22) |
|                                                        | →            | Cd           |                                                               | air                         | unspecified  |                |          | Cadmium                            | 3.65E-08 kg |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22) |
|                                                        | →            | Cu           |                                                               | air                         | unspecified  |                |          | Copper                             | 2.31E-07 kg |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,22) |
|                                                        | →            | HF           |                                                               | air                         | unspecified  |                |          | Hydrogen fluoride                  | 2.35E-06 kg |      | calculated after IPPC 2000 | 1    | 1.52     | (2,3,2,3,1,3,31) |
|                                                        | →            | HCl          |                                                               | air                         | unspecified  |                |          | Hydrogen chloride                  | 5.20E-06 kg |      | calculated after IPPC 2000 | 1    | 1.52     | (2,3,2,3,1,3,31) |
|                                                        | →            | SO2          |                                                               | air                         | unspecified  |                |          | Sulfur dioxide                     | 7.70E-05 kg |      | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,15) |
|                                                        | →            | NOx          |                                                               | air                         | unspecified  |                |          | Nitrogen oxides                    | 1.80E-04 kg |      | calculated after IPPC 2000 | 1    | 1.52     | (2,3,2,3,1,3,16) |
|                                                        | →            | CO           |                                                               | air                         | unspecified  |                |          | Carbon monoxide, fossil            | 2.32E-03 kg |      | calculated after IPPC 2000 | 1    | 5.01     | (2,3,2,3,1,3,17) |

Fig. 5.18 Flows for "steel, electric, chromium steel 18/8, at plant" and its representation in the ecoinvent database (cont.)

| General Flow information                               |              |                            |                                                                                                                | Representation in ecoinvent |                            |                |          |                                                                                  |             |      | Uncertainty                |      |          |                  |
|--------------------------------------------------------|--------------|----------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------|----------------------------|----------------|----------|----------------------------------------------------------------------------------|-------------|------|----------------------------|------|----------|------------------|
| Input                                                  | Process Name | Output                     | Remarks                                                                                                        | Category                    | Sub category               | Infrastructure | Location | Module name in ecoinvent                                                         | Mean value  | Unit | Source mean value          | Type | StdV 95% | General Comment  |
| steel, electric, chromium steel 18/8, at plant; part 3 | →            | TOC (total organic carbon) | as aromatic hydrocarbons (92% C); individually mentioned sources of organic carbon are subtracted              | air                         | unspecified                |                |          | Hydrocarbons, aromatic                                                           | 7.70E-05 kg |      | calculated after IPPC 2000 | 1    | 1.52     | (2,3,2,3,1,3,18) |
|                                                        | →            | benzene                    | (92% C)                                                                                                        | air                         | unspecified                |                |          | Benzene                                                                          | 2.29E-06 kg |      | calculated after IPPC 2000 | 1    | 2.01     | (2,3,2,3,1,3,23) |
|                                                        | →            | hexachlorobenzene          | (25 %C)                                                                                                        | air                         | unspecified                |                |          | Benzene, hexachloro-                                                             | 2.00E-08 kg |      | calculated after IPPC 2000 | 1    | 2.01     | (2,3,2,3,1,3,23) |
|                                                        | →            | PAH                        | (92%C); benzene and hexachlorobenzene are not included                                                         | air                         | unspecified                |                |          | PAH, polycyclic aromatic hydrocarbons                                            | 3.73E-08 kg |      | calculated after IPPC 2000 | 1    | 3.01     | (2,3,2,3,1,3,21) |
|                                                        | →            | PCB                        | (50%C)                                                                                                         | air                         | unspecified                |                |          | Polychlorinated biphenyls                                                        | 2.33E-08 kg |      | calculated after IPPC 2000 | 1    | 3.01     | (2,3,2,3,1,3,21) |
|                                                        | →            | dioxin (I-TEQ)             | (45% C)                                                                                                        | air                         | unspecified                |                |          | Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin                         | 4.54E-12 kg |      | calculated after IPPC 2000 | 1    | 3.01     | (2,3,2,3,1,3,21) |
|                                                        | →            | slag from furnace          | amount inventoried corresponds to the 53% of the slag that is landfilled. The rest is recycled or reused       | waste management            | residual material landfill | No             | CH       | disposal, slag, unalloyed electr. steel, 0% water, to residual material landfill | 6.63E-02 kg |      | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,6)  |
|                                                        | →            | slag from ladle            | amount inventoried corresponds to the 53% of the slag that is landfilled. The rest is recycled or reused       | waste management            | residual material landfill | No             | CH       | disposal, slag, unalloyed electr. steel, 0% water, to residual material landfill | 1.06E-02 kg |      | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,6)  |
|                                                        | →            | dust                       | amount inventoried corresponds to the 34% of the total dust that is landfilled. The rest is recycled or reused | waste management            | residual material landfill | No             | CH       | disposal, dust, alloyed EAF steel, 15.4% water, to residual material landfill    | 5.10E-03 kg |      | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,6)  |
|                                                        | →            | refractory material        | although partly recycled total amount is assumed to be disposed in inert material landfill.                    | waste management            | inert material landfill    | No             | CH       | disposal, inert waste, 5% water, to inert material landfill                      | 5.00E-03 kg |      | calculated after IPPC 2000 | 1    | 1.11     | (2,3,2,3,1,3,6)  |
|                                                        | →            | EAF chromium steel slab    |                                                                                                                | metals                      | extraction                 | No             | RER      | chromium steel 18/8, at plant                                                    | 1.00E+00 kg |      |                            |      |          |                  |

Fig. 5.18 Flows for "steel, electric, chromium steel 18/8, at plant" and its representation in the ecoinvent database (cont.)

| General Flow information                                       |                                |                                 |                                             | Representation in ecoinvent |                     |                |          |                                                                 |               |      | Uncertainty       |      |          |                                        |
|----------------------------------------------------------------|--------------------------------|---------------------------------|---------------------------------------------|-----------------------------|---------------------|----------------|----------|-----------------------------------------------------------------|---------------|------|-------------------|------|----------|----------------------------------------|
| Input                                                          | Process Name                   | Output                          | Remarks                                     | Category                    | Sub category        | Infrastructure | Location | Module name in ecoinvent                                        | Mean value    | Unit | Source mean value | Type | StdV 95% | General Comment                        |
| Total land use (transformation)                                | electric arc furnace converter |                                 |                                             | resource                    | land                |                |          | Transformation, from unknown                                    | 5.00E+05 m2   |      | estimated         | 1    | 5.00     | (5,nA,nA,nA,nA,nA,8); rough estimation |
| Land occupation for buildings and open storage area            |                                |                                 |                                             | resource                    | land                |                |          | Occupation, industrial area, built up                           | 2.00E+07 m2a  |      | estimated         | 1    | 5.00     | (5,nA,nA,nA,nA,nA,7); rough estimation |
| Land transformation to buildings and open bauxite storage area |                                |                                 | assumption: 80% of total area is built up   | resource                    | land                |                |          | Transformation, to industrial area, built up                    | 4.00E+05 m2   |      | estimated         | 1    | 5.00     | (5,nA,nA,nA,nA,nA,8); rough estimation |
| Land occupation for vegetation area                            |                                |                                 |                                             | resource                    | land                |                |          | Occupation, industrial area, vegetation                         | 5.00E+06 m2a  |      | estimated         | 1    | 5.00     | (5,nA,nA,nA,nA,nA,7); rough estimation |
| Land transformation to vegetation area                         |                                |                                 | assumption: 20% of total area is vegetation | resource                    | land                |                |          | Transformation, to industrial area, vegetation                  | 1.00E+05 m2   |      | estimated         | 1    | 5.00     | (5,nA,nA,nA,nA,nA,8); rough estimation |
| Building, multi story                                          |                                |                                 |                                             | construction processes      | buildings           | Yes            | RER      | building, multi-storey                                          | 2.00E+03 m3   |      | estimated         | 1    | 5.00     | (5,nA,nA,nA,nA,nA,9); rough estimation |
| Building, hall                                                 |                                |                                 |                                             | construction processes      | buildings           | Yes            | CH       | building, hall                                                  | 2.50E+05 m2   |      | estimated         | 1    | 5.00     | (5,nA,nA,nA,nA,nA,9); rough estimation |
| heavy machine, unspecified                                     |                                |                                 |                                             | construction processes      | machinery           | Yes            | RER      | industrial machine, heavy, unspecified, at plant                | 1.10E+07 kg   |      | estimated         | 1    | 5.00     | (5,nA,nA,nA,nA,nA,9); rough estimation |
| Poor concrete for additional foundation                        |                                |                                 |                                             | construction materials      | concrete            | No             | CH       | concrete, sole plate and foundation, at plant                   | 5.00E+03 m3   |      | estimated         | 1    | 5.00     | (5,nA,nA,nA,nA,nA,9); rough estimation |
| firebrick                                                      |                                |                                 |                                             | construction materials      | bricks              | No             | DE       | refractory, basic, packed, at plant                             | 1.00E+05 kg   |      | estimated         | 1    | 5.00     | (5,nA,nA,nA,nA,nA,9); rough estimation |
|                                                                |                                | foundation concrete to disposal |                                             | waste management            | building demolition | No             | CH       | disposal, building, concrete, not reinforced, to final disposal | 1.10E+07 kg   |      | estimated         | 1    | 5.00     | (5,nA,nA,nA,nA,nA,9); rough estimation |
|                                                                |                                | firebricks to disposal          |                                             | waste management            | building demolition | No             | CH       | disposal, building, brick, to final disposal                    | 5.00E+03 kg   |      | estimated         | 1    | 5.00     | (5,nA,nA,nA,nA,nA,9); rough estimation |
|                                                                |                                | EAF converter                   | Output: 500000000 kg/a, life time: 50 a     | metals                      | extraction          | Yes            | RER      | electric arc furnace converter                                  | 1.00E+00 unit |      |                   |      |          |                                        |

Fig. 5.19 Flows for "electric arc furnace converter" and its representation in the ecoinvent database



## 5.8 Cast iron

### 5.8.1 Production process and infrastructure

Cast iron is made by remelting pig iron in small furnaces (e.g. cupola furnaces or electric arc furnaces) and poured into moulds to make castings. It usually contains 2% to 6% carbon. Scrap iron or steel is often added to vary the composition. Cast iron is used extensively to make machine parts, engine cylinder blocks, stoves, pipes, steam radiators, and many other products. Grey cast iron, or grey iron, is produced when the iron in the mould is cooled slowly. Part of the carbon separates out in plates in the form of graphite but remains physically mixed in the iron. Grey iron is brittle but soft and easily machined. White cast iron, or white iron, which is harder and more brittle, is made by cooling the molten iron rapidly. The carbon remains distributed throughout the iron as cementite (iron carbide,  $\text{Fe}_3\text{C}$ ). A malleable cast iron can be made by annealing white iron castings in a special furnace. Some of the carbon separates from the cementite; it is much more finely divided than in grey iron. A ductile iron may be prepared by adding magnesium to the molten pig iron; when the iron is cast the carbon forms tiny spherical nodules around the magnesium. Ductile iron is strong, shock resistant, and easily machined.<sup>7</sup>

### 5.8.2 Emissions

#### Air

Air emissions depend mainly on the type of furnaces and the fuel used in them. Emission composition of the production in a cupola furnace is similar to that of the blast furnace while emissions of the cast iron production in electric arc furnaces is similar to those of the steel making in those furnaces.

#### Water

There are no reports of direct water emissions from these processes.

### 5.8.3 Waste

As for the steel making the main wastes from cast iron production is slag and filter dust. No specific information was found.

### 5.8.4 Cast iron in ecoinvent

The background information to the process "cast iron, at plant" are given in Tab. 5.23, and flows are shown in Fig. 5.20

For a lack of specific data the cast iron production is inventoried using the data of the EAF steel making but assuming a mix of iron input of 65% pig iron and 35% scrap. Since cast iron usually is un- or low-alloyed and melted in electric furnaces, this assumptions seem justified.

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<sup>7</sup> <http://www.infoplease.com/ce6/sci/A0858908.html>, accessed 12.02.03

**Tab. 5.23 Meta information for the cast iron production processes**

|                                       |                                                                                                                  |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Name                                  | cast iron, at plant                                                                                              |
| Location                              | RER                                                                                                              |
| Infrastructure Process                | 0                                                                                                                |
| Unit                                  | kg                                                                                                               |
| Data Set Version                      | 2.0                                                                                                              |
| Included Processes                    | Transports of metal and other input materials to electric arc furnace, melting and refining process and casting. |
| Amount                                | 1                                                                                                                |
| Local Name                            | Gusseisen, ab Werk                                                                                               |
| Synonyms                              | grey iron//white iron//Grauguss                                                                                  |
| General Comment to reference function | 35% scrap and 65% pig iron assumed as iron input                                                                 |
| Start Date                            | 2001                                                                                                             |
| End Date                              | 2002                                                                                                             |
| Data Valid For Entire Period          | 1                                                                                                                |
| Other Period Text                     |                                                                                                                  |
| Geography text                        | Data relate to plants in the EU                                                                                  |
| Technology text                       | Electric arc furnace for melting. Energy consumption and emissions from EAF steel making                         |
| Representativeness [%]                |                                                                                                                  |
| Production Volume                     | unknown                                                                                                          |
| Sampling Procedure                    | Internet search                                                                                                  |
| Extrapolations                        | see technology                                                                                                   |
| Uncertainty Adjustments               | none                                                                                                             |

**Data quality**

Data uncertainty is derived using the Pedigree matrix as described in (Frischknecht et al. (2003)) except for the infrastructure where this method would result in values far too low.

Data are estimates based on a technology that is similar to one of different technologies used in the cast iron production. Thus data uncertainty is rather high.

Data for the infrastructure is roughly estimated.

| General Flow information      |   |                             |  |                                         | Representation in ecoinvent             |                                                                      |                   |                  |                                       |                                     |                                                       | Uncertainty                |                            |                               |                  |                 |                      |
|-------------------------------|---|-----------------------------|--|-----------------------------------------|-----------------------------------------|----------------------------------------------------------------------|-------------------|------------------|---------------------------------------|-------------------------------------|-------------------------------------------------------|----------------------------|----------------------------|-------------------------------|------------------|-----------------|----------------------|
| Input                         |   | Process Name                |  | Output                                  | Remarks                                 | Cate gory                                                            | Sub category      | Infra struc ture | Loca tion                             | Modul name in ecoinvent             | Mean value                                            | Unit                       | Source mean value          | Typ e                         | StDv 95%         | General Comment |                      |
|                               |   |                             |  |                                         |                                         |                                                                      |                   |                  |                                       |                                     |                                                       |                            |                            |                               |                  |                 |                      |
| iron scrap                    | → | cast iron, at plant; part 1 |  |                                         | assumption: 35% scrap, 65% primary iron | metals                                                               | extraction        | No               | RER                                   | iron scrap, at plant                | 3.87E-01 kg                                           |                            | calculated after IPPC 2000 | 1                             | 1.23             | (2,3,2,3,3,3,3) |                      |
| pig iron                      | → |                             |  | assumption: 35% scrap, 65% primary iron | metals                                  | extraction                                                           | No                | GLO              | pig iron, at plant                    | 7.18E-01 kg                         |                                                       | calculated after IPPC 2000 | 1                          | 1.23                          | (2,3,2,3,3,3,3)  |                 |                      |
| lime                          | → |                             |  |                                         | construction materials                  | additives                                                            | No                | CH               | quicklime, in pieces, loose, at plant | 5.50E-02 kg                         |                                                       | calculated after IPPC 2000 | 1                          | 1.23                          | (2,3,2,3,3,3,4)  |                 |                      |
| coal                          | → |                             |  |                                         | hard coal                               | fuels                                                                | No                | UCTE             | hard coal mix, at regional storage    | 1.40E-02 kg                         |                                                       | calculated after IPPC 2000 | 1                          | 1.23                          | (2,3,2,3,3,3,11) |                 |                      |
| graphite electrode            | → |                             |  |                                         |                                         | metals                                                               | extraction        | No               | RER                                   | anode, aluminium electrolysis       | 3.00E-03 kg                                           |                            | calculated after IPPC 2000 | 1                             | 1.23             | (2,3,2,3,3,3,4) |                      |
| refractory materials (lining) | → |                             |  |                                         |                                         | construction materials                                               | bricks            | No               | DE                                    | refractory, basic, packed, at plant | 1.35E-02 kg                                           |                            | calculated after IPPC 2000 | 1                             | 1.23             | (2,3,2,3,3,3,4) |                      |
| oxygen (99%)                  | → |                             |  |                                         |                                         | density 1.429 kg/m3                                                  | chemicals         | inorganics       | No                                    | RER                                 | oxygen, liquid, at plant                              | 5.07E-02 kg                |                            | calculated after IPPC 2000    | 1                | 1.23            | (2,3,2,3,3,3,4)      |
| other energy                  | → |                             |  |                                         |                                         | all other energy assumed as natural gas                              | natural gas       | fuels            | No                                    | RER                                 | natural gas, high pressure, at consumer               | 9.75E-01 MJ                |                            | calculated after IPPC 2000    | 1                | 1.23            | (2,3,2,3,3,3,11)     |
| electricity                   | → |                             |  |                                         |                                         |                                                                      | electricity       | production mix   | No                                    | UCTE                                | electricity, medium voltage, production UCTE, at grid | 4.24E-01 kWh               |                            | calculated after IPPC 2000    | 1                | 1.23            | (2,3,2,3,3,3,2)      |
| transport road                | → |                             |  |                                         |                                         | scrap, pig iron, lime, refractories, coal & electrode: 100 km;       | transport systems | road             | No                                    | RER                                 | transport, lorry >16t, fleet average                  | 1.19E-01 tkm               |                            | estimated (standard distance) | 1                | 2.09            | (4,5,nA,nA,nA,nA,5)  |
| transport rail                | → |                             |  |                                         |                                         | scrap, pig iron, lime, refractories & coal: 200 km; electrode: 600km | transport systems | train            | No                                    | RER                                 | transport, freight, rail                              | 1.21E-01 tkm               |                            | estimated (standard distance) | 1                | 2.09            | (4,5,nA,nA,nA,nA,5)  |
| water                         | → |                             |  |                                         |                                         | closed cycle --> not inventoried                                     |                   |                  |                                       |                                     |                                                       | 0.00E+00                   |                            | IPPC 2000                     |                  |                 |                      |
| infrastructure                | → |                             |  |                                         |                                         | for EAF process, secondary metallurgy and casting                    | metals            | extraction       | Yes                                   | RER                                 | electric arc furnace converter                        | 4.00E-11 unit              |                            | estimated                     | 1                | 3.23            | (5,nA,nA,nA,nA,nA,9) |

Fig. 5.20 Flows for "cast iron, at plant" and its representation in the ecoinvent database

| General Flow information    |              |              |                                                               | Representation in ecoinvent |              |                |          |                                    |             |      |                            | Uncertainty |          |                  |  |
|-----------------------------|--------------|--------------|---------------------------------------------------------------|-----------------------------|--------------|----------------|----------|------------------------------------|-------------|------|----------------------------|-------------|----------|------------------|--|
| Input                       | Process Name | Output       | Remarks                                                       | Category                    | Sub category | Infrastructure | Location | Module name in ecoinvent           | Mean value  | Unit | Source mean value          | Type        | StdV 95% | General Comment  |  |
| cast iron, at plant; part 2 | →            | waste heat   | from coal, electrode (29 MJ/kg), other energy and electricity | air                         | unspecified  |                |          | Heat, waste                        | 3.01E+00 MJ |      | calculated                 | 1           | 1.22     | (1,3,2,3,3,2,13) |  |
|                             | →            | dust, coarse | 15% > 10 um                                                   | air                         | unspecified  |                |          | Particulates, > 10 um              | 5.86E-05 kg |      | calculated after IPPC 2000 | 1           | 1.57     | (2,3,2,3,3,3,25) |  |
|                             | →            | dust, medium | 42,5% between 2.5 um and 10 um                                | air                         | unspecified  |                |          | Particulates, > 2.5 um, and < 10um | 1.66E-04 kg |      | calculated after IPPC 2000 | 1           | 2.06     | (2,3,2,3,3,3,26) |  |
|                             | →            | dust, fine   | 42.5% < 2.5 um                                                | air                         | unspecified  |                |          | Particulates, < 2.5 um             | 1.66E-04 kg |      | calculated after IPPC 2000 | 1           | 3.06     | (2,3,2,3,3,3,27) |  |
|                             | →            | Hg           |                                                               | air                         | unspecified  |                |          | Mercury                            | 2.24E-06 kg |      | calculated after IPPC 2000 | 1           | 5.06     | (2,3,2,3,3,3,22) |  |
|                             | →            | Pb           |                                                               | air                         | unspecified  |                |          | Lead                               | 1.81E-06 kg |      | calculated after IPPC 2000 | 1           | 5.06     | (2,3,2,3,3,3,22) |  |
|                             | →            | Cr           |                                                               | air                         | unspecified  |                |          | Chromium                           | 1.25E-06 kg |      | calculated after IPPC 2000 | 1           | 5.06     | (2,3,2,3,3,3,22) |  |
|                             | →            | Ni           |                                                               | air                         | unspecified  |                |          | Nickel                             | 7.01E-07 kg |      | calculated after IPPC 2000 | 1           | 5.06     | (2,3,2,3,3,3,22) |  |
|                             | →            | Zn           |                                                               | air                         | unspecified  |                |          | Zinc                               | 2.29E-05 kg |      | calculated after IPPC 2000 | 1           | 5.06     | (2,3,2,3,3,3,22) |  |
|                             | →            | Cd           |                                                               | air                         | unspecified  |                |          | Cadmium                            | 3.65E-08 kg |      | calculated after IPPC 2000 | 1           | 5.06     | (2,3,2,3,3,3,22) |  |
|                             | →            | Cu           |                                                               | air                         | unspecified  |                |          | Copper                             | 2.31E-07 kg |      | calculated after IPPC 2000 | 1           | 5.06     | (2,3,2,3,3,3,22) |  |
|                             | →            | HF           |                                                               | air                         | unspecified  |                |          | Hydrogen fluoride                  | 2.35E-06 kg |      | calculated after IPPC 2000 | 1           | 1.57     | (2,3,2,3,3,3,31) |  |
|                             | →            | HCl          |                                                               | air                         | unspecified  |                |          | Hydrogen chloride                  | 5.20E-06 kg |      | calculated after IPPC 2000 | 1           | 1.57     | (2,3,2,3,3,3,31) |  |
|                             | →            | SO2          |                                                               | air                         | unspecified  |                |          | Sulfur dioxide                     | 7.70E-05 kg |      | calculated after IPPC 2000 | 1           | 1.23     | (2,3,2,3,3,3,15) |  |
|                             | →            | NOx          |                                                               | air                         | unspecified  |                |          | Nitrogen oxides                    | 1.80E-04 kg |      | calculated after IPPC 2000 | 1           | 1.57     | (2,3,2,3,3,3,16) |  |
|                             | →            | CO           |                                                               | air                         | unspecified  |                |          | Carbon monoxide, fossil            | 2.32E-03 kg |      | calculated after IPPC 2000 | 1           | 5.06     | (2,3,2,3,3,3,17) |  |

Fig. 5.20 Flows for "cast iron, at plant" and its representation in the ecoinvent database (cont.)

| General Flow information    |              |                            |                                                                                                                | Representation in ecoinvent |                            |                |          |                                                                                  |             |      | Uncertainty                |      |          |                  |
|-----------------------------|--------------|----------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------|----------------------------|----------------|----------|----------------------------------------------------------------------------------|-------------|------|----------------------------|------|----------|------------------|
| Input                       | Process Name | Output                     | Remarks                                                                                                        | Category                    | Sub category               | Infrastructure | Location | Module name in ecoinvent                                                         | Mean value  | Unit | Source mean value          | Type | StdV 95% | General Comment  |
| cast iron, at plant; part 3 | →            | TOC (total organic carbon) | as aromatic hydrocarbons (92% C); individually mentioned sources of organic carbon are subtracted              | air                         | unspecified                |                |          | Hydrocarbons, aromatic                                                           | 7.70E-05 kg |      | calculated after IPPC 2000 | 1    | 1.57     | (2,3,2,3,3,3,18) |
|                             | →            | benzene                    | (92% C)                                                                                                        | air                         | unspecified                |                |          | Benzene                                                                          | 2.29E-06 kg |      | calculated after IPPC 2000 | 1    | 2.06     | (2,3,2,3,3,3,23) |
|                             | →            | hexachlorobenzene          | (25 %C)                                                                                                        | air                         | unspecified                |                |          | Benzene, hexachloro-                                                             | 2.00E-08 kg |      | calculated after IPPC 2000 | 1    | 2.06     | (2,3,2,3,3,3,23) |
|                             | →            | PAH                        | (92%C); benzene and hexachlorobenzene are not included                                                         | air                         | unspecified                |                |          | PAH, polycyclic aromatic hydrocarbons                                            | 3.73E-08 kg |      | calculated after IPPC 2000 | 1    | 3.06     | (2,3,2,3,3,3,21) |
|                             | →            | PCB                        | (50%C)                                                                                                         | air                         | unspecified                |                |          | Polychlorinated biphenyls                                                        | 2.33E-08 kg |      | calculated after IPPC 2000 | 1    | 3.06     | (2,3,2,3,3,3,21) |
|                             | →            | dioxin (I-TEQ)             | (45% C)                                                                                                        | air                         | unspecified                |                |          | Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin                         | 4.54E-12 kg |      | calculated after IPPC 2000 | 1    | 3.06     | (2,3,2,3,3,3,21) |
|                             | →            | slag from furnace          | amount inventoried corresponds to the 64% of the slag that is landfilled. The rest is recycled or reused       | waste management            | residual material landfill | No             | CH       | disposal, slag, unalloyed electr. steel, 0% water, to residual material landfill | 8.00E-02 kg |      | calculated after IPPC 2000 | 1    | 1.23     | (2,3,2,3,3,3,6)  |
|                             | →            | slag from ladle            | amount inventoried corresponds to the 64% of the slag that is landfilled. The rest is recycled or reused       | waste management            | residual material landfill | No             | CH       | disposal, slag, unalloyed electr. steel, 0% water, to residual material landfill | 1.28E-02 kg |      | calculated after IPPC 2000 | 1    | 1.23     | (2,3,2,3,3,3,6)  |
|                             | →            | dust                       | amount inventoried corresponds to the 64% of the total dust that is landfilled. The rest is recycled or reused | waste management            | residual material landfill | No             | CH       | disposal, dust, unalloyed EAF steel, 15.4% water, to residual material landfill  | 9.60E-03 kg |      | calculated after IPPC 2000 | 1    | 1.23     | (2,3,2,3,3,3,6)  |
|                             | →            | refractory material        | although partly recycled total amount is assumed to be disposed in inert material landfill.                    | waste management            | inert material landfill    | No             | CH       | disposal, inert waste, 5% water, to inert material landfill                      | 5.00E-03 kg |      | calculated after IPPC 2000 | 1    | 1.23     | (2,3,2,3,3,3,6)  |
|                             | →            | cast iron                  |                                                                                                                | metals                      | extraction                 | No             | RER      | cast iron, at plant                                                              | 1.00E+00 kg |      |                            |      |          |                  |

Fig. 5.20 Flows for "cast iron, at plant" and its representation in the ecoinvent database (cont.)

## **5.9 Consumer mixes for carbon steel, low-alloy steel and stainless steel**

### **5.9.1 Steel in ecoinvent**

The different steel alloys inventoried in ecoinvent are modelled by mixtures of differently produced steel and alloy materials. In Europe in 2000 39.8% of the steel was produced in electric furnaces and the remaining 60.2% in basic oxygen furnaces. The world production mix was 58.5% basic oxygen furnace, 34% electric furnace and 4.4% open hearth furnace (IISI (2002)). For the consumption mix in Europe we assume 63% BOF and 37% EAF production.

Because steel for every application is hot rolled, this process is also included in the modules. For a description of the process see Part IXX Metalprocessing.

The background information to the processes is given in Tab. 5.24, and flows are shown in:

- reinforcing steel, at plant Fig. 5.21
- steel, low-alloyed, at plant Fig. 5.22
- chromium steel 18/8, at plant Fig. 5.23

### **Data quality**

Data uncertainty is derived using the Pedigree matrix as described in (Frischknecht et al. (2003)).

The shares of the different technologies stem from a statistical yearbook. Thus the data are reliable.

Tab. 5.24 Meta information for the consumer steels

|                                       |                                                                                                                         |                                                                                                                         |                                                                                                                         |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Name                                  | reinforcing steel, at plant                                                                                             | steel, low-alloyed, at plant                                                                                            | chromium steel 18/8, at plant                                                                                           |
| Location                              | RER                                                                                                                     | RER                                                                                                                     | RER                                                                                                                     |
| Infrastructure Process                | 0                                                                                                                       | 0                                                                                                                       | 0                                                                                                                       |
| Unit                                  | kg                                                                                                                      | kg                                                                                                                      | kg                                                                                                                      |
| Data Set Version                      | 2.0                                                                                                                     | 2.0                                                                                                                     | 2.0                                                                                                                     |
| Included Processes                    | Mix of differently produced steels and hot rolling                                                                      | Mix of differently produced steels and hot rolling                                                                      | Mix of differently produced steels and hot rolling                                                                      |
| Amount                                | 1                                                                                                                       | 1                                                                                                                       | 1                                                                                                                       |
| Local Name                            | Armierungsstahl, ab Werk                                                                                                | Stahl, niedriglegiert, ab Werk                                                                                          | Chromstahl 18/8, ab Werk                                                                                                |
| Synonyms                              | carbon steel//unalloyed steel                                                                                           |                                                                                                                         | high alloyed steel//stainless steel                                                                                     |
| General Comment to reference function | Represents Average of World and European production mix. This is assumed to correspond to the consumption mix in Europe | Represents Average of World and European production mix. This is assumed to correspond to the consumption mix in Europe | Represents Average of World and European production mix. This is assumed to correspond to the consumption mix in Europe |
| Start Date                            | 2000                                                                                                                    | 2000                                                                                                                    | 2000                                                                                                                    |
| End Date                              | 2002                                                                                                                    | 2002                                                                                                                    | 2002                                                                                                                    |
| Data Valid For Entire Period          | 1                                                                                                                       | 1                                                                                                                       | 1                                                                                                                       |
| Other Period Text                     |                                                                                                                         |                                                                                                                         |                                                                                                                         |
| Geography text                        | Data relate to plants in the EU                                                                                         | Data relate to plants in the EU                                                                                         | Data relate to plants in the EU                                                                                         |
| Technology text                       | technology mix                                                                                                          | technology mix                                                                                                          | technology mix                                                                                                          |
| Representativeness [%]                |                                                                                                                         |                                                                                                                         |                                                                                                                         |
| Production Volume                     | unknown                                                                                                                 | unknown                                                                                                                 | unknown                                                                                                                 |
| Sampling Procedure                    | literature, statistics                                                                                                  | literature, statistics                                                                                                  | literature, statistics                                                                                                  |
| Extrapolations                        |                                                                                                                         |                                                                                                                         |                                                                                                                         |
| Uncertainty Adjustments               | none                                                                                                                    | none                                                                                                                    | none                                                                                                                    |

| General Flow information             |                             |                                  |         | Representation in ecoinvent |              |                |          |                                                |             |      |                                | Uncertainty |          |                            |  |
|--------------------------------------|-----------------------------|----------------------------------|---------|-----------------------------|--------------|----------------|----------|------------------------------------------------|-------------|------|--------------------------------|-------------|----------|----------------------------|--|
| Input                                | Process Name                | Output                           | Remarks | Category                    | Sub category | Infrastructure | Location | Module name in ecoinvent                       | Mean value  | Unit | Source mean value              | Type        | StdV 95% | General Comment            |  |
| unalloyed basic oxygen furnace steel | reinforcing steel, at plant |                                  |         |                             |              |                |          |                                                |             |      |                                |             |          |                            |  |
| unalloyed electric arc furnace steel |                             |                                  |         | metals                      | extraction   | No             | RER      | steel, converter, unalloyed, at plant          | 6.30E-01 kg |      | estimation, based on IISI 2002 | 1           | 1.11     | (2,3,2,3,1,3,3)            |  |
| hot rolling                          |                             |                                  |         | metals                      | extraction   | No             | RER      | steel, electric, un- and low-alloyed, at plant | 3.70E-01 kg |      | estimation, based on IISI 2002 | 1           | 1.11     | (2,3,2,3,1,3,3)            |  |
|                                      |                             |                                  |         | metals                      | processing   | No             | RER      | hot rolling, steel                             | 1.00E+00 kg |      | calculated                     | 1           | 1.00     | relates directly to output |  |
|                                      |                             | reinforcing steel (carbon steel) |         | metals                      | extraction   | No             | RER      | reinforcing steel, at plant                    | 1.00E+00 kg |      |                                |             |          |                            |  |

Fig. 5.21 Flows for "reinforcing steel, at plant" and its representation in the ecoinvent database

| General Flow information               |                              |                   |         | Representation in ecoinvent |              |                |          |                                                |             |      |                                | Uncertainty |          |                            |  |
|----------------------------------------|------------------------------|-------------------|---------|-----------------------------|--------------|----------------|----------|------------------------------------------------|-------------|------|--------------------------------|-------------|----------|----------------------------|--|
| Input                                  | Process Name                 | Output            | Remarks | Category                    | Sub category | Infrastructure | Location | Module name in ecoinvent                       | Mean value  | Unit | Source mean value              | Type        | StdV 95% | General Comment            |  |
| low-alloyed basic oxygen furnace steel | steel, low-alloyed, at plant |                   |         |                             |              |                |          |                                                |             |      |                                |             |          |                            |  |
| low-alloyed electric arc furnace steel |                              |                   |         | metals                      | extraction   | No             | RER      | steel, converter, low-alloyed, at plant        | 6.30E-01 kg |      | estimation, based on IISI 2002 | 1           | 1.11     | (2,3,2,3,1,3,3)            |  |
| hot rolling                            |                              |                   |         | metals                      | extraction   | No             | RER      | steel, electric, un- and low-alloyed, at plant | 3.70E-01 kg |      | estimation, based on IISI 2002 | 1           | 1.11     | (2,3,2,3,1,3,3)            |  |
|                                        |                              |                   |         | metals                      | processing   | No             | RER      | hot rolling, steel                             | 1.00E+00 kg |      | calculated                     | 1           | 1.00     | relates directly to output |  |
|                                        |                              | low alloyed steel |         | metals                      | extraction   | No             | RER      | steel, low-alloyed, at plant                   | 1.00E+00 kg |      |                                |             |          |                            |  |

Fig. 5.22 Flows for "steel, low-alloyed, at plant" and its representation in the ecoinvent database



| General Flow information            |                               |                     |         | Representation in ecoinvent |              |                |          |                                                 |            |      | Uncertainty                    |      |            |                            |
|-------------------------------------|-------------------------------|---------------------|---------|-----------------------------|--------------|----------------|----------|-------------------------------------------------|------------|------|--------------------------------|------|------------|----------------------------|
| Input                               | Process Name                  | Output              | Remarks | Category                    | Sub category | Infrastructure | Location | Module name in ecoinvent                        | Mean value | Unit | Source mean value              | Type | StdDev 95% | General Comment            |
| basic oxygen furnace chromium steel | chromium steel 18/8, at plant |                     |         | metals                      | extraction   | No             | RER      | steel, converter, chromium steel 18/8, at plant | 6.30E-01   | kg   | estimation, based on IISI 2002 | 1    | 1.11       | (2,3,2,3,1,3,3)            |
| electric arc furnace chromium steel |                               |                     |         | metals                      | extraction   | No             | RER      | steel, electric, chromium steel 18/8, at plant  | 3.70E-01   | kg   | estimation, based on IISI 2002 | 1    | 1.11       | (2,3,2,3,1,3,3)            |
| hot rolling                         |                               |                     |         | metals                      | processing   | No             | RER      | hot rolling, steel                              | 1.00E+00   | kg   | calculated                     | 1    | 1.00       | relates directly to output |
|                                     |                               | chromium steel 18/8 |         | metals                      | extraction   | No             | RER      | chromium steel 18/8, at plant                   | 1.00E+00   | kg   |                                |      |            |                            |

Fig. 5.23 Flows for "chromium steel 18/8, at plant" and its representation in the ecoinvent database

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## Part III

# Copper, Molybdenum & Telluride and Silver containing byproducts

Data v2.1 (2009)

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## Comments on version 2.1

This part corresponds to the original report in Althaus et al. (2004) that has been updated with the changes reported in Frischknecht et al. (2006). The changes were implemented with v1.2 and affect mainly the datasets for primary and secondary copper by increasing the emission factors to air for dioxin, carbon monoxide, particulate matter and volatile organic compounds.

The inventories covering copper alloys (brass, bronze and brazing solder), which were treated in the now removed “Part VIII Other Metals” in Althaus et al. (2004), are now included in this Part III.

Additionally the generation of telluride and silver containing anode slime has been introduced in this version to answer the related demand from the inventories of gold and silver (Part IX) and tellurium (Part X). This extension was part of the mandate commissioned by the Swiss Federal Office for the Environment (FOEN) to improve the overall quality and applicability of the metals inventories.

In version 2.1 we corrected an error in the dataset “molybdenum, at regional storage (RER)”. See Fig. 5.21.

Chapter “6 Cumulative Results and Interpretation” of the original report has been deleted for the current update.

## Summary

The primary copper and molybdenum production is modelled for the global situation, based on data from various sources. Additionally, the situations for different concentrate-producing regions are modelled separately. Since most of tellurium and considerable parts of silver originate from copper production, a specific chapter treats the generation of silver and telluride containing anode slime.

The mining and beneficiation are modelled as coupled processes with copper and molybdenite concentrate as outputs.

For copper also the secondary production in Europe is inventoried. Finally a mix is generated to be used for copper consumed in Europe.

Some of the datasets described in this report make only sense in the context of the copper and molybdenum production and are therefore not intended for other uses.

The datasets to be chosen to represent copper in a LCA study are either one of the primary copper datasets, the secondary copper dataset or the European mix (at regional storage). The decision whether to use primary, secondary or a mix of the two depends on the goal and scope of the LCA to be conducted.

If molybdenum is needed as alloying element in an LCA study, molybdenite should be chosen as input. The metallic molybdenum is only used in very specific applications.



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# 1 Introduction

This chapter describes the production of copper from mining of the ore to consumer in Europe and the coupled production of molybdenum. Inventory data for the copper production is taken mainly from Krauss et al. (1999) together with background data from IPPC (2002), IPPC (2001), EEA (1999) and Ayres et al. (2002) (primary copper) and Rentz et al. (1999) and Bruch et al. (1995) (secondary copper).

Since copper is always associated with other metals, the production yields a fraction of other metals than copper, mainly nickel (Ni), molybdenum (Mo) and platinum group metals. The process of primary production described in this chapter is formally a multioutput-process with the coupled products Cu and Mo. No other potential co-products have been considered in this study. This makes it necessary to allocate the inventory to these products. The demand for copper-containing ore is allocated by mass. The rest of the inventory is allocated by revenue.

Copper is used as pure metal and as alloying element with other metals and in steels. On the other hand, molybdenum is almost exclusively used as alloying element in steels. Its marketed product form is molybdenum trioxide and ferromolybdenum.

According to EEA (1999), emissions of sulphur dioxide from non-ferrous metal production, particularly copper production, contribute less than 10% to the total global emissions of this pollutant. However, copper smelting can be the most important source of sulphur dioxide emissions in certain regions, such as the Kola Peninsula, the Urals, the Norilsk area, and the Fergana region in Russia, Lower Silesia in Poland, and the Gijon region in Spain. During the copper production process various trace elements originating from impurities in the ore are emitted to the atmosphere. This process is the major source of atmospheric arsenic and copper (about 50 % of the global emissions of the element), and indium (almost 90 %), and a significant source of atmospheric antimony, cadmium and selenium (ca. 30 %), and nickel and tin (ca. 10 %). The production of copper together with nickel is the major source of pollution from these and other elements in regions with non-ferrous metal production.

According to mythology, the goddess Venus (or Aphrodite) was born on the Mediterranean island of Cyprus, formerly Kypros (Greek), where copper was exploited millennia before Christ. Therefore, in early times the Romans named it cyprium, later called cuprum. This name is the origin of copper and of the corresponding words in most Romance and Germanic languages, e.g., cobre (Spanish and Portuguese), cuivre (French), Kupfer (German), koper (Dutch), and koppar (Swedish).

## 2 Reserves and Resources of Copper and Molybdenum

Metals are a special class of materials. They are not consumed but rather used during a product life cycle. When recovered the properties are theoretically identical to those in the raw material. The metal can be re-used in another product life cycle, provided that its concentration makes a recovery economic. The recovered metal is referred to as secondary metal, whereas the metal originating directly from the ore is referred to as primary metal. In LCIs these two types – primary and secondary metal – are often differentiated because they are assigned with different ecological burden.

Molybdenum and Copper are coexisting in porphyry deposits of the copper-molybdenum type, as molybdenite ( $\text{MoS}_2$ ) and chalcopyrite ( $\text{CuFeS}_2$ ). About half of the world-wide produced molybdenum is a co-product of the primary copper industry while for another substantial part copper is the co-product (Krauss et al. (1999) and Sebenik et al. (1997)). Hence almost all of the molybdenum is produced in a process similar to the copper primary production. Molybdenum secondary production – mainly from spent petroleum catalysts – is not remarkable, and no secondary production is considered in this study. USGS (2003) estimates that 30% of the molybdenum is re-used in the form of the molybdenum content steel alloys which are recycled to the foundries. Secondary production of copper from scrap plays an important role. The resources of primary copper are limited, a continuous depletion within 60 years is estimated (Ayres et al. (2002), p. 10).

### 2.1 Primary copper

Recoverable copper deposits are now mostly (90-95%) in the form of disseminated sulphide minerals, such as chalcopyrite ( $\text{CuFeS}_2$ ), chalcocite ( $\text{Cu}_2\text{S}$ ), bornite ( $\text{Cu}_5\text{FeS}_4$ ) and enargite ( $\text{Cu}_3\text{AsS}_4$ ). On average, a tonne of sulphur is associated with each tonne of copper. By far the most important deposits are of the disseminated porphyry copper type, in which the dominant mineral is chalcopyrite. Globally, 90%-95% of copper is derived from sulphide ores. Arsenic is associated with some porphyry copper deposits, such as Butte, Montana, and El Indio (Chile).

Another type of porphyry deposit is the copper-molybdenum type, in which the mineral molybdenite ( $\text{MoS}_2$ ) is interspersed with chalcopyrite, as in Henderson Colorado, Quartz Hill Canada and a few other locations in Chile and the Philippines. These ore bodies are mainly mined for molybdenum, with copper as a by-product.

So called oxide ores, which account for 5%-10% of global output, are weathered products of sulphide minerals, notably malachite ( $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$ ), azurite ( $2\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$ ), cuprite ( $\text{Cu}_2\text{O}$ ) and chrysocolla ( $\text{CuSiO}_3 \cdot 2\text{H}_2\text{O}$ ) (Ayres et al. (2002)).

Overall more than 200 minerals contain copper in definable amounts, but only ca. 20 are of importance as copper ores (Tab. 2.1) or as semiprecious stones (turquoise and malachite). Secondary minerals are formed in sulphide ore bodies near the earth's surface in two stages. In the oxidation zone, oxygen-containing water forms copper oxides, subsalts (subcarbonates and subsulphates), and silicates. In the deeper cementation zone, copper-bearing solutions from these salts are transformed into secondary copper sulphides (chalcocite and covellite) and even native copper of often high purity, e.g., in the Michigan copper district (Keweenaw Peninsula). The grade of copper ore at a mine or in a mining region decreases as ore is removed, sometimes dropping below that of gangue left from earlier mining. The grade of the copper ore mined and milled in the U.S. dropped from 3.5% in 1900 to nowadays under 0.7% which was the content of the 1900's tailings (Fig. 2.1, Graedel et al. (2002)).

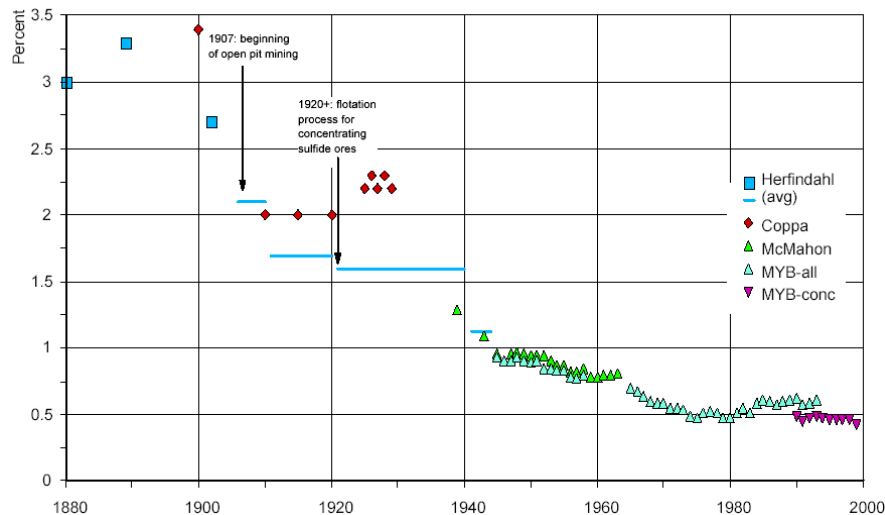


Fig. 2.1 Development of copper ore grade in U. S. since 1880 (Ayres et al. (2002)).

Other metallic elements frequently found in copper ores are iron, lead, zinc, antimony, and arsenic; less common are selenium, tellurium, bismuth, silver, and gold. Substantial enrichments sometimes occur in complex ores. For example, ores from Sudbury, Ontario, in Canada contain nickel and copper in nearly the same concentrations, as well as considerable amounts of platinum metals. The copper ores from Zaire and Zambia are useful sources of cobalt. Many porphyry copper ores in America contain significant amounts of molybdenum and are the most important single source of rhenium. The extraction of precious metals and other rare elements can be decisive for the profitability of copper mines, smelters, and refineries (Fabian (1997)).

Land-based resources are estimated to be 1.6 billion tons of copper, and resources in deep-sea nodules are estimated to be 700 million tons (USGS (2003)). A detailed overview over the global refinery production and a statistic on US use and production of copper is available in the online version of the USGS “Mineral Commodity Summary” (USGS (2003)). The world-wide mine production by country in 2002 and the estimated reserves are listed in Tab. 2.3, copper production over the last seven thousand years is shown in Fig. 2.2.

Exploitable reserves of recoverable copper were estimated at about 100 million tons in 1935; new discoveries raised this to 212 million tons in 1960. Reserves grew again sharply to 340 million metric tons (MMT) in 1984, but since then they have declined slowly to 321 MMT in 1990 and 310 MMT in 1994. Estimates vary according to prices and assumptions. Total potential resources have increased somewhat over the same period, from 500 MMT to around 590 MMT. As a matter of interest, cumulative global production of copper between 1970 and 1996 was 216 MMT (Ayres et al. (2002), p. 10). According to USGS in 2002 (Tab. 2.3) estimated resources amount to 480 MMT (“reserves”) and potential resources to 950 MMT (“reserve base”). With the actual mine production of around 13.5 MMT/a, the reserves would last 36 years and the reserve base 70 years.

Production of tellurium and silver are tightly related to the production of copper. Tellurium occurs in tellurium-bearing minerals such as sylvanite ((Au, Ag)<sub>2</sub>Te<sub>2</sub>) in copper and gold deposits. Silver and tellurium are contained in small traces in the copper ore and arise in the anode slime of the electrolytic

refining of copper. Whereas tellurium is won almost exclusively as by-product from the copper production<sup>1</sup>, copper ores contribute to 25-30% to the world silver supply (Klapwijk et al. (2006)).

Tab. 2.1 Resources. The most important copper minerals (Fabian (1997)).

| Mineral       | Formula                                                                 | Copper wt-% | Crystal system | Density, g/cm <sup>3</sup> |
|---------------|-------------------------------------------------------------------------|-------------|----------------|----------------------------|
| Native copper | Cu                                                                      | 99.92       | Cubic          | 8.9                        |
| Chalcosite    | Cu <sub>2</sub> S                                                       | 79.9        | Orthorhombic   | 5.5-5.8                    |
| Digenite      | Cu <sub>9</sub> S <sub>5</sub>                                          | 78.0        | cubic          | 5.6                        |
| Covellite     | CuS                                                                     | 66.5        | hexagonal      | 4.7                        |
| Chalcopyrite  | CuFeS <sub>2</sub>                                                      | 34.6        | tetragonal     | 4.1-4.3                    |
| Bornite       | Cu <sub>5</sub> FeS <sub>4</sub> /Cu <sub>3</sub> FeS <sub>3</sub>      | 55.5-69.7   | tetragonal     | 4.9-5.3                    |
| Tennantite    | Cu <sub>12</sub> As <sub>4</sub> S <sub>13</sub>                        | 42-52       | cubic          | 4.4-4.8                    |
| Tetraedrite   | Cu <sub>12</sub> Sb <sub>4</sub> S <sub>13</sub>                        | 30-45       | cubic          | 4.6-5.1                    |
| Enargite      | Cu <sub>3</sub> AsS <sub>4</sub>                                        | 48.4        | orthorhombic   | 4.4-4.5                    |
| Bourbonite    | CuPbSbS <sub>3</sub>                                                    | 13.0        | orthorhombic   | 5.7-5.9                    |
| Cuprite       | Cu <sub>2</sub> O                                                       | 88.8        | cubic          | 6.15                       |
| Tenorite      | CuO                                                                     | 79.9        | monoclinic     | 6.4                        |
| Malachite     | CuCO <sub>3</sub> · Cu(OH) <sub>2</sub>                                 | 57.5        | monoclinic     | 4.0                        |
| Azurite       | 2 CuCO <sub>3</sub> · Cu(OH) <sub>2</sub>                               | 55.3        | monoclinic     | 3.8                        |
| Chrysocolla   | CuSiO <sub>3</sub> · n H <sub>2</sub> O                                 | 30-36       | (amorphous)    | 1.9-2.3                    |
| Diopside      | Cu <sub>6</sub> [Si <sub>6</sub> O <sub>18</sub> ] · 6 H <sub>2</sub> O | 40.3        | rhombohedral   | 3.3                        |
| Brochantite   | CuSO <sub>4</sub> · 3 Cu(OH) <sub>2</sub>                               | 56.2        | monoclinic     | 4.0                        |
| Antlerite     | CuSO <sub>4</sub> · 2 Cu(OH) <sub>2</sub>                               | 53.8        | orthorhombic   | 3.9                        |
| Chalcanthite  | CuSO <sub>4</sub> · 5 H <sub>2</sub> O                                  | 25.5        | triclinic      | 2.2-2.3                    |
| Atacamite     | CuCl <sub>2</sub> · 3 Cu(OH) <sub>2</sub>                               | 59.5        | orthorhombic   | 3.75                       |

<sup>1</sup> <http://www.mmta.co.uk/economicsFacts/miningJournalReview.aspx>, "Tellurium" in the Mining Journal Review, The Minor Metals Trade Association, accessed in July 2007

Tab. 2.2 Occurrence. Typical copper contents of natural materials (Fabian (1997)).

| Mineral                 | Content (ppm) |
|-------------------------|---------------|
| Basalt                  | 85            |
| Diorite                 | 30            |
| Granite                 | 10            |
| Sandstone               | 1             |
| Copper ores (poor)      | 5'000         |
| Copper ores (rich)      | 50'000        |
| Native copper           | 950'000       |
| Seawater                | 0.003         |
| Deep-sea clays          | 200           |
| Manganese nodules       | 10'000        |
| Marine ore sludges      | 10'000        |
| Earth's crust (average) | 50            |
| Meteorites (average)    | 180           |

Tab. 2.3 Mine production and reserves of copper in 2001. "Reserves" is referred to as "the part of the reserve base which could be economically extracted or produced at the time of determination", while "Reserve base" denotes "that part of an identified resource that meets specified minimum (...) criteria related to current mining and production practices" (USGS (2003)).

|                              | Mine production  |                                 | Reserves | Reserve base |
|------------------------------|------------------|---------------------------------|----------|--------------|
|                              | 2001 [kt Cu / a] | 2002 [kt Cu / a]<br>(estimated) | [kt Cu]  | [kt Cu]      |
| United States                | 1340             | 1130                            | 35000    | 70000        |
| Australia                    | 869              | 850                             | 24000    | 43000        |
| Canada                       | 633              | 625                             | 10000    | 23000        |
| Chile                        | 4740             | 4450                            | 160000   | 370000       |
| China                        | 590              | 580                             | 26000    | 63000        |
| Indonesia                    | 1050             | 1100                            | 28000    | 38000        |
| Kazakhstan                   | 470              | 450                             | 14000    | 20000        |
| Mexico                       | 367              | 330                             | 27000    | 40000        |
| Peru                         | 722              | 850                             | 35000    | 60000        |
| Poland                       | 474              | 500                             | 31000    | 50000        |
| Russia                       | 620              | 680                             | 20000    | 30000        |
| Zambia                       | 300              | 320                             | 19000    | 35000        |
| Other countries              | 1510             | 1500                            | 60000    | 110000       |
| World total (may be rounded) | 13700            | 13400                           | 480000   | 950000       |

Tab. 2.4 World production of copper, silver and tellurium in tonnes per year, tpa. Data compiled after USGS (2003) through 2004 and USGS (2007) for 2005/6.

|      | Copper                    | Silver                        | Tellurium                     |
|------|---------------------------|-------------------------------|-------------------------------|
| year | tpa world mine production | tpa world refinery production | tpa world refinery production |
| 1986 | 7'940'000                 | 13'000                        | 85.4                          |
| 1987 | 8'240'000                 | 14'000                        | 73.8                          |
| 1988 | 8'720'000                 | 15'500                        | 69.3                          |
| 1989 | 9'040'000                 | 16'400                        | 65.0                          |
| 1990 | 9'200'000                 | 16'600                        | 67.4                          |
| 1991 | 9'330'000                 | 15'600                        | 82.9                          |
| 1992 | 9'470'000                 | 14'900                        | 97.6                          |
| 1993 | 9'490'000                 | 14'100                        | 88.2                          |
| 1994 | 9'500'000                 | 14'000                        | 117.0                         |
| 1995 | 10'000'000                | 14'900                        | 175.0                         |
| 1996 | 11'000'000                | 15'100                        | 122.0                         |
| 1997 | 11'500'000                | 16'500                        | 109.0                         |
| 1998 | 12'100'000                | 17'200                        | 123.0                         |
| 1999 | 12'800'000                | 17'600                        | 116.0                         |
| 2000 | 13'200'000                | 18'100                        | 110.0                         |
| 2001 | 13'700'000                | 18'900                        | 109.0                         |
| 2002 | 13'600'000                | 18'500                        | 89.0                          |
| 2003 | 13'700'000                | 18'400                        | 95.2                          |
| 2004 | 14'600'000                | 19'700                        | 92.7                          |
| 2005 | 15'000'000                | 19'300                        | 119.0                         |
| 2006 | 15'300'000                | 19'500                        | 128.0                         |

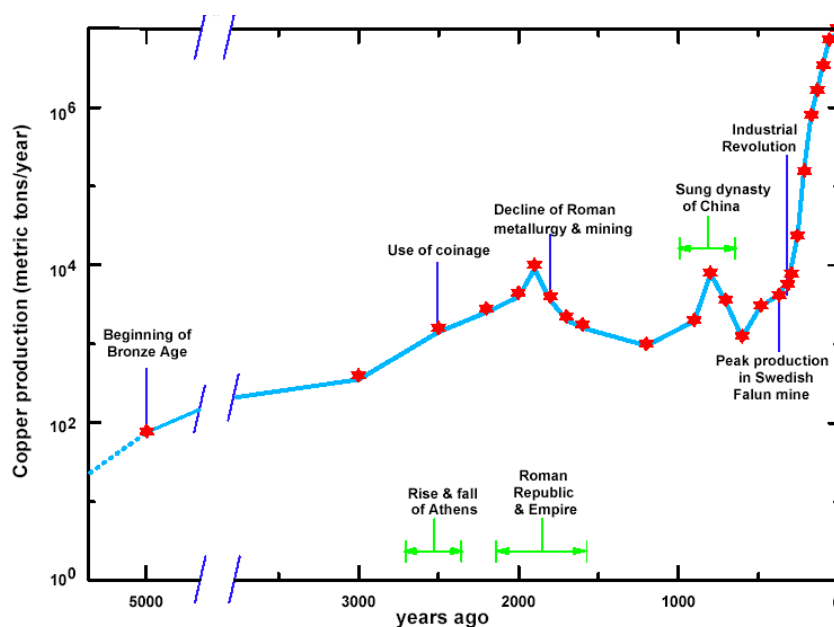


Fig. 2.2 History of copper production in logarithmic scale (Ayres et al. (2002)).

## 2.2 Primary Molybdenum

About the half of the world-wide produced molybdenum originates as co-product from the copper industry. The chief mineral values for the coupled production of molybdenum and copper are chalcopyrite ( $\text{CuFeS}_2$ ) and chalcocite ( $\text{Cu}_2\text{S}$ ), associated with pyrite ( $\text{FeS}_2$ ), and small amounts of other sulphides, including molybdenite. Generally there are five genetic types of molybdenum deposits (Sebenik et al. (1997)):

1. porphyry deposits in which metallic sulphides are disseminated throughout large volumes of altered and fractured rock,
2. contact-metamorphic zones and bodies in which silicated limestone is adjacent to intrusive granites,
3. quartz veins,
4. pegmatites, and
5. deposits bedded in sedimentary rocks.

Types (1) – (3) are of hydrothermal origin and provide almost all of the molybdenum ore currently mined. The last two are of no economic importance at present. Average molybdenite concentrations in primary porphyry deposits range from 0.05 to 0.25 %; in secondary copper-molybdenum porphyry deposits molybdenite concentrations are much lower (0.01 – 0.05 %) so that the mineral can only be recovered as a by-product.

Molybdenum is obtained commercially almost exclusively from molybdenite, which is either mined and concentrated as the primary product of the mine or open pit, or recovered as a concentrate during the processing of ore from a copper mine. In the latter case the molybdenite can be either a by- or a co-product, depending on its economic importance to the output of the mine. Molybdenum is also recovered from solutions obtained by leaching scheelite ores in the production of tungsten. It is extracted as a by- or co-product of copper production in Canada, Chile, Iran, Mexico, Papua New Guinea, Peru, Philippines, and in the United States (Arizona, Nevada, New Mexico, and Utah).

Molybdenum supply in the Western world can be broken down into four production segments (Sebenik et al. (1997)):

- 1) primary mine production (ca. 40 %),
- 2) by- or co-production output from copper and scheelite mines (ca. 55 %),
- 3) imports from the People's Republic of China (3 %), and
- 4) molybdenum recovered from the processing of spent petroleum catalyst (2 %).

The world mine production of molybdenum amounted to 130 kt in 2001 (USGS (2003)).

Reserves of molybdenite in the market economy countries have been estimated in a survey evaluating identified ore bodies, i.e., those which have been explored as well as those which have been exploited. The results indicate the following amounts of recoverable molybdenum: United States, 4100 kt; Chile, 1770 kt; Canada, 928 kt; Mexico, 306 kt; Peru, 288 kt; other countries, 356 kt. Ore bodies producing primarily molybdenum contain 55 % of the reserves identified; only 29 % of these ore bodies were being exploited at the time of the survey (January 1985). Operations producing molybdenum as a by-product contained the remaining 45 % of reserves and 67 % of them were producing molybdenite (Sebenik et al. (1997)). The estimated total recoverable molybdenum from primary and byproduct reserves is listed in Tab. 2.5.



Tab. 2.5 Global reserves of molybdenum. Total recoverable reserves from primary production and as by-product (Sebenik et al. (1997)).

| Source of reserve                 | Amount in [kt] |
|-----------------------------------|----------------|
| Primary producing properties      | 1'265          |
| Primary nonproducing properties   | 3'036          |
| Byproduct producing properties    | 2'316          |
| Byproduct nonproducing properties | 1'142          |

## 2.3 Secondary Copper

The recycling of copper is as old as the usage of the metal itself. World-wide refinery production in year 2000 was  $14.9 \cdot 10^6$  tonnes copper, of which  $1.92 \cdot 10^6$  tonnes copper (=13.2%) originated from secondary copper sources (USGS (2003)). In Germany in 1994,  $1.48 \cdot 10^6$  tonnes copper was consumed, of which  $0.65 \cdot 10^6$  tonnes were from secondary materials, of which again roughly 50% was re-cycled in the metallurgical step (Krauss et al. (1999)).

Scrap is classified as prompt scrap (new scrap, manufacturing scrap), which is generated during the fabrication of copper products, and old scrap, which is generated from products exiting the use phase. These differ greatly in their quality. Because of its high purity, most of the prompt scrap can be directly re-melted, while a good deal of old scrap has to return to the refinery and the smelter stage. About 36% of the copper in old scrap comes from pure copper products, and about 64% from alloy products (Ayres et al. (2002)). Bertram et al. (2002) suggest to assume that the old scrap consists of the waste categories ELV, C&D, and WEEE (Tab. 2.6).

Depending on the quality of the recycled material, the blending of primary and secondary substance flows takes place at different production stages. In general, secondary copper used in smelters and refineries is composed of 70% old and 30% prompt scrap. For the year 1994 Bertram et al. (2002) calculated an input of 540 kt of old scrap and 230 kt (=29.9%) of prompt scrap into the smelters / refineries and an input of 200 kt of old scrap and 750 kt (=78.9%) of prompt scrap into the production of semis (semi-fabricated products, such as shapes and ingots).

Tab. 2.6 Sources of copper waste for Europe in 1994 (Bertram et al. (2002)).

| Source                                 | Total waste |               |      | Copper content |                        |           |                   |
|----------------------------------------|-------------|---------------|------|----------------|------------------------|-----------|-------------------|
|                                        | Mt / a      | Uncertainty % | %    | Mt Cu / a      | % Cu in waste fraction | Range %   | % of copper waste |
| IW Industrial waste                    | 350         | ± 30          | 45.7 | 0.07           | 0.02                   | 0.01-0.05 | 7.6               |
| C&D Construction & demolition waste    | 184         | ± 100         | 24.0 | 0.129          | 0.07                   | 0.05-1    | 14                |
| MW Municipal solid waste               | 165         | ± 20          | 21.5 | 0.083          | 0.05                   | 0.04-0.06 | 9.0               |
| HW Hazardous waste                     | 30          | ± 10          | 3.9  | 0.015          | 0.05                   | 0.02-0.08 | 1.6               |
| ELV End of life vehicles               | 27          | ± 20          | 3.5  | 0.243          | 0.9                    | 0.8-1.0   | 26.4              |
| WEEE Electrical & electronic equipment | 2.9         | ± 40          | 0.4  | 0.377          | 13                     | 6-14      | 41                |
| SS Sewage sludge                       | 7.7         | ± 20          | 1.0  | 0.0028         | 0.037                  | 0.02-0.04 | 0.3               |
| Total                                  | 767         |               | 100  | 0.92           | 0.12                   |           | 100               |

## 3 Characterisation of the materials

### 3.1 Copper

Copper [7440-50-8], the red metal, apart from gold the only metallic element with a colour different from a grey tone, has been known since the early days of the human kind. It has always been one of the significant materials, and today it is the most frequently used heavy non-ferrous metal. The utility of pure copper is based on its physical and chemical properties, above all its electrical and thermal conductivity (exceeded only by silver), its outstanding ductility and thus excellent workability, and its corrosion resistance (a chemical behaviour making it a half-noble metal). Copper compounds and ores are distinguished by bright coloration, especially reds, greens, and blues. Copper in soil is an essential trace element for most creatures, including humans. Its common alloys, particularly brass and bronze, are of great practical importance. Commercial grades of cathode copper today range from 99.95% to 99.97% pure.

Most properties of metallic copper depend on the degree of purity and on the source of the metal. Variations in properties are caused by

- Grade of copper, i.e., the oxygen content: tough-pitch copper, deoxidised copper, oxygen-free copper;
- Content of native impurities (e.g., arsenic) or remnants of additives (e.g., phosphorus), which form solid solutions or separate phases at the grain boundaries;
- Thermal and mechanical pre-treatment of the metal, which lead to states such as cast copper, hot-rolled copper, cold-worked (hard) copper, annealed (soft) copper, and sintered copper.

The atomic number of copper is 29, and the atomic mass  $A_r$  is  $63.546 \pm 0.003$ . Copper consists of the two natural, stable isotopes  $^{63}\text{Cu}$  (68.94 %) and  $^{65}\text{Cu}$  (31.06 %). There are also nine synthetic radioactive isotopes with atomic masses between 58 and 68, of which  $^{67}\text{Cu}$  has the longest half-life, ca. 58.5 h.

The theoretical density at 20 °C, computed from lattice constant and atomic mass is  $8.93 \text{ g/cm}^3$ . The international standard was fixed at  $8.89 \text{ g/cm}^3$ . The density of commercial copper depends on its composition, especially the oxygen content, its mechanical and thermal pre-treatment, and the temperature. At 20 °C, a range of values from  $8.30 \text{ g/cm}^3$  to  $8.93 \text{ g/cm}^3$  are found.

The corresponding electrical resistivity ( $\rho$ ) is  $1.7241 \times 10^{-8} \text{ W}\cdot\text{cm}$ . The theoretical conductivity at 20 °C is nearly 60.0 MS/m or 103.4 % IACS, and today commercial oxygen-free copper (e.g., C10200 or Cu – OF) has a conductivity of 101 % IACS.

### 3.2 Copper alloys: brass bronze and brazing solders

This chapter is mainly based on Isbell (1997) and on a internet source<sup>2</sup>

There are scores of different copper alloys ranging from high copper alloys with copper content of >99% (w/w) to brasses with only about 50% (w/w) copper. A differentiation of wrought and cast copper alloys is common.

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<sup>2</sup> <http://www.stahl-info.de/deutsch/schriften/MB235-WeichundHartloeten.pdf>, accessed on 16.04.03

### 3.2.1 Brass

The series of alloys from which a majority of the industry's products are made are the copper-zinc alloys – the brasses. The most popular alloy of the group is cartridge brass (70/30 brass). Zinc is by far the most important alloying element in the copper alloys. Lead can be alloyed to brasses to enhance the machinability. If secondary material is used to produce copper alloys, contamination with unwanted heavy metals (e.g. lead, cadmium, beryllium...) can be a problem.

### 3.2.2 Bronze

Also of technical relevance are the copper-tin alloys – the tin bronzes. Beside about 5% (w/w) (maximum 10%) Sn, tin bronzes often contain some phosphorus and sometimes lead or silver. These alloys provide the best combination of strength, electrical conductivity, and resistance to corrosion, as well as great versatility.

The probably most important family of the speciality alloys is that of the copper-aluminium alloys - the aluminium bronzes. Aluminium bronzes contain between 2 and 15% (w/w) Al, possibly with other alloying elements. When cold finished products are to be manufactured, the aluminium content is limited to 9% (w/w). Aluminium bronze alloys may also contain up to 4.5% Fe, 2 % Sn, 6%, Ni, 3.5% Mn and 3% Si to achieve special properties. These alloys combine good formability, good conductivity and high strength.

### 3.2.3 Solders

Alloys or pure metals are used as solders or brazing alloys. When their liquidus temperature is  $<450\text{ }^{\circ}\text{C}$  the process is called soft soldering, and when it is  $>450\text{ }^{\circ}\text{C}$  it is known as brazing. In high-temperature brazing, the operation is carried out in the absence of a fluxing agent and air, and the bonding metal has a liquidus temperature  $>900\text{ }^{\circ}\text{C}$ .

The main elements used in solders are copper, tin, zinc, silver, lead, cadmium and sometimes nickel, manganese or cobalt. An important class of brazing solders are the "General-purpose, cadmium-free brazing alloys". Their melting point is between  $700\text{ }^{\circ}\text{C}$  and  $1100\text{ }^{\circ}\text{C}$ .

Fluxes are organic or inorganic substances that remove oxides and other harmful layers from the surface to be joined by soldering or brazing. They thereby provide the conditions that allow wetting by the molten filler metal. Copper, copper - tin, or silver can be brazed in air without fluxes if phosphorus-containing copper-based brazing alloys are used. The phosphorus has a similar effect to that of a flux.

## 3.3 Molybdenite concentrate and Molybdenum trioxide

Molybdenite concentrate is the purified product of the flotation of molybdenite ore and consists of 90 – 95 % Molybdenum disulphide. Molybdenum disulphide is the raw material for molybdenum trioxide, which is the starting point of most applications of molybdenum. Some properties of Mo-compounds are listed in Tab. 3.1.

Molybdenum disulphide (CAS 1317-33-5;  $\text{MoS}_2$ ) is a black crystalline solid that occurs in two forms: hexagonal, which is found in the mineral molybdenite (CAS 1309-56-4) and rhombohedral. The solid is diamagnetic and a semiconductor. Lubricant-grade molybdenum disulphide consists of 99 %  $\text{MoS}_2$ .

Molybdenum trioxide (CAS 1313-27-5;  $\text{MoO}_3$ ) is the ultimate product of the oxidation of molybdenum compounds. It forms white orthorhombic crystals that are photosensitive, i.e., turn blue in light. The compound is weakly paramagnetic and an n-type semiconductor. It is produced by roasting mo-

lybdenum disulphide in air at 600 °C and may be purified by sublimation. It can also be prepared by reacting ammonium molybdate at 550 °C with oxygen or by precipitation from an aqueous ammonium molybdate solution treated with concentrated nitric acid (Sebenik et al. (1997)).

Tab. 3.1 Properties of molybdenum trioxide and dioxide and molybdenum disulphide (Sebenik et al. (1997))

| Property                                                                                   | MoO <sub>3</sub>      | MoO <sub>2</sub>                | MoS <sub>2</sub> |
|--------------------------------------------------------------------------------------------|-----------------------|---------------------------------|------------------|
| Crystal structure                                                                          | orthorhombic          | monoclinic                      |                  |
| Mo coordination                                                                            | octahedral, distorted | octahedral, distorted<br>rutile |                  |
| $\Delta H_f^\circ$ , kJ/mol*                                                               | -745.2                | -588.7                          | -234.1           |
| $\Delta G_f^\circ$ , kJ/mol*                                                               | -667.9                | -533.1                          | -225.7           |
| $S^\circ$ J/K*                                                                             | 77.7                  | 47.2                            | 62.8             |
| Density, g/cm <sup>3</sup> (21°C)                                                          | 4.692                 | 6.29                            |                  |
| mp, °C                                                                                     | 801                   |                                 | >1600            |
| bp, °C                                                                                     | 1155                  |                                 |                  |
| Sublimation, °C                                                                            | ca. 700               | ca. 1100                        |                  |
| Solubility, g/L                                                                            |                       |                                 |                  |
| Water 28°C                                                                                 | 0.49                  | insoluble                       |                  |
| * Standard enthalpy and free energy of formation from the elements and entropy at 298.15 K |                       |                                 |                  |

## 4 Use / Application of Material

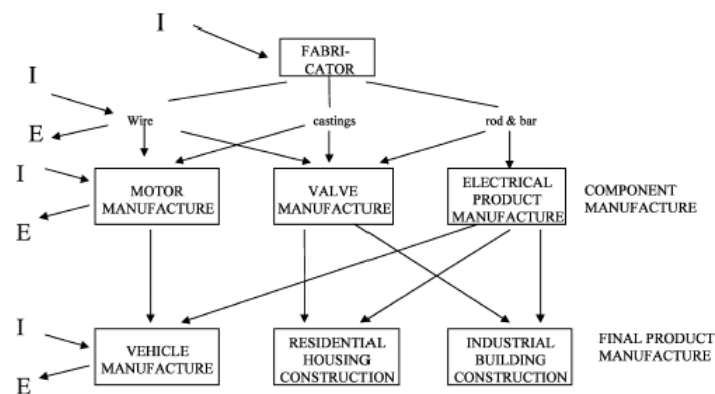
While copper keeps its chemical pure form until its end use as alloying element or final product in different applications, molybdenum is mainly won as molybdenite ( $\text{MoS}_2$ ) which is the purified concentrate that results from beneficiation. Molybdenite is the starting point for the production of technical-grade molybdenum trioxide and ferromolybdenum, which are used as alloying materials in the steel production.

### 4.1 Copper

Before end-use the pure metal produced in refineries or re-melting plants is manufactured into semi-fabricated products, such as shapes and ingots (known as “semis”). These manufacturing processes are subdivided into hot working, cold working and, if necessary, process annealing. These processes are dealt with in Part IXX - Metal Processing.

Few of the semis are ready for use as such. They more typically travel to first-stage manufacturers, who make metal rods, wire, and casting, and then to component manufacturers who produce such things as motors, valves, and insulated cables (Fig. 4.1). The components in turn are incorporated by final-stage manufacturers into a variety of finished products – automobiles, commercial buildings, residential housing, and so forth. These uses are shown in Tab. 4.1, they account for about 79% of the world-wide consumption of copper. The figures in Tab. 4.1 are useful guides to the relative use magnitudes, albeit these copper containing products are the basis for copper re-use. The relations between end-use, production and in- and export for Germany are given in Fig. 4.2.

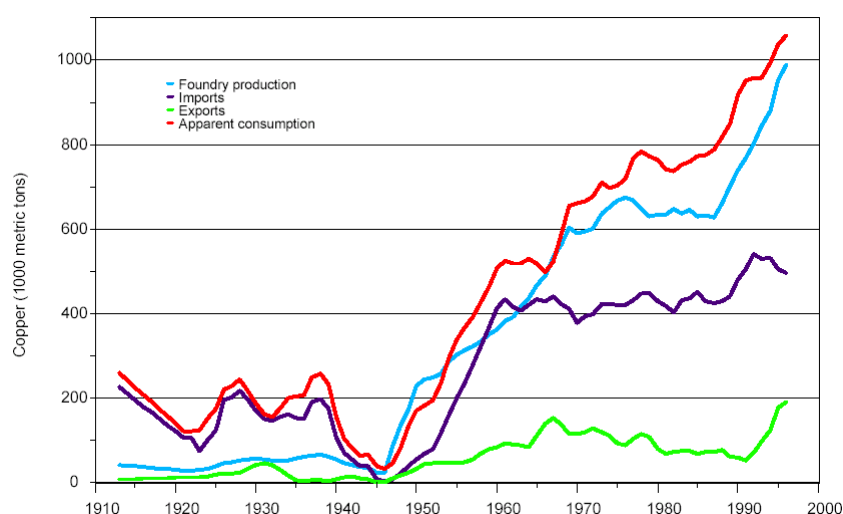
Generally it can be said that copper is losing some traditional markets to other materials. One of them is for telecommunications cables, mainly due to increasing use of glass fibres and wireless devices for local distribution – the so-called “last mile”. Other traditional copper markets, such as copper roofing, copper water pipe, and brass hardware of all kinds are under pressure from cheaper materials, including aluminium, galvanised iron and plastics. On the other hand, copper is virtually irreplaceable for local distribution of electric power, household wiring, motor-generator windings, electronic circuitry and (as alloy) for some kinds of heat exchangers. Further substitution of copper by aluminium, in particular, seems unlikely.



**Fig. 4.1** Copper “semis” production. An illustration of the transformation of copper from pure metal fabrication flows through component manufacture (intermediate products) to a selection of final products. I and E indicate import and export flows, respectively (Graedel et al. (2002))

**Tab. 4.1** Principal end-uses of copper in 1990. RT is the average residence time of copper in the respective use; S/W is the estimated distribution between scrap (recovering) and waste (disposal/incineration) at the end of use (Graedel et al. (2002)).

| #  | Category               | Use (%) | RT (years) | S/W   |
|----|------------------------|---------|------------|-------|
| 1  | Building wire          | 14      | 45         | 50/50 |
| 2  | Tube                   | 12      | 60         | 45/55 |
| 3  | Alloy rod              | 11      | 20         | 10/90 |
| 4  | Magnet wire            | 9       | 15         | 50/50 |
| 5  | Telecommunication wire | 8       | 50         | 25/75 |
| 6  | Power cable            | 8       | 40         | 60/40 |
| 7  | Copper sheet and strip | 8       | 50         | 60/40 |
| 8  | Alloy sheet and strip  | 7       | 25         | 20/80 |
| 9  | Casting alloys         | 6       | 30         | 50/50 |
| 10 | Motor vehicle wire     | 4       | 10         | 80/20 |
| 11 | Appliance wire         | 4       | 20         | 50/50 |
| 12 | Bare wire              | 3       | 10         | 0/100 |
| 13 | Copper rod             | 2       | 40         | 60/40 |
| 14 | Alloy tube             | 2       | 35         | 95/5  |
| 15 | Wire (other)           | 1       | 5          | 0/100 |
| 16 | Alloy wire             | 1       | 5          | 5/95  |
| 17 | Chemical and powder    | <1      | 1          | 5/95  |



**Fig. 4.2** Copper foundry production, imports, exports & apparent consumption in Germany, 1913 - 1998 (3 year moving averages) (Ayres et al. (2002)).

## 4.2 Molybdenite concentrate

Flotation remains the preferred and almost exclusive method of upgrading the molybdenite mineral to an industrial-grade molybdenite concentrate containing 90 – 95 %  $\text{MoS}_2$  [1317-33-5], the balance being gangue.

Some commercial-grade molybdenite concentrate is subjected to additional grinding and flotation to produce lubricant-grade molybdenum disulfide (99 % MoS<sub>2</sub>). The purified product is added to grease and oils as a solid lubricant. Molybdenite is converted to technical-grade MoO<sub>3</sub>.

### 4.3 Molybdenum trioxide and Ferromolybdenum

Molybdenum is mainly used as an alloying element in steel, cast iron, and superalloys to increase hardenability, strength, toughness, and corrosion resistance. Initially, molybdenum was used in the lamp industry. Nowadays it is an increasingly important material in a wide range of applications (Tab. 4.2). In high-temperature applications, molybdenum competes with iron-, nickel- and cobalt-base superalloys, ceramics, and other high-melting metals (tungsten, tantalum, and niobium).

Molybdenite concentrate is converted to technical-grade MoO<sub>3</sub> by roasting it in air in a multiple-hearth furnace. The principal commercial molybdenum product is technical-grade molybdenum trioxide (MoO<sub>3</sub>) [1313-27-5]. In addition to being the major compound for adding molybdenum to steel, it is also the starting compound for all other molybdenum products including ammonium dimolybdate, pure molybdenum trioxide, sodium molybdate, molybdenum metal, and finally ferromolybdenum.

In the steel and foundry industry molybdenum is added to melts, either as technical oxide or in the form of ferromolybdenum [11121-95-2]. Compared to pure molybdenum, ferromolybdenum dissolves much more easily in the steel melt and is cheaper to produce.

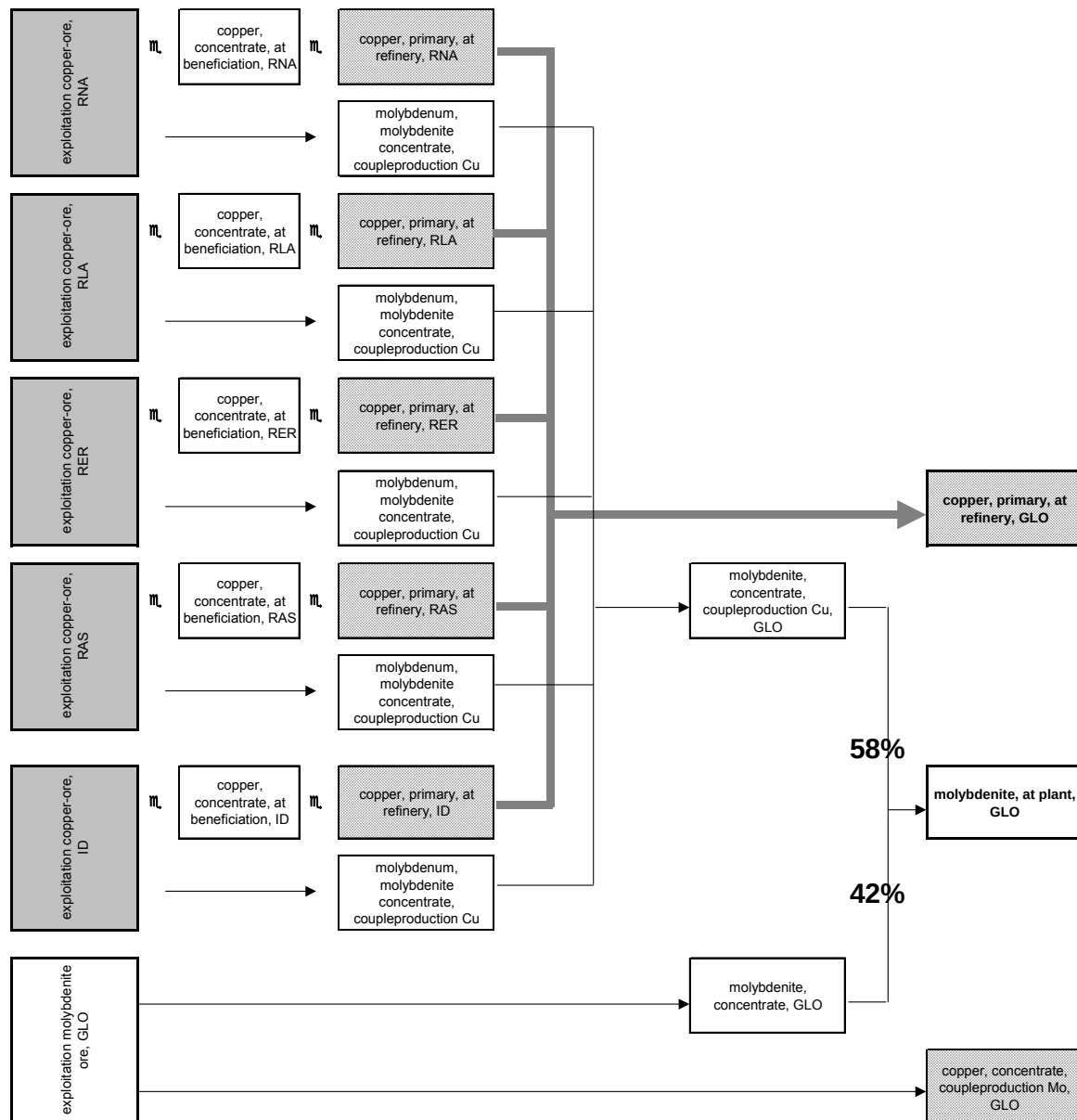
The principal raw materials used for ferromolybdenum production are molybdenum concentrates and technical-grade molybdenum trioxide. Technical-grade molybdenum trioxide is commonly used as the molybdenum raw material. Two principal processes are used for molybdenum oxide reduction: carbothermic reduction in a submerged arc furnace and metallothermic reduction. The most commonly applied method is the silicothermic reduction, a type of metallothermic reduction (Sebenik et al. (1997)).

Tab. 4.2 Uses of molybdenum and molybdenum alloys (Sebenik et al. (1997)).

| Area of application                              | Examples of uses                                                                                                                                   |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Lamp and lighting industries                     | support wires, sealing ribbons, dimming caps                                                                                                       |
| Electronic and semiconductor industries          | semiconductor base plates, heat sinks, contact pins, sputtering targets, control grids, klystron and travelling-wave tube components               |
| High-temperature and vacuum furnace construction | heating elements, thermal radiation shields, furnace ware                                                                                          |
| Glass and ceramic industries                     | glass-melting electrodes, installations in glass production tanks, drawing dies, crucibles for producing sapphire single crystals                  |
| Casting technology and metalworking              | forging dies for isothermal forging, extrusion dies, die-casting moulds, hot-galvanizing equipment                                                 |
| Coating                                          | spray wire, spray powder, evaporation boats, sputtering targets, components of chemical vapour deposition equipment                                |
| Nuclear technology                               | furnace parts and charging equipment for sintering UO <sub>2</sub> ; first wall, diverter, and limiter components for experimental fusion reactors |
| Medicine                                         | rotating X-ray anodes, collimators                                                                                                                 |

## 5 System Characterisation

In this study the production of copper and molybdenum is modelled as coupled production system. In a first step the ore – jointly containing copper and molybdenum – is exploited. This mining and beneficiation step is formally a multi-output process, yielding copper concentrate and molybdenite concentrate. Depending on whether the raw material was an ore from primary molybdenum porphyry or from another sulphide deposit, either copper is the co-product or molybdenite. Since the coupled produced copper from primary molybdenum production is of no further importance, the coupled produced molybdenite from primary copper production accounts to 55% of the molybdenum production worldwide. The global molybdenite mix is thus formed together with the 40% “pure” molybdenite production, accounting for 95% of the overall molybdenum production (Fig. 5.1).

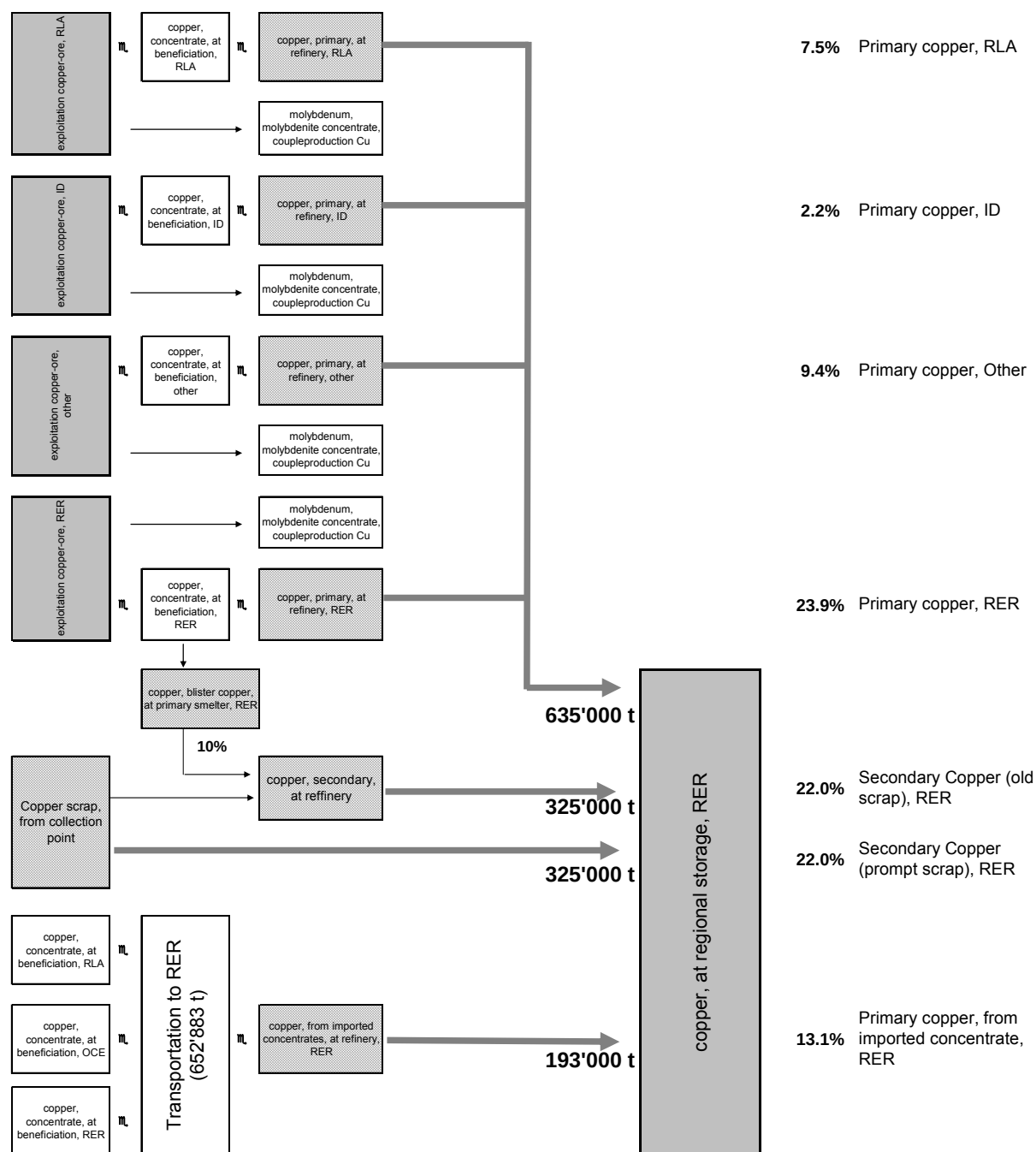


**Fig. 5.1** Schematic representation of the coupled production of copper and molybdenite in ecoinvent. RNA, RLA, RER, RAS, ID refer to the regions North America, South / Middle America, Europe, Russia / Asia, Indonesia / Australia; GLO is the global average.



The inventory representing the consumption of copper in Europe consists of a mix of imported primary metal from different parts of the world, imported copper concentrate which is refined in Europe itself, and secondary metal, which is partly recycled into foundries and partly into prior metallurgical steps (Fig. 5.2). As a proxy for Europe, the situation for Germany in 1994 is chosen.

The meta information on the modules described in this chapter are summarised in Tab. 5.1.



**Fig. 5.2** Schematic representation of the consumer mix of copper in RER in ecoinvent. As a proxy for Europe, the consumption pattern for Germany in 1994 is chosen.

**Tab. 5.1 Ecoinvent meta information for the processes for the production of primary copper and primary molybdenum.**

| Name                                  | mining copper ore                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | copper, primary, at refinery                                                                                                                                                                                                                                    |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location                              | GLO, RNA, RLA, RER, RAS, ID                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | GLO, RNA, RLA, RER, RAS, ID                                                                                                                                                                                                                                     |
| Infrastructure Process                | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0                                                                                                                                                                                                                                                               |
| Unit                                  | kg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | kg                                                                                                                                                                                                                                                              |
| Data Set Version                      | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2.0                                                                                                                                                                                                                                                             |
| Included Processes                    | The module includes a mining and a beneficiation step with the mining infrastructure and disposal of overburden and tailings (see part "Auxiliary Processes").                                                                                                                                                                                                                                                                                                                                                                                                                | The module includes the pre-treatment of the ore, the reduction and the refining copper to cathode copper, including the country-specific share in hydrometallurgically won copper (SX-EW)                                                                      |
| Amount                                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1                                                                                                                                                                                                                                                               |
| Local Name                            | Abbau Kupfererz                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Kupfer, primär, ab Raffinerie                                                                                                                                                                                                                                   |
| Synonyms                              | Copper concentrate // Kupfererz konzentrat // Molybdenite // Molybdänit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                 |
| General comment to reference function | The multi-output-process "mining copper-ore" delivers the co-products "copper concentrate, at beneficiation" and "molybdenum, molybdenite concentrate, couple production Cu". The product "copper concentrate" is intermediary required in the following copper production (5.2.5 Winning of primary copper in ecoinvent). The module "molybdenite concentrate" is designed for the use in Ecoinvent as starting material for the manufacturing of molybdenum trioxide and ferromolybdenum, both materials which used in the manufacturing of high quality steels and alloys. | The product "copper, primary, at refinery" is used as pure metal or as alloying element in various technical applications.                                                                                                                                      |
| CAS Number                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 7440-50-8                                                                                                                                                                                                                                                       |
| Start Date                            | 1994                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1994                                                                                                                                                                                                                                                            |
| End Date                              | 2003                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2003                                                                                                                                                                                                                                                            |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1                                                                                                                                                                                                                                                               |
| Other Period Text                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                 |
| Geography text                        | This module is valid for the whole corresponding region. Some agents used like lime or sodium cyanide originate from data sets valid for Switzerland or Europe. Processes like blasting and transport originate from data sets valid for Switzerland or Europe.                                                                                                                                                                                                                                                                                                               | This module is valid for the whole corresponding region. Some agents used like lime or sodium cyanide originate from data sets valid for Switzerland or Europe.                                                                                                 |
| Technology text                       | Mining is done 70% open pit and 30% underground, followed by joint beneficiation of copper and molybdenite through flotation, where considerable amounts of agents are added. Overburden and tailings are disposed near the mining site.                                                                                                                                                                                                                                                                                                                                      | The ore is pre-treated, reduced and refined according to the country specific mix of process alternatives: reverberatory furnace 23.7%; flash smelting furnaces 60.7%; other 6.2%; SX-EW 9.4%. An overall abatement for sulphur dioxide of 45.4% was estimated. |
| Representativeness [%]                | 84                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 84                                                                                                                                                                                                                                                              |
| Production Volume                     | 9.5 million tonnes in Cu in 1994                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 9.5 million tonnes in Cu in 1994                                                                                                                                                                                                                                |
| Sampling Procedure                    | literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | literature                                                                                                                                                                                                                                                      |
| Extrapolations                        | see Geography and Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | see Geography and Technology                                                                                                                                                                                                                                    |
| Uncertainty Adjustments               | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | none                                                                                                                                                                                                                                                            |

**Tab. 5.1** Ecoinvent meta information for the processes for the production of primary copper and primary molybdenum. (continued)

| Name                                  | copper, blister-copper, at primary smelter                                                                                                                                                                                                                                                                 | copper, SX-EW, at refinery                                                                                                                                                                                                                               |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location                              | RER                                                                                                                                                                                                                                                                                                        | GLO                                                                                                                                                                                                                                                      |
| Infrastructure Process                | 0                                                                                                                                                                                                                                                                                                          | 0                                                                                                                                                                                                                                                        |
| Unit                                  | kg                                                                                                                                                                                                                                                                                                         | kg                                                                                                                                                                                                                                                       |
| Data Set Version                      | 2.0                                                                                                                                                                                                                                                                                                        | 2.0                                                                                                                                                                                                                                                      |
| Included Processes                    | The module includes the same steps as the primary copper production without the refining.                                                                                                                                                                                                                  | The module includes the mining of the ore including the mining infrastructure, the leaching and extraction, the electro winning and the disposal of the leaching residues.                                                                               |
| Amount                                | 1                                                                                                                                                                                                                                                                                                          | 1                                                                                                                                                                                                                                                        |
| Local Name                            | Kupfer, Blisterkupfer, ab Verhüttung                                                                                                                                                                                                                                                                       | Kupfer, SX-EW, ab Raffinerie                                                                                                                                                                                                                             |
| Synonyms                              |                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                          |
| General comment to reference function | The product "blister copper", i.e. raw copper after converting with an average Cu content of 98%, is an intermediary product from primary copper production used uniquely in the secondary copper production. It makes 90% of the feed for secondary copper, whereas copper scrap makes the remaining 10%. | The product "copper, SX-EW, at refinery" is used in the production mix represented in module "copper, primary, at refinery" of the respective region.                                                                                                    |
| CAS Number                            |                                                                                                                                                                                                                                                                                                            | 7440-50-8                                                                                                                                                                                                                                                |
| Start Date                            | 1994                                                                                                                                                                                                                                                                                                       | 1994                                                                                                                                                                                                                                                     |
| End Date                              | 2003                                                                                                                                                                                                                                                                                                       | 2003                                                                                                                                                                                                                                                     |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                                                                                          | 1                                                                                                                                                                                                                                                        |
| Other Period Text                     |                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                          |
| Geography text                        | This module is valid for the whole corresponding region. Some agents used like lime or sodium cyanide originate from data sets valid for Switzerland or Europe.                                                                                                                                            | This module is valid for an average world-wide process.                                                                                                                                                                                                  |
| Technology text                       | The ore is pre-treated and reduced according to the country specific mix of process alternatives: reverberatory furnace 6.2%; flash smelting furnaces 76%; other 17.8%. An overall abatement for sulphur dioxide of 95% was estimated.                                                                     | Heap leaching was assumed as average process. The ore is mined in open cut technique, leached and extracted with sulphuric acid and reduced by electro winning. For electricity mix a mix of 60% hydropower and 40% electricity at grid UCTE was chosen. |
| Representativeness [%]                | 94                                                                                                                                                                                                                                                                                                         | 100                                                                                                                                                                                                                                                      |
| Production Volume                     | 2.3 million tonnes in Europe in 1994 , SX-EW copper excluded                                                                                                                                                                                                                                               | 870 thousand tonnes in Cu in 1994                                                                                                                                                                                                                        |
| Sampling Procedure                    | literature                                                                                                                                                                                                                                                                                                 | literature                                                                                                                                                                                                                                               |
| Extrapolations                        | see Geography and Technology                                                                                                                                                                                                                                                                               | see Geography and Technology                                                                                                                                                                                                                             |
| Uncertainty Adjustments               | none                                                                                                                                                                                                                                                                                                       | none                                                                                                                                                                                                                                                     |

**Tab. 5.1 Ecoinvent meta information for the processes for the production of primary copper and primary molybdenum. (continued)**

| <b>Name</b>                           | <b>copper, secondary, at refinery</b>                                                                                                                                                                                                                                                                                                                                              | <b>copper, from imported concentrates, at refinery</b>                                                                                                                                                                                                                                                     |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location                              | RER                                                                                                                                                                                                                                                                                                                                                                                | DE                                                                                                                                                                                                                                                                                                         |
| Infrastructure Process                | 0                                                                                                                                                                                                                                                                                                                                                                                  | 0                                                                                                                                                                                                                                                                                                          |
| Unit                                  | kg                                                                                                                                                                                                                                                                                                                                                                                 | kg                                                                                                                                                                                                                                                                                                         |
| Data Set Version                      | 2.0                                                                                                                                                                                                                                                                                                                                                                                | 2.0                                                                                                                                                                                                                                                                                                        |
| Included Processes                    | The module includes the collection and handling of the copper scrap, the refining and the processes for waste water treatment.                                                                                                                                                                                                                                                     | The module includes the pre-treatment of the ore, reduction and refining of copper cathodes in RER. It does NOT include the mining and beneficiation step.                                                                                                                                                 |
| Amount                                | 1                                                                                                                                                                                                                                                                                                                                                                                  | 1                                                                                                                                                                                                                                                                                                          |
| Local Name                            | Kupfer, sekundär, ab Raffinerie                                                                                                                                                                                                                                                                                                                                                    | Kupfer, aus importierten Konzentraten, ab Raffinerie                                                                                                                                                                                                                                                       |
| Synonyms                              |                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                            |
| General comment to reference function | The module describes the production of secondary copper in Germany with an output of 176'000 t copper cathodes in 1994. It is designed for the use of the metal as part of the supply mix "copper, at regional storage". It is not to be used if the impact of the secondary copper is considerable in a LCA study. In such cases, a more detailed assessment has to be conducted. | The product "copper, from imported concentrates, at refinery" is an auxiliary module used uniquely in the consumption mix in order to separate concentrate production and their transportation from different countries and the pyrolytical winning in RER. It does not contain any copper bearing inputs. |
| CAS Number                            |                                                                                                                                                                                                                                                                                                                                                                                    | 7440-50-8                                                                                                                                                                                                                                                                                                  |
| Start Date                            | 1994                                                                                                                                                                                                                                                                                                                                                                               | 1994                                                                                                                                                                                                                                                                                                       |
| End Date                              | 2003                                                                                                                                                                                                                                                                                                                                                                               | 2003                                                                                                                                                                                                                                                                                                       |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                                                                                                                                                                  | 1                                                                                                                                                                                                                                                                                                          |
| Other Period Text                     |                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                            |
| Geography text                        | This module is valid for one single very large plant, which is taken as proxy for production in RER.                                                                                                                                                                                                                                                                               | This module is valid for the whole corresponding region. Some agents used like lime or sodium cyanide originate from data sets valid for Switzerland or Europe.                                                                                                                                            |
| Technology text                       | In three different stages, different types of copper scrap and 10% of the feed of blister copper are refined to copper cathodes. Waste water is led to a communal treatment plant.                                                                                                                                                                                                 | The ore is pre-treated, reduced and refined according to the country specific mix of process alternatives: reverberatory furnace 6.2%; flash smelting furnaces 76%; other 17.8%. An overall abatement for sulphur dioxide of 95% was estimated.                                                            |
| Representativeness [%]                | 15.3                                                                                                                                                                                                                                                                                                                                                                               | 100                                                                                                                                                                                                                                                                                                        |
| Production Volume                     | Env. 50% of the 2.3 million tonnes = 1.2 million tonnes in STAF Europe in 1994.                                                                                                                                                                                                                                                                                                    | 193000 tonnes in 1994 from this source                                                                                                                                                                                                                                                                     |
| Sampling Procedure                    | literature                                                                                                                                                                                                                                                                                                                                                                         | literature                                                                                                                                                                                                                                                                                                 |
| Extrapolations                        | see Geography and Technology                                                                                                                                                                                                                                                                                                                                                       | see Geography and Technology                                                                                                                                                                                                                                                                               |
| Uncertainty Adjustments               | none                                                                                                                                                                                                                                                                                                                                                                               | none                                                                                                                                                                                                                                                                                                       |

**Tab. 5.1** Ecoinvent meta information for the processes for the production of primary copper and primary molybdenum. (continued)

| Name                                  | mining molybdenite                                                                                                                                                                                                                                                                                                                                                                                      | molybdenite, at plant                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location                              | GLO                                                                                                                                                                                                                                                                                                                                                                                                     | GLO                                                                                                                                                                                                                                                                                                                                                            |
| Infrastructure Process                | 0                                                                                                                                                                                                                                                                                                                                                                                                       | 0                                                                                                                                                                                                                                                                                                                                                              |
| Unit                                  | kg                                                                                                                                                                                                                                                                                                                                                                                                      | kg                                                                                                                                                                                                                                                                                                                                                             |
| Data Set Version                      | 2.0                                                                                                                                                                                                                                                                                                                                                                                                     | 2.0                                                                                                                                                                                                                                                                                                                                                            |
| Included Processes                    | The module includes a mining and a beneficiation step with the mining infrastructure and disposal of overburden and tailings (see part "Auxiliary Processes").                                                                                                                                                                                                                                          | All relevant processes in mining, beneficiation, metallurgy and disposal of the tailings are included.                                                                                                                                                                                                                                                         |
| Amount                                | 1                                                                                                                                                                                                                                                                                                                                                                                                       | 1                                                                                                                                                                                                                                                                                                                                                              |
| Local Name                            | Abbau Molybdänit                                                                                                                                                                                                                                                                                                                                                                                        | Molybdänit, ab Fabrik                                                                                                                                                                                                                                                                                                                                          |
| Synonyms                              | Molybdenite // Molybdänit                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                |
| General comment to reference function | The module describes the mining of molybdenite as main product. It is designed solely as part of the molybdenum production mix. The multi-output-process "mining of molybdenite" delivers the co-products "molybdenite concentrate, main product" and "copper, concentrate, couple production Mo". Molybdenite from coupled production module is designed for its use in the world-wide production-mix. | The module describes the production mix composed of molybdenite from coupled production in copper industry and molybdenite as main product. It is designed for the use in ecoinvent as starting material for the manufacturing of molybdenum trioxide and ferromolybdenum, both materials finally used in the manufacturing of high quality steels and alloys. |
| CAS Number                            |                                                                                                                                                                                                                                                                                                                                                                                                         | 1309-56-4                                                                                                                                                                                                                                                                                                                                                      |
| Start Date                            | 1994                                                                                                                                                                                                                                                                                                                                                                                                    | 1994                                                                                                                                                                                                                                                                                                                                                           |
| End Date                              | 2003                                                                                                                                                                                                                                                                                                                                                                                                    | 2003                                                                                                                                                                                                                                                                                                                                                           |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                                                                                                                                                                                       | 1                                                                                                                                                                                                                                                                                                                                                              |
| Other Period Text                     |                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                |
| Geography text                        | Mining is done 70% open pit and 30% underground, followed by joint beneficiation of copper and molybdenite through flotation, where considerable amounts of agents are added. Overburden and tailings are disposed near the mining site.                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                |
| Technology text                       | This module is valid for the whole corresponding region. Some agents used like lime or sodium cyanide originate from data sets valid for Switzerland or Europe. Processes like blasting and transport originate from data sets valid for Switzerland or Europe.                                                                                                                                         | This production mix is fully based on the related mining of copper ore.                                                                                                                                                                                                                                                                                        |
| Representativeness [%]                | 40                                                                                                                                                                                                                                                                                                                                                                                                      | 95                                                                                                                                                                                                                                                                                                                                                             |
| Production Volume                     | 52000 t Mo in Molybdenite                                                                                                                                                                                                                                                                                                                                                                               | 130000 t Mo in Molybdenite                                                                                                                                                                                                                                                                                                                                     |
| Sampling Procedure                    | literature                                                                                                                                                                                                                                                                                                                                                                                              | literature                                                                                                                                                                                                                                                                                                                                                     |
| Extrapolations                        | see Geography and Technology                                                                                                                                                                                                                                                                                                                                                                            | see Technology                                                                                                                                                                                                                                                                                                                                                 |
| Uncertainty Adjustments               | none                                                                                                                                                                                                                                                                                                                                                                                                    | none                                                                                                                                                                                                                                                                                                                                                           |

Tab. 5.1 Ecoinvent meta information for the processes for the production of primary copper and primary molybdenum. (continued)

| Name                                  | copper, at regional storage                                                                                                                                                                                                                 | molybdenum, at regional storage                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location                              | RER                                                                                                                                                                                                                                         | RER                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Infrastructure Process                | 0                                                                                                                                                                                                                                           | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Unit                                  | Kg                                                                                                                                                                                                                                          | kg                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Data Set Version                      | 2.0                                                                                                                                                                                                                                         | 2.1                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Included Processes                    | Transport of primary metal to Europe from the countries importing to Europe is included, as the import of concentrate, which is processed in RER. As import pattern the situation of Germany in 1994 was chosen.                            | The module includes the production of high grade primary molybdenum by pyro- and hydrometallurgical processes, the disposal of the slag, and the treatment of the waste water.                                                                                                                                                                                                                                                                                   |
| Amount                                | 1                                                                                                                                                                                                                                           | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Local Name                            | Kupfer, ab Regionallager                                                                                                                                                                                                                    | Molybdän, ab Regionallager                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Synonyms                              |                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| General comment to reference function | The module characterises the copper used in Germany in 1994, reflecting its origin and the fraction of secondary metal. It is designed for the use of the metal in various technical applications such as alloys and construction material. | Molybdenum is used e.g. in catalytic converters. It is not used for alloying steel, for which molybdenum trioxide or molybdenite are used. For lack of specific data, the metallurgical process of the zinc production is used as a proxy. Only the input of the concentrate is adapted. Although zinc and molybdenum concentrate are both sulfidic and the metal content is almost the same, this is a crude assumption and the data quality therefore is poor. |
| CAS Number                            |                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Start Date                            | 1994                                                                                                                                                                                                                                        | 2000                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| End Date                              | 2003                                                                                                                                                                                                                                        | 2003                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                           | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Other Period Text                     |                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Geography text                        | In this module the consumption pattern of Germany in 1994 is used as proxy for the situation in RER.                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Technology text                       | This consumption mix represents the technology used of the countries importing to Germany in 1994 and differentiates between pyrolytical processes, hydrolytical processes and secondary copper.                                            | A mix of 80% hydrometallurgical and 20% pyrometallurgical production of zinc is chosen to represent the molybdenum production.                                                                                                                                                                                                                                                                                                                                   |
| Representativeness [%]                | 100                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Production Volume                     | 1.5 million tonnes in Cu in 1994                                                                                                                                                                                                            | unknown                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Sampling Procedure                    | literature                                                                                                                                                                                                                                  | literature                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Extrapolations                        | see Geography and Technology                                                                                                                                                                                                                | see Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Uncertainty Adjustments               | none                                                                                                                                                                                                                                        | none                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## 5.1 Exploitation of copper ore

Copper ore is being mined for thousands of years. In the past decades the grades of worked copper ores continuously fell and dropped recently clearly below 1% world-wide (Fig. 2.1 for U.S. deposits). Nevertheless, mining and beneficiation techniques remain basically the same with open cut mining and flotation beneficiation. The hydrometallurgical process skips the beneficiation step in favour of a leaching step, which is followed by electro-winning.

### 5.1.1 Production process

**Mining.** Sulphidic copper ores are mined only 30% underground, the major part is mined in large open cut operations. According to (Ayres et al. (2002)) the ore mined in 1900 in the U.S. had a high content of 3.4% and was mined entirely underground. Open pit mining permits the use of very large equipment. Resulting economies of scale enable the exploitation of lower grade disseminated (porphyry) ores – the ores now mainly mined.

**Beneficiation.** After mining, the ore is first ground. In a next step it is subjected to gravity concentration to separate the metal-bearing particles from the unwanted minerals. After this first concentration step, flotation is carried out to remove the gangue from the sulphidic minerals. For neutralisation lime is added. In the flotation several organic chemicals are used as collector, frother, activator, depressor and flocculant. Sometimes cyanide is used as depressant for pyrite. Tailings usually are led to tailings heaps or ponds. As a result, copper concentrates containing around 30% Cu are produced. Molybdenite concentrate are further ground and purified. It leaves the process as co-product with a concentration of 90 – 95 % Molybdenum disulphide.

### 5.1.2 Emissions & Waste

#### Mining

The major emissions are due to mineral born pollutants in the effluents. Open cut mining generates large quantities of dust, which contains elevated contents of metals and sulphur. Rain percolates through overburden and accounts to metal emissions to groundwater. Overburden is deposited close to the mine.

#### Beneficiation

Ore handling and processing produce large amounts of dust, containing PM<sub>10</sub> and several metals from the ore itself. Flotation produces effluents containing several organic agents<sup>3</sup> used. Some of these chemicals evaporate and account for VOC emissions to air. Namely xanthates decompose hydrolytically to release carbon disulphide. Tailings effluent contains additional sulphuric acid from acid rock drainage.

Tailings are deposited as piles and in ponds. In the sulphidic tailings occurs acid rock drainage (ARD) over a long period of time. These effects are handled in the section “Non Ferrous Metal Winning – Auxiliary Processes”.

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<sup>3</sup> Floating agents consist of collectors (xanthate or aerofloat) and frothing reagent (eg. Methyl Isobutyl Carbinol). Xanthanes are a broad class of organic salts formed by treatment of an alcohol with carbon disulfide in the presence of an alkali, potassium ethyl xanthate (KC<sub>3</sub>H<sub>5</sub>S<sub>2</sub>O, CAS No: 140-89-6). Aerofloat is a salt of carbonyldithiophosphoric acids. A widely used frothing reagent is e.g. methyl isobutyl carbinol, CAS: 108-11-2, syn. Methyl Amyl Alcohol.

### 5.1.3 Exploitation of copper-ore in ecoinvent

The modules describe the coupled production of copper- and molybdenum-concentrate in 1994 for different regions of the world and the world-wide production mix. Please note that in the database distinct datasets for the regions Northern America (RNA), South America (RLA), Europe (RER), Asia and Russia (RAS), Indonesia & Australia (ID) and the world-wide average (GLO) exist. However, in this report only the global (GLO) module is discussed in detail, the regional modules do not differ in their structure but only in their values. This module yields copper concentrate, which is used in pyrometallurgical processes. Hydrometallurgical copper winning – e. g. SX-EW – is based on different beneficiation techniques. Following modules in ecoinvent are covered in this chapter:

- mining copper ore (GLO)  
with the outputs:
  - copper, concentrate, at beneficiation (GLO)
  - molybdenum, molybdenite concentrate, coupleproduction Cu (GLO)
- mining copper ore (RNA)  
with the outputs:
  - copper, concentrate, at beneficiation (RNA)
  - molybdenum, molybdenite concentrate, coupleproduction Cu (RNA)
- mining copper ore (RLA)  
with the outputs:
  - copper, concentrate, at beneficiation (RLA)
  - molybdenum, molybdenite concentrate, coupleproduction Cu (RLA)
- mining copper ore (RER)  
with the outputs:
  - copper, concentrate, at beneficiation (RER)
  - molybdenum, molybdenite concentrate, coupleproduction Cu (RER)
- mining copper ore (RAS)  
with the outputs:
  - copper, concentrate, at beneficiation (RAS)
  - molybdenum, molybdenite concentrate, coupleproduction Cu (RAS)
- mining copper ore (ID)  
with the outputs:
  - copper, concentrate, at beneficiation (ID)
  - molybdenum, molybdenite concentrate, coupleproduction Cu (ID)
- copper, concentrate, at beneficiation (RNA)
- copper, concentrate, at beneficiation (RLA)
- copper, concentrate, at beneficiation (RER)
- copper, concentrate, at beneficiation (RAS)
- copper, concentrate, at beneficiation (ID)

The layout of the modules with general flow information, remarks, sources, values and uncertainty information are shown in Fig. 5.3 and Fig. 5.4. Ecoinvent meta information for this process are summarised in Tab. 5.1 on page 17. The following paragraphs describe calculations, sources and assumptions chosen in this study.



## Allocation

The ecological burden of the copper ore mining is distributed among the coupled products copper concentrate and molybdenite concentrate. As allocation procedure the revenue is chosen. The molybdenum price is quoted as US\$ per kilogram molybdenum content in molybdenite concentrate. This price is stable since 1983 with a value of around US\$ 5.75 per kg Mo (USGS (2003)), this gives a revenue of US\$ 3190 per tonne molybdenite concentrate with an assumed composition of 92.5% MoS<sub>2</sub>. Copper concentrate is traded in large quantities. The price of copper concentrate could not yet be assessed; it is estimated with the final bids of three auctions held in 2000 on an online web-auction<sup>4</sup> resulting in an average price of 15.3 US cents per pound copper payable for refining and smelting. With a copper price of 0.95 US\$/kg a price of 522 US\$ per tonne of concentrate with 29.7% copper content is calculated. These assumptions lead to an average revenue of 17 US\$ and 1816 US\$ per tonne molybdenite and copper, respectively, contained in the ore – with the metallurgical yield included – and thus to an allocation factor of 1% and 99% for molybdenite and copper concentrate, respectively.

## Infrastructure and land use

The production infrastructure used directly for exploitation is inventoried within this module, whereas the infrastructure like buildings and land-use for the operation is inventoried within auxiliary modules, which comprises both mining and beneficiation facilities. These datasets are treated in the part auxiliary processes of this report. The mobile equipment used for exploration like drilling machines, scoop trams and underground trucks is inventoried with the diesel consumption of 69.2 kJ per tonne mined raw ore. The used module “diesel, burned in building machine” includes the infrastructure of the machines. Conveyor belts are further assumed with a capacity of 2000 t/h which yields 7.6E-5 m conveyor belt required per tonne mined raw ore (Tab. 5.2).

According to Part XX, Non Ferrous Metal Winning – Auxiliary Processes, land use was assessed with the value for open pit mining (0.0025 m<sup>2</sup>/t) and the respective share in the world copper (70 %) production, yielding a 0.00174 m<sup>2</sup>/t and 0.0522 m<sup>2</sup>/t respectively. For underground mining no direct land use is estimated. A lifetime of 30 years is assumed.

**Tab. 5.2 Assumptions for the infrastructure used in underground mining exploitation. As lifetime ten years are estimated, and 5000 working hours yearly.**

| action      | share | machine        | throughput   | unit       | infrastructure  |
|-------------|-------|----------------|--------------|------------|-----------------|
|             |       |                | t / lifetime | Length (m) | m / t ore mined |
| Open pit    | 70%   | Conveyor belts | 100'000'000  | 2000       | 1E-5            |
| Underground | 30%   | Conveyor belts | 100'000'000  | 2000       | 2E-5            |
| Mix         |       |                |              |            | 7.6E-5          |

## Mining

In the considered process the ore is mined mainly in open cut operations. According to Martens et al. (2002) 69.6 % of the copper world production originates from open pit mines, but no further differentiation in regions or countries was available. Hence the world average is used for all regions. The overburden is disposed in piles near the mine, the standard module for disposal of non-sulphidic over-

<sup>4</sup> <http://www.copperconcentrate.com/default.asp>, accessed on 01.03.2003

burden was chosen, which includes the land use and transformation through the disposal. No overburden is refilled according to Krauss et al. (1999), the amount is documented for each of the five world regions. Also water use is documented, but as overall value, which is comprised in the following beneficiation step. Concerning electricity supply mix, Hilbrans & Hinrichs (1999) assume for the nickel-production a share of 60% of hydropower due to the operations high and constant energy demand. This assumption is used also in this inventory.

### **Beneficiation**

Once the ore is mined, it is ground and conditioned for further concentration. Considerable amounts of lime and flotation agents are added. The separated gangue is disposed in tailings ponds; the concentrated ore is fed to the metallurgy, which is assumed to be on-site. The respective standard module “disposal, sulphidic tailings, off-site” is chosen. (see “Non Ferrous Metal Winning – Auxiliary Processes”, Chapter “Disposal of Tailings and Overburden”).

The infrastructure of beneficiation is included in the mining infrastructure. Dust emissions comprise additionally the dust emissions due to mining activities. Like in the mining step, a share of 60% hydroelectricity was chosen according to Hilbrans & Hinrichs (1999).

**Lime:** Lime input was documented in Krauss et al. (1999) for each process alternative with a range of 1200–1330 g / t ore.

**Steel:** The steel input for milling (substitution of components for abrasion) was documented and ranges from 690 to 920 g/t with an average of 765 g/t ground ore assumed herewith. The mill work consists of special austenitic manganese steel of Hadfield grade, which is inventoried with chromium steel as a proxy.

**Reagents:** No exact information on mass and nature of the floating agents used had been available, therefore general values were assessed in Krauss et al. (1999) based on values in year 1978. Newer values reported in IPPC (2001) have been taken in this study (Tab. 5.4). For copper mining a range of 25 – 300 g collector and 25 – 250 g frothing reagent per tonne sulphidic ore was recommended in (Krauss et al. (1999)). This is confirmed by the values in Tab. 5.4. Collector and frother are inventoried for ecoinvent as unspecified organic chemicals. According to NPI (2001), xanthates hydrolytically decompose to 0.53 mg carbon disulphide per mg xanthate. With 189 g xanthates per tonne ore from Tab. 5.4 this corresponds to 100 g carbon disulphide per t mined ore (Tab. 5.5).

Activator and Depressants – excluding sodium cyanate – are modelled as unspecific inorganic chemicals; their emission to water is covered by the emissions reported from mining sites, which are shown in Tab. 5.3.

For pH adjustment and leaching, the value for sulphuric acid was considered with 3.74 kg/t ore mined (Tab. 5.5). As sulphuric acid is a by-product from the metallurgy, its production is regarded as burden free. Requirement of other acids is neglected. Acid emissions to water due to the use of sulphuric acid are not accounted.

Since composition and environmental fate of activator, depressant and sulphuric acid is uncertain, reported data for total emissions shown in Tab. 5.4 are taken. Note, that some theoretical emissions due to the use of the reagents are higher than the reported ones in Tab. 5.3.

Sodium cyanide is widely used for extracting gold and other metals such as silver, copper, and zinc. It is used as leachate and as depressant for certain sulphidic minerals. A cyanide recovery of 80 – 90% is reported due to abatement and recovery in the Golden Cross mine in New Zealand (Anonymous (1998a)). Here a value of 80% is chosen for cyanide-abatement technology. With a reported use of 35 g NaCN per tonne ore mined (Tab. 5.4), this means a direct emission of 3.88 g HCN per tonne ore mined. An overview of the reagents accounted for is presented in Tab. 5.5.

**Water use & Effluents:** Water use was documented in Krauss et al. (1999) and has been inventoried in this study as originating from river without pumping activities. Additional to direct emissions due to

the use of chemicals, total emissions to water are chosen according to Tab. 5.3. These values are from base metal sites, and are thus only an approximation for the real effluent concentrations in the copper production. It is assumed here that the effluents also include acid rock drainage (ARD). ARD after mine closure is modelled in the module “disposal, sulphidic tailings, off-site”. Effluent volume was not documented in Krauss et al. (1999); for this study it has been assumed that the water requirements equal the effluents.

**Dust:** A case study in Anonymous (1998b) on a projected open cut gold-copper mining and processing facility in Australia predicts for dust emissions per tonne of ore mined values between 0.18 kg/t and 0.46 kg/t. The prediction was made using the Emission Estimation Technique Manual for Mining (NPI (2001)). The maximum value of 0.46 kg/t is chosen for total dust emissions. According to the IAEA<sup>5</sup> the dust composition is approximated with values of a specific sulphidic nickel ore in Tab. 5.6. For PM<sub>10</sub> values, the ratio TDS : PM<sub>10</sub> : PM<sub>2.5</sub> was approximated with values from copper ore mining 101.72 : 50 : 5 g / t ore taken from the CEMIP-database<sup>6</sup>, i.e. one tonne emitted dust corresponds to 491.5 kg PM<sub>10</sub> and 49.2 kg PM<sub>2.5</sub> respectively. The airborne emissions resulting from mining and beneficiation activities are summed up in Tab. 5.6. In ecoinvent, for particulates differential classes are used adding up to the total dust emission, i. e. 49.2 kg PM<2.5, 442 kg PM2.5-10 and 508 kg PM>10.

### Data quality

**Mining;** Data for major mass fluxes are satisfactory, while the emissions into water may be overestimated due to different system boundaries of reported effluent volume from (Krauss et al. (1999)) and total effluent emissions reported in (IPPC (2002)). High uncertainties are reflected by a high standard deviation (stdev95) of 10.

**Beneficiation;** Like in mining, the emissions to water can be overestimated. The emissions from tailings are assessed in an external module, which itself contains rough assumptions of a global overburden and tailings composition and a simplified disposal scenario. Emissions though can vary considerably from case to case.

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<sup>5</sup> The IAEA Database of Natural Matrix Reference Materials, <http://www-naweb.iaea.org/nahu/external/e4/nmrm/>, accessed on 01.03.2003

<sup>6</sup> [http://www.air.sk/tno/cepmeip/em\\_factors.php](http://www.air.sk/tno/cepmeip/em_factors.php), accessed 01.03.2003

**Tab. 5.3** Composition of effluents from mining activities; values are taken from total emissions to water from base metal sites in (IPPC (2002)).

| Compound        | Unit                    | Range         | This study<br>(geometric mean) | Number of<br>Samples |
|-----------------|-------------------------|---------------|--------------------------------|----------------------|
| Discharge       | Mio. m <sup>3</sup> / a | 0.08 - 22.9   | 3.65                           | 6                    |
| Ca              | g / m <sup>3</sup>      | 686           | 686                            | 1                    |
| SO <sub>4</sub> | g / m <sup>3</sup>      | 1750 - 3175   | 2357                           | 2                    |
| COD             | g / m <sup>3</sup>      | 2.24 - 48.5   | 10.4                           | 2                    |
| Solids          | g / m <sup>3</sup>      | 2.38 - 11.3   | 5.17                           | 4                    |
| Al              | mg / m <sup>3</sup>     | 69.3 - 108    | 86.3                           | 2                    |
| As              | mg / m <sup>3</sup>     | 0.264 - 14.1  | 2.95                           | 3                    |
| Cd              | mg / m <sup>3</sup>     | 0.0901 - 1.02 | 0.316                          | 4                    |
| Co              | mg / m <sup>3</sup>     | 0.742 - 0.823 | 0.781                          | 2                    |
| Cr              | mg / m <sup>3</sup>     | 0.0311 - 9.62 | 0.547                          | 2                    |
| Cu              | mg / m <sup>3</sup>     | 1.24 - 44.9   | 7.91                           | 5                    |
| Fe              | mg / m <sup>3</sup>     | 61.7 - 1326   | 291                            | 3                    |
| Mn              | mg / m <sup>3</sup>     | 24.7          | 24.7                           | 1                    |
| Hg              | mg / m <sup>3</sup>     | 0.0155 - 0.12 | 0.0377                         | 3                    |
| Ni              | mg / m <sup>3</sup>     | 0.792 - 1338  | 24.3                           | 3                    |
| Pb              | mg / m <sup>3</sup>     | 0.0155 - 20   | 2.80                           | 4                    |
| Zn              | mg / m <sup>3</sup>     | 5.37 - 225    | 75.9                           | 5                    |
| N               | g / m <sup>3</sup>      | 2.50 - 1786   | 22.8                           | 3                    |

**Tab. 5.4** Consumption of reagents in base metal mineral processing plants. Data taken from (IPPC (2002)).

| Reagents                                                                                 | Unit | Range       | This study<br>(geometric mean) | Number of<br>samples |
|------------------------------------------------------------------------------------------|------|-------------|--------------------------------|----------------------|
| Collectors – Xanthate                                                                    | g/t  | 110 - 300   | 188.6                          | 6                    |
| Frothers - Sylvamine, MIBC, Dowfroth                                                     | g/t  | 9.7 - 150   | 37.2                           | 5                    |
| Sodium cyanide                                                                           | g/t  | 4 - 310     | 35                             | 2                    |
| Depressants – Zinc sulphate, Iron sulphate, Acetic acid, Sodiumchromate, Dithiophosphate | g/t  | 35 - 432    | 231.5                          | 5                    |
| Activator - Copper sulphate                                                              | g/t  | 433 - 876   | 538                            | 4                    |
| Flotation agents                                                                         | g/t  | 19.0        | 19.0                           | 1                    |
| Flocculants - CMC                                                                        | g/t  | 1 - 100     | 11.1                           | 3                    |
| Sulphuric acid (pH)                                                                      | g/t  | 387 - 12000 | 3739                           | 4                    |
| Sodium hydroxide (pH)                                                                    | g/t  | 30 - 490    | 121                            | 2                    |
| Nitric acid (pH)                                                                         | g/t  | 150         | 150                            | 1                    |
| Hydrochloric acid (pH)                                                                   | g/t  | 1           | 1                              | 1                    |

**Tab. 5.5 Emissions and input due to chemical use in beneficiation in g per tonne treated ore. The values are the ones chosen in this study. For further details read in text.**

| Reagent              | Type of burden     | Unit | Chosen value                      |
|----------------------|--------------------|------|-----------------------------------|
| Collector            | input technosphere | g/t  | 189                               |
| Carbon disulfide     | air emission       | g/t  | 100                               |
| Frother              | input technosphere | g/t  | 37                                |
| Lime                 | input technosphere | g/t  | 2000                              |
| Carbon dioxide       | air emission       | g/t  | inventoried in metallurgical step |
| Activator/Depressant | input technosphere | g/t  | 828                               |
| Sulphuric acid       | input technosphere | g/t  | 3739                              |
| Sodium Cyanide       | input technosphere | g/t  | 35.2                              |
| Cyanide              | water emission     | g/t  | 3.88                              |

**Tab. 5.6 Emissions to air resulting from mining and beneficiation. Values for total dust emission are taken from NPI 2001**

| Emission to air |    | Unit         | Emission per tonne mined ore |
|-----------------|----|--------------|------------------------------|
| Antimony        | Sb | mg / mg dust | 0.0000002                    |
| Arsenic         | As | mg / mg dust | 0.0000015                    |
| Beryllium       | Be | mg / mg dust | 0.0000026                    |
| Boron           | Be | mg / mg dust | 0.00001                      |
| Cadmium         | Cd | mg / mg dust | 0.00000011                   |
| Chromium        | Cr | mg / mg dust | 0.0001                       |
| Cobalt          | Co | mg / mg dust | 0.00002                      |
| Copper          | Cu | mg / mg dust | 0.00005                      |
| Fluorine        | F  | mg / mg dust | 0.00095                      |
| Lead            | Pb | mg / mg dust | 0.000014                     |
| Manganese       | Mn | mg / mg dust | 0.00095                      |
| Mercury         | Hg | mg / mg dust | 0.00000005                   |
| Nickel          | Ni | mg / mg dust | 0.00008                      |
| Selenium        | Se | mg / mg dust | 0.00000005                   |
| Zinc            | Zn | mg / mg dust | 0.00019                      |
| PM<2.5          |    | mg / mg dust | 0.0492                       |
| PM2.5-10        |    | mg / mg dust | 0.442                        |
| PM>10           |    | mg / mg dust | 0.508                        |

## 5. System Characterisation

| General Flow Information                    |                                |        | Representation in ecoinvent                                                                                            |                                                                                       |                   |                            |          |                                                                                |                                       | Uncertainty Informations |                       |      |                    |                 |
|---------------------------------------------|--------------------------------|--------|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------|----------------------------|----------|--------------------------------------------------------------------------------|---------------------------------------|--------------------------|-----------------------|------|--------------------|-----------------|
| Input                                       | Process Name                   | Output | Remarks                                                                                                                | Category                                                                              | Sub-category      | Infra-structure            | Location | Modul name in ecoinvent                                                        | Mean value                            | Unit                     | Source for mean value | Type | Standard Deviation | General Comment |
| Copper in ground                            | mining copper ore, GLO; mining |        | 83.5% yield, 62.1 kg overburden, 35.9 kg ore, 35.1 kg tailings per kg Cu concentrate                                   | resource                                                                              | in ground         |                            |          | Copper, 0.99% in sulfide, Cu 0.36% and Mo 8.2E-3% in crude ore, in ground      | 3.56E-01 kg                           | (Krauss, 1999)           | 1                     | 1.13 | (2,2,2,1,1,4,12)   |                 |
| Molybdene in ground                         |                                |        | 80% yield. Only 36% of Mo mined together wit Cu is used --> 64% of resource use is lost and allocated to Cu production | resource                                                                              | in ground         |                            |          | Molybdenum, 0.022% in sulfide, Mo 8.2E-3% and Cu 0.36% in crude ore, in ground | 7.84E-03 kg                           | own assumptions          | 1                     | 1.13 | (2,2,2,1,1,4,12)   |                 |
| Land transformation from nature             |                                |        |                                                                                                                        | resource                                                                              | land              |                            |          | Transformation, from unknown                                                   | 6.25E-05 m2                           | (Martens 2002)           | 1                     | 2.01 | (2,2,2,1,1,2,8)    |                 |
| Land transformation to mine                 |                                |        |                                                                                                                        | resource                                                                              | land              |                            |          | Transformation, to mineral extraction site                                     | 6.25E-05 m2                           | (Martens 2002)           | 1                     | 2.01 | (2,2,2,1,1,2,8)    |                 |
| Land use for mining                         |                                |        |                                                                                                                        | resource                                                                              | land              |                            |          | Occupation, mineral extraction site                                            | 1.88E-03 m2a                          | (Martens 2002)           | 1                     | 1.51 | (2,2,2,1,1,2,7)    |                 |
| explosive                                   |                                |        |                                                                                                                        | construction processes                                                                | civil engineering | No                         | RER      | blasting                                                                       | 3.59E-02 kg                           | (Krauss, 1999)           | 1                     | 1.13 | (2,2,2,1,1,4,4)    |                 |
| diesel fuel                                 |                                |        |                                                                                                                        | construction processes                                                                | machinery         | No                         | GLO      | diesel, burned in building machine                                             | 2.83E-03 MJ                           | (Krauss, 1999)           | 1                     | 1.13 | (2,2,2,1,1,4,4)    |                 |
| electricity from grid, mining               |                                |        | Assumption: 40% electricity from grid                                                                                  | electricity                                                                           | production mix    | No                         | UCTE     | electricity, medium voltage, production UCTE, at grid                          | 4.22E-02 kWh                          | (Krauss, 1999)           | 1                     | 1.14 | (2,3,2,1,1,4,2)    |                 |
| hydropower, mining                          |                                |        | Assumption: 60% hydropower                                                                                             | hydro power                                                                           | power plants      | No                         | RER      | electricity, hydropower, at run of-river power plant                           | 6.33E-02 kWh                          | (Krauss, 1999)           | 1                     | 1.14 | (2,3,2,1,1,4,2)    |                 |
| exploitation infrastructure, conveyor belts |                                |        | exploration infrastructure, 2000 m belts assumed                                                                       | construction processes                                                                | machinery         | Yes                        | RER      | conveyor belt, at plant                                                        | 2.72E-06 m                            | own assumptions          | 1                     | 3.36 | (5,5,1,1,3,5,9)    |                 |
| infrastructure of mining (underground)      |                                |        | 30% underground; life time output of mine 2E10 kg)                                                                     | metals                                                                                | extraction        | Yes                        | GLO      | non-ferrous metal, mine, underground                                           | 1.57E-10 unit                         | own assumptions          | 1                     | 3.36 | (5,5,1,1,3,5,9)    |                 |
| infrastructure of mining (surface)          |                                |        | 70% open cut; life time output of mine 2E10 kg)                                                                        | metals                                                                                | extraction        | Yes                        | GLO      | non-ferrous metal, mine, surface                                               | 3.59E-10 unit                         | own assumptions          | 1                     | 3.36 | (5,5,1,1,3,5,9)    |                 |
|                                             |                                |        | overburden, disposed                                                                                                   |                                                                                       | waste management  | residual material landfill | No       | GLO                                                                            | disposal, sulfidic tailings, off-site | 6.21E+01 kg              | (Krauss, 1999)        | 1    | 1.13               | (2,2,2,1,1,4,6) |
|                                             |                                |        | overburden, refilled                                                                                                   | no overburden refilled according to (Krauss 1999)                                     |                   |                            |          |                                                                                |                                       | 0.00E+00 kg              | (Krauss, 1999)        | 1    | 1.13               | (2,2,2,1,1,4,6) |
|                                             |                                |        | effluents, mining                                                                                                      | included in beneficiation effluents, not evident if accounted in (Krauss 1999) or not |                   |                            |          |                                                                                |                                       | 0.00E+00 m3              | (Krauss, 1999)        | 1    | 1.13               | (2,2,2,1,1,4,6) |

Fig. 5.3 Flows for “Mining of copper-ore, GLO – part mining” and its representation in the ecoinvent database. Values correspond to the functional unit of 1 kg copper ore.

## 5. System Characterisation

| General Flow Information               |                                       |                            | Representation in ecoinvent                                                                                                                                                  |                  |                            |                 |                                                      |                                                           |                | Uncertainty Informations          |                       |                  |                    |                 |
|----------------------------------------|---------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------------|-----------------|------------------------------------------------------|-----------------------------------------------------------|----------------|-----------------------------------|-----------------------|------------------|--------------------|-----------------|
| Input                                  | Process Name                          | Output                     | Remarks                                                                                                                                                                      | Category         | Sub-category               | Infra-structure | Location                                             | Modul name in ecoinvent                                   | Mean value     | Unit                              | Source for mean value | Type             | Standard Deviation | General Comment |
| electricity, from grid, beneficiation  | mining copper ore, GLO; beneficiation |                            | Assumption: 40% electricity from grid                                                                                                                                        | electricity      | production mix             | No              | UCTE                                                 | electricity, medium voltage, production UCTE, at grid     | 2.51E-01 kWh   | (Krauss, 1999)                    | 1                     | 1.14             | (2,3,2,1,1,4,2)    |                 |
| hydropower, beneficiation              |                                       | Assumption: 60% hydropower | hydro power                                                                                                                                                                  | power plants     | No                         | RER             | electricity, hydropower, at run of-river power plant | 3.76E-01 kWh                                              | (Krauss, 1999) | 1                                 | 1.14                  | (2,3,2,1,1,4,2)  |                    |                 |
| fresh and processwater (beneficiation) |                                       |                            | resource                                                                                                                                                                     | in water         |                            |                 | Water, river                                         | 3.59E-02 m3                                               | (Krauss, 1999) | 1                                 | 1.13                  | (2,2,1,1,1,4,12) |                    |                 |
| lime                                   |                                       |                            | construction materials                                                                                                                                                       | others           | No                         | CH              | limestone, milled, packed, at plant                  | 4.28E-02 kg                                               | (Krauss, 1999) | 1                                 | 1.04                  | reported values  |                    |                 |
| steel                                  |                                       |                            | Abrasion masticator                                                                                                                                                          | metals           | extraction                 | No              | RER                                                  | chromium steel 18/8, at plant                             | 2.67E-02 kg    | (Krauss, 1999)                    | 1                     | 1.20             | reported values    |                 |
| floating agents, organic               |                                       |                            |                                                                                                                                                                              | chemicals        | organics                   | No              | GLO                                                  | chemicals organic, at plant                               | 8.11E-03 kg    | (IPPC, 2002)                      | 1                     | 2.00             | reported values    |                 |
| floating agents, inorganic             |                                       |                            |                                                                                                                                                                              | chemicals        | inorganics                 | No              | GLO                                                  | chemicals inorganic, at plant                             | 2.76E-02 kg    | (IPPC, 2002)                      | 1                     | 2.00             | reported values    |                 |
| Sodium cyanide                         |                                       |                            |                                                                                                                                                                              | chemicals        | inorganics                 | No              | RER                                                  | sodium cyanide, at plant                                  | 1.26E-03 kg    | (IPPC, 2002)                      | 1                     | 2.00             | reported values    |                 |
| Sulphuric Acid                         |                                       |                            | burden-free, since byproduct from metallurgy.                                                                                                                                |                  |                            |                 |                                                      |                                                           | 1.34E-01 kg    | (IPPC, 2002)                      | 1                     | 2.00             | reported values    |                 |
| infrastructure beneficiation           |                                       |                            | AIOH-production plant taken as proxy for beneficiation. A factor of 1.4 is applied to relate to bauxite input rather then AIOH output.                                       | metals           | extraction                 | Yes             | RER                                                  | aluminium hydroxide, plant                                | 3.02E-10 unit  | own assumptions                   | 1                     | 3.07             | (3,2,2,1,3,4,9)    |                 |
|                                        |                                       | tailings, disposed         | not clear, wether net disposal or partly re-filled in mine. Here assumed as net disposal                                                                                     | waste management | residual material landfill | No              | GLO                                                  | disposal, sulfidic tailings, off-site                     | 3.51E+01 kg    | (Krauss, 1999)                    | 1                     | 1.10             | (2,2,2,3,1,3,6)    |                 |
|                                        |                                       | tailings, refilled         | no tailings refilled according to (Krauss 1999)                                                                                                                              |                  |                            |                 |                                                      |                                                           | 0.00E+00 kg    | (Krauss, 1999)                    | 1                     | 1.13             | (2,2,2,1,1,4,6)    |                 |
|                                        |                                       | effluents, beneficiation   | composition in Tab. 5.3, high variability assumed. No treatment of waste water.                                                                                              |                  |                            |                 |                                                      |                                                           | 3.59E-02 m3    | (Krauss, 1999)                    | 1                     | 1.10             | (2,2,2,3,1,3,6)    |                 |
|                                        |                                       | Emission Cyanide           | From sodium cyanide, 80% control                                                                                                                                             | water            | river                      |                 |                                                      | Cyanide                                                   | 1.39E-04 kg    | (IPPC, 2002), (Anonymous 1998)    | 1                     | 1.84             | (4,2,2,3,4,4,33)   |                 |
|                                        |                                       | emission carbondisulfide   | 0.53 kg carbondisulfide out of 1 kg xanthate through hydrolytical decomposition                                                                                              | air              | low population density     |                 |                                                      | Carbon disulfide                                          | 3.59E-03 kg    | (NPI, 2001)                       | 1                     | 2.31             | (4,2,2,5,4,4,23)   |                 |
|                                        |                                       | emission carbon dioxide    | 0.44 kg CO2 per kg lime                                                                                                                                                      | air              | low population density     |                 |                                                      | Carbon dioxide, fossil                                    | 1.35E-02 kg    | (NPI, 2001)                       | 1                     | 2.30             | (5,3,3,6,5,5,24)   |                 |
|                                        |                                       | Dust                       | including emissions from mining activities. Dust is the sum of particulate emissions represented in fractions PM<2.5, PM2.5-10, PM>10, including metals and salts (Tab. 5.6) |                  |                            |                 |                                                      |                                                           | 1.65E-02 kg    | (NPI, 2001)                       | 1                     | 1.59             | (3,2,1,1,3,4,25)   |                 |
|                                        |                                       | PM<2.5                     | including emissions from mining activities. Sum up together with other PM fraction to total of dust                                                                          | air              | low population density     |                 |                                                      | Particulates, > 10 um                                     | 8.12E-04 kg    | (NPI, 2001), (naweb, 2003), cemip | 1                     | 2.69             | (3,2,1,1,5,4,19)   |                 |
|                                        |                                       | PM2.5-10                   | including emissions from mining activities. Sum up together with other PM fraction to total of dust                                                                          | air              | low population density     |                 |                                                      | Particulates, > 2.5 um, and < 10um                        | 7.31E-03 kg    | (NPI, 2001), (naweb, 2003), cemip | 1                     | 3.69             | (3,2,1,1,5,4,20)   |                 |
|                                        |                                       | PM>10                      | including emissions from mining activities. Sum up together with other PM fraction to total of dust                                                                          | air              | low population density     |                 |                                                      | Particulates, < 2.5 um                                    | 8.40E-03 kg    | (NPI, 2001), (naweb, 2003), cemip | 1                     | 2.69             | (3,2,1,1,5,4,23)   |                 |
|                                        |                                       | waste heat                 |                                                                                                                                                                              | air              | low population density     |                 |                                                      | Heat, waste                                               | 2.64E+00 MJ    |                                   | 1                     | 1.14             | (2,3,2,1,1,4,13)   |                 |
|                                        |                                       | Copper Concentrate         | 29.7% Cu                                                                                                                                                                     | metals           | extraction                 | No              | GLO                                                  | copper, concentrate, at beneficiation                     | 1.00E+00 kg    | (Krauss, 1999)                    |                       |                  |                    |                 |
|                                        |                                       | Molybdenite concentrate    | 55% Mo in concentrate (92.5% MoS2 in molybdenite)                                                                                                                            | metals           | extraction                 | No              | GLO                                                  | molybdenum, molybdenite concentrate, couple production Cu | 4.11E-03 kg    | own assumptions                   |                       |                  |                    |                 |

Fig. 5.4 Flows for “Mining of copper-ore, GLO – part mining” and its representation in the ecoinvent database. Values correspond to the functional unit of 1 kg copper ore.

## 5.2 Winning primary copper

The standard process for the pyrometallurgical production of primary copper in use since the late 19th century involves three major stages:

1. ore mining and beneficiation,
2. smelting and desulphurisation and
3. final refining.

The newer hydrometallurgical process, known as solvent extraction-electro-winning (SX-EW), is in a rapid growth stage and it is to some extent displacing the older pyrometallurgical approach. The pyrometallurgical approach is a net producer of by-product sulphuric acid, whereas the hydrometallurgical process is a net consumer of the acid. Thus, in the long run it is likely that the two will be increasingly combined.

In this study primary copper production is modelled as an aggregation of the pyrometallurgical and SX-EW production routes using the region-specific shares along with (Krauss et al. (1999)).

**Tab. 5.7 Overview of the shares of different production processes in different regions, in million tonnes, and the share of SX-EW (Krauss et al. (1999)). Note that the two regions North America and South America are not distinguished and therefore concerning sulphur dioxide emissions are treated as one in this study.**

| Production in MMT (10 <sup>6</sup> metric tonnes)                                                                                                                                                                                                                             | North America | South America | Euro pe | CIS <sup>7</sup> / Asia | Indonesia / Australia | World      | Com-ment |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------|---------|-------------------------|-----------------------|------------|----------|
| Region in ecoinvent                                                                                                                                                                                                                                                           | RNA           | RLA           | RER     | RAS                     | ID                    | GLO        |          |
| Total copper prod.                                                                                                                                                                                                                                                            | 3840          |               | 1185    | 2915                    | 295                   | 8760       |          |
| Reverberatory furnace                                                                                                                                                                                                                                                         | 895           |               | 70      | 595                     | 90                    | 2010       | A        |
| Electric furnace                                                                                                                                                                                                                                                              | 30            |               | 80      | 170                     |                       | 425        | B        |
| Outokumpu                                                                                                                                                                                                                                                                     | 995           |               | 775     | 1320                    | 40                    | 3160       | B        |
| Inco                                                                                                                                                                                                                                                                          | 350           |               |         | 90                      |                       | 440        | B        |
| Kivcet                                                                                                                                                                                                                                                                        |               |               |         | 40                      |                       | 40         | B        |
| Mitsubishi                                                                                                                                                                                                                                                                    | 100           |               |         | 200                     |                       | 300        | B        |
| Noranda                                                                                                                                                                                                                                                                       | 180           |               |         |                         | 40                    | 220        | B        |
| Teniente                                                                                                                                                                                                                                                                      | 415           |               |         |                         |                       | 415        | B        |
| Vanyukov                                                                                                                                                                                                                                                                      |               |               |         | 155                     |                       | 155        | B        |
| Other                                                                                                                                                                                                                                                                         | 200           |               | 200     | 40                      | 90                    | 530        | A        |
| SX-EW                                                                                                                                                                                                                                                                         | 675 (17.6%)   |               |         |                         | 25 (8.8%)             | 800 (9.4%) | C        |
| total considered                                                                                                                                                                                                                                                              | 3840          |               | 1125    | 2610                    | 285                   | 8495       |          |
| Abatement SO <sub>2</sub>                                                                                                                                                                                                                                                     | 50%           | 50%           | 95%     | 20%                     | 20%                   | 45%        |          |
| A furnace type with high SO <sub>2</sub> -emission (2.5 t / t) (Ayres et al. (2002))<br>B furnace type with moderate SO <sub>2</sub> -emission (0.15 t / t) (Ayres et al. (2002))<br>C furnace type with little SO <sub>2</sub> -emission (0.015 t / t) (Ayres et al. (2002)) |               |               |         |                         |                       |            |          |

### 5.2.1 Production process and infrastructure

There are many different process possibilities to win the metal. The chosen process depends on the composition of the ore, the local costs of energy supply, and the local legislation. Basically two differ-

<sup>7</sup> Commonwealth of Independent States



ent types can be distinguished: the hydrometallurgical and the pyrometallurgical process with a share of 90.6 % and 9.4 % of the world-wide copper produced in 2004. The hydrometallurgical process is described in detail later in Section 5.3 Hydrolytic winning of primary copper.

**Pyrometallurgy;** The pyrometallurgical treatment of copper concentrates includes three types of unit operation: drying & roasting (pre-treatment), smelting & converting (reduction), and finally refining. In the roasting step sulphur is driven off as sulphur dioxide and part of the iron is oxidised. In smelting, the roaster product is melted with a siliceous flux which combines with the oxidised iron to produce two immiscible phases, a liquid silicate slag which can be discarded, and a solution of molten sulphides which contains the metal values. In the converting operation on the sulphide melt, more sulphur is driven off as sulphur dioxide, and the remaining iron is oxidised and fluxed for removal as silicate slag, leaving a high-grade copper sulphide matte. In several modern operations the roasting step has been eliminated, and the copper sulphide concentrate is treated directly in the smelter. The converted copper is referred to as blister copper with a purity of 98%.

**Refining;** Conventional refining comprises three stages: (1) pyrometallurgical or fire refining, (2) electrolytic refining, and (3) remelting of cathodes and casting of shapes. Fire refining is applied to crude copper from converters and furnaces, cement copper from hydrometallurgy, anode scrap from electrolytic refining, and high-grade copper scrap, chiefly unalloyed wire scrap. About 80 % of the world copper production is refined by electrolysis. This treatment yields copper with high electrical conductivity and provides for separation of valuable impurities, especially precious metals. Copper cathodes must be remelted and cast to shapes because the structure of copper formed by electrocrystallization is not suitable for working to semi finished products. Cathodes are remelted in several types of special furnaces that perform the tasks of melting, post refining (if necessary), holding, and casting.

## 5.2.2 Emission

### Metallurgy

**Sulphur dioxide;** In every process steps sulphur dioxide is emitted to air. Recovery of sulphur dioxide is only economic for high concentrated off-gas. SO<sub>2</sub> emissions are by far the most important in the metallurgy.

**Carbon dioxide;** In the beneficiation step, considerable amounts of lime is added to the ore for pH-stabilisation. Lime forms later flux in the metallurgical step. Lime decomposes into CO<sub>2</sub> to form calcite.

**Dust;** Dust carries over from the roasting, smelting and converting processes. Particulate emissions to air consist of metals and thus are often returned to the leaching process after treatment.

**Emissions to water;** Metals and their compounds and substances in suspension are the main pollutants emitted to water. The metals concerned are Cu, Ni, Co, As, and Cr. Other significant substances are chlorides and sulphates. Wastewater from wet gas cleaning (if used) of the different metallurgical stages is the most important source. The leaching stages are usually operated on a closed circuit and drainage systems, and are therefore regarded as minor sources.

### Refining

Also in this step the combustion of sulphur leads to emissions of SO<sub>2</sub>. Nitrogen oxides are produced in significant amounts during acid digestion using nitric acid. Chlorine and HCl can be formed during a number of digestion, electrolytic, and purification processes. Dust and metals are generally emitted from incinerators and furnaces. Organic compounds, namely dioxins, can be emitted from smelting stages resulting from the poor combustion of oil and plastic in the feed material. All these emissions are subject to abatement technologies and controlling. Large quantities of effluents contain substantial amounts of metals and organic substances.

### 5.2.3 Waste

#### Metallurgy & Separation from Non Ferrous Metals

The production of metals is related to the generation of several co-products, residues and wastes, which are listed in the European Waste Catalogue. Some of the process specific residues can be reused or recovered in preliminary process steps (e. g. dross, filter dust) or construction (e. g. cleaned slag). Residues also arise from the treatment of liquid effluents, the main residue being gypsum waste and metal hydroxides from the wastewater neutralisation plant. These residuals have to be disposed, usually in lined ponds.

#### Refining

Like in the pyrometallurgical step, considerable quantities of solid residuals are generated, which are mostly recycled within the process or sent to other specialists to recover any precious metals. Final residues generally comprise hydroxide filter cakes (iron hydroxide, 60% water, cat I industrial waste).

### 5.2.4 Copper, blister copper, at primary smelter in ecoinvent

This module is uniquely used in the secondary copper production where, besides copper scrap, 10% of the feed consists of blister copper, i.e. raw copper after converting with an average content of 98%. The inventory of this module is identical to that of the primary production of copper in RER, with the only difference that the refining step is omitted and with 98% of the in- and outputs, since the metal content of blister copper is only 98%. The inventory has an output of 1 kg blister copper.

### 5.2.5 Winning of primary copper in ecoinvent

This part is mainly based on Krauss et al. (1999). The general process description was taken from Fabian (1997), whereas the process specific emissions are from IPPC (2001) and EEA (1999). In Krauss et al. (1999) data for the different process alternatives in copper production was collected in detail. The distinction there was made between procedures depending on different furnaces: Inco, Teniente, Outokumpu, Mitsubishi, Noranda, Vanyukov, reverberatory furnace, electric furnace, Kivcet. According to the share of the different procedures in the respective countries an aggregated production mix for a country was calculated (Tab. 5.7). Where applicable, the respective share in SX-EW – copper is included into the production mix.

Please note that in the database distinct datasets for the regions Northern America (RNA), South America (RLA), Europe (RER), Asia and Russia (RAS), Indonesia / Australia (ID), and the world-wide average (GLO) exist. However, in this report only the global (GLO) module is discussed, the regional modules do not differ in their structure but in their values according to Tab. 5.8.

The module describes the production mix for refined copper in 1994. It is designed for the use of the metal as raw material for the semis manufacturing or as alloying element. The modules with general flow information, remarks, sources, values and uncertainty information are shown in Fig. 5.5, Fig. 5.6 and Fig. 5.7. The meta information for this process in ecoinvent are summarised in Tab. 5.1 on page 17 ff. The following paragraphs describe calculations, sources and assumptions chosen in this study. The functional unit of this process is one kilogram of cathode copper.

#### Metallurgy

In this study all nine process alternatives from Tab. 5.7 have been considered according to their production share given in Krauss et al. (1999). These alternatives cover 84 % of the world production of primary copper. The inventory already is partly aggregated and reflects the mix of the different process alternatives within a certain process step, e.g. electricity consumption in the roasting step. For de-

tails on single process alternatives, e. g. the Outokumpu – process, please consult Krauss et al. (1999) or IPPC (2001).

**Pre-treatment;** Flux demand (quartz sand) is documented in Krauss et al. (1999). Additional 250 kg of limestone per tonne of copper has been assumed in this study according to Ayres et al. (2002). Drying is done by gas. Of the electricity 40% is assumed to originate from the grid while run-off plants deliver 60%. Sulphuric acid produced from off-gas is not inventoried (cut-off). All process specific emissions are inventoried in the smelting step.

**Smelting and Reduction;** 40% electricity is from grid, 60% hydropower assumed. A production of steam is reported which is not inventoried as by-product. Dross and slag are reported to be recycled, for this study however they are assumed to be disposed of in a landfill. In this step the major part of the sulphur dioxide is emitted, which represent the main impact besides the energy use. The values for emissions vary in a wide range from 4 kg to 2000 kg SO<sub>2</sub> per tonne metal produced (Anonymous (1998c)), but single values of much higher magnitude are also reported, e. g. nickel-copper smelting in Noril'sk (RUS) with 5300 kg SO<sub>2</sub> / t metal produced (Hochfeld (1997)). The target for “best available technique” in Europe is set to 6 – 16 kg SO<sub>2</sub> / t Cu (IPPC (2001)), the World Bank mentions 25 kg SO<sub>2</sub> / t Cu for “modern plants using good industrial practices”. SO<sub>2</sub> emissions depend basically on the possibility of economical recovering. Recovering of sulphur dioxide as sulphuric acid is economic if the SO<sub>2</sub> content in the off-gas is sufficient and therefore is dependent from the production process and the control-factor (abatement). In this study following emission factors are chosen relating to Tab. 5.7: reverbatroy furnace and undefined processes (“other”): 2500 kg / t Cu; all other alternatives: 150 kg / t Cu. The control-factors chosen are shown in Tab. 5.7, they are based on Ayres et al. (2002). For this calculation the regions North and South America are treated equally due to missing specific data. The resulting SO<sub>2</sub> emissions are shown in Tab. 5.8; they range from 36 kg / t Cu (RER) to 1296 kg / t Cu (ID) with a world average of 458 kg / t Cu. The calcination of previously added lime (pre-treatment and beneficiation, see Section 5.1.3 Exploitation of copper-ore in ecoinvent) releases 0.44 kg CO<sub>2</sub> per tonne lime. Several emissions to air are documented for European conditions in EEA (1999); the resulting emissions are chosen for this study (Tab. 5.9). Elaborate abatement technology – values of column “impact control” in Tab. 5.8 – are applied only for RNA and RER, whereas for the other regions values corresponding to limited control are chosen. VOC emissions are mentioned in EEA (1999) with a value of about 0.015 kg / t (unabated, including roasting and smelting). Dust emissions according to Anonymous (1998c) range between 0.1 kg / t Cu and 20 kg / t Cu, whereas IPPC (2001) mentions for modern European plants a range of 0.16 kg / t Cu to 1 kg / t Cu. In this study 0.4 kg / t Cu is chosen, which is assumed to apply for all covered processes. The distribution of particulate sizes is taken from CEPMEIP<sup>8</sup> (Tab. 5.10). For water use the value for primary nickel production of 1.4 m<sup>3</sup> per tonne concentrate worked is chosen (Hochfeld (1997)). The volume of the effluent is assessed with the value for water use and respective emissions into water are calculated according to the content in Tab. 5.11.

**Refining;** No further wastes are considered from this step. Region's specific production mix is compiled with the specific pyrolytical technologies and the specific share in SX-EW – copper.

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<sup>8</sup> [http://www.air.sk/tno/cepmeip/em\\_factors.php](http://www.air.sk/tno/cepmeip/em_factors.php), accessed on 01.03.2003

**Tab. 5.8 Pyrometallurgy of copper. Background data per tonne of copper produced for the different regions covered in this study after (Krauss et al. (1999)).**

| Step / Description                                                                                                                                                                                                                                                               |                 | Unit | RNA   | RLA   | RER  | RAS  | ID   | GLO  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|-------|-------|------|------|------|------|
| Input                                                                                                                                                                                                                                                                            | output          |      |       |       |      |      |      |      |
| Pre-treatment                                                                                                                                                                                                                                                                    |                 |      |       |       |      |      |      |      |
| Copper concentrate <sup>1)</sup>                                                                                                                                                                                                                                                 |                 | t    | 3.60  | 3.18  | 4.14 | 4.06 | 3.16 | 3.48 |
| Compressed Air                                                                                                                                                                                                                                                                   |                 | Nm3  | 1230  | 1370  | 1300 | 1600 | 1320 | 1450 |
| electricity                                                                                                                                                                                                                                                                      |                 | kWh  | 45    | 91    | 38   | 70   | 94   | 65   |
| electricity, grid                                                                                                                                                                                                                                                                |                 | kWh  | 18    | 36.4  | 15.2 | 28   | 37.6 | 26   |
| hydropower                                                                                                                                                                                                                                                                       |                 | kWh  | 27    | 54.6  | 22.8 | 42   | 56.4 | 39   |
| fossil energy (gas)                                                                                                                                                                                                                                                              |                 | MJ   | 1234  | 218   | 1379 | 1162 | 545  | 980  |
|                                                                                                                                                                                                                                                                                  | off-gas         | Nm3  | 1400  | 1580  | 1450 | 1900 | 1860 | 1650 |
|                                                                                                                                                                                                                                                                                  | sulphuric acid  | kg   | 140   | 380   | 130  | 220  | 380  | 220  |
| Reduction                                                                                                                                                                                                                                                                        |                 |      |       |       |      |      |      |      |
| electricity                                                                                                                                                                                                                                                                      |                 | kWh  | 145   | 52    | 200  | 205  | 86   | 155  |
| electricity, grid                                                                                                                                                                                                                                                                |                 | kWh  | 58    | 20.8  | 80   | 82   | 34.4 | 62   |
| hydropower                                                                                                                                                                                                                                                                       |                 | kWh  | 87    | 31.2  | 120  | 123  | 51.6 | 93   |
| fossil energy (oil heavy)                                                                                                                                                                                                                                                        |                 | MJ   | 4779  | 9641  | 4532 | 6592 | 9146 | 6592 |
| fossil energy (gas)                                                                                                                                                                                                                                                              |                 | MJ   | 72.6  |       |      | 72.6 |      | 36.3 |
| fossil energy (coke)                                                                                                                                                                                                                                                             |                 | kg   | 1     |       |      | 2    |      | 1    |
| Compressed air                                                                                                                                                                                                                                                                   |                 | Nm3  | 3600  | 3640  | 4400 | 4400 | 3500 | 3900 |
| Electrode graphite                                                                                                                                                                                                                                                               |                 | kg   | 1     | 0.1   | 1    | 1    | 0.3  | 1    |
| Oxygen                                                                                                                                                                                                                                                                           |                 | Nm3  | 363   | 152   | 300  | 310  | 208  | 290  |
| silica sand                                                                                                                                                                                                                                                                      |                 | kg   | 790   | 750   | 750  | 930  | 803  | 831  |
|                                                                                                                                                                                                                                                                                  | sulphuric acid  | t    | 3.6   | 1.95  | 3.6  | 2.7  | 1.94 | 2.52 |
|                                                                                                                                                                                                                                                                                  | slag            | t    | 1.78  | 1.755 | 1.85 | 1.9  | 1.65 | 180  |
|                                                                                                                                                                                                                                                                                  | off-gas         | Nm3  | 6500  | 8300  | 7000 | 7800 | 8300 | 7600 |
|                                                                                                                                                                                                                                                                                  | sulphur dioxide | kg   | 397   | 397   | 36   | 577  | 1297 | 458  |
| Refining                                                                                                                                                                                                                                                                         |                 |      |       |       |      |      |      |      |
| electricity                                                                                                                                                                                                                                                                      |                 | kWh  | 309   | 324   | 309  | 340  | 324  | 324  |
| electricity, grid                                                                                                                                                                                                                                                                |                 | kWh  | 124   | 130   | 124  | 136  | 130  | 130  |
| hydropower                                                                                                                                                                                                                                                                       |                 | kWh  | 185   | 194   | 185  | 204  | 194  | 194  |
| fossil energy (gas)                                                                                                                                                                                                                                                              |                 | MJ   | 11217 | 2468  | 2360 | 2541 | 2468 | 2468 |
|                                                                                                                                                                                                                                                                                  | off gas         | MJ   | 300   | 315   | 300  | 330  | 315  | 315  |
|                                                                                                                                                                                                                                                                                  | dross (dry)     | t    | 10    | 10    | 10   | 10   | 10   | 10   |
|                                                                                                                                                                                                                                                                                  | Compressed air  | t    | 300   | 315   | 300  | 330  | 315  | 315  |
| 1) Krauss et al. (1999) give different values for copper content in concentrates and for concentrate consumption (per t of copper) in the chapters for mining and for metallurgy. For consistency within the data, the values given in the mining chapter are used in ecoinvent. |                 |      |       |       |      |      |      |      |

**Tab. 5.9 Metal emission factors to air for primary copper production (EEA (1999)).** Chosen value is the average of the given range. Missing ranges – tagged with \* – have been extrapolated from limited / impact control with the average factor of 20. Values for “impact control” apply only to RNA and RER.

| Compound  | impact control (g / t Cu) |             | limited control (g / t Cu) |             |
|-----------|---------------------------|-------------|----------------------------|-------------|
| Arsenic   | 32.5                      | (15-50)     | 1000                       | (500-1500)  |
| Chromium  | 0.05                      |             | 1                          |             |
| Cadmium   | 6.5                       | (3-10)      | 350                        | (200-500)   |
| Copper    | 250                       | (200-300)   | 2750                       | (1500-4000) |
| Indium    | 0.125                     | (0.05-0.2)* | 2.5                        | (1-4)       |
| Manganese | 15                        | (5-25)*     | 300                        | (100-500)   |
| Nickel    | 55                        | (10-100)*   | 1950                       | (900-3000)  |
| Lead      | 150                       | (50-250)*   | 2500                       | (1000-4000) |
| Antimony  | 5.5                       | (1-10)      | 125                        | (50-200)    |
| Selenium  | 5.5                       | (1-10)      | 100                        | (50-150)    |
| Tin       | 6.25                      | (2.5-10)    | 125                        | (50-200)    |
| Vanadium  | 0.375                     | (0.25-0.5)  | 7.5                        | (5-10)      |
| Zinc      | 150                       | (100-200)   | 750                        | (500-1000)  |
| Bismuth   | 7.5                       | (5-10)*     | 150                        | (100-200)   |
| Mercury   | 0.1                       | (0.1-0.1)   | 2                          | (2-2)*      |

**Tab. 5.10 Emissions to air per tonne copper.** Data was taken from similar processes in nickel (A) and copper (B) metallurgy (IPPC (2001)). The distribution of the particle size is taken from CEPMEIP (C), values for VOC from EEA (1999) (D).

| Substance       | Amount in kg / t<br>(this study) | Source     |
|-----------------|----------------------------------|------------|
| Dust            | 0.41                             | A, B       |
| PM<2.5          | 0.0005                           | C          |
| PM2.5-10        | 0.3                              | C          |
| PM>10           | 0.10                             | C          |
| SO <sub>2</sub> | See Tab. 5.8                     | estimation |
| CO              | 0.03                             | A          |
| VOC             | 0.015                            | D          |
| TCDD            | 1.00E-08                         | B          |

**Tab. 5.11** Composition of different wastewater effluents after treatment. A simple average is chosen. Data from European copper operations (IPPC (2001)).

| Sub-stance | Process Water         | Surface run off       | Direct cooling water    | This study (average)  |
|------------|-----------------------|-----------------------|-------------------------|-----------------------|
|            | [g / m <sup>3</sup> ] | [g / m <sup>3</sup> ] | [g / m <sup>3</sup> ]   | [g / m <sup>3</sup> ] |
| Cu         | 0.045<br>(0.01-0.2)   | 0.063<br>(0.01-0.4)   | 0.05<br>(0.01-0.25)     | 0.0527                |
| Pb         | 0.006<br>(0.001-0.04) | 0.032<br>(0.005-0.2)  | 0.01<br>(0.001-0.1)     | 0.0160                |
| As         | 0.032<br>(0.01-0.1)   | 0.014<br>(0.003-0.07) | 0.01<br>(0.001-0.1)     | 0.0187                |
| Ni         | 0.024<br>(0.004-0.15) | 0.028<br>(0.002-0.4)  | 0.011<br>(0.002-0.06)   | 0.0212                |
| Cd         | 0.003<br>(0.0001-0.1) | 0.004<br>(0.0002-0.1) | 0.001<br>(0.0001-0.003) | 0.00273               |
| Zn         | 0.045<br>(0.01-0.2)   | 0.11<br>(0.03-0.4)    | 0.1<br>(0.02-0.5)       | 0.0848                |
| Cr         |                       |                       |                         | 0.029                 |
| Hg         |                       |                       |                         | 0.00029               |
| Sn         |                       |                       |                         | 0.029                 |

### Data quality

Process data is satisfactory and valid for over 84 % of the primary copper production. The main emissions dominating possibly the whole process is sulphur dioxide, which itself is not reported in Krauss et al. (1999). Reported values range in two orders of magnitude. Poor data is available on composition and fate of solid waste, for which the reuse rate was not documented. High uncertainties in volume and composition of the effluents are assumed.

## 5. System Characterisation

| General Flow Information  |                                                  |               | Representation in ecoinvent                                                             |                        |                 |                 |          |                                                      |            | Uncertainty Informations |                       |      |                    |                 |
|---------------------------|--------------------------------------------------|---------------|-----------------------------------------------------------------------------------------|------------------------|-----------------|-----------------|----------|------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------|
| Input                     | Process Name                                     | Output        | Remarks                                                                                 | Category               | Sub-category    | Infra-structure | Location | Modul name in ecoinvent                              | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment |
| Compressed Air            | copper, primary, at refinery, GLO; pre-treatment |               | not inventoried in Ecoinvent, energy consumption comprised in chosen value              |                        |                 |                 |          |                                                      | 1.31E+00   | Nm3                      | (Krauss, 1999)        | 1    | 1.10               | (2,2,2,3,1,3,4) |
| Copper concentrate        |                                                  |               | 90.6% of copper input (rest by copper from SX-EW); 96.8% yield, concentrate at 29.7% Cu | metals                 | extraction      | No              | GLO      | copper, concentrate, at beneficiation                | 3.15E+00   | kg                       | (Krauss, 1999)        | 1    | 1.10               | (2,2,2,3,1,3,4) |
| electricity, grid         |                                                  |               | 40% electricity from grid                                                               | electricity            | production mix  | No              | UCTE     | electricity, high voltage, production UCTE, at grid  | 2.36E-02   | kWh                      | (Krauss, 1999)        | 1    | 1.10               | (2,2,2,3,1,3,2) |
| hydropower                |                                                  |               | 60% electricity from hydropower                                                         | hydro power            | power plants    | No              | RER      | electricity, hydropower, at run-of-river power plant | 3.53E-02   | kWh                      | (Krauss, 1999)        | 1    | 1.10               | (2,2,2,3,1,3,2) |
| fossile energy (gas)      |                                                  |               |                                                                                         | natural gas            | heating systems | No              | RER      | natural gas, burned in industrial furnace >100kW     | 8.88E-01   | MJ                       | (Krauss, 1999)        | 1    | 1.10               | (2,2,2,3,1,3,1) |
| limestone                 |                                                  |               |                                                                                         | construction materials | others          | No              | CH       | limestone, milled, packed, at plant                  | 2.26E-01   | kg                       | (Ayres, 2002)         | 1    | 1.51               | (5,2,2,3,1,3,4) |
| infrastructure metallurgy |                                                  |               |                                                                                         | metals                 | extraction      | Yes             | GLO      | non-ferrous metal, smelter                           | 9.57E-12   | unit                     | own assumptions       | 1    | 3.07               | (3,2,2,1,3,4,9) |
|                           |                                                  | off-gas       | not inventoried, no data                                                                |                        |                 |                 |          |                                                      | 1.49E+00   | Nm3                      | (Krauss, 1999)        | 1    |                    |                 |
|                           |                                                  | sulfuric acid | not inventoried, assumed to be burden free (cut off)                                    |                        |                 |                 |          |                                                      | 1.99E-01   | kg                       | (Krauss, 1999)        | 1    | 1.10               | (2,2,2,3,1,3,4) |
|                           |                                                  |               |                                                                                         |                        |                 |                 |          |                                                      |            |                          |                       |      |                    |                 |

Fig. 5.5 Flows for “copper, primary, at refinery, GLO – part pre-treatment” and its representation in the ecoinvent database. Values correspond to the functional unit of 1 kg copper.

## 5. System Characterisation

| General Flow Information   |                                              |                                 | Representation in ecoinvent                                                                              |                        |                            |                 |                                                                  |                                                                        |            | Uncertainty Informations |                                  |      |                    |                  |
|----------------------------|----------------------------------------------|---------------------------------|----------------------------------------------------------------------------------------------------------|------------------------|----------------------------|-----------------|------------------------------------------------------------------|------------------------------------------------------------------------|------------|--------------------------|----------------------------------|------|--------------------|------------------|
| Input                      | Process Name                                 | Output                          | Remarks                                                                                                  | Category               | Sub-category               | Infra-structure | Location                                                         | Modul name in ecoinvent                                                | Mean value | Unit                     | Source for mean value            | Type | Standard Deviation | General Comment  |
| silica sand                | copper, primary, at refinery, GLO; reduction |                                 |                                                                                                          | construction materials | additives                  | No              | DE                                                               | silica sand, at plant                                                  | 7.53E-01   | kg                       | (Krauss, 1999)                   | 1    | 1.51               | (5,2,2,3,1,3,4)  |
| electricity, grid          |                                              | 40% electricity from grid       | electricity                                                                                              | production mix         | No                         | UCTE            | electricity, high voltage, production UCTE, at grid              | 5.62E-02                                                               | kWh        | (Krauss, 1999)           | 1                                | 1.10 | (2,2,2,3,1,3,2)    |                  |
| hydropower                 |                                              | 60% electricity from hydropower | hydro power                                                                                              | power plants           | No                         | RER             | electricity, hydropower, at run of-river power plant             | 8.42E-02                                                               | kWh        | (Krauss, 1999)           | 1                                | 1.10 | (2,2,2,3,1,3,2)    |                  |
| fossile energy (oil heavy) |                                              |                                 | oil                                                                                                      | heating systems        | No                         | RER             | heavy fuel oil, burned in industrial furnace 1MW, non-modulating | 5.97E+00                                                               | MJ         | (Krauss, 1999)           | 1                                | 1.10 | (2,2,2,3,1,3,1)    |                  |
| fossile energy (gas)       |                                              |                                 | natural gas                                                                                              | heating systems        | No                         | RER             | natural gas, burned in industrial furnace >100kW                 | 3.29E-02                                                               | MJ         | (Krauss, 1999)           | 1                                | 1.10 | (2,2,2,3,1,3,1)    |                  |
| fossile energy (coke)      |                                              |                                 | hard coal                                                                                                | heating systems        | No                         | RER             | hard coal, burned in industrial furnace 1-10MW                   | 2.59E-02                                                               | MJ         | (Krauss, 1999)           | 1                                | 1.10 | (2,2,2,3,1,3,2)    |                  |
| Compressed air             |                                              |                                 |                                                                                                          |                        |                            |                 |                                                                  | 3.53E+00                                                               | Nm3        | (Krauss, 1999)           | 1                                | 1.48 | (4,4,4,5,3,5,3)    |                  |
| Electrode graphite         |                                              |                                 | metals                                                                                                   | extraction             | No                         | RER             | anode, aluminium electrolysis                                    | 9.06E-04                                                               | kg         | (Krauss, 1999)           | 1                                | 2.07 | (5,5,5,6,4,6,4)    |                  |
| Oxygen                     |                                              |                                 | chemicals                                                                                                | inorganics             | No                         | RER             | oxygen, liquid, at plant                                         | 2.63E-01                                                               | kg         | (Krauss, 1999)           | 1                                | 1.10 | (2,2,2,3,1,3,4)    |                  |
| water use                  |                                              |                                 | for overall process                                                                                      | resource               | in water                   |                 |                                                                  | Water, river                                                           | 4.41E-03   | m3                       | (Krauss, 1999)                   | 1    | 1.13               | (2,2,1,1,1,4,12) |
|                            |                                              | Effluent                        | Inventoried as wastewater treatment w/o pollution. For overall process, waterborn emission see Tab. 5.11 | waste management       | wastewater treatment       | No              | CH                                                               | treatment, sewage, unpolluted, to wastewater treatment, class 3        | 4.41E-03   | m3                       | (Hochfeld, 1999)                 | 1    | 1.13               | (2,2,1,1,1,4,13) |
|                            |                                              | slag                            | 50% recycling assumed --> disposal 50%. Proxy for Ni smelter slag used                                   | waste management       | residual material landfill | No              | CH                                                               | disposal, nickel smelter slag, 0% water, to residual material landfill | 8.15E-01   | kg                       | (Krauss, 1999)                   | 1    | 1.10               | (2,2,2,3,1,3,6)  |
|                            |                                              | off-gas                         | not inventoried, no data                                                                                 |                        |                            |                 |                                                                  |                                                                        | 6.88E+00   | Nm3                      | (Krauss, 1999)                   |      |                    |                  |
|                            |                                              | sulphur dioxide in off-gas      | comprise emissions from drying and roasting                                                              | air                    | low population density     |                 |                                                                  | Sulfur dioxide                                                         | 4.15E-01   | kg                       | (Krauss, 1999), (Anonymous 2000) | 1    | 1.64               | (5,4,1,3,3,5,15) |
|                            |                                              | carbon dioxide (lime)           | 0.44 kg / kg lime                                                                                        | air                    | low population density     |                 |                                                                  | Carbon dioxide, fossil                                                 | 9.96E-02   | kg                       | (Krauss, 1999)                   | 1    | 1.89               | (5,4,1,3,3,5,7)  |
|                            |                                              | Total dust                      | composition in Tab 5.9 and Tab 5.10                                                                      |                        |                            |                 |                                                                  |                                                                        | 9.53E-03   | kg                       | (IPPC, 2002)                     |      |                    |                  |
|                            |                                              | PM<2.5                          | see Tab 5.9 and Tab 5.10                                                                                 | air                    | low population density     |                 |                                                                  | Particulates, < 2.5 um                                                 | 4.60E-07   | kg                       | (IPPC, 2002)                     | 1    | 3.07               | (3,2,1,1,3,4,27) |
|                            |                                              | PM2.5-10                        | see Tab 5.9 and Tab 5.10                                                                                 | air                    | low population density     |                 |                                                                  | Particulates, > 2.5 um, and < 10um                                     | 2.76E-04   | kg                       | (IPPC, 2002)                     | 1    | 1.34               | (3,2,1,1,3,4,28) |
|                            |                                              | PM>10                           | see Tab 5.9 and Tab 5.10                                                                                 | air                    | low population density     |                 |                                                                  | Particulates, > 10 um                                                  | 9.21E-05   | kg                       | (IPPC, 2002)                     | 1    | 1.50               | (3,2,1,1,3,4,29) |
|                            |                                              | Carbon monoxide                 | see Tab 5.9 and Tab 5.10                                                                                 | air                    | low population density     |                 |                                                                  | Carbon monoxide, fossil                                                | 2.72E-05   | kg                       | (IPPC, 2002)                     | 1    | 5.08               | (3,2,1,1,3,4,22) |
|                            |                                              | VOC                             | see Tab 5.9 and Tab 5.10                                                                                 | air                    | low population density     |                 |                                                                  | NMVOC, non-methane volatile organic compounds, unspecified origin      | 1.36E-05   | kg                       | (IPPC, 2002)                     | 1    | 1.59               | (3,2,1,1,3,4,16) |
|                            |                                              | TCDD                            | see Tab 5.9 and Tab 5.10                                                                                 | air                    | low population density     |                 |                                                                  | Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-n-dioxin               | 1.81E-12   | kg                       | (IPPC, 2002)                     | 1    | 3.07               | (3,2,1,1,3,4,21) |

Fig. 5.6 Flows for “copper, primary, at refinery, GLO – part reduction” and its representation in the ecoinvent database. Values correspond to the functional unit of 1 kg copper.



## 5. System Characterisation

| General Flow Information |                                             |                       | Representation in ecoinvent                          |             |                        |                 |          |                                                      |            | Uncertainty Informations |                       |      |                    |                  |
|--------------------------|---------------------------------------------|-----------------------|------------------------------------------------------|-------------|------------------------|-----------------|----------|------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|------------------|
| Input                    | Process Name                                | Output                | Remarks                                              | Category    | Sub-category           | Infra-structure | Location | Modul name in ecoinvent                              | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment  |
| electricity, grid        | copper, primary, at refinery, GLO; refining |                       | 40% electricity from grid                            | electricity | production mix         | No              | UCTE     | electricity, high voltage, production UCTE, at grid  | 1.17E-01   | kWh                      | (Krauss, 1999)        | 1    | 1.10               | (2,2,2,3,1,3,2)  |
| hydropower               |                                             |                       | 60% electricity from hydropower                      | hydro power | power plants           | No              | RER      | electricity, hydropower, at run of-river power plant | 1.76E-01   | kWh                      | (Krauss, 1999)        | 1    | 1.10               | (2,2,2,3,1,3,2)  |
| fossile energy (gas)     |                                             |                       |                                                      | natural gas | heating systems        | No              | RER      | natural gas, burned in industrial furnace >100KW     | 2.24E+00   | MJ                       | (Krauss, 1999)        | 1    | 1.10               | (2,2,2,3,1,3,1)  |
|                          |                                             | off gas               | not inventoried, no data                             |             |                        |                 |          |                                                      | 2.85E-01   | MJ                       | (Krauss, 1999)        | 1    | 1.10               | (2,2,2,3,1,3,4)  |
|                          |                                             | sulfuric acid         | not inventoried, assumed to be burden free (cut off) |             |                        |                 |          |                                                      | 2.28E+00   | kg                       | (Krauss, 1999)        | 1    | 1.10               | (2,2,2,3,1,3,4)  |
|                          |                                             | dross (dry)           | internal recovery, not inventoried                   |             |                        |                 |          |                                                      | 9.06E-03   | kg                       | (Krauss, 1999)        |      |                    |                  |
|                          |                                             | waste heat (over all) |                                                      | air         | low population density |                 |          | Heat, waste                                          | 1.77E+00   | MJ                       |                       | 1    | 1.14               | (2,3,2,1,1,4,13) |
| copper from SX-EW        |                                             |                       |                                                      | metals      | extraction             | No              | GLO      | copper, SX-EW, at refinery                           | 9.42E-02   | kg                       | (Krauss, 1999)        | 1    | 1.10               | (2,2,2,3,1,3,1)  |
|                          |                                             | Copper Cathode        |                                                      | metals      | extraction             | No              | GLO      | copper, primary, at refinery                         | 1.00E+00   | kg                       | (Krauss, 1999)        | 1    |                    |                  |

Fig. 5.7 Flows for “copper, primary, at refinery, GLO – part refining” and its representation in the ecoinvent database. Values correspond to the functional unit of 1 kg copper.

## 5.3 Hydrolytic winning of primary copper

This part is mainly based on (Krauss et al. (1999)). The general process description was taken from (Fabian (1997)). Process specific emissions derived from (IPPC (2002)) and (IPPC (2001))

### 5.3.1 Production process and infrastructure

Oxide ores and supergene sulphide ores (i.e. ores not containing iron) can be recovered most easily by hydro-metallurgical techniques. The general steps of mining and refining are identical to those described in Section 5.1 Exploitation of copper ore. The difference in this type of production is in the beneficiation and the metallurgical step. Hydrometallurgical process is only applicable to oxidic ore, sulphides have to be roasted prior to leaching.

Several hydrometallurgical processes are in commercial operation for the treatment of copper mattes to copper cathodes. The dominant process in the copper production is the solvent extraction-electrowinning (SX-EW), which essentially by-passes the beneficiation and smelting stages, substituting a leaching stage followed by cementation or electro-winning.

Hydrometallurgical processes can generally be subdivided into the following steps (Fabian (1997)):

1. Pre-treatment, physical (e.g., comminution, grinding) or chemical (e.g., roasting)
  2. Leaching, chiefly with sulphuric acid or ammoniacal solutions, sometimes under oxidising conditions. In copper production, leaching is used in a number of ways:
    - In situ leaching, applied in loose ore bodies. In situ leaching recovers copper from small oxide ore bodies (too small to mine economically) or worked-out underground mines. Dump leaching is used to recover copper from large quantities (millions of tonnes) of strip mine waste with a very low grade. Dilute sulphuric acid is trickled through the material. Once the process starts it continues naturally if water and air are circulated through the heap. The time required is typically measured in years. Sulphur dioxide is emitted during such operations. Soluble copper is then recovered from drainage tunnels and ponds. Copper recovery rates vary from 30% to 70%.
    - Dump or heap leaching of lumpy or ground material. Heap leaching is essentially the same as dump leaching, except that it is applied to oxide ores, which are previously ground or crushed and heaped on a prepared surface.
    - Vat or percolation leaching of small-sized material. In vat leaching, oxide ores are crushed and placed in large concrete vats (up to 18,000 tonnes capacity), then flooded by concentrated sulphuric acid, which converts most of the copper to copper sulphate. Extraction of one tonne of copper from ore with a grade of 1% requires 4400 litres of 96% sulphuric acid. Vat leaching typically recovers 75% of the copper. The process is fairly rapid (hours to days).
    - Agitation or slime leaching of fine-grade material without pressure
    - Pressure leaching of fine-grade material with or without oxygen
- In practice, only a few proven chemicals, lixivants, are used: Dilute sulphuric acid for basic carbonates and sulphates, and silicates; Solutions of iron (III) salts (sulphate) as oxidising agents in sulphuric acid for sulphidic ore; Ammoniacal solutions of ammonium salts (carbonate) for oxidised ores. Recently, the bacteria-assisted leaching of low-grade copper ores and industrial wastes has been developed. Certain autotrophic bacteria accelerate the oxidation of copper and iron sulphides. Typical species are *Thiobacillus thiooxidans*, the reaction takes place slowly in atmospheric oxygen (Acid Rock Drainage), but the bacteria catalyse it.

3. Solution cleaning, e.g. by precipitation of impurities and filtration or selective enrichment of copper by solvent extraction or ion exchange. The solvent extraction process comprises two steps: selective extraction of copper from an aqueous leach solution into an organic phase (extraction circuit) and the re-extraction or stripping of the copper into dilute sulphuric acid to give a solution suitable for electro winning (stripping circuit).
4. Separation, precipitation of copper metal or copper compounds such as  $\text{Cu}_2\text{O}$ ,  $\text{CuS}$ ,  $\text{CuCl}$ ,  $\text{CuI}$ ,  $\text{CuCN}$ , or  $\text{CuSO}_4 \cdot 5 \text{H}_2\text{O}$  (crystallisation)
5. Post-treatment, e.g., fire-refining of crude cement copper or thermal reduction of copper compounds

However, because solvent extraction-electro-winning (SX-EW) requires ten times as much electrical energy as classical electrolytic refining ( $2000 - 2600 \text{ kWh t}^{-1}$ ), and because established solvent extraction techniques do not work well on copper sulphide ores, electro-winning techniques have remained complementary to smelting. Today about 10 – 15% of the world copper production is from SX-EW of copper oxide ores (Graedel et al. (2002) and Krauss et al. (1999)).

### 5.3.2 Emissions and wastes

Several emissions to air and water occur in the mining step. These emissions are identical to those in the conventional mining process and are shortly discussed there.

**Pre-treatment;** In the eventual roasting step to remove sulphides, considerable amounts of sulphur dioxide may be emitted into the air, since the concentration of sulphur dioxide in the off-gas probably is too small for recovery as sulphidic acid. Contrary to the beneficiation step of the pyrometallurgical process, no tailings develop additionally to the mining.

**Leaching;** For in-situ and heap leaching considerable amounts of sulphuric acid and leaching agents emit into water and air. No figures are currently available on the dimension of such emissions. After the solvent-solvent extraction, considerable amounts of leaching residues remain, which consist of undissolved minerals and the remainders of leaching chemicals.

**Electro-winning;** Most of the electrolyte and the sulphuric acid is recovered into the leaching step. Presumably no considerable wastes and emissions occur, but there are no available data.

### 5.3.3 Hydrolytic winning of primary copper in ecoinvent.

This part is mainly based on (Krauss et al. (1999)). The general process description was taken from (Fabian (1997)), but no process specific emissions were documented there.

The module describes the production of copper with the SX-EW process. It is designed for the use of the metal as part of the production mix “copper, primary, at plant”.

The modules with general flow information, remarks, sources, values and uncertainty information are shown in Fig. 5.8 and Fig. 5.9. Ecoinvent meta information for this process is summarised in Tab. 5.1 on page 17 ff. The following paragraphs describe calculations, sources and assumptions made in this study. The functional unit of this process is one kilogram of cathode copper.

**Mining;** Mining is similar to what is described in Section 5.1 Exploitation of copper-ore, but no beneficiation occurs. No exact information on the mining step is available in Krauss et al. (1999), therefore a mining process described in the previous section has been chosen as proxy. This is done with specific values for diesel use, explosives, and energy according to open pit mining. For further details please consult the Section mentioned above. Data on ore input, overburden, and gangue are documented in Krauss et al. (1999). Since no further information is available on presumably used crushing

and grinding operations of the material, they have been estimated with additional energy and mechanical wear (milling equipment) from the conventional copper-beneficiation in RLA. No use of conditioning agents is to be expected.

**Pre-treatment: Leaching and extraction;** Following Krauss et al. (1999), heap leaching is considered as leaching process. Of the total electricity use of 600 kW /t Cu, 250 kWh/t Cu and 350 kWh/t Cu are required for leaching (sprinkling) and extracting (pumping), respectively. Krauss et al. (1999) state a loss of 171 m<sup>3</sup> / t Cu leachate through infiltration and evaporation, and 1.7 t / t Cu sulphuric acid and 276 t / t Cu leaching residues but their fate is not evident. Since no information on the magnitude of leachate loss into the ground exists, no emissions to water are considered. Thus additional emissions of heavy metals, and sulphate into the ground are to be expected. Leaching agents were inventoried with the inventory of ammonia production as proxy.

**Electro-winning and Refining;** Use of sulphuric acid was in no case inventoried, since it is treated in this study as by-product of pyrolytical winning of metals from sulphidic ores. No further emissions or wastes are considered important, since the electrolyte is fully recycled to the leaching stage. Losses in electrolyte and leachate are inventoried in the pre-treatment step.

### **Data quality**

There is not sufficient information for declaring this data set of good data quality. In particular, uncertainty for energy consumption in a likely additional crushing / milling step (not modelled herewith) and probable leaching to ground may lead to an underestimation of the environmental impacts. Since the energy consumption considered for electro-winning is much higher than in the pyrolytical processes, this will probably dominate the environmental impacts; thus, the disregard of beneficiation and leaching impacts may be compensated.

## 5. System Characterisation

| General Flow Information                    |                                                      |        | Representation in ecoinvent                                                           |                                                                      |                        |                            |          |                                                                           |                                                       | Uncertainty Informations |                       |      |                    |                  |                 |
|---------------------------------------------|------------------------------------------------------|--------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------|------------------------|----------------------------|----------|---------------------------------------------------------------------------|-------------------------------------------------------|--------------------------|-----------------------|------|--------------------|------------------|-----------------|
| Input                                       | Process Name                                         | Output | Remarks                                                                               | Category                                                             | Sub-category           | Infra-structure            | Location | Modul name in ecoinvent                                                   | Mean value                                            | Unit                     | Source for mean value | Type | Standard Deviation | General Comment  |                 |
| Copper in ground                            | copper, SX-EW, at refinery, GLO; mining and leaching |        | 70.0% yield, 554.0 kg overburden, 277.0 kg ore, 0.0 kg tailings per kg Cu concentrate | resource                                                             | in ground              |                            |          | Copper, 0.99% in sulfide, Cu 0.36% and Mo 8.2E-3% in crude ore, in ground | 1.43E+00 kg                                           | (Krauss, 1999)           |                       | 1    | 1.13               | (2,2,2,1,1,4,12) |                 |
| Land transformation from nature             |                                                      |        |                                                                                       | resource                                                             | land                   |                            |          | Transformation, from unknown                                              | 4.82E-04 m2                                           | (Martens 2002)           |                       | 1    | 2.06               | (3,2,2,1,3,2,8)  |                 |
| Land transformation to mine                 |                                                      |        |                                                                                       | resource                                                             | land                   |                            |          | Transformation, to mineral extraction site                                | 4.82E-04 m2                                           | (Martens 2002)           |                       | 1    | 2.06               | (3,2,2,1,3,2,8)  |                 |
| Land use for mining                         |                                                      |        |                                                                                       | resource                                                             | land                   |                            |          | Occupation, mineral extraction site                                       | 1.45E-02 m2a                                          | (Martens 2002)           |                       | 1    | 1.58               | (3,2,2,1,3,2,7)  |                 |
| explosive                                   |                                                      |        |                                                                                       | construction processes                                               | civil engineering      | No                         | RER      | blasting                                                                  | 2.77E-01 kg                                           | (Krauss, 1999)           |                       | 1    | 1.25               | (2,2,2,1,3,4,4)  |                 |
| diesel fuel (energy)                        |                                                      |        |                                                                                       | construction processes                                               | machinery              | No                         | GLO      | diesel, burned in building machine                                        | 8.31E+00 MJ                                           | (Krauss, 1999)           |                       | 1    | 1.25               | (2,2,2,1,3,4,4)  |                 |
| electricity from grid, mining               |                                                      |        |                                                                                       | Assumption: 40% electricity from grid                                | electricity            | production mix             | No       | UCTE                                                                      | electricity, medium voltage, production UCTE, at grid | 6.08E-01 kWh             | (Krauss, 1999)        |      | 1                  | 1.25             | (2,3,2,1,3,4,2) |
| hydropower, mining                          |                                                      |        |                                                                                       | Assumption: 60% hydropower                                           | hydro power            | power plants               | No       | RER                                                                       | electricity, hydropower, at run of-river power plant  | 9.13E-01 kWh             | (Krauss, 1999)        |      | 1                  | 1.25             | (2,3,2,1,3,4,2) |
| electricity, from grid, beneficiation       |                                                      |        |                                                                                       | Assumption: 40% electricity from grid, of total 15.1 kWh / t raw ore | electricity            | production mix             | No       | UCTE                                                                      | electricity, medium voltage, production UCTE, at grid | 1.67E+00 kWh             | assumption            |      | 1                  | 1.58             | (4,2,2,3,4,4,2) |
| hydropower, beneficiation                   |                                                      |        |                                                                                       | Assumption: 60% electricity from grid, of total 15.1 kWh / t raw ore | hydro power            | power plants               | No       | RER                                                                       | electricity, hydropower, at run of-river power plant  | 2.51E+00 kWh             | assumption            |      | 1                  | 1.58             | (4,2,2,3,4,4,2) |
| steel                                       |                                                      |        |                                                                                       | Abrasion masticator                                                  | metals                 | extraction                 | No       | RER                                                                       | chromium steel 18/8, at plant                         | 1.91E-01 kg              | assumption            |      | 1                  | 1.20             | (4,2,2,3,4,4,4) |
| exploitation infrastructure, conveyor belts |                                                      |        |                                                                                       | exploration infrastructure, 2000 m belts assumed                     | construction processes | machinery                  | Yes      | RER                                                                       | conveyor belt, at plant                               | 2.10E-05 m               | own assumptions       |      | 1                  | 1.58             | (4,2,2,3,4,4,4) |
| infrastructure of mining (surface)          |                                                      |        |                                                                                       | including crushing / grinding facilities                             | metals                 | extraction                 | Yes      | GLO                                                                       | non-ferrous metal, mine, surface                      | 1.39E-08 unit            | own assumptions       |      | 1                  | 3.07             | (3,2,2,1,3,4,9) |
|                                             |                                                      |        | overburden, disposed                                                                  |                                                                      | waste management       | residual material landfill | No       | GLO                                                                       | disposal, sulfidic tailings, off-site                 | 5.54E+02 kg              | (Krauss, 1999)        |      | 1                  | 1.13             | (2,2,2,1,1,4,6) |
|                                             |                                                      |        | overburden, refilled                                                                  | no overburden refilled according to (Krauss 1999)                    |                        |                            |          |                                                                           |                                                       | 0.00E+00 kg              | (Krauss, 1999)        |      | 1                  | 1.13             | (2,2,2,1,1,4,6) |
|                                             |                                                      |        | effluents, mining                                                                     | no data, effluents probable, not inventoried                         |                        |                            |          |                                                                           |                                                       | 0.00E+00 m3              |                       |      | 1                  | 1.13             | (2,2,2,1,1,4,6) |

Fig. 5.8 Flows for “copper, SX-EW, at refinery – part mining” and its representation in the ecoinvent database. Values correspond to the functional unit of 1 kg copper.

## 5. System Characterisation

| General Flow Information |                                                              |                            | Representation in ecoinvent                                        |             |                        |                 |          |                                                      |               | Uncertainty Informations |                       |      |                    |                  |
|--------------------------|--------------------------------------------------------------|----------------------------|--------------------------------------------------------------------|-------------|------------------------|-----------------|----------|------------------------------------------------------|---------------|--------------------------|-----------------------|------|--------------------|------------------|
| Input                    | Process Name                                                 | Output                     | Remarks                                                            | Category    | Sub-category           | Infra-structure | Location | Modul name in ecoinvent                              | Mean value    | Unit                     | Source for mean value | Type | Standard Deviation | General Comment  |
| electricity, grid        | copper, SX-EW, at refinery, GLO; pre-treatment and reduction |                            | 40% electricity from grid                                          | electricity | production mix         | No              | UCTE     | electricity, high voltage, production UCTE, at grid  | 2.40E-01 kWh  |                          | (Krauss, 1999)        | 1    | 1.10               | (2,2,2,3,1,3,2)  |
| hydropower               |                                                              |                            | 60% electricity from hydropower                                    | hydro power | power plants           | No              | RER      | electricity, hydropower, at run of-river power plant | 3.60E-01 kWh  |                          | (Krauss, 1999)        | 1    | 1.10               | (2,2,2,3,1,3,2)  |
| sulfuric acid (leaching) |                                                              |                            | not inventoried, assumed to be by-product from pyrolytical winning |             |                        |                 |          |                                                      | 1.70E+00 kg   |                          |                       |      |                    |                  |
| water use                |                                                              |                            |                                                                    | resource    | in water               |                 |          | Water, river                                         | 1.70E-01 m3   |                          | (Krauss, 1999)        | 1    | 1.13               | (2,2,1,1,1,4,12) |
| leaching agents          |                                                              |                            | approximated with ammonia inventory                                | chemicals   | inorganics             | No              | RER      | ammonia, liquid, at regional storehouse              | 1.70E-03 kg   |                          | (Hilbrans 1999)       | 1    | 1.10               | (2,2,2,3,1,3,4)  |
| infrastructure leaching  |                                                              |                            | chemical plant chosen as proxy                                     | chemicals   | organics               | Yes             | RER      | chemical plant, organics                             | 4.00E-10 unit |                          | own assumptions       | 1    | 3.07               | (3,2,2,1,3,4,9)  |
|                          |                                                              | leaching residues          | not inventoried, no data                                           |             |                        |                 |          |                                                      | 2.76E+02 kg   |                          | (Krauss, 1999)        | 1    |                    |                  |
| electricity, grid        |                                                              |                            | 40% electricity from grid                                          | electricity | production mix         | No              | UCTE     | electricity, high voltage, production UCTE, at grid  | 9.12E-01 kWh  |                          | (Krauss, 1999)        | 1    | 1.10               | (2,2,2,3,1,3,2)  |
| hydropower               |                                                              |                            | 60% electricity from hydropower                                    | hydro power | power plants           | No              | RER      | electricity, hydropower, at run of-river power plant | 1.37E+00 kWh  |                          | (Krauss, 1999)        | 1    | 1.10               | (2,2,2,3,1,3,2)  |
|                          |                                                              | waste heat (over all)      |                                                                    | air         | low population density |                 |          | Heat, waste                                          | 3.09E+01 MJ   |                          |                       | 1    | 1.14               | (2,3,2,1,1,4,13) |
|                          |                                                              | copper, SX-EW, at refinery |                                                                    | metals      | extraction             | No              | GLO      | copper, SX-EW, at refinery                           | 1.00E+00 kg   |                          | (Krauss, 1999)        |      |                    |                  |

Fig. 5.9 Flows for “copper, SX-EW, at refinery – part metallurgy” and its representation in the ecoinvent database. Values correspond to the functional unit of 1 kg copper.

## 5.4 Winning of secondary copper

Krauss et al. (1999) give a picture of imports of copper and copper concentrate to Germany. This mix is used in this study as proxy for Europe.

### 5.4.1 Production process and infrastructure

Secondary copper consists of various types of scrap. Prompt scrap is directly reused in foundries and is not further processed. Old scrap has to be treated in a secondary copper smelter, where a variety of metal values are recuperated. Depending on the chemical composition, the raw materials of a secondary copper smelter are processed in different types of furnaces, including:

- blast furnaces (up to 30% of Cu in the average charge),
- converters (about 75% Cu), and
- anode furnaces (about 95% Cu).

A scheme of the process considered is given in Fig. 5.10. The blast furnace metal (“black copper”) is treated in a converter; then, the converter metal is refined in an anode furnace. In each step additional raw material with corresponding copper content is added. In the blast furnace, a mixture of raw materials, iron scrap, limestone and sand as well as coke is charged at the top. Air that can be enriched with oxygen is blown through the tuyeres. The coke is burnt and the charge materials are smelted under reducing conditions. Black copper and slag are discharged from tapholes (EEA (1999)).

The converters used in primary copper smelting, working on mattes containing iron sulphide, generate surplus heat and additions of scrap copper are often used to control the temperature. The converter provides a convenient and cheap form of scrap treatment, but often with only moderately efficient gas cleaning. Alternatively, hydrometallurgical treatment of scrap, using ammonia leaching, yields to solutions which can be reduced by hydrogen to obtain copper powder (EEA (1999)).

Alternatively, these solutions can be treated by solvent extraction to produce feed to a copper-winning cell.

Converter copper is charged together with copper raw materials in anode furnace operation. For smelting the charge, oil or coal dust is used, mainly in reverberatory furnaces. After smelting, air is blown on the bath to oxidise the remaining impurities.

Leaded brasses, containing as much as 3% of lead, are widely used in various applications and recycling of their scrap waste is an important activity. Such scrap contains usually much swarf and turnings coated with lubricant and cutting oils. Copper-containing cables and motors contain plastic or rubber insulants, varnishes, and lacquers. In such cases, scrap needs pre-treatment to remove these non-metallic materials. The smaller sizes of scrap can be pre-treated thermally in a rotary kiln provided with an after-burner to consume smoke and oil vapours (so-called Intal process). There are also various techniques available to remove rubber and plastic insulations of electric cables.

### 5.4.2 Emissions and waste

The stages used for secondary copper production are generally similar to those for primary production but the raw material is usually oxidic or metallic and process conditions are therefore varied. Other than in the primary copper production elevated levels of halogenated organic compounds may arise, such as TCDD. Slags are usually used in construction.

### 5.4.3 Winning of secondary copper in ecoinvent

This part is mainly based on Rentz et al. (1999), who described one single large plant. These values are added with Bruch et al. (1995).

The module describes the production of secondary copper in Germany as proxy for Europe. It is designed for the use of the metal as part of the supply mix “copper, at regional storage”. It is not to be used if the impact of the secondary copper is expected to be considerable in a LCA study. In such cases, a more detailed analysis has to be conducted.

The modules with general flow information, remarks, sources, values and uncertainty information are shown in Fig. 5.11 and Fig. 5.12. Ecoinvent meta information for this process is summarised in Tab. 5.1 on page 17 ff.

The values for material input and output was taken from Rentz et al. (1999), which describe the worlds biggest plant for secondary production (Hüttenwerke Kayser AG, GER). No allocation to other by-products like copper sulphate has been considered, but the whole ecological burden is attributed to the main product secondary copper cathodes. Electricity demand for electrolysis was taken from Bruch et al. (1995), whose fossil energy demand for smelting (9.21 GJ / t Cu) is similar to this used in this study (overall 9.1 GJ / t Cu). Copper scrap was modelled with a proxy module for the collection and handling of iron scrap. Since scrap does not count as resource according to ecoinvent cut-off rule – i. e. no inventory entry –, the different scrap origin (iron instead of copper) does not play a role. About 10 % of blister copper is used from primary copper production into secondary production.

Emissions were taken from Rentz et al. (1999) and added with values reported in EEA (1999) (Tab. 5.12 and Tab. 5.13). Effluent volume was mentioned in Bruch et al. (1995) with 1 m<sup>3</sup> / t Cu. This volume is represented in the dataset as “treatment of clean waste water”, thus no emissions to water are inventoried. The emissions into water – i. e. after waste water treatment – are accounted for in the this dataset as direct emissions. The specific emission factor for TCDD (50 µg / t Cu) was reported in EC (2002).

#### Data quality

The overall quality is sufficient for the use proposed. But poor data exist for collection and beneficiation of old scrap and no data on leaching agents and emissions from Hydrolytic processes were available. However, the bias should be negligible because of high level of throughputs and energy demand.



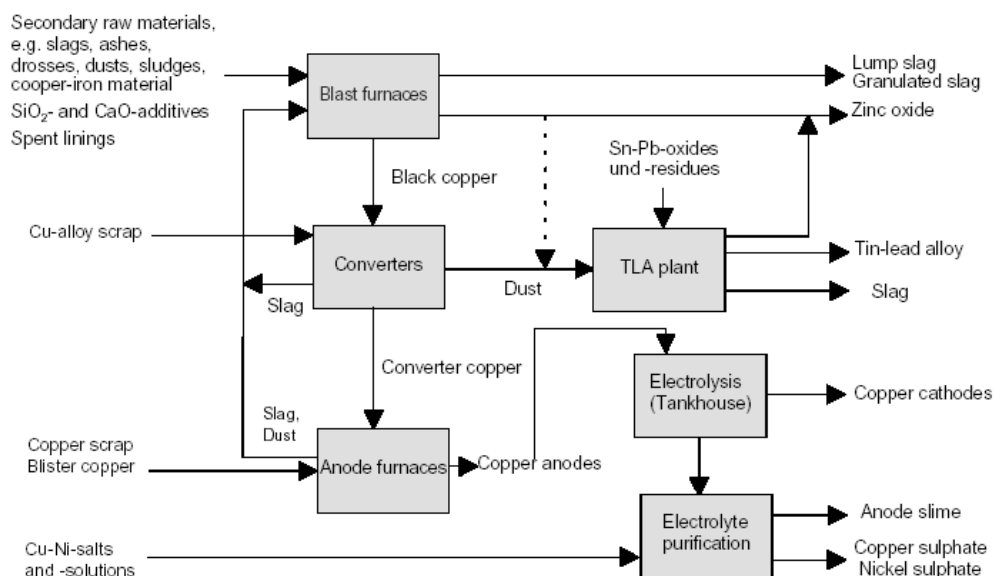


Fig. 5.10 Secondary copper production. Process scheme after Rentz et al. (1999). Scheme relates to the plant “Hüttenwerke Kayser AG”, Germany.

Tab. 5.12 Secondary copper production. Emissions to air from two different sources (EEA (1999) and Rentz et al. (1999)); the last column shows the value chosen in this study.

| Emission to air (g / t Cu) | (Rentz 1999) | (EEA 1999) | This study |
|----------------------------|--------------|------------|------------|
| Particulate matter         | 470          |            | 470        |
| PM<2.5                     | 282          |            | 282        |
| PM2.5-10                   | 94           |            | 94         |
| PM>10                      | 94           |            | 94         |
| SO2                        | 3000         |            | 3000       |
| NOx                        | 1000         |            | 1000       |
| CO                         | 2000         |            | 2000       |
| Arsenic                    | 1            | 2          | 2          |
| Antimony                   |              | 3          | 3          |
| Cadmium                    |              | 3          | 3          |
| Copper                     | 50           | 85         | 85         |
| Lead                       | 50           | 90         | 90         |
| Nickel                     |              | 1          | 1          |
| Zinc                       | 100          | 375        | 375        |

**Tab. 5.13** Secondary copper production. Emissions to water according to Rentz et al. (1999) for one big plant (Hüttenwerke Kayser AG).

| Emission to water after treatment. [g / t] |      |
|--------------------------------------------|------|
| Copper                                     | 1    |
| Lead                                       | 0.5  |
| Arsenic                                    | 0.1  |
| Nickel                                     | 0.3  |
| Cadmium                                    | 0.01 |
| Zinc                                       | 0.2  |

## 5. System Characterisation

| General Flow Information      |                                             |        | Representation in ecoinvent                                                        |                                |                 |                 |          |                                                                  |                            | Uncertainty Informations |                       |                 |                    |                  |                 |
|-------------------------------|---------------------------------------------|--------|------------------------------------------------------------------------------------|--------------------------------|-----------------|-----------------|----------|------------------------------------------------------------------|----------------------------|--------------------------|-----------------------|-----------------|--------------------|------------------|-----------------|
| Input                         | Process Name                                | Output | Remarks                                                                            | Category                       | Sub-category    | Infra-structure | Location | Modul name in ecoinvent                                          | Mean value                 | Unit                     | Source for mean value | Type            | Standard Deviation | General Comment  |                 |
|                               |                                             |        |                                                                                    |                                |                 |                 |          |                                                                  |                            |                          |                       |                 |                    |                  |                 |
| Copper scrap                  | copper, secondary, at refinery, RER; inputs |        | proxy for collecting scrap                                                         | metals                         | extraction      | No              | RER      | iron scrap, at plant                                             | 7.39E-01 kg                |                          | (Rentz 1999)          | 1               | 1.13               | (2,4,2,3,1,nA,4) |                 |
| Blister copper                |                                             |        |                                                                                    | metals                         | extraction      | No              | RER      | copper, blister-copper, at primary smelter                       | 1.14E-01 kg                |                          | (Rentz 1999)          | 1               | 1.13               | (2,4,2,3,1,nA,4) |                 |
| Copper alloy scrap            |                                             |        | proxy for collecting scrap                                                         | metals                         | extraction      | No              | RER      | iron scrap, at plant                                             | 1.99E-01 kg                |                          | (Rentz 1999)          | 1               | 1.13               | (2,4,2,3,1,nA,4) |                 |
| Ext. Intermeiate materials    |                                             |        | proxy for collecting scrap                                                         | metals                         | extraction      | No              | RER      | iron scrap, at plant                                             | 2.27E-01 kg                |                          | (Rentz 1999)          | 1               | 1.13               | (2,4,2,3,1,nA,4) |                 |
| Copper-iron material          |                                             |        | proxy for collecting scrap                                                         | metals                         | extraction      | No              | RER      | iron scrap, at plant                                             | 1.42E-01 kg                |                          | (Rentz 1999)          | 1               | 1.13               | (2,4,2,3,1,nA,4) |                 |
| limestone                     |                                             |        |                                                                                    | construction materials         | others          | No              | CH       | limestone, milled, packed, at plant                              | 7.39E-02 kg                |                          | (Rentz 1999)          | 1               | 1.13               | (2,4,2,3,1,nA,4) |                 |
| silica                        |                                             |        |                                                                                    | construction materials         | additives       | No              | DE       | silica sand, at plant                                            | 6.25E-02 kg                |                          | (Rentz 1999)          | 1               | 1.13               | (2,4,2,3,1,nA,4) |                 |
| coke                          |                                             |        | Hu=28.6 MJ/kg -> 4063 MJ / t Cu                                                    | hard coal                      | heating systems | No              | RER      | hard coal, burned in industrial furnace 1-10MW                   | 4.06E+00 MJ                |                          | (Rentz 1999)          | 1               | 1.13               | (2,4,2,3,1,nA,1) |                 |
| coal                          |                                             |        | Hu=28.9 MJ/kg -> 2463 MJ / t Cu                                                    | hard coal                      | heating systems | No              | RER      | hard coal, burned in industrial furnace 1-10MW                   | 2.46E+00 MJ                |                          | (Rentz 1999)          | 1               | 1.13               | (2,4,2,3,1,nA,1) |                 |
| Fuel (oil)                    |                                             |        | Hu=41.2 MJ/kg -> 2575 MJ / t Cu                                                    | oil                            | heating systems | No              | RER      | heavy fuel oil, burned in industrial furnace 1MW, non-modulating | 2.58E+00 MJ                |                          | (Rentz 1999)          | 1               | 1.13               | (2,4,2,3,1,nA,1) |                 |
| Electricity, grid             |                                             |        | 11.34 GJ / t Cu primary energy in electricity, effectiveness 0.35 = 110 kWh / t Cu | electricity                    | production mix  | No              | UCTE     | electricity, high voltage, production UCTE, at grid              | 1.10E+00 kWh               |                          | (Bruch 1995)          | 1               | 1.13               | (2,4,2,3,1,nA,2) |                 |
| infrastructure pyrometallurgy |                                             |        |                                                                                    |                                | metals          | extraction      | Yes      | GLO                                                              | non-ferrous metal, smelter | 9.57E-12 unit            |                       | own assumptions | 1                  | 3.07             | (3,2,2,1,3,4,9) |
| infrastructure leaching       |                                             |        |                                                                                    | chemical plant chosen as proxy | chemicals       | organics        | Yes      | RER                                                              | chemical plant, organics   | 4.00E-10 unit            |                       | own assumptions | 1                  | 3.07             | (3,2,2,1,3,4,9) |

Fig. 5.11 Flows for “copper, secondary, at refinery” (inputs) and its representation in the ecoinvent database. Values correspond to the functional unit of 1 kg copper.

## 5. System Characterisation

| General Flow Information |                                              |                    | Representation in ecoinvent                                                                                                                          |                  |                            |                 |          |                                                                        |             | Uncertainty Informations |                       |      |                    |                   |
|--------------------------|----------------------------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------------|-----------------|----------|------------------------------------------------------------------------|-------------|--------------------------|-----------------------|------|--------------------|-------------------|
| Input                    | Process Name                                 | Output             | Remarks                                                                                                                                              | Category         | Sub-category               | Infra-structure | Location | Modul name in ecoinvent                                                | Mean value  | Unit                     | Source for mean value | Type | Standard Deviation | General Comment   |
|                          | copper, secondary, at refinery, RER; outputs | Slag               | internal use in building, not inventoried.                                                                                                           |                  |                            |                 |          |                                                                        | 4.55E-01 kg |                          | (Rentz 1999)          | 1    | 1.13               | (2,4,2,3,1,nA,6)  |
|                          |                                              | Anode slime        |                                                                                                                                                      | waste management | residual material landfill | No              | CH       | disposal, nickel smelter slag, 0% water, to residual material landfill | 5.68E-03 kg |                          | (Rentz 1999)          | 1    | 1.13               | (2,4,2,3,1,nA,6)  |
|                          |                                              | Particulate matter | Consists of different fractions (PM2.5, PM10, TSP) and heavymetals. Composition see Tab. 5.12                                                        |                  |                            |                 |          |                                                                        | 4.70E-04 kg |                          | (Rentz 1999)          | 1    | 1.52               | (2,4,2,3,1,nA,25) |
|                          |                                              | TCDD               |                                                                                                                                                      | air              | low population density     |                 |          | Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin               | 5.00E-11 kg |                          | (EC 2002)             | 1    | 10.00              | (2,4,2,3,1,nA,21) |
|                          |                                              | Effluents          | Inventoried as wastewater treatment w/o pollution. Pollution after treatment is inventoried extra according Tab 5.13. 1 m3 / t Cu effluents assumed. | waste management | wastewater treatment       | No              | CH       | treatment, sewage, unpolluted, to wastewater treatment, class 3        | 1.00E-03 m3 |                          | (Bruch 1995)          | 1    | 1.13               | (2,4,2,3,1,nA,6)  |
|                          |                                              | Lead-tin alloys    | by-product, no burden                                                                                                                                |                  |                            |                 |          |                                                                        | 2.10E-02 kg |                          | (Rentz 1999)          |      |                    |                   |
|                          |                                              | Zinc oxides        | by-product, no burden                                                                                                                                |                  |                            |                 |          |                                                                        | 5.11E-02 kg |                          | (Rentz 1999)          |      |                    |                   |
|                          |                                              | Nickel sulphate    | by-product, no burden                                                                                                                                |                  |                            |                 |          |                                                                        | 1.36E-02 kg |                          | (Rentz 1999)          |      |                    |                   |
|                          |                                              | Copper sulphate    | by-product, no burden                                                                                                                                |                  |                            |                 |          |                                                                        | 1.25E-02 kg |                          | (Rentz 1999)          |      |                    |                   |
|                          |                                              | waste heat         | from electricity, 100% to air                                                                                                                        |                  |                            |                 |          |                                                                        | 3.97E+00 kg |                          | calculation           | 1    | 1.13               | (2,4,2,3,1,nA,13) |
|                          |                                              | Copper cathodes    |                                                                                                                                                      | metals           | extraction                 | No              | RER      | copper, secondary, at refinery                                         | 1.00E+00 kg |                          | (Rentz 1999)          |      |                    |                   |

Fig. 5.12 Flows for “copper, secondary, at refinery” (outputs) and its representation in the ecoinvent database. Values correspond to the functional unit of 1 kg copper.

## 5.5 Copper, at regional storage

### 5.5.1 Copper, from imported concentrates, at refinery in ecoinvent

This module is identical to “Copper, primary, at refinery, RER” but without neither any input of copper concentrate, nor any input of SX-EW copper. In this way it can be used to be combined with imported concentrates, whose production – mining and beneficiation – otherwise would be counted double.

### 5.5.2 Copper, at regional storage in ecoinvent

This module is based on information on the origin and type of imported copper for Germany in 1994 cited in Krauss et al. (1999). The layout relates to Fig. 5.2.

The module characterises the copper supply in Germany in 1994. It is designed for the use of the metal in various technical applications such as alloys and construction material in European production processes.

The layout of the modules with general flow information, remarks, sources, values and uncertainty information are shown in Fig. 5.13. Ecoinvent meta information for this process is summarised in Tab. 5.1 on page 17 ff.

Data on the amount and origin of the imported primary copper and concentrate is given in Tab. 5.14 and Tab. 5.15, from which the transport service in tkm results. Krauss et al. (1999) estimate the secondary copper use with 650'000 t in 1994, where 50% went directly into the foundries and 50% into the secondary copper production.

For copper and concentrate supplied by countries with no specification of the origin (“other”), the respective modules for South America is used, but for transport distance the resulting mean distance from RER, RLA, and ID has been chosen.

For prompt scrap, which is used directly in the foundries, the respective module for collection and handling of iron scrap has been chosen as proxy.

#### Data quality

The overall quality of this module is fair, depending on accuracy of the consumption values. The system of copper production in its different pathways is far more complex than pictured in this study. A more comprehensive approach as shown in Bertram et al. (2002) may enhance the quality of the data.

**Tab. 5.14 Import of copper concentrate to Germany in 1994. In the last column values for transport service in tonnes and kilometres (tkm) are given (Krauss et al. (1999)). Per kg “copper, from imported concentrates” (=13.1% in the supplier mix “copper, at regional storage”) 3.38 kg concentrates in average are required (=0.441 kg concentrates per kg “copper, at regional storage”).**

| Origin      | RER      | RLA      | ID       | other    | total        | Per kg concentrate |
|-------------|----------|----------|----------|----------|--------------|--------------------|
| concentrate | 197669 t | 206235 t | 164316 t | 84612 t  | 652832 t     | 1 kg               |
| ship sea    | 2750 km  | 14268 km | 18800 km | 11939 km | 7.59E+09 tkm | 11.62 tkm          |
| ship river  |          | 412 km   |          | 412 km   | 1.20E+08 tkm | 0.184 tkm          |
| train       | 190 km   |          |          | 190 km   | 5.36E+07 tkm | 0.082 tkm          |
| lorry       | 45 km    | 310 km   | 10 km    | 122 km   | 8.48E+07 tkm | 0.13 tkm           |

**Tab. 5.15** Import of primary copper (=56% in “copper, at regional storage”) to Germany in 1994. In the last column values for transport service in tonnes and kilometres (tkm) are given (Krauss et al. (1999)).

| Origin     | RER      | RLA      | ID       | other    | imported | total        | Per kg Primary | Per kg mix |
|------------|----------|----------|----------|----------|----------|--------------|----------------|------------|
| copper     | 353281 t | 111078 t | 32584 t  | 138979 t | 193000 t | 635922 t     | 1 kg           | 0.56 kg    |
| ship sea   | 3071 km  | 14100 km | 20000 km | 12390 km |          | 5.02E+09 tkm | 6.06 tkm       | 3.40 tkm   |
| ship river | 87 km    |          |          | 87 km    |          | 4.28E+07 tkm | 0.052 tkm      | 0.029 tkm  |
| train      | 2000 km  | 550 km   | 1050 km  | 1200 km  | 200 km   | 1.01E+09 tkm | 1.22 tkm       | 0.681 tkm  |
| lorry      | 15 km    | 200 km   |          | 108 km   | 100 km   | 6.18E+07 tkm | 0.075 tkm      | 0.042 tkm  |

**Tab. 5.16** Copper supply in the European region and the resulting fractions in the supply dataset “copper, at regional storage, RER” (column “% of total”). Copper supply from imported concentrates means that the copper is produced within Europe (Germany being reference region) from concentrates that are extracted elsewhere (see Tab. 5.17).

| Copper supply              | primary Copper | secondary copper | % of primary | % of total |
|----------------------------|----------------|------------------|--------------|------------|
| RER                        | 353281 t       |                  | 43%          | 23.9%      |
| RLA                        | 111078 t       |                  | 13%          | 7.51%      |
| ID                         | 32584 t        |                  | 4%           | 2.20%      |
| Other                      | 138979 t       |                  | 17%          | 9.40%      |
| from imported concentrates | 193000 t       |                  | 23%          | 13.1%      |
| total primary copper       | 828922 t       |                  | 100%         | 56.0%      |
|                            |                |                  |              |            |
| Prompt scrap               |                | 325000 t         |              | 22.0%      |
| Old scrap                  |                | 325000 t         |              | 22.0%      |
| total secondary            |                | 650000 t         |              | 44.0%      |
|                            |                |                  |              |            |
| total copper               |                | 1478922 t        |              | 100%       |

**Tab. 5.17** Concentrate imported for the production of primary copper within Europe (Germany being reference region). The resulting concentrate requirements in the dataset “copper, at regional storage” are displayed in the last column “kg / kg ‘copper, at regional storage’”.

| Concen-<br>trate                               | Amount   | % from imported<br>concentrates | kg / kg imported con-<br>centrates | kg / kg “copper, at<br>regional storage” |
|------------------------------------------------|----------|---------------------------------|------------------------------------|------------------------------------------|
| RER                                            | 197669 t | 30%                             | 1.02                               | 0.134                                    |
| RLA                                            | 206235 t | 32%                             | 1.07                               | 0.139                                    |
| ID                                             | 164316 t | 25%                             | 0.85                               | 0.111                                    |
| other                                          | 84612 t  | 13%                             | 0.44                               | 0.057                                    |
|                                                |          |                                 |                                    |                                          |
| total                                          | 652832 t | 100%                            | 3.38                               | 0.441                                    |
|                                                |          |                                 |                                    |                                          |
| copper pro-<br>duced from<br>imported<br>conc. | 193000 t |                                 |                                    | 13.1%                                    |
| average grade                                  | 29.6%    |                                 |                                    |                                          |

## 5. System Characterisation

| General Flow Information           |                                      |        | Representation in ecoinvent                              |                   |              |                 |          |                                                 | Uncertainty Informations |      |                       |      |                    |                 |
|------------------------------------|--------------------------------------|--------|----------------------------------------------------------|-------------------|--------------|-----------------|----------|-------------------------------------------------|--------------------------|------|-----------------------|------|--------------------|-----------------|
| Input                              | Process Name                         | Output | Remarks                                                  | Category          | Sub-category | Infra-structure | Location | Modul name in ecoinvent                         | Mean value               | Unit | Source for mean value | Type | Standard Deviation | General Comment |
| Primary copper from RLA            | copper, at regional storage          |        |                                                          | metals            | extraction   | No              | RLA      | copper, primary, at refinery                    | 7.51E-02                 | kg   | (Krauss, 1999)        | 1    | 1.07               | (2,1,1,3,1,1,4) |
| primary copper from ID             |                                      |        |                                                          | metals            | extraction   | No              | ID       | copper, primary, at refinery                    | 2.20E-02                 | kg   | (Krauss, 1999)        | 1    | 1.07               | (2,1,1,3,1,1,4) |
| primary copper from RER            |                                      |        |                                                          | metals            | extraction   | No              | RER      | copper, primary, at refinery                    | 2.39E-01                 | kg   | (Krauss, 1999)        | 1    | 1.07               | (2,1,1,3,1,1,4) |
| primary copper other countries     |                                      |        | No information on origin. Proxy production in RLA chosen | metals            | extraction   | No              | RLA      | copper, primary, at refinery                    | 9.40E-02                 | kg   | (Krauss, 1999)        | 1    | 1.07               | (2,1,1,3,1,1,4) |
| import primary copper (sea)        |                                      |        |                                                          | transport systems | ship         | No              | OCE      | transport, transoceanic freight ship            | 3.40E+00                 | tkm  | (Krauss, 1999)        | 1    | 2.00               | (2,1,1,3,1,1,5) |
| import primary copper (river)      |                                      |        |                                                          | transport systems | ship         | No              | RER      | transport, barge tanker                         | 2.90E-02                 | tkm  | (Krauss, 1999)        | 1    | 2.00               | (2,1,1,3,1,1,5) |
| import primary copper (train)      |                                      |        |                                                          | transport systems | train        | No              | RER      | transport, freight, rail                        | 6.81E-01                 | tkm  | (Krauss, 1999)        | 1    | 2.00               | (2,1,1,3,1,1,5) |
| import primary copper (lorry)      |                                      |        |                                                          | transport systems | road         | No              | RER      | transport, lorry >16t, fleet average            | 4.18E-02                 | tkm  | (Krauss, 1999)        | 1    | 2.00               | (2,1,1,3,1,1,5) |
| copper concentrate RLA             |                                      |        |                                                          | metals            | extraction   | No              | RLA      | copper concentrate, at beneficiation            | 1.39E-01                 | kg   | (Krauss, 1999)        | 1    | 1.07               | (2,1,1,3,1,1,4) |
| copper concentrate ID              |                                      |        |                                                          | metals            | extraction   | No              | ID       | copper concentrate, at beneficiation            | 1.11E-01                 | kg   | (Krauss, 1999)        | 1    | 1.07               | (2,1,1,3,1,1,4) |
| copper concentrate RER             |                                      |        |                                                          | metals            | extraction   | No              | RER      | copper concentrate, at beneficiation            | 1.34E-01                 | kg   | (Krauss, 1999)        | 1    | 1.07               | (2,1,1,3,1,1,4) |
| copper concentrate other countries |                                      |        | No information on origin. Proxy production in RLA chosen | metals            | extraction   | No              | RLA      | copper concentrate, at beneficiation            | 5.72E-02                 | kg   | (Krauss, 1999)        | 1    | 1.07               | (2,1,1,3,1,1,4) |
| import copper concentrate (sea)    |                                      |        |                                                          | transport systems | ship         | No              | OCE      | transport, transoceanic freight ship            | 5.13E+00                 | tkm  | (Krauss, 1999)        | 1    | 2.00               | (2,1,1,3,1,1,5) |
| import copper concentrate (river)  |                                      |        |                                                          | transport systems | ship         | No              | RER      | transport, barge tanker                         | 8.10E-02                 | tkm  | (Krauss, 1999)        | 1    | 2.00               | (2,1,1,3,1,1,5) |
| import copper concentrate (train)  |                                      |        |                                                          | transport systems | train        | No              | RER      | transport, freight, rail                        | 3.63E-02                 | tkm  | (Krauss, 1999)        | 1    | 2.00               | (2,1,1,3,1,1,5) |
| import copper concentrate (lorry)  |                                      |        |                                                          | transport systems | road         | No              | RER      | transport, lorry >16t, fleet average            | 5.73E-02                 | tkm  | (Krauss, 1999)        | 1    | 2.00               | (2,1,1,3,1,1,5) |
| primary production DE              |                                      |        |                                                          | metals            | extraction   | No              | DE       | copper, from imported concentrates, at refinery | 1.31E-01                 | kg   | (Krauss, 1999)        | 1    | 1.07               | (2,1,1,3,1,1,4) |
| prompt scrap from RER              |                                      |        | as proxy collection and handling of iron scrap used.     | metals            | extraction   | No              | RER      | iron scrap, at plant                            | 2.20E-01                 | kg   | (Krauss, 1999)        | 1    | 1.07               | (2,1,1,3,1,1,4) |
| secondary copper from RER          |                                      |        |                                                          | metals            | extraction   | No              | RER      | copper, secondary, at refinery                  | 2.20E-01                 | kg   | (Krauss, 1999)        | 1    | 1.07               | (2,1,1,3,1,1,4) |
|                                    |                                      |        |                                                          | metals            | extraction   | No              | RER      | copper, at regional storage                     | 1.00E+00                 | kg   | (Krauss, 1999)        |      |                    |                 |
|                                    | copper, primary, at regional storage |        |                                                          |                   |              |                 |          |                                                 |                          |      |                       |      |                    |                 |

Fig. 5.13 Flows for “copper, at regional storage” and its representation in the ecoinvent database. Values correspond to the functional unit of 1 kg copper.

## 5.6 Copper alloys: brass bronze and brazing solders

From the multitude of different copper alloys the two most important are modelled as brass and bronze:

- Brass in ecoinvent is cartridge brass (70/30 brass, C26000) composed of 70% (w/w) copper and 30% (w/w) zinc.
- Bronze in ecoinvent is CuSn4 (C51000) composed of 94.97% (w/w) Cu, 5% (w/w) Sn and 0.03% (w/w) P.

From the multitude of brazing solders the general purpose cadmium-free solders are often used. In this class, the CuZn40 is an important material that is chosen as "typical" brazing solder in this study.

Production of copper alloys is basically done by melting the different metals and casting of ingots. No specific data are available for the processes. It is assumed that due to the high vapour pressure of zinc and tin, parts of these inputs arise as air emissions. It is further assumed that filters are commonly used to abate these emissions. The filter dust containing mainly zinc or tin is assumed to be recycled.

The casting of brass and bronze to the final form needs the moulds and energy for melting the alloy. Moulds are either made of metal (steel) or of sand.

### 5.6.1 Copper alloy production in ecoinvent

The three modules "brass, at plant", "bronze, at plant", and "brazing solder, cadmium free, at plant" include the material and energy inputs for melting the metals. The energy is calculated using the heat capacity and the enthalpy of fusion of the constituents (see Tab. 5.18). A mix of 50% natural gas, 40% heavy fuel oil, and 10% electricity is assumed for the heat production. A metal loss of 1% is assumed. This metal is recycled.

Tin and zinc emissions to air (after abatement) are assumed 0.1% (w/w) of the respective input.

The standard distances according to Frischknecht et al. (2003) for the transportation of the metal inputs are used.

The infrastructure is approximated by the infrastructure for aluminium casting. Thereby it is assumed that the amount of infrastructure corresponds to the volume of material throughput and thus is recalculated according to the different densities of aluminium and copper alloys.

The ecoinvent meta-information for the copper alloy production is given in Tab. 5.19 and the in- and output flows for "brass, at plant", "bronze, at plant", and "brazing solder, cadmium free, at plant" are given in Fig. 5.14, Fig. 5.15 and Fig. 5.16 respectively. Those for "casting, brass" and "casting, bronze" are given in Fig. 5.17 and Fig. 5.18 respectively.



Tab. 5.18 Calculation of energy consumption for brass and bronze production

|                                           | Copper | Zinc  | Tin   | Brass | Bronze | Solder | Unit     | Source             |
|-------------------------------------------|--------|-------|-------|-------|--------|--------|----------|--------------------|
| thermal capacity                          | 383    | 385   | 227   |       |        |        | J/(kg*K) | Binz et al. (1984) |
| heat of fusion (*10 <sup>5</sup> )        | 2.05   | 1.11  | 0.596 |       |        |        | J/kg     |                    |
| melting point                             | 1083   | 419.5 | 231.9 | 950   | 1050   | 900    | °C       |                    |
| production temp.                          | 1200   | 1200  | 1200  | 1100  | 1100   | 1100   | °C       | estimate           |
| energy input (theoretical)                | 0.665  | 0.573 | 0.332 | 0.643 | 0.654  | 0.634  | MJ/kg    | calculation        |
| energy input (for ecoinvent) <sup>1</sup> | 0.731  | 0.630 | 0.365 | 0.708 | 0.720  | 0.698  | MJ/kg    |                    |
| fraction Cu <sup>2</sup>                  | 1      | 0     | 0     | 0.707 | 0.9595 | 0.606  | -        | definition         |
| fraction Zn <sup>2</sup>                  | 0      | 1     | 0     | 0.303 | 0      | 0.404  | -        |                    |
| fraction Sn <sup>2</sup>                  | 0      | 0     | 1     | 0     | 0.0505 | 0      | -        |                    |

1 10% heat loss during heating and melting assumed

2 including 10% material loss for brass, bronze and solder

### 5.6.2 Data quality

Data uncertainty is derived using the Pedigree matrix as described in (Frischknecht et al. (2003)).

All data for the brass and bronze production are assumptions based on qualitative information and theoretical models. Thus uncertainty is rather high. These data are of insufficient quality for LCA where these metals play an important role.

Tab. 5.19 Meta information for the production processes of the modelled copper alloys

| Name                                  | brass, at plant                                                                                                                              | bronze, at plant                                                                                                                             | brazing solder, cadmium free, at plant                                                                                                        |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Location                              | CH                                                                                                                                           | CH                                                                                                                                           | RER                                                                                                                                           |
| Infrastructure Process                | 0                                                                                                                                            | 0                                                                                                                                            | 0                                                                                                                                             |
| Unit                                  | kg                                                                                                                                           | kg                                                                                                                                           | kg                                                                                                                                            |
| Data Set Version                      | 2.0                                                                                                                                          | 2.0                                                                                                                                          | 2.0                                                                                                                                           |
| Included Processes                    | copper and zinc including their melting and casting of brass ingots                                                                          | copper and tin including their melting and casting of bronze ingots                                                                          | copper and tin including their melting and casting of solder                                                                                  |
| Amount                                | 1                                                                                                                                            | 1                                                                                                                                            | 1                                                                                                                                             |
| Local Name                            | Messing, ab Werk                                                                                                                             | Bronze, ab Werk                                                                                                                              | Hartlot, cadmiumfrei, ab Werk                                                                                                                 |
| Synonyms                              |                                                                                                                                              |                                                                                                                                              |                                                                                                                                               |
| General Comment to reference function | Stands for brass with 70% Cu and 30% Zn. All data are calculated based on assumptions and theoretical models. Their overall quality is poor. | Stands for bronze with 95% Cu and 5% Sn. All data are calculated based on assumptions and theoretical models. Their overall quality is poor. | Stands for solder with 60% Cu and 40% Zn. All data are calculated based on assumptions and theoretical models. Their overall quality is poor. |
| CAS Number                            |                                                                                                                                              |                                                                                                                                              |                                                                                                                                               |
| Start Date                            | 2000                                                                                                                                         | 2000                                                                                                                                         | 2000                                                                                                                                          |
| End Date                              | 2000                                                                                                                                         | 2000                                                                                                                                         | 2000                                                                                                                                          |
| Data Valid For Entire Period          | 1                                                                                                                                            | 1                                                                                                                                            | 1                                                                                                                                             |
| Other Period Text                     |                                                                                                                                              |                                                                                                                                              |                                                                                                                                               |
| Geography text                        | Production data relate to the European average. Transports of inputs relate to the consumption in Switzerland                                | Production data relate to the European average. Transports of inputs relate to the consumption in Switzerland                                | Production data relate to the European average. Transports of inputs relate to the consumption in Europe                                      |
| Technology text                       | Assumed energy mix for melting. Abatement of air emissions assumed                                                                           | Assumed energy mix for melting. Abatement of air emissions assumed                                                                           | Assumed energy mix for melting. Abatement of air emissions assumed                                                                            |
| Representativeness [%]                |                                                                                                                                              |                                                                                                                                              |                                                                                                                                               |
| Production Volume                     |                                                                                                                                              |                                                                                                                                              |                                                                                                                                               |
| Sampling Procedure                    | based on literature and internet                                                                                                             | based on literature and internet                                                                                                             | based on literature and internet                                                                                                              |
| Extrapolations                        | none                                                                                                                                         | none                                                                                                                                         | none                                                                                                                                          |
| Uncertainty Adjustments               | none                                                                                                                                         | none                                                                                                                                         | none                                                                                                                                          |

Tab. 5.19 Meta information for the production processes (cont.)

| Name                                  | casting, brass                                                                                      | casting, bronze                                                                                     |
|---------------------------------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Location                              | CH                                                                                                  | CH                                                                                                  |
| Infrastructure Process                | 0                                                                                                   | 0                                                                                                   |
| Unit                                  | kg                                                                                                  | kg                                                                                                  |
| Data Set Version                      | 2.0                                                                                                 | 2.0                                                                                                 |
| Included Processes                    | Melting of copper and zinc and casting of brass parts. Metal input is not included.                 | Melting of copper and tin and casting of bronze parts. Metal input is not included.                 |
| Amount                                | 1                                                                                                   | 1                                                                                                   |
| Local Name                            | Giessen, Messing                                                                                    | Giessen, Bronze                                                                                     |
| Synonyms                              |                                                                                                     |                                                                                                     |
| General Comment to reference function | All data are calculated based on assumptions and theoretical models. Their overall quality is poor. | All data are calculated based on assumptions and theoretical models. Their overall quality is poor. |
| CAS Number                            |                                                                                                     |                                                                                                     |
| Start Date                            | 1996                                                                                                | 1996                                                                                                |
| End Date                              | 2003                                                                                                | 2003                                                                                                |
| Data Valid For Entire Period          | 1                                                                                                   | 1                                                                                                   |
| Other Period Text                     |                                                                                                     |                                                                                                     |
| Geography text                        | Data relate to the European average.                                                                | Data relate to the European average.                                                                |
| Technology text                       | Assumed energy mix for melting. Abatement of air emissions assumed                                  | Assumed energy mix for melting. Abatement of air emissions assumed                                  |
| Representativeness [%]                |                                                                                                     |                                                                                                     |
| Production Volume                     |                                                                                                     |                                                                                                     |
| Sampling Procedure                    | based on literature and internet                                                                    | based on literature and internet                                                                    |
| Extrapolations                        | none                                                                                                | none                                                                                                |
| Uncertainty Adjustments               | none                                                                                                | none                                                                                                |

## 5. System Characterisation

| General Flow Information |                 |            | Representation in ecoinvent                                                                 |                   |                 |                 |          |                                                       |            | Uncertainty Informations |                          |      |                    |                             |
|--------------------------|-----------------|------------|---------------------------------------------------------------------------------------------|-------------------|-----------------|-----------------|----------|-------------------------------------------------------|------------|--------------------------|--------------------------|------|--------------------|-----------------------------|
| Input                    | Process Name    | Output     | Remarks                                                                                     | Category          | Sub-category    | Infra-structure | Location | Modul name in ecoinvent                               | Mean value | Unit                     | Source for mean value    | Type | Standard Deviation | General Comment             |
| copper                   | brass, at plant |            |                                                                                             | metals            | extraction      | No              | RER      | copper, at regional storage                           | 7.07E-01   | kg                       | calculation              | 1    | 1.70               | (4,5,3,5,4,5,3)             |
| zinc                     |                 |            |                                                                                             | metals            | extraction      | No              | RER      | zinc for coating, at regional storage                 | 3.03E-01   | kg                       | calculation              | 1    | 1.70               | (4,5,3,5,4,5,3)             |
| heat from natural gas    |                 |            |                                                                                             | natural gas       | heating systems | No              | RER      | heat, natural gas, at industrial furnace >100kW       | 3.54E-01   | MJ                       | calculation / estimation | 1    | 1.70               | (4,5,3,5,4,5,1)             |
| heat from heavy fuel oil |                 |            |                                                                                             | oil               | heating systems | No              | RER      | heat, heavy fuel oil, at industrial furnace 1MW       | 2.83E-01   | MJ                       | calculation / estimation | 1    | 1.70               | (4,5,3,5,4,5,1)             |
| electricity              |                 |            |                                                                                             | electricity       | production mix  | No              | UCTE     | electricity, medium voltage, production UCTE, at grid | 1.97E-02   | kWh                      | calculation / estimation | 1    | 1.70               | (4,5,3,5,4,5,2)             |
| transport rail           |                 |            | 600 km                                                                                      | transport systems | train           | No              | RER      | transport, freight, rail                              | 6.06E-01   | tkm                      | estimation               | 1    | 2.09               | (4,5,n.A.,n.A.,n.A.,n.A.,5) |
| transport road           |                 |            | 50 km                                                                                       | transport systems | road            | No              | CH       | transport, lorry 28t                                  | 5.05E-02   | tkm                      | estimation               | 1    | 1.30               | (4,5,n.A.,n.A.,n.A.,n.A.,6) |
| infrastructure           |                 |            | Infrastructure for aluminium casing, corrected by density (Al 2700 kg/m3, Brass 8470 kg/m3) | metals            | extraction      | Yes             | RER      | aluminium casting, plant                              | 4.90E-11   | unit                     | calculation / estimation | 1    | 3.39               | (4,5,3,5,4,5,9)             |
|                          |                 | waste heat | from electricity use                                                                        | air               | unspecified     |                 |          | Heat, waste                                           | 7.08E-02   | MJ                       | calculation              | 1    | 1.70               | (4,5,3,5,4,5,13)            |
|                          |                 | Zn         | 0.1% of Zn input to air                                                                     | air               | unspecified     |                 |          | Zinc                                                  | 3.03E-04   | kg                       | calculation / estimation | 1    | 1.95               | (4,5,3,5,4,5,31)            |
|                          |                 | waste      | dust from filters to recycling --> not inventoried                                          |                   |                 |                 |          |                                                       | 9.80E-03   |                          | calculation / estimation |      |                    |                             |
|                          |                 | brass      |                                                                                             | metals            | extraction      | No              | CH       | brass, at plant                                       | 1.00E+00   | kg                       |                          |      |                    |                             |

Fig. 5.14 Flows for "brass, at plant" and its representation in the ecoinvent database

## 5. System Characterisation

| General Flow Information |                  |            | Representation in ecoinvent                                                                 |                   |                 |                 |          |                                                       |               | Uncertainty Informations |                       |      |                    |                             |
|--------------------------|------------------|------------|---------------------------------------------------------------------------------------------|-------------------|-----------------|-----------------|----------|-------------------------------------------------------|---------------|--------------------------|-----------------------|------|--------------------|-----------------------------|
| Input                    | Process Name     | Output     | Remarks                                                                                     | Category          | Sub-category    | Infra-structure | Location | Modul name in ecoinvent                               | Mean value    | Unit                     | Source for mean value | Type | Standard Deviation | General Comment             |
| copper                   | bronze, at plant |            |                                                                                             | metals            | extraction      | No              | RER      | copper, at regional storage                           | 9.60E-01 kg   | calculation              |                       | 1    | 1.70               | (4,5,3,5,4,5,3)             |
| tin                      |                  |            |                                                                                             | metals            | extraction      | No              | RER      | tin, at regional storage                              | 5.05E-02 kg   | calculation              |                       | 1    | 1.70               | (4,5,3,5,4,5,3)             |
| heat from natural gas    |                  |            |                                                                                             | natural gas       | heating systems | No              | RER      | heat, natural gas, at industrial furnace >100kW       | 3.60E-01 MJ   | calculation / estimation |                       | 1    | 1.70               | (4,5,3,5,4,5,1)             |
| heat from heavy fuel oil |                  |            |                                                                                             | oil               | heating systems | No              | RER      | heat, heavy fuel oil, at industrial furnace 1MW       | 2.88E-01 MJ   | calculation / estimation |                       | 1    | 1.70               | (4,5,3,5,4,5,1)             |
| electricity              |                  |            |                                                                                             | electricity       | production mix  | No              | UCTE     | electricity, medium voltage, production UCTE, at grid | 2.00E-02 kWh  | calculation / estimation |                       | 1    | 1.70               | (4,5,3,5,4,5,2)             |
| transport rail           |                  |            | 600 km                                                                                      | transport systems | train           | No              | RER      | transport, freight, rail                              | 6.06E-01 tkm  | estimation               |                       | 1    | 2.09               | (4,5,n.A.,n.A.,n.A.,n.A.,5) |
| transport road           |                  |            | 50 km                                                                                       | transport systems | road            | No              | CH       | transport, lorry 28t                                  | 5.05E-02 tkm  | estimation               |                       | 1    | 1.30               | (4,5,n.A.,n.A.,n.A.,n.A.,6) |
| infrastructure           |                  |            | Infrastructure for aluminium casing, corrected by density (Al 2700 kg/m3, Brass 8470 kg/m3) | metals            | extraction      | Yes             | RER      | aluminium casting, plant                              | 4.90E-11 unit | calculation / estimation |                       | 1    | 3.39               | (4,5,3,5,4,5,9)             |
|                          |                  | waste heat | from electricity use                                                                        | air               | unspecified     |                 |          | Heat, waste                                           | 7.20E-02 MJ   | calculation              |                       | 1    | 1.70               | (4,5,3,5,4,5,13)            |
|                          |                  | Sn         | 0.1% of Sn input to air                                                                     | air               | unspecified     |                 |          | Tin                                                   | 5.05E-05 kg   | calculation / estimation |                       | 1    | 1.95               | (4,5,3,5,4,5,31)            |
|                          |                  | waste      | dust from filters to recycling --> not inventoried                                          |                   |                 |                 |          |                                                       | 1.00E-02      | calculation / estimation |                       |      |                    |                             |
|                          |                  | bronze     |                                                                                             | metals            | extraction      | No              | CH       | bronze, at plant                                      | 1.00E+00 kg   |                          |                       |      |                    |                             |

Fig. 5.15 Flows for "bronze, at plant" and its representation in the ecoinvent database

## 5. System Characterisation

| General Flow Information |                                        |                | Representation in ecoinvent                                                                 |                   |                 |                 |          |                                                       |            | Uncertainty Informations |                          |      |                    |                             |
|--------------------------|----------------------------------------|----------------|---------------------------------------------------------------------------------------------|-------------------|-----------------|-----------------|----------|-------------------------------------------------------|------------|--------------------------|--------------------------|------|--------------------|-----------------------------|
| Input                    | Process Name                           | Output         | Remarks                                                                                     | Category          | Sub-category    | Infra-structure | Location | Modul name in ecoinvent                               | Mean value | Unit                     | Source for mean value    | Type | Standard Deviation | General Comment             |
| copper                   | brazing solder, cadmium free, at plant |                |                                                                                             | metals            | extraction      | No              | RER      | copper, at regional storage                           | 6.06E-01   | kg                       | calculation              | 1    | 1.70               | (4,5,3,5,4,5,3)             |
| zinc                     |                                        |                |                                                                                             | metals            | extraction      | No              | RER      | zinc for coating, at regional storage                 | 4.04E-01   | kg                       | calculation              | 1    | 1.70               | (4,5,3,5,4,5,3)             |
| heat from natural gas    |                                        |                |                                                                                             | natural gas       | heating systems | No              | RER      | heat, natural gas, at industrial furnace >100kW       | 3.49E-01   | MJ                       | calculation / estimation | 1    | 1.70               | (4,5,3,5,4,5,1)             |
| heat from heavy fuel oil |                                        |                |                                                                                             | oil               | heating systems | No              | RER      | heat, heavy fuel oil, at industrial furnace 1MW       | 2.79E-01   | MJ                       | calculation / estimation | 1    | 1.70               | (4,5,3,5,4,5,1)             |
| electricity              |                                        |                |                                                                                             | electricity       | production mix  | No              | UCTE     | electricity, medium voltage, production UCTE, at grid | 1.94E-02   | kWh                      | calculation / estimation | 1    | 1.70               | (4,5,3,5,4,5,2)             |
| transport rail           |                                        |                | 200 km                                                                                      | transport systems | train           | No              | RER      | transport, freight, rail                              | 2.02E-01   | tkm                      | estimation               | 1    | 2.09               | (4,5,n.A.,n.A.,n.A.,n.A.,5) |
| transport road           |                                        |                | 100 km                                                                                      | transport systems | road            | No              | RER      | transport, lorry 32t                                  | 1.01E-01   | tkm                      | estimation               | 1    | 1.30               | (4,5,n.A.,n.A.,n.A.,n.A.,6) |
| infrastructure           |                                        |                | Infrastructure for aluminium casing, corrected by density (Al 2700 kg/m3, Brass 8470 kg/m3) | metals            | extraction      | Yes             | RER      | aluminium casting, plant                              | 4.90E-11   | unit                     | calculation / estimation | 1    | 3.39               | (4,5,3,5,4,5,9)             |
|                          |                                        | waste heat     | from electricity use                                                                        | air               | unspecified     |                 |          | Heat, waste                                           | 6.98E-02   | MJ                       | calculation              | 1    | 1.70               | (4,5,3,5,4,5,13)            |
|                          |                                        | Zn             | 0.1% of Zn input to air                                                                     | air               | unspecified     |                 |          | Zinc                                                  | 4.04E-04   | kg                       | calculation / estimation | 1    | 1.95               | (4,5,3,5,4,5,31)            |
|                          |                                        | waste          | dust from filters to recycling --> not inventoried                                          |                   |                 |                 |          |                                                       | 9.70E-03   |                          | calculation / estimation |      |                    |                             |
|                          |                                        | brazing solder |                                                                                             | metals            | extraction      | No              | RER      | brazing solder, cadmium free, at plant                | 1.00E+00   | kg                       |                          |      |                    |                             |

Fig. 5.16 Flows for "brazing solder, cadmium free, at plant" and its representation in the ecoinvent database

## 5. System Characterisation

| General Flow Information |                |                  | Representation in ecoinvent                                                                 |             |                 |                 |          |                                                       |               | Uncertainty Informations |                          |      |                    |                  |
|--------------------------|----------------|------------------|---------------------------------------------------------------------------------------------|-------------|-----------------|-----------------|----------|-------------------------------------------------------|---------------|--------------------------|--------------------------|------|--------------------|------------------|
| Input                    | Process Name   | Output           | Remarks                                                                                     | Category    | Sub-category    | Infra-structure | Location | Modul name in ecoinvent                               | Mean value    | Unit                     | Source for mean value    | Type | Standard Deviation | General Comment  |
| heat from natural gas    | casting, brass |                  |                                                                                             | natural gas | heating systems | No              | RER      | heat, natural gas, at industrial furnace >100kW       | 3.54E-01 MJ   |                          | calculation / estimation | 1    | 1.70               | (4,5,3,5,4,5,1)  |
| heat from heavy fuel oil |                |                  |                                                                                             | oil         | heating systems | No              | RER      | heat, heavy fuel oil, at industrial furnace 1MW       | 2.83E-01 MJ   |                          | calculation / estimation | 1    | 1.70               | (4,5,3,5,4,5,1)  |
| electricity              |                |                  |                                                                                             | electricity | production mix  | No              | UCTE     | electricity, medium voltage, production UCTE, at grid | 1.97E-02 kWh  |                          | calculation / estimation | 1    | 1.70               | (4,5,3,5,4,5,2)  |
| infrastructure           |                |                  | Infrastructure for aluminium casing, corrected by density (Al 2700 kg/m3, Brass 8470 kg/m3) | metals      | extraction      | Yes             | RER      | aluminium casting, plant                              | 4.90E-11 unit |                          | calculation / estimation | 1    | 3.39               | (4,5,3,5,4,5,9)  |
|                          |                | waste heat       | from electricity use                                                                        | air         | unspecified     |                 |          | Heat, waste                                           | 7.08E-02 MJ   |                          | calculation              | 1    | 1.70               | (4,5,3,5,4,5,13) |
|                          |                | Zn               | 0.1% of zn input to air                                                                     | air         | unspecified     |                 |          | Zinc                                                  | 3.03E-04 kg   |                          | calculation / estimation | 1    | 1.95               | (4,5,3,5,4,5,31) |
|                          |                | waste            | dust from filters to recycling --> not inventoried                                          |             |                 |                 |          |                                                       | 9.70E-03      |                          | calculation / estimation |      |                    |                  |
|                          |                | casting of brass |                                                                                             | metals      | processing      | No              | CH       | casting, brass                                        | 1.00E+00 kg   |                          |                          |      |                    |                  |

Fig. 5.17 Flows for "casting, brass" and its representation in the ecoinvent database

## 5. System Characterisation

| General Flow Information |                 |                   | Representation in ecoinvent                                                                 |                 |                 |                 |                                                       |                                                 |               | Uncertainty Informations |                          |      |                    |                  |
|--------------------------|-----------------|-------------------|---------------------------------------------------------------------------------------------|-----------------|-----------------|-----------------|-------------------------------------------------------|-------------------------------------------------|---------------|--------------------------|--------------------------|------|--------------------|------------------|
| Input                    | Process Name    | Output            | Remarks                                                                                     | Category        | Sub-category    | Infra-structure | Location                                              | Modul name in ecoinvent                         | Mean value    | Unit                     | Source for mean value    | Type | Standard Deviation | General Comment  |
| heat from natural gas    | casting, bronze |                   |                                                                                             | natural gas     | heating systems | No              | RER                                                   | heat, natural gas, at industrial furnace >100kW | 3.60E-01 MJ   |                          | calculation / estimation | 1    | 1.70               | (4,5,3,5,4,5,1)  |
| heat from heavy fuel oil |                 |                   | oil                                                                                         | heating systems | No              | RER             | heat, heavy fuel oil, at industrial furnace 1MW       | 2.88E-01 MJ                                     |               | calculation / estimation | 1                        | 1.70 | (4,5,3,5,4,5,1)    |                  |
| electricity              |                 |                   | electricity                                                                                 | production mix  | No              | UCTE            | electricity, medium voltage, production UCTE, at grid | 2.00E-02 kWh                                    |               | calculation / estimation | 1                        | 1.70 | (4,5,3,5,4,5,2)    |                  |
| infrastructure           |                 |                   | Infrastructure for aluminium casing, corrected by density (Al 2700 kg/m3, Brass 8470 kg/m3) | metals          | extraction      | Yes             | RER                                                   | aluminium casting, plant                        | 4.90E-11 unit |                          | calculation / estimation | 1    | 3.39               | (4,5,3,5,4,5,9)  |
|                          |                 | waste heat        | from electricity use                                                                        | air             | unspecified     |                 |                                                       | Heat, waste                                     | 7.20E-02 MJ   |                          | calculation              | 1    | 1.70               | (4,5,3,5,4,5,13) |
|                          |                 | Sn                | 0.1% of Sn input to air                                                                     | air             | unspecified     |                 |                                                       | Tin                                             | 5.05E-05 kg   |                          | calculation / estimation | 1    | 1.95               | (4,5,3,5,4,5,31) |
|                          |                 | waste             | dust from filters to recycling --> not inventoried                                          |                 |                 |                 |                                                       |                                                 | 9.95E-03      |                          | calculation / estimation |      |                    |                  |
|                          |                 | casting of bronze |                                                                                             | metals          | processing      | No              | CH                                                    | casting, bronze                                 | 1.00E+00 kg   |                          |                          |      |                    |                  |

Fig. 5.18 Flows for "casting, bronze" and its representation in the ecoinvent database



## 5.7 Telluride and Silver containing slime as by-product

The chief source of tellurium is a by-product of electrolytic copper refining, where it concentrates into the anode slimes. This source accounts for over 90% of the supply, although tellurium is also recovered from the flue dusts and gases generated during the smelting of copper, bismuth and lead ores, and from lead refinery skimmings<sup>9</sup>.

Like tellurium, also silver accumulates in the anode slimes. Different sources report that the share of silver from copper extraction declined from around 30% in 1996 to 25% today (JMB (2002); Klapwijk et al. (2006); Renner et al. (2002)).

Since the composition of the arising slimes chiefly depends on the composition of the copper anodes, the content of silver and tellurium varies considerably (Tab. 5.20).

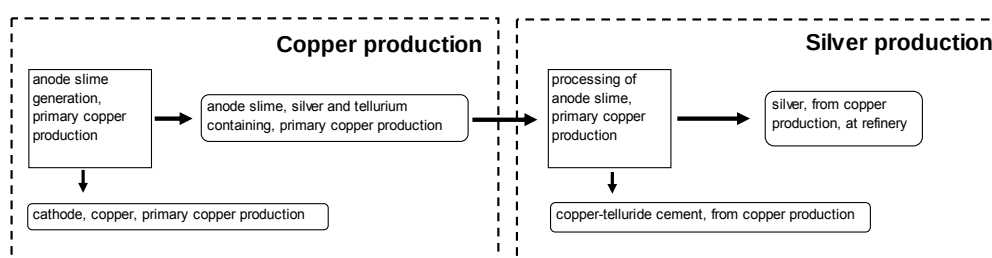
**Tab. 5.20** Characterisation of different anode slimes. Numbers are in %wt; span relates to reported minimal – maximal values; the mean value in Cooper (1990) is the median of the reported operations (n=18).

| Operations               | Ag                 | Te             | Se                | Cu                 | Source                 |
|--------------------------|--------------------|----------------|-------------------|--------------------|------------------------|
| Canadian smelter         | 0.2 - 3.5          | 4 - 25         | 4 - 15            | 16 - 24            | Chen & Dutrizac (2004) |
| Noranda slimes           | 0.52               | 10.9           | 21.3              | 20.3               | Knockaert (2002)       |
| Canadian copper refiners | 21.3               | 3.19           | 10.9              | 20.3               | Knockaert (2002)       |
| Nippon mining            | 20.58              | 3.64           | 15.23             | 4.73               | Knockaert (2002)       |
| Inco                     | 6                  | 2              | 7                 | 17                 | Knockaert (2002)       |
| 18 operations world-wide | 10.8<br>(4.1 - 62) | 1.1<br>(0 - 3) | 7.9<br>(0.6 - 46) | 21.3<br>(2.0 - 53) | Cooper (1990)          |

### 5.7.1 Anode slime generation in ecoinvent

The multi-output-process "anode slime generation, primary copper production" delivers the co-products "anode slime, silver and tellurium containing, primary copper production" and "cathode, copper, primary copper production". The product "cathode, ..." is an intermediary in the production of refined copper and is not to be used by the LCA practitioner. The flow "anode slime, ..." is a by-product of this process that receives part of the burdens allocated.. (Fig. 5.19).

The modules with general flow information, remarks, sources, values and uncertainty information are shown in Fig. 5.20. The ecoinvent meta information for this process is summarised in Tab. 5.21.



**Fig. 5.19** Process layout of the anode slime generation and its relation to other processes in ecoinvent.

<sup>9</sup> <http://www.mmta.co.uk/economicsFacts/miningJournalReview.aspx>, "Tellurium" in the Mining Journal Review, The Minor Metals Trade Association, accessed in July 2007.

Tab. 5.21 Meta information for the dataset anode slime generation, primary copper production.

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>                           | <b>anode slime generation, primary copper production</b>                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Location                              | GLO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Infrastructure Process                | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Unit                                  | kg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Data Set Version                      | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Included Processes                    | The module includes the upstream production of primary copper represented by the primary copper input and the resource depletion of silver and telluride found jointly with copper.                                                                                                                                                                                                                                                                                                                 |
| Amount                                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Local Name                            | Anodensumpf Erzeugung, primäre Kupferproduktion                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| General comment to reference function | The multi-output-process "anode slime generation, primary copper production" delivers the two co-products "anode slime, silver and tellurium containing, primary copper production" and "cathode, copper, primary copper production". The product "anode slime, ..." is an intermediary in the production of secondary gold from electric scrap and is not to be used by the LCA practitioner. The flow "cathode, ..." is a by-product of this process that receives part of the burdens allocated. |
| Start Date                            | 1993                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| End Date                              | 2003                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Geography text                        | Data refers to global copper production and hence reflects a global average.                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Technology text                       | Electrorefining of copper anode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Sampling Procedure                    | literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Extrapolations                        | see Geography and Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Uncertainty Adjustments               | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

Both reference flows are intended to be used as raw material in the production of the respective metals. For further information on process and inventory of the refined metals refer to the respective Parts within this report.

Based on anode slime compositions listed in Tab. 5.20, the anode slime is assumed to contain 20% residual copper, 20% silver and 3% tellurium. The nominal generation of this anode slime is harmonised with the world production average over the last three years (Tab. 2.4). Considering that 25% of silver (4'875 t/a average from 2004-2006) and 95% of tellurium (107.6 t/a average from 2004-2006) were supplied by the copper industry, 326 g refined silver and 7 g refined tellurium are produced per tonne copper that is extracted (Tab. 5.22). Given an extraction efficiency of 100% for silver (closed cycle) the amount of 1.63 kg anode slime (20% silver) per tonne cathode copper produced is calculated. Taking the Te content of 3% in the slime, 49 g Te per tonne copper extracted would then be co-extracted on a global average. However, this 49 g/t contrast with the 7 g/t Te that are estimated with a top-down calculation basing on mine production figures. This mismatch between bottom-up and top-down calculation can be explained by the fact, that in many cases the rarer but jointly extracted metals such as Tellurium are completely recovered on a global average. Many operations do not recover Tellurium at all. A second reason may be found in the limited extraction efficiency when leaching tellurium.

However, the allocation of the upstream burden from the copper production – which determines this inventory – is allocated by means of the effective produced amount of silver and tellurium from this source (top-down figures). Therefore the uncertainty about slime composition and completeness of the recovery of the resources in the ground do not affect this inventory model but the amount of silver and tellurium resources required (input flows from nature).

The upstream burden is reflected by the production of one kg copper on a global average; the associated burdens are allocated to the reference flows by their value. The elementary flows that represent

the depletion of the resource itself for tellurium and silver are introduced using the same extraction yields as for copper, since the metals are incorporated in the same mineral particles.

**Tab. 5.22 Production factors for the by-products silver and tellurium from copper production, three years average (USGS (2007)).**

| Year | Copper mine production, tpa | Silver from copper, tpa | Tellurium from copper, tpa | Silver per copper, t / t | Tellurium per copper, t / t |
|------|-----------------------------|-------------------------|----------------------------|--------------------------|-----------------------------|
| 2004 | 14'600'000                  | 4'925                   | 88.065                     | 3.37E-04                 | 6.03E-06                    |
| 2005 | 15'000'000                  | 4'825                   | 113.05                     | 3.22E-04                 | 7.54E-06                    |
| 2006 | 15'300'000                  | 4'875                   | 121.6                      | 3.19E-04                 | 7.95E-06                    |
| mean | 14'966'667                  | 4'875                   | 107.6                      | 3.26E-04                 | 7.17E-06                    |

### 5.7.2 Allocation and resource correction

The allocation split is done by the value represented by the reference flows “cathode copper” and “anode slime”. Since the anode slime has no market price as such, it is assessed by the marginal profit that is assumed to be in the order of the overall operations profit. A value of 10% of the price of the refined commodity (telluride and silver) has been chosen.

As explained in the preceding section the value of the anode slime is measured in the effective amount of silver and tellurium that is produced on a global average, including the extraction losses from the point where silver or tellurium is being extracted, i.e. CuTe precipitate. Therefore, and in accordance with Tab. 5.22, it is assumed that 326 g Ag and 7.97 g Te per tonne produced copper cathode are effectively extracted, assuming a closed-loop situation for silver (recycling the electrolyte to the process) and 90% for the preparation of CuTe as precipitate. The proceeds for refined silver and tellurium are at 269 \$/kg and 151 \$/kg (Tab. 5.23).

With this approach an internal value of 5.45 \$ per kg anode slime and 4.48 \$ per kg cathode copper and a respective allocation split of 0.2% of the burden to the 1.63 g of anode slime and the remaining 99.8% of the burden to the 1 kg of cathode copper.

Part of the copper resource that is represented by the upstream inventory for copper production is allocated by the economic procedure to the anode slime and has to be corrected.

The depletion of silver (326 g) and tellurium (49 g) per tonne of copper extracted is not accounted for in the upstream processes represented by the copper production. They therefore have to be introduced and are allocated fully to the anode slime.

Per kg copper produced, 1.12 kg copper resource is depleted. According to the allocation split 0.20% of this amount or 2.22E-3 kg is attributed to the anode slime and has to be subtracted and re-allocated to the cathode copper produced.

**Tab. 5.23** Average market price for copper (LME), silver and tellurium in \$ per tonne. Data compiled after USGS (2003) through 2004 and USGS (2007) for 2005/6.

|                                | <b>Silver</b> | <b>Copper</b> | <b>Tellurium</b> |
|--------------------------------|---------------|---------------|------------------|
|                                | \$ / t        | \$ / t        | \$ / t           |
| 1996                           | 166'900       | 2'404         | 46'300           |
| 1997                           | 157'200       | 2'358         | 41'900           |
| 1998                           | 178'000       | 1'734         | 39'700           |
| 1999                           | 169'000       | 1'674         | 33'100           |
| 2000                           | 161'000       | 1'944         | 30'900           |
| 2001                           | 140'000       | 1'694         | 33'100           |
| 2002                           | 148'000       | 1'671         | 35'300           |
| 2003                           | 157'000       | 1'879         | 37'500           |
| 2004                           | 207'000       | 2'910         | 22'000           |
| 2005                           | 238'880       | 3'704         | 211'644          |
| 2006                           | 360'088       | 6'812         | 220'462          |
| This study<br>(mean 2004-2006) | 268'656       | 4'475         | 151'369          |

### 5.7.3 Data quality

This inventory is mainly based on theoretical considerations to reflect the upstream burden and resources depletion in order to allocate them to the reference flow of anode slime. High uncertainties in the amount and allocation split are assumed.

## 5. System Characterisation

| General Flow Information     |                                                   |                                   | Representation in ecoinvent                                                                                                                   |          |              |                 |          |                                                                             |            | Uncertainty Informations |                       |      |                    |                 |
|------------------------------|---------------------------------------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------|-----------------|----------|-----------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------|
| Input                        | Process Name                                      | Output                            | Remarks                                                                                                                                       | Category | Sub-category | Infra-structure | Location | Modul name in ecoinvent                                                     | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment |
| Copper input                 | anode slime generation, primary copper production | by alloc split                    |                                                                                                                                               | metals   | extraction   | No              | GLO      | copper, primary, at refinery                                                | 1.00E+00   | kg                       |                       | 1    |                    |                 |
| Tellurium in copper resource |                                                   | to slime                          |                                                                                                                                               | resource | in ground    |                 |          | Tellurium, 0.5ppm in sulfide, Te 0.2ppm, Cu and Ag, in crude ore, in ground | 4.89E-05   | kg                       |                       | 1    | 1.09               | (2,2,1,2,1,3,3) |
| Silver in copper resource    |                                                   | to slime                          |                                                                                                                                               | resource | in ground    |                 |          | Silver, 3.2ppm in sulfide, Ag 1.2ppm, Cu and Te, in crude ore, in ground    | 3.26E-04   | kg                       |                       | 1    | 1.09               | (2,2,1,2,1,3,3) |
| Correction Copper            |                                                   | to slime                          | Necessary to correct the allocation of copper resource to the slime by economic allocation split. Subtract 0.20% of a total 1.12 kg required. | metals   | extraction   | No              | GLO      | resource correction, CuMo, copper, negative                                 | 2.22E-03   | kg                       |                       | 1    | 1.09               | (2,2,1,2,1,3,3) |
| Correction Copper            |                                                   | to copper                         | The subtracted amount is allocated to the cathode copper flow.                                                                                | metals   | extraction   | No              | GLO      | resource correction, CuMo, copper, positive                                 | 2.22E-03   | kg                       |                       | 1    | 1.09               | (2,2,1,2,1,3,3) |
|                              |                                                   | cathode, copper                   |                                                                                                                                               | metals   | extraction   | No              | GLO      | cathode, copper, primary copper production                                  | 1.00E+00   | kg                       |                       | 1    | 1.09               | (2,2,1,2,1,3,3) |
|                              |                                                   | Anode slime, from copper refining |                                                                                                                                               | metals   | extraction   | No              | GLO      | anode slime, silver and tellurium containing, primary copper production     | 1.63E-03   | kg                       |                       | 1    | 1.09               | (2,2,1,2,1,3,3) |
|                              |                                                   |                                   |                                                                                                                                               |          |              |                 |          |                                                                             |            |                          |                       |      |                    |                 |

Fig. 5.20 Flows for “anode slime generation, primary copper production” and its representation in the ecoinvent database. Values correspond to the functional unit of 1 kg copper.

## 5.8 Molybdenum

Molybdenite is the starting material for molybdenum trioxide and ferromolybdenum, which is the raw material for alloying steels with molybdenum. Since molybdenite is almost exclusively won from porphyry deposits, its production process is related to the exploitation of copper ores. Secondary porphyry deposits bear lower concentrations of molybdenum, in this case it is won only as co-product in the copper production (see Section 2.2 Primary Molybdenum on page 7). Molybdenum from co-production is modelled as co-product in Section 5.2 Winning primary copper. The ecological burden is distributed between molybdenite and copper concentrate according to their revenue (price times production).

According to the scheme in Fig. 5.1 a separate module for the exploitation of molybdenite as main product is necessary as well as a production mix module for molybdenite. These two modules are discussed in the following sections.

### 5.8.1 Exploitation of molybdenite in ecoinvent

The module describes the exploitation of molybdenite as main product. It is based entirely on the related process mining of copper ore. It is designed solely as part of a production mix. The multioutput-process “mining of molybdenite” delivers the two co-products “molybdenite concentrate, main product” and “copper, concentrate, coupleproduction Mo”.

The ecoinvent meta information for this process is summarised in Tab. 5.1 on page 17 ff.

The primary porphyry deposits are richest in molybdenum and are exploited for its molybdenum values. Hence the attribution of the ecological burden is different, since copper concentrate now is the by-product. Beside of the different resource demanded and the consequently different output of molybdenite – relating to the higher Mo content (0.041% in crude ore) – no other differences with respect to the Module “mining copper ore” (GLO) exist. Details are discussed in chapter 5.1 of this study.

#### Allocation

The ecological burden of the molybdenite mining is distributed among the coupled products copper concentrate and molybdenite concentrate. As allocation procedure the revenue is chosen. With a price of US\$ 5.75 per kg Mo (USGS (2003)) this gives a revenue of US\$ 3190 per tonne molybdenite concentrate with an assumed composition of 92.5% MoS<sub>2</sub>. Copper concentrate is traded in large quantities. However, the price of copper concentrate could not yet be assessed; it is estimated with the final bids of three auctions held in 2000 on an online web-auction<sup>10</sup> resulting in an average price of 15.3 US cents payable for refining and smelting per pound copper. With a copper price of 0.95 US\$/kg, a price of 522 US\$ per tonne of concentrate with 29.7% copper content is calculated. These assumptions and a production ratio of 74 kg molybdenite concentrate per 3.5 t copper concentrate lead to an average revenue of 236 US\$ and 1816 US\$ per tonne molybdenite and copper contained in the ore, respectively, – with the metallurgical yield included – and to an allocation factor of 11% and 89% for molybdenite and copper concentrate, respectively.

#### Data quality

The same constraints apply like for the module “mining of copper ore”. The assessment will highly vary with the price of copper concentrate and molybdenite. The yield of molybdenite is not evident; it is merely estimated based on the content of molybdenum in the ore.

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<sup>10</sup> <http://www.copperconcentrate.com/default.asp>, accessed at 01. 03. 2003

### 5.8.2 Molybdenite, at plant in ecoinvent

The module describes the production mix composed of molybdenite from coupled production in copper industry and molybdenite as main product.

Ecoinvent meta information for this process are summarised in Tab. 5.1 on page 17 ff.

Within the world-wide production of molybdenite a share of 58% originates from co-production in the copper industry, 42% originate from primary deposits. The molybdenite from scheelite leaching in the tungsten production is neglected in this study. Since molybdenite is a intermediary product, it is modelled as production mix, i.e. no transportation and energy use for the distribution is included in this module. Molybdenite from co-production is inventoried as “Molybdenum, molybdenite concentrate, couple-production Cu”, the main product molybdenite as “Molybdenum, molybdenite concentrate, main product”.

For the subsequent production of molybdenum oxide and later ferromolybdenum, transportation to the manufacturing plants and transport of the finished product to the user has to be considered.

#### Data quality

The main constraints in this module are missing information on the production process. The choice of copper production as proxy is a strong simplification. The overall assessment will highly vary with share – molybdenum as main- or by-product – and price of copper concentrate and molybdenite.

### 5.8.3 Molybdenum, at regional storage in ecoinvent

Molybdenum is produced from molybdenite by roasting to  $\text{MoO}_3$  and subsequent metallurgical steps. No process specific data for these steps have been found. Thus the roasting and metallurgical processes from the zinc production are used as a proxy. The concentrate of zinc and molybdenite are both sulphidic and the metal concentration is similar. Thus the inventory for the production of 1 kg "zinc, at regional storage" is used directly for describing the production of 1 kg molybdenum. To adjust the concentrate input, the amount of zinc concentrate that is needed in the production of zinc is subtracted and the molybdenite concentrate therefore inventoried. The amount of molybdenite concentrate is calculated assuming the same yield as for zinc concentrate to zinc. It is slightly lower than the amount of zinc concentrate, because the metal concentration is slightly higher (55% Mo vs. 53% Zn).

The layout of the modules with general flow information, remarks, sources, values and uncertainty information are shown in Fig. 5.21. The ecoinvent meta information for this process is summarised in Tab. 5.1 (page 17 ff).

#### Data quality

The very crude approximation of the previously described production of molybdenum with data from another metal implies a poor data quality. The data set must only be used if the relevance of molybdenum is low in a LCA study. If the focus of such a study would lie on molybdenum, the process would need to be further investigated.

## 5. System Characterisation

| General Flow Information                   |                                 |                  | Representation in ecoinvent                       |          |              |                 |          |                                              |            | Uncertainty Informations |                       |      |                    |                                             |
|--------------------------------------------|---------------------------------|------------------|---------------------------------------------------|----------|--------------|-----------------|----------|----------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|---------------------------------------------|
| Input                                      | Process Name                    | Output           | Remarks                                           | Category | Sub-category | Infra-structure | Location | Module name in ecoinvent                     | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                             |
| Molybdenite from coupled production        | molybdenite, at plant           | Molybdenite, mix |                                                   | metals   | extraction   | No              | GLO      | molybdenum concentrate, couple production Cu | 5.79E-01   | kg                       | (Krauss, 1999)        | 1    | 1.10               | (2,2,2,3,1,3,4)                             |
| Molybdenite from main production           |                                 |                  |                                                   | metals   | extraction   | No              | GLO      | Molybdenum concentrate, main product         | 4.21E-01   | kg                       | (Krauss, 1999)        | 1    | 1.10               | (2,2,2,3,1,3,2)                             |
|                                            |                                 |                  |                                                   | metals   | extraction   | No              | GLO      | molybdenite, at plant                        | 1.00E+00   | kg                       | (Krauss, 1999)        | 1    | 1.10               | (2,2,2,3,1,3,2)                             |
| Molybdenite                                | molybdenum, at regional storage | Molybdenum       | concentration 55%                                 | metals   | extraction   | No              | GLO      | molybdenite, at plant                        | 1.83E+00   | kg                       | calculated            | 1    | 2.15               | (4,5,12,5,5,12)                             |
| zinc                                       |                                 |                  |                                                   | metals   | extraction   | No              | RER      | zinc, primary, at regional storage           | 1.00E+00   | kg                       | definition            | 1    | 2.15               | (4,5,12,5,5,4)                              |
| zinc concentrate                           |                                 |                  | concentration 53%                                 | metals   | extraction   | No              | GLO      | zinc concentrate, at beneficiation           | -1.90E+00  | kg                       | calculated            | 1    | 1.00               | exact amount inventoried in zinc production |
| resource correction Cd in zinc concentrate |                                 |                  | 0.013004 kg Cd ore in 0.62565 kg Zn concentrate   | metals   | extraction   | No              | GLO      | resource correction, PbZn, cadmium, positive | 3.95E-02   | kg                       | calculated            | 1    | 1.00               | exact amount inventoried in zinc production |
| Resource correction In in zinc concentrate |                                 |                  | 0.00021673 kg In ore in 0.62565 kg Zn concentrate | metals   | extraction   | No              | GLO      | resource correction, PbZn, indium, positive  | 6.59E-04   | kg                       | calculated            | 1    | 1.00               | exact amount inventoried in zinc production |
|                                            |                                 |                  |                                                   | metals   | extraction   | No              | RER      | molybdenum, at regional storage              | 1.00E+00   | kg                       |                       |      |                    |                                             |
|                                            |                                 |                  |                                                   |          |              |                 |          |                                              |            |                          |                       |      |                    |                                             |

Fig. 5.21 Flows for “molybdenite, at plant” and “molybdenum, at regional storage” and their representation in ecoinvent.



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## Part IV

# Nickel

Data v2.1 (2009)

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## Comments on version 2.1

This part corresponds to the original report in Althaus et al. (2004) that has been updated with the changes reported in Frischknecht et al. (2006). The changes were implemented with v1.3 and affect mainly the amount of added lime and the herewith related emissions of carbon dioxide in the metallurgical step.

The amount of limestone input in the Ferronickel-Dataset was increased by factor 4. The CO<sub>2</sub> emission related to limestone decomposition in MO-Process (#35) is decreased a factor 16. The allocation factors applied on this emission (Values v1.2: 8.46% on Copper / 91.54% on Nickel) are corrected to 14% on Copper / 86% on Nickel. Further more. The amount of sulphidic tailings disposed in the MO-Process increased by 2/3.

In Version 2.1, the representation of part of the electricity input into the Ferronickel-dataset was corrected from UCTE mix to hydropower.

Chapter “6 Cumulative Results and Interpretation” of the original report has been deleted for the current update.

## Summary

The primary nickel and ferronickel production is modelled for the global situation.

Nickel is co-produced with Copper and the model is made accordingly. Allocation is made by revenue.

For a lack of data the secondary nickel production could not be inventoried. However, since most of the nickel is used in alloys it is not recycled separately but with the alloyed steel.

Mining, beneficiation and the different metallurgical processes are inventoried vertically aggregated because of difficulties to clearly allocate some of the flows to one single step and because the intermediate steps would not make sense in another context. The Figures in this report however contain the available information to the single steps.



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# 1 Introduction

This chapter describes the production of Nickel from cradle to consumer in Europe. Inventory data is taken mainly from (Hilbrans & Hinrichs (1999)) together with background data from IPPC (2002) and IPPC (2001). Market data on nickel and copper are provided by the London Metal Exchange<sup>1</sup>.

Since nickel always is associated with other metals, the production yields a fraction of other metals than nickel, mainly copper (Cu), cobalt (Co) and platinum group metals. The process of primary production described in this chapter is formally a multioutput-process with the coupled products Ni and Cu. This makes it necessary to allocate the inventory to these products. The demand for nickel-containing ore is allocated by mass. The rest of the inventory is allocated by revenue. The coupled production of cobalt is neglected because due to a lack of data no meaningful allocation factors could be established.

Known annual world production capacity of primary nickel products in the early 1990s was approximately 1240 kilotonnes, 485 of which were in former Soviet and East Bloc countries. In recent years, annual world production has averaged in excess of 900 kilotonnes. Primary nickel products fall broadly into two categories - Class I and Class II. Class I products contain at least 99.8% nickel whilst all other primary nickel products are Class II. Unlike Class I products, Class II products range widely in nickel content.

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<sup>1</sup> London Metal Exchange, daily and monthly stocks and prices under [http://www.lme.co.uk/data\\_prices/home.html](http://www.lme.co.uk/data_prices/home.html), accessed at 01.03.2003



## 2 Reserves and Resources of Nickel

Metals are a special class of materials. They are not consumed but rather used during a product life cycle. When recovered the properties are theoretically identical. The metal can be re-used in another product life cycle, provided that its concentration makes a recovery profitable. The recovered metal is referred to as secondary metal, whereas the metal originating directly from the ore is referred to as primary metal. In LCIs these two types – primary and secondary metal – are often differentiated because they bear a different ecological burden.

Since nickel is used in ecoinvent as intermediate product in different metal manufacturing industries, no secondary production is covered in this report. Secondary nickel is consumed directly in the form of re-melted scrap and other recycled products generally in the production of ferro-nickel and stainless steel. Other secondary materials such as catalysts and precipitator dusts are recovered in the primary smelting processes, usually in the slag furnace.

Nickel ores can be classified in a sulphidic type and an oxidic type. An overview over the different nickel bearing ores is given in Tab. 2.1, the composition of some ores in Tab. 2.2. Although 80 % of the world's known nickel resources on land are in oxide ore bodies, only 40 % of nickel production comes from this source. The greater part of the nickel produced is recovered from sulphide ores. The nickel content of sulphide ores can be concentrated relatively easily and cheaply by physical methods, while oxide ores must be treated in bulk. The potential for recovering the more numerous and valuable co-products also favours the exploitation of sulphide ore. However, the grades of sulphide ores are continually declining as the richer ores are consumed, and the cost of underground mining, particularly labour costs, is continually increasing. Inevitably therefore, oxide ore bodies, which are all relatively uniform in grade and can be surface mined, must in future support a larger share of nickel production (Kerfoot (1997)).

A detailed overview over the global factory production and a statistic on US use and production of nickel is available in the online version of the USGS “Mineral Commodity Summary” (USGS (2003)). The world-wide mine production by country in 2002 and the estimated reserves is listed in Tab. 2.3.

### 2.1 Sulfidic Ores

Sulphide nickel ores consist principally of nickeliferous pyrrhotite ( $\text{Fe}_7\text{S}_8$ ), pentlandite ( $(\text{Ni}, \text{Fe})_9\text{S}_8$ ), and chalcopyrite ( $\text{CuFeS}_2$ ). Pentlandite, the commonest of the sulphide minerals, probably accounts for 60 % of world nickel production. Nickeliferous pyrrhotite is normally the most abundant phase in a nickel ore. It contains nickel in solid solution (0.2 – 0.5 % Ni) in addition to very finely divided pentlandite inclusions. Chalcopyrite is the most important copper mineral, although cubanite is often present in lesser amounts. Sulphide ores typically grade 0.4 – 2.0 % nickel, 0.2 – 2.0 % copper, 10 – 30 % iron, and 5 – 20 % sulphur. The balance consists of silica, magnesia, alumina, and calcium oxide.

Although the major minerals are the same in all large sulphide ore bodies, the relative amounts of pyrrhotite, pentlandite, and chalcopyrite vary widely. The relative amounts of the three minerals largely determine the choice of processing route. In addition to nickel production, a large number of valuable co-products can be recovered from sulphide ores, including copper, cobalt, the platinum group metals, gold, silver, selenium, tellurium, sulphur, and iron. Cobalt occurs principally in solid solution in pentlandite, at concentrations ranging from 1 to 5 % of the nickel content.

Major sulphide ore bodies occur in Canada, the Soviet Union, the Republic of South Africa, Australia, Zimbabwe, and Finland. According to (Hilbrans & Hinrichs (1999)) 62 % of the world nickel production originates from sulfidic ore in 1994.

## 2.2 Laterite Ores: Limonite and Silicate

The oxidic ores of nickel are formed by a chemical concentration process that occurs as a result of the lateritic weathering of peridotite rock. The dissolved nickel and magnesium, and the colloidal silica, percolate downwards in the laterite deposit, where they precipitate as hydrated silicates.

Some of the nickel may in fact remain in the upper layer, which is thus enriched in iron and nickel, but depleted in magnesium and silica. Such material, consisting mainly of ferric oxide minerals, is described as limonitic. This type of ore is typically relatively rich in cobalt and chromium. By contrast, in the silicate minerals, the separation of iron and nickel is more complete, and the nickel is present in silicate minerals with high magnesia content. Both types of material are almost always found in a laterite deposit, but in widely differing proportions.

Oxide nickel mineral deposits are currently exploited in Australia, Brazil, Cuba, the Dominican Republic, Greece, Indonesia, the Philippines, New Caledonia, Yugoslavia, the United States, and the Soviet Union. The laterite deposits in Oregon and California were formed when the climate was sub-tropical in past geologic epochs.

Tab. 2.1 Principal nickel minerals (Kerfoot (1997)).

| Mineral                     | Ideal formula                                                 | Nickel content, % |
|-----------------------------|---------------------------------------------------------------|-------------------|
| <b>Sulfides</b>             |                                                               |                   |
| Pentlandite                 | $(\text{Ni,Fe})_9\text{S}_8$                                  | 34.22             |
| Millerite                   | $\text{NiS}$                                                  | 64.67             |
| Heazlewoodite               | $\text{Ni}_3\text{S}_2$                                       | 73.30             |
| Polydymite                  | $\text{Ni}_3\text{S}_4$                                       | 57.86             |
| Siegenite                   | $(\text{Co,Ni})_3\text{S}_4$                                  | 28.89             |
| Violarite                   | $\text{Ni}_2\text{FeS}_4$                                     | 38.94             |
| <b>Arsenides</b>            |                                                               |                   |
| Niccolite                   | $\text{NiAs}$                                                 | 43.92             |
| Rammelsbergite              | $\text{NiAs}_2$                                               | 28.15             |
| Gersdorffite                | $\text{NiAsS}$                                                | 35.42             |
| <b>Antimonides</b>          |                                                               |                   |
| Breithauptite               | $\text{NiSb}$                                                 | 32.53             |
| <b>Silicates and oxides</b> |                                                               |                   |
| Garnierite                  | $(\text{Ni,Mg})_6\text{Si}_4\text{O}_{10}(\text{OH})_8$       | $\leq 47\%$       |
| Nickeliferous limonite      | $(\text{Fe,Ni})\text{O}(\text{OH}) \cdot n\text{H}_2\text{O}$ | low               |

**Tab. 2.2 Composition of some nickel ores after (IPPC (2001)).**

| Ore                    | Type     | Ni%  | Cu%  | Co%  |
|------------------------|----------|------|------|------|
| Murrin Murrin          | Laterite | 1.25 |      | 0.08 |
| Cawse                  | Laterite | 1    |      | 0.07 |
| Cerro Matoso           | Laterite | 2.89 |      |      |
| Selebi-Phikwe          | Sulphide | 0.65 | 0.75 | 0.06 |
| Falconbridge, Sudbury  | Sulphide | 1.7  | 1.8  | 0.05 |
| Falconbridge, Raglan   | Sulphide | 3.2  | 0.9  | 0.05 |
| INCO, Copper Cliff     | Sulphide | 1.55 | 2    | 0.04 |
| Outokumpu, Silver Swan | Sulphide | 9.8  |      | 0.11 |
| Mount Keith            | Sulphide | 0.6  | 0    | 0.01 |

**Tab. 2.3 Mine production and reserves of nickel in 2001. “Reserves” is referred to as “the part of the reserve base which could be economically extracted or produced at the time of determination”, while “Reserve base” denotes “that part of an identified resource that meets specified minimum (...) criteria related to current mining and production practices” (USGS (2003)).**

|                       | Mine production  |                                 | Reserves | Reserve base |
|-----------------------|------------------|---------------------------------|----------|--------------|
|                       | 2001 [kT Ni / a] | 2002 [kT Ni / a]<br>(estimated) | [kT Ni]  | [kT Ni]      |
| Australia             | 197              | 186                             | 22000    | 27000        |
| Botswana              | 26.2             | 23.2                            | 490      | 920          |
| Brazil                | 45.4             | 44.9                            | 670      | 6000         |
| Canada                | 193.361          | 188                             | 5200     | 15000        |
| China                 | 51.5             | 55.7                            | 3600     | 7600         |
| Colombia              | 52.962           | 55.4                            | 900      | 1100         |
| Cuba                  | 70.662           | 73.1                            | 5600     | 23000        |
| Dominican Republic    | 31               | 24.3                            | 690      | 1000         |
| Greece                | 20.83            | 22.4                            | 490      | 900          |
| Indonesia             | 102              | 105                             | 3200     | 13000        |
| New Caledonia         | 117.554          | 98.2                            | 4400     | 12000        |
| Philippines           | 27.359           | 31.8                            | 940      | 5200         |
| Russia                | 325              | 328                             | 6600     | 9200         |
| South Africa          | 36.443           | 38                              | 3700     | 12000        |
| Venezuela             | 13.6             | 20.6                            | 610      | 610          |
| Zimbabwe              | 8.145            | 9.69                            | 15       | 260          |
| Other countries       | 9.24             | 13.7                            | 1300     | 5100         |
| World total (rounded) | 1330             | 1320                            | 61000    | 140000       |

## 3 Characterisation of the materials

Nickel is rarely used without further processing. Nickel is marketed solely on the basis of its chemical composition and according to its chemical purity as either a Class I or a Class II product. Typical compositions for Class I and II products are given in Tab. 3.2.

Class I includes the high-purity products, electrolytic cathode, carbonyl refined granules and powder, and hydrogen-reduced nickel in powder or briquette form.

Class II includes the various grades of ferronickel, nickel oxide, and metallised nickel oxide. In this study only ferronickel is of significance because of its use in the manufacturing of alloy steel and stainless steel.

However, Class I and Class II are intermediate market products. Most of it is used for alloying other metals. Stainless steels are particularly valued for their hygienic properties in hospitals and kitchens. In some applications, such as jet engines, nickel alloys are essential and cannot be substituted with other materials.

Inecoinvent only the intermediate nickel products “Class I Nickel” and “Ferronickel” are of importance. They are asked in the steel manufacturing and production of other alloys.

The reported effects of nickel and its compounds on humans are wide ranging, comprising effects that are both beneficial - the probable essentiality of nickel in humans - as well as harmful - skin allergy and, in certain circumstances, respiratory cancer (NIDIS (2003)).

Health Effects include

- Allergy of the skin to nickel - resulting in dermatitis. The following image shows a case of nickel eczema (dermatitis).
- Allergy of the lung to nickel - resulting in asthma
- In certain circumstances of exposure in refining, there may be a risk of cancer of the paranasal sinuses (close to the nose) and of cancer of the lung. This risk has now probably been completely removed in most modern plants.
- Nickel tetracarbonyl - an intermediate in the Mond process for refining nickel is very toxic to the lungs and heart.

### 3.1 Class I Nickel

Commercial nickel of this class (CAS 7440-02-0) is normally 99.7 % pure or better. Synonyms are “unwrought nickel, not alloyed”. Nickel is a silver-white metal. Factors that make nickel and its alloys commercially valuable commodities include strength, corrosion resistance, high ductility, good thermal and electrical conductivity, magnetic characteristics (below 340 °C) and catalytic properties. Some physical properties of pure nickel are given in Tab. 3.1. Nickel is not tarnished by air and is resistant to fluorine (F<sub>2</sub>) but is attacked by acids (except conc. HNO<sub>3</sub>). Class I nickel is marketed as cathodes, granules, powder, briquettes and rounds.

Tab. 3.1 Some physical properties of pure nickel (Kerfoot (1997)).

| property                                                   | value                                   |
|------------------------------------------------------------|-----------------------------------------|
| Melting Point (mp)                                         | 1455°C (1728 K)                         |
| Boiling Point (bp)                                         | 2730°C (3003 K)                         |
| Relative density (25°C)                                    | 8.9                                     |
| Volume increase on melting                                 | 4.5%                                    |
| Heat of fusion (mp)                                        | 302 J/g                                 |
| Heat of sublimation (25°C)                                 | 7317 J/g                                |
| Heat of vaporization ( $T_{crit}$ )                        | 6375 J/g                                |
| Standard entropy                                           | 29.81 J/K                               |
| Thermal conductivity (0-100°C)                             | 88.5 W m <sup>-1</sup> K <sup>-1</sup>  |
| Heat capacity (0-100°C)                                    | 0.452 J g <sup>-1</sup> K <sup>-1</sup> |
| Electrical resistivity (20°C)                              | 6.9 μΩ/cm                               |
| Temperature coefficient of electrical resistivity(0-100°C) | 6.8×10 <sup>-3</sup> K <sup>-1</sup>    |
| Thermal expansion coefficient (0-100°C)                    | 13.3×10 <sup>-6</sup> K <sup>-1</sup>   |
| Modulus of elasticity                                      | 199.5 GPa                               |
| Brinell hardness                                           | 85                                      |

## 3.2 Ferronickel

Ferronickel (CAS 11133-76-9) is a market product of the nickel industry. It consists of an alloy of iron with nickel, with a nickel content of 20 - 50 % (Tab. 3.2). It is used in the manufacturing of stainless steels.

Tab. 3.2 Commercial forms of nickel (Kerfoot (1997)).

| Type             | Composition, % |                    |       |        |         |        |       |
|------------------|----------------|--------------------|-------|--------|---------|--------|-------|
|                  | Ni             | Co                 | Cu    | Fe     | C       | S      | O     |
| <b>Class I:</b>  |                |                    |       |        |         |        |       |
| Cathodes         | >99.90         |                    | 0.005 | 0.002  | 0.01    | 0.001  |       |
| Pellets          | >99.97         | 5×10 <sup>-5</sup> | 0.001 | 0.0015 | <0.10   | 0.0003 |       |
| Powder           | 99.74          |                    |       | <0.10  | <0.10   | <0.010 | <0.15 |
| Briquettes       | 99.90          | 0.03               | 0.001 | 0.01   | 0.01    | 0.0035 |       |
| Rondelles        | 99.25          | 0.37               | 0.046 | 0.022  | 0.022   | 0.004  | 0.042 |
| <b>Class II:</b> |                |                    |       |        |         |        |       |
| Ferronickel      | 20-50          | 1-2                |       | rest   | 1.5-1.8 | <0.3   |       |
| Nickel oxide     | 76.0           | 1.0                | 0.75  | 0.30   |         | 0.006  | rest  |

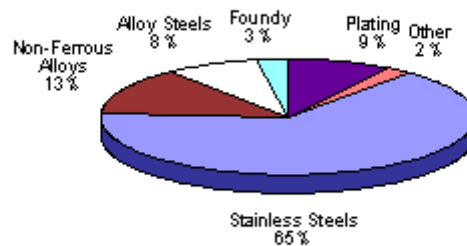
## 4 Use / Application of Material

Unlike other non-ferrous metals, nickel is rarely used by itself but is commonly mixed with other metals to produce alloys. There are many different alloys containing nickel, each providing a particular combination of technical properties (corrosion resistance, mechanical properties and service life). Thus the two forms of nickel – class I nickel and ferronickel – are to high degree intermediate products for the metal industry.

**Tab. 4.1 Production of different nickel products (USGS (2003)).**

|              | 2000    |     |
|--------------|---------|-----|
| Ferronickel  | 220000  | 20% |
| Metal        | 638000  | 57% |
| Oxide sinter | 98200   | 9%  |
| Chemicals    | 18000   | 2%  |
| Unspecified  | 143000  | 13% |
| Total        | 1117200 |     |

More than 80 % of primary nickel is consumed in some 3'000 different alloys, grouped generally as stainless steels, alloy steels, and non-ferrous alloys. About 65 % of primary nickel is used in stainless steels alone, as shown in Fig. 4.1.



**Fig. 4.1 Uses of nickel in different industries. Source: NiDI<sup>2</sup>, 1996**

Electroplating accounts for roughly 10 % of primary nickel consumption, with the balance consumed by foundries and a host of other uses including chemicals, catalysts, batteries, welding rods, coinage, pigments for enamels/glasses/ceramics, electronics, and printing inks.

Most of the plating, foundry and "other" applications are "end-uses" of nickel; that is to say, the products are used directly by the customer or "end-user". Steels and other nickel alloys, on the other hand, are "intermediate" products that must be further processed or "transformed" into end-use commercial products in a number of industrial applications. These applications are shown in Fig. 4.2.

The major end-use sectors for nickel shown in Fig. 4.2 reflect various applications for the many of known nickel alloys as well as the direct use of primary nickel for electroplated and other products noted above. The use therefore can be diverted into an intermediate use in the form of metallic nickel (class I) and Ferronickel and into the end-use as nickel alloys.

<sup>2</sup> Nickel Development Institute, <http://www.nidi.org/>, accessed at 01. 03. 2003.

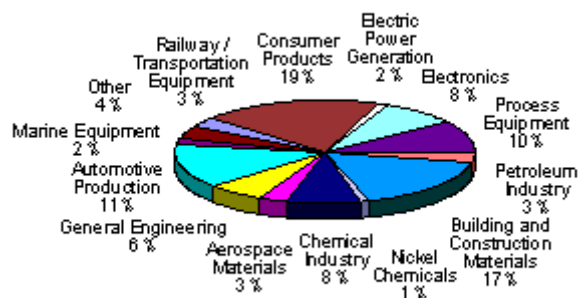


Fig. 4.2 Application of nickel Source: NiDl<sup>3</sup>, 1996

## 4.1 Nickel (class I)

Primary nickel metal powder is used in many applications including battery manufacture, coin production, welding products, fuel filters, brake linings, catalysts, cutting tools, paints, plastics, electronics, implants, and surface treatments by plasma spraying.

## 4.2 Ferronickel

Ferro-alloys are mainly used as master alloys in the iron and steel industry, because it is the most economic way to introduce an alloying element in the steel melt. Ferronickel in this case therefore is used as master alloy in the production of stainless steels. Further more it is used as starting materials in the preparation of chemicals and pure metal; as reducing agents (e.g., the use of ferrosilicon to reduce rich slag); as alloying elements in nonferrous alloys and as starting materials for special products such as amorphous metals.

## 4.3 Nickel Alloys

Nickel-bearing, "austenitic" stainless steels contain variable amounts of nickel, between 5 to 25 percent, and average about 10 % nickel. Commercial uses include chemical and food processing equipment, transportation equipment, structural claddings and other architectural applications, durable consumer goods such as cookware, and hospital and medical applications.

Nickel alloys can be hot or cold worked. Other processes include annealing, drawing, forging, spinning, and joining by soldering, brazing, gas welding, and resistance welding. Machining of most nickel alloys requires high-speed tools and flooding with cutting compounds.

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<sup>3</sup> Nickel Development Institute, <http://www.nidi.org/>, accessed at 01. 03. 2003.

**Tab. 4.2 Some of the more common nickel alloys.**

| <b>Alloy</b>     | <b>Constituents</b>                                |
|------------------|----------------------------------------------------|
| Monel metals     | combinations of nickel and copper                  |
| Inconel metals   | nickel alloys containing chromium and iron         |
| Hastelloy metals | combinations of chromium, molybdenum and iron      |
| Illium nickels   | nickel alloys with chromium, molybdenum and copper |

Nickel alloy steels are used in heavy machinery manufacturing, armoured plate, tools, and high-temperature equipment including gas turbines and scrubbers. Non-ferrous alloys include nickel-copper alloys, typically containing about 65% nickel, and copper-nickel alloys, typically containing about 10% nickel, used extensively in desalination plants for converting seawater into fresh water. Ni-Cu/Cu-Ni alloys are also employed in marine equipment and inorganic acid and alkali handling equipment. Nickel-chromium alloys, containing approximately 40-70% nickel, are used in corrosive chemical environments and high-temperature applications such as heating elements and gas turbine aero-engine components. Nickel-copper-zinc alloys, including nickel silver with less than 18% nickel, are used for decorative purposes, including jewellery.

Different alloy types are used in specific applications. The constitution of some common nickel alloys is given in Tab. 4.2.

- Corrosion-resistant alloys: nickel, monel, hastelloy, illium, nirolium for use in chemical and process plant, plating coinage, food processing and water treatment.
- High temperature alloys: inconel, incoloy, nimonic, nimocast used for furnace components, boiler components, heat-treatment equipment, power generation, heat exchangers.
- Electrical Alloys: nickel, brightray, nichrome, constantan for use in ferry electronics, heating elements, industrial furnace elements, resistors, thermocouples and spark plugs.
- Magnetic Alloys: nickel, permalloy, mumetal used in magnetostrictive devices, magnetic shields, HF transformation, loading cells, transformer and rotor laminations, temperature compensators.
- Controlled-expansion and constant modulus alloys: invar, nile, ni-Span, elinvar, bimetals, thermostats, glass-sealing and weighing machines.
- Miscellaneous alloys: nitinol, hy-stor employed in shape-memory devices, hydrogen storage and brazing alloys.

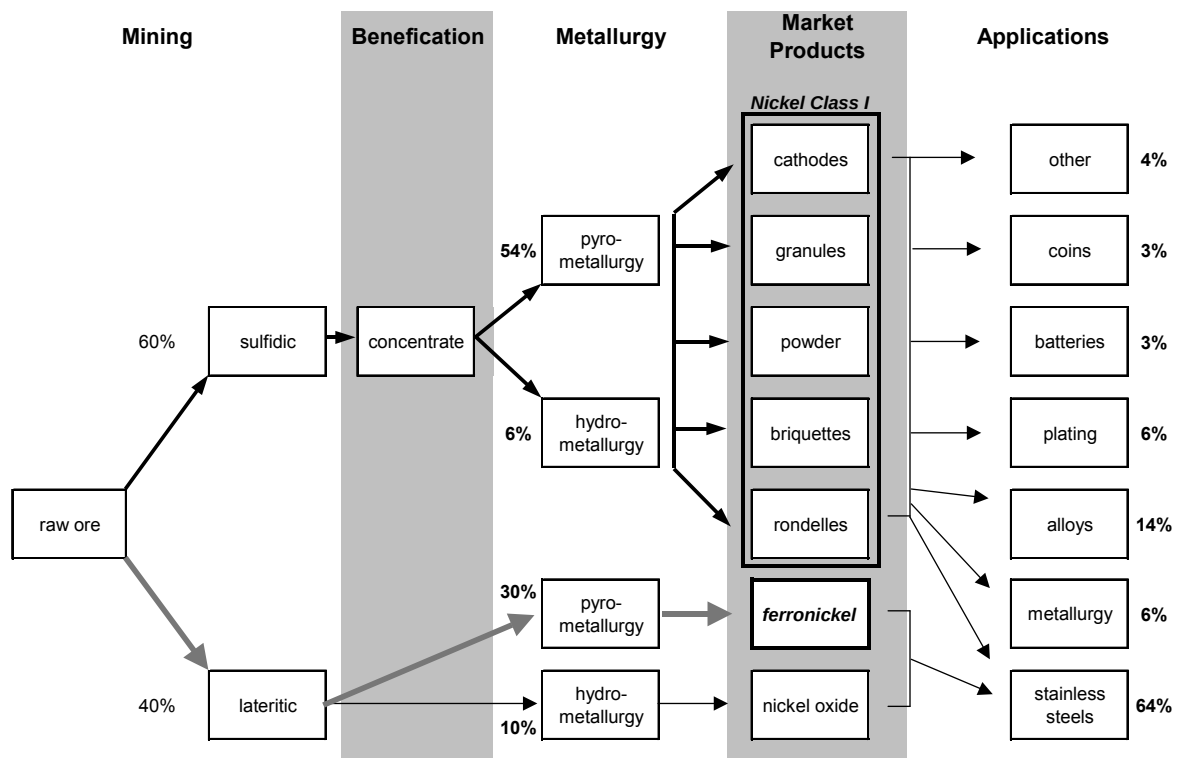


## 5 System Characterisation

The winning of nickel can be classified in two main process chains as shown in Fig. 5.1. The first process chain represents the winning of nickel from sulphidic ore, yielding Class I Nickel as market product. This class of nickel is used in a wide range of applications. The other path represents the winning of ferronickel from lateritic ore.

The aim of this study is to inventory the two intermediate products “Nickel Class I” and “Ferronickel” as raw materials for nickel alloys. These two modules represent the most important application of nickel. The system boundaries are set at the production gate, both modules represent the production mix, no transport to the desired region of the application is included.

The meta information of the modules are summarised in Tab. 5.1.



**Fig. 5.1** Winning of nickel from ore to application. Two main process chains can be identified, the sulphidic path yielding Class I nickel and the lateritic path yielding ferronickel. Ferronickel is mainly used in the production of stainless steel, whereas Class I nickel is used in a wide range of applications.

**Tab. 5.1** Ecoinvent meta information for the processes for the production of Class I nickel and ferronickel.

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>                           | <b>nickel production, sulphidic ore, primary</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Location                              | GLO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Infrastructure Process                | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Unit                                  | kg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Data Set Version                      | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Included Processes                    | The module includes a mining and a beneficiation step with the mining and beneficiation infrastructure and disposal of overburden and tailings. Subsequently it includes the metallurgy step with the disposal of slag, the infrastructure and the separation of the co-product copper, and the refining step yielding the desired Class I nickel inclusively the metallurgical infrastructure. Production, application and emissions of most agents and additives used in beneficiation and metallurgy are also included.                                                                                                                                                                                                                                                                                                                                                                                         |
| Amount                                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Local Name                            | Nickel-Produktion, sulfidisches Erz, primär                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Synonyms                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| General Comment to reference function | The module describes the global production mix for Class I nickel in 1994 with a coupled production of a certain amount of copper. The multioutput-process "nickel production, sulphidic ore, primary" delivers the co-products "nickel, 99.5%, at plant" and "copper, primary, coupleproduction nickel". It is designed for the use of the metal as intermediate product or as raw material in the plating industry or the manufacturing of stainless steels and alloys. It is not to be used if the impact of the Class I nickel is considered to be high. In such cases, a more detailed analysis depending on scope and allocation procedures has to be conducted. The data used is mainly based on a study of the energy and material streams resulting from the production of class I nickel. Lacking data mainly on direct process specific emissions were taken from similar processes for copper winning. |
| CAS Number                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Start Date                            | 1994                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| End Date                              | 2003                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Other Period Text                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Geography text                        | This module represents a global production mix. Some processes like blasting, lime and energy production and transport originate from data sets valid for Switzerland or Europe.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Technology text                       | Mining is done underground, followed by beneficiation through flotation where considerable amounts of agents and lime are added. Overburden and tailings are disposed on the mining site and partly re-filled. This module covers the five major production routes, aggregated according to their market share in 1994. Sulphur dioxide in the off-gas is recovered producing sulphuric acid. For electricity mix a mix of 60% hydropower and 40% electricity at grid was chosen.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Representativeness [%]                | 100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Production Volume                     | 547'684 t nickel in sulfidic raw ore in 1994                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Sampling Procedure                    | literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

Tab. 5.1 Ecoinvent meta information for the processes for the production of Class I nickel and ferronickel. (cont.)

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>                           | <b>ferronickel, 25% Ni, at plant</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Location                              | GLO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Infrastructure Process                | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Unit                                  | kg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Data Set Version                      | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Included Processes                    | The module includes a mining and a beneficiation step with the mining infrastructure. Subsequently it includes the metallurgy step with the disposal of slag and the metallurgical infrastructure. Production, application and emissions of most agents and additives used in beneficiation and metallurgy are also included.                                                                                                                                                                                                                                             |
| Amount                                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Local Name                            | Ferronickel, 25% Ni, ab Werk                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Synonyms                              | ferro-alloy//Eisenlegierung                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| General Comment to reference function | The module describes the global production mix for ferronickel in 1994. It is designed for the use of the metal as raw material in the manufacturing of stainless steels and alloys. It is not to be used if the impact of ferronickel is considered to be high. In such cases, a more detailed analysis has to be conducted. The data used is mainly based on a study of the energy and material streams resulting from the production of class I nickel. Lacking data mainly on direct process specific emissions were taken from similar processes for copper winning. |
| CAS Number                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Start Date                            | 1994                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| End Date                              | 2003                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Other Period Text                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Geography text                        | This module represents a global production mix. Some processes like blasting, lime and energy production and transport originate from data sets valid for Switzerland or Europe.                                                                                                                                                                                                                                                                                                                                                                                          |
| Technology text                       | Mining is done in open cuts, followed by beneficiation with a simple mechanical separation. Overburden and tailings are disposed inside the mining panels. This module covers the major production route for ferronickel. For electricity mix a mix of 60% hydropower and 40% electricity at grid was chosen.                                                                                                                                                                                                                                                             |
| Representativeness [%]                | 80                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Production Volume                     | 329'478 t nickel in lateritic raw ore in 1993                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Sampling Procedure                    | literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Extrapolations                        | see Geography and Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Uncertainty Adjustments               | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## 5.1 Winning of Class I Nickel

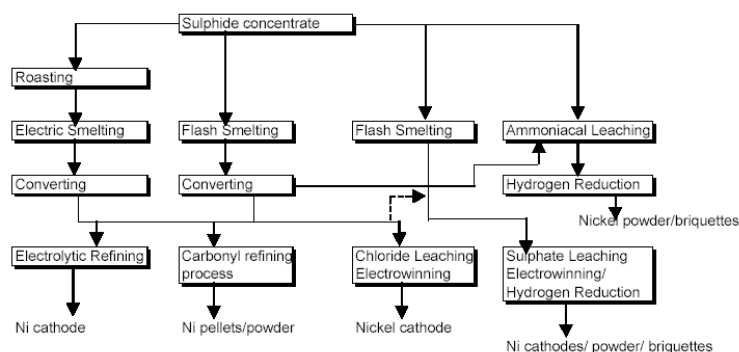
This part is mainly based on (Hilbrans & Hinrichs (1999)). General process description was taken from Kerfoot (1997). Allocation is done by revenue between the coupled products Nickel and Copper using the values of 6647\$ / t Nickel and 1986\$ / t Copper.

### 5.1.1 Production process and infrastructure

**Mining.** 95% of sulphidic nickel ores are mined underground in depths between 200m and 1800m, the ore is transferred to the beneficiation. Widening of the tunnels is mainly done by blasting. The overburden – material, which does not contain PGM-bearing ore – is deposited off-site and is partially re-filled into the tunnels.

**Beneficiation.** After mining, the ore is first ground. In a next step it is subjected to gravity concentration to separate the metallic particles from the PGM-bearing minerals. After this first concentration step, flotation is carried out to remove the gangue from the sulphidic minerals. For neutralisation lime is added. In the flotation several organic chemicals are used as collector, frother, activator, depressor and flocculant. Sometimes cyanide is used as depressant for pyrite. Tailings usually are led to tailing heaps or ponds. As a result, nickel concentrates containing 7 - 25% Ni are produced.

**Metallurgy.** There are many different process possibilities to win the metal. The chosen process depends on the composition of the ore, the local costs of energy carrier and the local legislation. Basically two different types can be distinguished: the hydrometallurgical and the pyrometallurgical process with a share of 6 % and 54 % of the worldwide nickel produced. An overview is given in Fig. 5.2.



**Fig. 5.2** Overview over some process possibilities for nickel winning. The chosen process depends on the composition of the ore, the local costs of energy carrier and the local legislation. Scheme from (IPPC (2001))

**Pyrometallurgy.** The pyrometallurgical treatment of nickel concentrates includes three types of unit operation: roasting, smelting, and converting. In the roasting step sulphur is driven off as sulphur dioxide and part of the iron is oxidised. In smelting, the roaster product is melted with a siliceous flux which combines with the oxidised iron to produce two immiscible phases, a liquid silicate slag which can be discarded, and a solution of molten sulphides which contains the metal values. In the converting operation on the sulphide melt, more sulphur is driven off as sulphur dioxide, and the remaining iron is oxidised and fluxed for removal as silicate slag, leaving a high-grade nickel – copper sulphide matte. In several modern operations the roasting step has been eliminated, and the nickel sulphide concentrate is treated directly in the smelter.

**Hydrometallurgy.** Several hydrometallurgical processes are in commercial operation for the treatment of nickel – copper mattes to produce separate nickel and copper products. In addition, the hydrometallurgical process developed by Sherritt Gordon in the early 1950s for the direct treatment of nickel sulphide concentrates, as an alternative to smelting, is still commercially viable and competitive, despite very significant improvements in the economics and energy efficiency of nickel smelting

technology. In a typical hydrometallurgical process, the concentrate or matte is first leached in a sulphate or chloride solution to dissolve nickel, cobalt, and some of the copper, while the sulphide is oxidised to insoluble elemental sulphur or soluble sulphate. Frequently, leaching is carried out in a two-stage countercurrent system so that the matte can be used to partially purify the solution, for example, by precipitating copper by cementation. In this way a nickel – copper matte can be treated in a two-stage leach process to produce a copper-free nickel sulphate or nickel chloride solution, and a leach residue enriched in copper.

**Refining.** In many applications, high-purity nickel is essential and Class I nickel products, which include electrolytic cathode, carbonyl powder, and hydrogen-reduced powder, are made by a variety of refining processes.

The carbonyl refining process uses the property of nickel to form volatile nickel-carbonyl compounds from which elemental nickel subsides to form granules. Electrolytic nickel refineries treat cast raw nickel anodes in a electrolyte. Under current the anode dissolves and pure nickel deposits on the cathode. This electrolytic refining process is obsolete because of high energy demand and the necessity of building the crude nickel anode by reduction with coke. It is still practised in Russia. Most refineries recover electrolytic nickel by direct electrowinning from purified solutions produced by the leaching of nickel or nickel – copper mattes. Some companies recover refined nickel powder from purified ammoniacal solution by reduction with hydrogen.

The multitude of processing possibilities was reduced in (Hilbrans & Hinrichs (1999)) to five major process routes for nickel winning from sulphidic ores. Their layout is given in Tab. 5.2. This classification will be used further.

**Tab. 5.2 Overview over the major process routes in the nickel winning from sulphidic ores. After (Hilbrans & Hinrichs (1999)).**

| Process              | Ore                       | Roasting              | Primary Smelting               | Matte converting | Separation co-product | Primary Refining    | Nickel Refining                        |
|----------------------|---------------------------|-----------------------|--------------------------------|------------------|-----------------------|---------------------|----------------------------------------|
| <b>alternative 1</b> | Nickel-concentrate        | fluidized bed furnace | Electric Furnace Smelting:     | Converter        |                       |                     | Matte refining (Electrorefining)       |
| <b>alternative 2</b> |                           |                       | Flash-Smelting                 | Converter        |                       |                     | Precipitation with hydrogen            |
| <b>alternative 3</b> | Nickel-Copper-Concentrate | fluidized bed furnace | Electric Furnace Smelting:     | Converter        | Wet                   | Roasting, reduction | Reductive electrolysis                 |
| <b>alternative 4</b> |                           |                       | Flash-Smelting                 | Converter        | Dry                   |                     | Electrolytic Refining (Electrowinning) |
| <b>alternative 5</b> |                           | Multiple hearth       | Reverberatory Furnace smelting | Converter        | Dry                   |                     | Carbonyl Process                       |

## 5.1.2 Emission

### Mining

The major emissions are due to mineral born pollutants in the effluents. The underground mining operations generate roughly 80 % of the dust emissions from open pit operations, since the major dust sources do not take place underground. Rain percolate through overburden and accounts to metal emissions to groundwater.

## Beneficiation

Ore handling and processing produce large amounts of dust, containing PM<sub>10</sub> and several metals from the ore itself. Flotation produce effluents containing several organic agents<sup>4</sup> used. Some of these chemicals evaporate and account for VOC emissions to air. Namely xanthates decompose hydrolytically to release carbon disulphide. Tailings effluent contains additional sulphuric acid from acid rock drainage.

## Metallurgy & Separation from Non Ferrous Metals

**Sulphur dioxide.** In all of the process steps sulphur dioxide is emitted to air. Recovery of sulphur dioxide is only economic for high concentrated off-gas. SO<sub>2</sub> emissions are by far the most important in the metallurgy.

**Carbon dioxide.** In the beneficiation step, considerable amounts of lime is added to the ore for pH-stabilisation, lime forms later flux in the metallurgical step. Lime decomposes into CO<sub>2</sub> to form calcite.

**Dust.** Dust carry over from the roasting, smelting and converting processes. Particulate emissions to the air consist of metals and thus are often returned to the leaching process after treatment.

**Chlorine.** Chlorine is used in some leaching stages and is produced during the subsequent electrolysis of chloride solution. The chlorine evolved is collected and re-used in the leach stage. The presence of chlorine in wastewater can lead to the formation of organic chlorine compounds (AOX) if solvents etc. are also present in a mixed wastewater.

**VOC.** VOCs can be emitted from the solvent extraction stages. A variety of solvents are used as they contain various complexing agents to form complexes with the desired metal that are soluble in the organic layer.

**Emissions to water.** Metals and their compounds and substances in suspension are the main pollutants emitted to water. The metals concerned are Cu, Ni, Co, As and Cr. Other significant substances are chlorides and sulphates. Wastewater from wet gas cleaning (if used) of the different metallurgical stages are the most important sources. The leaching stages are usually operated on a closed circuit and drainage systems, and are therefore regarded as minor sources.

## Refining

Also in this step the combustion of sulphur leads to emissions of SO<sub>2</sub>. Nitrogen oxides are produced in significant amounts during acid digestion using nitric acid. Chlorine and HCl can be formed during a number of digestion, electrolytic and purification processes. Chlorine is used extensively in the Miller process and in the dissolution stages using hydrochloric acid and chlorine mixtures respectively. Dust and metals are generally emitted from incinerators and furnaces. VOC can be emitted from solvent extraction processes, while organic compounds, namely dioxins, can be emitted from smelting stages resulting from the poor combustion of oil and plastic in the feed material. All these emissions are subject to abatement technologies and controlling. Large quantities of effluents contain amounts of metals and organic substances.

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<sup>4</sup> Floating agents consist of collectors (xanthate or aerofloat) and frothing reagent (eg. Methyl Isobutyl Carbinol). Xanthates are a broad class of organic salts formed by treatment of an alcohol with carbon disulfide in the presence of an alkali, potassium ethyl xanthate (KC<sub>3</sub>H<sub>5</sub>S<sub>2</sub>O, CAS No: 140-89-6). Aerofloat is a salt of carbonyldithiophosphoric acids. A widely used frothing reagent is e.g. methyl isobutyl carbinol, CAS: 108-11-2, syn. Methyl Amyl Alcohol.

### 5.1.3 Waste

#### Mining and Beneficiation

Overburden is deposited close to the mine. Tailings are deposited as piles and in ponds. In both the overburden and the tailings, acid rock drainage occurs over a long period of time. These effects are handled in section IX “Non Ferrous Metal Winning – Auxiliary Processes”, chapter “Disposal of Tailings and Overburden”.

#### Metallurgy & Separation from Non Ferrous Metals

The production of metals is related to the generation of several co-products, residues and wastes, which are listed in the European Waste Catalogue. Some of the process specific residues can be reused or recovered in preliminary process steps (e. g. dross, filter dust) or construction (e. g. cleaned slag). Residues also arise from the treatment of liquid effluents, the main residue being gypsum waste and metal hydroxides from the wastewater neutralisation plant. These residuals have to be disposed, usually in lined ponds.

#### Refining

Like in the pyro- and hydrometallurgical step quantities of solid residuals are generated, which are mostly recycled within the process or sent to other specialists to recover any precious metals. Final residues generally comprise hydroxide filter cakes (ironhydroxide, 60% water, cat I industrial waste).

### 5.1.4 Class I Nickel in ecoinvent

In Hilbrans & Hinrichs (1999) data for the different process alternatives shown in Tab. 5.2 was collected in detail. However, in this study an average was calculated based on the values from alternative 1 to 5, weighted with their share to the world-wide production. The resulting module thus represents an aggregated value for the world-wide production of Class I nickel.

The module describes the global production mix for Class I nickel in 1994 with a coupled production of a certain amount of copper. The multioutput-process “nickel production, sulphidic ore, primary” delivers the co-products “nickel, 99.5%, at plant” and “copper, primary, coupleproduction nickel”. It is designed for the use of the metal as intermediate product or as raw material in the plating industry or the manufacturing of stainless steels and alloys. The layout of the modules with general flow information, remarks, sources, values and uncertainty information are shown in Fig. 5.3, Fig. 5.4, Fig. 5.5, Fig. 5.6, Fig. 5.7, Fig. 5.8 and Fig. 5.9. The following paragraphs describe calculations, sources and assumptions chosen in this study. The functional unit of this process is one kilogram of Class I nickel.

#### Infrastructure and land use

The mobile equipment used for exploitation like drilling machines, scoop trams and underground trucks is inventoried with the diesel consumption stated in Hilbrans & Hinrichs (1999). The used module “diesel, burned in building machine” includes the infrastructure of the machines. 2000 m conveyor belts are further assumed with a capacity of 2000 t / h which gives 2E-5 m conveyor belt required per tonne mined raw ore. Since mining is underground and no heap leaching is done, no land use for exploitation is assumed.

Mining facilities and smeltery are inventoried with respective modules, beneficiation plant is approximated according to chapter II “Non Ferrous Metal Winning – Auxiliary Processes” with the module “aluminium hydroxide, plant”.

**Tab. 5.3 Infrastructure in underground mining, assumptions used. A lifetime of ten years are estimated, and 5000 working hours yearly.**

| action    | Machine        | throughput   | unit       | Infrastructure  |
|-----------|----------------|--------------|------------|-----------------|
|           |                | t / lifetime | Length (m) | m / t ore mined |
| Transport | Conveyor belts | 100'000'000  | 2000       | 2E-5            |

## Mining.

In the considered processes the ore is mined underground. The overburden is disposed in piles near the mine, the standard module for disposal of non-sulphidic overburden was chosen, which includes the land use and transformation through the disposal. Refilled overburden is assumed to be burden free, but cement and stowing material (assumption: sand) has to be added as stated in Hilbrans & Hinrichs (1999). The degree of the refilling is documented for each alternative in Hilbrans & Hinrichs (1999). Also water use and effluent from mining is documented. The use of freshwater was inventoried as river water, where as the pit water was assumed to be groundwater, which has to be pumped. The composition of the effluent was taken from IPPC (2002), its values are shown in Tab. 5.4. Concerning electricity mix Hilbrans & Hinrichs (1999) assumes a share of 60% of hydropower due to the operations high and constant energy demand. This assumption is used also in this inventory. Cement, explosives and sand are assumed to be transported by lorry over 50 km.

## Beneficiation

Once the ore is mined, it is ground and conditioned for further concentration. Considerable amounts of lime and flotation agents are added. The separated gangue is disposed in tailings ponds, the concentrated ore is fed to the metallurgy, which is assumed to be on-site.

Dust emissions comprise additionally the dust emissions due to mining activities. Like in the mining step, a share of 60% hydroelectricity was chosen according to Hilbrans & Hinrichs (1999).

**Lime.** Lime input was documented in Hilbrans & Hinrichs (1999) for each process alternative. The specific lime input ranges from 2.0 to 8.5 kg per t ore treated with a weighted mean of 5.5 kg / t ore.

**Reagents.** No information on mass and nature of the floating agents used are available. For copper mining a range of 25 – 300 g collector and 25 – 250 g frothing reagent per ton sulphidic ore is recommended (Krauss et al. (1999)). This is confirmed by the values in Tab. 5.5. Collector and frother are balanced in ecoinvent in their production as unspecified organic chemical. According to NPI (2001) xanthates hydrolytically decompose to 0.53 mg carbon disulphide per mg xanthate. With 189 mg xanthates from Tab. 5.5 this is 100 mg carbon disulphide per t mined ore (Tab. 5.6).

Activator and Depressants – excluding sodium cyanate – are modelled as unspecific inorganic chemicals, their emission to water is covered by reported emissions from mining sites shown in Tab. 5.4.

For pH adjustment and leaching the value for sulphuric acid was considered with 3.74 kg / t ore mined (Tab. 5.6). As sulphuric acid is a by-product from the metallurgy, it is regarded as burden free. Other acids are neglected.

Since composition and environmental fate of activator, depressant and sulphuric acid is uncertain, reported data for total emissions shown in Tab. 5.6 are taken. Note that some theoretical emissions due to the use of the reagents are higher than the reported ones in Tab. 5.4.

Sodium cyanide is widely used for extracting gold and other metals such as silver, copper and zinc. It is used as leachate and as depressant for certain sulphidic minerals. The environmental fate of cyanide compounds is complex and could not further be queried. A cyanide recovery of 80 – 90% is reported due to abatement and recovery in the Golden Cross mine in New Zealand (Anonymous (1998a)). Here a value of 80% is chosen for cyanide-abatement technology. With a reported use of 35 g NaCN per tonne ore mined (Tab. 5.5), this means a direct emission of 3.75 g CN<sup>-</sup> per tonne ore mined.



**Effluents.** Additional to direct emissions due to the use of chemicals, total emissions to water are chosen according to Tab. 5.4. These values are from base metal sites, thus only an approximation for the real effluent concentrations in the nickel production. It is assumed here that the effluents also include acid rock drainage (ARD). ARD after mine closure is modelled in the module “disposal, sulphidic tailings, off-site”.

**Dust.** A case study in Anonymous (1998b) about a projected open cut gold-copper mining and processing facility in Australia predict annual dust emissions between 0.18 kg/t and 0.46 kg/t of dust produced per tonne of ore mined. The prediction was made using the Emission Estimation Technique Manual for Mining (NPI (2001)). The maximum value of 0.46 kg/t is chosen for total dust emissions. Underground mining activities are assumed to generate 80 % of the dust emissions compared to open pit mining (Tab. 5.7) yielding a 0.37 kg dust per tonne mined ore. According to the IAEA<sup>5</sup> the dust composition is approximated with values of a specific sulphidic nickel ore in Tab. 5.8. For PM<sub>10</sub> values, the ratio TDS : PM<sub>10</sub> : PM<sub>2.5</sub> was approximated with values from copper ore mining 101.72 : 50 : 5 g / t ore taken from the CEMIP-database<sup>6</sup>, i. e. one tonne emitted dust corresponds to 491.5 kg PM<sub>10</sub> and 49.2 kg PM<sub>2.5</sub> respectively. The dust emissions resulting from underground activities are summed up in Tab. 5.8. In Ecoinvent, for particulates differential classes are used adding up to the total dust emission, i. e. 49.2 kg PM<2.5, 442 kg PM2.5-10 and 508 kg PM>10 per tonne dust

**Tab. 5.4 Mining & Beneficiation, emissions to water. Composition of effluents from mining activities, values are taken from total emissions to water from base metal sites in (IPPC (2002)).**

| Compound        | Unit                    | Range         | This study (geometric mean) | Number of Samples |
|-----------------|-------------------------|---------------|-----------------------------|-------------------|
| Discharge       | Mio. m <sup>3</sup> / a | 0.08 - 22.9   | 3.65                        | 6                 |
| Ca              | g / m <sup>3</sup>      | 686           | 686                         | 1                 |
| SO <sub>4</sub> | g / m <sup>3</sup>      | 1750 - 3175   | 2357                        | 2                 |
| COD             | g / m <sup>3</sup>      | 2.24 - 48.5   | 10.4                        | 2                 |
| Solids          | g / m <sup>3</sup>      | 2.38 - 11.3   | 5.17                        | 4                 |
| Al              | mg / m <sup>3</sup>     | 69.3 - 108    | 86.3                        | 2                 |
| As              | mg / m <sup>3</sup>     | 0.264 - 14.1  | 2.95                        | 3                 |
| Cd              | mg / m <sup>3</sup>     | 0.0901 - 1.02 | 0.316                       | 4                 |
| Co              | mg / m <sup>3</sup>     | 0.742 - 0.823 | 0.781                       | 2                 |
| Cr              | mg / m <sup>3</sup>     | 0.0311 - 9.62 | 0.547                       | 2                 |
| Cu              | mg / m <sup>3</sup>     | 1.24 - 44.9   | 7.91                        | 5                 |
| Fe              | mg / m <sup>3</sup>     | 61.7 - 1326   | 291                         | 3                 |
| Mn              | mg / m <sup>3</sup>     | 24.7          | 24.7                        | 1                 |
| Hg              | mg / m <sup>3</sup>     | 0.0155 - 0.12 | 0.0377                      | 3                 |
| Ni              | mg / m <sup>3</sup>     | 0.792 - 1338  | 24.3                        | 3                 |
| Pb              | mg / m <sup>3</sup>     | 0.0155 - 20   | 2.80                        | 4                 |
| Zn              | mg / m <sup>3</sup>     | 5.37 – 225    | 75.9                        | 5                 |
| N               | g / m <sup>3</sup>      | 2.50 - 1786   | 22.8                        | 3                 |

<sup>5</sup> The IAEA Database of Natural Matrix Reference Materials, <http://www-naweb.iaea.org/nahu/external/e4/nmrm/>, accessed 01.03.2003

<sup>6</sup> [http://www.air.sk/tno/cepmeip/em\\_factors.php](http://www.air.sk/tno/cepmeip/em_factors.php), accessed 01.03.2003

Tab. 5.5 Consumption of reagents in base metal mineral processing plants. Data taken from IPPC (2002).

| Reagents                                                                                        | Unit | Range     | This study<br>(geometric mean) | Number of samples |
|-------------------------------------------------------------------------------------------------|------|-----------|--------------------------------|-------------------|
| <b>Collectors</b> – Xanthate                                                                    | g/t  | 110 - 300 | 188.6                          | 6                 |
| <b>Frothers</b> - Sylvapine, MIBC, Dowfroth                                                     | g/t  | 9.7 - 150 | 37.2                           | 5                 |
| Sodium cyanide                                                                                  | g/t  | 4-310     | 35                             | 2                 |
| <b>Depressants</b> – Zinc sulphate, Iron sulphate, Acetic acid, Sodiumchromate, Dithiophosphate | g/t  | 35 - 432  | 231.5                          | 5                 |
| <b>Activator</b> - Copper sulphate                                                              | g/t  | 433 - 876 | 538                            | 4                 |
| <b>Flotation agents</b>                                                                         | g/t  | 19.0      | 19.0                           | 1                 |
| <b>Flocculants</b> - CMC                                                                        | g/t  | 1-100     | 11.1                           | 3                 |
| Sulphuric acid (pH)                                                                             | g/t  | 387-12000 | 3739                           | 4                 |
| Sodium hydroxide (pH)                                                                           | g/t  | 30-490    | 121                            | 2                 |
| Nitric acid (pH)                                                                                | g/t  | 150       | 150                            | 1                 |
| Hydrochloric acid (pH)                                                                          | g/t  | 1         | 1                              | 1                 |

Tab. 5.6 Emissions and input due to chemical use in beneficiation in g per tonne treated ore. The values are the ones chosen in this study. For further details read in text.

| Reagent              | Type of burden     | Unit | Chosen value                      |
|----------------------|--------------------|------|-----------------------------------|
| Collector            | input technosphere | g/t  | 189                               |
| Carbon disulfide     | air emission       | g/t  | 100                               |
| Frother              | input technosphere | g/t  | 37                                |
| Lime                 | input technosphere | kg/t | 2.0 – 8.5                         |
| Carbon dioxide       | air emission       | g/t  | inventoried in metallurgical step |
| Activator/Depressant | input technosphere | g/t  | 828                               |
| Sulphuric acid       | input technosphere | g/t  | 3739                              |
| Sodium Cyanide       | input technosphere | g/t  | 35.2                              |
| Cyanide              | water emission     | g/t  | 3.88                              |

**Tab. 5.7 Dust emissions from typical coal strip-mining operations (Anonymous (1998b)). After this table underground mining cause roughly 80% of the dust emissions of open pit mining.**

| Sources of dust       | Truck and shovel operation | Dragline operation | Occurrence in underground mining |
|-----------------------|----------------------------|--------------------|----------------------------------|
|                       | % of total dust emission   |                    |                                  |
| Dragline              | n.a                        | 27                 | Yes                              |
| Haul roads            |                            |                    |                                  |
| Overburden            | 35                         | -                  | Yes                              |
| Coal                  | 7                          | 42                 | Yes                              |
| Loading               |                            |                    |                                  |
| Overburden            | 12                         | -                  | Yes                              |
| Coal                  | 4                          | 5                  | Yes                              |
| Drilling              |                            |                    |                                  |
| (Overburden and coal) | 1                          | 2                  | No                               |
| Blasting              |                            |                    |                                  |
| (Overburden and coal) | 5                          | 1                  | No                               |
| Truck dumping         | 7                          | 5                  | Yes                              |
| Topsoil removal       | 18                         | 10                 | No                               |
| Exposed areas         | 6                          | 7                  | No                               |
| Haul-road repairs     | 5                          | 1                  | Yes                              |

**Tab. 5.8 Emissions to air resulting from mining activities per tonne mined ore. Values for total dust emission are taken from Anonymous (1998b), partition of fractions of particulate matter from the CEMIP-database<sup>7</sup>, the dust composition correspond to an specific nickel sulphide with values taken from the IEAE-database (see text).**

| Emission to air | Unit       | Emission per tonne mined ore |
|-----------------|------------|------------------------------|
| Cobalt          | mg / t ore | 151                          |
| Copper          | mg / t ore | 3680                         |
| Lead            | mg / t ore | 36.8                         |
| Nickel          | mg / t ore | 4526                         |
| Silver          | mg / t ore | 1.58                         |
| Palladium       | mg / t ore | 9.94E-03                     |
| Platinum        | mg / t ore | 0.151                        |
| Aluminium       | mg / t ore | 18400                        |
| Gold            | mg / t ore | 0.0736                       |
| Calcium         | mg / t ore | 12880                        |
| Magnesium       | mg / t ore | 11040                        |
| Sulphur         | mg / t ore | 36800                        |
|                 |            |                              |
| PM<2.5          | g / t ore  | 18.1                         |
| PM2.5-10        | g / t ore  | 162.8                        |
| PM>10           | g / t ore  | 187.1                        |
| total dust      | g / t ore  | 368                          |

### Metallurgy and Separation of Copper as Co-product.

As shown in Tab. 5.2 there exist a variety of different production routes in the nickel metallurgy. In this study all five process alternatives were considered according to their production share given in Hilbrans & Hinrichs (1999). These alternatives cover at least 79 % of the world production of Class I nickel. The values thus already are aggregated and reflect the mix of the different alternatives within a certain process step, e. g. the electricity consumption in the roasting step. Consequently allocation factors for nickel differ from step to step.

**Drying** (alternative 2 &4). Flux is assumed to consist of 50% quartzsand and 50% lime, whose combustion releases 0.44 kg CO<sub>2</sub> per tonne lime, inventoried later in “overall process”. Drying is done by gas. According to Hilbrans & Hinrichs (1999), 40% of the electricity consumed is from grid, 60% is hydropower.

**Roasting** (alternative 1, 3 & 5). Flux is assumed to consist of 50% quartzsand and 50% lime, whose combustion releases 0.44 kg CO<sub>2</sub> per tonne lime, inventoried later in “overall process”. Drying is done by oil. According to Hilbrans & Hinrichs (1999), 40% of the electricity consumed is from grid, 60% is hydropower. A production of steam is reported in (Hilbrans & Hinrichs (1999)), which can not be accounted for as benefit.

**Smelting** (all alternatives). Smelting is mainly done by oil (alternative 5, fluidized bed kiln) and in alternative 4 high amounts of coke are required. 40% electricity are from grid, 60% hydropower assumed. A production of steam is reported, which can not be accounted for as benefit. Dust emissions from smelting are inventoried later under “overall process”. Here the major part of the sulphur dioxide is emitted. The value of 1,5 t SO<sub>2</sub> per tonne metal produced includes also the sulphur dioxide emissions from drying and roasting. This value is reported at INCO’s environmental report for the sud-

<sup>7</sup> [http://www.air.sk/tno/cepmeip/em\\_factors.php](http://www.air.sk/tno/cepmeip/em_factors.php), accessed 01.03.2003

bury's operation<sup>8</sup>. It was chosen because it probably depicts the global situation more accurate than the much lower values reported in (IPPC (2001)) reflecting the European situation with 0.164 t SO<sub>2</sub> / t, or the much higher value from the Russian operation at Noril'sk (5.3 t SO<sub>2</sub>/ t) reported in (Hochfeld (1997)).

**Converting** (all alternatives). Converting is done by gas. According to Hilbrans & Hinrichs (1999), 40% of the electricity consumed is from grid, 60% is hydropower. A production of steam is reported in (Hilbrans & Hinrichs (1999)), which can not be accounted for as benefit. Reported dust emissions are not fugitive, they are caught in the wet scrubber and form a residue, which has to be disposed. Fugitive dust emissions are inventoried later under "overall process".

**Sulphuric acid production** (all alternatives). For production 40% electricity from grid, 60% hydropower assumed. The acid formed is not inventoried because it is assumed to be a by-product.

**Other processes** (all alternatives). Different electricity use from facilities etc. 40% of the electricity from grid, 60% hydropower assumed.

**Leaching / copper separation** (alternatives 3, 4 & 5). Pyrometallurgical separation is done by gas. 40% electricity from grid, 60% hydropower assumed. Leaching agents reported in (Hilbrans & Hinrichs (1999)) are approximated with the inventory for ammoniac production but no process emissions were considered due to lack of data. Ammonium sulphate as by-product from alternative 2 was not inventoried and assumed to be burden free (cut off).

**Overall process.** The emissions to air are taken from different sources as shown in Tab. 5.9. Effluents were approximated with the value of 1.4 m<sup>3</sup> per tonne concentrate worked from PGM-operations reported in (Hochfeld (1997)) (Tab. 5.10). The emissions due to the effluent are taken from (IPPC (2001)). Carbon dioxide from the calcination of lime added in several steps before is released at a rate of 0.44 t CO<sub>2</sub> per tonne lime added.

**Purification.** Until this step all inventory was allocated according the revenue obtained by co-production of copper in alternatives 3, 4 & 5 (The only exception is the resource that is allocated by mass (34% to copper)). In this purification step the inventory is fully allocated to nickel since no copper is present any more. Each alternative differs from each other. An overview over the different contributions is given in Fig. 5.7. No data on chlorine input and related emissions is available. For the reduction with hydrogen (alternative 2) a simple stoichiometry Ni:H of 59:1 yielding and a reaction efficiency of 90% was assumed, yielding a necessary input of 19.1 kg hydrogen per tonne nickel reduced. For all electricity use a share of 40% electricity from grid and 60% hydropower was assumed.

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<sup>8</sup> globally operating nickel producer, <http://www.inco.com>

**Tab. 5.9 Emissions to air per tonne produced metal.** Data was taken from similar processes in nickel (A) and copper (B) metallurgy and multiplied with the sum of produced metals (PGM, Ni & Cu). Data is taken from (IPPC (2001)), the distribution of the particle size is taken from CEPMEIP<sup>9</sup> (C). VOC emission from leaching (D). In this study values for SO<sub>2</sub> emissions are estimated roughly with 1.5 t / t Ni as stated in the environmental report of the sudbury operation.

| Substance       | Amount in kg / t | Source     |
|-----------------|------------------|------------|
| Pb              | 0.03             | B          |
| Zn              | 0.10             | B          |
| Sn              | 0.007            | B          |
| Cu              | 0.04             | B          |
| As              | 0.006            | B          |
| Total dust      | 0.41             | A          |
| PM<2.5          | 0.0005           | C          |
| PM2.5-10        | 0.3              | C          |
| PM>10           | 0.10             | C          |
| SO <sub>2</sub> | 1500             | estimation |
| Ni              | 0.03             | A          |
| Co              | 0.03             | A          |
| VOC             | 0.20             | A, D       |
| TCDD            | 1.00E-08         | B          |

**Tab. 5.10 Composition of different wastewater effluents after treatment.** A simple average from different reported values of European copper operations is chosen (IPPC (2001)). Values in the columns “process water”, “surface run off” and “direct cooling water” represent geometric means of the range reported. Values for Cr, Hg and Sn correspond to a copper semis production site, reported in (IPPC (2001)).

| Sub-<br>stance | Process<br>Water      | Surface run<br>off    | Direct coo-<br>ling water | This study<br>(arithmetic<br>mean) |
|----------------|-----------------------|-----------------------|---------------------------|------------------------------------|
|                | [g / m <sup>3</sup> ] | [g / m <sup>3</sup> ] | [g / m <sup>3</sup> ]     | [g / m <sup>3</sup> ]              |
| Cu             | 0.045<br>(0.01-0.2)   | 0.063<br>(0.01-0.4)   | 0.05<br>(0.01-0.25)       | 0.0527                             |
| Pb             | 0.006<br>(0.001-0.04) | 0.032<br>(0.005-0.2)  | 0.01<br>(0.001-0.1)       | 0.0160                             |
| As             | 0.032<br>(0.01-0.1)   | 0.014<br>(0.003-0.07) | 0.01<br>(0.001-0.1)       | 0.0187                             |
| Ni             | 0.024<br>(0.004-0.15) | 0.028<br>(0.002-0.4)  | 0.011<br>(0.002-0.06)     | 0.0212                             |
| Cd             | 0.003<br>(0.0001-0.1) | 0.004<br>(0.0002-0.1) | 0.001<br>(0.0001-0.003)   | 0.00273                            |
| Zn             | 0.045<br>(0.01-0.2)   | 0.11<br>(0.03-0.4)    | 0.1<br>(0.02-0.5)         | 0.0848                             |
| Cr             |                       |                       |                           | 0.029                              |
| Hg             |                       |                       |                           | 0.00029                            |
| Sn             |                       |                       |                           | 0.029                              |

<sup>9</sup> [http://www.air.sk/tno/cepmeip/em\\_factors.php](http://www.air.sk/tno/cepmeip/em_factors.php), accessed 01.03.2003

**Data quality**

**Mining.** Data for major mass fluxes are satisfactory, while the emissions in to water may be overestimated due to different system boundaries of reported effluent volume from Hilbrans & Hinrichs (1999) and total effluent emissions reported in IPPC (2002).

**Beneficiation.** Like in mining, the emissions to water can be overestimated. The emissions from tailings are assessed in an extern module, which itself contains rough assumptions of a global overburden and tailings composition and a simplified disposal scenario. Emissions though can vary considerably from case to case.

**Metallurgy.** Process data is satisfactory and valid for over 90 % of the Class I nickel production. The main emissions dominating possibly the whole process is sulphur dioxide, which is not reported in Hilbrans & Hinrichs (1999). The chosen value of a single operation is consistent with other reported values in IPPC (2001) and Hochfeld (1997). Poor data are available on composition and fate of solid waste, the reuse rate was documented.

**Separation of copper and further refining.** Main data on energy and mass fluxes is satisfactory, while data on direct emissions due to hydrometallurgical processes is poor. Namely chlorine emissions can be considerable, generating both HCl emissions to air and AOX emissions to water – emissions which are not assessed in this work. An other emission, which is not inventoried due to poor data, is carbondisulfide used in alternative 5 in considerable amounts.

| General Flow Information |                                                   |                                                |                        | Representation in ecoinvent                                                                                                                                                                                    |                        |                            |                 |          |                                                                         |               | Uncertainty Informations |                                        |      |                    |                  |
|--------------------------|---------------------------------------------------|------------------------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|----------------------------|-----------------|----------|-------------------------------------------------------------------------|---------------|--------------------------|----------------------------------------|------|--------------------|------------------|
| Input                    | Process Name                                      | Output                                         | Allocation coefficient | Remarks                                                                                                                                                                                                        | Category               | Sub-category               | Infra-structure | Location | Modul name in ecoinvent                                                 | Mean value    | Unit                     | Source for mean value                  | Type | Standard Deviation | General Comment  |
| mining                   | nickel production, sulphidic ore, primary, part I | Ni in ground                                   | 100% to Ni; mass       | 51 kg overburden and 110 kg crude ore per kg Class I Nickel mined (resulting in 97.2 kg tailings and 12.8 kg concentrate). 82% yield of mining/beneficiation, 97% yield of metallurgy --> 79.5% overall yield. | resource               | in ground                  |                 |          | Nickel, 1.13% in sulfide, Ni 0.76% and Cu 0.76% in crude ore, in ground | 1.26E+00 kg   |                          | (Hilbrans, 1999)                       | 1    | 1.13               | (2,2,2,1,1,4,12) |
|                          |                                                   | Cu in ground                                   | 100% to Cu; mass       |                                                                                                                                                                                                                | resource               | in ground                  |                 |          | Copper, 1.13% in sulfide, Cu 0.76% and Ni 0.76% in crude ore, in ground | 6.48E-01 kg   |                          | (Hilbrans, 1999)                       | 1    | 1.13               | (2,2,2,1,1,4,12) |
|                          |                                                   | explosive                                      | 27% to Cu; revenue     |                                                                                                                                                                                                                | construction processes | civil engineering          | No              | RER      | blasting                                                                | 1.66E-01 kg   |                          | (Hilbrans, 1999)                       | 1    | 1.13               | (2,2,2,1,1,4,4)  |
|                          |                                                   | diesel fuel (energy)                           | 27% to Cu; revenue     |                                                                                                                                                                                                                | construction processes | machinery                  | No              | GLO      | diesel, burned in building machine                                      | 1.12E+01 MJ   |                          | (Hilbrans, 1999)                       | 1    | 1.13               | (2,2,2,1,1,4,4)  |
|                          |                                                   | electricity from grid, mining                  | 27% to Cu; revenue     | Assumption: 40% electricity from grid                                                                                                                                                                          | electricity            | production mix             | No              | UCTE     | electricity, medium voltage, production UCTE, at grid                   | 1.74E+00 kWh  |                          | (Hilbrans, 1999)                       | 1    | 1.14               | (2,3,2,1,1,4,2)  |
|                          |                                                   | hydropower, mining                             | 27% to Cu; revenue     | Assumption: 60% hydropower                                                                                                                                                                                     | hydro power            | power plants               | No              | RER      | electricity, hydropower, at run of-river power plant                    | 2.62E+00 kWh  |                          | (Hilbrans, 1999)                       | 1    | 1.14               | (2,3,2,1,1,4,2)  |
|                          |                                                   | fresh and processwater (mining)                | 27% to Cu; revenue     | unspecified - because of the relative abundance of pumped ground water in the mining process, surface water is chosen here.                                                                                    | resource               | in water                   |                 |          | Water, river                                                            | 3.82E-02 m3   |                          | (Hilbrans, 1999)                       | 1    | 1.13               | (2,2,2,1,1,4,12) |
|                          |                                                   | stowing material, sand                         | 27% to Cu; revenue     | additional material, no excavation included                                                                                                                                                                    | construction materials | additives                  | No              | CH       | sand, at mine                                                           | 4.56E+01 kg   |                          | (Hilbrans, 1999)                       | 1    | 1.13               | (2,2,2,1,1,4,4)  |
|                          |                                                   | cement                                         | 27% to Cu; revenue     |                                                                                                                                                                                                                | construction materials | binder                     | No              | CH       | portland calcareous cement, at plant                                    | 3.62E+00 kg   |                          | (Hilbrans, 1999)                       | 1    | 1.13               | (2,2,2,1,1,4,4)  |
|                          |                                                   | pit water                                      | 27% to Cu; revenue     |                                                                                                                                                                                                                | resource               | in water                   |                 |          | Water, well, in ground                                                  | 6.79E-02 m3   |                          | (Hilbrans, 1999)                       | 1    | 1.13               | (2,2,2,1,1,4,12) |
|                          |                                                   | transports, lorry, mining                      | 27% to Cu; revenue     | transportation of cement, explosives and sand; 50km assumed.                                                                                                                                                   | transport systems      | road                       | No              | RER      | transport, lorry >16t, fleet average                                    | 2.47E+00 tkm  |                          | calculations based on (Hilbrans, 1999) | 1    | 2.11               | (3,2,2,1,3,5,5)  |
|                          |                                                   | exploration infrastructure, conveyor belts     | 27% to Cu; revenue     | exploration infrastructure, 2000 m belts assumed                                                                                                                                                               | construction processes | machinery                  | Yes             | RER      | conveyor belt, at plant                                                 | 4.20E-06 m    |                          | own assumptions                        | 1    | 1.26               | (3,2,2,1,3,4,4)  |
|                          |                                                   | infrastructure of underground mining operation | 27% to Cu; revenue     |                                                                                                                                                                                                                | metals                 | extraction                 | Yes             | GLO      | non-ferrous metal mine, underground                                     | 5.50E-09 unit |                          | own assumptions                        | 1    | 3.07               | (3,2,2,1,3,4,9)  |
|                          |                                                   | overburden, disposed                           | 27% to Cu; revenue     |                                                                                                                                                                                                                | waste management       | residual material landfill | No              | GLO      | disposal, sulfidic tailings, off-site                                   | 3.49E+01 kg   |                          | (Hilbrans, 1999)                       | 1    | 1.13               | (2,2,2,1,1,4,6)  |
|                          |                                                   | overburden, refilled                           | 27% to Cu; revenue     | not inventoried, no ecological burden                                                                                                                                                                          |                        |                            |                 |          |                                                                         | 1.58E+01 kg   |                          | (Hilbrans, 1999)                       | 1    | 1.13               | (2,2,2,1,1,4,6)  |
|                          |                                                   | off-gas from blasting                          | 27% to Cu; revenue     | included in module blasting (2941)                                                                                                                                                                             |                        |                            |                 |          |                                                                         |               |                          |                                        |      |                    |                  |
|                          |                                                   | off-gas from transports                        | 27% to Cu; revenue     | included in module diesel, burned in building machine (1350)                                                                                                                                                   |                        |                            |                 |          |                                                                         |               |                          |                                        |      |                    |                  |
|                          |                                                   | effluents, mining                              | 27% to Cu; revenue     | Detailed composition see Tab. 5.4. The emissions are inventoried.                                                                                                                                              |                        |                            |                 |          |                                                                         | 6.79E-02 m3   |                          | (Hilbrans, 1999)                       | 1    | 1.13               | (2,2,2,1,1,4,6)  |

**Fig. 5.3 Flows for the multi-output process “Nickel production” (part mining) and its representation in the ecoinvent database. Values correspond to the functional unit of 1 kg Class I nickel. Allocation values indicate the amount of the flow that is allocated to the co-product copper.**



| General Flow Information |                                                    |                                       |                        | Representation in ecoinvent                                                                                                                                                   |                        |                            |                 |          |                                       |            | Uncertainty Informations |                                |      |                    |                                  |
|--------------------------|----------------------------------------------------|---------------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|----------------------------|-----------------|----------|---------------------------------------|------------|--------------------------|--------------------------------|------|--------------------|----------------------------------|
| Input                    | Process Name                                       | Output                                | Allocation coefficient | Remarks                                                                                                                                                                       | Category               | Sub-category               | Infra-structure | Location | Modul name in ecoinvent               | Mean value | Unit                     | Source for mean value          | Type | Standard Deviation | General Comment                  |
| beneficiation            | nickel production, sulphidic ore, primary, part II | electricity, beneficiation            | 27% to Cu; revenue     | included in mining                                                                                                                                                            |                        |                            |                 |          |                                       |            |                          |                                |      |                    |                                  |
|                          |                                                    | fresh and processwater (benefication) | 27% to Cu; revenue     |                                                                                                                                                                               | resource               | in water                   |                 |          | Water, well, in ground                | 1.52E-01   | m3                       | (Hilbrans, 1999)               | 1    | 1.13               | (2,2,1,1,1,4,12)                 |
|                          |                                                    | lime                                  | 27% to Cu; revenue     |                                                                                                                                                                               | construction materials | others                     | No              | CH       | limestone, milled, packed, at plant   | 4.96E-02   | kg                       | (Krauss, 1999)                 | 1    | 2.00               | reported values                  |
|                          |                                                    | transports, lorry, beneficiation      | 27% to Cu; revenue     | transportation of lime, floating agents and sodium cyanide; 50 km assumed.                                                                                                    | transport systems      | road                       | No              | RER      | transport, lorry >16t, fleet average  | 8.15E-03   | tkm                      | calculation                    | 1    | 2.07               | (3,2,1,5,3,2,5); calculation     |
|                          |                                                    | infrastructure of beneficiation       | 27% to Cu; revenue     | AlOH-production plant chosen as proxy                                                                                                                                         | metals                 | extraction                 | Yes             | RER      | aluminium hydroxide, plant            | 9.24E-10   | unit                     | own assumptions                | 1    | 3.07               | (3,2,2,1,3,4,9)                  |
|                          |                                                    | floating agents, organic              | 27% to Cu; revenue     |                                                                                                                                                                               | chemicals              | organics                   | No              | GLO      | chemicals organic, at plant           | 2.48E-02   | kg                       | (IPPC, 2002)                   | 1    | 2.00               | reported values                  |
|                          |                                                    | floating agents, inorganic            | 27% to Cu; revenue     |                                                                                                                                                                               | chemicals              | inorganics                 | No              | GLO      | chemicals inorganic, at plant         | 8.47E-02   | kg                       | (IPPC, 2002)                   | 1    | 2.00               | reported values                  |
|                          |                                                    | Sodium cyanide                        | 27% to Cu; revenue     | assumed to be produced on site                                                                                                                                                | chemicals              | organics                   | No              | RER      | hydrogen cyanide, at plant            | 3.85E-03   | kg                       | (IPPC, 2002)                   | 1    | 2.00               | reported values                  |
|                          |                                                    | Sulphuric Acid                        | 27% to Cu; revenue     | burden-free, since byproduct from metallurgy                                                                                                                                  |                        |                            |                 |          |                                       | 2.54E-01   | kg                       | (IPPC, 2002)                   | 1    | 2.00               | reported values                  |
|                          |                                                    | tailings, disposed                    | 27% to Cu; revenue     | Tailings that are disposed of in ponds or other facilities.                                                                                                                   | waste management       | residual material landfill | No              | GLO      | disposal, sulfidic tailings, off-site | 6.50E+01   | kg                       | (Hilbrans, 1999)               | 1    | 1.10               | (2,2,2,3,1,3,6); reported values |
|                          |                                                    | tailings, refilled                    | 27% to Cu; revenue     | not inventoried, no ecological burden                                                                                                                                         |                        |                            |                 |          |                                       | 3.33E+01   | kg                       | (Hilbrans, 1999)               | 1    | 1.13               | (2,2,2,1,1,4,6)                  |
|                          |                                                    | effluents, beneficiation              | 27% to Cu; revenue     | composition in Tab. 5.4. The single emissions are inventoried with a variability of 10                                                                                        |                        |                            |                 |          |                                       | 1.52E-01   | m3                       | (Hilbrans, 1999)               | 1    | 1.10               | (2,2,2,3,1,3,6); reported values |
|                          |                                                    | Emission Cyanide                      | 27% to Cu; revenue     | From sodium cyanide (35g NaCN per t crude ore, 80% recovered --> 3.75 g CN- per t crude ore)                                                                                  | water                  | river                      |                 |          | Cyanide                               | 4.12E-04   | kg                       | (IPPC, 2002), (Anonymous 1998) | 1    | 1.84               | (4,2,2,3,4,4,33)                 |
|                          |                                                    | emission carbondisulfide              | 27% to Cu; revenue     | 0.53 kg carbondisulfide out of 1 kg xanthate through hydrolytical decomposition                                                                                               | air                    | low population density     |                 |          | Carbon disulfide                      | 1.10E-02   | kg                       | (NPI, 2001)                    | 1    | 2.31               | (4,2,2,5,4,4,23)                 |
|                          |                                                    | Dust                                  | 27% to Cu; revenue     | including emissions from mining activities. Dust is the sum of particulate emissions represented in fractions PM<2.5, PM2.5-10, PM>10, including metals and salts. (Tab. 5.8) |                        |                            |                 |          |                                       | 4.05E-02   | kg                       | (NPI, 2001)                    | 1    | 1.59               | (3,2,1,1,3,4,25)                 |

Fig. 5.4 Flows for the multi-output process “Nickel production” (part beneficiation) and its representation in the ecoinvent database. Values correspond to the functional unit of 1 kg Class I nickel. Allocation values indicate the amount of the flow that is allocated to the co-product copper.

| General Flow Information |                                                     |                             |                                     | Representation in ecoinvent       |                                 |                        |                 |          |                                                      |                                                                  | Uncertainty Informations |                       |                 |                    |                 |
|--------------------------|-----------------------------------------------------|-----------------------------|-------------------------------------|-----------------------------------|---------------------------------|------------------------|-----------------|----------|------------------------------------------------------|------------------------------------------------------------------|--------------------------|-----------------------|-----------------|--------------------|-----------------|
| Input                    | Process Name                                        | Output                      | Allocation coefficient              | Remarks                           | Category                        | Sub-category           | Infra-structure | Location | Modul name in ecoinvent                              | Mean value                                                       | Unit                     | Source for mean value | Type            | Standard Deviation | General Comment |
| drying                   | nickel production, sulphidic ore, primary, part III | limestone                   | 17% to Cu; revenue to Cu; revenue   | flux, 50% limestone assumed       | construction materials          | others                 | No              | CH       | limestone, milled, packed, at plant                  | 5.40E-01 kg                                                      | (Hilbrans 1999)          | 1                     | 1.51            | (5,2,2,3,1,3,4)    |                 |
|                          |                                                     | silica sand                 | 17% to Cu; revenue to Cu; revenue   | flux, 50% quartzsand assumed      | construction materials          | additives              | No              | DE       | silica sand, at plant                                | 5.40E-01 kg                                                      | (Hilbrans 1999)          | 1                     | 1.51            | (5,2,2,3,1,3,4)    |                 |
|                          |                                                     | electricity drying, grid    | 11% to Cu; revenue to Cu; revenue   | 40% electricity from grid         | electricity                     | production mix         | No              | UCTE     | electricity, high voltage, production UCTE, at grid  | 2.83E-02 kWh                                                     | (Hilbrans 1999)          | 1                     | 1.10            | (2,2,2,3,1,3,2)    |                 |
|                          |                                                     | hydropower drying           | 11% to Cu; revenue to Cu; revenue   | 60% electricity from hydropower   | hydro power                     | power plants           | No              | RER      | electricity, hydropower, at run of-river power plant | 4.24E-02 kWh                                                     | (Hilbrans 1999)          | 1                     | 1.10            | (2,2,2,3,1,3,2)    |                 |
|                          |                                                     | fossile energy drying (gas) | 11% to Cu; revenue to Cu; revenue   |                                   | natural gas                     | heating systems        | No              | RER      | natural gas, burned in industrial furnace >100kW     | 2.44E+00 MJ                                                      | (Hilbrans 1999)          | 1                     | 1.10            | (2,2,2,3,1,3,1)    |                 |
|                          |                                                     | roasting                    | limestone                           | 13% to Cu; revenue to Cu; revenue | flux, 50% limestone assumed     | construction materials | others          | No       | CH                                                   | limestone, milled, packed, at plant                              | 1.66E+00 kg              | (Hilbrans 1999)       | 1               | 1.51               | (5,2,2,3,1,3,4) |
|                          |                                                     |                             | silica sand                         | 13% to Cu; revenue to Cu; revenue | flux, 50% quartzsand assumed    | construction materials | additives       | No       | DE                                                   | silica sand, at plant                                            | 1.66E+00 kg              | (Hilbrans 1999)       | 1               | 1.51               | (5,2,2,3,1,3,4) |
|                          |                                                     |                             | electricity roasting, grid          | 11% to Cu; revenue to Cu; revenue | 40% electricity from grid       | electricity            | production mix  | No       | UCTE                                                 | electricity, high voltage, production UCTE, at grid              | 8.68E-02 kWh             | (Hilbrans 1999)       | 1               | 1.10               | (2,2,2,3,1,3,2) |
|                          |                                                     |                             | hydropower roasting                 | 11% to Cu; revenue to Cu; revenue | 60% electricity from hydropower | hydro power            | power plants    | No       | RER                                                  | electricity, hydropower, at run of-river power plant             | 1.30E-01 kWh             | (Hilbrans 1999)       | 1               | 1.10               | (2,2,2,3,1,3,2) |
|                          |                                                     |                             | fossile energy (oil)                | 20% to Cu; revenue to Cu; revenue |                                 | oil                    | heating systems | No       | RER                                                  | heavy fuel oil, burned in industrial furnace 1MW, non-modulating | 1.80E+00 MJ              | (Hilbrans 1999)       | 1               | 1.10               | (2,2,2,3,1,3,1) |
|                          | steam, credit                                       | 4% to Cu; revenue           | not inventoried, no credits assumed |                                   |                                 |                        |                 |          | -7.81E-01 MJ                                         | (Hilbrans 1999)                                                  | 1                        | 1.10                  | (2,2,2,3,1,3,4) |                    |                 |

Fig. 5.5 Flows for the multi-output process “Nickel production” (part drying/roasting) and its representation in the ecoinvent database. Values correspond to the functional unit of 1 kg Class I nickel. Allocation values indicate the amount of the flow that is allocated to the co-product copper.

| General Flow Information |                                                    |                              |                           | Representation in ecoinvent                                        |                  |                            |                 |          |                                                                        |            | Uncertainty Informations |                       |      |                    |                  |
|--------------------------|----------------------------------------------------|------------------------------|---------------------------|--------------------------------------------------------------------|------------------|----------------------------|-----------------|----------|------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|------------------|
| Input                    | Process Name                                       | Output                       | Allocation coefficient    | Remarks                                                            | Category         | Sub-category               | Infra-structure | Location | Modul name in ecoinvent                                                | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment  |
| smelting                 | nickel production, sulphidic ore, primary, part IV | electricity melting, grid    | 7% to Cu; revenue to Cu;  | 40% electricity from grid                                          | electricity      | production mix             | No              | UCTE     | electricity, high voltage, production UCTE, at grid                    | 1.16E+00   | kWh                      | (Hilbrans 1999)       | 1    | 1.10               | (2,2,2,3,1,3,2)  |
|                          |                                                    | hydropower melting           | 7% to Cu; revenue to Cu;  | 60% electricity from hydropower                                    | hydro power      | power plants               | No              | RER      | electricity, hydropower, at run of-river power plant                   | 1.74E+00   | kWh                      | (Hilbrans 1999)       | 1    | 1.10               | (2,2,2,3,1,3,2)  |
|                          |                                                    | fossile energy (gas)         | 9% to Cu; revenue to Cu;  |                                                                    | natural gas      | heating systems            | No              | RER      | natural gas, burned in industrial furnace >100kW                       | 2.80E+00   | MJ                       | (Hilbrans 1999)       | 1    | 1.10               | (2,2,2,3,1,3,1)  |
|                          |                                                    | fossile energy (coke)        | 15% to Cu; revenue to Cu; |                                                                    | hard coal        | heating systems            | No              | RER      | hard coal, burned in industrial furnace 1-10MW                         | 2.20E+00   | MJ                       | (Hilbrans 1999)       | 1    | 1.10               | (2,2,2,3,1,3,2)  |
|                          |                                                    | fossile energy melting (oil) | 20% to Cu; revenue to Cu; |                                                                    | oil              | heating systems            | No              | RER      | heavy fuel oil, burned in industrial furnace 1MW, non-modulating       | 2.48E+01   | MJ                       | (Hilbrans 1999)       | 1    | 1.10               | (2,2,2,3,1,3,2)  |
|                          |                                                    | infrastructure smeltery      | 27% to Cu; revenue to Cu; | including infrastructure for separation of copper                  | metals           | extraction                 | Yes             | GLO      | non-ferrous metal smelter                                              | 4.61E-11   | unit                     | assumption            | 1    | 3.30               | (4,2,1,5,4,4,9)  |
|                          |                                                    | steam, credit                | 15% to Cu; revenue to Cu; | not inventoried, no credits assumed                                |                  |                            |                 |          |                                                                        | -1.31E+01  | MJ                       | (Hilbrans 1999)       | 1    | 1.10               | (2,2,2,3,1,3,4)  |
|                          |                                                    | slag                         | 13% to Cu; revenue to Cu; | disposal                                                           | waste management | residual material landfill | No              | CH       | disposal, nickel smelter slag, 0% water, to residual material landfill | 1.05E+01   | kg                       | (Hilbrans 1999)       | 1    | 1.10               | (2,2,2,3,1,3,6)  |
|                          |                                                    | dust                         |                           | inventoried as overall emissions                                   |                  |                            |                 |          |                                                                        |            |                          |                       |      |                    |                  |
|                          |                                                    | sulphur dioxide in off-gas   | 10% to Cu; revenue to Cu; | comprise emissions from drying an roasting                         | air              | low population density     |                 |          | Sulfur dioxide                                                         | 1.50E+00   | kg                       | (Inco 2002)           | 1    | 1.64               | (5,4,1,3,3,5,15) |
| converting               | nickel production, sulphidic ore, primary, part IV | electricity, grid            | 15% to Cu; revenue to Cu; | 40% electricity from grid                                          | electricity      | production mix             | No              | UCTE     | electricity, high voltage, production UCTE, at grid                    | 5.66E-01   | kWh                      | (Hilbrans 1999)       | 1    | 1.10               | (2,2,2,3,1,3,2)  |
|                          |                                                    | hydropower                   | 15% to Cu; revenue to Cu; | 60% electricity from hydropower                                    | hydro power      | power plants               | No              | RER      | electricity, hydropower, at run of-river power plant                   | 8.49E-01   | kWh                      | (Hilbrans 1999)       | 1    | 1.10               | (2,2,2,3,1,3,2)  |
|                          |                                                    | fossile energy (gas)         | 12% to Cu; revenue to Cu; | energy carrier assumed to be natural gas                           | natural gas      | heating systems            | No              | RER      | natural gas, burned in industrial furnace >100kW                       | 3.05E+00   | MJ                       | (Hilbrans 1999)       | 1    | 1.10               | (2,2,2,3,1,3,1)  |
|                          |                                                    | steam, credit                | 13% to Cu; revenue to Cu; | not inventoried, no credits assumed                                |                  |                            |                 |          |                                                                        | -2.86E+00  | MJ                       | (Hilbrans 1999)       | 1    | 1.10               | (2,2,2,3,1,3,4)  |
|                          |                                                    | airborn dust                 | 12% to Cu; revenue to Cu; | residues in wet scrubber, approximated with slag disposal - module | waste management | residual material landfill | No              | CH       | disposal, nickel smelter slag, 0% water, to residual material landfill | 3.97E-01   | kg                       | (Hilbrans 1999)       | 1    | 1.10               | (2,2,2,3,1,3,6)  |
|                          |                                                    |                              |                           |                                                                    |                  |                            |                 |          |                                                                        |            |                          |                       |      |                    |                  |

Fig. 5.6 Flows for the multi-output process “Nickel production” (part smelting/converting) and its representation in the ecoinvent database. Values correspond to the functional unit of 1 kg Class I nickel. Allocation values indicate the amount of the flow that is allocated to the co-product copper.

|                             | General Flow Information           |                                                   |                 |                                                      | Representation in ecoinvent                                                                                                         |                                                            |                        |                 |                        |                                                      |                                                      | Uncertainty Informations |                       |                 |                                                    |                  |               |
|-----------------------------|------------------------------------|---------------------------------------------------|-----------------|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|------------------------|-----------------|------------------------|------------------------------------------------------|------------------------------------------------------|--------------------------|-----------------------|-----------------|----------------------------------------------------|------------------|---------------|
|                             | Input                              | Process Name                                      | Output          | Allocation coefficient                               | Remarks                                                                                                                             | Category                                                   | Sub-category           | Infra-structure | Location               | Modul name in ecoinvent                              | Mean value                                           | Unit                     | Source for mean value | Type            | Standard Deviation                                 | General Comment  |               |
| sulf. Acid                  | electricity, grid                  | nickel production, sulphidic ore, primary, part V | sulfuric acid   | 12%                                                  | to Cu; revenue                                                                                                                      | 40% electricity from grid                                  | electricity            | production mix  | No                     | UCTE                                                 | electricity, high voltage, production UCTE, at grid  | 4.99E-01                 | kWh                   | (Hilbrans 1999) | 1                                                  | 1.10             | (2,2,3,1,3,2) |
|                             | hydropower                         |                                                   |                 | 12%                                                  | to Cu; revenue                                                                                                                      | 60% electricity from hydropower                            | hydro power            | power plants    | No                     | RER                                                  | electricity, hydropower, at run of-river power plant | 7.49E-01                 | kWh                   | (Hilbrans 1999) | 1                                                  | 1.10             | (2,2,3,1,3,2) |
| other                       | electricity, grid                  |                                                   | waste heat      | 11%                                                  | to Cu; revenue                                                                                                                      | not inventoried, assumed to be burden free (cut off)       |                        |                 |                        |                                                      | 1.13E+01                                             | kg                       | (Hilbrans 1999)       | 1               | 1.10                                               | (2,2,3,1,3,4)    |               |
|                             | hydropower                         |                                                   |                 | 17%                                                  | to Cu; revenue                                                                                                                      | 40% electricity from grid                                  | electricity            | production mix  | No                     | UCTE                                                 | electricity, high voltage, production UCTE, at grid  | 7.40E-01                 | kWh                   | (Hilbrans 1999) | 1                                                  | 1.10             | (2,2,3,1,3,2) |
| leaching, copper separation | electricity, grid (copper removal) |                                                   | 17%             | to Cu; revenue                                       | 60% electricity from hydropower                                                                                                     | hydro power                                                | power plants           | No              | RER                    | electricity, hydropower, at run of-river power plant | 1.11E+00                                             | kWh                      | (Hilbrans 1999)       | 1               | 1.10                                               | (2,2,3,1,3,2)    |               |
|                             |                                    |                                                   | 0.03%           | to Cu; revenue                                       | from total electricity                                                                                                              | air                                                        | low population density |                 |                        | Heat, waste                                          | 5.27E+01                                             | MJ                       | (Hilbrans 1999)       | 1               | 1.10                                               | (2,2,3,1,3,2)    |               |
|                             | hydropower (copper removal)        |                                                   | 4%              | to Cu; revenue                                       | 40% electricity from grid                                                                                                           | electricity                                                | production mix         | No              | UCTE                   | electricity, high voltage, production UCTE, at grid  | 2.91E-01                                             | kWh                      | (Hilbrans 1999)       | 1               | 1.10                                               | (2,2,3,1,3,2)    |               |
|                             | fossile energy (copper removal)    |                                                   | 4%              | to Cu; revenue                                       | 60% electricity from hydropower                                                                                                     | hydro power                                                | power plants           | No              | RER                    | electricity, hydropower, at run of-river power plant | 4.37E-01                                             | kWh                      | (Hilbrans 1999)       | 1               | 1.10                                               | (2,2,3,1,3,2)    |               |
|                             | leaching agents                    |                                                   | 2%              | to Cu; revenue                                       | energy carrier assumed to be natural gas                                                                                            | natural gas                                                | heating systems        | No              | RER                    | natural gas, burned in industrial furnace >100kW     | 1.76E+00                                             | MJ                       | (Hilbrans 1999)       | 1               | 1.25                                               | (3,2,2,3,3,3,1)  |               |
|                             | ammoniac                           |                                                   | 0%              | to Cu; revenue                                       | approximated with ammonia inventory                                                                                                 | chemicals                                                  | inorganics             | No              | RER                    | ammonia, liquid, at regional storehouse              | 8.37E-02                                             | kg                       | (Hilbrans 1999)       | 1               | 1.10                                               | (2,2,3,1,3,4)    |               |
|                             | triisooctylamine (TIOA)            |                                                   | 0%              | to Cu; revenue                                       | no data available                                                                                                                   |                                                            |                        |                 |                        | n.a                                                  |                                                      |                          |                       |                 |                                                    |                  |               |
|                             | overall                            |                                                   | ammonium sulfat | 8%                                                   | to Cu; revenue                                                                                                                      | no data available                                          |                        |                 |                        |                                                      |                                                      | n.a                      |                       |                 |                                                    |                  |               |
|                             |                                    |                                                   |                 | 0%                                                   | to Cu; revenue                                                                                                                      | by-product from alternative (2), considered as burden free |                        |                 |                        |                                                      |                                                      | 2.99E-01                 | kg                    | (Hilbrans 1999) | 1                                                  | 1.10             | (2,2,3,1,3,6) |
|                             |                                    |                                                   |                 | 0%                                                   | to Cu; revenue                                                                                                                      | internal recovery, not inventoried                         |                        |                 |                        |                                                      |                                                      | 1.67E-01                 | kg                    | (Hilbrans 1999) | 1                                                  | 1.10             | (2,2,3,1,3,6) |
|                             |                                    | 0%                                                |                 | to Cu; revenue                                       | internal recovery, not inventoried                                                                                                  |                                                            |                        |                 |                        |                                                      | 3.11E-02                                             | kg                       | (Hilbrans 1999)       | 1               | 1.10                                               | (2,2,3,1,3,6)    |               |
|                             |                                    | 100%                                              |                 | to Cu; mass                                          | co-product, allocation by revenue                                                                                                   | metals                                                     | extraction             | No              | GLO                    | copper, primary, couple production nickel            | 5.15E-01                                             | kg                       | (Hilbrans 1999)       |                 |                                                    |                  |               |
|                             |                                    | 9%                                                |                 | to Cu; revenue                                       | 1.4m3 per t concentrate worked as reported from PGM operations. Single emissions inventoried according to composition in Tab. 5.10. |                                                            |                        |                 |                        |                                                      | 1.26E-02                                             | m3                       | (Hilbrans 1999)       | 1               | 2.15                                               | (4,5,1,3,5,5,6)  |               |
|                             |                                    | Dust emission for overall process                 | 9%              | to Cu; revenue                                       | Single emissions inventoried according to composition in Tab 5.9.                                                                   |                                                            |                        |                 |                        |                                                      | 4.07E-04                                             | kg                       | (IPPC, 2002)          | 1               | 1.59                                               | (3,2,1,1,3,4,25) |               |
| 14%                         |                                    |                                                   | to Cu; revenue  | lime addition from benefication, roasting and drying | air                                                                                                                                 | unspecified                                                |                        |                 | Carbon dioxide, fossil | 9.92E-01                                             | kg                                                   | (IPPC, 2002)             | 1                     | 1.58            | (4,2,1,1,4,4,14); excluding stdev of lime addition |                  |               |

Fig. 5.7 Flows for the multi-output process “Nickel production” (part sulphuric acid / copper separation) and its representation in the ecoinvent database. Values correspond to the functional unit of 1 kg Class I nickel. Allocation values indicate the amount of the flow that is allocated to the co-product copper.

| General Flow Information     |                                                    |                         |                        | Representation in ecoinvent              |                  |                            |                 |          |                                                                        |            | Uncertainty Informations |                       |      |                    |                 |
|------------------------------|----------------------------------------------------|-------------------------|------------------------|------------------------------------------|------------------|----------------------------|-----------------|----------|------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------|
| Input                        | Process Name                                       | Output                  | Allocation coefficient | Remarks                                  | Category         | Sub-category               | Infra-structure | Location | Modul name in ecoinvent                                                | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment |
| reduction<br>4/5             | nickel production, sulphidic ore, primary, part VI | electricity, grid       | 0% mass                | 40% electricity from grid                | electricity      | production mix             | No              | UCTE     | electricity, high voltage, production UCTE, at grid                    | 1.60E-02   | kWh                      | (Hilbrans 1999)       | 1    | 1.10               | (2,2,2,3,1,3,2) |
|                              |                                                    | steam, credit           | 0% mass                |                                          |                  |                            |                 |          |                                                                        | -2.40E-01  | MJ                       | (Hilbrans 1999)       |      |                    |                 |
|                              |                                                    | hydropower              | 0% mass                | 60% electricity from hydropower          | hydro power      | power plants               | No              | RER      | electricity, hydropower, at run of-river power plant                   | 2.40E-02   | kWh                      | (Hilbrans 1999)       | 1    | 1.10               | (2,2,2,3,1,3,2) |
|                              |                                                    | fossile energy          | 0% mass                |                                          | natural gas      | heating systems            | No              | RER      | natural gas, burned in industrial furnace >100kW                       | 3.87E+00   | MJ                       | (Hilbrans 1999)       | 1    | 1.10               | (2,2,2,3,1,3,1) |
|                              |                                                    | electricity, grid       | 0% mass                | 40% electricity from grid                | electricity      | production mix             | No              | UCTE     | electricity, high voltage, production UCTE, at grid                    | 5.57E-01   | kWh                      | (Hilbrans 1999)       | 1    | 1.10               | (2,2,2,3,1,3,2) |
|                              |                                                    | hydropower              | 0% mass                | 60% electricity from hydropower          | hydro power      | power plants               | No              | RER      | electricity, hydropower, at run of-river power plant                   | 8.35E-01   | kWh                      | (Hilbrans 1999)       | 1    | 1.10               | (2,2,2,3,1,3,2) |
|                              |                                                    | fossile energy (oil)    | 0% mass                |                                          | natural gas      | heating systems            | No              | RER      | natural gas, burned in industrial furnace >100kW                       | 9.93E-02   | MJ                       | (Hilbrans 1999)       | 1    | 1.10               | (2,2,2,3,1,3,1) |
|                              |                                                    | chlorine                | 0% mass                | not available                            |                  |                            |                 |          |                                                                        |            |                          |                       |      |                    |                 |
|                              |                                                    | chlorine emission (HCl) | 0% mass                | not available                            |                  |                            |                 |          |                                                                        |            |                          |                       |      |                    |                 |
|                              |                                                    | anode slime             | 0% mass                | approximated with slag disposal - module | waste management | residual material landfill | No              | CH       | disposal, nickel smelter slag, 0% water, to residual material landfill | 7.48E-02   | kg                       | (Hilbrans 1999)       | 1    | 2.15               | (4,5,1,3,5,5,6) |
|                              |                                                    | spoiled anodes          | 0% mass                | internal use, not inventoried            |                  |                            |                 |          |                                                                        | 1.82E-01   | kg                       | (Hilbrans 1999)       | 1    |                    |                 |
|                              |                                                    | electricity, grid       | 0% mass                | 40% electricity from grid                | electricity      | production mix             | No              | UCTE     | electricity, high voltage, production UCTE, at grid                    | 6.44E-03   | kWh                      | (Hilbrans 1999)       | 1    | 1.10               | (2,2,2,3,1,3,2) |
| purification of leachate (4) | nickel production, sulphidic ore, primary, part VI | hydropower              | 0% mass                | 60% electricity from hydropower          | hydro power      | power plants               | No              | RER      | electricity, hydropower, at run of-river power plant                   | 9.66E-03   | kWh                      | (Hilbrans 1999)       | 1    | 1.10               | (2,2,2,3,1,3,2) |
|                              |                                                    | fossile energy (oil)    | 0% mass                |                                          | oil              | heating systems            | No              | RER      | heavy fuel oil, burned in industrial furnace 1MW, non-modulating       | 1.07E-04   | MJ                       | (Hilbrans 1999)       | 1    | 1.10               | (2,2,2,3,1,3,1) |
|                              |                                                    | dross from leaching     | 0% mass                | internal recovery, not inventoried       |                  |                            |                 |          |                                                                        | 1.07E-03   | kg                       | (Hilbrans 1999)       | 1    |                    | (2,2,2,3,1,3,6) |
|                              |                                                    |                         |                        |                                          |                  |                            |                 |          |                                                                        |            |                          |                       |      |                    |                 |

Fig. 5.8 Flows for the multi-output process “Nickel production” (part nickel reduction / purification step) and its representation in the ecoinvent database. Values correspond to the functional unit of 1 kg Class I nickel. Allocation values indicate the amount of the flow that is allocated to the co-product copper.

| General Flow Information |                                                     |        |                        | Representation in ecoinvent              |                  |                            |                 |          |                                                                  |            |      | Uncertainty Informations |      |                    |                 |  |
|--------------------------|-----------------------------------------------------|--------|------------------------|------------------------------------------|------------------|----------------------------|-----------------|----------|------------------------------------------------------------------|------------|------|--------------------------|------|--------------------|-----------------|--|
| Input                    | Process Name                                        | Output | Allocation coefficient | Remarks                                  | Category         | Sub-category               | Infra-structure | Location | Modul name in ecoinvent                                          | Mean value | Unit | Source for mean value    | Type | Standard Deviation | General Comment |  |
| electricity, grid        | nickel production, sulphidic ore, primary, part VII | nickel | 0% mass                | 40% electricity from grid                | electricity      | production mix             | No              | UCTE     | electricity, high voltage, production UCTE, at grid              | 4.40E-02   | kWh  | (Hilbrans 1999)          | 1    | 1.10               | (2,2,2,3,1,3,2) |  |
| hydropower               |                                                     |        | 0% mass                | 60% electricity from hydropower          | hydro power      | power plants               | No              | RER      | electricity, hydropower, at run of-river power plant             | 6.60E-02   | kWh  | (Hilbrans 1999)          | 1    | 1.10               | (2,2,2,3,1,3,2) |  |
| fossile energy (gas)     |                                                     |        | 0% mass                |                                          | natural gas      | heating systems            | No              | RER      | natural gas, burned in industrial furnace >100kW                 | 1.37E+00   | MJ   | (Hilbrans 1999)          | 1    | 1.10               | (2,2,2,3,1,3,1) |  |
| hydrogen                 |                                                     |        | 0% mass                |                                          | chemicals        | inorganics                 | No              | RER      | hydrogen, liquid, at plant                                       | 4.56E-03   | kg   | (Hilbrans 1999)          | 1    | 1.51               | (5,2,2,3,1,3,4) |  |
| electricity, grid        |                                                     |        | 0% mass                | 40% electricity from grid                | electricity      | production mix             | No              | UCTE     | electricity, high voltage, production UCTE, at grid              | 1.11E-01   | kWh  | (Hilbrans 1999)          | 1    | 1.10               | (2,2,2,3,1,3,2) |  |
| hydropower               |                                                     |        | 0% mass                | 60% electricity from hydropower          | hydro power      | power plants               | No              | RER      | electricity, hydropower, at run of-river power plant             | 1.66E-01   | kWh  | (Hilbrans 1999)          | 1    | 1.10               | (2,2,2,3,1,3,2) |  |
| fossile energy (oil)     |                                                     |        | 0% mass                |                                          | oil              | heating systems            | No              | RER      | heavy fuel oil, burned in industrial furnace 1MW, non-modulating | 1.21E+00   | MJ   | (Hilbrans 1999)          | 1    | 1.10               | (2,2,2,3,1,3,1) |  |
|                          |                                                     |        | 0% mass                | approximated with slag disposal - module | waste management | residual material landfill | No              | CH       | 0% water, to residual material landfill                          | 6.91E-06   | kg   | (Hilbrans 1999)          | 1    | 2.15               | (4,5,1,3,5,5,6) |  |
|                          |                                                     |        | 0%                     |                                          | metals           | extraction                 | No              | GLO      | nickel, 99.5%, at plant                                          | 1.00E+00   | kg   | (Hilbrans 1999)          |      |                    |                 |  |
|                          |                                                     |        |                        |                                          |                  |                            |                 |          |                                                                  |            |      |                          |      |                    |                 |  |

**Fig. 5.9** Flows for the multi-output process “Nickel production” (part carbonyl and refining) and its representation in the ecoinvent database. Values correspond to the functional unit of 1 kg Class I nickel. Allocation values indicate the amount of the flow that is allocated to the co-product copper.

## 5.2 Ferronickel

This part is mainly based on Hilbrans & Hinrichs (1999). General process description was taken from Kerfoot (1997).

### 5.2.1 Production process and infrastructure

**Mining.** Nickel oxide ores occur in two types: a limonitic type which occurs in the upper zone of a lateritic deposit, in which ferric oxide materials are predominant, and a silicate, occurring at greater depth in the deposit, which has lower iron content, but is usually richer in nickel than the limonite, and contains high levels of magnesia and silica. Most deposits contain both types of ore. Usually it is possible to separate the ore by selective mining and screening into an iron-containing limonite fraction and a magnesium silicate fraction enriched in nickel.

**Beneficiation.** Two processes – the Nippon Yakin Oheyama and the MINPRO–PAMCO process – have been developed in which the nickel is first converted by high-temperature treatment to a metallic phase, which can then be separated from the bulk of the ore by standard mineral beneficiation techniques.

**Metallurgy.** The metallurgy of nickel oxide ores differs from that of sulphides in that oxide ores are not amenable to most of the standard mineral beneficiation methods due to the chemical dissemination of the nickel in the oxide minerals. Nickel laterite ores have high moisture contents (typically up to 45 %) as well as chemically bound water in the hydroxide form. For this, the ore has to be dried, usually direct-fired rotary units operating at about 250 °C, to 15 – 20 % moisture. Calcining to dehydrate the ore and prereduction prior to electric furnace smelting are generally carried out. Chemically bound water is released above ca. 400 °C, reduction of the oxides to metal starts at 500 – 600 °C. Dehydration and pre-reduction of the ore in the kiln prior to smelting, optimises the utilisation of energy available from the reductants and fuel, thus reducing energy consumption in the smelting operation.

Class II nickel products such as nickel oxide, metallised nickel oxide, and ferronickel, are produced directly by smelting and roasting. They are sufficiently pure without refining for many nickel applications such as stainless steel production. The rotary kiln – electric furnace smelting process is now used almost universally for the production of ferronickel from oxide ores. Also ores with a low Ni/Fe ratio can be smelted to yield an acceptable ferronickel grade.

**Ferronickel refining.** Crude ferronickel produced by the conventional rotary kiln – electric furnace process usually contains high levels of carbon and sulphur. These ferronickels are first treated under reducing conditions to remove sulphur and are then refined sequentially under oxidising conditions with suitable fluxes to remove carbon, silicon, and phosphorus. A variety of equipment is used for the refining of ferronickel, including electric arc furnaces, shaking ladles, and low-frequency induction furnaces for desulphurisation, and oxygen-blown converters for silicon, carbon, and phosphorus removal. Sulphur is generally removed under reducing conditions by adding soda ash, lime, or calcium carbide to the molten ferronickel. Good agitation of the melt is essential to ensure effective mixing of the reagent with the metal phase

### 5.2.2 Emission

#### Mining

The major emissions are due to mineral born pollutants in the effluents. Mining operations generate large amounts of dust. Rain percolate through overburden and accounts to metal emissions to groundwater.

## Beneficiation

Beneficiation is done by classification, no emissions arise from this step.

### Metallurgy & Separation from Non Ferrous Metals

**Carbon dioxide.** In the ore pre-treatment, considerable amounts of lime are added. Lime decomposes into CO<sub>2</sub> to form calcite.

**Dust.** Dust carry over from the roasting, smelting and converting processes. Particulate emissions to the air consist of metals and thus are often returned to the leaching process after treatment. Dust consists to a considerable part of heavy metals.

**Other emissions to air.** The most important pollutants from the production of ferro-alloys in general besides dust are SO<sub>2</sub>, NO<sub>x</sub>, CO, CO<sub>2</sub>, HF, PAH and VOC. Only few SO<sub>2</sub> coming from metallurgical coke is formed.

**Emissions to water.** Metals and their compounds and substances in suspension are the main pollutants emitted to water. The metals concerned are Zn, Fe, Co, Pb and Hg. Other significant substances are chlorides and sulphates. Wastewaters from wet gas cleaning (if used) of the different metallurgical stages are the most important sources. The leaching stages are usually operated on closed circuit and drainage systems, and are therefore regarded as minor sources.

## 5.2.3 Waste

### Mining and Beneficiation

Overburden is deposited close inside the mine panel. Since no further beneficiation takes place in the ferronickel production, no beneficiation tailings arise.

### Metallurgy & Separation from Non Ferrous Metals

The production of metals is related to the generation of several residues and wastes, which are listed in the European Waste Catalogue. The most specific residues are filter dusts, sludge from wet scrubbers, slag from the smelting process, used furnace linings and packaging material like drums or big-bags. These residues are partly sold as by-products, recycled to the process or in cases of wastes without economic utility transported to a deposit or a landfill.

Some of the process specific residues can be reused or recovered in preliminary process steps (e. g. dross, filter dust) or construction (e. g. cleaned slag). Residues also arise from the treatment of liquid effluents, the main residue being gypsum waste and metal hydroxides from the wastewater neutralisation plant. These residuals have to be disposed, usually in lined ponds.

## 5.2.4 Ferronickel in ecoinvent

The module describes the global ferronickel production in 1994. It is designed for the use of the alloy as raw material in the manufacturing of stainless steels and alloys. The layout of the modules with general flow information, remarks, sources, values and uncertainty information are shown in Fig. 5.10, Fig. 5.11 and Fig. 5.12. The following paragraphs describe calculations, sources and assumptions chosen in this study. The functional unit of this process is one kg of ferronickel. A general description of the module and a summary of its meta information can be found in Tab. 5.1.

### Infrastructure and land use

The mobile equipment used for exploitation like drilling machines, scoop trams and underground trucks is inventoried with the diesel consumption stated in Hilbrans & Hinrichs (1999). The used module “diesel, burned in building machine” includes the infrastructure of the machines. 10'000m conveyor belts are further assumed with a capacity of 2000 t / h which yields 2E-5 m conveyor belt required per tonne mined raw ore.



Transformation and occupation of land through direct exploitation are assumed with the standard value for non-ferrous metals without heap leaching processes of 0.0025 m<sup>2</sup>/t and 0.075 m<sup>2</sup>/t ore mined, that is 0.055 m<sup>2</sup>/t and 1.65 m<sup>2</sup>/t per tonne Nickel produced.

Mining facilities and smeltery are inventoried with respective modules according to chapter II “Non Ferrous Metal Winning – Auxiliary Processes”.

### **Mining.**

In the considered processes the ore is mined in open cuts. The overburden is disposed in piles inside the mining panel, therefore no disposal of overburden is assumed (Hilbrans & Hinrichs (1999)). Also water use and effluent from mining is documented in Hilbrans & Hinrichs (1999). The use of freshwater was inventoried as river water, where as the pit water was assumed to be groundwater, which has to be pumped. The composition of the effluent was taken from IPPC (2002), its values are shown in Tab. 5.4. Concerning electricity mix Hilbrans & Hinrichs (1999) assumes a share of 60% of hydropower due to the operations high and constant energy demand. This assumption is used also in this inventory. For laterites no further beneficiation is necessary and possible except classification. Therefore no tailings arise.

### **Metallurgy**

**Energy.** In Hilbrans & Hinrichs (1999) the energy demand for drying and roasting of the wet ore is stated, which contributes to a major extent to the overall energy demand in the ferronickel production (Fig. 5.11).

**Flux.** The input of limestone is estimated roughly with values reported from the related manufacturing processes for ferro-chrome and calcium silicate in IPPC (2001). Choosing a geometric mean of the values a value of 470 kg limestone per tonne ferro-alloy is required. Consequentially 0.44 kg CO<sub>2</sub> per tonne flux is emitted after the formula  $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$ .

**Dust.** The emission of dust is estimated roughly considering values from ferro-chrome production. There overall values ranging from 0.5 to 1 kg per tonne ferro-alloy is reported in IPPC (2001). In this study a value of 1 kg per tonne is chosen using a PM-partition reported in CEPMEIP<sup>10</sup>. Dust composition is approximated with the specific values from class I nickel production in this section, resulting in emission factors in Tab. 5.11.

**Effluent.** Specific value of 6.9 m<sup>3</sup> per tonne ferro-alloy chosen from IPPC (2001).

**Air emissions.** No VOC-emissions were assumed since no hydrometallurgical processes are considered. For TCDD the same values as in class I nickel metallurgy are chosen (IPPC (2001)).

### **Data quality**

**Mining.** Data for major mass fluxes are satisfactory, while the emissions in to water may be overestimated due to different system boundaries of reported effluent volume from Hilbrans & Hinrichs (1999) and total effluent emissions reported in IPPC (2002).

**Metallurgy.** Process data is satisfactory. Poor data are available on composition and fate of solid waste, which has to be seen as major impact within the overall process due to its big mass.

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<sup>10</sup> [http://www.air.sk/tno/cepmeip/em\\_factors.php](http://www.air.sk/tno/cepmeip/em_factors.php), accessed 01.03.2003

**Tab. 5.11 Emissions to air per tonne produced ferronickel. Data was taken as proxy from similar processes in ferrochrome (A) and copper metallurgy (B) reported in IPPC (2001), the distribution of the particle size is taken from CEPMEIP<sup>11</sup> (C).**

| Substance  | Amount in kg / t | Source |
|------------|------------------|--------|
| Pb         | 0.077            | B      |
| Zn         | 0.25             | B      |
| Sn         | 0.016            | B      |
| Cu         | 0.10             | B      |
| As         | 0.014            | B      |
| Total dust | 1.0              | A      |
| PM<2.5     | 0.0013           | C      |
| PM2.5-10   | 0.75             | C      |
| PM>10      | 0.25             | C      |
| Ni         | 0.03             | A      |
| Co         | 0.03             | A      |
| TCDD       | 1.00E-08         | B      |

<sup>11</sup> [http://www.air.sk/tno/cepmeip/em\\_factors.php](http://www.air.sk/tno/cepmeip/em_factors.php), accessed 01.03.2003

| General Flow Information                   |                                       |                      |                                                                                                                                                                                                                                                        | Representation in ecoinvent                                                               |                        |                   |                 |          |                                                           |            | Uncertainty Informations |                       |      |                    |                  |
|--------------------------------------------|---------------------------------------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|------------------------|-------------------|-----------------|----------|-----------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|------------------|
| Input                                      | Process Name                          | Output               | Allocation coefficient                                                                                                                                                                                                                                 | Remarks                                                                                   | Category               | Sub-category      | Infra-structure | Location | Modul name in ecoinvent                                   | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment  |
| Ni in ground                               | ferronickel, 25% Ni, at plant, part I |                      |                                                                                                                                                                                                                                                        | Overall 57.5% yield assumed. 18 kg overburden and 22 kg ore required for 1 kg ferronickel | resource               | in ground         |                 |          | Nickel, 1.98% in silicates, 1.04% in crude ore, in ground | 1.74E+00   | kg                       | (Hilbrans, 1999)      | 1    | 1.13               | (2,2,2,1,1,4,12) |
| explosive                                  |                                       |                      |                                                                                                                                                                                                                                                        |                                                                                           | construction processes | civil engineering | No              | RER      | blasting                                                  | 1.20E-03   | kg                       | (Hilbrans, 1999)      | 1    | 1.13               | (2,2,2,1,1,4,4)  |
| diesel fuel (energy)                       |                                       |                      |                                                                                                                                                                                                                                                        |                                                                                           | construction processes | machinery         | No              | GLO      | diesel, burned in building machine                        | 1.91E+00   | MJ                       | (Hilbrans, 1999)      | 1    | 1.13               | (2,2,2,1,1,4,4)  |
| electricity from grid, mining              |                                       |                      | 40% electricity from grid                                                                                                                                                                                                                              |                                                                                           | electricity            | production mix    | No              | UCTE     | electricity, high voltage, production UCTE, at grid       | 9.29E-02   | kWh                      | (Hilbrans 1999)       | 1    | 1.10               | (2,2,2,3,1,3,2)  |
| hydropower, mining                         |                                       |                      | 60% electricity from hydropower                                                                                                                                                                                                                        |                                                                                           | hydro power            | power plants      | No              | RER      | electricity, hydropower, at run of-river power plant      | 1.39E-01   | kWh                      | (Hilbrans 1999)       | 1    | 1.10               | (2,2,2,3,1,3,2)  |
| exploration infrastructure, conveyor belts |                                       |                      | exploration infrastructure, 10'000 m belts assumed                                                                                                                                                                                                     |                                                                                           | construction processes | machinery         | Yes             | RER      | conveyor belt, at plant                                   | 8.00E-07   | m                        | own assumptions       | 1    | 1.26               | (3,2,2,1,3,4,4)  |
| infrastructure mining                      |                                       |                      |                                                                                                                                                                                                                                                        |                                                                                           | metals                 | extraction        | Yes             | GLO      | non-ferrous metal mine, surface                           | 2.00E-09   | unit                     | own assumptions       | 1    | 3.07               | (3,2,2,1,3,4,9)  |
| Land trans-formation, ground               |                                       |                      | 0.0025 m2/t ore mined                                                                                                                                                                                                                                  |                                                                                           | resource               | land              |                 |          | Transformation, from unknown                              | 5.49E-05   | m2                       | (IPPC, 2002)          | 1    | 3.07               | (3,2,2,1,3,4,9)  |
| Land trans-ormation, mine                  |                                       |                      | 0.0025 m2/t ore mined                                                                                                                                                                                                                                  |                                                                                           | resource               | land              |                 |          | Transformation, to mineral extraction site                | 5.49E-05   | m2                       | (IPPC, 2002)          | 1    | 2.08               | (3,2,2,1,3,4,8)  |
| Land use for mining                        |                                       |                      | average lifetime of 30 a assumed                                                                                                                                                                                                                       |                                                                                           | resource               | land              |                 |          | Occupation, mineral extraction site                       | 1.65E-03   | m2a                      | (IPPC, 2002)          | 1    | 1.59               | (3,2,2,1,3,4,7)  |
|                                            |                                       |                      | no overburden disposed, but refilled in mining pit                                                                                                                                                                                                     |                                                                                           |                        |                   |                 |          |                                                           | 0.00E+00   | kg                       | (Hilbrans, 1999)      | 1    | 1.13               | (2,2,2,1,1,4,6)  |
|                                            |                                       |                      | not inventoried, no ecological burden                                                                                                                                                                                                                  |                                                                                           |                        |                   |                 |          |                                                           | 1.80E+01   | kg                       | (Hilbrans, 1999)      | 1    | 1.13               | (2,2,2,1,1,4,6)  |
|                                            |                                       |                      | Detailed composition see Tab. 5.4, same composition as underground mining effluents including emissions from mining activities. Dust is the sum of particulate emissions represented in fractions PM<2.5, PM2.5-10, PM>10, including metals and salts. |                                                                                           |                        |                   |                 |          |                                                           | 8.00E-03   | m3                       | (Hilbrans, 1999)      | 1    | 1.13               | (2,2,2,1,1,4,6)  |
|                                            |                                       |                      |                                                                                                                                                                                                                                                        |                                                                                           |                        |                   |                 |          |                                                           | 1.01E-02   | kg                       | (NPI, 2001)           | 1    | 1.59               | (3,2,1,1,3,4,25) |
|                                            |                                       | overburden, disposed |                                                                                                                                                                                                                                                        |                                                                                           |                        |                   |                 |          |                                                           |            |                          |                       |      |                    |                  |
|                                            |                                       | overburden, refilled |                                                                                                                                                                                                                                                        |                                                                                           |                        |                   |                 |          |                                                           |            |                          |                       |      |                    |                  |
|                                            |                                       | effluents, mining    |                                                                                                                                                                                                                                                        |                                                                                           |                        |                   |                 |          |                                                           |            |                          |                       |      |                    |                  |
|                                            |                                       | Dust                 |                                                                                                                                                                                                                                                        |                                                                                           |                        |                   |                 |          |                                                           |            |                          |                       |      |                    |                  |

Fig. 5.10 Flows for “Ferronickel, part mining” and its representation in the ecoinvent database. The values correspond to the functional unit of the production of 1 kg ferronickel

| General Flow Information |                                   |         |                                        | Representation in ecoinvent    |                                                                                        |                        |                            |          |                         |                                                                        | Uncertainty Informations |                       |                            |                    |                 |                                                    |
|--------------------------|-----------------------------------|---------|----------------------------------------|--------------------------------|----------------------------------------------------------------------------------------|------------------------|----------------------------|----------|-------------------------|------------------------------------------------------------------------|--------------------------|-----------------------|----------------------------|--------------------|-----------------|----------------------------------------------------|
| Input                    | Process Name                      | Output  | Allocation coefficient                 | Remarks                        | Category                                                                               | Sub-category           | Infra-structure            | Location | Modul name in ecoinvent | Mean value                                                             | Unit                     | Source for mean value | Type                       | Standard Deviation | General Comment |                                                    |
| drying                   | roasting                          | melting | ferronickel, 25% Ni, at plant, part II | electricity drying, grid       | 40% electricity from grid                                                              | electricity            | production mix             | No       | UCTE                    | electricity, high voltage, production UCTE, at grid                    | 2.60E-01                 | kWh                   | (Hilbrans 1999)            | 1                  | 1.10            | (2,2,2,3,1,3,2)                                    |
|                          |                                   |         |                                        | hydropower drying              | 60% electricity from hydropower                                                        | hydro power            | power plants               | No       | RER                     | electricity, hydropower, at run of-river power plant                   | 3.90E-01                 | kWh                   | (Hilbrans 1999)            | 1                  | 1.10            | (2,2,2,3,1,3,2)                                    |
|                          |                                   |         |                                        | fossile energy drying (gas)    |                                                                                        | natural gas            | heating systems            | No       | RER                     | natural gas, burned in industrial furnace >100kW                       | 2.88E+01                 | MJ                    | (Hilbrans 1999)            | 1                  | 1.10            | (2,2,2,3,1,3,1)                                    |
|                          |                                   |         |                                        | electricity roasting, grid     | 40% electricity from grid                                                              | electricity            | production mix             | No       | UCTE                    | electricity, high voltage, production UCTE, at grid                    | 3.20E-01                 | kWh                   | (Hilbrans 1999)            | 1                  | 1.10            | (2,2,2,3,1,3,2)                                    |
|                          |                                   |         |                                        | hydropower roasting            | 60% electricity from hydropower                                                        | hydro power            | power plants               | No       | RER                     | electricity, hydropower, at run of-river power plant                   | 4.80E-01                 | kWh                   | (Hilbrans 1999)            | 1                  | 1.10            | (2,2,2,3,1,3,2)                                    |
|                          |                                   |         |                                        | fossile energy roasting (coke) |                                                                                        | hard coal              | heating systems            | No       | RER                     | hard coal, burned in industrial furnace 1-10MW                         | 4.00E+01                 | MJ                    | (Hilbrans 1999)            | 1                  | 1.10            | (2,2,2,3,1,3,1)                                    |
|                          |                                   |         |                                        | electricity melting, grid      | 40% electricity from grid                                                              | electricity            | production mix             | No       | UCTE                    | electricity, high voltage, production UCTE, at grid                    | 2.72E+00                 | kWh                   | (Hilbrans 1999)            | 1                  | 1.10            | (2,2,2,3,1,3,2)                                    |
|                          |                                   |         |                                        | hydropower melting             | 60% electricity from hydropower                                                        | hydro power            | power plants               | No       | RER                     | electricity, hydropower, at run of-river power plant                   | 4.08E+00                 | kWh                   | (Hilbrans 1999)            | 1                  | 1.10            | (2,2,2,3,1,3,2)                                    |
|                          |                                   |         |                                        | fossile energy melting (oil)   |                                                                                        | oil                    | heating systems            | No       | RER                     | heavy fuel oil, burned in industrial furnace 1MW, non-modulating       | 3.75E+00                 | MJ                    | (Hilbrans 1999)            | 1                  | 1.10            | (2,2,2,3,1,3,1)                                    |
|                          |                                   |         |                                        | flux, ferronickel              | flux, limestone assumed                                                                | construction materials | others                     | No       | CH                      | limestone, milled, packed, at plant                                    | 4.69E-01                 | kg                    | (IPPC 2001)                | 1                  | 1.51            | (5,2,2,3,1,3,4)                                    |
|                          |                                   |         |                                        | infrastructure smeltery        |                                                                                        | metals                 | extraction                 | Yes      | GLO                     | non-ferrous metal smelter                                              | 6.48E-11                 | unit                  | own assumptions            | 1                  | 3.07            | (3,2,2,1,3,4,9)                                    |
|                          |                                   |         |                                        |                                | approximated with slag disposal - module                                               | waste management       | residual material landfill | No       | CH                      | disposal, nickel smelter slag, 0% water, to residual material landfill | 1.24E+01                 | kg                    | (Hilbrans 1999)            | 1                  | 2.15            | (4,5,1,3,5,5,6)                                    |
|                          |                                   |         |                                        |                                | in scrubber, approximated with slag disposal - module                                  | waste management       | residual material landfill | No       | CH                      | disposal, nickel smelter slag, 0% water, to residual material landfill | 3.00E-01                 | kg                    | (Hilbrans 1999)            | 1                  | 2.15            | (4,5,1,3,5,5,6)                                    |
|                          |                                   |         |                                        |                                | lime addition as melting flux                                                          | air                    | unspecified                |          |                         | Carbon dioxide, fossil                                                 | 2.06E-01                 | kg                    | calculations, (IPPC, 2002) | 1                  | 1.58            | (4,2,1,1,4,4,14); excluding stdev of lime addition |
|                          |                                   |         |                                        |                                | Composition as in Class I Nickel production (Tab 5.10). The emissions are inventoried. |                        |                            |          |                         |                                                                        | 6.90E-03                 | m3                    | (IPPC 2001)                | 1                  | 2.15            | (4,5,1,3,5,5,6)                                    |
|                          |                                   |         |                                        |                                | Composition in Tab 5.11. The emissions are inventoried.                                |                        |                            |          |                         |                                                                        | 1.00E-03                 | kg                    | (IPPC, 2002)               | 1                  | 1.59            | (3,2,1,1,3,4,25)                                   |
|                          | slag                              |         |                                        |                                |                                                                                        |                        |                            |          |                         |                                                                        |                          |                       |                            |                    |                 |                                                    |
|                          | dust                              |         |                                        |                                |                                                                                        |                        |                            |          |                         |                                                                        |                          |                       |                            |                    |                 |                                                    |
|                          | carbondioxide, from lime addition |         |                                        |                                |                                                                                        |                        |                            |          |                         |                                                                        |                          |                       |                            |                    |                 |                                                    |
|                          | effluent, general                 |         |                                        |                                |                                                                                        |                        |                            |          |                         |                                                                        |                          |                       |                            |                    |                 |                                                    |
|                          | dust, from overall process        |         |                                        |                                |                                                                                        |                        |                            |          |                         |                                                                        |                          |                       |                            |                    |                 |                                                    |

Fig. 5.11 Flows for “Ferronickel, part drying, roasting and melting” and its representation in the ecoinvent database. The values correspond to the functional unit of the production of 1 kg ferronickel

| General Flow Information                                                                                                                                                                      |                                                    |                       |                                 | Representation in ecoinvent |                 |                        |                 |                                                      |                         |                 | Uncertainty Informations |                       |                 |                    |                 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|-----------------------|---------------------------------|-----------------------------|-----------------|------------------------|-----------------|------------------------------------------------------|-------------------------|-----------------|--------------------------|-----------------------|-----------------|--------------------|-----------------|
| Input                                                                                                                                                                                         | Process Name                                       | Output                | Allocation coefficient          | Remarks                     | Category        | Sub-category           | Infra-structure | Location                                             | Modul name in ecoinvent | Mean value      | Unit                     | Source for mean value | Type            | Standard Deviation | General Comment |
| <div>other/transp</div> <div>refining</div> <div>electricity other, grid</div> <div>hydropower other</div> <div>electricity, grid</div> <div>hydropower</div> <div>fossile energy (gas)</div> | <div>ferronickel, 25% Ni, at plant, part III</div> | <div>waste heat</div> |                                 | from total electricity      | air             | low population density |                 |                                                      | Heat, waste             | 3.34E+01 MJ     |                          | (Hilbrans 1999)       | 1               | 1.10               | (2,2,2,3,1,3,2) |
|                                                                                                                                                                                               |                                                    |                       | 40% electricity from grid       | electricity                 | production mix  | No                     | UCTE            | electricity, high voltage, production UCTE, at grid  | 1.20E-01 kWh            | (Hilbrans 1999) | 1                        | 1.10                  | (2,2,2,3,1,3,2) |                    |                 |
|                                                                                                                                                                                               |                                                    |                       | 60% electricity from hydropower | hydro power                 | power plants    | No                     | RER             | electricity, hydropower, at run of-river power plant | 1.80E-01 kWh            | (Hilbrans 1999) | 1                        | 1.10                  | (2,2,2,3,1,3,2) |                    |                 |
|                                                                                                                                                                                               |                                                    |                       | 40% electricity from grid       | electricity                 | production mix  | No                     | UCTE            | electricity, high voltage, production UCTE, at grid  | 2.00E-01 kWh            | (Hilbrans 1999) | 1                        | 1.10                  | (2,2,2,3,1,3,2) |                    |                 |
|                                                                                                                                                                                               |                                                    |                       | 60% electricity from hydropower | hydro power                 | power plants    | No                     | RER             | electricity, hydropower, at run of-river power plant | 3.00E-01 kWh            | (Hilbrans 1999) | 1                        | 1.10                  | (2,2,2,3,1,3,2) |                    |                 |
|                                                                                                                                                                                               |                                                    |                       |                                 | natural gas                 | heating systems | No                     | RER             | natural gas, burned in industrial furnace >100kW     | 1.00E+00 MJ             | (Hilbrans 1999) | 1                        | 1.10                  | (2,2,2,3,1,3,1) |                    |                 |
|                                                                                                                                                                                               |                                                    |                       |                                 | metals                      | extraction      | No                     | GLO             | ferronickel, 25% Ni, at plant                        | 1.00E+00 kg             | (Hilbrans 1999) |                          |                       |                 |                    |                 |
|                                                                                                                                                                                               |                                                    |                       | <div>Ferronickel</div>          |                             |                 |                        |                 |                                                      |                         |                 |                          |                       |                 |                    |                 |

Fig. 5.12 Flows for “Ferronickel, parts others/transport and refining” and its representation in the ecoinvent database. The values correspond to the functional unit of the production of 1 kg ferronickel

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## Part V

# **Platinum Group Metals (PGM) Platinum, Palladium, Rhodium**

Data v2.1 (2009)

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Classen M., Althaus H.-J., Blaser S., Tuchscheid M., Jungbluth N., Doka G., Faist Emmenegger M. and Scharnhorst W. (2009) Life Cycle Inventories of Metals. Final report ecoinvent data v2.1, No 10. EMPA Dübendorf, Swiss Centre for Life Cycle Inventories, Dübendorf, CH, Online-Version under: [www.ecoinvent.ch](http://www.ecoinvent.ch).

Final report of updated Swiss National Life Cycle Inventory Database "ecoinvent", version 2.1.



## Comments on version 2.1

This part corresponds to the original report in Althaus et al. (2004) that has been updated with the changes reported in Frischknecht et al. (2006). The changes were implemented with v1.3 affect the co-product dataset of nickel due to an incorrect nickel price underlying the allocation factors.

The correction caused an increase of attributed exchanges for nickel and a decrease for the others, especially in the case of PGM-production in Russia, since there the co-product nickel has a considerable share in the output.

No changes were made in version 2.1 compared to version 2.0.

Chapter “6 Cumulative Results and Interpretation” of the original report has been deleted for the current update. Note that the flow names in the tables displaying unit processes have not been updated to reflect the name changes with version 2.0. This namely affects the designation of transportation service flows.

## Summary

The primary production of platinum group metals (PGM) is modelled for South Africa and Russia. Together these two countries produce more than 90% of the world’s PGM.

For the secondary production of PGM the recycling process of automotive catalysts is modelled.

The consumer mixes for PGM in Europe are finally modelled from the outputs of these three modules.

Mining, beneficiation and the different metallurgical processes are inventoried vertically aggregated because of difficulties to clearly allocate some of the flows to one single step and because the intermediate steps would not make sense in another context. The Figures in this report however contain the available information to the single steps.



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# 1 Introduction

This chapter describes the production of the platinum-group metals Platinum (Pt); Palladium (Pd) and Rhodium (Rh) from cradle to consumer in Europe. Inventory data is taken mainly from (Hochfeld (1997)) with background data from (IPPC (2002)) and (IPPC (2001)). Market data on PGM and non ferrous metals are provided by Johnson Matthey<sup>1</sup> and the London Metal Exchange respectively<sup>2</sup>.

Since the PGM are associated with other metals, their production always yields a fraction of other metals than PGM, mainly Nickel (Ni) and Copper (Cu). Hence the process of primary production described in this chapter is formally a multioutput-process with the coupled products Pt, Pd, Rh, Ni and Cu. This makes it necessary to allocate the inventory to these products. The demand of PGM-containing ore is allocated by mass. The rest of the inventory is allocated by revenue.

PGM are highly valuable, their occurrence is scarce. PGM-containing ores are mined only in few sites. The deposits in Noril'sk (RUS) and Bushveld (RSA) contribute over 90% to the global PGM-production. The data on these two PGM winning operations therefore represents the global primary production of PGM according to their market share.

The great value of the PGM makes recycling profitable to some extent. This secondary production is approximated with the declared amount of metal from the recycling of automotive catalysts. For Europe this assumption is reasonable, since automotive catalysts account for over 90% of the metal use.

The market for PGM is of great importance. On the one hand, market prices determine the profitability of the PGM winning, both for primary and secondary production. On the other hand the inventory is allocated by revenue to the different metals. An ecological assessment of PGM therefore is always dependent on the actual market situation. Further more, the producing countries hold stocks of PGM in order to influence market price and to protect their own PGM-demanding industries. As precious metal, Platinum in particular is dealt as investment.

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<sup>1</sup> Johnson Matthey's Website "Platinum Today", <http://www.platinum.matthey.com>, accessed at 01.03.2003

<sup>2</sup> London Metal Exchange, daily and monthly stocks and prices under [http://www.lme.co.uk/data\\_prices/home.html](http://www.lme.co.uk/data_prices/home.html), accessed at 01.03.2003

## 2 Reserves and Resources of PGM

Metals are a special class of materials. They are not consumed but rather used during a product life cycle. When recovered the properties are theoretically identical. The metal can be re-used in another product life cycle, provided that its concentration makes a recovery profitable. The recovered metal is referred to as secondary metal, whereas the metal originating directly from the ore is referred to as primary metal. In LCIs these two types – primary and secondary metal – are often differentiated because they bear a different ecological burden.

### 2.1 Primary PGM

The PGMs in primary deposits have been transferred from the earth's interior. Tectonic movements of the earth's crust, followed by the eruption of magma, have led to their presence in regions close to the surface. Almost invariably, the platinum group elements have separated from ultrabasic magmas. Norite (Mg – Fe – Ca – Al silicate) contains mainly sulphidic intrusions, whereas intrusions in dunite (Mg – Fe silicate) are mainly sulphide free.

When primary deposits are washed and weathered, secondary deposits are formed, also known as placers or alluvial deposits. Mechanical concentration of the heavy constituents by flowing water takes place, together with chemical dissolution and precipitation of the platinum group metals. Alluvial deposits usually originate from dunite. For easily washable sand, a PGM content of 0.05 ppm is economic.

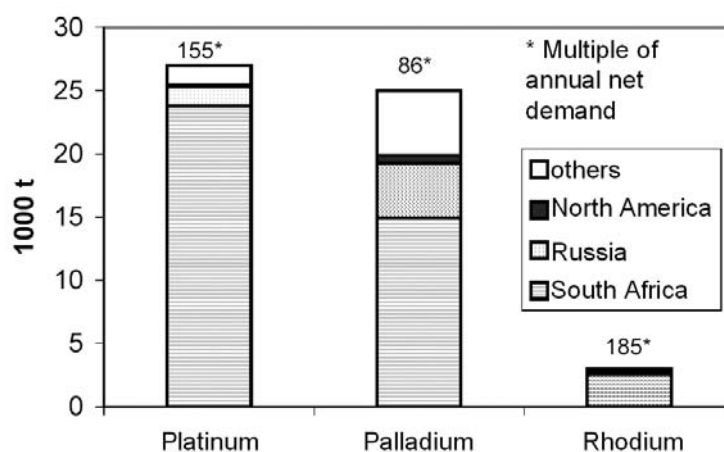
The PGMs occur in a large number of minerals. Workable ore deposits contain mainly sperrylite (PtAs<sub>2</sub>), cooperite (PtS), stibiopalladinite (Pd<sub>3</sub>Sb), laurite (RuS<sub>2</sub>), ferroplatinum (Fe – Pt), polyxene (Fe – Pt – other platinum metals), osmiridium (Os – Ir), and iridium platinum (Ir – Pt). These minerals are associated with particular carrier materials, e.g., iron pyrites, nickel iron pyrites, or chrome iron ore (Renner (1997)).

The ratio and concentration of the PGM and their accompanying metals vary from site to site (s. Tab. 2.1). PGM deposits can be classified into the types oxidic-silicatic associations (e. g. UG2) and sulphidic associations, which can further be divided into PGM-dominated (e. g. Merensky, Stillwater) and Ni-Cu-dominated (Sudbury, Norsil'sk) (Hochfeld (1997)). Of the estimated 112'900 t PGM-resources 55.6% are supposed to be found in South Africa, 15.9% in Russia, 12.3% in Finland, 8.9% in Zimbabwe and 6.0% in the USA (Hochfeld (1997)). However, the PGM-reserves – the profitable workable deposits – are mainly located in South Africa and Russia (s. Fig. 2.1).

**Tab. 2.1 Characteristic metal concentrations in the most important PGM-mining sites. Data from Hochfeld (1997)**

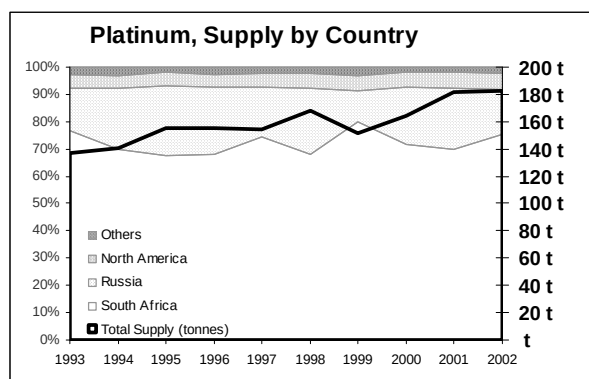
| Element |     | Merensky          | UG2               | Stillwater        | Sudbury           | Noril'sk           |
|---------|-----|-------------------|-------------------|-------------------|-------------------|--------------------|
|         |     | RSA               | RSA               | USA               | CAN               | RUS                |
| Ni      | %   | 0.28              | 0.16              | <0.1              | 1.4 – 1.5         | up to 3            |
| Cu      | %   | 0.17              | 0.03              | <0.1              | 1.1 – 1.3         | up to 4            |
| Pt      | ppm | 4.82              | 3.22              | 6.0               | 0.38              | 2.5                |
| Pd      | ppm | 2.04              | 3.24              | 20.0              | 0.39              | 7.3                |
| Rh      | ppm | 0.24              | 0.54              | 0.21              | 0.03              | 0.2                |
| Ru      | ppm | 0.44 <sup>2</sup> | 0.72 <sup>2</sup> | 0.36 <sup>2</sup> | 0.04 <sup>2</sup> | n. a. <sup>2</sup> |
| Ir      | ppm | 0.06 <sup>2</sup> | 0.11 <sup>2</sup> | 0.21 <sup>2</sup> | 0.01 <sup>2</sup> | n. a. <sup>2</sup> |
| Os      | ppm | 0.04 <sup>2</sup> | 0.07 <sup>2</sup> | - <sup>2</sup>    | 0.01 <sup>2</sup> | n. a. <sup>2</sup> |
| Au      | ppm | 0.26              | 0.07              | 0.21              | -                 | 0.25 <sup>2</sup>  |

<sup>2</sup> Values for Ru, Ir and Os are taken from Hagelücken (2001b).

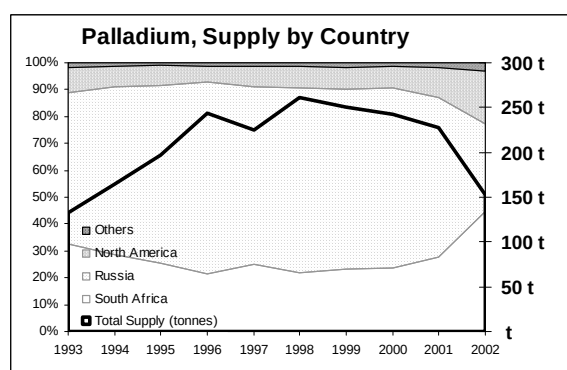


**Fig. 2.1 PGM-reserves by region.** Reserves are defined as at the present time profitably workable deposits. The global PGM-resources are estimated to be 112'900 t. "Multiple of annual demand" indicates the years until the actual resources are depleted assuming a constant demand.

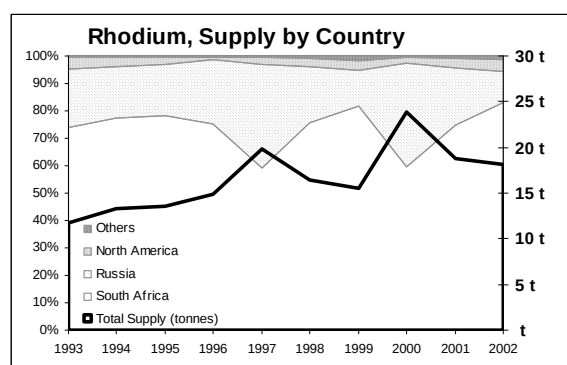
Today worked ores are from primary deposits, whereas earlier mainly placers in Russia (Ural) were exploited. Production mainly depends on the market price of the PGMs and the accompanying metals like Ni or Cu. As seen from geographical distribution of the main resources, major producers are located in South Africa (RSA) on the Bushveld Complex (Merensky-Reef and UG2) and in the Russian Federation (RUS) at Noril'sk-Talnakh. These two major sites account for over 90 % of the PGM production world-wide. The deposits in RSA are mined by seven big companies in RUS the company Norilsk Nickel is responsible for the whole exploitation. Fig. 2.2, Fig. 2.3 and Fig. 2.4 show the development of the production of platinum, palladium and rhodium respectively split by country. Other PGM winning countries are the USA (Stillwater), Canada (Sudbury) and Zimbabwe. They contribute only little to the global supply even though the capacity of the US palladium deposit has been greatly increased in recent years.



**Fig. 2.2** Platinum production by country. Bold line represents total supply in tonnes. Grey areas indicate the countries share in the global supply. Data taken from JohnsonMatthey (2003), grey areas correspond to the axis on the left (percentage of total supply), whereas the total supply correspond to the axis on the right (supply in tonnes).



**Fig. 2.3** Palladium production by country. Bold line represents total supply in tonnes. Grey areas indicate the countries share in the global supply. Data taken from (JohnsonMatthey (2003)), grey areas correspond to the axis on the left (percentage of total supply), whereas the total supply correspond to the axis on the right (supply in tonnes).



**Fig. 2.4** Rhodium production by country. Bold line represents total supply in tonnes. Grey areas indicate the countries share in the global supply. Data taken from JohnsonMatthey (2003), grey areas correspond to the axis on the left (percentage of total supply), whereas the total supply correspond to the axis on the right (supply in tonnes).

## 2.2 Secondary PGM

Another important source is the recovery of secondary PGM metals. Large amounts of PGMs are used in automotive catalysts. Due to the high value of the metals, automotive catalysts are recovered. They are dismantled, the ceramic carrier then is dissolved and the metals regained through hydrometallurgical processes. This source is still small compared to the total demand world-wide, but it is rising. In the USA already 11.5 t of total 40.1 t platinum used in 2001 originated from automotive catalysts. This value is higher than in Europe because of the earlier introduction of automotive catalysts. The pool of PGM in US automotive catalysts is stable. In Europe and Japan this pool still rises until steady-state is reached with a recovery of expected 52% of all used automotive catalysts for Germany (Hagelücken (2001b)). This future recovery quota is uncertain, because the PGM recovery depends mainly on the market price. Further development probably will lead to a higher demand and thus to a higher recovery quota. The actual recovery in Europe amounts 2.8 t, 1.4t and 1.1t for platinum, palladium and rhodium respectively (for calculations and comments see. 5.1.2 Secondary Production).

Large quantities of secondary materials are present in the form of used Pt-Rh gauze, used for catalytic formation of nitric acid. At the present, the pool of such gauze amounts about 50 t world-wide. Also the glass industry generates large quantities of defective components for recovery. Defective laboratory equipment – crucibles and dishes – makes a considerable contribution. Also large quantities of Pt- and Rh-containing thermocouple components are provided by the electrical industry (Renner (1997)). In addition, an increasing proportion of the cars currently being scrapped are fitted with catalysts – it was only in 1993 that all new gasoline cars in Europe were equipped with catalytic converters. This will lead consequently to a higher amount of PGM's recovered.

Often, the recovery operation is included in the sales contract. Materials are sent directly for recovery and do not appear on the raw materials market. Because of this, no data on the recovery of the existing PGM pool is available.



## 3 Characterisation of the PGMs

This chapter is mainly based on information provided by the International Platinum Association's website<sup>3</sup>. Some physical and chemical properties of the PGMs are summed up in Tab. 3.1.

### 3.1 Platinum

Platinum is a silvery-white metal and is probably best known for its use in jewellery. However, jewellery actually represents only about 38% of overall platinum use, an equivalent volume of platinum is used in catalytic converters, with the remaining 20% being used in other industrial applications.

Platinum is hard and extremely dense. Platinum and its relatives, iridium and osmium, are the most dense metals known (platinum is nearly twice as dense as lead and 11% more dense than gold). It has a high melting point and high temperature stability and corrosion resistance; and it is a good oxidation catalyst, conductive and oxidation resistant. It's biological compatibility makes it an important component in medical applications.

### 3.2 Palladium

Like platinum, palladium is silvery-white in colour. Its melting point is the lowest of all PMG, but it is still high compared to other popular metals. It is also the least dense of the PMGs. It has also high temperature stability and corrosion resistance. The rarest of all PGMs apart from iridium, palladium is also a good oxidation catalyst, conductive, oxidation resistant and ductile when annealed.

Palladium has the ability to absorb big amounts of hydrogen. This makes it an efficient and safe hydrogen storage medium and purifier. It is also used in chemical processes that require hydrogen exchange between two reactants, such as that which produces butadiene and cyclohexane, the raw materials for synthetic rubber and nylon.

Palladium's catalytic qualities make it necessary in catalytic converters and air purification equipment. Its chemical stability and electrical conductivity lead to more effective and durable plating than using gold in electronic components.

### 3.3 Rhodium

Its hardness makes it an alloying agent to harden platinum. With this hardness, along with its high melting point, high temperature stability and corrosion resistance, rhodium is used in many industrial processes such as glass, glass fibre and nitric acid production. Like other PGMs, Rhodium has catalytic qualities, and thus is used in catalytic converters.

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<sup>3</sup> <http://www.platinuminfo.net>. accessed at 01.03.2003

Tab. 3.1 Physical and chemical properties of the platinum group metals.

| Property                             | unit                                          | Rh                | Pd                | Pt     |
|--------------------------------------|-----------------------------------------------|-------------------|-------------------|--------|
| Atomic number                        |                                               | 45                | 46                | 78     |
| Atomic weight                        | g / mol                                       | 102.91            | 106.42            | 195.08 |
| Rarity in lithosphere                | ppm                                           | $2 \cdot 10^{-4}$ | $6 \cdot 10^{-4}$ | 0.001  |
| Density                              | g/cm <sup>3</sup>                             | 12.41             | 12.02             | 21.45  |
| Electrical conductivity              | $10^6 \text{ cm}^{-1} \Omega^{-1}$            | 0.211             | 0.095             | 0.0966 |
| Electrical resistivity               | $\mu \Omega \text{ cm}$                       | 4.33              | 9.93              | 9      |
| Thermal conductivity at 0°C          | $\text{W m}^{-1} \text{ }^\circ\text{C}^{-1}$ | 150               | 76                | 73     |
| Melting point                        | °C                                            | 1960              | 1554              | 1772   |
| Hardness (Brinell value)             | MN m <sup>-2</sup>                            | 1100              | 37.3              | 392    |
| Tensile strength, annealed condition | kg/mm2                                        | 71                | 17                | 14     |

## 4 Use / Application of PGM

Common to all of the PGMs is their use in electronics and as an element in catalytic converters. Catalytic converters are mainly used in automobiles, in the petroleum industry and hydrogen fuel cells. The use in automotive catalysts covers a good part of the overall PGM demand, in the case of Rhodium it amounts over 80%. The resource needed for automotive catalysts is covered partly from recycled automotive catalysts. This share differs greatly from country to country.

Second largest segment is the use of the metals in jewellery, namely in the case of Platinum. Another important field of application is the use in dental alloys.

Magnetic alloys containing platinum (mainly with cobalt) are components in the hard disks of 90% of the computers manufactured 2001. It is the largest single electronic application for platinum with rapidly increasing demand.

PGM catalysts for nitric acid production take the form of gauze made out of fine wire. Rhodium is added for strength and to reduce the amount of platinum loss during the conversion of the gas.

PGMs are widely used in medicine because they are inert and do not corrode inside the body. Allergic reactions to PGM are extremely rare. They have good electrical conductivity, which makes them an ideal electrode material.

Tab. 4.1 resumes the use of different PGMs, Fig. 4.1, Fig. 4.2 and Fig. 4.3 show the PGM demand by application in 2001.

The demand of the different PGM varies greatly depending on stocks and price. Overall, the platinum used equals in magnitude the demand of palladium (198 t Pt or 6'370 troz<sup>4</sup> and 4'880 troz or 152 t Pd respectively in 2002), whereas the demand of Rhodium of 18 t or 568 troz is markedly smaller (JohnsonMatthey (2003)).

**Tab. 4.1 Main applications of PGMs. Table taken from Johnson Matthey<sup>5</sup>**

| Application         |                                   | PGMs used      |
|---------------------|-----------------------------------|----------------|
| Automotive catalyst | control of vehicle pollution      | Pt, Pd, Rh     |
| Chemical            | catalysts for chemical synthesis  | Pt, Pd, Rh, Ru |
|                     | electrode coating                 | Ir, Ru         |
| Dental              | alloys for dental restorations    | Pt, Pd         |
| Electronics         | computer hard disks               | Pt             |
|                     | multi-layer ceramic capacitors    | Pd             |
|                     | crucibles to grow single crystals | Ir             |
| Fuel Cells          | catalysts for power generation    | Pt             |
| Glass               | glassmaking equipment             | Pt, Rh         |
| Investment          | coins and bars                    | Pt             |
| Jewellery           | wedding rings, fashion jewellery  | Pt             |
| Medical             | anti-cancer drugs, implants       | Pt             |
| Petroleum           | catalysts for gasoline refining   | Pt             |
| Sensors             | temperature & gas sensing         | Pt, Rh         |
| Other               | spark plugs, pollution control    | Pt, Ir         |

<sup>4</sup> troz = troy ounce, 1'000 troz correspond to 31.1 kg.

<sup>5</sup> Johnson Matthey's website "Platinum Today", <http://www.platinum.matthey.com>, accessed at 01.03.2003

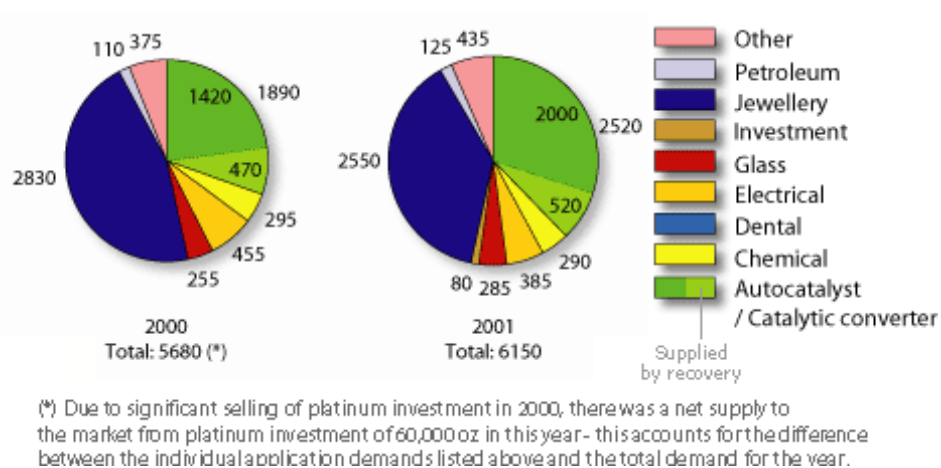


Fig. 4.1 Platinum, demand by application in 2001. Values in ounces (1000 troy ounces correspond to 31 kg), data taken from International Platinum Association<sup>6</sup>

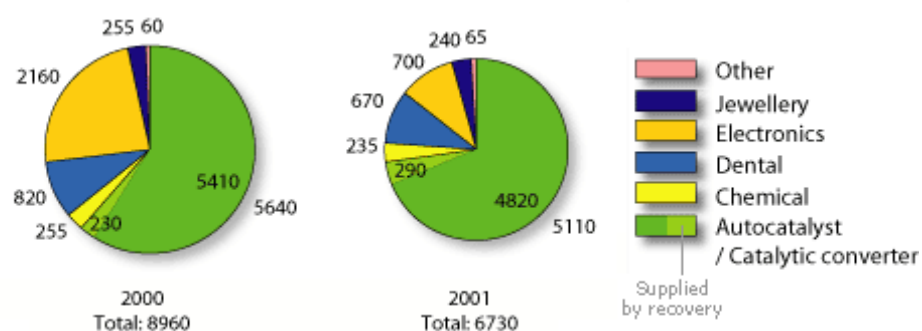


Fig. 4.2 Palladium, demand by application in 2001. Values in ounces (1000 troy ounces correspond to 31 kg), data taken from International Platinum Association<sup>6</sup>

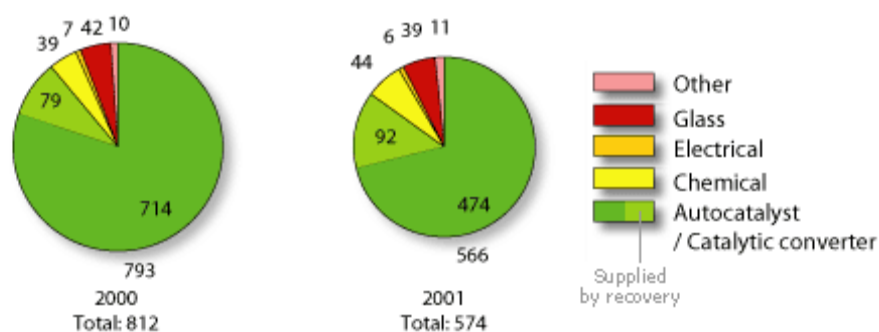


Fig. 4.3 Rhodium, demand by application in 2001. Values in ounces (1000 troy ounces correspond to 31 kg), data taken from International Platinum Association<sup>6</sup>

<sup>6</sup> International Platinum Association's website, <http://www.platinuminfo.net/>, accessed at 01.03.2003

## 5 System Characterisation

Platinum group metal, which is consumed in Europe, consists of a certain fraction of secondary materials originating from scrap. This fraction of secondary material is joined before consumption with primary material, originating from ore. Two single plants are responsible for around 90% of the primary platinum, palladium and rhodium production – these are the operations in Norsil'sk in Russia and in the Bushveld Complex in South Africa. The PGM are however not the only noteworthy products of these enterprises. Namely Noril'sk Nikkel production is characterised by the production of nickel and copper, PGM being a co-product. This fact makes it obvious that the ecological burden arising from the primary production has to be allocated to the primary products nickel, copper, platinum, palladium and rhodium. The same is true for the secondary production, where the metal values are recovered jointly. Fig. 5.1 gives an overview over the process scheme and the corresponding modules in ecoinvent.

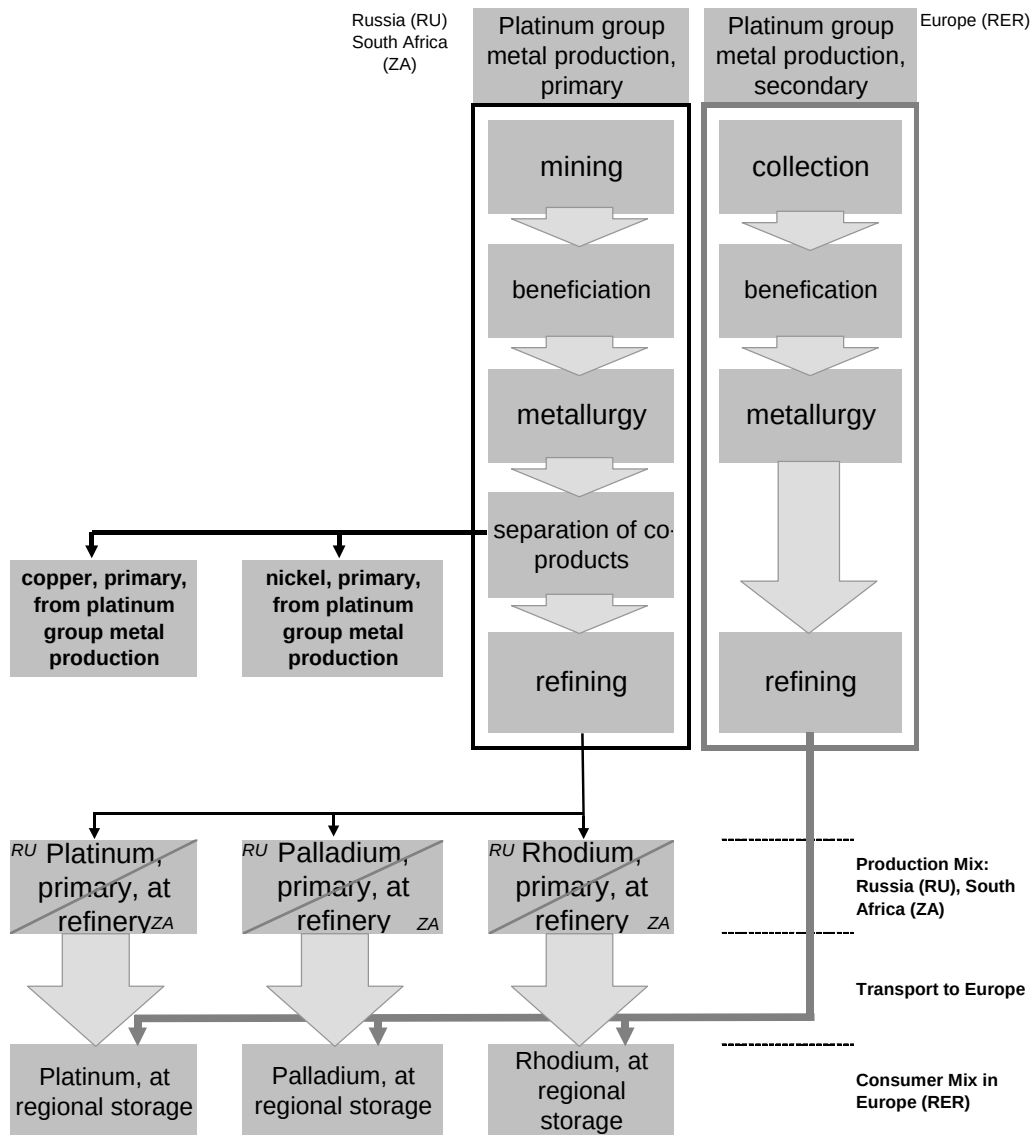


Fig. 5.1 Scheme of the PGM production in ecoinvent.

## 5.1 Allocation

The PGM production in ecoinvent is described formally as multi output process, yielding the coupled products nickel, copper and PGM as a mixture. The next chapter explains the allocation procedure in detail.

### 5.1.1 Primary Production

The exact ratio of the different PGMs present in this mixture is not reported in the literature used. It was assumed that the ratio of the PGM produced corresponds to the corresponding ratio in the raw ore (see Tab. 2.1).

For the coupled production of Ni, Cu and PGM the demand of ore is allocated according to the production output by mass Hochfeld (1997). The ore is handled in ecoinvent as a mixed ore with 4.7E-4% Pt, 3.1E-4% Pd, 0.2E-4% Rh in crude ore. The copper and nickel content in the ore depends on the deposit. For RU it is 3.2% and 2.3% respectively and for ZA the copper content is 0.052% and the

nickel content is 0.037%. As copper and nickel are leaving the PGM production before the final refining step, the burdens of refining are allocated only to PGM while the burdens of the other steps are allocated to copper, nickel and the PGMs. Thus different overall allocation factors for the different inventory flows are established depending on the flows occurrence and magnitude in the different production steps.

For all inventories other than ore an allocation by revenue is made, because this reflects economic reality of metal mining in a higher degree than an allocation by mass. Exploitation of an ore is always dependent on sufficient revenue expected. If a specific ore lacks sufficient revenue expected, it is not mined. Already mined ore with insufficient metal content usually is put aside for further beneficiation at a later time, at which market prices allow a profitable production of the metal. As ores contain different metals, the overall accounting of all metals produced is decisive. Even if the winning of a certain single metal is not profitable, it may be through cross-subsidising by another more profitable co-product. Thus the standard way chosen to allocate ecological burdens – except the demand of ore – is the allocation by revenue.

As PGM-mining companies have long term decision horizons, the average price of the last ten year period was chosen, this is for Pt, Pd and Rh US\$ 14'000, US\$ 9'420 and US\$ 30'700 per kilogram respectively<sup>7</sup>. The prices for Ni and Cu in this time period were US\$ 6.65 and US\$ 1.99 per kilogram respectively<sup>8</sup>. The fluctuation in price is documented in Fig. 5.2, Fig. 5.3, Fig. 5.4 and Fig. 5.5. Allocation and environmental burden may change depending on the period the price average is based on.

Tab. 5.1 gives a summary of the values chosen to calculate the allocation factors in the primary production. The resulting factors are summed up in Tab. 5.2

### 5.1.2 Secondary Production

Similar considerations are made for the allocation of secondary PGMs that originate from recovered automotive catalysts. It is assumed that all the secondary metal comes from automotive catalysts. Other sources of scrap were neglected. As in primary production, no data on the ratio of single PGM produced was available. This ratio differs greatly from scrap to scrap. Instead of the ore composition like in primary production, here the amount of annually recovered Pt, Pd and Rh respectively (in Europe) are chosen as proxy for the scrap composition. This composition ratio was used for allocation by mass. Multiplication with the market price gives the ratio used for the allocation by revenue.

Since no data on European rhodium recovery was available, it was approximated with the figure for global recovery rate of 15%<sup>9</sup> and a total annual Rh demand in Europe of 6.4 t. For the year 2002 this gives an amount of 1.11 t/a recovered Rh in Europe. The figures for Pt and Pd recovered in Europe are documented and amount 2.80 t/a Pt and 1.40 t/a Pd in the year 2002 (JohnsonMatthey (2003)).

This ratio of 1.11t/a : 1.40t/a : 2.80t/a for Rhodium : Palladium : Platinum corresponds to allocation factors summed up in Tab. 5.2.

<sup>7</sup> Data taken from Johnson Matthey's Website "Platinum Today", <http://www.platinum.matthey.com>, accessed at 01.03.2003

<sup>8</sup> Data taken from "London Metal Exchange", [http://www.lme.co.uk/data\\_prices/home.html](http://www.lme.co.uk/data_prices/home.html), accessed at 01.03.2003

<sup>9</sup> Total global rhodium demand of 20.7 t and an amount of 3.08 t recovered from automotive catalysts world-wide gives secondary rhodium a share of 15 % in the global demand.

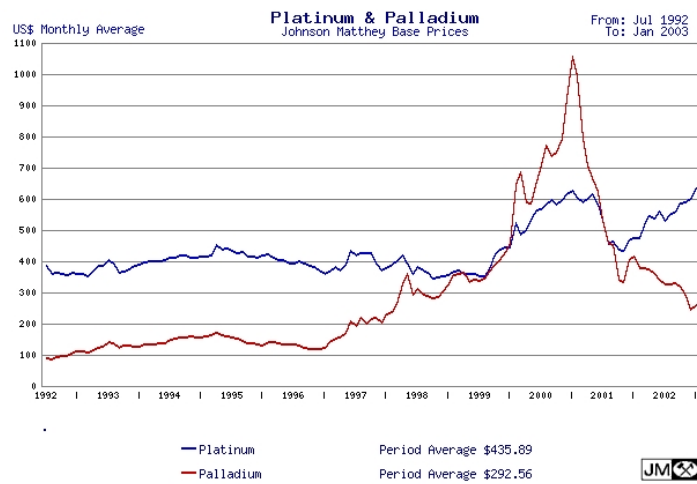


Fig. 5.2 Price development of platinum and palladium in the past ten years. Values in US\$ per troy ounce.

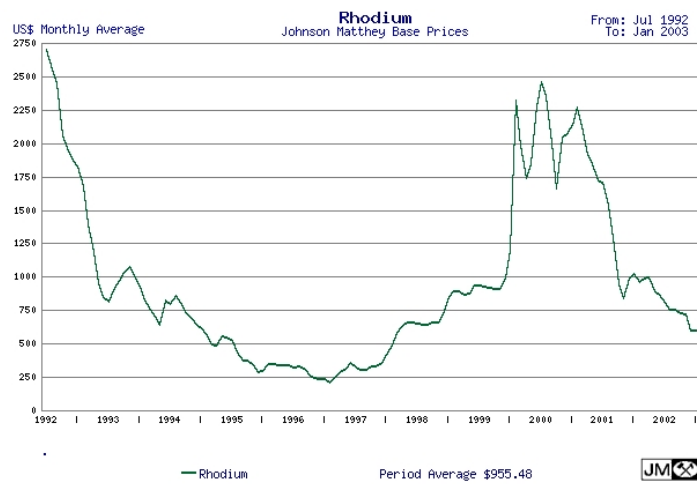


Fig. 5.3 Price development of rhodium in the past ten years. Values in US\$ per troy ounce.



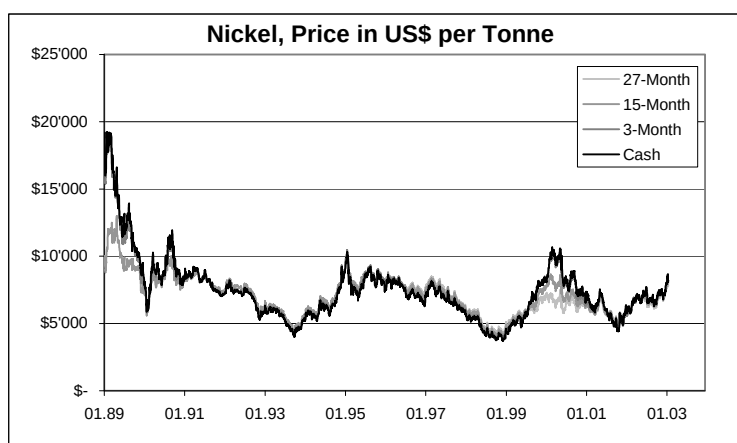


Fig. 5.4 Price development of nickel in the past ten years. Values in US\$ per tonne.

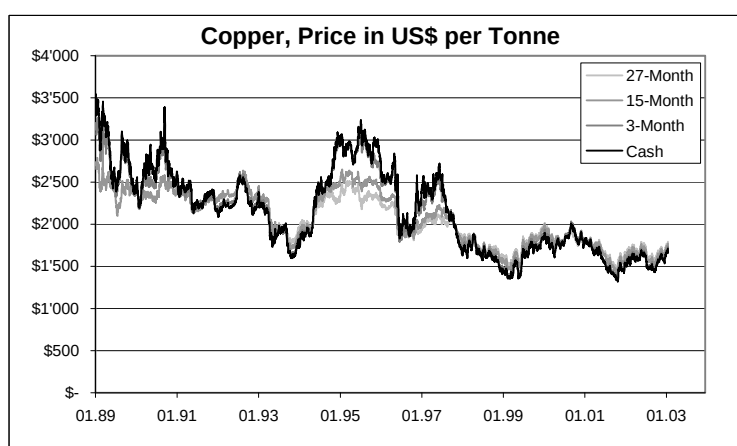


Fig. 5.5 Price development of copper in the past ten years. Values in US\$ per tonne.

Tab. 5.1 Assumptions made for allocation in the primary production. Details in the text above.

|                                     | Unit           | Primary PGM-<br>production (ZA) | Primary PGM-<br>production (RU) | global     |
|-------------------------------------|----------------|---------------------------------|---------------------------------|------------|
| functional unit, 1 t PGM            | t / t PGM      | 1                               | 1                               |            |
| Platinum in PGM-mix                 | % share in PGM | 68%                             | 25%                             |            |
| Palladium in PGM-mix                | % share in PGM | 29%                             | 73%                             |            |
| Rhodium in PGM-mix                  | % share in PGM | 3%                              | 2%                              |            |
| Nickel as co-product                | t / t PGM      | 147                             | 2'310                           |            |
| Copper as co-product                | t / t PGM      | 104                             | 3'200                           |            |
| Platinum, price on 10 year average  | US\$ / tonne   |                                 |                                 | 14'017'726 |
| Palladium, price on 10 year average | US\$ / tonne   |                                 |                                 | 9'420'169  |
| Rhodium, price on 10 year average   | US\$ / tonne   |                                 |                                 | 30'703'965 |
| Nickel, price on 10 year average    | US\$ / tonne   |                                 |                                 | 6'647      |
| Copper, price on 10 year average    | US\$ / tonne   |                                 |                                 | 1'986      |

**Tab. 5.2** Resulting factors for allocation based on values shown in Tab. 5.1. These factors allocate the ecological burden to the respective products. Allocation by mass reflects the PGM composition in the raw material (ore, scrap). Allocation by revenue is calculated by multiplication of the PGM content in the raw material with the respective market price. The allocation by mass is solely used for the allocation of PGM-ore to the respective products.

| Products  | Primary Production      |                         |                            |                            | Secondary Production      |                              |
|-----------|-------------------------|-------------------------|----------------------------|----------------------------|---------------------------|------------------------------|
|           | Allocation by mass (ZA) | Allocation by mass (RU) | Allocation by revenue (ZA) | Allocation by revenue (RU) | Allocation per mass (RER) | Allocation per revenue (RER) |
| <b>Pt</b> | 0.27%                   | 0.005%                  | 66%                        | 11%                        | 53%                       | 45%                          |
| <b>Pd</b> | 0.11%                   | 0.013%                  | 19%                        | 21%                        | 26%                       | 15%                          |
| <b>Rh</b> | 0.01%                   | 0.0004%                 | 7%                         | 2%                         | 21%                       | 39%                          |
| <b>Ni</b> | 41.27%                  | 41.88%                  | 7%                         | 47%                        | -                         | -                            |
| <b>Cu</b> | 58.33%                  | 58.10%                  | 1%                         | 19%                        | -                         | -                            |

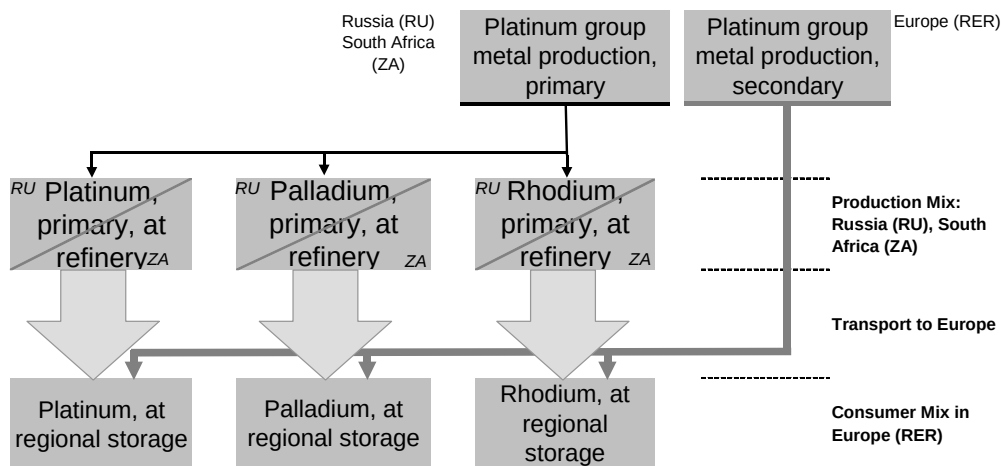
## 5.2 PGM consumer mix

In ecoinvent, following modules represent the use of platinum, palladium and rhodium in Europe (consumer mix):

- “platinum, at regional storage ”
- “palladium, at regional storage ”
- “rhodium, at regional storage ”

The consumer mix is an assembly of the primary PGM transported from ZA and RU to Europe and secondary PGM, which are recovered and consumed in Europe (Fig. 5.6). The modules for primary and secondary production are described in the sections 5.3 and 5.4 respectively.

The layout of the modules with general flow information, remarks and values are shown in Fig. 5.7, Fig. 5.8 and Fig. 5.9. An overview over the module’s general information and metadata is given in Tab. 5.3



**Fig. 5.6** Scheme of the PGM, at regional storage, in ecoinvent.

**Tab. 5.3 Ecoinvent meta information for the modules of consumption of platinum, palladium and rhodium.**

| <b>Name</b>                  | <b>palladium, at regional storage</b>                                                                                                                                                                              | <b>platinum, at regional storage</b>                                                                                                                                                                            | <b>rhodium, at regional storage</b>                                                                                                                                                                           |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location                     | RER                                                                                                                                                                                                                | RER                                                                                                                                                                                                             | RER                                                                                                                                                                                                           |
| InfrastructureProcess        | 0                                                                                                                                                                                                                  | 0                                                                                                                                                                                                               | 0                                                                                                                                                                                                             |
| Unit                         | kg                                                                                                                                                                                                                 | kg                                                                                                                                                                                                              | kg                                                                                                                                                                                                            |
| Dataset Version              | 2.0                                                                                                                                                                                                                | 2.0                                                                                                                                                                                                             | 2.0                                                                                                                                                                                                           |
| IncludedProcesses            | Primary PGM transported from ZA and RU for consumption in Europe and secondary PGM recovered and consumed in RER.                                                                                                  | Primary PGM transported from ZA and RU for consumption in Europe and secondary PGM recovered and consumed in RER.                                                                                               | Primary PGM transported from ZA and RU for consumption in Europe and secondary PGM recovered and consumed in RER.                                                                                             |
| Amount                       | 1                                                                                                                                                                                                                  | 1                                                                                                                                                                                                               | 1                                                                                                                                                                                                             |
| LocalName                    | Palladium, ab Regional-lager                                                                                                                                                                                       | Platin, ab Regionallager                                                                                                                                                                                        | Rhodium, ab Regional-lager                                                                                                                                                                                    |
| Synonyms                     | Platingruppenmetall//platinumgroup metal                                                                                                                                                                           | Platingruppenmetall//platinumgroup metal                                                                                                                                                                        | Platingruppenmetall//platinumgroup metal                                                                                                                                                                      |
| GeneralComment               | The consumer mix is an assembly of the primary palladium transported from ZA and RUS for consumption in Europe and secondary palladium recovered and consumed in RER according to the global market share in 2002. | The consumer mix is an assembly of the primary platinum transported from ZA and RU for consumption in Europe and secondary platinum recovered and consumed in RER according to the global market share in 2002. | The consumer mix is an assembly of the primary rhodium transported from ZA and RU for consumption in Europe and secondary rhodium recovered and consumed in RER according to the global market share in 2002. |
| StartDate                    | 2002                                                                                                                                                                                                               | 2002                                                                                                                                                                                                            | 2002                                                                                                                                                                                                          |
| EndDate                      | 2002                                                                                                                                                                                                               | 2002                                                                                                                                                                                                            | 2002                                                                                                                                                                                                          |
| Data valid for entire period | 1                                                                                                                                                                                                                  | 1                                                                                                                                                                                                               | 1                                                                                                                                                                                                             |
| Geography Text               | Global supply data for Europe used, no import statistic available. Supply per region is assumed to correspond to the import to Europe                                                                              | Global supply data for Europe used, no import statistic available. Supply per region is assumed to correspond to the import to Europe                                                                           | Global supply data for Europe used, no import statistic available. Supply per region is assumed to correspond to the import to Europe                                                                         |
| Technology Text              | European consumer mix                                                                                                                                                                                              | European consumer mix                                                                                                                                                                                           | European consumer mix                                                                                                                                                                                         |
| Representativeness           | 92%                                                                                                                                                                                                                | 92%                                                                                                                                                                                                             | 98%                                                                                                                                                                                                           |
| ProductionVolume             | 54.9 t / a                                                                                                                                                                                                         | 52.4 t / a                                                                                                                                                                                                      | 7.49 t / a                                                                                                                                                                                                    |
| SamplingProcedure            | Literature                                                                                                                                                                                                         | Literature                                                                                                                                                                                                      | Literature                                                                                                                                                                                                    |
| Extrapolations               | see Geography and Technology                                                                                                                                                                                       | see Geography and Technology                                                                                                                                                                                    | see Geography and Technology                                                                                                                                                                                  |
| UncertaintyAdjustments       | none                                                                                                                                                                                                               | none                                                                                                                                                                                                            | none                                                                                                                                                                                                          |

### 5.2.1 PGM, at regional storage, in Ecoinvent

These modules are valid for the consumption of the platinum group metals (PGM) Platinum, Palladium and Rhodium in Europe.

The secondary production is reflected by the recovery of automotive catalysts in Germany according to the actual recycling rate for Europe (Tab. 5.4). It is assumed to be valid for whole Europe.

The primary-production is reflected by the production processes of RSA and RUS according to their market-share. These two countries produce over 90 % of the PGMs. The mix of consumed PGM is assumed to correspond to the production mix shown in chapter 2.1 An average of the share in the global production over the last ten years was chosen (Tab. 5.5). Due to the importance of PGM, transport is assumed to take place by plane, the metal being protected with a case of 50% of its mass.

### Data quality

Allocations will shift with changing market situation. The recycling rate – i. e. the ratio of secondary platinum – may be underestimated, as most of the recovered material does not appear on the market.

**Tab. 5.4 Demand of secondary and primary PGM. Assessment of the share of the secondary production in the world wide demand. The recovery from automotive catalysts has to be added to the total demand to obtain the net demand. For Rh, no region specific figures are available, a world-wide recovery rate is chosen. Values are taken from JohnsonMatthey (2003). Note that the palladium demand in the US is exceptionally small in 2002, resulting in an overestimated ratio of secondary production, which was stable at around 5% of the total consumption until 2001. This is due to the use of stocks by US auto manufacturers in lieu of metal purchases.**

| <b>Values in t for 2002</b>         | <b>Pt</b> | <b>Pd</b>                      | <b>Rh</b>  |
|-------------------------------------|-----------|--------------------------------|------------|
| total global demand (primary)       | 198       | 152                            | 17.7       |
| total global recovery (secondary)   | 17.7      | 11.5                           | 3.1        |
| global net demand (total)           | 216       | 163                            | 20.7       |
| global ratio secondary              | 8%        | 7%                             | 15%        |
| total European demand (primary)     | 52.1      | 51.0                           | 6.4        |
| total European recovery (secondary) | 2.8       | 1.4                            | 1.1        |
| European net demand (total)         | 54.9      | 52.4                           | 7.5        |
| European ratio secondary            | <b>5%</b> | <b>3%</b>                      | <b>15%</b> |
| US ratio secondary                  | 28%       | 21% (2002)<br>5% (1993 – 2001) |            |
| Japanese ratio secondary            | 4.2%      | 3.0%                           |            |

**Tab. 5.5 Production of primary metal in South Africa and the Russian Federation. Ten year averages chosen from data in JohnsonMatthey (2003).**

| <b>Share in global primary production<br/>(ten year average)</b> | <b>Pt</b> | <b>Pd</b> | <b>Rh</b> |
|------------------------------------------------------------------|-----------|-----------|-----------|
| South Africa (ZA)                                                | 72.2%     | 26.3%     | 72.9%     |
| Russian Federation (RU)                                          | 20.1%     | 63.2%     | 23.3%     |
| Represented global market volume                                 | 92.3%     | 89.5%     | 96.2%     |
| South Africa (ZA), relating to total                             | 78.2%     | 29.4%     | 75.7%     |
| Russian Federation (RU), relating to total                       | 21.8%     | 70.6%     | 24.3%     |

| General Flow Information |   |                               |  |        |  | Representation in ecoinvent                                                                                                      |                   |             |                |          |                                  |                               | Uncertainty Informations |                         |      |                        |                 |
|--------------------------|---|-------------------------------|--|--------|--|----------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------|----------------|----------|----------------------------------|-------------------------------|--------------------------|-------------------------|------|------------------------|-----------------|
| Input                    |   | Process Name                  |  | Output |  | Remarks                                                                                                                          | Category          | Subcategory | Infrastructure | Location | Modul name in ecoinvent          | Mean value                    | Unit                     | Source for mean value   | Type | Standard Deviation 95% | General Comment |
| Platinum, RSA            | m | platinum, at regional storage |  |        |  | According to market share: 72% of global primary production                                                                      | metals            | extraction  | No             | ZA       | platinum, primary, at refinery   | 0.74                          | kg                       | (JohnsonMatthey , 2003) | 1    | 1.12                   | (1,2,1,1,1,4,3) |
| Platinum, RUS            | m |                               |  |        |  | According to market share: 20% of global primary production                                                                      | metals            | extraction  | No             | RU       | platinum, primary, at refinery   | 0.21                          | kg                       | (JohnsonMatthey , 2003) | 1    | 1.12                   | (1,2,1,1,1,4,3) |
| Platinum, recycling      | m |                               |  |        |  | Recycling solely from recovered autocatalysts: 5% of consumption                                                                 | metals            | extraction  | No             | RER      | platinum, secondary, at refinery | 0.05                          | kg                       | (JohnsonMatthey , 2003) | 1    | 1.12                   | (1,2,1,1,1,4,3) |
| Transportation           | m |                               |  |        |  | Only aircargo considered, packaging is roughly estimed to be 50% of the metal mass. Distance to RER: from ZA 15000km, RU 2500 km | transport systems | airplane    | No             | RER      | transport, aircraft, freight     | 17.49                         | tkm                      | default distances       | 1    | 1.12                   | (1,2,1,1,1,4,3) |
|                          |   |                               |  |        |  |                                                                                                                                  |                   | metals      | extraction     | No       | RER                              | platinum, at regional storage | 1.00                     | kg                      |      |                        |                 |
|                          |   |                               |  |        |  | Platinum, at consumer                                                                                                            |                   |             |                |          |                                  |                               |                          |                         |      |                        |                 |

Fig. 5.7 Flows for “Platinum, at regional storage” and its representation in the ecoinvent database

| General Flow Information |   |                                |  |        |  | Representation in ecoinvent                                                                                                      |                   |             |                |          |                                   |            | Uncertainty Informations |                        |      |                        |                 |
|--------------------------|---|--------------------------------|--|--------|--|----------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------|----------------|----------|-----------------------------------|------------|--------------------------|------------------------|------|------------------------|-----------------|
| Input                    |   | Process Name                   |  | Output |  | Remarks                                                                                                                          | Category          | Subcategory | Infrastructure | Location | Modul name in ecoinvent           | Mean value | Unit                     | Source for mean value  | Type | Standard Deviation 95% | General Comment |
| Palladium, RSA           | m | palladium, at regional storage |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
| Palladium, RUS           | m |                                |  |        |  |                                                                                                                                  | metals            | extraction  | No             | ZA       | palladium, primary, at refinery   | 0.29       | kg                       | (JohnsonMatthey, 2003) | 1    | 1.12                   | (1,2,1,1,1,4,3) |
| Palladium, recycling     | m |                                |  |        |  | Recycling solely from recovered autocatalysts: 3% of consumption                                                                 | metals            | extraction  | No             | RER      | palladium, secondary, at refinery | 0.03       | kg                       | (JohnsonMatthey, 2003) | 1    | 1.12                   | (1,2,1,1,1,4,3) |
| Transportation           | m |                                |  |        |  | only aircargo considered, packaging is roughly estimed to be 50% of the metal mass. Distance to RER: from ZA 15000km, RU 2500 km | transport systems | airplane    | No             | RER      | transport, aircraft, freight      | 8.98       | tkm                      | default distances      | 1    | 1.12                   | (1,2,1,1,1,4,3) |
|                          |   |                                |  |        |  |                                                                                                                                  | metals            | extraction  | No             | RER      | palladium, at regional storage    | 1.00       | kg                       |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
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|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |
|                          |   |                                |  |        |  |                                                                                                                                  |                   |             |                |          |                                   |            |                          |                        |      |                        |                 |

Fig. 5.8 Flows for “Palladium, at regional storage” and its representation in the ecoinvent database

| General Flow Information |   |                              |  |        |  | Representation in ecoinvent                                                                                                      |                   |             |                |          |                                 |                              | Uncertainty Informations |                         |      |                        |                 |
|--------------------------|---|------------------------------|--|--------|--|----------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------|----------------|----------|---------------------------------|------------------------------|--------------------------|-------------------------|------|------------------------|-----------------|
| Input                    |   | Process Name                 |  | Output |  | Remarks                                                                                                                          | Category          | Subcategory | Infrastructure | Location | Modul name in ecoinvent         | Mean value                   | Unit                     | Source for mean value   | Type | Standard Deviation 95% | General Comment |
| Rhodium, RSA             | m | rhodium, at regional storage |  |        |  | According to market share: 73% of global primary production                                                                      | metals            | extraction  | No             | ZA       | rhodium, primary, at refinery   | 0.64                         | kg                       | (JohnsonMatthey , 2003) | 1    | 1.12                   | (1,2,1,1,1,4,3) |
| Rhodium, RUS             | m |                              |  |        |  | According to market share: 23% of global primary production                                                                      | metals            | extraction  | No             | RU       | rhodium, primary, at refinery   | 0.21                         | kg                       | (JohnsonMatthey , 2003) | 1    | 1.12                   | (1,2,1,1,1,4,3) |
| Rhodium, recyding        | m |                              |  |        |  | Recycling solely from recovered autocatalysts: 15% of consumption                                                                | metals            | extraction  | No             | RER      | rhodium, secondary, at refinery | 0.15                         | kg                       | (JohnsonMatthey , 2003) | 1    | 1.12                   | (1,2,1,1,1,4,3) |
| Transportation           | m |                              |  |        |  | only aircargo considered, packaging is roughly estimed to be 50% of the metal mass. Distance to RER: from ZA 15000km, RU 2500 km | transport systems | airplane    | No             | RER      | transport, aircraft, freight    | 15.26                        | tkm                      | default distances       | 1    | 1.12                   | (1,2,1,1,1,4,3) |
|                          |   |                              |  |        |  |                                                                                                                                  |                   | metals      | extraction     | No       | RER                             | rhodium, at regional storage | 1.00                     | kg                      |      |                        |                 |

Fig. 5.9 Flows for “Rhodium, at regional storage” and its representation in the ecoinvent database

## 5.3 Winning of Primary PGM

This part is mainly based on Hochfeld (1997). Lacking data – mainly on direct emissions to air and to water – was taken from different sources where stated. Because of the big variance between the different production processes due to different ore composition, primary PGM production is represented by the two major sites in Bushveld (ZA) and Noril'sk (RU). The producing companies in these two deposits determine the PGM production within their countries and world-wide.

While the companies in RSA are mainly PGM-dominated, Norilsk Nickel in the Russian Federation earns a large share through Nickel and Copper. Also the ratio of the PGM produced differ between the production in RSA and the one in RUS (see chapter 5.1 above).

For this reasons, the production process in RSA and the one in RUS are treated separately. Their products – the pure metals Pt, Pd, Rh, Ni and Cu – are assembled later according to their market share.

### 5.3.1 Production process

The production process described is valid for both the operations in RSA and in RUS. Differences are stated.

#### Mining

The PGM containing ore is mined underground and transferred to the beneficiation. Widening of the tunnels is mainly done by blasting. The overburden – material, which does not contain PGM-bearing ore – is deposited off-site and is partly refilled into the tunnels.

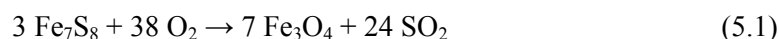
#### Beneficiation

After mining, the ore is first ground. In a next step it is subjected to gravity concentration to separate the metallic particles from the PGM-bearing minerals. After this first concentration step, flotation is carried out to remove the gangue from the sulfidic minerals. For neutralisation lime is added. In the flotation several organic chemicals are used as collector, frother, activator, depressor and flocculant. Sometimes cyanide is used as depressant for pyrite. Tailings usually are led to tailing heaps or ponds.

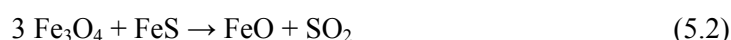
#### Metallurgy

The pyrometallurgical step may be divided in three substeps, which can be effected in combination depending on the procedure used. The process is basically the same as in the nickel or copper production. Emitted sulphur dioxide usually is recovered from the off-gas stream producing sulphuric acid.

**Roasting.** The sulphidic ore is roasted at 600 – 700 °C. This oxidises the iron present, dries and pre-heats the ore. Since iron has got a bigger affinity to oxygen than other metals, it gets oxidised to slag releasing the sulphur as sulphur dioxide (Formula 5.1). This reaction is strongly exothermic. Other metals associate again with sulphur to sulphur compounds.



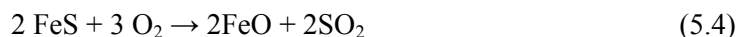
**Smelting.** In this step, the bigger part of the gangue and the iron sulphide is eliminated. The valuable metals concentrate in the matte. Smelting is done at temperatures of 1400 °C. Main reaction is the slagging of the prior formed magnetite ( $\text{Fe}_3\text{O}_4$ ) with the flux (calcite, silicate), according to the formulas 5.2 and 5.3.



The melt is parted from magnetite and now consists of a sulphidic and a oxidic phase. The matte is tapped at 1400 °C and processed in the converting step.



**Converting.** In the converting step, iron sulphide is removed from the low-grade matte by oxidation and slagging. The slag, which contains high levels of metals, is returned to the smelting step for recovery. The final product is high-grade, low-iron nickel – copper matte, which typically contains 20 % S, less than 1 % Fe, and also some precious metals. Converting is a batch operation. Air or oxygen-enriched air is blown through the molten matte to form iron oxides and remove sulphur as sulphur dioxide. The iron oxides combine with added silica flux to form an iron silicate slag (formulas 5.4 and 5.5).



In the processes regarded in this study Pierce-Smith-Converters are used.

### Separation from Non Ferrous Metals

The main accompanying metals in the PGM winning are Nickel and Copper. In the case of the Russian process the winning of these two metals is the main objective. The separation is done in a hydrometallurgical process. In a typical hydrometallurgical process, the concentrate or matte is first leached in a sulphate or chloride solution to dissolve the metals, while the sulphide is oxidised to insoluble elemental sulphur or soluble sulphate. In this way a nickel – copper matte can be treated in a two-stage leach process to produce a copper-free nickel sulphate or nickel chloride solution, and a leach residue enriched in copper. The copper in the copper-rich residue is solubilised by repeated leaching and recovered from solution electrolytically as cathode copper. Nickel is recovered from the purified nickel sulphate or chloride solution either electrolytically as a pure nickel cathode or by chemical reduction with hydrogen to give pure nickel powder. The remaining leach residue contains all PGM, which are extracted in the refining step.

### Refining

The usual starting point for separating platinum group metals or producing compounds and catalysts is an aqueous solution of PGM. Most raw materials can be dissolved in oxidising acids. Once dissolved, there are namely two processes for refining.

**Selective precipitation.** This is the classical process. Separation is based on the principles of analytical chemistry of PGM. In the crystallisation process, the solubility of chloro complexes of the PGM are very important, as is the possibility of altering this solubility. The temperature dependence of this solubility is generally quite large and can be utilised for precipitation. An even more useful phenomenon in separation technology is the effect on solubility of the addition of a common ion, which is utilised in precipitation crystallisation. Conversion of the precipitated salts to their metals is carried out mainly by thermal decomposition at 1000 °C.

**Solvent extraction.** This process is based on the ability of the metals to form together with organic molecules stable complexes. After total dissolution the liquid is treated with a cascade of different specific solvents. Liquid – liquid extraction is characterised by distribution coefficients that vary greatly from element to element. These can usually be further modified by chemical methods. Although separations are generally better than those effected by precipitation, coarse separation by solvent extraction must be further purified.

The principal raw materials are concentrates produced from ores, mattes and slimes from nickel and copper operations. Secondary materials such as spent catalyst or electronic component scrap are also significant sources. PGM refining is complex and individual process stages may have to be repeated to achieve the required purity. Precious metal refineries are complex networks of main and subsidiary processes. It is therefore not possible to identify single process steps and their contribution to emissions and consumptions. The product of this step is the pure platinum group metal.

### 5.3.2 Emissions

#### Mining

Major emissions are due to mineral born pollutants present in effluents. Being underground, the mining has less emissions to air than open pit mining. The major dust sources are on the surface therefore the dust emissions are in the same magnitude like in open pit mining. Rain percolate through overburden and account to metal emissions to groundwater.

#### Beneficiation

Ore handling and processing produce large amounts of dust, containing PM<sub>10</sub> and several metals from the ore itself. Flotation produce effluents containing several organic agents<sup>10</sup> used. Some of these chemicals evaporate and account for VOC emissions to air. Namely xanthates decompose hydrolytically to release carbon disulphide. Tailings effluent contain additional sulphuric acid from acid rock drainage.

#### Metallurgy & Separation from Non Ferrous Metals

**Sulphur dioxide.** In all of the process steps sulphur dioxide is emitted to air. Recovery of sulphur dioxide is only economic for high concentrated off-gas. Therefore recovery is not effected in all process alternatives. SO<sub>2</sub> emissions are by far the most important in the metallurgy.

**Carbon dioxide.** In the beneficiation step, considerable amounts of lime is added to the ore for pH-stabilisation, lime forms later flux in the metallurgical step. Lime decomposes into CO<sub>2</sub> to form calcite.

**Dust.** Dust carry over from the roasting, smelting and converting processes. Particulate emissions to the air consist of metals and thus are often returned to the leaching process after treatment.

**Chlorine.** Chlorine is used in some leaching stages and is produced during the subsequent electrolysis of chloride solution. The chlorine evolved is collected and re-used in the leach stage. The presence of chlorine in wastewater can lead to the formation of organic chlorine compounds (AOX) if solvents etc. are also present in a mixed wastewater.

**VOC.** VOCs can be emitted from the solvent extraction stages. A variety of solvents are used as they contain various complexing agents to form complexes with the desired metal that are soluble in the organic layer.

**Emissions to water.** Metals and their compounds and substances in suspension are the main pollutants emitted to water. The metals concerned are Cu, Ni, Co, As and Cr. Other significant substances are chlorides and sulphates. Wastewater from wet gas cleaning (if used) of the different metallurgical stages are the most important sources. The leaching stages are usually operated on a closed circuit and drainage systems, and are therefore regarded as minor sources.

#### Refining

Also in this step the combustion of sulphur leads to emissions of SO<sub>2</sub>. Nitrogen oxides are produced in significant amounts during acid digestion using nitric acid. Chlorine and HCl can be formed during a number of digestion, electrolytic and purification processes. Chlorine is used extensively in the Miller process and in the dissolution stages using hydrochloric acid and chlorine mixtures respectively. Dust

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<sup>10</sup> Floating agents consist of collectors (xanthate or aerofloat) and frothing reagent (eg. Methyl Isobutyl Carbinol). Xanthates are a broad class of organic salts formed by treatment of an alcohol with carbon disulfide in the presence of an alkali, potassium ethyl xanthate (KC<sub>3</sub>H<sub>5</sub>S<sub>2</sub>O, CAS No: 140-89-6). Aerofloat is a salt of carbonyldithiophosphoric acids. A widely used frothing reagent is e.g. methyl isobutyl carbinol, CAS: 108-11-2, syn. Methyl Amyl Alcohol.

and metals are generally emitted from incinerators and furnaces. VOC can be emitted from solvent extraction processes, while organic compounds, namely dioxins, can be emitted from smelting stages resulting from the poor combustion of oil and plastic in the feed material. All these emissions are subject to abatement technologies and controlling. Large quantities of effluents contain amounts of metals and organic substances.

### **5.3.3 Waste**

#### **Mining and Beneficiation**

Overburden are deposited close to the mine. Tailings are deposited as piles and in ponds. In both the overburden and the tailings occur acid rock drainage over a long period of time. These effects are handled in the section “Non Ferrous Metal Winning – Auxiliary Processes”, chapter “Disposal of Tailings and Overburden”.

#### **Metallurgy & Separation from Non Ferrous Metals**

The production of metals is related to the generation of several co-products, residues and wastes, which are listed in the European Waste Catalogue. Some of the process specific residues can be reused or recovered in preliminary process steps (e. g. dross, filter dust) or construction (e. g. cleaned slag). Residues also arise from the treatment of liquid effluents, the main residue being gypsum waste and metal hydroxides from the wastewater neutralisation plant. These residuals have to be disposed, usually in lined ponds.

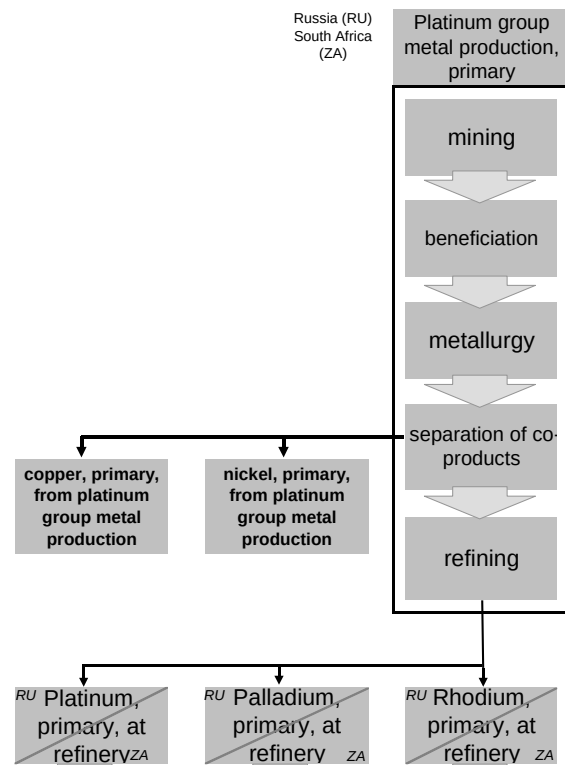
#### **Refining**

Like in the pyro- and hydrometallurgical step quantities of solid residuals are generated, which are mostly recycled within the process or sent to other specialists to recover any precious metals. Final residues generally comprise hydroxide filter cakes (ironhydroxide, 60% water, cat I industrial waste).

### **5.3.4 Primary PGM in ecoinvent**

This module is modelled in two different specifications: “PGM-Production, primary, RSA” for the South African operations and “ PGM-Production, primary, RUS” for the Russian operations. Both modules are similarly modelled, exceptions are mentioned in the text below. The scheme of this multi-output process is shown in Fig. 5.10.

The layout of the modules with general flow information, remarks, sources, values and uncertainty information are shown in Fig. 5.11 - Fig. 5.15. An overview over the module’s general information and metadata is given in Tab. 5.6. The following paragraphs describe calculations, sources and assumptions chosen in this study.



**Fig. 5.10** Scheme of the multi-output process “platinum group metal production, primary” in Ecoinvent. Its jointly produced co-products are “platinum, primary, at refinery”, “palladium, primary, at refinery”, “rhodium, primary, at refinery”, “nickel, primary, couple-production, platinum group metal production” and “copper, primary, couple-production, platinum group metal production”.

**Tab. 5.6 Ecoinvent meta information for the multi-output processes in the primary production of PGM in Russia and South Africa.**

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>                         | platinum group metal production, primary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | platinum group metal production, primary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Location</b>                     | ZA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | RU                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Infrastructure Process</b>       | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Unit</b>                         | kg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | kg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Dataset Version</b>              | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Included Processes</b>           | The module includes a mining and a beneficiation step with the mining infrastructure and disposal of overburden and tailings. Subsequently it includes the metallurgy step with the disposal of slag, the infrastructure and the separation of the co-products nickel and copper, and the refining step yielding the desired PGM-mix inclusively the refining infrastructure. Production, application and emissions of most agents used in beneficiation and metallurgy are also included.                                                                                                                                                                                                                                                                                                                                 | The module includes a mining and a beneficiation step with the mining infrastructure and disposal of overburden and tailings. Subsequently it includes the metallurgy step with the disposal of slag, the infrastructure and the separation of the co-products nickel and copper, and the refining step yielding the desired PGM-mix inclusively the refining infrastructure. Production, application and emissions of most agents used in beneficiation and metallurgy are also included.                                                                                                                                                                                                                                                                                                                                           |
| <b>Amount</b>                       | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Local Name</b>                   | Platingruppenmetalle-Produktion, primär                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Platingruppenmetalle-Produktion, primär                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Synonyms</b>                     | Platingruppenmetall//platinumgroup metal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Platingruppenmetall//platinumgroup metal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>General Comment</b>              | The multioutput-process "PGM-Production, primary" delivers the co-products "platinum, primary, at refinery", "palladium, primary, at refinery", "rhodium, primary, at refinery", "nickel, primary, from platinum group metal production" and "copper, primary, from platinum group metal production" in South Africa (ZA). The module is designed for the use of the metal in technical systems, where it plays a minor role like the use in manufacturing of electronic or technical chemistry using certain catalysts. It is not to be used if the impact of the PGM within the modelled process in scope is considered to be high. In such cases, a more detailed analysis depending on scope and allocation procedures has to be conducted. The data used is mainly based on a LCA study for autocatalysts in Germany. | The multioutput-process "PGM-Production, primary" delivers the co-products "platinum, primary, at refinery", "palladium, primary, at refinery", "rhodium, primary, at refinery", "nickel, primary, from platinum group metal production" and "copper, primary, from platinum group metal production" in the Russian Federation (RU). The module is designed for the use of the metal in technical systems, where it plays a minor role like the use in manufacturing of electronic or technical chemistry using certain catalysts. It is not to be used if the impact of the PGM within the modelled process in scope is considered to be high. In such cases, a more detailed analysis depending on scope and allocation procedures has to be conducted. The data used is mainly based on a LCA study for autocatalysts in Germany. |
| <b>Start Date</b>                   | 1995                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1995                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>End Date</b>                     | 2002                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2002                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Data valid for entire period</b> | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Geography Text</b>               | Some agents used like lime or sodium cyanide originate from data sets valid for Switzerland or Europe. Processes like blasting and transport originate from data sets valid for Switzerland or Europe.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Some agents used like lime or sodium cyanide originate from data sets valid for Switzerland or Europe. Processes like blasting and transport originate from data sets valid for Switzerland or Europe.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

**Tab. 5.6 Ecoinvent meta information for the multi-output processes in the primary production of PGM in Russia and South Africa. (cont.)**

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Technology Text         | Mining is done underground, followed by beneficiation through flotation where considerable amounts of agents are added. Overburden and tailings are disposed on the mining site and partly refilled. In ZA a electric arc furnace with a Söderberg electrode system and a Pierce-Smith-Converter is used. sulphur dioxide in the off-gas is recovered producing sulphuric acid. The separation of non ferrous metals is done hydrometallurgically, the refining by selective precipitation. Although electricity production in ZA is mainly coal based, the UCTE production mix is inventoried. | Mining is done underground, followed by beneficiation through flotation where considerable amounts of agents are added. Overburden and tailings are disposed on the mining site and partly refilled. In RU a Outokumpu flash furnace followed by a Pierce-Smith-Converter is used. The resulting nickel matte is treated by electroly refining. No recovery of sulphur dioxide in the off gas. The refining is done by selective precipitation, using data from the ZA process as proxy. For electricity mix the UCTE production mix is inventoried |
| Representativeness      | 63%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 23%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Production Volume       | 221 t/a                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 81 t/a                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Sampling Procedure      | Literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Extrapolations          | see Geography and Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | see Geography and Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Uncertainty Adjustments | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

### Infrastructure and land use

The production infrastructure used directly for exploration is inventoried within this module, whereas the infrastructure like buildings and land-use for the operation is inventoried in an extern module. The mobile equipment used for exploration like drilling machines, scoop trams and underground trucks is inventoried assessing a standard value for diesel, burned in building machine, i. e. 69.2 kJ per tonne mined raw ore. 2000 m conveyor belts are further assumed with a capacity of 2000 t / h which yields  $2 \cdot 10^{-5}$  m conveyor belt required per tonne mined raw ore. Mining facilities are inventoried as unspecific module according to part II “Non Ferrous Metal Winning – Auxiliary Processes”.

Transformation and occupation of land through direct exploration are assumed with the standard value for non-ferrous metals without heap leaching processes of 0.0025 m<sup>2</sup>/t and 0.075 m<sup>2</sup>/a/t ore mined.

### Mining

In the considered processes the ore is mined underground. The overburden is disposed in piles near the mine (see. Part II “Non Ferrous Metal Winning – Auxiliary Processes”). Based on IPCC (2002) a value of 50 % refilled overburden is assumed. Demand for water was not available for the process in RUS, it was approximated with the value for RSA (of 0.21m<sup>3</sup> / t mined ore). Two third of the water resource use is assumed to be ground water, the rest is surface water. No volume of mining effluents was available, values for water demand were taken. Composition of mining effluents is approximated with values taken from total emissions to water from base metal sites (s. Tab. 5.10). Transport from mine to beneficiation were neglected, it is assumed to take place in direct vicinity of the mine. For diesel use the value of 69.2 kJ per tonne mined raw ore for underground mining given in Hilbrans & Hinrichs (1999) was chosen.

Dust emissions from mining activities are comprised in the beneficiation step below.

Infrastructure of the mine is modelled with the unspecific module “mining infrastructure” from part II “Non Ferrous Metal Winning – Auxiliary Processes”, chapter “Disposal of Tailings and Overburden”. The electricity mix is approximated with the UCTE production mix. For the ore demand an overall

yield of 90 % is assumed, based on the 91 % primary yield in the nickel winning process reported in Hilbrans & Hinrichs (1999).

### Beneficiation

Once the ore is mined, it is ground and conditioned for further concentration. Considerable amounts of lime and flotation agents are added. The separated gangue is disposed in tailings ponds, the concentrated ore is fed to the metallurgy, which is assumed to be on site.

The infrastructure of beneficiation is included in mining infrastructure. Dust emissions comprise additionally the dust emissions due to mining activities. By approximation again electricity production mix for the UCTE is chosen. 50 % of the reported energy consumption is assumed to be used to dry the material, using natural gas, burned in industrial furnace >100kW.

**Lime.** The value for lime – geometric mean – is taken from copper mining where a range from 1 to 4 kg per tonne sulphidic ore is recommended (Krauss et al. (1999)). This value is confirmed by IPPC (2002), Tab. 5.11.

**Reagents.** No information on mass and nature of the floating agents used is available. For copper mining a range of 25 – 300 g collector and 25 – 250 g frothing reagent per ton sulphidic ore is recommended (Krauss et al. (1999)). This is confirmed by the values in Tab. 5.11. Collector and frother are balanced in ecoinvent in their production as unspecified organic chemical. According to NPI (2001) xanthates hydrolytically decompose to 0.53 mg carbon disulphide per mg xanthate. With 189 mg xanthates from Tab. 5.11 this is 102.4 mg carbon disulphide per tonne mined ore.

Activator and Depressants – excluding sodium cyanate – are modelled as unspecific inorganic chemicals, their emission to water is covered by reported emissions from mining sites shown in Tab. 5.10.

For pH adjustment and leaching the value for sulphuric acid was considered with 3.74 kg / t ore mined (Tab. 5.11). As sulphuric acid is an by-product from the metallurgy, it is regarded as burden free. Other acids are neglected.

Since composition and environmental fate of activator, depressant and sulphuric acid is uncertain, reported data for total emissions shown in Tab. 5.10 are taken.

Sodium cyanide is widely used for extracting gold and other metals such as silver, copper and zinc. It is used as leachate and as depressant for certain sulphidic minerals. The environmental fate of cyanide compounds is complex and could not further be quarried. A cyanide recovery of 80 – 90% is reported due to abatement and recovery in the Golden Cross mine in New Zealand (Anonymous (1998a)). Here, a value of 80% is chosen for cyanide-abatement technology. With a reported use of 35 g NaCN per tonne ore mined (Tab. 5.11), this means a direct emission of 3.88 g HCN per tonne ore mined. An overview of the reagents accounted for is presented in Tab. 5.12.

**Effluents.** For the beneficiation effluents of the Russian production no data was available, the same value like RSA are taken (0.35 m<sup>3</sup> / t mined ore). Additional to direct emissions due to the use of chemicals, total emissions to water are chosen according to Tab. 5.10. These values are from base metal sites, thus only an approximation for the real effluent concentrations in the PGM production. It is assumed here that the effluents also include acid rock drainage (ARD). ARD after mine closure is modelled in the module “disposal, sulphidic tailings, off-site”.

**Dust.** A case study in (Anonymous (1998b)) about a projected open cut gold-copper mining and processing facility in Australia predict annual dust emissions between 0.18 kg/t and 0.46 kg/t of dust produced per tonne of ore mined. The prediction was made using the Emission Estimation Technique Manual for Mining (NPI (2001)). The maximum value of 0.46 kg/t is chosen for total dust emissions. Underground mining activities are assumed to generate 80 % of the dust emissions compared to open pit mining (Tab. 5.8) yielding a 0.37 kg dust per t mined ore. According to NPI (2001) the dust com-

position is approximated with values in Tab. 5.7 the average earth's crust. For PM<sub>10</sub> values, the ratio TDS : PM<sub>10</sub> : PM<sub>2.5</sub> was approximated with values from copper ore mining 101.72 : 50 : 5 g / t ore taken from the CEMIP-database<sup>11</sup>, i. e. one tonne emitted dust corresponds to 491.5 kg PM<sub>10</sub> and 49.2 kg PM<sub>2.5</sub> respectively. The dust emissions resulting from underground activities are summed up in Tab. 5.9. In Ecoinvent, for particulates differential classes are used adding up to the total dust emission, i. e. 49.2 kg PM<2.5, 442 kg PM2.5-10 and 508 kg PM>10 per tonne dust.

**Tab. 5.7 Mineral born dust composition depending on rock. Data taken from (NPI (2001)). Values for earth's crust are chosen in this study.**

| Element   | Unit  | Symbol | Basalt | Granite | This study<br>(Earth's Crust) |
|-----------|-------|--------|--------|---------|-------------------------------|
| Antimony  | g / t | Sb     | 0.69   | 0.2     | 0.2                           |
| Arsenic   | g / t | As     | 1.5    | 1.5     | 1.5                           |
| Beryllium | g / t | Be     | 0.3    | 5       | 2.6                           |
| Boron     | g / t | Be     | 8      | 12      | 10                            |
| Cadmium   | g / t | Cd     | 0.13   | 0.09    | 0.11                          |
| Chromium  | g / t | Cr     | 200    | 4       | 100                           |
| Cobalt    | g / t | Co     | 35     | 1       | 20                            |
| Copper    | g / t | Cu     | 90     | 13      | 50                            |
| Fluorine  | g / t | F      | 510    | 1400    | 950                           |
| Lead      | g / t | Pb     | 3      | 24      | 14                            |
| Manganese | g / t | Mn     | 1500   | 400     | 950                           |
| Mercury   | g / t | Hg     | 0.12   | 0.08    | 0.05                          |
| Nickel    | g / t | Ni     | 150    | 0.5     | 80                            |
| Selenium  | g / t | Se     | 0.05   | 0.05    | 0.05                          |
| Zinc      | g / t | Zn     | 150    | 240     | 190                           |

<sup>11</sup> [http://www.air.sk/tno/cepmeip/em\\_factors.php](http://www.air.sk/tno/cepmeip/em_factors.php), accessed 01.03.2003



**Tab. 5.8 Dust emissions from typical coal strip-mining operations (Anonymous (1998b)). After this table underground mining cause roughly 80% of the dust emissions of open pit mining.**

| Sources of dust       | Truck and shovel operation | Dragline operation | Occurrence in underground mining |
|-----------------------|----------------------------|--------------------|----------------------------------|
|                       | % of total dust emission   |                    |                                  |
| Dragline              | n.a                        | 27                 | Yes                              |
| Haul roads            |                            |                    |                                  |
| Overburden            | 35                         | -                  | Yes                              |
| Coal                  | 7                          | 42                 | Yes                              |
| Loading               |                            |                    |                                  |
| Overburden            | 12                         | -                  | Yes                              |
| Coal                  | 4                          | 5                  | Yes                              |
| Drilling              |                            |                    |                                  |
| (Overburden and coal) | 1                          | 2                  | No                               |
| Blasting              |                            |                    |                                  |
| (Overburden and coal) | 5                          | 1                  | No                               |
| Truck dumping         | 7                          | 5                  | Yes                              |
| Topsoil removal       | 18                         | 10                 | No                               |
| Exposed areas         | 6                          | 7                  | No                               |
| Haul-road repairs     | 5                          | 1                  | Yes                              |

**Tab. 5.9 Emissions to air resulting from mining and beneficiation activities per tonne mined ore.**

|                           | Unit   | Emission per tonne mined ore |
|---------------------------|--------|------------------------------|
| Antimony                  | mg / t | 0.0736                       |
| Arsenic                   | mg / t | 0.552                        |
| Beryllium                 | mg / t | 0.96                         |
| Boron                     | mg / t | 3.68                         |
| Cadmium                   | mg / t | 0.04048                      |
| Chromium                  | mg / t | 36.8                         |
| Cobalt                    | mg / t | 7.36                         |
| Copper                    | mg / t | 18.4                         |
| Fluorine                  | mg / t | 349.6                        |
| Lead                      | mg / t | 5.152                        |
| Manganese                 | mg / t | 349.6                        |
| Mercury                   | mg / t | 0.0184                       |
| Nickel                    | mg / t | 29.44                        |
| Selenium                  | mg / t | 0.0184                       |
| Zinc                      | mg / t | 69.92                        |
| Particulate Matter <10µm  | g / t  | 181                          |
| Particulate Matter <2.5µm | g / t  | 18.1                         |

**Tab. 5.10 Composition of effluents from mining activities, values are taken from total emissions to water from different base metal sites in (IPPC (2002)).**

| Compound        | Unit                    | Range         | This study (geometric mean) | Number of Samples |
|-----------------|-------------------------|---------------|-----------------------------|-------------------|
| Discharge       | Mio. m <sup>3</sup> / a | 0.08 - 22.9   | 3.65                        | 6                 |
| Ca              | g / m <sup>3</sup>      | 686           | 686                         | 1                 |
| SO <sub>4</sub> | g / m <sup>3</sup>      | 1750 - 3175   | 2357                        | 2                 |
| COD             | g / m <sup>3</sup>      | 2.24 - 48.5   | 10.4                        | 2                 |
| Solids          | g / m <sup>3</sup>      | 2.38 - 11.3   | 5.17                        | 4                 |
| Al              | mg / m <sup>3</sup>     | 69.3 - 108    | 86.3                        | 2                 |
| As              | mg / m <sup>3</sup>     | 0.264 - 14.1  | 2.95                        | 3                 |
| Cd              | mg / m <sup>3</sup>     | 0.0901 - 1.02 | 0.316                       | 4                 |
| Co              | mg / m <sup>3</sup>     | 0.742 - 0.823 | 0.781                       | 2                 |
| Cr              | mg / m <sup>3</sup>     | 0.0311 - 9.62 | 0.547                       | 2                 |
| Cu              | mg / m <sup>3</sup>     | 1.24 - 44.9   | 7.91                        | 5                 |
| Fe              | mg / m <sup>3</sup>     | 61.7 - 1326   | 291                         | 3                 |
| Mn              | mg / m <sup>3</sup>     | 24.7          | 24.7                        | 1                 |
| Hg              | mg / m <sup>3</sup>     | 0.0155 - 0.12 | 0.0377                      | 3                 |
| Ni              | mg / m <sup>3</sup>     | 0.792 - 1338  | 24.3                        | 3                 |
| Pb              | mg / m <sup>3</sup>     | 0.0155 - 20   | 2.80                        | 4                 |
| Zn              | mg / m <sup>3</sup>     | 5.37 – 225    | 75.9                        | 5                 |
| N               | g / m <sup>3</sup>      | 2.50 - 1786   | 22.8                        | 3                 |

**Tab. 5.11 Consumption of reagents in base metal mineral processing plants. Data taken from (IPPC (2002)).**

| Reagents                                                                                        | Unit | Range       | This study (geometric mean) | Number of samples |
|-------------------------------------------------------------------------------------------------|------|-------------|-----------------------------|-------------------|
| <b>Collectors</b> – Xanthate                                                                    | g/t  | 110 - 300   | 188.6                       | 6                 |
| <b>Frothers</b> - Sylvapine, MIBC, Dowfroth                                                     | g/t  | 9.7 - 150   | 37.2                        | 5                 |
| Sodium cyanide                                                                                  | g/t  | 4-310       | 35                          | 2                 |
| <b>Depressants</b> – Zinc sulphate, Iron sulphate, Acetic acid, Sodiumchromate, Dithiophosphate | g/t  | 35 - 432    | 231.5                       | 5                 |
| <b>Activator</b> - Copper sulphate                                                              | g/t  | 433 - 876   | 538                         | 4                 |
| <b>Flotation agents</b>                                                                         | g/t  | 19.0        | 19.0                        | 1                 |
| <b>Lime</b>                                                                                     | g/t  | 350 - 43685 | 2000 <sup>1)</sup>          | 6                 |
| <b>Flocculants</b> - CMC                                                                        | g/t  | 1-100       | 11.1                        | 3                 |
| Sulphuric acid (pH)                                                                             | g/t  | 387-12000   | 3739                        | 4                 |
| Sodium hydroxide (pH)                                                                           | g/t  | 30-490      | 121                         | 2                 |
| Nitric acid (pH)                                                                                | g/t  | 150         | 150                         | 1                 |
| Hydrochloric acid (pH)                                                                          | g/t  | 1           | 1                           | 1                 |

<sup>1)</sup> Value taken from (Krauss et al. (1999)), which correspond to the geometric mean of 2304 g / t

**Tab. 5.12 Emissions and input due to chemical use in beneficiation in g per tonne treated ore. The values are the ones chosen in this study. For further details read in text.**

| Reagent              | Type of burden     | Unit | Chosen value                      |
|----------------------|--------------------|------|-----------------------------------|
| Collector            | input technosphere | g/t  | 189                               |
| Carbon disulfide     | air emission       | g/t  | 100                               |
| Frother              | input technosphere | g/t  | 37                                |
| Lime                 | input technosphere | g/t  | 2000                              |
| Carbon dioxide       | air emission       |      | inventoried in metallurgical step |
| Activator/Depressant | input technosphere | g/t  | 828                               |
| Sulphuric acid       | input technosphere | g/t  | 3739                              |
| Sodium Cyanide       | input technosphere | g/t  | 35.2                              |
| Cyanide              | water emission     | g/t  | 3.88                              |

### Metallurgy and Separation of the non-ferrous metals

The ore concentrate is worked in the typical process steps. In the RSA the sulphidic matte is separated from the slag with an electric arc furnace with a Söderberg electrode system. The ore is poorer in sulphur, resulting in a higher oxygen demand. Air and Oxygen is blown through the matte in a Pierce-Smith-Converter. The cooled matte is ground and magnetically separated. After the separation the PGM-containing part is leached and treated hydrometallurgically. After the electrolytic precipitation, the remaining PGM containing dross is treated in the refinery. The transport to the refinery is assumed to be short and therefore neglected.

In Russia the ore is treated in a Outokumpu flash furnace, where it is roasted and smelted in an auto-genous process (i.e. the heat of the exothermal oxidation of FeS is used in the process, s. 5.3.1). The matte is converted in a Pierce-Smith-Converter. The off-gases of the conversion are not treated. The copper-nickel matte is separated in nickel and copper concentrate, latter being processed further in the near copper facility. The nickel matte is treated with electrowinning (roasting and reduction with coke) and cast into anodes. The remaining PGM containing dross is transferred to the refinery in Krasno-jarsk.

In both processes lime was added in beneficiation which releases now fossil carbon dioxide. Flux is assumed to consist of 50% lime, which also releases carbon dioxide. 0.75 t flux per t matte is reported for copper in Krauss et al. (1999). Burning of 1 t lime ( $\text{CaCO}_3$ ) yields 0.44 t  $\text{CO}_2$ .

Effluent is approximated with demand on water, which was only available for RSA and extrapolated for RUS with 1.4 m<sup>3</sup> per t concentrate input. The effluents are treated. An overview of concentrations in different waste waters after treatment is shown in Tab. 5.14. The values are assembled from different processes in the nickel and copper metallurgy (IPPC (2001)). The data shows high variations, a geometric mean was built to calculate average emissions. To cover the uncertainties, a standard deviation of 10 was chosen for each emission.

The atmospheric emissions are also taken from relative processes from copper and nickel operations observed in IPPC (2001). These emissions are stated as mass per metal produced, therefore it is multiplied with the total metal production of the two considered processes. The distribution of particulate matter is taken from CEPMEIP<sup>12</sup>. The values for sulphur dioxide are stated in Hochfeld (1997). Emissions of sulphur dioxide are one key issue of the metallurgy. In IPPC (2001) much lower values are reported, this seems to be true for European operations. Inco Ltd. states in their environmental report

<sup>12</sup> <http://www.air.sk/tno/cepmeip/>, accessed 01.03.2003

Emissions of 1'500 kg SO<sub>2</sub> / t metal produced in the period between 1997 and 2001. The corresponding values in Hochfeld (1997) are 5'300 for the Russian and 760 kg SO<sub>2</sub> / t for the RSA operations.

In Hochfeld (1997) no specific values for the separation step are reported except for energy, water, dross and the coupled products nickel and copper.

**Tab. 5.13 Emissions to air per tonne produced metal.** Data was taken from similar processes in nickel (A) and copper (B) metallurgy and multiplied with the sum of produced metals (PGM, Ni & Cu). Data is taken from IPPC (2001), the distribution of the particle size is taken from CEPMEIP (C). VOC emission from leaching (D). In this study values for SO<sub>2</sub> emissions are taken from Hochfeld (1997) (E).

| Substance             | Amount in kg / t<br>(this study) | Source |
|-----------------------|----------------------------------|--------|
| Pb                    | 0.03                             | B      |
| Zn                    | 0.10                             | B      |
| Sn                    | 0.007                            | B      |
| Cu                    | 0.04                             | B      |
| As                    | 0.006                            | B      |
| Total dust            | 0.407                            | A      |
| PM<2.5                | 0.000508                         | C      |
| PM2.5-10              | 0.304                            | C      |
| PM>10                 | 0.102                            | C      |
| SO <sub>2</sub> , RSA | 760                              | E      |
| SO <sub>2</sub> , RUS | 5300                             | E      |
| Co                    | 0.03                             | A      |
| VOC                   | 0.20                             | A, D   |
| TCDD                  | 1.00E-08                         | B      |

**Tab. 5.14** Composition of different wastewater effluents after treatment. A simple average from different reported values of European copper operations is chosen (IPPC (2001)). Values in the columns “process water”, “surface run off” and “direct cooling water” represent geometric means of the range reported. Values for Cr, Hg and Sn correspond to a copper semis production site, reported in (IPPC (2001)).

| Sub-stance | Process Water         | Surface run off       | Direct cooling water    | This study (average)  |
|------------|-----------------------|-----------------------|-------------------------|-----------------------|
|            | [g / m <sup>3</sup> ] | [g / m <sup>3</sup> ] | [g / m <sup>3</sup> ]   | [g / m <sup>3</sup> ] |
| Cu         | 0.045<br>(0.01-0.2)   | 0.063<br>(0.01-0.4)   | 0.05<br>(0.01-0.25)     | 0.0527                |
| Pb         | 0.006<br>(0.001-0.04) | 0.032<br>(0.005-0.2)  | 0.01<br>(0.001-0.1)     | 0.0160                |
| As         | 0.032<br>(0.01-0.1)   | 0.014<br>(0.003-0.07) | 0.01<br>(0.001-0.1)     | 0.0187                |
| Ni         | 0.024<br>(0.004-0.15) | 0.028<br>(0.002-0.4)  | 0.011<br>(0.002-0.06)   | 0.0212                |
| Cd         | 0.003<br>(0.0001-0.1) | 0.004<br>(0.0002-0.1) | 0.001<br>(0.0001-0.003) | 0.00273               |
| Zn         | 0.045<br>(0.01-0.2)   | 0.11<br>(0.03-0.4)    | 0.1<br>(0.02-0.5)       | 0.0848                |
| Cr         |                       |                       |                         | 0.029                 |
| Hg         |                       |                       |                         | 0.00029               |
| Sn         |                       |                       |                         | 0.029                 |

### Refining of PGM

The refining step is covered only partially in Hochfeld (1997). Only for the operations in RSA electricity and gas consumption and water demand are reported. These values are added with specific emissions from five large precious metal refineries from IPPC (2001) in Tab. 5.15.. Values for BOD is based on the reported value for COD which is assumed to amount 150% of BOD (Zimmermann et al. (1996)). Values for TOC are calculated using the standard procedure approximating the organic matter with C<sub>6</sub>H<sub>12</sub>O<sub>6</sub> (Frischknecht et al. (2003b)). Particle distribution is assessed with values for open hearth exhausts from steel foundries after ESP control from EPA (1998). Transports are included in emissions to air.

Precious metal refineries recover simultaneously several precious metals from a variety of raw materials. No information about the way of allocation is available, an allocation by mass is assumed, i. e. emissions of a tonne precious metal correspond to 1 tonne PGM, notwithstanding any possible co-product.

For the operation in RUS the same burden is assumed. The PGM-containing matte has to be transported from Norsil'sk 1500 km up Yenisey river to Krasnojarsk for refining.

**Tab. 5.15** Reported emissions to air and water from 5 large refinery processes. Data taken from IPPC (2001). Annual operation hours assumed with 6000 h / a. Values for BOC, TOC and DOC calculated

| Parameter                 | Unit       | This study<br>(geometric mean) | Range        | No of samples |
|---------------------------|------------|--------------------------------|--------------|---------------|
| <b>Emissions to air</b>   |            |                                |              |               |
| Production                | tonnes / a | 940                            | 102 - 2500   | 5             |
| Dust                      | kg / t PGM | 16.4                           | 2 - 127      | 5             |
| PM<2.5                    | kg / t PGM | 8.1                            | 8.1          | 1             |
| PM2.5-10                  | kg / t PGM | 5.9                            | 5.9          | 1             |
| PM>10                     | kg / t PGM | 2.4                            | 2.4          | 1             |
| NOx                       | kg / t PGM | 35.2                           | 7 - 154      | 4             |
| SO2                       | kg / t PGM | 18.6                           | 3.1 - 232    | 3             |
| <b>Emissions to water</b> |            |                                |              |               |
| Ag                        | g / t PGM  | 3.7                            | 2.78 - 4.8   | 2             |
| Pb                        | g / t PGM  | 37                             | 3.51 - 12353 | 5             |
| Hg                        | g / t PGM  | 0.8                            | 0.05 - 12    | 4             |
| Cu                        | g / t PGM  | 141.5                          | 8.35 - 26961 | 5             |
| Ni                        | g / t PGM  | 65.0                           | 4.8 - 16078  | 5             |
| COD                       | kg / t PGM | 25.8                           | 11.1 - 60    | 2             |
| BOD                       | kg / t PGM | 17.2                           |              | calc.         |
| TOC                       | kg / t PGM | 10.1                           |              | calc.         |
| DOC                       | kg / t PGM | 10.1                           |              | calc.         |

### Data quality

The uncertainties are assessed using the pedigree matrix (see Frischknecht et al. (2003b)). For detailed results compare the different flows and the ecoinvent meta information.

**Mining.** Data for major mass fluxes are satisfactory, while the emissions in to water may be overestimated due to different system boundaries of reported effluent volume from Hochfeld (1997) and total effluent emissions reported in IPPC (2001).

**Beneficiation.** Like in mining, the emissions to water can be overestimated. The emissions from tailings are assessed in an extern module, which itself contains rough assumptions of a global overburden and tailings composition and a simplified disposal scenario. Emissions though can vary considerably from case to case, it is not evident, whether the calculated emissions meet the reality.

**Metallurgy.** Process data is satisfactory and valid for over 90 % of the PGM. The main emissions dominating possibly the whole PGM process is sulphur dioxide, which is reported for both processes and found to be consistent with other data Krauss et al. (1999), IPPC (2001) and Hilbrans & Hinrichs (1999). Poor data are available on composition and fate of solid waste, an overestimation may occur due to higher recycling and reuse rates than assumed in this work.

**Separation of the non-ferrous metals.** Main data on energy and mass fluxes is satisfactory, while data on direct emissions due to hydrometallurgical processes is poor. Namely chlorine emissions can be considerable, generating both HCl emissions to air and AOX emissions to water – emissions which are not assessed in this work due to lacking data.

**Refining.** No data on many specific emissions in to the air was available. Namely high-impact solvents and complexants are not registered. Due to the closed-loop strategies of such refineries, only small emissions of problematic substances are expected.

## 5 System Characterisation

| General Flow Information      |   |                                                   |                                                                        |                        | Representation in ecoinvent                                                                                                             |                  |                                                                                                                    |                         |                |                                       |                                                                                  |                                                       | Uncertainty Informations |                |                                  |                       |                 |                        |                  |
|-------------------------------|---|---------------------------------------------------|------------------------------------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------|-------------------------|----------------|---------------------------------------|----------------------------------------------------------------------------------|-------------------------------------------------------|--------------------------|----------------|----------------------------------|-----------------------|-----------------|------------------------|------------------|
| Input                         |   | Process Name                                      |                                                                        | Output                 |                                                                                                                                         | Remarks          | Category                                                                                                           | Subcategory             | Infrastructure | Location                              | Model name in ecoinvent                                                          | Mean value ZA                                         | Mean value RU            | Unit           |                                  | Source for mean value | Type            | Standard Deviation 95% | General Comment  |
| Ore (Pt)                      | ⌚ | platinum group metal production, primary (mining) | ore is allocated by mass to the coupled products Pt, Pd, Rh, Ni and Cu |                        | 90% yield assumed, RU: 89750 kg ore per kg PGM; ZA: 265'000 kg ore per kg PGM                                                           | resource         | in ground                                                                                                          |                         |                |                                       | Pt, Pt 4.8E-4%, Pd 2.0E-4%, Rh 2.4E-5%, Ni 3.7E-2%, Cu 5.2E-2% in ore, in ground | 7.53E-01                                              | 7.53E-01                 | kg             | (Krauss, 1999), (Hilbrans, 1999) | 1                     | 1.66            | reported values        |                  |
| Ore (Pd)                      | ⌚ |                                                   |                                                                        |                        | 90% yield assumed, RU: 89750 kg ore per kg PGM; ZA: 265'000 kg ore per kg PGM                                                           | resource         | in ground                                                                                                          |                         |                |                                       | Pd, Pd 2.0E-4%, Pt 4.8E-4%, Rh 2.4E-5%, Ni 3.7E-2%, Cu 5.2E-2% in ore, in ground | 3.20E-01                                              | 3.20E-01                 | kg             | (Krauss, 1999), (Hilbrans, 1999) | 1                     | 3.90            | reported values        |                  |
| Ore (Rh)                      | ⌚ |                                                   |                                                                        |                        | 90% yield assumed, RU: 89750 kg ore per kg PGM; ZA: 265'000 kg ore per kg PGM                                                           | resource         | in ground                                                                                                          |                         |                |                                       | Rh, Rh 2.4E-5%, Pt 4.8E-4%, Pd 2.0E-4%, Ni 3.7E-2%, Cu 5.2E-2% in ore, in ground | 3.77E-02                                              | 3.77E-02                 | kg             | (Krauss, 1999), (Hilbrans, 1999) | 1                     | 33.19           | reported values        |                  |
| Ore (Ni)                      | ⌚ |                                                   |                                                                        |                        | 90% yield assumed, RU: 89750 kg ore per kg PGM; ZA: 265'000 kg ore per kg PGM                                                           | resource         | in ground                                                                                                          |                         |                |                                       | Ni, Ni 3.7E-2%, Pt 4.8E-4%, Pd 2.0E-4%, Rh 2.4E-5%, Cu 5.2E-2% in ore, in ground | 1.64E+02                                              | 1.64E+02                 | kg             | (Krauss, 1999), (Hilbrans, 1999) | 1                     | 0.01            | reported values        |                  |
| Ore (Cu)                      | ⌚ |                                                   |                                                                        |                        | 90% yield assumed, RU: 89750 kg ore per kg PGM; ZA: 265'000 kg ore per kg PGM                                                           | resource         | in ground                                                                                                          |                         |                |                                       | Cu, Cu 5.2E-2%, Pt 4.8E-4%, Pd 2.0E-4%, Rh 2.4E-5%, Ni 3.7E-2% in ore, in ground | 1.15E+02                                              | 1.15E+02                 | kg             | (Krauss, 1999), (Hilbrans, 1999) | 1                     | 0.01            | reported values        |                  |
| Electricity, mining           | ⌚ |                                                   |                                                                        |                        | The rest of the inventory is allocated by revenue (s. chapter Allocation) as stated.                                                    |                  | The supply mix of the mining operations are assumed to be similar to the electricity mix of the aluminium industry | electricity             | production mix | No                                    | UCTE                                                                             | electricity, medium voltage, production UCTE, at grid | 9.81E+03                 | 7.48E+03       | kWh                              | (Hochfeld, 1997)      | 1               | 1.26                   | (3,2,1,1,3,4,2)  |
| Water, mining, ground         | ⌚ |                                                   |                                                                        |                        |                                                                                                                                         |                  | resource                                                                                                           | in water                |                |                                       |                                                                                  | Water, well, in ground                                | 3.71E+01                 | 1.26E+01       | m3                               | (Hochfeld, 1997)      | 1               | 1.26                   | (3,2,1,1,3,4,12) |
| Water, mining, fresh          | ⌚ |                                                   |                                                                        |                        |                                                                                                                                         |                  | resource                                                                                                           | in water                |                |                                       |                                                                                  | Water, river                                          | 1.86E+01                 | 6.28E+00       | m3                               | (Hochfeld, 1997)      | 1               | 1.26                   | (3,2,1,1,3,4,12) |
| Blasting                      | ⌚ |                                                   |                                                                        |                        |                                                                                                                                         |                  | construction processes                                                                                             | civil engineering       | No             | RER                                   | blasting                                                                         |                                                       | 1.33E-01                 | 4.49E-02       | kg                               | (Hochfeld, 1997)      | 1               | 1.26                   | (3,2,1,1,3,4,1)  |
| Diesel                        | ⌚ |                                                   |                                                                        |                        |                                                                                                                                         |                  | construction processes                                                                                             | machinery               | No             | GLO                                   | diesel, burned in building machine                                               |                                                       | 1.95E+04                 | 6.90E+03       | MJ                               | (Hochfeld, 1997)      | 1               | 2.33                   | (4,3,2,2,4,5,5)  |
| infrastructure, conveyor belt | ⌚ |                                                   |                                                                        | construction processes | machinery                                                                                                                               | Yes              | RER                                                                                                                | conveyor belt, at plant |                | 5.65E-03                              | 1.99E-03                                                                         | m                                                     | (Hochfeld, 1997)         | 1              | 3.93                             | (5,4,3,3,5,6,9)       |                 |                        |                  |
| Mine, infrastructure          | ⌚ |                                                   |                                                                        |                        | 50 a lifetime: 400'000 t / a -> 5e-11 units per kg ore mined                                                                            | metals           | extraction                                                                                                         | Yes                     | GLO            | non-ferrous metal, mine, underground  | 4.99E-06                                                                         | 1.41E-05                                              | unit                     | rough estimate | 1                                | 3.07                  | (3,2,1,1,3,4,9) |                        |                  |
|                               | ⌚ | Overburden, disposed                              |                                                                        |                        | not clear, whether net disposal or partly re-filled in mine. Assumption: 50% refilled.                                                  | waste management | residual material landfill                                                                                         | No                      | GLO            | disposal, sulfidic tailings, off-site | 8.75E+03                                                                         | 4.49E+03                                              | kg                       | (IPPC, 2002)   | 1                                | 1.13                  | (2,2,1,1,1,4,6) |                        |                  |
|                               | ⌚ | Overburden, refilled                              |                                                                        |                        | not clear, whether net disposal or partly re-filled in mine. Assumption: 50% refilled.                                                  |                  |                                                                                                                    |                         |                |                                       | 8.75E+03                                                                         | 4.49E+03                                              | kg                       | (IPPC, 2002)   | 1                                | 1.13                  | (2,2,1,1,1,4,6) |                        |                  |
|                               | ⌚ | Effluents, mining                                 |                                                                        |                        | For effluent composition see Tab. 5.14. Due to high heterogeneity of effluent, a standard deviation of 10 is assumed for concentrations |                  |                                                                                                                    |                         |                |                                       | 5.57E+01                                                                         | 1.88E+01                                              | m3                       | (IPPC, 2002)   | 1                                | 1.13                  | (2,2,1,1,1,4,6) |                        |                  |

**Fig. 5.11 Flows for “PGM-Production, primary – Part mining” and their representation in the ecoinvent database. Values in column “Mean value RUS” correspond to the Russian PGM production, the values in the column “Mean value RSA” correspond to the South African PGM production. Uncertainty assumptions are identical for both modules. Values relate to the production of 1 kg PGM.**

| General Flow Information   |   |                                                          |   |                                                                      |   | Representation in ecoinvent        |                                                                                                                                                                                                      |                  |                            |                                                       |                                     |                                       |               | Uncertainty Informations |                       |                  |                        |                  |                                                     |
|----------------------------|---|----------------------------------------------------------|---|----------------------------------------------------------------------|---|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------------|-------------------------------------------------------|-------------------------------------|---------------------------------------|---------------|--------------------------|-----------------------|------------------|------------------------|------------------|-----------------------------------------------------|
| Input                      |   | Process Name                                             |   | Output                                                               |   | Remarks                            | Category                                                                                                                                                                                             | Subcategory      | Infrastructure             | Location                                              | Modul name in ecoinvent             | Mean value ZA                         | Mean value RU | Unit                     | Source for mean value | Type             | Standard Deviation 95% | General Comment  |                                                     |
| Electricity, beneficiation | m | platinum group metal production, primary (beneficiation) | m | Inventory is allocated by revenue (s. chapter Allocation) as stated. |   | electricity                        | production mix                                                                                                                                                                                       | No               | UCTE                       | electricity, medium voltage, production UCTE, at grid | 4.02E+03                            | 1.18E+03                              | kWh           | (Hochfeld, 1997)         | 1                     | 1.13             | (2,2,1,1,1,4,2)        |                  |                                                     |
| Water, beneficiation       | m |                                                          |   |                                                                      |   | 0.35 m3 per t ore mined is assumed | resource                                                                                                                                                                                             | in water         |                            |                                                       | Water, river                        | 9.36E+01                              | 3.17E+01      | m3                       | (Hochfeld, 1997)      | 1                | 1.13                   | (2,2,1,1,1,4,12) |                                                     |
| Lime addition              | m |                                                          |   |                                                                      |   |                                    | construction materials                                                                                                                                                                               | others           | No                         | CH                                                    | limestone, milled, packed, at plant | 5.30E+02                              | 1.79E+02      | kg                       | (Krauss, 1999)        | 1                | 2.00                   | reported values  |                                                     |
| Floating agents, organic   | m |                                                          |   |                                                                      |   |                                    | chemicals                                                                                                                                                                                            | organics         | No                         | GLO                                                   | chemicals organic, at plant         | 5.98E+01                              | 2.03E+01      | kg                       | (IPPC, 2002)          | 1                | 2.00                   | reported values  |                                                     |
| Floating agents, inorganic | m |                                                          |   |                                                                      |   |                                    | chemicals                                                                                                                                                                                            | inorganics       | No                         | GLO                                                   | chemicals inorganic, at plant       | 2.04E+02                              | 6.91E+01      | kg                       | (IPPC, 2002)          | 1                | 2.00                   | reported values  |                                                     |
| Sodium Cyanide             | m |                                                          |   |                                                                      |   |                                    | chemicals                                                                                                                                                                                            | organics         | No                         | RER                                                   | hydrogen cyanide, at plant          | 9.33E+00                              | 3.16E+00      | kg                       | (IPPC, 2002)          | 1                | 2.00                   | reported values  |                                                     |
| Sulphuric Acid             | m |                                                          |   |                                                                      |   |                                    | burden-free, since byproduct from metallurgy                                                                                                                                                         |                  |                            |                                                       |                                     | 6.11E+02                              | 2.07E+02      | kg                       | (IPPC, 2002)          | 1                | 2.00                   | reported values  |                                                     |
|                            |   |                                                          |   |                                                                      | m | Effluents, beneficiation           | Assumed to equal the demand of water. Composition is given in Tab. 5.10                                                                                                                              |                  |                            |                                                       |                                     | 9.36E+01                              | 3.17E+01      | m3                       | Rough estimation      |                  |                        |                  |                                                     |
|                            |   |                                                          |   |                                                                      | m | Emission Cyanide                   | From sodium cyanide, 80% control                                                                                                                                                                     | water            | river                      |                                                       |                                     | Cyanide                               | 1.03E+00      | 3.48E-01                 | kg                    | (NPI, 2001)      | 1                      | 3.84             | (4,2,2,3,4,4,33); including stdev of sodium cyanide |
|                            |   |                                                          |   |                                                                      | m | Dust (air, beneficiation)          | including emissions from mining activities. Dust is the sum of particulate emissions represented in fractions PM<2.5, PM2.5-10, PM>10, including metals and salts. Composition is given in Tab. 5.9. |                  |                            |                                                       |                                     | 1.04E+02                              | 3.67E+01      | kg                       | (NPI, 2001)           | 1                | 1.59                   | (3,2,1,1,3,4,25) |                                                     |
|                            |   |                                                          |   |                                                                      | m | carbon disulfide                   | due to hydrolysis of xanthates                                                                                                                                                                       | air              | low population density     |                                                       |                                     | Carbon disulfide                      | 2.65E+01      | 8.97E+00                 | kg                    | (NPI, 2001)      | 1                      | 2.31             | (4,2,2,5,4,4,23)                                    |
|                            |   |                                                          |   |                                                                      | m | Tailings                           | not clear, whether net disposal or partly re-filled in mine. Here assumed as 50% disposal                                                                                                            | waste management | residual material landfill | No                                                    | GLO                                 | disposal, sulfidic tailings, off-site | 1.28E+05      | 3.94E+04                 | kg                    | (Hochfeld, 1997) | 1                      | 1.13             | (2,2,2,3,1,4,6)                                     |

**Fig. 5.12** Flows for “PGM-Production, primary – Part beneficiation” and their representation in the ecoinvent database. Values in column “Mean value RU” correspond to the Russian PGM production, the values in the column “Mean value ZA” correspond to the South African PGM production. Uncertainty assumptions are identical for both modules. Values relate to the production of 1 kg PGM.



| General Flow Information |   |                                                       |  |                                                           |  | Representation in ecoinvent                                   |                   |                            |                |                                                  |                                                                         |               |               | Uncertainty Informations |                       |      |                        |                                                    |
|--------------------------|---|-------------------------------------------------------|--|-----------------------------------------------------------|--|---------------------------------------------------------------|-------------------|----------------------------|----------------|--------------------------------------------------|-------------------------------------------------------------------------|---------------|---------------|--------------------------|-----------------------|------|------------------------|----------------------------------------------------|
| Input                    |   | Process Name                                          |  | Output                                                    |  | Remarks                                                       | Category          | Subcategory                | Infrastructure | Location                                         | Modul name in ecoinvent                                                 | Mean value ZA | Mean value RU | Unit                     | Source for mean value | Type | Standard Deviation 95% | General Comment                                    |
| Electricity, metallurgy  | m | platinum group metal production, primary (metallurgy) |  | Inventory is allocated by revenue (s. chapter Allocation) |  |                                                               | electricity       | production mix             | No             | UCTE                                             | electricity, medium voltage, production UCTE, at grid                   | 6.00E+03      | 1.16E+04      | kWh                      | (Hochfeld, 1997)      | 1    | 1.13                   | (2,2,2,3,1,4,2)                                    |
| Gas, metallurgy          | m |                                                       |  |                                                           |  | natural gas                                                   | heating systems   | No                         | RER            | natural gas, burned in industrial furnace >100kW | 0.00E+00                                                                | 1.25E+05      | MJ            | (Hochfeld, 1997)         | 1                     | 1.13 | (2,2,2,3,1,4,1)        |                                                    |
| water, metallurgy        | m |                                                       |  |                                                           |  | resource                                                      | in water          |                            |                | Water, river                                     | 1.28E+01                                                                | 7.14E+01      | m3            | (Hochfeld, 1997)         | 1                     | 1.13 | (2,2,2,3,1,4,12)       |                                                    |
| Flux                     | m |                                                       |  |                                                           |  | chemicals                                                     | inorganics        | No                         | GLO            | chemicals inorganic, at plant                    | 1.88E+02                                                                | 4.14E+03      | kg            | (Hochfeld, 1997)         | 1                     | 1.26 | (3,2,1,1,3,4,3)        |                                                    |
| infrastructure smelter   | m |                                                       |  |                                                           |  | metals                                                        | extraction        | Yes                        | GLO            | non-ferrous metal, smelter                       | 2.60E-08                                                                | 1.45E-07      | unit          | assumption               | 1                     | 3.30 | (4,2,1,5,4,4,9)        |                                                    |
|                          | m |                                                       |  | Carbondioxide, from limestone                             |  | limestone added in beneficiation                              | air               | unspecified                |                |                                                  | Carbon dioxide, fossil                                                  | 2.33E+02      | 7.90E+01      | kg                       | (Hochfeld, 1997)      | 1    | 1.58                   | (4,2,1,1,4,4,14); excluding stdev of lime addition |
|                          | m |                                                       |  | Carbondioxide, from flux                                  |  |                                                               | air               | unspecified                |                |                                                  | Carbon dioxide, fossil                                                  | 4.14E+01      | 9.10E+02      | kg                       | (Hochfeld, 1997)      | 1    | 1.58                   | (4,2,2,3,4,4,14)                                   |
|                          | m |                                                       |  | metallurgy effluent                                       |  | composition in Tab 5.14                                       |                   |                            |                |                                                  |                                                                         | 1.28E+01      | 7.14E+01      | m3                       | (Hochfeld, 1997)      | 1    | 1.58                   | (4,2,2,3,4,4,6)                                    |
|                          | m |                                                       |  | Sulfur dioxide (air, metallurgy)                          |  |                                                               | air               | low population density     |                |                                                  | Sulfur dioxide                                                          | 1.91E+02      | 2.95E+04      | kg                       | (Hochfeld, 1997)      | 1    | 1.38                   | (4,2,2,1,3,5,15)                                   |
|                          | m |                                                       |  | Dust (air, metallurgy)                                    |  | Composition in Tab 5.13                                       |                   |                            |                |                                                  |                                                                         | 1.02E-01      | 2.24E+00      | kg                       | (IPPC, 2002)          | 1    | 1.59                   | (3,2,1,1,3,4,25)                                   |
|                          | m |                                                       |  | VOC (air, metallurgy)                                     |  | in dust, composition in Tab 5.13                              | air               | low population density     |                |                                                  | NM VOC, non-methane volatile organic compounds, unspecified origin      | 5.01E-02      | 1.10E+00      | kg                       | (IPPC, 2002)          | 1    | 1.59                   | (3,2,1,1,3,4,16)                                   |
|                          | m |                                                       |  | TCDD (air, metallurgy)                                    |  | in dust, composition in Tab 5.13                              | air               | low population density     |                |                                                  | Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin                | 2.51E-09      | 5.52E-08      | kg                       | (IPPC, 2002)          | 1    | 3.07                   | (3,2,1,1,3,4,21)                                   |
|                          | m |                                                       |  | slag                                                      |  | value for RU including dross from follow ing separation step. | w aste management | residual material landfill | No             | CH                                               | disposal, nickel smelter slag, 0% w ater, to residual material landfill | 6.40E+02      | 4.14E+04      | kg                       | (Hochfeld, 1997)      | 1    | 1.13                   | (2,2,2,3,1,4,6)                                    |

**Fig. 5.13** Flows for “PGM-Production, primary – Part metallurgy” and their representation in the ecoinvent database. Values in column “Mean value RU” correspond to the Russian PGM production, the values in the column “Mean value ZA” correspond to the South African PGM production. Uncertainty assumptions are identical for both modules. Values relate to the production of 1 kg PGM.

| General Flow Information |   |                                                                               |   |                                                           |   |                      | Representation in ecoinvent                                                                                 |                            |                |                |                                                                        |                                                       |            |                                                       | Uncertainty Informations |                       |      |                        |                 |      |   |
|--------------------------|---|-------------------------------------------------------------------------------|---|-----------------------------------------------------------|---|----------------------|-------------------------------------------------------------------------------------------------------------|----------------------------|----------------|----------------|------------------------------------------------------------------------|-------------------------------------------------------|------------|-------------------------------------------------------|--------------------------|-----------------------|------|------------------------|-----------------|------|---|
| Input                    |   | Process Name                                                                  |   | Output                                                    |   |                      | Remarks                                                                                                     | Category                   | Subcategory    | Infrastructure | Location                                                               | Modul name in ecoinvent                               | Mean value |                                                       | Unit                     | Source for mean value | Type | Standard Deviation 95% | General Comment |      |   |
| Electricity, separation  | ⌚ | platinum group metal production, primary (separation from non-ferrous metals) | ⌚ | Inventory is allocated by revenue (s. chapter Allocation) | ⌚ |                      |                                                                                                             | electricity                | production mix | No             | UCTE                                                                   | electricity, medium voltage, production UCTE, at grid | 1.59E+03   | 5.56E+03                                              | kWh                      | (Hochfeld, 1997)      | 1    | 1.13                   | (2,2,2,3,1,4,2) |      |   |
| Gas, separation          | ⌚ |                                                                               |   |                                                           |   |                      | natural gas                                                                                                 | heating systems            | No             | RER            | natural gas, burned in industrial furnace >100kW                       | 3.98E+03                                              | 0.00E+00   | MJ                                                    | (Hochfeld, 1997)         | 1                     | 1.13 | (2,2,2,3,1,4,1)        |                 |      |   |
| water, separation        | ⌚ |                                                                               |   |                                                           |   |                      | resource                                                                                                    | in water                   |                |                | Water, river                                                           | 1.28E+01                                              | 8.27E+02   | m3                                                    | (Hochfeld, 1997)         | 1                     | 1.13 | (2,2,2,3,1,4,6)        |                 |      |   |
|                          |   |                                                                               |   | ⌚                                                         |   | dross                | waste management                                                                                            | residual material landfill | No             | CH             | disposal, nickel smelter slag, 0% water, to residual material landfill | 3.45E+00                                              | 6.35E+01   | kg                                                    | (Hochfeld, 1997)         | 1                     | 1.26 | (3,2,1,1,3,4,6)        |                 |      |   |
|                          |   |                                                                               |   | ⌚                                                         |   | nickel as co-product | As co-product, allocation by revenue: ZA 2%, RU 47%, geography corresponding to respective origin (ZA / RU) |                            |                | metals         | extraction                                                             | No                                                    |            | nickel, primary, from platinum group metal production | 1.47E+02                 | 2.31E+03              | kg   | (Hochfeld, 1997)       | 0               | 0.00 | 0 |
|                          |   |                                                                               |   | ⌚                                                         |   | copper as co-product | As co-product, allocation by revenue: ZA 1%, RU 19%, geography corresponding to respective origin (ZA / RU) |                            |                | metals         | extraction                                                             | No                                                    |            | copper, primary, from platinum group metal production | 1.04E+02                 | 3.21E+03              | kg   | (Hochfeld, 1997)       | 0               | 0.00 | 0 |
|                          |   |                                                                               |   |                                                           |   |                      |                                                                                                             |                            |                |                |                                                                        |                                                       |            |                                                       |                          |                       |      |                        |                 |      |   |

Fig. 5.14 Flows for “PGM-Production, primary – Part separation” and their representation in the ecoinvent database. Values in column “Mean value RU” correspond to the Russian PGM production, the values in the column “Mean value ZA” correspond to the South African PGM production. Uncertainty assumptions are identical for both modules. Values relate to the production of 1 kg PGM.

| General Flow Information                      |    |                                                     |                                                     |        |  | Representation in ecoinvent                             |                 |                        |                                                      |                                                  |                                                       |            | Uncertainty Informations |                                 |                                     |          |                        |                  |  |  |  |
|-----------------------------------------------|----|-----------------------------------------------------|-----------------------------------------------------|--------|--|---------------------------------------------------------|-----------------|------------------------|------------------------------------------------------|--------------------------------------------------|-------------------------------------------------------|------------|--------------------------|---------------------------------|-------------------------------------|----------|------------------------|------------------|--|--|--|
| Input                                         |    | Process Name                                        |                                                     | Output |  | Remarks                                                 | Category        | Subcategory            | Infrastructure                                       | Location                                         | Modul name in ecoinvent                               | Mean value |                          | Unit                            | Source for mean value               | Type     | Standard Deviation 95% | General Comment  |  |  |  |
| Electricity, refining                         | ML | platinum group metal production, primary (refining) | Total allocation to PGM.<br>Pt:72% - Pd:20% - Rh:8% | PM<2.5 |  |                                                         | electricity     | production mix         | No                                                   | UCTE                                             | electricity, medium voltage, production UCTE, at grid | 4.59E+02   | 4.59E+02                 | kWh                             | (Hochfeld, 1997)                    | 1        | 1.25                   | (2,2,2,1,3,4,2)  |  |  |  |
| Gas, refining                                 | ML |                                                     |                                                     |        |  | natural gas                                             | heating systems | No                     | RER                                                  | natural gas, burned in industrial furnace >100kW | 7.76E+02                                              | 7.76E+02   | MJ                       | (Hochfeld, 1997)                | 1                                   | 1.25     | (2,2,2,1,3,4,1)        |                  |  |  |  |
| Transportation, river infrastructure refinery | ML |                                                     |                                                     |        |  | only RU, 1500 km riverboat from noril'sl to krasnojarsk |                 |                        |                                                      |                                                  | -                                                     | 6.35E+04   |                          | (Hochfeld, 1997)                |                                     |          |                        |                  |  |  |  |
|                                               | ML |                                                     |                                                     |        |  |                                                         | chemicals       | organics               | Yes                                                  | RER                                              | chemical plant, organics                              | 4.00E-10   | 4.00E-10                 | unit                            | assumption                          | 1        | 3.30                   | (4,2,1,5,4,4,9)  |  |  |  |
|                                               | ML |                                                     |                                                     |        |  |                                                         | air             | low population density |                                                      |                                                  | Particulates, > 10 um                                 | 8.09E-03   | 8.09E-03                 | kg                              | (EPA, 1998), (IPPC, 2001)           | 1        | 2.08                   | (3,2,2,5,3,2,26) |  |  |  |
|                                               | ML |                                                     |                                                     |        |  |                                                         | air             | low population density |                                                      |                                                  | Particulates, > 2.5 um, and < 10um                    | 5.93E-03   | 5.93E-03                 | kg                              | (EPA, 1998), (IPPC, 2001)           | 1        | 3.07                   | (3,2,2,5,3,2,27) |  |  |  |
|                                               | ML |                                                     |                                                     |        |  |                                                         | air             | low population density |                                                      |                                                  | Particulates, < 2.5 um                                | 2.40E-03   | 2.40E-03                 | kg                              | (EPA, 1998), (IPPC, 2001)           | 1        | 1.34                   | (3,2,2,5,3,2,28) |  |  |  |
|                                               | ML |                                                     |                                                     |        |  |                                                         | air             | low population density |                                                      |                                                  | Nitrogen oxides                                       | 3.52E-02   | 3.52E-02                 | kg                              | (IPPC, 2001)                        | 1        | 1.59                   | (3,2,1,1,3,4,16) |  |  |  |
|                                               | ML |                                                     |                                                     |        |  |                                                         | air             | low population density |                                                      |                                                  | Sulfur dioxide                                        | 1.86E-02   | 1.86E-02                 | kg                              | (IPPC, 2001)                        | 1        | 1.26                   | (3,2,1,1,3,4,15) |  |  |  |
|                                               | ML |                                                     |                                                     |        |  |                                                         | water           | river                  |                                                      |                                                  | Lead                                                  | 3.73E-05   | 3.73E-05                 | kg                              | (IPPC, 2001)                        | 1        | 5.08                   | (3,2,1,1,3,4,35) |  |  |  |
|                                               | ML |                                                     |                                                     |        |  |                                                         | water           | river                  |                                                      |                                                  | Mercury                                               | 7.83E-07   | 7.83E-07                 | kg                              | (IPPC, 2001)                        | 1        | 5.08                   | (3,2,1,1,3,4,35) |  |  |  |
|                                               | ML |                                                     |                                                     |        |  |                                                         | water           | river                  |                                                      |                                                  | Copper, ion                                           | 1.41E-04   | 1.41E-04                 | kg                              | (IPPC, 2001)                        | 1        | 5.08                   | (3,2,1,1,3,4,35) |  |  |  |
|                                               | ML |                                                     |                                                     |        |  |                                                         | water           | river                  |                                                      |                                                  | Nickel, ion                                           | 6.50E-05   | 6.50E-05                 | kg                              | (IPPC, 2001)                        | 1        | 5.08                   | (3,2,1,1,3,4,35) |  |  |  |
|                                               | ML |                                                     |                                                     |        |  |                                                         | water           | river                  |                                                      |                                                  | COD, Chemical Oxygen Demand                           | 2.58E-02   | 2.58E-02                 | kg                              | (IPPC, 2001)                        | 1        | 1.59                   | (3,2,1,1,3,4,32) |  |  |  |
|                                               | ML |                                                     |                                                     |        |  |                                                         | water           | river                  |                                                      |                                                  | BOD5, Biological Oxygen Demand                        | 1.72E-02   | 1.72E-02                 | kg                              | assumption COD = 1.5 * BOD          | 1        | 1.84                   | (4,2,2,3,4,4,32) |  |  |  |
|                                               | ML |                                                     |                                                     |        |  |                                                         | water           | river                  |                                                      |                                                  | TOC, Total Organic Carbon                             | 1.01E-02   | 1.01E-02                 | kg                              | calculated using standard procedure | 1        | 1.59                   | (3,2,1,1,3,4,32) |  |  |  |
|                                               | ML |                                                     |                                                     |        |  |                                                         | water           | river                  |                                                      |                                                  | DOC, Dissolved Organic Carbon                         | 1.01E-02   | 1.01E-02                 | kg                              | Assumption DOC = TOC                | 1        | 1.59                   | (3,2,1,1,3,4,32) |  |  |  |
|                                               | ML |                                                     |                                                     |        |  |                                                         | Platinum        |                        | as co-product, allocation by revenue: ZA 66%, RU 11% | metals                                           | extraction                                            | No         | ZA                       | platinum, primary, at refinery  | 6.78E-01                            | 2.50E-01 | kg                     | (Hochfeld, 1997) |  |  |  |
|                                               | ML |                                                     |                                                     |        |  |                                                         | Palladium       |                        | as co-product, allocation by revenue: ZA 19%, RU 21% | metals                                           | extraction                                            | No         | ZA                       | palladium, primary, at refinery | 2.88E-01                            | 7.30E-01 | kg                     | (Hochfeld, 1997) |  |  |  |
|                                               | ML |                                                     |                                                     |        |  |                                                         | Rhodium         |                        | as co-product, allocation by revenue: ZA 7%, RU 2%   | metals                                           | extraction                                            | No         | ZA                       | rhodium, primary, at refinery   | 3.39E-02                            | 2.00E-02 | kg                     | (Hochfeld, 1997) |  |  |  |

**Fig. 5.15** Flows for “PGM-Production, primary – Part refining” and their representation in the ecoinvent database. Values in column “Mean value RU” correspond to the Russian PGM production, the values in the column “Mean value ZA” correspond to the South African PGM production. Uncertainty assumptions are identical for both modules. Values relate to the production of 1 kg PGM.

## 5.4 Winning of Secondary PGM

As shown in chapter 2, considerable amounts of PGMs originate from secondary production. Since the intended module is a consumer-mix valid for European consumers, the secondary production of PGM represents the collection, beneficiation and refining of PGM containing scrap in the European Region. The process in concern is mainly based on (Hochfeld (1997)) and describes the secondary production in Germany, which will stand for the whole European region. Further more only data on recycling of automotive catalysts are available. Since it can be assumed, that by far the most part of PGM recycling origins from recovered automotive catalysts, this study focuses solely on recycling of automotive catalysts and neglect other sources of secondary PGMs.

### 5.4.1 Production process and infrastructure

The production process can mainly be described in the three steps: Collection, Beneficiation and Refining. Information on production process is taken from (Hagelücken (2001a)).

**Collection.** Spoiled automotive catalysts are bought by specialised enterprises from different origins. Part of it originates from scrap dealer recycling end-of-life cars. Further more during the cars operating phase, defective catalysts are exchanged in garages. The third sources is the production waste, i. e. defective catalysts which do not reach market. In most cases, there are fix agreements between the different supplier and the intermediary trade. Although the trade with PGM containing scrap is risky. Motor car manufacturer built up their own internal recycling systems with their contractor garages and gained access to exchanged catalysts.

**Beneficiation.** The catalysts are dismantled and then sold to refining companies. Refineries too have long-term contracts with the intermediary trade.

**Refinery.** The scrap first is shredded and then hydrometallurgically or pyrometallurgical processed. The direct hydrometallurgical path dissolves the scrap with strong acids for following precipitation. This has the handicap of effluents and larger amounts of dross and PGM losses. Therefore it is rarely used nowadays. In the pyrometallurgical process, the scrap is smelted and the ceramic wafer is slagged. The PGMs are concentrated in the collector metal, usually copper. Low-content PGM scrap is often smelted together with other non-ferrous metal matte. This is cheap, but effects a high loss in PGM. The collector matte from the furnace then is treated hydrometallurgically by re-precipitation as described in 5.3.1. In this step usually production scrap from other industries (glass, chemical laboratories) is joined. The overall PGM-yield is 98 % for platinum and 85 % for rhodium – no data on recovery of palladium was available.

### 5.4.2 Emissions and Waste

#### Air

Emissions during collection and beneficiation are gases from transportation and dust from dismantling. Also in this step the combustion leads to emissions of SO<sub>2</sub>. Nitrogen oxides are produced in significant amounts during acid digestion using nitric acid. Chlorine and HCl can be formed during a number of digestion processes. Chlorine is used extensively in the dissolution stages using hydrochloric acid and chlorine mixtures respectively. Dust and metals are generally emitted from incinerators and furnaces. VOC can be emitted from solvent extraction processes, while organic compounds, namely dioxins, can be emitted from smelting stages resulting from the poor combustion of oil and plastic in the feed material. All these emissions are subject to abatement technologies and controlling.

**Water**

During collection and beneficiation no serious emissions are reported. Effluents from refining contain considerable amounts of metals and organic substances.

**Waste**

Solid residues from pyrometallurgical step are usually re-used in copper facilities, final residues generally comprise hydroxide filter cakes.

**5.4.3 Secondary PGM in ecoinvent**

This module in ecoinvent describes the collecting of automotive catalysts in Germany, the dismantling of the catalysts and the pyrometallurgical processing followed by a hydrometallurgical purification step. The data is mainly taken from Hochfeld (1997) and IPPC (2001).

The scrap is assumed to be transported over 600 km by lorry, further emissions are neglected. The origin of the automotive catalysts is burden-free. For dismantling an average electricity-input of 50 kWh/t is assumed in Hochfeld (1997).

The metallurgical step consists of an arc-furnace. Values for electricity-input and lime and copper addition are reported in Hochfeld (1997) for one big German enterprise, but no direct specific emissions. These values are taken from IPPC (2001), representing the same refining-step as in the primary production. Data on the hydrometallurgical electrolysis still lacks and is assessed with data on refining from the module “Primary Production of PGM”. The emissions from Tab. 5.15 used there and the respective energy use per tonne PGM from Hochfeld (1997) are also chosen for secondary production.

The layout of the modules with general flow information, remarks, sources, values uncertainty information are shown in Fig. 5.16. An overview over the meta information is given in Tab. 5.16.

**Data quality**

The uncertainties are assessed using the pedigree matrix (see Frischknecht et al. (2003b)). For detailed results compare the different flows and the ecoinvent meta information.

Based on the literature no satisfactory process data was available. The hydrometallurgical step is approximated with emissions and energy use of the refining in the primary production.

**Tab. 5.16 Ecoinvent meta information for the multi-output processes in the secondary production of PGM in Europe from recycled automotive catalysts.**

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>                  | <b>platinum group metal production, secondary</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Location                     | RER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| InfrastructureProcess        | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Unit                         | kg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Dataset Version              | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| IncludedProcesses            | The module includes a collection step, with the transportation of the scrap to the intermediary dealer, a beneficiation step and a metallurgical step. Production, application and emissions of the most important agents are also included.                                                                                                                                                                                                                                                                                                     |
| Amount                       | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| LocalName                    | Platingruppenmetalle-Produktion, sekundär                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Synonyms                     | Platingruppenmetall//platinumgroup metal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| GeneralComment               | This module describes the collecting of autocatalysts in Germany, the dismantling of the catalysts and the pyrometallurgical processing followed by a hydrometallurgical purification step. The multioutput-process "PGM-Production, secondary, RER" delivers the co-products "platinum, secondary, at refinery", "palladium, secondary, at refinery" and " rhodium, secondary, at refinery". The smelter slag disposal is inventoried as disposal of inert material, thus neglecting potentially important (but unknown) heavy metal emissions. |
| StartDate                    | 1995                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| EndDate                      | 2002                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Data valid for entire period | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Geography Text               | Data for Germany are used in this module for Europe. Some agents used like lime or transports originate from data sets that are valid only for Switzerland.                                                                                                                                                                                                                                                                                                                                                                                      |
| Technology Text              | After collection and transportation to the intermediary dealer, the scrap is ground in a mill. The resulting material is fed to specialised refineries. The metallurgical step consists of an arc-furnace. Same refining process as in primary production assumed (selective precipitation)                                                                                                                                                                                                                                                      |
| Representativeness           | 4.6%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| ProductionVolume             | 5.3 t / a                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| SamplingProcedure            | Literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Extrapolations               | see Geography and Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| UncertaintyAdjustments       | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

| General Flow Information   |   |                                            |   |                     |  | Representation in ecoinvent                                               |                                                                                              |                |                |          |                                                       | Uncertainty Informations                                    |        |                       |                  |                        |                  |                  |
|----------------------------|---|--------------------------------------------|---|---------------------|--|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------------|----------------|----------|-------------------------------------------------------|-------------------------------------------------------------|--------|-----------------------|------------------|------------------------|------------------|------------------|
| Input                      |   | Process Name                               |   | Output              |  | Remarks                                                                   | Category                                                                                     | Subcategory    | Infrastructure | Location | Modul name in ecoinvent                               | Mean value                                                  | Unit   | Source for mean value | Type             | Standard Deviation 95% | General Comment  |                  |
| spoiled converter          | m | platinum group metal production, secondary |   |                     |  | Not inventoried, assumed to be burden free. Average content of PGM: 0.25% |                                                                                              |                |                |          |                                                       | 400.00                                                      | kg     |                       | 1                | 1.08                   | (1,2,2,1,1,3,12) |                  |
| transport to beneficiation | m |                                            |   |                     |  | assumption: 600 km transport                                              | transport systems                                                                            | road           | No             | RER      | transport, lorry 32t                                  | 240.00                                                      | tkm    |                       | 1                | 2.01                   | (1,2,2,1,1,3,5)  |                  |
| Electricity (grinding)     | m |                                            |   |                     |  |                                                                           | electricity                                                                                  | production mix | No             | UCTE     | electricity, medium voltage, production UCTE, at grid | 20.00                                                       | kWh    |                       | 1                | 1.29                   | (3,2,1,5,3,4,2)  |                  |
| Electricity (smelting)     | m |                                            |   |                     |  |                                                                           | electricity                                                                                  | production mix | No             | UCTE     | electricity, medium voltage, production UCTE, at grid | 600.00                                                      | kWh    |                       | 1                | 1.29                   | (3,2,1,5,3,4,2)  |                  |
| Copper, loss to slag       | m |                                            |   |                     |  |                                                                           | metals                                                                                       | extraction     | No             | RER      | copper, at regional storage                           | 1.00                                                        | kg     |                       | 1                | 1.60                   | (4,2,1,5,4,4,4)  |                  |
| lime                       | m |                                            |   |                     |  |                                                                           | construction materials                                                                       | binder         | No             | CH       | lime, hydrated, loose, at plant                       | 80.00                                                       | kg     |                       | 1                | 1.60                   | (4,2,1,5,4,4,4)  |                  |
|                            |   |                                            | m | CO2                 |  |                                                                           | due to lime addition                                                                         | air            | unspecified    |          |                                                       | Carbon dioxide, fossil                                      | 35.20  | kg                    |                  | 1                      | 1.60             | (4,2,1,5,4,4,14) |
|                            |   |                                            | m | Slag                |  |                                                                           |                                                                                              |                |                | No       |                                                       | disposal, inert waste, 5% water, to inert material landfill | 445.80 | kg                    | (Hochfeld, 1997) | 1                      | 1.29             | (3,2,1,5,3,4,6)  |
|                            |   |                                            | m | Platinum, secondary |  |                                                                           | Ratio reflecting the reported average PGM recycling in RER. Allocation of 45% of the burden. | metals         | extraction     | No       | RER                                                   | platinum, secondary, at refinery                            | 0.53   | kg                    |                  |                        |                  |                  |
|                            |   |                                            | m | Platinum, secondary |  |                                                                           | Ratio reflecting the reported average PGM recycling in RER. Allocation of 15% of the burden. | metals         | extraction     | No       | RER                                                   | palladium, secondary, at refinery                           | 0.26   | kg                    |                  |                        |                  |                  |
|                            |   |                                            | m | Platinum, secondary |  |                                                                           | Ratio reflecting the reported average PGM recycling in RER. Allocation of 39% of the burden. | metals         | extraction     | No       | RER                                                   | rhodium, secondary, at refinery                             | 0.21   | kg                    |                  |                        |                  |                  |

Fig. 5.16 Flows for “PGM-production, secondary, RER ” and its representation in the ecoinvent database

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## Part VI

# Chromium

Data v2.1 (2009)

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## Comments on version 2.0

This part corresponds to the original report in Althaus et al. (2004) that has been updated with the changes reported in Frischknecht et al. (2006). The changes were implemented with v1.2 which corrected the amount of resource required in the chromite dataset by factor two. Additionally to the reported errors the emissions of solids, PAH and chromium to water had been corrected by a factor 1E-3 in the dataset “ferrochromium, at plant” as compared to the data in version 1.3.

The flow names in the tables displaying unit process data have not been updated consistently to reflect the name changes with version 2.0.

No changes were made in version 2.1 compared to version 2.0.

Chapter “6 Cumulative Results and Interpretation” of the original report has been deleted for the current update.

## Summary

This part describes the production of chromite, ferrochromium and metallic chromium from cradle to consumer in Europe. Chromite is used for the production of chemicals and refractories. Ferrochromium is used as alloy for steel and chromium is mainly used for special iron free alloys (superalloys), e.g. for high-temperature exposed components in chemical and petrochemical industries.

Mining, beneficiation and the different metallurgical processes are inventoried vertically aggregated because of difficulties to clearly allocate some of the flows to one single step and because the intermediate steps would not make sense in another context. The Figures in this report however contain the available information to the single steps.

Since no chromium is recycled within the chromium industry but directly to the foundries of the stainless steel industry, no secondary chromium is considered in this study.



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# 1 Introduction

This part describes the production of ferrochromium and metallic chromium from cradle to consumer in Europe. Inventory data for the ferrochromium production is taken mainly from Adelhardt & Antrekowitsch (1998) together with background data from IPPC (2002), IPPC (2001), Papp (1994) and EEA (1999).

Like many metals used as alloying element, chromium is recycled from steel-scrap in foundries directly as secondary alloys. Depending on their composition, these secondary chromium containing materials are used together with ferrochromium as secondary source for chromium. In this study, only the primary material ferrochromium and metallic chromium is considered. Secondary chromium sources – i.e. alloys from recycled steel scrap – have to be inventoried in the respective production module, e.g. in the stainless steel production.

The main environmental impact by producing refractory metals is metal containing dust and hard metal powder as well as fume from the smelting processes for instance by producing chromium metal. Dust emissions occur from storage, handling of raw materials and products and the furnace operation, where both stack and fugitive emissions play an important role. Hydrogen fluoride, which is used in several processes, is highly toxic and needs therefore to be handled very carefully to avoid any health problems for the plant staff. Another environmental impact of the production of hard metals is the toxicity of metal compounds such as chromium.

Over the last 15 to 20 years, the world pattern of the ferro-alloy market has deeply changed:

- The consumption of the developing countries has vastly increased with the development of their steel production;
- Their production has even more increased because they have taken a growing share of the traditional markets of the industrialised countries, where the steel production was stagnant or growing at a slow rate.
- The ferro-alloy industry is facing a growing proportion of imports, at first from the new industrialised countries and in recent years from the countries of East Europe and the CIS (former Soviet Union).

As a consequence, EU ferro-alloy production has been submitted to a difficult competition that results in a decreasing trend of the total amount of ferro-alloys produced. Although the last years showed some stabilisation, even a slightly increase of the production, the European industry remains very sensitive.

## 2 Reserves and Resources of Chromium

Metals are a special class of materials. They are not consumed but rather used during a product life cycle. When recovered the properties are theoretically identical to those in the raw material. The metal can be re-used in another product life cycle, provided that its concentration makes a recovery economic. The recovered metal is referred to as secondary metal, whereas the metal originating directly from the ore is referred to as primary metal. In LCIs these two types – primary and secondary metal – are often differentiated because they bear a different ecological burden.

### 2.1 Primary Chromium

Of the many minerals that contain chromium only the chromium spinels are of economic importance. Chromite is found in stratiform and podiform deposits.

- Stratiform deposits (seam-like) Layers up to a meter thick; e.g. Bushveld Igneous Complex (South Africa), the Great Dyke (Zimbabwe) and the Stillwater Complex (United States).
- Podiform deposits (shaped like sacks or tubes). Range in size; Ural Mountains, Albania, Zimbabwe, and the Philippines.

Generally, richer lumpy Cr bearing ores have been preferred for smelting, whereas those with  $\text{Cr}_2\text{O}_3$  content less than 40 % have been used in refractories. The internal texture of the chromite ore bodies varies widely. The closest chromite crystal packing results in the formation of massive ores containing 75 - 85 vol% of chromite.

Production and resources are presented in Tab. 2.1. The largest chromite-ore-producing countries (India, Kazakhstan, and South Africa) accounted for about 76% of world production. South Africa alone accounts for more than 45% of world production and has been the major supplier of chromium in the form of chromite ore and ferrochromium to western industrialised countries (USGS (2003)).

### 2.2 Secondary Chromium

The major end use of chromium is in stainless steel, and this is the major form in which chromium is recycled. Ferrochromium is added to iron at steel-producing plants to make the iron-chromium alloy stainless steel.

Stainless steel scrap can substitute for ferrochromium as a source of chromium. Scrap is generated during the manufacturing process (new scrap) and as a result of recycling obsolete equipment (old scrap). Scrap from these sources is collected and sorted by grade (i.e., chemical composition) in scrap yards. Scrap brokers play a role in moving material from where it is recovered to where it is consumed. The steel industry consumes stainless steel scrap as a source of chromium and nickel. Thus chromium units are recycled when stainless steel is reused. A study of domestic stainless steel found that its average chromium content is about 17% (USGS (2003)). Papp (2001) reports a recycling rate of 20% for the US in 1998 with a domestic (primary) production of 144 kt and a consumption of old (75.3 kt) and new scrap (28.6 kt) of total 103.9 kt, imports of 263 kt and releases of stock of 37.2 kt to meet the consumption of chromium in the fabrication of stainless steel of 522 kt and exports of 22.5 kt (all values in chromium contained).

The typical raw materials for the production of secondary ferro alloys are the following residues generated mainly within the production of stainless steel:

- Electric arc furnace (EAF) filter dust
- Converter filter dust
- All kind of scale fines
- Shot blasting dust
- Grinding dust

Since no chromium is recycled within the chromium industry but directly to the foundries of the stainless steel industry, no secondary chromium is considered in this study.

**Tab. 2.1 Mine production and reserves of chromium in 2001. “Reserves” is referred to as “the part of the reserve base which could be economically extracted or produced at the time of determination”, while “Reserve base” denotes “that part of an identified resource that meets specified minimum (...) criteria related to current mining and production practices” (USGS (2003)).**

|                              | Mine production  |                                 | Reserves | Reserve base |
|------------------------------|------------------|---------------------------------|----------|--------------|
|                              | 2001 [kt Cr / a] | 2002 [kt Cr / a]<br>(estimated) | [kt Cr]  | [kt Cr]      |
| United States                | —                | —                               | —        | 7000         |
| India                        | 1680             | 1900                            | 18000    | 39000        |
| Kazakhstan                   | 2050             | 2300                            | 410000   | 410000       |
| South Africa                 | 5500             | 5800                            | 790000   | 5500000      |
| Other countries              | 2900             | 3000                            | 420000   | 1100000      |
| World total (may be rounded) | 12100            | 13000                           | 1600000  | 7100000      |



## 3 Characterisation of the Materials

Based on the raw material chromite, various chromium containing materials are produced for the use in a broad field of applications.

### 3.1 Chromite ore

Chromite [53293-42-8] is a spinel  $\text{FeO} \cdot \text{Cr}_2\text{O}_3$ . In nature it is a mixture described by the formula  $(\text{Fe}^{2+}, \text{Mg})\text{O} \cdot (\text{Cr}, \text{Al}, \text{Fe}^{3+})_2\text{O}_3$ . The theoretical  $\text{Cr}_2\text{O}_3$  content in chromite would be 67.9% (46.5% Cr), but in reality this value ranges between 25% and 60%  $\text{Cr}_2\text{O}_3$  (17% - 40.8% Cr). The chromium content in chromite is  $2 \cdot 52 / 152 = 68.4\%$ , Adelhardt & Antrekowitsch (1998) calculated a world-wide average content of 37.3% Chromite (25.5% Cr) in raw ore. Chromite ore rarely contains more than 50 %  $\text{Cr}_2\text{O}_3$ ; other minerals such as  $\text{SiO}_2$  can also be present. Typical composition of some important metallurgical chromite ores are summarised in Tab. 3.1.

A high Cr : Fe ratio is advantageous to produce an alloy with high chromium content. Chromite ores are classified as follows:

- Ores rich in chromium:  $> 46\% \text{Cr}_2\text{O}_3$ , Cr : Fe  $> 2 : 1$ ; for the production of ferrochromium
- Ores rich in iron:  $40 - 46\% \text{Cr}_2\text{O}_3$ , Cr : Fe  $< 2 : 1$ ; for the production of charge chrome and for the chemical industry
- Ores rich in aluminium:  $> 60\% (\text{Cr}_2\text{O}_3 + \text{Al}_2\text{O}_3)$ ,  $> 20\% \text{Al}_2\text{O}_3$ ; for refractories

Metallurgical-grade chromite ores are classified as hard lumpy or friable lump types, fines, and concentrates. Concentrates are produced by mechanical upgrading of lean ores or fines. In the production of high-carbon ferrochromium, generally a lumpy type of chromite ore is necessary. About 80 % of chromite ores in the western world are fines ( $< 10 \text{ mm}$ ). Therefore, efforts have been made to agglomerate these. Fines of chromite ores can be used to produce low-carbon ferrochromium.

### 3.2 Chromium Metal

At room temperature Chromium [7440-47-3] is resistant to ordinary corrosive agents, which explains its use as an electroplated, protective coating. It dissolves in non-oxidising mineral acids, such as hydrochloric and sulphuric acids, but not in cold aqua regia or nitric acid, which passivate the metal. At elevated temperature it reacts with halogens, silicon, boron, nitrogen, oxygen, and carbon. Chromium and chromium-rich alloys are brittle at room temperature (Downing et al. (1997)).

Chromium metal is a metallurgical industry product. It is produced by one of two processes: electrolysis or aluminothermic reduction. Chromium metal produced by these processes is essentially the same differing only in amount of various trace chemical constituents. The largest production of chromium metal is by the aluminothermic reduction process (ICDA).

Selected physical properties of elemental chromium are given in Tab. 3.2.

### 3.3 Ferrochromium

In the metallurgical industry, the principal product is ferrochromium, a mixture of chromium and iron with varying amounts of other chemical constituents. Ferrochromium is further classified by its carbon content as high-carbon, medium-carbon, or low-carbon ferrochromium. By far, the largest production

of ferrochromium is the high-carbon variety. It is the major source of chromium for the production of steel alloys.<sup>1</sup>

Ferrochromium is a master alloy of iron and chromium, containing 45 – 95 % Cr and various amounts of iron, carbon, and other elements. The ferrochromium alloys are classified by their carbon content and are known by their French names because basic work in this field was carried out mainly in France:

- High-carbon ferrochromium (HC ferrochromium) with 4 – 10 % C, "ferrochrome carbure"
- Medium-carbon ferrochromium (MC ferrochromium) with 0.5 – 4 % C, "ferrochrome affiné"
- Low-carbon ferrochromium (LC ferrochromium) with 0.01 – 0.5 % C, "ferrochrome suraffiné"

The mechanical and chemical properties of steel can be improved by alloying it with ferrochromium. Chromium combined with nickel gives stainless steel excellent chemical resistance (Downing et al. (1997)). Some physical properties of different ferrochromium alloys are given in Tab. 3.3

Tab. 3.1 Analyses of some chromite ores, metallurgical grade (Downing et al. (1997))

| Compound                       | Chromite ore, wt% |                  |           |             |              |
|--------------------------------|-------------------|------------------|-----------|-------------|--------------|
|                                | Transvaal lump    | Zimbabwe friable | USSR lump | Turkey lump | Albania lump |
| Cr <sub>2</sub> O <sub>3</sub> | 42.55             | 49.53            | 53.73     | 47.58       | 40.5         |
| FeO                            | 21.85             | 11.6             | 8.5       | 9.45        | 11.4         |
| Fe <sub>2</sub> O <sub>3</sub> | 4.85              | 2.2              | 4.3       | 3.1         | –            |
| MgO                            | 9.26              | 17.52            | 17.3      | 18.7        | 23.3         |
| Al <sub>2</sub> O <sub>3</sub> | 15.5              | 7.3              | 9.4       | 8.8         | 7.3          |
| SiO <sub>2</sub>               | 5.54              | 5.14             | 4.3       | 8.33        | 12.6         |
| L.O.I.e                        | 0.2               | 2.3              | 2.1       | 2.25        | 3.4          |
| Cr : Fe                        | 1.43              | 3.20             | 3.81      | 3.42        | 3.13         |

<sup>1</sup> ICDA, International Chromium Development Association, <http://www.chromium-asoc.com/>, accessed 01.03.2003

Tab. 3.2 Some physical properties of chromium (Downing et al. (1997)).

|                                                  | Chromium                                  |
|--------------------------------------------------|-------------------------------------------|
| $A_r$                                            | 51.996 I                                  |
| Atomic number                                    | 24                                        |
| mp                                               | 1857 °C                                   |
| bp                                               | 2672 °C                                   |
| Density $\rho$ at 20 °C                          | 7.19 g/cm <sup>3</sup>                    |
| Crystal structure                                | cubic, body centered                      |
| Specific heat at 25 °C                           | 23.25 J mol <sup>-1</sup> K <sup>-1</sup> |
| Molar entropy $S^\circ_{298}$                    | 23.64 J mol <sup>-1</sup> K <sup>-1</sup> |
| Heat of fusion                                   | 16.93 kJ/mol                              |
| Latent heat of vaporization at bp                | 344.3 kJ/mol                              |
| Linear coefficient of thermal expansion at 20 °C | $6.2 \times 10^{-6}$                      |
| Resistivity at 20 °C                             | $12.9 \times 10^{-8} \Omega \text{ m}$    |
| Thermal conductivity at 20 °C                    | $67 \text{ W m}^{-1} \text{ K}^{-1}$      |

Tab. 3.3 Some physical properties of ferrochromium and ferrosilicochromium (Downing et al. (1997)).

| Alloy                                               | Density ( $\rho$ ),<br>g/cm <sup>3</sup> | Melting range, °C |         |
|-----------------------------------------------------|------------------------------------------|-------------------|---------|
|                                                     |                                          | Liquidus          | Solidus |
| Chromium metal (electrolytic)                       | 7.2                                      | 1900              |         |
| Chromium metal (aluminothermic)                     | 7.2                                      | 1850              |         |
| Low-carbon ferrochromium (72% Cr, 0.01% C)          | 7.35                                     | 1690              | 1660    |
| Low-carbon ferrochromium (72% Cr, 0.05% C)          | 7.35                                     | 1670              | 1639    |
| Low-carbon ferrochromium (69% Cr, 0.1% C)           | 7.35                                     | 1604              | 1343    |
| High-carbon ferrochromium (69% Cr, 4–6% C, 1% Si)   | 7.2                                      | 1500              | 1350    |
| High-carbon ferrochromium (64% Cr, 5% C, 1% Si)     | 7.1                                      | 1450              | 1340    |
| Ferrochromium, charge grade (63% Cr, 5.5% C, 7% Si) | 6.7                                      | 1500              | 1400    |
| Ferrochromium, charge grade (56% Cr, 6% C, 5% Si)   | 6.8                                      | 1493              | 1660    |
| Ferrochromium silicon (36% Cr, 40% Si)              | 5.3                                      | 1388              | 1360    |

## 4 Use / Application of Material

Chromium has a wide range of uses in metals, chemicals, and refractories. However, in this study only ferrochromium is of further importance. A schematic overview is given in the following Fig. 4.1.

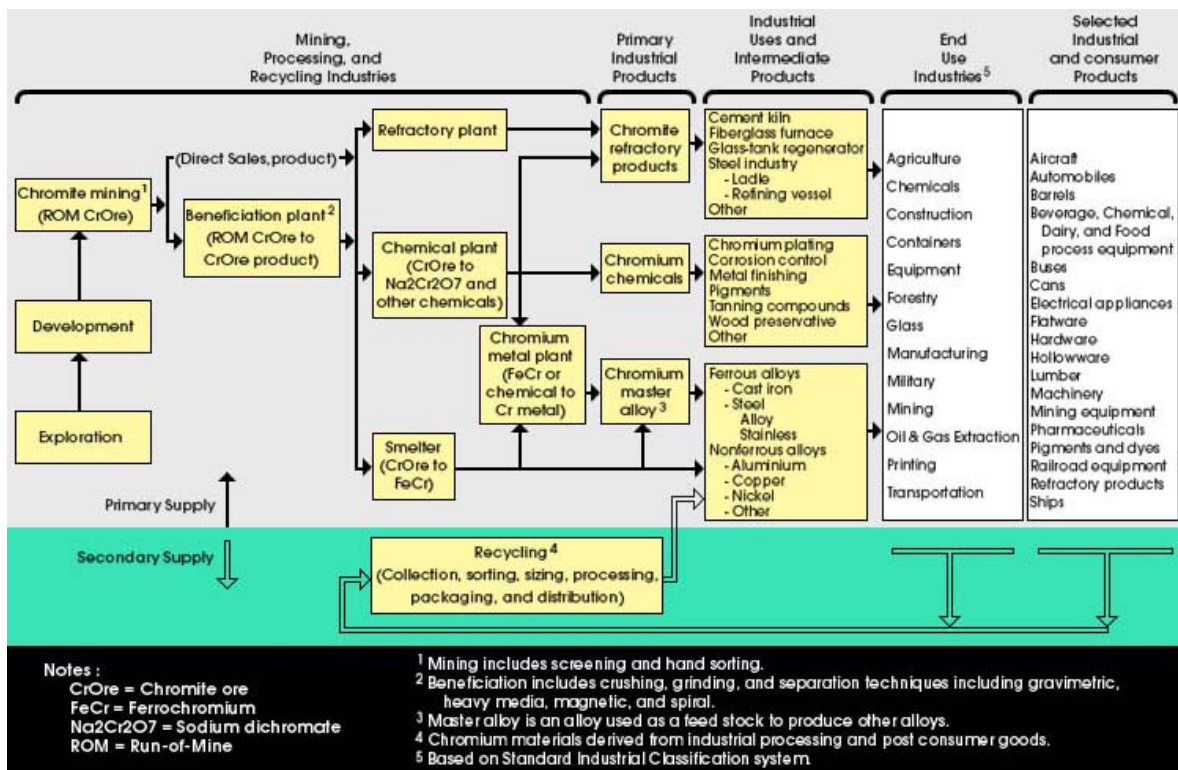


Fig. 4.1 The chromium life cycle process from mining to commercial, scheme taken from ICRA's homepage at <http://www.chromium-asoc.com/>

### 4.1 Chromite ore

Chromite ore is the central intermediary product in the chromium industry. From it the three main product groups are made: ferrochromium, chromium chemicals and refractories. Chromite is used directly in the refractory industry to make bricks, mortar, and ramming and gunning mixes. Chromite enhances thermal shock and slag resistance, volume stability, strength and is used mainly in the ferrous and non-ferrous industry, cement industry and glass manufacturing. In foundries for certain applications chromite is used as moulding sand, for which chromite sand with more specific properties is applied.

In 1994 around 8% of the chromium produced was used in the refractory industry, while 80% went into the production of ferrochromium and 12% into the chemical industry (Adelhardt & Antrekowitsch (1998)).

In the refractory industry, the principal products are shaped and unshaped refractories. Refractories are used to line metallurgical furnaces that produce steel or copper alloys. They are also used in the heat exchange section of glass making furnaces and in cement kilns. These refractories are consumed in the process of making steel, copper, glass, and cement, but they are not part of those products. The annual

chromite production for 2002 is given in Fig. 4.2<sup>2</sup>. Further details on use of refractories in the glass industry can be found in Hischier (2003)

The consumption of chromium was reported in Papp (1994) to be 30 t Cr / Mt cement and 450 t Cr / Mt copper in the cement and copper industry respective. Adelhardt & Antrekowitsch (1998) reports the use of 2 and 5 kg refractories per tonne of converter and electric steel respectively.

After use, refractories are recuperated in order to remove hexavalent chromium, which is built under high temperature and oxidising conditions (Adelhardt & Antrekowitsch (1998)).

## 4.2 Chromium Chemicals

In the chemical industry, the principal product is sodium dichromate. Sodium dichromate is used both as feed material for the production of other chromium chemicals and as process material for the surface treatment of metals. The use of chromite by the chemical industry is substantially smaller than that of the metallurgical industry. Chemicals made from or along with sodium dichromate, are industrial products including chromic acid, chromic oxide, and potassium dichromate. The uses of chromium chemicals are quite diverse and include metal finishing, leather tanning, pigment production, and wood preservation<sup>2</sup>.

Chromium compounds are used in numerous fields. In addition to the applications mentioned, chromates have long been used in printing as an aid in photomechanical reproduction. For some time, chromium dioxide has been a component of magnetic tapes for information storage. About 12 % of the chromium used world-wide is in the form of chromium chemicals (Adelhardt & Antrekowitsch (1998)).

Directly or via several intermediate stages, sodium dichromate ( $\text{Na}_2\text{Cr}_2\text{O}_7 \cdot 2 \text{H}_2\text{O}$ ; CAS 7789-12-0), is the starting material for the production of all chromium compounds and pure chromium metal. For further details on use and production see Althaus et al. (2003)

Chromium(III) oxide (Chromium Hydroxide) ( $\text{Cr}_2\text{O}_3$ ; CAS 1308-38-9), finds widespread application as a green pigment resistant to atmospheric conditions and heat. In addition, it is used as a colorant in glass products and printing inks, as a vitrifiable pigment in the ceramics industry, and as a polishing agent because of its considerable hardness.

Chromium(IV) Oxide (Chromium Dioxide) ( $\text{CrO}_2$ ; CAS 12018-01-8) is used as a magnetic pigment. So far no other uses have achieved any significance.

Chromium trioxide ( $\text{CrO}_3$ ; CAS 1333-82-0), also known as chromic acid anhydride or chromic acid, is made by the reaction of sodium dichromate with sulphuric acid. Electroplating is the most important field of application of chromium(VI) oxide. Numerous mixtures containing chromium trioxide are on the market; these "compounds" often contain hexafluorosilicates which improve the properties of the chromium coatings. Chromic acid solutions are also used for passivating zinc, aluminium, cadmium, and brass. Proprietary mixtures predominantly contain additions of fluoride, nitrate, and phosphate ions. Other uses for chromic acid are in the production of chromium dioxide and in wood preservation.

## 4.3 Chromium Metal

Pure chromium metal, produced by electrolytic or aluminothermic processes, is used for alloying non-ferrous engineering materials. Chromium metal is a major source of chromium for non-ferrous alloys that exclude iron, such as certain superalloys. The most common materials are nickel-based and cobalt-based alloys, most of which are used as high temperature alloys (Downing et al. (1997)).

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<sup>2</sup> <http://www.chromium-asoc.com/publications/frame.html>

The aerospace industry is the major source of demand for chromium metal in the form of superalloys, which are manufactured and consumed mostly in North America and Europe. World chromium metal demand was increasing by 1'000 t/yr over the last years. Chromium metal production is estimated at 22'000 t in 2001; 21'000 t in 2000; and 20'000 t in 1999. The trend is expected to continue. Besides aerospace industry, the major end use industries are gas turbine, and petrochemical (USGS (2003)).

## 4.4 Ferrochromium

Ferrochromium is used principally as a master alloy to produce chromium-containing steel and cast iron. Chromium in steel imparts corrosion resistance because of the formation of a thin, continuous, impervious chromium oxide film on the steel surface.

Most ferrochromium is used to manufacture stainless steel. Martensitic grades with ca. 13 % Cr are used for applications such as knives, whereas ferritic grades with 18 – 22 % Cr are used as deep-drawing sheets. Austenitic CrNi steels exhibit especially good corrosion (acid) resistance and are used for equipment in the chemical industry as well as for food-processing machinery. The annual production of ferrochromium dropped from 5'720'000 t in 2000 to 4'680'000 in 2001 (USGS (2003)).

Ferrochromium is classified usually into two major groups: low-carbon and high-carbon ferrochromium. It is used as a source of chromium by the steel industry and as a source of chromium for non-ferrous alloys that permit some iron content.

High-carbon ferrochromium is one of the chromium industry's most important products with a share of 91% in the production. Low-carbon ferrochromium production accounts to 9% of the total and dropped from 792'738 t in 1988 to 413'079 t in 2001; an annual compound growth rate of 0.95 over the time period (USGS (2003)).

Technological changes in the consuming industry permit the substitution of high-carbon ferrochromium for low-carbon ferrochromium. The substitution is desirable because low-carbon ferrochromium production is more expensive than high-carbon ferrochromium production.

The annual ferrochromium production for 2002 is given in Fig. 4.3.

## 4.5 Chromium Alloys

Stainless steel is composed of two broad categories – austenitic and ferritic. The names are related to the molecular structure of the steel but also identify which grades require nickel (austenitic) and which do not (ferritic).

The only true chromium-based alloys that have been developed and used commercially are a few chromium – nickel alloys. These materials contain 50 – 60 % chromium; the remaining percentage is nickel, with either niobium (columbium) or titanium specified as carbon and nitrogen scavengers. These alloys, often referred to as superalloys, are used in the construction of high-temperature exposed components in chemical and petrochemical industries (Downing et al. (1997)).

The most widely known chromium containing alloy is the stainless steel AISI 302 (Fe/Cr18/Ni8), which contains 18 % Cr and 8 % Ni. A small amount of molybdenum further improves the acid resistance. Heat-resistant steel (24 – 26 % Cr and 19 – 21 % Ni) is used in power stations for steam boilers and heat exchangers; superalloys based on nickel or cobalt (ca. 30 % Cr) are used for gas turbine parts.

The annual production of stainless steels for 2002 is given in Fig. 4.4.

Tab. 4.1 Uses of chromium compounds (Anger et al. (1997))

| Branch of industry    | Product                           | Use                                                    |
|-----------------------|-----------------------------------|--------------------------------------------------------|
| Building industry     | chromium(III) oxide               | pigment for coloring building materials                |
| Chemical industry     | dichromates, chromium(VI) oxide   | oxidation of organic compounds, bleaching of montan    |
|                       |                                   | waxes, manufacture of chromium complex dyes            |
|                       | chromium(III) oxide               | catalysts                                              |
| Printing industry     | dichromates                       | photomechanical reproduction processes                 |
|                       | chromium(VI) oxide                | chromium plating of printing cylinders                 |
| Petroleum industry    | chromates(VI)                     | corrosion protection                                   |
| Paints and lacquers   | chromates, chromium(III) oxide    | pigments                                               |
| Refractory industry   | chromium(III) oxide               | additive for increasing slag resistance                |
| Electroplating        | chromium(VI) oxide                | bright and hard chromium plating                       |
| Wood industry         | chromates, chromium(VI) oxide     | in mixtures of salts for protecting wood against fungi |
|                       |                                   | and insects                                            |
| Leather industry      | basic chromium(III) sulfates      | tanning of smoothed skins                              |
| Metal industry        | chromium boride, chromium carbide | flame sprays                                           |
|                       | chromium(III) oxide               | polishing agents                                       |
| Metallurgy            | chromium(III) oxide               | aluminothermic extraction of pure chromium metal       |
| Textile industry      | dichromates                       | dyeing with chrome dyes                                |
|                       | basic chromium(III) acetates and  | mordanting of textiles                                 |
|                       | chromium(III) fluorides           |                                                        |
| Recording industry    | chromium(VI) oxide                | magnetic information storage                           |
| Pyrotechnics industry | dichromates                       | additive to igniting mixtures                          |

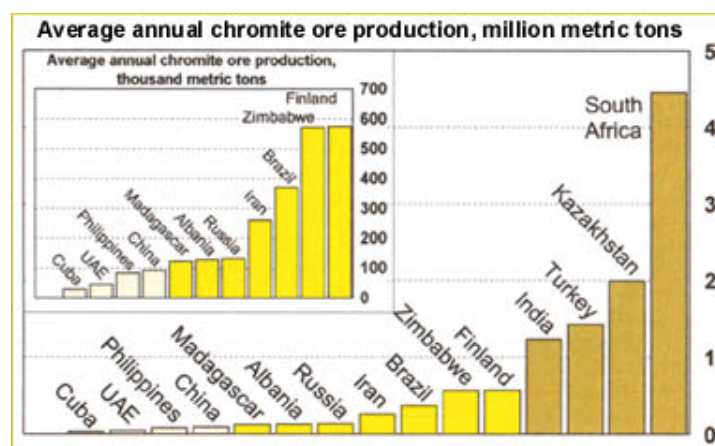


Fig. 4.2 Annual chromite ore production in 2001.

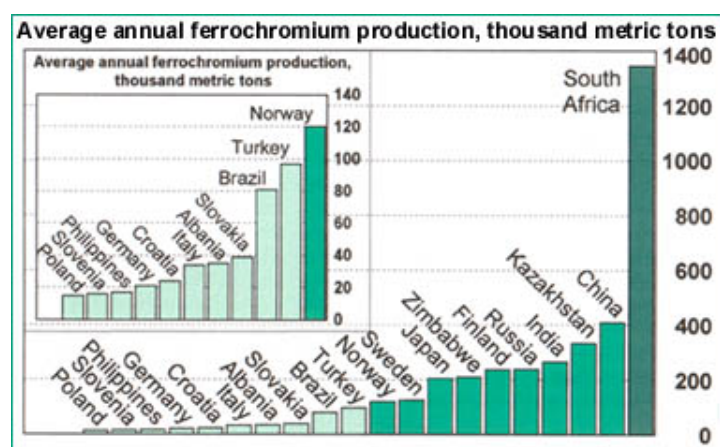


Fig. 4.3 Annual ferrochromium production in 2001.

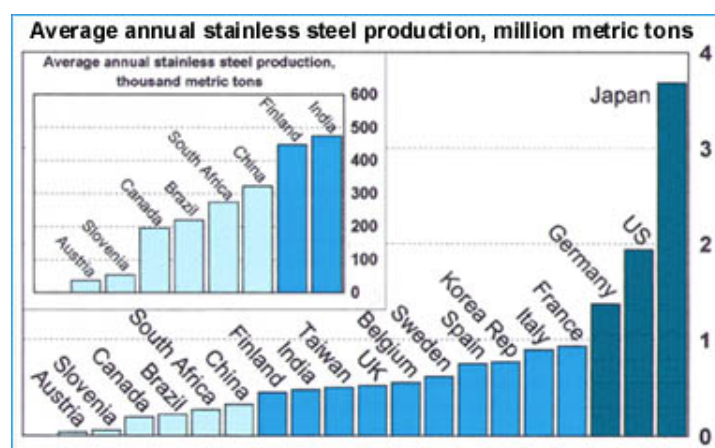
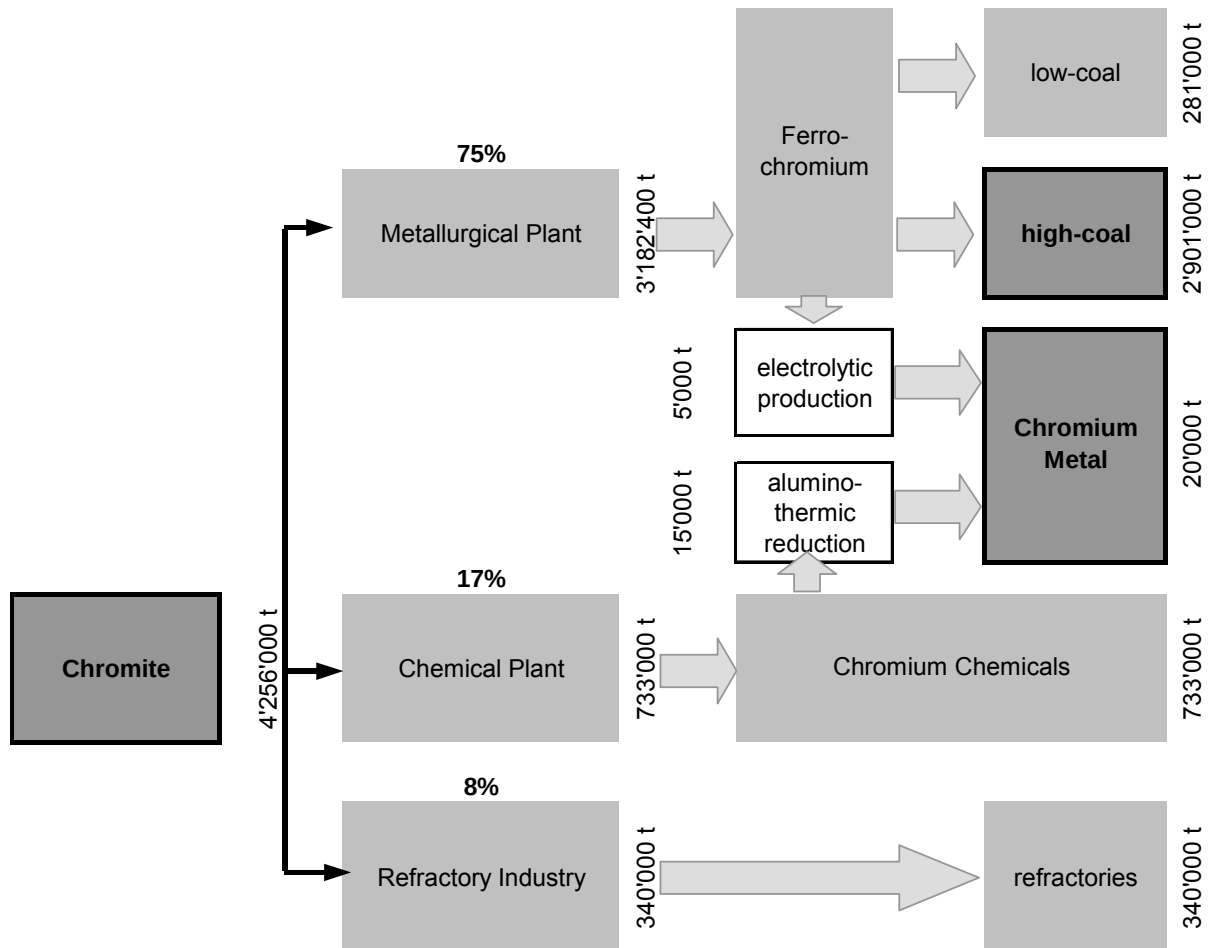


Fig. 4.4 Annual production of stainless steel in 2001, the main demand in ferrochromium.



## 5 System Characterisation

This study focuses on the main processes in the chromium production as shown in Fig. 4.1. As shown in chapter 4 Use / Application of Material no secondary production of chromium has to be considered, since the recycling of chromium occurs directly into the foundries in the form of re-melted alloys from chromium containing scrap.



**Fig. 5.1** Scheme of the chromium production system considered. The study focuses on the production of high-carbon ferrochromium, which is based on chromite. Chromite itself is used in different other modules in ecoinvent like “portachrom, at plant” Hischier (2003), “sodium dichromate, at plant” Althaus et al. (2003) and in the cement production. Metallic chromium is won from two different processes. Values shown are in tonnes of chromium contained and are based on USGS (2003) and Adelhardt & Antrekowitsch (1998).

Tab. 5.1 Ecoinvent meta information for the processes for the production of chromite and ferrochromium.

| Name                         | chromite, ore concentrate, at beneficiation                                                                                                                                                                                                                                                                                                                                                                                                             | ferrochromium, high-carbon, 68% Cr, at plant                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location                     | GLO                                                                                                                                                                                                                                                                                                                                                                                                                                                     | GLO                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Infrastructure Process       | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Unit                         | kg                                                                                                                                                                                                                                                                                                                                                                                                                                                      | kg                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Dataset Version              | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Included Processes           | The module includes a mining and a beneficiation step with the mining infrastructure and disposal of overburden and tailings (see part II "Auxiliary Processes").                                                                                                                                                                                                                                                                                       | The module includes the pre-treatment of the ore and the reduction to high-carbon ferrochromium. No secondary production considered, since secondary chromium is recycled to the foundries and has to be inventoried in the respective module, i.e. when casting the final raw material in the stainless steel production.                                                                                                                                         |
| Amount                       | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Local Name                   | Chromit, Erz, ab Aufbereitung                                                                                                                                                                                                                                                                                                                                                                                                                           | Ferrochrom, hochgekohlt, 68% Cr, ab Werk                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Synonyms                     | chromeisenstein                                                                                                                                                                                                                                                                                                                                                                                                                                         | ferro-chrome//ferrochrome//chromeisen                                                                                                                                                                                                                                                                                                                                                                                                                              |
| General Comment              | The module describes the production of chromite in 1994 for a world-wide production average. It is designed for the use of chromite as intermediary product in the production of ferrochromium, dichromate and refractories. Process data is taken mainly from a material flow analysis for the consumption of chromium compounds in Germany. The ore concentrate produced has a content of 30.7% Cr (44.9% Cr <sub>2</sub> O <sub>3</sub> ) by weight. | The module describes the production of chromite in 1994 for a world-wide production average. It is designed for the use of chromite as intermediary product in the production of ferrochromium, dichromate and refractories. This module does not consider secondary sources of chromium. Process data is taken mainly from a material flow analysis for the consumption of chromium compounds in Germany. Results are used as a proxy for the European situation. |
| Start Date                   | 1994                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1994                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| End Date                     | 2003                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2003                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Data valid for entire period | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Geography Text               | This module relates to the global situation. Some agents used like lime originate from data sets valid for Switzerland or Europe. Processes like blasting and transport originate from data sets valid for Switzerland or Europe                                                                                                                                                                                                                        | This module relates to the world wide production situation. Some Processes originate from data sets valid for Switzerland or Europe.                                                                                                                                                                                                                                                                                                                               |
| Technology Text              | Mining is done 21% open pit and 79% underground, followed by a beneficiation of the ore through classification. Overburden and tailings are disposed near the mining site.                                                                                                                                                                                                                                                                              | Direct reduction of chromite ores with carbon in three-phase submerged arc furnaces. 50% ordinary technology, 50% new processes like Outokumpu, Showa Denko, Krupp-CODIR & plasma furnace.                                                                                                                                                                                                                                                                         |
| Representativeness           | 7400%                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 7400%                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Production Volume            | 9.7 million tonnes in 1994                                                                                                                                                                                                                                                                                                                                                                                                                              | 2.2 million tonnes chromium contained in ferrochrome in 1994                                                                                                                                                                                                                                                                                                                                                                                                       |
| Sampling Procedure           | literature                                                                                                                                                                                                                                                                                                                                                                                                                                              | literature                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Extrapolations               | see Geography and Technology                                                                                                                                                                                                                                                                                                                                                                                                                            | see Geography and Technology                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Uncertainty Adjustments      | none                                                                                                                                                                                                                                                                                                                                                                                                                                                    | none                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

Tab. 5.1 Ecoinvent meta information for the processes for the production of chromite and ferrochromium. (cont.)

| Name                         | ferrochromium, high-carbon, 68% Cr, at regional storage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | chromium, at regional storage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location                     | RER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | RER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Infrastructure Process       | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Unit                         | kg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | kg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Dataset Version              | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Included Processes           | The module includes the transportation of high-carbon ferrochromium to Europe. No secondary production considered, since secondary chromium is recycled to the foundries and has to be inventoried in the respective module, i.e. when casting the final raw material in the stainless steel production.                                                                                                                                                                                                                                                                                                       | Both aluminothermic production and electrolysis are aggregated in this module. It comprises the production of Cr <sub>2</sub> O <sub>3</sub> as intermediary product, the aluminothermic process, the transport of the raw materials to Europe and the electrolysis. No emissions are taken into account. No secondary production considered, since secondary chromium is recycled to the foundries and has to be inventoried in the respective module, i.e. when casting the final raw material in the stainless steel production. |
| Amount                       | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Local Name                   | Ferrochrom, hochgekohlt, 68% Cr, ab Regionallager                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Chrom, ab Regionallager                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Synonyms                     | ferro-chrome//ferrochrome//chromeisen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | chrome                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| General Comment              | The module describes the consumption of chromite in Europe in 1994. It is designed for the use of chromite as intermediary product in the production of ferrochromium, dichromate and refractories. This module does not consider secondary sources of chromium. Process data is taken mainly from a material flow analysis for the consumption of chromium compounds in Germany. Results are used as a proxy for the European situation. Disposal of smelter slag is inventoried as disposal of inert material, thus neglecting the potentially important (but quantitatively unknown) heavy metal emissions. | The module describes the chromium metal consumed Europe in 1994. It is designed solely for the use of the metal as alloying element in super-alloys (Ni-Cr). Usually stainless steels are made from ferrochrome and not from chromium metal. Chromium is considered to be made by aluminothermic process (75%) and electrolysis (25%). This module does not consider secondary sources of chromium. The general data quality is not satisfying, due to a lack of information.                                                       |
| Start Date                   | 1994                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1994                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| End Date                     | 2003                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2003                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Data valid for entire period | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Geography Text               | This module relates to the consumption in Europe, although the import mix for Germany is chosen. Production relates to a global average.                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | World-wide production mix is chosen as proxy for the consumption mix in Europe.                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Technology Text              | Most transport by ship                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Metallic chromium is produced by aluminothermic process (75%) and electrolysis of dissolved ferrochromium (25%)                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Representativeness           | 0%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 10000%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| ProductionVolume             | n.a. 350'000 t in GER (1994), no info on consumpt. in RER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 22'000 t prod glob (2000), RER roughly 20% (4'300 t). No data on consumpt.                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Sampling Procedure           | literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Extrapolations               | see Geography and Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | see Geography and Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Uncertainty Adjustments      | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

## 5.1 Exploitation of chromite

### 5.1.1 Production process and infrastructure

**Mining.** Chromite ores are usually mined underground. Adelhardt & Antrekowitsch (1998) report an estimated share of 71% underground mining activities in the mine production in 1994 considered in their study, representing 78% of the total production. Of a surveyed part of 52% of the total world production, 69% of the chromite originated from stratiform deposits.

**Beneficiation.** After mining, the ore is first crushed in several stages with jaw and / or cone crushers, and then subsequently ground with rod and or ball mills and finally screened for classification. In a second step the classified material is subjected to gravity concentration to separate the metal-bearing particles from the unwanted minerals. For this drum separators and de-watering screens for lumps are used and cone separators and a high-gradient magnetic separator for fine material. No flotation is done. Chromite yields vary in a range from 65% to 85% (Adelhardt & Antrekowitsch (1998)).

### 5.1.2 Emissions & Waste

#### Mining

The major emissions are due to mineral born pollutants in the effluents. Open cut mining generates large quantities of dust, which contains elevated contents of metals. Rain percolates through overburden and leads to metal emissions to groundwater. Overburden is deposited close to the mine.

#### Beneficiation

Ore handling and processing produce large amounts of dust, containing PM<sub>10</sub> and several metals from the ore itself. Tailings are deposited as piles and in ponds. Since the tailings are not sulphidic, no acid rock drainage (ARD) occurs. In the tailings material the most significant contents are Cr and Ni, which occur as insoluble compounds and are considered not to cause any negative effects IPPC (2002).

### 5.1.3 “chromite, at beneficiation” in ecoinvent

The module describes the world-wide average production of chromite in 1994. It is designed for the use of chromite as intermediary product in the production of ferrochromium, dichromate and refractories. Process data is taken mainly from Adelhardt & Antrekowitsch (1998), which conducted a material flow analysis for the consumption of chromium compounds in Germany. Results are used as a proxy for the European situation. Emission factors are mainly taken from IPPC (2002) and Anonymous (1998b).

The layout of the modules with general flow information, remarks, sources, values and uncertainty information are shown in Fig. 5.2 and Fig. 5.3. Ecoinvent meta information for this process are summarised in Tab. 5.1 on page 13. The following paragraphs describe calculations, sources and assumptions chosen in this study.

#### Infrastructure and resource use

The mobile equipment used for exploration like drilling machines, scoop trams and underground trucks is inventoried with the diesel consumption reported in Adelhardt & Antrekowitsch (1998). The used module “diesel, burned in building machine” includes the infrastructure of the machines. Use of conveyor belt was assessed according Tab. 5.2.

The world average chromite content after beneficiation amounts to 30.7% (44.9% Cr<sub>2</sub>O<sub>3</sub>), a yield from mining and beneficiation of 81% can be calculated (Adelhardt & Antrekowitsch (1998)).

According to part IX, chapter 2.1 of this report (Non Ferrous Metal Winning – Auxiliary Processes) no direct land use for underground mining (71%) is inventoried. Land use for open pit mining (29%) per tonne of ore is assessed with  $0.0025 \text{ m}^2/\text{t}$ . Thus the average land transformation for mining (underground and open pit) is  $0.000725 \text{ m}^2/\text{t}$  ore and the occupation is  $0.0218 \text{ m}^2/\text{a/t}$  ore (a lifetime of 30 years is assumed). Mining facilities are inventoried as unspecific module according to part IX (Non Ferrous Metal Winning – Auxiliary Processes).

**Tab. 5.2 Assumptions for infrastructure used in underground mining exploration. As lifetime ten years are estimated, and 5000 working hours yearly.**

| Action      | Share | Machine        | Throughput   | Unit       | Infrastructure       |
|-------------|-------|----------------|--------------|------------|----------------------|
|             |       |                | t / lifetime | Length (m) | m / t ore mined      |
| Open pit    | 29 %  | Conveyor belts | 100'000'000  | 10'000     | $1 \cdot 10^{-4}$    |
| Underground | 71 %  | Conveyor belts | 100'000'000  | 2'000      | $2 \cdot 10^{-5}$    |
| Mix         |       |                |              |            | $4.32 \cdot 10^{-5}$ |

## Mining

According to Adelhardt & Antrekowitsch (1998) 71 % of the chromite world production originates from underground mines. The overburden is disposed in piles near the mine, 1.77 kg per kg ore mined. The standard module for disposal of non-sulphidic overburden was chosen, which includes the land use and transformation through the disposal. No overburden is refilled according to Adelhardt & Antrekowitsch (1998). In the world-wide average  $0.35 \text{ m}^3/\text{t}$  crude ore of the pit water is used for the beneficiation. The remaining pit water is not inventoried as output but its emissions are inventoried together with the emissions of the beneficiation step.

## Beneficiation

Once the ore is mined, it is ground and classified. No flotation is needed in beneficiation. The separated gangue is disposed in tailings ponds, the concentrated ore is fed to the metallurgy, which is assumed to be on-site. The respective standard module “disposal, non-sulphidic tailings, off-site” is chosen (c.f. part IX (Non Ferrous Metal Winning – Auxiliary Processes), chapter 2.3 (Disposal of Tailings and Overburden)).

The beneficiation infrastructure is approximated with a similar facility from the aluminium production (c.f. part 1 (Aluminium)). Dust emissions comprise additionally the dust emissions due to mining activities. Hilbrans & Hinrichs (1999) assume for the nickel mining a share of 60% hydroelectricity due to the operations high and constant energy demand. The same assumption is made for chromite. The value comprises the electricity demand of the mining, which is expected to be small compared to the beneficiation.

**Steel:** The steel input (abrasion) for milling is documented in IPPC (2002) for a single Finnish chromite mine (Kula) and amounts to  $300 \text{ g} / \text{t}$  ground ore. The mill (jaws, balls, etc.) consists of special austenitic manganese steel of Hadfield grade, which is approximated with chromium steel in the inventory.

**Water use & effluents:** Water use is documented in Adelhardt & Antrekowitsch (1998) and is inventoried in this study as originating from river without pumping activities. Total emissions to water are chosen according to Tab. 5.3. These values are reported from a chromite site in Finland (Kula) and recommended values. The reported value of  $150 \text{ mg} / \text{l}$  COD is assumed to amount to 150% of BOD (Zimmermann et al. (1996)). Values for TOC are calculated using the standard procedure in

Frischknecht et al. (2003) approximating the unknown organic matter with the simplified formula for organic matter  $C_6H_{12}O_6$ . This gives 58.7 mg / l TOC and DOC respectively.

**Dust:** A case study in (Anonymous (1998a)) on a projected open cut gold-copper mining and processing facility in Australia predict dust emissions between 0.18 kg/t and 0.46 kg/t of dust produced per tonne of ore mined. The prediction was made using the Emission Estimation Technique Manual for Mining (NPI (2001)). The maximum value of 0.46 kg/t is chosen for total dust emissions, the value for underground mining is assessed with 80% of this amount. The dust composition is approximated with the chromite ore composition from Tab. 3.1. For  $PM_{10}$  values, the ratio TSP :  $PM_{10}$  :  $PM_{2.5}$  was approximated with values from manganese ore mining 101.72 : 50 : 5 g / t ore taken from the CEMIP-database<sup>3</sup>, i.e. one tonne emitted dust corresponds to 492 kg  $PM_{10}$  and 49 kg  $PM_{2.5}$  respectively. The airborne emissions resulting from mining and beneficiation activities are summed up in Tab. 5.4. In ecoinvent, for particulates differential classes are used adding up to the total dust emission, i.e. 49 kg  $PM_{<2.5}$ , 442 kg  $PM_{2.5-10}$  and 508 kg  $PM_{>10}$  per tonne of total dust.

### Data quality

**Mining:** Data for major mass fluxes are satisfactory, while the emissions into water may be overestimated due to different system boundaries of reported effluent volume from Adelhardt & Antrekowitsch (1998) and total effluent emissions reported in IPPC (2002) and Anonymous (1998b).

**Beneficiation:** Like in mining, the emissions to water can be overestimated. No emissions from the disposal of the tailings are modelled since no information about such emissions is available.

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<sup>3</sup> [http://www.air.sk/tno/cepmeip/em\\_factors.php](http://www.air.sk/tno/cepmeip/em_factors.php), accessed 01.03.2003

**Tab. 5.3 Effluent levels to be achieved during operation and after mine closure according to Anonymous (1998b) (Maximum value) and reported values from Kemi site (Finland) after IPPC (2002) (Reported value).**

| Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Maximum value (mg/l) | Reported value (mg/l) | This Study (mg/l) |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-----------------------|-------------------|
| Mining Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Base metal           | Chromite ore          |                   |
| pH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 6-9                  |                       |                   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                      |                       |                   |
| TSS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 50                   | 20.6                  | 50                |
| Oil and grease                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 10                   |                       | 10                |
| Cyanide                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1.0                  |                       | - <sup>b)</sup>   |
| Free                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.1                  |                       | - <sup>b)</sup>   |
| Weak acid dissociable (WAD)                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 0.5                  |                       | - <sup>b)</sup>   |
| COD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 150                  |                       | 150               |
| Arsenic                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0.1                  |                       | - <sup>c)</sup>   |
| Cadmium                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0.1                  |                       | - <sup>c)</sup>   |
| Chromium (hexavalent)                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0.1                  |                       | - <sup>a)</sup>   |
| Chromium, ion                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                      |                       | 0.1 <sup>a)</sup> |
| Chromium (in solid)                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                      | 0.05                  | - <sup>c)</sup>   |
| Copper                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0.5                  |                       | - <sup>c)</sup>   |
| Iron                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 3.5                  | 6.9                   | 6.9               |
| Lead                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.2                  |                       | - <sup>c)</sup>   |
| Mercury                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0.01                 |                       | - <sup>c)</sup>   |
| Nickel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0.5                  |                       | 0.5               |
| Zinc                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2                    |                       | - <sup>c)</sup>   |
| Total metals                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 10                   |                       | -                 |
| <sup>a)</sup> Inventoried in Ecoinvent as trivalent chromium ion. Hexavalent chromium is assumed to oxidise quickly to trivalent chromium in the course of abatement and aeration.<br><sup>b)</sup> chromite is no sulphidic ore, therefore no use of this substance and consequently no emission is assumed<br><sup>c)</sup> Maximum values relates to the whole range of base metal extraction. This specific emission is not assumed to be of importance in chromite exploitation. |                      |                       |                   |

**Tab. 5.4 Emissions to air resulting from mining activities per tonne mined ore. Values for total dust emission are taken from Anonymous (1998a), partition of fractions of particulate matter from the CEMIP-database (mining of manganese ore), the dust composition correspond to the average chromite ore from "3.1 Chromite ore" in this study.**

| Emission to air | Unit      | Emission per tonne mined ore |
|-----------------|-----------|------------------------------|
| Iron            | g / t ore | 48.5                         |
| Silicon         | g / t ore | 13.3                         |
| Chromium        | g / t ore | 126                          |
| Aluminium       | g / t ore | 20.2                         |
| Magnesium       | g / t ore | 41.0                         |
| PM<2.5          | g / t ore | 19.4                         |
| PM2.5-10        | g / t ore | 175                          |
| PM>10           | g / t ore | 201                          |
| total dust      | g / t ore | 395                          |

| General Flow Information                   |              |                      | Representation in ecoinvent                                                                                                                                                                                                                                 |                        |                            |                 |          |                                                           |            | Uncertainty Informations |                              |      |                    |                                                      |
|--------------------------------------------|--------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|----------------------------|-----------------|----------|-----------------------------------------------------------|------------|--------------------------|------------------------------|------|--------------------|------------------------------------------------------|
| Input                                      | Process Name | Output               | Remarks                                                                                                                                                                                                                                                     | Category               | Sub-category               | Infra-structure | Location | Modul name in ecoinvent                                   | Mean value | Unit                     | Source for mean value        | Type | Standard Deviation | General Comment                                      |
| Chromium in ground                         | mining       |                      | 81% yield assumed (19% Cr loss in mining / beneficiation), 1.48kg ore and 1.74kg overburden per kg chromite                                                                                                                                                 | resource               | in ground                  |                 |          | Chromium, 25.5 in chromite, 11.6% in crude ore, in ground | 0.38       | kg                       | (Adelhardt 1998)             | 1    | 1.13               | (2,2,3,1,1,2,12)                                     |
| diesel fuel (energy)                       |              |                      |                                                                                                                                                                                                                                                             | construction processes | machinery                  | No              | GLO      | diesel, burned in building machine                        | 0.081      | MJ                       | (Adelhardt 1998)             | 1    | 1.13               | (2,2,3,1,1,2,1)                                      |
| explosive                                  |              |                      |                                                                                                                                                                                                                                                             | construction processes | civil engineering          | No              | RER      | blasting                                                  | 1.5E-3     | kg                       | (Adelhardt 1998)             | 1    | 1.13               | (2,2,3,1,1,2,4)                                      |
| exploration infrastructure, conveyor belts |              |                      | exploration infrastructure, 2000 m belts assumed                                                                                                                                                                                                            | construction processes | machinery                  | Yes             | RER      | conveyor belt, at plant                                   | 6.4E-8     | m                        | estimation                   | 1    | 3.32               | (5,5,1,1,1,5,9); rough estimate                      |
| infrastructure of mining (underground)     |              |                      | 50 a lifetime; 400'000 t / a -> 5e-11 units per kg ore mined                                                                                                                                                                                                | metals                 | extraction                 | Yes             | GLO      | non-ferrous metal mine, underground                       | 5.3E-11    | unit                     | estimation                   | 1    | 3.32               | (5,5,1,1,1,5,9); rough estimate                      |
| infrastructure of mining (surface)         |              |                      | 51 a lifetime; 400'000 t / a -> 5e11 units per kg ore mined                                                                                                                                                                                                 | metals                 | extraction                 | Yes             | GLO      | non-ferrous metal mine, surface                           | 2.2E-11    | unit                     | estimation                   | 1    | 3.32               | (5,5,1,1,1,5,9); rough estimate                      |
| Land transformation from nature            |              |                      | direct land use, represents specific share of surface mines                                                                                                                                                                                                 | resource               | land                       |                 |          | Transformation, from unknown                              | 1.1E-6     | m2                       | (Martens 2002), calculations | 1    | 2.03               | (3,2,3,1,1,2,8); calculation based on reported value |
| Land transformation to mine                |              |                      | direct land use, represents specific share of surface mines                                                                                                                                                                                                 | resource               | land                       |                 |          | Transformation, to mineral extraction site                | 1.1E-6     | m2                       | (Martens 2002), calculations | 1    | 2.03               | (3,2,3,1,1,2,8); calculation based on reported value |
| Land use for mining                        |              |                      | direct land use, represents specific share of surface mines                                                                                                                                                                                                 | resource               | land                       |                 |          | Occupation, mineral extraction site                       | 3.2E-5     | m2a                      | (Martens 2002), calculations | 1    | 1.53               | (3,2,3,1,1,2,7); calculation based on reported value |
|                                            |              |                      | no overburden assumed to be refilled                                                                                                                                                                                                                        | waste management       | residual material landfill | No              | GLO      | disposal, non-sulfidic overburden, off-site               | 1.77       | kg                       | (Adelhardt 1998)             | 1    | 1.16               | (3,2,3,1,1,2,6)                                      |
|                                            |              | overburden, disposed |                                                                                                                                                                                                                                                             |                        |                            |                 |          |                                                           |            |                          |                              |      |                    |                                                      |
|                                            |              | effluents, mining    | (1.5 - 0.25) m3/t pit water per crude ore (0.25 m3/t used in beneficiation) without treatment. The composition is given in Tab. 5.3. The emissions are inventoried with the emissions of beneficiation. The uncertainties of the single emissions are high. |                        |                            |                 |          |                                                           | 1.9E-3     | m3                       | (Adelhardt 1998)             | 1    | 1.13               | (2,2,3,1,1,2,6)                                      |

Fig. 5.2 Flows for “chromite, at beneficiation – part mining” and their representation in the ecoinvent database. Values correspond to the functional unit of 1 kg of chromite.



| General Flow Information                |               |                          | Representation in ecoinvent                                                                                                                                                                                                                |                  |                            |                 |          |                                                       |            | Uncertainty Informations |                       |      |                    |                                  |
|-----------------------------------------|---------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------------|-----------------|----------|-------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|----------------------------------|
| Input                                   | Process Name  | Output                   | Remarks                                                                                                                                                                                                                                    | Category         | Sub-category               | Infra-structure | Location | Modul name in ecoinvent                               | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                  |
| electricity, from grid<br>beneficiation | beneficiation | waste heat               | Comprises electricity demand of mining. Assumption: 40% electricity from grid                                                                                                                                                              | electricity      | production mix             | No              | UCTE     | electricity, medium voltage, production UCTE, at grid | 0.0150     | kWh                      | (Adelhardt 1998)      |      |                    | (2,2,3,1,1,2,2)                  |
| hydropower                              |               | waste heat               | Comprises electricity demand of mining. Assumption: 60% hydropower                                                                                                                                                                         | hydropower       | power plants               | No              | RER      | electricity, hydropower, at run-of-river power plant  | 0.0225     | kWh                      | (Adelhardt 1998)      |      |                    | (3,2,3,1,1,2,2)                  |
| fresh and processwater (beneficiation)  |               |                          |                                                                                                                                                                                                                                            | resource         | in water                   |                 |          | Water, river                                          | 2.7E-3     | m3                       | (Adelhardt 1998)      | 1    | 1.13               | (2,2,3,1,1,2,12)                 |
| steel                                   |               |                          | Abrasion in mill (steel balls)                                                                                                                                                                                                             | metals           | extraction                 | No              | RER      | chromium steel 18/8, at plant                         | 4.5E-4     | kg                       | (IPPC, 2002)          | 1    | 1.2                | (2,3,2,3,1,3,4)                  |
| infrastructure of beneficiation         |               |                          | Aluminiumhydroxide production as proxy                                                                                                                                                                                                     | metals           | extraction                 | Yes             | RER      | aluminium hydroxide, plant                            | 3.1E-15    | unit                     | estimation            |      |                    | (5,5,1,1,1,5,9); rough estimate  |
|                                         |               | tailings, disposed       | no tailings refilled according to (Adelhardt. 1998)                                                                                                                                                                                        | waste management | residual material landfill | No              | GLO      | disposal, non-sulfidic tailings, off-site             | 0.49       | kg                       | (Adelhardt 1998)      | 1    | 1.16               | (3,2,3,1,1,2,6); reported values |
|                                         |               | effluents, beneficiation | Emissions inventoried according to composition in Tab. 5.3, they are not shown in this figure. Emissions have high uncertainties. No treatment of waste water.                                                                             |                  |                            |                 |          |                                                       | 2.3E-3     | m3                       | (Adelhardt 1998)      | 1    | 1.13               | (2,2,3,1,1,2,6); reported values |
|                                         |               | Dust                     | Including emissions from mining activities. Dust is the sum of particulate emissions represented in fractions PM<2.5, PM2.5-10, PM>10, including metals. These emissions are inventoried as shown in Tab 5.4 but not shown in this figure. |                  |                            |                 |          |                                                       | 5.9E-4     | kg                       | (NPI, 2001)           | 1    | 1.59               | (3,2,1,1,3,4,25)                 |
|                                         |               | Chromite                 | concentrate, 44.9% Chromium contained                                                                                                                                                                                                      | metals           | extraction                 | No              | GLO      | chromite, ore concentrate, at beneficiation           | 1          | kg                       | (Adelhardt 1998)      |      |                    |                                  |

Fig. 5.3 Flows for “chromite, at beneficiation – part beneficiation” and their representation in the ecoinvent database. Values correspond to the functional unit of 1 kg chromite.

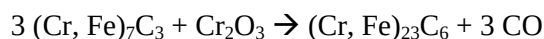
## 5.2 Winning of high-carbon ferrochromium

As seen in Fig. 5.1, high-carbon ferrochromium accounts for the major part of chromium consumption. Although the traditional type for steel manufacturing was the high-chrome low-carbon ferrochrome. Much more economic and hence of higher importance for the industry nowadays is high-carbon ferrochrome. Since the introduction of the AOD process (Argon-Oxygen-Decarburisation) the steel industry prefers high-carbon ferrochrome, which originates predominantly from stratiform deposits rich in iron (Adelhardt & Antrekowitsch (1998)).

### 5.2.1 Production process and infrastructure

High-carbon ferrochromium is produced almost exclusively by direct reduction of chromite ores with carbon (coke, coal, or charcoal) in large, three-phase submerged arc furnaces with 10 - 50 MVA capacity, corresponding to 15 000 – 60 000 t/a of ferrochromium production.

The oxides of iron and chromium present in the chromite can be readily reduced at high temperature with carbon. Because of the tendency of chromium to form carbides, a carbon-containing alloy is always obtained. Carbides with higher carbon content formed initially at lower temperature react at higher temperature with  $\text{Cr}_2\text{O}_3$  and form carbides with lower carbon content; finally, reduction of  $\text{SiO}_2$  – in the case of silicochromium production – starts at higher temperature. The basic reaction can be formalised as follows:



Electric arc furnaces work continuously with an input of chromite, coke, dolomite, silica and bauxite or other  $\text{Al}_2\text{O}_3$  containing materials. Metal and slag are tapped in regular intervals. The submerged arc furnaces use Söderberg electrodes where the electrode may be formed by hot paste, briquettes, blocks or paste cylinders.

The coke rate is calculated on the basis of the stoichiometric requirement of the oxides and on the amount of dissolved carbon in the alloy; allowance is made for some combustion at the top of the furnace and for reaction with moisture. The slag composition is important to produce metal of desired quality and to maintain smooth furnace operation.

Because more than 80 % of chromite ore mined in the western world is finer than 10 mm, agglomeration processes have been adopted to provide good burden porosity in the electric arc furnaces. In some ferrochromium plants, fines and lumpy ores are blended in ratios between 1 : 1 and 4 : 1. (Downing et al. (1997)).

In older plants with quite small furnaces the off-gas containing considerable amounts of CO is burned in the top and cleaned thereafter. Modern plants with semi-closed furnaces do not need to burn the CO. This results in a lower off-gas volume – factor 10 to 20 – and the possibility of re-using the gas as secondary fuel in other processes. By using a closed, sealed furnace the volume of the off-gas can be reduced by even a factor of 50 – 75. The smelting charge can be pre-heated in a shaft or rotary kiln by using the CO-gas from the smelting process. The yield of chromium amounts from 85% to 95%.

Triggered by enhanced demand of chromium in the metallurgical industry, bigger plants and new processes recently evolved. These processes - Outokumpu, Showa Denko, Krupp-CODIR and plasma - are discussed in detail in Adelhardt & Antrekowitsch (1998). They feature much lower specific energy demand and higher yield (around 90%).

Fluxing agents, e.g., quartzite or alumina (corundum or bauxite) and lime, are charged with the burden for slag formation.

Slag that contains metal is also crushed and the metal content recovered by means of slag-metal separation. Recovered slag-free lump metal can be sold, while slag contaminated metal and fines are generally recycled to the smelting process.

An example of a high-carbon ferrochromium production process using a closed submerged electric arc furnace is shown in Fig. 5.4.

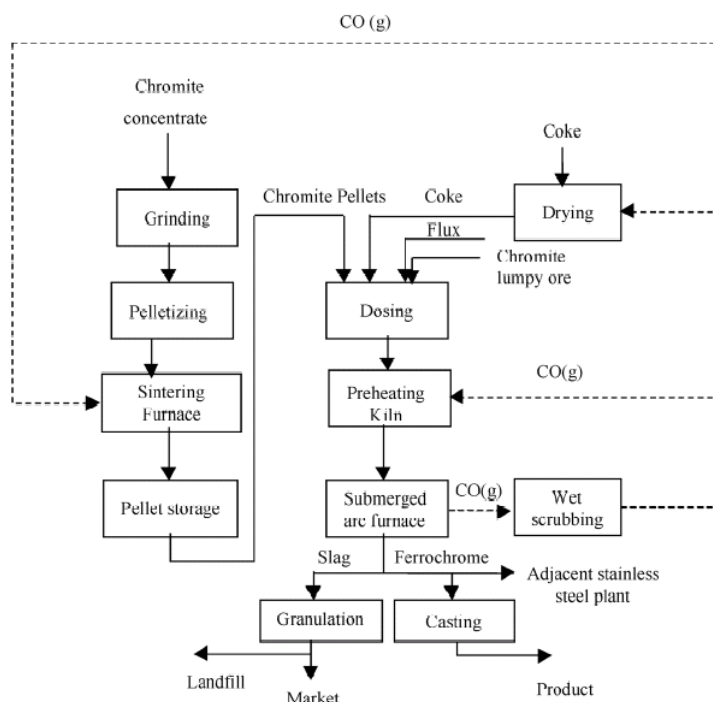


Fig. 5.4 Example of a High-carbon ferrochrome production process using a closed submerged electric arc furnace (IPPC (2001)).

## 5.2.2 Emission

**Air:** The main environmental impact by producing ferro-alloys is the emission of dust and fume from the smelting processes. Dust emissions occur as well from storage, handling and the pre-treatment of raw materials where fugitive dust emissions play an important role. Depending on the raw material and the process used other emissions to air are  $\text{SO}_2$ ,  $\text{NO}_x$ ,  $\text{CO}$ -gas,  $\text{CO}_2$ , polycyclic aromatic hydrocarbon (PAH), VOCs, and volatile metals. The formation of dioxins in the combustion zone and in the cooling part of the off-gas treatment system (de-novo synthesis) may be possible.

**Water:** Water emissions are produced from cooling, granulation and other processes and site related effluents. The wastewater generated by a wet cleaning abatement system plays also an important role.

## 5.2.3 Waste

Process residues and by-products with a significant amount is slag, filter dust and sludge and spent refractories. These materials are already recycled and reused to a large extent where it is possible. Rich slag, that means slag with a relatively high proportion of metal oxides, is used as raw material in other ferro-alloy productions.

## 5.2.4 Ferrochromium, high-carbon, 68% Cr, at plant in ecoinvent

This part is mainly based on Adelhardt & Antrekowitsch (1998). General process description was taken from (Fabian (1997)), process specific emissions from IPPC (2001), IPCC (1996), EPA (1998) and EC (2002).

Adelhardt & Antrekowitsch (1998) differentiates between old (submerged electric arc furnace) processes and the newer (Outokumpu, Showa Denko, Krupp-CODIR and plasma), and assumes a share of 50% each.

The module describes the world average production mix for ferrochromium in 1994 based on Adelhardt & Antrekowitsch (1998). It is designed for the use of the metal as alloying element. The layout of the modules with general flow information, remarks, sources, values and uncertainty information are shown in Fig. 5.5 (inputs) and Fig. 5.6 (output). Ecoinvent meta information for this process are summarised in Tab. 5.1 on page 13. The following paragraphs describe calculations, sources and assumptions chosen in this study. The functional unit of this process is one kg of ferrochrome with an average chromium content of 68%.

**Infrastructure and transport:** It is assumed that the bauxite used as flux is transported by ship over 20'000 km, the additives chromite, bauxite, silica, coke, slag and the disposals slag and cake by lorry over 50 km.

**Energy:** Adelhardt & Antrekowitsch (1998) calculates an average gross electricity demand of 3533 kWh containing smelting (4000 kWh and 2500 kWh for old and new process respectively), de-dusting (74.8 kWh), crushing & screening (0.68 kWh) and production of the electrode (207 kWh), assuming the new and the old process to be equal in production volume. In this study, the production of the electrodes is inventoried in a special module and has to be subtracted from this value, resulting in an average electricity consumption of 3325 kWh / t Cr. An overview over different in- and output values from different sources is given in Tab. 5.5

**Emissions to air:** Air emissions are taken from various sources, they are summarised and discussed in Tab. 5.6.

**Water use and emissions:** Water use for European ferrochrome production is reported in IPPC (2001), the average of 20 m<sup>3</sup> per tonne is chosen resulting in 10 m<sup>3</sup> of effluent (volume of process water). The effluent has to be treated in a communal waste-water treatment plant, a process which is inventoried separately in the chosen module "treatment, sewage, unpolluted, to wastewater treatment, class 3". Emissions to water inventoried are based on reported values after internal process water treatment and before waste water treatment plant (Tab. 5.7).

Tab. 5.5 Production of ferrochrome. In- and outputs for the production of 1 t ferrochrome.

| Input                                                                                                                                                                                                                                                                                                                                                                                                                        | Unit           | Adelhardt & Antrekowitsch (1998) | IPPC (2001) |      | This study              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----------------------------------|-------------|------|-------------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                              |                |                                  | low         | high |                         |
| Chromite                                                                                                                                                                                                                                                                                                                                                                                                                     | kg             | 2.43                             | 2.4         | 3    | <b>2.43</b>             |
| Electricity total                                                                                                                                                                                                                                                                                                                                                                                                            | kWh            | 3325                             | 3800        | 4500 | <b>3325</b>             |
| smelting (old)                                                                                                                                                                                                                                                                                                                                                                                                               | kWh            | 4000                             |             |      | <b>4000</b>             |
| smelting (new)                                                                                                                                                                                                                                                                                                                                                                                                               | kWh            | 2500                             |             |      | <b>2500</b>             |
| smelting (avg)                                                                                                                                                                                                                                                                                                                                                                                                               | kWh            | 3250                             |             |      | <b>3250</b>             |
| de-dust                                                                                                                                                                                                                                                                                                                                                                                                                      | kWh            | 74.8                             |             |      | <b>74.8</b>             |
| crushing                                                                                                                                                                                                                                                                                                                                                                                                                     | kWh            | 0.68                             |             |      | <b>0.68</b>             |
| coke                                                                                                                                                                                                                                                                                                                                                                                                                         | kg             | 340                              | 120         | 225  | <b>340<sup>a)</sup></b> |
| reductant                                                                                                                                                                                                                                                                                                                                                                                                                    |                |                                  | 550         | 700  | <b>-<sup>a)</sup></b>   |
| slag                                                                                                                                                                                                                                                                                                                                                                                                                         | t              | 0.80                             |             |      | <b>0.80</b>             |
| electrode                                                                                                                                                                                                                                                                                                                                                                                                                    | kg             | 19.99                            | 7           | 20   | <b>20.0</b>             |
| silica                                                                                                                                                                                                                                                                                                                                                                                                                       | kg             | 46.2                             |             |      | <b>46.2</b>             |
| fluxes                                                                                                                                                                                                                                                                                                                                                                                                                       | kg             |                                  | 100         | 400  | <b>204<sup>b)</sup></b> |
| process water                                                                                                                                                                                                                                                                                                                                                                                                                | m <sup>3</sup> |                                  | 5           | 15   | <b>10</b>               |
| cooling water                                                                                                                                                                                                                                                                                                                                                                                                                | m <sup>3</sup> |                                  | 5           | 15   | <b>10</b>               |
| <b>Output</b>                                                                                                                                                                                                                                                                                                                                                                                                                |                |                                  |             |      |                         |
| Cr in ferrochromium                                                                                                                                                                                                                                                                                                                                                                                                          | t              | 0.68                             |             |      |                         |
| ferrochromium                                                                                                                                                                                                                                                                                                                                                                                                                | t              | 1                                | 1           | 1    |                         |
| slag                                                                                                                                                                                                                                                                                                                                                                                                                         | t              | 2.65                             | 1           | 1.7  | <b>2.65</b>             |
| dust in scrubber                                                                                                                                                                                                                                                                                                                                                                                                             | kg             | 47.6                             | 20          | 80   | <b>47.6</b>             |
| <sup>a)</sup> Not evident, whether what reductant comprises coke or not. In this study it is assumed to be the sum containing coke and other fuels. Therefore the reported value for coke in Adelhardt & Antrekowitsch (1998) is taken as mass for reductant and inventoried as coke.<br><sup>b)</sup> Derived from the average flux of 250 kg less the 46.2 kg reported silica; this amount will be inventoried as bauxite. |                |                                  |             |      |                         |

**Tab. 5.6 Emissions to air in the production of 1 tonne ferrochrome. Values are from different sources: BREF (IPPC (2001)), IPCC (IPCC (1996)), EPA (EPA (1998)) and EEA (EC (2002)). Chosen values for this study are the average of the given range, for contradictory values see notes below.**

| Emission to Air   | Unit   | BREF  |       | IPPC            |      | EPA               |                    | EEA             | This study                 |
|-------------------|--------|-------|-------|-----------------|------|-------------------|--------------------|-----------------|----------------------------|
|                   |        | Low   | High  | Low             | High | Low <sup>b)</sup> | High <sup>a)</sup> |                 |                            |
| NO <sub>x</sub>   | kg / t | 0.5   | 1.5   | 0.05            | 11.7 |                   |                    |                 | <b>5.88 <sup>c)</sup></b>  |
| NM VOC            | kg / t |       |       | 1.7             |      |                   |                    |                 | <b>1.7</b>                 |
| CO                | kg / t |       |       | 1 <sup>g)</sup> |      |                   |                    |                 | <b>1</b>                   |
| PM<2.5            | kg / t |       |       |                 |      | 0.96              | 49                 |                 | <b>25.0</b>                |
| PM2.5-10          | kg / t |       |       |                 |      | 0.24              | 22                 |                 | <b>11.1</b>                |
| PM>10             | kg / t |       |       |                 |      | 0                 | 7                  |                 | <b>3.5</b>                 |
| TSP               | kg / t | 0.5   | 1     |                 |      | 1.2               | 78                 |                 | <b>39.6 <sup>d)</sup></b>  |
| SO <sub>2</sub>   | kg / t | 0.2   | 3     |                 |      |                   |                    |                 | <b>1.6</b>                 |
| PCDD              | µg / t |       |       |                 |      |                   |                    | 1 <sup>f)</sup> | <b>1</b>                   |
| Cr <sup>III</sup> | kg / t | 1%    | 15%   |                 |      |                   |                    |                 | <b>8% <sup>e)</sup></b>    |
| Cr <sup>VI</sup>  | kg / t | 0.10% | 0.30% |                 |      |                   |                    |                 | <b>0.20% <sup>e)</sup></b> |
| CN                | kg / t | 0.02  | 0.05  |                 |      |                   |                    |                 | <b>0.035</b>               |

<sup>a)</sup> No abatement  
<sup>b)</sup> ESP facility  
<sup>c)</sup> The average of the IPCC study is preferred to the value in BREF  
<sup>d)</sup> Averages of EPA values are chosen for world wide mix.  
<sup>e)</sup> Values given as percent in dust (TSP), resulting in emissions of 3.17 kg Cr<sup>III</sup> and 79.2 g Cr<sup>VI</sup> per tonne ferrochrome.  
<sup>f)</sup> SNAP-Code 040309, „Other - Installations for smelting and casting of non ferrous metals“  
<sup>g)</sup> Value taken from ferrosilicon production

**Tab. 5.7 Emissions to water in the production of 1 tonne ferrochromium. The reported values are before final waste-water treatment and after internal abatement including lamella thickener, vacuum filter and a sand and carbon filter. Based on the chosen values, final emissions into water are calculated.**

| Emission to Water<br>(before treatment) | Unit   | BREF |      | This Study        |
|-----------------------------------------|--------|------|------|-------------------|
|                                         |        | Low  | High |                   |
| TSS                                     | kg / t | 5    | 10   | 7.5 <sup>a)</sup> |
| Cr                                      | mg / t | 10   | 20   | 15 <sup>b)</sup>  |
| PAH                                     | mg / t | 300  | 1000 | 650 <sup>c)</sup> |
| dry cake                                | kg / t | 200  | 400  | 300 <sup>d)</sup> |

<sup>b)</sup> Zimmermann et al. (1996) reports a transfer-coefficient for chromium into water of 51.4%. This gives an emission after treatment of 7.7 mg Cr / t  
<sup>a)</sup> An elimination of 95% TSS assumed, resulting in an emission after treatment of 375 g TSS / t  
<sup>c)</sup> Based on Zimmermann et al. (1996) transfer coefficients of 10% to water is assumed, resulting in emissions of 65 mg PAH / t. Emissions to air as CO<sub>2</sub> and NMVOC are neglected.  
<sup>d)</sup> Dry cake results from internal waste-water treatment. In this study it is assumed to be disposed as inert waste to an inert material landfill.

## Data quality

Process data from Adelhardt & Antrekowitsch (1998) is satisfactory and valid for over 74 % of the world-wide primary chromium production. Process specific emissions to air and water had to be taken from another source covering another geography and technology. Nevertheless the overall quality of the modules is fair.

## 5. System Characterisation

| General Flow Information  |                                              |            | Representation in ecoinvent                          |                        |                 |                 |          |                                                       |            | Uncertainty Informations |                               |      |                    |                                                   |
|---------------------------|----------------------------------------------|------------|------------------------------------------------------|------------------------|-----------------|-----------------|----------|-------------------------------------------------------|------------|--------------------------|-------------------------------|------|--------------------|---------------------------------------------------|
| Input                     | Process Name                                 | Output     | Remarks                                              | Category               | Sub-category    | Infra-structure | Location | Modul name in ecoinvent                               | Mean value | Unit                     | Source for mean value         | Type | Standard Deviation | General Comment                                   |
| Chromite                  | ferrochromium, high-carbon, 68% Cr, at plant |            |                                                      |                        |                 |                 |          |                                                       |            |                          |                               |      |                    |                                                   |
| slag                      |                                              |            |                                                      |                        |                 |                 |          |                                                       |            |                          |                               |      |                    |                                                   |
| electricity, grid         |                                              | waste heat | 40% electricity from grid                            | electricity            | production mix  | No              | UCTE     | electricity, medium voltage, production UCTE, at grid | 1.33       | kWh                      | (Adelhardt 1998), estimations | 1    | 1.16               | (3,2,3,1,1,2,2); estimate based on reported value |
| hydropower                |                                              | waste heat | 60% electricity from hydropower                      | hydro power            | power plants    | No              | RER      | electricity, hydropower, at run-of-river power plant  | 2.00       | kWh                      | (Adelhardt 1998), estimations | 1    | 1.16               | (3,2,3,1,1,2,2); estimate based on reported value |
| fossile energy (coke)     |                                              |            |                                                      | hard coal              | heating systems | No              | RER      | hard coal, burned in industrial furnace 1-10MW        | 9.724      | MJ                       | (Adelhardt 1998)              | 1    | 1.13               | (2,2,3,1,1,2,1)                                   |
| silica sand               |                                              |            |                                                      | construction materials | additives       | No              | DE       | silica sand, at plant                                 | 0.0462     | kg                       | (Adelhardt 1998)              | 1    | 1.13               | (2,2,3,1,1,2,4)                                   |
| fluxes                    |                                              |            |                                                      | metals                 | extraction      | No              | GLO      | bauxite, at mine                                      | 0.204      | kg                       | (IPPC, 2001), estimations     | 1    | 1.74               | reported values                                   |
| Electrode graphite        |                                              |            |                                                      | metals                 | extraction      | No              | RER      | anode, aluminium electrolysis                         | 0.0200     | kg                       | (Adelhardt 1998)              | 1    | 1.13               | (2,2,3,1,1,2,3)                                   |
| process water             |                                              |            |                                                      | resource               | in water        |                 |          | Water, river                                          | 0.0100     | m3                       | (IPPC, 2001)                  | 1    | 1.11               | (2,3,2,3,1,3,12)                                  |
| cooling water             |                                              |            |                                                      | resource               | in water        |                 |          | Water, cooling, unspecified natural origin            | 0.0100     | m3                       | (IPPC, 2001)                  | 1    | 1.11               | (2,3,2,3,1,3,2)                                   |
| transport (sea)           |                                              |            | 20000 km for bauxite                                 | transport systems      | ship            | No              | OCE      | transport, transoceanic tanker                        | 4.08       | tkm                      | estimate                      | 1    | 2.11               | (4,4,1,1,1,5,5); calculation based on assumptions |
| transport (road)          |                                              |            | 50 km for chromite, bauxit, silica, coke, slag, cake | transport systems      | road            | No              | RER      | transport, lorry >16t, fleet average                  | 0.341      | tkm                      | estimate                      | 1    | 2.11               | (4,4,1,1,1,5,5); calculation based on assumptions |
| production infrastructure |                                              |            |                                                      | metals                 | extraction      | Yes             | GLO      | non-ferrous metal smelter                             | 7.2E-12    | unit                     | estimate                      | 1    | 3.32               | (5,5,1,1,1,5,9); rough estimate                   |

Fig. 5.5 Flows for “ferrochromium – inputs” and their representation in the ecoinvent database. Values correspond to the functional unit of 1 kg ferrochromium.

| General Flow Information                     |              |                  | Representation in ecoinvent                                                                                                                                                      |                  |                         |                 |          |                                                                   |            | Uncertainty Informations |                       |      |                    |                          |
|----------------------------------------------|--------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------------------|-----------------|----------|-------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|--------------------------|
| Input                                        | Process Name | Output           | Remarks                                                                                                                                                                          | Category         | Sub-category            | Infra-structure | Location | Modul name in ecoinvent                                           | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment          |
| ferrochromium, high-carbon, 68% Cr, at plant | m            | Effluents        | Inventoried as wastewater treatment w/o pollution. For overall process, waterborn emission see Tab. 5.7                                                                          | waste management | wastewater treatment    | No              | CH       | treatment, sewage, unpolluted, to wastewater treatment, class 3   | 0.010      | m3                       | (Adelhardt 1998)      | 1    | 1.13               | (2,2,3,1,1,2,6)          |
|                                              | m            | dry cake         | due to effluent, Tab 5.7                                                                                                                                                         | waste management | inert material landfill | No              | CH       | disposal, inert waste, 5% water, to inert material landfill       | 0.300      | kg                       | (IPPC, 2001)          | 1    | 1.33               | reported values          |
|                                              | m            | NOx              |                                                                                                                                                                                  | air              | low population density  |                 |          | Nitrogen oxides                                                   | 5.9E-3     | kg                       | (IPPC, 2001)          | 1    | 1.99               | reported values          |
|                                              | m            | NMVOG            |                                                                                                                                                                                  | air              | low population density  |                 |          | NMVOC, non-methane volatile organic compounds, unspecified origin | 1.7E-3     | kg                       | (IPPC, 2001)          | 1    | 1.52               | (2,3,2,3,1,3,16)         |
|                                              | m            | Dust (TSP)       | Dust consists of PM<2.5, PM2.5-10, PM>10, including Cr III, Cr VI and CN These specific emissions are inventoried according to table 5.6, but they are not shown in this figure. |                  |                         |                 |          |                                                                   | 4.0E-2     | kg                       | (IPPC, 2001)          | 1    | 1.97               | based on reported values |
|                                              | m            | SO2              |                                                                                                                                                                                  | air              | low population density  |                 |          | Sulfur dioxide                                                    | 1.6E-3     | kg                       | (IPPC, 2001)          | 1    | 1.88               | based on reported values |
|                                              | m            | CO               |                                                                                                                                                                                  | air              | low population density  |                 |          | Carbon monoxide, fossil                                           | 1.0E-3     | kg                       | (IPPC, 2001)          | 1    | 5.01               | (2,3,2,3,1,3,17)         |
|                                              | m            | PCDD             | de-novo synthesis                                                                                                                                                                | air              | low population density  |                 |          | Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin          | 1.0E-12    | kg                       | (IPPC, 2001)          | 1    | 3.01               | (2,3,2,3,1,3,21)         |
|                                              | m            | slag, disposal   | inventoried as disposal of inert material, thus neglecting the potentially important (but quantitatively unknown) heavy metal emissions                                          | waste management | inert material landfill | No              | CH       | disposal, inert waste, 5% water, to inert material landfill       | 2.65       | kg                       | (Adelhardt 1998)      | 1    | 1.1                | (2,2,2,3,1,3,6)          |
|                                              | m            | dust in scrubber | from wet scrubber, deposited in landfill                                                                                                                                         | waste management | inert material landfill | No              | CH       | disposal, inert waste, 5% water, to inert material landfill       | 0.048      | kg                       | (Adelhardt 1998)      | 1    | 1.11               | (2,3,2,3,1,3,6)          |
|                                              | m            | ferrochromium    |                                                                                                                                                                                  | metals           | extraction              | No              | GLO      | ferrochromium, high-carbon, 68% Cr, at plant                      | 1          | kg                       | (Adelhardt 1998)      | 1    |                    |                          |

Fig. 5.6 Flows for “ferrochromium – outputs” and their representation in the ecoinvent database. Values correspond to the functional unit of 1 kg ferrochromium.



### 5.2.5 Ferrochromium, high-carbon, 68% Cr, at regional storage in ecoinvent

The module describes the ferrochromium consumed Europe in 1994. The situation for Germany described in Adelhardt & Antrekowitsch (1998) is taken as proxy for Europe. It is designed for the use of the metal as alloying element. The layout of the modules with general flow information, remarks, sources, values and uncertainty information are shown in Fig. 5.7. Ecoinvent meta information for this process are summarised in Tab. 5.1 on page 13. The following paragraphs describe calculations, sources and assumptions chosen in this study. The functional unit of this process is one tonne of ferrochrome consumed in Europe with an average chromium content of 68%.

Adelhardt & Antrekowitsch (1998) examined the transport means and distances of ferrochromium from different supplying countries to Germany (Tab. 5.8 and Tab. 5.9). Since imported chromite is determined for the production of low-carbon ferrochrome, no import of chromite is considered for the production of high-carbon ferrochrome, i. e. all of the high-carbon ferrochrome consumed in Europe is considered to be imported. Besides transport services no further impacts – such as energy consumption through reloading and transferring – are inventoried.

**Tab. 5.8 Chromite transports to Germany for low-carbon ferrochrome production in 1994. Mass, distance and resulting transport intensity. Data from Adelhardt & Antrekowitsch (1998).**

|          | Unit | South Africa | Turkey | Brasilia | other | total            |
|----------|------|--------------|--------|----------|-------|------------------|
| chromite | t    | 128880       | 25540  | 10700    | 7410  | 172530           |
| ship sea | km   | 15500        |        | 8900     | 12200 | <b>12654 tkm</b> |
| train    | km   | 1200         | 300    | 300      | 600   | <b>985 tkm</b>   |
| lorry    | km   |              | 300    | 300      | 300   | <b>76 tkm</b>    |

**Tab. 5.9 Ferrochromium transports to Germany in 1994. Mass, distance and resulting transport intensity. Data from Adelhardt & Antrekowitsch (1998).**

|                                                                                                                                                                                                                                                                                                                                                                                                          | unit | South Africa | Russia             | Norway              | other  | total            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--------------|--------------------|---------------------|--------|------------------|
| ferrochromium                                                                                                                                                                                                                                                                                                                                                                                            | t    | 88670        | 85470              | 45040               | 135480 | 354660           |
| ship sea                                                                                                                                                                                                                                                                                                                                                                                                 | km   | 15500        | 3159               | 17900 <sup>a)</sup> | 12186  | <b>11565 tkm</b> |
| train                                                                                                                                                                                                                                                                                                                                                                                                    | km   | 900          | 5300 <sup>c)</sup> | 2800 <sup>b)</sup>  | 3000   | <b>3004 tkm</b>  |
| <sup>a)</sup> Value consists of 2'400 km direct transport from Norway and additional 15'500 km previous transportation to Norway<br><sup>b)</sup> Value consists of 2'400 km direct transport from Norway and additional 2'500 km previous transportation to Norway.<br><sup>c)</sup> Value consists of 2'500 km direct transport from Russia and additional 2'500 km previous transportation to Russia. |      |              |                    |                     |        |                  |

| General Flow Information         |                                                         |                                    | Representation in ecoinvent     |                   |              |                 |          |                                                         |            | Uncertainty Informations |                       |      |                    |                                                   |
|----------------------------------|---------------------------------------------------------|------------------------------------|---------------------------------|-------------------|--------------|-----------------|----------|---------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|---------------------------------------------------|
| Input                            | Process Name                                            | Output                             | Remarks                         | Category          | Sub-category | Infra-structure | Location | Modul name in ecoinvent                                 | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                                   |
| ferrochromium production         | ferrochromium, high-carbon, 68% Cr, at regional storage |                                    | global average                  | metals            | extraction   | No              | GLO      | ferrochromium, high-carbon, 68% Cr, at plant            | 1          | kg                       |                       | 1    | 1.13               | (2,2,3,1,1,2,4)                                   |
| Transport ferrochromium over sea |                                                         |                                    | distances and means see tab 5.9 |                   | ship         | No              | OCE      | transport, transoceanic tanker                          | 11.6       | tkm                      | (Adelhardt 1998)      | 1    | 2.02               | (2,2,3,1,1,2,5); calculation based on assumptions |
| Transport ferrochromium by train |                                                         |                                    | distances and means see tab 5.9 | transport systems | train        | No              | RER      | transport, freight, rail                                | 3.00       | tkm                      | (Adelhardt 1998)      | 1    | 2.02               | (2,2,3,1,1,2,5); calculation based on assumptions |
|                                  |                                                         | ferrochromium, at regional storage |                                 | metals            | extraction   | No              | RER      | ferrochromium, high-carbon, 68% Cr, at regional storage | 1          | kg                       |                       |      |                    |                                                   |

Fig. 5.7 Flows for “Ferrochromium, at regional storage” and its representation in the ecoinvent database. Values correspond to the functional unit of 1 kg ferrochromium.

## 5.3 Production of metallic chromium

Chromium metal can either be made from chromite ore and concentrates by carbo- and metallothermic reductions or by electrolysis from chromic acid. In order to get chromic acid the chromite ore is roasted with sodium carbonate followed by a leaching process with sulphuric acid to produce sodium chromate that can further be transformed to chromic acid IPPC (2001).

### Aluminothermic process

The usual method of reduction by carbon is inapplicable to the reduction of refractory oxides and ores. The products commonly covered under this title are usually required to be low in carbon and they are avid carbide formers. Also, no conventional refractories would stand up to the long reduction times at the high temperatures required. For such products, advantage is taken of the high temperatures achieved and the rapidity of metallothermic reactions using reactive metals, notably Al (Hall (1997)).

The thermic process uses aluminium as a reducing agent for chromium hydroxide. The charge is weighed and loaded into a bin, which is taken to an enclosed room to mix the contents. The firing pot is prepared by ramming refractory sand mixed with water around a central former. After ramming the firing pot, the inner surface is coated with a weak binder solution and dried under a gas fired hood before being transferred to the firing station. The raw material mix is automatically fed at a controlled rate into the firing pot, where the exothermic reaction takes place. When the metal has solidified following the reaction, the firing pot is removed and transferred by crane to a cooling conveyor. On removal from the cooling conveyor (by crane), the firing pot is placed on a stripping bogie for transfer to a stripping booth. Inside the closed booth, the pot casing is hoisted off the solidified metal/slag. The slag is separated from the Chromium metal “button” and sent to a despatch storage area. Water is used to reduce button temperature to below 100 °C. After cooling the metal button is transferred to other departments on site for cleaning, breaking, crushing and grinding to achieve the desired product size (IPPC (2001)).

### Electrolysis

In the electrolytic process normally high carbon ferrochrome is used as the feed material which is then converted into chromium alum by dissolution with sulphuric acid at temperatures at about 200 °C. After several process steps using crystallisation filtration ageing, a second filtration and a clarifying operation the alum becomes the electrolyte for a diaphragm cell. Chromium is plated onto stainless steel cathodes until it attains a thickness of ca. 3 mm. The process is very sensitive. The additional degassing (heating at 420 °C) stage is necessary because the carbon content of the electrolytic chromium is sometimes too high for further industrial applications. The cooled chromium metal is fragmented with a breaker prior to crushing and drumming. The generated slag can be reused as refractory lining or sold as abrasive or refractory material. IPPC (2001).

### 5.3.1 Emissions & Wastes

The general manufacturing of refractory metals normally takes place at high temperatures. The environmental impact on air, water and land can be classified as follows.

**Emissions to air:** Dust and fume emissions from smelting, hard metal and carbide production; other emissions to air are ammonia (NH<sub>3</sub>), acid fume (HCl), hydrogen fluoride (HF), VOC's and heavy metals.

**Emissions to water:** Overflow water from wet scrubbing systems; Wastewater from slag and metal granulation; Blow down from cooling water cycles.

**Wastes:** Dust, fume and sludge; Slag.

### 5.3.2 “Chromium, at regional storage” in Ecoinvent

The module describes the chromium metal consumed in Europe in 1994. The situation for Germany described in Adelhardt & Antrekowitsch (1998) is taken as proxy for Europe. It is designed solely for the use of the metal as alloying element in superalloys (Ni-Cr). The layout of the modules with general flow information, remarks, sources, values and uncertainty information are shown in Fig. 5.8. Ecoinvent meta information for this process are summarised in Tab. 5.1 on page 13. The following paragraphs describe calculations, sources and assumptions chosen in this study. The functional unit of this process is one tonne of chromium consumed in Europe.

11'150 t chromium metal was produced 1989 by aluminothermic processes, whereas 4'350 t came from electrolytic winning (Adelhardt & Antrekowitsch (1998)). The share in electrolytic processes is not rising significantly; a production level of 5'000 t in 2003 is assumed. This gives a share of the preferred aluminothermic process of 75 %.

#### Aluminothermic process

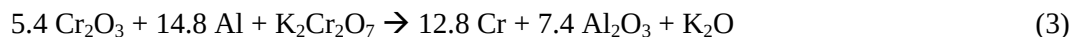
The aluminothermic process uses chromium hydroxide ( $\text{Cr}_2\text{O}_3$ ) as raw material and aluminium powder as reducing agent. It is the preferred production process with an assumed share of 75 %. Hall (1997) describes the fundamental reaction practice as follows:



Heat required for products at 2000 °C amounts to 540 kJ.  $\text{Al}_2\text{O}_3$  (H2273-H298) is 356 kJ, 2 Cr (H2273-H298) is 184 kJ. According to this, there is a heat balance, so there should be a satisfactory result. However, it is found in practice that 125 kJ extra heat are required, provided by Al plus  $\text{K}_2\text{Cr}_2\text{O}_7$ . The extra heat reaction can be formalised as follows



The energy required to heat the additional  $\text{K}_2\text{O}$  is assumed to be 200 kJ, which leads to a stoichiometric factor of  $(1768 - (184 + 2 \cdot 356 + 200))/125 = 5.4$ . An addition of  $5.4 \cdot (1) + (2)$  gives:



which gives the final stoichiometry. Hence per kg chromium 442 g  $\text{K}_2\text{Cr}_2\text{O}_7$ , 600 g Al and 1.23 kg  $\text{Cr}_2\text{O}_3$  is necessary. Adelhardt & Antrekowitsch (1998) gives a yield of 89 % - 92 %, an average of 90.5 % is chosen.

Assuming a  $\text{Cr}_2\text{O}_3$  content of 44.9% in the chromite concentrate and a yield of 90% in the  $\text{Cr}_2\text{O}_3$  production from chromite, for the 1.23 kg  $\text{Cr}_2\text{O}_3$  3.05 kg chromite concentrate are required.

$\text{CaO}$  is not used as flux in Cr manufacture as it spoils the slag for use in refractories and calcium chromite is a stable compound.  $\text{CaF}_2$  often is used as its low melting point of 1418 °C assists in promoting a smooth reaction. The amount used is often 1 – 2 % of the total charge, never more than 5 % (Hall (1997)). However, HF emissions are not considered in this study due to the lack of data.

For transportation the information in Adelhardt & Antrekowitsch (1998) is used. It is assumed, that the chromium is produced in Europe and the chromite is imported in the same transportation mix as illustrated in Tab. 5.8.

Adelhardt & Antrekowitsch (1998) states that the energy needed for the production of  $\text{Cr}_2\text{O}_3$  is 85 MJ per tonne chromium content, 250 MJ / t-Cr for the aluminothermic winning and 25 MJ / t-Cr for the refining. The 250 MJ / t-Cr comprise the production of the aluminium powder, which amounts to 185 MJ / t-Al according to EAA (2000). With this, a sum of 317 MJ per tonne metallic chromium is required: 153 MJ for the  $\text{Cr}_2\text{O}_3$ -production (1.23 kg with 68% chromium contained), 139 MJ for the aluminothermic process and finally 25 MJ for the refining. The production of aluminium and dichromate are not included in this figures and inventoried in the respective modules. The energy in the

$\text{Cr}_2\text{O}_3$ -production is assumed to originate from natural gas. For the aluminothermic process and refining a share of 60% hydroelectricity and 40% UCTE-mix is assumed.

### **Electrolytic process**

The rest of chromium (25 %) is still produced from ferrochrome in an electrolytic process. No information on material input and output were available. The energy demand is stated in Adelhardt & Antrekowitsch (1998) to be 18 kWh, whereas Papp (1994) reports 22.7 kWh per tonne chromium citing industry sources in 1993. In this study the higher value of 22.7 kWh (=81.8 MJ) was chosen. Considerable amounts of hydrogen and ammonia ought to be used for reduction and leaching, but no detailed information is available. Hence none of these emissions are taken in account.

| General Flow Information                                        |                               |                                    | Representation in ecoinvent                                     |                   |                 |                 |          |                                                       |            | Uncertainty Informations |                       |      |                    |                                                       |
|-----------------------------------------------------------------|-------------------------------|------------------------------------|-----------------------------------------------------------------|-------------------|-----------------|-----------------|----------|-------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-------------------------------------------------------|
| Input                                                           | Process Name                  | Output                             | Remarks                                                         | Category          | Sub-category    | Infra-structure | Location | Module name in ecoinvent                              | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                                       |
| chromite as raw material                                        | chromium, at regional storage | waste heat                         |                                                                 | metals            | extraction      | No              | GLO      | chromite, ore concentrate, at beneficiation           | 2.53       | kg                       | Calculation           | 1    | 1.57               | (5,2,2,1,1,5,4); estimate based on assumptions        |
| Dichromate                                                      |                               |                                    |                                                                 | chemicals         | inorganics      | No              | RER      | sodium dichromate, at plant                           | 0.366      | kg                       | Calculation           | 1    | 1.57               | (5,2,2,1,1,5,4); estimate based on assumptions        |
| Aluminium powder                                                |                               | waste heat                         |                                                                 | metals            | extraction      | No              | RER      | aluminium, primary, at plant                          | 0.498      | kg                       | Calculation           | 1    | 1.57               | (5,2,2,1,1,5,4); estimate based on assumptions        |
| Energy for chromium hydroxide                                   |                               |                                    | assumed to be from gas                                          | natural gas       | heating systems | No              | RER      | natural gas, burned in industrial furnace >100kW      | 103.0      | MJ                       | (Adelhardt 1998)      | 1    | 1.13               | (2,2,3,1,1,2,1)                                       |
| Hydropower for aluminothermic proc. & refining                  |                               | waste heat                         | assumed to be 60% of aluminothermic process and refining        | hydropower        | power plants    | No              | RER      | electricity, hydropower, at run-of-river power plant  | 22.6       | kWh                      | (Adelhardt 1998)      | 1    | 1.16               | (3,2,3,1,1,2,2)                                       |
| electricity for aluminothermic proc. & refining                 |                               | waste heat                         | assumed to be 40% of aluminothermic process and refining        | electricity       | production mix  | No              | UCTE     | electricity, medium voltage, production UCTE, at grid | 15.09      | kWh                      | (Adelhardt 1998)      | 1    | 1.16               | (3,2,3,1,1,2,2)                                       |
| Transport chromite over sea                                     |                               |                                    |                                                                 | transport systems | ship            | No              | OCE      | transport, transoceanic tanker                        | 12.7       | tkm                      | (Adelhardt 1998)      | 1    | 2.03               | (3,2,3,1,1,2,5); calculation based on reported values |
| Transport chromite by train                                     |                               |                                    |                                                                 | transport systems | train           | No              | RER      | transport, freight, rail                              | 0.99       | tkm                      | (Adelhardt 1998)      | 1    | 2.03               | (3,2,3,1,1,2,5); calculation based on reported values |
| Transport chromite by road                                      |                               | waste heat                         |                                                                 | transport systems | road            | No              | RER      | transport, lorry 32t                                  | 0.076      | tkm                      | (Adelhardt 1998)      | 1    | 2.03               | (3,2,3,1,1,2,5); calculation based on reported values |
| Ferrochromium for electrolysis                                  |                               |                                    |                                                                 | metals            | extraction      | No              | GLO      | ferrochromium, high-carbon, 68% Cr, at plant          | 0.250      | kg                       | (Adelhardt 1998)      | 1    | 1.13               | (2,2,3,1,1,2,4)                                       |
| Energy for electrolysis                                         |                               |                                    |                                                                 | electricity       | production mix  | No              | UCTE     | electricity, medium voltage, production UCTE, at grid | 5.68       | kWh                      | (Papp 1994)           | 1    | 1.13               | (2,2,3,1,1,2,2)                                       |
| Infrastructure, unspecified for aluminothermic and electrolysis |                               | ferrochromium, at regional storage | smeltery taken as proxy, all mass input (2.84kg) accounted for. | metals            | extraction      | Yes             | GLO      | non-ferrous metal, smelter                            | 1.08E-11   | unit                     | rough estimation      | 1    | 3.32               | (5,5,1,1,1,5,9); rough estimate                       |
|                                                                 |                               |                                    | global average                                                  | metals            | extraction      | No              | RER      | chromium, at regional storage                         | 1.00       | kg                       |                       |      |                    |                                                       |

Fig. 5.8 Flows for “Chromium, at regional storage” and its representation in the ecoinvent database. Values correspond to the functional unit of 1 kg ferrochromium.

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## Part VII

# Manganese

Data v2.1 (2009)

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## Comments on version 2.0

This part corresponds to the original report in Althaus et al. (2004). None of the changes reported in Frischknecht et al. (2006) regarded the datasets described in this part. Therefore no alteration of the original text was made for version 2.0.

No changes were made in version 2.1 compared to version 2.0.

Chapter “6 Cumulative Results and Interpretation” of the original report has been deleted for the current update. Note that the flow names in the tables displaying unit processes have not been updated to reflect the name changes with version 2.0. This namely affects the designation of transportation service flows.

## Summary

This chapter describes the production of ferromanganese and metallic manganese from cradle to consumer in Europe. Inventory data for the production is taken mainly from Adelhardt & Saiger (1999) together with background data from IPPC (2002), IPPC (2001) and Wellbeloved et al. (1997).

Only the primary production is inventoried because manganese is mainly used in alloys and recycled in them.

Ferromanganese is to be used for alloying steel while manganese is used for alloying aluminium or for direct use (e.g. sputtering)



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# 1 Introduction

This chapter describes the production of ferromanganese and metallic manganese from cradle to consumer in Europe. Inventory data for the production is taken mainly from Adelhardt & Saiger (1999) together with background data from IPPC (2002), IPPC (2001) and Wellbeloved et al. (1997).

Like many metals used as alloying element, manganese is recycled from steel-scrap in foundries directly as secondary alloys. Depending on their composition, these secondary manganese containing materials are used together with ferromanganese – and manganese ore – as secondary source for manganese. In this study, only the primary material ferromanganese and metallic manganese is considered. Secondary manganese sources – i.e. alloys from recycled steel scrap – have to be inventoried in the respective production module, e.g. in the steel production.

Over the last 15 to 20 years, the world pattern of the ferro-alloy market has deeply changed:

- The consumption of the developing countries has vastly increased with the development of their steel production;
- Their production has even more increased because they have taken a growing share of the traditional markets of the industrialised countries, where the steel production was stagnant or growing at a slow rate.
- The ferro-alloy industry is facing a growing proportion of imports, at first from the new industrialised countries and in recent years from the countries of East Europe and the CIS (former Soviet Union).

As a consequence, EU ferro-alloy production has been submitted to a difficult competition that results in a decreasing trend of the total amount of ferro-alloys produced. Although the last years showed some stabilisation, even a slightly increase of the production, the European industry remains very sensitive.

## 2 Reserves and Resources of Manganese

Metals are a special class of materials. They are not consumed but rather used during a product life cycle. When recovered the properties are theoretically identical to those in the former raw material. The metal can be re-used in another product life cycle, provided that its concentration makes a recovery economic. The recovered metal is referred to as secondary metal, whereas the metal originating directly from the ore is referred to as primary metal. In LCIs these two types – primary and secondary metal – are often differentiated because they bear a different ecological burden.

### 2.1 Primary Manganese

Manganese is contained in many minerals, of which approximately 250 can be regarded as true manganese minerals. The major ores, generally with manganese contents above 35 %, are hydrated or anhydrous oxides. The silicates and carbonates occur to a lesser extent. Economically significant manganese deposits are all of sedimentary origin, having been dissolved from crystalline rocks and deposited as the oxide, hydroxide, or carbonate. The major manganese minerals are listed in Tab. 2.1 (Wellbeloved et al. (1997)). Wad is the term given to the soft, earthy, hydrous mixture of manganese and iron oxides of variable manganese content.

The most important manganese ore deposits are located in the Republic of South Africa, the Soviet Union, Australia, Gabon in West Africa, and Brazil. The deposits in India and Mexico are of declining importance, and the majority of the production is consumed domestically (Tab. 2.2).

Another potentially important source of manganese are the deep-sea nodules located in certain areas of the ocean floor. Its exploitation has not yet been proven economic.

### 2.2 Secondary Manganese

The major end use of manganese is in iron and steel, and this is the major form in which manganese is recycled (ISS, iron and steel scrap). A smaller fraction ends up as aluminium beverage cans, which are recycled too. On a weight basis, aluminium beverage cans typically consisted 75% of bodies made from alloy 3004 and 22% of lids made from alloy 5182 in the US in 1994. The nominal manganese content of alloy 3004 is 1.1%, and that of alloy 5182 is 0.35% (USGS (1998)).

Manganese scrap enters the production process in the furnaces of the steel industry in the form of alloys with a specific composition. USGS (1998) reports a recycling rate of 37% and a manganese loss in the recycling process of 52% for the US in 1998.

Since no manganese is recycled within the primary industry but directly to the foundries of the iron and steel industry, no secondary manganese is considered in this study.

Tab. 2.1 Manganese minerals of economic importance (Wellbeloved et al. (1997))

| Mineral       | Formula                                                                        | Theoretical Mn content, % | Relative density |
|---------------|--------------------------------------------------------------------------------|---------------------------|------------------|
| Pyrolusite    | $\beta\text{-MnO}_2$                                                           | 63.2                      | 5                |
| Braunite      | $\text{Mn}^{2+}\text{Mn}_6^{3+}\text{SiO}_{12}$                                | 66.6                      | 4.7–4.9          |
| Manganite     | $\gamma\text{-MnOOH}$                                                          | 62                        | 4.3              |
| Psilomelane   | $(\text{Ba}, \text{Mn}^{2+})_3(\text{O}, \text{OH})_6\text{Mn}_8\text{O}_{16}$ | 45–60                     | 4.4–4.7          |
| Cryptomelane  | $\text{KMn}_8\text{O}_{16}$                                                    | 62                        | 4.3              |
| Hausmannite   | $\text{Mn}_2\text{MnO}_4$                                                      | 73                        | 4.7–5.0          |
| Jacobsite     | $\text{Fe}_2\text{MnO}_4$                                                      | 23.8                      | 4.8              |
| Bixbyite      | $(\text{Mn}, \text{Fe})_2\text{O}_3$                                           | 30–40                     | 5                |
| Rhodonite     | $(\text{Mn}, \text{Fe}, \text{Ca})\text{SiO}_3$                                | 42                        | 3.4–3.6          |
| Rhodochrosite | $\text{MnCO}_3$                                                                | 47.6                      | 3.3–3.6          |
| Bementite     | $\text{Mn}_3\text{Si}_2\text{O}_5(\text{OH})_4$                                | 43.2                      | 3.5              |

Tab. 2.2 Mine production and reserves of manganese in 2001. “Reserves” is referred to as “the part of the reserve base which could be economically extracted or produced at the time of determination”, while “Reserve base” denotes “that part of an identified resource that meets specified minimum (...) criteria related to current mining and production practices” (USGS (2003)).

|                        | Mine production   |                               | Reserves | Reserve base |
|------------------------|-------------------|-------------------------------|----------|--------------|
|                        | 2001 [kT Mn / a]  | 2002 [kT Mn / a] <sup>e</sup> | [kT Mn]  | [kT Mn]      |
| Australia              | 948               | 890                           | 32000    | 82000        |
| Brazil                 | 1430 <sup>e</sup> | 1500                          | 25000    | 52000        |
| China                  | 500 <sup>e</sup>  | 500                           | 40000    | 100000       |
| Gabon                  | 830 <sup>e</sup>  | 860                           | 20000    | 160000       |
| India                  | 600 <sup>e</sup>  | 630                           | 15000    | 33000        |
| Mexico                 | 100               | 100                           | 4000     | 9000         |
| South Africa           | 1479              | 1300                          | 18000    | 114000000    |
| Ukraine                | 930 <sup>e</sup>  | 960                           | 140000   | 520000       |
| Other countries        | 750 <sup>e</sup>  | 860                           | Small    | Small        |
| World total (rounded)  | 7600 <sup>e</sup> | 7600                          | 300000   | 5000000      |
| <sup>e</sup> estimated |                   |                               |          |              |

## 3 Characterisation of the Materials

Manganese ore is the raw material for several manganese alloys and manganese chemicals and finally manganese metal. Generally, high-carbon ferromanganese and silicomanganese are produced from a blend of manganese-containing ores, and in the case of silicomanganese, slags and silica are added.

### 3.1 Manganese ore

Typically, only ores containing a minimum of 35 % Mn are regarded as manganese ores. Ores containing 10 – 35 % Mn are categorised as ferruginous manganese ores, and iron ores with 5 – 10 % Mn are referred to as manganiferous ores (Wellbeloved et al. (1997)). Pyrolusite (a mineral form of manganese dioxide), braunite (an oxysilicate), and rhodochrosite (a manganese carbonate) are among the minerals more commonly found in manganese ores (USGS (1998)).

The three most important areas of application of manganese ores and the specific requirements, are as follows (Wellbeloved et al. (1997)):

1. Metallurgical grade ore for the iron and steel industry, which generally contains 38 – 55 % Mn. The phosphorus content should preferably be below 0.1 %, and the concentrations of  $\text{Al}_2\text{O}_3$ ,  $\text{SiO}_2$ ,  $\text{CaO}$ ,  $\text{MgO}$ , and S are important. The manganese/iron ratio is critical; a 7.5 : 1 ratio, for example, is required for a standard ferromanganese alloy with 78 % Mn.
2. Battery grade ore, containing 70 – 85 %  $\text{MnO}_2$  (44 – 54 % Mn). The ore should generally contain less than 0.05 % of metals more electronegative than zinc, such as copper, nickel, cobalt, and arsenic. The suitability of manganese dioxide for use in batteries depends on a number of factors, including the crystal structure, surface area, pore size distribution, particle shape and size, electrical conductivity, surface conditions, chemical composition, and structure defects.
3. Chemical grade ore whose specifications vary considerably depending on the end use. Included in this category are feed stocks for electrolytic manganese and manganese dioxide, manganese chemicals, colorants and, in the Republic of South Africa, an oxidant in uranium extraction.

Compositions of ores of some major operations are given in Tab. 3.1 and Tab. 3.2.

### 3.2 Manganese Metal

Manganese is a silver-grey metal, resembling iron. It is hard and very brittle, and its primary uses in a metallic form are as an alloying, desulphurising, and deoxidising agent for steel, cast iron, and non-ferrous metals. Due to its high tendency to oxidise, annealing colours are often evident. Manganese belongs to group 7 and period 4 of the periodic table with an atomic number of 25 and atomic mass of 54.94. Six isotopes have been produced, of which only  $^{55}\text{Mn}$  is stable. The other isotopes,  $^{51}\text{Mn}$ ,  $^{52}\text{Mn}$ ,  $^{54}\text{Mn}$ ,  $^{56}\text{Mn}$ , and  $^{57}\text{Mn}$ , have half-lives ranging from 46 min to 310 d. Manganese exists in four allotropic forms:  $\alpha$ ,  $\beta$ ,  $\gamma$ , and  $\delta$ . It has a low vapour pressure (Wellbeloved et al. (1997)).

Selected physical properties of manganese are given in Tab. 3.3.

### 3.3 Ferromanganese

A number of manganese-containing ferroalloys are manufactured which are used largely in the mild steel, foundry, and stainless steel industries. The principal manganese ferroalloys and their typical components are high-carbon ferromanganese (78% manganese, 7% carbon, balance mostly iron) and silicomanganese (66% manganese, 17% silicon, 2% carbon). The names and typical compositions of these alloys are given in Tab. 3.4.



**Tab. 3.1 Analyses of some manganese ores, metallurgical grade (Wellbeloved et al. (1997))**

| Producer                  | Product              | Mn    | Fe      | P         | SiO <sub>2</sub> | Al <sub>2</sub> O <sub>3</sub> | CaO     | MgO     |
|---------------------------|----------------------|-------|---------|-----------|------------------|--------------------------------|---------|---------|
| Samancor, RSA             | Mamatwan Lumpy       | 37–38 | 4–6     | 0.08      | 4–6              | 0.5                            | 14–16   | 2.4     |
| Samancor, RSA             | Mamatwan Ferruginous | 32–34 | 4–6     | 0.08      | 4–6              | 0.5                            | 14–16   | 3.5     |
| Samancor, RSA             | Wessels High Grade   | 50–51 | 9–11    | 0.08      | 5–7              | 0.5                            | 4–6     | 0.6     |
| Samancor, RSA             | Wessels Low Grade    | 38–40 | 18–20   | 0.08      | 5–7              | 0.5                            | 4–6     | 1.2     |
| Groote Eylandt, Australia | Metallurgical Lump   | 49.0  | 2.5     | 0.06      | 6.0              | 4.0                            | 0.05    | 0.10    |
| Groote Eylandt, Australia | Siliceous Lump       | 44.0  | 2.5     | 0.06      | 15.0             | 4.0                            | 0.05    | 0.10    |
| Amapa, Brazil             | Bitolado 48          | 48–49 | 5.5–6.9 | 0.08–0.11 | 2–3              | 3.7–5.5                        |         |         |
| Amapa, Brazil             | Carbonate Grosso     | 33–37 | 2.5–4   | 0.04–0.10 | 10–14            | 2.0–3.8                        | 3.3–5.2 | 2.2–4.5 |
| CVRD, Brazil              | Lumpy                | 45.0  | 3.7     | 0.08      | 3.6              | 9.8                            | 0.25    | 0.30    |
| Comilog, Gabon            | Gabon Lumpy          | 48–51 | 3–5     | 0.11–0.14 | 2.5–4            | 6                              | 0.11    | 0.07    |
| Soviet Union              | Tchiaturi Fines      | 48.2  | 1.1     | 0.142     | 9.2              | 1.6                            | 2.26    | 0.50    |

**Tab. 3.2 Typical compositions of some important battery grade manganese ores (chemical content on % dry basis, Wellbeloved et al. (1997))**

| Producer       | MnO <sub>2</sub> | Mn   | Fe  | Pb    | Cu    | Ni    | As    | Co    |
|----------------|------------------|------|-----|-------|-------|-------|-------|-------|
| CVRD, Brazil   | 76.5             | 51.0 | 3.2 | 0.035 | 0.050 | 0.080 | 0.004 | 0.035 |
| Comilog, Gabon | 83.6             |      | 1.8 |       | 0.03  | 0.05  |       | 0.095 |

Tab. 3.3 Some physical properties of metallic manganese (Wellbeloved et al. (1997)).

| Property                                        | $\alpha$ -Mn                     | $\beta$ -Mn                    | $\gamma$ -Mn                    | $\delta$ -Mn                           | Molten Mn                                                    |
|-------------------------------------------------|----------------------------------|--------------------------------|---------------------------------|----------------------------------------|--------------------------------------------------------------|
| Atomic number                                   | 25                               |                                |                                 |                                        |                                                              |
| Atomic mass                                     | 54.938                           |                                |                                 |                                        |                                                              |
| Crystal structure                               | complex cubic                    | complex cubic                  | fcc                             | bcc                                    |                                                              |
| Lattice parameter, nm                           | 0.8894                           | 0.6290                         | 0.3774                          | 0.3080                                 |                                                              |
| Transition temperature, °C                      | $\alpha \leftrightarrow \beta$   | $\beta \leftrightarrow \gamma$ | $\gamma \leftrightarrow \delta$ | $\delta \leftrightarrow \text{liquid}$ | liquid $\leftrightarrow$ gas                                 |
|                                                 | 727                              | 1095                           | 1104                            | 1244                                   | 2032                                                         |
| Standard electrode potential, V                 | 1.134                            |                                |                                 |                                        |                                                              |
| Magnetic susceptibility, m <sup>3</sup> /kg     | $1.21 \times 10^{-7}$            |                                |                                 |                                        |                                                              |
| Density, g/cm <sup>3</sup>                      | 7.44                             | 7.29                           | 7.21                            |                                        | $7.55\text{--}1.2 \times 10^{-3} \text{ t}$<br>(1244–1500°C) |
| Brinell hardness, MPa                           | 3923–4119                        |                                |                                 |                                        |                                                              |
| Hardness, Mohs scale                            | 5                                |                                |                                 |                                        |                                                              |
| Volume increase on transition, %                | $\alpha \leftrightarrow \beta$   | $\beta \leftrightarrow \gamma$ | $\gamma \leftrightarrow \delta$ | $\delta \leftrightarrow \text{liquid}$ |                                                              |
|                                                 | 3.4                              | 0.8                            | 0.8                             | 2.9                                    |                                                              |
| Resistivity, $\Omega \text{ cm}$                | $150\text{--}260 \times 10^{-6}$ | $90 \times 10^{-6}$            | $40 \times 10^{-6}$             |                                        |                                                              |
| Compressibility, m <sup>2</sup> /N              | $8.4 \times 10^{-7}$             |                                |                                 |                                        |                                                              |
| Coef. of linear expansion, K <sup>-1</sup>      | $22.3 \times 10^{-6}$            | $24.9 \times 10^{-6}$          | $14.8 \times 10^{-6}$           | $41.6 \times 10^{-6}$                  |                                                              |
| Latent heat of fusion, J/g                      | 244                              |                                |                                 |                                        |                                                              |
| Latent heat of vaporization, J/g                | 4020                             |                                |                                 |                                        |                                                              |
| Heat of transition, J/mol                       | 2240                             | 2282                           | 1800                            | 14655                                  | 219901                                                       |
| Specific heat, Jg <sup>-1</sup> K <sup>-1</sup> | 0.477                            | 0.482                          | 0.502                           |                                        |                                                              |
| Enthalpy, J/mol                                 | 4999                             |                                | 5112                            |                                        |                                                              |
| Entropy, Jmol <sup>-1</sup> K <sup>-1</sup>     | 32.0                             |                                | 32.3                            |                                        | 173.8                                                        |

Tab. 3.4 Types of ferromanganese and their general compositions (Wellbeloved et al. (1997)).

| Alloy                                 | Composition, % |          |         |
|---------------------------------------|----------------|----------|---------|
|                                       | Manganese      | Carbon   | Silicon |
| High-carbon ferromanganese (carburé)  | 72-80          | 7.5      | <1.25   |
| Medium-carbon ferromanganese (affiné) | 75-85          | <2.0     |         |
| Low-carbon ferromanganese (suraffiné) | 76-92          | 0.5-0.75 |         |
| Silicomanganese                       | 65-75          | <2.5     | 15-25   |
| Ferromanganese silicon                | 58-72          | 0.08     | 23-35   |
| Spiegeleisen                          | 16-28          | <6.5     | 11-45   |

## 4 Use / Application of Material

Metallurgical applications account for most manganese consumption, of which 85% to 90% in the US has been going to steel-making and about 8% to the manufacture of dry cell batteries. The preponderance of the manganese used for making batteries is now electrolytic manganese dioxide (EMD) because usage of natural battery ore has declined greatly. The manufacture of manganese chemicals, such as potassium permanganate, and agricultural use of manganese in animal feed and plant fertiliser as oxide, sulphate, and oxysulphate together account for another 5% of use. These patterns of domestic use are typical for other industrialised countries having well-developed steel industries (Fig. 4.1).

In non-metallurgical uses, the most common valences for manganese are two and four, and oxygen is the main element with which manganese is combined. Accordingly, the mineral and commodity chemistry of manganese centers on such compounds as manganous oxide ( $\text{MnO}$ ), manganese dioxide ( $\text{MnO}_2$ ), manganese carbonate ( $\text{MnCO}_3$ ), and manganese sulphate ( $\text{MnSO}_4$ ) (USGS (1998)). The predominant products and their production route is illustrated in Fig. 4.2.

### 4.1 Manganese ore

Manganese ore is the central intermediary product in the manganese industry. From it the three main product groups are made: ferromanganese for the use in the steel industry (over 90%) and manganese compounds for the use in the chemical industry and battery manufacture. As illustrated in Fig. 4.1, battery or chemical grade ore are processed into manganese metal or manganese oxides, whereas from metallurgical grade ores manganese alloys are made. Silicomanganese is made predominantly from high-silica ore.

### 4.2 Manganese Metal

The largest application of manganese metal in the metallurgical sector is in aluminium alloying, particularly the alloys from which items such as beverage cans and siding are made. The predominant form in which manganese is added to molten aluminium is briquette (typically 75% manganese, 25% aluminium) made by compacting powders of the two metals. In 1990, the amount of manganese metal used by the steel industry was about one-third as much as by the aluminium industry. Because use of manganese metal, as opposed to ferroalloy, is a relatively expensive way of adding manganese, the metal is added in steel making as a “trimming addition” to adjust final composition, particularly when an increase in carbon content is not permissible. Today the world market is in excess of  $140 \times 10^3$  t/a. The compositions of some alloys are given in Tab. 4.1.

### 4.3 Ferromanganese and Silicomanganese

Generally, high-carbon ferromanganese and silicomanganese are produced from a blend of manganese-containing ores, and in the case of silicomanganese, slag and silica are added. USGS (2003) estimates that manganese unit consumption in steel making in the US in 2001 amounts to 5.4 kg manganese per tonne raw steel. The use of manganese alloys depends strongly on the steel industry. Tab. 4.2 gives an overview over the end uses of ferromanganese in the U.S. in 2001.

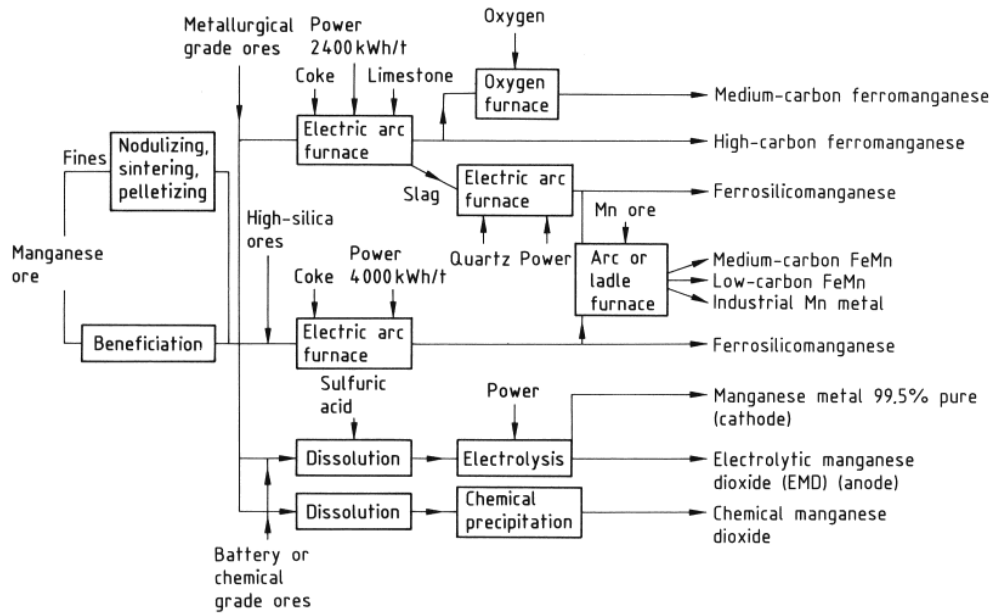


Fig. 4.1 Summary of manganese products and their process routes (Wellbeloved et al. (1997)).

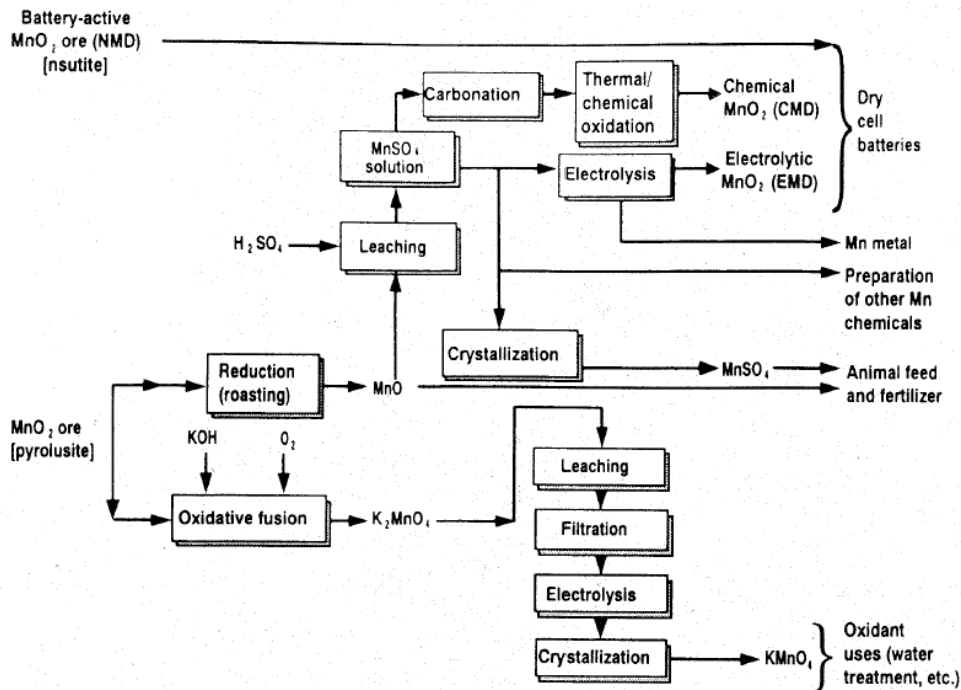


Fig. 4.2 Simplified diagram for the production of key manganese chemicals (Jones (1994)).

**Tab. 4.1 Manganese in alloys. Composition of typical alloys containing manganese (Wellbeloved et al. (1997)).**

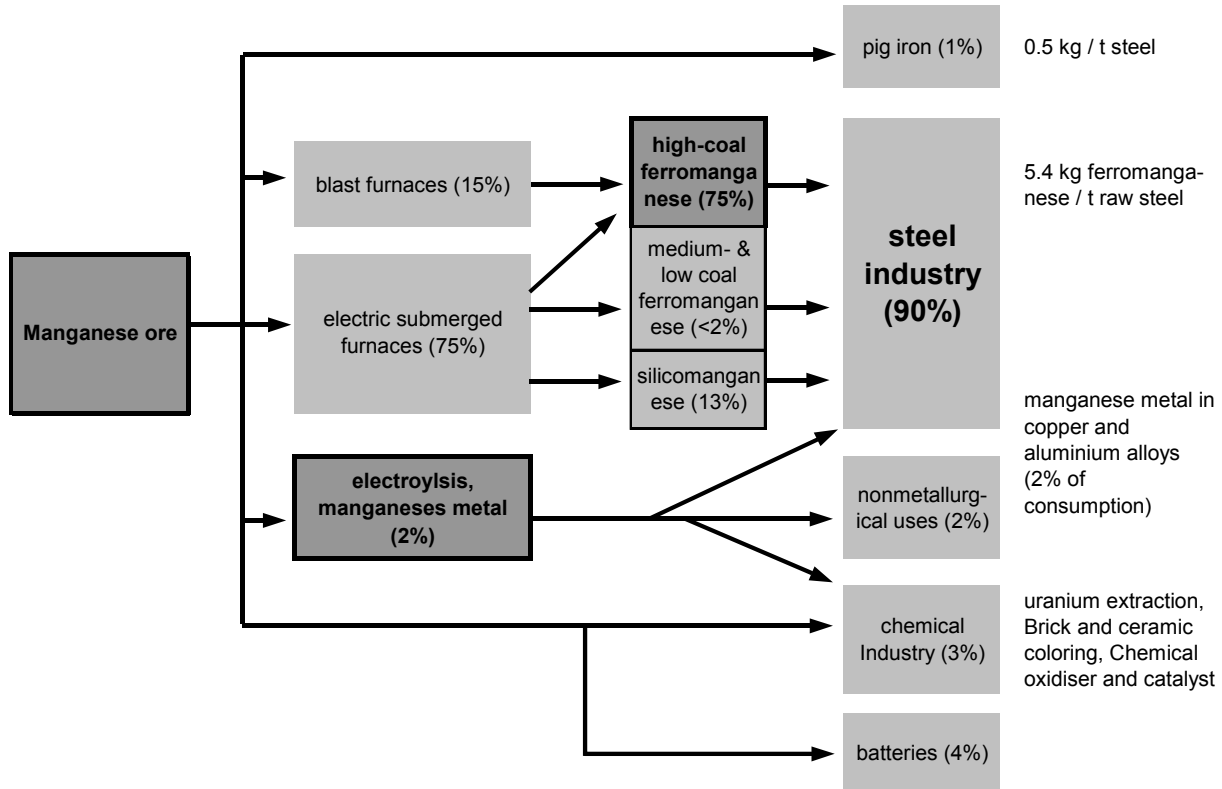
| <b>Alloy</b>                     | <b>Mn, %</b> | <b>Cu, %</b> | <b>Ni, %</b> | <b>Zn, %</b> | <b>Pb, %</b> | <b>Al, %</b> | <b>Fe, %</b> |
|----------------------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| Copper manganese                 | 10-40        | 56-70        |              |              |              |              |              |
| Manganese bronze                 | 5-15         |              |              |              |              |              |              |
| Resistin                         | 15           |              |              |              |              |              |              |
| Hard brass                       | 1.5          | 56           |              | 42           |              |              |              |
| Special brass 58 Al <sub>2</sub> | 2            | 58           | 3            | 37           | 0.5          | 1.6          |              |
| Special brass 59                 | 2.5          | 58           | 3            |              |              | 1.5          |              |
| Special brass 64                 | 4            | 65           |              |              |              | 7            | 4            |
| Manganin                         | 12           | 83           | 3-5          | 0.2          |              |              |              |

**Tab. 4.2 End uses of ferromanganese, silicomanganese and manganese metal in the U.S in 2001 (USGS (1998)).**

| End use                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Ferromanganese |                       |         | Silico-Manganese    | Manganese metal     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------------------|---------|---------------------|---------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | High carbon    | Medium and low carbon | Total   |                     |                     |
| Steel:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                |                       |         |                     |                     |
| Carbon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 122'000        | 85'300                | 207'000 | 61'200              | 621                 |
| High-strength, low-alloy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 16'900         | 3'230                 | 20'100  | 2'620               | 2                   |
| Stainless and heat-resisting                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 9'680          | 2                     | 9'680   | 4'180               | 1'210               |
| Full alloy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 14'300         | 4'080                 | 18'400  | 16'900              | 2                   |
| Unspecified <sup>3</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 998            | 292                   | 1'290   | 678                 | 1'120               |
| Total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 163'000        | 92'900                | 256'000 | 85'500              | 2'950               |
| Cast irons                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 8'100          | 454                   | 8'560   | 1'040               | 5                   |
| Superalloys                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | W              | W                     | W       | --                  | W                   |
| Alloys (excluding alloy steels and superalloys)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 902            | 436                   | 1'340   | 4                   | 16'100 <sup>5</sup> |
| Miscellaneous and unspecified                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | --             | W                     | W       | 4                   | W                   |
| Total consumption                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 172'000        | 93'800                | 266'000 | 86'500 <sup>6</sup> | 19'100              |
| Total manganese content <sup>7</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 134'000        | 75'300                | 210'000 | 59'900              | 19'100              |
| Stocks, consumers and producers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 9'240          | 15'700                | 24'900  | 16'200              | 2'210               |
| <p>W Withheld to avoid disclosing company proprietary data; included with "Alloys (excluding alloy steels and superalloys)."</p> <p>-- Zero.</p> <p><sup>1</sup> Data are rounded to no more than three significant digits; may not add to totals shown.</p> <p><sup>2</sup> Withheld to avoid disclosing company proprietary data; included with "Steel: Unspecified."</p> <p><sup>3</sup> Includes electrical and tool steel, and items indicated by footnote <sup>2</sup>.</p> <p><sup>4</sup> Withheld to avoid disclosing company proprietary data.</p> <p><sup>5</sup> Approximately 85% of this combined total was for consumption in aluminum alloys.</p> <p><sup>6</sup> Internal evaluation indicates that silicomanganese consumption is considerably understated.</p> <p><sup>7</sup> Estimated based on typical percent manganese content.</p> |                |                       |         |                     |                     |

## 5 System Characterisation

This study focuses to the main processes in the manganese production. As shown in chapter 4 “Use / Application of Material” no secondary production of manganese has to be considered, since the recycling of manganese occurs directly into the foundries in the form of re-melted iron and steel scrap.



**Fig. 5.1 Scheme of the manganese production system considered and applications of its products after Adelhardt & Saiger (1999). Percentages relate to the total manganese contained. In this study the processes for ore exploitation, production of high-coal ferromanganese and manganese metal are considered.**

According to Fig. 5.1, the following modules cover the manganese production system for the use of manganese in its main application areas “steel industry” as ferronickel and as manganese metal in different specific uses. They are summarised in Tab. 5.1.

- “manganese, concentrate, at beneficiation” Fig. 5.2, Fig. 5.3
- “ferromanganese, high-coal, 74% Mn, at regional storage” Fig. 5.4, Fig. 5.5
- “manganese, pure metal, at regional storage” Fig. 5.6

Tab. 5.1 Ecoinvent meta information for the processes for the production of ferromanganese and manganese metal.

| Name                         | manganese, concentrate, at beneficiation                                                                                                                                                                                                                                                                                      | ferromanganese, high-coal, 74.5% Mn, at regional storage                                                                                                                                                                                                                                                                                                                                                                                | manganese, pure metal, at regional storage                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location                     | GLO                                                                                                                                                                                                                                                                                                                           | RER                                                                                                                                                                                                                                                                                                                                                                                                                                     | RER                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Infrastructure Process       | 0                                                                                                                                                                                                                                                                                                                             | 0                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Unit                         | kg                                                                                                                                                                                                                                                                                                                            | kg                                                                                                                                                                                                                                                                                                                                                                                                                                      | kg                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Dataset Version              | 2.0                                                                                                                                                                                                                                                                                                                           | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Included Processes           | The module includes a mining and a beneficiation step with the mining infrastructure and disposal of overburden and tailings (see part II "Auxiliary Processes").                                                                                                                                                             | The module includes the reduction to high-carbon ferromanganese. No secondary production considered, since secondary manganese is recycled to the foundries and has to be inventoried in the respective module, i. e. when casting the final raw material in the steel production.                                                                                                                                                      | The module production by electrolysis from ore and by electrothermic process from ferromanganese and slag. Transportation to Europe is only partially considered.                                                                                                                                                                                                                                                              |
| Amount                       | 1                                                                                                                                                                                                                                                                                                                             | 1                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Local Name                   | Mangan, Erzkonzentrat, ab Aufbereitung                                                                                                                                                                                                                                                                                        | Ferromangan, hochgekohlt, 74.5% Mn, ab Regionallager                                                                                                                                                                                                                                                                                                                                                                                    | Mangan, ab Regionallager                                                                                                                                                                                                                                                                                                                                                                                                       |
| Synonyms                     |                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                         | Manganese metal // metallisches Mangan                                                                                                                                                                                                                                                                                                                                                                                         |
| General Comment              | The module describes the production of manganese ore in 1994 for a world-wide production average. It is designed for the use of manganese ore as intermediary product in the production of ferromanganese and manganese metal. For the use of this module in the battery manufacture, the data quality may not be sufficient. | The module describes the consumption of ferromanganese in Europe in 1995. It is designed for the use of the metal as alloying element. This module does not consider secondary sources of manganese. Process data is taken mainly from a material flow analysis for the consumption of chromium in Germany. Results are used as a proxy for the European situation. This module may be used as proxy for the silicomanganese production | The module describes the consumption of manganese metal in Europe in 1994. It is designed solely for the use of the metal in special applications like sputtering or as alloying element. This module is explicitly not to be used as alloying element in bulk iron or steel industry. It does not consider secondary sources of manganese. Process data is based on rough estimations; the overall data quality is very poor. |
| Start Date                   | 1994                                                                                                                                                                                                                                                                                                                          | 1994                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2003                                                                                                                                                                                                                                                                                                                                                                                                                           |
| End Date                     | 2003                                                                                                                                                                                                                                                                                                                          | 2003                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2003                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Data valid for entire period | 1                                                                                                                                                                                                                                                                                                                             | 1                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Geography Text               | This module relates to the global situation. Some modules used like lime, blasting and transport originate from data sets valid for Switzerland or Europe                                                                                                                                                                     | This module relates to the consumption in Europe, although the import mix for Germany is chosen.                                                                                                                                                                                                                                                                                                                                        | No information were available on production or consumption mix.                                                                                                                                                                                                                                                                                                                                                                |

| <b>Name (continued)</b> | <b>manganese, concentrate, at beneficiation</b>                                                                                                                                                    | <b>ferromanganese, high-coal, 74.5% Mn, at regional storage</b>                                                                           | <b>manganese, pure metal, at regional storage</b>                                                                                                                      |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Technology Text         | Mining is done 70% open pit and 30% underground, followed by a beneficiation of the ore through classification. Overburden and tailings are non-sulphidic, they are disposed near the mining site. | The ore is processed in blast furnaces (20%), electric arc furnaces without flux (27%), electric arc furnaces with calcareous flux (53%). | The metal is won by electrolysis (assumption: 25%) and electrothermic processes (assumption: 75%). No detailed information available, mainly based on rough estimates. |
| Representativeness      | 55%                                                                                                                                                                                                | 88%                                                                                                                                       | n. a.                                                                                                                                                                  |
| Production Volume       | 23.5 million tonnes in 1995                                                                                                                                                                        | 3.4 million tonnes manganese contained in ferromanganese in 2001                                                                          | 140'000 t in 2000                                                                                                                                                      |
| Sampling Procedure      | Literature                                                                                                                                                                                         | literature                                                                                                                                | literature                                                                                                                                                             |
| Extrapolations          | see Geography and Technology                                                                                                                                                                       | see Geography and Technology                                                                                                              | see Geography and Technology                                                                                                                                           |
| Uncertainty Adjustments | none                                                                                                                                                                                               | none                                                                                                                                      | none                                                                                                                                                                   |



## 5.1 Exploitation of manganese ore

### 5.1.1 Production process

**Mining:** Manganese ores are usually mined in open pits. Adelhardt & Saiger (1999) reports an estimated share of 70% surface mining activities in the mine production in 1994. The deposits are of various nature and therefore the average manganese concentration differs greatly, from Asia with 24%<sub>wt</sub> to Australia with 48.5%<sub>wt</sub> and a world-wide average of 35.7%<sub>wt</sub> manganese content in the ore.

**Beneficiation:** After mining, the ore is first crushed in several stages with jaw and / or cone crushers, and then subsequently ground with rod and or ball mills and finally screened for classification. In a second step the classified material it is subjected to gravity concentration to separate the metal-bearing particles from the unwanted minerals. For this drum separators and de-watering screens for lumps are used and cone separators and a high-gradient magnetic separator for fine material. No flotation is done. Recovery of Manganese vary in a range from 65% to 75% (Adelhardt & Saiger (1999)). Various intermediate treatment processes can be applied to manganese natural ores or concentrates prior to smelting to a ferroalloy or direct addition to a steel making process. These processes are employed for one or both of the following two reasons: 1) to convert a fine ore to a more suitable size, and 2) to calcine and/or reduce the ore to a more suitable final product. Agglomeration processes include nodulizing, pelletizing, and briquetting. Reduction can be carried out in a rotary kiln or sinter plant. Sintering has become the preferred process for intermediate treatment (Wellbeloved et al. (1997)).

### 5.1.2 Emissions & Waste

#### Mining

The major emissions are due to mineral born pollutants in the effluents. Open cut mining generates large quantities of dust, which contains elevated contents of metals. Rain percolate through overburden and accounts to metal emissions to groundwater. Overburden is deposited close to the mine.

#### Beneficiation

Ore handling and processing produce large amounts of dust, containing PM<sub>10</sub> and several metals from the ore itself. Tailings are deposited as piles and in ponds. Since the tailings are not sulphidic, no acid rock drainage (ARD) occurs.

### 5.1.3 “Manganese, concentrate, at beneficiation” in ecoinvent

The module describes the production of manganese ore in 1994 for a world-wide production average. It is designed for the use of manganese ore as intermediary product in the production of ferromanganese, manganese metal and for the use of the ore in the manufacturing of batteries. Process data is taken mainly from Adelhardt & Saiger (1999), which conducted a material flow analysis for the consumption of manganese compounds in Germany.

The layout of the modules with general flow information, remarks, sources, values and uncertainty information are shown in Fig. 5.2 and Fig. 5.3. Ecoinvent meta information for this process are summarised in Tab. 5.1 on page 11. The following paragraphs describe calculations, sources and assumptions chosen in this study.

#### Infrastructure and resource use

The mobile equipment used for exploration like drilling machines, scoop trams and underground trucks is inventoried with the diesel consumption, which is reported in Adelhardt & Saiger (1999). The

used module “diesel, burned in building machine” includes the infrastructure of these machines. Use of conveyor belt was assessed according Tab. 5.2.

The world average manganese content after beneficiation amounts to 42.4%, a yield from mining and beneficiation of 69.2% can be calculated (Adelhardt & Saiger (1999)).

According to part “Non Ferrous Metal Winning – Auxiliary Processes” no direct land use for underground mining (30%) is inventoried, land use for open pit mining (70%) is assessed with  $0.0025 \text{ m}^2/\text{t}$ , yielding a  $1.75 \cdot 10^{-3} \text{ m}^2/\text{t}$  and  $52.5 \cdot 10^{-3} \text{ m}^2/\text{t}$  respectively. A lifetime of 30 years is assumed. Mining facilities are inventoried as unspecific module according to part IX “Non Ferrous Metal Winning – Auxiliary Processes”.

**Tab. 5.2 Assumptions for infrastructure used in underground mining exploration. As lifetime ten years are estimated, and 5000 working hours yearly.**

| Action      | Share | Machine                     | Throughput   | Unit       | Infrastructure  |
|-------------|-------|-----------------------------|--------------|------------|-----------------|
|             |       |                             | t / lifetime | Length (m) | m / t ore mined |
| Open pit    | 70 %  | Conveyor belts (2000 t / h) | 100'000'000  | 10'000     | 1E-4            |
| Underground | 30%   | Conveyor belts (2000 t / h) | 100'000'000  | 2'000      | 2E-5            |
| Mix         |       |                             |              |            | 7.6E-5          |

## Mining

According to Adelhardt & Saiger (1999) 70 % of the manganese ore world production originates from surface mining operations. The overburden – 2.61 kg per kg ore concentrate extracted – is disposed in piles near the mine, the standard module for disposal of non-sulphidic overburden was chosen, which includes the land use and transformation through the disposal. Refilled overburden is assumed to be burden free. No overburden is refilled according to Adelhardt & Saiger (1999). In the world-wide average  $0.25 \text{ m}^3/\text{t}$  crude ore of the pit water is used for the beneficiation. The remaining pit water is not inventoried as output but its emissions are inventoried together with the emissions of the beneficiation step. The specific use of explosives vary between 0.1 kg (South America) and 0.3 kg (Africa) per tonne raw ore with a world-wide average of  $0.26 \text{ kg} / \text{t}$ . Diesel use ranges from 0.7 (South America) to 2,3 (Australia) with an average of 1.2 litre per tonne ore exploited. Electricity demand is comprised in beneficiation.

## Beneficiation

Once the ore is mined, it is ground and classified, no flotation occurs. The separated gangue is disposed on heaps, the concentrated ore is fed to the metallurgy, which is assumed to be on-site. The respective standard module “disposal, non-sulphidic tailings, off-site” is chosen. (s. “Non Ferrous Metal Winning – Auxiliary Processes”, chapter “Disposal of Tailings and Overburden”).

The infrastructure of beneficiation is included in mining infrastructure. Dust emissions comprise additionally the dust emissions due to mining activities. Hilbrans & Hinrichs (1999) assume for the nickel mining a share of 60% hydroelectricity due to the operations high and constant energy demand. The value comprises the electricity demand of the mining, which is expected to be small compared to the beneficiation.

**Steel:** The steel input (abrasion) for chromite milling is documented in IPPC (2002) for a single Finnish chromite mine (Kemi site) and amounts to  $300 \text{ g} / \text{t}$  ground ore. Since non information on steel consumption in the beneficiation of manganese ore is available and the beneficiation process works analogous to chromite winning, this value is also taken in this study. The mill (rods, balls, etc.) consists of special austenitic manganese steel of Hadfield grade, which is inventoried as chromium steel as a proxy.

**Water use & Effluents:** Water use for mining and beneficiation is documented in Adelhardt & Saiger (1999) and is inventoried in this study as originating from river without pumping activities. As no data on pit water use are given, the specific water use for chromite exploitation of  $1.5 \text{ m}^3 / \text{t}$  raw ore mined was chosen. The water input for beneficiation is given as  $0.8 \text{ m}^3 / \text{t}$  raw ore mined. As no data on emissions to water could be found, the recommendations of the world bank for base metal sites was used (Tab. 5.3). These recommended maximum concentrations are multiplied with the sum of effluent volumes from mining and beneficiation ( $3.5 \text{ m}^3/\text{t}$  ore). The value of  $150 \text{ mg} / \text{l}$  COD is assumed to amount to 150% of BOD (Zimmermann et al. (1996)). Values for TOC are calculated using the standard procedure in Frischknecht et al. (2003) approximating the unknown organic matter with the simplified formula for organic matter  $\text{C}_6\text{H}_{12}\text{O}_6$ . This gives  $58.7 \text{ mg} / \text{l}$  TOC and DOC respectively.

**Dust:** A case study in Anonymous (1998a) on a projected open cut gold-copper mining and processing facility in Australia predict annual dust emissions between  $0.18 \text{ kg/t}$  and  $0.46 \text{ kg/t}$  of dust produced per tonne of ore mined. The prediction was made using the Emission Estimation Technique Manual for Mining (NPI (2001)). The maximum value of  $0.46 \text{ kg/t}$  is chosen for total dust emissions, the value for underground mining is assessed with 80% of this amount. The dust composition is approximated with the manganese ore composition from Tab. 3.1 and the demands on quality for metallurgical grade ore. For  $\text{PM}_{10}$  values, the ratio TSP :  $\text{PM}_{10}$  :  $\text{PM}_{2.5}$  of manganese ore mining  $101.72 : 50 : 5 \text{ g} / \text{t}$  ore taken from the CEMIP-database<sup>1</sup>, i. e. one tonne emitted dust corresponds to  $492 \text{ kg PM}_{10}$  and  $49 \text{ kg PM}_{2.5}$  respectively. The airborne emissions resulting from mining and beneficiation activities are summed up in Tab. 5.4. In Ecoinvent, for particulates differential classes are used adding up to the total dust emission, i. e.  $49 \text{ kg PM}<2.5$ ,  $442 \text{ kg PM}_{2.5-10}$  and  $508 \text{ kg PM}>10$  per tonne of total dust.

### Data quality

Data for major mass fluxes are satisfactory, while the emissions to water and to air are based on assumption. The emissions to water may additionally be overestimated due to different system boundaries of reported effluent volume from Adelhardt & Saiger (1999) and recommendations in Anonymous (1998b). Furthermore, high uncertainties are associated with the chosen level of impurities of Pb, As, Cu and Zn in the manganese ore resulting in emissions to water and to air. The emissions from tailings and overburden are assessed in another module, which itself contains rough assumptions of a global overburden and tailings composition and a simplified disposal scenario. Emissions though can vary considerably from case to case.

Overall, the data quality is not satisfactory for this module. For specific purposes such as the use as battery grade ore in the battery manufacture further investigations concerning process specific emissions are necessary.

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<sup>1</sup> [http://www.air.sk/tno/cepmeip/em\\_factors.php](http://www.air.sk/tno/cepmeip/em_factors.php), SNAP-code 040616, accessed 01.03.2003

**Tab. 5.3 Effluent levels to be achieved during operation and after mine closure according to World Bank Group, Anonymous (1998b) (Maximum).**

| Parameter                                                                                                                                                            | Maximum (mg/l) | This Study (mg/l) |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------|
| Mining Type                                                                                                                                                          | Base metal     |                   |
| pH                                                                                                                                                                   | 6–9            |                   |
|                                                                                                                                                                      |                |                   |
| TSS                                                                                                                                                                  | 50             | 50                |
| Oil and grease                                                                                                                                                       | 10             | 10                |
| Cyanide                                                                                                                                                              | 1.0            | - <sup>b</sup>    |
| Free                                                                                                                                                                 | 0.1            | - <sup>b</sup>    |
| Weak acid dissociable (WAD)                                                                                                                                          | 0.5            | - <sup>b</sup>    |
| COD                                                                                                                                                                  | 150            | 150               |
| Arsenic                                                                                                                                                              | 0.1            | 0.1 <sup>a</sup>  |
| Cadmium                                                                                                                                                              | 0.1            | - <sup>c</sup>    |
| Chromium (hexavalent)                                                                                                                                                | 0.1            | - <sup>c</sup>    |
| Chromium, ion                                                                                                                                                        |                | - <sup>c</sup>    |
| Chromium (in solid)                                                                                                                                                  |                | - <sup>c</sup>    |
| Copper                                                                                                                                                               | 0.5            | 0.1 <sup>a</sup>  |
| Iron                                                                                                                                                                 | 3.5            | 3.5               |
| Lead                                                                                                                                                                 | 0.2            | 0.1 <sup>a</sup>  |
| Mercury                                                                                                                                                              | 0.01           | - <sup>c</sup>    |
| Nickel                                                                                                                                                               | 0.5            | - <sup>c</sup>    |
| Zinc                                                                                                                                                                 | 2              | 0.1 <sup>a</sup>  |
| Total metals                                                                                                                                                         | 10             | -                 |
| <sup>a</sup> Metals, which are frequent impurities in manganese ore. Rough assumption.                                                                               |                |                   |
| <sup>b</sup> manganese ore is not sulphidic, therefore no use of this substance and consequently no emission is assumed.                                             |                |                   |
| <sup>c</sup> Maximum values relates to the whole range of base metal extraction. This specific emission is not assumed to be of importance in manganese extraction.. |                |                   |

**Tab. 5.4 Emissions to air resulting from mining activities per tonne mined ore. Values for total dust emission are taken from Anonymous (1998a), partition of fractions of particulate matter from the CEMIP-database (mining of manganese ore), the dust composition correspond to the average manganese ore from “3.1 Manganese ore” in this study. The impurities As, Pb, Cu and Zn are based on quality demands for metallurgical manganese ore: Mn  $\geq$  48%, Fe  $\leq$  6%, As  $\leq$  0.18%, Cu+Pb+Zn  $\leq$  0.3% (Adelhardt & Saiger (1999)), assuming single emission factors for Cu  $\leq$  0.1%, Pb  $\leq$  0.1%, Zn  $\leq$  0.1%.**

| <b>Emission to air</b> | <b>Unit</b> | <b>Emission per tonne mined ore</b> |
|------------------------|-------------|-------------------------------------|
| Iron                   | g / t ore   | 24.5                                |
| Manganese              | g / t ore   | 185.2                               |
| Phosphorous            | g / t ore   | 0.4                                 |
| Aluminium              | g / t ore   | 7.1                                 |
| Magnesium              | g / t ore   | 2.8                                 |
| Silica                 | g / t ore   | 13.5                                |
| Calcium                | g / t ore   | 13.0                                |
| Arsenic                | g / t ore   | 0.4                                 |
| Lead                   | g / t ore   | 0.4                                 |
| Copper                 | g / t ore   | 0.4                                 |
| Zinc                   | g / t ore   | 0.4                                 |
| PM<2.5                 | g / t ore   | 21.3                                |
| PM2.5-10               | g / t ore   | 191                                 |
| PM>10                  | g / t ore   | 220                                 |
| total dust             | g / t ore   | 432                                 |

| General Flow Information                   |              |                      | Representation in ecoinvent                                                                                                                                                                                                                                                           |                        |                            |                 |          |                                                                        |            | Uncertainty Informations |                              |      |                    |                                                      |
|--------------------------------------------|--------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|----------------------------|-----------------|----------|------------------------------------------------------------------------|------------|--------------------------|------------------------------|------|--------------------|------------------------------------------------------|
| Input                                      | Process Name | Output               | Remarks                                                                                                                                                                                                                                                                               | Category               | Sub-category               | Infra-structure | Location | Modul name in ecoinvent                                                | Mean value | Unit                     | Source for mean value        | Type | Standard Deviation | General Comment                                      |
| Manganese in ground                        | mining       |                      | 69% yield assumed (31% Mn loss in mining / beneficiation), 1.72kg crude ore and 2.61kg overburden per kg ore concentrate                                                                                                                                                              | resource               | in ground                  |                 |          | Manganese, 35.7% in sedimentary deposit, 14.2% in crude ore, in ground | 6.13E-01   | kg                       | (Adelhardt 1999)             | 1    | 1.13               | (2,2,3,1,1,2,4)                                      |
| diesel fuel (energy)                       |              |                      | 1.2 l / raw ore. Only machines, electricity demand inventoried in beneficiation.                                                                                                                                                                                                      | construction processes | machinery                  | No              | GLO      | diesel, burned in building machine                                     | 7.97E-02   | MJ                       | (Adelhardt 1999)             | 1    | 2.08               | reported values w / range                            |
| explosive                                  |              |                      | 0.26 kg explosives / t raw ore                                                                                                                                                                                                                                                        | construction processes | civil engineering          | No              | RER      | blasting                                                               | 4.52E-04   | kg                       | (Adelhardt 1999)             | 1    | 1.15               | reported values w / range                            |
| exploration infrastructure, conveyor belts |              |                      | exploration infrastructure, 2000 m belts assumed                                                                                                                                                                                                                                      | construction processes | machinery                  | Yes             | RER      | conveyor belt, at plant                                                | 1.30E-07   | m                        | estimation                   | 1    | 3.32               | (5,5,1,1,1,5,9); rough estimate                      |
| infrastructure of mining (underground)     |              |                      |                                                                                                                                                                                                                                                                                       | metals                 | extraction                 | Yes             | GLO      | non-ferrous metal, mine, underground                                   | 8.30E-12   | unit                     | estimation                   | 1    | 3.32               | (5,5,1,1,1,5,9); rough estimate                      |
| infrastructure of mining (surface)         |              |                      |                                                                                                                                                                                                                                                                                       | metals                 | extraction                 | Yes             | GLO      | non-ferrous metal, mine, surface                                       | 1.94E-11   | unit                     | estimation                   | 1    | 3.32               | (5,5,1,1,1,5,9); rough estimate                      |
| Land transformation from nature            |              |                      |                                                                                                                                                                                                                                                                                       | resource               | land                       |                 |          | Transformation, from unknown                                           | 3.00E-06   | m <sup>2</sup>           | (Martens 2002), calculations | 1    | 2.03               | (3,2,3,1,1,2,8); calculation based on reported value |
| Land transformation to mine                |              |                      |                                                                                                                                                                                                                                                                                       | resource               | land                       |                 |          | Transformation, to mineral extraction site                             | 3.00E-06   | m <sup>2</sup>           | (Martens 2002), calculations | 1    | 2.03               | (3,2,3,1,1,2,8); calculation based on reported value |
| Land use for mining                        |              |                      |                                                                                                                                                                                                                                                                                       | resource               | land                       |                 |          | Occupation, mineral extraction site                                    | 9.01E-05   | m <sup>2a</sup>          | (Martens 2002), calculations | 1    | 1.53               | (3,2,3,1,1,2,7); calculation based on reported value |
|                                            |              |                      |                                                                                                                                                                                                                                                                                       |                        |                            |                 |          |                                                                        |            |                          |                              |      |                    |                                                      |
|                                            |              | overburden, disposed | no overburden assumed to be refilled                                                                                                                                                                                                                                                  | waste management       | residual material landfill | No              | GLO      | disposal, non-sulfidic overburden, off-site                            | 2.61E+00   | kg                       | (Adelhardt 1999)             | 1    | 1.16               | (3,2,3,1,1,2,6)                                      |
|                                            |              | effluents, mining    | (1.5 - 0.15) m <sup>3</sup> /t pit water per crude ore (0.15 m <sup>3</sup> /t used in beneficiation) without treatment. The composition is given in Tab. 5.3. The emissions are inventoried with the emissions of beneficiation. The uncertainties of the single emissions are high. |                        |                            |                 |          |                                                                        | 2.32E-03   | m <sup>3</sup>           | (Adelhardt 1998), estimation | 1    | 1.13               | (2,2,3,1,1,2,4)                                      |

Fig. 5.2 Flows for “manganese concentrate, at beneficiation – part mining” and its representation in the ecoinvent database. Values correspond to the functional unit of 1 kg of manganese concentrate.

## 5. System Characterisation

| General Flow Information                |               |                          | Representation in ecoinvent                                                                                                                                                                                                                      |                  |                            |                 |          |                                                       |            | Uncertainty Informations |                       |      |                    |                                  |
|-----------------------------------------|---------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------------|-----------------|----------|-------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|----------------------------------|
| Input                                   | Process Name  | Output                   | Remarks                                                                                                                                                                                                                                          | Category         | Sub-category               | Infra-structure | Location | Module name in ecoinvent                              | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                  |
| electricity, from grid, total           | beneficiation | waste heat               | Assumption: 40% electricity from grid. Value comprising electricity demand of mining.                                                                                                                                                            | electricity      | production mix             | No              | UCTE     | electricity, medium voltage, production UCTE, at grid | 3.02E-03   | kWh                      | (Adelhardt 1999)      | 1    | 1.16               | (3,2,3,1,1,2,2)                  |
| hydropower, total                       |               | waste heat               | Assumption: 60% hydropower. Value comprising electricity demand of mining.                                                                                                                                                                       | hydropower       | power plants               | No              | RER      | electricity, hydropower, at run-of-river power plant  | 4.53E-03   | kWh                      | (Adelhardt 1999)      | 1    | 1.16               | (3,2,3,1,1,2,2)                  |
| fresh and process water (beneficiation) |               |                          | 0.8 m <sup>3</sup> /t crude ore (not including the recycled pit water)                                                                                                                                                                           | resource         | in water                   |                 |          | Water, river                                          | 1.37E-03   | m <sup>3</sup>           | (Adelhardt 1999)      | 1    | 1.13               | (2,2,3,1,1,2,12)                 |
| steel                                   |               |                          | abrasion in mill (steel balls, etc.)                                                                                                                                                                                                             | metals           | extraction                 | No              | RER      | chromium steel 18/8, at plant                         | 5.15E-04   | kg                       | (IPPC, 2002)          | 1    | 1.62               | (5,5,1,1,1,5,4)                  |
| infrastructure beneficiation            |               |                          | Aluminiumhydroxide production as proxy                                                                                                                                                                                                           | metals           | extraction                 | Yes             | RER      | aluminium hydroxide, plant                            | 1.44E-11   | unit                     | rough estimate        | 1    | 3.32               | (5,5,1,1,1,5,9)                  |
|                                         |               | tailings, disposed       | no tailings assumed to be refilled                                                                                                                                                                                                               | waste management | residual material landfill | No              | GLO      | disposal, non-sulfidic tailings, off-site             | 7.10E-01   | kg                       | (Adelhardt 1999)      | 1    | 1.16               | (3,2,3,1,1,2,6); reported values |
|                                         |               | effluents, beneficiation | composition in Tab. 5.3, emissions with high uncertainties. No treatment of waste water.                                                                                                                                                         |                  |                            |                 |          |                                                       | 1.20E-03   | m <sup>3</sup>           | (Adelhardt 1999)      | 1    | 1.16               | (3,2,3,1,1,2,6); estimated       |
|                                         |               | Dust                     | Including emissions from mining activities. Dust is the sum of particulate emissions represented in fractions PM<2.5, PM2.5-10, PM>10, including metals. These emissions are inventoried as shown in Tab 5.4, they are not shown in this figure. |                  |                            |                 |          |                                                       | 7.42E-04   | kg                       | (NPI, 2001)           | 1    | 1.59               | (3,2,1,1,3,4,25)                 |
|                                         |               |                          |                                                                                                                                                                                                                                                  |                  |                            |                 |          |                                                       |            |                          |                       |      |                    |                                  |
|                                         |               | manganese concentrate    | 42.4% Mn                                                                                                                                                                                                                                         | metals           | extraction                 | No              | GLO      | manganese, concentrate, at beneficiation              | 1.00E+00   | kg                       | (Adelhardt 1999)      |      |                    |                                  |

**Fig. 5.3** “manganese concentrate, at beneficiation – part beneficiation” and its representation in the ecoinvent database. Values correspond to the functional unit of 1 kg manganese concentrate.

## 5.2 Winning of high carbon ferromanganese

Generally, high-carbon ferromanganese and silicomanganese are produced from a blend of manganese-containing ores, and in the case of silicomanganese, slag and silica are added. High-carbon ferromanganese can be converted to medium-carbon manganese by an oxygen blowing process, and silicomanganese can be further refined into medium- or low-carbon ferromanganese as well as manganese metal.

### 5.2.1 Production process and infrastructure

Ferromanganese can be produced in blast furnaces in a manner similar to pig iron and spiegeleisen; however, in the Western world only four producers employ this method. In the former Soviet Union, the majority of the high-carbon ferromanganese is produced in blast furnaces. The choice of the use of blast furnaces over electric furnaces is based on the relative price of coke and electricity. Blast furnaces are usually used where the cost of power is high in relation to coke. In blast furnaces, coke is used both as a reductant and as the energy source. The coke rate in blast furnaces is higher than in submerged arc furnaces, which use electricity as the power source. The product produced from blast furnaces generally contains 76 % Mn and 16 % Fe; the ferromanganese produced in the former Soviet Union is generally of a lower grade

According to Adelhardt & Saiger (1999), 15% (related to the manganese contained) of the manganese production was done by blast furnace and 75% by electric arc furnaces to produce silico- or ferromanganese, the rest being done mainly by electrolysis for other purposes.

The raw materials required for the production of high-carbon ferromanganese are manganese ores, fluxes such as limestone, dolomite, or silica, and solid fuels and reductants such as coke.

In order to produce ferromanganese of the required grade a single ore is seldom suitable because the desired Mn/Fe ratio of the charge determines the Mn content of the final product. Ores from various sources are therefore blended to achieve the ideal ratio and to limit the contents of the deleterious components silica, alumina (bauxite), and phosphorus in the raw material mix.

#### Blast furnace

In comparison to iron making, high gas temperatures are required in ferromanganese production because the reduction of manganese(II) oxide takes place at a higher temperature than is required for the reduction of wustite. This is achieved by oxygen enrichment of the hot blast. The recovery of manganese in the alloy is usually 75 – 85 %. This is influenced by the MnO content of the slag, the slag-to-metal ratio, and losses in the flue gases. The MnO content of the slag is highly dependent on the basicity ratio  $(\text{CaO} + \text{MgO})/\text{SiO}_2$ .

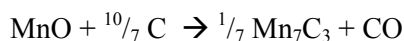
The reduction of the higher manganese oxides to manganese(II) oxide takes place in the upper zone of the shaft (reduction of manganese oxides with carbon monoxide and carbon). These generally occur below 900 °C and are indirect. The reactions are exothermic, and the heat generated causes high top temperatures and necessitates water cooling of the furnace top. The reduction of manganese(II) oxide  $\text{MnO} + \text{C} \rightarrow \text{Mn} + \text{CO}$  is highly endothermic, in contrast to the weakly endothermic reduction of wustite. This requires higher temperatures and, consequently, higher coke rates are required for the smelting of ferromanganese in blast furnaces.

#### Electric arc furnaces

The majority of producers of ferromanganese in the Western world use submerged arc electric furnaces. Although electric furnaces have lower capacities than blast furnaces they have the advantage that the heat requirement is provided by electricity, and coke and coal are added to the feed only as reductants. Recently built electric furnaces have capacities of 75 – 90 MVA. Smaller furnaces are still popular with producers because they offer flexibility in that they can be switched more easily between different products than their larger counterparts. Electric furnaces used are generally circular and have



three electrodes. The reduction of manganese(II) oxide occurs by the contact of carbon with the molten oxide in the slag phase. The overall reaction is :



Large electric furnaces are usually completely closed at the top, and the CO-rich gas leaves the furnace at approximately 290 °C and is cleaned in cyclones and venturi scrubbers. The gas is then either flamed off to the atmosphere or, more recently, is used to generate electricity. Smaller furnaces may be open, the gas then is usually withdrawn by fans and cleaned in a bag-filter plant. In this case the gases are completely burnt in the furnace and have no commercial value.

After the metal is tapped from the furnace it is cast into moulds formed from ferromanganese fines or cast iron and allowed to solidify. The alloy is then removed, crushed, and screened into various size fractions, depending on the requirements of the user. An alternative to this practice is the use of a casting machine. In this case the metal is tapped directly onto a moving train consisting of small molds. The metal then solidifies and is ejected from the mould at the end of the strand.

Adelhardt & Saiger (1999) distinguishes between a electric arc furnace process with calcareous flux and one without flux, where the energy consumption is smaller due to reduction of the ore by excess CO. The yield of the process without flux is smaller with 70% compared with 75% of the process with calcareous flux, due to a slag richer in MnO. However, the manganese rich slag – sometimes over 25% manganese – of the process without flux is used in the production of silicomanganese, enhancing the overall yield considerably.

### **Silicomanganese**

The production of silicomanganese is very similar to the ferromanganese production. The main difference is in the type of flux – silica instead of lime – and of course the composition of the product. Also the energy demand is some what higher than in the production of ferromanganese. Many plants produce alternately ferrosilicomanganese and then – processing the developed slag – silicomanganese. However, for purposes of inventorying silicomanganese it is suitable to take ferromanganese as proxy.

## **5.2.2 Emission**

**Air:** The main environmental impact by producing ferro-alloys is the emission of dust and fume from the smelting processes. Dust emissions occur as well from storage, handling and the pre-treatment of raw materials where fugitive dust emissions play an important role. Depending on the raw material and the process used other emissions to air are SO<sub>2</sub>, NO<sub>x</sub>, CO-gas, CO<sub>2</sub>, polycyclic aromatic hydrocarbon (PAH), VOCs, and volatile metals. The formation of dioxins in the combustion zone and in the cooling part of the off-gas treatment system (de-novo synthesis) may be possible.

**Water:** Water emissions are produced from cooling, granulation and other processes and site related effluents. The wastewater generated by a wet cleaning abatement system plays also an important role.

## **5.2.3 Waste**

Process residues and by-products with a significant amount is slag, filter dust and sludge and spent refractories. These materials are already recycled and reused to a large extent where it is possible. Rich slag, that means slag with a relatively high proportion of metal oxides, is used as raw material in other ferro-alloy productions.

### 5.2.4 Ferromanganese in ecoinvent

This part is mainly based on Adelhardt & Saiger (1999). General process description was taken from Wellbeloved et al. (1997), process specific emissions from IPPC (2001), IPCC (1996), EPA (1998) and EC (2002).

Adelhardt & Saiger (1999) differentiates between a blast furnace process (15% of the production respective to manganese contained) and the two processes with electric arc furnaces (with calcareous flux and without flux, total of 60% of the production) and finally the production of silicomanganese in electric arc furnaces (13%), totalling to 88% of the total manganese processed to high-carbon ferromanganese. In this study an aggregated mix is used, which represent an average production of ferromanganese and silicomanganese. 2% is processed to low-carbon ferromanganese, while 10% is used in the non-ferrous metallurgy, the chemical industry and the battery manufacture (Fig. 5.1). It is suitable also for the production of silicomanganese, which is produced in a similar way, although using silica instead of lime and about 10% more energy respective to the manganese contained (Tab. 5.5).

The module describes the world average consumption mix for ferromanganese in Europe in 1994. The situation for Germany described in Adelhardt & Saiger (1999) is taken as proxy for Europe. It is designed for the use of the metal as alloying element. The layout of the modules with general flow information, remarks, sources, values and uncertainty information are shown in Fig. 5.4 (inputs) and Fig. 5.5 (output). Ecoinvent meta information for this process are summarised in Tab. 5.1 on page 11. The following paragraphs describe calculations, sources and assumptions chosen in this study. The functional unit of this process is one kg of ferromanganese with an average manganese content of 74.5%.

**Infrastructure and transport:** The production infrastructure is represented by the non-ferrous metal smeltery that is describes in part “Non ferrous metal winning, auxiliary processes” of this report. It is assumed that the raw materials manganese ore, silica, dolomite, lime, coke and the disposals slag and cake are transported by lorry over 50 km.

**Energy, raw-materials and output:** A weighted average of the different inventoried processes – blast furnace and electric arc furnace- is calculated. An overview over different in- and output values from different sources is given in Tab. 5.5. Slag is assumed to be sold to silicomanganese production, dust as reported in Adelhardt & Saiger (1999) is assumed to build process residue, which has to be disposed into a landfill. This is inventoried in this study with the disposal of non-ferrous slag as proxy.

**Emissions to air:** Air emissions are taken from various sources, they are summarised and discussed in Tab. 5.6.

**Water use and emissions:** Water use for European ferrochrome production is reported in IPPC (2001), the average of 20 m<sup>3</sup> (10 m<sup>3</sup> – 30 m<sup>3</sup>) per tonne is chosen as an approximation, resulting in 10 m<sup>3</sup> of effluents (volume of process water). The effluent has to be treated in a communal waste-water treatment plant, a process which is inventoried separately in the chosen module “treatment, sewage, unpolluted, to wastewater treatment, class 3”. Emissions to water are inventoried separately each as entry according to reported values of effluents after internal process water treatment and before waste water treatment plant (Tab. 5.7).

**Production and import to Europe.** As proxy for the consumption pattern the imports to Germany in 1995 are chosen (Tab. 5.8). Only the ferromanganese imports were considered, assuming, that the imported manganese ore is destined mainly to the chemical industry and the battery manufacture and only a small amount is used for blending in the domestic ferromanganese production. This results in a transport intensity of 16 tkm ship and 2 tkm freight train per kg ferromanganese (Tab. 5.9).

### Data quality

Process data is satisfactory and valid for the whole world-wide primary ferromanganese production. Process specific emissions to air and water had to be taken from an other source, which cover other

geography and technology. Nevertheless the overall quality of the module is fair. The import pattern may have changed in the recent years with a trend to longer import distances.

**Tab. 5.5 Production of ferromanganese and silicomanganese. In- and outputs for the production of ferromanganese with different processes. The last two columns relate to the production of 1 kg manganese alloy using a weighted mix for ferromanganese according to the share in global production. (Adelhardt & Saiger (1999))**  
Data for silicomanganese show, that the production of ferromanganese is suitable as a proxy inventory.

| Inventory                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Unit           | Ferromanganese    |                      |            | Silico-manganese     | Ferromanganese <sup>e)</sup> | Silico-manganese |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------------------|----------------------|------------|----------------------|------------------------------|------------------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                | Blast furnace     | Electric arc furnace |            | Electric arc furnace | Total, per 1 kg              | Total, per 1 kg  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |                   | No flux              | Calc. Flux |                      |                              |                  |
| Share in global production                                                                                                                                                                                                                                                                                                                                                                                                                                       |                | 17%               | 23%                  | 45%        | 15%                  |                              |                  |
| <b>Inputs:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                |                   |                      |            |                      |                              |                  |
| Manganese concentrate                                                                                                                                                                                                                                                                                                                                                                                                                                            | t              | 2.8               | 3.4                  | 3.1        | 3.0                  | 2.33E-03                     | 1.98E-03         |
| Electricity                                                                                                                                                                                                                                                                                                                                                                                                                                                      | kWh            | 26                | 3784                 | 4581       | 4844                 | 2.57                         | 3.25             |
| Process water <sup>a)</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                      | m <sup>3</sup> | 10                | 10                   | 10         | 10                   | 0.0075                       | 0.0067           |
| Cooling water <sup>a)</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                      | m <sup>3</sup> | 10                | 10                   | 10         | 10                   | 0.0075                       | 0.0067           |
| Fossil energy (coke)                                                                                                                                                                                                                                                                                                                                                                                                                                             | kg             | 2000              | 400                  | 180        | 500                  | 0.45                         | 0.34             |
| Electrode graphite                                                                                                                                                                                                                                                                                                                                                                                                                                               | kg             |                   | 23                   | 30         | 38                   | 1.66E-02                     | 2.55E-02         |
| Silica sand                                                                                                                                                                                                                                                                                                                                                                                                                                                      | kg             |                   |                      |            | 500                  |                              | 0.34             |
| Dolomite                                                                                                                                                                                                                                                                                                                                                                                                                                                         | kg             | 100 <sup>c)</sup> |                      |            |                      | 0.015                        |                  |
| Lime                                                                                                                                                                                                                                                                                                                                                                                                                                                             | kg             | 100 <sup>c)</sup> | 60                   | 520        |                      | 0.23                         |                  |
| <b>Outputs:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                |                   |                      |            |                      |                              |                  |
| Slag                                                                                                                                                                                                                                                                                                                                                                                                                                                             | kg             | 1440              | 2100                 | 2320       | 1950                 | 1.55                         | 1.31             |
| Dust                                                                                                                                                                                                                                                                                                                                                                                                                                                             | kg             | 130               | 10                   | 10         | 10                   | 0.025                        | 6.71E-03         |
| Effluents <sup>b)</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                          | m <sup>3</sup> | 10                | 10                   | 10         | 10                   | 7.45E-03                     | 6.71E-03         |
| SO <sub>2</sub> <sup>d)</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                    | kg             | 1.74              | 6.78E-02             | 6.78E-02   | 7.60E-02             | 2.99E-04                     | 5.10E-05         |
| Ferromanganese                                                                                                                                                                                                                                                                                                                                                                                                                                                   | kg             | 1390              | 1330                 | 1330       | 1490                 | 1.00                         | 1.00             |
| Manganese concentration                                                                                                                                                                                                                                                                                                                                                                                                                                          | %              | 72%               | 75%                  | 75%        | 67%                  | 72.5%                        | 73.4%            |
| <sup>a)</sup> values for process water are taken from IPPC (2001) for the similar process of ferrochrome production<br><sup>b)</sup> volume for effluent equals the process water used.<br><sup>c)</sup> only sum dolomite plus lime reported, a split 50/50 is assumed.<br><sup>d)</sup> emission factor for SO <sub>2</sub> are taken from IPPC (2001) (see Tab. 5.6)<br><sup>e)</sup> weighted average of blast furnace and electric arc furnace production.. |                |                   |                      |            |                      |                              |                  |

**Tab. 5.6 Emissions to air in the production of 1 tonne ferromanganese. Values are from different sources: BREF (IPPC (2001)), IPCC (IPCC (1996)), EPA (EPA (1998)) and EEA (EC (2002)). Chosen values for this study are the average of the given range.**

| Emission to air | Unit   | BREF               |                    | IPPC            |      | EPA               |                    | EEA             | This study         |
|-----------------|--------|--------------------|--------------------|-----------------|------|-------------------|--------------------|-----------------|--------------------|
|                 |        | Low                | High               | Low             | High | Low <sup>b)</sup> | High <sup>a)</sup> |                 |                    |
| NO <sub>x</sub> | kg / t |                    |                    | 0.05            | 11.7 |                   |                    |                 | <b>5.88</b>        |
| NM VOC          | kg / t |                    |                    | 1.7             |      |                   |                    |                 | <b>1.7</b>         |
| CO              | kg / t |                    |                    | 1 <sup>e)</sup> |      |                   |                    |                 | <b>1</b>           |
| PM<2.5          | kg / t |                    |                    |                 |      | 0.24              | 17                 |                 | <b>8.62</b>        |
| PM2.5-10        | kg / t |                    |                    |                 |      | 0.16              | 7                  |                 | <b>3.58</b>        |
| PM>10           | kg / t |                    |                    |                 |      | 0.08              | 4                  |                 | <b>2.04</b>        |
| TSP             | kg / t |                    |                    |                 |      | 0.48              | 28                 |                 | 14.2 <sup>c)</sup> |
| SO <sub>2</sub> | kg / t | 0.05 <sup>f)</sup> | 1.25 <sup>f)</sup> |                 |      |                   |                    |                 | 0.3 <sup>f)</sup>  |
| PCDD            | µg / t |                    |                    |                 |      |                   |                    | 1 <sup>d)</sup> | <b>1</b>           |

<sup>a)</sup> No abatement  
<sup>b)</sup> Bag house  
<sup>c)</sup> Average of EPA values are chosen for world wide mix.  
<sup>d)</sup> SNAP-Code 040309, „Other - Installations for smelting and casting of non ferrous metals“  
<sup>e)</sup> Value taken from ferrosilicon production  
<sup>f)</sup> values for Electric arc furnace (low) and blast furnace (high) resulting in a weighted average value (see Tab. 5.5)  
 CO<sub>2</sub> emission is not inventoried separately as it is included in the emissions of the "hard coal, burned..." inputs

**Tab. 5.7 Emissions to water in the production of 1 tonne ferromanganese. The values are reported values for ferro-chromium production (IPPC (2001)) chosen as approximation. Emissions are before final waste-water treatment and after internal abatement including lamella thickener, vacuum filter and a sand and carbon filter. Based on the chosen values, final emissions into water are calculated.**

| Emission to Water<br>(before treatment)  | Unit   | IPPC (2001) |      | This Study        | Resulting values after treatment |
|------------------------------------------|--------|-------------|------|-------------------|----------------------------------|
|                                          |        | Low         | High |                   |                                  |
| TSS                                      | kg / t | 5           | 10   | 7.5 <sup>a)</sup> | 0.375                            |
| PAH                                      | mg / t | 300         | 1000 | 650 <sup>b)</sup> | 65                               |
| waste from internal wastewater treatment |        |             |      |                   |                                  |
| dry cake                                 | kg / t | 200         | 400  | 300 <sup>c)</sup> | 300                              |

<sup>a)</sup> An elimination of 95% TSS assumed, resulting in an emission after treatment of 375 g TSS / t  
<sup>b)</sup> Based on Zimmermann et al. (1996) transfer coefficients of 10% to water is assumed, resulting in emissions of 65 mg PAH / t. Resulting emissions to air due to metabolism as CO<sub>2</sub> and NM VOC are neglected.  
<sup>c)</sup> Dry cake results from internal waste-water treatment. In this study it is assumed to be disposed as inert waste to an inert material landfill.

**Tab. 5.8 Manganese import pattern for Germany in 1995 (Adelhardt & Saiger (1999)).**

| <b>Manganese consumption pattern for Germany 1995</b> |         |     |
|-------------------------------------------------------|---------|-----|
| ore & concentrate                                     | 11'355  | 3%  |
| manganese metal                                       | 24'433  | 7%  |
| Ferromanganaese                                       | 198'334 | 53% |
| silicomanganese                                       | 135'336 | 36% |
| other                                                 | 5'942   | 2%  |
| Total imports                                         | 375'400 |     |

**Tab. 5.9 Import of manganese into Germany for different products and resulting transport intensity in [tkm]. Mix is taken as proxy for the European situation. Values are from (Adelhardt & Saiger (1999)) and estimations based on distances provided by "dataloy systems"<sup>2</sup>.**

| <b>Manganese ore</b>                                                                                                |    | <b>Australia</b>     | <b>South Africa</b> | <b>Morocco</b>       | <b>other</b> | <b>total</b>          |
|---------------------------------------------------------------------------------------------------------------------|----|----------------------|---------------------|----------------------|--------------|-----------------------|
| Manganese ore                                                                                                       | t  | 3'123                | 2'702               | 2'157                | 3'372        | 11'355                |
| ship sea                                                                                                            | km | 15'500               | 15'500              | 1'700                | 10'900       | <b>11'512 tkm / t</b> |
| train                                                                                                               | km | 1'200                | 1'400 <sup>b)</sup> | 1'500                | 1'367        | <b>1'354 tkm / t</b>  |
| <b>Ferromanganese</b>                                                                                               |    | <b>France</b>        | <b>South Africa</b> | <b>Norway</b>        | <b>other</b> | <b>total</b>          |
| Ferromanganese                                                                                                      | t  | 60'680               | 40'280              | 30'440               | 66'930       | 198'330               |
| ship sea                                                                                                            | km | 16'100 <sup>a)</sup> | 15'500              | 16'300 <sup>a)</sup> | 15'967       | <b>15'964 tkm / t</b> |
| train                                                                                                               | km | 2'300 <sup>c)</sup>  | 1'400 <sup>b)</sup> | 2'300 <sup>c)</sup>  | 2'000        | <b>2'016 tkm / t</b>  |
| <b>Manganese metal</b>                                                                                              |    | <b>China</b>         | <b>Ukraine</b>      | <b>South Africa</b>  | <b>other</b> | <b>total</b>          |
| Manganese                                                                                                           | t  | 11'923               | 6'792               | 3'616                | 2'101        | 24'433                |
| ship sea                                                                                                            | km | 15'500               | 3'000               | 16'300               | 11'600       | <b>11'768 tkm / t</b> |
| train                                                                                                               | km | 2'500                | 300                 | 2'300                | 1'700        | <b>1'753 tkm / t</b>  |
| <sup>a)</sup> 15'500 km sea transport included to include bias for previously transported raw materials             |    |                      |                     |                      |              |                       |
| <sup>b)</sup> 800 km transport by freight train included to include bias for previously transported raw materials   |    |                      |                     |                      |              |                       |
| <sup>c)</sup> 2'000 km transport by freight train included to include bias for previously transported raw materials |    |                      |                     |                      |              |                       |

<sup>2</sup> <http://www.dataloy.com/>, accessed 01.03.2003

| General Flow Information          |                                                          |            | Representation in ecoinvent                               |                        |                 |                 |          |                                                       |            | Uncertainty Informations |                               |      |                    |                                                   |
|-----------------------------------|----------------------------------------------------------|------------|-----------------------------------------------------------|------------------------|-----------------|-----------------|----------|-------------------------------------------------------|------------|--------------------------|-------------------------------|------|--------------------|---------------------------------------------------|
| Input                             | Process Name                                             | Output     | Remarks                                                   | Category               | Sub-category    | Infra-structure | Location | Module name in ecoinvent                              | Mean value | Unit                     | Source for mean value         | Type | Standard Deviation | General Comment                                   |
| manganese concentrate             | ferromanganese, high-coal, 74.5% Mn, at regional storage | waste heat |                                                           | metals                 | extraction      | No              | GLO      | manganese, concentrate, at beneficiation              | 2.33E+00   | kg                       | (Adelhardt 1998)              | 1    | 1.13               | (2,2,3,1,1,2,4)                                   |
| electricity, grid                 |                                                          |            | 40% electricity from grid                                 | electricity            | production mix  | No              | UCTE     | electricity, medium voltage, production UCTE, at grid | 1.03E+00   | kWh                      | (Adelhardt 1998), estimations | 1    | 1.16               | (3,2,3,1,1,2,2); estimate based on reported value |
| hydropower                        |                                                          | waste heat | 60% electricity from hydropower                           | hydropower             | power plants    | No              | RER      | electricity, hydropower, at run-of-river power plant  | 1.54E+00   | kWh                      | (Adelhardt 1998), estimations | 1    | 1.16               | (3,2,3,1,1,2,2); estimate based on reported value |
| fossil energy (coke)              |                                                          |            |                                                           | hard coal              | heating systems | No              | RER      | hard coal, burned in industrial furnace 1-10MW        | 4.50E-01   | MJ                       | (Adelhardt 1998)              | 1    | 1.13               | (2,2,3,1,1,2,3)                                   |
| lime                              |                                                          |            |                                                           | construction materials | others          | No              | CH       | limestone, milled, packed, at plant                   | 2.32E-01   | kg                       | (IPPC, 2001), estimations     | 1    | 1.52               | (5,2,3,1,1,2,4); estimate                         |
| dolomite                          |                                                          |            |                                                           | construction materials | others          | No              | RER      | dolomite, at plant                                    | 1.49E-02   | kg                       | (IPPC, 2001), estimations     | 1    | 1.52               | (5,2,3,1,1,2,4); estimate                         |
| Electrode graphite                |                                                          |            |                                                           | metals                 | extraction      | No              | RER      | anode, aluminium electrolysis                         | 1.66E-02   | kg                       | (Adelhardt 1998)              | 1    | 1.13               | (2,2,3,1,1,2,3)                                   |
| process water                     |                                                          |            |                                                           | resource               | in water        |                 |          | Water, well, in ground                                | 7.45E-03   | m <sup>3</sup>           | (IPPC, 2001)                  | 1    | 1.11               | (2,3,2,3,1,3,12)                                  |
| cooling water                     |                                                          |            |                                                           | resource               | in water        |                 |          | Water, cooling, unspecified natural origin            | 7.45E-03   | m <sup>3</sup>           | (IPPC, 2001)                  | 1    | 1.11               | (2,3,2,3,1,3,2)                                   |
| transport (resources)             |                                                          |            | 50 km for concentrate, silica, lime, dolomite, coke, slag | transport systems      | road            | No              | RER      | transport, lorry 32t                                  | 2.30E-01   | tkm                      | estimate                      | 1    | 2.11               | (4,4,1,1,1,5,5); calculation based on assumptions |
| Transport ferromanganese over sea |                                                          |            |                                                           | transport systems      | ship            | No              | OCE      | transport, transoceanic tanker                        | 1.60E+01   | tkm                      | (Adelhardt 1999)              | 1    | 2.02               | (2,2,3,1,1,2,5); calculation based on assumptions |
| Transport ferromanganese by train |                                                          |            |                                                           | transport systems      | train           | No              | RER      | transport, freight, rail                              | 2.02E+00   | tkm                      | (Adelhardt 1999)              | 1    | 2.02               | (2,2,3,1,1,2,5); calculation based on assumptions |
| production infrastructure         |                                                          |            |                                                           | metals                 | extraction      | Yes             | GLO      | non-ferrous metal, smelter                            | 6.88E-12   | unit                     | rough estimate                | 1    | 3.32               | (5,5,1,1,1,5,9)                                   |

Fig. 5.4 Flows for “ferromanganese – inputs” and its representation in the ecoinvent database. Values correspond to the functional unit of 1 kg ferromanganese.

| General Flow Information                                 |              |                                       | Representation in ecoinvent                                                                                                                                                                          |                  |                         |                 |          |                                                                    |            | Uncertainty Informations |                       |      |                    |                          |
|----------------------------------------------------------|--------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------------------|-----------------|----------|--------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|--------------------------|
| Input                                                    | Process Name | Output                                | Remarks                                                                                                                                                                                              | Category         | Sub-category            | Infra-structure | Location | Modul name in ecoinvent                                            | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment          |
| ferromanganese, high-coal, 74.5% Mn, at regional storage | m            | Effluents                             | Inventoried as wastewater treatment w/o pollution. For overall process, wastewater emission see Tab. 5.7                                                                                             | waste management | wastewater treatment    | No              | CH       | treatment, sewage, unpolluted, to wastewater treatment, class 3    | 7.45E-03   | m <sup>3</sup>           | (Adelhardt 1998)      | 1    | 1.13               | (2,2,3,1,1,2,6)          |
|                                                          | m            | dust in wet scrubber                  | from wet scrubber, deposited in landfill                                                                                                                                                             | waste management | inert material landfill | No              | CH       | disposal, inert waste, 5% water, to inert material landfill        | 2.53E-02   | kg                       | (Adelhardt 1999)      | 1    | 1.13               | (2,2,3,1,1,2,6)          |
|                                                          | m            | NOx                                   |                                                                                                                                                                                                      | air              | low population density  |                 |          | Nitrogen oxides                                                    | 5.88E-03   | kg                       | (IPPC, 2001)          | 1    | 1.99               | reported values          |
|                                                          | m            | NM VOC                                |                                                                                                                                                                                                      | air              | low population density  |                 |          | NM VOC, non-methane volatile organic compounds, unspecified origin | 1.70E-03   | kg                       | (IPPC, 2001)          | 1    | 1.52               | (2,3,2,3,1,3,16)         |
|                                                          | m            | carbon dioxide from dolomite and lime | from calcination of dolomite and lime. CO2 from burning of fuels is inventoried in the "heat at..." module                                                                                           | air              | low population density  |                 |          | Carbon dioxide, fossil                                             | 1.09E-01   | kg                       | calculation           | 1    | 1.3                | (4,1,1,1,1,5,14)         |
|                                                          | m            | Dust (TSP)                            | Dust is the sum of particulate emissions represented in fractions PM<2.5, PM2.5-10, PM>10, including metals. These emissions are inventoried as shown in Tab 5.7, they are not shown in this figure. |                  |                         |                 |          |                                                                    | 1.42E-02   | kg                       | (IPPC, 2001)          | 1    | 1.97               | based on reported values |
|                                                          | m            | SO2                                   |                                                                                                                                                                                                      | air              | low population density  |                 |          | Sulfur dioxide                                                     | 3.00E-04   | kg                       | (IPPC, 2001)          | 1    | 4.17               | based on reported values |
|                                                          | m            | CO                                    |                                                                                                                                                                                                      | air              | low population density  |                 |          | Carbon monoxide, fossil                                            | 1.00E-03   | kg                       | (IPPC, 2001)          | 1    | 5.01               | (2,3,2,3,1,3,17)         |
|                                                          | m            | PCDD                                  | de-novo synthesis                                                                                                                                                                                    | air              | low population density  |                 |          | Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin           | 1.00E-12   | kg                       | (IPPC, 2001)          | 1    | 3.01               | (2,3,2,3,1,3,21)         |
|                                                          | m            | slag                                  | sold to silicomanganese production, not inventoried                                                                                                                                                  |                  |                         |                 |          |                                                                    | 1.55E+00   | kg                       | (Adelhardt 1998)      | 1    | 1.13               | (2,2,3,1,1,2,6)          |
|                                                          | m            | ferromanganese                        |                                                                                                                                                                                                      | metals           | extraction              | No              | RER      | ferromanganese, high-coal, 74.5% Mn, at regional storage           | 1.00E+00   | kg                       | (Adelhardt 1998)      |      |                    |                          |

Fig. 5.5 Flows for "ferromanganese – outputs" and its representation in the ecoinvent database. Values correspond to the functional unit of 1 kg ferromanganese.

### 5.3 Production of metallic manganese

Manganese metal is mainly used in the iron, steel and aluminium industry. In aluminium production for instance manganese is used as an alloying element improving the product hardness. For the production of pure manganese metal several processes are known (Wellbeloved et al. (1997), IPPC (2001)):

- Electrolysis of aqueous manganese salts
- Electrothermal decomposition of manganese ores
- Silico-thermic reduction of manganese ores or slag
- Alumino-thermic reduction of manganese ores or slag
- Distillation of ferro-manganese
- Electrolysis of fused salts

From the above listed processes the first two are the most important ones. The other processes are not economically significant.

**Electrolysis of aqueous manganese salts:** The production of manganese metal by the electrolysis of aqueous manganese salts requires at first a milling of the manganese ore. Milling increases the active surface and ensures sufficient reactivity in both the reduction and the subsequent leaching steps. After milling the manganese ore is fed to a rotary kiln where the reduction and calcination takes place. This process is carried out at about 850 - 1000 °C in a reducing atmosphere. As a reducing agent, several carbon sources can be used e.g. anthracite, coal, charcoal and hydrocarbon oil or natural gas. The calcined ore needs to be cooled below 100 °C to avoid a further re-oxidation. The subsequent leaching process is carried out with recycled electrolyte, mainly sulphuric acid. After leaching and filtration the iron content is removed from the solution by oxidative precipitation and the nickel and cobalt are removed by sulphide precipitation. The purified electrolyte is then treated with SO<sub>2</sub> in order to ensure plating of γ-Mn during electrolysis. Electrolysis is carried out in diaphragm cells. The cathode is normally made of stainless steel or titanium. For the anode lead-calcium or lead-silver alloy can be used. After an appropriate reaction time the cathodes are removed from the electrolysis bath. The manganese that is deposited on the cathode starter-sheet is stripped off mechanically and then washed and dried. The metal is crushed to produce metal flakes or powder or granulated, depending on the end use.

**Electrothermal decomposition of manganese ores:** The electrothermal process is the second important process to produce manganese metal in an industrial scale. The electrothermal process takes place as a multistage process. In the first stage manganese ore is smelted with only a small amount of reductant in order to reduce mostly the iron oxide. This produces a low-grade ferro-manganese and a slag that is rich in Mn-oxide. The slag is then smelted in the second stage with silicon to produce silico-manganese. The molten silicomanganese can be treated with liquid slag from the first stage to obtain relatively pure manganese metal. For the last step a ladle or shaking ladle can be used. The manganese metal produced by the electrothermal process contains up to 98% of Mn.

#### 5.3.1 Emissions & Wastes

The general manufacturing of refractory metals normally takes place at high temperatures. The environmental impact on air, water and land can be classified as follows.

**Emissions to air:** Dust and fume emissions from smelting, hard metal and carbide production; Other emissions to air are ammonia (NH<sub>3</sub>), acid fume (HCl), hydrogen fluoride (HF), VOC and heavy metals.

**Emissions to water:** Overflow water from wet scrubbing systems; Wastewater from slag and metal granulation; Blow down from cooling water cycles.



**Wastes:** Dust, fume and sludge; Slag.

### 5.3.2 “manganese, pure metal, at regional storage” in ecoinvent

The module describes the manganese metal consumed in Europe in 1994. It is designed solely for the use of the metal in special applications like sputtering or alloying element. This module is explicitly not to be used as alloying element in bulk iron or steel industry. The overall quality of this module is poor due to a lack of data.

The layout of the module with general flow information, remarks, sources, values and uncertainty information is shown in Fig. 5.6. Ecoinvent meta information for this process are summarised in Tab. 5.1 on page 11. The following paragraphs describe calculations, sources and assumptions chosen in this study. The functional unit of this process is one tonne of manganese consumed in Europe. No information is available on the share of the two major production processes, a ratio electrothermal/electrolysis of 75/25 is assumed.

#### **Electrolytic process**

No information on material input and output were available. The energy demand is stated in Wellbeloved et al. (1997) to be 9 - 12 kWh per kg manganese. Considerable amounts of chemicals ought to be used for reduction and leaching, but no detailed information is available. Hence none of these emissions are taken in account. For grinding prior to electrolysis, a value of 20 kWh electrical energy per tonne processed ore is assumed, reported in IPPC (2002) for base metal sites. A yield of 60 % is roughly estimated.

#### **Electrothermal process**

The production of bulk manganese by this method is a well protected secret and little information appears in the literature. The process is a refinement of the silicothermic reduction of manganese. It resembles that of low-carbon ferromanganese in electric arc furnaces of 1 – 3 MW power consumption. The raw material used is either low-iron manganese ore or a manganese slag concentrate. The silicon is in the form of a special silicomanganese which is made from a manganese slag concentrate or from pure ores. Reduction as complete as possible is ensured by the addition of lime, which forms a basic slag (Wellbeloved et al. (1997)).

In this study it is assumed, that 75 % of the Mn contained in the raw material originates from ore with a special grade, which is processed in a way similar to high-coal ferromanganese, the rest being slag, which is burden free. No information on the further refining process of manganese is available and no estimations are made in this study. Adelhardt & Saiger (1999) reports efficiencies between 60 % and 65 %, for this study the average of 62.5% was chosen.

**Import to Europe:** Since some transport intensity is already included in the 1.2 kg ferromanganese used for the electrothermic process, no further transportation is inventoried within this module. This approach is obviously not correct, but it will not influence the data quality of the inventoried process, which is already poor.

#### **Data quality**

This process is mainly based on rough estimates. The process specific emissions could be considerable but due to lack of data none are considered. This module only is to be used, where it plays a minor role, like in sputtering. If the impact of the modules use is considered to be high, more specific investigations have to be conducted.

| General Flow Information               |                                |                 | Representation in ecoinvent |             |                        |                 |          |                                                          |            | Uncertainty Informations |                       |      |                    |                                 |
|----------------------------------------|--------------------------------|-----------------|-----------------------------|-------------|------------------------|-----------------|----------|----------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|---------------------------------|
| Input                                  | Process Name                   | Output          | Remarks                     | Category    | Sub-category           | Infra-structure | Location | Module name in ecoinvent                                 | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                 |
| ferromanganese                         | manganese, at regional storage |                 |                             | metals      | extraction             | No              | RER      | ferromanganese, high-coal, 74.5% Mn, at regional storage | 1.21E+00   | kg                       | Calculation           | 1    | 1.62               | (5,5,1,1,1,5,4); rough estimate |
| high silica slag                       |                                |                 | not inventoried             |             |                        |                 |          |                                                          | 9.68E-01   | kg                       | Calculation           | 1    | 1.62               | (5,5,1,1,1,5,4); rough estimate |
| manganese concentrate for electrolysis |                                |                 |                             | metals      | extraction             | No              | GLO      | manganese, concentrate, at beneficiation                 | 9.83E-01   | kg                       | Calculation           | 1    | 1.62               | (5,5,1,1,1,5,4); rough estimate |
| Energy for electrolysis                |                                |                 |                             | electricity | production mix         | No              | UCTE     | electricity, medium voltage, production UCTE, at grid    | 2.63E+00   | kWh                      | (Wellbeloved 197)     | 1    | 1.14               | based on theoretical values     |
| Infrastructure, unspecified            |                                |                 |                             | metals      | extraction             | Yes             | GLO      | non-ferrous metal, smelter                               | 1.06E-14   | unit                     | rough estimate        | 1    | 3.32               | (5,5,1,1,1,5,9)                 |
|                                        |                                | waste heat      |                             | air         | low population density |                 |          | Heat, waste                                              | 9.47E+00   | MJ                       |                       | 1    | 1.16               | (3,2,3,1,1,2,13)                |
|                                        |                                | manganese metal | global average              | metals      | extraction             | No              | RER      | manganese, at regional storage                           | 1.00E+00   | kg                       |                       |      |                    |                                 |

Fig. 5.6 Flows for “Manganese, pure metal, at regional storage” and its representation in the ecoinvent database. Values correspond to the functional unit of 1 kg manganese metal.

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## Part VIII

# **Lead, Zinc and the Co-Products Cadmium and Indium**

Data v2.1 (2009)

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## Comments on this part

The production of lead, zinc, cadmium and indium is inventoried in the modules described in this report. Data reported are mainly based on literature and theoretical models.

The Swiss Federal Office for the Environment (FOEN) commissioned Empa in 2005 to identify the datasets that have to be improved in order to improve the overall quality and applicability of metals inventories in the metal processing and electronic industry. Based on that preparatory study the significance of the metals extracted from combined lead and zinc ores was identified. In order to improve and extend datasets the original parts have been reworked completely for ecoinvent version 2.0. It has been extended by a series of metals that are mainly produced from intermediate raw materials originating from lead-zinc extraction: cadmium, silver and indium.

This part corresponds to the original report in Althaus et al. (2004) on lead and zinc. The inventory modelling has been done consistently with multi output modules using economic allocation. All the errors reported in previous versions 1.x as documented in Frischknecht et al. (2006) have been corrected. Notably the emission factors to air for primary zinc production have been corrected and updated with the revised source (EEA, 2003). Likewise the values in Tab. 6.19 have been corrected as reported in the changes to v1.1.

Chapter “6 Cumulative Results and Interpretation” of the original report has been deleted for the current update.



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**Oxide Ores**

**Sulfide and Oxide Ores**

**Sulfide Ores**

**Interconnected Carrier Metal Cycles**

**Legend:**

- Carrier metals, bulk metals, generally of lower value.
- Co-elements that also have considerable production infrastructure. Valuable to high economic value; some used in high-tech applications.
- Co-elements that have no, or limited considerable production infrastructure. Mostly highly valuable, high-tech metals (e.g., essential in electronics).
- Co-elements that end up in residues, or as emissions. Costly because of waste management or end-of-pipe measures.

ecoinvent v2.1 report No. 10

## 2 Reserves and Resources of Materials

### 2.1 Lead

Lead, like copper and zinc, is a markedly chalcophilic element. Its mean abundance in the Earth's crust is 16 g/t (0.0016 %). Enrichments occur only in the pneumatolytic and especially the hydrothermal sulphide phase in the form of vein (lode), impregnated, and particularly metasomatic deposits. Sedimentary enrichments are of little importance.

Galena or lead sulphide [CAS 1314-87-0], PbS, theoretically 86.6 % Pb, is by far the most important lead mineral and the sole primary one. Other common forms of naturally occurring lead are usually associated with or derived from galena deposits. Frequent impurities in lead minerals are zinc, copper, arsenic, tin, antimony, silver, gold, and bismuth. Of the total world lead and zinc mining output, mixed lead-zinc ore deposits are the most important and account for ca. 70 % of the total production of both metals. In second place are deposits that contain predominantly lead ores (ca. 20% of total production), and the remainder (ca. 10%) is obtained as a by-product from zinc, copper – zinc, and other deposits (Sutherland et al., 1997). For sulphidic deposits of the zinc-lead-copper type Krauss et al. (1999) reports an average composition in parts of metal by the mass of 2.4 Zn : 1.4 Pb : 1 Cu. Mined lead ores contain approximately 6 % Pb in average (Ayres et al., 2002).

**Tab. 2.1** Mine production and reserves of lead in 2005. “Reserves” is referred to as “the part of the reserve base which could be economically extracted or produced at the time of determination”, while “Reserve base” denotes “that part of an identified resource that meets specified minimum (...) criteria related to current mining and production practices” (USGS, 2006b).

| Lead                     | Mine production  |                                 | Reserves | Reserve base |
|--------------------------|------------------|---------------------------------|----------|--------------|
|                          | 2005 [kt Pb / a] | 2006 [kt Pb / a]<br>(estimated) | [kt Pb]  | [kt Pb]      |
| United States            | 426              | 430                             | 8100     | 20000        |
| Australia                | 776              | 780                             | 15000    | 28000        |
| Canada                   | 73               | 79                              | 2000     | 9000         |
| China                    | 1000             | 1050                            | 11000    | 36000        |
| India                    | 58               | 60                              | NA       | NA           |
| Ireland                  | 64               | 65                              | NA       | NA           |
| Kazakhstan               | 44               | 55                              | 5000     | 7000         |
| Mexico                   | 130              | 140                             | 1500     | 2000         |
| Morocco                  | 31               | 42                              | 500      | 1000         |
| Peru                     | 319              | 320                             | 3500     | 4000         |
| Poland                   | 48               | 60                              | NA       | 5400         |
| South Africa             | 42               | 50                              | 400      | 700          |
| Sweden                   | 61               | 61                              | 500      | 1000         |
| Other countries          | 198              | 170                             | 19000    | 30000        |
| World total<br>(rounded) | 3270             | 3360                            | 67000    | 140000       |

## 2.2 Lead as secondary resource

Recycling of lead is performed at a high level world wide (USGS, 2006b). The production of secondary lead, for example in the US, has risen from about 23'131 tonnes in 1907<sup>1</sup> (USGS, 2006c) to 1'100'000 tonnes in 2005 (Gabby, 2006.) At present it is estimated that about 85.0% lead used can economical be recovered (Thornton et al., 2001). Because of the permanently increasing demand (demand is higher than lead is available), the price of lead raised comparably (as general tendency for large price fluctuations can also be observed in Fig. 1).

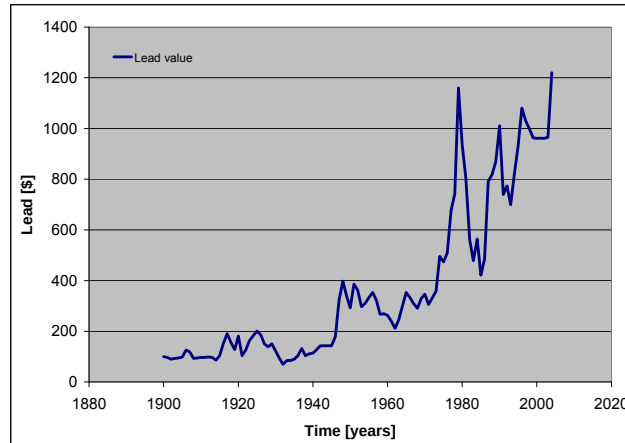


Fig. 2.1 Value of lead (adopted from DiFrancesco et al., 2006).

On a global level, the refining of secondary lead has increased from ~ 1'500'000 tons in 1976 to approximately 2'800'000 tons in 1998 (Dzioubinski & Chipman, 1999). Key source for secondary lead are lead-acid batteries contributing with 92.0% to the total secondary lead production of the US (Gabby (2004)). The treatment of such batteries is enforced by adequate governmental directives, for instance in France (100.0%), Denmark (99.9%) and Portugal (75.0%) (Thornton et al., 2001). Beside automotive batteries, secondary lead is recovered from used lead-acid rechargeable batteries as for instance applied for back-up power supply in antenna stations of mobile phone networks (Scharnhorst, 2005).

In recent years, the primary production has seen a shift from pure primary production to more flexible technologies that are able to treat increasingly also secondary materials (QSL furnace at Stolberg facility, Kivcet furnace at Trail). Hence, the distinction between processing primary and secondary lead is more and more blurred. The UK based consultancy CHR Metals estimate that the amount of secondary lead production has raised to 58% of the world lead supply (Roberts, 2003). The ILZSG-statistics indicate a further decrease in the share of primary (mine) production compared to the total reported metal production, viz. an increase in secondary (post consumer) production (Tab. 2.2).

<sup>1</sup> The earliest documented statistics on secondary lead.

Tab. 2.2 Lead production and consumption worldwide. Statistics taken from ILZSG (2007) and USGS (2006b).

|                | Mine Production     | Metal Production | Metal Consumption | Ratio mine production / metal production |
|----------------|---------------------|------------------|-------------------|------------------------------------------|
| Year           | Thousand tonnes     | Thousand tonnes  | Thousand tonnes   | %                                        |
| 2000           | 3100 *              |                  |                   |                                          |
| 2001           | 3100 *              |                  |                   |                                          |
| 2002           | 2910 *              |                  |                   |                                          |
| 2003           | 3111 (2950 *)       | 6764             | 6835              | 46%                                      |
| 2004           | 3129 (3150 *)       | 6957             | 7279              | 45%                                      |
| 2005           | 3415 (3270 *)       | 7637             | 7800              | 45%                                      |
| 2006           | 3492 (3360, est. *) | 8024             | 8031              | 44%                                      |
| * USGS (2006b) |                     |                  |                   |                                          |

## 2.3 Zinc

With an average concentration in the earth's crust of 65 g/t (0.0065 %), zinc is the 24th element in order of abundance. It occurs only in the chemically combined state. Like copper and lead, zinc is a strongly chalcophilic element. It usually occurs as sulphide, mostly deposited from hydrothermal solutions and simic and sialic magmas. Sedimentary deposits are of less importance.

By far the most important zinc mineral today, and probably also the only primary ore, is zinc blende. The other minerals listed here are of only local importance. Native zinc is practically nonexistent. Isomorphous components of zinc blende such as cadmium, gallium, germanium, indium, and thallium do not form deposits of their own. Some of these metals are recovered during zinc ore processing.

Zinc blende (or sphalerite, ZnS) has the theoretical composition 67.09 % Zn, 32.91 % S. It is light yellow to black in colour, usually black to brown depending on the iron content, crystallises in the regular tetrahedral system, and has a Mohs hardness of 3 – 4 and a density of 3.9 – 4.1 g/cm<sup>3</sup>. The most important impurity is FeS, which is always present at concentrations between 0.3 and >20 %. Sulphides of Pb, Cd, Mn, and Cu are often present. Zinc blende also often contains small amounts of As, Sn, Bi, Co, Hg, In, Tl, Ga, and Ge, and nearly always contains Ag and often Au. The hexagonal modification of Zinc blende is known as Wurzite (conchoidal or radiating blende). Its density is 3.98 g/cm<sup>3</sup>. Other less important minerals are: Smithsonite (zinc spar, calamine) ZnCO<sub>3</sub>; Hemimorphite, Zn<sub>2</sub>SiO<sub>4</sub> · H<sub>2</sub>O; Willemite, Zn<sub>2</sub>SiO<sub>4</sub>; Franklinite, (Zn,Fe,Mn)O · (Fe<sub>2</sub>,Mn<sub>2</sub>)O<sub>3</sub>, zinc bloom (hydrozincite), ZnCO<sub>3</sub> · n Zn(OH)<sub>2</sub>; red zinc ore (zincite), ZnO; and troostite (bainite), (Zn,Mn)<sub>2</sub>SiO<sub>4</sub> (Graf & Schleining, 1997).

According to Ayres et al. (2002), the average concentration of the material currently mined is 3-11%. The annual mine production and world-wide reserves are listed in Tab. 2.3.

Approximately 30% of the yearly zinc consumption in Europe is secondary or recycled zinc. Approximately 50% of this secondary zinc is recycled within the industry. This is particularly true in the galvanising and brass sector. Scrap arising from the production or processing of products can be recycled almost immediately. Residues and scrap, which are relevant and significant to the secondary zinc industry, include dust from copper alloy making, residues from the die casting industry, ashes, bottom and top drosses from the galvanising industry, old roofing and other sheet materials, non-ferrous fraction from the shredding of old cars and of other mainly steel containing products, dust from electric arc steel making and cast iron making and residues from chemical uses of zinc and burnt tyres (IPPC, 2001).

**Tab. 2.3** Mine production and reserves of zinc in 2006. “Reserves” is referred to as “the part of the reserve base which could be economically extracted or produced at the time of determination”, while “Reserve base” denotes “that part of an identified resource that meets specified minimum (...) criteria related to current mining and production practices” (USGS, 2006b).

| Zinc                     | Mine production  |                                 | Reserves | Reserve base |
|--------------------------|------------------|---------------------------------|----------|--------------|
|                          | 2005 [kt Zn / a] | 2006 [kt Zn / a]<br>(estimated) | [kt Zn]  | [kt Zn]      |
| Australia                | 1'330            | 1'400                           | 33'000   | 80'000       |
| Canada                   | 755              | 725                             | 11'000   | 31'000       |
| China                    | 2'450            | 2'500                           | 33'000   | 92'000       |
| Kazakhstan               | 400              | 450                             | 30'000   | 35'000       |
| Mexico                   | 470              | 450                             | 8'000    | 25'000       |
| Peru                     | 1'200            | 1'210                           | 16'000   | 20'000       |
| Other countries          | 2'400            | 2'500                           | 59'000   | 87'000       |
| World total<br>(rounded) | 9'800            | 10'000                          | 220'000  | 460'000      |

## 2.4 Cadmium

Cadmium is widely distributed. It occurs in the earth's crust with a content estimated to be between 0.08 and 0.5 ppm. In top soil cadmium content usually lies between 0.1 and 1 ppm. None of known cadmium minerals such as cadmium sulfide (greenockite, 77.6 % Cd), cadmium carbonate (61.5 % Cd) and cadmium oxide (87.5 % Cd) is of industrial importance because the deposits are too small. Only zinc minerals in which cadmium is found as an isomorphic component, with concentrations ranging from 0.05 to 0.8 %, averaging about 0.2 % have economic significance for cadmium recovery. In addition, lead and copper ores contain small amounts of cadmium, which can be separated during the roasting and smelting processes (Schulte-Schrepping & Piscator, 2002).

The bulk of the cadmium being recovered is associated with ores of sphalerite (ZnS) that is extracted for its zinc content. Cadmium therefore is a by-product from the zinc extraction where it accumulates in flue dust (pyrometallurgical processes) and purification precipitates (hydrometallurgical processes). Estimated world resources of cadmium were about 6 million tons in 2006, based on identified zinc resources of 1.9 billion tons containing about 0.3% cadmium (USGS, 2007).

The recycling of cadmium is still on a low level. It basically occurs via the recovery of EAF dusts from zinc manufacturing. Hawkins et al. (2006) estimate that in the U.S. less than 20% of the cadmium in the product waste stream is currently collected for recycling. On a world wide average this number appears to be lower, a level of 10% seems to be a reasonable assumption.

This consideration, together with the assertion that 80% of the primary cadmium produced originates from zinc extraction (USGS, 2007) and under the assumption of stable stocks, leads to the estimate that 72% (=80% primary from zinc \* 90% primary in supply) of the cadmium supply may be related to the primary zinc production.

## 2.5 Indium

Indium does not occur as pure metal. Although widely distributed, it generally occurs in low concentrations. The content of indium in the earth crust is estimated to be 0.1 ppm. Indium is found as a trace element in many minerals and is accumulated during the extraction of the main metal values such as zinc, lead and copper.

### 2.5.1 Primary sources: Indium ores

Sphalerite is the most important indium-containing mineral followed by lead and copper sulfides. The amount of indium in sphalerite may vary widely even within a deposit. The indium concentration is typically 10 – 20 ppm, but may be as high as 1 % (Felix, 1996). An overview over different deposit types and ore grade / tonnage relationship is given in Fig. 2.2.

Due to its low concentrations indium is not mined for itself but is recovered as an important by-product of zinc metal processing operations. It accumulates during extraction in various intermediate products and waste. Due to the complex technology required to recover indium, only a few operations are able to produce the metal. The main producers of indium (Umicore, Dowa Mining, Nippon Mining, Sumitomo Metal Mining, Teck Cominco, Falconbridge) are all into zinc processing. However, Schwarz-Schampera & Herzig (2002) stress that the resource base is underestimated when only looking at zinc bearing minerals, since noteworthy deposits combined with tin (cassiterite) exist.

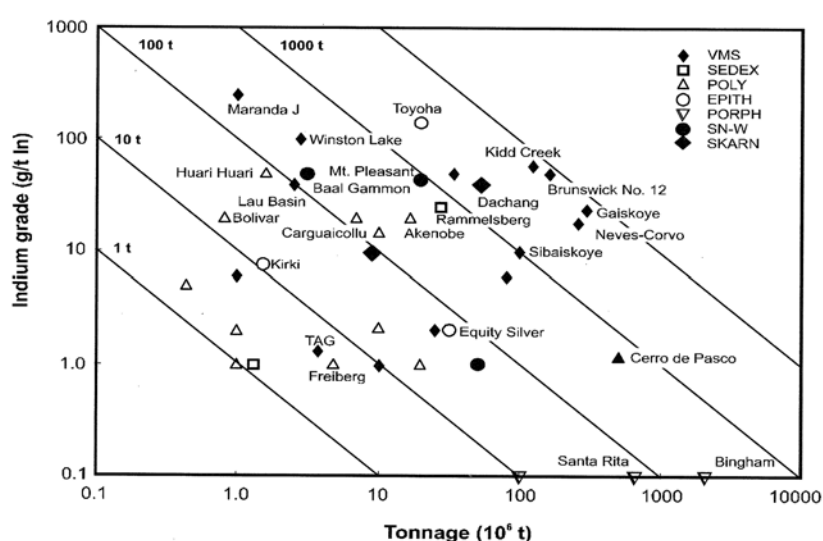


Fig. 2.2 Distribution of ore grade and reserve for different indium bearing deposits (Schwarz-Schampera & Herzig, 2002).

### 2.5.2 Secondary sources

Worldwide only small amounts of Indium (mainly Indium Tin Oxide (ITO) scrap) are recycled. The reason for the low recycling rate is the lack of infrastructure for collecting indium-containing products, although recycling is supposed to expand significantly if the current price of indium is sustained. Indium tin oxide (ITO) use is highly inefficient, as only about 15% of ITO is utilised in the LCD production process; the rest is scrap. The major problem in recycling the ITO scrap is the high cost associated with the process. The process takes about 12 weeks from collection of scrap to fabrication of secondary indium products. A large increase in ITO scrap could be difficult for the recycling industry to handle because of large capital costs, environmental restrictions, and storage space (USGS, 2006a).

Taking an average concentration of 2 ppm of ITO in collected backlights<sup>2</sup>, 90% recovery rate and 95% reprocessing efficiency, the required amount of scrap backlights to gain a kg indium would amount to 585 tonnes. Even with prices as high as 1'000 \$ per kg ITO a global collection system is not probable to evolve, and therefore no post consumer scrap recycling may be considered.

<sup>2</sup> Content in 17" backlight (1.8 kg) 1.76 ppm, in 15" backlight (1.1 kg) 2.05 ppm for total film thickness of 500 nm.

### 2.5.3 Indium Supply

The supply of indium mainly depends on a few zinc extracting operations in Canada, Russia, China and Japan (Stevens, 2007). Due to a sharp rise in ITO demand in LCD fabs the primary producer may not meet the demand. This development led to a dramatic increase of the price starting in 2003 (Fig. 2.3). The supply gap is filled with releases of strategic stocks and the processing of new scrap. In 2004 Teckcominco (2005) estimated that almost 50% of the supply was met by secondary (industry scrap) sources (Fig. 2.4). Stevens (2007) estimates that existing potentials for future indium supply (“x-factors”) may be activated by enhancing the recovery efficiency by 5%, treatment of secondary materials and a slowing down of the demand by LCD fabs. He estimates a share in reclaimed secondary (prompt scrap) sources of 60% in 2008 (Fig. 2.5).

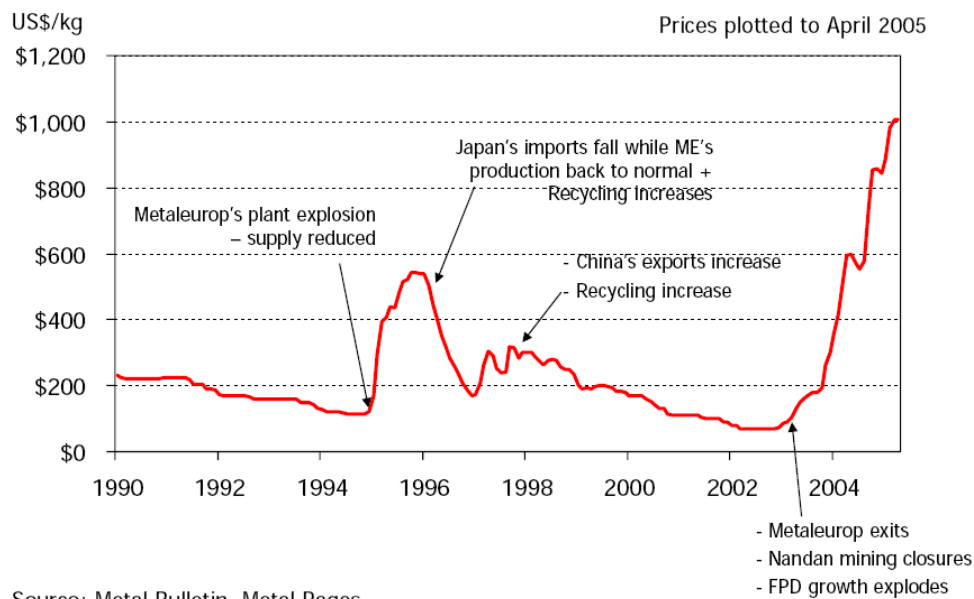


Fig. 2.3 Indium prices with comments (Teckcominco, 2005; Metal Bulletin, 2006).



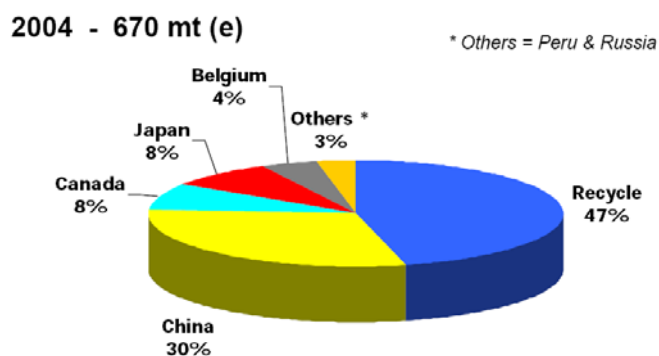


Fig. 2.4 Supply of Indium in 2004. Recycling of prompt scrap (ITO targets and the like) plays a vital role in the indium supply (Teckcominco, 2005).

| indium (metric tons) |                | 2004 | 2005 | 2006 | 2007 | 2008 |
|----------------------|----------------|------|------|------|------|------|
| sources              | Virgin         | 321  | 424  | 464  | 510  | 610  |
|                      | Reclaim        | 250  | 357  | 541  | 729  | 911  |
| uses                 | FPD (25% CAGR) | 416  | 595  | 833  | 1041 | 1302 |
|                      | Other          | 159  | 169  | 172  | 177  | 218  |
| x-factor net         |                | -4   | 17   | 0    | 21   | 1    |

Fig. 2.5 Estimated sources of Indium. Assumption that all potentials for indium recovery are seized and that the demand will slow down (Stevens, 2007). FPD = Flat panel display, CAGR = compound annual growth rate.

### 3 Coupled Production

The metals lead, zinc, cadmium, silver (contained in the parkes process crust of primary lead production) and indium are mainly found coupled in deposits. A compilation of world production is given in Tab. 3.1.

Namely the cadmium metal is almost exclusively from Zinc ores. Today approximately 80% of world cadmium production (=90% of the supply) was derived from mining, smelting, and refining of zinc, and the remaining 20% came from copper and lead smelting and the recycling of cadmium products (USGS, 2007). The ratio between the world mine production of Cadmium and Zinc is quite stable over the last 20 years at about 2 kg Cd produced per tonne extracted Zinc. This corresponds to the average concentration of Cd found in worked Zn ores of about 0.3%.

**Tab. 3.1 World production in metric tonnes per year compiled from Kelly & Matos (2007) through 2004, for 2005 and 2006 the respective mineral commodity summaries from USGS (2006b) were taken.**

|      | Lead      | Silver | Zinc       | Cadmium  | Indium   | Cd / Zn | In / Zn  | Ag / Pb  |
|------|-----------|--------|------------|----------|----------|---------|----------|----------|
|      |           |        | Mine       | Refinery | Refinery |         |          |          |
| year | t/a       | t/a    | t/a        | t/a      | t/a      | Ratio   | Ratio    | ratio    |
| 1986 | 3'240'000 | 13'000 | 6'840'000  | 19'100   | 41       | 0.28%   | 6.01E-06 | 4.01E-03 |
| 1987 | 3'430'000 | 14'000 | 7'190'000  | 19'000   | 54       | 0.26%   | 7.44E-06 | 4.08E-03 |
| 1988 | 3'420'000 | 15'500 | 6'770'000  | 21'900   | 106      | 0.32%   | 1.57E-05 | 4.53E-03 |
| 1989 | 3'400'000 | 16'400 | 6'820'000  | 21'400   | 115      | 0.31%   | 1.69E-05 | 4.82E-03 |
| 1990 | 3'370'000 | 16'600 | 7'150'000  | 20'200   | 118      | 0.28%   | 1.65E-05 | 4.93E-03 |
| 1991 | 3'260'000 | 15'600 | 7'270'000  | 20'900   | 140      | 0.29%   | 1.93E-05 | 4.79E-03 |
| 1992 | 3'200'000 | 14'900 | 7'250'000  | 19'600   | 140      | 0.27%   | 1.93E-05 | 4.66E-03 |
| 1993 | 2'900'000 | 14'100 | 6'910'000  | 18'700   | 140      | 0.27%   | 2.03E-05 | 4.86E-03 |
| 1994 | 2'800'000 | 14'000 | 7'050'000  | 18'200   | 145      | 0.26%   | 2.06E-05 | 5.00E-03 |
| 1995 | 2'710'000 | 14'900 | 7'280'000  | 20'100   | 239      | 0.28%   | 3.28E-05 | 5.50E-03 |
| 1996 | 2'920'000 | 15'100 | 7'480'000  | 18'900   | 200      | 0.25%   | 2.67E-05 | 5.17E-03 |
| 1997 | 3'100'000 | 16'500 | 7'540'000  | 20'300   | 230      | 0.27%   | 3.05E-05 | 5.32E-03 |
| 1998 | 3'060'000 | 17'200 | 7'570'000  | 20'200   | 230      | 0.27%   | 3.04E-05 | 5.62E-03 |
| 1999 | 3'080'000 | 17'600 | 7'960'000  | 20'000   | 215      | 0.25%   | 2.70E-05 | 5.71E-03 |
| 2000 | 3'200'000 | 18'100 | 8'770'000  | 20'300   | 335      | 0.23%   | 3.82E-05 | 5.66E-03 |
| 2001 | 3'120'000 | 18'900 | 8'910'000  | 20'000   | 408      | 0.22%   | 4.58E-05 | 6.06E-03 |
| 2002 | 2'870'000 | 18'500 | 8'520'000  | 18'000   | 366      | 0.21%   | 4.30E-05 | 6.45E-03 |
| 2003 | 3'140'000 | 18'400 | 9'600'000  | 18'500   | 256      | 0.19%   | 2.67E-05 | 5.86E-03 |
| 2004 | 3'110'000 | 19'700 | 9'600'000  | 18'700   | 406      | 0.19%   | 4.23E-05 | 6.33E-03 |
| 2005 | 3'270'000 | 19'300 | 9'800'000  | 19'400   | 500      | 0.20%   | 5.10E-05 | 5.90E-03 |
| 2006 | 3'360'000 | 19'500 | 10'000'000 | 20'900   | 480      | 0.21%   | 4.80E-05 | 5.80E-03 |

## 4 Characterisation of the materials

### 4.1 Lead

The element lead (Pb, CAS 7439-92-1) is a lustrous metal. When freshly cut, surfaces are bluish white but oxidise readily to the familiar dull grey colour. Lead has four naturally occurring isotopes:  $^{208}\text{Pb}$  (51 – 53 %),  $^{206}\text{Pb}$  (23.5 – 27 %),  $^{207}\text{Pb}$  (20.5 – 23 %) and  $^{204}\text{Pb}$  (1.35 – 1.5 %). Three of the natural isotopes are radioactive ( $^{208}\text{Pb}$ , end product of thorium decay;  $^{206}\text{Pb}$  and  $^{207}\text{Pb}$ , from uranium series). A number of unstable radioactive isotopes of Pb also exist. Lead is a preferred shielding material against gamma rays and X-rays; if sufficiently pure, it does not become radioactive when used for shielding against neutron sources (Sutherland et al., 1997). Some physical properties are listed in Tab. 4.1.

Tab. 4.1 Some physical properties of lead (Sutherland et al., 1997).

| Physical properties of lead                                    |         |
|----------------------------------------------------------------|---------|
| Melting point °C                                               | 327.4   |
| Latent heat of fusion, J/g                                     | 23.4    |
| Boiling point °C                                               | 1741    |
| Latent heat of vaporisation, J/g                               | 862     |
| Vapour pressure, kPa 980 °C                                    | 0.133   |
| Density, g/cm <sup>3</sup> 20 °C                               | 11.336  |
| Mean specific heat, J g <sup>-1</sup> K <sup>-1</sup> 0–100 °C | 0.131   |
| Specific electrical resistance, Ω/cm 20 °C                     | 20.65   |
| Brinell hardness (20 °C)                                       | 2.5–3.0 |

### 4.2 Zinc

The remarkably low boiling point of 906 °C is important in pyrometallurgical zinc production processes. Zinc can be worked at temperatures as low as 120 °C, and its good deformability is the basis for some of its uses. Zinc is resistant to air because of the self-protective coating formed. It is also resistant to halogens, but is rapidly corroded by HCl gas. Because of its high surface area, zinc dust is much more reactive and can even be pyrophoric, e.g., reacting vigorously at elevated temperatures with the elements oxygen, chlorine, and sulphur (Graf & Schleining, 1997). Zinc is traded in different purity grades, which are specified by different authorities. In Northern America, SHG is the highest purity zinc grade, followed by HG and finally PW. Comparable specifications for primary zinc grades can be found in Europe (European Standard EN 1179, Tab. 4.2), in Japan (Japanese Industrial Standard JIS H 2107), and in Australia (Australian Standard AS 1242).

Tab. 4.2 Composition of different zinc grades in Europe, data provided by IZA<sup>3</sup>.

| EN 1179 Grade Designation | Pb    | Fe    | Cd    | Al    | Cu    | Sn | Total non-zinc | Zinc, min |
|---------------------------|-------|-------|-------|-------|-------|----|----------------|-----------|
| Z1                        | 0.003 | 0.002 | 0.003 | 0.001 | 0.001 | 0  | 0.005          | 100       |
| Z2                        | 0.005 | 0.003 | 0.005 | -     | 0.002 | 0  | 0.01           | 100       |
| Z3                        | 0.03  | 0.02  | 0.01  | -     | 0.002 | 0  | 0.05           | 100       |
| Z4                        | 0.45  | 0.05  | 0.01  | -     | -     | -  | 0.5            | 99.5      |
| Z5                        | 1.4   | 0.05  | 0.01  | -     | -     | -  | 1.5            | 98.5      |

Tab. 4.3 Some physical properties of zinc, data provided by IZA<sup>3</sup>.

| Physical properties of zinc                                        |                                                                    |
|--------------------------------------------------------------------|--------------------------------------------------------------------|
| Atomic number                                                      | 30                                                                 |
| Atomic weight                                                      | 65.37                                                              |
| Density (at 25°C)                                                  | 7140 kg/m <sub>3</sub>                                             |
| Melting point                                                      | 419.5°C (692.7 K)                                                  |
| Boiling point (760 mm Hg)                                          | 907°C (1180 K)                                                     |
| Modulus of elasticity                                              | 70'000 MN/m <sub>2</sub>                                           |
| Specific heat (20°C)                                               | 0.382 kJ/kg*K                                                      |
| Latent heat of fusion (419.5°C)                                    | 100.9 kJ/kg                                                        |
| Latent heat of vaporisation (906°C)                                | 1.782 MJ/kg                                                        |
| Heat capacity                                                      |                                                                    |
| - solid                                                            | $C_p = 22.40 + 10.5 \times 1/1000 \text{ 1/TJmol (298 - 692.7 K)}$ |
| - liquid                                                           | $C_p = 31.40 \text{ 1/Jmol}$                                       |
| - gas                                                              | $C_p = 20.80 \text{ 1/Jmol}$                                       |
| Linear coefficient of thermal expansion (polycrystalline 20-250°C) | 39.7 µm/m*K                                                        |
| Volume coefficient of thermal expansion (20 - 400°C)               | 0.89 x 1/1000000/K                                                 |
| Thermal conductivity: solid (18°C)                                 | 113W/m*K                                                           |
| Electrical resistivity (polycrystalline at 20°C)                   | 59.0 nΩ·m                                                          |
| Standard electrode potential (against H <sub>2</sub> electrode)    | -0.762 V                                                           |

### 4.3 Cadmium

Features of Cadmium metal are compiled in Tab. 4.4. Due to its toxicity the dissipative use has been cut down and regulated.

Tab. 4.4 Properties of cadmium<sup>4</sup>.

| Natural cadmium levels in the environment |                                 |
|-------------------------------------------|---------------------------------|
| Atmosphere                                | 0.1 to 5 ng/m <sup>3</sup>      |
| Earth's crust                             | 0.1 to 0.5 µg/g                 |
| Marine sediment                           | ~1 µg/g                         |
| Sea-water                                 | ~0.1 µg/l                       |
| Cadmium physical properties               |                                 |
| Form                                      | white, soft metal               |
| Characteristics                           | malleable, ductile and flexible |
| Melting point                             | 321°C                           |
| Atomic number                             | 48                              |
| Atomic weight                             | 112.41                          |
| Density                                   | 8.64 g/cm <sup>3</sup>          |

<sup>3</sup> <http://www.iza.com/properties.html>, access date may 2007

<sup>4</sup> <http://www.cadmium.org/>, access date april 2007

## 4.4 Indium

Indium is a very soft, silvery-white metal that has a bright luster. As a pure metal indium emits a high-pitched "cry" when it is bent. Both gallium and indium are able to wet glass. Indium has a low melting point (429.75 K) but a high boiling point (2353.15 K). Indium becomes superconducting at 3.37 K. Metallic indium is not oxidized by air or oxygen at room temperature. It reacts directly with arsenic, antimony, the halogens, oxygen, phosphorus, sulfur, selenium, and tellurium when heated. Indium dissolves only slowly in cold dilute mineral acids, but more readily in hot dilute or concentrated acids. Alkalis do not attack the massive metal. Indium forms alloys with most other metals. Extensive solid solutions are formed with lead, thallium, and mercury. Some properties of indium are compiled in Tab. 4.5 (Wikipedia Contributors, 2007; Felix, 1996).

Tab. 4.5 Some properties of the element indium (Wikipedia Contributors, 2007; Felix, 1996).

| Properties                                             | Unit                                              | Value                                                 |
|--------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------|
| Density (near r.t.)                                    | $\text{g}\cdot\text{cm}^{-3}$                     | 7.31                                                  |
| Liquid density at m.p.                                 | $\text{g}\cdot\text{cm}^{-3}$                     | 7.02                                                  |
| Melting point                                          | $^{\circ}\text{C}$                                | 156.6                                                 |
| Boiling point                                          | $^{\circ}\text{C}$                                | 2072                                                  |
| Heat of fusion                                         | $\text{kJ}\cdot\text{mol}^{-1}$                   | 3.281                                                 |
| Heat of vaporization                                   | $\text{kJ}\cdot\text{mol}^{-1}$                   | 231.8                                                 |
| Heat capacity (25 $^{\circ}\text{C}$ )                 | $\text{J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$  | 26.74                                                 |
| Standard atomic weight                                 | $\text{g}\cdot\text{mol}^{-1}$                    | 114.818(3)                                            |
| Electron configuration                                 |                                                   | [Kr] 4d <sup>10</sup> 5s <sup>2</sup> 5p <sup>1</sup> |
| Electrons per shell                                    |                                                   | 2, 8, 18, 18, 3                                       |
| Electrical resistivity (20 $^{\circ}\text{C}$ )        | $\text{n}\ \Omega\cdot\text{m}$                   | 83.7                                                  |
| Thermal conductivity (300 K)                           | $\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$    | 81.8                                                  |
| Thermal expansion (25 $^{\circ}\text{C}$ )             | $\mu\text{m}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ | 32.1                                                  |
| Young's modulus                                        | GPa                                               | 11                                                    |
| Mohs hardness                                          |                                                   | 1.2                                                   |
| Brinell hardness                                       | MPa                                               | 8.83                                                  |
| CAS registry number                                    |                                                   | 7440-74-6                                             |
| Standard electrode potentials                          |                                                   |                                                       |
| $\text{In}^{3+} + 3\text{e} \rightarrow \text{In}$     | V                                                 | -0.338                                                |
| $\text{In}^{3+} + 2\text{e} \rightarrow \text{In}^{+}$ |                                                   | -0.40                                                 |
| $\text{In}^{2+} + \text{e} \rightarrow \text{In}^{+}$  |                                                   | -0.40                                                 |
| $\text{In}^{3+} + \text{e} \rightarrow \text{In}^{2+}$ |                                                   | -0.49                                                 |

## 5 Use / Application of Material

### 5.1 Lead

Generally toxicity concerns and substitution by plastics and copper have reduced the market considerably. Main applications of lead are the following (Sutherland et al., 1997):

**Batteries:** The battery market is by far the largest end use of lead. Batteries for starting, lighting and ignition (SLI) represent the predominant use of lead. Traction batteries are used to power electric vehicles. Stationary batteries are used for standby and emergency power supply.

**Pigments and Lead Compounds:** After batteries, the next largest use of refined lead is in glass making, ceramic glazes, plastic stabilisers, and paints. The lead pigment market is now very small.

**Semi finished Products:** Rolled and extruded lead, mainly sheet and pipe, account for about 8 % of total lead use.

**Cable Sheathing:** A major area of lead use was in the protection of electrical and telecommunications cables, but this market has declined drastically due to substitution by aluminium and plastics.

**Lead Alloys:** Solders, type metal, bearing alloys containing lead still account for about 4 % of the lead consumption.

**Gasoline Additives:** In 1972 the use of tetraethyl lead for improving the octane rating of gasoline was nearly 400'000 t/a. Since then, this sector has declined by 70 % due to restrictions through environmental legislation.

**Shot and Ammunition:** Shot and ammunition account for about 2 % of lead consumption and represent a steady market.

**Miscellaneous:** Uses that do not fall into the above categories account for about 4 % of lead use, and include products featuring the high density of lead such as wheel weights, yacht keels, ornamental items, stained glass, and massive radiation shielding.

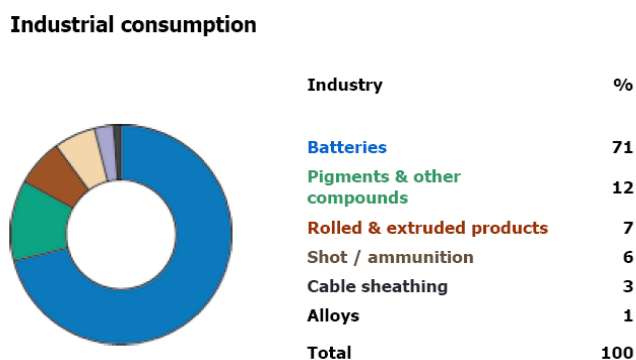


Fig. 5.1 Applications of lead in 2001 (LME website<sup>5</sup>)

<sup>5</sup> [http://www.lme.com/lead\\_industryusage.asp](http://www.lme.com/lead_industryusage.asp), accessed in may 2007

## 5.2 Zinc

Due to its manifold properties, Zinc is used in a broad field of applications (Tab. 5.1). The quantities of first- and end-use are given in Fig. 5.2. Main uses are as follows (Graf & Schleining, 1997):

In the industrialised countries, almost 50 % of the zinc consumed is used for corrosion protection. Commercial zinc (98.5 – 99.5 %) is usually used for hot-dip galvanising in production and jobbing plants, and high-purity zinc (99.95 %) for the continuous galvanising of steel strip or wire. The metallurgical processes used to produce the less pure grades of zinc means that lead is the main impurity present. This is an advantage, as the presence of 1 % lead reduces the surface tension of zinc by ca. 40 % compared with that of the high-grade product. The components being galvanised are therefore more effectively wetted by the molten zinc.

Pressure die casting is another major area of use and is dominated by alloys of the type ZnAl4Cu1 and modifications thereof. Approximately 50 % of all zinc pressure die castings go to the automobile industry.

Zinc-based chemicals, including zinc dust, account for ca. 7 % of the worldwide zinc consumption. Zinc oxide is quantitatively by far the most important zinc-based chemical product, followed by zinc dust, zinc sulphate, and zinc chloride in order of importance and quantity. All other zinc compounds are of minor importance.

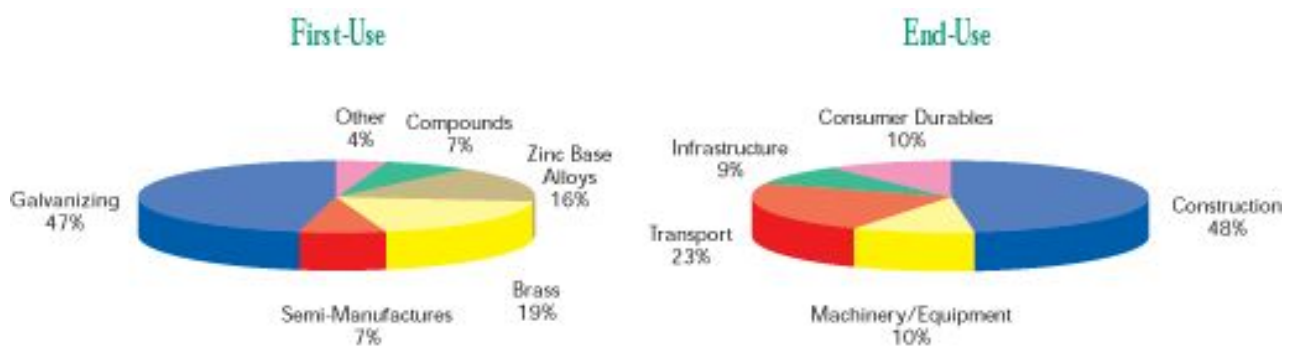


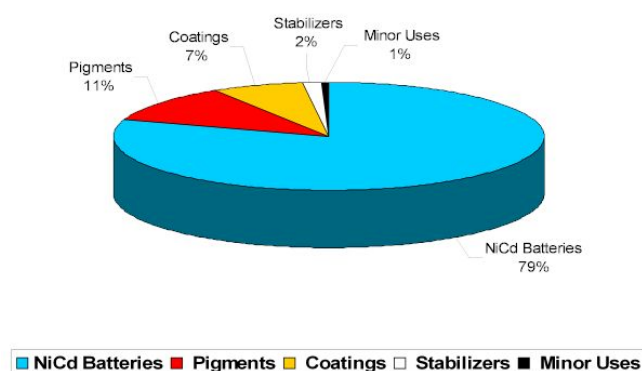
Fig. 5.2 Zinc consumption in 2000: first-use and end-use.

Tab. 5.1 Properties, first- and end-use of zinc, data provided by IZA<sup>6</sup>.

| Property                                                                            | First-Use                                                                                   | End-Use                                                                                  |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| - Reactivity with iron<br>- Corrosion resistance<br>- Electrochemical               | Corrosion protection (galvanising, zinc thermal spraying, electroplating, zinc-rich paints) | Building/construction, energy/power, street furniture, agriculture, automotive/transport |
| - Low melting point<br>- Fluidity<br>- Capacity for surface treatment<br>- Strength | Die casting and gravity casting                                                             | Automotive equipment, household appliances, fittings, toys, tools, etc.                  |
| - Alloying characteristics                                                          | Brass (copper-zinc alloy), aluminium alloys, magnesium alloys                               | Building/construction, fittings, automotive and electrical components, etc.              |
| - Formability<br>- Resistance to corrosion                                          | Rolled zinc                                                                                 | Building/construction                                                                    |
| - Electrochemical                                                                   | Batteries                                                                                   | Automotive/transport, computers, medical equipment, consumer products                    |
| - Chemical                                                                          | Zinc oxide, zinc stearate                                                                   | Tyres, all rubber goods, paint pigments, ceramic glazes, electrostatic copying paper     |
| - Essential nutrient                                                                | Zinc compounds                                                                              | Food industry, animal feed, fertilisers                                                  |
| - Healing                                                                           | Zinc compounds                                                                              | Pharmaceutical industry, cosmetics industry                                              |

### 5.3 Cadmium

In recent years, the consumption pattern of cadmium in its various applications has shifted away from the market areas of pigments, stabilisers and coatings to rapidly growing applications in Ni-Cd batteries<sup>7</sup>.

Fig. 5.3 Uses of Cadmium in different applications (reference year 2003<sup>7</sup>)

<sup>6</sup> [http://www.iza.com/zwo\\_org/](http://www.iza.com/zwo_org/), access date 01.03.03

<sup>7</sup> <http://www.cadmium.org>, accessed in April 2007



Description of the different uses of Cadmium <sup>7</sup>:

**Nickel-Cadmium Batteries.** Cadmium hydroxide is utilised as one of the two principal electrode materials in Ni-Cd batteries which have extensive applications in the railroad and aircraft industry for starting and emergency power and in consumer applications such as cordless power tools, cellular telephones, camcorders, portable computers, portable household appliances and toys.

**Cadmium Pigments.** Cadmium sulphide and cadmium sulposelenide are utilised as bright yellow to deep red pigments in plastics, ceramics, glasses, enamels and artists colours. They are used in applications where high temperature or high pressure processing is required.

**Cadmium Stabilisers.** Cadmium-bearing stabilisers retard the degradation processes in polyvinylchloride (PVC) which occur upon exposure to heat and ultraviolet light. These stabilisers contain organic cadmium salts, usually carboxylates such as cadmium laurate or cadmium stearate, which are incorporated into PVC before processing.

**Cadmium Coatings.** Cadmium coatings are utilised on steel, aluminium, and certain other non-ferrous metal fasteners and moving parts to provide the best available combination of corrosion resistance, particularly in salt and alkali media, and lubricity or low coefficient of friction. They are also employed in many electrical or electronic applications where a combination of corrosion resistance and low electrical resistivity are required. In addition, cadmium coatings exhibit excellent plating characteristics on a wide variety of substrates, have good galvanic comparability with aluminium, and are readily solderable.

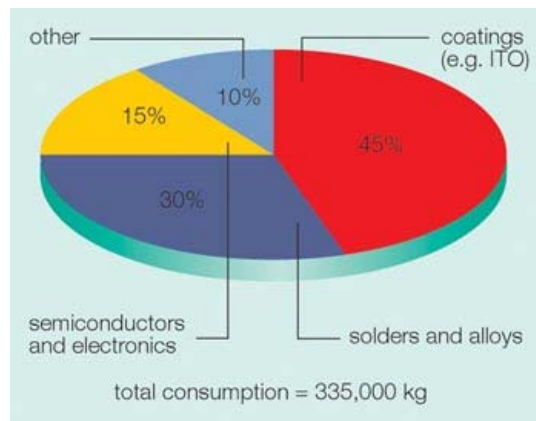
**Alloys and Minor Uses.** Cadmium alloys include (a) electrical conductivity alloys, (b) heat conductivity alloys, and (c) electrical contact alloys. Other minor uses of cadmium include cadmium telluride and cadmium sulphide in solar cells, and other semiconducting cadmium compounds in a variety of electronic applications.

## 5.4 Indium

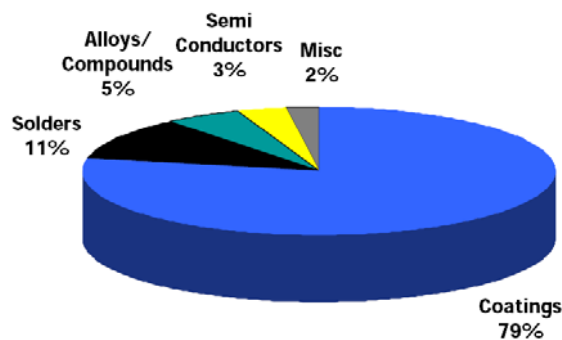
Today virtually all of the LCD produced use ITO as conducting layer. ITO is the transparent conductor of choice in terms of high transmissivity of visible light with low electric resistivity compared to other transparent conductors, such as tin oxide (SnO<sub>2</sub>) and zinc oxide (ZnO) (Granqvist & Hultaker, 2002). The sputtering of the substrate is done by the display manufacturers. Due to the very large cost of building TFT factories, the number of major panel vendors is small. Several of the best known are Sharp Corporation, Samsung Electronics, LG.Philips LCD and AU Optronics. As of April 2006, the three biggest manufacturers are LG.Philips and Samsung with 22% of the market, followed by AU Optronics with a 19% share (Wikipedia Contributors, 2006).

Jansseune (2003) estimates that in 2002 45% of the indium supplied went as ITO into coating, whereas 2004 an estimated 72% of the indium was used for ITO coatings (Jorgenson & George, 2004). End 2006 Hewett (2006) estimated the share of ITO in the total indium supply to 80%. In 2004 perhaps 65 %of the indium used in coating goes into display devices, and the remainder, into transparent heat reflection. ITO coatings also are used in architectural glass and solar panel applications (high infrared reflectance), cathode-ray tubes (CRTs), low-pressure sodium lamps, and windshield glass. (Jorgenson & George, 2004).

In 2005 the USGS estimates that 70% of the Indium was used in coatings, electrical components and semiconductors, 12%; solders and alloys, 12%; and research and other, 6% (USGS, 2006a).



**Fig. 5.4** In 2002, the semiconductor industry accounted for 15% of indium consumption, equivalent to around 50,000 kg. Credit: USGS (Jansseune, 2003).



**Fig. 5.5** In 2004 already almost 80% of the indium went into ITO coatings (Teckcominco, 2005).

In 2004 Teckcominco (2005) estimated about 640 tonnes of indium in the end-use, of which ITO coatings amounted to 79% (Fig. 5.5).

### Coatings

Indium's largest use is for coatings. Metallic indium ingots are processed into an oxide and then mixed with tin-oxide to form indium-tin-oxide (ITO). This material is used to produce sputtering targets. These targets are placed in a vacuum and the ITO is "sputtered" onto glass for use in display devices such as laptop computer screens, LCD watches, appliance display panels and portable phones.

ITO is also used in traditional applications such as low-pressure sodium lamps, architectural glass and oven doors (heat reflector applications).

New applications for indium coatings include high definition televisions and thin screen computer monitors (Teckcominco, 2006).

Application of ITO coatings is a wasteful process and only around 20-30% of the material actually ends up on the substrate (Jansseune, 2005). The rest is recycled as prompt scrap.

### Alloys

Indium has a relatively low melting point at 157°C. This enables it to be utilized in alloy production such as AgInSbTe sputtering target manufacturing for the electronics industry (Teckcominco, 2006).

### **Electronics & Semiconductors**

Indium is used as a doping agent for germanium (Ge) and gallium arsenide (GaAs) semiconductors.

The largest use for high purity indium is for the production of indium phosphide (InP) for use in the semiconductor industry. InP is also used in photonic applications, such as lasers, photo detectors, optical modulators and amplifiers, and in electronic applications, such as, HEMT's (high electron mobility transistors), HBT's (heterojunction bipolar transistors), and FET's (field effect transistors), millimeter wave and microwave circuits, and high-speed digital integrated circuits. (Teckcominco, 2006)

## 6 System Characterisation

This chapter describes the mining of sulphidic zinc-lead deposits and the related extraction of lead and zinc but also minor metal values (metals contained in the ore, whose extraction is profitable) such as silver, cadmium and indium for which these deposits represent an important source for worldwide primary production (Silver: 20%, Cadmium: 80%, Indium: ~100%). Therefore, the inventory modelling involves a series of multi-output processes in which all burdens are attributed by economic allocation procedures based on Guinée et al. (2004) and Althaus & Classen (2005) using the average prices from 2004 through 2006 reported by the U.S. Geological Survey (see Tab. 3.1):

- 6.1. The module “mining, zinc-lead-deposit” delivers the flows
  - “lead, concentrate, at beneficiation” (containing silver) and
  - “zinc, concentrate, at beneficiation” (containing cadmium and indium)
- 6.2. The module “smelting, primary lead production” processes the concentrate to the flows
  - “lead, primary, at plant” and
  - “parkes process crust, from desilverising of lead” as an intermediate step for the silver chain
- 6.3. “lead, secondary, at plant”
- 6.4. “lead, at regional storage” (a blend of primary and secondary lead)
- 6.5. The module “smelting, primary zinc production” processes the concentrate to the flows
  - “zinc, primary, at regional storage” (primary production)
  - “cadmium sludge, from zinc electrolysis, at plant” as raw material for cadmium production, and
  - “leaching residues, indium rich, from zinc circuit, at smelter” as raw material for indium production.
- 6.6. “cadmium, primary, at plant”
- 6.7. “indium, at regional storage”

A graphic representation is shown in Fig. 6.1, the boxes with rounded corners represent reference flows from multi-output processes, bold lines denote datasets to be used by the LCA practitioner.

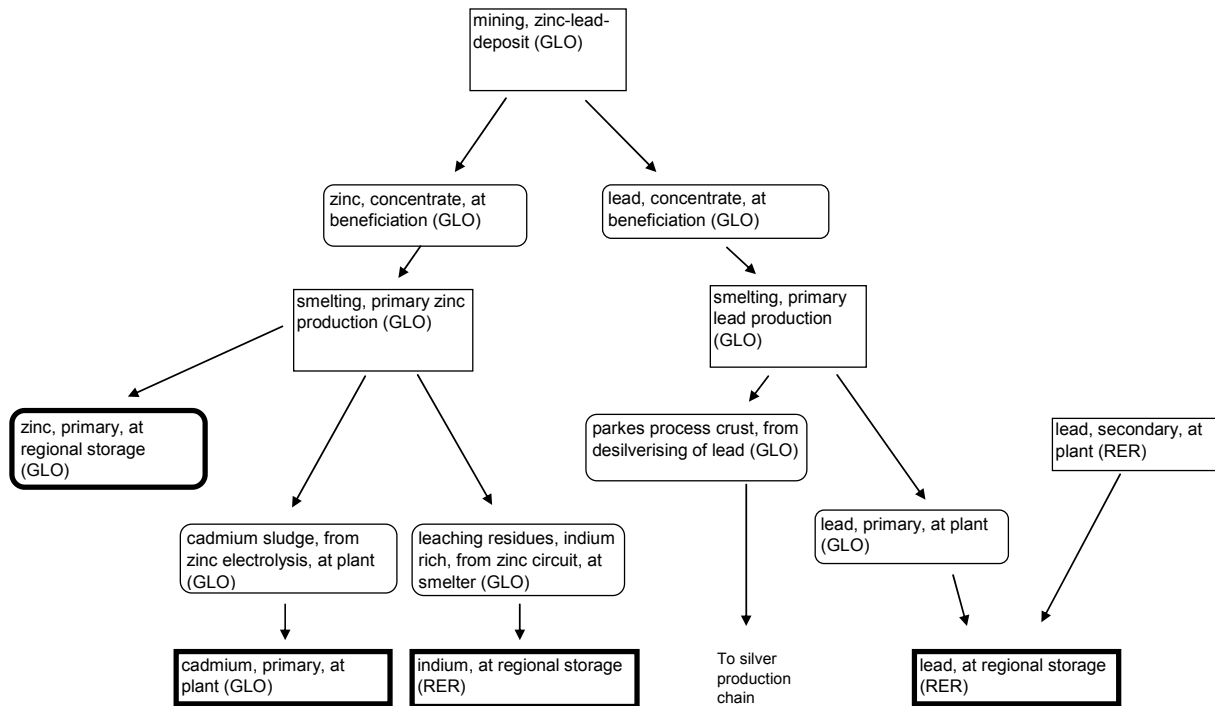


Fig. 6.1 Graphic representation of the modules and flows resulting from the mining of zinc-lead deposits.

Tab. 6.1 Meta information for the mining of zinc and lead bearing ore.

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>                           | <b>mining, zinc-lead-deposit</b>                                                                                                                                                                                                                                                                                                                                                                           |
| Location                              | GLO                                                                                                                                                                                                                                                                                                                                                                                                        |
| Infrastructure Process                | 0                                                                                                                                                                                                                                                                                                                                                                                                          |
| Unit                                  | kg                                                                                                                                                                                                                                                                                                                                                                                                         |
| Data Set Version                      | 2.0                                                                                                                                                                                                                                                                                                                                                                                                        |
| Included Processes                    | The module includes mining and beneficiation steps with the mining infrastructure and disposal of overburden and tailings (see part "Auxiliary Processes").                                                                                                                                                                                                                                                |
| Amount                                | 1                                                                                                                                                                                                                                                                                                                                                                                                          |
| Local Name                            | Abbau Zink-Blei-Lagerstätte                                                                                                                                                                                                                                                                                                                                                                                |
| Synonyms                              |                                                                                                                                                                                                                                                                                                                                                                                                            |
| General Comment to reference function | The module describes the global production mix for jointly produced zinc- and lead-concentrates from sulphidic deposits. The multi output-process "exploitation, zinc-lead-deposit" delivers the co-products "lead, concentrate, at beneficiation" and "zinc, concentrate, at beneficiation". It is designed for the use of the concentrates as raw material for the ultimate production of lead and zinc. |
| Start Date                            | 1994                                                                                                                                                                                                                                                                                                                                                                                                       |
| End Date                              | 2003                                                                                                                                                                                                                                                                                                                                                                                                       |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                                                                                                                                                                                          |
| Other Period Text                     |                                                                                                                                                                                                                                                                                                                                                                                                            |
| Geography text                        | This module represents a global production mix. Some agents used like lime or sodium cyanide originate from data sets valid for Switzerland or Europe. Processes like blasting and transport originate from data sets valid for Switzerland or Europe.                                                                                                                                                     |
| Technology text                       | This module is approximated with values from nickel-mining. Mining is assumed to be underground, followed by froth beneficiation. Overburden and tailings are disposed inside the mining panels. For electricity mix a mix of 25% coal and 75% gas fired plants was chosen.                                                                                                                                |
| Representativeness [%]                |                                                                                                                                                                                                                                                                                                                                                                                                            |
| Production Volume                     |                                                                                                                                                                                                                                                                                                                                                                                                            |
| Sampling Procedure                    | literature                                                                                                                                                                                                                                                                                                                                                                                                 |
| Extrapolations                        | see Geography and Technology                                                                                                                                                                                                                                                                                                                                                                               |
| Uncertainty Adjustments               | see Geography and Technology                                                                                                                                                                                                                                                                                                                                                                               |

Tab. 6.2 Meta information for the lead production.

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                             |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name                                  | smelter, primary lead production                                                                                                                                                                                                                                                                                                                                                                                          | lead, secondary, at plant                                                                                                                                                                                                                                  | lead, at regional storage                                                                                                                                                                                                                                                                                   |
| Location                              | GLO                                                                                                                                                                                                                                                                                                                                                                                                                       | RER                                                                                                                                                                                                                                                        | RER                                                                                                                                                                                                                                                                                                         |
| Unit                                  | kg                                                                                                                                                                                                                                                                                                                                                                                                                        | kg                                                                                                                                                                                                                                                         | kg                                                                                                                                                                                                                                                                                                          |
| Data Set Version                      | 2                                                                                                                                                                                                                                                                                                                                                                                                                         | 2.0                                                                                                                                                                                                                                                        | 2                                                                                                                                                                                                                                                                                                           |
| Included Processes                    | The module includes the production of primary lead with the sinter/blast furnace (ISP) and direct smelting process, the disposal of slag and final refining of lead                                                                                                                                                                                                                                                       | Collection, sorting and remelting of the lead contained in lead acid batteries.                                                                                                                                                                            | The module includes the production of primary and secondary lead for Europe.                                                                                                                                                                                                                                |
| Local Name                            | Verhüttung, Primärblei Produktion                                                                                                                                                                                                                                                                                                                                                                                         | Blei, sekundär, ab Werk                                                                                                                                                                                                                                    | Blei, ab Regionallager                                                                                                                                                                                                                                                                                      |
| Synonyms                              |                                                                                                                                                                                                                                                                                                                                                                                                                           | Recycling of lead acid (PbA) batteries // Recycling von Batterien                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                             |
| General Comment to reference function | The module describes the primary production of lead in Europe. The multi-output-process "Smelter, primary lead production" delivers the co-products "Lead, primary, at plant" and "Parkes process crust, from desilverising of lead".. The flow "Lead, ..." is part of the respective European supply mix. The by-product "Parkes process crust, ..." receives part of the burden and enters the silver production chain. | The module describes the production of secondary lead in Europe. The feed of secondary material consists of scrap lead acid (PbA) batteries from automobiles. The data refers to one big operation in Europe that operates with representative technology. | This dataset describes the supply mix of lead used in Europe. It is modelled with a blend of 25% primary and 75% secondary production, complemented with the transport service required to transport the material to the regional storage. In the global average the secondary production amounts only 65%. |
| Start Date                            | 2000                                                                                                                                                                                                                                                                                                                                                                                                                      | 2000                                                                                                                                                                                                                                                       | 2000                                                                                                                                                                                                                                                                                                        |
| End Date                              | 2005                                                                                                                                                                                                                                                                                                                                                                                                                      | 2005                                                                                                                                                                                                                                                       | 2005                                                                                                                                                                                                                                                                                                        |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                                                                                                                                                                                                         | 1                                                                                                                                                                                                                                                          | 1                                                                                                                                                                                                                                                                                                           |
| Geography text                        | This module represents the production of primary lead on a global average.                                                                                                                                                                                                                                                                                                                                                | This module represents the European production based on a single major site in Belgium.                                                                                                                                                                    | This module represents the consumption mix of lead in Europe.                                                                                                                                                                                                                                               |
| Technology text                       | A mix of 56% direct smelting and 44% sinter/blast furnace (ISP) is chosen. For emission control 56% improved and 44% limited control is chosen.                                                                                                                                                                                                                                                                           | The referred operation uses a shaft furnace with post combustion, which is the usual technology for secondary smelters.                                                                                                                                    | A mix of 25% primary and 75% secondary production.                                                                                                                                                                                                                                                          |
| Representativeness [%]                |                                                                                                                                                                                                                                                                                                                                                                                                                           | 0.8 %                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                             |
| Production Volume                     | 1'398'000 t in 1994                                                                                                                                                                                                                                                                                                                                                                                                       | 5'000'000 t in 2005                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                             |
| Sampling Procedure                    | literature                                                                                                                                                                                                                                                                                                                                                                                                                | literature                                                                                                                                                                                                                                                 | literature                                                                                                                                                                                                                                                                                                  |
| Extrapolations                        | see Geography and Technology                                                                                                                                                                                                                                                                                                                                                                                              | see Geography and Technology                                                                                                                                                                                                                               | see Geography and Technology                                                                                                                                                                                                                                                                                |
| Uncertainty Adjustments               | none                                                                                                                                                                                                                                                                                                                                                                                                                      | none                                                                                                                                                                                                                                                       | none                                                                                                                                                                                                                                                                                                        |

Tab. 6.3 Meta information for the zinc production process

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>                           | <b>smelting, primary zinc production</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Location                              | GLO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Infrastructure Process                | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Unit                                  | kg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Data Set Version                      | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Included Processes                    | The module includes the production of high grade (SHG) primary zinc by pyro- and hydrometallurgical processes, the disposal of the slag and the treatment of the waste water. No secondary production - which amounts to 30% in Europe - is considered.                                                                                                                                                                                                                                                                       |
| Amount                                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Local Name                            | Verhüttung, Primärzink Produktion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Synonyms                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| General Comment to reference function | The module describes the primary production of zinc in Europe. The multi-output-process "smelting, primary zinc production" delivers the three co-products "zinc, primary, at regional storage", "cadmium sludge, from zinc electrolysis, at plant" and "leaching residues, indium rich, from zinc circuit, at smelter". The flow "zinc, ..." represents the European supply mix of primary zinc. The by-products "leaching residues, ..." and "zinc, ..." receive part of the burden and enter the primary production chain. |
| CAS Number                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Start Date                            | 1994                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| End Date                              | 2003                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Other Period Text                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Geography text                        | This module represents the consumption mix of primary zinc in Europe. It is assumed that 75% of the concentrates are imported, while all zinc is produced within Europe.                                                                                                                                                                                                                                                                                                                                                      |
| Technology text                       | A mix of 80% hydrometallurgical and 20% pyrometallurgical production is chosen. For emission control 80% improved (hydro) and 20% limited control (pyro) is chosen.                                                                                                                                                                                                                                                                                                                                                           |
| Representativeness [%]                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Production Volume                     | 10'000'000 t in 2006                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Sampling Procedure                    | literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Extrapolations                        | see Geography and Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Uncertainty Adjustments               | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |



Tab. 6.4 Meta information for the production of primary cadmium, at regional storage.

| Name                                  | cadmium, primary production, at regional storage                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location                              | RER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Infrastructure Process                | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Unit                                  | kg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Data Set Version                      | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Included Processes                    | The module includes the treatment of cadmium sludge from hydro-metallurgical zinc operations and subsequent refining by electrolysis. Transports and infrastructure are considered.                                                                                                                                                                                                                                                                                                                       |
| Amount                                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Local Name                            | Cadmium, Primärproduktion, ab Regionallager                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Synonyms                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| General Comment to reference function | The module describes the primary production of cadmium metal (99.9%) on a global average. No secondary sources from post consumer scrap are considered. Only the most important source, the cadmium sludges from hydrometallurgical zinc operations, is reflected with the dataset. For the use in applications where cadmium is a major constituent – e.g. in NiCd battery systems – the dataset should be complemented with additional specific emission factors and the secondary production process.. |
| CAS Number                            | 7440-43-9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Start Date                            | 2000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| End Date                              | 2005                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Other Period Text                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Geography text                        | This module represents the production of primary cadmium on a global average. The inventory however is compiled with European average data, e.g. for electricity production.                                                                                                                                                                                                                                                                                                                              |
| Technology text                       | Cadmium sludge formed in hydrometallurgical zinc operations by purification of zinc solution. Subsequent purification and oxidation of the sponge. Cadmium cathodes formed by electrowinning. Other sources from pyrometallurgical zinc wastes and post consumer scrap (batteries) disregarded.                                                                                                                                                                                                           |
| Representativeness [%]                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Production Volume                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Sampling Procedure                    | literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Extrapolations                        | see Geography and Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Uncertainty Adjustments               | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

Tab. 6.5 Meta-information for the dataset "indium, at regional storage"

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>                           | <b>indium, at regional storage</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Location                              | RER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Infrastructure Process                | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Unit                                  | kg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Data Set Version                      | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Included Processes                    | The module includes the production of indium from zinc extraction operations. Infrastructure and transports to Europe are considered. Most process specific emissions are missing.                                                                                                                                                                                                                                                                                                                                                              |
| Amount                                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Local Name                            | Indium, ab Regionallager                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Synonyms                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| General Comment to reference function | The module describes the primary production of indium metal (99.999%, 5N) on a European average. No secondary sources from post consumer scrap are considered. Only the most important source of indium raw material, the hydrometallurgical zinc operations, is reflected with the dataset. This dataset is based on the respective flow from multi-output module "smelting, primary zinc production". This dataset is not to be used when indium is a substantial part of the system modelled, viz. for comparisons of special solder pastes. |
| CAS Number                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Start Date                            | 2000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| End Date                              | 2005                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Other Period Text                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Geography text                        | This module represents the production of primary indium for the consumption in Europe. The inventory however is compiled with global average data, e.g. for zinc-production.                                                                                                                                                                                                                                                                                                                                                                    |
| Technology text                       | Indium containing residues from hydrometallurgical zinc operations are leached. Subsequent solvent extraction and precipitation of sponge indium. Indium cathodes are cast and electrorefined, followed by vacuum refining to obtain 5N quality. Solvent-extraction and electrowinning has been extrapolated from copper refining.                                                                                                                                                                                                              |
| Representativeness [%]                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Production Volume                     | 480 t in 2006                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Sampling Procedure                    | literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Extrapolations                        | see Geography and Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Uncertainty Adjustments               | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

## 6.1 Mining of zinc-lead-deposits

About 70 % of zinc and the same amount of lead is won from a sulphidic ore, where the elements occur jointly. They are also jointly extracted.

### 6.1.1 Production process and infrastructure

Zinc-lead ores are assumed to be mined mostly underground. The ore is transferred to the beneficiation. Widening of the tunnels is mainly done by blasting. The overburden – material, which does not contain metal-bearing ore – is partly deposited off-site and partly refilled into the tunnels.

After mining, the ore is firstly ground. In a next step it goes through gravity concentration to separate the metallic particles from the metal bearing minerals. After this first concentration step, flotation is carried out to remove the gangue from the sulphidic minerals. For neutralisation lime is added. In the flotation several organic chemicals are used as collector, frother, activator, depressor and flocculant. Sometimes cyanide is used as depressant. Tailings are usually transferred to tailing heaps or ponds. Concentrates are produced containing around 50% zinc and 50% lead.

### 6.1.2 Waste

The overburden is disposed of close to the mine. Tailings are deposited as piles and in ponds. In both the overburden and the tailings acid rock drainage occurs over a long period of time. These effects are handled in the part IX (Non Ferrous Metal Winning – Auxiliary Processes, chapter Disposal of Tailings and Overburden) of this report.

### 6.1.3 Emission

The major emissions are mineral born pollutants in effluents and dust. The prevalently underground mining operations generate roughly 80 % of the dust emissions from surface operations, since the major dust sources do not take place underground. Rain percolates through the overburden and generates metal leachates to groundwater.

Ore handling and processing produce large amounts of dust, containing PM<sub>10</sub> and several metals from the ore itself. Flotation produces effluents containing several organic agents<sup>8</sup> used. Some of these chemicals evaporate and account for VOC emissions to air. In particular, xanthates decompose hydrolytically to release carbon disulphide. Tailings effluent contains additional sulphuric acid from acid rock drainage.

### 6.1.4 Mining, zinc-lead deposits in ecoinvent

Since the extraction of zinc and lead is very similar to the extraction of other sulphidic deposits like nickel and copper bearing ores, and no information on the specific mining processes for zinc-lead deposits is available, the data for the exploitation of nickel provided in Hilbrans & Hinrichs (1999) was used. Mass flows were adjusted to values specific to zinc-lead extraction, according to information given in Ayres et al. (2002).

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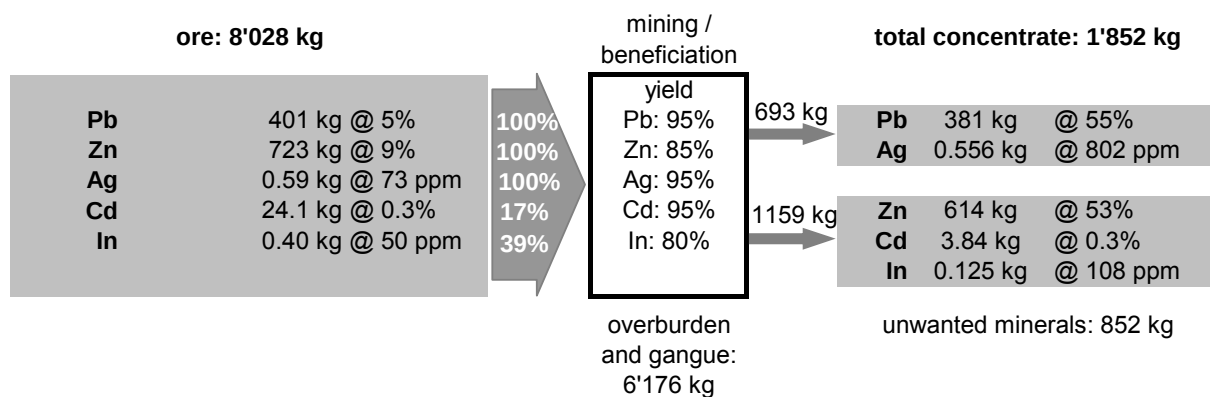
<sup>8</sup> Floating agents consist of collectors (xanthate or aerofloat) and frothing reagent (eg. Methyl Isobutyl Carbinol). Xanthates are a broad class of organic salts formed by treatment of an alcohol with carbon disulfide in the presence of an alkali, potassium ethyl xanthate (KC<sub>3</sub>H<sub>5</sub>S<sub>2</sub>O, CAS No: 140-89-6). Aerofloat is a salt of carbonyldithiophosphoric acids. A widely used frothing reagent is e.g. methyl isobutyl carbinol, CAS: 108-11-2, syn. Methyl Amyl Alcohol.

The module describes the global production mix for jointly produced zinc- and lead-concentrates from sulphidic deposits. The multioutput-process “exploitation, zinc-lead-deposit” delivers the co-products “lead, concentrate, at beneficiation” and “zinc, concentrate, at beneficiation”. It is designed for the use of the concentrates as raw material for the production of lead and zinc. The layout of the modules with general flow information, remarks, sources, values and uncertainty information is shown in Fig. 6.3 and Fig. 6.4. The following paragraphs describe calculations, sources and assumptions chosen in this study. The functional unit of this process is one kg of added zinc- and lead-concentrates. The meta-information for this dataset is compiled in Tab. 6.1

Fig. 6.2 gives an overview of the general assumptions made in the mass flows of this module. It is assumed, that two virtually distinctive concentrate products are produced. This of course is not true in real production, but it is a formal necessity for the further use in ecoinvent in the modules for zinc and lead production.

The range of concentrate composition described varies considerably. Depending on the processing technology, a high degree (60-75% of a single metal, lead or zinc) concentrate may be required. Most smelters accept lower grade concentrates that range 50-60% Pb (e.g. Trail, Mt. Isa). In this study a content of 55% Pb in the concentrate is chosen. The range of zinc content reported in zinc concentrates is much narrower than for lead. A compilation of concentrates of 80 mines in (Rodier, 1990) showed a range between 45 and 60% Zn, with a mean value of 53%. Reuter (2005) states an average zinc concentrate to have 50% Zn and uses 54% in his calculations.

In order to comply with the global production figures the assumption is made that not all of the minor metals cadmium and indium are not entirely recovered. Only a share of the extracted volume is worked for these metals, the other operations discard these metal values with the gangue. This is represented by the percentages in white colour in the input arrow in Fig. 6.2.



**Fig. 6.2** Basic assumptions for the mining process of zinc- and lead ore from sulphidic deposits. Values relate to the production of the sum of 1 tonne metals contained in the concentrate, they are mainly taken from Ayres et al. (2002) and IPPC (2001).

### Infrastructure and land use

The infrastructure used for ore exploitation is inventoried within this module, whereas the infrastructure like buildings and land-use for the operation is inventoried in another appropriate module, comprising both mining and beneficiation facilities. The mobile equipment used for exploitation like drilling machines, scoop trams and underground trucks is inventoried through their diesel consumption. The module used for this purpose “diesel, burned in building machine” includes the infrastructure of the machines. Additionally, 2000 m conveyor belts are assumed with a capacity of 2000 t/h which yields  $2 \cdot 10^{-5}$  m conveyor belt required per tonne mined raw ore.

Since mining is underground and no heap leaching is done, no land use for exploitation is assumed.

**Tab. 6.6** Infrastructure in underground mining, assumptions used. A lifetime of ten years and 5000 working hours yearly are estimated.

| action    | machine        | throughput lifetime | unit       | Infrastructure    |
|-----------|----------------|---------------------|------------|-------------------|
|           |                | Mass (t)            | Length (m) | m/t ore mined     |
| Transport | Conveyor belts | 100'000'000         | 2000       | $2 \cdot 10^{-5}$ |

## Mining

In the considered processes the ore is mined underground. The overburden is disposed in piles near the mine, for which the standard module for disposal of sulphidic overburden has been chosen, which includes the land use and land transformation by the disposal. Overburden refilled into the mine is assumed to be burden free. The degree of the refilling is documented for extraction of nickel-ores in Hilbrans & Hinrichs (1999). This source is also used for water use and effluent volume from mining. The use of freshwater was inventoried as river water, whereas the pit water was assumed to be groundwater, which has to be pumped out. The composition of a generic effluent from mines was taken from IPPC (2002), its values are shown in Tab. 6.8. No waste-water treatment is considered. No information on the energy production in mining is available. In many cases the electricity used in the often remote mining operations is made on site by gas or coal fired power plants. In this study a split of 75% gas fired and 25% coal fired plants was assumed based on own estimations. Specific values and corresponding inventory flows are shown in Tab. 6.7. Cement, explosives and sand are assumed to be transported by lorry over 100 km.

For refilling the pit, apart from overburden additional stowing material – mostly sand – and cement are used.

## Beneficiation

Once the ore is mined, it is ground and conditioned for subsequent concentration. Considerable amounts of lime and flotation agents are added. The separated gangue is disposed in tailings ponds, the concentrated ore is fed to the metallurgy, which is assumed to be on-site.

The infrastructure of beneficiation is included in mining infrastructure. Dust emissions comprise additionally the dust emissions due to mining activities. Like in the mining step, a share of 60% hydroelectricity was chosen according to Hilbrans & Hinrichs (1999).

**Lime:** Lime input is documented in Hilbrans & Hinrichs (1999) for the beneficiation of lead-/zinc-sulphide, the input of 554 g / t ore is taken from the specific input in nickel beneficiation.

**Reagents:** No information on the nature and mass of the floating agents used is available. For copper mining a range of 25 – 300 g collector and 25 – 250 g frothing reagent per ton sulphidic ore is recommended (Krauss et al., 1999). This is confirmed by the values in Tab. 6.9. Collector and frother are balanced in ecoinvent as unspecified organic chemical. According to NPI (2001) xanthates hydrolytically decompose to 0.53 mg carbon disulphide per mg xanthate. With 189 mg xanthates from Tab. 6.9 this is about 100 mg carbon disulphide per t mined ore. This again is modelled as unspecific organic chemical.

Activator and Depressants – excluding sodium cyanate – are modelled as unspecific inorganic chemicals. Their emission to water is covered by reported emissions from mining sites shown in Tab. 6.8.

For pH adjustment and leaching, sulphuric acid is considered with 3.74 kg / t ore mined (Tab. 6.10). As sulphuric acid is a by-product from the metallurgy, its production is regarded in ecoinvent as burden free. Other acids are neglected.

Since composition and environmental fate of activator, depressant and sulphuric acid are uncertain, reported data for total effluents shown in Tab. 6.8 are taken.

Sodium cyanide is widely used for extracting gold and other metals such as silver, copper and zinc. It is used as leachate and as depressant for certain sulphidic minerals. A cyanide recovery of 80 – 90% is reported due to abatement and recovery in the Golden Cross mine in New Zealand (Anonymous, 1998). Here a value of 80% is chosen for cyanide-abatement technology. With a reported use of 35 g NaCN per tonne ore mined (Tab. 6.9) this means a direct emission of 3.75 g CN<sup>-</sup> per tonne ore mined. An overview of the reagents accounted for is presented in Tab. 6.9.

**Effluents:** Additional to direct emissions due to the use of chemicals, emissions to water are chosen according to Tab. 6.8. These values are from base metal sites, thus only an approximation for the real effluent concentrations in the zinc / lead extraction. It is assumed here that the effluents also include acid rock drainage (ARD). ARD after mine closure is modelled in the module “disposal, sulphidic tailings, off-site”.

**Dust:** The dust emissions are approximated in both amount and particle distribution approximated with specific values from copper mining from the CEPMEIP-Database<sup>9</sup>. According to IAEA<sup>10</sup> the dust composition is approximated with concentrations of the zinc-lead-ore (Tab. 6.11). The dust emissions are summed up in Tab. 6.11. In ecoinvent, for particulates three different classes are used, whose total corresponds to the total dust emission.

**Tab. 6.7** specific input and effluent values used and their corresponding value per t concentrate established in this inventory. The specific values are approximated by values from sulphidic nickel deposits (Hilbrans & Hinrichs, 1999).

|                                 |     | per t ground<br>(ore plus o-<br>verburden) | per t ore | per t con-<br>centrate |
|---------------------------------|-----|--------------------------------------------|-----------|------------------------|
| Amount ground treated           | t   |                                            |           | 7.30                   |
| Amount ore retrieved            | t   |                                            |           | 4.33                   |
| stowing material, sand          | t   | 0.284                                      |           | 2.07                   |
| cement                          | t   | 0.0225                                     |           | 0.165                  |
| explosive                       | kg  | 1.56                                       |           | 11.4                   |
| diesel fuel (energy)            | MJ  | 69.8                                       |           | 510                    |
| el. Energie, total              | kWh |                                            | 39.6      | 172                    |
| fresh and processwater (mining) | m3  |                                            | 0.347     | 1.51                   |
| effluents, mining               | m3  | 0.423                                      |           | 3                      |
| effluents, beneficiation        | m3  |                                            | 1.38      | 6.00                   |

<sup>9</sup> [http://www.air.sk/tno/cepmeip/em\\_factors.php](http://www.air.sk/tno/cepmeip/em_factors.php), accessed in 2003

<sup>10</sup> The IAEA Database of Natural Matrix Reference Materials, <http://www-naweb.iaea.org/nahu/external/e4/nmrml/>, accessed 01.03.2003

**Tab. 6.8 Emissions to water from mining and beneficiation. Composition of effluents from mining activities, values are approximations from total emissions to water from base metal sites in (IPPC, 2002).**

| Compound        | Unit                | Range         | This study<br>(geometric mean) | Number of<br>Samples |
|-----------------|---------------------|---------------|--------------------------------|----------------------|
| Ca              | g / m <sup>3</sup>  | 686           | 686                            | 1                    |
| SO <sub>4</sub> | g / m <sup>3</sup>  | 1750 - 3175   | 2357                           | 2                    |
| COD             | g / m <sup>3</sup>  | 2.24 - 48.5   | 10.4                           | 2                    |
| Solids          | g / m <sup>3</sup>  | 2.38 - 11.3   | 5.17                           | 4                    |
| Al              | mg / m <sup>3</sup> | 69.3 - 108    | 86.3                           | 2                    |
| As              | mg / m <sup>3</sup> | 0.264 - 14.1  | 2.95                           | 3                    |
| Cd              | mg / m <sup>3</sup> | 0.0901 - 1.02 | 0.316                          | 4                    |
| Co              | mg / m <sup>3</sup> | 0.742 - 0.823 | 0.781                          | 2                    |
| Cr              | mg / m <sup>3</sup> | 0.0311 - 9.62 | 0.547                          | 2                    |
| Cu              | mg / m <sup>3</sup> | 1.24 - 44.9   | 7.91                           | 5                    |
| Fe              | mg / m <sup>3</sup> | 61.7 - 1326   | 291                            | 3                    |
| Mn              | mg / m <sup>3</sup> | 24.7          | 24.7                           | 1                    |
| Hg              | mg / m <sup>3</sup> | 0.0155 - 0.12 | 0.0377                         | 3                    |
| Ni              | mg / m <sup>3</sup> | 0.792 - 1338  | 24.3                           | 3                    |
| Pb              | mg / m <sup>3</sup> | 0.0155 - 20   | 2.80                           | 4                    |
| Zn              | mg / m <sup>3</sup> | 5.37 – 225    | 75.9                           | 5                    |
| N               | g / m <sup>3</sup>  | 2.50 - 1786   | 22.8                           | 3                    |

**Tab. 6.9 Consumption of reagents in base metal mineral processing plants. Data taken from (IPPC, 2002).**

| Reagents                                                                                        | Unit | Range     | This study<br>(geometric mean) | Number of<br>samples |
|-------------------------------------------------------------------------------------------------|------|-----------|--------------------------------|----------------------|
| <b>Collectors</b> – Xanthate                                                                    | g/t  | 110 - 300 | 188.6                          | 6                    |
| <b>Frothers</b> - Sylvapine, MIBC, Dowfroth                                                     | g/t  | 9.7 - 150 | 37.2                           | 5                    |
| Sodium cyanide                                                                                  | g/t  | 4-310     | 35                             | 2                    |
| <b>Depressants</b> – Zinc sulphate, Iron sulphate, Acetic acid, Sodiumchromate, Dithiophosphate | g/t  | 35 - 432  | 231.5                          | 5                    |
| <b>Activator</b> - Copper sulphate                                                              | g/t  | 433 - 876 | 538                            | 4                    |
| <b>Flotation agents</b>                                                                         | g/t  | 19.0      | 19.0                           | 1                    |
| <b>Flocculants</b> - CMC                                                                        | g/t  | 1-100     | 11.1                           | 3                    |
| Sulphuric acid (pH)                                                                             | g/t  | 387-12000 | 3739                           | 4                    |
| Sodium hydroxide (pH)                                                                           | g/t  | 30-490    | 121                            | 2                    |
| Nitric acid (pH)                                                                                | g/t  | 150       | 150                            | 1                    |

**Tab. 6.10 Emissions and input due to chemical use in beneficiation in g per tonne treated ore. The values are the ones chosen in this study. For further details read in text and Tab. 6.9.**

| Reagent              | Type of burden     | Unit | Chosen value |
|----------------------|--------------------|------|--------------|
| Collector            | input technosphere | g/t  | 189          |
| Carbon disulfide     | air emission       | g/t  | 100          |
| Frother              | input technosphere | g/t  | 37           |
| Lime                 | input technosphere | g/t  | 554          |
| Carbon dioxide       | air emission       | g/t  | 244          |
| Activator/Depressant | input technosphere | g/t  | 828          |
| Sulphuric acid       | input technosphere | g/t  | 3739         |
| Sodium Cyanide       | input technosphere | g/t  | 35.2         |
| Cyanide              | water emission     | g/t  | 3.75         |

**Tab. 6.11 Emissions to air resulting from mining activities per tonne mined ore. Approximated values derived from reported emission factors from copper extraction and a single ore composition. Values for total dust emission and, partition of fractions of particulate matter are taken from the CEMIP-database, the dust composition correspond to a specific nickel sulphide with values taken from the IAEA-database (see text).**

| Emission to air | Unit      | Emission / t ore |
|-----------------|-----------|------------------|
| Cadmium         | g / t ore | 0.102            |
| Lead            | g / t ore | 5.1              |
| Zinc            | g / t ore | 9.2              |
| Sulphur         | g / t ore | 10.2             |
| PM<2.5          | g / t ore | 5.0              |
| PM2.5-10        | g / t ore | 45               |
| PM>10           | g / t ore | 52               |
| total dust      | g / t ore | 101.72           |

## Allocation

According to the allocation principle pursued throughout the inventory modelling in this report, an economic allocation for the joint by-products lead and zinc concentrate is envisioned. Since beside the main constituents lead and zinc, also silver, cadmium and indium are present in the products, also these values have to be considered. Cadmium is considered contaminant and therefore does not affect the allocation procedure, since it does not receive a value within the zinc production. The cadmium containing wastes become valuable only after refining.



**Tab. 6.12 Overview over the relevant parameters to determine the allocation split: composition of the by-products and value of the metals contained.**

|                       | <b>Zinc concentrate</b> | <b>Lead Concentrate</b> | <b>Value in \$ / kg</b> |
|-----------------------|-------------------------|-------------------------|-------------------------|
| feed in kg            | 0.699                   | 0.301                   |                         |
| Pb in kg              | -                       | 0.166                   | 1.26                    |
| Zn in kg              | 0.370                   | -                       | 3.09                    |
| Ag in kg              | -                       | 3.29E-04                | 360                     |
| Cd in kg              | 0.0023                  | -                       | contaminant             |
| In in kg              | 6.03E-05                | -                       | 855                     |
| Value contained in \$ | 1.19                    | 0.33                    |                         |
| Allocation by value   | 79%                     | 21%                     |                         |
| Allocation by mass    | 70%                     | 30%                     |                         |

### 6.1.5 Data quality

Data uncertainty is derived using the pedigree matrix as described in Frischknecht et al. (2004). The inventory is fully based on values from nickel-mining reported in Hilbrans & Hinrichs (1999), which may eventually bias the assertions. For specific conclusions further investigation especially on the effluents composition and tailings emissions has to be done. The uncertainty of emissions that are adopted from nickel mining is arbitrarily reflected by a geometric standard deviation of 10. The choice of electricity mix bases on assumptions that have to be improved with further investigations.

## 6. System Characterisation

| General Flow Information                       |              |                      | Representation in ecoinvent                                                                            |                        |                            |                 |          |                                                                 |                                          | Uncertainty Informations |                       |                                        |                    |                  |                 |
|------------------------------------------------|--------------|----------------------|--------------------------------------------------------------------------------------------------------|------------------------|----------------------------|-----------------|----------|-----------------------------------------------------------------|------------------------------------------|--------------------------|-----------------------|----------------------------------------|--------------------|------------------|-----------------|
| Input                                          | Process Name | Output               | Remarks                                                                                                | Category               | Sub-category               | Infra-structure | Location | Modul name in ecoinvent                                         | Mean value                               | Unit                     | Source for mean value | Type                                   | Standard Deviation | General Comment  |                 |
| Zinc, in sulphidic ore                         | mining       |                      | 85% yield                                                                                              | resource               | in ground                  |                 |          | Zinc, 9.0% in sulfide, Zn 5.3%, Pb, Ag, Cd, In, in ground       | 3.90E-01                                 | kg                       | own assumptions       | 1                                      | 1.58               | (4,2,2,1,4,4,12) |                 |
| Lead, in sulphidic ore                         |              |                      | 95% yield                                                                                              | resource               | in ground                  |                 |          | Lead, 5.0% in sulfide, Pb 3.0%, Zn, Ag, Cd, In, in ground       | 2.17E-01                                 | kg                       | own assumptions       | 1                                      | 1.58               | (4,2,2,1,4,4,12) |                 |
| Silver, in sulphidic ore                       |              |                      | 95% yield                                                                                              | resource               | in ground                  |                 |          | Silver, 0.007% in sulfide, Ag 0.004%, Pb, Zn, Cd, In, in ground | 3.16E-04                                 | kg                       | own assumptions       | 1                                      | 1.58               | (4,2,2,1,4,4,12) |                 |
| Cadmium, in sulphidic ore                      |              |                      | 95% yield, in only 17% of the oredeposits Cd is extracted                                              | resource               | in ground                  |                 |          | Cadmium, 0.30% in sulfide, Cd 0.18%, Pb, Zn, Ag, In, in ground  | 1.30E-02                                 | kg                       | own assumptions       | 1                                      | 1.58               | (4,2,2,1,4,4,12) |                 |
| Indium, in sulphidic ore                       |              |                      | 80% yield, in only 39% of the oredeposits In is extracted                                              | resource               | in ground                  |                 |          | Indium, 0.005% in sulfide, In 0.003%, Pb, Zn, Ag, Cd, in ground | 2.17E-04                                 | kg                       | own assumptions       | 1                                      | 1.58               | (4,2,2,1,4,4,12) |                 |
| fresh and processwater (mining)                |              |                      | surface water is chosen.                                                                               | resource               | in water                   |                 |          | Water, river                                                    | 2.54E-03                                 | m3                       | (Hilbrans, 1999)      | 1                                      | 1.53               | (2,2,2,1,4,4,12) |                 |
| pit water                                      |              |                      |                                                                                                        | resource               | in water                   |                 |          | Water, well, in ground                                          | 3.09E-03                                 | m3                       | (Hilbrans, 1999)      | 1                                      | 1.53               | (2,2,2,1,4,4,12) |                 |
| Electricity from coal                          |              |                      | Assumption: 25% electricity from coal                                                                  | hard coal              | power plants               |                 | No       | US                                                              | electricity, hard coal, at power plant   | 4.29E-02                 | kWh                   | (Hilbrans, 1999)                       | 1                  | 1.58             | (4,2,2,1,4,4,2) |
| Electricity from gas                           |              |                      | Assumption: 75% electricity from gas                                                                   | natural gas            | power plants               |                 | No       | US                                                              | electricity, natural gas, at power plant | 1.29E-01                 | kWh                   | (Hilbrans, 1999)                       | 1                  | 1.58             | (4,2,2,1,4,4,2) |
| diesel fuel (energy)                           |              |                      |                                                                                                        | construction processes | machinery                  |                 | No       | GLO                                                             | diesel, burned in building machine       | 5.10E-01                 | MJ                    | (Hilbrans, 1999)                       | 1                  | 1.53             | (2,2,2,1,4,4,4) |
| stowing material, sand                         |              |                      | additional material, no excavation included                                                            | construction materials | additives                  |                 | No       | CH                                                              | sand, at mine                            | 2.07E+00                 | kg                    | (Hilbrans, 1999)                       | 1                  | 1.53             | (2,2,2,1,4,4,4) |
| cement                                         |              |                      |                                                                                                        | construction materials | binder                     |                 | No       | CH                                                              | portland calcareous cement, at plant     | 1.65E-01                 | kg                    | (Hilbrans, 1999)                       | 1                  | 1.53             | (2,2,2,1,4,4,4) |
| explosive                                      |              |                      |                                                                                                        | construction processes | civil engineering          |                 | No       | RER                                                             | blasting                                 | 1.14E-02                 | kg                    | (Hilbrans, 1999)                       | 1                  | 1.53             | (2,2,2,1,4,4,4) |
| transports, lorry, mining                      |              |                      | transportation of cement, explosives and sand 100km assumed.                                           | transport systems      | road                       |                 | No       | RER                                                             | transport, lorry >16t, fleet average     | 2.25E-01                 | tkm                   | calculations based on (Hilbrans, 1999) | 1                  | 2.11             | (4,2,2,1,3,4,5) |
| exploitation infrastructrue, conveyor belts    |              |                      | exploitation infrastructure, 2000 m belts assumed                                                      | construction processes | machinery                  |                 | Yes      | RER                                                             | conveyor belt, at plant                  | 1.46E-07                 | m                     | own assumptions                        | 1                  | 1.33             | (4,2,2,1,3,4,4) |
| infrastructure of underground mining operation |              |                      | including beneficiation facilities                                                                     | metals                 | extraction                 |                 | Yes      | GLO                                                             | non-ferrous metal mine, underground      | 2.17E-10                 | unit                  | own assumptions                        | 1                  | 3.32             | (5,5,1,1,1,5,9) |
|                                                |              | overburden, disposed |                                                                                                        | waste management       | residual material landfill | No              | GLO      | disposal, sulfidic tailings, off-site                           | 2.04E+00                                 | kg                       | (Hilbrans, 1999)      | 1                                      | 1.53               | (2,2,2,1,4,4,6)  |                 |
|                                                |              | overburden, refilled | not inventoried, no ecological burden                                                                  |                        |                            |                 |          |                                                                 | 9.26E-01                                 | kg                       | (Hilbrans, 1999)      | 1                                      | 1.53               | (2,2,2,1,4,4,6)  |                 |
|                                                |              | effluents, mining    | Detailed composition see Tab. 5.10, p. 30 (variability of the single emissions has with stdev95 of 10) |                        |                            |                 |          |                                                                 | 3.09E-03                                 | m3                       | (Hilbrans, 1999)      | 1                                      | 1.53               | (2,2,2,1,4,4,6)  |                 |

Fig. 6.3 Flows for "exploitation, zinc-lead deposit" (GLO) (part mining) and its representation in the ecoinvent database

## 6. System Characterisation

| General Flow Information              |               |                          | Representation in ecoinvent                                                                                                               |                        |                            |                 |          |                                       |            | Uncertainty Informations |                                   |      |                    |                                                                     |
|---------------------------------------|---------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------|----------------------------|-----------------|----------|---------------------------------------|------------|--------------------------|-----------------------------------|------|--------------------|---------------------------------------------------------------------|
| Input                                 | Process Name  | Output                   | Remarks                                                                                                                                   | Category               | Sub-category               | Infra-structure | Location | Modul name in ecoinvent               | Mean value | Unit                     | Source for mean value             | Type | Standard Deviation | General Comment                                                     |
| electricity, beneficiation            | beneficiation |                          | included in mining                                                                                                                        |                        |                            |                 |          |                                       |            |                          |                                   |      |                    |                                                                     |
| fresh and processwater (benefication) |               |                          |                                                                                                                                           | resource               | in water                   |                 |          | Water, well, in ground                | 6.00E-03   | m3                       | (Hilbrans, 1999)                  | 1    | 1.53               | (2,2,2,1,4,4,12)                                                    |
| lime                                  |               |                          |                                                                                                                                           | construction materials | others                     | No              | CH       | limestone, milled, packed, at plant   | 2.40E-03   | kg                       | (Krauss, 1999)                    | 1    | 2                  | reported values                                                     |
| transports, lorry, beneficiation      |               |                          | transportation of lime and chemicals, 100km assumed.                                                                                      | transport systems      | road                       | No              | RER      | transport, lorry >16t, fleet average  | 1.88E-03   | tkm                      | calculation                       | 1    | 2.11               | (4,2,2,1,3,4,5); calculation based on values for lime and chemicals |
| infrastruktüre beneficiation          |               |                          | included in infrastructure for mining                                                                                                     |                        |                            |                 |          |                                       |            |                          |                                   |      |                    |                                                                     |
| floating agents, organic              |               |                          | Collector and frother                                                                                                                     | chemicals              | organics                   | No              | GLO      | chemicals organic, at plant           | 9.79E-04   | kg                       | (IPPC, 2002)                      | 1    | 2                  | reported values                                                     |
| floating agents, inorganic            |               |                          | Depressant and activator                                                                                                                  | chemicals              | inorganics                 | No              | GLO      | chemicals inorganic, at plant         | 3.34E-03   | kg                       | (IPPC, 2002)                      | 1    | 2                  | reported values                                                     |
| Sodium cyanide                        |               |                          |                                                                                                                                           | chemicals              | organics                   | No              | RER      | hydrogen cyanide, at plant            | 1.52E-04   | kg                       | (IPPC, 2002)                      | 1    | 2                  | reported values                                                     |
| Sulphuric Acid                        |               |                          | burden-free, since byproduct from metallurgy                                                                                              |                        |                            |                 |          |                                       | 1.62E-02   | kg                       | (IPPC, 2002)                      | 1    | 2                  | reported values                                                     |
|                                       |               | tailings, disposed       | not clear, wether net disposal or partly re-filled in mine. Here assumed as net disposal                                                  | waste management       | residual material landfill | No              | GLO      | disposal, sulfidic tailings, off-site | 2.21E+00   | kg                       | (Hilbrans, 1999)                  | 1    | 1.53               | (2,2,2,1,4,4,6); reported values                                    |
|                                       |               | tailings, refilled       | not inventoried, no ecological burden                                                                                                     |                        |                            |                 |          |                                       | 1.13E+00   | kg                       | (Hilbrans, 1999)                  | 1    | 1.53               | (2,2,2,1,4,4,6)                                                     |
|                                       |               | effluents, beneficiation | Detailed composition see Tab. 6.9, p. 30 (variability of the single emissions has with stdev95 of 10)                                     |                        |                            |                 |          |                                       | 6.00E-03   | m3                       | (Hilbrans, 1999)                  | 1    | 1.53               | (2,2,2,1,4,4,6); reported values                                    |
|                                       |               | Emission Cyanide         | From sodium cyanide, 80% control                                                                                                          | water                  | river                      |                 |          | Cyanide                               | 1.62E-05   | kg                       | (IPPC, 2002), (Anonymous 1998)    | 1    | 1.79               | (2,2,2,3,4,4,33)                                                    |
|                                       |               | emission carbondisulfide | 0.53 kg carbondisulfide out of 1 kg xanthate through hydrolytical decomposition                                                           | air                    | low population density     |                 |          | Carbon disulfide                      | 4.33E-04   | kg                       | (NPI, 2001)                       | 1    | 2.26               | (2,2,2,5,4,4,23)                                                    |
|                                       |               | Dust                     | including emissions from mining activities. Inventoried are the emissions given in Tab. 6.12, p 31                                        |                        |                            |                 |          |                                       | 4.41E-04   | kg                       | (NPI, 2001)                       | 1    | 1.59               | (3,2,1,1,3,4,25)                                                    |
|                                       |               | Lead concentrate         | Concentrate contains 55% lead (0.21 kg Pb, yield = 95%) and 0.08% silver (0.30 g Ag, yield = 95%)                                         | metals                 | extraction                 | No              | GLO      | lead concentrate, at beneficiation    | 3.74E-01   | kg                       | (NPI, 2001), (naweb, 2003), cemip |      |                    |                                                                     |
|                                       |               | Zinc concentrate         | Concentrate contains 53% zinc (0.33 kg Zn, yield = 85%), 0.33% cadmium (2.07 g Cd, yield = 95%) and 0.01% indium (0.07 g In, yield = 80%) | metals                 | extraction                 | No              | GLO      | zinc concentrate, at beneficiation    | 6.26E-01   | kg                       | (NPI, 2001), (naweb, 2003), cemip |      |                    |                                                                     |

Fig. 6.4 Flows for "exploitation, zinc-lead deposit" (part beneficiation) and its representation in the ecoinvent database

## 6.2 Primary Lead Production Process and Infrastructure

There are two basic pyrometallurgical processes available for the production of lead from lead or mixed lead-zinc-sulphide concentrates: sinter oxidation / blast furnace reduction route or Direct Smelting Reduction Processes.

### 6.2.1 Production process and infrastructure

Both processes may also be used for concentrates mixed with secondary raw materials. Both processes are followed by a refining step to produce the final product with the required purity (Sutherland et al., 1997; IPPC, 2001).

The sinter oxidation / blast furnace reduction involves two steps:

- 1) a sintering oxidative roast to remove sulphur with production of PbO; and,
- 2) blast furnace reduction of the sinter product.

The objective of sintering lead concentrates is to remove as much sulphur as possible from the galena and the accompanying iron, zinc, and copper sulphides, while producing lump agglomerate with appropriate properties for subsequent reduction in the blast furnace (a type of a shaft furnace).

As raw material feed, lead concentrates are blended with recycled sinter fines, secondary material and other process materials and pelletised in rotating drums. Pellets are fed onto sinter machine and ignited. The burning pellets are conveyed over a series of wind-boxes through which air is blown. Sulphur is oxidised to sulphur dioxide and the reaction generates enough heat to fuse and agglomerate the pellets.

Sinter is charged to the blast furnace with metallurgical coke. Air and/or oxygen enriched air is injected and reacts with the coke to produce carbon monoxide. This generates sufficient heat to melt the charge. The gangue content of the furnace charge combines with the added fluxes or reagents to form a slag.

For smelting bulk lead-zinc-concentrates and secondary material, frequently the Imperial Smelting Furnace is used. Here, hot sinter and pre-heated coke as well as hot briquettes are charged. Hot air is injected. The reduction of the metal oxides not only produces lead and slag but also zinc, which is volatile at the furnace operating temperature and passes out of the ISF with the furnace off-gases. The gases also contain some cadmium and lead. The furnace gases pass through a splash condenser in which a shower of molten lead quenches them and the metals are absorbed into the liquid lead, the zinc is refined by distillation.

The Direct Smelting Reduction Process does not carry out the sintering stage separately. Lead sulphide concentrates and secondary materials are charged directly to a furnace and are then melted and oxidised. Sulphur dioxide is formed and is collected, cleaned and converted to sulphuric acid. Carbon (coke or gas) and fluxing agents are added to the molten charge and lead oxide is reduced to lead, a slag is formed. Some zinc and cadmium are “fumed” off in the furnace, their oxides are captured in the abatement plant and recovered.

Several processes are used for direct smelting of lead concentrates and some secondary material to produce crude lead and slag. Both smelting processes are used: the ISA Smelt/Ausmelt furnaces (sometimes in combination with blast furnaces), Kaldo (TBRC) and QSL integrated processes are used in EU and Worldwide. The Kivcet integrated process is also used and is a flash smelting process. The ISA Smelt/Ausmelt furnaces and the QSL take moist, pelletised feed and the Kaldo and Kivcet use dried feed.

**Refining:** Lead bullion may contain varying amounts of copper, silver, bismuth, antimony, arsenic and tin. Lead recovered from secondary sources may contain similar impurities, but generally antimony and calcium dominate.

There are two methods of refining crude lead: electrolytic refining and pyrometallurgical refining. Electrolytic refining uses anodes of de-copperised lead bullion and starter cathodes of pure lead. This is a high-cost process and is used infrequently. A pyrometallurgical refinery consists of a series of kettles, which are indirectly heated by oil or gas. Over a series of separation processes impurities and metal values are separated from the lead bouillon.

### 6.2.2 Waste

The production of metals is related to the generation of several by-products, residues and wastes, which are also listed in the European Waste Catalogue (Council Decision 94/3/EEC). The ISF or direct smelting furnaces also are significant sources of solid slag. This slag has been subjected to high temperatures and generally contains low levels of leachable metals, consequently it may be used in construction. Solid residues also arise as the result of the treatment of liquid effluents. The main waste stream is gypsum waste ( $\text{CaSO}_4$ ) and metal hydroxides that are produced at the wastewater neutralisation plant. These wastes are considered to be a cross-media effect of these treatment techniques but many are recycled to pyrometallurgical process to recover the metals. Dust or sludge from the treatment of gases are used as raw materials for the production of other metals such as Ge, Ga, In and As, etc or can be returned to the smelter or into the leach circuit for the recovery of lead and zinc. Hg/Se residues arise at the pre-treatment of mercury or selenium streams from the gas cleaning stage. This solid waste stream amounts to approximately 40 - 120 t/y in a typical plant. Hg and Se can be recovered from these residues depending on the market for these metals.

### 6.2.3 Emission

The main emissions to air from zinc and lead production are sulphur dioxide, other sulphur compounds and acid mists; nitrogen oxides and other nitrogen compounds, metals and their compounds; dust; VOC and dioxins. Other pollutants are considered to be of negligible importance for the industry, partly because they are not present in the production process and partly because they are immediately neutralised (e.g. chlorine) or occur in very low concentrations. Emissions are to a large extent bound to dust (except cadmium, arsenic and mercury that can be present in the vapour phase).

Metals and their compounds and materials in suspension are the main pollutants emitted to water. The metals concerned are Zn, Cd, Pb, Hg, Se, Cu, Ni, As, Co and Cr. Other significant substances are fluorides, chlorides and sulphates. Wastewater from the gas cleaning of the smelter and fluid-bed roasting stages are the most important sources.

### 6.2.4 Smelting, primary lead production in ecoinvent

No contiguous study on the lead production is yet available, therefore this module is based partly on values reported in Ayres et al. (2002), IPPC (2001), Xiao et al. (2003), and EEA (1999a).

The module describes the primary production of lead in Europe. The multi-output-process "smelting, primary lead production" delivers the co-products "lead, primary, at plant" and "parkes process crust, from desilverising of lead". The flow "lead, ..." is part of the respective European supply mix. The by-product "parkes process crust, ..." receives part of the burden and enters the silver production chain.

The modules including general flow information, remarks, sources, values, and uncertainty information are shown in Fig. 6.5 (inputs) and Fig. 6.6 (outputs). The following sections describe calculations, sources and assumptions chosen in this study. The meta-information for this dataset is compiled in Tab. 6.2.

The smelting process is modelled as a combination of the two predominant processes

- Sinter oxidation / blast furnace reduction (44% of world production, Tab. 6.13), and the newer

- Direct smelting process (QSL, Kivcet, etc., 56% of world production, Tab. 6.14)

The inputted raw materials and the emission factors are combined according to the production share in order to for the inventory in ecoinvent (Fig. 6.5 and Fig. 6.6). No quantitative compilation of electricity production is available, therefore the UCTE production mix was chosen as approximation (year 2000: nuclear 15%; hydro 4%; fossil 81%<sup>11</sup>).

The two predominant processes differ also in emission factors per tonne lead produced. Where the older sinter oxidation / blast furnace processes exhibit limited abatement possibilities, higher emissions occur from these (Tab. 6.15). The more recent direct smelting processes in contrast have improved emission control (Tab. 6.16). The combination of the two process technologies combined according to their production shares gives the emission factors used in this inventory (Tab. 6.17). Some of the emissions generally reported in literature and stated in the previous section – Ge, Ga, In, Se, acid mists, VOC (air); fluorides, chlorides and sulphates (water) – are not covered by this inventory because no emission factors could be found. The dioxin emissions are approximated with the reported emission factor of 50 µg-TEQ / t secondary zinc production in EC (2002).

**Tab. 6.13 Lead production: Sinter oxidation / blast furnace reduction. Input / output and energy demand reported from various sources. ISF = Imperial smelting furnace. 44 % of the lead production, emission factors in this table are combined with 56 % direct smelting in Tab. 6.14**

| Source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     | Ayres et al.<br>(2002) | IPPC<br>(2001)    | IPPC (2001)         | IPPC (2001)          | This study         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------------------------|-------------------|---------------------|----------------------|--------------------|
| Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     | Unknown                | ISF <sup>4)</sup> | Shaft<br>furn.(min) | Shaft furn.<br>(max) |                    |
| <b>Inputs</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |                        |                   |                     |                      |                    |
| Sec. materials                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | kg  |                        | 926               |                     |                      |                    |
| Concentrate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | kg  | 3312                   | 926               |                     |                      |                    |
| Total raw material                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | kg  | 3312                   | 1852              |                     |                      | 1855 <sup>1)</sup> |
| Fluxes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | kg  | 2691                   | n. a.             | n. a.               | n. a.                | 2691 <sup>1)</sup> |
| Lime                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | kg  | 92                     | n. a.             | n. a.               | n. a.                | 92 <sup>1)</sup>   |
| Coking Coal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | kg  | 667                    | 741               | 150                 | 225                  | 446 <sup>2)</sup>  |
| Natural Gas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | MJ  |                        |                   | 1815                | 2541                 | 1089               |
| Electricity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | kWh |                        |                   | 180                 | 300                  | 120                |
| Iron                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | kg  | n. a.                  | n. a.             | 120                 | 170                  | 145                |
| Sulphur                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | kg  | 3                      | n. a.             | n. a.               | n. a.                | 3                  |
| Air (Oxygen source)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | kg  | 12334                  | n. a.             | n. a.               | n. a.                | 12334              |
| <b>Outputs</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |                        |                   |                     |                      |                    |
| Slag                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | kg  | 3661                   | 519               |                     |                      | 519 <sup>3)</sup>  |
| Sulfuric acid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | kg  | 2021                   | 1296              |                     |                      | 2021               |
| Lead, refined                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | kg  | 1000                   | 259               | 1000                | 1000                 | 1000               |
| Lead+Zinc                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | kg  |                        | 1000              |                     |                      |                    |
| <sup>1)</sup> Corresponds to the raw material composition given by preceding inventory “mining, zinc-lead deposits”.<br><sup>2)</sup> Arithmetic mean, processes assumed to be evenly partitioned.<br><sup>3)</sup> value of IPPC (2001) taken, assuming that this figure represents internal recycling in contrary to the total value calculated in Ayres et al. (2002)<br><sup>4)</sup> Values normalised to the sum of 1 t metal (lead and zinc) produced.<br>n. a. = no values were reported, but not zero. |     |                        |                   |                     |                      |                    |

<sup>11</sup> <http://www.ucte.org/>

**Tab. 6.14 Lead production: Direct smelting. Input / output and energy demand per tonne lead reported in IPPC (2001). 56 % of the lead production, emission factors are combined with 44 % blast furnace / ISF in Tab. 6.13**

| Type                                                                                                                                                                                                                                                                                                                                                                                                                                  |     | QSL-1 | QSL-2              | Kivcet | TBRC | ISA  | This study         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------|--------------------|--------|------|------|--------------------|
| <b>Inputs</b>                                                                                                                                                                                                                                                                                                                                                                                                                         |     |       |                    |        |      |      |                    |
| Raw material                                                                                                                                                                                                                                                                                                                                                                                                                          | kg  | n. a  | 1530               | n. a   | n. a | 1356 | 1855 <sup>1)</sup> |
| Fluxes                                                                                                                                                                                                                                                                                                                                                                                                                                | kg  | n. a  | 235                | n. a   | n. a | 39   | 137 <sup>2)</sup>  |
| Coal                                                                                                                                                                                                                                                                                                                                                                                                                                  | kg  | 100   | 141                | 105    | 4    | 79   | 77 <sup>2)</sup>   |
| Natural Gas                                                                                                                                                                                                                                                                                                                                                                                                                           | MJ  | 726   | 695                | 0      | 0    | 0    | 178 <sup>2)</sup>  |
| Fuel Oil                                                                                                                                                                                                                                                                                                                                                                                                                              | MJ  | 0     | 0                  | 1030   | 1236 | 0    | 567 <sup>2)</sup>  |
| Electricity                                                                                                                                                                                                                                                                                                                                                                                                                           | kWh | 0     | 0                  | 250    | 500  | 0    | 188 <sup>2)</sup>  |
| Oxygen                                                                                                                                                                                                                                                                                                                                                                                                                                | kg  | 471   | 541                | 642    | 200  | 148  | 374 <sup>2)</sup>  |
| Nitrogen                                                                                                                                                                                                                                                                                                                                                                                                                              | kg  | n.a.  | 147                | 0      | 0    | 0    | 37 <sup>2)</sup>   |
| <b>Outputs</b>                                                                                                                                                                                                                                                                                                                                                                                                                        |     |       |                    |        |      |      |                    |
| Slag                                                                                                                                                                                                                                                                                                                                                                                                                                  | kg  | n. a  | 589                | n. a   | n. a | 111  | 350 <sup>2)</sup>  |
| Sulphuric acid                                                                                                                                                                                                                                                                                                                                                                                                                        | kg  | n. a  | 706                | n. a   | n. a | 278  | 492 <sup>2)</sup>  |
| Lead, bullion                                                                                                                                                                                                                                                                                                                                                                                                                         | kg  |       | 1059 <sup>3)</sup> |        |      |      |                    |
| Lead, refined                                                                                                                                                                                                                                                                                                                                                                                                                         | kg  | 1000  | 1000               | 1000   | 1000 | 1000 | 1000               |
| <sup>1)</sup> Corresponds to the raw material composition given by proceeding inventory "mining, zinc-lead deposits".<br><sup>2)</sup> Arithmetic mean of the two QSL-processes (one mean value) and the other three processes<br><sup>3)</sup> Only the output of lead bullion was reported, which was normalised to the amount of refined lead, assuming a lead concentration in the bullion of 94.5% and a refining yield of 100%. |     |       |                    |        |      |      |                    |

**Tab. 6.15 Lead: Emissions to air per tonne lead produced – limited control. Values are collected from various sources; they correspond to the conventional sinter-oxidation / blast furnace reduction process.**

| Source                                                                                                                                                                                                                            |        | Reported emissions in IPPC (2001), tab. 5.17 | Proposed emission factor in EEA (1999a) | Reported emission factor in EEA (1999a) <sup>1)</sup> | This study |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------------------------------------|-----------------------------------------|-------------------------------------------------------|------------|
| Sulphur dioxide                                                                                                                                                                                                                   | kg / t | 73.2                                         |                                         |                                                       | 73.2       |
| Lead                                                                                                                                                                                                                              | g / t  | 256.31                                       | 800                                     | 573                                                   | 573        |
| Cadmium                                                                                                                                                                                                                           | g / t  | 5.91                                         | 10                                      | 6.67                                                  | 6.67       |
| Antimony                                                                                                                                                                                                                          | g / t  | 4.76                                         |                                         |                                                       | 4.76       |
| Arsenic                                                                                                                                                                                                                           | g / t  | 2.26                                         | 3                                       | 8.87                                                  | 8.87       |
| Mercury                                                                                                                                                                                                                           | g / t  | 0.177                                        | 3 <sup>2)</sup>                         |                                                       | 0.177      |
| <sup>1)</sup> Geometric mean of a range of values reported in Poland and the value of a plant in Sweden.<br><sup>2)</sup> Value based on one single value of unknown abatement efficiency, therefore value of IPPC (2001) chosen. |        |                                              |                                         |                                                       |            |

**Tab. 6.16 Lead: Emissions to air per tonne lead produced – improved control. Values are collected from various sources. They correspond to the direct smelting process.**

| Source im-<br>proved abate-<br>ment                                                                                   |        | Reported emissions in<br>IPPC (2001), tab. 5.17 | Proposed<br>emission fac-<br>tor in EEA<br>(1999a) | Reported<br>emission fac-<br>tor in EEA<br>(1999a) 1) | This study        |
|-----------------------------------------------------------------------------------------------------------------------|--------|-------------------------------------------------|----------------------------------------------------|-------------------------------------------------------|-------------------|
| Sulphur dioxide                                                                                                       | kg / t | 1.61                                            |                                                    |                                                       | 1.61              |
| Lead                                                                                                                  | g / t  | 16.7                                            | 200                                                | 200                                                   | 108 <sup>1)</sup> |
| Cadmium                                                                                                               | g / t  | 0.0466                                          | 1                                                  | 0.6                                                   | 0.6               |
| Antimony                                                                                                              | g / t  | 0.317                                           |                                                    |                                                       | 0.317             |
| Arsenic                                                                                                               | g / t  | 0.0642                                          | 0.3                                                |                                                       | 0.3               |
| Mercury                                                                                                               | g / t  | 0.0100                                          | 3 <sup>2)</sup>                                    |                                                       | 0.01              |
| <sup>1)</sup> Mean of considered processes in IPPC and EAA.                                                           |        |                                                 |                                                    |                                                       |                   |
| <sup>2)</sup> Value based on one single value of unknown abatement efficiency, therefore value of IPPC (2001) chosen. |        |                                                 |                                                    |                                                       |                   |

**Tab. 6.17 Emission factors per tonne lead produced used in this inventory. Combination of limited and improved emission control according to the technologies and their production share (see Tab. 6.15 and Tab. 6.16).**

|                        |        | Limited control                     | Improved control | This Study |
|------------------------|--------|-------------------------------------|------------------|------------|
|                        |        | Sinter oxidation /<br>blast furnace | Direct smelting  | Mix        |
| Share world production |        | 44%                                 | 56%              | mix        |
| Sulphur dioxide        | kg / t | 73.2                                | 1.61             | 33         |
| Lead                   | g / t  | 573                                 | 60               | 312        |
| Cadmium                | g / t  | 6.67                                | 0.60             | 3.26       |
| Antimony               | g / t  | 4.76                                | 0.317            | 2.26       |
| Arsenic                | g / t  | 8.87                                | 0.300            | 4.06       |
| Mercury                | g / t  | 0.177                               | 0.010            | 0.083      |
| Dioxin                 | µg / t | 50                                  | 50               | 50         |

**Tab. 6.18 Emission to air: solid particles. Values taken from CEPMEIP-Database (SNAP 030304). Values might be overestimated considering that TSP from fuels burned in the smelting process are included in the shown figures. Additionally, TSP is accounted for in the respective modules for burning of fuels whose requirements are in Tab. 6.13 and Tab. 6.14.**

| Solid Particle Emission                       | Unit  | Medium / high emission<br>(sinter / blast furnace) | Low emission<br>(direct smelting) |
|-----------------------------------------------|-------|----------------------------------------------------|-----------------------------------|
| TSP                                           | g / t | 600                                                | 120                               |
| PM10                                          | g / t | 500                                                | 114                               |
| PM2.5                                         | g / t | 300                                                | 60                                |
| <b>Differential classes used in ecoinvent</b> |       |                                                    |                                   |
| PM<2.5                                        | g / t | 300                                                | 60                                |
| PM2.5-10                                      | g / t | 200                                                | 54                                |
| PM>10                                         | g / t | 100                                                | 6                                 |



**Tab. 6.19 Emissions to water.** Due to lack of data the values for a Chinese zinc smelter reported in Xiao et al. (2003) are used.

| Emission to water | Unit  | Value (unspecific) |
|-------------------|-------|--------------------|
| Hg                | g / t | 0.043              |
| Cd                | g / t | 2.6                |
| As                | g / t | 1                  |
| Pb                | g / t | 18                 |
| Cu                | g / t | 5                  |
| COD               | g / t | 583                |

### 6.2.5 By-products and allocation

Argentiferous lead and lead-zinc ores, which often occur in association, are distributed widely. In almost all silver-producing countries, they constitute an important fraction of the silver produced by mining and account for ca. 50 % of world silver production (Renner et al., 2002).

Mine production remains by far the largest contribution to silver supply, normally accounting for around two-thirds of the total. Beside mine production, post consumer scrap, disinvestment, government sales and producer hedging are sources for supply. Old scrap normally makes up a little over a fifth of the supply (Klapwijk et al., 2006).

The ratio between the world supply of silver and the world mine production of lead amounted in the last twenty years between four to six kilograms silver per tonne lead mined (s. Tab. 3.1). This leads to the assumption that in a world average per tonne lead mined 2.3 kg silver is jointly extracted and refined.

By the so-called Parkes process a silver containing crust arises where silver is accumulated and subsequently refined further (Renner et al., 2002). A metallurgical process Ag yield of 97.4% is assumed (Reuter, 2005). For more details on the process, see the respective chapter on Silver in this report.

However, the allocation of the subprocesses lead- and silver production is not evident to solve, since only the final commodity prices are known but no in-firm prices for the crust and for the raw lead bullion, on which the allocation would rely. Also the share of the products in total costs – gross sales method as proposed in Guinée et al. (2004) and Guinée et al. (2001) – is not easy to estimate. The by-product allocation problem – the joint production of silver and lead – is tackled here by a subdivision of the sub-processes (according to Ekvall & Finnveden, 2001). The silver production from the parkes process crust is treated in a distinct inventory. Hence the burdens have to be allocated between the lead and the silver containing crust.

As starting point for an estimation it is assumed that the profit margin for the production of silver corresponds to the company's performance. Taking an arbitrary performance value of 10% - which reflects the general profit expectations of major enterprises – the value of the crust corresponds consequently to 10% of the proceeds from the silver sales. Therefore 10 % of the silver price is taken to estimate the value of the silver containing Parkes process crust.

The composition and the corresponding values contained in the by-products result in a split of 4% to the silver and 96% to the lead (Tab. 6.20).

However, the resources (elementary flows) have to be allocated by mass to the respective by-products, in order to keep up with the resource balance in the final commodity. Since the concentrate required by this multi-output-process is allocated also by revenue, the therein contained resources are not correctly attributed and have to be corrected with a corrective exchange flow (Tab. 6.21).

**Tab. 6.20 By-products of primary lead production: Composition, flows and value to derive the revenue-based allocation factors (Values are the year averages from 2004 through 2006, reported by the U.S. Geological Survey)**

| Lead / Silver bearing concentrate | Process Yield | lead concentrate | parkes process crust | lead   | Value in \$ / kg |
|-----------------------------------|---------------|------------------|----------------------|--------|------------------|
| composition                       |               | input            | output               | output |                  |
| Pb                                | 98.0%         | 55%              | 39%                  | 100%   | 1.15             |
| Ag                                | 97.4%         | 0.08%            | 18%                  | 0%     | 27               |
| Amount                            | kg            | 1.86             | 0.0081               | 1      |                  |
| Value contained in \$             | \$            |                  | 0.04                 | 1.15   |                  |
| Allocation by Value               |               |                  | 3%                   | 97%    |                  |

**Tab. 6.21 Individual correction factors to of the respective by-products in the lead production.**

| correction from economic to per mass |    |         | according to economic allocation |                      |
|--------------------------------------|----|---------|----------------------------------|----------------------|
| Resources demand in the feed         |    |         | Lead                             | Parkes process crust |
| Concentrate                          | kg | 1.86    | 97%                              | 3%                   |
| Pb                                   | kg | 1.074   | 1.039                            | 0.0354               |
| Ag                                   | kg | 0.00157 | 0.00152                          | 0.000052             |
| Resources attributed in by-product   |    |         | according to per mass allocation |                      |
| Pb                                   |    |         | 1.074                            |                      |
| Ag                                   |    |         |                                  | 0.00157              |
| Resource exchange flow               |    |         | difference to be corrected       |                      |
| Pb                                   |    |         | 0.03536                          | -0.03536             |
| Ag                                   |    |         | -0.001515                        | 0.001515             |

## 6.2.6 Data quality

Data uncertainty is derived using the Pedigree matrix as described in Frischknecht et al. (2004). Some of the emissions that are described in the literature as typical for lead, could not be covered in this inventory due to missing emission factors, namely Ge, Ga In and Se into air and , Zn, Se, Cu, Ni, Co and Cr to water. Also salts – fluorides, chlorides and sulphates – are not included due to missing data. Further investigations on these issues have been done to improve the quality of this dataset. However, for the purpose of a background database the quality is sufficient. The choice of electricity mix bases on assumptions that have to be improved with further investigations.

## 6. System Characterisation

| General Flow Information    |                                            |        | Representation in ecoinvent                                                                       |                        |                 |                 |          |                                                                  |            | Uncertainty Informations |                       |      |                    |                 |
|-----------------------------|--------------------------------------------|--------|---------------------------------------------------------------------------------------------------|------------------------|-----------------|-----------------|----------|------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------|
| Input                       | Process Name                               | Output | Remarks                                                                                           | Category               | Sub-category    | Infra-structure | Location | Modul name in ecoinvent                                          | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment |
| Concentrate                 | smelting, primary lead production (inputs) |        | Concentrate contains 55% lead (1.02 kg Pb, yield = 98%) and 0.08% silver (1.49 g Ag, yield = 97%) | metals                 | extraction      | No              | GLO      | lead concentrate, at beneficiation                               | 1.86E+00   | kg                       | Calculation           | 1    | 1.33               | (3,3,2,1,3,5,4) |
| Correction Pb in Conc -> Pb |                                            |        | Pb resource in concentrate fully allocated to the Pb bearing by-product. Correction Lead          | metals                 | extraction      | No              | GLO      | resource correction, PbZn, lead, positive                        | 3.54E-02   | kg                       | calculation           | 1    | 1.33               | (3,3,2,1,3,5,4) |
| Correction Pb in Conc -> Ag |                                            |        | Pb resource in concentrate fully allocated to the Pb bearing by-product. Correction Silver.       | metals                 | extraction      | No              | GLO      | resource correction, PbZn, lead, negative                        | 3.54E-02   | kg                       | calculation           | 1    | 1.33               | (3,3,2,1,3,5,4) |
| Correction Ag in Conc -> Pb |                                            |        | Ag resource in concentrate fully allocated to the Ag bearing by-product. Correction Lead          | metals                 | extraction      | No              | GLO      | resource correction, PbZn, silver, negative                      | 1.52E-03   | kg                       | calculation           | 1    | 1.33               | (3,3,2,1,3,5,4) |
| Correction Ag in Conc -> Ag |                                            |        | Ag resource in concentrate fully allocated to the Ag bearing by-product. Correction Silver        | metals                 | extraction      | No              | GLO      | resource correction, PbZn, silver, positive                      | 1.52E-03   | kg                       | calculation           | 1    | 1.33               | (3,3,2,1,3,5,4) |
| Lime (total)                |                                            |        | lime + 50% of "Fluxes"                                                                            | construction materials | others          | No              | CH       | limestone, milled, packed, at plant                              | 6.69E-01   | kg                       | (Ayres, 2002)         | 1    | 1.25               | (3,3,2,1,3,3,4) |
| silica sand                 |                                            |        | 50% of "Fluxes"                                                                                   | construction materials | additives       | No              | DE       | silica sand, at plant                                            | 6.28E-01   | kg                       | (Ayres, 2002)         | 1    | 1.25               | (3,3,2,1,3,3,4) |
| Electricity from Grid       |                                            |        | UCTE mix: nuclear 15%; hydro 4%; fossil 81%                                                       | electricity            | production mix  | No              | UCTE     | electricity, medium voltage, production UCTE, at grid            | 1.58E-01   | kWh                      | (IPPC, 2002)          | 1    | 1.33               | (3,3,2,1,3,5,2) |
| Natural Gas                 |                                            |        |                                                                                                   | natural gas            | heating systems | No              | RER      | natural gas, burned in industrial furnace >100kW                 | 5.77E-01   | MJ                       | (IPPC, 2002)          | 1    | 1.25               | (3,3,2,1,3,3,1) |
| Sulfur                      |                                            |        | not inventoried, negligible                                                                       |                        |                 |                 |          |                                                                  | 1.32E-03   | kg                       | (Ayres, 2002)         | 1    | 1.25               | (3,3,2,1,3,3,4) |
| Oxygen                      |                                            |        |                                                                                                   | chemicals              | inorganics      | No              | RER      | oxygen, liquid, at plant                                         | 2.10E-01   | kg                       | (IPPC, 2002)          | 1    | 1.25               | (3,3,2,1,3,3,4) |
| Nitrogen                    |                                            |        |                                                                                                   | chemicals              | inorganics      | No              | RER      | nitrogen, liquid, at plant                                       | 2.07E-02   | kg                       | (IPPC, 2002)          | 1    | 1.25               | (3,3,2,1,3,3,2) |
| iron                        |                                            |        | Iron oxide minerals added, inventoried as iron ore concentrate                                    | metals                 | extraction      | No              | GLO      | iron ore, 65% Fe, at beneficiation                               | 6.36E-02   | kg                       | (IPPC, 2002)          | 1    | 1.25               | (3,3,2,1,3,3,2) |
| Fuel Oil                    |                                            |        |                                                                                                   | oil                    | heating systems | No              | RER      | heavy fuel oil, burned in industrial furnace 1MW, non-modulating | 2.84E-01   | MJ                       | (IPPC, 2002)          | 1    | 1.25               | (3,3,2,1,3,3,1) |
| Coal                        |                                            |        | 28.6 MJ energy contained per kg coal                                                              | hard coal              | heating systems | No              | RER      | hard coal, burned in industrial furnace 1-10MW                   | 6.97E+00   | MJ                       | (IPPC, 2002)          | 1    | 1.25               | (3,3,2,1,3,3,2) |
| Transport by train          |                                            |        | Assumption - 600km : concentrate, coal; 100km : lime, silica, sulphur, oxygen, nitrogen, iron     | transport systems      | train           | No              | RER      | transport, freight, rail                                         | 5.46E+00   | tkm                      | standard distances    | 1    | 1.31               | (4,3,1,1,3,3,2) |
| Transport by lorry (32t)    |                                            |        | Assumption - 100km : concentrate, coal; 50km : lime, silica, sulphur, oxygen, nitrogen, iron      | transport systems      | road            | No              | RER      | transport, lorry >16t, fleet average                             | 9.62E-01   | tkm                      | standard distances    | 1    | 1.31               | (4,3,1,1,3,3,2) |

Fig. 6.5 Flows for "Smelting, primary lead production" (inputs) and its representation in the ecoinvent database

## 6. System Characterisation

| General Flow Information                    |              |                                   | Representation in ecoinvent                     |                  |                            |                 |          |                                                                        |            | Uncertainty Informations |                                   |      |                    |                                      |
|---------------------------------------------|--------------|-----------------------------------|-------------------------------------------------|------------------|----------------------------|-----------------|----------|------------------------------------------------------------------------|------------|--------------------------|-----------------------------------|------|--------------------|--------------------------------------|
| Input                                       | Process Name | Output                            | Remarks                                         | Category         | Sub-category               | Infra-structure | Location | Modul name in ecoinvent                                                | Mean value | Unit                     | Source for mean value             | Type | Standard Deviation | General Comment                      |
| smelting, primary lead production (outputs) | m            | Slag                              | Proxy for disposal of lead smelter slag.        | waste management | residual material landfill | No              | CH       | disposal, nickel smelter slag, 0% water, to residual material landfill | 4.24E-01   | kg                       | (IPPC, 2002)                      | 1    | 1.25               | (3,3,2,1,3,3,6)                      |
|                                             |              | Sulphuric Acid                    | used, e.g. in beneficiation --> not inventoried |                  |                            |                 |          |                                                                        | 1.16E+00   | kg                       | (IPPC, 2002)                      | 1    | 1.25               | (3,3,2,1,3,3,4)                      |
|                                             |              | Cu                                |                                                 | water            | river                      |                 |          | Copper, ion                                                            | 5.00E-06   | kg                       | (Xiao, 2003)                      | 1    | 5.07               | (1,3,1,1,3,3,4,3,3), values from Zn- |
|                                             |              | Pb                                |                                                 | water            | river                      |                 |          | Lead                                                                   | 1.80E-05   | kg                       | (Xiao, 2003)                      | 1    | 5.07               | (1,3,1,1,3,3,4,3,3), values from Zn- |
|                                             |              | As                                |                                                 | water            | river                      |                 |          | Arsenic, ion                                                           | 1.00E-06   | kg                       | (Xiao, 2003)                      | 1    | 5.07               | (1,3,1,1,3,3,4,3,3), values from Zn- |
|                                             |              | Hg                                |                                                 | water            | river                      |                 |          | Mercury                                                                | 4.30E-08   | kg                       | (Xiao, 2003)                      | 1    | 5.07               | (1,3,1,1,3,3,4,3,3), values from Zn- |
|                                             |              | Cd                                |                                                 | water            | river                      |                 |          | Cadmium, ion                                                           | 2.60E-06   | kg                       | (Xiao, 2003)                      | 1    | 5.07               | (1,3,1,1,3,3,4,3,3), values from Zn- |
|                                             |              | COD                               |                                                 | water            | river                      |                 |          | COD, Chemical Oxygen Demand                                            | 5.83E-04   | kg                       | (Xiao, 2003)                      | 1    | 1.58               | (1,3,1,1,3,3,4,3,3), values from Zn- |
|                                             |              | BSB5                              |                                                 | water            | river                      |                 |          | BOD5, Biological Oxygen Demand                                         | 3.89E-04   | kg                       | calculated, based on (Xiao, 2003) | 1    | 1.58               | (1,3,1,1,3,3,4,3,3), values from Zn- |
|                                             |              | TOC                               |                                                 | water            | river                      |                 |          | TOC, Total Organic Carbon                                              | 2.28E-04   | kg                       | calculated, based on (Xiao, 2003) | 1    | 1.58               | (1,3,1,1,3,3,4,3,3), values from Zn- |
|                                             |              | DOC                               |                                                 | water            | river                      |                 |          | DOC, Dissolved Organic Carbon                                          | 2.28E-04   | kg                       | calculated, based on (Xiao, 2003) | 1    | 1.58               | (1,3,1,1,3,3,4,3,3), values from Zn- |
|                                             |              | Lead                              |                                                 | air              | low population density     |                 |          | Lead                                                                   | 3.12E-04   | kg                       | (IPPC, 2002)                      | 1    | 5.04               | (3,4,2,1,1,4,22)                     |
|                                             |              | Antimony                          |                                                 | air              | low population density     |                 |          | Antimony                                                               | 2.26E-06   | kg                       | (IPPC, 2002)                      | 1    | 5.04               | (3,4,2,1,1,4,22)                     |
|                                             |              | Mercury                           |                                                 | air              | low population density     |                 |          | Mercury                                                                | 8.31E-08   | kg                       | (IPPC, 2002)                      | 1    | 5.04               | (3,4,2,1,1,4,22)                     |
|                                             |              | Arsenic                           |                                                 | air              | low population density     |                 |          | Arsenic                                                                | 4.06E-06   | kg                       | (IPPC, 2002)                      | 1    | 5.04               | (3,4,2,1,1,4,22)                     |
|                                             |              | PM<2.5                            |                                                 | air              | low population density     |                 |          | Particulates, < 2.5 um                                                 | 1.65E-04   | kg                       | (IPPC, 2002)                      | 1    | 3.04               | (3,4,2,1,1,4,27)                     |
|                                             |              | PM2.5-10                          |                                                 | air              | low population density     |                 |          | Particulates, > 2.5 um, and < 10um                                     | 1.18E-04   | kg                       | (IPPC, 2002)                      | 1    | 1.28               | (3,4,2,1,1,4,28)                     |
|                                             |              | PM>10                             |                                                 | air              | low population density     |                 |          | Particulates, > 10 um                                                  | 4.72E-05   | kg                       | (IPPC, 2002)                      | 1    | 1.46               | (3,4,2,1,1,4,29)                     |
|                                             |              | Cadmium                           |                                                 | air              | low population density     |                 |          | Cadmium                                                                | 3.26E-06   | kg                       | (IPPC, 2002)                      | 1    | 1.55               | (3,4,2,1,1,4,16)                     |
|                                             |              | TCDD                              |                                                 | air              | low population density     |                 |          | Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin               | 5.00E-11   | kg                       | European Dioxin Inventory         | 1    | 10                 | reported                             |
|                                             |              | Carbon dioxide from lime addition | 0.44 kg CO2 / kg lime                           | air              | unspecified                |                 |          | Carbon dioxide, fossil                                                 | 2.94E-01   | kg                       | (IPPC, 2002)                      | 1    | 1.58               | (4,2,2,3,4,4,14)                     |
|                                             |              | Sulphur Dioxide                   |                                                 | air              | low population density     |                 |          | Sulfur dioxide                                                         | 3.30E-02   | kg                       | (IPPC, 2002)                      | 1    | 1.19               | (3,4,2,1,1,4,15)                     |
|                                             |              |                                   |                                                 |                  |                            |                 |          |                                                                        |            |                          |                                   |      |                    |                                      |
|                                             |              | Parkes Process crust              | Contains 18% Ag                                 | metals           | extraction                 | No              | GLO      | parkes process crust, from desilverising of lead                       | 8.05E-03   | kg                       |                                   |      |                    |                                      |
|                                             |              | Refined lead                      |                                                 | metals           | extraction                 | No              | GLO      | lead, primary, at plant                                                | 1.00E+00   | kg                       |                                   |      |                    |                                      |

Fig. 6.6 Flows for "Smelting, primary lead production " (outputs) and its representation in the ecoinvent database

## 6.3 Secondary Lead Production process and infrastructure

Secondary lead originates to a predominant share from spent lead acid (PbA) batteries. Gabby (2004) estimates 92% for this source in the U.S. in 2004. Lead acid batteries' recycling is very similar to the primary lead production process. The main differences are in material preparation before reduction, which affects plant size, since there is no need for sintering (Espinosa et al., 2004).

The processing of this sort of scrap can be done by different technologies. The classical way makes use of short or long rotary furnace (Various Authors, 1996). More recently the convergence of primary and secondary producer has led to hybrid systems that process simultaneously both lead ore and metallic post consumer scrap lead. One example is the QSL process at the Bleihütte Binsfelderhammer in Stolberg, Germany (Berzelius, 1993).

### 6.3.1 Lead, secondary, at plant in ecoinvent

This dataset describes the recovery of secondary lead from spent lead acid batteries as done at Campine, Belgium (Campine NV, 2006). The company uses a shaft furnace with post combustion (Quirijnen, 1999). The compilation of the unit process LCI as applied in ecoinvent is displayed in Tab. 6.22. The meta-information is reported in Tab. 6.2, at the beginning of this chapter.

Typically this technology produces 5000 t / a sulphuric acid (15% concentration), 25'000 t lead bullion (98% Pb), 1200 t / a slags (1% Pb) and 3000 t / a raw lead matte (10% Pb) to be shipped to primary smelters. Overall Pb yield is typically 98.8% at the plant level and 99.8% after reworking the matte (Quirijnen, 1999).

The operation treats junk batteries and plates but also lead cable sheathing, drosses and sludges, leaded glass and balancing weights. From this feed it manufactures mainly antimonial lead up to 10% Sb, calcium-aluminium lead alloys with or without tin and soft lead with low and high copper content. All these products are the result of a refining and alloying step to meet the compliance with the designations desired. For the purpose of this LCI-module all these alloys are subsumed under the reference flow "lead, secondary, at plant". The company deliveries amounted in 2004 and 2005 around 40'000 t lead per year, which represents ca. 0.8% of the world secondary lead production (estimated 5'000'000 t in 2005 (USGS, 2006b)).

The inventory data for the process has been adopted from the Battery LCA commissioned by the UK Department for Environment Food and Rural Affairs (Defra), prepared by Fisher et al. (2006). It represents data for Campine Ltd from 2004 (Tab. 6.22).

The values for the energy carriers have been converted in effective heat delivered with the respective energy content and energy efficiency as given in ecoinvent Centre (2005) and Frischknecht et al. (2004). The datasets consider the system specific combustion emissions, therefore the originally reported CO<sub>2</sub> emissions – mainly due to the combustion – are omitted for LCI calculation. It is not clear whether the emissions from SO<sub>2</sub> emissions are due to the combustion. Assuming a heavy fuel burner in Jungbluth (2003) with quite high SO<sub>2</sub> emissions (0.005 kg SO<sub>2</sub> per kg CO<sub>2</sub> emitted in off-gas.), only 35 % of the reported SO<sub>2</sub> emissions may be explained. Therefore it is assumed, that the SO<sub>2</sub> emissions originate from a process other than combustion and is counted for as indicated in Fisher et al. (2006).

The infrastructure has been approximated with the generic module for non-ferrous metal operations with 2'650'000 t/a output and 50 a life time. Transportation of the raw material is considered in the dataset "iron scrap, at plant" that is used as approximation for collection and preparation of the battery scrap. The rest of the auxiliary material, energy carrier, as well as the disposal of the slag is assumed to occur by lorry over 100 km distance.

Tab. 6.22 Unit process data for “lead, secondary, at plant” as given in Fisher et al. (2006).

| Flow                              | Quantity  | Unit | Comments / Inventory used                                                   |
|-----------------------------------|-----------|------|-----------------------------------------------------------------------------|
| <b>INPUTS</b>                     |           |      |                                                                             |
| Lead acid batteries (incl acid)   | 1000      | kg   | Approximated with the dataset for collection and preparation of iron scrap. |
| Limestone                         | 5.8       | kg   | Limestone, milled.                                                          |
| Iron scrap                        | 4.0       | kg   | Iron scrap.                                                                 |
| NaOH                              | 350       | kg   | 50% NaOH.                                                                   |
| Sodium nitrate                    | 0.4       | kg   | Generic inorganic chemicals.                                                |
| Sulphur                           | 0.9       | kg   | Secondary Sulphur.                                                          |
| Iron chloride                     | 0.9       | kg   | Iron (III) chloride (30%).                                                  |
| Slag                              | 150       | kg   | Reused from process                                                         |
| Electricity                       | 35.2      | kWh  | Grid Electricity, Medium Voltage.                                           |
| Natural gas                       | 16.2      | kg   | 45.4 MJ per kg energy content, energy efficiency 95%.                       |
| Coke                              | 20.0      | kg   | Hard coal coke with 28.6 MJ per kg energy content, energy efficiency 70%.   |
| Process water                     | 770       | kg   | Treated rainwater, reused through process.                                  |
| <b>OUTPUTS</b>                    |           |      |                                                                             |
| Flue dust for internal reuse      | 13.6      | kg   | Reused in process                                                           |
| Return slag for internal reuse    | 150       | kg   | Reused in process                                                           |
| Sulphuric acid for internal reuse | 71.0      | kg   | Sulphuric acid. ecoinvent, Europe,                                          |
| SO <sub>2</sub>                   | 7.1       | kg   | -                                                                           |
| CO <sub>2</sub> (fuel combustion) | 500       | kg   | Contained in dataset for gas combustion.                                    |
| Pb                                | 0.00127   | kg   | -                                                                           |
| Sb                                | 0.0000056 | kg   | -                                                                           |
| Excess slag to landfill           | 44.0      | kg   | Disposal to landfill.                                                       |
| Lead to processor                 | 650       | kg   | Reference Flow                                                              |

### 6.3.2 Data quality

Data uncertainty is derived using the Pedigree matrix as described in Frischknecht et al. (2004). As there is one facility that has to stand for a variety of technologies applied, the reliability and correctness of the data set is limited and may not be applicable to life cycle assessment studies of products where the production of secondary lead becomes important.

## 6. System Characterisation

| General Flow Information      |                                   |        | Representation in ecoinvent                                                                                   |                        |                            |                |          |                                                                        |            | Uncertainty Informations |                       |      |                    |                    |
|-------------------------------|-----------------------------------|--------|---------------------------------------------------------------------------------------------------------------|------------------------|----------------------------|----------------|----------|------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|--------------------|
| Input                         | Process Name                      | Output | Remarks                                                                                                       | Category               | Sub-category               | Infrastructure | Location | Modul name in ecoinvent                                                | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment    |
| Lead acid batteries           | lead, secondary, at plant (input) |        | Approximated with the dataset for collection and preparation of iron scrap.                                   | metals                 | extraction                 | No             | RER      | iron scrap, at plant                                                   | 1.54E+00   | kg                       | Fisher et al (2006)   | 2    | 0.08               | normal dist +/- 5% |
| Limestone                     |                                   |        |                                                                                                               | construction materials | others                     | No             | CH       | limestone, milled, packed, at plant                                    | 8.92E-03   | kg                       | Fisher et al (2006)   | 1    | 1.13               | (1,4,1,3,1,3,3)    |
| Iron scrap                    |                                   |        |                                                                                                               | metals                 | extraction                 | No             | RER      | iron scrap, at plant                                                   | 6.15E-03   | kg                       | Fisher et al (2006)   | 1    | 1.13               | (1,4,1,3,1,3,4)    |
| NaOH                          |                                   |        | Corresponds to 0.538 kg NaOH                                                                                  | chemicals              | inorganics                 | No             | RER      | sodium hydroxide, 50% in H2O, production mix, at plant                 | 2.69E-01   | kg                       | Fisher et al (2006)   | 1    | 1.13               | (1,4,1,3,1,3,4)    |
| Sodium nitrate                |                                   |        |                                                                                                               | chemicals              | inorganics                 | No             | RER      | sodium sulphate, powder, production mix, at plant                      | 6.15E-04   | kg                       | Fisher et al (2006)   | 1    | 1.13               | (1,4,1,3,1,3,4)    |
| Sulphur                       |                                   |        |                                                                                                               | chemicals              | inorganics                 | No             | RER      | secondary sulphur, at refinery                                         | 1.38E-03   | kg                       | Fisher et al (2006)   | 1    | 1.13               | (1,4,1,3,1,3,4)    |
| Iron chloride                 |                                   |        | Corresponds to 1.4E-3 kg iron chloride                                                                        | chemicals              | inorganics                 | No             | CH       | iron (III) chloride, 40% in H2O, at plant                              | 3.20E-04   | kg                       | Fisher et al (2006)   | 1    | 1.13               | (1,4,1,3,1,3,4)    |
| Slag                          |                                   |        | Reused from process, not inventoried.                                                                         |                        |                            |                |          |                                                                        | 2.31E-01   | kg                       | Fisher et al (2006)   | 1    |                    |                    |
| Electricity                   |                                   |        |                                                                                                               | electricity            | production mix             | No             | UCTE     | electricity, medium voltage, production UCTE, at grid                  | 5.42E-02   | kWh                      | Fisher et al (2006)   | 1    | 1.13               | (1,4,1,3,1,3,2)    |
| Natural gas                   |                                   |        | 45.4 MJ per kg energy content, energy efficiency 95%                                                          | natural gas            | heating systems            | No             | RER      | heat, natural gas, at industrial furnace >100kW                        | 1.08E+00   | MJ                       | Fisher et al (2006)   | 1    | 1.13               | (1,4,1,3,1,3,1)    |
| Coke                          |                                   |        | Hard coal coke with 28.6 MJ per kg energy content, energy efficiency 70%                                      | hard coal              | heating systems            | No             | RER      | heat, hard coal coke, at stove 5-15kW                                  | 6.15E-01   | MJ                       | Fisher et al (2006)   | 1    | 1.13               | (1,4,1,3,1,3,1)    |
| Process water                 |                                   |        | Treated rainwater, reused through process, not inventoried.                                                   |                        |                            |                |          |                                                                        | 1.18E+00   | kg                       | Fisher et al (2006)   |      |                    |                    |
| Transport auxiliaries         |                                   |        | Transport of the auxiliary materials and disposed waste (total 0.61 kg) over a distance of 1 km.              | transport systems      | road                       | No             | RER      | transport, lorry >16t, fleet average                                   | 6.13E-01   | tkm                      | Calculations          | 1    | 2.16               | (4,4,1,3,3,5,5)    |
| Infrastructure of facilities. |                                   |        | Approximated with generic module for non-ferrous metal operations with 2'650'000 t output and 50 a life time. | chemicals              | inorganics                 | No             | RER      | sulphuric acid, liquid, at plant                                       | 7.55E-12   | kg                       | Assumptions           | 1    | 3.15               | (4,4,1,3,3,5,9)    |
| Excess slag to landfill       |                                   |        | Proxy for disposal of lead smelter slag.                                                                      | waste management       | residual material landfill | No             | CH       | disposal, nickel smelter slag, 0% water, to residual material landfill | 6.77E-02   | kg                       | Fisher et al (2006)   | 1    | 1.4                | (4,4,1,3,3,5,6)    |

Fig. 6.7 Unit process inventories of the data set “Lead, secondary, at plant (inputs)”.

## 6. System Characterisation

| General Flow Information            |              |                                   | Representation in ecoinvent                                                      |          |                         |                 |          |                           |            | Uncertainty Informations |                       |      |                    |                  |
|-------------------------------------|--------------|-----------------------------------|----------------------------------------------------------------------------------|----------|-------------------------|-----------------|----------|---------------------------|------------|--------------------------|-----------------------|------|--------------------|------------------|
| Input                               | Process Name | Output                            | Remarks                                                                          | Category | Sub-category            | Infra-structure | Location | Modul name in ecoinvent   | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment  |
| lead, secondary, at plant (outputs) | M            | waste heat                        | Calculated from electricity use.                                                 | air      | high population density | No              | RER      | Heat, waste               | 1.95E-01   | MJ                       | Calculations          | 1    | 1.13               | (1,4,1,3,1,3,13) |
|                                     | M            | Flue dust for internal reuse      | not inventoried                                                                  |          |                         |                 |          |                           | 2.09E-02   | kg                       | Fisher et al (2006)   |      |                    |                  |
|                                     | M            | Return slag for internal reuse    | not inventoried                                                                  |          |                         |                 |          |                           | 2.31E-01   | kg                       | Fisher et al (2006)   |      |                    |                  |
|                                     | M            | Sulphuric acid for internal reuse | not inventoried                                                                  |          |                         |                 |          |                           | 1.09E-01   | kg                       | Fisher et al (2006)   |      |                    |                  |
|                                     | M            | SO2                               |                                                                                  | air      | high population density | No              | RER      | Sulfur dioxide            | 1.09E-02   | kg                       | Fisher et al (2006)   | 1    | 1.52               | (1,4,1,3,1,3,31) |
|                                     | M            | CO2 (fuel combustion)             | Already obtained in datasets for gas and coke combustion, hence not inventoried. |          |                         |                 |          |                           | 7.69E-01   | kg                       | Fisher et al (2006)   |      |                    |                  |
|                                     | M            | Pb                                |                                                                                  | air      | high population density |                 |          | Lead                      | 1.95E-06   | kg                       | Fisher et al (2006)   | 1    | 1.52               | (1,4,1,3,1,3,31) |
|                                     | M            | Sb                                |                                                                                  | air      | high population density |                 |          | Antimony                  | 8.62E-09   | kg                       | Fisher et al (2006)   | 1    | 1.52               | (1,4,1,3,1,3,31) |
|                                     | M            | Lead to processor                 |                                                                                  | metals   | extraction              | No              | RER      | lead, secondary, at plant | 1.00E+00   | kg                       | Fisher et al (2006)   | 1    |                    |                  |

Fig. 6.7 (Continued) Unit process inventories of the data set “Lead, secondary, at plant (outputs)”.



## 6.4 Lead, at regional storage in ecoinvent

This dataset describes the supply-mix of lead used in Europe. Following the recycled content approach it consists of a blend of the datasets for primary and secondary production, complemented with the transport service required to transport the material to the regional storage. The meta-information is compiled on Tab. 6.2 at the beginning of this chapter.

As indicated in Section 2.2 *Lead as secondary resource* the primary production of lead has seen a decline in recent years due to more stringent regulations in the use of lead. Since no actual (2007) figures are available, the value of 58% secondary lead in 2001 is taken as starting point. CHR Metals assumed in 2001 the decline in global lead mine production with a small increase thereafter and a respectively higher secondary production that satisfies the gap in supply (Roberts, 2003). The mine production data shows such a small decline in 2002 and a slow increase after that year. On the other hand the reported metal production increased faster, indicating that – under the assumption that no considerable stocks are being consumed, viz. steady state situation – also the secondary production has risen moderately. This reasoning leads to an estimated 65% secondary lead in the global supply mix, compared to 70 – 80% as reported for the U.S. in the years from 2003 – 2006 (USGS, 2006b). A value of 75% is adopted for the European dataset (country-code RER).

Average transport distances have been assumed to depict the European situation (Tab. 6.23).

**Tab. 6.23 Transport of primary and secondary lead to Europe. Assumptions.**

|       |    | <b>Primary lead</b> | <b>Secondary lead</b> |
|-------|----|---------------------|-----------------------|
| Lorry | km | 100                 | 100                   |
| Train | km | 200                 | 200                   |
| Ship  | km | 5000                | 0                     |

## 6. System Characterisation

| General Flow Information |                                                                                                                                      |                           | Representation in ecoinvent                                |                   |              |                 |          |                                      |            | Uncertainty Informations |                       |      |                    |                 |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------|------------------------------------------------------------|-------------------|--------------|-----------------|----------|--------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------|
| Input                    | Process Name                                                                                                                         | Output                    | Remarks                                                    | Category          | Sub-category | Infra-structure | Location | Modul name in ecoinvent              | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment |
| Primary Lead             | <div> <div>lead, at regional storage</div> <div> <div>ML</div> <div>ML</div> <div>ML</div> <div>ML</div> <div>ML</div> </div> </div> |                           | A global share of 25% primary production assumed.          | metals            | extraction   | No              | GLO      | lead, primary, at plant              | 2.50E-01   | kg                       | Roberts (2003)        | 4    |                    | Uniform dist    |
| Secondary Lead           |                                                                                                                                      |                           | A global share of 75% secondary production assumed.        | metals            | extraction   | No              | RER      | lead, secondary, at plant            | 7.50E-01   | kg                       | Roberts (2003)        | 4    |                    | Uniform dist    |
| Transport lorry          |                                                                                                                                      |                           | Assumed distance: 100 km for primary, 100 km for secondary | transport systems | road         | No              | RER      | transport, lorry >16t, fleet average | 1.00E-01   | tkm                      | Assumptions           | 1    | 2.32               | (5,5,1,1,1,5,5) |
| Transport Train          |                                                                                                                                      |                           | Assumed distance: 200 km for primary, 200 km for secondary | transport systems | train        | No              | RER      | transport, freight, rail             | 2.00E-01   | tkm                      | Assumptions           | 1    | 2.32               | (5,5,1,1,1,5,5) |
| Transport Ship           |                                                                                                                                      |                           | Assumed distance: 5000 km for primary, 0 km for secondary  | transport systems | ship         | No              | OCE      | transport, transoceanic freight ship | 1.25E+00   | tkm                      | Assumptions           | 1    | 2.32               | (5,5,1,1,1,5,5) |
|                          |                                                                                                                                      | lead, at regional storage |                                                            | metals            | extraction   | No              | RER      | lead, at regional storage            | 1.00E+00   | kg                       | Assumptions           | 1    |                    |                 |

Fig. 6.8 Unit process inventories of the data set Lead, at regional storage.

## 6.5 Primary Zinc Production Process

Zinc can be produced from primary raw materials by pyrometallurgical or hydrometallurgical methods (Fig. 6.9). Pyrometallurgical methods are used in other parts of the world but have gradually lost their importance and are not used in EU for simple zinc concentrates. Determining factors for this tendency are the need for an extra distillation stage to obtain high-grade zinc and the relatively low zinc extraction efficiency. The pyrometallurgical Imperial Smelting Furnace process (ISF) is however still of importance in EU because it enables complex lead-zinc concentrates and secondary material to be treated simultaneously, yielding saleable lead and zinc. It can also consume residues from other processes (IPPC, 2001).

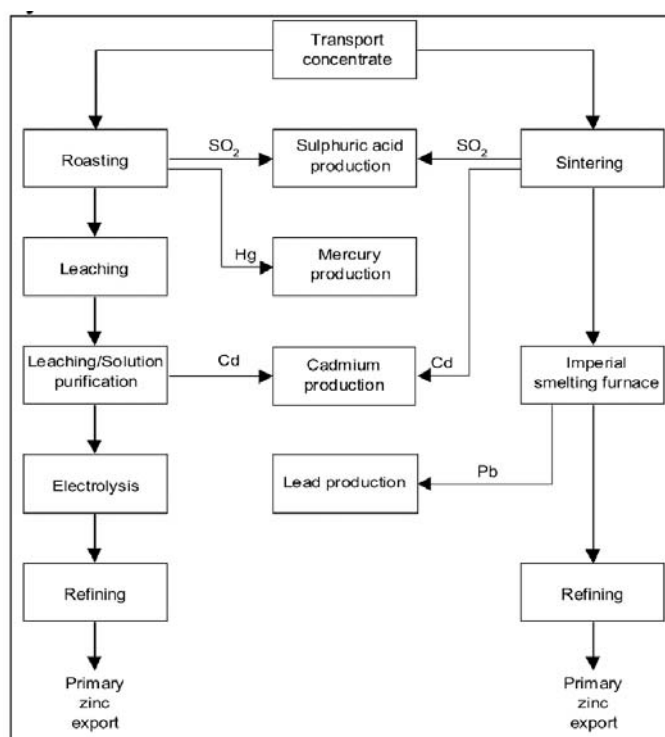
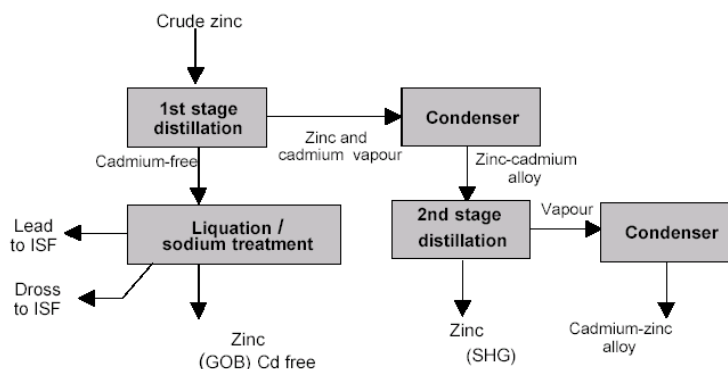


Fig. 6.9 General scheme of zinc production (Dove & Boustead, 1998). The pyrometallurgical and hydrometallurgical production processes account for 20% and 80% of the world production, respectively (IPPC, 2001).

### 6.5.1 Production process and infrastructure

The pyrometallurgical route is used for feed of mixed zinc/lead concentrates and secondary material and uses the Imperial Smelting Furnace (ISF), a special form of a blast furnace. The electrothermic furnace is used in other parts of the world. Zinc produced in the Imperial Smelting Furnace may contain varying amounts of cadmium, lead, copper, arsenic, antimony and iron. The process uses a refining stage. Zinc from the ISF is refined by reflux distillation in columns containing a large number of refractory trays (New Jersey Distillation). The lower ends of the columns are heated externally by natural gas. The upper ends are not heated and run cool enough to reflux the higher boiling point metals before vapours pass to a condenser. The New Jersey distillation column is also used for secondary zinc materials. Distillation proceeds in two stages; first the separation of zinc and cadmium from lead and then separation of cadmium from zinc. In the first stage, molten zinc is fed into a column where all the cadmium and a high proportion of the zinc are distilled. The mixture is condensed and fed directly to a second column. This column is operated at a slightly lower temperature to distil mainly cadmium, which is condensed as a zinc-cadmium alloy. The alloy is transferred to a cadmium refinery. The

metal run-off from the bottom of the second column is high-grade zinc (SHG) of 99.995% purity. The run-off metal from the first stage is zinc with lead, tin, arsenic, iron, antimony and copper impurities. This alloy is cooled to separate lead, which is recycled to the ISF splash condenser and an inter-metallic compound of iron, zinc and arsenic, which is recycled to the ISF itself. The zinc is then treated with sodium to remove residual arsenic and antimony as sodium arsenides and antimonides, respectively, which are also recycled to the ISF. The zinc produced in this way is of a lower grade (GOB), but free of cadmium, and is used mainly for galvanising (IPPC, 2001).



**Fig. 6.10** Process scheme, part of the pyrometallurgical process route (rectification). The residues of the first stage distillation are processed to GOB Zinc (lowest quality, “Good Ordinary Brand”), which is used for galvanising, whereas the SHG Zinc (highest quality, “Special High Grade”) is won by a second distillation (IPPC, 2001)

The hydrometallurgical route is used for zinc sulphide, oxide, carbonate or silicate concentrates, making about 80% of the total world output. The majority of the EU production facilities use the hydrometallurgical process, with a total production capacity of 1'665'000 t/a in 1997. Sulphide concentrates are roasted first in fluidised bed roasters to produce zinc oxide and sulphur dioxide. Roasting is an exothermic process and no additional fuel is used to sustain the reaction, the heat generated is recovered. The zinc oxide (calcine) passes from the furnace and is collected and cooled. Roaster gases are treated in hot electric precipitators to remove dust (which is passed to the calcine). Other dust and volatile metals such as Hg and Se are removed in a gas cleaning train that incorporates scrubbing systems and wet EPs. The sulphur dioxide is then converted to sulphuric acid in a conventional recovery system. Leaching of the calcine is carried out in a number of successive stages using a gradually increasing strength of hot sulphuric acid. The initial stages do not dissolve significant amounts of iron but the later ones do. The leaching process is carried out in a variety of reactors using open tanks. Leaching may be stopped after the neutral leach. The leach residue is sent to an ISF and added to the sinter feed. Zinc, lead and silver are recovered as metals, sulphur as H<sub>2</sub>SO<sub>4</sub>. Instead of an ISF, a Waelz Kiln may be used but SO<sub>2</sub> absorption is necessary in such a case. Other metals are also dissolved during the process and are removed after leaching. Iron is the major impurity, which is precipitated in three basic forms; Jarosite, Goethite or Haemetite (IPPC, 2001). Zinc produced hydrometallurgically does not need to be refined (Graf & Schleining, 1997).

### 6.5.2 Waste

The production of metals is related to the generation of several by-products, residues and wastes, which are also listed in the European Waste Catalogue (Council Decision 94/3/EEC). The electrolytic production of zinc is one of the main sources of solid waste in the non-ferrous industry. Relatively

large quantities of iron based solids are generated by the leaching process. Jarosite and goethite are classified as hazardous waste because of the content of leachable elements such as Cd, Pb and As. The leaching and purification processes and electrolysis of zinc and the refining stages of lead also generate other metal rich solids. These are usually rich in a specific metal and are recycled to the appropriate production process. The ISF or direct smelting furnaces also are significant sources of solid slag. This slag has been subjected to high temperatures and generally contains low levels of leachable metals, consequently then may be used in construction. Solid residues also arise as the result of the treatment of liquid effluents. The main waste stream is gypsum waste ( $\text{CaSO}_4$ ) and metal hydroxides that are produced at the wastewater neutralisation plant. Some of these wastes are recycled to pyrometallurgical process to recover the metals. Dust or sludge from the treatment of gases are used as raw materials for the production of other metals such as Ge, Ga, In and As etc or can be returned to a smelter or into the leach circuit for the recovery of lead and zinc. Hg/Se residues arise at the pre-treatment of mercury or selenium streams from the gas cleaning stage. This solid waste stream amounts to approximately 40 - 120 t/y in a typical plant. Hg and Se can be recovered from these residues depending on the market demand for these metals (IPPC, 2001).

### 6.5.3 Emission

The emissions to air can escape the process either as stack emissions or as fugitive emissions depending on the age of the plant and the technology used. Stack emissions are normally monitored continuously or periodically and reported. The main emissions to air from zinc production are sulphur dioxide ( $\text{SO}_2$ ), other sulphur compounds and acid mists; oxides of nitrogen ( $\text{NO}_x$ ) and other nitrogen compounds; metals and their compounds; dust; VOC and dioxins (IPPC, 2001).

Main emissions to water are metals and their compounds and materials in suspension are the main pollutants emitted to water. The metals concerned are Zn, Cd, Pb, Hg, Se, Cu, Ni, As, Co and Cr. Other significant substances are fluorides, chlorides and sulphates (IPPC, 2001).

### 6.5.4 Smelting, primary zinc production in ecoinvent

No contiguous study on the zinc production is yet available, therefore this module is based in parts on values reported in Ayres et al. (2002), Xiao et al. (2003), IPPC (2001) and EEA (1999b).

The multi-output-process "smelting, primary zinc production" delivers the three co-products "zinc, primary, at regional storage", "cadmium sludge, from zinc electrolysis, at plant" and "leaching residues, indium rich, from zinc circuit, at smelter". The flow "zinc, ..." represents the European supply mix of primary zinc. The by-products "leaching residues, ..." and "zinc, ..." receive part of the burden and enter the primary production chain.

The module describes the production of primary SHG zinc used for galvanising in Europe, it does not consider secondary production, which accounts in Europe for 30% of the metal consumed (IPPC, 2001). The dataset is designed for the use of zinc in coating processes, and is not suitable for comparisons of products containing zinc in a high degree.

The meta-information to the multi-output process "smelting, primary zinc production" is given in Tab. 6.3. The layout of the modules with general flow information, remarks, sources, values and uncertainty information are shown in Fig. 6.12 (inputs) and Fig. 6.13 (outputs). The following paragraphs describe calculations, sources and assumptions chosen in this study. The functional unit of this process is one kg of zinc. The meta-information is compiled above in Tab. 6.3 in the beginning of this chapter.

**Consumption mix:** The consumption mix – i.e. the pattern of the production processes producing the zinc actually used for galvanising – is approximated with the production mix world-wide of 80% hydrometallurgical and 20% pyrometallurgical route. Low quality zinc (GOB) is mainly produced via the pyrometallurgical route without a final refining step (Graf & Schleining, 1997). According to IPPC (2001) 75% of the concentrates used in European production are imported, for which a distance of

5000 km transport by ship is assumed. Since Europe is the world leader in zinc production, no imports of refined zinc is assumed.

**Demand in materials and energy:** The input of zinc concentrate was taken from Ayres et al. (2002) to be in line with the material flow. Most other values are taken from Xiao et al. (2003), which inventoried one hydro- and one pyrometallurgical smelter in China. Electricity values had to be taken from IPPC (2001) since the detailed values presented in Xiao et al. (2003) could not be reproduced. Nevertheless they are shown together with the actually chosen values and the resulting mix in Tab. 6.24. Also the coal gas consumption reported by Xiao et al. (2003) seems to be too high compared to the values for natural gas reported by IPPC (2001). Since no information on the composition and heating value of the coal gas was available, the numbers for natural gas given by IPPC were taken in this study. There are indications that the total energy consumption indicated by IPPC (2001) is 27% and 31% of the values for the Hydro and the Pyro process reported in Xiao et al. (2003). However, since Xiao et al. (2003) does not characterise energy carrier and end energy uses sufficiently his results remain ambiguous and are not taken. No quantitative compilation of electricity production is available, therefore for pyrometallurgical processes the UCTE production mix was chosen as approximation (year 2000: nuclear 15%; hydro 4%; fossil 81%<sup>12</sup>). Since the electrolysis plants depend on cheap and readily available electricity, they often are located near hydropower dams often owned by the metallurgical enterprise. A compilation of electricity sources for the aluminium electrolysis is given in the part “Aluminium” of this report, where a specific dataset for the electricity production for the aluminium electrolysis is established. This dataset is chosen to approximate the zinc electrolysis.

**Emissions and disposal.** Wastewater is reflected by the direct emissions and the volume reported in Xiao et al. (2003) and IPPC (2001). An additional process covers solely the treatment process itself – infrastructure, mixing and pumping, without accounting for specific emissions from wastewater – as if the treated wastewater was unpolluted (Doka (2003)). Emissions are summed up in Tab. 6.25 (particulate emission to air) and Tab. 6.26 (other emissions to air and to water). The dioxin emissions are approximated with the reported emission factor of 50 µg-TEQ / t secondary zinc production in EC (2002).

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<sup>12</sup> <http://www.ucte.org/>

**Tab. 6.24 Zinc production: input and output according to different sources. The values chosen in this study reflect the production split of 80% hydro- and 20% pyrometallurgical production.**

| Type                                                                                                                                                                                                                                                                                                                                            |                | Hydro                 | Zinc Elec-<br>trolysis<br>(Hydro) | Pyro (ISP)            | ISP         | This study, split<br>80 hydro : 20 py-<br>ro |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------------------|-----------------------------------|-----------------------|-------------|----------------------------------------------|
| Source                                                                                                                                                                                                                                                                                                                                          |                | Xiao et al.<br>(2003) | IPPC (2001)                       | Xiao et al.<br>(2003) | IPPC (2001) |                                              |
|                                                                                                                                                                                                                                                                                                                                                 | unit           | per t Zn              | per t Zn                          | per t Zn              | per t Zn    | per t Zn                                     |
| concentrate                                                                                                                                                                                                                                                                                                                                     | kg             |                       |                                   |                       |             | 1855 <sup>1)</sup>                           |
| Water                                                                                                                                                                                                                                                                                                                                           | t              | 16.2                  |                                   | 130                   |             | 39.0                                         |
| Diesel oil                                                                                                                                                                                                                                                                                                                                      | t              | 6.0E-4                |                                   | 2.5E-3                |             | 6.4E-4                                       |
| steam                                                                                                                                                                                                                                                                                                                                           | t              | 1.24                  |                                   | 0.53                  |             | 1.09                                         |
| Oxygen                                                                                                                                                                                                                                                                                                                                          | t              |                       |                                   | 0.54                  |             | 0.108                                        |
| Jarosite                                                                                                                                                                                                                                                                                                                                        | t              |                       | 0.350                             |                       |             | 0.280                                        |
| Solid wastes                                                                                                                                                                                                                                                                                                                                    | t              | 0.323                 |                                   | 0.582                 |             | 0.375 <sup>3)</sup>                          |
| Total coal                                                                                                                                                                                                                                                                                                                                      | t              | 0.801                 |                                   | 2.82                  | 1.1         | 0.22 <sup>2)</sup>                           |
| Coal gas                                                                                                                                                                                                                                                                                                                                        | m <sup>3</sup> | 769                   |                                   | 3247                  |             | Not defined                                  |
| Natural gas                                                                                                                                                                                                                                                                                                                                     | m <sup>3</sup> |                       |                                   |                       | 160         | 32.0 <sup>2)</sup>                           |
| total fuel<br>(coal/gas)                                                                                                                                                                                                                                                                                                                        | MJ             | 40274                 |                                   | 148309                |             |                                              |
| Electricity                                                                                                                                                                                                                                                                                                                                     | kWh            | 3835                  | 4100                              | 918                   | 1050        | 3490 <sup>2)</sup>                           |
| total energy                                                                                                                                                                                                                                                                                                                                    | MJ             | 54080                 |                                   | 151614                |             |                                              |
| <sup>1)</sup> Corresponds to the chosen content (55%) and the process yield of 98%.<br><sup>2)</sup> Energy values given in Xiao et al. (2003) could not be reproduced on the basis of the publication; the values reported from IPPC (2001) were taken instead.<br><sup>3)</sup> Including indium and cadmium containing residues and sludges. |                |                       |                                   |                       |             |                                              |

**Tab. 6.25 Zinc: solid particulates released to air. Values taken from CEPMEIP-Database <sup>Fehler! Textmarke nicht definiert.</sup>. Values might be overestimated – TSP through fuels burned in the smelting process are included in this figures, but are also accounted for in the respective modules for fuel burning.**

| Solid Particle Emission                       | Unit     | Medium / high emission<br>(pyrometallurgy, 80%) | Low emission<br>(hydrometallurgy, 20%) |
|-----------------------------------------------|----------|-------------------------------------------------|----------------------------------------|
| TSP                                           | g / t Zn | 500                                             | 200                                    |
| PM10                                          | g / t Zn | 400                                             | 180                                    |
| PM2.5                                         | g / t Zn | 300                                             | 160                                    |
| <b>Differential classes used in ecoinvent</b> |          |                                                 |                                        |
| PM<2.5                                        | g / t Zn | 300                                             | 160                                    |
| PM2.5-10                                      | g / t Zn | 100                                             | 20                                     |
| PM>10                                         | g / t Zn | 100                                             | 20                                     |

**Tab. 6.26 Zinc: Emissions to water and air, except particulates & emissions from fuel combustion. Values correspond to the production of 1 t zinc with hydro- and pyrometallurgical processes. The values chosen in this study reflect the production split of 80% hydro- and 20% pyrometallurgical production. Some values are not reported by Xiao et al. (2003); for calculations they are approximated with zero.**

|                          | Unit | Hydro              | Electrolysis            | Pyro (ISP)         | ISP        | This study, split 80 hydro : 20 pyro |
|--------------------------|------|--------------------|-------------------------|--------------------|------------|--------------------------------------|
| Source                   |      | Xiao et al. (2003) | EEA (2003), EEA (1999b) | Xiao et al. (2003) | EEA (2003) |                                      |
| <b>Emission to air</b>   |      |                    |                         |                    |            |                                      |
| Zn                       | kg   |                    | 0.1                     |                    | 10.0       | <b>2.080</b>                         |
| SO <sub>2</sub>          | kg   | 19.1               |                         | 11.0               |            | <b>17.5</b>                          |
| Pb                       | kg   | 0.097              | 0.005                   |                    | 0.5        | <b>0.104</b>                         |
| As                       | kg   | 1.58E-2            |                         |                    |            | <b>1.26E-2</b>                       |
| Hg                       | kg   |                    |                         | 3.47E-2            | 2.00E-2    | <b>4.0E-3</b>                        |
| TCDD                     | kg   |                    |                         |                    |            | <b>5E-11</b>                         |
| <b>Emission to water</b> |      |                    |                         |                    |            |                                      |
| COD                      | kg   | 0.583              |                         | Not reported       |            | <b>0.466</b>                         |
| BSB5                     | kg   | 0.389              |                         | Not reported       |            | <b>0.311</b>                         |
| TOC                      | kg   | 0.228              |                         | Not reported       |            | <b>0.182</b>                         |
| DOC                      | kg   | 0.228              |                         | Not reported       |            | <b>0.182</b>                         |
| Hg                       | kg   | 4.30E-5            |                         | 6.90E-4            |            | <b>1.72E-4</b>                       |
| Cd                       | kg   | 2.60E-3            |                         | 8.00E-3            |            | <b>3.68E-3</b>                       |
| As                       | kg   | 1.00E-3            |                         | 1.90E-3            |            | <b>1.18E-3</b>                       |
| Pb                       | kg   | 1.80E-2            |                         | 1.38E-1            |            | <b>4.20E-2</b>                       |
| Cu                       | kg   | 5.00E-3            |                         |                    |            | <b>4.00E-3</b>                       |
| Zn                       | kg   | Not reported       |                         | 2.29E-1            |            | <b>4.58E-2</b>                       |
| Fluoride                 | kg   |                    |                         | 1.49E-1            |            | <b>2.98E-2</b>                       |

### 6.5.5 By-products and allocation

Of the many by-products that are jointly extracted with the zinc, indium and cadmium are modelled within this inventory. For these metals the zinc extraction is the dominant source for raw-material. In order to attribute a share of the environmental burden from the joint extraction, an allocation factor for the three products zinc, indium and cadmium has to be determined. In accordance with Althaus & Classen (2005) an allocation procedure by revenue is applied.

However, the allocation of the subprocesses indium-, cadmium and zinc-production is not evident to solve, since only the final commodity prices are known but no in-firm prices for the metal containing residues and for the raw zinc, on which the allocation would rely. Also no share of the products in total costs – gross sales method as proposed in Guinée et al. (2004) and Guinée et al. (2001) – is easy to estimate. The by-product allocation problem – the joint production of cadmium, indium and zinc – is reduced here by a subdivision of the sub-processes (according to Ekvall & Finnveden (2001)). The indium production from the leaching residues and the cadmium production from sludge respectively is treated in a distinct inventory. Hence it is the task to allocate the burdens between the zinc, the sludge and the leaching residues.

As a starting point for an estimation it is assumed, that the profit margin for the production of silver corresponds to the company's performance. Taking a performance value of 10%, the value of the indium containing residues corresponds consequently to 10% of the proceeds from the indium sales. Therefore 10 % of the indium price is taken to estimate the value of the residues. Since Cadmium is to



be regarded as contaminant that has to be isolated from the main product anyway, the cadmium containing sludge is regarded as burden free (cut-off).

### **By-product Cadmium in flue dust and purification precipitate**

There are mainly two different mechanisms how cadmium accumulates during the zinc production process. During roasting of the concentrate in the pyro-route the cadmium vaporises and accumulates in the flue dust that is collected in scrubbers or ESP. The second source is the cadmium sponge that is precipitated in the course of zinc solution purification prior to electrowinning. In both cases cadmium arises as unwanted element that has to be separated from the main product (hydro) and has to be retained for controlled handling (pyro) due to its toxicity. Due to the marginal position of the pyrometallurgical route, here only the cadmium sludge from zinc solution purification is modelled in this multi-output module.

With worldwide roughly 20'000 tons refined cadmium per year in 2006 (Tab. 3.1), the assumption that 72% of the cadmium refinery output originates from zinc extraction (see Reserves and Resources of Materials) would mean 13'500 tonnes that are direct by-products from the 10'000'000 tonnes primary zinc produced. In terms of mass this means that for every tonne of zinc extracted 1.5 kg cadmium is refined. This results in a back-calculated value of 0.29% Cd in a concentrate of 53% Zn content.

According to Fthenakis (2004) the cadmium content of the zinc concentrate is around 0.3% to 0.5%. This value is in the same order as the concentrate in the raw ore and implies that only in 20% of the co-extracted zinc is processed to ingots. The remaining 80% is supposed to be discarded, e.g. due to low concentration or missing markets.

The resulting cadmium sludge from zinc solution purification is assumed to consist of 50% water, 45% cemented cadmium and the rest of zinc dust. It is further processed in a cadmium plant, where the sludge is dewatered, oxidised and refined by means of electrowinning (see 6.6 Cadmium Production process and infrastructure).

Despite of Cadmium being a commodity traded at international exchanges, it has to be regarded as contaminant at the stage where it arises in the zinc metallurgy. According to the common allocation practice therefore no upstream burdens are to be attributed to the by-product cadmium (Guinée et al., 2004).

### **By-product Indium in purification solutions**

Indium is produced by a few companies only, all of which are in the zinc production. In those deposit where indium is present the concentration ranges between 10 and 100 ppm with 50 ppm being a common value of those operations recovering indium (e.g. Brunswick).

It is very uncertain to which amount secondary indium is recovered, because until 2002 it was not a technological issue due to favourable indium prices (Jansseune, 2003). The sky-rocketing prices and the connected ramping up of production facilities will have changed this picture by 2007 but no assured estimate is available in the open literature. Therefore, an amount of 30% secondary production is arbitrarily chosen to assess the coupling of indium production to the zinc extraction.

The trend for increasing indium recovery within the zinc industry is displayed in Fig. 6.11. Based on the last years, a ratio of 45 g refined Indium per tonne extracted zinc may be established assuming steady state stocks and optimal recovery measures. Under the assumption of correspondingly 70% indium from primary production, 3.4E-05 kg indium per kg zinc was chosen as joint production factor.

The indium containing intermediate product is a solution from wet-chemical purification processes. In the hydrometallurgical zinc recovery process at Akita mine (Dowa Mining Co, Ltd.), gallium and indium are recovered by three stages of ion exchange columns. The feed solution to the third ion exchange column contains the relative concentrations Ga: 0.26, In: 0.11, Zn: 1.0 g / l (Teramoto et al., 1993). A value of 0.1 g / l is chosen for further calculations.

Given an extraction yield of 60% and under the assumption, the indium contained in the residue amounts to 10% of the sales proceeds, a share of 0.09% of the burden is allocated to the indium containing by-product.

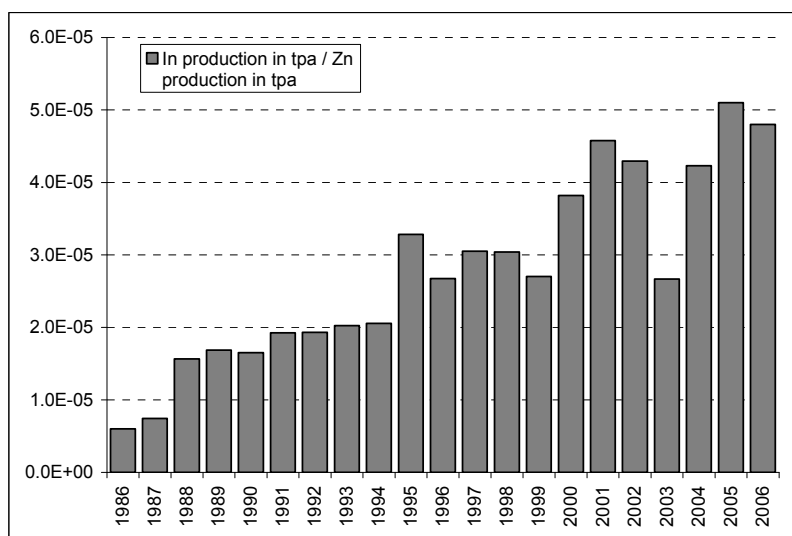


Fig. 6.11 Quotient of worldwide indium refinery production and zinc mine production as base for the assessment of the multi-output relation in ecoinvent USGS, 2006b). For explanations see in text above.

Tab. 6.27 By-products of primary zinc production: Composition, flows and value to derive the value-based allocation factors

| Metal                   | Process Yield | zinc concentrate (input) | cadmium sludge (output) | leaching residues, In (output) | Zinc, primary (output) | Value   |
|-------------------------|---------------|--------------------------|-------------------------|--------------------------------|------------------------|---------|
|                         | %             | Content                  | Content                 | Content                        | Content                | \$ / kg |
| Zn                      | 97 %          | 53%                      |                         |                                | 100%                   | 1.88    |
| Cd                      | 99 %          |                          | 45%                     |                                |                        | -       |
| In                      | 60 %          |                          |                         | 0.01%                          |                        | 77.5    |
| Amount in kg            |               | 1.91                     | 0.00338                 | 0.34                           | 1.00                   |         |
| Value contained (in \$) |               |                          | -                       | 0.003                          | 1.88                   |         |
| Allocation by Value     |               |                          | -                       | 0.14%                          | 99.86%                 |         |

### Correction of resource flow allocation

Even though overall allocation is performed according to economic revenue, the resources are allocated by mass to the respective by-product to balance the mass flow. Since the concentrate required by this multi-output-process is allocated also by revenue, the therein contained resources are not correctly attributed and have to be corrected with a corrective exchange flow (Tab. 6.28).

**Tab. 6.28 Individual correction factors to allocate the resource depletion to the respective by-products in order to balance the mass flow.**

| correction from economic to per mass |    |          | according to economic allocation |             |           |
|--------------------------------------|----|----------|----------------------------------|-------------|-----------|
| Resources attributed for in the feed |    |          | 0%                               | 0.14%       | 99.86%    |
| Concentrate                          | kg | 1.91     | Cd-sludge                        | In-leachate | Zinc      |
| Zn                                   | kg | 1.19     | -                                | 0.00166     | 1.19      |
| Cd                                   | kg | 0.0396   | -                                | 5.55E-05    | 0.0395    |
| In                                   | kg | 6.60E-04 | -                                | 9.24E-07    | 6.59E-04  |
| Resources attributed in by-product   |    |          | according to per mass allocation |             |           |
| Zn                                   |    |          |                                  |             | 1.19      |
| Cd                                   |    |          | 0.0396                           |             |           |
| In                                   |    |          |                                  | 6.60E-04    |           |
| Resource exchange flow               |    |          | difference to be corrected       |             |           |
| Zn                                   |    |          | -                                | -0.00166    | 1.66E-03  |
| Cd                                   |    |          | 0.0396                           | -5.55E-05   | -3.95E-02 |
| In                                   |    |          | -                                | 6.59E-04    | -6.59E-04 |

### 6.5.6 Data quality

Data uncertainty is derived using the Pedigree matrix as described in Frischknecht et al. (2004). Some of the less important emissions that are described in the literature as typical for lead, could not be covered in this inventory due to missing emission factors, viz. Ge, Ga In and Se into air and Se, Ni, Co and Cr to water. Also some salts –chlorides and sulphates – are not included due to missing data. Further investigations on these issues have been done to improve the quality of this dataset. However, for the purpose of a background database the quality is sufficient.

The inventory does not reflect the portion of secondary material in the zinc used. For specific conclusions further investigation has to be done. Further more, the choice of electricity mix bases on approximations that have to be improved with further investigations.

## 6. System Characterisation

| General Flow Information    |                                            |        | Representation in ecoinvent                                                                                                               |                        |                  |                 |          |                                                                 |            | Uncertainty Informations |                                    |      |                    |                 |
|-----------------------------|--------------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------------------|-----------------|----------|-----------------------------------------------------------------|------------|--------------------------|------------------------------------|------|--------------------|-----------------|
| Input                       | Process Name                               | Output | Remarks                                                                                                                                   | Category               | Sub-category     | Infra-structure | Location | Modul name in ecoinvent                                         | Mean value | Unit                     | Source for mean value              | Type | Standard Deviation | General Comment |
| Concentrate                 | smelting, primary zinc production (inputs) |        | Concentrate contains 53% zinc (1.01 kg Zn, yield = 99%), 0.33% cadmium (1.74 g Cd, yield = 99%) and 0.01% indium (0.06 g In, yield = 60%) | metals                 | extraction       | No              | GLO      | zinc concentrate, at beneficiation                              | 1.91E+00   | kg                       | Calculation                        | 1    | 1.33               | (3,3,2,1,3,5,4) |
| Correction Cd in Conc -> In |                                            |        | Cd resource in concentrate fully allocated to the Cd bearing by-product. Correction Indium.                                               | resource               | in ground        |                 |          | Cadmium, 0.30% in sulfide, Cd 0.18%, Pb, Zn, Ag, In, in ground  | -5.55E-05  | kg                       | Calculation from concentrate input | 1    | 1.33               |                 |
| Correction Cd in Conc -> Zn |                                            |        | Cd resource in concentrate fully allocated to the Cd bearing by-product. Correction Zinc                                                  | resource               | in ground        |                 |          | Cadmium, 0.30% in sulfide, Cd 0.18%, Pb, Zn, Ag, In, in ground  | -3.95E-02  | kg                       | Calculation from concentrate input | 1    | 1.33               |                 |
| Correction Zn in Conc -> Cd |                                            |        | Zn resource in concentrate fully allocated to Zinc. Correction Cadmium.                                                                   | resource               | in ground        |                 |          | Zinc, 9.0% in sulfide, Zn 5.3%, Pb, Ag, Cd, In, in ground       | -          | kg                       | Calculation from concentrate input | 1    | 1.33               |                 |
| Correction Zn in Conc -> In |                                            |        | Zn resource in concentrate fully allocated to Zinc. Correction Indium                                                                     | resource               | in ground        |                 |          | Zinc, 9.0% in sulfide, Zn 5.3%, Pb, Ag, Cd, In, in ground       | -1.66E-03  | kg                       | Calculation from concentrate input | 1    | 1.33               |                 |
| Correction Zn in Conc -> Zn |                                            |        | Zn resource in concentrate fully allocated to Zinc. Correction Zinc.                                                                      | resource               | in ground        |                 |          | Zinc, 9.0% in sulfide, Zn 5.3%, Pb, Ag, Cd, In, in ground       | 1.66E-03   | kg                       | Calculation from concentrate input | 1    | 1.33               |                 |
| Correction In in Conc -> Cd |                                            |        | In resource in concentrate fully allocated to Indium. Correction Cadmium.                                                                 | resource               | in ground        |                 |          | Indium, 0.005% in sulfide, In 0.003%, Pb, Zn, Ag, Cd, in ground | -          | kg                       | Calculation from concentrate input | 1    | 1.33               |                 |
| Diesel and Gasoil           |                                            |        | inventoried as diesel in building machine                                                                                                 | construction processes | machinery        | No              | GLO      | diesel, burned in building machine                              | 4.19E-02   | MJ                       | (Xiao, 2003)                       | 1    | 1.25               | (3,3,2,1,3,3,2) |
| steam                       |                                            |        |                                                                                                                                           | washing agents         | auxiliary agents | No              | RER      | steam, for chemical processes, at plant                         | 1.09E+00   | kg                       | (Xiao, 2003)                       | 1    | 1.25               | (3,3,2,1,3,3,1) |
| Total coal                  |                                            |        |                                                                                                                                           | hard coal              | heating systems  | No              | RER      | hard coal, burned in industrial furnace 1-10MW                  | 6.93E+00   | MJ                       | (IPPC, 2002)                       | 1    | 1.25               | (3,3,2,1,3,3,1) |
| Electricity Pyrometallurgy  |                                            |        | UCTE mix: nuclear 15%; hydro 4%; fossil 81%                                                                                               | electricity            | production mix   | No              | UCTE     | electricity, medium voltage, production UCTE, at grid           | 2.10E-01   | kWh                      | (IPPC, 2002)                       | 1    | 2.07               | (3,3,2,1,5,5,2) |
| Electricity Electrolysis    |                                            |        | Approximated with aluminium electrolysis                                                                                                  | electricity            | supply mix       | No              | GLO      | electricity, medium voltage, aluminium industry, at grid        | 3.28E+00   | kWh                      | (IPPC, 2002)                       | 1    | 2.07               | (3,3,2,1,5,5,2) |
| Natural Gas                 |                                            |        |                                                                                                                                           | natural gas            | heating systems  | No              | RER      | natural gas, burned in industrial furnace >100kW                | 1.60E+00   | MJ                       | (IPPC, 2002)                       | 1    | 1.25               | (3,3,2,1,3,3,1) |
| Water                       |                                            |        |                                                                                                                                           | resource               | in water         |                 |          | Water, river                                                    | 3.90E-02   | m3                       | (Xiao, 2003)                       | 1    | 1.25               | (3,3,2,1,3,3,2) |
| Oxygen                      |                                            |        |                                                                                                                                           | chemicals              | inorganics       | No              | RER      | oxygen, liquid, at plant                                        | 1.08E-01   | kg                       | (Xiao, 2003)                       | 1    | 1.25               | (3,3,2,1,3,3,4) |
| Transport by train          |                                            |        | Assumption - 600km : concentrate, coal, zinc; 100km : jarosite, oxygen                                                                    | transport systems      | train            | No              | RER      | transport, freight, rail                                        | 1.88E+00   | tkm                      | standard distances                 | 1    | 1.31               | (4,3,1,1,3,3,2) |
| Transport by lorry (32t)    |                                            |        | Assumption - 100km : concentrate, coal, zinc; 50km : jarosite, oxygen, solid waste                                                        | transport systems      | road             | No              | RER      | transport, lorry >16t, fleet average                            | 3.18E-01   | tkm                      | standard distances                 | 1    | 1.31               | (4,3,1,1,3,3,2) |

Fig. 6.12 Flows for "Zinc for coating, at regional storage" (inputs) and its representation in the ecoinvent

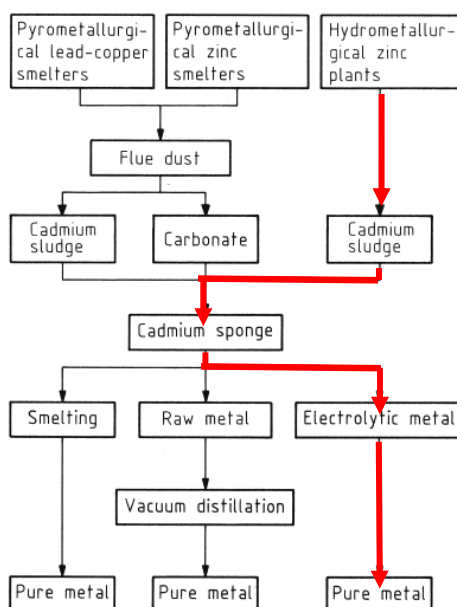
## 6. System Characterisation

| General Flow Information                    |              |                             | Representation in ecoinvent                                                               |                  |                         |                 |          |                                                                 |            | Uncertainty Informations |                           |      |                    |                  |
|---------------------------------------------|--------------|-----------------------------|-------------------------------------------------------------------------------------------|------------------|-------------------------|-----------------|----------|-----------------------------------------------------------------|------------|--------------------------|---------------------------|------|--------------------|------------------|
| Input                                       | Process Name | Output                      | Remarks                                                                                   | Category         | Sub-category            | Infra-structure | Location | Modul name in ecoinvent                                         | Mean value | Unit                     | Source for mean value     | Type | Standard Deviation | General Comment  |
| smelting, primary zinc production (outputs) | m            | Solid wastes                | Portion of the waste fraction that is not processed for byproduct metals but disposed of. | waste management | inert material landfill | No              | CH       | disposal, inert waste, 5% water, to inert material landfill     | 3.20E-02   | kg                       | (Xiao, 2003)              | 1    | 1.25               | (3,3,2,1,3,3,6)  |
|                                             |              | Effluents                   |                                                                                           | waste management | wastewater treatment    | No              | CH       | treatment, sewage, unpolluted, to wastewater treatment, class 3 | 3.90E-02   | m3                       |                           | 1    | 1.25               | (3,3,2,1,3,3,6)  |
|                                             |              | Cu (water)                  |                                                                                           | water            | river                   |                 |          | Copper, ion                                                     | 4.00E-06   | kg                       | (Xiao, 2003)              | 1    | 5.02               | (1,3,1,3,1,4,35) |
|                                             |              | Zn (water)                  |                                                                                           | water            | river                   |                 |          | Zinc, ion                                                       | 4.58E-05   | kg                       | (Xiao, 2003)              | 1    | 5.02               | (1,3,1,3,1,4,35) |
|                                             |              | Pb (water)                  |                                                                                           | water            | river                   |                 |          | Lead                                                            | 4.20E-05   | kg                       | (Xiao, 2003)              | 1    | 5.02               | (1,3,1,3,1,4,35) |
|                                             |              | As (water)                  |                                                                                           | water            | river                   |                 |          | Arsenic, ion                                                    | 1.18E-06   | kg                       | (Xiao, 2003)              | 1    | 5.02               | (1,3,1,3,1,4,35) |
|                                             |              | Hg (water)                  |                                                                                           | water            | river                   |                 |          | Mercury                                                         | 1.72E-07   | kg                       | (Xiao, 2003)              | 1    | 5.02               | (1,3,1,3,1,4,35) |
|                                             |              | Cd (water)                  |                                                                                           | water            | river                   |                 |          | Cadmium, ion                                                    | 3.68E-06   | kg                       | (Xiao, 2003)              | 1    | 5.02               | (1,3,1,3,1,4,35) |
|                                             |              | Fluoride (water)            |                                                                                           | water            | river                   |                 |          | Fluoride                                                        | 2.98E-05   | kg                       | (Xiao, 2003)              | 1    | 1.58               | (1,3,1,3,3,4,32) |
|                                             |              | COD (water)                 |                                                                                           | water            | river                   |                 |          | COD, Chemical Oxygen Demand                                     | 4.66E-04   | kg                       | (Xiao, 2003)              | 1    | 1.6                | (3,3,1,3,3,4,32) |
|                                             |              | BSB                         |                                                                                           | water            | river                   |                 |          | BOD5, Biological Oxygen Demand                                  | 3.11E-04   | kg                       | (Xiao, 2003)              | 1    | 1.65               | (3,5,1,3,3,4,32) |
|                                             |              | TOC                         |                                                                                           | water            | river                   |                 |          | TOC, Total Organic Carbon                                       | 1.82E-04   | kg                       | (Xiao, 2003)              | 1    | 1.65               | (3,5,1,3,3,4,32) |
|                                             |              | DOC                         |                                                                                           | water            | river                   |                 |          | DOC, Dissolved Organic Carbon                                   | 1.82E-04   | kg                       | (Xiao, 2003)              | 1    | 1.65               | (3,5,1,3,3,4,32) |
|                                             |              | Pb (air)                    |                                                                                           | air              | low population density  |                 |          | Lead                                                            | 1.04E-04   | kg                       | (Xiao, 2003)              | 1    | 5.04               | (3,4,2,1,1,4,22) |
|                                             |              | Zn (air)                    |                                                                                           | air              | low population density  |                 |          | Zinc                                                            | 2.08E-03   | kg                       | (Xiao, 2003)              | 1    | 5.04               | (3,4,2,1,1,4,22) |
|                                             |              | Hg (air)                    |                                                                                           | air              | low population density  |                 |          | Mercury                                                         | 2.29E-05   | kg                       | (Xiao, 2003)              | 1    | 5.04               | (3,4,2,1,1,4,22) |
|                                             |              | As (air)                    |                                                                                           | air              | low population density  |                 |          | Arsenic                                                         | 1.26E-05   | kg                       | (Xiao, 2003)              | 1    | 5.04               | (3,4,2,1,1,4,22) |
|                                             |              | PM<2.5                      |                                                                                           | air              | low population density  |                 |          | Particulates, < 2.5 um                                          | 1.88E-04   | kg                       | CEPMEIP                   | 1    | 3.04               | (3,4,2,1,1,4,27) |
|                                             |              | PM2.5-10                    |                                                                                           | air              | low population density  |                 |          | Particulates, > 2.5 um, and < 10um                              | 3.60E-05   | kg                       | CEPMEIP                   | 1    | 1.28               | (3,4,2,1,1,4,28) |
|                                             |              | PM>10                       |                                                                                           | air              | low population density  |                 |          | Particulates, > 10 um                                           | 3.60E-05   | kg                       | CEPMEIP                   | 1    | 1.46               | (3,4,2,1,1,4,29) |
|                                             |              | TCDD                        |                                                                                           | air              | low population density  |                 |          | Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin        | 5.00E-11   | kg                       | European Dioxin Inventory | 1    | 10                 | reported         |
|                                             |              | SO2 (air)                   |                                                                                           | air              | low population density  |                 |          | Sulfur dioxide                                                  | 1.75E-02   | kg                       | (Xiao, 2003)              | 1    | 1.19               | (3,4,2,1,1,4,15) |
|                                             |              | waste heat                  | Calculated from electricity use.                                                          | air              | high population density | No              | RER      | Heat, waste                                                     | 1.26E+01   | MJ                       | Calculations              | 1    | 1.13               | (1,4,1,3,1,3,13) |
|                                             |              | Residues, indium containing | Contains 0.010% In                                                                        | metals           | extraction              | No              | GLO      | leaching residues, indium rich, from zinc circuit, at smelter   | 3.38E-03   | kg                       | Calculations              |      |                    |                  |
|                                             |              | Cadmium sludge              | Contains 45% Cd                                                                           | metals           | extraction              | No              | GLO      | cadmium sludge, from zinc electrolysis, at plant                | 3.39E-01   | kg                       | Calculations              |      |                    |                  |
|                                             |              | Refined zinc                |                                                                                           | metals           | extraction              | No              | RER      | zinc, primary, at regional storage                              | 1.00E+00   | kg                       |                           |      |                    |                  |

Fig. 6.13 Flows for "zinc for coating, at regional storage" (outputs) and its representation in the ecoinvent database

## 6.6 Cadmium Production process and infrastructure

In hydrometallurgical zinc refining, cadmium is removed from the feed together with copper by reduction with zinc dust to give a metallic sludge. Similar cadmium sludge forms in the production of zinc sulfide pigments or zinc carbonate. These are the most important sources for cadmium refining today (Schulte-Schrepping & Piscator, 2002).



**Fig. 6.14 Sources and process routes for cadmium production. The most common production path is the electrolysis from processed cadmium sludge from hydrometallurgical Zinc operations (Schulte-Schrepping & Piscator, 2002), modelled in this study.**

Further secondary starting materials mentioned above may be added to potential primary raw materials without causing large alterations in the refining process. The different raw materials are dissolved in a solution.

Cadmium is extracted by oxidizing the enriched starting materials solutions. In order to oxidize the cadmium, either air is blown in or manganese dioxide is added. Cadmium is precipitated from the solution as a metal sponge by reduction with zinc dust. The precipitated sponge contains 90 % cadmium and only 2.5 % zinc.

Today, the largest amounts of cadmium are recovered electrolytically. The cadmium sponge is precipitated directly in the tank with the electrolytic acid. The anode consists of lead, which does not dissolve, whereas the cathode is aluminum sheet. The current yield is ca. 93 %; the energy consumption is 1250 kWh per tonne of cadmium cathode sheet deposited. Typical impurity levels in electrolytic cadmium sheet are 5 – 10 ppm Cu, 15 – 20 ppm Pb, 5 – 10 ppm Tl, and 5 ppm Zn.

### 6.6.1 Cadmium, primary, at regional storage in ecoinvent

This dataset gives only a rough estimate on the production of primary cadmium from leaching residues from hydrometallurgical zinc operations by means of electrolytic refining. It is mainly based on literature data (Schulte-Schrepping & Piscator, 2002) and is designed for the use of cadmium in electric devices, namely semiconductor applications such as CdTe-films.

The layout of the modules, general flow information, remarks, sources, values and uncertainty information are shown in

Fig. 6.15. The following paragraphs describe calculations, sources and assumptions chosen in this study. The functional unit of this process is one kg of refined cadmium sheet. Tab. 6.4 in the beginning of this chapter summarises the meta-information for this dataset.

**Precipitation.** The zinc dust used for cementation of the cadmium sponge is recovered later in the sludge of the electrolysis and will be returned into the zinc cycle. Therefore no account for zinc dust is done.

**Oxidation.** The oxidation of the sponge is often done with air, therefore no chemicals consumption is considered in this step. Also no additional heat for drying is assumed.

**Electrolysis.** The consumption of 1.25 kWh electricity per kg sheet is considered in this inventory as reported in Schulte-Schrepping & Piscator (2002). Taking copper electrowinning as approximation, 0.2 m<sup>3</sup> electrolyte per tonne is bled (drained) and purified for recycling into electrolytic refining (Davenport et al., 2002). It is assumed that all of this volume is replenished with fresh electrolyte (180 kg H<sub>2</sub>SO<sub>4</sub> per m<sup>3</sup>), since no process data on the purification of spent electrolyte are available.

Because of the lack of a complete and contiguous LCI study that treats the production of cadmium and the absence of problematic processes and auxiliaries employed, no emissions to air and water could be inventoried.

### 6.6.2 Data Quality

Data uncertainty is derived using the Pedigree matrix as described in Frischknecht et al. (2004). This dataset is comparably coarse since it disregards all but the most important environmental impacts attributable to this process. However, since cadmium is not used in applications as pure metal, but diluted in alloys or chemicals, this dataset is of sufficient quality. For the use of this inventory in applications where cadmium is a major constituent – e.g. in NiCd battery systems – the dataset must be complemented with additional specific emission factors and the secondary production process.

## 6. System Characterisation

| General Flow Information     |                            |                 | Representation in ecoinvent                                                                             |                   |                         |                 |          |                                                       |            | Uncertainty Informations |                           |      |                    |                     |
|------------------------------|----------------------------|-----------------|---------------------------------------------------------------------------------------------------------|-------------------|-------------------------|-----------------|----------|-------------------------------------------------------|------------|--------------------------|---------------------------|------|--------------------|---------------------|
| Input                        | Process Name               | Output          | Remarks                                                                                                 | Category          | Sub-category            | Infra-structure | Location | Modul name in ecoinvent                               | Mean value | Unit                     | Source for mean value     | Type | Standard Deviation | General Comment     |
| Cadmium starting material    | cadmium, primary, at plant |                 |                                                                                                         | metals            | extraction              | No              | GLO      | cadmium sludge, from zinc electrolysis, at plant      | 2.24E+00   | kg                       | Calculations              | 4    |                    | uniform dist +/- 5% |
| Electricity for electrolysis |                            |                 |                                                                                                         | electricity       | production mix          | No              | UCTE     | electricity, medium voltage, production UCTE, at grid | 1.25E+00   | kWh                      | Schulte-Schrepping (2002) | 4    | 1.13               | (1,2,3,3,1,3,2)     |
| Transport lorry              |                            |                 | Transport over 250 km to regional storage.                                                              | transport systems | road                    | No              | RER      | transport, lorry >16t, fleet average                  | 2.50E-01   | tkm                      | Assumptions               | 1    | 2.16               | (4,4,1,3,3,5,5)     |
| Electrolyte                  |                            |                 | 0.2 m3 bled per tonne, 180 kg H2SO4 per m3 electrolyte                                                  | chemicals         | inorganics              | No              | RER      | sulphuric acid, liquid, at plant                      | 3.60E-02   | kg                       | Assumptions               | 1    | 1.33               | (4,4,1,3,3,3,4)     |
| Facilities                   |                            |                 | Approximated with "chemical plant, organic", which as an output of 50'000 t per year (25 year lifetime) | chemicals         | organics                | Yes             | RER      | chemical plant, organics                              | 8.00E-10   | unit                     | Assumptions               | 1    | 3.15               | (4,4,1,3,3,5,9)     |
|                              |                            | waste heat      | Calculated from electricity use.                                                                        | air               | high population density | No              | RER      | Heat, waste                                           | 4.50E+00   | MJ                       | Calculations              | 1    | 1.13               | (1,2,3,3,1,3,13)    |
|                              |                            | Cadmium Primary |                                                                                                         | metals            | extraction              | No              | GLO      | cadmium, primary, at plant                            | 1.00E+00   | kg                       |                           | 1    |                    |                     |

Fig. 6.15 Input and output flows for "cadmium, primary production, at regional storage"



## 6.7 Indium Production Process and Infrastructure

Industrially, indium is produced mainly by recovering from by-products of zinc smelting such as fumes, flue dust, jarosite or leach solution (Kammer et al., 2006). These by-products contain small amounts of indium together with other accompanying elements. Therefore a complex hydrometallurgical process is required for isolating indium from impurities such as arsenic, zinc and cadmium. These processes highly depend on the composition given by the specific deposit.

Since the recovery processes are optimised to the main intended products – zinc, lead – the indium recovery is subject to quite a poor efficiency. Indium concentrates generally at two points in the hydrometallurgical zinc extraction process, while roasting the concentrate and in the neutral leach residue.

Jorgenson & George (2004) report that a traditionally low 30 % yield of available indium in treatable residues has been improved to nowadays 80%. Within the jarosite residues a maximum of 60% can be dissolved from the neutral leach residue, which leads to an estimated overall extraction efficiency of around 50% of the indium, i.e. 50% of the indium present in the ore is transferred to the pregnant solution.

If the indium content in the ore is back calculated based on process efficiencies and the indium refinery production and compared to the indium contained in the extracted lead-zinc ore with 50ppm In, it can be estimated that only 40% of the indium extracted is treated for recovery. Thus an overall yield of 19% may be estimated which corresponds to an overall yield of 20% indicated in Stevens (2007).

Generally, indium is recovered by removing impurities through various chemical purification methods such as acid and alkali leaching, solvent extraction, neutralization, sulphidising and cementation are combined to gain a high indium content solution. From this solution indium is either directly electro-won (e.g. Okamoto & Takebayashi, 1996) or precipitated with zinc as a refined sponge indium that is cast into anodes and electro-refined (Felix, 1996; Power Porto, 1990). To overcome undesirable chlorine gas generation while electrowinning, a two-compartment cell is required (Okamoto & Takebayashi, 1996).

### 6.7.1 Concentration of Indium in the solution

In this first step indium is concentrated from around 500 ppm in byproducts and 50 mg / l in a first leach stage to 30 g / l by hydrometallurgical means.

Generally indium containing byproducts arise in the extraction of zinc, lead and tin. The lead and tin metallurgy are quantitatively of minor importance in terms of mine production, although some big single operations are based on lead-zinc and tin deposits, e.g. Cominco at Trail, Canada. Among the indium producers in tin circuits, Capper Pass in Hull, U. K. is mentioned (Alfantazi & Moskalyk, 2003; Felix, 1996; Power Porto, 1990). However, this operation has been – last owned by Rio Tinto Zinc Ltd. – closed down in 1991 and decommissioned. Since then the tin circuits are of minor importance as source for indium bearing raw materials.

In technical literature quite a few different purification processes are described that follow a row of leaching, neutralisation and precipitation stages. However, the more recent patent literature gives strong indications that solvent extraction (SX) techniques are favoured as e.g. proposed by Tomii & Tsuchida (1981) and studied by Power Porto (1990) and Fortes et al. (2003). This development is linked to a shift in the zinc extraction technology from pyrolytical processes to hydrometallurgical processes, where indium accumulates in the leach solution. In a review article Paiva (2001) states, that commercial grade indium is industrially produced using leaching, cementation and electrowinning technologies. SX is often used to recover indium from leach residues. More recent work confirms this tendency and indicates the introduction of more specific ion exchange and extraction resins (Liu et al., 2006; Zhou & Ruan, 2006).

Based on the above mentioned sources it can be concluded that most of the indium produced today is won in hydrometallurgical zinc operations like the Akita refinery in Akita, JP or the Zhouzhou smelter in Hunan Province, China, Cominco's Trail operation (30 tonnes per year<sup>13</sup>) or Falconbridge's Kidd Creek Refinery, Canada. These operations process ores from close-by mines and indium bearing intermediate products, prompt scrap and post consumer scrap.

A typical process scheme for this type of operations is given in Fig. 6.16 on the example of the Kidd Creek refinery. The process is similar to the descriptions given in Kudo & Maruyama (2001) and Power Porto (1990).

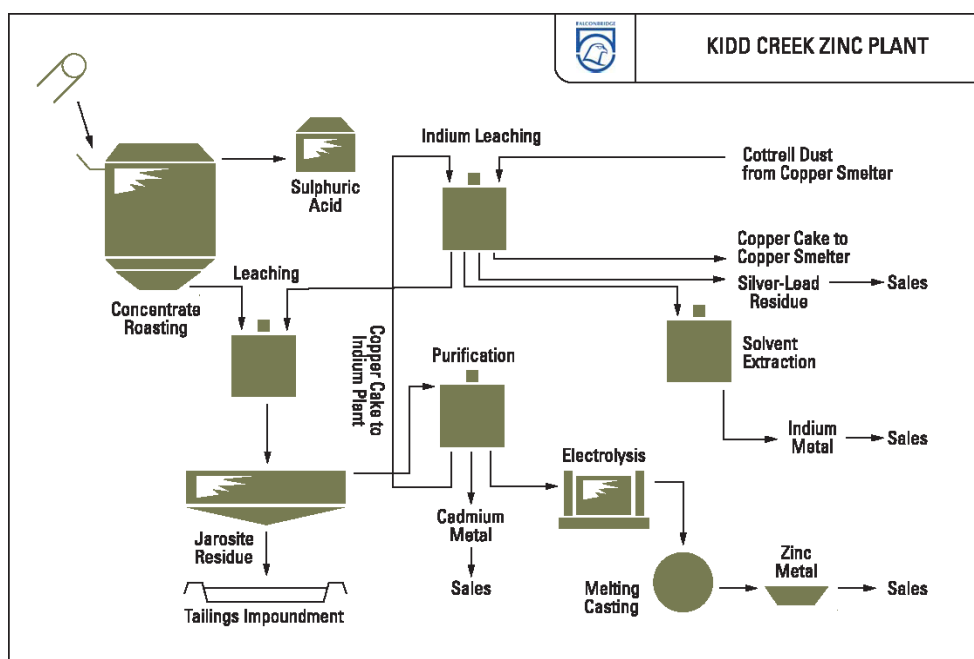


Fig. 6.16 Process scheme of Falconbridge's Kidd Creek refinery<sup>14</sup>

The leach solution which carries indium has usually a concentration of 120 ppm in Zinc residue (Akita, Alfantazi & Moskalyk, 2003), 50-70 mg / L after first leaching (Akita, Felix, 1996) 690 mg / L (typical zinc plant after concentration, Fortes et al., 2003), 200 mg / L (Laboratory setting, Liu et al., 2006, 0.4% (in secondary oxide, 0.68% after preleaching Felix, 1996), 2.7% chloride slag (copper pass, Felix, 1996). Stevens (2007) indicates a range of 300-500 ppm for high indium concentrates.

This solution is concentrated either by SX or a row of neutralisation-precipitation steps to ca. 1 g / L (Power Porto, 1990), or 2 g / L (Akita; Felix, 1996). From this solution sponge indium is precipitated either on aluminium sheets or with zinc powder. The resulting sponge is cast to raw indium anodes with a purity of 95-99.5 %.

As Tomii & Tsuchida (1981) put forward, a main challenge of the neutralisation process is the removal of accompanying contaminants, therefore this quite unspecific procedure is highly complicated, costly and does not yield the purity required for subsequent refining to 5N (five nines = 99.999%) quality. Within the different concentration techniques SX today is regarded as one of the most promising separation operations (Paiva, 2001).

<sup>13</sup> In 2005, <http://www.teckcominco.com/operations/trail/production.htm>, accessed in april 2007

<sup>14</sup> <http://www.xstrata.com/>, accessed in april 2007

However, because of the presence of ions with similar extraction tendency as indium, it is difficult to prepare high purity indium (>4N) only by electrowinning unless several combinations of purification methods are applied (Zhou & Ruan, 2006).

Therefore a sequence of (i) solvent extraction (ii) cementation with zinc powder (iii) anode casting is considered to be the most appropriate process for gaining raw indium that meets the requirements for further refining to over 5N quality. This process corresponds to the study set-up in Power Porto (1990) that is shown in Fig. 6.17.

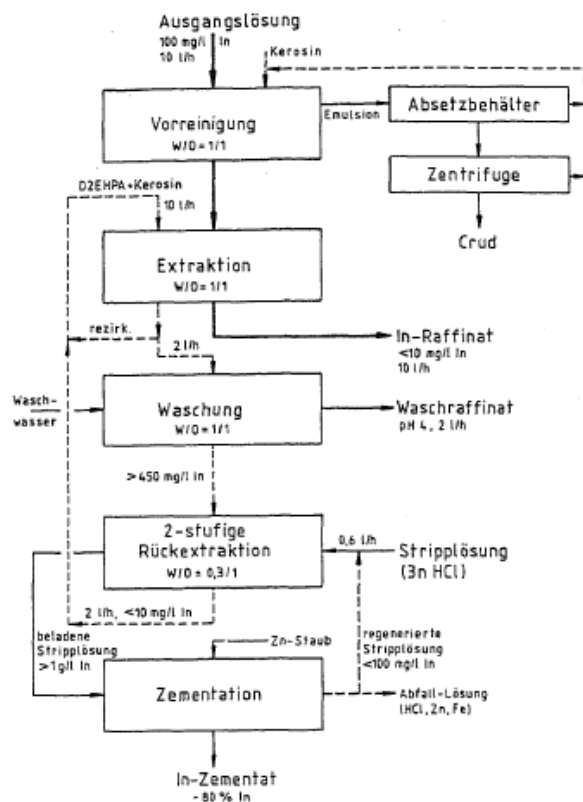


Fig. 6.17 Study set-up of a semi industrial extraction facility by Power Porto (1990) that reflects an up-to-date solvent extraction process scheme. Process covers solvent extraction and cementation; the resulting cementate ("In-Zementat") is cast further into anodes, which is not depicted in this figure.

## 6.7.2 Refining of Raw Indium

There are basically three possibilities described for refining raw sponge indium (99.5%) to the required 5N purity: Vacuum refining, electrorefining zone refining and electrowinning or a combination of all of these. Kammer et al. (2006) additionally mention distillation of  $\text{InCl}_3$  and vacuum distillation. The main challenge is to remove the accompanying contaminants that have a negative effect on the properties of the semiconductor made from indium. Namely thallium and cadmium have detrimental effects on the semiconducting properties.

Purification of metals by vacuum refining is widely practiced. Generally vacuum refining is employed after the electrolytic step to further reduce the level of volatile impurities in the indium metal. Indium metal has low melting point and high boiling point. Therefore, it is possible to remove the impurity metals with high vapour pressure by distilling indium metal at relatively low temperature under vac-

uum. The impurity metals, which are not well removed by chemical methods, are easily removed by vacuum refining at relatively low temperatures (Lee & Sohn, 2003).

### 6.7.3 Indium production in ecoinvent

The module describes the primary production of indium metal (99.999%, 5N purity) on a European average. It is based on the respective flow from the multi-output module "smelting, primary zinc production". Meta-information for this dataset are given in Tab. 6.5, an overview over the inventory is given in Fig. 6.18

Due to the strategic importance of the material and because of the heterogeneous raw materials, no contiguous information on the most common recovery processes are available for LCI modelling. However, there exist many procedures common with the solvent extraction / electrowinning process used in copper production. There also pregnant leach solution is subjected to a solvent-solvent extraction. Based on the process descriptions above, following general steps are covered in this inventory:

- Leaching and solvent extraction with D2EHPA
- Cementation with zinc powder and anode casting
- Electrefining
- Vacuum distillation

#### Leaching and solvent extraction (SX)

The inventory bases on the leaching residues from hydrometallurgical zinc extraction as raw materials. Davenport et al. (2002) states that in the case of SX in copper production 35 g Cu / L are required for electrowinning. SX has to yield this concentration starting from leach solutions not richer than 1 - 6 g / L. This concentration factor of around ten times is comparable with the SX of indium, where a solution of 50 – 200 mg / L is concentrated to 1 to 2 g / l.

In this study it is assumed that the lower concentration level in the case of solvent extraction of indium leads to a proportional increase of throughput compared to SX of copper. The leach solution of indium is 20 to 30 times less concentrated than in the copper extraction. It is assumed in this study, that the required accumulation of indium requires consequently 20 to 30 times the volume put through. Hence the specific emissions as well as energy and material consumption documented for the solvent extraction of copper are multiplied by 25 and are taken as approximation for the SX of indium. The values for electricity consumption for pumping, mixing, etc. as well as the "crud" generation (solid particles associated with oil/water dispersions that are disposed of) are taken from Krauss et al. (1999). Direct emissions were estimated based on reported yearly air emissions of an U.S. SX/EW copper operation in Arizona (ADEQ, 2006) and the reported copper output of the facility<sup>15</sup>.

#### Cementation and anode casting

Crude indium is formed by the precipitation of a sponge. Here, it is assumed that the sponge contains 20%wt moisture and is dried prior to anode casting. Drying is assumed to be effected in a spray dryer used in ceramic industry (38kW at 250 kg per hour Timokhova (2002), Melting is assessed using the heat of fusion and assuming a system efficiency of 70%.

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<sup>15</sup> 2006 BHP Billiton Annual Report, Information on the Company: Pinto Valley Operation produced 8'500 t in 2006

### Electrorefining

No information on electrorefining of indium was available. However, this process – where the anode dissolves anodically – usually ranges between 0.2 to 0.5 kWh per kg. In this inventory a value as required in copper refining is chosen. In Davenport et al. (2002, Tab 16.4) for electrorefining of copper a range of 0.25 to 0.35 kWh / kg is given. Taking the upper value, an electricity consumption of 0.35 kWh per kg indium cathode sheet is assumed.

Belsky et al. (1980) state a current density of 0.015 to 0.03 A / cm<sup>2</sup> (150 to 300 A / m<sup>2</sup>) for the electrorefining of indium, which is comparable to copper (200 to 300 A / m<sup>2</sup>). Zhou et al. (2005) determine optimal conditions of 0.2-0.3 V bath voltage but at lower current densities of 50-80 A / m<sup>2</sup>.

In Davenport et al. (2002) it is stated that 0.1 to 0.2 m<sup>3</sup> electrolyte is bled per tonne of copper, i.e. recirculated for purification. In Zhou et al. (2005) the electrolyte is a low concentration solution of sulphuric acid. We assume a value of 0.15 kg 25% H<sub>2</sub>SO<sub>4</sub> per kg indium refined.

### Vacuum refining

A continuous process is described in McNamara et al. (1989) on behalf of Indium Corporation of America Inc. In the proposed installation liquid indium is heated under vacuum (10<sup>-7</sup> to 10<sup>-8</sup> torr) to 1'400 °C in order to volatilise remaining impurities. The original patent by Belsky et al. (1980) remains less specific but claims the process duration of two heating phases of 1-5 h at 850-940°C followed by 0.5-2 h at 950-1100°C. The long duration of this refining step is also confirmed by Lee & Sohn (2003) who found slow evaporation rates. In this study we assume that a RF induction heated Vacuum furnace is employed that uses 0.5 kW per kg load. This assumption corresponds to a batch of 30 kg indium that is heated for 4 hours in a 15kW furnace as often used in lab-scale. Commercial furnaces<sup>16</sup> exhibit energy supply from 36 kW (0.01 m<sup>3</sup> volume) to 90 kW (0.127 m<sup>3</sup> volume). Assuming that one third of the heated volume can catch the indium load (7.3 g / cm<sup>3</sup>) a somewhat higher span of 0.5 – 0.3 kW per kg batch size results. Inductotherm Corp. reports 0.15 kW per kg for its Acutrac (R) furnace<sup>17</sup>. However, bearing in mind that the metal is very costly and no large batch sizes are supposed to be processed, the upper value of this span is justified.

Additional power consumption for vacuum preparation is disregarded due to a lack of reliable sources and because this particular process is expected to be of minor importance compared to the overall impacts. Also the inert gas atmosphere is supposed to be of minor importance for the environmental impacts and is therefore not inventoried.

### Transport and Infrastructure

The indium metal is assumed to be transported by lorry to and from the airport (250 km each), by plane over 5'000 km to Europe. Additional weight due to packaging is accounted for with additional 50%.

The infrastructure is estimated based on the module "chemical plant, organics", that has a built area of about 4.2 ha and an average output of 50'000 t/a (Althaus et al., 2003).

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<sup>16</sup> <http://www.vacuumfurnace.com/>, accessed in may 2007

<sup>17</sup> <http://www.inductothermgroup.com/>, accessed in may 2007 - 68 -

## 6. System Characterisation

| General Flow Information        |                             |                 | Representation in ecoinvent                                                                             |                   |                         |                 |          |                                                                   |            | Uncertainty Informations |                       |      |                    |                  |
|---------------------------------|-----------------------------|-----------------|---------------------------------------------------------------------------------------------------------|-------------------|-------------------------|-----------------|----------|-------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|------------------|
| Input                           | Process Name                | Output          | Remarks                                                                                                 | Category          | Sub-category            | Infra-structure | Location | Module name in ecoinvent                                          | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment  |
| Leach solution                  | indium, at regional storage |                 |                                                                                                         | metals            | extraction              | No              | GLO      | leaching residues, indium rich, from zinc circuit, at smelter     | 1.00E+04   | kg                       |                       | 1    | 1.32               | (3,2,1,2,3,5,3)  |
| Electricity for mixing (SX)     |                             |                 | Based on copper SX                                                                                      | electricity       | production mix          | No              | UCTE     | electricity, medium voltage, production UCTE, at grid             | 8.75E+00   | kWh                      | Krauss et al 1999     | 1    | 1.38               | (4,2,1,2,3,5,2)  |
| Organic Chemicals               |                             |                 | Crud formation, based on copper SX                                                                      | chemicals         | organics                | No              | GLO      | chemicals organic, at plant                                       | 4.25E-02   | kg                       | Krauss et al 1999     | 1    | 1.3                | (4,2,1,2,1,5,3)  |
|                                 |                             | VOCs            | based on a copper SX operation                                                                          | air               | high population density |                 |          | NMVOC, non-methane volatile organic compounds, unspecified origin | 9.17E-02   | kg                       |                       | 1    | 1.62               | (4,2,1,2,1,5,16) |
|                                 |                             | H2SO4 mist      | based on a copper SX operation, approx with HCl (Electrolyte)                                           | air               | low population density  |                 |          | Sulfuric acid                                                     | 4.34E-02   | kg                       |                       | 1    | 1.62               | (4,2,1,2,1,5,18) |
| drying energy                   |                             |                 | Spray dryer 38 kW @ 250 kg / h                                                                          | electricity       | production mix          | No              | UCTE     | electricity, medium voltage, production UCTE, at grid             | 1.57E-01   | kWh                      | Timokhova 2002        | 1    | 1.38               | (4,2,1,2,3,5,2)  |
| melting energy                  |                             |                 | heat of fusion 3.281 kJ / mol, 70% efficiency                                                           | electricity       | production mix          | No              | UCTE     | electricity, medium voltage, production UCTE, at grid             | 9.92E-03   | kWh                      |                       | 1    | 1.3                | (4,2,1,2,1,5,2)  |
| sulphuric acid                  |                             |                 | Replenishment of electrolyte, H2SO4, 25% conc.                                                          | chemicals         | inorganics              | No              | RER      | sulphuric acid, liquid, at plant                                  | 3.75E-02   | kg                       | Davenport 2002        | 1    | 1.38               | (4,2,1,2,3,5,3)  |
| Electrowinning electricity      |                             |                 | Based on Copper Electrorefining                                                                         | electricity       | production mix          | No              | UCTE     | electricity, medium voltage, production UCTE, at grid             | 3.50E-01   | kWh                      | Davenport 2002        | 1    | 1.38               | (4,2,1,2,3,5,2)  |
| Vacuum distillation electricity |                             |                 | Based on different furnace types from 0.7 to 0.15 kW per kg.                                            | electricity       | production mix          | No              | UCTE     | electricity, medium voltage, production UCTE, at grid             | 2.00E+00   | kWh                      | Estimate              | 1    | 1.3                | (4,2,1,2,1,5,2)  |
| Transport by truck to dealer    |                             |                 | 2 x 250 km distance, with 50%wt additional packaging                                                    | transport systems | road                    | No              | RER      | transport, lorry >16t, fleet average                              | 7.50E-01   | tkm                      | Assumption            | 1    | 2.33               | (5,3,1,1,3,5,5)  |
| Transport by plane to region    |                             |                 | 5'000 km, with 50%wt additional packaging                                                               | transport systems | airplane                | No              | RER      | transport, aircraft, freight                                      | 7.50E+00   | tkm                      | Assumption            | 1    | 2.33               | (5,3,1,1,3,5,5)  |
| Facilities                      |                             |                 | Approximated with "chemical plant, organic", which as an output of 50'000 t per year (25 year lifetime) | chemicals         | organics                | Yes             | RER      | chemical plant, organics                                          | 8.00E-10   | unit                     | Assumption            | 1    | 3.32               | (5,1,1,1,3,5,9)  |
|                                 |                             | waste heat      | Calculated from electricity use.                                                                        | air               | high population density | No              | RER      | Heat, waste                                                       | 4.06E+01   | MJ                       |                       | 1    | 1.05               | (1,1,1,1,1,1,3)  |
|                                 |                             | Indium metal 5N |                                                                                                         | metals            | extraction              | No              | RER      | indium, at regional storage                                       | 1.00E+00   | kg                       |                       | 1    |                    |                  |

Fig. 6.18 Flows for "Indium, at regional storage" and their representation in the ecoinvent database

#### **6.7.4 Data Quality Considerations**

The production process is rather new and not much is known from public sources about quantitative process measures. Therefore the data has been established based on theoretical reasoning and rough assumptions, extrapolating the data from copper refining. Except for the sub-process of solvent extraction which employs organic extraction solution and emits VOC and H<sub>2</sub>SO<sub>4</sub> mist, no direct emissions from other sub-processes have been considered. Infrastructure has been approximated with the related technology of chemical plants. The uncertainty of the used environmental flows has been calculated with the pedigree approach. The data does comply with the demands for the use in electronic devices. This dataset is not to be used when indium is a substantial part of the system modelled (e.g. for comparisons of special solder paste).

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## Part IX

# Gold and Silver

Data v2.1 (2009)

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Final report of updated Swiss National Life Cycle Inventory Database "ecoinvent", version 2.1

## Comments on version 2.1

This part corresponds to version the part in version 2.0 without changes.

## Comments on this part

The Swiss Federal Office for the Environment (FOEN) commissioned Empa in 2005 to identify the datasets that have to be improved in order to improve the overall quality and applicability of metals inventories in the metal processing and electronic industry. Based on that preparatory study Gold and Silver were identified as widely applied metals with potentially high impact and therefore of high relevance.

## Summary

This part documents the inventories for primary and secondary silver and gold, respectively, as well as the production of copper telluride cement, which is a by-product of the treatment of copper anode slimes in order to recover silver. The inventories for primary gold are compiled country specifically according to the environmental reports of the world leading gold producers. These inventories also cover the production of silver from auriferous ores in a multi-output module. The rest of the primary silver production is reflected by the processing of silver bearing by-products of lead / zinc (Parkes crust) and copper (anode slimes) production. Secondary gold and silver production is reflected by the recovery processes in complex integrated smelters (Boliden, SE).





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# 1 Introduction

This life cycle inventory (LCI) of gold and silver is meant to be used as background data sets for the preparation of life cycle assessments (LCA) of technical products such as electronic components. This LCI includes data sets of gold mining and refining in Australia, Canada, Chile, Peru, Papua New Guinea, South Africa, Sweden, Tanzania and the USA, as well as of the gold supply to Europe. It also includes data sets of silver as a by-product of gold production in Chile, Peru, Papua New Guinea and Sweden.

Data in this report are based mostly on environmental reports of large multinational companies. However, it must be assumed that these sources represent rather the best practices for gold mining. Various NGOs like Berne Declaration, Greenpeace<sup>1</sup> etc. report on severe impacts of gold mining on environment, local communities or human rights concerning projects or mines of important companies like Barrick Gold and Newmont Mining, the two largest gold companies of the world today. Controversial issues are for example a mine project in Chile involving relocating/destroying of three glaciers in the Andes which are of crucial importance to the Pascua Lama Highlands, or the development of a mine in Nevada, USA in the Western Shoshone Lands, which is expected to have a negative effect on the region's scarce water resources and to destroy Shoshone sacred places. The nodirtygold campaign<sup>2</sup> denounces among others the problem of tailings dam failures which release large amounts of cyanide-laden tailings in rivers. Such impacts of the gold mining are very difficult to quantify and include in LCI due lacking of data and also because of the punctual character of the events. Therefore it must be assumed that the environmental impacts of gold mining are rather underestimated in the following data sets, which consider only regular operation.

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<sup>1</sup> See also <http://www.evb.ch/researchcs>, <http://www.evb.ch/researchubs>, <http://www.greenpeace.ch>, etc.

<sup>2</sup> See <http://www.nodirtygold.org>

## 2 Reserves and Resources of Materials

### 2.1 Gold

#### 2.1.1 Gold from primary resources

“Gold abundance in the Earth’s crust is 0.002 per 10<sup>6</sup> atoms of silicon. Gold average economic abundance in the ore is today in the range of 0.1 to 0.2 oz/ton or 3.42-6.84 gram per ton of ore, but also deposits with a tenth of those values, are exploited.” (Gasparrini, 1993)

In the last years gold was produced in following countries:

**Tab. 2.1: Annual share of gold mine production<sup>3</sup>.**

|                                                     | 2000             | 2001             | 2002             | 2003             | 2004             | Average          |
|-----------------------------------------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| South Africa                                        | 16.8%            | 15.4%            | 15.5%            | 14.7%            | 14.0%            | 15.3%            |
| Russia                                              | 11.5%            | 12.1%            | 12.5%            | 12.9%            | 13.9%            | 12.5%            |
| USA                                                 | 13.7%            | 13.1%            | 11.7%            | 10.9%            | 10.6%            | 12.0%            |
| Australia                                           | 11.5%            | 11.1%            | 10.7%            | 11.1%            | 10.7%            | 11.0%            |
| Indonesia&PG                                        | 7.8%             | 9.1%             | 8.1%             | 8.0%             | 6.8%             | 8.0%             |
| China                                               | 7.0%             | 7.2%             | 7.5%             | 8.0%             | 8.8%             | 7.7%             |
| Africa without South Africa                         | 6.1%             | 6.5%             | 7.0%             | 6.6%             | 6.7%             | 6.6%             |
| Peru                                                | 5.2%             | 5.4%             | 6.2%             | 6.7%             | 7.1%             | 6.1%             |
| Canada                                              | 6.1%             | 6.2%             | 6.0%             | 5.5%             | 5.3%             | 5.8%             |
| South and Central America without Peru/Brazil/Chile | 5.5%             | 5.3%             | 5.1%             | 5.8%             | 5.7%             | 5.5%             |
| Asia                                                | 2.3%             | 2.2%             | 2.6%             | 2.5%             | 2.6%             | 2.5%             |
| Brazil                                              | 2.0%             | 1.7%             | 1.6%             | 1.6%             | 1.7%             | 1.7%             |
| Chile                                               | 2.1%             | 1.7%             | 1.5%             | 1.5%             | 1.6%             | 1.7%             |
| Tanzania                                            | 0.6%             | 1.2%             | 1.7%             | 1.9%             | 2.1%             | 1.5%             |
| Europe (incl. Turkey)                               | 0.9%             | 1.1%             | 1.2%             | 1.2%             | 1.4%             | 1.2%             |
| New Zealand                                         | 0.4%             | 0.4%             | 0.4%             | 0.4%             | 0.3%             | 0.4%             |
| Middle East                                         | 0.2%             | 0.2%             | 0.2%             | 0.4%             | 0.4%             | 0.3%             |
| India                                               | 0.2%             | 0.1%             | 0.1%             | 0.1%             | 0.2%             | 0.2%             |
| Pacific                                             | 0.2%             | 0.2%             | 0.2%             | 0.1%             | 0.2%             | 0.2%             |
| <b>World Production (kg)</b>                        | <b>2'567'348</b> | <b>2'563'867</b> | <b>2'548'471</b> | <b>2'548'693</b> | <b>2'431'367</b> | <b>2'532'671</b> |

The total volume of gold production is rather stable. However, substantial changes occurred by the country. South Africa’s production diminished between 2000 and 2004, whereas production in Russia, China and Peru increased considerably.

This life cycle inventory is based on data from Australia, Canada, Chile, Peru, Papua New Guinea, South Africa, Sweden, Tanzania and the United States of America. The data set „gold primary, at refinery“ takes into account the average production shares 2000-2004 of these countries (Tab. 2.2). Production in African countries is approximated by South African and Tanzanian production, production in South and Central America by Chilean and Peruvian production, production in Asia (incl. Middle East) and Pacific by production in Papua New Guinea, and production in Europe by Swedish production. The production from Russia / former USSR is approximated with Canadian production, due to similar geographical and meteorological conditions.

<sup>3</sup> [http://www.indexmundi.com/en/commodities/minerals/gold/gold\\_table08.html](http://www.indexmundi.com/en/commodities/minerals/gold/gold_table08.html).

Tab. 2.2: Shares of the countries in mine production and the respective share in data set „gold primary, at refinery“

|                   | Average<br>2000-2004 | This project | Remark                                              |
|-------------------|----------------------|--------------|-----------------------------------------------------|
| South Africa      | 15.3%                | <b>18.6%</b> | incl. half of African countries                     |
| USA               | 12.0%                | <b>12.0%</b> |                                                     |
| Australia         | 11.0%                | <b>11.4%</b> | incl. New Zealand                                   |
| Canada            | 5.8%                 | <b>18.4%</b> | incl. Russia                                        |
| Papua New Guinea  | 2.7%                 | <b>18.8%</b> | Pacific, Asia, Middle East, excluding Russia        |
| Chile             | 1.7%                 | <b>5.3%</b>  | incl. half of South- and Central American countries |
| Peru              | 6.1%                 | <b>9.7%</b>  | incl. half of South- and Central American countries |
| Tanzania          | 1.5%                 | <b>4.8%</b>  | incl. half of African countries                     |
| Sweden            | 0.2%                 | <b>1.2%</b>  | Europe (incl. Turkey)                               |
| Rest of the World | 46%                  |              |                                                     |
| Total             | 100%                 | <b>100%</b>  |                                                     |

### 2.1.2 Gold from secondary resources

Among other precious metals, such as silver, palladium, platinum, etc., gold is contained in varying amounts in metallic scrap of any kind. Depending on the application, for instance printed wiring board assemblies may contain between 0.03 wt% (Goosey & Kellner, 2002) and 0.089 wt% (Motorola, 2005). Other authors report gold contents ranging from 0.035 wt% up to 0.18 wt% (Zhang et al., 2000). Other sources of secondary gold are: scrap jewellery, dental applications, plated rejected components, porcelain scrap, and mirrors (George, 2006; Syed, 2006).

In the U.S. the production of secondary gold has increased from 16.1 tonnes in 1964 to 45.0 tonnes in 2004 and thus accounts for about 14.8 % of the total gold production (Geoffroy & Cardarelli, 2005; George, 2006). Due to the scarcity of the metal and because of the drastically increased demand, the price of gold (and secondary gold) has risen dramatically since the late 1970ies.

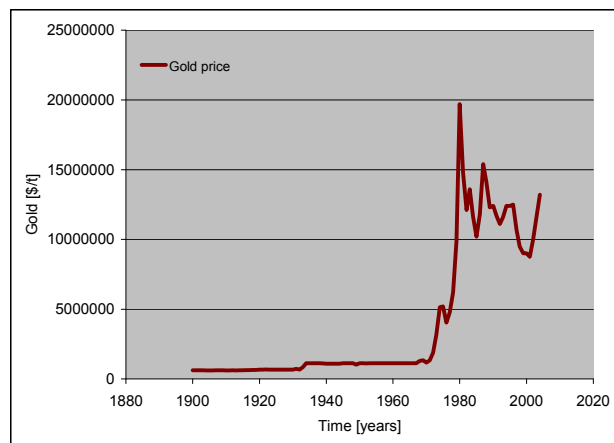


Fig. 1 Value of gold (adapted from George (2006)).

Key sources for the refinement of secondary gold, however, are new and old metallic scrap. In 2000 it was estimated that about 280 tonnes of gold were present in electronic goods (Holliday & Goodman, 2002). Gold can be extracted from the scrap in a number of subsequent steps (depending on the required degree of purity) with a purity between 90.0 % (Doré) and 99.99 % (Boliden Contech A.B., 1995). Again, depending on the intended application of the final product, the recovered gold is either added as bullion or higher grade secondary gold to primary gold. Along with the extraction of gold

from new and old electronic scrap, typically silver, palladium and (in case) selenium are co-extracted (Boliden, 2000; Boliden, 2002a; Boliden Contech A.B., 1995).

## 2.2 Silver

The extraction of silver from true silver ores – in which silver is practically the only valuable metal – is by far not the biggest source for supply. Depending on the price of silver, it can be cost-effective to process materials containing 0.01 % (100 ppm) silver, or even less. The most important primary sources of silver are traditionally argentiferous lead and lead-zinc ores that are widely distributed. They constitute the most important fraction of the mine production. However, since the mine production of lead declined in the last years, these deposits became less important for the worldwide supply. The separation and recovery of silver during lead production is profitable even for a silver content of a few ppm. Argentiferous copper ores, which sometimes also contain nickel and cobalt, satisfy another important share of the world silver demand. Even very low contents of silver can be recovered profitably from copper ores, since almost all the copper is electrorefined, and the silver remains in the anode slimes. The largest argentiferous copper reserves in the world are in Canada (Sudbury, Ontario, Noranda-Rouyn, Quebec). Extraction of silver from gold ores is carried out mainly in the Republic of South Africa, and also in Canada, Alaska, and Sweden. The quantity obtained from these deposits is around 10-20% of the world mine production. Silver-containing tin deposits are concentrated in Bolivia (Cerro de Pasco, Potosi) but also occur in Burma and Argentina. Silver produced from tin ores amounts to only 2 % of world silver production. (Renner et al., 2002b).

### 2.2.1 Silver as by-product of lead/zinc, copper and gold production

Most of the world silver is produced as by-product from lead/zinc, copper and gold bearing deposits. The share of the different ore deposits that constitute the world mine production changes over the years (Tab. 2.3). Recently, the Silver Institute (2002, p. 26) estimates the breakdown by ore type as follows: 25 % silver ores, 15% gold ores, 24 % copper ores, 35% lead, zinc, and lead-zinc ores and 1% other ores (Butterman & Hilliard, 2004, JMB, 2002) Nigel H Maund estimated the supply in 2004 as follows<sup>4</sup>: Lead - Zinc deposits (mostly Mexico and Cannington, and smaller Chinese contributions) 31%; Copper deposits (Kupferschiefer of Poland and porphyry copper mines) 25%; Gold deposits (in particular Carlin type) 14% Primary silver deposits (in particular "Idaho type") 28%.

This LCI corresponds to the available data from Klapwijk ((2006, p.87)) as shown in Tab. 2.4, that specifies the share of the origin countries for the data set „ silver mix, from combined gold / silver production, at refinery“.

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<sup>4</sup> “Silver – An Economic Geologist Perspective”, [http://www.gold-eagle.com/editorials\\_04/nmaund072004.html](http://www.gold-eagle.com/editorials_04/nmaund072004.html)

Tab. 2.3 Silver supply differentiated into sources.

| Source      | Renner et al. (2002b) | JMB (2002) (Silver institute) | Nigel H Maund <sup>5</sup> | Klapwijk et al. (2006) | Primary production assumed in this study | Mix primary and secondary |
|-------------|-----------------------|-------------------------------|----------------------------|------------------------|------------------------------------------|---------------------------|
| year        | 1996                  | 2002                          | 2004                       | 2006                   |                                          |                           |
| Lead / Zinc | 50%                   | 35%                           | 31%                        |                        | 31%                                      | 24%                       |
| Copper      | 30%                   | 24%                           | 25%                        |                        | 25%                                      | 20%                       |
| Silver      | 10%                   | 25%                           | 28%                        |                        | 28%                                      | 22%                       |
| Gold        | 10%                   | 15%                           | 14%                        |                        | 14%                                      | 11%                       |
| other       |                       | 1%                            | 2%                         |                        | 2%                                       | 2%                        |
| Secondary   |                       |                               |                            | 21%                    |                                          | 21%                       |
| Mine        |                       |                               |                            | 70%                    |                                          |                           |
| Hedging     |                       |                               |                            | 2%                     |                                          |                           |
| Gov sales   |                       |                               |                            | 7%                     |                                          |                           |

Tab. 2.4 Shares of the countries in the data set „silver mix, from combined gold / silver production, at regional storage“, Source: Klapwijk et al. (2006)

|                         |        | %        | This project: |                  | Remark                                       |
|-------------------------|--------|----------|---------------|------------------|----------------------------------------------|
| Central & South America | Chile  | 28.8%    | <b>32.0%</b>  | Chile            | incl. Central & South america excluding Peru |
|                         | Peru   | 13.1%    | <b>31.3%</b>  | Peru             | incl. half of USA/Canada                     |
|                         | others | 3.2%     |               |                  |                                              |
| Asia                    |        | 14.8%    | <b>14.8%</b>  | Papua New Guinea | incl. Asia                                   |
| North America           | USA    | 14.4%    |               |                  |                                              |
|                         | Canada | 22.0%    |               |                  |                                              |
| others                  |        | 3.7%     | <b>21.9%</b>  | Sweden           | incl. half of USA/Canada                     |
| World production (kg)   |        | 2.36E+06 | <b>100.0%</b> |                  |                                              |

## 2.2.2 Silver from old scrap

According to the Silver Institute 21% of the silver demand was satisfied from old scrap (Klapwijk et al., 2006) (as Secondary in Tab. 2.3).

## 2.3 Allocation for Metals

Metal prices are used as the allocation parameter for polymetallic mines. Prices in USD of copper, lead, zinc, gold and silver stem from Boliden for the year of 1996 and 2001, the prices for 2005 are from USGS (2006). The prices per tonne in the yearly average 2004-2006 are compiled from Kelly & Matos (2007) through 2004 and the respective 2007 mineral commodity summaries from USGS (2006) for values after 2004. These prices have been used in this project for the allocation of different metals.

<sup>5</sup> “Silver – An Economic Geologist Perspective”, [http://www.gold-eagle.com/editorials\\_04/nmaund072004.html](http://www.gold-eagle.com/editorials_04/nmaund072004.html)

Prices were relatively stable between 1996 and 2001, whereas a strong demand in the years 2005 to 2006 lead to a sharp increase in 2006 (see Tab. 2.5). Relative shares remained about constant.

**Tab. 2.5: Price of gold, silver, copper, lead and zinc 1996, 2001, 2005 and the average prices 2004-2006, used in this project (sources see text).**

| Prices of metals | 1996      | 2001      | 2005      | Average 2004-2006, used<br>in this project |
|------------------|-----------|-----------|-----------|--------------------------------------------|
|                  | US\$ / kg | US\$ / kg | US\$ / kg | US\$ / kg                                  |
| Prices of metals | 10'714    | 8'682     | 14'295    | 15'682                                     |
| Silver           | 157       | 140       | 239       | 269                                        |
| Gold             | 2.1       | 0.8       | 1.4       | 1.9                                        |
| Silver           | 0.7       | 1.5       | 3.7       | 4.5                                        |
| Zinc             | 1.1       | 0.5       | 1.0       | 1.2                                        |



## 3 Characterisation of the materials

For gold and silver the term precious metals is used because of their rarity and corrosion resistance. Gold and silver are often mined together in polymetallic mines.

### 3.1 Gold

Apart from copper, Gold is the only metallic element with a colour different from a grey tone. It is a soft yellow metal with a moderately high melting point (1064°C) and a high electrical and thermal conductivity exceeded only among the elements by silver and copper. It is the most malleable and ductile of the metals. One ounce of gold can be beaten in to foil some 300 square feet in area. Tab. 3.1 displays some physical properties of the element gold.

**Tab. 3.1** Some physical properties of gold (Renner et al., 2002a).

| Property                      | Value                                  |
|-------------------------------|----------------------------------------|
| Melting Point                 | 1064.43°C                              |
| Boiling Point                 | 2808°C                                 |
| Density at 20°C               | 19.32 g/cm <sup>3</sup>                |
| Density at 1065°C             | 17.32 g/cm <sup>3</sup>                |
| Vapor pressure at 1064°C      | 0.002 Pa                               |
| Atomic volume at 20°C         | 10.21 cm <sup>3</sup> /mol             |
| Electrical resistivity at 0°C | 2.06×10 <sup>-6</sup> Ω cm             |
| Thermal conductivity at 0°C   | 3.14 Wcm <sup>-1</sup> K <sup>-1</sup> |
| Specific heat                 | 0.138 Jg <sup>-1</sup> K <sup>-1</sup> |
| Enthalpy of fusion            | 12.77 kJ/mol                           |
| Enthalpy of vaporization      | 324.4 kJ/mol                           |
| Tensile strength              | 127.5 N/mm <sup>2</sup>                |

### 3.2 Silver

“Silver has a moderately high melting point (961.9°C) and the highest electrical and thermal conductivity of all metals. When in the metallic form, silver is unaffected by water or oxygen, but is altered by ozone with the formation of silver oxide, and by sulphur and other sulphur compounds with the formation of silver sulphide” (Gasparrini, 1993). Tab. 3.2 summarises some physical properties of elemental silver.

**Tab. 3.2** Some physical properties of silver (Renner et al., 2002b).

| Physical Property               | Value                                   |
|---------------------------------|-----------------------------------------|
| Melting point                   | 961.9 °C                                |
| Boiling point                   | 2212 °C                                 |
| Specific heat capacity at 25 °C | 0.23 J kg <sup>-1</sup> K <sup>-1</sup> |
| Thermal conductivity            | 418 W m <sup>-1</sup> K <sup>-1</sup>   |
| Vapor pressure at 1030 °C       | 1.33 Pa                                 |
| Density at 20 °C                | 10.49 g/cm <sup>3</sup>                 |
| Density of liquid at mp         | 9.35 g/cm <sup>3</sup>                  |
| Brinell hardness                | 26                                      |
| Modulus of elasticity           | 82 000 MPa                              |
| Tensile strength                | 140 MPa                                 |
| Resistivity at 0 °C             | 1.465 Ωcm                               |

## 4 Use / Application of Material

Gold is used mainly for jewellery, electronics and other industrial and decorative appliance (Tab. 4.1). In electric/electronic devices, it is mostly used as connector or as solder paste (Goosey & Kellner, 2002). The latter one is typically a gold alloy incorporating Si (98%Au, 2%Si), Ge (87.5%Au, 12.5%Ge), In (82%Au, 18%In) or Sn (80%Au, 20%Sn). The principal users of silver are industrial applications, photography and jewellery (Fig. 4.1).

Tab. 4.1 World Gold demand in different application fields (World Gold Council<sup>6</sup>).

| Tonnes                                         | 2005    | 2006    | % ch<br>2006<br>vs<br>2005 | Q1'06 | Q2'06 | Q3'06 | Q4'06   | Q1'07 <sup>2</sup> | % ch<br>Q1'07<br>vs<br>Q1'06 |
|------------------------------------------------|---------|---------|----------------------------|-------|-------|-------|---------|--------------------|------------------------------|
| <b>Jewellery consumption</b>                   | 2,707.2 | 2,279.3 | -16                        | 488.0 | 522.8 | 557.9 | 710.7   | 572.8              | 17                           |
| <b>Industrial &amp; dental</b>                 | 426.7   | 451.5   | 6                          | 110.7 | 113.6 | 113.5 | 113.6   | 111.8              | 1                            |
| Electronics                                    | 279.2   | 304.4   | 9                          | 74.8  | 77.3  | 77.0  | 75.3    | 75.9               | 2                            |
| Other Industrial                               | 85.1    | 86.4    | 2                          | 20.6  | 21.1  | 21.6  | 23.1    | 21.2               | 3                            |
| Dentistry                                      | 62.4    | 60.7    | -3                         | 15.3  | 15.3  | 14.9  | 15.2    | 14.7               | -4                           |
| <b>Identifiable Investment</b>                 | 594.8   | 649.6   | 9                          | 199.1 | 138.7 | 124.1 | 187.7   | 147.1              | -26                          |
| <b>Net retail investment</b>                   | 386.7   | 389.5   | 1                          | 86.2  | 89.8  | 104.9 | 108.6   | 110.7              | 28                           |
| Bar Hoarding                                   | 262.5   | 225.9   | -14                        | 43.5  | 43.7  | 64.7  | 74.0    | 61.2               | 41                           |
| Official Coin                                  | 110.9   | 129.1   | 16                         | 33.5  | 42.5  | 30.1  | 22.9    | 34.4               | 3                            |
| Medals/Imitation Coin                          | 37.0    | 58.9    | 59                         | 11.7  | 11.0  | 15.3  | 21.0    | 20.2               | 72                           |
| Other identified retail invest. <sup>1</sup>   | -23.7   | -24.5   | ...                        | -2.5  | -7.4  | -5.2  | -9.4    | -5.1               | ...                          |
| <b>ETFs &amp; similar products<sup>4</sup></b> | 208.1   | 260.2   | 25                         | 112.9 | 48.9  | 19.2  | 79.1    | 36.4               | -68                          |
| <b>Total identifiable demand</b>               | 3,728.7 | 3,380.4 | -9                         | 797.8 | 775.1 | 795.5 | 1,012.0 | 831.7              | 4                            |

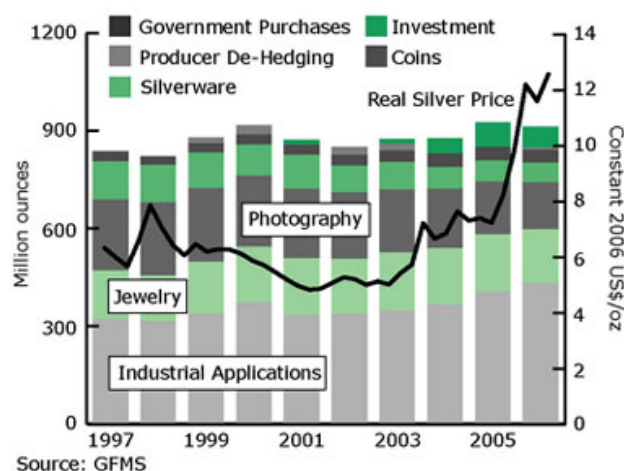


Fig. 4.1 World silver demand in different application fields (Silver Institute<sup>7</sup>).

<sup>6</sup> <http://www.gold.org/> accessed in july 2007

<sup>7</sup> <http://www.silverinstitute.org/supply/>, accessed in july 2007

## 5 System Characterisation

The production of silver is heavily interlaced with the production of other metals, namely lead and zinc, copper and gold. This is taken into account by establishing a complex LCI model (Fig. 5.1, for supply mix see chapter 5.6 *Silver, at regional storage*). However, not every source contributing to the worldwide production of gold and silver could be evaluated in ecoinvent version 2.0.

Due to a considerable lack of systematic information on processes in the artisanal mining of gold, no inventory on these processes could be established. The production of silver as main product – which accounts for 22% of the world supply – has been disregarded in this study for two reasons: first, a coverage of 78% seems fair enough to represent silver. Second, the production of silver as main-product occurs with the same technologies as the production of silver as by-product and thus can be approximated by these datasets. An overview over the system boundaries chosen for the inventories in this chapter is given in Fig. 5.2.

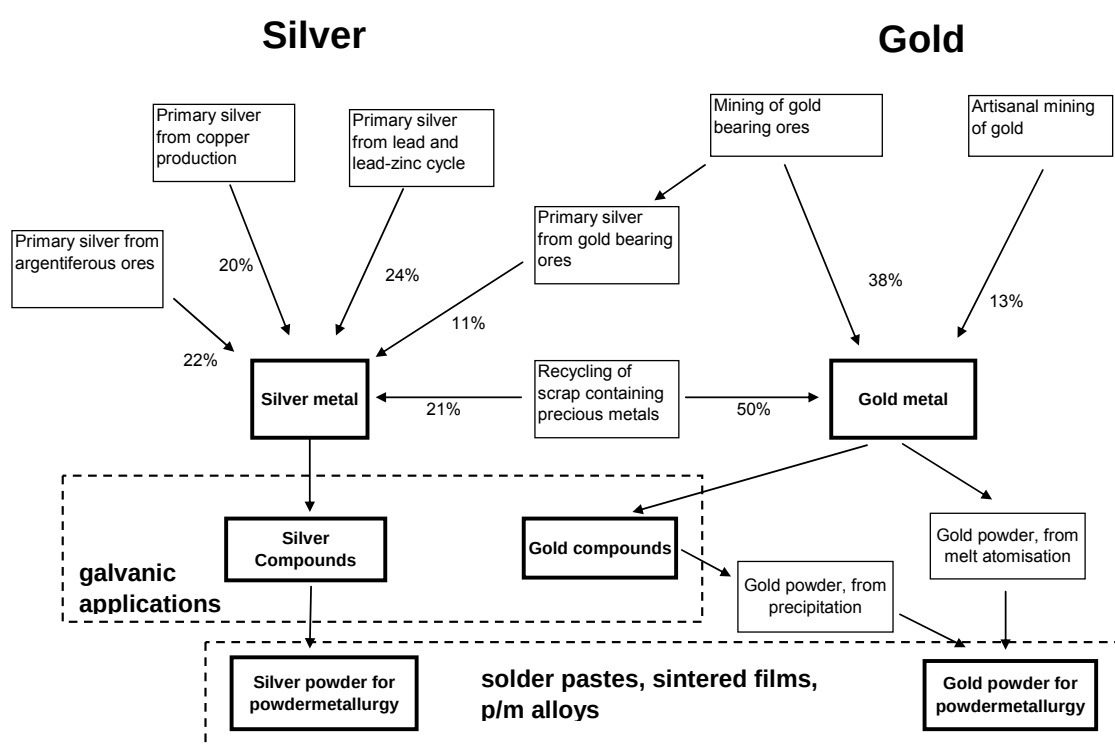


Fig. 5.1 General overview of Silver and Gold production.

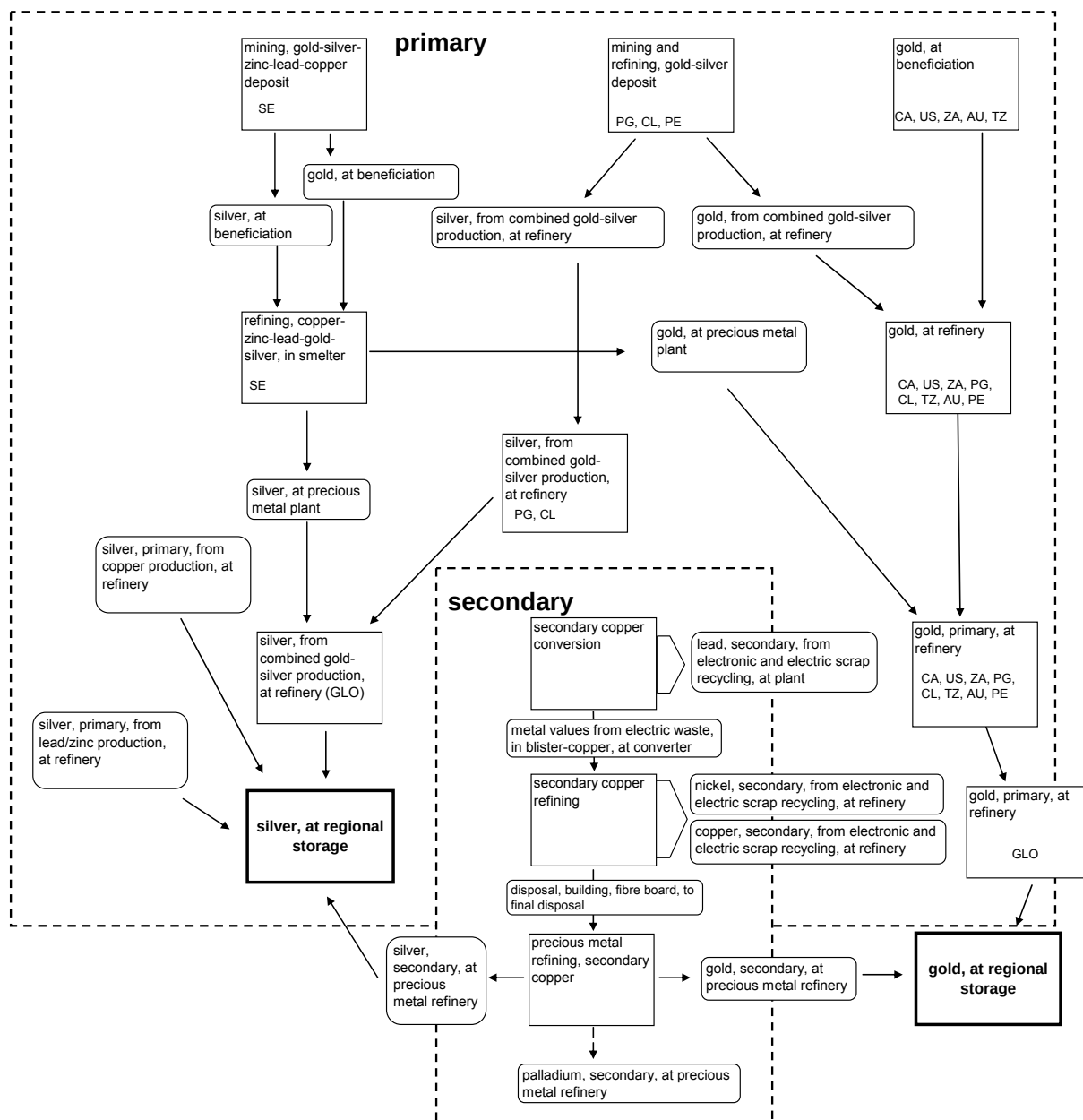


Fig. 5.2 System lay-out in ecoinvent data v2.0. Boxes with rounded edges indicate reference flows from multi-output processes.

## **5.1 Production of gold and silver from auriferous ores**

### **5.1.1 Gold and silver mining**

The description of gold production is based on Newmont (2004).

#### **Exploration**

The gold mining process begins with geologists discovering a deposit. After finding indications of gold, drill rigs are used to obtain samples from below the surface. These samples are analyzed for their gold content. If there is enough gold in the deposit to be produced economically, a mine is permitted and developed, if environmental regulations are fulfilled.

#### **Open Pit Mining**

The ore is mined in four steps: drilling, blasting, loading and hauling. In the case of a surface mine, a pattern of holes is drilled in the pit and filled with explosives. The explosives are detonated in order to break up the ground so large shovels or front-end loaders can load it into haul trucks.

#### **Underground Mining**

Some ore bodies are more economically mined underground. In this case, a tunnel called an adit or a shaft is dug into the earth. Sort tunnels leading from the adit or shaft, called stopes, are dug to access the ore. The surface containing the ore, called a face, is drilled and loaded with explosives. Following blasting, the broken ore is loaded onto electric trucks and taken to the surface. Once mining is completed in a particular stope, it is backfilled with a cement compound.

#### **Ore and Waste Haulage**

The haul trucks transport the ore to various areas for processing. The grade and type of ore determine the processing method used. Higher-grade ores are taken to a mill. Lower grade ores are taken to leach pads. Some ores may be stockpiled for later processing.

#### **Heap leaching**

##### *Heap leaching*

The ore is crushed or placed directly on lined leach pads where a dilute cyanide solution is applied to the surface of the heap. The solution percolates down through the ore, where it leaches the gold and flows to a central collection location. The solution is recovered in this closed system. The pregnant leach solution is fed to electrowinning cells and undergoes the same steps as described below from Electro-winning.

#### **Ore Processing**

##### *Milling*

The ore is fed into a series of grinding mills where steel balls grind the ore to a fine slurry or powder.

##### *Oxidization and leaching*

Some types of ore require further processing before gold is recovered. In this case, the slurry is pressure-oxidized in an autoclave before going to the leaching tanks or a dry powder is fed through a roaster in which it is oxidized using heat before being sent to the leaching tanks as a slurry. The slurry is thickened and runs through a series of leaching tanks. The gold in the slurry adheres to carbon in the tanks.

##### *Stripping*

The carbon is then moved into a stripping vessel where the gold is removed from the carbon by pumping a hot caustic solution through the carbon. The carbon is later recycled.

*Electro-winning*

The gold-bearing solution is pumped through electro-winning cells or through a zinc precipitation circuit where the gold is recovered from the solution.

*Smelting*

The gold is then melted in a furnace at about 1'064°C and poured into moulds, creating doré bars. Doré bars are unrefined gold bullion bars containing between 60% and 95% gold.

**Refining**

The bars are sent to a refinery for further processing into pure gold.

**5.1.2 Gold and silver production, Sweden (SE)**

The following data on gold and silver origin from the reports of the Swedish company Boliden (2006a), that operates mines and smelters in Sweden, Finland, Norway, Ireland and the Netherlands (copper, zinc, lead, gold and silver production). This LCI includes the data of the Swedish mines Aitik, Boliden and Garpenberg, which produce gold, silver and other metals.

**Mining infrastructure**

This data set includes land use as well as rough estimates on buildings and machines necessary in gold mining. The open pit mine “Aitik” produces the large part the ore. Therefore data for buildings and machines are based on the (generic) data set “non-ferrous metal, mine, surface” (and not “non-ferrous metal, mine, surface underground”). For details about infrastructure in mining, see the corresponding section of this report. It is assumed, that the quantities of machinery as well as surface and volume of buildings are equal to the generic mine. This approach is rough, and the uncertainty is rather high. As the mined ore from the generic mine is some orders of magnitude lower than the mine in focus, a proportional scaling by milled ore will lead to a higher mistake (Tab. 5.1). The lifetime of mines is assumed to be 50 years.

**Tab. 5.1: Quantity of machinery as well as surface and volume of buildings for a “non-ferrous metal mine, surface”, Source: see corresponding section in this report**

| Dataset                                          | Unit           | Quantity per mine (lifetime 50y) |
|--------------------------------------------------|----------------|----------------------------------|
| <i>infrastructure, mine, surface</i>             |                |                                  |
| building, multi-storey                           | m <sup>3</sup> | 2.75E+5                          |
| building, hall                                   | m <sup>2</sup> | 1.25E+4                          |
| industrial machine, heavy, unspecified, at plant | kg             | 8.50E+5                          |
| conveyor belt, at plant                          | m              | 6.00E+2                          |

Land use is based on the aggregated figures from the environmental report (Tab. 5.2). Not the entire area of the mine is in active use at one time; the value in Tab. 5.2 is assumed to be the average share of the mine area in use (see line “Disturbed area in active use”). The total area is occupied during an average of about 11 years according to the following formula:

$$\frac{\text{Disturbed area in active use}}{\text{Area disturbed during life of mine}} * \text{lifetime of mine} = \frac{4.65 * 10^7}{1.98 * 10^8} * 50y = 11.6y$$
, the occupation of the mine is calculated as Disturbed area in active use \* 50y = Occupation, mineral extraction site

Tab. 5.2 Land use of the Aitik, Boliden Area and Garpenberg Mines in 2005 (Boliden, 2006d)

| Land use                           |                | Aitik    | Boliden Area             | Garpenberg  | Sweden   | Remark:                                                                     |
|------------------------------------|----------------|----------|--------------------------|-------------|----------|-----------------------------------------------------------------------------|
| Type of mine:                      |                | open pit | underground,<br>open pit | underground |          |                                                                             |
| Area disturbed during life of mine | m <sup>2</sup> | n.a.     | n.a.                     | n.a.        | 1.98E+08 | Transformation, from unknown and Transformation, to mineral extraction site |
| Disturbed area in active use       | m <sup>2</sup> | n.a.     | n.a.                     | n.a.        | 4.65E+07 | Occupation, mineral extraction site, during 50 years                        |
| Area available for revegetation    | m <sup>2</sup> | n.a.     | n.a.                     | n.a.        | 4.53E+07 | added to "Transformation, to unknown"                                       |
| Area revegetated                   | m <sup>2</sup> | n.a.     | n.a.                     | n.a.        | 1.19E+06 | Transformation, to heterogeneous, agricultural                              |

Tab. 5.3 Unit process raw data of the polymetallic mines in Sweden

|                | Name                                             | Location | Infrastructure | Process | Unit | mine, gold-silver-zinc-lead-copper | Uncertainty | Type | Standard Deviation 95%                            | General Comment |
|----------------|--------------------------------------------------|----------|----------------|---------|------|------------------------------------|-------------|------|---------------------------------------------------|-----------------|
|                | Location<br>Infrastructure<br>Process<br>Unit    |          |                |         |      | SE<br>1<br>unit                    |             |      |                                                   |                 |
| product        | mine, gold-silver-zinc-lead-copper               | SE       | 1              | unit    |      | 1.00E+0                            |             |      |                                                   |                 |
| technosphere   | building, multi-storey                           | RER      | 1              | m3      |      | 2.75E+5                            | 1           | 2.39 | (4,5,5,5,5,5); Rough estimates                    |                 |
|                | building, hall                                   | CH       | 1              | m2      |      | 1.25E+4                            | 1           | 2.39 | (4,5,5,5,5,5); Rough estimates                    |                 |
|                | industrial machine, heavy, unspecified, at plant | RER      | 1              | kg      |      | 8.50E+5                            | 1           | 2.39 | (4,5,5,5,5,5); Rough estimates                    |                 |
|                | conveyor belt, at plant                          | RER      | 1              | m       |      | 6.00E+2                            | 1           | 2.39 | (4,5,5,5,5,5); Rough estimates                    |                 |
| resource, land | Transformation, from unknown                     | -        | -              | m2      |      | 1.98E+8                            | 1           | 2.01 | (1,3,2,1,1,3); Environmental report, Boliden 2005 |                 |
|                | Transformation, to mineral extraction site       | -        | -              | m2      |      | 1.98E+8                            | 1           | 2.01 | (1,3,2,1,1,3); Environmental report, Boliden 2005 |                 |
|                | Occupation, mineral extraction site              | -        | -              | m2a     |      | 2.33E+9                            | 1           | 1.51 | (1,3,2,1,1,3); Environmental report, Boliden 2005 |                 |
|                | Transformation, from mineral extraction site     | -        | -              | m2      |      | 1.98E+8                            | 1           | 2.01 | (1,3,2,1,1,3); Environmental report, Boliden 2005 |                 |
|                | Transformation, to heterogeneous, agricultural   | -        | -              | m2      |      | 1.19E+6                            | 1           | 2.01 | (1,3,2,1,1,3); Environmental report, Boliden 2005 |                 |
|                | Transformation, to unknown                       | -        | -              | m2      |      | 1.97E+8                            | 1           | 2.01 | (1,3,2,1,1,3); Environmental report, Boliden 2005 |                 |

## Mining

The energy and water requirements as well as the emissions are allocated to the different metals on the basis of their economic value calculated with the prices shown in Tab. 2.5. Tab 5.4 summarizes the concentrate production (expressed in metal quantity) of the considered plants for the year 2005.

**Tab 5.4: Production (quantity of metals) in year 2005 of the mines Aitik, Bolden and Garpenberg mines (Boliden, 2006a)**

| Production                       |           | Aitik    | Boliden Area | Garpenberg | Sweden          |                                  |
|----------------------------------|-----------|----------|--------------|------------|-----------------|----------------------------------|
| Ore:                             | kg        | 2.04E+10 | 1.69E+09     | 1.10E+09   | <b>2.32E+10</b> |                                  |
| Production Gold                  | kg        | 1.84E+03 | 2.43E+03     | 2.03E+02   | <b>4.47E+03</b> | Price per kg: 15682.1 US \$      |
| Production Silver                | kg        | 4.13E+04 | 8.72E+04     | 9.76E+04   | <b>2.26E+05</b> | Price per kg: 268.7 US \$        |
| Production Zinc                  | kg        |          | 8.73E+07     | 5.84E+07   | <b>1.46E+08</b> | Price per kg: 1.9 US \$          |
| Production Copper                | kg        | 6.56E+07 | 2.07E+07     | 5.65E+05   | <b>8.69E+07</b> | Price per kg: 4.5 US \$          |
| Production Lead                  | kg        |          | 3.04E+06     | 2.07E+05   | <b>3.25E+06</b> | Price per kg: 1.2 US \$          |
| Head grade                       |           |          |              |            |                 |                                  |
| Gold                             | kg/kg ore | 2.02E-06 | 5.17E-05     | 8.87E-05   | <b>1.93E-07</b> |                                  |
| Silver                           | kg/kg ore | 5.44E-04 | 1.39E-02     | 2.38E-02   | <b>9.75E-06</b> |                                  |
| Zinc                             | kg/kg ore | n.a.     | 5.17E-02     | 5.31E-02   | <b>6.28E-03</b> |                                  |
| Copper                           | kg/kg ore | 3.22E-03 | 1.23E-02     | 5.14E-04   | <b>3.75E-03</b> |                                  |
| Lead                             | kg/kg ore | n.a.     | 1.80E-03     | 1.88E-04   | <b>1.40E-04</b> |                                  |
| Recovery rate of metals from ore |           |          |              |            | <b>90%</b>      | Assumption, based on global data |

The recovery rate (proportion of valuable material physically recovered in the processing of ore<sup>8</sup>) is taken into account in order to define the resource extraction. It is estimated 90% on the basis of global data. The average concentration (head grade, expressed in kg metal per kg ore) of metals in ore for Sweden is calculated with the quantity of ore milled and the corresponding production rate.

The turnover of the production sites (Table 5.5) is calculated with the metal quantity multiplied with the prices (2005) in Tab. 2.5. The share in turnover is used as allocation factor for the respective metal.

**Table 5.5: Calculated turnover (prices 2005) and derived allocation factors for the Swedish mines**

|                   |       | Aitik           | Boliden Area    | Garpenberg      | Sweden          | Allocation factor SE |
|-------------------|-------|-----------------|-----------------|-----------------|-----------------|----------------------|
| Production Gold   | US \$ | 2.89E+07        | 3.81E+07        | 3.18E+06        | <b>7.01E+07</b> | 8.8 %                |
| Production Silver | US \$ | 1.11E+07        | 2.34E+07        | 2.62E+07        | <b>6.07E+07</b> | 7.6 %                |
| Production Zinc   | US \$ |                 | 1.64E+08        | 1.10E+08        | <b>2.73E+08</b> | 34.3 %               |
| Production Copper | US \$ | 2.94E+08        | 9.28E+07        | 2.53E+06        | <b>3.89E+08</b> | 48.8 %               |
| Production Lead   | US \$ |                 | 3.50E+06        | 2.38E+05        | <b>3.74E+06</b> | 0.5 %                |
| Total             | US \$ | <b>3.34E+08</b> | <b>3.22E+08</b> | <b>1.42E+08</b> | <b>7.97E+08</b> | 100 %                |

Energy and water requirements as well as emissions of the considered mines are summed up in Table 5.6. Water emissions are assumed to be emitted to rivers.

<sup>8</sup> The recovery rate is generally stated as a percentage of the material recovered compared to the total material originally present ([Barrick, 2007 #2767]).



**Table 5.6: Total energy, material requirements, and emissions of the Swedish multi-metal mines in 2005 (Boliden, 2006a); n.a. = not available**

| <b>Energy</b>                    |     | <b>Aitik</b> | <b>Boliden Area</b> | <b>Garpenberg</b> | <b>Sweden</b>   | <b>Remark:</b>                                                                                        |
|----------------------------------|-----|--------------|---------------------|-------------------|-----------------|-------------------------------------------------------------------------------------------------------|
| Energy consumption               | kWh | 4.32E+08     | 2.41E+08            | 1.23E+08          | <b>7.97E+08</b> | Share is calculated with global percentage of Boliden-Group                                           |
| Electricity                      | kWh | n.a.         | n.a.                | n.a.              | <b>5.74E+08</b> | SE, electricity mix                                                                                   |
| Coal/Coke                        | MJ  | n.a.         | n.a.                | n.a.              | <b>3.15E+08</b> | in LCI: hard coal, burned in industrial furnace                                                       |
| Diesel (mobile equipment)        | MJ  | n.a.         | n.a.                | n.a.              | <b>1.72E+08</b> | Diesel, burned in building machine                                                                    |
| Natural Gas                      | MJ  | n.a.         | n.a.                | n.a.              | <b>8.60E+07</b> | in LCI: natural gas, burned in industrial furnace, >100kW                                             |
| Oil                              | MJ  | n.a.         | n.a.                | n.a.              | <b>1.15E+08</b> | heavy fuel oil, burned in industrial furnace 1MW, non-modulating                                      |
| <b>Materials</b>                 |     |              |                     |                   |                 |                                                                                                       |
| Cyanide                          | kg  | n.a.         | n.a.                | n.a.              | <b>2.32E+05</b> | estimated with US-Data, based on processed Gold, in LCI as sodium cyanide                             |
| Lime                             | kg  | n.a.         | n.a.                | n.a.              | <b>5.18E+05</b> | estimated from Steward (1999) by processed gold and silver, accounted for as limestone, milled        |
| Caustic                          | kg  | n.a.         | n.a.                | n.a.              | <b>9.54E+04</b> | estimated from Steward (1999) by processed gold and silver, accounted for as sodium hydroxide         |
| Carbon                           | kg  | n.a.         | n.a.                | n.a.              | <b>1.39E+05</b> | estimated from Steward (1999) by processed gold and silver, accounted for as Charcoal, at plant       |
| Zinc                             | kg  | n.a.         | n.a.                | n.a.              | <b>2.37E+04</b> | Scaled from Steward (1999) by processed gold, accounted for as zinc, primary, at regional storage     |
| Sulphuric Acid                   | kg  | n.a.         | n.a.                | n.a.              | <b>5.45E+03</b> | estimated from Steward (1999) by processed gold and silver                                            |
| Hydrochloric Acid                | kg  | n.a.         | n.a.                | n.a.              | <b>1.12E+05</b> | estimated from Steward (1999) by processed gold and silver                                            |
| Explosives                       | kg  | n.a.         | n.a.                | n.a.              | <b>2.07E+07</b> | Share of Tara-mines is excluded, calculated by milled ore, partly ANFO explosives, in LCI: "blasting" |
| <b>Water resources</b>           |     |              |                     |                   |                 |                                                                                                       |
| Water use                        | m3  | 1.00E+06     | 5.25E+06            | 6.27E+05          | <b>6.87E+06</b> | Share is calculated with global percentage of Boliden-Group                                           |
| Seawater                         | m3  | n.a.         | n.a.                | n.a.              | <b>3.51E+06</b> | In LCI as water, salt, ocean                                                                          |
| Surface Water Use                | m3  | n.a.         | n.a.                | n.a.              | <b>1.58E+06</b> | In LCI as Water, river                                                                                |
| Groundwater Use                  | m3  | n.a.         | n.a.                | n.a.              | <b>2.75E+05</b> | In LCI as Water, well, in ground                                                                      |
| Municipal Water                  | m3  | n.a.         | n.a.                | n.a.              | <b>6.87E+04</b> | in LCI as tap, water                                                                                  |
| Recirculated Water               | m3  | n.a.         | n.a.                | n.a.              | <b>1.44E+06</b> | recycled Water, not accounted for in inventory                                                        |
| <b>Emissions to Water</b>        |     |              |                     |                   |                 |                                                                                                       |
| Zinc                             | kg  | 8.80E+01     | 1.70E+03            | 3.76E+02          | <b>2.16E+03</b> |                                                                                                       |
| Copper                           | kg  | 8.20E+01     | 1.55E+02            | 1.60E+01          | <b>2.53E+02</b> |                                                                                                       |
| Lead                             | kg  | 8.00E-01     | 9.00E+00            | 1.26E+02          | <b>1.36E+02</b> |                                                                                                       |
| <b>Emissions to Air</b>          |     |              |                     |                   |                 |                                                                                                       |
| Zinc                             | kg  | 4.50E+00     | n.a.                | 0.00E+00          | <b>5.11E+00</b> | Total based on Aitik-Mine, scaled by total milled ore                                                 |
| Copper                           | kg  | 1.63E+01     | n.a.                | 0.00E+00          | <b>1.85E+01</b> | Total based on Aitik-Mine, scaled by total milled ore                                                 |
| Lead                             | kg  | 5.30E+00     | n.a.                | 0.00E+00          | <b>6.02E+00</b> | Total based on Aitik-Mine, scaled by total milled ore                                                 |
| Carbon Dioxide (from explosives) | kg  | n.a.         | n.a.                | n.a.              | <b>3.30E+06</b> | Calculation, based on data from Newmont (2005)                                                        |

The unit process raw data of metal mining in Sweden are summarized in Tab. 5.7 and Tab. 5.8. Cyanide consumption is estimated with the average of US-Data from Barrick (2006b) and Newmont (2005b) (scaled by the produced gold), other material requirements are estimated with Australian data (scaled by the produced gold and silver). Material requirements are estimated only for the gold and silver production. Since the module “blasting” does not include the CO<sub>2</sub> emissions from explosives, they are calculated with the factor of the mines of Newmont (2005b) (0.16 kg CO<sub>2</sub> per kg explosive). Transport of materials is calculated on the base of standard distances (Frischknecht et al., 2004). Data about tailings disposal is approximated with data from Papua New Guinea.

Tab. 5.7: Unit process raw data of the metal mining in Sweden

|                                      | Name                                                                             | Location | Infrastructure | Process | Unit    | mining, gold-silver-zinc-lead-copper deposit | Uncertainty Type | Standard Deviation % | General Comment                                                                                       |
|--------------------------------------|----------------------------------------------------------------------------------|----------|----------------|---------|---------|----------------------------------------------|------------------|----------------------|-------------------------------------------------------------------------------------------------------|
|                                      | Location<br>Infrastructure<br>Unit                                               |          |                |         |         | SE<br>0<br>a                                 |                  |                      |                                                                                                       |
| allocated products                   | gold, from combined metal production, at beneficiation                           | SE       | 0              | kg      | 4.47E+3 |                                              |                  |                      |                                                                                                       |
|                                      | silver, from combined metal production, at beneficiation                         | SE       | 0              | kg      | 2.26E+5 |                                              |                  |                      |                                                                                                       |
|                                      | copper, from combined metal production, at beneficiation                         | SE       | 0              | kg      | 8.69E+7 |                                              |                  |                      |                                                                                                       |
|                                      | lead, from combined metal production, at beneficiation                           | SE       | 0              | kg      | 3.25E+6 |                                              |                  |                      |                                                                                                       |
|                                      | zinc, from combined metal production, at beneficiation                           | SE       | 0              | kg      | 1.46E+8 |                                              |                  |                      |                                                                                                       |
| technosphere                         | electricity, medium voltage, at grid                                             | SE       | 0              | kWh     | 5.74E+8 |                                              | 1                | 1.14                 | (2,4,1,3,1,3); Environmental report                                                                   |
|                                      | hard coal, burned in industrial furnace 1-10MW                                   | RER      | 0              | MJ      | 3.15E+8 |                                              | 1                | 1.14                 | (2,4,1,3,1,3); Environmental report                                                                   |
|                                      | diesel, burned in building machine                                               | GLO      | 0              | MJ      | 1.72E+8 |                                              | 1                | 1.14                 | (2,4,1,3,1,3); Environmental report                                                                   |
|                                      | natural gas, burned in industrial furnace >100kW                                 | RER      | 0              | MJ      | 8.60E+7 |                                              | 1                | 1.14                 | (2,4,1,3,1,3); Environmental report                                                                   |
|                                      | heavy fuel oil, burned in industrial furnace 1MW, non-modulating                 | RER      | 0              | MJ      | 1.15E+8 |                                              | 1                | 1.14                 | (2,4,1,3,1,3); Environmental report                                                                   |
|                                      | sodium cyanide, at plant                                                         | RER      | 0              | kg      | 2.32E+5 |                                              | 1                | 1.27                 | (1,3,3,5,3,3); Calculation, based about average US-data                                               |
|                                      | limestone, milled, packed, at plant                                              | CH       | 0              | kg      | 5.18E+5 |                                              | 1                | 1.27                 | (1,3,3,5,3,3); Calculation, based about thesis on Australian precious metal production, data for 1999 |
|                                      | sodium hydroxide, 50% in H <sub>2</sub> O, production mix, at plant              | RER      | 0              | kg      | 9.54E+4 |                                              | 1                | 1.27                 | (1,3,3,5,3,3); Calculation, based about thesis on Australian precious metal production, data for 1999 |
|                                      | zinc, primary, at regional storage                                               | RER      | 0              | kg      | 2.37E+4 |                                              | 1                | 1.27                 | (1,3,3,5,3,3); Calculation, based about thesis on Australian precious metal production, data for 1999 |
|                                      | charcoal, at plant                                                               | GLO      | 0              | kg      | 1.39E+5 |                                              | 1                | 1.27                 | (1,3,3,5,3,3); Calculation, based about thesis on Australian precious metal production, data for 1999 |
|                                      | sulphuric acid, liquid, at plant                                                 | RER      | 0              | kg      | 5.45E+3 |                                              | 1                | 1.27                 | (1,3,3,5,3,3); Calculation, based about thesis on Australian precious metal production, data for 1999 |
|                                      | blasting                                                                         | RER      | 0              | kg      | 2.07E+7 |                                              | 1                | 1.14                 | (2,4,1,3,1,3); Annual report                                                                          |
|                                      | hydrochloric acid, 30% in H <sub>2</sub> O, at plant                             | RER      | 0              | kg      | 1.12E+5 |                                              | 1                | 1.27                 | (1,3,3,5,3,3); Calculation, based about thesis on Australian precious metal production, data for 1999 |
|                                      | transport, lorry >16t, fleet average                                             | RER      | 0              | tkm     | 2.35E+6 |                                              | 1                | 2.09                 | (4,5,na,na,na,na); Calculation, based about standard distances                                        |
|                                      | transport, freight, rail                                                         | RER      | 0              | tkm     | 1.92E+7 |                                              | 1                | 2.09                 | (4,5,na,na,na,na); Calculation, based about standard distances                                        |
|                                      | mine, gold-silver-zinc-lead-copper                                               | SE       | 1              | unit    | 2.00E-2 |                                              | 1                | 3.02                 | (2,4,1,3,1,3); Environmental report                                                                   |
|                                      | tap water, at user                                                               | RER      | 0              | kg      | 6.87E+4 |                                              | 1                | 1.14                 | (2,4,1,3,1,3); Environmental report                                                                   |
|                                      | disposal, sulfidic tailings, off-site                                            | GLO      | 0              | kg      | 7.28E+8 |                                              | 1                | 1.14                 | (2,4,1,3,1,3); Environmental report                                                                   |
| emission air, low population density | Copper                                                                           | -        | -              | kg      | 1.85E+1 |                                              | 1                | 5.02                 | (2,4,1,3,1,3); Environmental report                                                                   |
|                                      | Carbon dioxide, fossil                                                           | -        | -              | kg      | 3.30E+6 |                                              | 1                | 1.14                 | (2,4,1,3,1,3); Environmental report                                                                   |
|                                      | Lead                                                                             | -        | -              | kg      | 6.02E+0 |                                              | 1                | 5.02                 | (2,4,1,3,1,3); Environmental report                                                                   |
|                                      | Zinc                                                                             | -        | -              | kg      | 5.11E+0 |                                              | 1                | 5.02                 | (2,4,1,3,1,3); Environmental report                                                                   |
|                                      | Heat, waste                                                                      | -        | -              | MJ      | 2.06E+9 |                                              | 1                | 1.14                 | (2,4,1,3,1,3); Environmental report, Calculation                                                      |
| emission water, river                | Copper, ion                                                                      | -        | -              | kg      | 2.53E+2 |                                              | 1                | 3.02                 | (2,4,1,3,1,3); Environmental report                                                                   |
|                                      | Lead                                                                             | -        | -              | kg      | 1.36E+2 |                                              | 1                | 5.02                 | (2,4,1,3,1,3); Environmental report                                                                   |
|                                      | Zinc, ion                                                                        | -        | -              | kg      | 2.16E+3 |                                              | 1                | 5.02                 | (2,4,1,3,1,3); Environmental report                                                                   |
| resource, in water                   | Water, salt, ocean                                                               | -        | -              | m3      | 3.51E+6 |                                              | 1                | 1.14                 | (2,4,1,3,1,3); Environmental report                                                                   |
|                                      | Water, well, in ground                                                           | -        | -              | m3      | 2.75E+5 |                                              | 1                | 1.14                 | (2,4,1,3,1,3); Environmental report                                                                   |
|                                      | Water, river                                                                     | -        | -              | m3      | 1.58E+6 |                                              | 1                | 1.14                 | (2,4,1,3,1,3); Environmental report                                                                   |
| resource, in ground                  | Gold, Au 9.7E-4%, Ag 9.7E-4%, Zn 0.63%, Cu 0.38%, Pb 0.014%, in ore, in ground   | -        | -              | kg      | 4.97E+3 |                                              | 1                | 1.14                 | (2,4,1,3,1,3); Environmental report                                                                   |
|                                      | Silver, Ag 9.7E-4%, Au 9.7E-4%, Zn 0.63%, Cu 0.38%, Pb 0.014%, in ore, in ground | -        | -              | kg      | 2.51E+5 |                                              | 1                | 1.14                 | (2,4,1,3,1,3); Environmental report                                                                   |
|                                      | Zinc, Zn 0.63%, Au 9.7E-4%, Ag 9.7E-4%, Cu 0.38%, Pb 0.014%, in ore, in ground   | -        | -              | kg      | 9.66E+7 |                                              | 1                | 1.14                 | (2,4,1,3,1,3); Environmental report                                                                   |
|                                      | Copper, Cu 0.38%, Au 9.7E-4%, Ag 9.7E-4%, Zn 0.63%, Pb 0.014%, in ore, in ground | -        | -              | kg      | 3.61E+6 |                                              | 1                | 1.14                 | (2,4,1,3,1,3); Environmental report                                                                   |
|                                      | Lead, Pb 0.014%, Au 9.7E-4%, Ag 9.7E-4%, Zn 0.63%, Cu 0.38%, in ore, in ground   | -        | -              | kg      | 1.62E+8 |                                              | 1                | 1.14                 | (2,4,1,3,1,3); Environmental report                                                                   |
|                                      |                                                                                  |          |                |         |         |                                              |                  |                      |                                                                                                       |

**Tab. 5.8: Allocation factors applied on mining data of the different metals. Material requirements were calculated on the basis of average data for gold and silver processing and extrapolated with the mined quantity of gold and silver. They are therefore only allocated to these two metals**

|                                      | Name                                                                             | Location | InfrastructureProcess | Unit | mining, gold-silver-zinc-lead-copper deposit | gold, from combined metal production, at beneficiation | silver, from combined metal production, at beneficiation | copper, from combined metal production, at beneficiation | lead, from combined metal production, at beneficiation | zinc, from combined metal production, at beneficiation |
|--------------------------------------|----------------------------------------------------------------------------------|----------|-----------------------|------|----------------------------------------------|--------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------|
|                                      |                                                                                  |          |                       |      | SE 0 kg                                      | SE 0 kg                                                | SE 0 kg                                                  | SE 0 kg                                                  | SE 0 kg                                                | SE 0 kg                                                |
| allocated products                   | gold, from combined metal production, at beneficiation                           | SE       | 0                     | kg   | 4.47E+3                                      | 100                                                    | -                                                        | -                                                        | -                                                      | -                                                      |
|                                      | silver, from combined metal production, at beneficiation                         | SE       | 0                     | kg   | 2.28E+5                                      | -                                                      | 100                                                      | -                                                        | -                                                      | -                                                      |
|                                      | copper, from combined metal production, at beneficiation                         | SE       | 0                     | kg   | 8.69E+7                                      | -                                                      | -                                                        | 100                                                      | -                                                      | -                                                      |
|                                      | lead, from combined metal production, at beneficiation                           | SE       | 0                     | kg   | 3.25E+6                                      | -                                                      | -                                                        | -                                                        | 100                                                    | -                                                      |
| technosphere                         | zinc, from combined metal production, at beneficiation                           | SE       | 0                     | kg   | 1.48E+8                                      | -                                                      | -                                                        | -                                                        | -                                                      | 100                                                    |
|                                      | electricity, medium voltage, at grid                                             | SE       | 0                     | kWh  | 5.74E+8                                      | 8.8                                                    | 7.6                                                      | 48.8                                                     | 0.5                                                    | 34.3                                                   |
|                                      | hard coal, burned in industrial furnace 1-10MW                                   | RER      | 0                     | MJ   | 3.15E+8                                      | 8.8                                                    | 7.6                                                      | 48.8                                                     | 0.5                                                    | 34.3                                                   |
|                                      | diesel, burned in building machine                                               | GLO      | 0                     | MJ   | 1.72E+8                                      | 8.8                                                    | 7.6                                                      | 48.8                                                     | 0.5                                                    | 34.3                                                   |
|                                      | natural gas, burned in industrial furnace >100kW                                 | RER      | 0                     | MJ   | 8.60E+7                                      | 8.8                                                    | 7.6                                                      | 48.8                                                     | 0.5                                                    | 34.3                                                   |
|                                      | heavy fuel oil, burned in industrial furnace 1MW, non-modulating                 | RER      | 0                     | MJ   | 1.15E+8                                      | 8.8                                                    | 7.6                                                      | 48.8                                                     | 0.5                                                    | 34.3                                                   |
|                                      | sodium cyanide, at plant                                                         | RER      | 0                     | kg   | 2.32E+5                                      | 54.0                                                   | 46.0                                                     | -                                                        | -                                                      | -                                                      |
|                                      | limestone, milled, packed, at plant                                              | CH       | 0                     | kg   | 5.18E+5                                      | 54.0                                                   | 46.0                                                     | -                                                        | -                                                      | -                                                      |
|                                      | sodium hydroxide, 50% in H <sub>2</sub> O, production mix, at plant              | RER      | 0                     | kg   | 9.54E+4                                      | 54.0                                                   | 46.0                                                     | -                                                        | -                                                      | -                                                      |
|                                      | zinc, primary, at regional storage                                               | RER      | 0                     | kg   | 2.37E+4                                      | 54.0                                                   | 46.0                                                     | -                                                        | -                                                      | -                                                      |
|                                      | charcoal, at plant                                                               | GLO      | 0                     | kg   | 1.39E+5                                      | 54.0                                                   | 46.0                                                     | -                                                        | -                                                      | -                                                      |
|                                      | sulphuric acid, liquid, at plant                                                 | RER      | 0                     | kg   | 5.45E+3                                      | 54.0                                                   | 46.0                                                     | -                                                        | -                                                      | -                                                      |
|                                      | blasting                                                                         | RER      | 0                     | kg   | 2.07E+7                                      | 8.8                                                    | 7.6                                                      | 48.8                                                     | 0.5                                                    | 34.3                                                   |
|                                      | hydrochloric acid, 30% in H <sub>2</sub> O, at plant                             | RER      | 0                     | kg   | 1.12E+5                                      | 54.0                                                   | 46.0                                                     | -                                                        | -                                                      | -                                                      |
|                                      | transport, lorry >16t, fleet average                                             | RER      | 0                     | tkm  | 2.35E+6                                      | 8.8                                                    | 7.6                                                      | 48.8                                                     | 0.5                                                    | 34.3                                                   |
|                                      | transport, freight, rail                                                         | RER      | 0                     | tkm  | 1.92E+7                                      | 8.8                                                    | 7.6                                                      | 48.8                                                     | 0.5                                                    | 34.3                                                   |
|                                      | mine, gold-silver-zinc-lead-copper                                               | SE       | 1                     | unit | 2.00E-2                                      | 8.8                                                    | 7.6                                                      | 48.8                                                     | 0.5                                                    | 34.3                                                   |
|                                      | tap water, at user                                                               | RER      | 0                     | kg   | 6.87E+4                                      | 8.8                                                    | 7.6                                                      | 48.8                                                     | 0.5                                                    | 34.3                                                   |
|                                      | disposal, sulfidic tailings, off-site                                            | GLO      | 0                     | kg   | 7.28E+8                                      | 8.8                                                    | 7.6                                                      | 48.8                                                     | 0.5                                                    | 34.3                                                   |
| emission air, low population density | Copper                                                                           | -        | -                     | kg   | 1.85E+1                                      | 8.8                                                    | 7.6                                                      | 48.8                                                     | 0.5                                                    | 34.3                                                   |
|                                      | Carbon dioxide, fossil                                                           | -        | -                     | kg   | 3.30E+6                                      | 8.8                                                    | 7.6                                                      | 48.8                                                     | 0.5                                                    | 34.3                                                   |
|                                      | Lead                                                                             | -        | -                     | kg   | 6.02E+0                                      | 8.8                                                    | 7.6                                                      | 48.8                                                     | 0.5                                                    | 34.3                                                   |
|                                      | Zinc                                                                             | -        | -                     | kg   | 5.11E+0                                      | 8.8                                                    | 7.6                                                      | 48.8                                                     | 0.5                                                    | 34.3                                                   |
|                                      | Heat, waste                                                                      | -        | -                     | MJ   | 2.06E+9                                      | 8.8                                                    | 7.6                                                      | 48.8                                                     | 0.5                                                    | 34.3                                                   |
| emission water, river                | Copper, ion                                                                      | -        | -                     | kg   | 2.53E+2                                      | 8.8                                                    | 7.6                                                      | 48.8                                                     | 0.5                                                    | 34.3                                                   |
|                                      | Lead                                                                             | -        | -                     | kg   | 1.36E+2                                      | 8.8                                                    | 7.6                                                      | 48.8                                                     | 0.5                                                    | 34.3                                                   |
|                                      | Zinc, ion                                                                        | -        | -                     | kg   | 2.16E+3                                      | 8.8                                                    | 7.6                                                      | 48.8                                                     | 0.5                                                    | 34.3                                                   |
| resource, in water                   | Water, salt, ocean                                                               | -        | -                     | m3   | 3.51E+6                                      | 8.8                                                    | 7.6                                                      | 48.8                                                     | 0.5                                                    | 34.3                                                   |
|                                      | Water, well, in ground                                                           | -        | -                     | m3   | 2.75E+5                                      | 8.8                                                    | 7.6                                                      | 48.8                                                     | 0.5                                                    | 34.3                                                   |
|                                      | Water, river                                                                     | -        | -                     | m3   | 1.58E+6                                      | 8.8                                                    | 7.6                                                      | 48.8                                                     | 0.5                                                    | 34.3                                                   |
| resource, in ground                  | Gold, Au 9.7E-4%, Ag 9.7E-4%, Zn 0.63%, Cu 0.38%, Pb 0.014%, in ore, in ground   | -        | -                     | kg   | 4.97E+3                                      | 100.0                                                  | -                                                        | -                                                        | -                                                      | -                                                      |
|                                      | Silver, Ag 9.7E-4%, Au 9.7E-4%, Zn 0.63%, Cu 0.38%, Pb 0.014%, in ore, in ground | -        | -                     | kg   | 2.51E+5                                      | -                                                      | 100.0                                                    | -                                                        | -                                                      | -                                                      |
|                                      | Zinc, Zn 0.63%, Au 9.7E-4%, Ag 9.7E-4%, Cu 0.38%, Pb 0.014%, in ore, in ground   | -        | -                     | kg   | 9.86E+7                                      | -                                                      | -                                                        | 100.0                                                    | -                                                      | -                                                      |
|                                      | Copper, Cu 0.38%, Au 9.7E-4%, Ag 9.7E-4%, Zn 0.63%, Pb 0.014%, in ore, in ground | -        | -                     | kg   | 3.61E+6                                      | -                                                      | -                                                        | -                                                        | 100.0                                                  | -                                                      |
|                                      | Lead, Pb 0.014%, Au 9.7E-4%, Ag 9.7E-4%, Zn 0.63%, Cu 0.38%, in ore, in ground   | -        | -                     | kg   | 1.62E+8                                      | -                                                      | -                                                        | -                                                        | -                                                      | 100.0                                                  |
|                                      |                                                                                  | -        | -                     |      |                                              |                                                        |                                                          |                                                          |                                                        |                                                        |

## Refining

The total production in 2005 of the smelter in Rönnskär amounted to 223'400 tonnes of copper, 27'000 tonnes of lead, 35'000 tonnes of zinc, 430 tonnes of silver and 17 tonnes of gold. A high share of the production is recycled metal, e.g. 74% for gold. Since the production of secondary gold and silver from scrap is investigated in a dedicated chapter of this report, only material requirement and emissions from the treatment of primary material is considered (see Tab. 5.10). Gold and Silver are treated in the PGM plant in Rönnskär, the other metals undergo other treatments.

Tab. 5.9 Metal production volume 2005 from primary material (Boliden, 2006b)

| Production              |    | from Scrap |     | from Mines   | Remark:                                                                                  |
|-------------------------|----|------------|-----|--------------|------------------------------------------------------------------------------------------|
| concentrate from Mines  | kg |            |     | 6.47E+08 82% | from Boliden (2005)                                                                      |
| Secondary Raw Materials | kg | 1.41E+08   | 18% |              | from Boliden (2005)                                                                      |
| contained lead          | kg | n.a.       |     | 2.22E+07     | calculated from output with share of concentrate input, in LCI: lead, at beneficiation   |
| contained copper        | kg | n.a.       |     | 1.83E+08     | calculated from output with share of concentrate input, in LCI: copper, at beneficiation |
| contained Zinc          | kg | n.a.       |     | 2.87E+07     | calculated from output with share of concentrate input, in LCI: zinc, at beneficiation   |
| contained gold          | kg | 1.25E+04   |     | 4.47E+03     | from Boliden(2005), in LCI: gold, at beneficiation                                       |
| contained silver        | kg | 2.04E+05   |     | 2.26E+05     | from Boliden (2005), in LCI: silver, at beneficiation                                    |

The allocation on the basis of the revenue (estimated with the prices of Tab. 2.5) results in 54% of the requirements and emissions attributed to gold and 46% attributed to silver for the PM-plant (Precious metal plant). The allocation for the refining of the other metals results in 6% attributed to zinc, 91% to copper and 3% to lead (see Tab. 5.10).

Tab. 5.10: Metal production volume in 2005, calculated turnover and shares in turnover of the different products in the Rönnskär Smelter (Boliden, 2006b). The revenue is calculated on the basis of the metal prices in Tab. 2.5 (average from the years 2004-2006). The PM-Plant is a sub process of the refinery, dedicated to the refinement of gold and silver.

| Allocation of metals |    | refining metals | Calc. Turnover  |             |                                                                  |
|----------------------|----|-----------------|-----------------|-------------|------------------------------------------------------------------|
| Production Gold      | kg | 4.47E+03        | 7.01E+07        | 6.8%        | refining gold-silver (PM-plant) 53.6% Price per kg: 15682.1 US\$ |
| Production Silver    | kg | 2.26E+05        | 6.07E+07        | 5.9%        | 46.4% Price per kg: 268.7 US\$                                   |
| Production Zinc      | kg | 2.87E+07        | 5.39E+07        | 5.2%        | refining, copper-zinc-lead 6.0% Price per kg: 1.9 US\$           |
| Production Copper    | kg | 1.83E+08        | 8.21E+08        | 79.6%       | 91.2% Price per kg: 4.5 US\$                                     |
| Production Lead      | kg | 2.22E+07        | 2.55E+07        | 2.5%        | 2.8% Price per kg: 1.2 US\$                                      |
| <b>Total</b>         |    |                 | <b>1.03E+09</b> | <b>100%</b> |                                                                  |

The total annual energy and material requirements as well as emissions of the refinery are summarized in Tab. 5.11. LPG use is approximated with the data set “natural gas, burned in industrial furnace”. The SO<sub>2</sub> emissions of the combustion of heavy fuel oil and coal are already included in the data set “heavy fuel oil, burned in industrial furnace 1MW, non-modulating” and “hard coal, burned in industrial furnace 1-10MW”, respectively. Particulates fractions are determined with standard data for non-ferrous metal industry<sup>9</sup>. Transport requirements are estimated to be 250 tkm lorry and 250 tkm train (average distance from the mines to the smelter, according to environmental report), for fossil fuels an average energy density of 40 MJ/kg is used. Where available, data from the PGM-plant is used to determine the requirement of gold- and silver treatment (Boliden, 2004). In this case, the energy requirements of the PGM-Plant are indicated in Tab. 5.11 as share of the whole refining process.

<sup>9</sup> PM2.5 82%, PM10 92% ([http://www.iiasa.ac.at/~rains/cgi-bin/rains\\_pm](http://www.iiasa.ac.at/~rains/cgi-bin/rains_pm), retrieved 6.4.2004).

**Tab. 5.11 Annual energy consumption, material requirements and emissions of the Rönnskär smelter (Boliden, 2004; Boliden, 2006b)**

| Energy                   |                 | refining copper-zinc-lead, Boliden (2006) |        | refining gold-silver, Boliden (2004) |       | Remark:                                                                                        |
|--------------------------|-----------------|-------------------------------------------|--------|--------------------------------------|-------|------------------------------------------------------------------------------------------------|
| Electricity              | kWh             | 4.95E+08                                  | 99.67% | 1.62E+06                             | 0.33% | SE, electricity mix, Source: Boliden (2004), (2006)                                            |
| Oil                      | MJ              | 3.76E+08                                  | 99.76% | 9.06E+05                             | 0.24% | In LCI as: heavy fuel oil, burned in industrial furnace, Source: Boliden (2004), (2006)        |
| Coke                     | MJ              |                                           | 0.00%  | 7.55E+04                             | 100%  | in LCI as: hard coal, burned in industrial furnace 1-10MW, Source: Boliden (2004), (2006)      |
| Natural Gas              | MJ              | 1.48E+07                                  | 99.96% | 5.59E+03                             | 0.04% | in LCI: natural gas, burned in industrial furnace, >100kW, Source: Boliden (2004), (2006)      |
| <b>Materials</b>         |                 |                                           |        |                                      |       |                                                                                                |
| Lime                     | kg              | 1.89E+06                                  |        |                                      |       | no lime is used for gold/silver purification, in LCI as "limestone, milled"                    |
| Caustic                  | kg              | 1.89E+07                                  |        |                                      |       | no caustic soda is used for gold/silver purification, in LCI as "sodium hydroxid"              |
| Carbon                   | kg              | 4.48E+07                                  |        |                                      |       | no carbon is used for gold/silver purification, in LCI as "charcoal"                           |
| Ammonia                  | kg              | 1.40E+06                                  |        |                                      |       | no ammonia is used for gold/silver purification, in LCI as "ammonia, liquid"                   |
| Oxygen                   | Nm <sup>3</sup> | 6.90E+07                                  |        |                                      |       | Source: Boliden (2006)                                                                         |
| Oxygen                   | kg              | 9.31E+07                                  |        |                                      |       | Recalculation with an average density of 1.35 kg/m <sup>3</sup>                                |
| Seawater                 | m <sup>3</sup>  |                                           |        | 8.17E+02                             | 100%  | data for PGM-Plant, allocated to gold and silver, in LCI as water, salt, ocean (Boliden, 2004) |
| <b>Emission to Water</b> |                 |                                           |        |                                      |       |                                                                                                |
| Copper                   | kg              | 3.86E+02                                  |        |                                      |       | Source: Boliden (2006)                                                                         |
| Lead                     | kg              | 4.19E+02                                  |        |                                      |       | Source: Boliden (2006)                                                                         |
| Zinc                     | kg              | 1.97E+03                                  |        |                                      |       | Source: Boliden (2006)                                                                         |
| Cadmium                  | kg              | 4.93E+01                                  |        |                                      |       | Source: Boliden (2006)                                                                         |
| Arsenic                  | kg              | 9.86E+02                                  |        |                                      |       | Source: Boliden (2006)                                                                         |
| Mercury                  | kg              | 1.15E+02                                  |        |                                      |       | Source: Boliden (2006)                                                                         |
| Nickel                   | kg              | 1.56E+02                                  |        |                                      |       | Source: Boliden (2006)                                                                         |
| <b>Emissions to air</b>  |                 |                                           |        |                                      |       |                                                                                                |
| Dust                     | kg              | 2.63E+04                                  | 99.75% | 6.45E+01                             | 0.25% | Source: Boliden (2004), (2006)                                                                 |
| Copper                   | kg              | 1.15E+03                                  |        |                                      |       | Source: Boliden (2006)                                                                         |
| Lead                     | kg              | 2.79E+04                                  |        |                                      |       | Source: Boliden (2006)                                                                         |
| Zinc                     | kg              | 4.19E+03                                  |        |                                      |       | Source: Boliden (2006)                                                                         |
| Cadmium                  | kg              | 5.75E+01                                  |        |                                      |       | Source: Boliden (2006)                                                                         |
| Arsenic                  | kg              | 1.89E+02                                  |        |                                      |       | Source: Boliden (2006)                                                                         |
| Mercury                  | kg              | 1.07E+02                                  |        |                                      |       | Source: Boliden (2006)                                                                         |
| Sulphur Dioxide          | kg              | 3.16E+06                                  |        |                                      |       | not accounted in LCI, as the emissions are included in the datasets of diesel and naturalgas   |
| Selenium                 | kg              | 1.61E+01                                  |        |                                      |       | data for PGM-Plant, allocated to gold and silver, Source: Boliden (2004)                       |
| Dioxin                   | kg              | 1.81E-03                                  |        |                                      |       | Source: Boliden (2006)                                                                         |
| <b>Waste</b>             |                 |                                           |        |                                      |       |                                                                                                |
| combustible Waste        | kg              | 1.72E+05                                  |        |                                      |       | disposal, municipal solid waste, 22.9% water, to municipal incineration                        |
| Wood waste               | kg              | 3.90E+05                                  |        |                                      |       | disposal, wood untreated, 20% water, to municipal incineration                                 |
| Metals                   | kg              | 2.79E+05                                  |        |                                      |       | not accounted in LCI, assumed to be recycled                                                   |
| Paper                    | kg              | 5.75E+03                                  |        |                                      |       | disposal, paper, 11.2% water, to municipal incineration                                        |
| Corrugated cardboard     | kg              | 3.61E+04                                  |        |                                      |       | disposal, packaging cardboard, 19.6% water, to municipal incineration                          |
| Landfill waste           | kg              | 2.09E+05                                  |        |                                      |       | disposal, sulfidic tailings, off-site                                                          |
| Oil/water                | kg              | 1.85E+05                                  |        |                                      |       | disposal, refinery sludge, 89.5% water, to hazardous waste incineration                        |

## 5. System Characterisation

Tab. 5.12: Unit Raw data for “refining, copper-zinc-lead-gold-silver in smelter”

|                                      | Name                                                                    | Location | Infrastructure | Process | Unit    | refining, copper-zinc-lead-gold-silver, in smelter | UncertaintyType | StandardDeviation65 | %                                                       | GeneralComment |
|--------------------------------------|-------------------------------------------------------------------------|----------|----------------|---------|---------|----------------------------------------------------|-----------------|---------------------|---------------------------------------------------------|----------------|
|                                      | Location<br>Infrastructure<br>Unit                                      |          |                |         |         | SE<br>0<br>a                                       |                 |                     |                                                         |                |
| allocated products                   | zinc, from combined metal production, at refinery                       | SE       | 0              | kg      | 2.87E+7 |                                                    |                 |                     |                                                         |                |
|                                      | copper, from combined metal production, at refinery                     | SE       | 0              | kg      | 1.83E+8 |                                                    |                 |                     |                                                         |                |
|                                      | lead, from combined metal production, at refinery                       | SE       | 0              | kg      | 2.22E+7 |                                                    |                 |                     |                                                         |                |
|                                      | gold, from combined metal production, at refinery                       | SE       | 0              | kg      | 4.47E+3 |                                                    |                 |                     |                                                         |                |
|                                      | silver, from combined metal production, at refinery                     | SE       | 0              | kg      | 2.26E+5 |                                                    |                 |                     |                                                         |                |
| technosphere                         | zinc, from combined metal production, at beneficiation                  | SE       | 0              | kg      | 2.87E+7 | 1                                                  | 1.13            | (1,4,1,3,1,3);      | Environmental report                                    |                |
|                                      | copper, from combined metal production, at beneficiation                | SE       | 0              | kg      | 1.83E+8 | 1                                                  | 1.13            | (1,4,1,3,1,3);      | Environmental report                                    |                |
|                                      | lead, from combined metal production, at beneficiation                  | SE       | 0              | kg      | 2.22E+7 | 1                                                  | 1.13            | (1,4,1,3,1,3);      | Environmental report                                    |                |
|                                      | gold, from combined metal production, at beneficiation                  | SE       | 0              | kg      | 4.47E+3 | 1                                                  | 1.13            | (1,4,1,3,1,3);      | Environmental report                                    |                |
|                                      | silver, from combined metal production, at beneficiation                | SE       | 0              | kg      | 2.26E+5 | 1                                                  | 1.13            | (1,4,1,3,1,3);      | Environmental report                                    |                |
|                                      | electricity, medium voltage, at grid                                    | SE       | 0              | kWh     | 4.95E+8 | 1                                                  | 1.13            | (1,4,1,3,1,3);      | Environmental report, Boliden 2004 / 2006               |                |
|                                      | heavy fuel oil, burned in industrial furnace 1MW, non-modulating        | RER      | 0              | MJ      | 3.76E+8 | 1                                                  | 1.13            | (1,4,1,3,1,3);      | Environmental report, Boliden 2004 / 2006               |                |
|                                      | hard coal, burned in industrial furnace 1-10MW                          | RER      | 0              | MJ      | 3.76E+8 | 1                                                  | 1.13            | (1,4,1,3,1,3);      | Environmental report, Boliden 2004 / 2006               |                |
|                                      | natural gas, burned in industrial furnace >100kW                        | RER      | 0              | MJ      | 1.48E+7 | 1                                                  | 1.13            | (1,4,1,3,1,3);      | Environmental report, Boliden 2004 / 2006               |                |
|                                      | limestone, milled, packed, at plant                                     | CH       | 0              | kg      | 1.89E+6 | 1                                                  | 1.13            | (1,4,1,3,1,3);      | Environmental report, Boliden 2006                      |                |
|                                      | sodium hydroxide, 50% in H2O, production mix, at plant                  | RER      | 0              | kg      | 1.89E+7 | 1                                                  | 1.13            | (1,4,1,3,1,3);      | Environmental report, Boliden 2006                      |                |
|                                      | charcoal, at plant                                                      | GLO      | 0              | kg      | 4.48E+7 | 1                                                  | 1.13            | (1,4,1,3,1,3);      | Environmental report, Boliden 2006                      |                |
|                                      | ammonia, liquid, at regional storehouse                                 | RER      | 0              | kg      | 1.40E+6 | 1                                                  | 1.13            | (1,4,1,3,1,3);      | Environmental report, Boliden 2006                      |                |
|                                      | oxygen, liquid, at plant                                                | RER      | 0              | kg      | 9.31E+7 | 1                                                  | 1.13            | (1,4,1,3,1,3);      | Environmental report, Boliden 2004 / 2006               |                |
|                                      | facilities anode refinery, secondary copper                             | SE       | 1              | unit    | 2.67E-2 | 1                                                  | 3.55            | (5,5,2,2,4,5);      | Rough Estimation                                        |                |
|                                      | facilities precious metal refinery                                      | SE       | 1              | unit    | 7.52E-2 | 1                                                  | 3.55            | (5,5,2,2,4,5);      | Rough Estimation                                        |                |
|                                      | transport, lorry >16t, fleet average                                    | RER      | 0              | tkm     | 2.61E+7 | 1                                                  | 2.09            | (4,5,na,na,na,na);  | Estimation                                              |                |
|                                      | transport, freight, rail                                                | RER      | 0              | tkm     | 2.61E+7 | 1                                                  | 2.09            | (4,5,na,na,na,na);  | Estimation                                              |                |
| resource, in water                   | Water, salt, ocean                                                      | -        | -              | m3      | 8.17E+2 | 1                                                  | 1.30            | (4,5,na,na,na,na);  | Estimation                                              |                |
| emission air, low population density | Particulates, < 2.5 um                                                  | -        | -              | kg      | 2.16E+4 | 1                                                  | 3.02            | (1,4,1,3,1,3);      | Calculation based on environmental report, Boliden 2004 |                |
|                                      | Particulates, > 2.5 um, and < 10um                                      | -        | -              | kg      | 2.63E+3 | 1                                                  | 2.02            | (1,4,1,3,1,3);      | Calculation based on environmental report, Boliden 2004 |                |
|                                      | Particulates, > 10 um                                                   | -        | -              | kg      | 2.10E+3 | 1                                                  | 1.52            | (1,4,1,3,1,3);      | Calculation based on environmental report, Boliden 2004 |                |
|                                      | Copper                                                                  | -        | -              | kg      | 1.15E+3 | 1                                                  | 5.02            | (1,4,1,3,1,3);      | Environmental report, Boliden 2006                      |                |
|                                      | Lead                                                                    | -        | -              | kg      | 2.79E+4 | 1                                                  | 5.02            | (1,4,1,3,1,3);      | Environmental report, Boliden 2006                      |                |
|                                      | Zinc                                                                    | -        | -              | kg      | 4.19E+3 | 1                                                  | 5.02            | (1,4,1,3,1,3);      | Environmental report, Boliden 2006                      |                |
|                                      | Cadmium                                                                 | -        | -              | kg      | 5.75E+1 | 1                                                  | 5.02            | (1,4,1,3,1,3);      | Environmental report, Boliden 2006                      |                |
|                                      | Arsenic                                                                 | -        | -              | kg      | 1.89E+2 | 1                                                  | 5.02            | (1,4,1,3,1,3);      | Environmental report, Boliden 2006                      |                |
|                                      | Mercury                                                                 | -        | -              | kg      | 1.07E+2 | 1                                                  | 5.02            | (1,4,1,3,1,3);      | Environmental report, Boliden 2006                      |                |
|                                      | Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin                | -        | -              | kg      | 1.81E-3 | 1                                                  | 3.02            | (1,4,1,3,1,3);      | Environmental report, Boliden 2006                      |                |
|                                      | Selenium                                                                | -        | -              | kg      | 1.61E+1 | 1                                                  | 5.02            | (1,4,1,3,1,3);      | Environmental report, Boliden 2004 / 2006               |                |
|                                      | Heat, waste                                                             | -        | -              | MJ      | 1.78E+9 | 1                                                  | 1.13            | (1,4,1,3,1,3);      | Calculation                                             |                |
| emission water, ocean                | Copper, ion                                                             | -        | -              | kg      | 3.86E+2 | 1                                                  | 3.02            | (1,4,1,3,1,3);      | Environmental report, Boliden 2006                      |                |
|                                      | Lead                                                                    | -        | -              | kg      | 4.19E+2 | 1                                                  | 5.02            | (1,4,1,3,1,3);      | Environmental report, Boliden 2006                      |                |
|                                      | Zinc, ion                                                               | -        | -              | kg      | 1.97E+3 | 1                                                  | 5.02            | (1,4,1,3,1,3);      | Environmental report, Boliden 2006                      |                |
|                                      | Cadmium, ion                                                            | -        | -              | kg      | 4.93E+1 | 1                                                  | 3.02            | (1,4,1,3,1,3);      | Environmental report, Boliden 2006                      |                |
|                                      | Arsenic, ion                                                            | -        | -              | kg      | 9.86E+2 | 1                                                  | 5.02            | (1,4,1,3,1,3);      | Environmental report, Boliden 2006                      |                |
|                                      | Mercury                                                                 | -        | -              | kg      | 1.15E+2 | 1                                                  | 5.02            | (1,4,1,3,1,3);      | Environmental report, Boliden 2006                      |                |
|                                      | Nickel, ion                                                             | -        | -              | kg      | 1.56E+2 | 1                                                  | 5.02            | (1,4,1,3,1,3);      | Environmental report, Boliden 2006                      |                |
| technosphere                         | disposal, municipal solid waste, 22.9% water, to municipal incineration | CH       | 0              | kg      | 1.72E+5 | 1                                                  | 1.13            | (1,4,1,3,1,3);      | Environmental report, Boliden 2006                      |                |
|                                      | disposal, wood untreated, 20% water, to municipal incineration          | CH       | 0              | kg      | 3.90E+5 | 1                                                  | 1.13            | (1,4,1,3,1,3);      | Environmental report, Boliden 2006                      |                |
|                                      | disposal, paper, 11.2% water, to municipal incineration                 | CH       | 0              | kg      | 5.75E+3 | 1                                                  | 1.13            | (1,4,1,3,1,3);      | Environmental report, Boliden 2006                      |                |
|                                      | disposal, packaging cardboard, 19.6% water, to municipal incineration   | CH       | 0              | kg      | 3.61E+4 | 1                                                  | 1.13            | (1,4,1,3,1,3);      | Environmental report, Boliden 2006                      |                |
|                                      | disposal, sulfidic tailings, off-site                                   | GLO      | 0              | kg      | 2.09E+5 | 1                                                  | 1.13            | (1,4,1,3,1,3);      | Environmental report, Boliden 2006                      |                |
|                                      | disposal, refinery sludge, 89.5% water, to hazardous waste incineration | CH       | 0              | kg      | 1.85E+5 | 1                                                  | 1.13            | (1,4,1,3,1,3);      | Environmental report, Boliden 2006                      |                |

**Tab. 5.13 Allocation factors of the data set “refining, copper-zinc-lead-gold-silver in smelter” (Boliden, 2004; Boliden, 2006b)**

|                                                         | Name                                                                    | Location | InfrastructureProcess | Unit | refining, copper-zinc-lead-gold-silver, in smelter | zinc, from combined metal production, at refinery | copper, from combined metal production, at refinery | lead, from combined metal production, at refinery | gold, from combined metal production, at refinery | silver, from combined metal production, at refinery |
|---------------------------------------------------------|-------------------------------------------------------------------------|----------|-----------------------|------|----------------------------------------------------|---------------------------------------------------|-----------------------------------------------------|---------------------------------------------------|---------------------------------------------------|-----------------------------------------------------|
|                                                         |                                                                         |          |                       |      | SE                                                 | SE                                                | SE                                                  | SE                                                | SE                                                | SE                                                  |
|                                                         |                                                                         |          |                       |      | 0<br>a                                             | 0<br>kg                                           | 0<br>kg                                             | 0<br>kg                                           | 0<br>kg                                           | 0<br>kg                                             |
| allocated products                                      | zinc, from combined metal production, at refinery                       | SE       | 0                     | kg   | 2.87E+7                                            | 100                                               | -                                                   | -                                                 | -                                                 | -                                                   |
|                                                         | copper, from combined metal production, at refinery                     | SE       | 0                     | kg   | 1.83E+8                                            | -                                                 | 100                                                 | -                                                 | -                                                 | -                                                   |
|                                                         | lead, from combined metal production, at refinery                       | SE       | 0                     | kg   | 2.22E+7                                            | -                                                 | -                                                   | 100                                               | -                                                 | -                                                   |
|                                                         | gold, from combined metal production, at refinery                       | SE       | 0                     | kg   | 4.47E+3                                            | -                                                 | -                                                   | -                                                 | 100                                               | -                                                   |
|                                                         | silver, from combined metal production, at refinery                     | SE       | 0                     | kg   | 2.26E+5                                            | -                                                 | -                                                   | -                                                 | -                                                 | 100                                                 |
| technosphere                                            | zinc, from combined metal production, at beneficiation                  | SE       | 0                     | kg   | 2.87E+7                                            | 100.0                                             | -                                                   | -                                                 | -                                                 | -                                                   |
|                                                         | copper, from combined metal production, at beneficiation                | SE       | 0                     | kg   | 1.83E+8                                            | -                                                 | 100.0                                               | -                                                 | -                                                 | -                                                   |
|                                                         | lead, from combined metal production, at beneficiation                  | SE       | 0                     | kg   | 2.22E+7                                            | -                                                 | -                                                   | 100.0                                             | -                                                 | -                                                   |
|                                                         | gold, from combined metal production, at beneficiation                  | SE       | 0                     | kg   | 4.47E+3                                            | -                                                 | -                                                   | -                                                 | 100.0                                             | -                                                   |
|                                                         | silver, from combined metal production, at beneficiation                | SE       | 0                     | kg   | 2.26E+5                                            | -                                                 | -                                                   | -                                                 | -                                                 | 100.0                                               |
|                                                         | electricity, medium voltage, at grid                                    | SE       | 0                     | kWh  | 4.95E+8                                            | 6.0                                               | 90.9                                                | 2.8                                               | 0.18                                              | 0.15                                                |
|                                                         | heavy fuel oil, burned in industrial furnace 1MW, non-modulating        | RER      | 0                     | MJ   | 3.76E+8                                            | 6.0                                               | 91.0                                                | 2.8                                               | 0.1                                               | 0.1                                                 |
|                                                         | hard coal, burned in industrial furnace 1-10MW                          | RER      | 0                     | MJ   | 3.76E+8                                            | -                                                 | -                                                   | -                                                 | 53.6                                              | 46.4                                                |
|                                                         | natural gas, burned in industrial furnace >100kW                        | RER      | 0                     | MJ   | 1.48E+7                                            | 6.0                                               | 91.1                                                | 2.8                                               | 0.0                                               | 0.0                                                 |
|                                                         | limestone, milled, packed, at plant                                     | CH       | 0                     | kg   | 1.89E+6                                            | 6.0                                               | 91.2                                                | 2.8                                               | -                                                 | -                                                   |
|                                                         | sodium hydroxide, 50% in H2O, production mix, at plant                  | RER      | 0                     | kg   | 1.89E+7                                            | 6.0                                               | 91.2                                                | 2.8                                               | -                                                 | -                                                   |
|                                                         | charcoal, at plant                                                      | GLO      | 0                     | kg   | 4.48E+7                                            | 6.0                                               | 91.2                                                | 2.8                                               | -                                                 | -                                                   |
|                                                         | ammonia, liquid, at regional storehouse                                 | RER      | 0                     | kg   | 1.40E+6                                            | 6.0                                               | 91.2                                                | 2.8                                               | -                                                 | -                                                   |
|                                                         | oxygen, liquid, at plant                                                | RER      | 0                     | kg   | 9.31E+7                                            | 5.2                                               | 79.6                                                | 2.5                                               | 6.8                                               | 5.9                                                 |
|                                                         | facilities anode refinery, secondary copper                             | SE       | 1                     | unit | 2.67E+2                                            | 5.2                                               | 79.6                                                | 2.5                                               | 6.8                                               | 5.9                                                 |
|                                                         | facilities precious metal refinery                                      | SE       | 1                     | unit | 7.52E+2                                            | 5.2                                               | 79.6                                                | 2.5                                               | 6.8                                               | 5.9                                                 |
|                                                         | transport, lorry >16t, fleet average                                    | RER      | 0                     | tkm  | 2.61E+7                                            | 5.2                                               | 79.6                                                | 2.5                                               | 6.8                                               | 5.9                                                 |
|                                                         | transport, freight, rail                                                | RER      | 0                     | tkm  | 2.61E+7                                            | 5.2                                               | 79.6                                                | 2.5                                               | 6.8                                               | 5.9                                                 |
|                                                         | Water, salt, ocean                                                      | -        | -                     | m3   | 8.17E+2                                            | -                                                 | -                                                   | -                                                 | 53.6                                              | 46.4                                                |
| resource, in water emission air, low population density | Particulates, < 2.5 um                                                  | -        | -                     | kg   | 2.16E+4                                            | 6.0                                               | 91.0                                                | 2.8                                               | 0.1                                               | 0.1                                                 |
|                                                         | Particulates, > 2.5 um, and < 10um                                      | -        | -                     | kg   | 2.63E+3                                            | 6.0                                               | 91.0                                                | 2.8                                               | 0.1                                               | 0.1                                                 |
|                                                         | Particulates, > 10 um                                                   | -        | -                     | kg   | 2.10E+3                                            | 6.0                                               | 91.0                                                | 2.8                                               | 0.1                                               | 0.1                                                 |
|                                                         | Copper                                                                  | -        | -                     | kg   | 1.15E+3                                            | 5.2                                               | 79.6                                                | 2.5                                               | 6.8                                               | 5.9                                                 |
|                                                         | Lead                                                                    | -        | -                     | kg   | 2.79E+4                                            | 5.2                                               | 79.6                                                | 2.5                                               | 6.8                                               | 5.9                                                 |
|                                                         | Zinc                                                                    | -        | -                     | kg   | 4.19E+3                                            | 5.2                                               | 79.6                                                | 2.5                                               | 6.8                                               | 5.9                                                 |
|                                                         | Cadmium                                                                 | -        | -                     | kg   | 5.75E+1                                            | 5.2                                               | 79.6                                                | 2.5                                               | 6.8                                               | 5.9                                                 |
|                                                         | Arsenic                                                                 | -        | -                     | kg   | 1.89E+2                                            | 5.2                                               | 79.6                                                | 2.5                                               | 6.8                                               | 5.9                                                 |
|                                                         | Mercury                                                                 | -        | -                     | kg   | 1.07E+2                                            | 5.2                                               | 79.6                                                | 2.5                                               | 6.8                                               | 5.9                                                 |
|                                                         | Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin                | -        | -                     | kg   | 1.81E-3                                            | 5.2                                               | 79.6                                                | 2.5                                               | 6.8                                               | 5.9                                                 |
|                                                         | Selenium                                                                | -        | -                     | kg   | 1.61E+1                                            | -                                                 | -                                                   | -                                                 | 53.6                                              | 46.4                                                |
|                                                         | Heat, waste                                                             | -        | -                     | MJ   | 1.78E+9                                            | 5.2                                               | 79.6                                                | 2.5                                               | 6.8                                               | 5.9                                                 |
|                                                         | Copper, ion                                                             | -        | -                     | kg   | 3.86E+2                                            | 5.2                                               | 79.6                                                | 2.5                                               | 6.8                                               | 5.9                                                 |
|                                                         | Lead                                                                    | -        | -                     | kg   | 4.19E+2                                            | 5.2                                               | 79.6                                                | 2.5                                               | 6.8                                               | 5.9                                                 |
|                                                         | Zinc, ion                                                               | -        | -                     | kg   | 1.97E+3                                            | 5.2                                               | 79.6                                                | 2.5                                               | 6.8                                               | 5.9                                                 |
|                                                         | Cadmium, ion                                                            | -        | -                     | kg   | 4.93E+1                                            | 5.2                                               | 79.6                                                | 2.5                                               | 6.8                                               | 5.9                                                 |
|                                                         | Arsenic, ion                                                            | -        | -                     | kg   | 9.86E+2                                            | 5.2                                               | 79.6                                                | 2.5                                               | 6.8                                               | 5.9                                                 |
|                                                         | Mercury                                                                 | -        | -                     | kg   | 1.15E+2                                            | 5.2                                               | 79.6                                                | 2.5                                               | 6.8                                               | 5.9                                                 |
|                                                         | Nickel, ion                                                             | -        | -                     | kg   | 1.56E+2                                            | 5.2                                               | 79.6                                                | 2.5                                               | 6.8                                               | 5.9                                                 |
| technosphere                                            | disposal, municipal solid waste, 22.9% water, to municipal incineration | CH       | 0                     | kg   | 1.72E+5                                            | 5.2                                               | 79.6                                                | 2.5                                               | 6.8                                               | 5.9                                                 |
|                                                         | disposal, wood untreated, 20% water, to municipal incineration          | CH       | 0                     | kg   | 3.90E+5                                            | 5.2                                               | 79.6                                                | 2.5                                               | 6.8                                               | 5.9                                                 |
|                                                         | disposal, paper, 11.2% water, to municipal incineration                 | CH       | 0                     | kg   | 5.75E+3                                            | 5.2                                               | 79.6                                                | 2.5                                               | 6.8                                               | 5.9                                                 |
|                                                         | disposal, packaging cardboard, 19.6% water, to municipal incineration   | CH       | 0                     | kg   | 3.61E+4                                            | 5.2                                               | 79.6                                                | 2.5                                               | 6.8                                               | 5.9                                                 |
|                                                         | disposal, sulfidic tailings, off-site                                   | GLO      | 0                     | kg   | 2.09E+5                                            | 5.2                                               | 79.6                                                | 2.5                                               | 6.8                                               | 5.9                                                 |
|                                                         | disposal, refinery sludge, 89.5% water, to hazardous waste incineration | CH       | 0                     | kg   | 1.85E+5                                            | 5.2                                               | 79.6                                                | 2.5                                               | 6.8                                               | 5.9                                                 |

Tab. 5.14 Ecospold Meta information of metal production in Sweden

| Name                       | mine, gold-silver-zinc-lead-copper                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | mining, gold-silver-zinc-lead-copper deposit                                                                                | refining, copper-zinc-lead-gold-silver, in smelter       |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| Location                   | SE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | SE                                                                                                                          | SE                                                       |
| InfrastructureProcess      | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 0                                                                                                                           | 0                                                        |
| Unit                       | unit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | a                                                                                                                           | a                                                        |
| Type                       | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 5                                                                                                                           | 5                                                        |
| Version                    | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2.0                                                                                                                         | 2.0                                                      |
| energyValues               | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 0                                                                                                                           | 0                                                        |
| LanguageCode               | en                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | en                                                                                                                          | en                                                       |
| LocalLanguageCode          | de                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | de                                                                                                                          | de                                                       |
| Person                     | 44                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 44                                                                                                                          | 44                                                       |
| QualityNetwork             | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1                                                                                                                           | 1                                                        |
| DataSetRelatesToProduct    | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1                                                                                                                           | 1                                                        |
| IncludedProcesses          | <p>Land use, buildings and machinery of a combined metal mine</p> <p>This data set includes combined mining of gold, silver, copper, zinc and lead in Sweden incl. energy and material use, water and air emissions of metals and land use.</p> <p>This data set includes the extraction of metals and by products from concentrates and recycled raw materials. It includes energy and water consumption, concentrate inputs, transports, waste disposal, infrastructure use as well as emissions into air and water.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                             |                                                          |
| Amount                     | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1                                                                                                                           | 1                                                        |
| LocalName                  | Mine, Gold-Silber-Zink-Blei-Kupfer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Abbau Gold-Silber-Zink-Blei-Kupfer-Lagerstätte                                                                              | Raffination, Kupfer-Zink-Blei-Gold-Silber in Metallhütte |
| Synonyms                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                             |                                                          |
| GeneralComment             | <p>This data set includes land use of the Swedish considered mines incl. revegetated areas as well as generic estimates of buildings and machinery of the mine based on the data set non-ferrous metal, mine, surface/underground.</p> <p>This multi-output process "mining, gold-silver-zinc-lead-copper deposit" delivers the five co-products "gold/silver/copper/zinc/lead, from combined metal production, at beneficiation". The metals consisting of 1kg metal in the product, further refining is required for pure metal.</p> <p>This data set includes the mining in Sweden in open pit and underground mines incl. land use. Allocation is based on economic value.</p> <p>The multi-output process "refining, copper-zinc-lead-gold-silver in smelter" delivers the five co-products "copper/zinc/lead/gold/silver from combined metal production, at refinery". This data set describes the raffination of gold and silver in a Swedish smelter. Allocation is based on economic value.</p> |                                                                                                                             |                                                          |
| InfrastructureIncluded     | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1                                                                                                                           | 1                                                        |
| Category                   | metals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | metals                                                                                                                      | metals                                                   |
| SubCategory                | extraction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | extraction                                                                                                                  | extraction                                               |
| LocalCategory              | Metalle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Metalle                                                                                                                     | Metalle                                                  |
| LocalSubCategory           | Gewinnung                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Gewinnung                                                                                                                   | Gewinnung                                                |
| CASNumber                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                             |                                                          |
| StartDate                  | 2004                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2004                                                                                                                        | 2004                                                     |
| EndDate                    | 2006                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2006                                                                                                                        | 2006                                                     |
| DataValidForEntirePeriod   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1                                                                                                                           | 1                                                        |
| OtherPeriodText            | Period of public environmental report                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Period of public environmental report                                                                                       | Period of public environmental report                    |
| Text                       | Land use for Sweden, other inputs based on rough assumptions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Material requirements for gold and silver are estimated with Australian data, cyanide consumption is estimated with US-Data | Energy use and emissions are from Swedish mines.         |
| Text                       | Assumptions for buildings and machinery based on the data set "non-ferrous metal, mine, surface/underground with 50 years life time.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | open pit mines and underground mines                                                                                        | modern swedish refinement plant                          |
| Percent                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 20                                                                                                                          |                                                          |
| ProductionVolume           | >2.32E7 t ore                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | >2.32E7 t ore                                                                                                               | >2.36E+5 t of metals                                     |
| SamplingProcedure          | environmental report for landuse only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | environmental report                                                                                                        | environmental report                                     |
| Extrapolations             | Buildings and machinery are based on global assumptions. The figures were extrapolated on the basis of the ore production (2.32E10 kg per year)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | See Geography, air emissions are estimated with data of one mine                                                            | none                                                     |
| UncertaintyAdjustments     | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | none                                                                                                                        | none                                                     |
| Person                     | 43                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 43                                                                                                                          | 43                                                       |
| DataPublishedIn            | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2                                                                                                                           | 2                                                        |
| ReferenceToPublishedSource | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 10                                                                                                                          | 10                                                       |
| Copyright                  | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1                                                                                                                           | 1                                                        |
| AccessRestrictedTo         | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 0                                                                                                                           | 0                                                        |
| CompanyCode                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                             |                                                          |
| CountryCode                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                             |                                                          |
| PageNumbers                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                             |                                                          |
| Validator                  | 51                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 51                                                                                                                          | 51                                                       |
| Details                    | automatic validation in Excel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | automatic validation in Excel                                                                                               | automatic validation in Excel                            |
| OtherDetails               | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | none                                                                                                                        | none                                                     |



### 5.1.3 Gold and silver production, Papua New Guinea (PG)

Data stem from the environmental report of Placer Dome Inc. (Placer Dome Inc., 2002b; Placer Dome Inc., 2003b) and Barrick (2006b) for the Misima Mine and the Porgera Mine. Both are open pit mines. The Misima Mine was in operation until 2004 and produced gold and silver. In January 2006, the Porgera Mine was sold to the Canadian company of Barrick. Actual data (2006) from Barrick is used where available, no environmental report has been published in the years 2004 and 2005. Since no actual production takes place in the Misima mine, the data of the last year of full activity (year 2000) is used as one production site of gold in Papua New Guinea (see Tab. 5.15 for production values).

The Porgera Mine is located in the Enga Province of the Papua New Guinea highlands about 550 kilometres north west of the capital Port Moresby and produces only gold.

The Misima mine is situated on an island 190 km east from the main island of Papua New Guinea. The distance to the capital Port Moresby is about 650 km. The tailings of Misima are disposed of with a pipeline in deep-sea (Placer Dome Inc., 2000).

Tab. 5.15 shows the production volume in 2000/2006, calculated turnover and shares in turnover of the different products of the Misima and the Porgera Mine.

The recovery rate (proportion of valuable material physically recovered in the processing of ore) is taken into account in order to define the resource extraction. The average concentration (head grade, expressed in kg metal per kg ore) of metals in ore for Papua New Guinea is calculated with the quantity of ore milled and the corresponding production rate.

**Tab. 5.15: Production volume 2000 / 2006, calculated turnover and shares in turnover of the different products of the Misima and the Porgera Mine (Placer Dome Inc., 2001, Barrick, 2006a), Basis: prices 2004-2006 according to Tab. 2.5**

| Production           |           | Misima<br>2000 | Porgera<br>2006 | Papua New<br>Guinea |                                             |
|----------------------|-----------|----------------|-----------------|---------------------|---------------------------------------------|
| Ore milled           | kg        | 6.10E+09       | 5.23E+09        | <b>1.13E+10</b>     |                                             |
| Production Gold      | kg        | 6.77E+03       | 1.68E+04        | <b>2.36E+04</b>     |                                             |
| Production Silver    | kg        | 2.40E+04       |                 | <b>2.40E+04</b>     |                                             |
| Value Gold           | US\$      | 1.06E+08       | 2.63E+08        | <b>2.63E+08</b>     | Value of 1kg Gold: 15680 US\$               |
| Value Silver         | US\$      | 6.45E+06       |                 | <b>6.45E+06</b>     | Value of 1kg Silver: 269 US\$               |
| Allocation Gold      |           |                |                 | <b>97.6%</b>        |                                             |
| Allocation Silver    |           |                |                 | <b>2.4%</b>         |                                             |
| <i>Head grade</i>    |           |                |                 |                     |                                             |
| Gold                 | kg/kg ore | 1.11E-06       | 3.21E-06        | <b>2.08E-06</b>     |                                             |
| Silver               | kg/kg ore | 3.93E-06       |                 | <b>2.12E-06</b>     |                                             |
| <i>Recovery rate</i> |           |                |                 | <b>90%</b>          | based on reports of Placer-Dome Inc. (2002) |

### Mining infrastructure

This data set includes land use as well as rough estimates on buildings and machines necessary in gold mining. It is assumed, that the quantities of machinery as well as surface and volume of buildings are equal to the generic mine. This approach is rough, and the uncertainty is rather high. As the mined ore from the generic mine is some orders of magnitude lower than the mine in focus, a proportional scaling by milled ore will lead to a higher mistake (see Tab. 5.1: Gold and Silver production in Sweden). The lifetime of mines is assumed to be 50 years.

Land use is based on the figures from the environmental report (Tab. 5.16). Not all the area of the mine is in active use (disturbed area) at one time; the value in Tab. 5.16 is assumed to be the average share of the mine area in use. Therefore, the total area is occupied during an average of about 25 years according to the following formula:

$$\frac{\text{Disturbed area in active use}}{\text{Area disturbed during life of mine}} * \text{lifetime of mine} = \frac{1.17 * 10^7}{2.32 * 10^8} * 50y = 25y$$
 , the occupation of the mine is calculated as Disturbed area in active use \* 50y = Occupation, mineral extraction site

Tab. 5.16: Land use of the Misima and Porgera Mines in 2002 (Placer Dome Inc., 2002a)

| Land use                           |                | Misima 2000 | Porgera 2006 | Papua New Guinea | Remark                                                                       |
|------------------------------------|----------------|-------------|--------------|------------------|------------------------------------------------------------------------------|
| Area disturbed during life of mine | m <sup>2</sup> | 6.11E+06    | 1.71E+07     | <b>2.32E+07</b>  | Transformation, from unknown and Transformation, to mineral extraction site, |
| Disturbed area in active use       | m <sup>2</sup> | 2.53E+06    | 9.21E+06     | <b>1.17E+07</b>  | Occupation, mineral extraction site during 50 years                          |
| Area available for revegetation    | m <sup>2</sup> | 1.80E+05    | 1.74E+06     | <b>1.92E+06</b>  | added in "Transformation, to unknown"                                        |
| Area revegetated                   | m <sup>2</sup> | 3.40E+06    | 6.16E+06     | <b>9.56E+06</b>  | Transformation, to heterogeneous, agricultural                               |

Tab. 5.17: Unit process raw data of the gold and silver mine in Papua New Guinea

|                | Name                                             | Location | Infrastructure | Process | Unit | mine, gold and silver | Uncertainty | Type | Standard deviation | 95 %                                        | GeneralComment |
|----------------|--------------------------------------------------|----------|----------------|---------|------|-----------------------|-------------|------|--------------------|---------------------------------------------|----------------|
|                |                                                  |          |                |         |      |                       |             |      |                    |                                             |                |
|                | Location InfrastructureProcess Unit              |          |                |         |      | PG 1 unit             |             |      |                    |                                             |                |
| product        | mine, gold and silver                            | PG       | 1              | unit    |      | 1.00E+0               |             |      |                    |                                             |                |
| technosphere   | building, multi-storey                           | RER      | 1              | m3      |      | 2.75E+5               | 1           | 2.39 | (4,5,5,5,5,5);     | Rough estimates                             |                |
|                | building, hall                                   | CH       | 1              | m2      |      | 1.25E+4               | 1           | 2.39 | (4,5,5,5,5,5);     | Rough estimates                             |                |
|                | industrial machine, heavy, unspecified, at plant | RER      | 1              | kg      |      | 8.50E+5               | 1           | 2.39 | (4,5,5,5,5,5);     | Rough estimates                             |                |
|                | conveyor belt, at plant                          | RER      | 1              | m       |      | 6.00E+2               | 1           | 2.39 | (4,5,5,5,5,5);     | Rough estimates                             |                |
| resource, land | Transformation, from unknown                     | -        | -              | m2      |      | 2.32E+7               | 1           | 2.01 | (1,3,2,1,1,3);     | Environmental report, Placer Dome Inc. 2003 |                |
|                | Transformation, to mineral extraction site       | -        | -              | m2      |      | 2.32E+7               | 1           | 2.01 | (1,3,2,1,1,3);     | Environmental report, Placer Dome Inc. 2003 |                |
|                | Occupation, mineral extraction site              | -        | -              | m2a     |      | 5.87E+8               | 1           | 1.51 | (1,3,2,1,1,3);     | Environmental report, Placer Dome Inc. 2003 |                |
|                | Transformation, from mineral extraction site     | -        | -              | m2      |      | 2.32E+7               | 1           | 2.01 | (1,3,2,1,1,3);     | Environmental report, Placer Dome Inc. 2003 |                |
|                | Transformation, to heterogeneous, agricultural   | -        | -              | m2      |      | 9.56E+6               | 1           | 2.01 | (1,3,2,1,1,3);     | Environmental report, Placer Dome Inc. 2003 |                |
|                | Transformation, to unknown                       | -        | -              | m2      |      | 1.37E+7               | 1           | 2.01 | (1,3,2,1,1,3);     | Environmental report, Placer Dome Inc. 2003 |                |

## Mining and refining

The recovery processes of the Misima Mine are cyanide leach and carbon in pulp (CIP), in the Porgera Mine pressure oxidation and cyanide leach. Both mines produce doré.

The following table (Tab. 5.18) sums up the total requirements and emissions of the Misima and Porgera Mines. For the Porgera Mine data from 2006 is used where available, the consumption of some material requirements is scaled with the weight of processed gold and data from the year 2002. For the Misima Mine data from 2000 is used, when the Misima Mine was still producing (Placer Dome Inc., 2001). Since the module “blasting” does not include the CO<sub>2</sub> emissions from explosives, they are calculated with the factor of the mines of Newmont (2005b) (0.16 kg CO<sub>2</sub> per kg explosive). The emissions to air and in water from the Misima and Porgera mines are as declared in the environmental reports, only liquid effluents have been modelled. Emissions to water head from the Misima-Mine to the ocean, whereas from Porgera they are into the river.

**Tab. 5.18: Material and energy requirements of the Misima Mine in 2000 (Placer Dome Inc., 2001) and the Porgera Mine in 2002 / 2006 (Placer Dome Inc., 2002b, Barrick, 2006b)**

| Energy                           |     | Misima<br>2000 | Porgera<br>2006 | Papua New<br>Guinea | Remark                                                                                                                                                                               |
|----------------------------------|-----|----------------|-----------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Electricity for mining           | kWh | 1.14E+08       |                 | 1.14E+08            | Electricity is produced by the mine on site or with own power stations, environmental impacts are included in diesel-electric moduls <sup>1)</sup>                                   |
| Electricity for refining         | kWh | n.a.           | n.a.            | 5.28E+04            | assumed with Möbius / Wohlwill-electrolysis, electricity is produced by the mine on site, added to "Diesel, burned in diesel-electric set" with thermal-electrical efficiency of 35% |
| Diesel (generators)              | MJ  | 2.77E+07       |                 | 2.77E+07            | Diesel, burned in diesel-electric generating set <sup>2)</sup>                                                                                                                       |
| Diesel (mobile equipment)        | MJ  | 7.15E+08       | 2.96E+09        | 3.68E+09            | Diesel, burned in building machine                                                                                                                                                   |
| Petrol                           | l   | 5.28E+04       |                 | 5.28E+04            | converted in transport, van <3.5t (0.38l/tkm)                                                                                                                                        |
| Oil                              | MJ  | 9.87E+08       |                 | 9.87E+08            | heavy fuel oil, burned in industrial furnace 1MW, non-modulating                                                                                                                     |
| LPG                              | MJ  | 3.43E+05       | 5.33E+09        | 5.33E+09            | natural gas, burned in industrial furnace                                                                                                                                            |
| <b>Materials</b>                 |     |                |                 |                     |                                                                                                                                                                                      |
| Acetylen                         | m3  | 2.69E+03       |                 | 2.69E+03            | Density at 15°C: 1.095 kg/m3                                                                                                                                                         |
| Cyanide                          | kg  | 2.37E+06       | 6.05E+05        | 2.98E+06            | In LCI as sodium cyanide                                                                                                                                                             |
| Lime                             | kg  | 1.60E+07       | 1.34E+07        | 2.94E+07            | Value of Porgera scaled from consumption in 2003 by processed gold.                                                                                                                  |
| Caustic                          | kg  | 7.88E+04       | 1.96E+05        | 2.74E+05            | Scaled from Steward (1999) by processed gold, accounted for sodium hydroxide                                                                                                         |
| Flocculent / Flotation chemicals | kg  | 1.76E+05       | 5.25E+05        | 7.01E+05            | approximated as organic chemicals                                                                                                                                                    |
| Carbon                           | kg  | 2.36E+05       | 5.53E+05        | 7.89E+05            | Value of Porgera scaled from consumption in 2003 by processed gold.                                                                                                                  |
| Zinc                             | kg  | 1.96E+04       | 4.86E+04        | 6.82E+04            | Scaled from Steward (1999) by processed gold, accounted for as zinc, primary, at regional storage                                                                                    |
| Sulphuric Acid                   | kg  |                | 8.74E+04        | 8.74E+04            | Value of Porgera scaled from consumption in 2003 by processed gold.                                                                                                                  |
| Hydrochloric Acid                | kg  | 2.20E+05       | 8.74E+04        | 3.07E+05            | Value of Porgera scaled from consumption in 2003 by processed gold.                                                                                                                  |
| Steel grinding media             | kg  | 2.83E+06       | 3.96E+06        | 6.79E+06            | Value of Porgera scaled from consumption in 2003 by processed gold.                                                                                                                  |
| Explosives                       | kg  | 1.56E+06       | 3.43E+06        | 4.99E+06            | Partly ANFO explosives, in LCI: "blasting"                                                                                                                                           |
| <b>Water resources</b>           |     |                |                 |                     |                                                                                                                                                                                      |
| Water from bore fields           | m3  | 3.30E+06       |                 | 3.30E+06            | in LCI: Water, unspecified natural origin <sup>3)</sup>                                                                                                                              |
| Groundwater                      | m3  |                | 2.85E+07        | 2.85E+07            | in LCI: Water, well, in ground <sup>4)</sup>                                                                                                                                         |
| <b>Emissions to Water</b>        |     |                |                 |                     |                                                                                                                                                                                      |
| Arsenic                          | kg  | 2.40E+01       | 4.87E+02        | 5.11E+02            | Corrected value for Misima <sup>5)</sup>                                                                                                                                             |
| Cadmium                          | kg  | 9.50E+01       | 2.30E+03        | 2.39E+03            | Value of Porgera scaled from emission in 2003 by milled ore                                                                                                                          |
| Copper                           | kg  | 9.76E+04       | 3.52E+04        | 1.33E+05            | Value of Porgera scaled from emission in 2003 by milled ore                                                                                                                          |
| Cyanide                          | kg  | 1.70E+05       | 4.57E+04        | 2.16E+05            | Value of Porgera scaled from emission in 2003 by milled ore                                                                                                                          |
| Lead                             | kg  | 2.83E+02       | 6.77E+01        | 3.51E+02            | Value of Porgera scaled from emission in 2003 by milled ore                                                                                                                          |
| Mercury                          | kg  | 0.00E+00       | 5.43E+00        | 5.43E+00            | Value of Porgera scaled from emission in 2003 by milled ore                                                                                                                          |
| Nickel                           | kg  | 6.67E+02       | 3.11E+04        | 3.18E+04            | Value of Porgera scaled from emission in 2003 by milled ore                                                                                                                          |
| Zinc                             | kg  | 6.36E+03       | 8.41E+05        | 8.47E+05            | Value of Porgera scaled from emission in 2003 by milled ore                                                                                                                          |
| <b>Emissions to Air</b>          |     |                |                 |                     |                                                                                                                                                                                      |
| Carbon Dioxide (from explosives) | kg  | 2.50E+05       | 5.49E+05        | 7.99E+05            | Calculation, based on data from Newmont (2005)                                                                                                                                       |
| Zinc                             | kg  | n.a.           | n.a.            | 2.50E+00            | based on Swedish data, scaled by milled ore                                                                                                                                          |
| Copper                           | kg  | n.a.           | n.a.            | 9.05E+00            | based on Swedish data, scaled by milled ore                                                                                                                                          |
| Lead                             | kg  | n.a.           | n.a.            | 2.94E+00            | based on Swedish data, scaled by milled ore                                                                                                                                          |
| Disposal                         | kg  | 6.09E+09       | 1.96E+09        | 8.05E+09            | Placer Dome Inc. (2002), in LCI: Disposal, sulphidic tailings                                                                                                                        |

<sup>1</sup> Electricity in Misima is produced from diesel and heavy fuel oil. Porgera's principal source of power is supplied by a 73-kilometre transmission line from the gas fired and PJV-owned Hides Power Station. The station has a total output of 62 megawatts ("MW"). A back up diesel power station is located at

the mine and has an output of 13MW. The average power requirement of the mine is about 60 MW. The environmental impacts of electricity use in Papua New Guinea are considered in the modules “diesel, burned in diesel-electric generating set” and “heavy fuel oil, burned in industrial furnace”.

- <sup>2</sup> An 18 MW diesel fired power station supplies electrical power. Diesel was used in the station due to the unavailability of previously supplied heavy fuel oil.
- <sup>3</sup> Process water is supplied from pit dewatering bores and in-pit water. Potable water is sourced from boreholes in the coastal limestone. The figure in the environmental report is 3.33E+12 l. However, as the discharge in 2002 is about 7.0E+9 l, the value in the report is assumed to be 1000 times too high and corrected to 3.33E+09 l.
- <sup>4</sup> The main water supply of the mine is the Waile Creek Dam, located approximately 7 kilometres from the mine. The reservoir has a capacity of approximately 717, 000 m<sup>3</sup> of water. Water for the grinding circuit is also extracted from Kogai Creek, which is located adjacent to the grinding circuit. The mine operates four water treatment plants for potable water and five sewage treatment plants.
- <sup>5</sup> The figure in the environmental report from 2000 was 0.24 kg. However, 2001-2003, the values are 23-47kg, so the figure for 2000 is assumed to be 100 times too small and is corrected to 24 kg.

The figures are allocated on the basis of the allocation factors shown in Tab. 5.15. Air emissions (zinc, copper and lead) are approximated with Swedish data. As Swedish data stem from a combined process, they were allocated to gold and silver for Sweden and scaled by the milled ore for Papua New Guinea. Transport distances are estimated to be 600 km for the materials on land (corresponds to the average distance from capital Port Moresby to the mines).

Since the mines produce doré gold (unrefined gold bullion bars containing between 60% and 95% gold), the refinery process is less complex than the refinery process of Rönnskär in Sweden. It is assumed, that the gold doré-bars undergo the treatment of Wohlwill electrolysis (for details see 5.2.2.1, “Gold / Silver, secondary, from electronic scrap in ecoinvent”). Due to the lack of specific data, the figure of refinement from electronic scrap is used: 1.63 kWh per kg gold (Renner et al., 2002a). The refinery of silver requires 0.6 kWh per kg of silver (Möbius electrolysis). No emissions were assumed, since on the one hand the high value of the processed material requires an efficient emission control and on the other hand, refinement is a hydrometallurgical process with low impact. The infrastructure is approximated with the facility of precious metal refinery. The unit process raw data are summarized in Tab. 5.19.

Tab. 5.19: Unit process raw data of gold and silver mining and refining in Papua New Guinea

|                                      | Name                                                             | Location | Infrastructure | Process | Unit    | mining and refining, gold-silver deposit | Uncertainty | Type                                                                                                  | Standard Deviation 95% | General Comment |
|--------------------------------------|------------------------------------------------------------------|----------|----------------|---------|---------|------------------------------------------|-------------|-------------------------------------------------------------------------------------------------------|------------------------|-----------------|
|                                      | Location<br>InfrastructureProcess<br>Unit                        |          |                |         |         | PG<br>0<br>a                             |             |                                                                                                       |                        |                 |
| allocated products                   | gold, from combined gold-silver production, at refinery          | PG       | 0              | kg      | 2.36E+4 |                                          |             |                                                                                                       |                        |                 |
|                                      | silver, from combined gold-silver production, at refinery        | PG       | 0              | kg      | 2.40E+4 |                                          |             |                                                                                                       |                        |                 |
| technosphere                         | diesel, burned in diesel-electric generating set                 | GLO      | 0              | MJ      | 2.82E+7 | 1                                        | 1.14        | (2,4,2,3,1,3); Environmental report                                                                   |                        |                 |
|                                      | diesel, burned in building machine                               | GLO      | 0              | MJ      | 3.68E+9 | 1                                        | 1.14        | (2,4,2,3,1,3); Environmental report                                                                   |                        |                 |
|                                      | transport, van <3.5t                                             | RER      | 0              | tkm     | 1.51E+5 | 1                                        | 2.02        | (2,4,2,3,1,3); Environmental report                                                                   |                        |                 |
|                                      | heavy fuel oil, burned in industrial furnace 1MW, non-modulating | RER      | 0              | MJ      | 9.87E+8 | 1                                        | 1.14        | (2,4,2,3,1,3); Environmental report                                                                   |                        |                 |
|                                      | natural gas, burned in industrial furnace >100kW                 | RER      | 0              | MJ      | 5.33E+9 | 1                                        | 1.14        | (2,4,2,3,1,3); Environmental report                                                                   |                        |                 |
|                                      | acetylene, at regional storehouse                                | CH       | 0              | kg      | 2.78E+3 | 1                                        | 1.14        | (2,4,2,3,1,3); Environmental report                                                                   |                        |                 |
|                                      | sodium cyanide, at plant                                         | RER      | 0              | kg      | 2.98E+6 | 1                                        | 1.14        | (2,4,2,3,1,3); Environmental report                                                                   |                        |                 |
|                                      | limestone, milled, packed, at plant                              | CH       | 0              | kg      | 2.94E+7 | 1                                        | 1.14        | (2,4,2,3,1,3); Environmental report                                                                   |                        |                 |
|                                      | sodium hydroxide, 50% in H2O, production mix, at plant           | RER      | 0              | kg      | 2.74E+5 | 1                                        | 1.15        | thesis about Australian precious metal production, data for 1999                                      |                        |                 |
|                                      | chemicals organic, at plant                                      | GLO      | 0              | kg      | 7.01E+5 | 1                                        | 1.15        | (2,3,2,5,1,3); Environmental report, Placer Dome Inc. 2003                                            |                        |                 |
|                                      | charcoal, at plant                                               | GLO      | 0              | kg      | 7.89E+5 | 1                                        | 1.15        | (2,3,2,5,1,3); Calculation, based on thesis about Australian precious metal production, data for 1999 |                        |                 |
|                                      | zinc, primary, at regional storage                               | RER      | 0              | kg      | 6.82E+4 | 1                                        | 1.15        | (2,3,2,5,1,3); Calculation, based on thesis about Australian precious metal production, data for 1999 |                        |                 |
|                                      | sulphuric acid, liquid, at plant                                 | RER      | 0              | kg      | 8.74E+4 | 1                                        | 1.15        | (2,3,2,5,1,3); Calculation, based on thesis about Australian precious metal production, data for 1999 |                        |                 |
|                                      | hydrochloric acid, 30% in H2O, at plant                          | RER      | 0              | kg      | 3.07E+5 | 1                                        | 1.11        | (2,3,2,3,1,3); Environmental report, Placer Dome Inc. 2003                                            |                        |                 |
|                                      | steel, converter, chromium steel 18/8, at plant                  | RER      | 0              | kg      | 6.79E+6 | 1                                        | 1.11        | (2,3,2,3,1,3); Environmental report, Placer Dome Inc. 2003                                            |                        |                 |
|                                      | blasting                                                         | RER      | 0              | kg      | 4.99E+6 | 1                                        | 1.11        | (2,3,2,3,1,3); Environmental report, Placer Dome Inc. 2003, Barrick 2006                              |                        |                 |
|                                      | transport, lorry >16t, fleet average                             | RER      | 0              | tkm     | 2.79E+7 | 1                                        | 2.02        | (2,4,na,na,na,na); Calculation, based on estimated distance of 600km                                  |                        |                 |
|                                      | disposal, sulfidic tailings, off-site                            | GLO      | 0              | kg      | 8.05E+9 | 1                                        | 1.30        | (4,5,na,na,na,na); Environmental report                                                               |                        |                 |
| infrastructure                       | mine, gold and silver                                            | PG       | 1              | unit    | 2.00E-2 | 1                                        | 3.01        | (2,3,2,3,1,3); Environmental report                                                                   |                        |                 |
|                                      | facilities precious metal refinery                               | SE       | 1              | unit    | 7.83E-3 | 1                                        | 3.09        | (4,5,na,na,na,na); Environmental report                                                               |                        |                 |
| emission air, low population density | Copper                                                           | -        | -              | kg      | 9.05E+0 | 1                                        | 5.02        | (2,4,2,3,1,3); Calculation, based on Swedish production                                               |                        |                 |
|                                      | Lead                                                             | -        | -              | kg      | 2.94E+0 | 1                                        | 5.02        | (2,4,2,3,1,3); Calculation, based on Swedish production                                               |                        |                 |
|                                      | Zinc                                                             | -        | -              | kg      | 2.50E+0 | 1                                        | 5.02        | (2,4,2,3,1,3); Calculation, based on Swedish production                                               |                        |                 |
|                                      | Carbon dioxide, fossil                                           | -        | -              | kg      | 7.99E+5 | 1                                        | 1.14        | (2,4,2,3,1,3); Environmental report                                                                   |                        |                 |
| emission water, ocean                | Arsenic, ion                                                     | -        | -              | kg      | 2.40E+1 | 1                                        | 5.02        | (2,4,2,3,1,3); Environmental report                                                                   |                        |                 |
|                                      | Cadmium, ion                                                     | -        | -              | kg      | 9.50E+1 | 1                                        | 3.02        | (2,4,2,3,1,3); Environmental report                                                                   |                        |                 |
|                                      | Copper, ion                                                      | -        | -              | kg      | 9.76E+4 | 1                                        | 3.02        | (2,4,2,3,1,3); Environmental report                                                                   |                        |                 |
|                                      | Cyanide                                                          | -        | -              | kg      | 1.70E+5 | 1                                        | 3.02        | (2,4,2,3,1,3); Environmental report                                                                   |                        |                 |
|                                      | Lead                                                             | -        | -              | kg      | 2.83E+2 | 1                                        | 5.02        | (2,4,2,3,1,3); Environmental report                                                                   |                        |                 |
|                                      | Mercury                                                          | -        | -              | kg      | 0       | 1                                        | 5.02        | (2,4,2,3,1,3); Environmental report                                                                   |                        |                 |
|                                      | Nickel, ion                                                      | -        | -              | kg      | 6.67E+2 | 1                                        | 5.02        | (2,4,2,3,1,3); Environmental report                                                                   |                        |                 |
|                                      | Zinc, ion                                                        | -        | -              | kg      | 6.36E+3 | 1                                        | 5.02        | (2,4,2,3,1,3); Environmental report                                                                   |                        |                 |
| emission water, river                | Arsenic, ion                                                     | -        | -              | kg      | 4.87E+2 | 1                                        | 5.02        | (2,4,2,3,1,3); Environmental report                                                                   |                        |                 |
|                                      | Cadmium, ion                                                     | -        | -              | kg      | 2.30E+3 | 1                                        | 3.02        | (2,4,2,3,1,3); Environmental report                                                                   |                        |                 |
|                                      | Copper, ion                                                      | -        | -              | kg      | 3.52E+4 | 1                                        | 3.02        | (2,4,2,3,1,3); Environmental report                                                                   |                        |                 |
|                                      | Cyanide                                                          | -        | -              | kg      | 4.57E+4 | 1                                        | 3.02        | (2,4,2,3,1,3); Environmental report                                                                   |                        |                 |
|                                      | Lead                                                             | -        | -              | kg      | 6.77E+1 | 1                                        | 5.02        | (2,4,2,3,1,3); Environmental report                                                                   |                        |                 |
|                                      | Mercury                                                          | -        | -              | kg      | 5.43E+0 | 1                                        | 5.02        | (2,4,2,3,1,3); Environmental report                                                                   |                        |                 |
|                                      | Nickel, ion                                                      | -        | -              | kg      | 3.11E+4 | 1                                        | 5.02        | (2,4,2,3,1,3); Environmental report                                                                   |                        |                 |
|                                      | Zinc, ion                                                        | -        | -              | kg      | 8.41E+5 | 1                                        | 5.02        | (2,4,2,3,1,3); Environmental report                                                                   |                        |                 |
| resource, in water                   | Water, well, in ground                                           | -        | -              | m3      | 2.85E+7 | 1                                        | 1.14        | (2,4,2,3,1,3); Environmental report                                                                   |                        |                 |
|                                      | Water, unspecified natural origin                                | -        | -              | m3      | 3.30E+6 | 1                                        | 1.14        | (2,4,2,3,1,3); Environmental report                                                                   |                        |                 |
| resource, in ground                  | Gold, Au 2.1E-4%, Ag 2.1E-4%, in ore, in ground                  | -        | -              | kg      | 2.62E+4 | 1                                        | 1.14        | (2,4,2,3,1,3); Estimation                                                                             |                        |                 |
|                                      | Silver, Ag 2.1E-4%, Au 2.1E-4%, in ore, in ground                | -        | -              | kg      | 2.67E+4 | 1                                        | 1.14        | (2,4,2,3,1,3); Estimation                                                                             |                        |                 |

Tab. 5.20 Allocation factors of the data set “mining and refining, gold-silver deposit”

|                                      | Name                                                             | Location | Infrastructure | Process | Unit | mining and refining, gold-silver deposit | gold, from combined gold-silver production, at refinery | silver, from combined gold-silver production, at refinery |
|--------------------------------------|------------------------------------------------------------------|----------|----------------|---------|------|------------------------------------------|---------------------------------------------------------|-----------------------------------------------------------|
|                                      | Location<br>Infrastructure<br>Process<br>Unit                    |          |                |         |      | PG<br>0<br>a                             | PG<br>0<br>kg                                           | PG<br>0<br>kg                                             |
| allocated products                   | gold, from combined gold-silver production, at refinery          | PG       | 0              | kg      |      | 2.36E+4                                  | 100                                                     | -                                                         |
|                                      | silver, from combined gold-silver production, at refinery        | PG       | 0              | kg      |      | 2.40E+4                                  | -                                                       | 100                                                       |
| technosphere                         | diesel, burned in diesel-electric generating set                 | GLO      | 0              | MJ      |      | 2.82E+7                                  | 97.6                                                    | 2.4                                                       |
|                                      | diesel, burned in building machine                               | GLO      | 0              | MJ      |      | 3.68E+9                                  | 97.6                                                    | 2.4                                                       |
|                                      | transport, van <3.5t                                             | RER      | 0              | tkm     |      | 1.51E+5                                  | 97.6                                                    | 2.4                                                       |
|                                      | heavy fuel oil, burned in industrial furnace 1MW, non-modulating | RER      | 0              | MJ      |      | 9.87E+8                                  | 97.6                                                    | 2.4                                                       |
|                                      | natural gas, burned in industrial furnace >100kW                 | RER      | 0              | MJ      |      | 5.33E+9                                  | 97.6                                                    | 2.4                                                       |
|                                      | acetylene, at regional storehouse                                | CH       | 0              | kg      |      | 2.78E+3                                  | 97.6                                                    | 2.4                                                       |
|                                      | sodium cyanide, at plant                                         | RER      | 0              | kg      |      | 2.98E+6                                  | 97.6                                                    | 2.4                                                       |
|                                      | limestone, milled, packed, at plant                              | CH       | 0              | kg      |      | 2.94E+7                                  | 97.6                                                    | 2.4                                                       |
|                                      | sodium hydroxide, 50% in H2O, production mix, at plant           | RER      | 0              | kg      |      | 2.74E+5                                  | 97.6                                                    | 2.4                                                       |
|                                      | chemicals organic, at plant                                      | GLO      | 0              | kg      |      | 7.01E+5                                  | 97.6                                                    | 2.4                                                       |
|                                      | charcoal, at plant                                               | GLO      | 0              | kg      |      | 7.89E+5                                  | 97.6                                                    | 2.4                                                       |
|                                      | zinc, primary, at regional storage                               | RER      | 0              | kg      |      | 6.82E+4                                  | 97.6                                                    | 2.4                                                       |
|                                      | sulphuric acid, liquid, at plant                                 | RER      | 0              | kg      |      | 8.74E+4                                  | 97.6                                                    | 2.4                                                       |
|                                      | hydrochloric acid, 30% in H2O, at plant                          | RER      | 0              | kg      |      | 3.07E+5                                  | 97.6                                                    | 2.4                                                       |
|                                      | steel, converter, chromium steel 18/8, at plant                  | RER      | 0              | kg      |      | 6.79E+6                                  | 97.6                                                    | 2.4                                                       |
|                                      | blasting                                                         | RER      | 0              | kg      |      | 4.99E+6                                  | 97.6                                                    | 2.4                                                       |
|                                      | transport, lorry >16t, fleet average                             | RER      | 0              | tkm     |      | 2.79E+7                                  | 97.6                                                    | 2.4                                                       |
|                                      | disposal, sulfidic tailings, off-site                            | GLO      | 0              | kg      |      | 8.05E+9                                  | 97.6                                                    | 2.4                                                       |
| infrastructure                       | mine, gold and silver                                            | PG       | 1              | unit    |      | 2.00E-2                                  | 97.6                                                    | 2.4                                                       |
|                                      | facilities precious metal refinery                               | SE       | 1              | unit    |      | 7.83E-3                                  | 97.6                                                    | 2.4                                                       |
| emission air, low population density | Copper                                                           | -        | -              | kg      |      | 9.05E+0                                  | 97.6                                                    | 2.4                                                       |
|                                      | Lead                                                             | -        | -              | kg      |      | 2.94E+0                                  | 97.6                                                    | 2.4                                                       |
|                                      | Zinc                                                             | -        | -              | kg      |      | 2.50E+0                                  | 97.6                                                    | 2.4                                                       |
|                                      | Carbon dioxide, fossil                                           | -        | -              | kg      |      | 7.99E+5                                  | 97.6                                                    | 2.4                                                       |
| emission water, ocean                | Arsenic, ion                                                     | -        | -              | kg      |      | 2.40E+1                                  | 97.6                                                    | 2.4                                                       |
|                                      | Cadmium, ion                                                     | -        | -              | kg      |      | 9.50E+1                                  | 97.6                                                    | 2.4                                                       |
|                                      | Copper, ion                                                      | -        | -              | kg      |      | 9.76E+4                                  | 97.6                                                    | 2.4                                                       |
|                                      | Cyanide                                                          | -        | -              | kg      |      | 1.70E+5                                  | 97.6                                                    | 2.4                                                       |
|                                      | Lead                                                             | -        | -              | kg      |      | 2.83E+2                                  | 97.6                                                    | 2.4                                                       |
|                                      | Mercury                                                          | -        | -              | kg      |      | 0                                        | 97.6                                                    | 2.4                                                       |
|                                      | Nickel, ion                                                      | -        | -              | kg      |      | 6.67E+2                                  | 97.6                                                    | 2.4                                                       |
|                                      | Zinc, ion                                                        | -        | -              | kg      |      | 6.36E+3                                  | 97.6                                                    | 2.4                                                       |
| emission water, river                | Arsenic, ion                                                     | -        | -              | kg      |      | 4.87E+2                                  | 97.6                                                    | 2.4                                                       |
|                                      | Cadmium, ion                                                     | -        | -              | kg      |      | 2.30E+3                                  | 97.6                                                    | 2.4                                                       |
|                                      | Copper, ion                                                      | -        | -              | kg      |      | 3.52E+4                                  | 97.6                                                    | 2.4                                                       |
|                                      | Cyanide                                                          | -        | -              | kg      |      | 4.57E+4                                  | 97.6                                                    | 2.4                                                       |
|                                      | Lead                                                             | -        | -              | kg      |      | 6.77E+1                                  | 97.6                                                    | 2.4                                                       |
|                                      | Mercury                                                          | -        | -              | kg      |      | 5.43E+0                                  | 97.6                                                    | 2.4                                                       |
|                                      | Nickel, ion                                                      | -        | -              | kg      |      | 3.11E+4                                  | 97.6                                                    | 2.4                                                       |
|                                      | Zinc, ion                                                        | -        | -              | kg      |      | 8.41E+5                                  | 97.6                                                    | 2.4                                                       |
| resource, in water                   | Water, well, in ground                                           | -        | -              | m3      |      | 2.85E+7                                  | 97.6                                                    | 2.4                                                       |
|                                      | Water, unspecified natural origin                                | -        | -              | m3      |      | 3.30E+6                                  | 97.6                                                    | 2.4                                                       |
| resource, in ground                  | Gold, Au 2.1E-4%, Ag 2.1E-4%, in ore, in ground                  | -        | -              | kg      |      | 2.62E+4                                  | 100.0                                                   | -                                                         |
|                                      | Silver, Ag 2.1E-4%, Au 2.1E-4%, in ore, in ground                | -        | -              | kg      |      | 2.67E+4                                  | -                                                       | 100.0                                                     |

Tab. 5.21 Ecospold Meta information of gold and silver production in Papua New Guinea

|                            |                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name                       | mine, gold and silver                                                                                                                                                                                           | mining and refining, gold-silver deposit                                                                                                                                                                                                                                                                                                          |
| Location                   | PG                                                                                                                                                                                                              | PG                                                                                                                                                                                                                                                                                                                                                |
| InfrastructureProcess      | 1                                                                                                                                                                                                               | 0                                                                                                                                                                                                                                                                                                                                                 |
| Unit                       | unit                                                                                                                                                                                                            | a                                                                                                                                                                                                                                                                                                                                                 |
| Type                       | 1                                                                                                                                                                                                               | 5                                                                                                                                                                                                                                                                                                                                                 |
| Version                    | 2.0                                                                                                                                                                                                             | 2.0                                                                                                                                                                                                                                                                                                                                               |
| energyValues               | 0                                                                                                                                                                                                               | 0                                                                                                                                                                                                                                                                                                                                                 |
| LanguageCode               | en                                                                                                                                                                                                              | en                                                                                                                                                                                                                                                                                                                                                |
| LocalLanguageCode          | de                                                                                                                                                                                                              | de                                                                                                                                                                                                                                                                                                                                                |
| Person                     | 44                                                                                                                                                                                                              | 44                                                                                                                                                                                                                                                                                                                                                |
| QualityNetwork             | 1                                                                                                                                                                                                               | 1                                                                                                                                                                                                                                                                                                                                                 |
| DataSetRelatesToProduct    | 1                                                                                                                                                                                                               | 1                                                                                                                                                                                                                                                                                                                                                 |
| IncludedProcesses          | Land use, buildings and machinery of a gold and silver mine                                                                                                                                                     | This data set includes the combined mining and refining of gold and silver in open pit mines in Papua New Guinea incl. energy and material use, water and air emissions of metals and land use.                                                                                                                                                   |
| Amount                     | 1                                                                                                                                                                                                               | 1                                                                                                                                                                                                                                                                                                                                                 |
| LocalName                  | Mine, Gold und Silber                                                                                                                                                                                           | Abbau und Raffination, Gold-Silber-Lagerstätte                                                                                                                                                                                                                                                                                                    |
| Synonyms                   |                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                   |
| GeneralComment             | This data set includes land use of the considered mines incl. revegetated areas as well as estimates of buildings and machinery of the mine based on the data set non-ferrous metal, mine, surface/underground. | This multi-output process 'mining and refining, gold-silver deposit' delivers the two co-products 'gold, from combined gold-silver-production, at refinery' and 'silver, from combined gold-silver-production, at refinery'. This data set includes the mining in Papua New Guinea in open pit mines incl. land use. Air emissions are estimated. |
| InfrastructureIncluded     | 1                                                                                                                                                                                                               | 1                                                                                                                                                                                                                                                                                                                                                 |
| Category                   | metals                                                                                                                                                                                                          | metals                                                                                                                                                                                                                                                                                                                                            |
| SubCategory                | extraction                                                                                                                                                                                                      | extraction                                                                                                                                                                                                                                                                                                                                        |
| LocalCategory              | Metalle                                                                                                                                                                                                         | Metalle                                                                                                                                                                                                                                                                                                                                           |
| LocalSubCategory           | Gewinnung                                                                                                                                                                                                       | Gewinnung                                                                                                                                                                                                                                                                                                                                         |
| Formula                    |                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                   |
| StatisticalClassification  |                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                   |
| CASNumber                  |                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                   |
| StartDate                  | 2000                                                                                                                                                                                                            | 2000                                                                                                                                                                                                                                                                                                                                              |
| EndDate                    | 2006                                                                                                                                                                                                            | 2006                                                                                                                                                                                                                                                                                                                                              |
| DataValidForEntirePeriod   | 1                                                                                                                                                                                                               | 1                                                                                                                                                                                                                                                                                                                                                 |
| OtherPeriodText            | Period of public environmental report                                                                                                                                                                           | Period of two public environmental report                                                                                                                                                                                                                                                                                                         |
| Text                       | Land use of Papua New Guinea-plant, other inputs based on rough assumptions.                                                                                                                                    | Material requirements are estimated with Australian data, Air emissions are estimated with Swedish data.                                                                                                                                                                                                                                          |
| Text                       | Assumptions for buildings and machinery based on the data set "non-ferrous metal, mine, surface/underground with 50 years life time.                                                                            |                                                                                                                                                                                                                                                                                                                                                   |
| Percent                    |                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                   |
| ProductionVolume           | 11.3 Mill. t ore per year                                                                                                                                                                                       | 23.57 t gold and 24 t silver per year                                                                                                                                                                                                                                                                                                             |
| SamplingProcedure          | environmental report for landuse in the year 2000 and one mine in 2006                                                                                                                                          | environmental report of one mine in the year 2000 and one mine in 2006                                                                                                                                                                                                                                                                            |
| Extrapolations             | Buildings and machinery are based on global assumptions. The figures were extrapolated on the basis of the ore production.                                                                                      | Emissions to water are calculated with older data of one mine                                                                                                                                                                                                                                                                                     |
| UncertaintyAdjustments     | none                                                                                                                                                                                                            | none                                                                                                                                                                                                                                                                                                                                              |
| Person                     | 44                                                                                                                                                                                                              | 44                                                                                                                                                                                                                                                                                                                                                |
| DataPublishedIn            | 2                                                                                                                                                                                                               | 2                                                                                                                                                                                                                                                                                                                                                 |
| ReferenceToPublishedSource | 10                                                                                                                                                                                                              | 10                                                                                                                                                                                                                                                                                                                                                |
| Copyright                  | 1                                                                                                                                                                                                               | 1                                                                                                                                                                                                                                                                                                                                                 |
| AccessRestrictedTo         | 0                                                                                                                                                                                                               | 0                                                                                                                                                                                                                                                                                                                                                 |
| CompanyCode                |                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                   |
| CountryCode                |                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                   |
| PageNumbers                |                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                   |
| Validator                  | 51                                                                                                                                                                                                              | 51                                                                                                                                                                                                                                                                                                                                                |
| Details                    | automatic validation in Excel                                                                                                                                                                                   | automatic validation in Excel                                                                                                                                                                                                                                                                                                                     |
| OtherDetails               | none                                                                                                                                                                                                            | none                                                                                                                                                                                                                                                                                                                                              |

### 5.1.4 Gold and silver production, Chile (CL)

Data for the Mine “La Coipa” stem from the environmental report of Placer Dome Inc. (2002b). Since the mine has been sold in 2006 to Kinross Gold Corporation and Goldcorp Inc., no more recent data is available. Both enterprises do not publish their environmental data. Therefore, the older data from 2002 is used to model the gold- and silver production in Chile.

The La Coipa Mine produces gold and silver and is an open pit mine. Tab. 5.22 shows the production volume 2002 and calculated turnover of the La Coipa Mine. The recovery rate (proportion of valuable material physically recovered in the processing of ore) is taken into account in order to define the resource extraction. The average concentration (head grade, expressed in kg metal per kg ore) of metals in ore for Chile is calculated with the quantity of ore milled and the corresponding production rate.

**Tab. 5.22: Production volume 2002, calculated turnover and shares in turnover of the different products of the La Coipa Mine (Placer Dome Inc., 2003a). Basis: prices 2005 according to Tab. 2.5.**

| Production        |       | La Coipa |                               |
|-------------------|-------|----------|-------------------------------|
| Ore milled        | kg    | 3.17E+09 |                               |
| Production Gold   | kg    | 2.99E+03 |                               |
| Production Silver | kg    | 1.12E+05 |                               |
| Value Gold        | US\$  | 4.69E+07 | Value of 1kg Gold: 15680 US\$ |
| Value Silver      | US\$  | 3.01E+07 | Value of 1kg Silver: 269 US\$ |
| Allocation Gold   |       | 60.9%    |                               |
| Allocation Silver |       | 39.1%    |                               |
| Head grade        |       |          |                               |
| Gold              | kg/kg | 1.11E-06 |                               |
| Silver            | kg/kg | 4.16E-05 |                               |
| Recovery rate     |       | 85%      |                               |

### Mining infrastructure

This data set includes land use as well as rough estimates on buildings and machines necessary in gold mining. It is assumed, that the quantities of machinery as well as surface and volume of buildings are equal to the generic mine. This approach is rough, and the uncertainty is rather high. As the mined ore from the generic mine is some orders of magnitude lower than the mine in focus, a proportional scaling by milled ore will lead to a higher mistake (see Tab. 5.1: Gold and Silver production in Sweden). The lifetime of mines is assumed to be 50 years.

Land use is based on the figures from the environmental report (Tab. 5.23). Not all the area of the mine is in active use at one time; the value in Tab. 5.23 is assumed to be the average share of the mine area in use. Therefore, the total area is occupied during an average of about 43 years, according to formula:

$$\frac{\text{Disturbed area in active use}}{\text{Area disturbed during life of mine}} * \text{lifetime of mine} = \frac{1.16 * 10^7}{1.34 * 10^8} * 50y = 43y$$
 , the occupation of the mine is calculated as Disturbed area in active use \* 50y = Occupation, mineral extraction site

**Tab. 5.23: Land use of the La Coipa Mine in 2002 (Placer Dome Inc., 2004).**

| Land use                                   |                | La Coipa | Remark                                                                       |
|--------------------------------------------|----------------|----------|------------------------------------------------------------------------------|
| Area disturbed during life of mine         | m <sup>2</sup> | 1.34E+07 | Transformation, from unknown and Transformation, to mineral extraction site, |
| Disturbed area in active use <sup>1)</sup> | m <sup>2</sup> | 1.16E+07 | Occupation, mineral extraction site, during 50 years                         |
| Area available for revegetation            | m <sup>3</sup> | 1.72E+06 | Transformation to unknown                                                    |
| Area revegetated                           | m <sup>2</sup> | 3.00E+04 | added in "Transformation, to heterogeneous, agricultural"                    |

<sup>1</sup> This figure is 0 in the original data, which is assumed to be a mistake. The area in active use is therefore calculated as the area disturbed during life of mine minus area available for revegetation and area revegetated.



Tab. 5.24: Unit process raw data for the gold and silver mine in Chile.

|                | Name                                             | Location | Infrastructure | Process | Unit | mine, gold and silver | Uncertainty Type | Standard Deviation 95% | General Comment                                            |
|----------------|--------------------------------------------------|----------|----------------|---------|------|-----------------------|------------------|------------------------|------------------------------------------------------------|
|                |                                                  |          |                |         |      |                       |                  |                        |                                                            |
|                | Location<br>InfrastructureProcess<br>Unit        |          |                |         |      | CL<br>1<br>unit       |                  |                        |                                                            |
| product        | mine, gold and silver                            | CL       | 1              | unit    |      | 1.00E+0               |                  |                        |                                                            |
| technosphere   | building, multi-storey                           | RER      | 1              | m3      |      | 2.75E+5               | 1                | 2.39                   | (4,5,5,5,5,5); Rough estimates                             |
|                | building, hall                                   | CH       | 1              | m2      |      | 1.25E+4               | 1                | 2.39                   | (4,5,5,5,5,5); Rough estimates                             |
|                | industrial machine, heavy, unspecified, at plant | RER      | 1              | kg      |      | 8.50E+5               | 1                | 2.39                   | (4,5,5,5,5,5); Rough estimates                             |
|                | conveyor belt, at plant                          | RER      | 1              | m       |      | 6.00E+2               | 1                | 2.39                   | (4,5,5,5,5,5); Rough estimates                             |
| resource, land | Transformation, from unknown                     | -        | -              | m2      |      | 1.34E+7               | 1                | 2.01                   | (1,3,2,1,1,3); Environmental report, Placer Dome Inc. 2002 |
|                | Transformation, to mineral extraction site       | -        | -              | m2      |      | 1.34E+7               | 1                | 2.01                   | (1,3,2,1,1,3); Environmental report, Placer Dome Inc. 2003 |
|                | Occupation, mineral extraction site              | -        | -              | m2a     |      | 5.80E+8               | 1                | 1.51                   | (1,3,2,1,1,3); Environmental report, Placer Dome Inc. 2004 |
|                | Transformation, from mineral extraction site     | -        | -              | m2      |      | 1.34E+7               | 1                | 2.01                   | (1,3,2,1,1,3); Environmental report, Placer Dome Inc. 2005 |
|                | Transformation, to heterogeneous, agricultural   | -        | -              | m2      |      | 3.00E+4               | 1                | 2.01                   | (1,3,2,1,1,3); Environmental report, Placer Dome Inc. 2006 |
|                | Transformation, to unknown                       | -        | -              | m2      |      | 1.34E+7               | 1                | 2.01                   | (1,3,2,1,1,3); Environmental report, Placer Dome Inc. 2007 |

### Mining and refining

The processing plant consists of primary crushing, a pre-crushing circuit, (semi autogenous ball mill crushing) grinding, leaching, filtering and washing, Merrill-Crowe plant and doré refinery. The Merrill-Crowe metal recovery circuit is better than a carbon-in-pulp system for the high-grade silver material. Tailings are filtered to recover excess water as well as residual cyanide and metals. A dry tailings disposal system was preferred to a conventional wet tailings impoundment because of site-specific environmental considerations. The following table (Table 5.25) sums up the total energy and material requirements of the La Coipa Mine in 2002.

Table 5.25: Energy and material requirements of the La Coipa Mine in 2002 (Placer Dome Inc., 2002b).

| Energy                           |                | La Coipa | Remark                                                                                            |
|----------------------------------|----------------|----------|---------------------------------------------------------------------------------------------------|
| Electricity for mining           | kWh            | 1.60E+08 | Chilean mix 1) approximated with the UCTE mix                                                     |
| Electricity for refining         | kWh            | 7.21E+04 | assumed with Möbius / Wohlwill-electrolysis, Chilean mix 1) approximated with the UCTE mix        |
| Diesel (generators)              | MJ             | 4.18E+08 | Diesel, burned in diesel-electric generating set                                                  |
| Diesel (mobile equipment)        | l              | 9.44E+07 | Diesel, burned in building machine                                                                |
| <b>Materials</b>                 |                |          |                                                                                                   |
| Cyanide                          | kg             | 2.14E+06 | In LCI as sodium cyanide                                                                          |
| Lime                             | kg             | 1.89E+05 | Scaled from Steward (1999) by processed gold                                                      |
| Caustic                          | kg             | 3.48E+04 | Scaled from Steward (1999) by processed gold, accounted for sodium hydroxide                      |
| Carbon                           | kg             | 5.07E+04 | Scaled from Steward (1999) by processed gold, accounted for as charcoal                           |
| Zinc                             | kg             | 8.65E+03 | Scaled from Steward (1999) by processed gold, accounted for as zinc, primary, at regional storage |
| Sulphuric Acid                   | kg             | 1.99E+03 | Scaled from Steward (1999) by processed gold                                                      |
| Hydrochloric Acid                | kg             | 4.08E+04 | Scaled from Steward (1999) by processed gold                                                      |
| Explosives                       | kg             | 1.72E+06 | Scaled from Peruvian Mine by milled ore, in LCI: "blasting"                                       |
| <b>Water resources</b>           |                |          |                                                                                                   |
| Water resources                  | m <sup>3</sup> | 2.87E+04 | Scaled from Peruvian Mine by milled ore, in LCI: Water, unspecified natural origin                |
| <b>Emissions to Water</b>        |                |          |                                                                                                   |
| Arsenic, ion                     | kg             | 1.24E+02 | estimated with Data from PG, scaled by milled ore                                                 |
| Cadmium, ion                     | kg             | 5.80E+02 | estimated with Data from PG, scaled by milled ore                                                 |
| Copper, ion                      | kg             | 3.21E+04 | estimated with Data from PG, scaled by milled ore                                                 |
| Cyanide                          | kg             | 5.22E+04 | estimated with Data from PG, scaled by milled ore                                                 |
| Lead                             | kg             | 8.49E+01 | estimated with Data from PG, scaled by milled ore                                                 |
| Mercury                          | kg             | 1.32E+00 | estimated with Data from PG, scaled by milled ore                                                 |
| Nickel, ion                      | kg             | 7.70E+03 | estimated with Data from PG, scaled by milled ore                                                 |
| Zinc, ion                        | kg             | 2.05E+05 | estimated with Data from PG, scaled by milled ore                                                 |
| <b>Emissions to Air</b>          |                |          |                                                                                                   |
| Carbon Dioxide (from explosives) | kg             | 2.75E+05 | Calculation, based on data from Newmont (2005)                                                    |
| Zinc                             | kg             | 6.99E-01 | estimated with Data from SE, scaled by milled ore                                                 |
| Copper                           | kg             | 2.53E+00 | estimated with Data from SE, scaled by milled ore                                                 |
| Lead                             | kg             | 8.24E-01 | estimated with Data from SE, scaled by milled ore                                                 |
| <b>Disposal</b>                  | kg             | 2.25E+09 | estimated with data from PG, in LCI as: Disposal, sulphidic tailings                              |

<sup>1</sup> Chilean electricity mix in 2005 (Nationmaster, 2004): fossil fuel 47%, hydro 51.5%, other 1.5%.

There are existing facilities for water extraction in the Salar de Maricunga, which is located 50 kilometres southeast of La Coipa. There is a water pump system that collects the water from five wells. The water is accounted for as ground water (water, well, in ground).

Water emissions are approximated with figures of the mines in Papua New Guinea, material use (with exception of cyanide) with figures of Australian data for gold and silver production (Stewart, 1999), see also chapter 5.1.9), air emissions (zinc, copper and lead) with Swedish data. As Swedish data stand for a combined process, they are allocated to gold and silver for Sweden and used in this form for Chile. Cyanide is accounted for as sodium cyanide, which is commonly used in the gold mining. Since the module "blasting" does not include the CO<sub>2</sub> emissions from explosives, they are calculated with the factor of the mines of Newmont (2005b) (0.16 kg CO<sub>2</sub> per kg explosive). Water emissions are assumed to be emitted to rivers. Transports of materials are estimated to be 600 km by truck. Data on the amount of tailings disposal is approximated with data of mines in Papua New Guinea.

Since the mines produce doré gold (unrefined gold bullion bars containing between 60% and 95% gold), the refinery process is less complex than the refinery process of Rönnskär in Sweden. It is assumed, that the gold doré-bars undergo the treatment of Wohlwill electrolysis (for details see 5.2.2.1, "Gold / Silver, secondary, from electronic scrap inecoinvent"). Due to the lack of specific data about primary refinement, the figure of refinement from electronic scrap is used: 1.63 kWh per kg gold

(Renner et al., 2002a). The refinery of silver requires 0.6 kWh per kg of silver (Möbius electrolysis). No emissions were assumed, since on the one hand the high value of the processed material requires an efficient emission control and on the other hand, refinement is a hydrometallurgical process with low impact. The infrastructure is approximated with the facility of precious metal refinery. The unit process raw data are summarized in Tab. 5.26.

**Tab. 5.26: Unit process raw data of gold and silver mining and refining in Chile**

|                                      | Name                                                      | Location | InfrastructureProcess | Unit | mining and refining, gold-silver deposit | UncertaintyType | StandardDeviation95 | GeneralComment                                                                                     |
|--------------------------------------|-----------------------------------------------------------|----------|-----------------------|------|------------------------------------------|-----------------|---------------------|----------------------------------------------------------------------------------------------------|
|                                      | Location<br>InfrastructureProcess<br>Unit                 |          |                       |      | CL<br>0<br>a                             |                 |                     |                                                                                                    |
| allocated                            | gold, from combined gold-silver production, at refinery   | CL       | 0                     | kg   | 2.99E+3                                  |                 |                     |                                                                                                    |
| products                             | silver, from combined gold-silver production, at refinery | CL       | 0                     | kg   | 1.12E+5                                  |                 |                     |                                                                                                    |
| technosphere                         | electricity, medium voltage, production UCTE, at grid     | UCTE     | 0                     | kWh  | 1.60E+8                                  | 1               | 1.14                | (2,4,2,3,1,3); Environmental report (Placer-Dome, 2002)                                            |
|                                      | diesel, burned in diesel-electric generating set          | GLO      | 0                     | MJ   | 4.18E+8                                  | 1               | 1.14                | (2,4,2,3,1,3); Environmental report (Placer-Dome, 2002)                                            |
|                                      | diesel, burned in building machine                        | GLO      | 0                     | MJ   | 9.44E+7                                  | 1               | 1.14                | (2,4,2,3,1,3); Environmental report (Placer-Dome, 2002)                                            |
|                                      | sodium cyanide, at plant                                  | RER      | 0                     | kg   | 2.14E+6                                  | 1               | 1.14                | (2,4,2,3,1,3); Calculation, based on thesis on Australian precious metal production, data for 1999 |
|                                      | limestone, milled, packed, at plant                       | CH       | 0                     | kg   | 1.89E+5                                  | 1               | 1.14                | (2,4,2,3,1,3); Calculation, based on thesis on Australian precious metal production, data for 1999 |
|                                      | sodium hydroxide, 50% in H2O, production mix, at plant    | RER      | 0                     | kg   | 3.48E+4                                  | 1               | 1.15                | (2,3,2,5,1,3); Calculation, based on thesis on Australian precious metal production, data for 1999 |
|                                      | charcoal, at plant                                        | GLO      | 0                     | kg   | 5.07E+4                                  | 1               | 1.15                | (2,3,2,5,1,3); Calculation, based on thesis on Australian precious metal production, data for 1999 |
|                                      | zinc, primary, at regional storage                        | RER      | 0                     | kg   | 8.65E+3                                  | 1               | 1.15                | (2,3,2,5,1,3); Calculation, based on thesis on Australian precious metal production, data for 1999 |
|                                      | sulphuric acid, liquid, at plant                          | RER      | 0                     | kg   | 1.99E+3                                  | 1               | 1.15                | (2,3,2,5,1,3); Calculation, based on thesis on Australian precious metal production, data for 1999 |
|                                      | hydrochloric acid, 30% in H2O, at plant                   | RER      | 0                     | kg   | 4.08E+4                                  | 1               | 1.11                | (2,3,2,3,1,3); Calculation, based on thesis on Australian precious metal production, data for 1999 |
|                                      | blasting                                                  | RER      | 0                     | kg   | 1.72E+6                                  | 1               | 1.11                | (2,3,2,3,1,3); Estimation, based on peruvian Data                                                  |
|                                      | transport, lorry >16t, fleet average                      | RER      | 0                     | tkm  | 3.10E+8                                  | 1               | 2.02                | (2,4,na,na,na,na); Calculation, based on estimated distance of 600km                               |
|                                      | disposal, sulfidic tailings, off-site                     | GLO      | 0                     | kg   | 2.25E+9                                  | 1               | 1.11                | (2,3,2,3,1,3); Estimation                                                                          |
| infrastructure                       | facilities precious metal refinery                        | SE       | 1                     | unit | 1.89E-2                                  | 1               | 3.09                | (4,5,na,na,na,na); Environmental report                                                            |
|                                      | mine, gold and silver                                     | CL       | 1                     | unit | 2.00E-2                                  | 1               | 3.01                | (2,3,2,3,1,3); Environmental report                                                                |
| emission air, low population density | Copper                                                    | -        | -                     | kg   | 2.53E+0                                  | 1               | 5.02                | (2,4,2,3,1,3); Estimation, based on Swedish Data                                                   |
|                                      | Lead                                                      | -        | -                     | kg   | 8.24E-1                                  | 1               | 5.02                | (2,4,2,3,1,3); Estimation, based on Swedish Data                                                   |
|                                      | Zinc                                                      | -        | -                     | kg   | 6.99E-1                                  | 1               | 5.02                | (2,4,2,3,1,3); Estimation, based on Swedish Data                                                   |
|                                      | Carbon dioxide, fossil                                    | -        | -                     | kg   | 2.75E+5                                  | 1               | 1.14                | (2,4,2,3,1,3); Estimation, based on Swedish Data                                                   |
|                                      | Heat, waste                                               | -        | -                     | MJ   | 5.76E+8                                  | 1               | 1.14                | (2,4,2,3,1,3); Calculation                                                                         |
| emission water, ocean                | Arsenic, ion                                              | -        | -                     | kg   | 1.24E+2                                  | 1               | 5.02                | (2,4,2,3,1,3); Estimation, based on PG-Data                                                        |
|                                      | Cadmium, ion                                              | -        | -                     | kg   | 5.80E+2                                  | 1               | 3.02                | (2,4,2,3,1,3); Estimation, based on PG-Data                                                        |
|                                      | Copper, ion                                               | -        | -                     | kg   | 3.21E+4                                  | 1               | 3.02                | (2,4,2,3,1,3); Estimation, based on PG-Data                                                        |
|                                      | Cyanide                                                   | -        | -                     | kg   | 5.22E+4                                  | 1               | 3.02                | (2,4,2,3,1,3); Estimation, based on PG-Data                                                        |
|                                      | Lead                                                      | -        | -                     | kg   | 8.49E+1                                  | 1               | 5.02                | (2,4,2,3,1,3); Estimation, based on PG-Data                                                        |
|                                      | Mercury                                                   | -        | -                     | kg   | 1.32E+0                                  | 1               | 5.02                | (2,4,2,3,1,3); Estimation, based on PG-Data                                                        |
|                                      | Nickel, ion                                               | -        | -                     | kg   | 7.70E+3                                  | 1               | 5.02                | (2,4,2,3,1,3); Estimation, based on PG-Data                                                        |
|                                      | Zinc, ion                                                 | -        | -                     | kg   | 2.05E+5                                  | 1               | 5.02                | (2,4,2,3,1,3); Estimation, based on PG-Data                                                        |
| resource, in water                   | Water, unspecified natural origin                         | -        | -                     | m3   | 2.87E+4                                  | 1               | 1.14                | (2,4,2,3,1,3); Estimation, based on PG-Data                                                        |
| resource, in ground                  | Gold, Au 1.1E-4%, Ag 4.2E-3%, in ore, in ground           | -        | -                     | kg   | 3.52E+3                                  | 1               | 1.14                | (2,4,2,3,1,3); Estimation                                                                          |
|                                      | Silver, Ag 4.2E-3%, Au 1.1E-4%, in ore, in ground         | -        | -                     | kg   | 1.32E+05                                 | 1               | 1.14                | (2,4,2,3,1,3); Estimation                                                                          |

Tab. 5.27 Allocation factors of the data set “mining and refining, gold-silver deposit”

|                                      | Name                                                      | Location | Infrastructure | Process | Unit | mining and refining, gold-silver deposit | gold, from combined gold-silver production, at refinery | silver, from combined gold-silver production, at refinery |
|--------------------------------------|-----------------------------------------------------------|----------|----------------|---------|------|------------------------------------------|---------------------------------------------------------|-----------------------------------------------------------|
|                                      | Location<br>Infrastructure<br>Unit                        |          |                |         |      | CL<br>0<br>a                             | CL<br>0<br>kg                                           | CL<br>0<br>kg                                             |
| allocated                            | gold, from combined gold-silver production, at refinery   | CL       | 0              | kg      |      | 2.99E+3                                  | 100                                                     | -                                                         |
| products                             | silver, from combined gold-silver production, at refinery | CL       | 0              | kg      |      | 1.12E+5                                  | -                                                       | 100                                                       |
| technosphere                         | electricity, medium voltage, production UCTE, at grid     | UCTE     | 0              | kWh     |      | 1.60E+8                                  | 60.9                                                    | 39.1                                                      |
|                                      | diesel, burned in diesel-electric generating set          | GLO      | 0              | MJ      |      | 4.18E+8                                  | 60.9                                                    | 39.1                                                      |
|                                      | diesel, burned in building machine                        | GLO      | 0              | MJ      |      | 9.44E+7                                  | 60.9                                                    | 39.1                                                      |
|                                      | sodium cyanide, at plant                                  | RER      | 0              | kg      |      | 2.14E+6                                  | 60.9                                                    | 39.1                                                      |
|                                      | limestone, milled, packed, at plant                       | CH       | 0              | kg      |      | 1.89E+5                                  | 60.9                                                    | 39.1                                                      |
|                                      | sodium hydroxide, 50% in H2O, production mix, at plant    | RER      | 0              | kg      |      | 3.48E+4                                  | 60.9                                                    | 39.1                                                      |
|                                      | charcoal, at plant                                        | GLO      | 0              | kg      |      | 5.07E+4                                  | 60.9                                                    | 39.1                                                      |
|                                      | zinc, primary, at regional storage                        | RER      | 0              | kg      |      | 8.65E+3                                  | 60.9                                                    | 39.1                                                      |
|                                      | sulphuric acid, liquid, at plant                          | RER      | 0              | kg      |      | 1.99E+3                                  | 60.9                                                    | 39.1                                                      |
|                                      | hydrochloric acid, 30% in H2O, at plant                   | RER      | 0              | kg      |      | 4.08E+4                                  | 60.9                                                    | 39.1                                                      |
|                                      | blasting                                                  | RER      | 0              | kg      |      | 1.72E+6                                  | 60.9                                                    | 39.1                                                      |
|                                      | transport, lorry >16t, fleet average                      | RER      | 0              | tkm     |      | 3.10E+8                                  | 60.9                                                    | 39.1                                                      |
|                                      | disposal, sulfidic tailings, off-site                     | GLO      | 0              | kg      |      | 2.25E+9                                  | 60.9                                                    | 39.1                                                      |
|                                      | facilities precious metal refinery                        | SE       | 1              | unit    |      | 1.89E-2                                  | 97.6                                                    | 2.4                                                       |
| infrastructure                       | mine, gold and silver                                     | CL       | 1              | unit    |      | 2.00E-2                                  | 60.9                                                    | 39.1                                                      |
| emission air, low population density | Copper                                                    | -        | -              | kg      |      | 2.53E+0                                  | 60.9                                                    | 39.1                                                      |
|                                      | Lead                                                      | -        | -              | kg      |      | 8.24E-1                                  | 60.9                                                    | 39.1                                                      |
|                                      | Zinc                                                      | -        | -              | kg      |      | 6.99E-1                                  | 60.9                                                    | 39.1                                                      |
|                                      | Carbon dioxide, fossil                                    | -        | -              | kg      |      | 2.75E+5                                  | 60.9                                                    | 39.1                                                      |
|                                      | Heat, waste                                               | -        | -              | MJ      |      | 5.76E+8                                  | 60.9                                                    | 39.1                                                      |
| emission water, ocean                | Arsenic, ion                                              | -        | -              | kg      |      | 1.24E+2                                  | 60.9                                                    | 39.1                                                      |
|                                      | Cadmium, ion                                              | -        | -              | kg      |      | 5.80E+2                                  | 60.9                                                    | 39.1                                                      |
|                                      | Copper, ion                                               | -        | -              | kg      |      | 3.21E+4                                  | 60.9                                                    | 39.1                                                      |
|                                      | Cyanide                                                   | -        | -              | kg      |      | 5.22E+4                                  | 60.9                                                    | 39.1                                                      |
|                                      | Lead                                                      | -        | -              | kg      |      | 8.49E+1                                  | 60.9                                                    | 39.1                                                      |
|                                      | Mercury                                                   | -        | -              | kg      |      | 1.32E+0                                  | 60.9                                                    | 39.1                                                      |
|                                      | Nickel, ion                                               | -        | -              | kg      |      | 7.70E+3                                  | 60.9                                                    | 39.1                                                      |
|                                      | Zinc, ion                                                 | -        | -              | kg      |      | 2.05E+5                                  | 60.9                                                    | 39.1                                                      |
| resource, in water                   | Water, unspecified natural origin                         | -        | -              | m3      |      | 2.87E+4                                  | 60.9                                                    | 39.1                                                      |
| resource, in ground                  | Gold, Au 1.1E-4%, Ag 4.2E-3%, in ore, in ground           | -        | -              | kg      |      | 3.52E+3                                  | 100.0                                                   | -                                                         |
|                                      | Silver, Ag 4.2E-3%, Au 1.1E-4%, in ore, in ground         | -        | -              | kg      |      | 1.32E+05                                 | -                                                       | 100.0                                                     |

Tab. 5.28 Ecospold Meta information of metal production in Chile

|                            |                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                       |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name                       | mine, gold and silver                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                       |
| Location                   | CL                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                       |
| InfrastructureProcess      | 1                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                       |
| Unit                       | unit                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                       |
| Type                       | 1                                                                                                                                                                                                               | 5                                                                                                                                                                                                                                                                                                                                     |
| Version                    | 2.0                                                                                                                                                                                                             | 2.0                                                                                                                                                                                                                                                                                                                                   |
| energyValues               | 0                                                                                                                                                                                                               | 0                                                                                                                                                                                                                                                                                                                                     |
| LanguageCode               | en                                                                                                                                                                                                              | en                                                                                                                                                                                                                                                                                                                                    |
| LocalLanguageCode          | de                                                                                                                                                                                                              | de                                                                                                                                                                                                                                                                                                                                    |
| Person                     | 44                                                                                                                                                                                                              | 44                                                                                                                                                                                                                                                                                                                                    |
| QualityNetwork             | 1                                                                                                                                                                                                               | 1                                                                                                                                                                                                                                                                                                                                     |
| DataSetRelatesToProduct    | 1                                                                                                                                                                                                               | 1                                                                                                                                                                                                                                                                                                                                     |
| IncludedProcesses          | Land use, buildings and machinery for mining                                                                                                                                                                    | This data set includes the combined mining and refining of gold and silver in an open pit mine in Chile incl. energy and material use, water and air emissions of metals and land use.                                                                                                                                                |
| Amount                     | 1                                                                                                                                                                                                               | 1                                                                                                                                                                                                                                                                                                                                     |
| LocalName                  | Mine, Gold und Silber                                                                                                                                                                                           | Abbau und Raffination, Gold-Silber-Lagerstätte                                                                                                                                                                                                                                                                                        |
| Synonyms                   |                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                       |
| GeneralComment             | This data set includes land use of the considered mines incl. revegetated areas as well as estimates of buildings and machinery of the mine based on the data set non-ferrous metal, mine, surface/underground. | This multi-output process 'mining and refining, gold-silver deposit' delivers the two co-products 'gold, from combined gold-silver-production, at refinery' and 'silver, from combined gold-silver-production, at refinery'. This data set includes the mining in Chile in open pit mine incl. land use. Air emissions are estimated. |
| InfrastructureIncluded     | 1                                                                                                                                                                                                               | 1                                                                                                                                                                                                                                                                                                                                     |
| Category                   | metals                                                                                                                                                                                                          | metals                                                                                                                                                                                                                                                                                                                                |
| SubCategory                | extraction                                                                                                                                                                                                      | extraction                                                                                                                                                                                                                                                                                                                            |
| LocalCategory              | Metalle                                                                                                                                                                                                         | Metalle                                                                                                                                                                                                                                                                                                                               |
| LocalSubCategory           | Gewinnung                                                                                                                                                                                                       | Gewinnung                                                                                                                                                                                                                                                                                                                             |
| Formula                    |                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                       |
| StatisticalClassification  |                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                       |
| CASNumber                  |                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                       |
| StartDate                  | 2002                                                                                                                                                                                                            | 2002                                                                                                                                                                                                                                                                                                                                  |
| EndDate                    | 2006                                                                                                                                                                                                            | 2006                                                                                                                                                                                                                                                                                                                                  |
| DataValidForEntirePeriod   | 1                                                                                                                                                                                                               | 1                                                                                                                                                                                                                                                                                                                                     |
| OtherPeriodText            | Period of public environmental report                                                                                                                                                                           | Period of public environmental report                                                                                                                                                                                                                                                                                                 |
| Text                       | Land use of Chilean-plant, other inputs based on rough assumptions.                                                                                                                                             | Material requirements are estimated with Australian data                                                                                                                                                                                                                                                                              |
| Text                       | Assumptions for buildings and machinery based on the data set "non-ferrous metal, mine, surface/underground with 50 years life time.                                                                            | Refinement is estimated with electrolysis-data.                                                                                                                                                                                                                                                                                       |
| Percent                    |                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                       |
| ProductionVolume           | 3.2 Mill. t ore per year                                                                                                                                                                                        | 2.99 t gold and 112 t silver per year                                                                                                                                                                                                                                                                                                 |
| SamplingProcedure          | environmental report                                                                                                                                                                                            | environmental report                                                                                                                                                                                                                                                                                                                  |
| Extrapolations             | Buildings and machinery are based on global assumptions. The figures were extrapolated on the basis of the ore production.                                                                                      | See Geography                                                                                                                                                                                                                                                                                                                         |
| UncertaintyAdjustments     | none                                                                                                                                                                                                            | none                                                                                                                                                                                                                                                                                                                                  |
| Person                     | 44                                                                                                                                                                                                              | 44                                                                                                                                                                                                                                                                                                                                    |
| DataPublishedIn            | 2                                                                                                                                                                                                               | 2                                                                                                                                                                                                                                                                                                                                     |
| ReferenceToPublishedSource | 10                                                                                                                                                                                                              | 10                                                                                                                                                                                                                                                                                                                                    |
| Copyright                  | 1                                                                                                                                                                                                               | 1                                                                                                                                                                                                                                                                                                                                     |
| AccessRestrictedTo         | 0                                                                                                                                                                                                               | 0                                                                                                                                                                                                                                                                                                                                     |
| CompanyCode                |                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                       |
| CountryCode                |                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                       |
| PageNumbers                |                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                       |
| Validator                  | 51                                                                                                                                                                                                              | 51                                                                                                                                                                                                                                                                                                                                    |
| Details                    | automatic validation in Excel                                                                                                                                                                                   | automatic validation in Excel                                                                                                                                                                                                                                                                                                         |
| OtherDetails               | none                                                                                                                                                                                                            | none                                                                                                                                                                                                                                                                                                                                  |

### 5.1.5 Gold and silver production, Peru (PE)

Yanacocha is a gold mine in northern Peru, considered one of the biggest and most profitable in the world. The Yanacocha-Mine is located in the Cajamarca Province of the Peruvian highlands about 450 kilometres north of the capital Lima and produces gold and silver. Data for the Mine "Yanacocha" stem from the environmental reports of Newmont (2005a) and Arana (2002). Newmont is the major shareholder together with Buenaventura, a Peruvian company. In 2005, Yanacocha produced 103'000

kg of gold. Minera Yanacocha consists of five open pit mines, four leach pads, two gold recovery plants, and a crushing and agglomeration facility.

The recovery rate (proportion of valuable material physically recovered in the processing of ore) is taken into account in order to define the resource extraction. The average concentration (head grade, expressed in kg metal per kg ore) of metals in ore for Peru is calculated with the quantity of ore milled, the corresponding production rate and the recovery rate.

**Tab. 5.29 Production, Allocation and Recovery rate of the various metals produced in 2005 by the Yanacocha-Mines (Newmont, 2005a; Arana, 2002), the prices are from Tab. 2.5.**

| <b>Yanacocha</b>  |           |          |                               |
|-------------------|-----------|----------|-------------------------------|
| Production Gold   | kg        | 1.03E+05 | Value of 1kg Gold: 15680 US\$ |
| Production Silver | kg        | 3.67E+04 | Value of 1kg Silver: 269 US\$ |
| Allocation Gold   |           | 99.4%    |                               |
| Allocation Silver |           | 0.6%     |                               |
| Ore milled        | kg        | 7.85E+10 |                               |
| <i>Head grade</i> |           |          |                               |
| Gold              | kg/kg ore | 1.54E-06 |                               |
| Silver            | kg/kg ore | 5.49E-07 |                               |
| Recovery rate     |           | 85%      |                               |

### Mining infrastructure

This data set includes land use as well as rough estimates on buildings and machines necessary in gold mining. It is assumed, that the quantities of machinery as well as surface and volume of buildings are equal to the generic mine. This approach is rough, and the uncertainty is rather high. As the mined ore from the generic mine is some orders of magnitude lower than the mine in focus, a proportional scaling by milled ore will lead to a higher mistake (see Tab. 5.1: Gold and Silver production in Sweden). The lifetime of mines is assumed to be 50 years.

Land use is based on the figures from the environmental report (Tab. 5.30). Not all the area of the mine is in active use at one time; the value in Tab. 5.30 is assumed to be the average share of the mine area in use. Therefore, the total area is occupied during an average of about 42 years, according to the following formula:

$$\frac{\text{Disturbed area in active use}}{\text{Area disturbed during life of mine}} * \text{lifetime of mine} = \frac{2.68 * 10^7}{3.19 * 10^8} * 50y = 42y$$
 , the occupation of the mine is calculated as  $\text{Disturbed area in active use} * 50y = \text{Occupation, mineral extraction site}$

**Tab. 5.30: Land use of the Yanacocha-Mine in 2005 (Newmont, 2005a)**

| <b>Land use</b>                    |                |          | <b>Remark</b>                                                                |
|------------------------------------|----------------|----------|------------------------------------------------------------------------------|
| Area disturbed during life of mine | m <sup>2</sup> | 3.19E+07 | Transformation, from unknown and Transformation, to mineral extraction site, |
| Disturbed area in active use       | m <sup>2</sup> | 2.68E+07 | Occupation, mineral extraction site, during 50 years                         |
| Area available for revegetation    | m <sup>2</sup> | 1.80E+06 | Transformation, to unknown                                                   |
| Area revegetated                   | m <sup>2</sup> | 3.37E+06 | Transformation, to heterogeneous, agricultural                               |

Tab. 5.31: Unit process raw data of the gold and silver mine in Peru.

|                | Name                                             | Location | Infrastructure | Process | Unit | mine, gold and silver | Uncertainty Type | Standard Deviation 95% | GeneralComment                                    |
|----------------|--------------------------------------------------|----------|----------------|---------|------|-----------------------|------------------|------------------------|---------------------------------------------------|
|                | Location<br>Infrastructure<br>Process<br>Unit    |          |                |         |      | PE<br>1<br>unit       |                  |                        |                                                   |
| product        | mine, gold and silver                            | PE       | 1              | unit    |      | 1.00E+0               |                  |                        |                                                   |
| technosphere   | building, multi-storey                           | RER      | 1              | m3      |      | 2.75E+5               | 1                | 2.39                   | (4,5,5,5,5,5); Rough estimates                    |
|                | building, hall                                   | CH       | 1              | m2      |      | 1.25E+4               | 1                | 2.39                   | (4,5,5,5,5,5); Rough estimates                    |
|                | industrial machine, heavy, unspecified, at plant | RER      | 1              | kg      |      | 8.50E+5               | 1                | 2.39                   | (4,5,5,5,5,5); Rough estimates                    |
|                | conveyor belt, at plant                          | RER      | 1              | m       |      | 6.00E+2               | 1                | 2.39                   | (4,5,5,5,5,5); Rough estimates                    |
| resource, land | Transformation, from unknown                     | -        | -              | m2      |      | 3.19E+7               | 1                | 2.01                   | (1,3,1,1,1,3); Environmental report, Newmont 2005 |
|                | Transformation, to mineral extraction site       | -        | -              | m2      |      | 3.19E+7               | 1                | 2.01                   | (1,3,1,1,1,3); Environmental report, Newmont 2005 |
|                | Occupation, mineral extraction site              | -        | -              | m2a     |      | 2.68E+7               | 1                | 1.51                   | (1,3,1,1,1,3); Environmental report, Newmont 2005 |
|                | Transformation, from mineral extraction site     | -        | -              | m2      |      | 3.19E+7               | 1                | 2.01                   | (1,3,1,1,1,3); Environmental report, Newmont 2005 |
|                | Transformation, to heterogeneous, agricultural   | -        | -              | m2      |      | 3.37E+6               | 1                | 2.01                   | (1,3,1,1,1,3); Environmental report, Newmont 2005 |
|                | Transformation, to unknown                       | -        | -              | m2      |      | 2.86E+7               | 1                | 2.01                   | (1,3,1,1,1,3); Environmental report, Newmont 2005 |

## Mining and refining

The ore mined at Minera Yanacocha is broken up into small pieces and placed in layers, forming a hill of ore known as a leach pad. A diluted sodium-cyanide solution is dripped onto the leach pad to extract the gold. The gold-laden solution is collected at the bottom of the pad for processing.

Table 5.25 displays the total energy and material requirements of the Yanagocha-Mines in 2005.

Tab. 5.32: Energy and material requirements of the Yanacocha-Mines in 2005 (Newmont, 2005a)

| Yanagocha                         |                |          | Remark                                                                                                                    |
|-----------------------------------|----------------|----------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Energy</b>                     |                |          |                                                                                                                           |
| Electricity for mining            | kWh            | 3.37E+08 | Environmental report, Peruvian mix 1) approximated with the Brazilian mix                                                 |
| Electricity for refining          | kWh            | 1.89E+05 | assumed with Möbius / Wohlwill-electrolysis, Peruvian mix 1) approximated with the Brazilian mix                          |
| Diesel (generators)               | MJ             | 7.88E+06 | Diesel, burned in diesel-electric generating set                                                                          |
| Diesel (mobile equipment)         | MJ             | 4.86E+09 | Diesel, burned in building machine                                                                                        |
| Petrol                            | l              | 4.38E+04 | converted in transport, van <3.5t (0.38l/tkm)                                                                             |
| Oil                               | MJ             | 1.41E+08 | heavy fuel oil, burned in industrial furnace 1MW, non-modulating                                                          |
| <b>Materials</b>                  |                |          |                                                                                                                           |
| Cyanide                           | kg             | 4.40E+06 | In LCI as sodium cyanide                                                                                                  |
| Lime                              | kg             | 1.19E+08 | Environmental report                                                                                                      |
| Caustic                           | kg             | 1.20E+06 | Scaled from Australian value by processed gold and silver, accounted for as Sodium Hydroxid                               |
| Carbon                            | kg             | 1.75E+06 | Scaled from Australian value by processed gold & silver, accounted for as Charcoal, at plant                              |
| Zinc                              | kg             | 2.99E+05 | Scaled from Steward (1999) by processed gold and silver, accounted for as zinc, primary, at regional storage              |
| Sulphuric Acid                    | kg             | 5.43E+05 | Acids (hydrochloric, sulphamic and sulphuric) consumption is divided in half sulphuric, half hydrochloric (Newmont, 2006) |
| Hydrochloric Acid                 | kg             | 5.43E+05 | Environmental report                                                                                                      |
| Explosives                        | kg             | 4.26E+07 | Partly ANFO explosives, approximates with Tovex                                                                           |
| <b>Water resources</b>            |                |          |                                                                                                                           |
| Surface Water Use                 | m <sup>3</sup> | 4.40E+05 | in LCI: Water, river                                                                                                      |
| Groundwater Use                   | m <sup>3</sup> | 2.72E+05 | in LCI: Water, well, in ground                                                                                            |
| Mine dewatering Use               | m <sup>3</sup> | 2.77E+06 | recycled Water, not accounted for in inventory                                                                            |
| <b>Emissions to Water</b>         |                |          |                                                                                                                           |
| Arsenic, ion                      | kg             | 3.07E+03 | Estimation with PG-Data, scaled by milled ore                                                                             |
| Cadmium, ion                      | kg             | 1.44E+04 | Estimation with PG-Data, scaled by milled ore                                                                             |
| Copper, ion                       | kg             | 7.96E+05 | Estimation with PG-Data, scaled by milled ore                                                                             |
| Cyanide                           | kg             | 1.29E+06 | Estimation with PG-Data, scaled by milled ore                                                                             |
| Lead                              | kg             | 2.10E+03 | Estimation with PG-Data, scaled by milled ore                                                                             |
| Mercury                           | kg             | 3.26E+01 | Estimation with PG-Data, scaled by milled ore                                                                             |
| Nickel, ion                       | kg             | 1.91E+05 | Estimation with PG-Data, scaled by milled ore                                                                             |
| Zinc, ion                         | kg             | 5.07E+06 | Estimation with PG-Data, scaled by milled ore                                                                             |
| <b>Emissions to Air</b>           |                |          |                                                                                                                           |
| Arsenic                           | kg             | 9.07E+00 | Environmental Report                                                                                                      |
| Carbon Monoxide                   | kg             | 9.60E+04 | Environmental Report, not accounted for                                                                                   |
| Carbon Dioxide (from explosives)  | kg             | 7.00E+06 | Environmental Report                                                                                                      |
| Lead                              | kg             | 4.54E+01 | Environmental Report                                                                                                      |
| Mercury                           | kg             | 3.45E+02 | Environmental Report                                                                                                      |
| Nitrogen Oxide (NOx)              | kg             | 1.50E+05 | Environmental Report, not accounted for                                                                                   |
| Particulates                      | kg             | 1.31E+04 | Environmental Report, not accounted for                                                                                   |
| Sulfur Dioxide (SO <sub>2</sub> ) | kg             | 6.04E+04 | Environmental Report, not accounted for                                                                                   |

- 1 Peruvian electricity mix in 2005 (Nationmaster, 2004): fossil fuel 14.5%, hydro 84.7% others: 0.8%, Brazilian electricity mix: fossil fuel 9.9%, hydro: 83.7, others: 6.4%

Since no reliable measures or values are available, water emissions are approximated with the figures of the mines in Papua New Guinea. The use of charcoal and zinc is estimated with figures of Australian data for gold and silver production (Stewart, 1999, see also chapter 5.1.9). Airborne emissions as carbon monoxide, nitrogen oxide, particulates and sulfur dioxide due to burning of fossil fuels are already inventoried and therefore not accounted in this LCI. The disposal of charcoal has been neglected. Since the module “blasting” does not include the CO<sub>2</sub> emissions from explosives, they are calculated with the factor of the mines of Newmont (2005b) (0.16 kg CO<sub>2</sub> per kg explosive). Cyanide is accounted for as sodium cyanide, which is commonly used in the gold mining. Water emissions are assumed to be emitted to rivers. Transports of materials are estimated to be 600 km per truck. Data



about amounts of tailings disposal is approximated with data of gold/silver mines in Papua New Guinea.

Since the mines produce doré gold (unrefined gold bullion bars containing between 60% and 95% gold), the refinery process is less complex than the refinery process of Rönnskär in Sweden. It is assumed, that the gold doré-bars undergo the treatment of Wohlwill electrolysis (for details see 5.2.2.1, “Gold / Silver, secondary, from electronic scrap in ecoinvent”). Due to the lack of specific data, the figure of refinement from electronic scrap is used: 1.63 kWh per kg gold (Renner et al., 2002a). The refinery of silver requires 0.6 kWh per kg of silver (Möbius electrolysis). No emissions were assumed, since on the one hand the high value of the processed material requires an efficient emission control and on the other hand, refinement is a hydrometallurgical process with low impact. The infrastructure is approximated with the facility of precious metal refinery. The unit process raw data are summarized in Tab. 5.33 .

Tab. 5.33: Unit process raw data of gold and silver mining and refining in Peru

|                                      | Name                                                      | Location | Infrastructure | Process | Unit     | mining and refining, gold-silver deposit | UncertaintyType | StandardDeviation95% | GeneralComment                                                                      |
|--------------------------------------|-----------------------------------------------------------|----------|----------------|---------|----------|------------------------------------------|-----------------|----------------------|-------------------------------------------------------------------------------------|
|                                      | Location<br>InfrastructureProcess<br>Unit                 |          |                |         |          | PE<br>0<br>a                             |                 |                      |                                                                                     |
| allocated                            | gold, from combined gold-silver production, at refinery   | PE       | 0              | kg      | 1.03E+5  |                                          |                 |                      |                                                                                     |
| products                             | silver, from combined gold-silver production, at refinery | PE       | 0              | kg      | 3.67E+4  |                                          |                 |                      |                                                                                     |
| technosphere                         | electricity, medium voltage, at grid                      | BR       | 0              | kWh     | 3.37E+08 | 1                                        | 1.20            | (3,4,1,5,1,3);       | Environmental report                                                                |
|                                      | diesel, burned in diesel-electric generating set          | GLO      | 0              | MJ      | 7.88E+06 | 1                                        | 1.16            | (3,4,1,2,1,3);       | Environmental report                                                                |
|                                      | diesel, burned in building machine                        | GLO      | 0              | MJ      | 4.86E+09 | 1                                        | 1.16            | (3,4,1,2,1,3);       | Environmental report                                                                |
|                                      | transport, van <3.5t                                      | RER      | 0              | tkm     | 1.15E+05 | 1                                        | 2.24            | (5,3,1,1,1,3);       | Estimations                                                                         |
|                                      | sodium cyanide, at plant                                  | RER      | 0              | kg      | 4.40E+06 | 1                                        | 1.14            | (3,3,1,1,1,3);       | Environmental report                                                                |
|                                      | limestone, milled, packed, at plant                       | CH       | 0              | kg      | 1.19E+08 | 1                                        | 1.14            | (3,3,1,1,1,3);       | Environmental report                                                                |
|                                      | sodium hydroxide, 50% in H2O, production mix, at plant    | RER      | 0              | kg      | 1.20E+06 | 1                                        | 1.20            | (3,3,3,5,1,3);       | Calculation, based on thesis on Australian precious metal production, data for 1999 |
|                                      | hydrochloric acid, 30% in H2O, at plant                   | RER      | 0              | kg      | 5.43E+05 | 1                                        | 1.14            | (3,3,1,1,1,3);       | Environmental report                                                                |
|                                      | zinc, primary, at regional storage                        | RER      | 0              | kg      | 2.99E+05 | 1                                        | 1.20            | (3,3,3,5,1,3);       | Calculation, based on thesis on Australian precious metal production, data for 1999 |
|                                      | charcoal, at plant                                        | GLO      | 0              | kg      | 1.75E+06 | 1                                        | 1.20            | (3,3,3,5,1,3);       | Calculation, based on thesis on Australian precious metal production, data for 1999 |
|                                      | sulphuric acid, liquid, at plant                          | RER      | 0              | kg      | 5.43E+05 | 1                                        | 1.14            | (3,3,1,1,1,3);       | Environmental report                                                                |
|                                      | blasting                                                  | RER      | 0              | kg      | 4.26E+07 | 1                                        | 1.09            | (1,3,1,1,1,3);       | Environmental report                                                                |
|                                      | transport, lorry >16t, fleet average                      | RER      | 0              | tkm     | 1.76E+08 | 1                                        | 2.24            | (5,3,1,1,1,3);       | Estimations                                                                         |
|                                      | disposal, sulfidic tailings, off-site                     | GLO      | 0              | kg      | 7.85E+10 | 1                                        | 1.17            | (3,3,1,5,1,3);       | Environmental report                                                                |
| infrastructure                       | facilities precious metal refinery                        | SE       | 1              | unit    | 2.29E-02 | 1                                        | 3.09            | (4,5,na,na,na,na);   | Environmental report                                                                |
|                                      | mine, gold and silver                                     | PE       | 1              | unit    | 2.00E-02 | 1                                        | 3.05            | (1,3,1,2,3,3);       | Annual report                                                                       |
| emission air, low population density | Arsenic                                                   | -        | -              | kg      | 9.07E+00 | 1                                        | 5.04            | (3,3,1,5,1,3);       | Environmental report                                                                |
|                                      | Carbon dioxide, fossil                                    | -        | -              | kg      | 7.00E+06 | 1                                        | 1.14            | (3,3,1,1,1,3);       | Environmental report                                                                |
|                                      | Lead                                                      | -        | -              | kg      | 4.54E+01 | 1                                        | 5.02            | (3,3,1,1,1,3);       | Environmental report                                                                |
|                                      | Mercury                                                   | -        | -              | kg      | 3.45E+02 | 1                                        | 5.02            | (3,3,1,1,1,3);       | Environmental report                                                                |
|                                      | Heat, waste                                               | -        | -              | MJ      | 1.21E+09 | 1                                        | 1.14            | (3,3,1,1,1,3);       | Calculation                                                                         |
| emission water, river                | Arsenic, ion                                              | -        | -              | kg      | 3.07E+03 | 1                                        | 5.04            | (3,3,2,5,1,3);       | Estimated with data from combined production, PG                                    |
|                                      | Cadmium, ion                                              | -        | -              | kg      | 1.44E+04 | 1                                        | 3.03            | (3,3,2,5,1,3);       | Estimated with data from combined production, PG                                    |
|                                      | Copper, ion                                               | -        | -              | kg      | 7.96E+05 | 1                                        | 3.03            | (3,3,2,5,1,3);       | Estimated with data from combined production, PG                                    |
|                                      | Cyanide                                                   | -        | -              | kg      | 1.29E+06 | 1                                        | 3.03            | (3,3,2,5,1,3);       | Estimated with data from combined production, PG                                    |
|                                      | Lead                                                      | -        | -              | kg      | 2.10E+03 | 1                                        | 5.04            | (3,3,2,5,1,3);       | Estimated with data from combined production, PG                                    |
|                                      | Mercury                                                   | -        | -              | kg      | 3.26E+01 | 1                                        | 5.04            | (3,3,2,5,1,3);       | Estimated with data from combined production, PG                                    |
|                                      | Nickel, ion                                               | -        | -              | kg      | 1.91E+05 | 1                                        | 5.04            | (3,3,2,5,1,3);       | Estimated with data from combined production, PG                                    |
|                                      | Zinc, ion                                                 | -        | -              | kg      | 5.07E+06 | 1                                        | 5.04            | (3,3,2,5,1,3);       | Estimated with data from combined production, PG                                    |
| resource, in water                   | Water, well, in ground                                    | -        | -              | m3      | 4.40E+05 | 1                                        | 1.17            | (3,3,1,5,1,3);       | Environmental report                                                                |
|                                      | Water, river                                              | -        | -              | m3      | 2.72E+05 | 1                                        | 1.17            | (3,3,1,5,1,3);       | Environmental report                                                                |
| resource, in ground                  | Gold, Au 1.3E-4%, Ag 4.6E-5%, in ore, in ground           | -        | -              | kg      | 1.21E+05 | 1                                        | 1.17            | (3,3,1,5,1,3);       | Environmental report                                                                |
|                                      | Silver, Ag 4.6E-5%, Au 1.3E-4%, in ore, in ground         | -        | -              | kg      | 4.31E+04 | 1                                        | 1.17            | (3,3,1,5,1,3);       | Environmental report                                                                |

Tab. 5.34 Allocation factors of the data set “mining, gold-silver deposit”

|                                      | Name                                                      | Location | Infrastructure | Process | Unit     | mining and refining, gold-silver deposit | gold, from combined gold-silver production, at refinery | silver, from combined gold-silver production, at refinery |
|--------------------------------------|-----------------------------------------------------------|----------|----------------|---------|----------|------------------------------------------|---------------------------------------------------------|-----------------------------------------------------------|
|                                      | Location<br>InfrastructureProcess<br>Unit                 |          |                |         |          | PE<br>0<br>a                             | PE<br>0<br>kg                                           | PE<br>0<br>kg                                             |
| allocated                            | gold, from combined gold-silver production, at refinery   | PE       | 0              | kg      | 1.03E+5  |                                          | 100                                                     | 0                                                         |
| products                             | silver, from combined gold-silver production, at refinery | PE       | 0              | kg      | 3.67E+4  |                                          | 0                                                       | 100                                                       |
| technosphere                         | electricity, medium voltage, at grid                      | BR       | 0              | kWh     | 3.37E+08 |                                          | 99.42                                                   | 0.6                                                       |
|                                      | diesel, burned in diesel-electric generating set          | GLO      | 0              | MJ      | 7.88E+06 |                                          | 99.42                                                   | 0.6                                                       |
|                                      | diesel, burned in building machine                        | GLO      | 0              | MJ      | 4.86E+09 |                                          | 99.42                                                   | 0.6                                                       |
|                                      | transport, van <3.5t                                      | RER      | 0              | tkm     | 1.15E+05 |                                          | 97.6                                                    | 2.4                                                       |
|                                      | sodium cyanide, at plant                                  | RER      | 0              | kg      | 4.40E+06 |                                          | 99.42                                                   | 0.6                                                       |
|                                      | limestone, milled, packed, at plant                       | CH       | 0              | kg      | 1.19E+08 |                                          | 99.42                                                   | 0.6                                                       |
|                                      | sodium hydroxide, 50% in H2O, production mix, at plant    | RER      | 0              | kg      | 1.20E+06 |                                          | 99.42                                                   | 0.6                                                       |
|                                      | hydrochloric acid, 30% in H2O, at plant                   | RER      | 0              | kg      | 5.43E+05 |                                          | 99.42                                                   | 0.6                                                       |
|                                      | zinc, primary, at regional storage                        | RER      | 0              | kg      | 2.99E+05 |                                          | 99.42                                                   | 0.6                                                       |
|                                      | charcoal, at plant                                        | GLO      | 0              | kg      | 1.75E+06 |                                          | 99.42                                                   | 0.6                                                       |
|                                      | sulphuric acid, liquid, at plant                          | RER      | 0              | kg      | 5.43E+05 |                                          | 99.42                                                   | 0.6                                                       |
|                                      | blasting                                                  | RER      | 0              | kg      | 4.26E+07 |                                          | 99.42                                                   | 0.6                                                       |
|                                      | transport, lorry >16t, fleet average                      | RER      | 0              | tkm     | 1.76E+08 |                                          | 99.42                                                   | 0.6                                                       |
|                                      | disposal, sulfidic tailings, off-site                     | GLO      | 0              | kg      | 7.85E+10 |                                          | 99.42                                                   | 0.6                                                       |
| infrastructure                       | facilities precious metal refinery                        | SE       | 1              | unit    | 2.29E-02 |                                          | 97.6                                                    | 2.4                                                       |
|                                      | mine, gold and silver                                     | PE       | 1              | unit    | 2.00E-02 |                                          | 99.42                                                   | 0.6                                                       |
| emission air, low population density | Arsenic                                                   | -        | -              | kg      | 9.07E+00 |                                          | 99.42                                                   | 0.6                                                       |
|                                      | Carbon dioxide, fossil                                    | -        | -              | kg      | 7.00E+06 |                                          | 99.42                                                   | 0.6                                                       |
|                                      | Lead                                                      | -        | -              | kg      | 4.54E+01 |                                          | 99.42                                                   | 0.6                                                       |
|                                      | Mercury                                                   | -        | -              | kg      | 3.45E+02 |                                          | 99.42                                                   | 0.6                                                       |
|                                      | Heat, waste                                               | -        | -              | MJ      | 1.21E+09 |                                          | 99.42                                                   | 0.6                                                       |
| emission water, river                | Arsenic, ion                                              | -        | -              | kg      | 3.07E+03 |                                          | 99.42                                                   | 0.6                                                       |
|                                      | Cadmium, ion                                              | -        | -              | kg      | 1.44E+04 |                                          | 99.42                                                   | 0.6                                                       |
|                                      | Copper, ion                                               | -        | -              | kg      | 7.96E+05 |                                          | 99.42                                                   | 0.6                                                       |
|                                      | Cyanide                                                   | -        | -              | kg      | 1.29E+06 |                                          | 99.42                                                   | 0.6                                                       |
|                                      | Lead                                                      | -        | -              | kg      | 2.10E+03 |                                          | 99.42                                                   | 0.6                                                       |
|                                      | Mercury                                                   | -        | -              | kg      | 3.26E+01 |                                          | 99.42                                                   | 0.6                                                       |
|                                      | Nickel, ion                                               | -        | -              | kg      | 1.91E+05 |                                          | 99.42                                                   | 0.6                                                       |
|                                      | Zinc, ion                                                 | -        | -              | kg      | 5.07E+06 |                                          | 99.42                                                   | 0.6                                                       |
| resource, in water                   | Water, well, in ground                                    | -        | -              | m3      | 4.40E+05 |                                          | 99.42                                                   | 0.6                                                       |
|                                      | Water, river                                              | -        | -              | m3      | 2.72E+05 |                                          | 99.42                                                   | 0.6                                                       |
| resource, in ground                  | Gold, Au 1.3E-4%, Ag 4.6E-5%, in ore, in ground           | -        | -              | kg      | 1.21E+05 |                                          | 100.00                                                  | -                                                         |
|                                      | Silver, Ag 4.6E-5%, Au 1.3E-4%, in ore, in ground         | -        | -              | kg      | 4.31E+04 |                                          | -                                                       | 100.0                                                     |

## 5.35 EcoSpold Meta information of gold / silver mining and refining in Peru

|                            |                                                                                                                                                                                                                 |  |                                                                                                                                                                                                                                                                                                                                       |  |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Name                       | mine, gold and silver                                                                                                                                                                                           |  | mining and refining, gold-silver deposit                                                                                                                                                                                                                                                                                              |  |
| Location                   | PE                                                                                                                                                                                                              |  | PE                                                                                                                                                                                                                                                                                                                                    |  |
| InfrastructureProcess      | 1                                                                                                                                                                                                               |  | 0                                                                                                                                                                                                                                                                                                                                     |  |
| Unit                       | unit                                                                                                                                                                                                            |  | a                                                                                                                                                                                                                                                                                                                                     |  |
| Type                       | 1                                                                                                                                                                                                               |  | 5                                                                                                                                                                                                                                                                                                                                     |  |
| Version                    | 2.0                                                                                                                                                                                                             |  | 2.0                                                                                                                                                                                                                                                                                                                                   |  |
| energyValues               | 0                                                                                                                                                                                                               |  | 0                                                                                                                                                                                                                                                                                                                                     |  |
| LanguageCode               | en                                                                                                                                                                                                              |  | en                                                                                                                                                                                                                                                                                                                                    |  |
| LocalLanguageCode          | de                                                                                                                                                                                                              |  | de                                                                                                                                                                                                                                                                                                                                    |  |
| Person                     | 44                                                                                                                                                                                                              |  | 44                                                                                                                                                                                                                                                                                                                                    |  |
| QualityNetwork             | 1                                                                                                                                                                                                               |  | 1                                                                                                                                                                                                                                                                                                                                     |  |
| DataSetRelatesToProduct    | 1                                                                                                                                                                                                               |  | 1                                                                                                                                                                                                                                                                                                                                     |  |
| IncludedProcesses          | Land use, buildings and machinery for mining                                                                                                                                                                    |  | This data set includes the combined mining and refining of gold and silver in open pit mines in Peru incl. energy and material use, water and air emissions of metals and land use.                                                                                                                                                   |  |
| Amount                     | 1                                                                                                                                                                                                               |  | 1                                                                                                                                                                                                                                                                                                                                     |  |
| LocalName                  | Mine, Gold und Silber                                                                                                                                                                                           |  | Abbau und Raffination, Gold-Silber-Lagerstätte                                                                                                                                                                                                                                                                                        |  |
| GeneralComment             | This data set includes land use of the considered mines incl. revegetated areas as well as estimates of buildings and machinery of the mine based on the data set non-ferrous metal, mine, surface/underground. |  | This multi-output process 'mining and refining, gold-silver deposit' delivers the two co-products 'gold, from combined gold-silver-production, at refinery' and 'silver, from combined gold-silver-production, at refinery'. This data set includes the mining in Peru in open pit mines incl. land use. Air emissions are estimated. |  |
| InfrastructureIncluded     | 1                                                                                                                                                                                                               |  | 1                                                                                                                                                                                                                                                                                                                                     |  |
| Category                   | metals                                                                                                                                                                                                          |  | metals                                                                                                                                                                                                                                                                                                                                |  |
| SubCategory                | extraction                                                                                                                                                                                                      |  | extraction                                                                                                                                                                                                                                                                                                                            |  |
| LocalCategory              | Metalle                                                                                                                                                                                                         |  | Metalle                                                                                                                                                                                                                                                                                                                               |  |
| LocalSubCategory           | Gewinnung                                                                                                                                                                                                       |  | Gewinnung                                                                                                                                                                                                                                                                                                                             |  |
| StartDate                  | 2004                                                                                                                                                                                                            |  | 2004                                                                                                                                                                                                                                                                                                                                  |  |
| EndDate                    | 2006                                                                                                                                                                                                            |  | 2006                                                                                                                                                                                                                                                                                                                                  |  |
| DataValidForEntirePeriod   | 1                                                                                                                                                                                                               |  | 1                                                                                                                                                                                                                                                                                                                                     |  |
| OtherPeriodText            | Period of public environmental report                                                                                                                                                                           |  | Period of public environmental report                                                                                                                                                                                                                                                                                                 |  |
| Text                       | Land use of Peruvian-plant, other inputs based on rough assumptions.                                                                                                                                            |  | Material requirements are estimated with Australian data, water emissions are estimated with Data from Papua New Guinea                                                                                                                                                                                                               |  |
| Text                       | Assumptions for buildings and machinery based on the data set "non-ferrous metal, mine, surface/underground with 50 years life time.                                                                            |  | The mining of ore from open pit mines is considered.                                                                                                                                                                                                                                                                                  |  |
| Percent                    |                                                                                                                                                                                                                 |  |                                                                                                                                                                                                                                                                                                                                       |  |
| ProductionVolume           | 78.5 Mill. t ore per year                                                                                                                                                                                       |  | 102.64 t gold and 36.66 t silver per year                                                                                                                                                                                                                                                                                             |  |
| SamplingProcedure          | environmental report                                                                                                                                                                                            |  | environmental report                                                                                                                                                                                                                                                                                                                  |  |
| Extrapolations             | Buildings and machinery are based on global assumptions. The figures were extrapolated on the basis of the ore production.                                                                                      |  | See Geography, estimations have been made for refinement of Gold                                                                                                                                                                                                                                                                      |  |
| UncertaintyAdjustments     | none                                                                                                                                                                                                            |  | none                                                                                                                                                                                                                                                                                                                                  |  |
| Person                     | 44                                                                                                                                                                                                              |  | 44                                                                                                                                                                                                                                                                                                                                    |  |
| DataPublishedIn            | 2                                                                                                                                                                                                               |  | 2                                                                                                                                                                                                                                                                                                                                     |  |
| ReferenceToPublishedSource | 10                                                                                                                                                                                                              |  | 10                                                                                                                                                                                                                                                                                                                                    |  |
| Copyright                  | 1                                                                                                                                                                                                               |  | 1                                                                                                                                                                                                                                                                                                                                     |  |
| AccessRestrictedTo         | 0                                                                                                                                                                                                               |  | 0                                                                                                                                                                                                                                                                                                                                     |  |
| CompanyCode                |                                                                                                                                                                                                                 |  |                                                                                                                                                                                                                                                                                                                                       |  |
| CountryCode                |                                                                                                                                                                                                                 |  |                                                                                                                                                                                                                                                                                                                                       |  |
| PageNumbers                |                                                                                                                                                                                                                 |  |                                                                                                                                                                                                                                                                                                                                       |  |
| Validator                  | 51                                                                                                                                                                                                              |  | 51                                                                                                                                                                                                                                                                                                                                    |  |
| Details                    | automatic validation in Excel                                                                                                                                                                                   |  | automatic validation in Excel                                                                                                                                                                                                                                                                                                         |  |
| OtherDetails               | none                                                                                                                                                                                                            |  | none                                                                                                                                                                                                                                                                                                                                  |  |

## 5.1.6 Gold production, Canada (CA)

Data stem from the environmental report of Placer Dome Inc. (2002b) for the Campbell, Musselwhite and Porcupine Mines. Since the mines have been sold in 2006 to Kinross Gold Corporation and Goldcorp Inc., no actual data is available. Both enterprises do not publish their environmental data. Therefore, data from 2002 is used to model the gold production in Canada.

The mines produce only gold. Campbell and Musselwhite are underground, Porcupine has underground and open pit mines. Tab. 5.36 shows the production volume in year 2002 of the Campbell, Musselwhite and Porcupine Mines. The recovery rate (proportion of valuable material physically recovered in the processing of ore) is taken into account in order to define the resource extraction. The average concentration (head grade, expressed in kg metal per kg ore) of metals in ore for Canada is calculated with the quantity of ore milled, the corresponding production rate and the recovery rate.

**Tab. 5.36: Production volume in 2002 of the Campbell, Musselwhite and Porcupine Mines (Placer Dome Inc., 2003a).**

| Production      |       | Campbell | Musselwhite | Porcupine | Canada          |
|-----------------|-------|----------|-------------|-----------|-----------------|
| Ore milled      | kg    | 3.57E+08 | 7.87E+08    | 1.10E+09  | <b>2.24E+09</b> |
| Production Gold | kg    | 6.01E+03 | 4.43E+03    | 3.17E+03  | <b>1.36E+04</b> |
| Head grade      |       |          |             |           |                 |
| Gold            | kg/kg | 1.98E-05 | 6.62E-06    | 3.39E-06  | <b>7.14E-06</b> |
| Recovery rate   |       |          |             |           | <b>85%</b>      |

### Mining infrastructure

This data set includes land use as well as rough estimates on buildings and machines necessary in gold mining. It is assumed, that the quantities of machinery as well as surface and volume of buildings are equal to the generic mine. This approach is rough, and the uncertainty is rather high. As the mined ore from the generic mine is some orders of magnitude lower than the mine in focus, a proportional scaling by milled ore will lead to a higher mistake (see Tab. 5.1: Gold and Silver production in Sweden). The lifetime of mines is assumed to be 50 years.

**Tab. 5.37: Land use of the Campbell, Musselwhite and Porcupine Mines in 2002 (Placer Dome Inc., 2002b).**

| Land use                           |                | Campbell | Musselwhite | Porcupine | Canada          | Remark                                                                       |
|------------------------------------|----------------|----------|-------------|-----------|-----------------|------------------------------------------------------------------------------|
| Area disturbed during life of mine | m <sup>2</sup> | 1.60E+06 | 4.44E+06    | 1.08E+08  | <b>1.14E+08</b> | Transformation, from unknown and Transformation, to mineral extraction site, |
| Disturbed area in active use       | m <sup>2</sup> | 6.80E+05 | 2.91E+06    | 4.70E+07  | <b>5.06E+07</b> | Occupation, mineral extraction site, during 50 years                         |
| Area available for revegetation    | m <sup>3</sup> | 1.50E+05 | 1.31E+06    | 4.80E+07  | <b>4.95E+07</b> | Transformation to unknown                                                    |
| Area revegetated                   | m <sup>2</sup> | 4.70E+05 | 8.00E+04    | 1.08E+07  | <b>1.14E+07</b> | added in "Transformation, to heterogeneous, agricultural"                    |
| Area that meets final objectives   | m <sup>3</sup> | 3.00E+05 | 0.00E+00    | 2.20E+06  | <b>2.50E+06</b> | added in "Transformation, to heterogeneous, agricultural"                    |

Land use is based on the figures from the environmental report (Tab. 5.37). Not all the area of the mine is in active use at one time; the value in Tab. 5.37 is assumed to be the average share of the mine area in use. Therefore, the total area is occupied during an average of about 22 years, according to the formula

$$\frac{\text{Disturbed area in active use}}{\text{Area disturbed during life of mine}} * \text{lifetime of mine} = \frac{5.06 * 10^7}{1.14 * 10^8} * 50y = 22y$$
 , the occupation of the mine is calculated as Disturbed area in active use \* 50y = Occupation, mineral extraction site

Tab. 5.38: Unit process raw data for the gold and silver mine in Canada.

|                | Name                                             | Location | Infrastructure | Process | Unit | mine, gold      | Uncertainty<br>Type | Standard<br>Deviation95% | GeneralComment                                             |
|----------------|--------------------------------------------------|----------|----------------|---------|------|-----------------|---------------------|--------------------------|------------------------------------------------------------|
|                |                                                  |          |                |         |      |                 |                     |                          |                                                            |
|                | Location<br>InfrastructureProcess<br>Unit        |          |                |         |      | CA<br>1<br>unit |                     |                          |                                                            |
| product        | mine, gold                                       | CA       | 1              | unit    |      | 1.00E+0         |                     |                          |                                                            |
| technosphere   | building, multi-storey                           | RER      | 1              | m3      |      | 2.75E+5         | 1                   | 2.39                     | (4,5,5,5,5,5); Rough estimates                             |
|                | building, hall                                   | CH       | 1              | m2      |      | 1.25E+4         | 1                   | 2.39                     | (4,5,5,5,5,5); Rough estimates                             |
|                | industrial machine, heavy, unspecified, at plant | RER      | 1              | kg      |      | 8.50E+5         | 1                   | 2.39                     | (4,5,5,5,5,5); Rough estimates                             |
|                | conveyor belt, at plant                          | RER      | 1              | m       |      | 6.00E+2         | 1                   | 2.39                     | (4,5,5,5,5,5); Rough estimates                             |
| resource, land | Transformation, from unknown                     | -        | -              | m2      |      | 1.14E+8         | 1                   | 2.01                     | (1,3,2,1,1,3); Environmental report, Placer Dome Inc. 2002 |
|                | Transformation, to mineral extraction site       | -        | -              | m2      |      | 1.14E+8         | 1                   | 2.01                     | (1,3,2,1,1,3); Environmental report, Placer Dome Inc. 2002 |
|                | Occupation, mineral extraction site              | -        | -              | m2a     |      | 2.53E+9         | 1                   | 1.51                     | (1,3,2,1,1,3); Environmental report, Placer Dome Inc. 2002 |
|                | Transformation, from mineral extraction site     | -        | -              | m2      |      | 1.14E+8         | 1                   | 2.01                     | (1,3,2,1,1,3); Environmental report, Placer Dome Inc. 2002 |
|                | Transformation, to heterogeneous, agricultural   | -        | -              | m2      |      | 1.39E+7         | 1                   | 2.01                     | (1,3,2,1,1,3); Environmental report, Placer Dome Inc. 2002 |
|                | Transformation, to unknown                       | -        | -              | m2      |      | 1.00E+8         | 1                   | 2.01                     | (1,3,2,1,1,3); Environmental report, Placer Dome Inc. 2002 |

### Mining and refining

In the Porcupine Mines, gold is recovered using a combination of gravity concentration, milling and cyanidation techniques. The milling process consists of primary crushing, secondary crushing, rod/ball mill grinding, gravity concentration, cyanide leaching, carbon-in-pulp gold recovery, stripping, electrowinning and refining.

In the Campbell Mine, the ore from the mine, after crushing and grinding, is processed by gravity separation, flotation, pressure oxidation, cyanidation and carbon-in-pulp process followed by electrowinning and gold refining to doré on site.

The Musselwhite Mine uses gravity separation, carbon in pulp, electrowinning and gold refining to doré on site. The following table (Tab. 5.39) shows the total requirements and emissions of the Campbell, Musselwhite and Porcupine Mines in 2002 as well as the total for Canada.

**Tab. 5.39: Annual requirements and emissions of the Campbell, Musselwhite and Porcupine Mines in 2002 (Placer Dome Inc., 2002b).**

| <b>Energy</b>                    |                | <b>Campbell</b> | <b>Musselwhite</b> | <b>Porcupine</b> | <b>Canada</b>   | <b>Remark</b>                                                                                     |
|----------------------------------|----------------|-----------------|--------------------|------------------|-----------------|---------------------------------------------------------------------------------------------------|
| Electricity for mining           | kWh            | 8.08E+07        | 6.17E+07           | 1.64E+08         | <b>3.07E+08</b> | Canadian mix approximated with the NORDEL mix <sup>1)</sup>                                       |
| Electricity for refining         | kWh            | n.a.            | n.a.               | n.a.             | <b>2.22E+04</b> | assumed with Möbius / Wohlwill-electrolysis                                                       |
| Diesel (generators)              | MJ             | 7.42E+05        | 2.56E+06           | 6.70E+06         | <b>1.00E+07</b> | Diesel, burned in diesel-electric generating set <sup>2)</sup>                                    |
| Diesel (mobile equipment)        | l              | 1.02E+07        | 3.53E+07           | 9.24E+07         | <b>1.38E+08</b> | Diesel, burned in building machine <sup>2)</sup>                                                  |
| Propane (LPG)                    | MJ             | 7.10E+07        | 2.45E+08           | 6.41E+08         | <b>9.57E+08</b> | <sup>2)</sup>                                                                                     |
| Transport, van <3.5t             | tkm            | 1.20E+05        | 3.32E+05           | 8.69E+05         | <b>1.32E+06</b> | Approximation for gasoline, calculated with specific fuel consumption (0.35l / tkm) <sup>2)</sup> |
| <b>Materials</b>                 |                |                 |                    |                  |                 |                                                                                                   |
| Cyanide                          | kg             | 2.49E+05        | 4.15E+05           | 1.19E+06         | <b>1.85E+06</b> | In LCI as sodium cyanide                                                                          |
| Lime                             | kg             | n.a.            | n.a.               | n.a.             | <b>8.60E+05</b> | Scaled from Steward (1999) by processed gold                                                      |
| Caustic                          | kg             | n.a.            | n.a.               | n.a.             | <b>1.58E+05</b> | Scaled from Steward (1999) by processed gold, accounted for sodium hydroxide                      |
| Carbon                           | kg             | n.a.            | n.a.               | n.a.             | <b>2.31E+05</b> | Scaled from Steward (1999) by processed gold, accounted for as charcoal                           |
| Zinc                             | kg             | n.a.            | n.a.               | n.a.             | <b>3.94E+04</b> | Scaled from Steward (1999) by processed gold, accounted for as zinc, primary, at regional storage |
| Sulphuric Acid                   | kg             | n.a.            | n.a.               | n.a.             | <b>9.06E+03</b> | Scaled from Steward (1999) by processed gold                                                      |
| Hydrochloric Acid                | kg             | n.a.            | n.a.               | n.a.             | <b>1.86E+05</b> | Scaled from Steward (1999) by processed gold                                                      |
| Explosives                       | kg             | n.a.            | n.a.               | n.a.             | <b>2.16E+05</b> | Scaled from US-Mines by milled ore, in LCI: "blasting"                                            |
| <b>Water resources</b>           |                |                 |                    |                  |                 |                                                                                                   |
| Groundwater                      | m <sup>3</sup> | 9.55E+05        | 2.68E+06           | 1.10E+06         | <b>4.74E+06</b> | in LCI: Water, well, in ground                                                                    |
| Water from onsite impoundments   |                | 5.80E+05        | 0                  | 0                | <b>5.80E+05</b> | Recycled Water: not accounted for in inventory                                                    |
| <b>Emissions to Water</b>        |                |                 |                    |                  |                 |                                                                                                   |
| Arsenic, ion                     | kg             | 8.00E+00        | 4.10E+00           | 5.10E+01         | <b>6.31E+01</b> | Environmental Report (Placer Dome, 2002)                                                          |
| Cadmium, ion                     | kg             | 0.00E+00        | 5.00E-01           | 1.40E+01         | <b>1.45E+01</b> | Environmental Report (Placer Dome, 2002)                                                          |
| Copper, ion                      | kg             | 1.30E+01        | 7.60E+00           | 1.52E+02         | <b>1.73E+02</b> | Environmental Report (Placer Dome, 2002)                                                          |
| Cyanide                          | kg             | 1.80E+01        | 4.20E+01           | 3.21E+02         | <b>3.81E+02</b> | Environmental Report (Placer Dome, 2002)                                                          |
| Lead                             | kg             | 7.00E-01        | 4.90E+00           | 2.60E+01         | <b>3.16E+01</b> | Environmental Report (Placer Dome, 2002)                                                          |
| Mercury                          | kg             | 1.00E-02        | 2.00E-02           | 1.50E-01         | <b>1.80E-01</b> | Environmental Report (Placer Dome, 2002)                                                          |
| Nickel, ion                      | kg             | 4.20E+01        | 2.10E+00           | 1.69E+02         | <b>2.13E+02</b> | Environmental Report (Placer Dome, 2002)                                                          |
| Zinc, ion                        | kg             | 2.30E+01        | 9.70E+00           | 5.20E+01         | <b>8.47E+01</b> | Environmental Report (Placer Dome, 2002)                                                          |
| <b>Emissions to Air</b>          |                |                 |                    |                  |                 |                                                                                                   |
| Carbon Dioxide (from explosives) | kg             | n.a.            | n.a.               | n.a.             | <b>3.45E+04</b> | Calculation, based on data from Newmont (2005)                                                    |
| Zinc                             | kg             | n.a.            | n.a.               | n.a.             | <b>4.95E-01</b> | estimated with Data from SE, scaled by milled ore                                                 |
| Copper                           | kg             | n.a.            | n.a.               | n.a.             | <b>1.79E+00</b> | estimated with Data from SE, scaled by milled ore                                                 |
| Lead                             | kg             | n.a.            | n.a.               | n.a.             | <b>5.83E-01</b> | estimated with Data from SE, scaled by milled ore                                                 |
| Disposal                         | kg             | n.a.            | n.a.               | n.a.             | <b>1.59E+09</b> | estimated with data from PG, in LCI as: Disposal, sulphidic tailings                              |

<sup>1</sup> Electricity mix Canada in 2000 (GENI (2004)): fossil 26%, hydro 61%, nuclear 12%, other 2%. Campbell mine uses only hydro power.

<sup>2</sup> The consumption of the different energy sources in Musselwhite and Porcupine is calculated on the basis of their share in the figures of the Campbell mine.

Material use is approximated with figures from Australian data for gold and silver production (Stewart, 1999), whereas air emissions (zinc, copper and lead) are modelled with Swedish data. As Swedish data stem for a combined process, they are allocated to gold and silver for Sweden and used

in this form for Canada. Electricity supply mix of Canada is approximated with the NORDEL electricity mix (Nordic countries power association) that has a related share of hydroelectric power production (The higher part of fossil electricity generation in NORDEL is somewhat compensated by the use of exclusively hydro by the Campbell-Mine). Explosive use is estimated with Data from US. Since the module “blasting” not includes the CO<sub>2</sub> emissions from explosives, they are calculated with the factor of the mines of Newmont (2005b) (0.16 kg CO<sub>2</sub> per kg explosive). Water emissions are assumed to be emitted to rivers. Estimation of transport is based on standard distances. Data about amounts of tailings disposal is approximated with data of mines in Papua New Guinea.

Since the mines produce doré gold (unrefined gold bullion bars containing between 60% and 95% gold), the refinery process is less complex than the refinery process of Rönnskär in Sweden. It is assumed, that the gold doré-bars undergo the treatment of Wohlwill electrolysis (for details see 5.2.2.1, “Gold / Silver, secondary, from electronic scrap in ecoinvent”). Due to the lack of specific data, the figure of refinement from electronic scrap is used: 1.63 kWh per kg gold (Renner et al., 2002a). No emissions were assumed, since on the one hand the high value of the processed material requires an efficient emission control and on the other hand, refinement is a hydrometallurgical process with low impact. The infrastructure is approximated with the facility of precious metal refinery. The unit process raw data are summarized in Tab. 5.40.



## 5. System Characterisation

Tab. 5.40: Unit process raw data of gold mining and refining in Canada.

|                                      | Name                                                    | Location | Infrastructure | Process | Unit | gold, at refinery |   |      | Uncertainty        | Type                                                                                | StandardDeviation | 95% | GeneralComment |
|--------------------------------------|---------------------------------------------------------|----------|----------------|---------|------|-------------------|---|------|--------------------|-------------------------------------------------------------------------------------|-------------------|-----|----------------|
|                                      | Location<br>InfrastructureProcess<br>Unit               |          |                |         |      | CA                | 0 | kg   |                    |                                                                                     |                   |     |                |
| product                              | gold, at refinery                                       | CA       | 0              | kg      |      | 1.00E+0           |   |      |                    |                                                                                     |                   |     |                |
| technosphere                         | electricity, medium voltage, production NORDEL, at grid | NORDEL   | 0              | kWh     |      | 2.25E+4           | 1 | 1.14 | (2,4,2,3,1,3);     | Environmental report (Placer-Dome, 2002)                                            |                   |     |                |
|                                      | diesel, burned in diesel-electric generating set        | GLO      | 0              | MJ      |      | 7.35E+2           | 1 | 1.14 | (2,4,2,3,1,3);     | Environmental report (Placer-Dome, 2002)                                            |                   |     |                |
|                                      | diesel, burned in building machine                      | GLO      | 0              | MJ      |      | 1.01E+4           | 1 | 1.14 | (2,4,2,3,1,3);     | Environmental report (Placer-Dome, 2002)                                            |                   |     |                |
|                                      | natural gas, burned in industrial furnace >100kW        | RER      | 0              | MJ      |      | 7.03E+4           | 1 | 1.14 | (2,4,2,3,1,3);     | Calculation, based on thesis on Australian precious metal production, data for 1999 |                   |     |                |
|                                      | transport, van <3.5t                                    | RER      | 0              | tkm     |      | 9.70E+1           | 1 | 2.02 | (2,4,2,3,1,3);     | Calculation, based on thesis on Australian precious metal production, data for 1999 |                   |     |                |
|                                      | sodium cyanide, at plant                                | RER      | 0              | kg      |      | 1.36E+2           | 1 | 1.14 | (2,4,2,3,1,3);     | Calculation, based on thesis on Australian precious metal production, data for 1999 |                   |     |                |
|                                      | limestone, milled, packed, at plant                     | CH       | 0              | kg      |      | 6.32E+1           | 1 | 1.14 | (2,4,2,3,1,3);     | Calculation, based on thesis on Australian precious metal production, data for 1999 |                   |     |                |
|                                      | sodium hydroxide, 50% in H2O, production mix, at plant  | RER      | 0              | kg      |      | 1.16E+1           | 1 | 1.15 | (2,3,2,5,1,3);     | Calculation, based on thesis on Australian precious metal production, data for 1999 |                   |     |                |
|                                      | charcoal, at plant                                      | GLO      | 0              | kg      |      | 1.70E+1           | 1 | 1.15 | (2,3,2,5,1,3);     | Calculation, based on thesis on Australian precious metal production, data for 1999 |                   |     |                |
|                                      | zinc, primary, at regional storage                      | RER      | 0              | kg      |      | 2.89E+0           | 1 | 1.15 | (2,3,2,5,1,3);     | Calculation, based on thesis on Australian precious metal production, data for 1999 |                   |     |                |
|                                      | sulphuric acid, liquid, at plant                        | RER      | 0              | kg      |      | 6.65E-1           | 1 | 1.15 | (2,3,2,5,1,3);     | Calculation, based on thesis on Australian precious metal production, data for 1999 |                   |     |                |
|                                      | hydrochloric acid, 30% in H2O, at plant                 | RER      | 0              | kg      |      | 1.36E+1           | 1 | 1.11 | (2,3,2,3,1,3);     | Calculation, based on thesis on Australian precious metal production, data for 1999 |                   |     |                |
|                                      | blasting                                                | RER      | 0              | kg      |      | 1.59E+1           | 1 | 1.11 | (2,3,2,3,1,3);     | Estimation, based on US-Data                                                        |                   |     |                |
|                                      | transport, lorry >16t, fleet average                    | RER      | 0              | tkm     |      | 2.29E+2           | 1 | 2.02 | (2,4,na,na,na,na); | Calculation, based on standard distances                                            |                   |     |                |
|                                      | transport, freight, rail                                | RER      | 0              | tkm     |      | 4.58E+2           | 1 | 2.02 | (2,4,na,na,na,na); | Calculation, based on standard distances                                            |                   |     |                |
| infrastructure                       | disposal, sulfidic tailings, off-site                   | GLO      | 0              | kg      |      | 1.17E+5           | 1 | 1.14 | (2,4,2,3,1,3);     | Estimation                                                                          |                   |     |                |
|                                      | facilities precious metal refinery                      | SE       | 1              | unit    |      | 1.65E-7           | 1 | 3.02 | (2,4,2,3,1,3);     | Estimation                                                                          |                   |     |                |
|                                      | mine, gold                                              | CA       | 1              | unit    |      | 1.47E-6           | 1 | 3.01 | (2,3,2,3,1,3);     | Environmental report                                                                |                   |     |                |
| emission air, low population density | Copper                                                  | -        | -              | kg      |      | 1.32E-4           | 1 | 5.02 | (2,4,2,3,1,3);     | Estimation, based on Swedish Data                                                   |                   |     |                |
|                                      | Lead                                                    | -        | -              | kg      |      | 4.28E-5           | 1 | 5.02 | (2,4,2,3,1,3);     | Estimation, based on Swedish Data                                                   |                   |     |                |
|                                      | Zinc                                                    | -        | -              | kg      |      | 3.64E-5           | 1 | 5.02 | (2,4,2,3,1,3);     | Estimation, based on Swedish Data                                                   |                   |     |                |
|                                      | Carbon dioxide, fossil                                  | -        | -              | kg      |      | 2.54E+0           | 1 | 1.14 | (2,4,2,3,1,3);     | Estimation, based on Swedish Data                                                   |                   |     |                |
|                                      | Heat, waste                                             | -        | -              | MJ      |      | 8.11E+4           | 1 | 1.14 | (2,4,2,3,1,3);     | Calculation                                                                         |                   |     |                |
| emission water, ocean                | Arsenic, ion                                            | -        | -              | kg      |      | 4.64E-3           | 1 | 5.01 | (1,3,2,3,1,3);     | Environmental report (Placer-Dome, 2002)                                            |                   |     |                |
|                                      | Cadmium, ion                                            | -        | -              | kg      |      | 1.07E-3           | 1 | 3.01 | (1,3,2,3,1,3);     | Environmental report (Placer-Dome, 2002)                                            |                   |     |                |
|                                      | Copper, ion                                             | -        | -              | kg      |      | 1.27E-2           | 1 | 3.01 | (1,3,2,3,1,3);     | Environmental report (Placer-Dome, 2002)                                            |                   |     |                |
|                                      | Cyanide                                                 | -        | -              | kg      |      | 2.80E-2           | 1 | 3.01 | (1,3,2,3,1,3);     | Environmental report (Placer-Dome, 2002)                                            |                   |     |                |
|                                      | Lead                                                    | -        | -              | kg      |      | 2.32E-3           | 1 | 5.01 | (1,3,2,3,1,3);     | Environmental report (Placer-Dome, 2002)                                            |                   |     |                |
|                                      | Mercury                                                 | -        | -              | kg      |      | 1.32E-5           | 1 | 5.01 | (1,3,2,3,1,3);     | Environmental report (Placer-Dome, 2002)                                            |                   |     |                |
|                                      | Nickel, ion                                             | -        | -              | kg      |      | 1.57E-2           | 1 | 5.01 | (1,3,2,3,1,3);     | Environmental report (Placer-Dome, 2002)                                            |                   |     |                |
|                                      | Zinc, ion                                               | -        | -              | kg      |      | 6.22E-3           | 1 | 5.01 | (1,3,2,3,1,3);     | Environmental report (Placer-Dome, 2002)                                            |                   |     |                |
|                                      |                                                         | -        | -              | kg      |      | 6.22E-3           | 1 | 5.01 | (1,3,2,3,1,3);     | Environmental report (Placer-Dome, 2002)                                            |                   |     |                |
|                                      |                                                         | -        | -              | kg      |      | 6.22E-3           | 1 | 5.01 | (1,3,2,3,1,3);     | Environmental report (Placer-Dome, 2002)                                            |                   |     |                |
| resource, in water                   | Water, well, in ground                                  | -        | -              | m3      |      | 3.48E+2           | 1 | 1.10 | (1,3,2,3,1,3);     | Environmental report (Placer-Dome, 2002)                                            |                   |     |                |
| resource, in ground                  | Gold, Au 7.1E-4%, in ore, in ground                     | -        | -              | kg      |      | 1.18E+0           | 1 | 1.10 | (1,3,2,3,1,3);     | Estimation                                                                          |                   |     |                |

Tab. 5.41 Ecospol Meta information of gold production in Canada

|                            |                                                                                                                                                                                                                                            |                                                                                                                                                                                         |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name                       | mine, gold                                                                                                                                                                                                                                 | gold, at refinery                                                                                                                                                                       |
| Location                   | CA                                                                                                                                                                                                                                         | CA                                                                                                                                                                                      |
| InfrastructureProcess      | 1                                                                                                                                                                                                                                          | 0                                                                                                                                                                                       |
| Unit                       | unit                                                                                                                                                                                                                                       | kg                                                                                                                                                                                      |
| Type                       | 1                                                                                                                                                                                                                                          | 1                                                                                                                                                                                       |
| Version                    | 2.0                                                                                                                                                                                                                                        | 2.0                                                                                                                                                                                     |
| energyValues               | 0                                                                                                                                                                                                                                          | 0                                                                                                                                                                                       |
| LanguageCode               | en                                                                                                                                                                                                                                         | en                                                                                                                                                                                      |
| LocalLanguageCode          | de                                                                                                                                                                                                                                         | de                                                                                                                                                                                      |
| Person                     | 44                                                                                                                                                                                                                                         | 44                                                                                                                                                                                      |
| QualityNetwork             | 1                                                                                                                                                                                                                                          | 1                                                                                                                                                                                       |
| DataSetRelatesToProduct    | 1                                                                                                                                                                                                                                          | 1                                                                                                                                                                                       |
| IncludedProcesses          | Land use, buildings and machinery for mining                                                                                                                                                                                               | This data set describes the mining of gold concentrate in Canada. It includes energy and water consumption, land use, transports, waste disposal as well as emissions to air and water. |
| Amount                     | 1                                                                                                                                                                                                                                          | 1                                                                                                                                                                                       |
| LocalName                  | Mine, Gold                                                                                                                                                                                                                                 | Gold, ab Raffinerie                                                                                                                                                                     |
| Synonyms                   |                                                                                                                                                                                                                                            |                                                                                                                                                                                         |
| GeneralComment             | This data set includes land use of the considered mines (open pit and underground) incl. revegetated areas as well as estimates of buildings and machinery of the mine based on the data set non-ferrous metal, mine, surface/underground. | This data set describes the beneficiation, concentration and refinement of gold.                                                                                                        |
| InfrastructureIncluded     | 1                                                                                                                                                                                                                                          | 1                                                                                                                                                                                       |
| Category                   | metals                                                                                                                                                                                                                                     | metals                                                                                                                                                                                  |
| SubCategory                | extraction                                                                                                                                                                                                                                 | extraction                                                                                                                                                                              |
| LocalCategory              | Metalle                                                                                                                                                                                                                                    | Metalle                                                                                                                                                                                 |
| LocalSubCategory           | Gewinnung                                                                                                                                                                                                                                  | Gewinnung                                                                                                                                                                               |
| Formula                    |                                                                                                                                                                                                                                            |                                                                                                                                                                                         |
| StatisticalClassification  |                                                                                                                                                                                                                                            |                                                                                                                                                                                         |
| CASNumber                  |                                                                                                                                                                                                                                            | 007440-57-5                                                                                                                                                                             |
| StartDate                  | 2003                                                                                                                                                                                                                                       | 2003                                                                                                                                                                                    |
| EndDate                    | 2006                                                                                                                                                                                                                                       | 2006                                                                                                                                                                                    |
| DataValidForEntirePeriod   | 1                                                                                                                                                                                                                                          | 1                                                                                                                                                                                       |
| OtherPeriodText            | Period of public environmental report                                                                                                                                                                                                      | Period of public environmental report                                                                                                                                                   |
| Text                       | Land use of Canadian mine, other inputs based on rough assumptions.                                                                                                                                                                        | Material requirements for gold concentration are estimated with Australian data, air emission are estimated with Swedish data.                                                          |
| Text                       | Assumptions for buildings and machinery based on the data set "non-ferrous metal, mine, surface/underground with 50 years life time.                                                                                                       | The mining of ore from open pit and underground mines is considered.                                                                                                                    |
| Percent                    |                                                                                                                                                                                                                                            |                                                                                                                                                                                         |
| ProductionVolume           | 2.2 Mill. t ore                                                                                                                                                                                                                            | 13.61 t gold                                                                                                                                                                            |
| SamplingProcedure          | environmental report                                                                                                                                                                                                                       | environmental report                                                                                                                                                                    |
| Extrapolations             | Buildings and machinery are based on global assumptions. The figures are extrapolated on the basis of the ore production.                                                                                                                  | See Geography, estimation of conversion of transport fuel to tkm                                                                                                                        |
| UncertaintyAdjustments     | none                                                                                                                                                                                                                                       | none                                                                                                                                                                                    |
| Person                     | 44                                                                                                                                                                                                                                         | 44                                                                                                                                                                                      |
| DataPublishedIn            | 2                                                                                                                                                                                                                                          | 2                                                                                                                                                                                       |
| ReferenceToPublishedSource | 10                                                                                                                                                                                                                                         | 10                                                                                                                                                                                      |
| Copyright                  | 1                                                                                                                                                                                                                                          | 1                                                                                                                                                                                       |
| AccessRestrictedTo         | 0                                                                                                                                                                                                                                          | 0                                                                                                                                                                                       |
| CompanyCode                |                                                                                                                                                                                                                                            |                                                                                                                                                                                         |
| CountryCode                |                                                                                                                                                                                                                                            |                                                                                                                                                                                         |
| PageNumbers                |                                                                                                                                                                                                                                            |                                                                                                                                                                                         |
| Validator                  | 51                                                                                                                                                                                                                                         | 51                                                                                                                                                                                      |
| Details                    | automatic validation in Excel                                                                                                                                                                                                              | automatic validation in Excel                                                                                                                                                           |
| OtherDetails               | none                                                                                                                                                                                                                                       | none                                                                                                                                                                                    |

### 5.1.7 Gold production, United States of America (US)

Data stem from the environmental report of Barrick (2006b) for the Bald Mountain, Cortez, Golden Sunlight, Goldstrike and Round Mountain Mines. Since the mines have been sold in 2006 from the company of Placer Dome to Barrick, previous environmental reports of Placer Dome Inc. (2002b) are used as well.

The data about the mines of Nevada and Golden Giant are from Newmont (2005b). The mines are either underground or open pits. All mines produce only gold. Tab. 5.42 shows the production volume 2005 of the considered US Mines. The recovery rate (proportion of valuable material physically recovered in the processing of ore) is taken into account in order to define the resource extraction. The average concentration (head grade, expressed in kg metal per kg ore) of metals in ore for United States is calculated with the quantity of ore milled, the corresponding production rate and the recovery rate.

### Mining infrastructure

This data set includes land use as well as rough estimates on buildings and machines necessary in gold mining. It is assumed, that the quantities of machinery as well as surface and volume of buildings are equal to the generic mine. This approach is rough, and the uncertainty is rather high. As the mined ore from the generic mine is some orders of magnitude lower than the mine in focus, a proportional scaling by milled ore will lead to a higher mistake (see Tab. 5.1: Gold and Silver production in Sweden). The lifetime of mines is assumed to be 50 years.

Land use is based on the figures from the environmental report of Newmont (2005b) and the previous environmental report of Placer Dome Inc. (2002b). Not all the area of the mine is in active use at one time; the value in Tab. 5.56 is assumed to be the average share of the mine area in use. Therefore, the total area is occupied during an average of about 43 years, according to the formula:

$$\frac{\text{Disturbed area in active use}}{\text{Area disturbed during life of mine}} * \text{lifetime of mine} = \frac{7.04 * 10^8}{8.14 * 10^8} * 50y = 43y$$
, the occupation of the mine is calculated as  $\text{Disturbed area in active use} * 50y = \text{Occupation, mineral extraction site}$

**Tab. 5.42** Production volume 2005 of the Golden Giant, Nevada, Bald Mountain, Cortez, Golden Sunlight, Goldstrike, and Round Mountain mines (Barrick, 2006b; Newmont, 2005b)

|                   |           | GoldenGiant | Nevada      | Bald Mountain | Cortez   | Golden Sunlight | Goldstrike | Round Mountain | United States   | Remark                                                                                   |
|-------------------|-----------|-------------|-------------|---------------|----------|-----------------|------------|----------------|-----------------|------------------------------------------------------------------------------------------|
| Production Gold   | kg        | 5.04E+03    | 7.46E+04    | 8.65E+03      | 1.31E+04 | 2.89E+03        | 5.79E+04   | 2.12E+04       | <b>1.83E+05</b> |                                                                                          |
| Recovery rate     |           | <i>n.a.</i> | <i>n.a.</i> | 91%           | 89%      | 78%             | 88%        | 88%            | <b>88%</b>      | Recovery rate, average value for the United States, based on data of Placer-Dome, (2002) |
| Ore milled        | kg        | 8.35E+08    | 2.32E+11    | 6.04E+09      | 3.08E+10 | 3.08E+09        | 1.28E+10   | 6.60E+10       | <b>3.52E+11</b> |                                                                                          |
| <i>Head grade</i> |           |             |             |               |          |                 |            |                |                 |                                                                                          |
| Gold              | kg/kg ore | 6.85E-06    | 3.66E-07    | 1.79E-06      | 5.43E-07 | 1.37E-06        | 5.84E-06   | 4.14E-07       | <b>5.93E-07</b> |                                                                                          |

**Tab. 5.43** Land use of the Golden Giant, Nevada, Bald Mountain, Cortez, Golden Sunlight, Goldstrike and Round Mountain mines (Barrick, 2006b; Newmont, 2005b)

| Land use                           |                | GoldenGiant | Nevada   | Bald Mountain | Cortez   | Golden Sunlight | Goldstrike | Round Mountain | United States   | Remark                                                                       |
|------------------------------------|----------------|-------------|----------|---------------|----------|-----------------|------------|----------------|-----------------|------------------------------------------------------------------------------|
| Area disturbed during life of mine | m <sup>2</sup> | 2.72E+06    | 1.33E+08 | 1.50E+07      | 2.89E+07 | 9.49E+06        | 3.80E+08   | 2.50E+08       | <b>8.19E+08</b> | Transformation, from unknown and Transformation, to mineral extraction site. |
| Disturbed area in active use       | m <sup>2</sup> | 2.12E+06    | 9.87E+07 | 8.63E+06      | 2.25E+07 | 4.14E+06        | 3.40E+08   | 2.28E+08       | <b>7.04E+08</b> | Occupation, mineral extraction site, during 50 years                         |
| Area available for revegetation    | m <sup>2</sup> | 2.00E+04    | 7.69E+06 | 1.30E+06      | 1.05E+06 | 1.28E+06        | 3.15E+07   | 2.80E+07       | <b>7.08E+07</b> | Transformation, to unknown                                                   |
| Area revegetated                   | m <sup>2</sup> | 5.80E+05    | 2.43E+07 | 5.04E+06      | 5.30E+06 | 2.30E+05        | 0.00E+00   | 0.00E+00       | <b>3.55E+07</b> | Transformation, to heterogeneous, agricultural                               |

Tab. 5.44: Unit process raw data for the gold and silver mine in the USA.

|                | Name                                             | Location | Infrastructure | Process | Unit | mine, gold | Uncertainty Type | Standard Deviation 95% | General Comment                                   |
|----------------|--------------------------------------------------|----------|----------------|---------|------|------------|------------------|------------------------|---------------------------------------------------|
|                | Location Infrastructure Process Unit             |          |                |         |      | US 1 unit  |                  |                        |                                                   |
| product        | mine, gold                                       | US       | 1              | unit    |      | 1.00E+0    |                  |                        |                                                   |
| technosphere   | building, multi-storey                           | RER      | 1              | m3      |      | 2.75E+5    | 1                | 2.39                   | (4,5,5,5,5,5); Rough estimates                    |
|                | building, hall                                   | CH       | 1              | m2      |      | 1.25E+4    | 1                | 2.39                   | (4,5,5,5,5,5); Rough estimates                    |
|                | industrial machine, heavy, unspecified, at plant | RER      | 1              | kg      |      | 8.50E+5    | 1                | 2.39                   | (4,5,5,5,5,5); Rough estimates                    |
|                | conveyor belt, at plant                          | RER      | 1              | m       |      | 6.00E+2    | 1                | 2.39                   | (4,5,5,5,5,5); Rough estimates                    |
| resource, land | Transformation, from unknown                     | -        | -              | m2      |      | 8.19E+8    | 1                | 2.01                   | (1,3,1,1,1,3); Environmental report, Newmont 2005 |
|                | Transformation, to mineral extraction site       | -        | -              | m2      |      | 8.19E+8    | 1                | 2.01                   | (1,3,1,1,1,3); Environmental report, Newmont 2005 |
|                | Occupation, mineral extraction site              | -        | -              | m2a     |      | 3.52E+10   | 1                | 1.51                   | (1,3,1,1,1,3); Environmental report, Newmont 2005 |
|                | Transformation, from mineral extraction site     | -        | -              | m2      |      | 8.19E+8    | 1                | 2.01                   | (1,3,1,1,1,3); Environmental report, Newmont 2005 |
|                | Transformation, to heterogeneous, agricultural   | -        | -              | m2      |      | 3.55E+7    | 1                | 2.01                   | (1,3,1,1,1,3); Environmental report, Newmont 2005 |
|                | Transformation, to unknown                       | -        | -              | m2      |      | 7.83E+8    | 1                | 2.01                   | (1,3,1,1,1,3); Environmental report, Newmont 2005 |

## Mining and refining

In the Bald Mountain Mines, the ore treatment method is based on conventional heap leaching technology followed by carbon absorption. The loaded carbon is stripped and refined in the newly commissioned refinery on site. Water is supplied by wells located on the mine property. Grid power was brought to Bald Mountain Mine in 1996. For this purpose, one 27-kilometre 69 KVA power line was constructed from the Alligator Ridge Mine substation to the grid.

At the Golden Sunlight Mines, the ore treatment plant is based on conventional carbon-in-pulp technology, with the addition of a Sand Tailings Retreatment (STR) gold recovery plant to recover gold that would otherwise be lost to tailings. The STR circuit removes the heavier gold bearing pyrite from the sand portion of the tailings by gravity separation. The gold is refined into doré at the mine. Tailing from the mill is discharged to an impoundment area where the solids are allowed to settle so the water can be reused. A cyanide recovery/destruction process was commissioned in 1998. It eliminates the hazard posed to wildlife at the tailings impoundment by lowering cyanide concentrations below 20 mg/l. Fresh water for ore processing, dust suppression, and fire control is supplied from the Jefferson Slough, which is an old natural channel of the Jefferson River. Ore processing also uses water pumped from the tailings impoundment. Pit water is treated in a facility located in the mill complex prior to disposal or for use in dust control. Drinking water is made available by filtering fresh water through an on-site treatment plant. Electric power is provided from a substation at the south property boundary. North-Western Energy supplies electricity the substation. Small diesel generators are used for emergency lighting. A natural gas pipeline supplies gas for heating buildings, a crusher, air scrubber, boiler, carbon reactivation kiln, and refining furnaces.

At the Cortez Mine, three different metallurgical processes are employed for the recovery of gold. The process used for a particular ore is determined based on grade and metallurgical character of that ore. Lower grade oxide ore is heap leached, while higher-grade non-refractory ore is treated in a conventional mill using cyanidation and a carbon-in-leach ("CIL") process. When carbonaceous ore is processed by Barrick, it is first dry ground, and then oxidized in a circulating fluid bed roaster, followed by CIL recovery. In 2002 a new leach pad and process plant was commissioned. This facility will be capable of processing 164 million tonnes of heap leach ore over the life of the asset. Heap leach ore production is hauled directly to heap leach pads for gold recovery. Water for process use is supplied from the open pit dewatering system. Approximately 90 litres per second of the pit dewatering volume is diverted for plant use. Electric power is supplied by Sierra Pacific Power Company ("SPPC") through a 73 kilometre, 120 kV transmission line. A long-term agreement is in place with SPPC to provide

power through the regulated power system. The average power requirement of the mine is about 160 GWh/year.

Further information about the mines and their infrastructure can be found in the corresponding chapters of Barrick (2006b); Newmont (2005b) and Barrick (2006b). The following table (Tab. 5.45) shows the requirements and emissions of the considered mines as well as the US total.

Material use of zinc and carbon is approximated with figures from Australian data for gold and silver production (Stewart, 1999). The consumption of Lime and caustic (Sulphuric Acid / Hydrochloric Acid) was taken from the environmental reports (where available), other ways estimated with the Australian values (Stewart, 1999).

The water emissions are estimated with data of Australia, it is assumed that they are emitted to rivers. Since the module “blasting” does not include the CO<sub>2</sub> emissions from explosives, they are calculated with the factor of the mines of Newmont (2005b) (0.16 kg CO<sub>2</sub> per kg explosive). The transport of materials is calculated with ecoinvent standard distances for Europe. Data of tailing disposal are approximated with data from Papua New Guinea.

Since the mines produce doré gold (unrefined gold bullion bars containing between 60% and 95% gold), the refinery process is less complex than the refinery process of Rönnskär in Sweden. It is assumed, that the gold doré-bars undergo the treatment of Wohlwill electrolysis (for details see 5.2.2.1, “Gold / Silver, secondary, from electronic scrap in ecoinvent”). Due to the lack of specific data, the figure of refinement from electronic scrap is used: 1.63 kWh per kg gold (Renner et al., 2002a). No emissions were assumed, since on the one hand the high value of the processed material requires an efficient emission control and on the other hand, refinement is a hydrometallurgical process with low impact. The infrastructure is approximated with the facility of precious metal refinery. The unit process raw data are summarized in Tab. 5.47.

Tab. 5.45 Annual requirements of or attributed to the Golden Giant, Nevada, Bald Mountain, Cortez, Golden Sunlight, Goldstrike and Round Mountain mines and total for US

| Energy                    |                | GoldenGiant | Nevada   | Bald Mountain | Cortez   | Golden Sunlight | Goldstrike | Round Mountain | United States | Remark                                                                                                                                                                    |
|---------------------------|----------------|-------------|----------|---------------|----------|-----------------|------------|----------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Electricity for mining    | kWh            | 1.14E+08    | 1.37E+09 | 1.44E+06      | 3.45E+07 | 2.77E+07        | 2.04E+08   | 3.28E+07       | 1.79E+09      | Electricity, medium voltage, US, share for Barrick-Mines is based on actual total energy consumption for older share of electricity (Placer Dome, 2002)                   |
| Electricity for refining  | kWh            | n.a.        | n.a.     | n.a.          | n.a.     | n.a.            | n.a.       | n.a.           | 2.99E+05      | assumed with Wohlwill-electrolysis                                                                                                                                        |
| Diesel (generators)       | MJ             | 0.00E+00    | 3.20E+04 | 4.66E+07      | 6.43E+08 | 1.67E+08        | 3.04E+09   | 4.90E+08       | 4.39E+09      | Diesel, burned in diesel-electric generating set, share for Barrick-Mines is based on actual total energy consumption for older Diesel in generators (Placer Dome, 2002)  |
| Diesel (mobile equipment) | MJ             | 2.01E+07    | 5.27E+09 | 4.74E+08      | 1.71E+09 | 3.26E+08        | 4.14E+09   | 1.66E+09       | 1.36E+10      | Diesel, burned in building machine, share for Barrick-Mines is based on actual total energy consumption for older share of diesel in mobile equipment (Placer Dome, 2002) |
| Petrol                    | l              | 2.57E+03    | 1.20E+05 | n.a.          | n.a.     | n.a.            | n.a.       | n.a.           | 1.23E+05      | converted in transport, van <3.5t (0.38l/tkm), consumption in Barrick-Mines is included in Diesel (mobile equipment)                                                      |
| Oil                       | MJ             | 2.66E+06    | 1.33E+08 | n.a.          | n.a.     | n.a.            | n.a.       | n.a.           | 1.36E+08      | heavy fuel oil, burned in industrial furnace 1MW, non-modulating, consumption in Barrick-Mines is included in Diesel (mobile equipment)                                   |
| <b>Materials</b>          |                |             |          |               |          |                 |            |                |               |                                                                                                                                                                           |
| Cyanide                   | kg             | 1.10E+05    | 9.38E+06 | 1.35E+06      | 4.52E+06 | 1.97E+06        | 4.91E+06   | 1.15E+07       | 9.49E+06      | In LCI as sodium cyanide                                                                                                                                                  |
| Lime                      | kg             | 5.09E+05    | 2.01E+08 | 5.47E+05      | 8.29E+05 | 1.83E+05        | 3.66E+06   | 1.34E+06       | 2.08E+08      | Data for Golden Giant and Nevada from environmental report, other data estimated from Steward (1999) by processed gold                                                    |
| Caustic                   | kg             | 0.00E+00    | 3.29E+04 | 1.01E+05      | 1.53E+05 | 3.37E+04        | 6.74E+05   | 2.46E+05       | 1.24E+06      | Data for Golden Giant and Nevada from environmental report, other data estimated from Steward (1999) by processed gold                                                    |
| Carbon                    | kg             | 8.55E+04    | 1.27E+06 | 1.47E+05      | 2.23E+05 | 4.91E+04        | 9.82E+05   | 3.59E+05       | 3.11E+06      | Scaled from Steward (1999) by processed gold, accounted for as Charcoal, at plant                                                                                         |
| Zinc                      | kg             | 1.46E+04    | 2.16E+05 | 2.50E+04      | 3.80E+04 | 8.37E+03        | 1.67E+05   | 6.12E+04       | 5.31E+05      | Scaled from Steward (1999) by processed gold, accounted for as zinc, primary, at regional storage                                                                         |
| Sulphuric Acid            | kg             | 2.08E+04    | 3.59E+07 | 5.75E+03      | 8.73E+03 | 1.92E+03        | 3.85E+04   | 1.41E+04       | 3.60E+07      | Data for Golden Giant and Nevada from environmental report, other data estimated from Steward (1999) by processed gold                                                    |
| Hydrochloric Acid         | kg             | 2.08E+04    | 3.59E+07 | 1.18E+05      | 1.79E+05 | 3.95E+04        | 7.89E+05   | 2.89E+05       | 3.73E+07      | Data for Golden Giant and Nevada from environmental report, other data estimated from Steward (1999) by processed gold                                                    |
| Explosives                | kg             | 2.39E+05    | 3.35E+07 | 5.38E+03      | 1.52E+04 | 4.27E+03        | 1.94E+04   | 7.21E+03       | 3.38E+07      | Partly ANFO explosives, in LCI: "blasting"                                                                                                                                |
| <b>Water resources</b>    |                |             |          |               |          |                 |            |                |               |                                                                                                                                                                           |
| Surface Water Use         | m <sup>3</sup> | 1.01E+05    | 0.00E+00 | n.a.          | n.a.     | n.a.            | n.a.       | n.a.           | 1.01E+05      | in LCI:Water, river                                                                                                                                                       |
| Groundwater Use           | m <sup>3</sup> | 0.00E+00    | 8.45E+06 | 2.09E+05      | 3.68E+06 | 5.65E+05        | 1.23E+07   | 6.74E+06       | 3.19E+07      | in LCI:Water, well, in ground                                                                                                                                             |
| Mine dewatering Use       | m <sup>3</sup> | 1.90E+04    | 2.48E+07 | n.a.          | n.a.     | n.a.            | n.a.       | n.a.           | 2.48E+07      | recycled Water, not accounted for in inventory                                                                                                                            |

Tab. 5.46 Annual emissions of or attributed to the Golden Giant, Nevada, Bald Mountain, Cortez, Golden Sunlight, Goldstrike and Round Mountain mines and total for US

| Emissions to Water               |    | GoldenGiant | Nevada   | Bald Mountain | Cortez   | Golden Sunlight | Goldstrike | Round Mountain | United States |                                                                                                                    |
|----------------------------------|----|-------------|----------|---------------|----------|-----------------|------------|----------------|---------------|--------------------------------------------------------------------------------------------------------------------|
| Arsenic                          | kg | n.a.        | n.a.     | n.a.          | n.a.     | n.a.            | n.a.       | n.a.           | 9.80E+03      | Total based on australian value, scaled on milled ore                                                              |
| Cadmium                          | kg | n.a.        | n.a.     | n.a.          | n.a.     | n.a.            | n.a.       | n.a.           | 4.42E+03      | Total based on australian value, scaled on milled ore                                                              |
| Copper                           | kg | n.a.        | n.a.     | n.a.          | n.a.     | n.a.            | n.a.       | n.a.           | 8.01E+04      | Total based on australian value, scaled on milled ore                                                              |
| Cyanide                          | kg | n.a.        | n.a.     | n.a.          | n.a.     | n.a.            | n.a.       | n.a.           | 1.31E+05      | Total based on australian value, scaled on milled ore                                                              |
| Lead                             | kg | n.a.        | n.a.     | n.a.          | n.a.     | n.a.            | n.a.       | n.a.           | 2.70E+05      | Total based on australian value, scaled on milled ore                                                              |
| Mercury                          | kg | n.a.        | n.a.     | n.a.          | n.a.     | n.a.            | n.a.       | n.a.           | 4.91E+03      | Total based on australian value, scaled on milled ore                                                              |
| Nickel                           | kg | n.a.        | n.a.     | n.a.          | n.a.     | n.a.            | n.a.       | n.a.           | 4.42E+04      | Total based on australian value, scaled on milled ore                                                              |
| Zinc                             | kg | n.a.        | n.a.     | n.a.          | n.a.     | n.a.            | n.a.       | n.a.           | 4.91E+05      | Total based on australian value, scaled on milled ore                                                              |
| <b>Emissions to Air</b>          |    |             |          |               |          |                 |            |                |               |                                                                                                                    |
| Antimony                         | kg | 2.63E+02    | n.a.     | n.a.          | n.a.     | n.a.            | n.a.       | n.a.           | 1.11E+05      | Total based on Golden Giant, environmental report of Newmont (2005)                                                |
| Arsenic                          | kg | 9.07E+00    | n.a.     | n.a.          | n.a.     | n.a.            | n.a.       | n.a.           | 3.82E+03      | Total based on Golden Giant, environmental report of Newmont (2005)                                                |
| Beryllium                        | kg | 0.00E+00    | n.a.     | n.a.          | n.a.     | n.a.            | n.a.       | n.a.           | 1.02E+03      | Total based on australian value, scaled by milled ore                                                              |
| Cadmium                          | kg | n.a.        | n.a.     | n.a.          | n.a.     | n.a.            | n.a.       | n.a.           | 5.42E+01      | Total based on australian value, scaled by milled ore                                                              |
| Carbon Monoxide                  | kg | 1.40E+03    | 2.41E+05 | n.a.          | n.a.     | n.a.            | n.a.       | n.a.           | -             | environmental report of Newmont (2005), not accounted for in LCI as emissions are already included in fossil fuels |
| Carbon Dioxide (from explosives) | kg | 7.00E+06    | 7.00E+06 | 2.16E+05      | 7.23E+05 | 3.14E+05        | 7.86E+05   | 1.84E+06       | 1.40E+07      | Total based on Nevada, environmental report of Newmont (2005)                                                      |
| Chromium                         | kg | n.a.        | n.a.     | n.a.          | n.a.     | n.a.            | n.a.       | n.a.           | 8.86E+03      | Total based on australian value, scaled by milled ore                                                              |
| Cobalt                           | kg | n.a.        | n.a.     | n.a.          | n.a.     | n.a.            | n.a.       | n.a.           | 1.53E+03      | Total based on australian value, scaled by milled ore                                                              |
| Lead                             | kg | 9.07E+00    | n.a.     | n.a.          | n.a.     | n.a.            | n.a.       | n.a.           | 3.82E+03      | Total based on Golden Giant, environmental report of Newmont (2005)                                                |
| Manganese                        | kg | n.a.        | n.a.     | n.a.          | n.a.     | n.a.            | n.a.       | n.a.           | 4.80E+04      | Total based on australian value, scaled by milled ore                                                              |
| Mercury                          | kg | 9.07E+00    | 3.90E+02 | n.a.          | n.a.     | n.a.            | n.a.       | n.a.           | 5.91E+02      | Total based on Nevada, environmental report of Newmont (2005)                                                      |
| Nitrogen Oxide (NOx)             | kg | 4.96E+03    | 1.18E+05 | n.a.          | n.a.     | n.a.            | n.a.       | n.a.           | -             | environmental report of Newmont (2005), not accounted for in LCI as emissions are already included in fossil fuels |
| Particulates (PM10)              | kg | 5.21E+04    | 2.80E+05 | n.a.          | n.a.     | n.a.            | n.a.       | n.a.           | -             | environmental report of Newmont (2005), not accounted for in LCI as emissions are already included in fossil fuels |
| Selenium                         | kg | n.a.        | n.a.     | n.a.          | n.a.     | n.a.            | n.a.       | n.a.           | 2.93E+04      | Total based on australian value, scaled by milled ore                                                              |
| Sulfur Dioxide (SO2)             | kg | 9.98E+01    | 9.79E+04 | n.a.          | n.a.     | n.a.            | n.a.       | n.a.           | -             | environmental report of Newmont (2005), not accounted for in LCI as emissions are already included in fossil fuels |



## 5. System Characterisation

Tab. 5.47: Unit process raw data of gold mining and refining in USA.

|                                      | Name                                                             | Location | Infrastructure | Process | Unit     | gold, at refinery | Uncertainty | Type | Standard Deviation | on95% | GeneralComment                                                                                     |
|--------------------------------------|------------------------------------------------------------------|----------|----------------|---------|----------|-------------------|-------------|------|--------------------|-------|----------------------------------------------------------------------------------------------------|
|                                      | Location<br>Infrastructure<br>Process<br>Unit                    |          |                |         |          | US<br>0<br>kg     |             |      |                    |       |                                                                                                    |
| product                              | gold, at refinery                                                | US       | 0              | kg      | 1.00E+00 |                   |             |      |                    |       |                                                                                                    |
| technosphere                         | electricity, medium voltage, at grid                             | US       | 0              | kWh     | 9.76E+03 | 1                 | 1.29        |      |                    |       | (3,4,1,5,3,3); Environmental report                                                                |
|                                      | diesel, burned in diesel-electric generating set                 | GLO      | 0              | MJ      | 2.39E+04 | 1                 | 1.16        |      |                    |       | (3,4,1,2,1,3); Environmental report                                                                |
|                                      | diesel, burned in building machine                               | GLO      | 0              | MJ      | 7.42E+04 | 1                 | 1.16        |      |                    |       | (3,4,1,2,1,3); Environmental report                                                                |
|                                      | heavy fuel oil, burned in industrial furnace 1MW, non-modulating | RER      | 0              | MJ      | 7.40E+02 | 1                 | 1.16        |      |                    |       | (3,4,1,2,1,3); Environmental report                                                                |
|                                      | transport, van <3.5t                                             | RER      | 0              | tkm     | 1.76E+00 | 1                 | 2.03        |      |                    |       | (3,4,1,2,1,3); Environmental report                                                                |
|                                      | sodium cyanide, at plant                                         | RER      | 0              | kg      | 5.18E+01 | 1                 | 1.14        |      |                    |       | (3,3,1,3,1,3); Environmental report                                                                |
|                                      | limestone, milled, packed, at plant                              | CH       | 0              | kg      | 1.14E+03 | 1                 | 1.17        |      |                    |       | (3,3,1,5,1,3); Environmental report                                                                |
|                                      | sodium hydroxide, 50% in H2O, production mix, at plant           | RER      | 0              | kg      | 6.76E+00 | 1                 | 1.17        |      |                    |       | (3,3,3,3,1,3); Calculation, based on thesis on Australian precious metal production, data for 1999 |
|                                      | hydrochloric acid, 30% in H2O, at plant                          | RER      | 0              | kg      | 2.04E+02 | 1                 | 1.14        |      |                    |       | (3,3,1,3,1,3); Environmental report                                                                |
|                                      | zinc, primary, at regional storage                               | RER      | 0              | kg      | 2.89E+00 | 1                 | 1.17        |      |                    |       | (3,3,3,3,1,3); Calculation, based on thesis on Australian precious metal production, data for 1999 |
|                                      | charcoal, at plant                                               | GLO      | 0              | kg      | 1.70E+01 | 1                 | 1.17        |      |                    |       | (3,3,3,2,1,3); Calculation, based on thesis on Australian precious metal production, data for 1999 |
|                                      | sulphuric acid, liquid, at plant                                 | RER      | 0              | kg      | 1.96E+02 | 1                 | 1.14        |      |                    |       | (3,3,1,3,1,3); Environmental report                                                                |
|                                      | blasting                                                         | RER      | 0              | kg      | 1.84E+02 | 1                 | 1.14        |      |                    |       | (1,3,1,5,1,3); Environmental report                                                                |
|                                      | transport, lorry >16t, fleet average                             | RER      | 0              | tkm     | 1.76E+00 | 1                 | 2.24        |      |                    |       | (5,3,1,3,1,3); Estimations                                                                         |
|                                      | disposal, sulfidic tailings, off-site                            | GLO      | 0              | kg      | 1.92E+06 | 1                 | 1.14        |      |                    |       | (3,3,1,2,1,3); Environmental report                                                                |
| infrastructure                       | facilities precious metal refinery                               | SE       | 1              | unit    | 1.65E-07 | 1                 | 3.05        |      |                    |       | (1,3,1,1,3,3); Annual report                                                                       |
|                                      | mine, gold                                                       | US       | 1              | unit    | 1.09E-07 | 1                 | 3.05        |      |                    |       | (1,3,1,1,3,3); Annual report                                                                       |
| emission air, low population density | Arsenic                                                          | -        | -              | kg      | 2.08E-02 | 1                 | 5.03        |      |                    |       | (3,4,1,1,1,3); Environmental report                                                                |
| emission air, low population density | Antimony                                                         | -        | -              | kg      | 6.04E-01 | 1                 | 5.03        |      |                    |       | (3,4,1,1,1,3); Environmental report                                                                |
|                                      | Carbon dioxide, fossil                                           | -        | -              | kg      | 7.63E+01 | 1                 | 1.14        |      |                    |       | (3,3,1,1,1,3); Environmental report                                                                |
|                                      | Beryllium                                                        | -        | -              | kg      | 5.58E-03 | 1                 | 5.04        |      |                    |       | (3,3,1,5,1,3); Calculation, based on Australian data                                               |
|                                      | Cadmium                                                          | -        | -              | kg      | 2.96E-04 | 1                 | 5.04        |      |                    |       | (3,3,1,5,1,3); Calculation, based on Australian data                                               |
|                                      | Lead                                                             | -        | -              | kg      | 2.08E-02 | 1                 | 5.03        |      |                    |       | (3,4,1,1,1,3); Environmental report                                                                |
|                                      | Chromium                                                         | -        | -              | kg      | 4.83E-02 | 1                 | 5.04        |      |                    |       | (3,3,1,5,1,3); Calculation, based on Australian data                                               |
|                                      | Cobalt                                                           | -        | -              | kg      | 8.37E-03 | 1                 | 5.04        |      |                    |       | (3,3,1,5,1,3); Calculation, based on Australian data                                               |
|                                      | Mercury                                                          | -        | -              | kg      | 3.22E-03 | 1                 | 5.03        |      |                    |       | (3,4,1,1,1,3); Environmental report                                                                |
|                                      | Manganese                                                        | -        | -              | kg      | 2.62E-01 | 1                 | 5.04        |      |                    |       | (3,3,1,5,1,3); Calculation, based on Australian data                                               |
|                                      | Selenium                                                         | -        | -              | kg      | 1.54E-01 | 1                 | 5.04        |      |                    |       | (3,3,1,5,1,3); Calculation, based on Australian data                                               |
|                                      | Heat, waste                                                      | -        | -              | MJ      | 3.51E+04 | 1                 | 1.22        |      |                    |       | (4,3,1,1,1,3); Calculation                                                                         |
| emission water, river                | Arsenic, ion                                                     | -        | -              | kg      | 5.35E-02 | 1                 | 5.04        |      |                    |       | (3,3,1,5,1,3); Calculation, based on Australian data                                               |
|                                      | Cadmium, ion                                                     | -        | -              | kg      | 2.41E-02 | 1                 | 3.03        |      |                    |       | (3,3,1,5,1,3); Calculation, based on Australian data                                               |
|                                      | Copper, ion                                                      | -        | -              | kg      | 4.37E-01 | 1                 | 3.03        |      |                    |       | (3,3,1,5,1,3); Calculation, based on Australian data                                               |
|                                      | Cyanide                                                          | -        | -              | kg      | 7.14E-01 | 1                 | 3.03        |      |                    |       | (3,3,1,5,1,3); Calculation, based on Australian data                                               |
|                                      | Lead                                                             | -        | -              | kg      | 1.47E+00 | 1                 | 5.04        |      |                    |       | (3,3,1,5,1,3); Calculation, based on Australian data                                               |
|                                      | Mercury                                                          | -        | -              | kg      | 2.68E-02 | 1                 | 5.04        |      |                    |       | (3,3,1,5,1,3); Calculation, based on Australian data                                               |
|                                      | Nickel, ion                                                      | -        | -              | kg      | 2.41E-01 | 1                 | 5.04        |      |                    |       | (3,3,1,5,1,3); Calculation, based on Australian data                                               |
|                                      | Zinc, ion                                                        | -        | -              | kg      | 2.68E+00 | 1                 | 5.04        |      |                    |       | (3,3,1,5,1,3); Calculation, based on Australian data                                               |
| resource, in water                   | Water, well, in ground                                           | -        | -              | m3      | 5.51E-01 | 1                 | 1.14        |      |                    |       | (3,3,1,1,1,3); Environmental report                                                                |
|                                      | Water, river                                                     | -        | -              | m3      | 1.74E+02 | 1                 | 1.14        |      |                    |       | (3,3,1,1,1,3); Environmental report                                                                |
| resource, in ground                  | Gold, Au 1.4E-4%, in ore, in ground                              | -        | -              | kg      | 1.14E+00 | 1                 | 1.14        |      |                    |       | (3,3,1,1,1,3); Environmental report                                                                |

Tab. 5.48 Ecospol Meta information of gold production in United States

| Name                       | mine, gold                                                                                                                                                                                                                                 | gold, at refinery                                                                                                                              |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Location                   | US                                                                                                                                                                                                                                         | US                                                                                                                                             |
| InfrastructureProcess      | 1                                                                                                                                                                                                                                          | 0                                                                                                                                              |
| Unit                       | unit                                                                                                                                                                                                                                       | kg                                                                                                                                             |
| Type                       | 1                                                                                                                                                                                                                                          | 1                                                                                                                                              |
| Version                    | 2.0                                                                                                                                                                                                                                        | 2.0                                                                                                                                            |
| energyValues               | 0                                                                                                                                                                                                                                          | 0                                                                                                                                              |
| LanguageCode               | en                                                                                                                                                                                                                                         | en                                                                                                                                             |
| LocalLanguageCode          | de                                                                                                                                                                                                                                         | de                                                                                                                                             |
| Person                     | 44                                                                                                                                                                                                                                         | 44                                                                                                                                             |
| QualityNetwork             | 1                                                                                                                                                                                                                                          | 1                                                                                                                                              |
| DataSetRelatesToProduct    | 1                                                                                                                                                                                                                                          | 1                                                                                                                                              |
| IncludedProcesses          | This data set describes the mining of gold concentrate in United States. It includes energy and water consumption, land use, transports, waste disposal as well as emissions to air and water.                                             |                                                                                                                                                |
| Amount                     | 1                                                                                                                                                                                                                                          | 1                                                                                                                                              |
| LocalName                  | Mine, Gold                                                                                                                                                                                                                                 | Gold, ab Raffinerie                                                                                                                            |
| Synonyms                   |                                                                                                                                                                                                                                            |                                                                                                                                                |
| GeneralComment             | This data set includes land use of the considered mines (open pit and underground) incl. revegetated areas as well as estimates of buildings and machinery of the mine based on the data set non-ferrous metal, mine, surface/underground. |                                                                                                                                                |
| InfrastructureIncluded     | 1                                                                                                                                                                                                                                          | 1                                                                                                                                              |
| Category                   | metals                                                                                                                                                                                                                                     | metals                                                                                                                                         |
| SubCategory                | extraction                                                                                                                                                                                                                                 | extraction                                                                                                                                     |
| LocalCategory              | Metalle                                                                                                                                                                                                                                    | Metalle                                                                                                                                        |
| LocalSubCategory           | Gewinnung                                                                                                                                                                                                                                  | Gewinnung                                                                                                                                      |
| Formula                    |                                                                                                                                                                                                                                            |                                                                                                                                                |
| StatisticalClassification  |                                                                                                                                                                                                                                            |                                                                                                                                                |
| CASNumber                  |                                                                                                                                                                                                                                            | 007440-57-5                                                                                                                                    |
| StartDate                  | 2003                                                                                                                                                                                                                                       | 2003                                                                                                                                           |
| EndDate                    | 2006                                                                                                                                                                                                                                       | 2006                                                                                                                                           |
| DataValidForEntirePeriod   | 1                                                                                                                                                                                                                                          | 1                                                                                                                                              |
| OtherPeriodText            |                                                                                                                                                                                                                                            |                                                                                                                                                |
|                            | Period of public environmental report                                                                                                                                                                                                      | Period of public environmental report                                                                                                          |
| Text                       | Land use of US-mine, other inputs based on rough assumptions.                                                                                                                                                                              |                                                                                                                                                |
| Text                       | Assumptions for buildings and machinery based on the data set "non-ferrous metal, mine, surface/underground with 50 years life time.                                                                                                       | Material requirements for gold concentration are estimated with Australian data, emissions to water and air are estimated with Australian data |
| Percent                    |                                                                                                                                                                                                                                            |                                                                                                                                                |
| ProductionVolume           | 351.6 Mill. t ore                                                                                                                                                                                                                          | 183.34 t gold                                                                                                                                  |
| SamplingProcedure          | environmental report                                                                                                                                                                                                                       | environmental report                                                                                                                           |
| Extrapolations             | Buildings and machinery are based on global assumptions. The figures are extrapolated on the basis of the ore production.                                                                                                                  | See Geography, several values have been scaled from one mine                                                                                   |
| UncertaintyAdjustments     | none                                                                                                                                                                                                                                       | none                                                                                                                                           |
| Person                     | 44                                                                                                                                                                                                                                         | 44                                                                                                                                             |
| DataPublishedIn            | 2                                                                                                                                                                                                                                          | 2                                                                                                                                              |
| ReferenceToPublishedSource | 10                                                                                                                                                                                                                                         | 10                                                                                                                                             |
| Copyright                  | 1                                                                                                                                                                                                                                          | 1                                                                                                                                              |
| AccessRestrictedTo         | 0                                                                                                                                                                                                                                          | 0                                                                                                                                              |
| CompanyCode                |                                                                                                                                                                                                                                            |                                                                                                                                                |
| CountryCode                |                                                                                                                                                                                                                                            |                                                                                                                                                |
| PageNumbers                |                                                                                                                                                                                                                                            |                                                                                                                                                |
| Validator                  | 51                                                                                                                                                                                                                                         | 51                                                                                                                                             |
| Details                    | automatic validation in Excel                                                                                                                                                                                                              | automatic validation in Excel                                                                                                                  |
| OtherDetails               | none                                                                                                                                                                                                                                       | none                                                                                                                                           |

### 5.1.8 Gold production, South Africa (ZA)

Data stem from the environmental report of Western Areas Limited (2006) for the South Deep Mine. The mine produces only gold and is an underground mine. Tab. 5.49 shows the production volume 2005 of the South Deep Mine. The recovery rate (proportion of valuable material physically recovered in the processing of ore) is taken into account in order to define the resource extraction. The average concentration (head grade, expressed in kg metal per kg ore) of metals in ore for South Africa is calculated with the quantity of ore milled, the corresponding production rate and the recovery rate.

Tab. 5.49: Production volume of the South Deep Mine in 2005 (Western Areas Limited, 2006).

| South Deep      |           |          |
|-----------------|-----------|----------|
| Production Gold | kg        | 1.43E+04 |
| Ore milled      | kg        | 2.14E+09 |
| Head grade      |           |          |
| Gold            | kg/kg ore | 6.92E-06 |
| Recovery rate   |           | 97%      |

### Mining infrastructure

This data set includes land use as well as rough estimates on buildings and machines necessary in gold mining. It is assumed, that the quantities of machinery as well as surface and volume of buildings are equal to the generic mine. This approach is rough, and the uncertainty is rather high. As the mined ore from the generic mine is some orders of magnitude lower than the mine in focus, a proportional scaling by milled ore will lead to a higher mistake (see Tab. 5.1: Gold and Silver production in Sweden). The lifetime of mines is assumed to be 50 years.

Land use is based on the figures from the previous environmental report of Placer Dome Inc. (2002b) see Tab. 5.50).

Tab. 5.50: Land use of the South Deep Mine in 2002 (Placer Dome Inc., 2002b)

| Land use                           |                |          | Remark                                                                       |
|------------------------------------|----------------|----------|------------------------------------------------------------------------------|
| Area disturbed during life of mine | m <sup>2</sup> | 4.61E+06 | Transformation, from unknown and Transformation, to mineral extraction site, |
| Disturbed area in active use       | m <sup>2</sup> | 4.61E+06 | Occupation, mineral extraction site, during 50 years                         |
| Area available for revegetation    | m <sup>2</sup> | 0.00E+00 | Transformation, to unknown                                                   |
| Area revegetated                   | m <sup>2</sup> | 0.00E+00 | Transformation, to heterogeneous, agricultural                               |

Tab. 5.51: Unit process raw data of the gold mine in South Africa.

|                | Name                                             | Location | Infrastructure | Process | Unit | mine, gold      | Uncertainty | Type | StandardDeviation95%                                  | GeneralComment |
|----------------|--------------------------------------------------|----------|----------------|---------|------|-----------------|-------------|------|-------------------------------------------------------|----------------|
|                | Location<br>InfrastructureProcess<br>Unit        |          |                |         |      | ZA<br>1<br>unit |             |      |                                                       |                |
| product        | mine, gold                                       | ZA       | 1              | unit    |      | 1.00E+0         |             |      |                                                       |                |
| technosphere   | building, multi-storey                           | RER      | 1              | m3      |      | 2.75E+5         | 1           | 2.39 | (4,5,5,5,5,5); Rough estimates                        |                |
|                | building, hall                                   | CH       | 1              | m2      |      | 1.25E+4         | 1           | 2.39 | (4,5,5,5,5,5); Rough estimates                        |                |
|                | industrial machine, heavy, unspecified, at plant | RER      | 1              | kg      |      | 8.50E+5         | 1           | 2.39 | (4,5,5,5,5,5); Rough estimates                        |                |
|                | conveyor belt, at plant                          | RER      | 1              | m       |      | 6.00E+2         | 1           | 2.39 | (4,5,5,5,5,5); Rough estimates                        |                |
| resource, land | Transformation, from unknown                     | -        | -              | m2      |      | 4.61E+6         | 1           | 2.01 | (1,3,2,1,1,3); Environmental report, Placer-Dome 2002 |                |
|                | Transformation, to mineral extraction site       | -        | -              | m2      |      | 4.61E+6         | 1           | 2.01 | (1,3,2,1,1,3); Environmental report, Placer-Dome 2002 |                |
|                | Occupation, mineral extraction site              | -        | -              | m2a     |      | 2.31E+8         | 1           | 1.51 | (1,3,2,1,1,3); Environmental report, Placer-Dome 2002 |                |
|                | Transformation, from mineral extraction site     | -        | -              | m2      |      | 4.61E+6         | 1           | 2.01 | (1,3,2,1,1,3); Environmental report, Placer-Dome 2002 |                |
|                | Transformation, to heterogeneous, agricultural   | -        | -              | m2      |      | 0               | 1           | 2.01 | (1,3,2,1,1,3); Environmental report, Placer-Dome 2002 |                |
|                | Transformation, to unknown                       | -        | -              | m2      |      | 4.61E+6         | 1           | 2.01 | (1,3,2,1,1,3); Environmental report, Placer-Dome 2002 |                |

### Mining and refining

The ore is mined by conventional underground long wall mining methods (50%) and mechanized drift and fill trackless methods (50%). The conventional stope ore is drilled, blasted and scraped or washed into a series of gullies that feed an ore pass system. The trackless ore is drilled, blasted and loaded with scoop trams to internal ore passes. The ore is then transported by rail to a shaft loading system and hoisted to surface. During June 2002 the new gold extraction plant was commissioned. It includes a two stage milling circuit (SAG Mill and Ball Mill), a gravity gold recovery circuit inclusive of leach reactors and a pump cell CIP plant. It has a planned capacity of 7,200 tonnes per day. It will be capable of producing at low cost and high extraction efficiency after an increased steady state mill feed is obtained. ESKOM is the supplier of electrical energy in the country; electricity at 132 kilovolts (kV) voltage level is supplied to South Deep from the national grid. The annual power requirement of the mine is approximately 610 GWh. The underground feed is presently 6.6kV with a plan for 11kV feed down the new South Deep Main Shaft complex.

Since the module “blasting” does not include the CO<sub>2</sub> emissions from explosives, they are calculated with the factor of the mines of Newmont (2005b) (0.16 kg CO<sub>2</sub> per kg explosive). Air emissions are approximated with Australian data, based on produced gold. Water emissions are estimated on the previous environmental report of Placer Dome Inc. (2002b) and assumed to be emitted to rivers. Transport of materials is calculated on the basis of standard distances. Data about amounts of tailing disposal is approximated with data from Papua New Guinea.

The following table (Tab. 5.52) sums up the total requirements and emissions of the South Deep Mine in 2005.

## 5. System Characterisation

Tab. 5.52: Requirements and emissions of the South Deep Mine in 2002 (Placer Dome Inc., 2002b).

| Energy                           |                |          | Remark                                                                                                                               |
|----------------------------------|----------------|----------|--------------------------------------------------------------------------------------------------------------------------------------|
| Electricity for mining           | kWh            | 1.68E+08 | South African electricity <sup>1)</sup> mix is approximated with the CENTREL mix, because of its high share of fossil energy sources |
| Electricity for refining         | kWh            | 2.34E+04 | assumed with Wohlwill-electrolysis                                                                                                   |
| Diesel (mobile equipment)        | MJ             | 9.22E+07 | Diesel, burned in building machine                                                                                                   |
| Petrol                           | l              | 2.38E+03 | converted in transport, van <3.5t (0.38l/tkm)                                                                                        |
| Materials                        |                |          |                                                                                                                                      |
| Cyanide                          | kg             | 2.46E+06 | In LCI as sodium cyanide                                                                                                             |
| Lime                             | kg             | 5.01E+06 | In LCI as Limestone, milled                                                                                                          |
| Caustic                          | kg             | 5.83E+05 | In LCI as sodium hydroxide                                                                                                           |
| Carbon                           | kg             | 5.50E+04 | from environmental report, in LCI as charcoal, at plant                                                                              |
| Zinc                             | kg             | 4.15E+04 | Scaled from Steward (1999) by processed gold, accounted for as zinc, primary, at regional storage                                    |
| Organic compounds                | kg             | 6.60E+04 | organic compounds as flocculent / flotation / magnafloc 351, in LCI as fatty alcohol                                                 |
| Hydrochloric Acid                | kg             | 1.11E+03 | from environmental report                                                                                                            |
| Explosives                       | kg             | 1.07E+06 | estimated with 0.5kg per ton of ore, approximated with Tovex                                                                         |
| Water resources                  |                |          |                                                                                                                                      |
| Groundwater Use                  | m <sup>3</sup> | 2.19E+06 | in LCI: Water, well, in ground                                                                                                       |
| Mine dewatering Use              | kg             | 0.00E+00 | recycled Water, not accounted for in inventory                                                                                       |
| Emissions to Water               |                |          |                                                                                                                                      |
| Arsenic                          | kg             | 7.94E+01 | based on older environmental report, scaled by milled ore (Placer-Dome, 2002)                                                        |
| Cadmium                          | kg             | 8.90E+00 | based on older environmental report, scaled by milled ore (Placer-Dome, 2002)                                                        |
| Copper                           | kg             | 4.40E+02 | based on older environmental report, scaled by milled ore (Placer-Dome, 2002)                                                        |
| Cyanide                          | kg             | 4.40E+03 | based on older environmental report, scaled by milled ore (Placer-Dome, 2002)                                                        |
| Lead                             | kg             | 5.29E+01 | based on older environmental report, scaled by milled ore (Placer-Dome, 2002)                                                        |
| Mercury                          | kg             | 8.90E+00 | based on older environmental report, scaled by milled ore (Placer-Dome, 2002)                                                        |
| Nickel                           | kg             | 1.01E+03 | based on older environmental report, scaled by milled ore (Placer-Dome, 2002)                                                        |
| Zinc                             | kg             | 6.61E+02 | based on older environmental report, scaled by milled ore (Placer-Dome, 2002)                                                        |
| Emissions to Air                 |                |          |                                                                                                                                      |
| Antimony                         | kg             | 2.31E+00 | based on Australian value, scaled by milled ore                                                                                      |
| Arsenic                          | kg             | 1.06E+02 | based on Australian value, scaled by milled ore                                                                                      |
| Beryllium                        | kg             | 6.22E+00 | based on Australian value, scaled by milled ore                                                                                      |
| Cadmium                          | kg             | 3.30E-01 | based on Australian value, scaled by milled ore                                                                                      |
| Carbon Dioxide (from explosives) | kg             | 1.71E+05 | Calculation, based on data from Newmont (2005)                                                                                       |
| Chromium                         | kg             | 5.39E+01 | based on Australian value, scaled by milled ore                                                                                      |
| Cobalt                           | kg             | 9.33E+00 | based on Australian value, scaled by milled ore                                                                                      |
| Lead                             | kg             | 2.62E+00 | based on Australian value, scaled by milled ore                                                                                      |
| Manganese                        | kg             | 2.92E+02 | based on Australian value, scaled by milled ore                                                                                      |
| Mercury                          | kg             | 1.50E+02 | based on Australian value, scaled by milled ore                                                                                      |
| Selenium                         | kg             | 1.78E+02 | based on Australian value, scaled by milled ore                                                                                      |

<sup>1</sup> South African electricity mix in 2007 (Nationmaster, 2004): fossil fuel 93.5 %, hydro 1.1%, nuclear 5.5%.

Since the mines produce doré gold (unrefined gold bullion bars containing between 60% and 95% gold), the refinery process is less complex than the refinery process of Rönnskär in Sweden. It is assumed, that the gold doré-bars undergo the treatment of Wohlwill electrolysis (for details see 5.2.2.1, “Gold / Silver, secondary, from electronic scrap inecoinvent”). Due to the lack of specific data, the figure of refinement from electronic scrap is used: 1.63 kWh per kg gold (Renner et al., 2002a). No emissions were assumed, since on the one hand the high value of the processed material requires an efficient emission control and on the other hand, refinement is a hydrometallurgical process with low

impact. The infrastructure is approximated with the facility of precious metal refinery. The unit process raw data are summarized in Tab. 5.53.

Tab. 5.53: Unit process raw data of gold mining and refining in South Africa

|                                      | Name                                                   | Location | Infrastructure | Unit | gold, at refinery | Uncertainty | StandardDeviation95% | GeneralComment                                                                                                           |
|--------------------------------------|--------------------------------------------------------|----------|----------------|------|-------------------|-------------|----------------------|--------------------------------------------------------------------------------------------------------------------------|
|                                      | Location<br>Infrastructure<br>Unit                     |          |                |      | ZA<br>0<br>kg     |             |                      |                                                                                                                          |
| product                              | gold, at refinery                                      | ZA       | 0              | kg   | 1.00E+00          |             |                      |                                                                                                                          |
| technosphere                         | electricity, medium voltage, production                | CENTREL  | 0              | kWh  | 1.17E+04          | 1           | 1.29                 | (3,4,1,5,3,3); Environmental report                                                                                      |
|                                      | CENTREL, at grid                                       | GLO      | 0              | MJ   | 6.43E+03          | 1           | 1.16                 | (3,4,1,2,1,3); Environmental report                                                                                      |
|                                      | diesel, burned in building machine                     | RER      | 0              | tkm  | 4.36E-01          | 1           | 2.03                 | (3,4,1,2,1,3); Environmental report                                                                                      |
|                                      | transport, van <3.5t                                   | RER      | 0              | kg   | 1.71E+02          | 1           | 1.14                 | (3,3,1,3,1,3); Environmental report                                                                                      |
|                                      | sodium cyanide, at plant                               | CH       | 0              | kg   | 3.49E+02          | 1           | 1.17                 | (3,3,1,5,1,3); Environmental report                                                                                      |
|                                      | limestone, milled, packed, at plant                    | RER      | 0              | kg   | 4.06E+01          | 1           | 1.17                 | (3,3,3,3,1,3); Environmental report                                                                                      |
|                                      | sodium hydroxide, 50% in H2O, production mix, at plant | RER      | 0              | kg   | 7.72E-02          | 1           | 1.14                 | (3,3,1,3,1,3); Environmental report                                                                                      |
|                                      | hydrochloric acid, 30% in H2O, at plant                | RER      | 0              | kg   | 2.89E+00          | 1           | 1.17                 | (3,3,3,3,1,3); Calculation, based on thesis on Australian precious metal production, data for 1999                       |
|                                      | zinc, primary, at regional storage                     | GLO      | 0              | kg   | 3.83E+00          | 1           | 1.17                 | (3,3,3,2,1,3); Environmental report                                                                                      |
|                                      | charcoal, at plant                                     | RER      | 0              | kg   | 4.60E+00          | 1           | 1.14                 | (3,3,1,3,1,3); Environmental report                                                                                      |
|                                      | fatty alcohol, petrochemical, at plant                 | RER      | 0              | kg   | 7.45E+01          | 1           | 1.14                 | (1,3,1,5,1,3); Environmental report                                                                                      |
|                                      | blasting                                               | RER      | 0              | tkm  | 4.36E-01          | 1           | 2.24                 | (5,3,1,3,1,3); Estimations                                                                                               |
|                                      | transport, lorry >16t, fleet average                   | GLO      | 0              | kg   | 1.49E+05          | 1           | 1.14                 | (3,3,1,2,1,3); Environmental report                                                                                      |
|                                      | disposal, sulfidic tailings, off-site                  | SE       | 1              | unit | 1.65E-07          | 1           | 3.06                 | (1,3,1,5,3,3); Annual report                                                                                             |
| infrastructure                       | facilities precious metal refinery                     | ZA       | 1              | unit | 1.39E-06          | 1           | 3.05                 | (1,3,1,1,3,3); Annual report                                                                                             |
| emission air, low population density | mine, gold                                             | -        | -              | kg   | 7.40E-03          | 1           | 5.07                 | (4,3,1,5,1,3); Estimation, based on Australian Data                                                                      |
|                                      | Arsenic                                                | -        | -              | kg   | 1.19E+01          | 1           | 1.25                 | (4,3,1,5,1,3); Estimation, based on Australian Data                                                                      |
|                                      | Carbon dioxide, fossil                                 | -        | -              | kg   | 1.83E-04          | 1           | 5.07                 | (4,3,1,5,1,3); Estimation, based on Australian Data                                                                      |
|                                      | Lead                                                   | -        | -              | kg   | 2.03E-02          | 1           | 5.04                 | (3,3,1,5,1,3); Calculation, based on Australian data                                                                     |
|                                      | Manganese                                              | -        | -              | kg   | 2.30E-05          | 1           | 5.04                 | (3,3,1,5,1,3); Calculation, based on Australian data                                                                     |
|                                      | Cadmium                                                | -        | -              | kg   | 1.61E-04          | 1           | 5.03                 | (3,4,1,1,1,3); Environmental report                                                                                      |
|                                      | Antimony                                               | -        | -              | kg   | 4.34E-04          | 1           | 5.04                 | (3,3,1,5,1,3); Calculation, based on Australian data                                                                     |
|                                      | Beryllium                                              | -        | -              | kg   | 3.76E-03          | 1           | 5.04                 | (3,3,1,5,1,3); Calculation, based on Australian data                                                                     |
|                                      | Chromium                                               | -        | -              | kg   | 6.51E-04          | 1           | 5.04                 | (3,3,1,5,1,3); Calculation, based on Australian data                                                                     |
|                                      | Cobalt                                                 | -        | -              | kg   | 1.20E-02          | 1           | 5.04                 | (3,3,1,5,1,3); Calculation, based on Australian data                                                                     |
|                                      | Selenium                                               | -        | -              | kg   | 1.05E-02          | 1           | 5.07                 | (4,3,1,5,1,3); Estimation, based on Australian Data                                                                      |
|                                      | Mercury                                                | -        | -              | kg   | 4.21E+04          | 1           | 1.22                 | (3,3,2,1,3,3); Estimation, based on previous environmental report (Placer-Dome, 2002)                                    |
|                                      | Heat, waste                                            | -        | -              | MJ   | 5.53E-03          | 1           | 5.07                 | (3,3,2,1,3,3); Estimation, based on previous environmental report (Placer-Dome, 2002)                                    |
|                                      | Arsenic, ion                                           | -        | -              | kg   | 6.20E-04          | 1           | 3.07                 | (3,3,2,1,3,3); Estimation, based on previous environmental report (Placer-Dome, 2002)                                    |
| emission water, river                | Cadmium, ion                                           | -        | -              | kg   | 3.07E-02          | 1           | 3.07                 | (3,3,2,1,3,3); Estimation, based on previous environmental report (Placer-Dome, 2002)                                    |
|                                      | Copper, ion                                            | -        | -              | kg   | 3.07E-01          | 1           | 3.07                 | (3,3,2,1,3,3); Estimation, based on previous environmental report (Placer-Dome, 2002)                                    |
|                                      | Cyanide                                                | -        | -              | kg   | 3.69E-03          | 1           | 5.07                 | (3,3,2,1,3,3); Estimation, based on previous environmental report (Placer-Dome, 2002)                                    |
|                                      | Lead                                                   | -        | -              | kg   | 6.20E-04          | 1           | 5.07                 | (3,3,2,1,3,3); Estimation, based on previous environmental report (Placer-Dome, 2002)                                    |
|                                      | Mercury                                                | -        | -              | kg   | 7.01E-02          | 1           | 5.07                 | (3,3,2,1,3,3); Estimation, based on previous environmental report (Placer-Dome, 2002)                                    |
|                                      | Nickel, ion                                            | -        | -              | kg   | 4.61E-02          | 1           | 5.06                 | (Estimated with data from other mine, 3,2,1,3,3); Estimation, based on previous environmental report (Placer-Dome, 2002) |
|                                      | Zinc, ion                                              | -        | -              | kg   | 1.53E+02          | 1           | 1.14                 | (3,3,1,1,1,3); Environmental report                                                                                      |
|                                      | Water, well, in ground                                 | -        | -              | m3   | 0.00E+00          | 1           | 1.14                 | (3,3,1,1,1,3); Environmental report                                                                                      |
| resource, in water                   | Water, river                                           | -        | -              | m3   | 1.03E+00          | 1           | 1.14                 | (3,3,1,1,1,3); Environmental report                                                                                      |
| resource, in ground                  | Gold, Au 6.7E-4%, in ore, in ground                    | -        | -              | kg   |                   |             |                      |                                                                                                                          |

Tab. 5.54 Ecospold Meta information of gold production in South Africa

|                            |                                                                                                                                                                                                                                |                                                                                                                                                                                               |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name                       | mine, gold                                                                                                                                                                                                                     | gold, at refinery                                                                                                                                                                             |
| Location                   | ZA                                                                                                                                                                                                                             | ZA                                                                                                                                                                                            |
| InfrastructureProcess      | 1                                                                                                                                                                                                                              | 0                                                                                                                                                                                             |
| Unit                       | unit                                                                                                                                                                                                                           | kg                                                                                                                                                                                            |
| Type                       | 1                                                                                                                                                                                                                              | 1                                                                                                                                                                                             |
| Version                    | 2.0                                                                                                                                                                                                                            | 2.0                                                                                                                                                                                           |
| energyValues               | 0                                                                                                                                                                                                                              | 0                                                                                                                                                                                             |
| LanguageCode               | en                                                                                                                                                                                                                             | en                                                                                                                                                                                            |
| LocalLanguageCode          | de                                                                                                                                                                                                                             | de                                                                                                                                                                                            |
| Person                     | 44                                                                                                                                                                                                                             | 44                                                                                                                                                                                            |
| QualityNetwork             | 1                                                                                                                                                                                                                              | 1                                                                                                                                                                                             |
| DataSetRelatesToProduct    | 1                                                                                                                                                                                                                              | 1                                                                                                                                                                                             |
| IncludedProcesses          | Land use, buildings and machinery for mining                                                                                                                                                                                   | This data set describes the mining of gold concentrate in South Africa. It includes energy and water consumption, land use, transports, waste disposal as well as emissions to air and water. |
| Amount                     | 1                                                                                                                                                                                                                              | 1                                                                                                                                                                                             |
| LocalName                  | Mine, Gold                                                                                                                                                                                                                     | Gold, ab Raffinerie                                                                                                                                                                           |
| Synonyms                   |                                                                                                                                                                                                                                |                                                                                                                                                                                               |
| GeneralComment             | This data set includes land use of the considered mines ( underground) incl. revegetated areas as well as estimates of buildings and machinery of the mine based on the data set non-ferrous metal, mine, surface/underground. | This data set describes the beneficiation, concentration and refinement of gold.                                                                                                              |
| InfrastructureIncluded     | 1                                                                                                                                                                                                                              | 1                                                                                                                                                                                             |
| Category                   | metals                                                                                                                                                                                                                         | metals                                                                                                                                                                                        |
| SubCategory                | extraction                                                                                                                                                                                                                     | extraction                                                                                                                                                                                    |
| LocalCategory              | Metalle                                                                                                                                                                                                                        | Metalle                                                                                                                                                                                       |
| LocalSubCategory           | Gewinnung                                                                                                                                                                                                                      | Gewinnung                                                                                                                                                                                     |
| Formula                    |                                                                                                                                                                                                                                |                                                                                                                                                                                               |
| StatisticalClassification  |                                                                                                                                                                                                                                |                                                                                                                                                                                               |
| CASNumber                  |                                                                                                                                                                                                                                | 007440-57-5                                                                                                                                                                                   |
| StartDate                  | 2002                                                                                                                                                                                                                           | 2002                                                                                                                                                                                          |
| EndDate                    | 2006                                                                                                                                                                                                                           | 2006                                                                                                                                                                                          |
| DataValidForEntirePeriod   | 1                                                                                                                                                                                                                              | 1                                                                                                                                                                                             |
| OtherPeriodText            | Period of public environmental report                                                                                                                                                                                          | Period of public environmental report                                                                                                                                                         |
| Text                       | Land use of South-African mine, other inputs based on rough assumptions.                                                                                                                                                       | Material requirements for gold concentration as well as emissions to air are estimated with Australian data.                                                                                  |
| Text                       | Assumptions for buildings and machinery based on the data set "non-ferrous metal, mine, surface/underground with 50 years life time.                                                                                           | The mining of ore from underground mines is considered.                                                                                                                                       |
| Percent                    |                                                                                                                                                                                                                                |                                                                                                                                                                                               |
| ProductionVolume           | 2.1 Mill. t ore                                                                                                                                                                                                                | 14.34 t gold                                                                                                                                                                                  |
| SamplingProcedure          | environmental report                                                                                                                                                                                                           | environmental report                                                                                                                                                                          |
| Extrapolations             | Buildings and machinery are based on global assumptions. The figures are extrapolated on the basis of the ore production.                                                                                                      | See Geography, electricity mix is extrapolated                                                                                                                                                |
| UncertaintyAdjustments     | none                                                                                                                                                                                                                           | none                                                                                                                                                                                          |
| Person                     | 44                                                                                                                                                                                                                             | 44                                                                                                                                                                                            |
| DataPublishedIn            | 2                                                                                                                                                                                                                              | 2                                                                                                                                                                                             |
| ReferenceToPublishedSource | 10                                                                                                                                                                                                                             | 10                                                                                                                                                                                            |
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| PageNumbers                |                                                                                                                                                                                                                                |                                                                                                                                                                                               |
| Validator                  | 51                                                                                                                                                                                                                             | 51                                                                                                                                                                                            |
| Details                    | automatic validation in Excel                                                                                                                                                                                                  | automatic validation in Excel                                                                                                                                                                 |
| OtherDetails               | none                                                                                                                                                                                                                           | none                                                                                                                                                                                          |

### 5.1.9 Gold production, Australia (AU)

Data stem from the environmental report of Newmont (2005b) for the Kalgoorlie, Jundee, Tanami and Pajingo Mines. The mines produce only gold. Tanami and Pajingo are underground mines, Kalgoorlie open pit, and Jundee open pit and underground. The average concentration (head grade, expressed in kg metal per kg ore) of metals in ore for Australia is calculated with the quantity of ore milled, the corresponding production rate and the recovery rate.

Tab 5.55 shows the production volume in 2005 of the considered Australian mines. The recovery rate (proportion of valuable material physically recovered in the processing of ore) is taken into account in order to define the resource extraction. The average concentration (head grade, expressed in kg metal per kg ore) of metals in ore for Australia is calculated with the quantity of ore milled, the corresponding production rate and the recovery rate.

**Tab 5.55: Production volume 2005 of the Kalgoorlie, Jundee, Tanami and Pajingo Mines (Newmont, 2005b).**

| Australia         |           | Kalgoorlie | Jundee   | Tanami   | Pajingo  | Australia | Remark                                                            |
|-------------------|-----------|------------|----------|----------|----------|-----------|-------------------------------------------------------------------|
| Production Gold   | kg        | 2.55E+04   | 1.06E+04 | 1.54E+04 | 5.97E+03 | 5.74E+04  |                                                                   |
| Ore milled        | kg        | 9.84E+10   | 1.40E+10 | 4.77E+09 | 1.28E+09 | 1.18E+11  |                                                                   |
| <i>Head grade</i> |           |            |          |          |          |           |                                                                   |
| Gold              | kg/kg ore | 2.81E-07   | 8.25E-07 | 3.50E-06 | 5.06E-06 | 5.27E-07  |                                                                   |
| Recovery rate     |           | n.a.       | n.a.     | n.a.     | n.a.     | 92%       | average value for Australia, based on data of Placer-Dome, (2002) |

### Mining infrastructure

This data set includes land use as well as rough estimates on buildings and machines necessary in gold mining. It is assumed, that the quantities of machinery as well as surface and volume of buildings are equal to the generic mine. This approach is rough, and the uncertainty is rather high. As the mined ore from the generic mine is some orders of magnitude lower than the mine in focus, a proportional scaling by milled ore will lead to a higher mistake (see Tab. 5.1: Gold and Silver production in Sweden). The lifetime of mines is assumed to be 50 years.

Land use is based on the figures from the environmental report (Tab. 5.56). Not all the area of the mine is in active use at one time; the value in Tab. 5.56 is assumed to be the average share of the mine area in use. Therefore, the total area is occupied during an average of about 28 years, according to the formula:

$$\frac{\text{Disturbed area in active use}}{\text{Area disturbed during life of mine}} * \text{lifetime of mine} = \frac{3.33 * 10^7}{6.00 * 10^8} * 50y = 28y, \text{ the occupation of the mine is calculated as } \text{Disturbed area in active use} * 50y = \text{Occupation, mineral extraction site}$$

**Tab. 5.56: Land use of the Kalgoorlie, Jundee, Tanami and Pajingo Mines (Newmont, 2005b)**

| Land use                           |                | Kalgoorlie | Jundee   | Tanami   | Pajingo  | Australia | Remark                                                                       |
|------------------------------------|----------------|------------|----------|----------|----------|-----------|------------------------------------------------------------------------------|
| Area disturbed during life of mine | m <sup>2</sup> | 2.01E+07   | 1.97E+07 | 1.67E+07 | 3.52E+06 | 6.00E+07  | Transformation, from unknown and Transformation, to mineral extraction site, |
| Disturbed area in active use       | m <sup>2</sup> | 1.09E+07   | 1.01E+07 | 1.05E+07 | 1.76E+06 | 3.33E+07  | Occupation, mineral extraction site, during 50 years                         |
| Area available for revegetation    | m <sup>2</sup> | 4.01E+05   | 5.00E+06 | 2.67E+05 | 1.42E+05 | 5.81E+06  | added in "Transformation, to unknown"                                        |
| Area revegetated                   | m <sup>2</sup> | 8.77E+06   | 4.54E+06 | 5.89E+06 | 1.62E+06 | 2.08E+07  | Transformation, to heterogeneous, agricultural                               |



Tab. 5.57: Unit process raw data of the gold mine in Australia

|                | Name                                             | Location | Infrastructure | Process | Unit | mine, gold      | Uncertainty | Type | StandardDeviation95%                              | GeneralComment |
|----------------|--------------------------------------------------|----------|----------------|---------|------|-----------------|-------------|------|---------------------------------------------------|----------------|
|                | Location<br>InfrastructureProcess<br>Unit        |          |                |         |      | AU<br>1<br>unit |             |      |                                                   |                |
| product        | mine, gold                                       | AU       | 1              | unit    |      | 1.00E+0         |             |      |                                                   |                |
| technosphere   | building, multi-storey                           | RER      | 1              | m3      |      | 2.75E+5         | 1           | 2.39 | (4,5,5,5,5,5); Rough estimates                    |                |
|                | building, hall                                   | CH       | 1              | m2      |      | 1.50E+4         | 1           | 2.39 | (4,5,5,5,5,5); Rough estimates                    |                |
|                | industrial machine, heavy, unspecified, at plant | RER      | 1              | kg      |      | 6.50E+5         | 1           | 2.39 | (4,5,5,5,5,5); Rough estimates                    |                |
|                | conveyor belt, at plant                          | RER      | 1              | m       |      | 6.00E+3         | 1           | 2.39 | (4,5,5,5,5,5); Rough estimates                    |                |
| resource, land | Transformation, from unknown                     | -        | -              | m2      |      | 6.00E+7         | 1           | 2.01 | (1,3,1,1,1,3); Environmental report, Newmont 2005 |                |
|                | Transformation, to mineral extraction site       | -        | -              | m2      |      | 6.00E+7         | 1           | 2.01 | (1,3,1,1,1,3); Environmental report, Newmont 2005 |                |
|                | Occupation, mineral extraction site              | -        | -              | m2a     |      | 1.67E+9         | 1           | 1.51 | (1,3,1,1,1,3); Environmental report, Newmont 2005 |                |
|                | Transformation, from mineral extraction site     | -        | -              | m2      |      | 6.00E+7         | 1           | 2.01 | (1,3,1,1,1,3); Environmental report, Newmont 2005 |                |
|                | Transformation, to heterogeneous, agricultural   | -        | -              | m2      |      | 2.08E+7         | 1           | 2.01 | (1,3,1,1,1,3); Environmental report, Newmont 2005 |                |
|                | Transformation, to unknown                       | -        | -              | m2      |      | 3.91E+7         | 1           | 2.01 | (1,3,1,1,1,3); Environmental report, Newmont 2005 |                |

## Mining and refining

Although the production of gold and silver was assessed in an older detailed study (Stewart, 1999), the production of silver is neglected due to the low production (see Tab. 5.58) and the absence of actual data. Nevertheless, the data from the mentioned study has been used as a proxy value for material requirements, where actual data is lacking. The allocation for gold and silver (see Tab. 5.58) is done by economical values with prices of Tab. 2.5.

Tab. 5.58: Production of gold / silver and material requirement in gold / silver mining in the year 1999 (Stewart, 1999)

| Production        | Australia | Allocation | per kg gold | per kg silver |
|-------------------|-----------|------------|-------------|---------------|
| Gold              | 3.00E+05  | 99.81%     |             |               |
| Silver            | 3.40E+04  | 0.19%      |             |               |
| Material Use      |           |            |             |               |
| Lime              | 1.90E+7   |            | 6.32E+01    | 1.04E+00      |
| Sodium hydroxide  | 3.50E+6   |            | 1.16E+01    | 1.92E-01      |
| Hydrochloric acid | 4.10E+6   |            | 1.36E+01    | 2.25E-01      |
| Zinc              | 8.70E+5   |            | 2.89E+00    | 4.77E-02      |
| Carbon            | 5.10E+6   |            | 1.70E+01    | 2.79E-01      |
| Sulphuric acid    | 2.00E+5   |            | 6.65E-01    | 1.10E-02      |

The following table (Tab. 5.59) sums up the total requirements and emissions of the Kalgoorlie, Jun-dee, Tanami and Pajingo Mines in 2005. Information stems from the environmental reports. Carbon and zinc consumption is calculated with data from Stewart (1999).

Tab. 5.59: Energy and material requirements of the Kalgoorlie, Jundee, Tanami and Pajingo Mines (Newmont, 2005b)

| Australia                 |                |            |          |          |          |           |                                                                                                                                   |
|---------------------------|----------------|------------|----------|----------|----------|-----------|-----------------------------------------------------------------------------------------------------------------------------------|
| Energy                    |                | Kalgoorlie | Jundee   | Tanami   | Pajingo  | Australia | Remark                                                                                                                            |
| Electricity for mining    | kWh            | 2.00E+08   | -        | 9.76E+07 | 5.40E+07 | 3.51E+08  | Australian electricity <sup>1)</sup> mix is approximated with the CENTREL mix, because of its high share of fossil energy sources |
| Electricity for refining  | kWh            | n.a.       | n.a.     | n.a.     | n.a.     | 9.36E+04  | assumed with Wohlwill-electrolysis                                                                                                |
| Diesel (generators)       | MJ             | -          | -        | 6.98E+07 | -        | 6.98E+07  | Diesel, burned in diesel-electric generating set                                                                                  |
| Diesel (mobile equipment) | MJ             | 2.09E+09   | 4.77E+08 | 1.43E+09 | 1.21E+08 | 4.12E+09  | Diesel, burned in building machine                                                                                                |
| Petrol                    | l              | 2.70E+03   | -        | 1.12E+06 | 4.52E+01 | 1.13E+06  | converted in transport, van <3.5t (0.38l/tkm)                                                                                     |
| Oil                       | MJ             | 3.23E+07   | 1.66E+07 | 1.44E+07 | 8.08E+06 | 7.14E+07  | heavy fuel oil, burned in industrial furnace 1MW, non-modulating                                                                  |
| LPG                       | MJ             | -          | 4.66E+08 | -        | -        | 4.66E+08  | natural gas, burned in industrial furnace                                                                                         |
| Materials                 |                |            |          |          |          |           |                                                                                                                                   |
| Cyanide                   | kg             | 3.94E+06   | 5.18E+05 | 7.85E+05 | 1.41E+05 | 5.38E+06  | In LCI as sodium cyanide                                                                                                          |
| Lime                      | kg             | 6.53E+07   | 8.20E+06 | 3.71E+06 | 6.02E+05 | 7.78E+07  | in LCI as limestone, milled, packed, at plant                                                                                     |
| Caustic                   | kg             | 5.11E+06   | 4.30E+05 | 6.70E+05 | 3.62E+05 | 6.57E+06  | in LCI as sodium hydroxide                                                                                                        |
| Carbon                    | kg             | 4.32E+05   | 1.80E+05 | 2.61E+05 | 1.01E+05 | 9.74E+05  | Scaled from Steward (1999) by processed gold, accounted for as Charcoal, at plant                                                 |
| Zinc                      | kg             | 7.37E+04   | 3.07E+04 | 4.44E+04 | 1.73E+04 | 1.66E+05  | Scaled from Steward (1999) by processed gold, accounted for as zinc, primary, at regional storage                                 |
| Sulphuric Acid            | kg             | 8.96E+05   | 2.86E+05 | 8.17E+04 | 6.91E+04 | 1.33E+06  | Acids (hydrochloric, sulphamic and sulphuric) consumption is divided in half sulphuric, half hydrochloric                         |
| Hydrochloric Acid         | kg             | 8.96E+05   | 2.86E+05 | 8.17E+04 | 6.91E+04 | 1.33E+06  | see Sulphuric Acid                                                                                                                |
| Explosives                | kg             | 1.81E+07   | 2.10E+06 | 8.48E+05 | 9.25E+05 | 2.20E+07  | Partly ANFO explosives, in LCI: "blasting"                                                                                        |
| Water resources           |                |            |          |          |          |           |                                                                                                                                   |
| Surface Water Use         | m <sup>3</sup> | 1.80E+06   | -        | 2.54E+06 | 3.21E+05 | 4.66E+06  | in LCI: Water, river                                                                                                              |
| Groundwater Use           | m <sup>3</sup> | 4.59E+06   | 1.66E+06 | -        | 2.57E+04 | 6.27E+06  | in LCI: Water, well, in ground                                                                                                    |
| Mine dewatering Use       | m <sup>3</sup> | 3.70E+05   | 1.87E+05 | -        | 1.50E+04 | 5.71E+05  | recycled Water, not accounted for in inventory                                                                                    |

<sup>1)</sup> Australian electricity mix (Nationmaster, 2004): fossil fuel 90.8%, hydro 8.3%, other 0.9%)

Water emissions are taken from data of the Granny Smith Mine (Placer Dome Inc., 2003a) for arsenic, and from data of the Henty Mine (Placer Dome Inc., 2003a) for the other water emissions. They are assumed to be released into rivers. Data about amounts of tailings disposal is approximated with data from Papua New Guinea. Since the module "blasting" does not include the CO<sub>2</sub> emissions from explosives, they are calculated with the factor of the mines of Newmont (2005b) (0.16 kg CO<sub>2</sub> per kg explosive).

Since the mines produce doré gold (unrefined gold bullion bars containing between 60% and 95% gold), the refinery process is less complex than the refinery process of Rönnskär in Sweden. It is assumed, that the gold doré-bars undergo the treatment of Wohlwill electrolysis (for details see 5.2.2.1, "Gold / Silver, secondary, from electronic scrap inecoinvent"). Due to the lack of specific data, the figure of refinement from electronic scrap is used: 1.63 kWh per kg gold (Renner et al., 2002a). No emissions were assumed, since on the one hand the high value of the processed material requires an efficient emission control and on the other hand, refinement is a hydrometallurgical process with low impact. The infrastructure is approximated with the facility of precious metal refinery.

Tab. 5.60 Emissions of the Kalgoorlie, Jundee, Tanami and Pajingo Mines (Newmont, 2005b)

| Emissions to Water               |    | Kalgoorlie | Jundee   | Tanami   | Pajingo  | Australia | Remark                                                                                                             |
|----------------------------------|----|------------|----------|----------|----------|-----------|--------------------------------------------------------------------------------------------------------------------|
| Arsenic                          | kg | 2.75E+03   | 3.90E+02 | 1.33E+02 | 3.58E+01 | 3.30E+03  | Emission based on data for Mine "Granny Smith", Placer Dome Inc. (2002), scaled by milled ore                      |
| Cadmium                          | kg | 1.24E+03   | 1.75E+02 | 5.99E+01 | 1.61E+01 | 1.49E+03  | Emission based on data for Mine "Henty", Placer Dome Inc. (2002), scaled by milled ore                             |
| Copper                           | kg | 2.24E+04   | 3.18E+03 | 1.09E+03 | 2.92E+02 | 2.70E+04  | Emission based on data for Mine "Henty", Placer Dome Inc. (2002), scaled by milled ore                             |
| Cyanide                          | kg | 3.66E+04   | 5.20E+03 | 1.78E+03 | 4.77E+02 | 4.41E+04  | Emission based on data for Mine "Henty", Placer Dome Inc. (2002), scaled by milled ore                             |
| Lead                             | kg | 7.56E+04   | 1.07E+04 | 3.66E+03 | 9.84E+02 | 9.09E+04  | Emission based on data for Mine "Henty", Placer Dome Inc. (2002), scaled by milled ore                             |
| Mercury                          | kg | 1.37E+03   | 1.95E+02 | 6.66E+01 | 1.79E+01 | 1.65E+03  | Emission based on data for Mine "Henty", Placer Dome Inc. (2002), scaled by milled ore                             |
| Nickel                           | kg | 1.24E+04   | 1.75E+03 | 5.99E+02 | 1.61E+02 | 1.49E+04  | Emission based on data for Mine "Henty", Placer Dome Inc. (2002), scaled by milled ore                             |
| Zinc                             | kg | 1.37E+05   | 1.95E+04 | 6.66E+03 | 1.79E+03 | 1.65E+05  | Emission based on data for Mine "Henty", Placer Dome Inc. (2002), scaled by milled ore                             |
| Emissions to Air                 |    |            |          |          |          |           |                                                                                                                    |
| Antimony                         | kg | 8.16E+01   | 4.54E+01 | 0.00E+00 | 1.06E+00 | 1.28E+02  | Data for Pajingo estimated with data of Kalgoorlie, Environmental Report, Newmont 2005                             |
| Arsenic                          | kg | 2.83E+03   | 2.68E+03 | 3.54E+02 | 1.81E+01 | 5.88E+03  | Environmental Report, Newmont 2005                                                                                 |
| Beryllium                        | kg | 1.81E+01   | 0.00E+00 | 0.00E+00 | 3.27E+02 | 3.45E+02  | Data for Pajingo estimated with data of Kalgoorlie, Environmental Report, Newmont 2005                             |
| Cadmium                          | kg | 9.07E+00   | 9.07E+00 | 0.00E+00 | 1.18E-01 | 1.83E+01  | Data for Pajingo estimated with data of Kalgoorlie, Environmental Report, Newmont 2005                             |
| Carbon Monoxide                  | kg | 6.80E+05   | 2.02E+05 | 5.16E+05 | 3.73E+04 | -         | environmental report of Newmont (2005), not accounted for in LCI as emissions are already included in fossil fuels |
| Carbon Dioxide (from explosives) | kg | 2.90E+06   | 3.37E+05 | 1.36E+05 | 1.48E+05 | 3.52E+06  | Environmental Report, Newmont 2005                                                                                 |
| Chromium                         | kg | 2.30E+03   | 4.81E+02 | 1.72E+02 | 2.72E+01 | 2.98E+03  | Environmental Report, Newmont 2005                                                                                 |
| Cobalt                           | kg | 4.63E+02   | 3.63E+01 | 9.07E+00 | 9.07E+00 | 5.17E+02  | Environmental Report, Newmont 2005                                                                                 |
| Lead                             | kg | 1.09E+02   | 9.07E+00 | 1.81E+01 | 9.07E+00 | 1.45E+02  | Environmental Report, Newmont 2005                                                                                 |
| Manganese                        | kg | 1.32E+04   | 1.24E+03 | 1.22E+03 | 5.26E+02 | 1.62E+04  | Environmental Report, Newmont 2005                                                                                 |
| Mercury                          | kg | 8.31E+03   | 0.00E+00 | 9.07E+00 | 0.00E+00 | 8.32E+03  | Environmental Report, Newmont 2005                                                                                 |
| Nitrogen Oxide (NOx)             | kg | 1.70E+06   | 4.64E+05 | 1.18E+06 | 4.50E+04 | -         | environmental report of Newmont (2005), not accounted for in LCI as emissions are already included in fossil fuels |
| Particulates                     | kg | 3.80E+06   | 1.33E+06 | 5.77E+05 | 2.61E+05 | -         | environmental report of Newmont (2005), not accounted for in LCI as emissions are already included in fossil fuels |
| Selenium                         | kg | 8.20E+03   | 1.16E+03 | 2.72E+01 | 1.07E+02 | 9.50E+03  | Data for Pajingo / Jundee estimated with data of Kalgoorlie, Environmental Report, Newmont 2005                    |
| Sulfur Dioxide (SO2)             | kg | 1.40E+08   | 1.82E+04 | 2.58E+04 | 3.44E+03 | -         | environmental report of Newmont (2005), not accounted for in LCI as emissions are already included in fossil fuels |

The unit process raw data are summarized in Tab. 5.61.

## 5. System Characterisation

Tab. 5.61: Unit process raw data of gold mining and refining in Australia.

|                                       | Name                                                             | Location | Infrastructure | Process  | Unit     | gold, at refinery | Uncertainty    | Type                 | Standard Deviation 95%                                                              | General Comment |
|---------------------------------------|------------------------------------------------------------------|----------|----------------|----------|----------|-------------------|----------------|----------------------|-------------------------------------------------------------------------------------|-----------------|
|                                       | Location<br>Infrastructure<br>Unit                               |          |                |          |          | AU<br>0<br>kg     |                |                      |                                                                                     |                 |
| product                               | gold, at refinery                                                | AU       | 0              | kg       | 1.00E+00 |                   |                |                      |                                                                                     |                 |
| technosphere                          | electricity, medium voltage, production CENTREL, at grid         | CENTREL  | 0              | kWh      | 6.12E+03 | 1                 | 1.29           | (3,4,1,5,3,3);       | Environmental report                                                                |                 |
|                                       | diesel, burned in diesel-electric generating set                 | GLO      | 0              | MJ       | 1.22E+03 | 1                 | 1.16           | (3,4,1,2,1,3);       | Environmental report                                                                |                 |
|                                       | diesel, burned in building machine                               | GLO      | 0              | MJ       | 7.18E+04 | 1                 | 1.16           | (3,4,1,2,1,3);       | Environmental report                                                                |                 |
|                                       | transport, van <3.5t                                             | RER      | 0              | tkm      | 5.16E+01 | 1                 | 2.03           | (3,4,1,2,1,3);       | Environmental report                                                                |                 |
|                                       | heavy fuel oil, burned in industrial furnace 1MW, non-modulating | RER      | 0              | MJ       | 1.24E+03 | 1                 | 1.16           | (3,4,1,2,1,3);       | Environmental report                                                                |                 |
|                                       | natural gas, burned in industrial furnace >100kW                 | RER      | 0              | MJ       | 8.11E+03 | 1                 | 1.16           | (3,4,1,2,1,3);       | Environmental report                                                                |                 |
|                                       | sodium cyanide, at plant                                         | RER      | 0              | kg       | 9.38E+01 | 1                 | 1.14           | (3,3,1,3,1,3);       | Environmental report                                                                |                 |
|                                       | limestone, milled, packed, at plant                              | CH       | 0              | kg       | 1.35E+03 | 1                 | 1.17           | (3,3,1,5,1,3);       | Environmental report                                                                |                 |
|                                       | sodium hydroxide, 50% in H2O, production mix, at plant           | RER      | 0              | kg       | 1.14E+02 | 1                 | 1.17           | (3,3,3,3,1,3);       | Environmental report                                                                |                 |
|                                       | hydrochloric acid, 30% in H2O, at plant                          | RER      | 0              | kg       | 2.32E+01 | 1                 | 1.14           | (3,3,1,3,1,3);       | Environmental report                                                                |                 |
|                                       | zinc, primary, at regional storage                               | RER      | 0              | kg       | 2.89E+00 | 1                 | 1.17           | (3,3,3,2,1,3);       | Calculation, based on thesis on Australian precious metal production, data for 1999 |                 |
|                                       | charcoal, at plant                                               | GLO      | 0              | kg       | 1.70E+01 | 1                 | 1.17           | (3,3,3,2,1,3);       | Calculation, based on thesis on Australian precious metal production, data for 1999 |                 |
|                                       | sulphuric acid, liquid, at plant                                 | RER      | 0              | kg       | 2.32E+01 | 1                 | 1.14           | (3,3,1,3,1,3);       | Environmental report                                                                |                 |
|                                       | blasting                                                         | RER      | 0              | kg       | 3.84E+02 | 1                 | 1.14           | (1,3,1,5,1,3);       | Environmental report                                                                |                 |
|                                       | transport, lorry >16t, fleet average                             | RER      | 0              | tkm      | 5.16E+01 | 1                 | 2.24           | (5,3,1,3,1,3);       | Estimations                                                                         |                 |
| disposal, sulfidic tailings, off-site | GLO                                                              | 0        | kg             | 2.06E+06 | 1        | 1.14              | (3,3,1,2,1,3); | Environmental report |                                                                                     |                 |
| infrastructure                        | mine, gold                                                       | AU       | 1              | unit     | 3.48E-07 | 1                 | 3.05           | (1,3,1,1,3,3);       | Annual report                                                                       |                 |
|                                       | facilities precious metal refinery                               | SE       | 1              | unit     | 1.65E-07 | 1                 | 3.05           | (1,3,1,1,3,3);       | Annual report                                                                       |                 |
| emission air, low population density  | Manganese                                                        | -        | -              | kg       | 2.82E-01 | 1                 | 5.04           | (3,3,1,5,1,3);       | Environmental report                                                                |                 |
|                                       | Cadmium                                                          | -        | -              | kg       | 3.18E-04 | 1                 | 5.04           | (3,3,1,5,1,3);       | Environmental report                                                                |                 |
|                                       | Antimony                                                         | -        | -              | kg       | 2.23E-03 | 1                 | 5.03           | (3,4,1,1,1,3);       | Environmental report                                                                |                 |
|                                       | Beryllium                                                        | -        | -              | kg       | 6.01E-03 | 1                 | 5.04           | (3,3,1,5,1,3);       | Environmental report                                                                |                 |
|                                       | Chromium                                                         | -        | -              | kg       | 5.20E-02 | 1                 | 5.04           | (3,3,1,5,1,3);       | Environmental report                                                                |                 |
|                                       | Cobalt                                                           | -        | -              | kg       | 9.01E-03 | 1                 | 5.04           | (3,3,1,5,1,3);       | Environmental report                                                                |                 |
|                                       | Selenium                                                         | -        | -              | kg       | 1.65E-01 | 1                 | 5.04           | (3,3,1,5,1,3);       | Environmental report                                                                |                 |
|                                       | Arsenic                                                          | -        | -              | kg       | 1.02E-01 | 1                 | 5.02           | (3,3,1,1,1,3);       | Environmental report                                                                |                 |
|                                       | Carbon dioxide, fossil                                           | -        | -              | kg       | 6.14E+01 | 1                 | 1.14           | (3,3,1,1,1,3);       | Environmental report                                                                |                 |
|                                       | Lead                                                             | -        | -              | kg       | 2.53E-03 | 1                 | 5.02           | (3,3,1,1,1,3);       | Environmental report                                                                |                 |
|                                       | Mercury                                                          | -        | -              | kg       | 1.45E-01 | 1                 | 5.02           | (3,3,1,1,1,3);       | Environmental report                                                                |                 |
|                                       | Heat, waste                                                      | -        | -              | MJ       | 2.20E+04 | 1                 | 1.22           | (4,3,1,1,1,3);       | Calculation                                                                         |                 |
|                                       | Arsenic, ion                                                     | -        | -              | kg       | 5.76E-02 | 1                 | 5.09           | (3,4,2,1,3,3);       | Environmental report                                                                |                 |
|                                       | Cadmium, ion                                                     | -        | -              | kg       | 2.59E-02 | 1                 | 3.07           | (3,4,2,1,3,3);       | Environmental report                                                                |                 |
|                                       | Copper, ion                                                      | -        | -              | kg       | 4.70E-01 | 1                 | 3.07           | (3,4,2,1,3,3);       | Environmental report                                                                |                 |
| Cyanide                               | -                                                                | -        | kg             | 7.68E-01 | 1        | 3.07              | (3,4,2,1,3,3); | Environmental report |                                                                                     |                 |
| Lead                                  | -                                                                | -        | kg             | 1.58E+00 | 1        | 5.09              | (3,4,2,1,3,3); | Environmental report |                                                                                     |                 |
| Mercury                               | -                                                                | -        | kg             | 2.88E-02 | 1        | 5.09              | (3,4,2,1,3,3); | Environmental report |                                                                                     |                 |
| Nickel, ion                           | -                                                                | -        | kg             | 2.59E-01 | 1        | 5.09              | (3,4,2,1,3,3); | Environmental report |                                                                                     |                 |
| Zinc, ion                             | -                                                                | -        | kg             | 2.88E+00 | 1        | 5.07              | (3,3,2,1,3,3); | Environmental report |                                                                                     |                 |
| resource, in water                    | Water, well, in ground                                           | -        | -              | m3       | 8.12E+01 | 1                 | 1.14           | (3,3,1,1,1,3);       | Environmental report                                                                |                 |
|                                       | Water, river                                                     | -        | -              | m3       | 1.09E+02 | 1                 | 1.14           | (3,3,1,1,1,3);       | Environmental report                                                                |                 |
| resource, in ground                   | Gold, Au 4.9E-5% in ore, in ground                               | -        | -              | kg       | 1.09E+00 | 1                 | 1.14           | (3,3,1,1,1,3);       | Environmental report                                                                |                 |

Tab. 5.62 Ecospol Meta information of gold production in Australia

|                            |                                                                                                                                                                                                                                            |                                                                                                                                                                                            |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name                       | mine, gold                                                                                                                                                                                                                                 | gold, at refinery                                                                                                                                                                          |
| Location                   | AU                                                                                                                                                                                                                                         | AU                                                                                                                                                                                         |
| InfrastructureProcess      | 1                                                                                                                                                                                                                                          | 0                                                                                                                                                                                          |
| Unit                       | unit                                                                                                                                                                                                                                       | kg                                                                                                                                                                                         |
| Type                       | 1                                                                                                                                                                                                                                          | 1                                                                                                                                                                                          |
| Version                    | 2.0                                                                                                                                                                                                                                        | 2.0                                                                                                                                                                                        |
| energyValues               | 0                                                                                                                                                                                                                                          | 0                                                                                                                                                                                          |
| LanguageCode               | en                                                                                                                                                                                                                                         | en                                                                                                                                                                                         |
| LocalLanguageCode          | de                                                                                                                                                                                                                                         | de                                                                                                                                                                                         |
| Person                     | 44                                                                                                                                                                                                                                         | 44                                                                                                                                                                                         |
| QualityNetwork             | 1                                                                                                                                                                                                                                          | 1                                                                                                                                                                                          |
| DataSetRelatesToProduct    | 1                                                                                                                                                                                                                                          | 1                                                                                                                                                                                          |
| IncludedProcesses          | Land use, buildings and machinery of mining                                                                                                                                                                                                | This data set describes the mining of gold concentrate in Australia. It includes energy and water consumption, land use, transports, waste disposal as well as emissions to air and water. |
| Amount                     | 1                                                                                                                                                                                                                                          | 1                                                                                                                                                                                          |
| LocalName                  | Mine, Gold                                                                                                                                                                                                                                 | Gold, ab Raffinerie                                                                                                                                                                        |
| Synonyms                   |                                                                                                                                                                                                                                            |                                                                                                                                                                                            |
| GeneralComment             | This data set includes land use of the considered mines (open pit and underground) incl. revegetated areas as well as estimates of buildings and machinery of the mine based on the data set non-ferrous metal, mine, surface/underground. | This data set describes the beneficiation, concentration and refinement of gold.                                                                                                           |
| InfrastructureIncluded     | 1                                                                                                                                                                                                                                          | 1                                                                                                                                                                                          |
| Category                   | metals                                                                                                                                                                                                                                     | metals                                                                                                                                                                                     |
| SubCategory                | extraction                                                                                                                                                                                                                                 | extraction                                                                                                                                                                                 |
| LocalCategory              | Metalle                                                                                                                                                                                                                                    | Metalle                                                                                                                                                                                    |
| LocalSubCategory           | Gewinnung                                                                                                                                                                                                                                  | Gewinnung                                                                                                                                                                                  |
| Formula                    |                                                                                                                                                                                                                                            |                                                                                                                                                                                            |
| StatisticalClassification  |                                                                                                                                                                                                                                            |                                                                                                                                                                                            |
| CASNumber                  |                                                                                                                                                                                                                                            | 007440-57-5                                                                                                                                                                                |
| StartDate                  | 2001                                                                                                                                                                                                                                       | 2001                                                                                                                                                                                       |
| EndDate                    | 2006                                                                                                                                                                                                                                       | 2006                                                                                                                                                                                       |
| DataValidForEntirePeriod   | 1                                                                                                                                                                                                                                          | 1                                                                                                                                                                                          |
| OtherPeriodText            | Period of public environmental report                                                                                                                                                                                                      | Period of public environmental report                                                                                                                                                      |
| Text                       | Land use of Australian mine, other inputs based on rough assumptions.                                                                                                                                                                      | the electricity mix has been estimated                                                                                                                                                     |
| Text                       | Assumptions for buildings and machinery based on the data set "non-ferrous metal, mine, surface/underground with 50 years life time.                                                                                                       | The mining of ore from open pit and underground mines is considered.                                                                                                                       |
| Percent                    |                                                                                                                                                                                                                                            |                                                                                                                                                                                            |
| ProductionVolume           | 118.5 Mill. t ore                                                                                                                                                                                                                          | 57.4 t gold                                                                                                                                                                                |
| SamplingProcedure          | environmental report                                                                                                                                                                                                                       | environmental report                                                                                                                                                                       |
| Extrapolations             | Buildings and machinery are based on global assumptions. The figures are extrapolated on the basis of the ore production.                                                                                                                  | Material requirements for gold concentration are estimated with older data from an Australian thesis, several values are scaled from one mine                                              |
| UncertaintyAdjustments     | none                                                                                                                                                                                                                                       | none                                                                                                                                                                                       |
| Person                     | 44                                                                                                                                                                                                                                         | 44                                                                                                                                                                                         |
| DataPublishedIn            | 2                                                                                                                                                                                                                                          | 2                                                                                                                                                                                          |
| ReferenceToPublishedSource | 10                                                                                                                                                                                                                                         | 10                                                                                                                                                                                         |
| Copyright                  | 1                                                                                                                                                                                                                                          | 1                                                                                                                                                                                          |
| AccessRestrictedTo         | 0                                                                                                                                                                                                                                          | 0                                                                                                                                                                                          |
| CompanyCode                |                                                                                                                                                                                                                                            |                                                                                                                                                                                            |
| CountryCode                |                                                                                                                                                                                                                                            |                                                                                                                                                                                            |
| PageNumbers                |                                                                                                                                                                                                                                            |                                                                                                                                                                                            |
| Validator                  | 51                                                                                                                                                                                                                                         | 51                                                                                                                                                                                         |
| Details                    | automatic validation in Excel                                                                                                                                                                                                              | automatic validation in Excel                                                                                                                                                              |
| OtherDetails               | none                                                                                                                                                                                                                                       | none                                                                                                                                                                                       |

### 5.1.10 Gold production, Tanzania (TZ)

Data stem from the environmental report of Placer Dome Inc. (2005) and Barrick (2006b) for the North Mara Mine. The mine produces only gold and is an open pit mine. Tab 5.63 shows the production volume in 2005 of the North Mara Mine. The recovery rate (proportion of valuable material physically recovered in the processing of ore) is taken into account in order to define the resource extraction. The average concentration (head grade, expressed in kg metal per kg ore) of metals in ore for Tanzania is calculated with the quantity of ore milled, the corresponding production rate and the recovery rate.

Tab 5.63: Production volume in 2004 of the North Mara Mine (Placer Dome Inc., 2005).

| Production      |           | North Mara |                                             |
|-----------------|-----------|------------|---------------------------------------------|
| Production Gold | kg        | 1.16E+04   |                                             |
| Ore milled      | kg        | 2.87E+09   |                                             |
| Head grade      |           |            |                                             |
| Gold            | kg/kg ore | 4.38E-06   |                                             |
| Recovery rate   |           | 92%        | based on reports of Placer-Dome Inc. (2005) |

### Mining infrastructure

This data set includes land use as well as rough estimates on buildings and machines necessary in gold mining. It is assumed, that the quantities of machinery as well as surface and volume of buildings are equal to the generic mine. This approach is rough, and the uncertainty is rather high. As the mined ore from the generic mine is some orders of magnitude lower than the mine in focus, a proportional scaling by milled ore will lead to a higher mistake (see Tab. 5.1: Gold and Silver production in Sweden). The lifetime of mines is assumed to be 50 years.

Tab, 5.64: Land use of North Mara Mine in 2004 (Placer Dome Inc., 2005).

| Land use                           |                | North Mara | Remark                                                                       |
|------------------------------------|----------------|------------|------------------------------------------------------------------------------|
| Area disturbed during life of mine | m <sup>2</sup> | 3.72E+06   | Transformation, from unknown and Transformation, to mineral extraction site, |
| Disturbed area in active use       | m <sup>2</sup> | 2.92E+06   | Occupation, mineral extraction site, during 50 years                         |
| Area revegetated                   | m <sup>2</sup> | 3.00E+04   | added in "Transformation, to heterogeneous, agricultural"                    |

Land use is based on the figures from the environmental report (Tab. 5.56). Not all the area of the mine is in active use at one time; the value in Tab. 5.56 is assumed to be the average share of the mine area in use. Therefore, the total area is occupied during an average of about 39 years, according to the formula:

$$\frac{\text{Disturbed area in active use}}{\text{Area disturbed during life of mine}} * \text{lifetime of mine} = \frac{2.92 * 10^6}{3.72 * 10^6} * 50y = 39y$$

, the occupation of the mine is calculated as Disturbed area in active use \* 50y = Occupation, mineral extraction site

Tab. 5.65: Unit process raw data of the gold mine in Tanzania.

|                | Name                                             | Location | Infrastructure | Process | Unit    | mine, gold      | Uncertainty Type | Standard Deviation 95%                                | General Comment |
|----------------|--------------------------------------------------|----------|----------------|---------|---------|-----------------|------------------|-------------------------------------------------------|-----------------|
|                |                                                  |          |                |         |         |                 |                  |                                                       |                 |
|                | Location<br>InfrastructureProcess<br>Unit        |          |                |         |         | TZ<br>1<br>unit |                  |                                                       |                 |
| product        | mine, gold                                       | TZ       | 1              | unit    | 1.00E+0 |                 |                  |                                                       |                 |
| technosphere   | building, multi-storey                           | RER      | 1              | m3      | 2.75E+5 | 1               | 2.39             | (4,5,5,5,5,5); Rough estimates                        |                 |
|                | building, hall                                   | CH       | 1              | m2      | 1.25E+4 | 1               | 2.39             | (4,5,5,5,5,5); Rough estimates                        |                 |
|                | industrial machine, heavy, unspecified, at plant | RER      | 1              | kg      | 8.50E+5 | 1               | 2.39             | (4,5,5,5,5,5); Rough estimates                        |                 |
|                | conveyor belt, at plant                          | RER      | 1              | m       | 6.00E+2 | 1               | 2.39             | (4,5,5,5,5,5); Rough estimates                        |                 |
| resource, land | Transformation, from unknown                     | -        | -              | m2      | 3.72E+6 | 1               | 2.02             | (1,3,3,1,1,3); Environmental report, Placer Dome 2003 |                 |
|                | Transformation, to mineral extraction site       | -        | -              | m2      | 3.72E+6 | 1               | 2.02             | (1,3,3,1,1,3); Environmental report, Placer Dome 2004 |                 |
|                | Occupation, mineral extraction site              | -        | -              | m2a     | 1.46E+8 | 1               | 1.53             | (1,3,3,1,1,3); Environmental report, Placer Dome 2005 |                 |
|                | Transformation, from mineral extraction site     | -        | -              | m2      | 3.72E+6 | 1               | 2.02             | (1,3,3,1,1,3); Environmental report, Placer Dome 2006 |                 |
|                | Transformation, to heterogeneous, agricultural   | -        | -              | m2      | 3.00E+4 | 1               | 2.02             | (1,3,3,1,1,3); Environmental report, Placer Dome 2007 |                 |
|                | Transformation, to unknown                       | -        | -              | m2      | 3.69E+6 | 1               | 2.02             | (1,3,3,1,1,3); Environmental report, Placer Dome 2008 |                 |

### Mining and refining

Ore is fed via a single jaw crusher into a secondary crusher and then onto a crushed ore stockpile. The crushed ore is reclaimed from the stockpile and conveyed into an open circuit mill before passing to a closed circuit ball mill regrind. A gravity separation circuit is used to scalp free gold from regrind circuit. The ore is then processed via cyanidation and carbon in leach, followed by electrowinning and gold refining to doré on site.

Water for processing is pumped from the Mara River, which is located three kilometres from the processing plant site. The water is pumped into a raw water dam which can store up to three months water demand. A plant at the mine treats water for own use. Placer Dome Tanzania has a license to extract 4,200 cubic metres of water daily from the river. The following table (Tab. 5.66) sums up the total requirements and emissions of the North Mara Mine in 2005.

Material use (except for cyanide and explosives) is approximated with figures from Australian data for gold mining (Stewart, 1999), air emissions with Australian Data. Since the module “blasting” does not include the CO<sub>2</sub> emissions from explosives, they are calculated with the factor of the mines of Newmont (2005b) (0.16 kg CO<sub>2</sub> per kg explosive). Water emissions are assumed to be emitted to rivers and are estimated on the base of data for South Africa. Estimation of transport is based on standard distances for Europe. Data about amounts of tailings disposal are approximated with data of mines in Papua New Guinea.

Tab. 5.66: Requirements and emissions of the North Mara Mine in 2005 (Placer Dome Inc., 2005).

| Energy                           |                | North Mara | Remark                                                                                                             |
|----------------------------------|----------------|------------|--------------------------------------------------------------------------------------------------------------------|
| Diesel (generators)              | MJ             | 1.00E+09   | Diesel, burned in diesel-electric generating set                                                                   |
| Diesel (mobile equipment)        | MJ             | 5.31E+08   | Diesel, burned in building machine                                                                                 |
| Petrol                           | l              | 2.90E+05   | calculate with the specific fuel consumption of delivery van, converted in transport, van <3.5t (0.38l/tkm)        |
| Electricity for refining         | kWh            | 1.89E+04   | assumed with Wohlwill-electrolysis, electricity demand for refining is approximated with CENTREL-mix <sup>1)</sup> |
| Materials                        |                |            |                                                                                                                    |
| Cyanide                          | kg             | 1.15E+06   | In LCI as sodium cyanide                                                                                           |
| Lime                             | kg             | 7.31E+05   | Scaled from Steward (1999) by processed gold                                                                       |
| Caustic                          | kg             | 1.35E+05   | Scaled from Steward (1999) by processed gold, accounted for sodium hydroxide                                       |
| Carbon                           | kg             | 1.96E+05   | Scaled from Steward (1999) by processed gold, accounted for as charcoal                                            |
| Zinc                             | kg             | 3.35E+04   | Scaled from Steward (1999) by processed gold, accounted for as zinc, primary, at regional storage                  |
| Sulphuric Acid                   | kg             | 7.70E+03   | Scaled from Steward (1999) by processed gold                                                                       |
| Hydrochloric Acid                | kg             | 1.58E+05   | Scaled from Steward (1999) by processed gold                                                                       |
| Explosives                       | kg             | 1.55E+06   | Partly ANFO explosives, in LCI: "blasting"                                                                         |
| Water resources                  |                |            |                                                                                                                    |
| Water from river                 | m <sup>3</sup> | 9.25E+05   | in LCI: Water, unspecified natural origin                                                                          |
| Emissions to Water <sup>2)</sup> |                |            |                                                                                                                    |
| Arsenic                          | kg             | 4.43E+01   | estimated with ZA-data, scaled by milled ore                                                                       |
| Cadmium                          | kg             | 4.97E+00   | estimated with ZA-data, scaled by milled ore                                                                       |
| Copper                           | kg             | 2.46E+02   | estimated with ZA-data, scaled by milled ore                                                                       |
| Cyanide                          | kg             | 2.46E+03   | estimated with ZA-data, scaled by milled ore                                                                       |
| Lead                             | kg             | 2.96E+01   | estimated with ZA-data, scaled by milled ore                                                                       |
| Mercury                          | kg             | 4.97E+00   | estimated with ZA-data, scaled by milled ore                                                                       |
| Nickel                           | kg             | 5.62E+02   | estimated with ZA-data, scaled by milled ore                                                                       |
| Zinc                             | kg             | 3.70E+02   | estimated with ZA-data, scaled by milled ore                                                                       |
| Emissions to Air <sup>3)</sup>   |                |            |                                                                                                                    |
| Carbon Dioxide (from explosives) | kg             | 2.48E+05   | Calculation, based on data from Newmont (2005)                                                                     |
| Antimony                         | kg             | 3.11E+00   | estimated with AU-data, scaled by milled ore                                                                       |
| Arsenic                          | kg             | 1.43E+02   | estimated with AU-data, scaled by milled ore                                                                       |
| Beryllium                        | kg             | 8.36E+00   | estimated with AU-data, scaled by milled ore                                                                       |
| Cadmium                          | kg             | 4.43E-01   | estimated with AU-data, scaled by milled ore                                                                       |
| Chromium                         | kg             | 7.24E+01   | estimated with AU-data, scaled by milled ore                                                                       |
| Cobalt                           | kg             | 1.25E+01   | estimated with AU-data, scaled by milled ore                                                                       |
| Lead                             | kg             | 3.52E+00   | estimated with AU-data, scaled by milled ore                                                                       |
| Manganese                        | kg             | 3.92E+02   | estimated with AU-data, scaled by milled ore                                                                       |
| Mercury                          | kg             | 2.02E+02   | estimated with AU-data, scaled by milled ore                                                                       |
| Selenium                         | kg             | 2.30E+02   | estimated with AU-data, scaled by milled ore                                                                       |
| Disposal                         | kg             | 2.04E+09   | estimated with data of PG, in LCI as: Disposal, sulphidic tailings                                                 |

- 1 Neither a Tanzanian electricity mix nor the exact place of refinement are available, therefore we assumed the use of CENTREL-electricity mix for its high share of fossil electricity generation.
- 2 Estimation, based on emissions in South Africa, scaled by processed gold (Placer-Dome, 2002).
- 3 Estimation, based on emissions in Australia, scaled by processed gold.

Since the mines produce doré gold (unrefined gold bullion bars containing between 60% and 95% gold), the refinery process is less complex than the refinery process of Rönnskär in Sweden. It is assumed, that the gold doré-bars undergo the treatment of Wohlwill electrolysis (for details see 5.2.2.1, "Gold / Silver, secondary, from electronic scrap inecoinvent"). Due to the lack of specific data, the figure of refinement from electronic scrap is used: 1.63 kWh per kg gold (Renner et al., 2002a). No emissions were assumed, since on the one hand the high value of the processed material requires an efficient emission control and on the other hand, refinement is a hydrometallurgical process with low impact. The infrastructure is approximated with the facility of precious metal refinery. The unit process raw data are summarized in Tab. 5.67.



## 5. System Characterisation

Tab. 5.67: Unit process raw data of gold mining and refining in Tanzania

|                                      | Name                                                     | Location | Infrastructure | Process | Unit     | gold, at refinery | Uncertainty | Type               | StandardDeviation                                                                   | 95 | GeneralComment |
|--------------------------------------|----------------------------------------------------------|----------|----------------|---------|----------|-------------------|-------------|--------------------|-------------------------------------------------------------------------------------|----|----------------|
|                                      | Location<br>InfrastructureProcess<br>Unit                |          |                |         |          | TZ<br>0<br>kg     |             |                    |                                                                                     |    |                |
| product                              | gold, at refinery                                        | TZ       | 0              | kg      | 1.00E+00 |                   |             |                    |                                                                                     |    |                |
| technosphere                         | diesel, burned in diesel-electric generating set         | GLO      | 0              | MJ      | 8.64E+04 | 1                 | 1.16        | (3,4,1,2,1,3);     | Environmental report                                                                |    |                |
|                                      | diesel, burned in building machine                       | GLO      | 0              | MJ      | 4.59E+04 | 1                 | 1.16        | (3,4,1,2,1,3);     | Environmental report                                                                |    |                |
|                                      | transport, van <3.5t                                     | RER      | 0              | tkm     | 6.60E+01 | 1                 | 2.03        | (3,4,1,2,1,3);     | Environmental report                                                                |    |                |
|                                      | electricity, medium voltage, production CENTREL, at grid | CENTREL  | 0              | kWh     | 1.63E+00 | 1                 | 1.16        | (3,4,1,2,1,3);     | Environmental report                                                                |    |                |
|                                      | sodium cyanide, at plant                                 | RER      | 0              | kg      | 9.97E+01 | 1                 | 1.16        | (3,4,1,2,1,3);     | Environmental report                                                                |    |                |
|                                      | limestone, milled, packed, at plant                      | CH       | 0              | kg      | 6.32E+01 | 1                 | 1.20        | (3,3,3,3,1,3);     | Calculation, based on thesis on Australian precious metal production, data for 1999 |    |                |
|                                      | sodium hydroxide, 50% in H2O, production mix, at plant   | RER      | 0              | kg      | 1.16E+01 | 1                 | 1.17        | (3,3,3,2,1,3);     | Calculation, based on thesis on Australian precious metal production, data for 1999 |    |                |
|                                      | charcoal, at plant                                       | GLO      | 0              | kg      | 1.70E+01 | 1                 | 1.17        | (3,3,3,3,1,3);     | Calculation, based on thesis on Australian precious metal production, data for 1999 |    |                |
|                                      | zinc, primary, at regional storage                       | RER      | 0              | kg      | 2.89E+00 | 1                 | 1.17        | (3,3,3,3,1,3);     | Calculation, based on thesis on Australian precious metal production, data for 1999 |    |                |
|                                      | sulphuric acid, liquid, at plant                         | RER      | 0              | kg      | 6.65E-01 | 1                 | 1.17        | (3,3,3,3,1,3);     | Calculation, based on thesis on Australian precious metal production, data for 1999 |    |                |
|                                      | hydrochloric acid, 30% in H2O, at plant                  | RER      | 0              | kg      | 1.36E+01 | 1                 | 1.17        | (3,3,3,3,1,3);     | Calculation, based on thesis on Australian precious metal production, data for 1999 |    |                |
|                                      | blasting                                                 | RER      | 0              | kg      | 1.34E+02 | 1                 | 1.16        | (3,4,1,2,1,3);     | Environmental report                                                                |    |                |
|                                      | transport, lorry >16t, fleet average                     | RER      | 0              | tkm     | 1.33E+04 | 1                 | 2.09        | (4,5,na,na,na,na); | Estimation, based on standard distances                                             |    |                |
|                                      | disposal, sulfidic tailings, off-site                    | GLO      | 0              | kg      | 1.76E+05 | 1                 | 1.16        | (3,4,1,2,1,3);     | Estimation                                                                          |    |                |
|                                      | facilities precious metal refinery                       | SE       | 1              | unit    | 1.65E-07 | 1                 | 3.03        | (3,4,1,2,1,3);     | Environmental report                                                                |    |                |
| infrastructure                       | mine, gold                                               | TZ       | 1              | unit    | 1.73E-06 | 1                 | 3.03        | (3,4,1,2,1,3);     | Environmental report                                                                |    |                |
| emission air, low population density | Antimony                                                 | -        | -              | kg      | 2.68E-04 | 1                 | 5.05        | (3,3,3,5,1,3);     | Estimated with data from AU                                                         |    |                |
|                                      | Arsenic                                                  | -        | -              | kg      | 1.23E-02 | 1                 | 5.05        | (3,3,3,5,1,3);     | Estimated with data from AU                                                         |    |                |
|                                      | Beryllium                                                | -        | -              | kg      | 7.22E-04 | 1                 | 5.05        | (3,3,3,5,1,3);     | Estimated with data from AU                                                         |    |                |
|                                      | Cadmium                                                  | -        | -              | kg      | 3.83E-05 | 1                 | 5.05        | (3,3,3,5,1,3);     | Estimated with data from AU                                                         |    |                |
|                                      | Carbon dioxide, fossil                                   | -        | -              | kg      | 2.14E+01 | 1                 | 1.20        | (3,3,3,5,1,3);     | Estimated with data from AU                                                         |    |                |
|                                      | Chromium                                                 | -        | -              | kg      | 1.08E-03 | 1                 | 5.05        | (3,3,3,5,1,3);     | Estimated with data from AU                                                         |    |                |
|                                      | Cobalt                                                   | -        | -              | kg      | 3.04E-04 | 1                 | 5.05        | (3,3,3,5,1,3);     | Estimated with data from AU                                                         |    |                |
|                                      | Lead                                                     | -        | -              | kg      | 3.39E-02 | 1                 | 5.05        | (3,3,3,5,1,3);     | Estimated with data from AU                                                         |    |                |
|                                      | Mercury                                                  | -        | -              | kg      | 1.99E-02 | 1                 | 5.05        | (3,3,3,5,1,3);     | Estimated with data from AU                                                         |    |                |
|                                      | Manganese                                                | -        | -              | kg      | 1.74E-02 | 1                 | 5.05        | (3,3,3,5,1,3);     | Estimated with data from AU                                                         |    |                |
|                                      | Selenium                                                 | -        | -              | kg      | 0.00E+00 | 1                 | 5.05        | (3,3,3,5,1,3);     | Estimated with data from AU                                                         |    |                |
|                                      | Arsenic, ion                                             | -        | -              | kg      | 3.83E-03 | 1                 | 5.10        | (3,3,3,5,3,3);     | Estimated with data from ZA                                                         |    |                |
| emission water, river                | Cadmium, ion                                             | -        | -              | kg      | 4.30E-04 | 1                 | 3.08        | (3,3,2,5,3,3);     | Estimated with data from ZA                                                         |    |                |
|                                      | Copper, ion                                              | -        | -              | kg      | 2.13E-02 | 1                 | 3.08        | (3,3,2,5,3,3);     | Estimated with data from ZA                                                         |    |                |
|                                      | Cyanide                                                  | -        | -              | kg      | 2.13E-01 | 1                 | 3.08        | (3,3,2,5,3,3);     | Estimated with data from ZA                                                         |    |                |
|                                      | Lead                                                     | -        | -              | kg      | 2.56E-03 | 1                 | 5.09        | (3,3,2,5,3,3);     | Estimated with data from ZA                                                         |    |                |
|                                      | Mercury                                                  | -        | -              | kg      | 4.30E-04 | 1                 | 5.09        | (3,3,2,5,3,3);     | Estimated with data from ZA                                                         |    |                |
|                                      | Nickel, ion                                              | -        | -              | kg      | 4.85E-02 | 1                 | 5.09        | (3,3,2,5,3,3);     | Estimated with data from ZA                                                         |    |                |
|                                      | Zinc, ion                                                | -        | -              | kg      | 3.19E-02 | 1                 | 5.09        | (3,3,2,5,3,3);     | Estimated with data from ZA                                                         |    |                |
|                                      | Water, river                                             | -        | -              | m3      | 7.99E+01 | 1                 | 1.14        | (3,3,1,1,1,3);     | Environmental report                                                                |    |                |
| resource, in water                   | Gold, Au 4.3E-4%, in ore, in ground                      | -        | -              | kg      | 1.09E+00 | 1                 | 1.14        | (3,3,1,1,1,3);     | Environmental report                                                                |    |                |
| resource, in ground                  |                                                          |          |                |         |          |                   |             |                    |                                                                                     |    |                |

Tab. 5.68 Ecospol Meta information of gold production in Tanzania

|                            |                                                                                                                                                                                                                                            |                                                                                                                                                                                           |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name                       | mine, gold                                                                                                                                                                                                                                 | gold, at refinery                                                                                                                                                                         |
| Location                   | TZ                                                                                                                                                                                                                                         | TZ                                                                                                                                                                                        |
| InfrastructureProcess      | 1                                                                                                                                                                                                                                          | 0                                                                                                                                                                                         |
| Unit                       | unit                                                                                                                                                                                                                                       | kg                                                                                                                                                                                        |
| Type                       | 1                                                                                                                                                                                                                                          | 1                                                                                                                                                                                         |
| Version                    | 2.0                                                                                                                                                                                                                                        | 2.0                                                                                                                                                                                       |
| energyValues               | 0                                                                                                                                                                                                                                          | 0                                                                                                                                                                                         |
| LanguageCode               | en                                                                                                                                                                                                                                         | en                                                                                                                                                                                        |
| LocalLanguageCode          | de                                                                                                                                                                                                                                         | de                                                                                                                                                                                        |
| Person                     | 44                                                                                                                                                                                                                                         | 44                                                                                                                                                                                        |
| QualityNetwork             | 1                                                                                                                                                                                                                                          | 1                                                                                                                                                                                         |
| DataSetRelatesToProduct    | 1                                                                                                                                                                                                                                          | 1                                                                                                                                                                                         |
| IncludedProcesses          | Land use, buildings and machinery of a gold and silver mine                                                                                                                                                                                | This data set describes the mining of gold concentrate in Tanzania. It includes energy and water consumption, land use, transports, waste disposal as well as emissions to air and water. |
| Amount                     | 1                                                                                                                                                                                                                                          | 1                                                                                                                                                                                         |
| LocalName                  | Mine, Gold                                                                                                                                                                                                                                 | Gold, ab Raffinerie                                                                                                                                                                       |
| Synonyms                   |                                                                                                                                                                                                                                            |                                                                                                                                                                                           |
| GeneralComment             | This data set includes land use of the considered mines (open pit and underground) incl. revegetated areas as well as estimates of buildings and machinery of the mine based on the data set non-ferrous metal, mine, surface/underground. | This data set describes the beneficiation, concentration and refinement of gold.                                                                                                          |
| InfrastructureIncluded     | 1                                                                                                                                                                                                                                          | 1                                                                                                                                                                                         |
| Category                   | metals                                                                                                                                                                                                                                     | metals                                                                                                                                                                                    |
| SubCategory                | extraction                                                                                                                                                                                                                                 | extraction                                                                                                                                                                                |
| LocalCategory              | Metalle                                                                                                                                                                                                                                    | Metalle                                                                                                                                                                                   |
| LocalSubCategory           | Gewinnung                                                                                                                                                                                                                                  | Gewinnung                                                                                                                                                                                 |
| Formula                    |                                                                                                                                                                                                                                            |                                                                                                                                                                                           |
| StatisticalClassification  |                                                                                                                                                                                                                                            |                                                                                                                                                                                           |
| CASNumber                  |                                                                                                                                                                                                                                            | 007440-57-5                                                                                                                                                                               |
| StartDate                  | 2003                                                                                                                                                                                                                                       | 2003                                                                                                                                                                                      |
| EndDate                    | 2006                                                                                                                                                                                                                                       | 2006                                                                                                                                                                                      |
| DataValidForEntirePeriod   | 1                                                                                                                                                                                                                                          | 1                                                                                                                                                                                         |
| OtherPeriodText            | Period of public environmental report                                                                                                                                                                                                      | Period of public environmental report                                                                                                                                                     |
| Text                       | Land use of Tanzanian mine, other inputs based on rough assumptions.                                                                                                                                                                       | Material requirements for gold concentration are estimated with Australian data. Emissions to water have been estimated with South-African, emissions to air with Australian data.        |
| Text                       | Assumptions for buildings and machinery based on the data set "non-ferrous metal, mine, surface/underground with 50 years life time.                                                                                                       | The mining of ore from open pit and underground mines is considered.                                                                                                                      |
| Percent                    |                                                                                                                                                                                                                                            |                                                                                                                                                                                           |
| ProductionVolume           | 2.9 Mill. t ore                                                                                                                                                                                                                            | 11.57 t gold                                                                                                                                                                              |
| SamplingProcedure          | environmental report                                                                                                                                                                                                                       | environmental report                                                                                                                                                                      |
| Extrapolations             | Buildings and machinery are based on global assumptions. The figures are extrapolated on the basis of the ore production.                                                                                                                  | See Geography, electricity mix has been estimated with foreign mix                                                                                                                        |
| UncertaintyAdjustments     | none                                                                                                                                                                                                                                       | none                                                                                                                                                                                      |
| Person                     | 44                                                                                                                                                                                                                                         | 44                                                                                                                                                                                        |
| DataPublishedIn            | 2                                                                                                                                                                                                                                          | 2                                                                                                                                                                                         |
| ReferenceToPublishedSource | 10                                                                                                                                                                                                                                         | 10                                                                                                                                                                                        |
| Copyright                  | 1                                                                                                                                                                                                                                          | 1                                                                                                                                                                                         |
| AccessRestrictedTo         | 0                                                                                                                                                                                                                                          | 0                                                                                                                                                                                         |
| CompanyCode                |                                                                                                                                                                                                                                            |                                                                                                                                                                                           |
| CountryCode                |                                                                                                                                                                                                                                            |                                                                                                                                                                                           |
| PageNumbers                |                                                                                                                                                                                                                                            |                                                                                                                                                                                           |
| Validator                  | 51                                                                                                                                                                                                                                         | 51                                                                                                                                                                                        |
| Details                    | automatic validation in Excel                                                                                                                                                                                                              | automatic validation in Excel                                                                                                                                                             |
| OtherDetails               | none                                                                                                                                                                                                                                       | none                                                                                                                                                                                      |

### 5.1.11 Data quality for primary gold / silver production

The datasets regarding the primary gold and silver production based mostly on environmental reports of mine corporations. Concerning energy, cyanide and explosive consumption, the data quality is good. Further material requirements have been estimated and scaled of Australian data by the production of gold.

Only a few companies show their emissions into air and water, thus in several of the country-specific datasets, emissions have been estimated by scaled ore. Information about tailings referred to Papua New-Guinea, where the only available data stems from. Therefore the data quality for emissions into air / water, as well as disposal of tailings is rather low.

A comparison of the datasets for gold and silver production on a country level can only be done with respect to the above mentioned limitations: A better performance of a country can be explained with better technology or with other assumptions on the emission level.

### 5.1.12 Primary gold, production mix (GLO)

These data sets describe an approximated average origin of gold. Shares of the different countries are specified in Tab. 2.2. No transports are considered, since these are included in the dataset “gold, at regional storage”.

Tab. 5.69: Unit process raw data for „gold, primary, at refinery“, Source: Tab. 2.2

|              | Name                                                    | Location | Infrastructure | Process | Unit | gold, primary, at refinery | Uncertainty Type | StandardDeviation95% | GeneralComment                                                                           |
|--------------|---------------------------------------------------------|----------|----------------|---------|------|----------------------------|------------------|----------------------|------------------------------------------------------------------------------------------|
|              | Location<br>InfrastructureProcess<br>Unit               |          |                |         |      | GLO<br>0<br>kg             |                  |                      |                                                                                          |
| product      | gold, primary, at refinery                              | GLO      | 0              | kg      |      | 1.00E+0                    |                  |                      |                                                                                          |
| technosphere | gold, at refinery                                       | ZA       | 0              | kg      |      | 1.86E-1                    | 1                | 1.08                 | (1,1,2,1,1,3); average gold production 2000-2004 & approximation for remaining countries |
|              | gold, at refinery                                       | US       | 0              | kg      |      | 1.20E-1                    | 1                | 1.08                 | (1,1,2,1,1,3); average gold production 2000-2004 & approximation for remaining countries |
|              | gold, at refinery                                       | AU       | 0              | kg      |      | 2.39E-1                    | 1                | 2.00                 | (1,1,2,1,1,3); average gold production 2000-2004 & approximation for remaining countries |
|              | gold, at refinery                                       | CA       | 0              | kg      |      | 5.81E-2                    | 1                | 1.08                 | (1,1,2,1,1,3); average gold production 2000-2004 & approximation for remaining countries |
|              | gold, from combined gold-silver production, at refinery | PG       | 0              | kg      |      | 1.88E-1                    | 1                | 1.51                 | (1,1,2,1,1,3); average gold production 2000-2004 & approximation for remaining countries |
|              | gold, from combined gold-silver production, at refinery | CL       | 0              | kg      |      | 5.28E-2                    | 1                | 2.00                 | (1,1,2,1,1,3); average gold production 2000-2004 & approximation for remaining countries |
|              | gold, from combined gold-silver production, at refinery | PE       | 0              | kg      |      | 9.69E-2                    | 1                | 2.00                 | (1,1,2,1,1,3); average gold production 2000-2004 & approximation for remaining countries |
|              | gold, at refinery                                       | TZ       | 0              | kg      |      | 4.75E-2                    | 1                | 1.08                 | (1,1,2,1,1,3); average gold production 2000-2004 & approximation for remaining countries |
|              | gold, from combined metal production, at refinery       | SE       | 0              | kg      |      | 1.17E-2                    | 1                | 2.00                 | (1,1,2,1,1,3); average gold production 2000-2004 & approximation for remaining countries |

### 5.1.13 Production of primary silver from auriferous ores, production mix

Shares of the different countries are specified for silver in Tab. 2.4. No transports are considered, since this takes place in the dataset “silver, at regional storage”.

Tab. 5.70: Unit process raw data for „silver, from combined gold-silver production, at refinery“, Source: Tab. 2.4.

|              | Name                                                              | Location | Infrastructure | Process | Unit | silver, from combined gold-silver production, at regional storage | Uncertainty Type | StandardDeviation95% | GeneralComment                                                                             |
|--------------|-------------------------------------------------------------------|----------|----------------|---------|------|-------------------------------------------------------------------|------------------|----------------------|--------------------------------------------------------------------------------------------|
|              | Location<br>InfrastructureProcess<br>Unit                         |          |                |         |      | RER<br>0<br>kg                                                    |                  |                      |                                                                                            |
| product      | silver, from combined gold-silver production, at regional storage | RER      | 0              | kg      |      | 1.00E+0                                                           |                  |                      |                                                                                            |
| technosphere | silver, from combined gold-silver production, at refinery         | PE       | 0              | kg      |      | 3.13E-1                                                           | 1                | 1.08                 | (1,1,2,1,1,3); average silver production 2000-2004 & approximation for remaining countries |
|              | silver, from combined gold-silver production, at refinery         | CL       | 0              | kg      |      | 3.20E-1                                                           | 1                | 1.08                 | (1,1,2,1,1,3); average silver production 2000-2004 & approximation for remaining countries |
|              | silver, from combined gold-silver production, at refinery         | PG       | 0              | kg      |      | 1.48E-1                                                           | 1                | 2.00                 | (1,1,2,1,1,3); average silver production 2000-2004 & approximation for remaining countries |
|              | silver, from combined gold-silver production, at refinery         | SE       | 0              | kg      |      | 2.19E-1                                                           | 1                | 1.08                 | (1,1,2,1,1,3); average silver production 2000-2004 & approximation for remaining countries |

Tab. 5.71 Ecospol Meta information of gold &amp; and silver mixes in ecoinvent v2.0

| Name                       | gold, primary, at refinery                                                                                                                       | silver, from combined gold-silver production, at regional storage                                                                                |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Location                   | GLO                                                                                                                                              | RER                                                                                                                                              |
| InfrastructureProcess      | 0                                                                                                                                                | 0                                                                                                                                                |
| Unit                       | kg                                                                                                                                               | kg                                                                                                                                               |
| Type                       | 1                                                                                                                                                | 1                                                                                                                                                |
| Version                    | 2.0                                                                                                                                              | 2.0                                                                                                                                              |
| energyValues               | 0                                                                                                                                                | 0                                                                                                                                                |
| LanguageCode               | en                                                                                                                                               | en                                                                                                                                               |
| LocalLanguageCode          | de                                                                                                                                               | de                                                                                                                                               |
| Person                     | 44                                                                                                                                               | 44                                                                                                                                               |
| QualityNetwork             | 1                                                                                                                                                | 1                                                                                                                                                |
| DataSetRelatesToProduct    | 1                                                                                                                                                | 1                                                                                                                                                |
| IncludedProcesses          | none                                                                                                                                             | none                                                                                                                                             |
| Amount                     | 1                                                                                                                                                | 1                                                                                                                                                |
| LocalName                  | Gold, aus Primärproduktion, ab Raffinerie                                                                                                        | Silber, aus kombinierter Gold-Silber-Produktion, ab Regionallager                                                                                |
| Synonyms                   |                                                                                                                                                  |                                                                                                                                                  |
| GeneralComment             | This data set describes the composition of primary gold production from at a global level.                                                       | This data set describes the composition of silver production from auriferous ores at a global level.                                             |
| InfrastructureIncluded     | 1                                                                                                                                                | 1                                                                                                                                                |
| Category                   | metals                                                                                                                                           | metals                                                                                                                                           |
| SubCategory                | extraction                                                                                                                                       | extraction                                                                                                                                       |
| LocalCategory              | Metalle                                                                                                                                          | Metalle                                                                                                                                          |
| LocalSubCategory           | Gewinnung                                                                                                                                        | Gewinnung                                                                                                                                        |
| Formula                    |                                                                                                                                                  |                                                                                                                                                  |
| StatisticalClassification  |                                                                                                                                                  |                                                                                                                                                  |
| CASNumber                  | 007440-57-5                                                                                                                                      | 007440-22-4                                                                                                                                      |
| StartDate                  | 2000                                                                                                                                             | 2000                                                                                                                                             |
| EndDate                    | 2004                                                                                                                                             | 2004                                                                                                                                             |
| DataValidForEntirePeriod   | 1                                                                                                                                                | 1                                                                                                                                                |
| OtherPeriodText            | Period of public statistics                                                                                                                      | Period of public statistics                                                                                                                      |
| Text                       | global data                                                                                                                                      | global data                                                                                                                                      |
| Text                       | Assumption on classification of the production countries has been made, based on global production statistics.                                   | Assumption on classification of the production countries has been made, based on global production statistics.                                   |
| Percent                    |                                                                                                                                                  |                                                                                                                                                  |
| ProductionVolume           | 2533 t gold                                                                                                                                      | 2362 t silver from auriferous ores                                                                                                               |
| SamplingProcedure          | Production statistics, based on average production rate 2000-2004                                                                                | Production statistics, based on average production rate 2000-2004                                                                                |
| Extrapolations             | The energy use, materials, emissions and land use of the mining and beneficiation are considered in the corresponding datasets at country level. | The energy use, materials, emissions and land use of the mining and beneficiation are considered in the corresponding datasets at country level. |
| UncertaintyAdjustments     | none                                                                                                                                             | none                                                                                                                                             |
| Person                     | 44                                                                                                                                               | 44                                                                                                                                               |
| DataPublishedIn            | 2                                                                                                                                                | 2                                                                                                                                                |
| ReferenceToPublishedSource | 10                                                                                                                                               | 10                                                                                                                                               |
| Copyright                  | 1                                                                                                                                                | 1                                                                                                                                                |
| AccessRestrictedTo         | 0                                                                                                                                                | 0                                                                                                                                                |
| CompanyCode                |                                                                                                                                                  |                                                                                                                                                  |
| CountryCode                |                                                                                                                                                  |                                                                                                                                                  |
| PageNumbers                |                                                                                                                                                  |                                                                                                                                                  |
| Validator                  | 51                                                                                                                                               | 51                                                                                                                                               |
| Details                    | automatic validation in Excel                                                                                                                    | automatic validation in Excel                                                                                                                    |
| OtherDetails               | none                                                                                                                                             | none                                                                                                                                             |



site-chromium), horizontally aligned converter unit fed with hydrocarbon fuel, industrial oxygen and calcium oxide (Fig. 5.4, Davenport et al., 2002; Outokumpu, 2006).

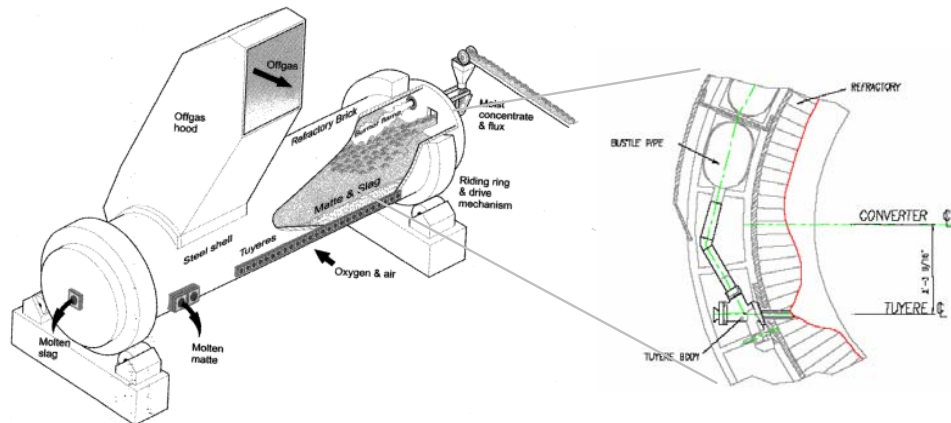


Fig. 5.4 Detail of a converter unit (adopted from Davenport et al., 2002).

**Refining.** The processing of the electronic scrap in the Aisle converter is followed by a further refinement in a) the anode casting plant and b) the electrolytic refinery. At BOLIDEN's Rönnskär facility a new Twin M-16 anode casting facility (Fig. 5.5) is deployed to cast the hot blister copper into anodes. The facility consists of an inlet through which the hot blister copper is transported to the loader. Increasing the processing efficiency, the loader alternately feeds the blister copper into two wheel caster (Outokumpu, 2004).

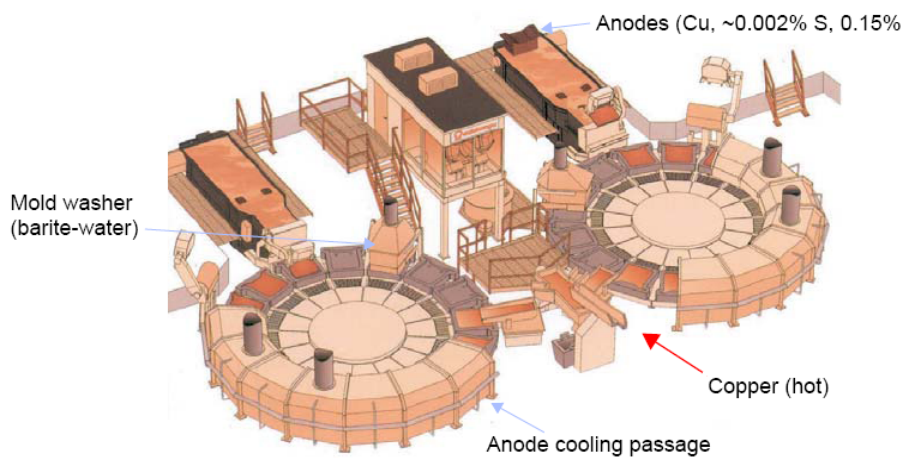


Fig. 5.5 Sketch of a twin-wheel anode caster (adopted from Outokumpu, 2004) and the key infrastructure materials.

Once cast, the anodes are then finished and stored in the anode container. The anode sheets typically have a size of 1.00 m \* 1.40 m (approximated based upon Outokumpu, 2004) and consist of 99.5% copper including, however, impurities of sulphur, oxygen and traces of precious metals (Davenport et al., 2002). In the next step the copper anode is processed in the electrolytic refinery plant. This facility is a large hall with a large number of cells.



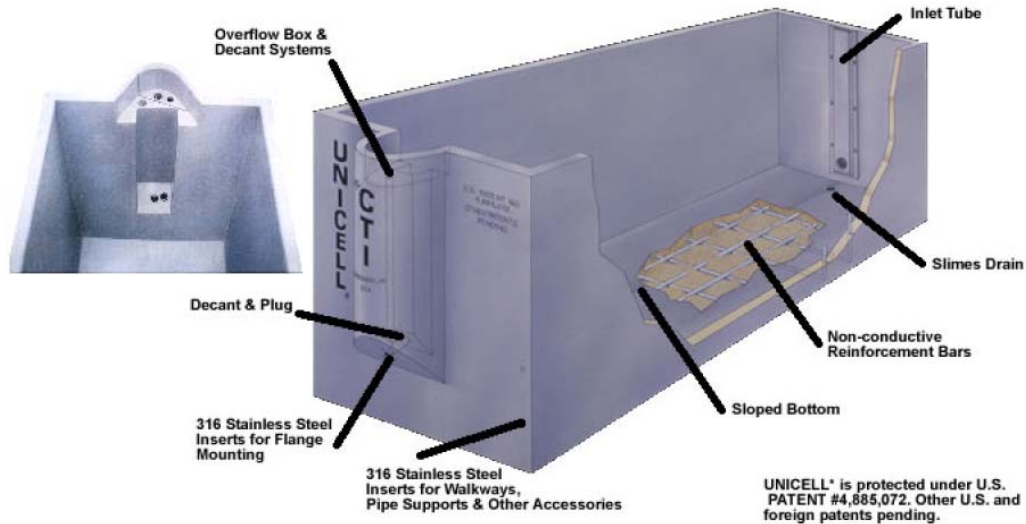


Fig. 5.6 Sketch of an electrolytic cell (adopted from Corrosion Technology International, 2006) and the key infrastructure materials

In 1999, the Rönnskär plant was equipped with 644 electrorefining cells and with 2 circulation tanks (Boliden, 2006c; Corrosion Technology International, 2006). The cells are made of stainless steel and reinforcement of glass fibre. The volume of the cells can vary considerably depending upon the process it is intended to be applied. The cells are continuously circulated by water spiked with copper, sulphuric acid and chloride (Davenport et al., 2002). Typically, the circulation time is about 0.02 m<sup>3</sup>/min and the power consumption ranges from less than 300 kWh/t material to nearly 400 kWh/t material (Davenport et al., 2002). The final products of the electrolytic step are copper and nickel. As a residual, precious metal containing anode slime is generated, which is finally treated in the precious metals plant in order to recover secondary gold, silver, palladium, and platinum.

**Anode slime treatment.** In a first step the precious metals containing anode slime is decopperised. The remaining residues are dried and processed by pyrometallurgical means. There also exists a hydrometallurgical route (Pesl & Anzinger, 2002). Whereas the pyrometallurgical route is apt to process large quantities of well defined anode slime from primary production, the hydrometallurgical route is more flexible to changes in feed composition as it occurs in the processing of slimes from secondary copper smelters. However, in Boliden the pyrometallurgical route is applied (Ludvigsson & Larsson, 2003). After de-copperising and precipitation of copper telluride the residue is dried and pyrolytically refined to silver rich dorée metal, which is cast into anodes for a subsequent electrorefining of silver (Moebius electrolysis) and Gold (Wohlwill electrolysis). The remaining slime consists mainly of PGM (platinum group metal) that is recovered.

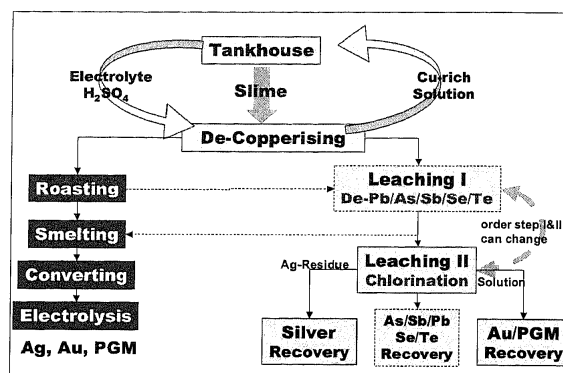


Fig. 5.7 Diagram of pyro- and hydrometallurgical process routes for anode slime treatment (Pesi &amp; Anzinger, 2002).

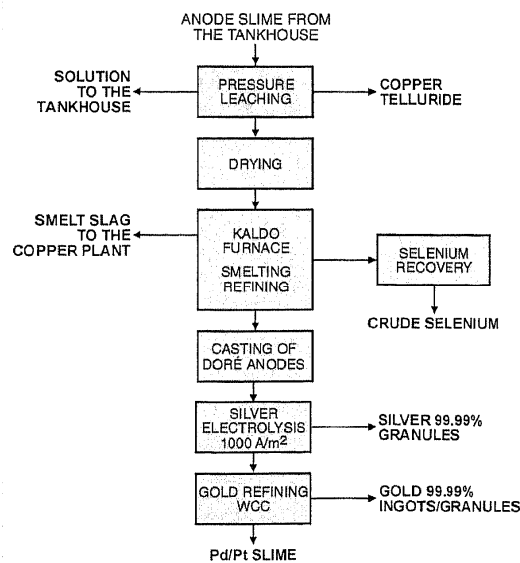


Fig. 5.8 Sketch of the anode slimes treatment and precious metal recovery approach applied at Rönnskär (Ludvigsson &amp; Larsson, 2003).

### 5.2.1 Waste

The recovery of precious metals, such as gold, is associated with the generation of a number of by-products off-gases, sludge, slag, waste heat and dust (Umweltbundesamt, 2001). Most of the by-products are captured by appropriate filters (Umweltbundesamt, 2001). From an environmental perspective, most important, is the treatment of the residuals of the cyanide-containing sludge generated while concentrating gold from doré-grade gold. This sludge is treated applying sulphur-dioxide and air before it is dumped into tailing ponds (Boliden, 2002b). Concerning the other by-products and their treatment, no utilisable information could be compiled.

### 5.2.2 Emissions

Despite of the application of modern filter technologies, the precious metals recovery process is also associated with the generation of emissions to the environment (HELCOM, 2002). In particular the following emissions to air are considered problematic: dust, copper, lead, zinc, cadmium, arsenic, mercury, sulphur dioxide, nitrogen oxide, fluoride, chloride and PCDD/F. Emissions to water are include copper, lead, zinc, cadmium, arsenic, mercury (Mark & Lehner, 2002 and Umweltbundesamt, 2001)

#### 5.2.2.1 Gold / Silver, secondary, from electronic scrap in ecoinvent

No explicit LCA study covering the extraction of gold as secondary material from electronic scrap has been performed so far. As mentioned in the beginning of this chapter the recovery process of secondary gold is represented by the recovery of electronic scrap by integrated smelters as conducted by Boliden in the Rönneskär site. This inventory is solely based upon available industry information and literature data. In cases where no explicit gold recovery information was available, information on the extraction of copper was used as proxy.

The precious metals in the scrap traverse a lead smelting step ("Kaldo plant") and a subsequent conversion of the precious metals containing copper matte. The resulting blister copper is cast into anodes that are further refined, whereas the precious metals are transferred into the anode slime. The precious



metals are extracted from the anode slime in the final step. The production of secondary gold and silver therefore consists of a series of three consecutive multi-output datasets:

#### Secondary Copper Conversion

- lead, secondary, from electronic and electric scrap recycling, at plant
- metal values from electric waste, in blister-copper, at converter

#### Secondary Copper Refining

- PM from electric waste, in anode slime, at refinery
- copper, secondary, from electronic and electric scrap recycling, at refinery
- nickel, secondary, from electronic and electric scrap recycling, at refinery

#### Secondary Copper Precious Metal Refining

- silver, secondary, at precious metal refinery
- gold, secondary, at precious metal refinery
- palladium, secondary, at precious metal refinery

The meta information for these processes is compiled in Tab. 5.74, the reference flows for the multi-input processes are displayed in Fig. 5.10, Fig. 5.11 and Fig. 5.12.

Based on process descriptions and corporate reports a mass flow balance has herewith been established that serves as skeleton for the three models of the three multi-output processes (Fig. 5.9). The mixed feed of the copper refinery considered does not exclusively consist of electronic scrap. The intermediate reference flows “metal values from electric waste, in blister-copper, at converter” and “PM from electric waste, in anode slime, at refinery” therefore relate only to that part of the metals contained in the blister copper and the anode slime that is attributable to the electronic scrap in the input feed (Tab. 5.72). The reference flows do not reflect the intermediary, but the intended metals contained in the intermediary. This is reflected by the naming of these flows (in analogy to the naming of elementary flows), denoting the content and not the container.

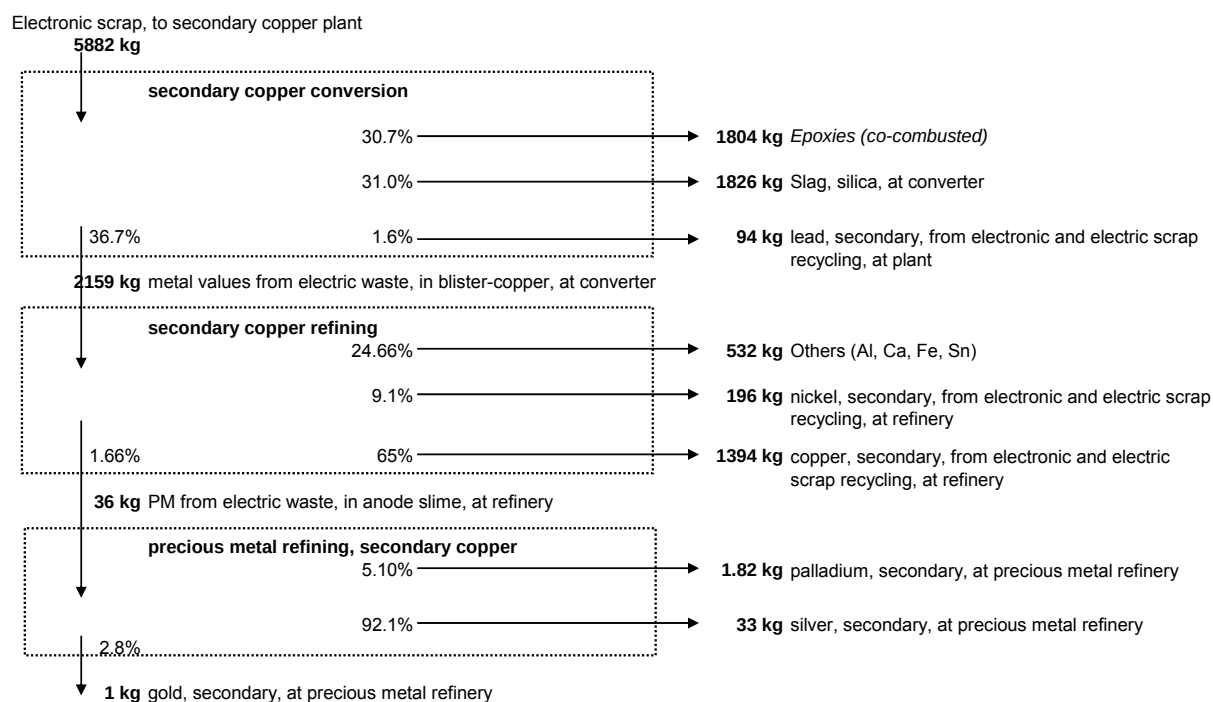


Fig. 5.9 Mass Balance of the tree consecutive multi-output processes for Gold / Silver, secondary from electronic scrap. Mass is scaled to 1 kg gold output.

Tab. 5.72 Constituents of the intermediate reference flows in the system model (Fig. 5.9).

|                    | Scrap input                                 | into blister-copper                                               | To cathode copper                                                            | PGM in anode slime                                  |
|--------------------|---------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------|-----------------------------------------------------|
| Corresponding flow | Electronic scrap, to secondary copper plant | metal values from electric waste, in blister-copper, at converter | copper, secondary, from electronic and electric scrap recycling, at refinery | PM from electric waste, in anode slime, at refinery |
| Mass               | 5882 kg                                     | 2159 kg                                                           | 1394kg                                                                       | 36 kg                                               |
| Cu                 | 23.67%                                      | 64.5%                                                             | 99.9%                                                                        |                                                     |
| Pb                 | 1.60%                                       |                                                                   |                                                                              |                                                     |
| Ni                 | 3.34%                                       | 9.1%                                                              |                                                                              |                                                     |
| Ag                 | 0.560%                                      | 1.53%                                                             |                                                                              | 92.1%                                               |
| Au                 | 0.049%                                      | 0.13%                                                             | 0.1%                                                                         | 2.8%                                                |
| Pd                 | 0.030%                                      | 0.08%                                                             |                                                                              | 5.1%                                                |
| Epoxies            | 30.7%                                       |                                                                   |                                                                              |                                                     |
| Silica             | 31.0%                                       |                                                                   |                                                                              |                                                     |
| other              | 9.05%                                       | 25%                                                               |                                                                              |                                                     |

### Allocation procedures

Based on the amount, the composition and the market prices of the materials involved, the burdens are allocated to the coupled reference flows based on their value (Tab. 5.73). The values contained were assessed according to the three years average from 2004 through 2006 based on USGS (2007).

**Tab. 5.73 Allocation of the burdens to the coupled multi-output reference flows according to their values depending on composition (Tab. 5.72) and the values (three years average from 2004 through 2006 based on USGS, 2007).**

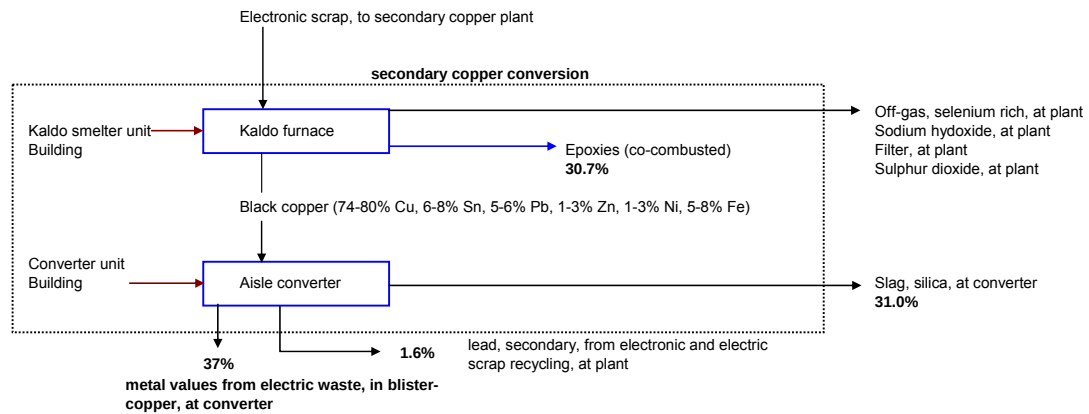
|                                  | <b>\$ per kg</b> | <b>% wt product</b> | <b>% burden</b> |
|----------------------------------|------------------|---------------------|-----------------|
| lead, secondary                  | 1.15             | 4.2%                | 0.1%            |
| metal values from electric waste | 36.1             | 95.8%               | 99.9%           |
|                                  |                  |                     |                 |
| PM from electric waste           | 1105             | 1.66%               | 81.5%           |
| copper, secondary                | 4.47             | 65%                 | 12.9%           |
| nickel, secondary                | 13.8             | 9.1%                | 5.6%            |
|                                  |                  |                     |                 |
| silver, secondary                | 269              | 92.1%               | 22.4%           |
| gold, secondary                  | 15682            | 2.8%                | 39.7%           |
| palladium, secondary             | 8214             | 5.10%               | 37.9%           |

Tab. 5.74 Meta information for the modelled production processes for secondary gold from electric scrap.

| Name                                  | secondary copper conversion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | secondary copper refining                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | precious metal refining, secondary copper                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location                              | SE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | SE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | SE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Infrastructure Process                | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Unit                                  | kg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | kg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | kg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Data Set Version                      | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Included Processes                    | Inculdes the processing of e-scrap in the Boliden process by the Kaldo plant and the converter aisle.                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Includes the refining of the copper anode to cathode copper.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Inculdes the refining of gold and silver.                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Amount                                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Local Name                            | Sekundär Kupfer Verhüttung                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Sekundär Kupfer Raffinierung                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Sekundär Kupfer Hütte, Edelmetallgewinnung                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Synonyms                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| General Comment to reference function | The multi-output-process "secondary copper conversion" delivers the two co-products "metal values from electric waste, in blister-copper, at converter" and "lead, secondary, from electronic and electric scrap recycling, at plant". The product "metal values from electric waste, ..." is an intermediary in the production of secondary gold from electric scrap and is not to be used by the LCA practitioner. The flow "lead, ..." is a by-product of this process that receives part of the burdens allocated. It must not be used to reflect lead. | The multi-output-process "secondary copper refining" delivers the three co-products "PM from electric waste, in anode slime, at refinery", "copper, secondary, from electronic and electric scrap recycling, at refinery", and "nickel, secondary, from electronic and electric scrap recycling, at refinery". The product "PM from electric waste, ..." is an intermediary in the production of secondary gold from electric scrap and is not to be used by the LCA practitioner. The flows "copper, ..." and "nickel, ..." are by-products of this process that receive part of the burdens allocated. They must not be used. | The multi-output-process "precious metal refining, secondary copper" delivers the co-products "silver, secondary, at precious metal refinery", "gold, secondary, at precious metal refinery" and "palladium, secondary, at precious metal refinery". The flows "silver, ..." and "gold, ..." are part of the global supply mix of gold and silver modelled in this part. The by-product "palladium, ..." receives its part of the burdens, but it is not aimed to be used by the data-user. |
| Start Date                            | 2002                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2002                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2002                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| End Date                              | 2005                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2005                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2005                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Geography text                        | Data describes a Swedish plant that is singular in the region and ranges worldwide among the big producer. It may be taken as approximation for the global situation                                                                                                                                                                                                                                                                                                                                                                                        | Data describes a Swedish plant that is singular in the region and ranges worldwide among the big producer. It may be taken as approximation for the global situation                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Data describes a Swedish plant that is singular in the region and ranges worldwide among the big producer. It may be taken as approximation for the global situation                                                                                                                                                                                                                                                                                                                        |
| Technology text                       | Conversion of Copper in a Kaldo Converter and treatment in converter aisle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Production of cathode copper by electrolytic refining.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Anode slime treatment by pressure leaching and top blown rotary converter. Production of Silver by Möbius Electrolysis, Gold by Wohlwill electrolysis, Palladium to further processing                                                                                                                                                                                                                                                                                                      |
| Uncertainty Adjustments               | See Geography and Technology.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## Secondary Copper Conversion

The energy and auxiliary requirements have been calculated on a per year basis from the respective patent and literature. 40'000 t annual electronic scrap treatment (Boliden, 2006c), 300 days actual annual operation time and 50 year service time were assumed to calculate process inputs and emissions per kg electronic scrap treated (Tab. 5.75). According to Boliden (2006c) and the assumed composition of the electronic scrap based on Motorola (2005), it can be calculated that 2.37% of the copper produced stems from electronic scrap (640 t of 27'000 t Pb per year) and 6.58% of the copper production, respectively (14'680 t of 223'000 t Cu per year). The Kaldo process uses the plastic in the scrap as fuel and as a reducing agent at the same time as organic compounds are oxidised. The caloric value in the electronic scrap is assumed to make up for the energy required to reduce the metals in the scrap. Therefore no additional demand in energy and oxygen is accounted for melting. For rotation of the furnace and ventilation in the converter electricity is required. No specific information was available. The electricity demand is stated in Boliden (2006c) to be 603 GWh per year for the whole site. Assuming an arbitrary share 5% that is attributable to the Kaldo plant, and disregarding the electricity for ventilation, 1.12 kWh per kg lead produced in the Kaldo furnace is calculated.



**Fig. 5.10 Unit process presentation of the converter step within the secondary copper plant. This multi-output process yields the flows “secondary copper conversion”.**

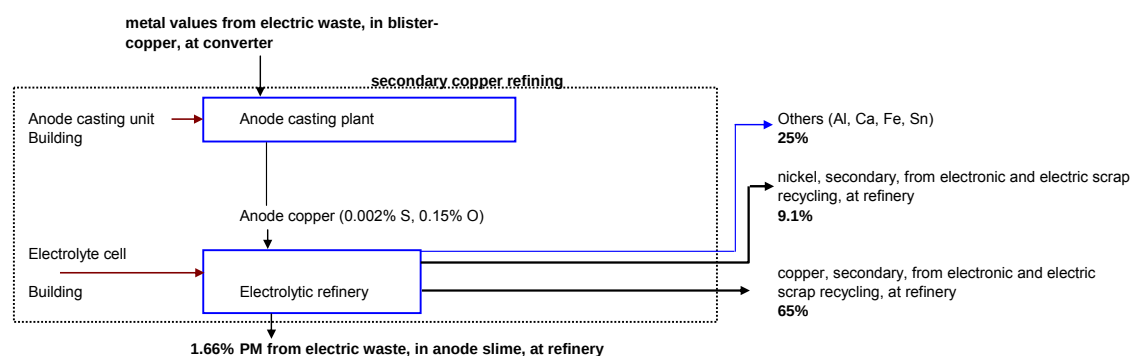
The arising slag is calculated from the content of ceramics in the scrap feed. Of the electronic scrap 31% consists of ceramics (Motorola, 2005). The yearly emissions for the converter aisle (“anode furnace” in source) and for the Kaldo plant have been taken from Umweltbundesamt (2001) and Mark & Lehner (2002), respectively. The distribution of differential PM classes have been adopted from copper smelting and amount  $PM_{<2.5} : PM_{2.5-10} : PM_{>10} = 3 : 2 : 1$  (CEPMEIP-Database, SNAP code 030304). The emissions have been allocated to electronic scrap processing according to their share in the production of lead and copper respectively (Tab. 5.75). The infrastructure is housed on an area of 1'200 m<sup>2</sup> (Boliden, 2000). The infrastructure was considered with the respective module “Facilities anode production, secondary copper” (Tab. 5.78). A service life time of 50 years and yearly production of 250'000 tons of combined copper and lead per year.

**Tab. 5.75** Emissions to air and water during the converter step. Respective Data for Kaldo plant taken from Umweltbundesamt (2001) and Mark & Lehner (2002) for converter aisle. Calculation bases on 27'000 t of lead and 223'000 t of copper, of which 2.37% and 6.58% originate from electronic scrap.

|                        |    | Kaldo    | Converter Aisle | Total                    |
|------------------------|----|----------|-----------------|--------------------------|
| Emissions to air       |    | per year | per year        | per kg e-scrap processed |
| Attribution to e-scrap | %  | 2.37%    | 6.58%           |                          |
| Dust                   | kg | 1780     | 1200            | 3.03E-06                 |
| PM >10 to air          | kg |          |                 | 5.05E-07                 |
| PM 2.5-10to air        | kg |          |                 | 1.01E-06                 |
| PM <2.5 to air         | kg |          |                 | 1.51E-06                 |
| Copper                 | kg | 30       | 34              | 7.37E-08                 |
| Lead                   | kg | 828      | 9               | 5.05E-07                 |
| Zinc                   | kg | 111      |                 | 6.58E-08                 |
| Cadmium                | kg | 2        |                 | 1.19E-09                 |
| Arsenic                | kg |          | 0.02            | 3.29E-11                 |
| Mercury                | kg | 4        | 0               | 2.37E-09                 |
| Sulphur dioxide        | kg | 9000     | 85600           | 1.46E-04                 |
| Nitrogen oxide         | kg | 116      | 0               | 6.87E-06                 |
| Fluor                  | kg | 34       | 0               | 2.01E-08                 |
| Chlorid                | kg | 4240     | 0               | 2.51E-06                 |
| PCDD/F                 | kg | 0.084    | 0               | 4.98E-11                 |
| Emissions to Water     |    | per year | per year        |                          |
| Copper                 | kg | n.a.     | 8               | 1.32E-08                 |
| Lead                   | kg | n.a.     | 2.8             | 4.61E-09                 |
| Zinc                   | kg | n.a.     | 2.6             | 4.28E-09                 |
| Cadmium                | kg | n.a.     | 0.02            | 3.29E-11                 |
| Arsenic                | kg | n.a.     | 0.23            | 3.79E-10                 |
| Mercury                | kg | n.a.     | 0.06            | 9.87E-11                 |

## Secondary Copper Refining

In this multi-output process the cast anodes are refined by electrolytic refining. Based on the assumed constitution of the electric scrap a process flow balance was established (Fig. 5.11). Whereas the winning of cathode copper may be considered as the intended process, in the context of the secondary gold production the arising of anode slime is of importance. The process burdens are allocated to the three products anode slime, nickel sulphate, and cathode copper as described above. The values for the calculations are taken from Davenport et al. (2002).



**Fig. 5.11 Unit process representation of the electro refining step within the secondary copper plant. This multi-output process yields the three (main) flows “Precious metals (PM) from electric waste”, “copper, secondary” and “nickel, secondary”.**

The electrorefining consumes 320 kWh per t of copper cathode. An amount of 0.2 m<sup>3</sup> electrolyte per tonne is bled and purified for recycling into electrolytic refining. It is assumed that all of this volume is replenished with fresh electrolyte. The main constituent in the electrolyte is 180 kg sulphuric acid per m<sup>3</sup>. Other additives like HCl, bone glue, thiourea, and avitane are disregarded, since they all range from 20 to 100 g / m<sup>3</sup>. The 45 kg Copper per m<sup>3</sup> electrolyte is assumed to originate from the anode material or salts from within the operation. It is thus not inventoried. To heat the electrolyte from 20°C to 65 °C no additional energy requirements are assumed, since the electrolysis heats up the electrolyte. No heat recovery between circulations is assumed.

The data for the emissions (Tab. 5.76) originate from HELCOM (2002) and Umweltbundesamt (2001).

The infrastructure is reflected with the specific inventory (see section *Infrastructure for Secondary gold production*). There the lifetime production of the respective unit is said to amount to 6'900'000 t cathode copper output. The infrastructure requirements in this dimensioned according to the copper produced

**Tab. 5.76 Emissions to air and water during the refining of the copper anode. Respective data taken from Umweltbundesamt (2001) and Mark & Lehner (2002) Calculation bases on 223'000 t/a cathode copper and .**

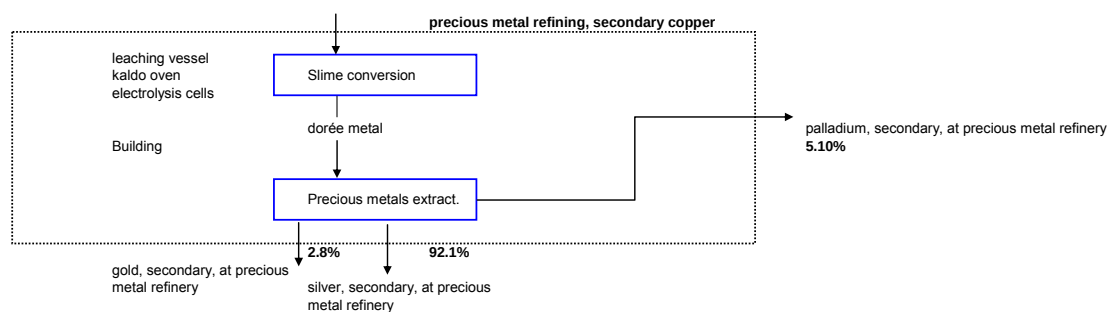
| Emission               | unit | annual release | Per t cathode copper | per kg anode slime |
|------------------------|------|----------------|----------------------|--------------------|
| Copper to water, lake  | kg   | 311.3          | 1.40E-03             | 5.44E-05           |
| Lead to water, lake    | kg   | 16.8           | 7.53E-05             | 2.94E-06           |
| Zinc to water, lake    | kg   | 16.8           | 7.53E-05             | 2.94E-06           |
| Cadmium to water, lake | kg   | 7.1            | 3.18E-05             | 1.24E-06           |
| Arsenic to water, lake | kg   | 8.04           | 3.61E-05             | 1.41E-06           |
| Mercury to water, lake | kg   | 0.09           | 4.04E-07             | 1.57E-08           |
| Nickel to water        | kg   | 28.7           | 1.29E-04             | 5.02E-06           |
| Dust to air            | kg   | 1300           | 5.83E-03             | 2.27E-04           |
| Arsenic to air         | kg   | 14             | 6.28E-05             | 2.45E-06           |
| Copper to air          | kg   | 24             | 1.08E-04             | 4.20E-06           |
| Lead to air            | kg   | 53             | 2.38E-04             | 9.27E-06           |
| Nickel to air          | kg   | 9              | 4.04E-05             | 1.57E-06           |

### Precious metal refining, secondary copper

The processing of anode slimes from secondary copper production takes place in four steps:

- Pressure leaching and Drying
- Smelting and converting of the residue
- Electrowinning of Silver
- Electrowinning of Gold

A diagram of the multi-output process is given in Fig. 5.12



**Fig. 5.12** Unit process representation of the precious metal refinery within the secondary copper plant. This multi-output process yields the flows “gold”, “silver” and “palladium, secondary, from electronic and electric scrap recycling”.

**Pressure leaching.** The slime is leached with electrolyte taken from the tankhouse under pressure (3-7 bar) and at elevated temperature of 115 °C. In Chen & Dutrizac (1990) we learn that the leaching can occur in a single stage for 5 h at 100 psi O<sub>2</sub> in a solution of 250 g/l H<sub>2</sub>SO<sub>4</sub> and 10% NaCl to prevent Ag from dissolution. This reduces the mass of the slime to 77% of the initial value. Assuming a volume of the leaching vessel of 10 m<sup>3</sup> and a reaction mass of 5 m<sup>3</sup> and a batch leaching procedure of 2'500 kg anode slime, 23 kg O<sub>2</sub> per batch or 0.00925 kg O<sub>2</sub> / kg slime are required. The sulphuric acid is assumed to originate from the electrolyte bleed and the NaCl is regarded not to be consumed by precipitation reactions. To heat up the 5'000 l electrolyte from 65 °C to 115 °C an additional 0.17 kWh/kg slime is needed, assuming 70% heat conversion efficiency.

**Drying.** 0.1 kWh electricity per kg are assumed to be required for drying.

**Smelting / Refining.** For fuel oil and requirements of oxygen as reducing agent the specific values for the conversion step in the Kaldo oven is taken as a proxy. Also the specific emission factors are taken (Kaldo furnace, Tab. 5.75) that derive from dividing the yearly emissions of the Kaldo furnace by the produced 27'000 t lead per year and scaling it to the metal produced by this process. According to Peterson & Lundquist (1977), the Kaldo process consumes 52.9 l oil and 123.5 Nm<sup>3</sup> of oxygen per tonne of charge (45% lead) or 4 MJ light fuel oil and 0.369 kg oxygen per kg lead bullion produced. These figures are multiplied with the total metal output of this inventory.

**Möbius electrolysis.** The silver concentration in the electrolyte is ca. 50 g/l. The free HNO<sub>3</sub> is 10 g/l and the anodic current density is 400 – 500 A/m<sup>2</sup>. The cell voltage is 2.0 – 2.5 V. The anodes and cathodes are usually arranged in parallel in the cell, and the energy consumption is ca. 0.6 kWh per kg of pure silver. To compensate for the anodically dissolved base metals that are not removed at the cathode (mainly copper), silver nitrate solution is fed into the electrolyte to prevent depletion of silver (Renner et al., 2002b). However, no additive consumption and replenishment of the electrolyte was considered. Likewise no emissions were assumed because of the purity and the high value of the material processed.

**Wohlwill electrolysis.** This process uses an electrolyte containing 2.5 mol/l of HCl and 2 mol/l of HAuCl<sub>4</sub> acid. Electrolysis is carried out with agitation at 65 – 75 °C. The raw gold is introduced as cast anode plates. The cathodes, on which the pure gold is deposited, were for many years made of fine gold of 0.25 mm thickness. These have now largely been replaced by sheet titanium or tantalum cathodes, from which the thick layer of fine gold can be peeled off. In a typical electrolysis cell, gold anodes weighing 12 kg and having dimensions 280×230×12 mm (0.138 m<sup>2</sup> surface) are used. Opposite to them are conductively connected cathode plates, arranged by two or three on a support rail. One cell normally contains five or six cathode units and four or five anodes. The maximum cell voltage [V] is 1.5 V and the maximum anodic current density [A] 1500 A/m<sup>2</sup> (Renner et al., 2002a). The South African Rand refinery (Auerswald & Radcliffe, 2005) gives a specific gold production rate of 0.2 kg per hour Wohlwill electrolysis. Assuming a current efficiency of 95% the energy consumption is [V] x [A] / 0.2 [kg/h] = 1.63 kWh per kg gold refined. No emissions were assumed because of the purity and the



high value of the material processed. The resulting sludge contains the PGM present in the electric scrap and is sold for further processing.

## 5. System Characterisation

| General Flow information                      |   |                             |  |                |  | Representation in ecoinvent                                                                                                  |                        |                                                       |                  |                            |                                                                  |            |                                                                        |          |                                        | Uncertainty information |                                        |                       |      |                       |
|-----------------------------------------------|---|-----------------------------|--|----------------|--|------------------------------------------------------------------------------------------------------------------------------|------------------------|-------------------------------------------------------|------------------|----------------------------|------------------------------------------------------------------|------------|------------------------------------------------------------------------|----------|----------------------------------------|-------------------------|----------------------------------------|-----------------------|------|-----------------------|
| Input                                         |   | Process Name                |  | Output         |  | Remarks                                                                                                                      | Category               | Sub-category                                          | Infra-structure  | Loca-tion                  | Modul name in ecoinvent                                          | Mean value | Mean value                                                             | Unit     | Source                                 | Type                    | StDv 95%                               | General Comment       |      |                       |
| Light fuel oil, burned                        | m | secondary copper conversion |  |                |  | Not inventoried because of calorific value contained in electronic scrap                                                     | oil                    | heating systems                                       | No               | RER                        | light fuel oil, burned in industrial furnace 1MW, non-modulating | 0.00E+00   |                                                                        | MJ       | (Pettersen, 1977)                      | 1                       | 1.51                                   | (2,3,5,2,1,1,1)       |      |                       |
| Natural gas, burned                           | m |                             |  |                |  | Not inventoried because of calorific value contained in electronic scrap                                                     | natural gas            | heating systems                                       | No               | RER                        | natural gas, burned in industrial furnace >100kW                 | 0.00E+00   |                                                                        | MJ       | (Davenport, 2002)                      | 1                       | 1.10                                   | (2,3,1,2,1,3,1)       |      |                       |
| Pulverised lignite, burned                    | m |                             |  |                |  | Not inventoried because of calorific value contained in electronic scrap                                                     | lignite                | fuels                                                 | No               | DE                         | pulverised lignite, at plant                                     | 0.00E+00   |                                                                        | MJ       | (Pettersen, 1977)                      | 1                       | 1.51                                   | (2,3,5,2,1,1,1)       |      |                       |
| Oxygen, liquid, at plant                      | m |                             |  |                |  | Not inventoried due to reducing abilities of electronic scrap.                                                               | chemicals              | inorganics                                            | No               | RER                        | oxygen, liquid, at plant                                         | 0.00E+00   |                                                                        | kg       | (Pettersen, 1977)                      | 1                       | 1.53                                   | (3,3,5,2,1,1,4)       |      |                       |
| Electricity, high voltage                     | m |                             |  |                |  | to operate the Kaldo Smelter: 1.12 kWh per kg lead produced.                                                                 | electricity            | production mix                                        | No               | UCTE                       | electricity, production mix UCTE                                 | 1.26E-01   |                                                                        | kWh      |                                        | 1                       | 1.14                                   | (3,3,1,2,1,3,2)       |      |                       |
| Quicklime, milled, loose, at plant            | m |                             |  |                |  | to operate the converter Aisle                                                                                               | construction materials | additives                                             | No               | CH                         | quicklime, milled, loose, at plant                               | 1.02E+00   |                                                                        | kg       | (Davenport, 2002)                      | 1                       | 1.10                                   | (2,3,1,2,1,3,4)       |      |                       |
| Infrastructure Kaldo furnace, Aisle Converter | m |                             |  |                |  | Assumption: Service time 50 years, 250'000 t metal produced per year; infrastructure scaled to Cu+Pb output of this process. | metals                 | extraction                                            | Yes              | SE                         | facilities blister-copper conversion, secondary copper           | 8.35E-11   |                                                                        | unit     | (BOLIDEN, 2006)                        | 1                       | 3.01                                   | (2,3,1,2,1,3,9)       |      |                       |
|                                               |   |                             |  |                |  | Slag, to further treatment                                                                                                   |                        | By product of Kaldo plant/Converter Aisle, burdenfree | waste management | residual material landfill | No                                                               | CH         | disposal, nickel smelter slag, 0% water, to residual material landfill | 1.68E+00 |                                        | kg                      | Mass balance                           | 1                     | 1.52 | (2,3,1,2,4,3,6)       |
|                                               |   |                             |  |                |  | PM >10 to air                                                                                                                |                        | See table and text in the respective section          | air              | low population density     |                                                                  |            | Particulates, > 10 um                                                  | 1.38E-06 |                                        | kg                      | APME 2000; Questionnaire 1999; CEPMEIP | 1                     | 5.00 | uncertainty estimated |
|                                               |   |                             |  |                |  | PM 2.5-10to air                                                                                                              |                        | See table and text in the respective section          | air              | low population density     |                                                                  |            | Particulates, > 2.5 um, and < 10um                                     | 2.75E-06 |                                        | kg                      | APME 2000; Questionnaire 1999; CEPMEIP | 1                     | 5.00 | uncertainty estimated |
|                                               |   |                             |  | PM <2.5 to air |  | See table and text in the respective section                                                                                 | air                    | low population density                                |                  |                            | Particulates, < 2.5 um                                           | 4.13E-06   |                                                                        | kg       | APME 2000; Questionnaire 1999; CEPMEIP | 1                       | 5.00                                   | uncertainty estimated |      |                       |

Fig. 5.13 Flows for the multi-output process "secondary copper conversion" and its representation in the ecoinvent database (part I, continued on next page)

## 5. System Characterisation

| General Flow information |   |                             |                                  |                                                                                         | Representation in ecoinvent |                        |                 |           |                                                                         |            |            |      | Uncertainty information                    |      |          |                       |
|--------------------------|---|-----------------------------|----------------------------------|-----------------------------------------------------------------------------------------|-----------------------------|------------------------|-----------------|-----------|-------------------------------------------------------------------------|------------|------------|------|--------------------------------------------|------|----------|-----------------------|
| Input                    |   | Process Name                | Output                           | Remarks                                                                                 | Category                    | Sub-category           | Infra-structure | Loca-tion | Modul name in ecoinvent                                                 | Mean value | Mean value | Unit | Source                                     | Type | StDv 95% | General Comment       |
| ▼ continued              | m | secondary copper conversion | Copper to air                    | See table and text in the respective section                                            | air                         | low population density |                 |           | Copper                                                                  | 2.01E-07   |            | kg   | APME 2000; Questionnaire 1999              | 1    | 5.00     | uncertainty estimated |
|                          |   |                             | Copper to water, lake            | See table and text in the respective section                                            | water                       | lake                   |                 |           | Copper, ion                                                             | 3.59E-08   |            | kg   | HELCOM 2002; Questionnaire 1999            | 1    | 5.00     | uncertainty estimated |
|                          |   |                             | Lead to air                      | See table and text in the respective section                                            | air                         | low population density |                 |           | Lead                                                                    | 1.38E-06   |            | kg   | APME 2000; Questionnaire 1999              | 1    | 5.00     | uncertainty estimated |
|                          |   |                             | Lead to water, lake              | See table and text in the respective section                                            | water                       | lake                   |                 |           | Lead                                                                    | 1.26E-08   |            | kg   | HELCOM 2002; Questionnaire 1999            | 1    | 5.00     | uncertainty estimated |
|                          |   |                             | Zinc to air                      | See table and text in the respective section                                            | air                         | low population density |                 |           | Zinc                                                                    | 1.79E-07   |            | kg   | APME 2000                                  | 1    | 5.00     | uncertainty estimated |
|                          |   |                             | Zinc to water, lake              | See table and text in the respective section                                            | water                       | lake                   |                 |           | Zinc, ion                                                               | 1.17E-08   |            | kg   | HELCOM 2002; Questionnaire 1999            | 1    | 5.00     | uncertainty estimated |
|                          |   |                             | Cadmium to air                   | See table and text in the respective section                                            | air                         | low population density |                 |           | Cadmium                                                                 | 3.23E-09   |            | kg   | APME 2000                                  | 1    | 5.00     | uncertainty estimated |
|                          |   |                             | Cadmium to water, lake           | See table and text in the respective section                                            | water                       | lake                   |                 |           | Cadmium, ion                                                            | 8.97E-11   |            | kg   | HELCOM 2002; Questionnaire 1999            | 1    | 5.00     | uncertainty estimated |
|                          |   |                             | Arsen to air                     | See table and text in the respective section                                            | air                         | low population density |                 |           | Arsenic                                                                 | 8.97E-11   |            | kg   | APME 2000; Questionnaire 1999              | 1    | 5.00     | uncertainty estimated |
|                          |   |                             | Arsen to water, lake             | See table and text in the respective section                                            | water                       | lake                   |                 |           | Arsenic, ion                                                            | 1.03E-09   |            | kg   | HELCOM 2002; Questionnaire 1999            | 1    | 5.00     | uncertainty estimated |
|                          |   |                             | Mercury to air                   | See table and text in the respective section                                            | air                         | low population density |                 |           | Mercury                                                                 | 6.46E-09   |            | kg   | APME 2000                                  | 1    | 5.00     | uncertainty estimated |
|                          |   |                             | Mercury to water, lake           | See table and text in the respective section                                            | water                       | lake                   |                 |           | Mercury                                                                 | 2.69E-10   |            | kg   | HELCOM 2002; Questionnaire 1999            | 1    | 5.00     | uncertainty estimated |
|                          |   |                             | Sulphur dioxide to air           | See table and text in the respective section                                            | air                         | low population density |                 |           | Sulfur dioxide                                                          | 3.98E-04   |            | kg   | APME 2000; Questionnaire 1999; HELCOM 2002 | 1    | 5.00     | uncertainty estimated |
|                          |   |                             | Nitrogen oxide to air            | See table and text in the respective section                                            | air                         | low population density |                 |           | Nitrogen oxides                                                         | 1.87E-05   |            | kg   | HELCOM 2002                                | 1    | 5.00     | uncertainty estimated |
|                          |   |                             | Fluor to air                     | See table and text in the respective section                                            | air                         | low population density |                 |           | Fluorine                                                                | 5.49E-08   |            | kg   | APME 2000                                  | 1    | 5.00     | uncertainty estimated |
|                          |   |                             | Chlorid to air                   | See table and text in the respective section                                            | air                         | low population density |                 |           | Chlorine                                                                | 6.85E-06   |            | kg   | APME 2000                                  | 1    | 5.00     | uncertainty estimated |
|                          |   |                             | PCDD /F to air                   | See table and text in the respective section                                            | air                         | low population density |                 |           | Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin                | 1.36E-10   |            | kg   | APME 2000                                  | 1    | 5.00     | uncertainty estimated |
|                          |   |                             | Waste, heat, emissions to air    |                                                                                         | air                         | unspecified            |                 |           | Heat, waste                                                             | 4.53E-01   |            | MJ   | calculation linked to electricity demand   | 1    | 1.14     | calculation           |
|                          |   |                             | Lead bullion                     | By-product of Kaldo plant /Converter Aisle at 1.2 \$ / kg, recieves 0.14% of the burden | metals                      | extraction             | No              | SE        | lead, secondary, from electronic and electric scrap recycling, at plant | 4.36E-02   |            | kg   |                                            |      |          |                       |
|                          |   |                             | Blister Copper, to anode casting | Key output of the process at 36.1 \$ / kg, recieves 99.86% of the burden                | metals                      | extraction             | No              | SE        | waste, in blister-copper, at converter                                  | 1.00E+00   |            | kg   |                                            |      |          |                       |

Fig. 5. Flows for the multi-output process "secondary copper conversion" and its representation in the ecoinvent database (part II, continued from previous page)

## 5. System Characterisation

| General Flow information                      |   |                           |                                              |                                                                                                                    | Representation in ecoinvent |                        |                 |          |                                                                              |            |            |      | Uncertainty information         |      |          |                              |
|-----------------------------------------------|---|---------------------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------------|------------------------|-----------------|----------|------------------------------------------------------------------------------|------------|------------|------|---------------------------------|------|----------|------------------------------|
| Input                                         |   | Process Name              | Output                                       | Remarks                                                                                                            | Category                    | Sub-category           | Infra-structure | Location | Modul name in ecoinvent                                                      | Mean value | Mean value | Unit | Source                          | Type | StDv 95% | General Comment              |
| Metal values from e-scrap, in blistercopper   | m | secondary copper refining |                                              | 64% copper from electric waste contained, corresponds to a feed of 39 kg blister copper.                           | metals                      | extraction             | No              | SE       | metal values from electric waste, in blister-copper, at converter            | 6.04E+01   |            | kg   | Calculations                    | 1    | 3.02     | (3,3,2,1,1,1,9)              |
| Infrastructure for anode casting and refining | m |                           |                                              | Lifetime production: 230000 t per year during 30 years, normalised to 60.4 kg metal produced                       | metals                      | extraction             | Yes             | SE       | facilities anode refinery, secondary copper                                  | 5.65E-09   |            | unit | Isaksson and Lehner, 2000       | 1    | 3.02     | (3,3,2,1,1,1,9)              |
| Electricity, high voltage                     | m |                           |                                              | 320 kWh per t of copper cathode required                                                                           | electricity                 | production mix         | No              | UCTE     | electricity, production mix UCTE                                             | 1.25E+01   |            | kWh  | Davenport 2002                  | 2    | 1.95     | (300 to 400 kWh / t cath Cu) |
| Water, electrolyte                            | m |                           |                                              | Assumption: 100% of the electrolyte bled (0.2kg / kg cathode copper) is refreshed                                  | water supply                | production             | No              | RER      | water, completely softened, at plant                                         | 7.80E+00   |            | kg   |                                 | 1    | 2.56     | (5,5,5,5,5,4)                |
| Sulfuric acid, electrolyte                    | m |                           |                                              | Concentration in the electrolyte 180 kg/m3. Other additives (HCl, bone glue, thiourea 20 to 100 g/m3) disregarded. | chemicals                   | inorganics             | No              | RER      | sulphuric acid, liquid, at plant                                             | 1.40E+00   |            | kg   |                                 | 1    | 2.56     | (5,5,5,5,5,4)                |
|                                               |   |                           |                                              | 0.2 m3 electrolyte per tonne is bled and purified for recycling.                                                   |                             |                        |                 |          |                                                                              | 7.80E-03   |            | m3   | Davenport 2002                  |      |          |                              |
| Heat, unspecified, in chemical plant          | m |                           |                                              | To heat the electrolyte from 20 °C to 65 °C. Circulated electrolyte (bleed), no heat recovery assumed.             | natural gas                 | heating systems        | No              | RER      | heat, natural gas, at industrial furnace >100kW                              | 1.47E+00   |            | MJ   |                                 | 1    | 1.40     | (4,5,3,1,3,3,1)              |
|                                               |   |                           | PM >10 to air                                |                                                                                                                    | air                         | low population         |                 |          | Particulates, > 10 um                                                        | 3.79E-05   |            | kg   | Questionnaire 1999,             | 1    | 2.24     | (1,2,1,3,5,3,25)             |
|                                               |   |                           | PM 2.5-10to air                              |                                                                                                                    | air                         | low population         |                 |          | Particulates, > 2.5 um, and <                                                | 7.58E-05   |            | kg   | Questionnaire 1999,             | 1    | 2.67     | (1,2,1,3,5,3,26)             |
|                                               |   |                           | PM <2.5 to air                               |                                                                                                                    | air                         | low population         |                 |          | Particulates, < 2.5 um                                                       | 1.14E-04   |            | kg   | Questionnaire 1999,             | 1    | 3.67     | (1,2,1,3,5,3,27)             |
|                                               |   |                           | Copper to water, lake                        | Calculated for Anode Caster/Electrolytic Refinery Plant                                                            | water                       | lake                   |                 |          | Copper, ion                                                                  | 5.44E-05   |            | kg   | HELCOM 2002; Questionnaire 1999 | 1    | 5.07     | (4,4,4,5,5,2,12)             |
|                                               |   |                           | Lead to water, lake                          | Calculated for Anode Caster/Electrolytic Refinery Plant                                                            | water                       | lake                   |                 |          | Lead                                                                         | 2.94E-06   |            | kg   | HELCOM 2002; Questionnaire 1999 | 1    | 5.07     | (4,4,4,5,5,2,12)             |
|                                               |   |                           | Zinc to water, lake                          | Calculated for Anode Caster/Electrolytic Refinery Plant                                                            | water                       | lake                   |                 |          | Zinc, ion                                                                    | 2.94E-06   |            | kg   | HELCOM 2002; Questionnaire 1999 | 1    | 5.07     | (4,4,4,5,5,2,12)             |
|                                               |   |                           | Cadmium to water, lake                       | Calculated for Anode Caster/Electrolytic Refinery Plant                                                            | water                       | lake                   |                 |          | Cadmium, ion                                                                 | 1.24E-06   |            | kg   | HELCOM 2002; Questionnaire 1999 | 1    | 5.07     | (4,4,4,5,5,2,12)             |
|                                               |   |                           | Arsen to water, lake                         | Calculated for Anode Caster/Electrolytic Refinery Plant                                                            | water                       | lake                   |                 |          | Arsenic, ion                                                                 | 1.41E-06   |            | kg   | HELCOM 2002; Questionnaire 1999 | 1    | 5.07     | (4,4,4,5,5,2,12)             |
|                                               |   |                           | Mercury to water, lake                       | Calculated for Anode Caster/Electrolytic Refinery Plant                                                            | water                       | lake                   |                 |          | Mercury                                                                      | 1.57E-08   |            | kg   | HELCOM 2002; Questionnaire 1999 | 1    | 5.07     | (4,4,4,5,5,2,12)             |
|                                               |   |                           | Nickel to water                              | Calculated for Anode Caster/Electrolytic Refinery Plant                                                            | water                       | lake                   |                 |          | Nickel, ion                                                                  | 5.02E-06   |            | kg   | Questionnaire 1999              | 1    | 5.07     | (4,4,4,5,5,2,12)             |
|                                               |   |                           | Arsen to air                                 | Calculated for Anode Caster/Electrolytic Refinery Plant                                                            | air                         | low population density |                 |          | Arsenic                                                                      | 2.45E-06   |            | kg   | HELCOM 2002; Questionnaire 1999 | 1    | 1.58     | (4,4,4,5,5,2,12)             |
|                                               |   |                           | Copper to air                                | Calculated for Anode Caster/Electrolytic Refinery Plant                                                            | air                         | low population density |                 |          | Copper                                                                       | 4.20E-06   |            | kg   | HELCOM 2002; Questionnaire 1999 | 1    | 1.58     | (4,4,4,5,5,2,12)             |
|                                               |   |                           | Lead to air                                  | Calculated for Anode Caster/Electrolytic Refinery Plant                                                            | air                         | low population density |                 |          | Lead                                                                         | 9.27E-06   |            | kg   | HELCOM 2002; Questionnaire 1999 | 1    | 1.58     | (4,4,4,5,5,2,12)             |
|                                               |   |                           | Nickel to air                                | Calculated for Anode Caster/Electrolytic Refinery Plant                                                            | air                         | low population density |                 |          | Nickel                                                                       | 1.57E-06   |            | kg   | HELCOM 2002; Questionnaire 1999 | 1    | 1.58     | (4,4,4,5,5,2,12)             |
|                                               |   |                           | Heat, waste, Nickel, precipitated from bleed | based on electricity consumption                                                                                   | air                         | unspecified            |                 |          | Heat, waste                                                                  | 4.49E+01   |            | MJ   | Calculation                     |      |          |                              |
|                                               |   |                           | Copper cathodes                              | By-product at 4.5 \$ / kg, recieves 12.86% of the burden                                                           | metals                      | extraction             | No              | SE       | nickel, secondary, from electronic and electric scrap recycling, at refinery | 5.49E+00   |            | kg   |                                 |      |          |                              |
|                                               |   |                           | Anode slime, at refinery                     | main at 13.8 \$ / kg, recieves 5.60% of the burden                                                                 | metals                      | extraction             | No              | SE       | copper, secondary, from electronic and electric scrap recycling, at refinery | 3.90E+01   |            | kg   |                                 |      |          |                              |
|                                               |   |                           |                                              | By-product at 1105 \$ / kg, recieves 81.54% of the burden                                                          | metals                      | extraction             | No              | SE       | precious metals from electric waste, in anode slime, at refinery             | 1.00E+00   |            | kg   |                                 |      |          |                              |

Fig. 5.14 Flows for the multi-output process "secondary copper refining" and its representation in the ecoinvent database

## 5. System Characterisation

| General Flow information            |   |                                           |                       |                                                                                                                                                                                  | Representation in ecoinvent |                        |                 |          |                                                                                                      |            |            |      | Uncertainty information              |      |          |                       |
|-------------------------------------|---|-------------------------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|------------------------|-----------------|----------|------------------------------------------------------------------------------------------------------|------------|------------|------|--------------------------------------|------|----------|-----------------------|
| Input                               |   | Process Name                              | Output                | Remarks                                                                                                                                                                          | Category                    | Sub-category           | Infra-structure | Location | Modul name in ecoinvent                                                                              | Mean value | Mean value | Unit | Source                               | Type | StDv 95% | General Comment       |
| Precious metals in anode slime      | m | precious metal refining, secondary copper |                       | 25% content of precious metals in slime, corresponds to 143.1 kg slime feed.                                                                                                     | metals                      | extraction             | No              | SE       | precious metals from electric waste, in anode slime, at refinery                                     | 3.58E+01   |            | kg   |                                      | 1    | 1.09     | (2,2,1,2,1,3,3)       |
| Infrastructure                      | m |                                           |                       |                                                                                                                                                                                  | metals                      | extraction             | Yes             | SE       | facilities precious metal refinery                                                                   | 5.89E-06   |            | unit |                                      | 1    | 3.01     | (2,2,1,2,1,3,9)       |
| electricity for leaching and drying | m |                                           |                       | 10 m3 reaction vessel with 5 m3 reaction mass. Leaching a 2500 kg slime batch at 100 psi O2 and 115°C. 0.17 kWh/kg slime. 0.1 kWh per dry product assumed for subsequent drying. | electricity                 | production mix         | No              | UCTE     | electricity, high voltage, production UCTE, at grid                                                  | 2.73E+01   |            | kWh  | Gürmen et al., 2004; Davenport, 2002 | 1    | 1.31     | (4,2,1,2,3,3,2)       |
| oxygen                              | m |                                           |                       |                                                                                                                                                                                  | chemicals                   | inorganics             | No              | RER      | oxygen, liquid, at plant                                                                             | 1.32E+00   |            | kg   | Aylmore, 2001                        | 1    | 1.31     | (4,2,1,2,3,3,3)       |
| Light fuel oil, burned              | m |                                           |                       | 52.9 l oil per t charge, 45% lead content = 4 MJ per kg lead bullion                                                                                                             | oil                         | heating systems        | No              | RER      | light fuel oil, burned in industrial furnace 1MW, non-modulating                                     | 6.44E+00   |            | MJ   | Peterson 1977                        | 1    | 5.00     | estimation            |
| Oxygen, liquid, at plant            | m |                                           |                       | 123.5 kg O2 per t charge, 45% lead content, 1.34 kg/m3 O2 = 0.369 per kg lead bullion                                                                                            | chemicals                   | inorganics             | No              | RER      | oxygen, liquid, at plant                                                                             | 5.50E-01   |            | kg   | Peterson 1977                        | 1    | 5.00     | estimation            |
| Electricity use Möbius (Silver)     | m |                                           |                       |                                                                                                                                                                                  | electricity                 | production mix         | No              | UCTE     | electricity, high voltage, production UCTE, at grid                                                  | 1.98E+01   |            | kWh  | Kennethcott, 2004                    | 1    | 2.10     | (4,2,1,2,5,5,2)       |
| Electricity use Wohlwill (Gold)     | m |                                           |                       |                                                                                                                                                                                  | electricity                 | production mix         | No              | UCTE     | electricity, high voltage, production UCTE, at grid                                                  | 1.63E+00   |            | kWh  | Mech, 2005                           | 1    | 2.10     | (4,2,1,2,5,5,2)       |
|                                     | m |                                           | PM >10                | specific for Kaldo, no emissions from electrolysis. Particle distribution from specific values for Kaldo oven, no emissions from electrolysis considered                         | air                         | low population density |                 |          | Particulates, > 10 um                                                                                | 3.93E-04   |            | kg   | Questionnaire, 1999, CEPMEIP         | 1    | 5.00     | uncertainty estimated |
|                                     | m |                                           | Copper (air)          | specific values for Kaldo oven, no emissions from electrolysis considered                                                                                                        | air                         | low population density |                 |          | Copper                                                                                               | 3.97E-05   |            | kg   | Questionnaire, 1999                  | 1    | 5.00     | uncertainty estimated |
|                                     | m |                                           | Lead (air)            | specific values for Kaldo oven, no emissions from electrolysis considered                                                                                                        | air                         | low population density |                 |          | Lead                                                                                                 | 1.10E-03   |            | kg   | Questionnaire, 1999                  | 1    | 5.00     | uncertainty estimated |
|                                     | m |                                           | Zinc (air)            | specific values for Kaldo oven, no emissions from electrolysis considered                                                                                                        | air                         | low population density |                 |          | Zinc                                                                                                 | 1.47E-04   |            | kg   | HELCOM 2002                          | 1    | 5.00     | uncertainty estimated |
|                                     | m |                                           | Cadmium (air)         | specific values for Kaldo oven, no emissions from electrolysis considered                                                                                                        | air                         | low population density |                 |          | Cadmium                                                                                              | 2.65E-06   |            | kg   | Questionnaire, 1999                  | 1    | 5.00     | uncertainty estimated |
|                                     | m |                                           | Mercury (air)         | specific values for Kaldo oven, no emissions from electrolysis considered                                                                                                        | air                         | low population density |                 |          | Mercury                                                                                              | 5.30E-06   |            | kg   | HELCOM 2002                          | 1    | 5.00     | uncertainty estimated |
|                                     | m |                                           | Sulphur dioxide (air) | specific values for Kaldo oven, no emissions from electrolysis considered                                                                                                        | air                         | low population density |                 |          | Sulfur dioxide                                                                                       | 1.19E-02   |            | kg   | HELCOM 2002                          | 1    | 5.00     | uncertainty estimated |
|                                     | m |                                           | Nitrogen oxide (air)  | specific values for Kaldo oven, no emissions from electrolysis considered                                                                                                        | air                         | low population density |                 |          | Nitrogen oxides                                                                                      | 1.54E-02   |            | kg   | HELCOM 2002                          | 1    | 5.00     | uncertainty estimated |
|                                     | m |                                           | Fluor (air)           | specific values for Kaldo oven, no emissions from electrolysis considered                                                                                                        | air                         | low population density |                 |          | Fluorine                                                                                             | 4.50E-05   |            | kg   | HELCOM 2002                          | 1    | 5.00     | uncertainty estimated |
|                                     | m |                                           | Chlorid (air)         | specific values for Kaldo oven, no emissions from electrolysis considered                                                                                                        | air                         | low population density |                 |          | Chlorine                                                                                             | 5.62E-03   |            | kg   | HELCOM 2002                          | 1    | 5.00     | uncertainty estimated |
|                                     | m |                                           | PCDD/F (air)          | specific values for Kaldo oven, no emissions from electrolysis considered                                                                                                        | air                         | low population density |                 |          | Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin gold, secondary, at precious metal refinery | 1.11E-07   |            | kg   | Lindström, 2001, HELCOM, 2002        | 1    | 5.00     | uncertainty estimated |
|                                     | m |                                           | Gold, secondary       | By-product at 15682 \$ / kg, receives 39.7% of the burden                                                                                                                        | metals                      | extraction             | No              | SE       | gold, secondary, at precious metal refinery                                                          | 1.00E+00   |            | kg   |                                      | 1    |          |                       |
|                                     | m |                                           | Silver, secondary     | By-product at 269 \$ / kg, receives 22.4% of the burden                                                                                                                          | metals                      | extraction             | No              | SE       | silver, secondary, at precious metal refinery                                                        | 3.29E+01   |            | kg   |                                      | 1    |          |                       |
|                                     | m |                                           | Palladium, secondary  | By-product at 8214 \$ / kg, receives 37.9% of the burden                                                                                                                         | metals                      | extraction             | No              | SE       | palladium, secondary, at precious metal refinery                                                     | 1.82E+00   |            | kg   |                                      | 1    |          |                       |

Fig. 5.15 Flows for the multi-output process "secondary copper, precious metal refinery" and its representation in the ecoinvent database

## Infrastructure for Secondary gold production

This sub-chapter presents the datasets describing the facilities used in the production of secondary gold. For each of the three steps conversion, refinery, and precious metal plant a corresponding infrastructure inventory was established. The respective meta-data are shown in Tab. 5.77, whereas the flows and representations can be found in Fig. 5.16, Fig. 5.17 and Fig. 5.18. For more detailed description and use of the facilities refer to the previous process description in this chapter.

**Tab. 5.77** Meta information for the infrastructure in the production of secondary gold from electric waste.

| Name                                  | facilities copper conversion, secondary copper                                                                                                                                                                                                    | facilities anode refinery, secondary copper                                                                                                                                                                                                                           | facilities precious metal refinery                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location                              | SE                                                                                                                                                                                                                                                | SE                                                                                                                                                                                                                                                                    | SE                                                                                                                                                                                                                                                                                                                                                   |
| Infrastructure Process                | 1                                                                                                                                                                                                                                                 | 1                                                                                                                                                                                                                                                                     | 1                                                                                                                                                                                                                                                                                                                                                    |
| Unit                                  | unit                                                                                                                                                                                                                                              | unit                                                                                                                                                                                                                                                                  | unit                                                                                                                                                                                                                                                                                                                                                 |
| Data Set Version                      | 2.0                                                                                                                                                                                                                                               | 2.0                                                                                                                                                                                                                                                                   | 2.0                                                                                                                                                                                                                                                                                                                                                  |
| Included Processes                    | Construction of facilities and housing, and their disposal. No transports and construction work considered.                                                                                                                                       | Construction of facilities and housing, and their disposal. No transports and construction work considered.                                                                                                                                                           | Construction of facilities and housing, and their disposal. No transports and construction work considered.                                                                                                                                                                                                                                          |
| Amount                                | 1                                                                                                                                                                                                                                                 | 1                                                                                                                                                                                                                                                                     | 1                                                                                                                                                                                                                                                                                                                                                    |
| Local Name                            | Anlagen Blisterkupfer Verhüttung, sekundär Kupfer                                                                                                                                                                                                 | Anlagen Anoden Raffinierung, Sekundär Kupfer                                                                                                                                                                                                                          | Anlagen Edelmetall Hütte                                                                                                                                                                                                                                                                                                                             |
| Synonyms                              | Kaldo Converter                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                      |
| General Comment to reference function | Production of Kaldo converter and converter aisle: chromium steel, refractory bricks and concrete foundation. Building hall houses the facilities, covering 1200m <sup>2</sup> . Service lifetime 50 year, 250'000 t of metals produced per year. | Production of twin anode casting wheel and electrolytic cells: chromium steel, refractory bricks and concrete foundation. Building hall houses the facilities, covering 6'883m <sup>2</sup> . Service lifetime 30 years, yearly capacity of 230'000 t copper refined. | The precious metal plant is equipped as follows: 1 Electrolyte cell, 1 Elution cell, 5 Electrowinning cell, 1 Calcination furnace, 1 Smelting furnace, 1 Electrorefining cell. A building hall houses the equipment covering 5'000m <sup>2</sup> . The service life time assumed is 25 years, the yearly capacity of anode slime treatment is 243 t. |
| Start Date                            | 2002                                                                                                                                                                                                                                              | 2002                                                                                                                                                                                                                                                                  | 2002                                                                                                                                                                                                                                                                                                                                                 |
| End Date                              | 2005                                                                                                                                                                                                                                              | 2005                                                                                                                                                                                                                                                                  | 2005                                                                                                                                                                                                                                                                                                                                                 |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                                 | 1                                                                                                                                                                                                                                                                     | 1                                                                                                                                                                                                                                                                                                                                                    |
| Geography text                        | Data describes a Swedish plant that is singular in the region and ranges worldwide among the big producer. It may be taken as approximation for the global situation                                                                              | Data describes a Swedish plant that is singular in the region and ranges worldwide among the big producer. It may be taken as approximation for the global situation                                                                                                  | Data describes a Swedish plant that is singular in the region and ranges worldwide among the big producer. It may be taken as approximation for the global situation                                                                                                                                                                                 |
| Technology text                       | Due to high temperature chromium steel and refractory materials are used.                                                                                                                                                                         | Due to high temperature chromium steel and refractory materials are used.                                                                                                                                                                                             | Due to high temperature chromium steel and refractory materials are used.                                                                                                                                                                                                                                                                            |
| Extrapolations                        | see Geography and Technology                                                                                                                                                                                                                      | see Geography and Technology                                                                                                                                                                                                                                          | see Geography and Technology                                                                                                                                                                                                                                                                                                                         |
| Uncertainty Adjustments               | see Geography and Technology                                                                                                                                                                                                                      | see Geography and Technology                                                                                                                                                                                                                                          | see Geography and Technology                                                                                                                                                                                                                                                                                                                         |

## Facilities copper conversion, secondary copper

For the furnaces a concrete foundation of 130 m<sup>3</sup> was assumed. The converter aisle and the Kaldo plant modelling was based on official commercial descriptions (Tab. 5.78). The housing was estimated to be 1'200 m<sup>2</sup> large, based on the corporate environmental report. The foundation was assessed with product descriptions to amount 130 m<sup>3</sup>, of which one third consists of exacting concrete, the rest of sole plate concrete. The disposal of the refractory lining is approximated with the disposal of nickel smelter slag. The steel is assumed to be recycled completely at the end of life of the facilities. Following the general modelling principles of ecoinvent, the steel therefore leaves the system burden-free and no disposal processes for steel are considered. The complete inventory is shown in Fig. 5.16.

**Tab. 5.78** Calculations for the infrastructure of the converter; 50 years service lifetime, 250'000 t production of total metals per year.

| Materialisation Converter                                              |    | Kaldo    | Converter Aisle | Total    |
|------------------------------------------------------------------------|----|----------|-----------------|----------|
| Chromium steel 18/8, at plant                                          | kg | 1.99E+03 | 5.96E+04        | 6.16E+04 |
| Section bar rolling, steel                                             | kg | 1.99E+03 | 5.22E+04        | 5.42E+04 |
| Drawing of pipes, steel                                                | kg |          | 7.44E+03        | 7.44E+03 |
| Refractory, basic, packed, at plant                                    | kg | 1.02E+05 | 3.22E+06        | 3.32E+06 |
| Disposal, nickel smelter slag, 0% water, to residual material landfill | kg | 1.02E+05 | 3.22E+06        | 3.32E+06 |

### Facilities anode refinery, secondary copper

This dataset comprises the anode casting unit and an array of 674 electrolytic cells with an estimated lifetime of 50 years and an annual copper production capacity of 230'000 t (Isaakson & Lehner, 2000). The anode casting unit (Fig. 5.5) and the electrolytic cells (Fig. 5.6) were herewith modelled based on official commercial descriptions (Tab. 5.79). The moulds of the anode casting wheels are lined with 0.002 m graphite. The walls of the electrolytic cells are made of stainless steel and reinforced with glass fibres. The disposal of the cell is approximated with the disposal of refractory. The steel is assumed to be recycled completely at the end of the facilities life time. Following the general modelling principles of ecoinvent, the steel therefore leaves the system burden free and no disposal processes for steel are considered. In the context of dataset's use within the production of secondary gold, 14'680 t blister copper from electronic scrap is yearly processed by the facilities.

**Tab. 5.79** Calculations for infrastructure of the anode refinery. 30 years service lifetime, 230'000 t copper per year.

| Materialisation anode refinery                                                   |       | Casting unit | Electrolyte cell | Total    |
|----------------------------------------------------------------------------------|-------|--------------|------------------|----------|
| Requiriements                                                                    | units | 1            | 674              |          |
| Chromium steel 18/8, at plant                                                    | kg    | 1.47E+05     | 2.97E+03         | 2.15E+06 |
| Section bar rolling, steel                                                       | kg    | 4.40E+04     |                  | 4.40E+04 |
| Sheet rolling, chromium steel                                                    | kg    | 1.03E+05     | 2.97E+03         | 2.10E+06 |
| Glass fibre reinforced plastic, polyamide, injection moulding, at plant          | kg    |              | 7.41E+02         | 4.99E+05 |
| Graphite, at plant                                                               | kg    | 7.49E+03     |                  | 7.49E+03 |
| Electronics for control units                                                    | kg    | 3.50E+01     |                  | 3.50E+01 |
| Disposal, refractory SPL, Al elec.lysis, 0% water, to residual material landfill | kg    |              | 3.71E+03         | 2.50E+06 |

### Precious metal refinery, secondary copper

This dataset describes the infrastructure operated in the precious metal refinery. The precious metal refinery stretches over 2'000 m<sup>2</sup> and houses following equipment: 1 pressure leaching vessel, 5 elec-

trourefining / electrowinning cells, 1 Kaldo furnace. The material consumption and disposal of the equipment is displayed in Tab. 5.80. The land use was assessed calculating with the 2'000 m<sup>2</sup> and the considered 25 year life time.

The electrowinning cell was reflected with the required steel for the vessel, the production of the steel wool (electrode, approximated with wire drawing) and a lead electrode, according to a operation for copper electrowinning described in Davenport et al. (2002). The other equipment parts have been materialised based on estimations and datasheets of the producer. The steel is assumed to be recycled completely at the end of the facilities life time. Following the general modelling principles of ecoinvent, the steel therefore leaves the system burden free and no disposal processes for steel are considered. The facilities produce 243 tonnes precious metals – silver, gold and palladium – per year.

**Tab. 5.80 Calculations for equipment of the precious metal plant. 25 years service lifetime, 243 t PM per year.**

|                                            |      | Pressure<br>leaching<br>vessel | Electro-<br>winning<br>cell | Kaldo<br>oven | Electro-<br>refining<br>cell | Total    |
|--------------------------------------------|------|--------------------------------|-----------------------------|---------------|------------------------------|----------|
| Requirements                               | unit | 1                              | 5                           | 1             | 5                            |          |
| Chromium steel 18/8, at plant              | kg   | 3.45E+03                       | 6.39E+01                    | 1.99E+03      | 5.63E+02                     | 8.58E+03 |
| Section bar rolling, steel                 | kg   |                                |                             | 1.99E+03      | 1.11E+02                     | 2.55E+03 |
| Sheet rolling, chromium steel              | kg   | 3.45E+03                       | 4.21E+02                    |               | 5.63E+02                     | 8.37E+03 |
| Wire drawing, steel                        | kg   |                                | 6.39E+01                    |               |                              | 3.19E+02 |
| Glass fibre reinforced plastic             | kg   | 7.41E+02                       | 2.09E+02                    |               |                              | 1.79E+03 |
| Concrete, exacting, at plant               | m3   |                                | 7.24E-01                    |               |                              | 3.62E+00 |
| Refractory, basic, packed, at plant        | kg   |                                |                             | 1.02E+05      | 2.33E+03                     | 1.14E+05 |
| Lead, at regional storage                  | kg   |                                | 5.41E+02                    |               |                              | 2.70E+03 |
| Tin, at regional storage                   | kg   |                                | 5.29E+00                    |               |                              | 2.64E+01 |
| Calcium carbide, technical grade, at plant | kg   |                                | 7.47E-02                    |               |                              | 3.74E-01 |
| Disposal, refractories                     | kg   |                                |                             | 1.02E+05      |                              | 1.02E+05 |
| Disposal, El. Cells                        | kg   | 3.71E+03                       |                             |               |                              | 3.71E+03 |
| Disposal, concrete                         | kg   |                                | 7.24E-01                    |               |                              | 3.62E+00 |



## 5. System Characterisation

| Input                                         |   | Process Name                                   |  | Output |  | Remarks                                                                                                                                                               | Category               | Sub-category               | Infra-structure | Location | Modul name in ecoinvent                                                | Mean value | Mean value | Unit | Source | Type | StDv 95% | General Comment |
|-----------------------------------------------|---|------------------------------------------------|--|--------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|----------------------------|-----------------|----------|------------------------------------------------------------------------|------------|------------|------|--------|------|----------|-----------------|
| Chromium steel 18/8, at plant                 | m | facilities copper conversion, secondary copper |  |        |  |                                                                                                                                                                       | metals                 | extraction                 | No              | RER      | chromium steel 18/8, at plant                                          | 6.16E+04   |            | kg   |        | 1    | 1.07     | (2,1,1,2,1,1,3) |
| Section bar rolling, steel                    | m |                                                |  |        |  |                                                                                                                                                                       | metals                 | processing                 | No              | RER      | section bar rolling, steel                                             | 5.42E+04   |            | kg   |        | 1    | 1.07     | (2,1,1,2,1,1,3) |
| Drawing of pipes, steel                       | m |                                                |  |        |  |                                                                                                                                                                       | metals                 | processing                 | No              | RER      | drawing of pipes, steel                                                | 7.44E+03   |            | kg   |        | 1    | 1.07     | (2,1,1,3,1,1,3) |
| Refractory, basic, packed, at plant           | m |                                                |  |        |  |                                                                                                                                                                       | construction materials | bricks                     | No              | DE       | refractory, basic, packed, at plant                                    | 3.32E+06   |            | kg   |        | 1    | 1.07     | (2,1,1,2,1,1,3) |
| Building, hall, steel construction            | m |                                                |  |        |  | Hall housing the kaldo oven and the converter aisle on 1'200 m2.<br>Only real industry facilities taken into account, no other areas (e.g. parking places) considered | construction processes | buildings                  | Yes             | CH       | building, hall, steel construction                                     | 1.20E+03   |            | m2   |        | 1    | 1.21     | (4,1,1,3,1,1,3) |
| Transformation, from unknown                  | m |                                                |  |        |  | the area on which the Kaldo oven and the converter aisle are build, calculated as see above                                                                           | resource               | land                       |                 |          | Transformation, from unknown                                           | 1.20E+03   |            | m2   |        | 1    | 2.00     | (1,1,1,1,1,1,8) |
| Transformation, to industrial area, built up  | m |                                                |  |        |  |                                                                                                                                                                       | resource               | land                       |                 |          | Transformation, to industrial area, built up                           | 1.20E+03   |            | m2   |        | 1    | 2.00     | (1,1,1,1,1,1,8) |
| Occupation, industrial area, built up         | m |                                                |  |        |  | the area occupied by the Kaldo oven and the converter aisle during 50 years                                                                                           | resource               | land                       |                 |          | Occupation, industrial area, built up                                  | 6.01E+04   |            | m2a  |        | 1    | 1.52     | (3,1,1,1,1,1,7) |
| Concrete, sole plate and foundation, at plant | m |                                                |  |        |  | 70% of the total 130 m3 concrete foundation                                                                                                                           | construction materials | concrete                   | No              | CH       | concrete, sole plate and foundation, at plant                          | 8.72E+01   |            | m3   |        | 1    | 1.12     | (3,1,1,3,1,1,3) |
| Concrete, exacting, at plant                  | m |                                                |  |        |  | 30% of the total 130 m3 concrete foundation                                                                                                                           | construction materials | concrete                   | No              | CH       | concrete, exacting, at plant                                           | 4.36E+01   |            | m3   |        | 1    | 1.12     | (3,1,1,3,1,1,3) |
|                                               |   |                                                |  |        |  | Proxy for the disposal of refractory bricks                                                                                                                           | waste management       | residual material landfill | No              | CH       | disposal, nickel smelter slag, 0% water, to residual material landfill | 3.32E+06   |            | kg   |        | 1    | 1.07     | (2,1,1,3,1,1,6) |
|                                               |   |                                                |  |        |  | Recycled, not inventoried due to cut off.                                                                                                                             |                        |                            |                 |          |                                                                        | 6.16E+04   |            | kg   |        | 1    | 1.07     | (2,1,1,3,1,1,6) |
|                                               |   |                                                |  |        |  |                                                                                                                                                                       | waste management       | recycling                  | No              | CH       | disposal, building, concrete, not reinforced, to sorting plant         | 1.31E+02   |            | kg   |        | 1    | 1.12     | (3,1,1,3,1,1,6) |
|                                               |   |                                                |  |        |  |                                                                                                                                                                       | electronics            | Metals                     | Yes             | SE       | facilities copper conversion, secondary copper                         | 1.00E+00   |            | unit |        |      |          |                 |
|                                               |   |                                                |  |        |  |                                                                                                                                                                       |                        |                            |                 |          |                                                                        |            |            |      |        |      |          |                 |

Fig. 5.16 Flows for "facilities copper conversion" and its representation in the ecoinvent database.

## 5. System Characterisation

| General Flow information                      |   |                                             |  |        | Representation in ecoinvent                                                                                                        |                        |                            |                 |          |                                                                                  |            |            |      | Uncertainty information      |      |          |                 |
|-----------------------------------------------|---|---------------------------------------------|--|--------|------------------------------------------------------------------------------------------------------------------------------------|------------------------|----------------------------|-----------------|----------|----------------------------------------------------------------------------------|------------|------------|------|------------------------------|------|----------|-----------------|
| Input                                         |   | Process Name                                |  | Output | Remarks                                                                                                                            | Category               | Sub-category               | Infra-structure | Location | Modul name in ecoinvent                                                          | Mean value | Mean value | Unit | Source                       | Type | StDv 95% | General Comment |
| Chromium steel 18/8, at plant                 | m | facilities anode refinery, secondary copper |  |        |                                                                                                                                    | metals                 | extraction                 | No              | RER      | chromium steel 18/8, at plant                                                    | 2.15E+06   |            | kg   |                              | 1    | 1.07     | (2,1,1,2,1,1,3) |
| Section bar rolling, steel                    | m |                                             |  |        |                                                                                                                                    | metals                 | processing                 | No              | RER      | section bar rolling, steel                                                       | 4.40E+04   |            | kg   |                              | 1    | 1.07     | (2,1,1,2,1,1,3) |
| Sheet rolling, chromium steel                 | m |                                             |  |        |                                                                                                                                    | metals                 | processing                 | No              | RER      | sheet rolling, chromium steel                                                    | 2.10E+06   |            | kg   |                              | 1    | 1.07     | (2,1,1,2,1,1,3) |
| Glass fibre reinforced plastic                | m |                                             |  |        |                                                                                                                                    | plastics               | others                     | No              | RER      | glass fibre reinforced plastic, polyamide, injection moulding, at plant          | 4.99E+05   |            | kg   |                              | 1    | 1.07     | (2,1,1,3,1,1,3) |
| Graphite, at plant                            | m |                                             |  |        |                                                                                                                                    | chemicals              | inorganics                 | No              | RER      | graphite, at plant                                                               | 7.49E+03   |            | kg   |                              | 1    | 1.07     | (2,1,1,2,1,1,3) |
| Electronics for control units                 | m |                                             |  |        |                                                                                                                                    | construction processes | buildings                  | No              | RER      | electronics for control units                                                    | 3.50E+01   |            | kg   |                              | 1    | 1.07     | (2,1,1,3,1,1,3) |
| Building, hall, steel construction            | m |                                             |  |        | To house the Anode casting plant and the Electrolytic Refinery Plant.                                                              | construction processes | buildings                  | Yes             | CH       | building, hall, steel construction                                               | 6.88E+03   |            | m2   | Davenport 2002, Boliden 2004 | 1    | 1.21     | (4,1,1,3,1,1,3) |
| Transformation, from unknown                  | m |                                             |  |        | Only real industry facilities taken into account, no other areas (e.g. parking places) considered                                  | resource               | land                       |                 |          | Transformation, from unknown                                                     | 6.88E+03   |            | m2   |                              | 1    | 2.00     | (1,1,1,1,1,1,8) |
| Transformation, to industrial area, built up  | m |                                             |  |        | the area on which the Anode casting plant and the Electrolytic Refinery Plant are build, calculated as see above                   | resource               | land                       |                 |          | Transformation, to industrial area, built up                                     | 6.88E+03   |            | m2   |                              | 1    | 2.00     | (1,1,1,1,1,1,8) |
| Occupation, industrial area, built up         | m |                                             |  |        | the area occupied by the Anode casting plant and the Electrolytic Refinery Plant during 30 years                                   | resource               | land                       |                 |          | Occupation, industrial area, built up                                            | 2.06E+05   |            | unit |                              | 1    | 1.52     | (3,1,1,1,1,1,7) |
| Concrete, sole plate and foundation, at plant | m |                                             |  |        | Foundation for Anode casting plant and Electrolytic Refinery Plant, assuming a fundament thickness of 1.2 m                        | construction materials | concrete                   | No              | CH       | concrete, sole plate and foundation, at plant                                    | 6.73E+03   |            | m3   |                              | 3    | 1.12     | (3,1,1,3,1,1,3) |
| Concrete, exacting, at plant                  | m |                                             |  |        | Foundation for Anode casting plant and Electrolytic Refinery Plant, assuming an additional concrete mass of 10% of sole and plate. | construction materials | concrete                   | No              | CH       | concrete, exacting, at plant                                                     | 1.55E+02   |            | m3   |                              | 4    | 1.12     | (3,1,1,3,1,1,3) |
|                                               | m |                                             |  |        | Approx with disposal Al Electrolysis                                                                                               | waste management       | residual material landfill | No              | CH       | disposal, refractory SPL, Al elec.lysis, 0% water, to residual material landfill | 2.50E+06   |            | kg   |                              | 1    | 1.51     | (2,1,1,3,4,1,6) |
|                                               | m |                                             |  |        | Recycled, not inventoried due to cut off.                                                                                          |                        |                            |                 |          |                                                                                  | 1.47E+05   |            | kg   |                              | 1    | 1.07     | (2,1,1,3,1,1,6) |
|                                               | m |                                             |  |        | Sum of concrete input                                                                                                              | waste management       | recycling                  | No              | CH       | disposal, building, concrete, not reinforced, to sorting plant                   | 6.88E+03   |            | kg   |                              | 1    | 1.07     | (2,1,1,3,1,1,6) |
|                                               | m |                                             |  |        |                                                                                                                                    | electronics            | Metals                     | Yes             | SE       | facilities anode refinery, secondary copper                                      | 1.00E+00   |            | unit |                              |      |          |                 |
|                                               | m |                                             |  |        | Disposal, El. Cells                                                                                                                |                        |                            |                 |          |                                                                                  |            |            |      |                              |      |          |                 |
|                                               | m |                                             |  |        | Disposal, steel, anode cast                                                                                                        |                        |                            |                 |          |                                                                                  |            |            |      |                              |      |          |                 |
|                                               | m |                                             |  |        | Disposal, concrete                                                                                                                 |                        |                            |                 |          |                                                                                  |            |            |      |                              |      |          |                 |
|                                               | m |                                             |  |        | Facilities (refining)                                                                                                              |                        |                            |                 |          |                                                                                  |            |            |      |                              |      |          |                 |

Fig. 5.17 Flows for "facilities anode refinery" and its representation in the ecoinvent database.

## 5. System Characterisation

| General Flow information                     |   |                                    |  |        | Representation in ecoinvent                                                                       |                        |                            |                 |          |                                                                                  |            |            |      | Uncertainty information |      |          |                 |
|----------------------------------------------|---|------------------------------------|--|--------|---------------------------------------------------------------------------------------------------|------------------------|----------------------------|-----------------|----------|----------------------------------------------------------------------------------|------------|------------|------|-------------------------|------|----------|-----------------|
| Input                                        |   | Process Name                       |  | Output | Remarks                                                                                           | Category               | Sub-category               | Infra-structure | Location | Modul name in ecoinvent                                                          | Mean value | Mean value | Unit | Source                  | Type | StDv 95% | General Comment |
| Chromium steel 18/8, at plant                | m | facilities precious metal refinery |  |        |                                                                                                   | metals                 | extraction                 | No              | RER      | chromium steel 18/8, at plant                                                    | 8.58E+03   |            | kg   |                         | 1    | 1.07     | (2,1,1,2,1,1,3) |
| Section bar rolling, steel                   | m |                                    |  |        |                                                                                                   | metals                 | processing                 | No              | RER      | section bar rolling, steel                                                       | 2.55E+03   |            | kg   |                         | 1    | 1.07     | (2,1,1,2,1,1,3) |
| Sheet rolling, chromium steel                | m |                                    |  |        |                                                                                                   | metals                 | processing                 | No              | RER      | sheet rolling, chromium steel                                                    | 8.37E+03   |            | kg   |                         | 1    | 1.07     | (2,1,1,2,1,1,3) |
| Glass fibre reinforced plastic               | m |                                    |  |        |                                                                                                   | plastics               | others                     | No              | RER      | glass fibre reinforced plastic, polyamide, injection moulding, at plant          | 1.79E+03   |            | kg   |                         | 1    | 1.07     | (2,1,1,3,1,1,3) |
| Concrete, exacting, at plant                 | m |                                    |  |        |                                                                                                   | construction materials | concrete                   | No              | CH       | concrete, exacting, at plant                                                     | 3.62E+00   |            | m3   |                         | 1    | 1.07     | (2,1,1,3,1,1,3) |
| Refractory, basic, packed, at plant          | m |                                    |  |        |                                                                                                   | construction materials | bricks                     | No              | DE       | refractory, basic, packed, at plant                                              | 1.14E+05   |            | kg   |                         | 1    | 1.07     | (2,1,1,3,1,1,3) |
| Lead, at regional storage                    | m |                                    |  |        |                                                                                                   | metals                 | extraction                 | No              | RER      | lead, at regional storage                                                        | 2.70E+03   |            | kg   |                         | 1    | 1.51     | (2,1,1,3,4,1,3) |
| Tin, at regional storage                     | m |                                    |  |        |                                                                                                   | metals                 | extraction                 | No              | RER      | tin, at regional storage                                                         | 2.64E+01   |            | kg   |                         | 1    | 1.51     | (2,1,1,3,4,1,3) |
| Calcium carbide, technical grade, at plant   | m |                                    |  |        |                                                                                                   | chemicals              | inorganics                 | No              | RER      | calcium carbide, technical grade, at plant                                       | 3.74E-01   |            | kg   |                         | 1    | 1.51     | (2,1,1,3,4,1,3) |
| Building, hall, steel construction           | m |                                    |  |        | To house the precious metal plant.                                                                | construction processes | buildings                  | Yes             | CH       | building, hall, steel construction                                               | 2.00E+03   |            | m2   |                         | 1    | 1.21     | (4,1,1,3,1,1,3) |
| Transformation, from unknown                 | m |                                    |  |        | Only real industry facilities taken into account, no other areas (e.g. parking places) considered | resource               | land                       |                 |          | Transformation, from unknown                                                     | 2.00E+03   |            | m2   |                         | 1    | 2.00     | (1,1,1,1,1,1,8) |
| Transformation, to industrial area, built up | m |                                    |  |        |                                                                                                   | resource               | land                       |                 |          | Transformation, to industrial area, built up                                     | 2.00E+03   |            | m2   |                         | 1    | 2.00     | (1,1,1,1,1,1,8) |
|                                              | m |                                    |  |        | Approximated with Nickel smelter slag disposal                                                    | waste management       | residual material landfill | No              | CH       | disposal, nickel smelter slag, 0% water, to residual material landfill           | 1.02E+05   |            | kg   |                         | 1    | 1.78     | (5,1,1,3,4,1,6) |
|                                              | m |                                    |  |        |                                                                                                   | waste management       | residual material landfill | No              | CH       | disposal, refractory SPL, Al elec.lysis, 0% water, to residual material landfill | 3.71E+03   |            | kg   |                         | 1    | 1.78     | (5,1,1,3,4,1,6) |
|                                              | m |                                    |  |        | Recycled, not inventoried due to cut off.                                                         |                        |                            |                 |          |                                                                                  | 5.62E+03   |            | kg   |                         | 1    | 1.07     | (2,1,1,3,1,1,6) |
|                                              | m |                                    |  |        | Sum of concrete input                                                                             | waste management       | recycling                  | No              | CH       | disposal, building, concrete, not reinforced, to sorting plant                   | 3.62E+00   |            | kg   |                         | 1    | 1.07     | (2,1,1,3,1,1,6) |
|                                              | m |                                    |  |        |                                                                                                   | electronics            | Metals                     | Yes             | SE       | facilities precious metal refinery                                               | 1.00E+00   |            | Unit |                         |      |          |                 |
|                                              | m |                                    |  |        | Disposal, refractories                                                                            |                        |                            |                 |          |                                                                                  |            |            |      |                         |      |          |                 |
|                                              | m |                                    |  |        | Disposal, El. Cells                                                                               |                        |                            |                 |          |                                                                                  |            |            |      |                         |      |          |                 |
|                                              | m |                                    |  |        | Disposal, steel                                                                                   |                        |                            |                 |          |                                                                                  |            |            |      |                         |      |          |                 |
|                                              | m |                                    |  |        | Disposal, concrete                                                                                |                        |                            |                 |          |                                                                                  |            |            |      |                         |      |          |                 |
|                                              | m |                                    |  |        | Facilities (Precious metal)                                                                       |                        |                            |                 |          |                                                                                  |            |            |      |                         |      |          |                 |

Fig. 5.18 Flows for "facilities precious metal refinery" and its representation in the ecoinvent database.

### 5.2.3 Data quality

Data uncertainty is derived using the Pedigree matrix as described in Frischknecht et al. (2004). The uncertainty for the inventory items related to the Kaldor process and the conversion are underestimated by the Pedigree approach, therefore the standard deviation of the flows is set to 5 to reflect their speculative nature. As there was only a limited access to information of the industry and as data partly dated back more than ten years ago, the reliability and correctness of the data set is limited. Due to limitation in time for this LCI study, only the Boliden process for the production of secondary gold was analysed. That may further limit the explanatory power when it comes to LCA studies of products where the production of secondary gold becomes important. However, since the concentration of precious metals in the electronic scrap feed is high compared to natural ores, the resulting exchanges related to the production of secondary gold and silver are low compared to gold and silver from ores. Therefore the quality is sufficient for the use as background data in a supply mix of primary and secondary gold.

## 5.3 Production of Silver from Lead-Zinc ores

Today, the silver in bullion lead, together with gold and platinum metals, is concentrated by the Parkes process. The process involves the production of a solid alloy of lead, zinc, and silver by stirring zinc into molten impure lead. On cooling a crust is formed that floats on the surface of the molten metal (450 °C) where it is skimmed off (Renner et al., 2002b). The composition of this crust is determined by the laws of solid solution of the ternary alloy system Ag-Pb-Zn. The composition of the crust lies between 5 to 10% Ag, 15 to 30% Zn, the remainder being lead with a small amount of oxygen and other contaminants such as copper (Evans et al., 1993). Reuter (2005) indicates in the calculations made a composition of 18% Ag, 43% Zn with 38% lead, and 1% antimony. This so-called parkes de-silvering crust is processed to raw silver by following steps:

- The adhering liquid lead is removed by **hot pressing**.
- The zinc is distilled in a **retort or vacuum furnace**. The residue consists of a lead – noble-metal alloy with up to 50 % silver, ca. 1 % zinc, and some copper, arsenic, antimony, and bismuth.
- It is **oxidized by cupellation** to obtain crude silver and oxidised lead (litharge) with remaining contaminants.
- The resulting raw silver is electrolytically refined (electrorefined) by the Möbius electrolysis.

### 5.3.1 Silver from lead production in ecoinvent

No explicit study was available that covers the life cycle inventory of the production of silver from lead. Thus this inventory is solely based upon available information from literature. In cases where no explicit silver recovery information was available, information on the refining of zinc and lead was used as proxy.

The meta information for these processes are compiled in Tab. 5.81, the reference flows for the multi-input processes are displayed in Fig. 5.19.

Tab. 5.81 Meta information for the production processes of silver from lead production.

|                                       |                                                                                                                                                        |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>                           | <b>silver, from lead production, at plant</b>                                                                                                          |
| Location                              | GLO                                                                                                                                                    |
| Infrastructure Process                | 0                                                                                                                                                      |
| Unit                                  | kg                                                                                                                                                     |
| Data Set Version                      | 2.0                                                                                                                                                    |
| Included Processes                    | Processing of raw material from lead desilvering, infrastructure, electricity and process emissions to air.                                            |
| Amount                                | 1                                                                                                                                                      |
| Local Name                            | Silber, aus Bleiproduktion, ab Werk                                                                                                                    |
| Synonyms                              |                                                                                                                                                        |
| General Comment to reference function | Production of refined silver (99.9%) from Parkes crust from lead desilvering.                                                                          |
| Start Date                            | 2002                                                                                                                                                   |
| End Date                              | 2005                                                                                                                                                   |
| Data Valid For Entire Period          | 1                                                                                                                                                      |
| Geography text                        | Data plant on average global context.                                                                                                                  |
| Technology text                       | Processing of Parkes desilvering crust by hot pressing, dezincing (vacuum distillation), cupellation of lead and moebius electrolysis (electrowinning) |
| Extrapolations                        | see Geography and Technology                                                                                                                           |
| Uncertainty Adjustments               | see Geography and Technology                                                                                                                           |

### Raw and auxiliary materials, waste

Assuming a silver concentration in the Parkes crust of 18% and an overall recovery yield of 97.3% (Reuter, 2005) an amount of 5.77 kg crust is fed to obtain 1 kg refined silver.

To form a Parkes crust, zinc is stirred into the molten lead. This added zinc however goes to 97% to the lead that is pressed out of the Parkes crust and is returned to the lead processing. The remainder is recovered to 98.2% by vacuum distillation and is returned to form the Parkes crust. Only the remaining 1.8% reports to the intermediate litharge (Reuter, 2005). All the intermediate products as well as the litharge are returned into the process. No further auxiliary materials such as fluxing agents are considered. All employed materials are supposed to be recycled into the production process, therefore no waste fraction is considered.

### Energy demand

Electricity demand for the different process stages is reported in Reuter (2005) per kg silver as follows: Vacuum Distillation: 1.44 kWh; Cupellation: 4.53 kWh; Moebius Electrolysis: 0.60 kWh.

### Process emissions

Rentz et al. (1999) report process emissions related to different processes in the lead and zinc industry in Germany. Since no emission factors for the silver production are available, values for comparable best available technologies (Remelting plant and New Jersey Retort) are taken as approximation. The distribution of differential PM classes have been adopted from copper smelting and amount PM<sub><2.5</sub> : PM<sub>2.5-10</sub> : PM<sub>>10</sub> = 3 : 2 : 1 (CEPMEIP-Database, SNAP code 030304). No emissions to water are considered.

### Transport and infrastructure

Since the parkes crust is supposed to be processed on site and only a small part of the feed is assumed to be acquired from supply external to the facility, no transports of the feed are considered. The infrastructure is reflected with the generic dataset for a metal smelter that is based on inventories from aluminium and iron industry. Lifetime: 50a, output: 2'650'000 t metal per year, reporting to the total amount of silver, zinc and lead intermediate produced.

**Tab. 5.82 Emissions to air from parkes crust processing. Approximation from BAT processes in German Zinc industry (Rentz et al., 1999), geometric mean of the lowest and highest value.**

| Emission to air | CBAT Remelting plant | CBAT New Jersey Retort | low                 | high                | geometric mean (this study) |
|-----------------|----------------------|------------------------|---------------------|---------------------|-----------------------------|
|                 | g / t Zinc produced  | g / t Zinc produced    | g / t Zinc produced | g / t Zinc produced | g / t Zinc produced         |
| Dust            | 25 - 90              | 190                    | 25                  | 190                 | 68.9                        |
| Zn              | 15 - 50              | 80                     | 15                  | 80                  | 34.6                        |
| Cd              | 0.03                 | 0.05                   | 0.03                | 0.05                | 0.04                        |
| Pb              | 2 - 8                | 6                      | 2                   | 8                   | 4                           |

### 5.3.2 Data quality

The emissions are approximated with generic processes in zinc and lead production. Data uncertainty is derived using the Pedigree matrix as described in Frischknecht et al. (2004). Quality is sufficient to figure as part in the supplymix of silver.

## 5. System Characterisation

| General Flow information |   |                                        |   |                         | Representation in ecoinvent |                                                                                                                                                                                                                       |                                                                                      |                        |                 |          |                                                       |                                        |            | Uncertainty information |                           |      |          |                                                     |
|--------------------------|---|----------------------------------------|---|-------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|------------------------|-----------------|----------|-------------------------------------------------------|----------------------------------------|------------|-------------------------|---------------------------|------|----------|-----------------------------------------------------|
| Input                    |   | Process Name                           |   | Output                  |                             | Remarks                                                                                                                                                                                                               | Category                                                                             | Sub-category           | Infra-structure | Location | Modul name in ecoinvent                               | Mean value                             | Mean value | Unit                    | Source                    | Type | StDv 95% | General Comment                                     |
| Parkes crust feed        | m | silver, from lead production, at plant |   |                         |                             | 18.1% Silver contained, overall recovery yield of 97.4%                                                                                                                                                               | metals                                                                               | extraction             | No              | GLO      | parkes process crust, from desilverising of lead      | 5.77E+00                               |            | kg                      | Reuter 2005               | 1    | 1.10     | Variability based on values reported in literature. |
| Total electricity demand | m |                                        |   |                         |                             | Hot pressing: 0.00 kWh; Vacuum Distillation: 1.44 kWh; Cupellation: 4.53 kWh; Moebius Electrolysis: 0.60 kWh; Based on inventories from aluminium and iron industry. Lifetime: 50a, output: 2650'000 t metal per year | electricity                                                                          | production mix         | No              | UCTE     | electricity, medium voltage, production UCTE, at grid | 6.57E+00                               |            | kWh                     | Reuter 2005               | 1    | 1.24     | (3,1,2,1,1,5,2)                                     |
| Infrastructure           | m |                                        |   |                         |                             |                                                                                                                                                                                                                       | metals                                                                               | extraction             | Yes             | GLO      | non-ferrous metal, smelter                            | 3.84E-11                               |            | unit                    | assumptions               | 1    | 3.32     | (4,1,2,1,4,5,9)                                     |
|                          |   |                                        | m | PM >10 to air           |                             | 17% share in the total dust emissions of 69 g per tonne crust processed                                                                                                                                               | air                                                                                  | low population density |                 |          | Particulates, > 10 um                                 | 6.63E-05                               |            | kg                      | Rentz et al 1999, CEPMEIP | 1    | 2.24     | (1,2,1,3,5,3,25)                                    |
|                          |   |                                        | m | PM 2.5-10to air         |                             | 33% share in the total dust emissions of 69 g per tonne crust processed                                                                                                                                               | air                                                                                  | low population density |                 |          | Particulates, > 2.5 um, and < 10um                    | 1.33E-04                               |            | kg                      | Rentz et al 1999, CEPMEIP | 1    | 2.67     | (1,2,1,3,5,3,26)                                    |
|                          |   |                                        | m | PM <2.5 to air          |                             | 50% share in the total dust emissions of 69 g per tonne crust processed                                                                                                                                               | air                                                                                  | low population density |                 |          | Particulates, < 2.5 um                                | 1.99E-04                               |            | kg                      | Rentz et al 1999, CEPMEIP | 1    | 3.67     | (1,2,1,3,5,3,27)                                    |
|                          |   |                                        | m | Zinc emission to air    |                             |                                                                                                                                                                                                                       | air                                                                                  | low population density |                 |          | Zinc                                                  | 3.46E-05                               |            | kg                      | Rentz et al 1999          | 1    | 2.24     | (1,2,1,3,5,3,31)                                    |
|                          |   |                                        | m | Cadmium emission to air |                             |                                                                                                                                                                                                                       | air                                                                                  | low population density |                 |          | Cadmium                                               | 3.87E-08                               |            | kg                      | Rentz et al 1999          | 1    | 2.24     | (1,2,1,3,5,3,31)                                    |
|                          |   |                                        | m | Lead emission to air    |                             |                                                                                                                                                                                                                       | air                                                                                  | low population density |                 |          | Lead                                                  | 3.46E-06                               |            | kg                      | Rentz et al 1999          | 1    | 2.24     | (1,2,1,3,5,3,31)                                    |
|                          |   |                                        | m | Waste heat              |                             |                                                                                                                                                                                                                       | air                                                                                  | low population density |                 |          | Heat, waste                                           | 2.37E+01                               |            | MJ                      | Calculation               | 1    | 1.06     | (1,2,1,3,1,1,13)                                    |
|                          |   |                                        | m | Zinc intermediate       |                             |                                                                                                                                                                                                                       | Internally returned to metallurgical productio process for recovery, not inventoried |                        |                 |          |                                                       | 2.28E+00                               |            | kg                      | Reuter 2005               | 1    | 1.24     | (3,1,1,1,1,5,3)                                     |
|                          |   |                                        | m | Lead intermediate       |                             |                                                                                                                                                                                                                       | Internally returned to metallurgical productio process for recovery, not inventoried |                        |                 |          |                                                       | 1.81E+00                               |            | kg                      | Reuter 2005               | 1    | 1.24     | (3,1,1,1,1,5,3)                                     |
|                          |   |                                        | m | Litharge                |                             |                                                                                                                                                                                                                       | Internally returned to metallurgical productio process for recovery, not inventoried |                        |                 |          |                                                       | 6.86E-01                               |            | kg                      | Reuter 2005               | 1    | 1.24     | (3,1,1,1,1,5,3)                                     |
|                          |   |                                        | m | Silver                  |                             |                                                                                                                                                                                                                       |                                                                                      | metals                 | extraction      | No       | GLO                                                   | silver, from lead production, at plant | 1.00E+00   |                         | kg                        |      |          |                                                     |

Fig. 5.19 Flows for "silver, from lead production, at plant" and its representation in the ecoinvent database

## 5.4 Production of silver and raw copper telluride from copper ores

No explicit study covering the extraction of silver as by-product from copper production could be found. This inventory is based on available industry information and literature data.

The processing of anode slimes from copper production takes place in four steps:

- Pressure leaching of slimes
- Precipitating of copper telluride from solution
- Drying, smelting and converting of the residue
- Electrowinning of silver

In the decopperising leaching process of copper refinery slimes, a considerable amount of tellurium dissolves (70-80% and 95% under atmospheric pressure leaching and pressurized leaching, respectively). The precipitation of telluride is done by copper choppings that are arranged in a column. The leach solution is led through the column and telluride precipitate on the surface (Shibasaki et al., 1992). Fig. 5.20 shows the process as done in Mitsubishi's Naoshima refinery.

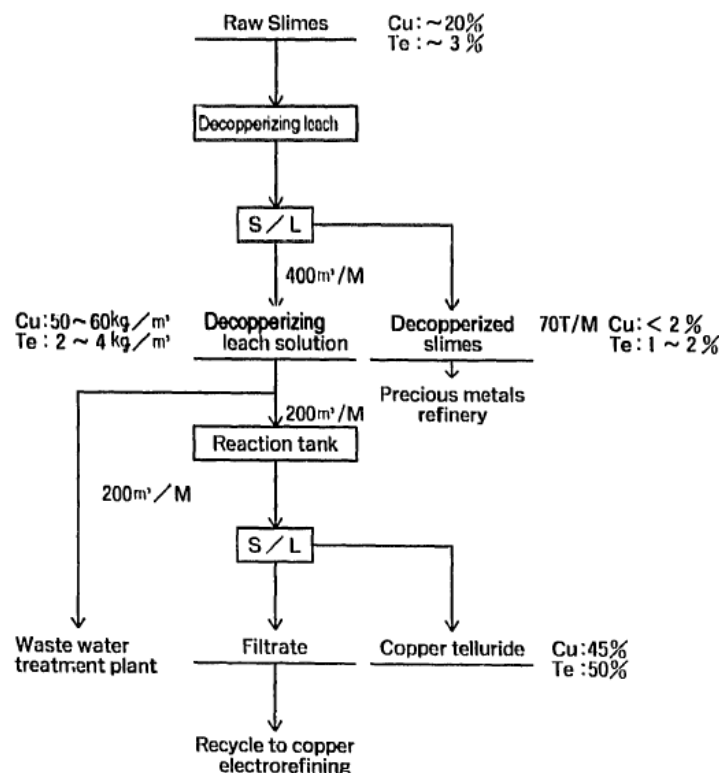


Fig. 5.20 Processing of anode slimes to retract silver and tellurium at Naoshima refinery, Mitsubishi Materials Corp. (Shibasaki et al., 1992).

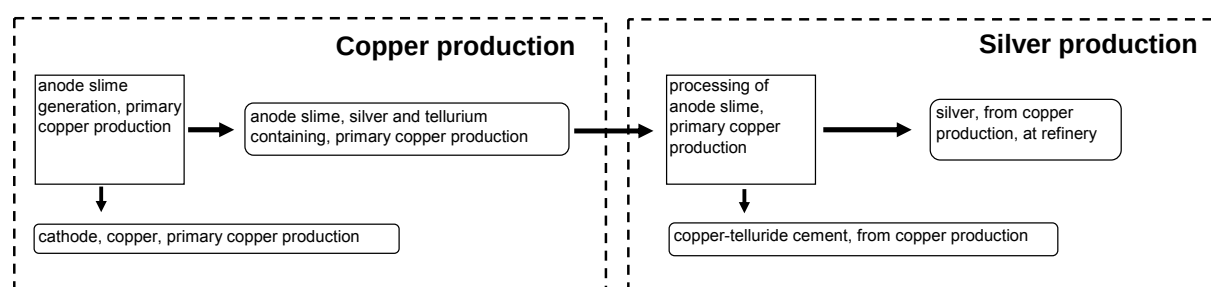
### 5.4.1 Processing of anode slime, primary copper production, in ecoinvent

This module consists of a multi-output dataset (see Fig. 5.22 and Tab. 5.83). The multi-output-process "processing of anode slime, primary copper production" delivers the two co-products "copper-telluride



cement, from copper production" and "silver, from copper production, at refinery". The product "copper-telluride cement, ..." is an intermediary in the production of telluride and is not intended to be used by the LCA practitioner. The flow "silver, ..." is a by-product of this process that receives part of the burdens. It must not be used to reflect silver. For silver and telluride, please refer to the respective adequate datasets.

The raw anode slime is inventoried by the by-product dataset "anode slime, silver and tellurium containing, primary copper production" (Fig. 5.21). The slime represented contains 20% Silver and 3% Tellurium which in turn reflects the coupled production ratio of silver and tellurium from copper production industry on a global average (see Part III "Copper" in this report).



**Fig. 5.21** Model overview and interrelation of silver production with copper extraction. For detailed information see Part III "Copper" in this report.

The anode slime processing corresponds to a part of similar processes described in the chapter 5.2.2.1 Gold / Silver, secondary, from electronic scrap in ecoinvent. Hence, the respective values are adapted for the processing of silver and telluride containing slimes.

**Tab. 5.83** Meta information for the production processes of silver from copper production.

| Name                                  | processing of anode slime, primary copper production                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location                              | GLO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Infrastructure Process                | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Unit                                  | kg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Data Set Version                      | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Included Processes                    | Generation of anode slime from primary copper production (input), leaching and precipitation of copper telluride, melting and electrolysis of silver                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Amount                                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Local Name                            | Anodensumpf-Aufbereitung, primäre Kupferproduktion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| General Comment to reference function | The multi-output-process "processing of anode slime, primary copper production" delivers the two co-products "copper-telluride cement, from copper production" and "silver, from copper production, at refinery". The product "copper-telluride cement, ..." is an intermediary in the production of telluride and is not to be used by the LCA practitioner. The flow "silver, ..." is a by-product of this process that receives part of the burdens allocated. The data set must not be used to reflect silver! For tellurium and silver the respective, adequate end-user datasets have to be chosen. |
| Geography text                        | Data on average global context.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Technology text                       | Leaching of anode slime from primary copper production and precipitation of copper telluride on copper choppings, melting of residues with subsequent Moebius electrolysis.                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Extrapolations                        | see Geography and Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Uncertainty Adjustments               | see Geography and Technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

**Pressure leaching.** The slime (5 kg) is leached with electrolyte taken from the tankhouse under pressure (3-7 bar) and at elevated temperature of 115 °C. From Chen & Dutrizac (1990) we learn that the

leaching can occur in a single stage for 5 h at 100 psi  $O_2$  in a solution of 250 g / l  $H_2SO_4$  and 10% NaCl to prevent Ag from dissolution. This reduces the mass of the slime to 77% of the initial value. Assuming a volume of the leaching vessel of 10 m<sup>3</sup> and a reaction volume of 5 m<sup>3</sup> and a batch leaching procedure of 2'500 kg anode slime, 23 kg  $O_2$  per batch or 0.00925 kg  $O_2$  / kg slime are required (see calculations in 5.2.2.1 *Gold / Silver, secondary, from electronic scrap in ecoinvent*). The sulphuric acid is assumed to originate from the electrolyte bleed and the NaCl is regarded not to be consumed by precipitation reactions. To heat up the 5'000 l electrolyte from 65 °C to 115 °C an additional 0.17 kWh/kg slime is needed, assuming 70% heat conversion efficiency. A process efficiency of 90% is assumed, the rest of the Tellurium reports to the residual solution.

**Precipitation.** The copper telluride is precipitated from the leach solution by solid copper. This precipitation – or stripping – is spontaneous given the optimal reaction time. Hence only the metallic copper choppings are accounted for, that are used to strip the raw copper telluride. They are contained in the produced raw copper telluride by 50%<sub>wt</sub>.

**Drying.** 0.1 kWh electricity per kg dry product is assumed to be required for drying.

**Smelting / Refining.** For fuel oil and requirements of oxygen as reducing agent the same specific values as for the conversion step in the Kaldo oven is taken as a proxy. Also the specific emission factors are taken (Kaldo furnace, Tab. 5.75) that derive from dividing the yearly emissions of the Kaldo furnace by the produced 27'000 t lead per year and scaling it to the metal produced by this process. According to Peterson & Lundquist (1977), the Kaldo process consumes 52.9 l oil and 123.5 Nm<sup>3</sup> of oxygen per tonne of charge (45% lead) or 4 MJ light fuel oil and 0.369 kg oxygen per kg lead bullion produced. These figures are multiplied with the total metal output of this inventory. Eventual electricity consumption for furnace operation is assumed to be low and therefore is disregarded in this step.

**Möbius electrolysis.** The silver concentration in the electrolyte is ca. 50 g/L. The free  $HNO_3$  is 10 g/L and the anodic current density is 400 – 500 A/m<sup>2</sup>. The cell voltage is 2.0 – 2.5 V. The anodes and cathodes are usually arranged in parallel in the cell, and the energy consumption is ca. 0.6 kWh per kg of pure silver. To compensate for the anodically dissolved base metals that are not removed at the cathode (mainly copper), silver nitrate solution is fed into the electrolyte to prevent depletion of silver (Renner et al., 2002b). However, no additive consumption and replenishment of the electrolyte was considered. Likewise no emissions were assumed because of the purity and the high value of the material processed.

**Allocation procedure.** The allocation of the subprocesses copper telluride and silver formation is not evident to solve, since only the final commodity prices are known but no in-firm price for the CuTe cement, on which the allocation would rely. Also the share of the products in total costs (gross sales method as proposed in Guinée et al. (2004) and Guinée et al. (2001)) is not easy to estimate. The by-product allocation problem – the joint production of silver and copper telluride – is reduced here by a subdivision of the sub-processes (according to Ekvall & Finnveden, 2001). The tellurium production from the raw copper telluride is treated in a distinct inventory. Hence it is the task to allocate the burdens between the silver and the CuTe cementate, whereas for latter no published price is available.

As starting point for an estimation it is assumed that the profit margin for the production of copper telluride cement corresponds to the company's performance. Taking an arbitrary performance value of 10% - which reflects the general profit expectations of major enterprises – the value of the cement corresponds consequently to 10% of the proceeds from sale of copper telluride. Therefore 10 % of the copper telluride price is taken to estimate the value of the cement.

The composition and the corresponding values in the by-products result in a split of 8% to the copper telluride cementate and 92% to the silver (Tab. 5.84). The values contained were assessed according to the three years average from 2004 through 2006 based on USGS (2007).

Tab. 5.84 Allocation factors for copper telluride and silver using economic criteria.

| Reference flows                |    | Amount | \$ / kg | Value | Split |
|--------------------------------|----|--------|---------|-------|-------|
| Copper Telluride, by-product   | kg | 0.0489 | 7.57    | 0.370 | 8%    |
| Silver, from copper production | kg | 1.0    | 4.48    | 4.48  | 92%   |

### 5.4.2 Data quality

The dataset bases to large parts on extrapolations and approximations. Data uncertainty is derived using the Pedigree matrix as described in Frischknecht et al. (2004). The uncertainty for the inventory items related to the Kaldo process and the conversion are underestimated by the Pedigree approach, therefore the standard deviation of the flows is set to 5 to reflect their speculative nature. However, the dataset is sufficient to serve as upstream process of silver and telluride used in background data. The dataset must not be used to make assertions regarding different refining processes.

## 5. System Characterisation

| General Flow information            |   |                                                      |                                |                                                                                                                                                                                  | Representation in ecoinvent |                        |                 |          |                                                                         |            |            |      | Uncertainty information              |      |          |                       |
|-------------------------------------|---|------------------------------------------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|------------------------|-----------------|----------|-------------------------------------------------------------------------|------------|------------|------|--------------------------------------|------|----------|-----------------------|
| Input                               |   | Process Name                                         | Output                         | Remarks                                                                                                                                                                          | Category                    | Sub-category           | Infra-structure | Location | Modul name in ecoinvent                                                 | Mean value | Mean value | Unit | Source                               | Type | Stdv 95% | General Comment       |
| Anode Slime from Copper refining    | m | processing of anode slime, primary copper production |                                | 25% content of precious metals in slime, corresponds to 143.1 kg slime feed.                                                                                                     | metals                      | extraction             | No              | GLO      | anode slime, silver and tellurium containing, primary copper production | 5.00E+00   |            | kg   |                                      | 1    | 1.09     | (2,2,1,2,1,3,3)       |
| Infrastructure                      | m |                                                      |                                |                                                                                                                                                                                  | metals                      | extraction             | Yes             | SE       | facilities precious metal refinery                                      | 1.73E-07   |            | unit |                                      | 1    | 3.01     | (2,2,1,2,1,3,9)       |
| electricity for leaching and drying | m |                                                      |                                | 10 m3 reaction vessel with 5 m3 reaction mass. Leaching a 2500 kg slime batch at 100 psi O2 and 115°C. 0.17 kWh/kg slime. 0.1 kWh per dry product assumed for subsequent drying. | electricity                 | production mix         | No              | UCTE     | electricity, high voltage, production UCTE, at grid                     | 9.35E-01   |            | kWh  | Gürmen et al., 2004; Davenport, 2002 | 1    | 2.06     | (4,2,1,2,5,3,2)       |
| oxygen                              | m |                                                      |                                |                                                                                                                                                                                  | chemicals                   | inorganics             | No              | RER      | oxygen, liquid, at plant                                                | 4.63E-02   |            | kg   | Aylmore, 2001                        | 1    | 2.06     | (4,2,1,2,5,3,3)       |
| Light fuel oil, burned              | m |                                                      |                                | 52.9 l oil per t charge, 45% lead content = 4 MJ per kg lead bullion                                                                                                             | oil                         | heating systems        | No              | RER      | light fuel oil, burned in industrial furnace 1MW, non-modulating        | 1.89E-01   |            | MJ   | Peterson 1977                        | 1    | 5.00     | uncertainty estimated |
| Oxygen, liquid, at plant            | m |                                                      |                                | 123.5 kg O2 per t charge, 45% lead content, 1.34 kg/m3 O2 = 0.369 per kg lead bullion                                                                                            | chemicals                   | inorganics             | No              | RER      | oxygen, liquid, at plant                                                | 1.61E-02   |            | kg   | Peterson 1977                        | 1    | 5.00     | uncertainty estimated |
| Electricity use Möbius (Silver)     | m |                                                      |                                |                                                                                                                                                                                  | electricity                 | production mix         | No              | UCTE     | electricity, high voltage, production UCTE, at grid                     | 6.00E-01   |            | kWh  | Kennethcott, 2004                    | 1    | 2.10     | (4,2,1,2,5,5,2)       |
| Copper Choppings                    | m |                                                      |                                | A RER specific mix of secondary copper, primary pyro and SXEW copper                                                                                                             | metals                      | extraction             | No              | RER      | copper, at regional storage                                             | 2.45E-02   |            | kg   | Kennethcott, 2004                    | 1    | 1.24     | (3,2,1,1,1,5,3)       |
|                                     | m |                                                      | PM >10                         | specific for Kald, no emissions from electrolysis. Particle distribution from CEPMEIP Database for copper dust.                                                                  | air                         | low population density |                 |          | Particulates, > 10 um                                                   | 1.15E-05   |            | kg   | Questionnaire, 1999, CEPMEIP         | 1    | 5.00     | uncertainty estimated |
|                                     | m |                                                      | PM 2.5-10                      |                                                                                                                                                                                  | air                         | low population density |                 |          | Particulates, > 2.5 um, and < 10um                                      | 2.31E-05   |            | kg   | Questionnaire, 1999, CEPMEIP         | 1    | 5.00     | uncertainty estimated |
|                                     | m |                                                      | PM <2.5                        |                                                                                                                                                                                  | air                         | low population density |                 |          | Particulates, < 2.5 um                                                  | 3.46E-05   |            | kg   | Questionnaire, 1999, CEPMEIP         | 1    | 5.00     | uncertainty estimated |
|                                     | m |                                                      | Copper (air)                   | specific values for Kald oven, no emissions from electrolysis considered                                                                                                         | air                         | low population density |                 |          | Copper                                                                  | 1.17E-06   |            | kg   | Questionnaire, 1999                  | 1    | 5.00     | uncertainty estimated |
|                                     | m |                                                      | Lead (air)                     | specific values for Kald oven, no emissions from electrolysis considered                                                                                                         | air                         | low population density |                 |          | Lead                                                                    | 3.22E-05   |            | kg   | Questionnaire, 1999                  | 1    | 5.00     | uncertainty estimated |
|                                     | m |                                                      | Copper (air)                   | specific values for Kald oven, no emissions from electrolysis considered                                                                                                         | air                         | low population density |                 |          | Copper                                                                  | 1.17E-06   |            | kg   | Questionnaire, 1999                  | 1    | 5.00     | uncertainty estimated |
|                                     | m |                                                      | Lead (air)                     | specific values for Kald oven, no emissions from electrolysis considered                                                                                                         | air                         | low population density |                 |          | Lead                                                                    | 3.22E-05   |            | kg   | Questionnaire, 1999                  | 1    | 5.00     | uncertainty estimated |
|                                     | m |                                                      | Zinc (air)                     | specific values for Kald oven, no emissions from electrolysis considered                                                                                                         | air                         | low population density |                 |          | Zinc                                                                    | 4.31E-06   |            | kg   | HELCOM 2002                          | 1    | 5.00     | uncertainty estimated |
|                                     | m |                                                      | Cadmium (air)                  | specific values for Kald oven, no emissions from electrolysis considered                                                                                                         | air                         | low population density |                 |          | Cadmium                                                                 | 7.77E-08   |            | kg   | Questionnaire, 1999                  | 1    | 5.00     | uncertainty estimated |
|                                     | m |                                                      | Mercury (air)                  | specific values for Kald oven, no emissions from electrolysis considered                                                                                                         | air                         | low population density |                 |          | Mercury                                                                 | 1.55E-07   |            | kg   | HELCOM 2002                          | 1    | 5.00     | uncertainty estimated |
|                                     | m |                                                      | Sulphur dioxide (air)          | specific values for Kald oven, no emissions from electrolysis considered                                                                                                         | air                         | low population density |                 |          | Sulfur dioxide                                                          | 3.50E-04   |            | kg   | HELCOM 2002                          | 1    | 5.00     | uncertainty estimated |
|                                     | m |                                                      | Nitrogen oxide (air)           | specific values for Kald oven, no emissions from electrolysis considered                                                                                                         | air                         | low population density |                 |          | Nitrogen oxides                                                         | 4.51E-04   |            | kg   | HELCOM 2002                          | 1    | 5.00     | uncertainty estimated |
|                                     | m |                                                      | Fluor (air)                    | specific values for Kald oven, no emissions from electrolysis considered                                                                                                         | air                         | low population density |                 |          | Fluorine                                                                | 1.32E-06   |            | kg   | HELCOM 2002                          | 1    | 5.00     | uncertainty estimated |
|                                     | m |                                                      | Chlorid (air)                  | specific values for Kald oven, no emissions from electrolysis considered                                                                                                         | air                         | low population density |                 |          | Chlorine                                                                | 1.65E-04   |            | kg   | HELCOM 2002                          | 1    | 5.00     | uncertainty estimated |
|                                     | m |                                                      | PCDD/F (air)                   | specific values for Kald oven, no emissions from electrolysis considered                                                                                                         | air                         | low population density |                 |          | Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin                | 3.26E-09   |            | kg   | Lindström, 2001, HELCOM, 2002        | 1    | 5.00     | uncertainty estimated |
|                                     | m |                                                      | Silver, from copper production | Main Product at 269 \$ / kg, recieves 99.9% of the burden                                                                                                                        | metals                      | extraction             | No              | GLO      | silver, from copper production, at refinery                             | 1.00E+00   |            | kg   |                                      |      |          |                       |
|                                     | m |                                                      | Copper Telluride, by-product   | Main Product at 8 \$ / kg, recieves 0.1% of the burden                                                                                                                           | metals                      | extraction             | No              | GLO      | copper telluride cement, from copper production                         | 4.89E-02   |            | kg   |                                      |      |          |                       |

Fig. 5.22 Flows for "Silver, from copper production, at plant" and its representation in the ecoinvent database

## 5.5 Gold, at regional storage

A large amount of gold is obtained from secondary materials. Renner et al. (2002a) estimates that around 50% of the world gold supply originates from secondary production. The world mine production in 2005 has been 2'518 tonnes, the supply of gold from scrap was reported to be 0.9 tonnes, or 26%<sup>10</sup>. GFSM reports for 2006 an increased scrap supply (1'100 tonnes) while the mine production dropped to 2.44 tonnes. (minus 3%), which corresponds to a secondary supply of 31%. In this inventory a share of 30 % secondary production is assumed.

Today, small-scale mining might account for one-quarter of the world gold output<sup>11</sup>. However, these processes were not inventoried in this study due to lack of information.

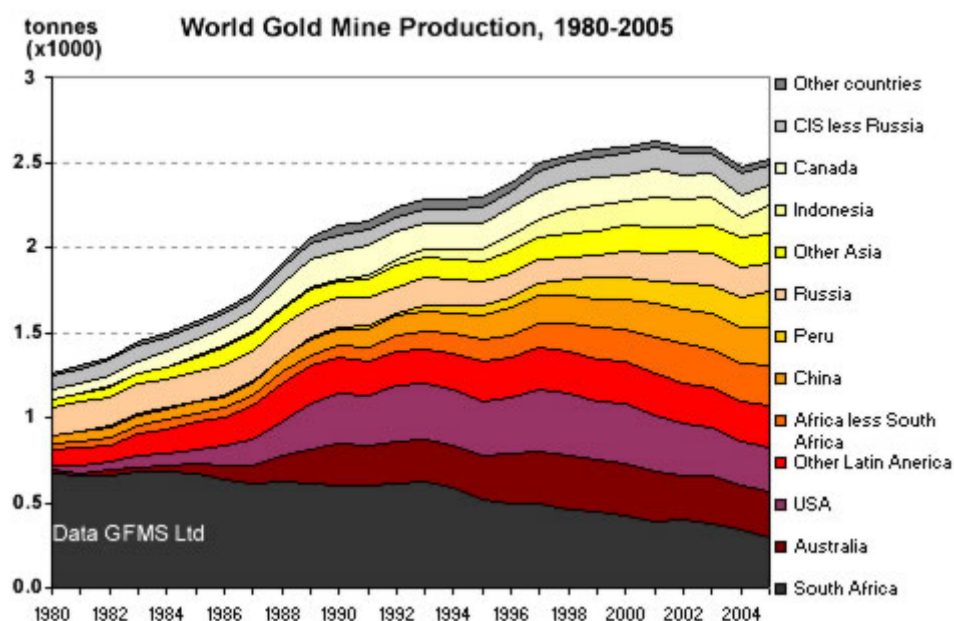


Fig. 5.23 World mine production (Data: GFSM).

A default distance of 250 km from the refinery to the airport and 250 km further to the regional dealer is assumed. Transport distances are estimated on bee-line measures (see Tab. 5.85) and reflect the mine production mix in Tab. 5.69. The resulting inventory is displayed in Fig. 5.25

## 5.6 Silver, at regional storage

This dataset represents the use of silver in Europe. The total mine production in 2005 amounted to 70% of the demand, whereas 21% of the silver was obtained from secondary sources (scrap), 7% from governmental sales and 2% from hedging (Klapwijk et al., 2006).

For primary production, only silver production as by-product from lead / zinc, copper, and gold extraction is available in the ecoinvent Database. These supply sources contribute for 76% of the total primary supply which is regarded sufficient to reflect the production of silver on a global scale. The shares of the available sources (lead/zinc, copper, gold) are scaled for the respective dataset to 100% (see Tab. 2.3). No detailed statistics on silver import to Europe was available, the transports of the primary silver are estimated on the supply statistics provided by the silver institute on a global average

<sup>10</sup> [http://www.gold.org/value/markets/supply\\_demand/recycled.html](http://www.gold.org/value/markets/supply_demand/recycled.html), accessed in july 2007.

<sup>11</sup> Christian Beinhoff (Unido, 2005) [http://www.unido.org/file-storage/download/?file\\_id=10644](http://www.unido.org/file-storage/download/?file_id=10644), accessed in may 2007.

(Tab. 5.85). A default distance of 250 km from the refinery to the airport and 250 km further to the regional dealer is assumed. The resulting inventory is displayed in Fig. 5.24.

**Tab. 5.85 Silver mine production in 2005, country share on total of 20'000 t<sup>12</sup> and distance to Europe<sup>13</sup>.**

| Country       | Mine production | Share | Distance, bee-line |
|---------------|-----------------|-------|--------------------|
|               | Tonnes          |       | km                 |
| Peru          | 3206            | 16%   | 11000              |
| Mexico        | 2884            | 14%   | 10000              |
| Australia     | 2419            | 12%   | 18000              |
| China         | 2022            | 10%   | 8000               |
| Chile         | 1384            | 7%    | 12000              |
| Russia        | 1319            | 7%    | 2000               |
| Poland        | 1266            | 6%    | 250                |
| United States | 1225            | 6%    | 8000               |
| Canada        | 1066            | 5%    | 8000               |
| Kazakhstan    | 809             | 4%    | 5000               |
| Bolivia       | 400             | 2%    | 10000              |
| Indonesia     | 309             | 2%    | 11000              |
| Sweden        | 284             | 1%    | 1300               |
| Morocco       | 231             | 1%    | 1500               |
| Argentina     | 163             | 1%    | 12000              |
| Turkey        | 163             | 1%    | 1500               |
| South Africa  | 88              | 0%    | 5000               |
| Iran          | 81              | 0%    | 4000               |
| Uzbekistan    | 69              | 0%    | 5000               |
| India         | 66              | 0%    | 6000               |

<sup>12</sup> <http://www.silverinstitute.org/>, accessed in may 2007

<sup>13</sup> <http://www.nucleus2000.de/>, accessed in may 2007

## 5. System Characterisation

| General Flow information |   |                             |  |        |  | Representation in ecoinvent                                                                                                             |                   |              |                 |           |                                                           |            |            | Uncertainty information |                       |      |          |                 |
|--------------------------|---|-----------------------------|--|--------|--|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------|--------------|-----------------|-----------|-----------------------------------------------------------|------------|------------|-------------------------|-----------------------|------|----------|-----------------|
| Input                    |   | Process Name                |  | Output |  | Remarks                                                                                                                                 | Category          | Sub-category | Infra-structure | Loca-tion | Modul name in ecoinvent                                   | Mean value | Mean value | Unit                    | Source                | Type | StDv 95% | General Comment |
| Silver from Lead / Zinc  | m | silver, at regional storage |  |        |  | 31% share in world mine production, which in turn accounts for 70%                                                                      | metals            | extraction   | No              | GLO       | silver, from lead production, at refinery                 | 3.21E-01   |            | kg                      | Silver institute 2005 | 1    | 1.05     | (1,1,1,1,1,1,3) |
| Silver from Copper       | m |                             |  |        |  | 25% share in world mine production, which in turn accounts for 70%                                                                      | metals            | extraction   | No              | GLO       | silver, from copper production, at refinery               | 2.59E-01   |            | kg                      | Silver institute 2005 | 1    | 1.05     | (1,1,1,1,1,1,3) |
| Silver from Gold         | m |                             |  |        |  | 14% share in world mine production, which in turn accounts for 70%                                                                      | metals            | extraction   | No              | GLO       | silver, from combined gold-silver production, at refinery | 1.45E-01   |            | kg                      | Silver institute 2005 | 1    | 1.05     | (1,1,1,1,1,1,3) |
| Secondary Silver         |   |                             |  |        |  | Accounts for 21% of the world supply                                                                                                    | metals            | extraction   | No              | SE        | silver, secondary, at precious metal refinery             | 2.75E-01   |            | kg                      | Klapwijk 2006         | 1    | 1.05     | (1,1,1,1,1,1,3) |
| Plane                    |   |                             |  |        |  | Based on share of countries supplying primary silver from Pb/Zn, Cu and Au production, beeline. 50% additional weight due to packaging. | transport systems | airplane     | No              | RER       | transport, aircraft, freight                              | 8.89E+00   |            | tkm                     | calculations          | 1    | 2.00     | (2,1,1,1,1,2,5) |
| Lorry                    |   |                             |  |        |  | 2 x 250 km; 50% additional weight due to packaging.                                                                                     | transport systems | road         | No              | RER       | transport, lorry >16t, fleet average                      | 7.50E-01   |            | tkm                     | calculations          | 1    | 2.00     | (2,1,1,1,1,2,5) |
|                          |   |                             |  |        |  |                                                                                                                                         | metals            | extraction   | No              | RER       | silver, at regional storage                               | 1.00E+00   |            | kg                      |                       |      |          |                 |

Fig. 5.24 Flows for "Silver, at regional storage" and its representation in the ecoinvent database.

| General Flow information     |   |                           |  |        |  | Representation in ecoinvent                                                             |                   |              |                 |           |                                             |                           |            |      |              |      |          | Uncertainty information |  |  |
|------------------------------|---|---------------------------|--|--------|--|-----------------------------------------------------------------------------------------|-------------------|--------------|-----------------|-----------|---------------------------------------------|---------------------------|------------|------|--------------|------|----------|-------------------------|--|--|
| Input                        |   | Process Name              |  | Output |  | Remarks                                                                                 | Category          | Sub-category | Infra-structure | Loca-tion | Modul name in ecoinvent                     | Mean value                | Mean value | Unit | Source       | Type | StDv 95% | General Comment         |  |  |
| Primary gold, production mix | m | gold, at regional storage |  |        |  | According to the producing countries                                                    | metals            | extraction   | No              | GLO       | gold, primary, at refinery                  | 6.89E-01                  |            | kg   | Estimation   | 1    | 1.22     | (4,3,1,1,1,1,3)         |  |  |
| Secondary gold               |   |                           |  |        |  | Assumption: from within Europe (Boliden, Umicore, Norddeutsche Affinerie, etc.)         | metals            | extraction   | No              | SE        | gold, secondary, at precious metal refinery | 3.11E-01                  |            | kg   | Estimation   | 1    | 1.22     | (4,3,1,1,1,1,3)         |  |  |
| Air freight                  |   |                           |  |        |  | Based on share of supplying countries, beeline. 50% additional weight due to packaging. | transport systems | airplane     | No              | RER       | transport, aircraft, freight                | 1.43E+01                  |            | tkm  | calculations | 1    | 2.00     | (2,1,1,1,1,2,5)         |  |  |
| Lorry                        |   |                           |  |        |  | 2 x 250 km; 50% additional weight due to packaging.                                     | transport systems | road         | No              | RER       | transport, lorry >16t, fleet average        | 7.50E-01                  |            | tkm  | calculations | 1    | 2.00     | (2,1,1,1,1,2,5)         |  |  |
|                              |   |                           |  |        |  |                                                                                         |                   | metals       | extraction      | No        | RER                                         | gold, at regional storage | 1.00E+00   |      | kg           |      |          |                         |  |  |

Fig. 5.25 Flows for "Gold, at regional storage" and its representation in the ecoinvent database.

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## Part X

# **Semiconductors: Tellurium, Gallium and Cadmium Compounds**

Data v2.1 (2009)

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Classen M., Althaus H.-J., Blaser S., Tuchschrnid M., Jungbluth N., Doka G., Faist Emmenegger M. and Scharnhorst W. (2009) Life Cycle Inventories of Metals. Final report ecoinvent data v2.1, No 10. EMPA Dübendorf, Swiss Centre for Life Cycle Inventories, Dübendorf, CH, Online-Version under: [www.ecoinvent.ch](http://www.ecoinvent.ch).

Final report of updated Swiss National Life Cycle Inventory Database "ecoinvent", version 2.1

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## Comments on version 2.1

This version corresponds to version 2.0 with changes of the errors detected in the meantime.

The cadmium emissions to air were too high due to a mistake regarding the filtration of airborne emissions in the cadmium purification process. The correct values are given in Tab. 1.

**Tab. 1**      **Corrected values of cadmium emissions ( to air/high population density) of the “cadmium, semiconductor grade” datasets**

| Dataset ID | Name                                                       | Value in ecoinvent Data v2.01 | Correct value | Unit  |
|------------|------------------------------------------------------------|-------------------------------|---------------|-------|
| DS-ID 6902 | cadmium, semiconductor-grade, at plant, US, [kg];          | 1.62E-04                      | 6.0E-06       | Kg/kg |
| DS-ID 6905 | cadmium chloride, semiconductor-grade, at plant, US, [kg]  | 4.4572E-5                     | 3.876E-06     | Kg(kg |
| DS-ID 6906 | cadmium telluride, semiconductor-grade, at plant, US, [kg] | 1.5E-5                        | 6.0E-06       | Kg/kg |

The waterborne cadmium emissions in the dataset “cadmium sulphide, semiconductor-grade, at plant, US, [k]”are mismatched as “from nature”. However, they are emissions “to nature”.



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# 1 Introduction

Several metals feature semi conductive behaviour. For this they have to be purified to a high degree and arranged in layers onto a carrier. This part describes the extraction and purification of some of the most important semiconductor materials:

- Tellurium *raw material from copper production (part III)*
- Gallium, semiconductor grade *raw material from aluminium production (part I)*
- Cadmium, semiconductor grade *raw material of lead-zinc production (part VIII)*
- Cadmium Telluride, semiconductor grade *raw material of lead-zinc production (part VIII)*
- Cadmium Chloride, semiconductor grade *raw material of lead-zinc production (part VIII)*
- Cadmium Sulphide, semiconductor grade *raw material of lead-zinc production (part VIII)*

The raw materials for these metals often are by-products of base metal production. For information on these, please refer to the respective part within this report.

## Tellurium

### Acknowledgement

We thank Vasilis Fthenakis and Olliver Krahl for the information provided in a personal communication.

### 2.1 Introduction

Tellurium (CAS-Number: 13494-80-9) is a metallic element that is often extracted as a by-product of processing copper, lead, gold, and bismuth ores. Tellurium has 8 natural stable isotopes and some 21 additional artificial unstable isotopes are known (Knockaert (2002)). A Hungarian scientist discovered tellurium in 1782. For a long time, only minor amounts were extracted. One reason for the increasing demand of tellurium is the application in the new field of thin film solar cells. Other common applications are thermoelectric devices and as an alloy for the production of free-machining steel.

### 2.2 Reserves and resources of tellurium

The main source of tellurium is the copper anode slime, which is a by-product of the electro winning-process from copper-treatment (Fthenakis (2004)).

The principal source of tellurium is anode sludge, produced during the electrolytic refining of blister copper. Tellurium is produced mainly in the United States, Canada, Peru and Japan. The recovery of tellurium from secondary materials accounts for 10-20% of world tellurium production. As copper ores are the main sources for tellurium, public statistics are based on the copper industry. The following numbers are from the former “United States Bureau of Mines” and are an estimation about the reserves, based on a fixed recovery factor of 0.065 kg Te per tonne of copper (Knockaert (2002)).

According to Fthenakis (2000) a recycling of the metals from the thin-film solar cell can be done, but is not accounted here.

Tab. 2.1 Reserves and resources of tellurium (t), Source: Guiling (1999d))

|               |               |               |       |
|---------------|---------------|---------------|-------|
| North America | 11'000        | United States | 6'000 |
|               |               | Canada        | 2'000 |
|               |               | Others        | 3'000 |
| South America | 9'000         | Chile         | 6'000 |
|               |               | Peru          | 2'000 |
|               |               | Others        | 1'000 |
| Europe        | 5'000         |               |       |
| Africa        | 5'000         |               |       |
| Asia          | 2'000         |               |       |
| Oceania       | 2'000         |               |       |
| <b>World</b>  | <b>34'000</b> |               |       |

### 2.3 Characterisation of tellurium

Tellurium is a relatively rare element, in the same chemical family as oxygen, sulphur, and selenium. Tellurium has a metallic glance in its pure state, and is relatively soft (Knockaert (2002)). For a long time, it was more a waste-product of metallurgy, than a raw material for industrial purposes.

Tab. 2.2 Chemical Characterisation of tellurium, Source: Knockaert (2002)

|                     |                                                    |
|---------------------|----------------------------------------------------|
| Chemical Symbol     | Te                                                 |
| Atomic mass         | 127.60 g/mol                                       |
| Density             | 6.24 g·cm <sup>-3</sup>                            |
| Melting point       | 449.5° C                                           |
| Boiling point       | 988 °C                                             |
| Heat capacity       | 25.73 J·mol <sup>-1</sup> ·K <sup>-1</sup> (25 °C) |
| Mohs hardness       | 2.25                                               |
| CAS registry number | 13494-80-9                                         |

## 2.4 Use and application of material

According to Georg (2004)) there are two main producers of refined tellurium. One is ASARCO from the United States and the other is Umicore in Belgium. Guilinger (1999d) states a list of 29 producers all around the world that fabricate three different categories of tellurium:

- tellurium dioxides – sludge and precipitates of variable purity around 35%
- commercial grade tellurium (99.7% purity)
- high purity tellurium (99.9% to 99.9999%)

The world production per year is in the range of 200 to 300 metric tons. Knockaert (2002) states a broad trade of concentrate, blister and anode copper. This complicates a detailed estimation about the global refinery production. The different market sectors are given in Georg (2004). Some further details are described below:

- Improvement of steel by addition of small quantities of tellurium (up to 0.1%), It helps to control the chill depth of rapidly cooled surfaces and produces hard, wear-resistant top layers.
- Tellurium is used in combination with other components in many catalysts (chemical reactions: oxidation, ammonisation, (de-) hydrogenation, (de-) halogenations, etc.)
- In electronics, tellurium is a well-known additive in selenium-photoreceptors. Tellurium increases the sensitivity and broadens the spectral response of the photoreceptor.
- Polycrystalline cadmium telluride is used in photovoltaic solar cells; CdTe has a direct band gap of 1.5eV and a high optical absorption coefficient.

Tab. 2.3 Market share of tellurium (data for United States), Source: Georg (2004)

|                                                                        |      |
|------------------------------------------------------------------------|------|
| Alloying element to improve the properties of iron and steel           | 40%  |
| Catalyst and chemicals                                                 | 24%  |
| Photoreceptors and thermoelectric devices                              | 22%  |
| Additive to non-ferrous-metals                                         | 8%   |
| Digital video discs and other minor uses (incl. thin-film solar cells) | 6%   |
| Total                                                                  | 100% |

The price for tellurium has increased rapidly in the last 2 years, mainly due to the increased demands from China and solar cell manufacturers worldwide. In November 2004 the price was about US\$ 50 per kilogram (Georg (2004)). In 2005 the price ranged from US\$ 110 to US\$ 150 per kg tellurium (Metal Bulletin (2006)), respectively US\$ 100 to 140 US\$ according to PPM Pure metals.<sup>1</sup> Knockaert

<sup>1</sup> Personal communication: Olliver Krahle, PPM Pure Metals GmbH, Germany (24<sup>th</sup> of January 2007)

(2002) reports that tellurium 99.99% (4N) is normally 40-50% more expensive than tellurium 99.9% (3N) and tellurium 99.999% (5N) is more than double price of the standard-grad tellurium (3N).

## 2.5 Production process

This part is mainly based on the information of Knockaert (2002) and Fthenakis (2004). After pyrometallurgic separations and electrolytic refining, the copper can be collected on the cathode with a purity of 99.95%. The metallic impurities form the anode slime, which is the main source for the production of tellurium (Fthenakis (2004). Further details and information about the copper process can be found in the corresponding section of this report.

The slime contains tellurium, beside various metals like Bi, Sb, Se, Pb, Au, Ag, As or Ni (see Tab. 2.4.). Therefore, the separation process is usually done for all metals at one place. Fig. 2.1 shows an example of a flow sheet for anode slime treatment.

Tab. 2.4 Composition of several copper anode slimes, Source: Knockaert (2002)

|                 | Canadian Copper refiners | Nippon Mining | Inco-Mine     | Geometric mean |
|-----------------|--------------------------|---------------|---------------|----------------|
| Silver          | 21.3%                    | 20.6%         | 6.0%          | 13.80%         |
| Copper          | 20.3%                    | 4.7%          | 25.0%         | 13.39%         |
| Selenium        | 10.9%                    | 15.2%         | 7.0%          | 10.51%         |
| Lead            | 8.5%                     | 6.5%          | 1.0%          | 3.82%          |
| Tellurium       | <b>3.2%</b>              | <b>3.6%</b>   | <b>2.0%</b>   | <b>2.85%</b>   |
| Arsenic         | 1.8%                     | 1.6%          | 0.8%          | 1.33%          |
| Nickel          | 0.5%                     | 0.0%          | 26.0%         | 0.74%          |
| Bismuth         | 0.4%                     | 1.6%          | 0.1%          | 0.39%          |
| Antimony        | 1.0%                     | 1.0%          | 0.1%          | 0.36%          |
| Gold            | 0.0%                     | 1.0%          | 0.1%          | 0.05%          |
| <b>Total</b>    | <b>67.9%</b>             | <b>55.9%</b>  | <b>68.1%</b>  |                |
| <i>Residual</i> | <i>32.10%</i>            | <i>44.10%</i> | <i>31.90%</i> |                |

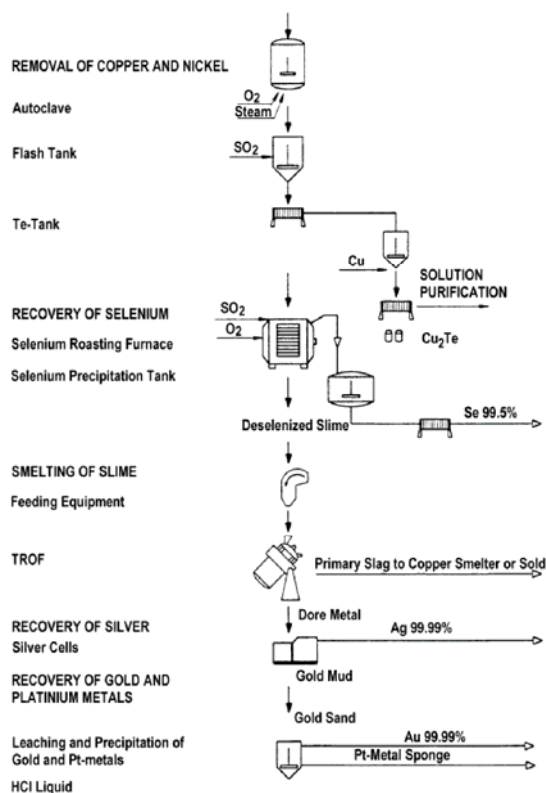


Fig. 2.1 Example flow sheet for anode slime treatment, Source: IPPC (2001)

The tellurium content in the copper varies. The mentioned “United States Bureau of Mines” estimates an average tellurium content of about 0.065 kg per ton of copper (Knockaert (2002)). The tellurium content in the sludge varies from 0.5% to 10%. Three copper-producers stated a typical tellurium concentration of 2%, 3.2% and 3.64%, with a geometrical mean of 2.85% (Tab. 2.4).

As the tellurium is separated from the slime as copper-telluride-cementate, the life cycle assessment starts with the separation of tellurium from the cementate, the Pre-treatment is not assessed in this dataset. The following processes are considered (see also Fig. 2.2):

- (i) Dissolving of copper-cementate with sulphuric acid
- (ii) Electrolysis of tellurium from the solution
- (iii) Purification of cathode-tellurium with Zone melting

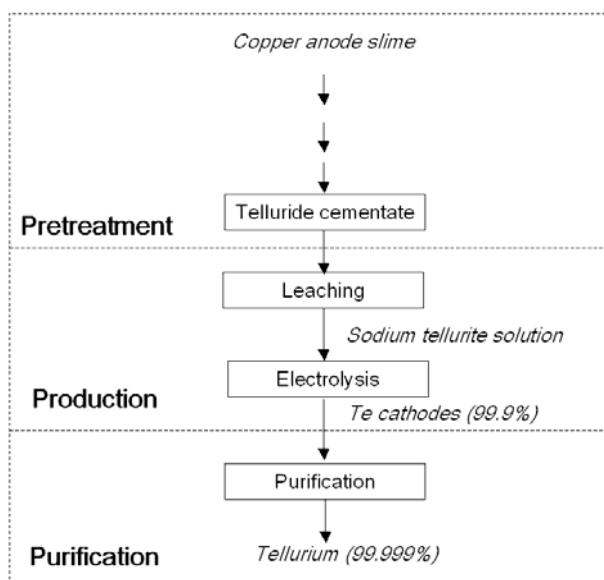


Fig. 2.2 Production process of tellurium

**Dissolving cementate and electrolysis:** The cementate of copper-telluride is treated with sulphuric acid. The next step is the electrolysis of the solution to extract the tellurium at the cathode. Further details and chemical reactions can be found in Knockaert (2002).

**Purification:** According to Fthenakis (2004), high purity tellurium is produced by electrolytic purification and subsequent melting and atomization or by vacuum-distillation. According to industrial sources in the mentioned report, electrolytic purification does not produce any emissions and all waste is recycled. On the other hand, it is mentioned that for the production of 0.6 kg tellurium (99.999% or 5N) 0.75 kg tellurium (99.99% or 4N) is required.<sup>2</sup> This represents a tellurium loss of 20% during this phase of purification.

## 2.6 Life cycle inventory

No LCA-study on tellurium production is available<sup>3</sup>. This module is based on reported values from Knockaert (2002) and (Fthenakis & Kim (2005); Fthenakis (2004)). The functional unit of this process is 1 kilogram of tellurium with a purity of 99.999% (5N). This purity allows a direct use for the production of solar cells and electronic devices. This dataset describes the production and purification of tellurium in Europe, starting from the anode sludge from copper purification.

**Transports:** Knockaert (2002) reported a wide trade with concentrate, blister and anode copper, so an average transport distance of 4000 km by ship was assumed for the anode slime with a 3% concentration of tellurium. The allocation for this transport was done by weight (see Tab. 2.4). The transport to the plant was assumed with standard transport distances.

**Demand in materials and energy:** The input feed is copper-telluride cementate, that is derived from the primary copper production cycle (See Part “Copper” in the corresponding chapter). The telluride

<sup>2</sup> Personal communication: Email from V. Fthenakis to N. Jungbluth, Nov. 2006

<sup>3</sup> Steinberger (1997) analysed the life cycle of CdTe-solar cells, though the assumptions for the production and purification of tellurium were too rough. Therefore no data from the mentioned thesis has been used.

content in the cementate is 50% (Shibasaki et al. 1992), an overall process yield of 90% is assumed. The arising copper rich solution is recycled to the copper electrowinning cells and leaves the system burden free.

During the process of leaching, for each tonne of copper-telluride 150 kg of H<sub>2</sub>SO<sub>4</sub> (25%) is used (Davenport et al. 2002). The demand of energy is reported by Fthenakis (2004). It is assumed, that the energy carrier is electricity. Since there is no specific process-data available for the tellurium production, the data from the corresponding chapters about the hydrometallurgical zinc-production was taken instead for water-use. However, the uncertainty is fairly high.

**Emissions:** Fthenakis (2004) states that “emissions in hydrometallurgical / electrolytic plants are likely to be negligible unless the sulphuric-acid tanks are open to the atmosphere.” For emissions, proxy-data of zinc-production has been used: Assuming a production with 50% open tanks, a value of 9.55g sulphuric acid per kg tellurium was chosen (Xiao et al. (2003), cited in the chapter about zinc-production)

**Infrastructure:** The infrastructure of a chemical plant, organics has been taken as an assumption with the mentioned output of 50'000 tons per year and a lifespan of 50 years.

Tab. 2.5 Demands in Materials and Energy / Emissions, according to literature

|                                                                                               | Unit                                                         | Minimal / Maximal Value | Used Value in this study<br>(geometrical mean) |      |
|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------|-------------------------|------------------------------------------------|------|
| Input                                                                                         |                                                              |                         |                                                |      |
| cadmium-telluride cement, from copper production                                              | kg                                                           | 1 / 0.5 / 90% = 2.22    | 2.22                                           |      |
| cadmium-telluride cement, from copper production, including a loss of 20% during purification | kg                                                           | 2.22/0.8 = 2.775        | 2.775                                          |      |
| electrolysis of tellurium, Electricity<br>Personal communication V. Fthenakis (2006)          | kWh                                                          | 6.04 - 8.22             | 7.13                                           |      |
| Purification of tellurium, Electricity<br>Personal communication V. Fthenakis (2006)          | kWh                                                          | 3.47 - 4.22             | 3.84                                           |      |
| Water, Xiao et al. (2003), cited in the chapter for zinc production                           | kg                                                           | 16.2                    | 16.2                                           |      |
| Emissions to air                                                                              |                                                              |                         |                                                |      |
| SO <sub>2</sub>                                                                               | Xiao et al. (2003), cited in the chapter for zinc production | g                       | 9.55                                           | 9.55 |

Tab. 2.6 Unit process raw data for tellurium

|                                         | Name                                                  | Location | Infrastructure | Process | Unit     | tellurium,<br>semiconductor-<br>grade, at plant | Uncertainty/Type<br>Standard Deviation<br>95% | GeneralComment                                                                            |
|-----------------------------------------|-------------------------------------------------------|----------|----------------|---------|----------|-------------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------|
|                                         |                                                       |          |                |         |          |                                                 |                                               |                                                                                           |
|                                         | Location<br>InfrastructureProcess<br>Unit             |          |                |         |          | GLO<br>0<br>kg                                  |                                               |                                                                                           |
| product                                 | tellurium, semiconductor-grade, at plant              | GLO      | 0              | kg      | 1.00E+0  |                                                 |                                               |                                                                                           |
| technosphere                            | copper-telluride cementate, from copper production    | GLO      | 0              | kg      | 2.78E+0  | 1                                               | 1.17                                          | (2,3,3,1,1,4); Literature (Shibasaki, 1992)                                               |
|                                         | electricity, medium voltage, production UCTE, at grid | UCTE     | 0              | kWh     | 1.09E+1  | 1                                               | 1.38                                          | (4,3,1,3,3,5); Literature, personal communication (Fthenakis)                             |
|                                         | sulphuric acid, liquid, at plant                      | RER      | 0              | kg      | 3.75E-2  | 1                                               | 2.15                                          | (4,5,2,3,5,5); Estimation                                                                 |
|                                         | chemical plant, organics                              | RER      | 1              | unit    | 4.00E-10 | 1                                               | 3.81                                          | (4,5,2,3,5,5); Estimation                                                                 |
|                                         | transport, lorry >16t, fleet average                  | RER      | 0              | tkm     | 1.39E-1  | 1                                               | 2.09                                          | (4,5,na,na,na,na); standard distances (100km), allocation of copper-cementate by weight   |
|                                         | transport, freight, rail                              | RER      | 0              | tkm     | 2.78E-1  | 1                                               | 2.09                                          | (4,5,na,na,na,na); standard distances (200km), allocation of copper-cementate by weight   |
|                                         | transport, transoceanic freight ship                  | OCE      | 0              | tkm     | 5.55E+0  | 1                                               | 2.09                                          | (4,5,na,na,na,na); estimated distances (4000km), allocation of copper-cementate by weight |
|                                         | treatment, sewage, to wastewater treatment, class 5   | CH       | 0              | m3      | 1.62E-2  | 1                                               | 1.60                                          | (2,4,3,3,4,5); Literature, Xiao in Althaus 2004 (extrapolation from zinc)                 |
| resource, in water                      | Water, river                                          | -        | -              | m3      | 1.62E-2  | 1                                               | 3.30                                          | (2,4,3,1,4,5); Literature, Xiao in Althaus 2004 (extrapolation from zinc)                 |
| emission air, low<br>population density | Sulfur dioxide                                        | -        | -              | kg      | 9.55E-3  | 1                                               | 1.60                                          | (2,4,3,1,4,5); Literature, Xiao in Althaus 2004 (extrapolation from zinc)                 |
|                                         | Heat, waste                                           | -        | -              | MJ      | 3.92E+1  | 1                                               | 1.22                                          | (2,3,1,1,1,5); Calculation                                                                |



Tab. 2.7 Ecospold Meta information of "tellurium, semi-conductor-grade, at plant, GLO"

|                    |                            |                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ReferenceFunction  | Name                       | tellurium, semiconductor-grade, at plant                                                                                                                                                                                                                                                                                                                                                                      |
| Geography          | Location                   | GLO                                                                                                                                                                                                                                                                                                                                                                                                           |
| ReferenceFunction  | InfrastructureProcess      | 0                                                                                                                                                                                                                                                                                                                                                                                                             |
| ReferenceFunction  | Unit                       | kg                                                                                                                                                                                                                                                                                                                                                                                                            |
| DataSetInformation | Type                       | 1                                                                                                                                                                                                                                                                                                                                                                                                             |
|                    | Version                    | 2                                                                                                                                                                                                                                                                                                                                                                                                             |
|                    | energyValues               | 0                                                                                                                                                                                                                                                                                                                                                                                                             |
|                    | LanguageCode               | en                                                                                                                                                                                                                                                                                                                                                                                                            |
|                    | LocalLanguageCode          | de                                                                                                                                                                                                                                                                                                                                                                                                            |
| DataEntryBy        | Person                     | 44                                                                                                                                                                                                                                                                                                                                                                                                            |
|                    | QualityNetwork             | 1                                                                                                                                                                                                                                                                                                                                                                                                             |
| ReferenceFunction  | DataSetRelatesToProduct    | 1                                                                                                                                                                                                                                                                                                                                                                                                             |
|                    | IncludedProcesses          | This datasets describes the production of high-grade tellurium (99.999%) from copper-telluride cementate by hydrometallurgical processes. The energy consumption, emissions as well as transports are considered.                                                                                                                                                                                             |
|                    | Amount                     | 1                                                                                                                                                                                                                                                                                                                                                                                                             |
|                    | LocalName                  | Tellur, Halbleiter, ab Werk                                                                                                                                                                                                                                                                                                                                                                                   |
|                    | Synonyms                   |                                                                                                                                                                                                                                                                                                                                                                                                               |
|                    | GeneralComment             | This datasets describes the production of high-grade tellurium (99.999%) from copper-telluride cementate by hydrometallurgical processes. These hydrometallurgical processes are the dissolving of copper-telluride cementate with sulphuric acid, the following electrolysis and subsequent purification with zone-melting. This purity of tellurium can be used in the production of thin film solar cells. |
|                    | InfrastructureIncluded     | 1                                                                                                                                                                                                                                                                                                                                                                                                             |
|                    | Category                   | metals                                                                                                                                                                                                                                                                                                                                                                                                        |
|                    | SubCategory                | extraction                                                                                                                                                                                                                                                                                                                                                                                                    |
|                    | LocalCategory              | Metalle                                                                                                                                                                                                                                                                                                                                                                                                       |
|                    | LocalSubCategory           | Gewinnung                                                                                                                                                                                                                                                                                                                                                                                                     |
|                    | Formula                    | Te                                                                                                                                                                                                                                                                                                                                                                                                            |
|                    | StatisticalClassification  |                                                                                                                                                                                                                                                                                                                                                                                                               |
|                    | CASNumber                  | 13494-80-9                                                                                                                                                                                                                                                                                                                                                                                                    |
| TimePeriod         | StartDate                  | 1992                                                                                                                                                                                                                                                                                                                                                                                                          |
|                    | EndDate                    | 2005                                                                                                                                                                                                                                                                                                                                                                                                          |
|                    | DataValidForEntirePeriod   | 1                                                                                                                                                                                                                                                                                                                                                                                                             |
|                    | OtherPeriodText            | based on literature and patents                                                                                                                                                                                                                                                                                                                                                                               |
| Geography          | Text                       | Production in GLO.                                                                                                                                                                                                                                                                                                                                                                                            |
| Technology         | Text                       | Production of tellurium from copper-telluride cementate (from copper production), purification to 99.999% (5N)                                                                                                                                                                                                                                                                                                |
| Representativeness | Percent                    |                                                                                                                                                                                                                                                                                                                                                                                                               |
|                    | ProductionVolume           | 200 - 300 t in 2005                                                                                                                                                                                                                                                                                                                                                                                           |
|                    | SamplingProcedure          | Analysis based on literature.                                                                                                                                                                                                                                                                                                                                                                                 |
|                    | Extrapolations             | Some data available only for zinc are used as proxy                                                                                                                                                                                                                                                                                                                                                           |
|                    | UncertaintyAdjustments     | none                                                                                                                                                                                                                                                                                                                                                                                                          |
| DataGenerator      | Person                     | 44                                                                                                                                                                                                                                                                                                                                                                                                            |
| AndPublication     | DataPublishedIn            | 2                                                                                                                                                                                                                                                                                                                                                                                                             |
|                    | ReferenceToPublishedSource | 10                                                                                                                                                                                                                                                                                                                                                                                                            |
|                    | Copyright                  | 1                                                                                                                                                                                                                                                                                                                                                                                                             |
|                    | AccessRestrictedTo         | 0                                                                                                                                                                                                                                                                                                                                                                                                             |
|                    | CompanyCode                |                                                                                                                                                                                                                                                                                                                                                                                                               |
|                    | CountryCode                |                                                                                                                                                                                                                                                                                                                                                                                                               |
|                    | PageNumbers                | Tellurium                                                                                                                                                                                                                                                                                                                                                                                                     |
| ProofReading       | Validator                  | 58                                                                                                                                                                                                                                                                                                                                                                                                            |
|                    | Details                    | automatic validation in Excel                                                                                                                                                                                                                                                                                                                                                                                 |
|                    | OtherDetails               | none                                                                                                                                                                                                                                                                                                                                                                                                          |

## 3 Gallium, semiconductor-grade

### 3.1 Introduction

Gallium (CAS-Number: 7440-55-3) is a metallic element, similar in many characteristics to aluminium. Gallium is an excellent conductor of heat and electricity and has a melting point of 29.8° C. It can be extracted from bauxite ore by several different processes. Gallium is widely used in the field of optoelectronics (e.g. LEDs), telecommunication, aerospace and many commercial and household items such as alloys, computers and DVDs (Moskalyk 2003).

### 3.2 Resources and production of gallium

Gallium is widely spread on Earth, but is rarely found in concentrations greater than 0.1%. Gallium-containing minerals do not have an economic significance. Gallium can be extracted as a trace component in bauxite, coal or several minerals. Gutiérrez et al. (1997) investigated the extraction of gallium from coal fly ash from power plants. Occasionally, high concentrations of gallium are found in association with polymetallic deposits (e.g. copper, zinc, silver).

However, the dominant world production of gallium is as a by-product from the production of aluminium oxide. Most gallium is recovered from the crude aluminium hydroxide (Guilinger 1999c). The U.S. Geological Survey estimates the world gallium reserves in known reserves of bauxite (concentration of 50 ppm) and zinc ores of about 1 million-ton gallium. France is the largest producer of refined gallium in the world (Moskalyk 2003).

Tab. 3.1 Production capacity and production in 2005, Source: Guilinger (1999c)

| Country         | Production capacity<br>t | Production 2005 <sup>4</sup><br>t | Market Share |
|-----------------|--------------------------|-----------------------------------|--------------|
| Australia       | 50                       | 21.3                              | 34%          |
| Germany         | 20                       | 8.5                               | 14%          |
| Kazakhstan      | 20                       | 8.5                               | 14%          |
| Russia          | 19                       | 8.1                               | 13%          |
| China           | 8                        | 3.4                               | 5%           |
| Hungary         | 8                        | 3.4                               | 5%           |
| Slovakia        | 8                        | 3.4                               | 5%           |
| Japan           | 7                        | 3.0                               | 5%           |
| Ukraine         | 5                        | 2.1                               | 3%           |
| United States   | 3                        | 1.3                               | 2%           |
| Worldwide       | 148                      | 63                                | 100%         |
| Company average | 16 companies             | 3.9 t / a                         |              |

### 3.3 Characterisation of gallium

Before its discovery, most of the properties of gallium have been predicted and described by Mendeleev on the basis of its position in his periodic table. The French Lecoq de Boisbaudran discovered

<sup>4</sup> Estimation (based on the world production in 2005 of 63 tons of gallium, assuming that each enterprise has the same ratio of production / production capacity)

gallium spectroscopically in 1875 (Latin *Gallia* meaning Gaul, essentially modern France) (Guilinger 1999c).

**Tab. 3.2 Physical properties of gallium, Source: Greber (2005) and Moskalyk (2003)**

| Chemical Symbol     | Ga                                                                                          |
|---------------------|---------------------------------------------------------------------------------------------|
| Atomic mass         | 69.723 g/mol                                                                                |
| Density             | 5.91 g·cm <sup>-3</sup>                                                                     |
| Melting point       | 29.76°C                                                                                     |
| Boiling point       | 2204°C                                                                                      |
| Heat capacity       | 25.86 J·mol <sup>-1</sup> ·K <sup>-1</sup> = 370 J·kg <sup>-1</sup> ·K <sup>-1</sup> (25°C) |
| Mohs hardness       | 1.5                                                                                         |
| Heat of fusion      | 5.59 kJ/mol                                                                                 |
| CAS registry number | 7440-55-3                                                                                   |

### 3.4 Use and application of material

About 95% of the gallium consumed annually (57 t) is in the form of gallium arsenide (GaAs) compounds, which are used in optoelectronic devices (e.g. LEDs = light emitting diodes, laser diodes, photo detectors and solar cells) or in integrated circuits. The current price for 1 kg gallium of 99.9999% purity was at the end of the year 2005 about 512 US\$ (Kramer 2005).

**Tab. 3.3 Market share of gallium in United States, Source: Kramer (2005)**

|                                        | Gallium (kg) | %     |
|----------------------------------------|--------------|-------|
| Optoelectronic devices                 |              |       |
| Laser diodes and light-emitting diodes | 3880         | 20.6% |
| Photo detectors and solar cells        | 219          | 1.2%  |
| Integrated circuits:                   |              |       |
| Analogue                               | 11700        | 62.3% |
| Digital                                | 115          | 0.6%  |
| Research and development               | 2800         | 14.9% |
| Other                                  | 76           | 0.4%  |
| Total                                  | 18700        | 100%  |

### 3.5 Production process

#### 3.5.1 Extraction from aluminium oxide liquor

This part is mainly based on Greber (2005) and Kramer (2005). The most important process for gallium production is the extraction of the metal from the Bayer liquor in aluminium oxide plants. The liquor contains 70 – 150 mg / litre of gallium, depending on the bauxite and the concentration of the caustic liquor. Selvi et al. (2004) reported a typical Bayer liquor as shown in Tab. 3.4.

Tab. 3.4. Composition of Bayer liquor, according to Selvi et al. (2004)

|                         |                        |
|-------------------------|------------------------|
| Sodium Hydroxide (NaOH) | 220 g / litre          |
| Aluminium               | 60 g / litre           |
| Gallium                 | 145 mg / litre         |
| Vanadium                | 130 mg / litre         |
| Viscosity               | 5.75 Cp (at 28°C)      |
| Specific Gravity        | 1.35 g/cm <sup>3</sup> |

The gallium extraction plant is usually located within or very near to the aluminium oxide plants. Fig. 3.1. shows the gallium extraction within the chain of aluminium production.

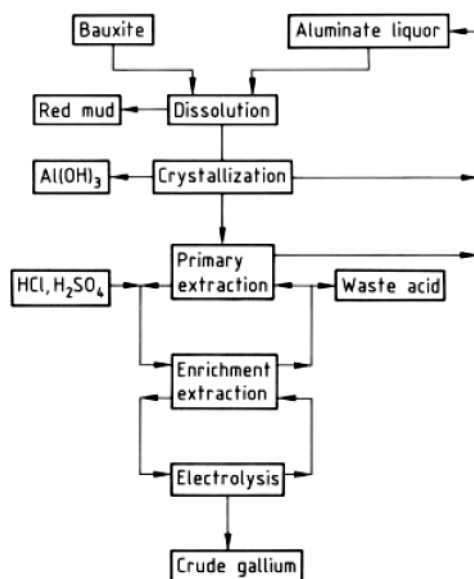


Fig. 3.1 Flow diagram showing gallium extraction by Rhône-Poulenc-Process, Source: Greber (2005)

Three types of processes are mainly used to extract the gallium. Below is a short summary of each process. A description in more details can be found in Greber (2005) and Power Porto (1990).

(1) **extraction with chelating agents (the Rhône-Poulenc-Process)**

In 1974, it was discovered, that hydroxyquinolines could be used to extract gallium from sodium aluminates liquors. The aluminate liquor is brought into contact with a solution of Kelex 100<sup>5</sup> in kerosene. Gallium is thereby extracted from the liquor along with aluminium and sodium. Most of the aluminium and sodium is scrubbed from the extraction by a treatment with dilute acid. The gallium itself is then stripped with concentrated sulphuric acid. The metal can be isolated from this concentrate by direct electrolysis. Although the process solutions are recycled, considerable quantities of waste acid is generated (Greber 2005). 35 to 40 tons of gallium is produced via this route.

(2) **electrolytic processes (the Alusuisse-Process)**

The electrolytic process depends on the fact that mercury forms an amalgam with gallium

<sup>5</sup> Kelex100 is a well-known solvent extraction product, chemical name: 7-(4-ethyl-1-methyloctyl)quinolin-8-ol CAS-Number: 29171-27-5

from which the gallium may be extracted with a caustic soda solution. The mercury is recycled to produce again sodium amalgam. Hence the mercury circulates in a closed system. This process requires special conditions to reach economically acceptable values (like > 150 g/l sodium and very low concentrations of vanadium, tungsten and molybdenum). The environmental problems associated with mercury further discourage the use of electrolytic processes.

**(3) fractional precipitation (the P  chiney-Process)**

The gallium-containing sodium aluminate solution is treated with carbon dioxide to precipitate its pure aluminium hydroxide. This increases the gallium content of the liquor. Further treatment with fractional precipitation is used to separate the gallium. Finally metallic gallium is obtained by electrolysis. This process is labour and energy-intensive. China produces gallium by this process.

Izumi (2004) describes some laboratory-scale processes about the re-extraction and recycling of gallium from electronic scrap. However, no recycling processes are taken into account since there is no data available.

Since the Rh  ne-Poulenc-Process dominates the production and is the least-cost effective, the focus of the following life-cycle inventory was set on this process (Moskalyk 2003).

### **3.5.2 Purification**

Different procedures can be used to remove the impurities such as mercury and zinc:

- vacuum distillation
- washing with aqueous acids and alkalis
- fractional distillation of liquid gallium compounds
- fractional crystallization
- single crystal growth

The purest gallium is obtained by fractional crystallization (Meijer et al. 2003) or single crystal growth with a purity of 99.999999% (Power Porto 1990). The fractional crystallization process purifies a raw gallium material utilizing segregation. Since solid gallium is lighter than liquid gallium, the solid gallium floats in the form of a small mass on the surface of the liquid gallium. Seed crystals are present in a melted raw gallium material and at surface. The melted raw gallium material is cooled with a gas to a temperature less than its melting point. The gallium material is then solidified on the seed crystals as nuclei with unidirect solidification (Maeda & Kadokura 1993). Bautista (2003) gives an overview about the different purification methods.

## **3.6 Life cycle inventories**

### **3.6.1 Gallium, in Bayer liquor from aluminium production, at plant**

The production process of gallium starts from the liquid and crystallized Bayer-Liquor. Although there is a production capacity of 160 t available, only 60 t are produced annually. This indicates a low economic interest of gallium production at the moment. Therefore, no upstream processes are taken into account. This is in line with the proposed methodology of (Guin  e et al. 2004).

### **3.6.2 Gallium, semiconductor-grade**

**Production (Accumulation):** The following data from Bautista (2003) is from a laboratory-scale installation. Selvi et al. (2004) reports that chelating agents can be reused more than 30 times in laboratory scale installations. In other settings (different types of chelating agents) more than 10 cycles have been found. This study assumes, that the chelating agent (Kelex 100) can be used 30 times and other

organic solvents can be re-used for one year. Since the average production is about 3.9 t per company and year, the calculation has been made with a recycling-rate of 3900 times at a production rate of 1 kg per cycle. It is assumed, that sulphuric acid is taken for stripping.

Krauss et al. (1999) reports for the “mixing and pumping of diluted leach, extracting agent and old electrolyte” in the copper-extraction an energy consumption of 350 kWh per t. Therefore this value has been taken as a proxy.

**Electrolysis (Reduction):** The resulting liquor after solvent extraction has an average gallium concentration of 145 g / litre. From this solvent, the gallium can be directly extracted via electrolysis. Since of missing data for electro winning, the related value for copper has been used: 2.22 kWh per kilogram of Gallium (Dresher 2001).

Meijer et al. (2003) assumed a material efficiency of 90% for electrolysis and purification (including partial reuse of waste), therefore an overall efficiency of 81% was assumed. However, the uncertainty is relatively high.

**Tab. 3.5 Key-figures of solvent extraction**

|                                | Unit        | Amount | Remark                                              |
|--------------------------------|-------------|--------|-----------------------------------------------------|
| <i>Bayer Liquor</i>            |             |        |                                                     |
| Gallium concentration          | g Ga/l      | 0.145  | Selvi et al. (2004)                                 |
| Process efficiency             | %           | 81%    | Meijer et al. (2003)                                |
| Bayer Liquor consumption       | l/kg Ga     | 8514   |                                                     |
| <i>Solvents</i>                |             |        |                                                     |
| Bayer Liquor to solvent ratio  |             | 1:1    | Selvi et al. (2004)                                 |
| Solvent consumption            | l/kg Ga     | 8514   |                                                     |
| <i>Production</i>              |             |        |                                                     |
| Average production per company | kg/a        | 3900   | Guilinger (1999c)                                   |
| Reuse of solvents              | cycles      | 3900   | Solvents are reused for one year (except Kelex 100) |
| Reuse of Kelex 100             | cycles      | 30     | Selvi et al. (2004)                                 |
| Average production per cycle   | kg Ga/Cycle | 1      | Assumption                                          |

**Tab. 3.6 Consumption of solvent and acid, Source: Selvi et al. (2004)**

|                                      |          |              | Re-use     | per kg Gallium |
|--------------------------------------|----------|--------------|------------|----------------|
| Total of solvent                     | 100 vol% | 8514 litre   | -          | -              |
| Kelex 100                            | 10 vol%  | (851 litre)  | 30-times   | 28.4 litre     |
| Versatic 10                          | 5 vol%   | (426) litre  | 3900 times | 0.11 litre     |
| n-decanol                            | 8 vol%   | (681) litre  | 3900 times | 0.17 litre     |
| Kerosene (as diluent)                | 77 vol%  | (6556) litre | 3900 times | 1.68 litre     |
| Acid: H <sub>2</sub> SO <sub>4</sub> | 100 vol% | (8514) litre | 3900 times | 2.18 litre     |

**Transports:** As the installations for gallium extraction are within the aluminium-plants, no transport of the Bayer liquor has been accounted for. Since it is assumed, that the production takes place in Europe, the transport of solvents and limestone is calculated with the standard distance from ecoinvent (see Tab. 3.7).

Tab. 3.7 Standard distances for transports of solvents and acid

| Type of transport | Distance                      | Source:                    |
|-------------------|-------------------------------|----------------------------|
| Railway           | 200 km in Europe (production) | Frischknecht et al. (2004) |
| Lorry             | 100 km in Europe (production) | Frischknecht et al. (2004) |

**Purification:** Meijer et al. (2003) describes the process of fractional crystallization for producing semiconductor grade gallium. The theoretical values for fractional crystallization are estimated with the energy needed for heating the metal from room temperature to melting-temperature and the heat of fusion. It is assumed that this process has an energy efficiency of 50%. No other purification methods have been taken into account.

Tab. 3.8 Physical properties of gallium to calculate the process of purification

|                        |                                                                                                                                                  |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Heat of fusion per mol | 5,59 kJ/mol                                                                                                                                      |
| Heat of fusion per kg  | 80.2 kJ/kg                                                                                                                                       |
| Heat Capacity per mol  | 25.86 J·mol <sup>-1</sup> ·K <sup>-1</sup> (25°C)                                                                                                |
| Weight                 | 69.7 g/mol                                                                                                                                       |
| Heat Capacity per mol  | 371 J·kg <sup>-1</sup> ·K <sup>-1</sup> = 0.371 kJ·kg <sup>-1</sup> ·K <sup>-1</sup>                                                             |
| Melting point          | 29.7° C                                                                                                                                          |
| Room temperature       | 20° C                                                                                                                                            |
| Calculation            | $(9.7^{\circ}\text{C} \cdot 0.371\text{ kJ}\cdot\text{kg}^{-1}\cdot\text{K}^{-1} + 80.2\text{ kJ}\cdot\text{kg}^{-1}) / 0.5 = 0.0465\text{ kWh}$ |

**Infrastructure:** As there is no specific data about the gallium production available, proxy-data has been used: For the processes of enrichment and extraction, the infrastructure of “chemical plant, organics” is considered, the electrolysis takes place in the plant of “aluminium electrolysis”.

**Waste and Emissions:** No specific data is available. Richardson (2002) reports a typical loss of 0.05% of feedstock on refinery storages. Therefore a yearly loss of 0.05% \* 8520 l = 4.26 litre of organic solvent is emitted to the air. It is assumed, that 50% of this stream consists of NMVOC. The sulphuric acid is neutralized with calcium carbonate, each Mol of calcium carbonate neutralize 1 mol of sulphuric acid. The worst-case scenario represents the fact, that carbonate and sulphate are emitted as ions into the water. It is more likely, that they will be precipitated as salt.

It is assumed, that solvents as Kelex 100, Versatic 10, n-decanol or Kerosene, which have to be replaced, are sent to hazardous waste incineration.

### 3.6.3 Gallium, at regional storage

Since the production of gallium take place all around the world, the transport distance to regional storage has been estimated on the base of Kazakhstan (Russia and Kazakhstan produce 27% of worldwide gallium production) to Europe. Gallium is with an average price of 500US\$ per kilogram a precious metal, therefore the transport was considered with airplane over a distance of 4300km (Distance: Kazakhstan-Moskau-Germany)

Tab. 3.9 Standard distances for transports of solvents and acid

| Type of transport | Distance                               | Source:                                 |
|-------------------|----------------------------------------|-----------------------------------------|
| Lorry             | 300 km in Europe (to regional storage) | Adapted from Frischknecht et al. (2004) |
| Airplane          | 4300km from Kazakhstan to Germany      | Estimation                              |

#### **3.6.4 Data Quality**

Data for the steps in the gallium production are roughly estimated. The values for the production and purification steps stem from older experimental sources. The proxy chosen to represent the electrolysis however is decreasing the overall quality.



Tab. 3.10 Unit process raw data for gallium, in Bayer liquor from aluminium production, at plant

|                     | Name                                      | Location | Category | SubCategory | Infrastructure | Unit | gallium, in Bayer liquor from aluminium production, at plant | Uncertainty Type | Standard Deviation | GeneralComment                                    |
|---------------------|-------------------------------------------|----------|----------|-------------|----------------|------|--------------------------------------------------------------|------------------|--------------------|---------------------------------------------------|
|                     | Location<br>InfrastructureProcess<br>Unit |          |          |             |                |      | GLO<br>0<br>kg                                               |                  |                    |                                                   |
| product             | gallium, in Bayer liquor from aluminium   | GLO      | -        | -           | 0              | kg   | 1                                                            |                  |                    |                                                   |
| resource, in ground | gallium, 0.014% in bauxite, in ground     | -        | resource | in ground   | -              | kg   | 1.00E+0                                                      | 1                | 1.05               | (1,1,1,1,1,1); data from literature (Selvi, 2004) |

Tab. 3.11 Unit process raw data for gallium, semiconductor-grade, at plant

|                                       | Name                                                                     | Location | Category | SubCategory             | InfrastructureProcess | Unit | gallium, semiconductor-grade, at plant | Uncertainty Type | Standard Deviation | GeneralComment                                                             |
|---------------------------------------|--------------------------------------------------------------------------|----------|----------|-------------------------|-----------------------|------|----------------------------------------|------------------|--------------------|----------------------------------------------------------------------------|
|                                       | Location<br>InfrastructureProcess<br>Unit                                |          |          |                         |                       |      | GLO<br>0<br>kg                         |                  |                    |                                                                            |
| product                               | gallium, semiconductor-grade, at plant                                   | GLO      | -        | -                       | 0                     | kg   | 1.00E+0                                |                  |                    |                                                                            |
| technosphere                          | gallium, in Bayer liquor from aluminium production, at plant             | GLO      | -        | -                       | 0                     | kg   | 1.23E+0                                | 1                | 1.59               | (2,4,2,1,4,5); Literature (Selvi, 2004)                                    |
|                                       | electricity, medium voltage, production UCTE, at grid                    | UCTE     | -        | -                       | 0                     | kWh  | 2.63E+0                                | 1                | 1.64               | (4,4,2,2,4,5); Literature (Dresher 2001) & (Meijer 2003)                   |
|                                       | sulphuric acid, liquid, at plant                                         | RER      | -        | -                       | 0                     | kg   | 2.18E+0                                | 1                | 1.84               | (5,4,2,2,4,5); Literature (Bautista 2003)                                  |
|                                       | lime, hydrated, packed, at plant                                         | CH       | -        | -                       | 0                     | kg   | 2.24E+0                                | 1                | 1.33               | (3,3,2,2,3,5); Calculation (for neutralization of sulphuric acid)          |
|                                       | kerosene, at regional storage                                            | RER      | -        | -                       | 0                     | kg   | 1.68E+0                                | 1                | 1.84               | (5,4,2,2,4,5); Literature (Bautista 2003)                                  |
|                                       | fatty alcohol, petrochemical, at plant                                   | RER      | -        | -                       | 0                     | kg   | 1.70E-1                                | 1                | 1.64               | (4,4,2,2,4,5); Literature (Bautista 2003), similar to n-decanol            |
|                                       | EDTA, ethylenediaminetetraacetic acid, at plant                          | RER      | -        | -                       | 0                     | kg   | 2.84E+1                                | 1                | 1.84               | (5,4,2,2,4,5); Literature (Bautista 2003), sas a proxy value to Kelex 100  |
|                                       | adipic acid, at plant                                                    | RER      | -        | -                       | 0                     | kg   | 1.10E-1                                | 1                | 1.84               | (5,4,2,2,4,5); Literature (Bautista 2003), as a proxy value to Versatic 10 |
|                                       | chemical plant, organics                                                 | RER      | -        | -                       | 1                     | unit | 4.00E-10                               | 1                | 3.76               | (4,3,2,2,5,5); Estimation                                                  |
|                                       | aluminium electrolysis, plant                                            | RER      | -        | -                       | 1                     | unit | 7.69E-9                                | 1                | 3.76               | (4,3,2,2,5,5); Estimation                                                  |
|                                       | transport, freight, rail                                                 | RER      | -        | -                       | 0                     | tkm  | 6.96E+0                                | 1                | 2.09               | (4,5,na,na,na,na); standard distances (200km)                              |
|                                       | transport, lorry >16t, fleet average                                     | RER      | -        | -                       | 0                     | tkm  | 3.48E+0                                | 1                | 2.09               | (4,5,na,na,na,na); standard distances (100km)                              |
|                                       | disposal, solvents mixture, 16.5% water, to hazardous waste incineration | CH       | -        | -                       | 0                     | kg   | 3.04E+1                                | 1                | 1.33               | (3,3,2,1,3,5); Calculation                                                 |
| emission air, high population density | Heat, waste                                                              | -        | air      | high population density | -                     | MJ   | 9.46E+0                                | 1                | 1.33               | (3,3,2,1,3,5); Calculation                                                 |
|                                       | NMVOC, non-methane volatile organic compounds, unspecified origin        | -        | air      | high population density | -                     | kg   | 1.09E-3                                | 1                | 1.33               | (3,3,2,1,3,5); Estimation, Literature (Richardson 2002)                    |
| emission water, river                 | Calcium, ion                                                             | -        | water    | river                   | -                     | kg   | 1.12E+0                                | 1                | 1.33               | (3,3,2,1,3,5); Calculation                                                 |
|                                       | Sulfate                                                                  | -        | water    | river                   | -                     | kg   | 2.18E+0                                | 1                | 1.33               | (3,3,2,1,3,5); Calculation                                                 |

Tab. 3.12 Unit process raw data for gallium, semiconductor-grade, at regional storage

| product      | Name                                              | Location | Category | SubCategory | InfrastructureProcess | Unit | gallium, semiconductor-grade, at regional storage |   |      | UncertaintyType                                                     |  | StandardDeviation |  | GeneralComment |  |
|--------------|---------------------------------------------------|----------|----------|-------------|-----------------------|------|---------------------------------------------------|---|------|---------------------------------------------------------------------|--|-------------------|--|----------------|--|
|              | Location<br>InfrastructureProcess<br>Unit         |          |          |             |                       |      | RER                                               | 0 | kg   |                                                                     |  |                   |  |                |  |
| technosphere | gallium, semiconductor-grade, at regional storage | RER      | -        | -           | 0                     | kg   | 1.00E+0                                           |   |      |                                                                     |  |                   |  |                |  |
|              | gallium, semiconductor-grade, at plant            | GLO      | -        | -           | 0                     | kg   | 1.00E+0                                           | 1 | 1.21 | (1,1,2,2,1,5); Calculation                                          |  |                   |  |                |  |
|              | transport, lorry >16t, fleet average              | RER      | -        | -           | 0                     | tkm  | 3.00E-1                                           | 1 | 2.09 | (4,5,na,na,na,na); estimated distances, based on standard distances |  |                   |  |                |  |
|              | transport, aircraft, freight                      | RER      | -        | -           | 0                     | tkm  | 4.30E+0                                           | 1 | 2.28 | (5,5,na,na,na,na); Estimation (4300km)                              |  |                   |  |                |  |

Tab. 3.13 EcoSpold Meta Information of gallium production

| ReferenceFunction  | Name                       | gallium, in Bayer liquor from aluminium production, at plant                                                                                                                                                                                                                                      | gallium, semiconductor-grade, at plant                                                                                                        | gallium, semiconductor-grade, at regional storage                                                                            |
|--------------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Geography          | Location                   | GLO                                                                                                                                                                                                                                                                                               | GLO                                                                                                                                           | RER                                                                                                                          |
| ReferenceFunction  | InfrastructureProcess      | 0                                                                                                                                                                                                                                                                                                 | 0                                                                                                                                             | 0                                                                                                                            |
| ReferenceFunction  | Unit                       | kg                                                                                                                                                                                                                                                                                                | kg                                                                                                                                            | kg                                                                                                                           |
| DataSetInformation | Type                       | 1                                                                                                                                                                                                                                                                                                 | 1                                                                                                                                             | 1                                                                                                                            |
|                    | Version                    | 2.0                                                                                                                                                                                                                                                                                               | 2.0                                                                                                                                           | 2                                                                                                                            |
|                    | energyValues               | 0                                                                                                                                                                                                                                                                                                 | 0                                                                                                                                             | 0                                                                                                                            |
|                    | LanguageCode               | en                                                                                                                                                                                                                                                                                                | en                                                                                                                                            | en                                                                                                                           |
|                    | LocalLanguageCode          | de                                                                                                                                                                                                                                                                                                | de                                                                                                                                            | de                                                                                                                           |
| DataEntryBy        | Person                     | 44                                                                                                                                                                                                                                                                                                | 44                                                                                                                                            | 44                                                                                                                           |
|                    | QualityNetwork             | 1                                                                                                                                                                                                                                                                                                 | 1                                                                                                                                             | 1                                                                                                                            |
| ReferenceFunction  | DataSetRelatesToProduct    | 1                                                                                                                                                                                                                                                                                                 | 1                                                                                                                                             | 1                                                                                                                            |
|                    | IncludedProcesses          | This data set describes the simultaneous extraction of gallium together with aluminium in the Bayer process of the aluminium production.                                                                                                                                                          | The energy consumption and auxiliary materials are considered for, emissions are estimated.                                                   | The module includes the production from gallium as well as the transport of gallium from global production places to europe. |
|                    | Amount                     | 1                                                                                                                                                                                                                                                                                                 | 1                                                                                                                                             | 1                                                                                                                            |
|                    | LocalName                  | Gallium, in Bayer-Liquor während Aluminiumherstellung, ab Werk                                                                                                                                                                                                                                    | Gallium, Halbleiter, ab Werk                                                                                                                  | Gallium, Halbleiter, ab Regionallager                                                                                        |
|                    | Synonyms                   |                                                                                                                                                                                                                                                                                                   |                                                                                                                                               |                                                                                                                              |
|                    | GeneralComment             | This dataset is designed for accounting for resource depletion in the gallium production. As gallium is assumed to be a by-product of the aluminium production, no upstream burden is allocated to the gallium solved in the bayer liquor. Do not use this dataset for the production of gallium. | Production of gallium from Bayer-liquor, purification to 99.9999% (6N)                                                                        | Transport of gallium to european regional storage, this module calls the module "gallium, semiconductor-grade, at plant".    |
|                    | InfrastructureIncluded     | 1                                                                                                                                                                                                                                                                                                 | 1                                                                                                                                             | 1                                                                                                                            |
|                    | Category                   | metals                                                                                                                                                                                                                                                                                            | metals                                                                                                                                        | metals                                                                                                                       |
|                    | SubCategory                | extraction                                                                                                                                                                                                                                                                                        | extraction                                                                                                                                    | extraction                                                                                                                   |
|                    | LocalCategory              | Metalle                                                                                                                                                                                                                                                                                           | Metalle                                                                                                                                       | Metalle                                                                                                                      |
|                    | LocalSubCategory           | Gewinnung                                                                                                                                                                                                                                                                                         | Gewinnung                                                                                                                                     | Gewinnung                                                                                                                    |
|                    | Formula                    | Ga                                                                                                                                                                                                                                                                                                | Ga                                                                                                                                            | Ga                                                                                                                           |
|                    | StatisticalClassification  |                                                                                                                                                                                                                                                                                                   |                                                                                                                                               |                                                                                                                              |
|                    | CASNumber                  | 7440-55-3                                                                                                                                                                                                                                                                                         | 7440-55-3                                                                                                                                     | 7440-55-3                                                                                                                    |
| TimePeriod         | StartDate                  | 2005                                                                                                                                                                                                                                                                                              | 1999                                                                                                                                          | 2005                                                                                                                         |
|                    | EndDate                    | 2005                                                                                                                                                                                                                                                                                              | 2005                                                                                                                                          | 2005                                                                                                                         |
|                    | DataValidForEntirePeriod   | 1                                                                                                                                                                                                                                                                                                 | 1                                                                                                                                             | 1                                                                                                                            |
|                    | OtherPeriodText            |                                                                                                                                                                                                                                                                                                   | patents and literatur                                                                                                                         |                                                                                                                              |
| Geography          | Text                       | Data from Literature                                                                                                                                                                                                                                                                              | Production in GLO.                                                                                                                            | RER                                                                                                                          |
| Technology         | Text                       | dissolution of gallium during the Bayer-Process for aluminium production                                                                                                                                                                                                                          | The module includes the production and purification of high grade gallium (99.9999%) by hydrometallurgical processes (Rhône-Poulenc-Process). | Transport                                                                                                                    |
| Representativeness | Percent                    |                                                                                                                                                                                                                                                                                                   | 63.00                                                                                                                                         |                                                                                                                              |
|                    | ProductionVolume           | unknown                                                                                                                                                                                                                                                                                           | 60 t in 2005                                                                                                                                  | 60 t in 2005                                                                                                                 |
|                    | SamplingProcedure          | estimation                                                                                                                                                                                                                                                                                        | Analysis based on literature.                                                                                                                 | Analysis based on literature.                                                                                                |
|                    | Extrapolations             | none                                                                                                                                                                                                                                                                                              | For the electrolysis, the proxy-value of copper is used, the infrastructure is estimated with related chemical plants                         | none                                                                                                                         |
|                    | UncertaintyAdjustments     | none                                                                                                                                                                                                                                                                                              | none                                                                                                                                          | none                                                                                                                         |
| DataGenerator      | Person                     | 44                                                                                                                                                                                                                                                                                                | 44                                                                                                                                            | 44                                                                                                                           |
| AndPublication     | DataPublishedIn            | 2                                                                                                                                                                                                                                                                                                 | 2                                                                                                                                             | 2                                                                                                                            |
|                    | ReferenceToPublishedSource | 10                                                                                                                                                                                                                                                                                                | 10                                                                                                                                            | 10                                                                                                                           |
|                    | Copyright                  | 1                                                                                                                                                                                                                                                                                                 | 1                                                                                                                                             | 1                                                                                                                            |
|                    | AccessRestrictedTo         | 0                                                                                                                                                                                                                                                                                                 | 0                                                                                                                                             | 0                                                                                                                            |
|                    | CompanyCode                |                                                                                                                                                                                                                                                                                                   |                                                                                                                                               |                                                                                                                              |
|                    | CountryCode                |                                                                                                                                                                                                                                                                                                   |                                                                                                                                               |                                                                                                                              |
|                    | PageNumbers                | Gallium                                                                                                                                                                                                                                                                                           | Gallium                                                                                                                                       | Gallium                                                                                                                      |
| ProofReading       | Validator                  | 58                                                                                                                                                                                                                                                                                                | 58                                                                                                                                            | 58                                                                                                                           |
|                    | Details                    | automatic validation in Excel                                                                                                                                                                                                                                                                     | automatic validation in Excel                                                                                                                 | automatic validation in Excel                                                                                                |
|                    | OtherDetails               | none                                                                                                                                                                                                                                                                                              | none                                                                                                                                          | none                                                                                                                         |

## 4 Cadmium purification and cadmium compounds

### Acknowledgement

We thank Vasilis Fthenakis for the information provided in a personal communication.

### 4.1 Introduction and Overview

In thin-film solar cells, cadmium compounds like cadmium telluride, cadmium sulphide or cadmium chloride are used for different tasks (photovoltaic junction, buffer-layer or back contact). For a more in-depth explanation about the use of cadmium compounds and the physical principles in a thin-film solar cell, see Karpov et al. (2005). Cadmium compounds are also used in other fields, mainly for electronic devices, detectors and pigments for painting.

Fig. 4.1 shows an overview of the inventoried cadmium compounds for solar industry, including the purification of cadmium as an intermediate step. All compounds are of high-purity (99.99 % to 99.9999 %), according to the requirements of the thin-film-solar cell manufacturers.

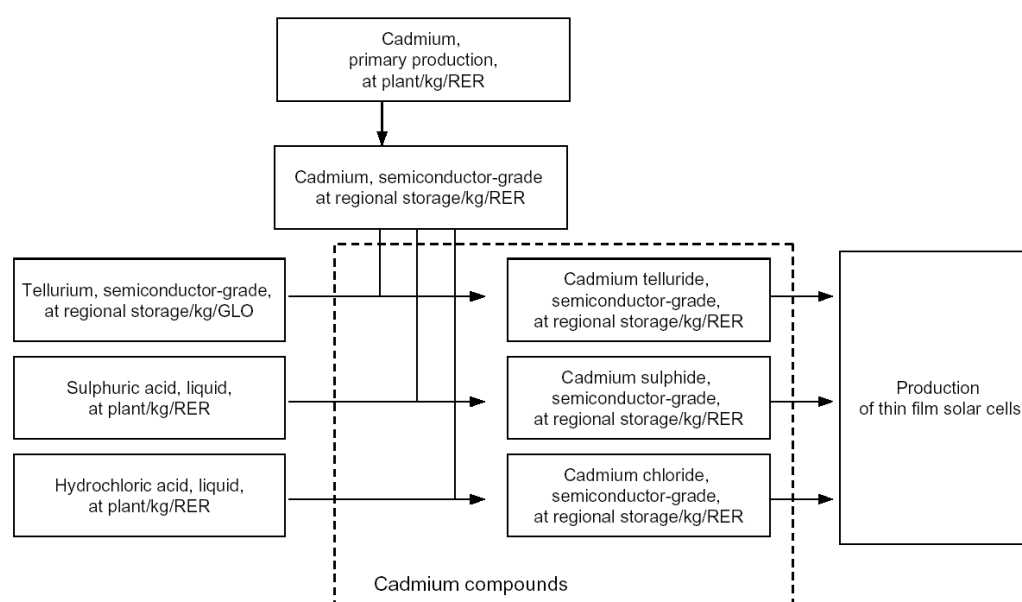


Fig. 4.1 Overview of the inventoried cadmium compounds

The cadmium compounds consist of the elementary materials of cadmium, tellurium, sulphur and chlor. Therefore no reserves or resources are mentioned here, estimations about the resources can be found in the corresponding chapters.

### 4.2 Characterisation of cadmium compounds

The appearance of cadmium compounds is in the range of dark black, grey-metallic up to yellow-orange, depending on the bounded element to the cadmium. All cadmium compounds are toxic.

**Tab. 4.1 Physical properties of Cadmium compounds, Source: ESPI Metals (2007) and Schulte-Schrepping & Piscator (2002)**

| Name                   | Cadmium                                              | Cadmium telluride                                    | Cadmium sulphide       | Cadmium chloride               |
|------------------------|------------------------------------------------------|------------------------------------------------------|------------------------|--------------------------------|
| Molecular formula      | Cd                                                   | CdTe                                                 | CdS                    | CdCl <sub>2</sub>              |
| Appearance             | silvery grey metallic                                | Black cubic crystalline powder                       | Yellow-orange powder   | white solid hygroscopic powder |
| Melting Point          | 321 °C                                               | 1041 °C                                              | 1750°C (100 bar)       | 564°C                          |
| Boiling Point          | 767 °C                                               | 1130 °C                                              | 980°C                  | 960°C                          |
| Specific Gravity       | 8.65 g/cm <sup>3</sup>                               | 6.2 g/cm <sup>3</sup>                                | 4.82 g/cm <sup>3</sup> | 4.08 g/cm <sup>3</sup>         |
| CAS-Number             | 744-43-9                                             | 1306-25-8                                            | 1306-23-6              | 10108-64-2                     |
| Molecular Weight       | 112.4 g/mol                                          | 240 g/mol                                            | 144.5 g/mol            | 183.3 g/mol                    |
| Chemical Nature        | metallic                                             | inorganic                                            | inorganic              | inorganic                      |
| Solubility in Water    | insoluble                                            | insoluble                                            | insoluble              | in hydrochloric acid           |
| Specific heat capacity | 213 J kg <sup>-1</sup> K <sup>-1</sup><br>(at 293 K) | 210 J kg <sup>-1</sup> K <sup>-1</sup><br>(at 293 K) | -                      | -                              |
| Enthalpy               | 99.87 kJ·mol <sup>-1</sup>                           |                                                      |                        |                                |

## 4.3 Production and Use of cadmium compounds

### 4.3.1 Cadmium, semi-conductor grade

High purity cadmium metal is essential for the preparation of compound semiconductors such as CdTe, HgCdTe, CdZnTe, CdS or CdSe. These semiconductor devices are primarily used as solar cells, IR detectors, imaging devices, electro-optic modulators, fluorescence and thermoelectric coolers. The production of semiconductor compounds requires high-purified (6N and above) cadmium metal (Ali et al. 2004).

### 4.3.2 Cadmium telluride (CdTe)

CdTe is a basic component of solar cells and is used as a several micrometers thin coating (Guilinger (1999b)). Another application is the use of CdTe for optical infrared devices (alloyed with mercury) or a solid-state x-ray and gamma ray detector (alloyed with zinc).

Menezes (2001), Izumi (2004) and Fthenakis (2000) describe the recycling of CdTe from photovoltaic-waste. However, cadmium compounds are not recycled on an industrial-scale and are, therefore, not accounted for here.

Four companies in the United States, two in Europe and one in Japan produce CdTe currently. Their monthly output varies between 45 kg to 270 kg (Guilinger 1999b). The shape of the produced CdTe is either in form of fragments (<1.5mm) or powder (<0.5mm). The typical purity is 99.999% CdTe with some ppm-parts of Copper, Iron, Lead, Bismuth, Nickel and Silicon.

### 4.3.3 Cadmium chloride ( $\text{CdCl}_2$ )

Cadmium chloride is used as "Cadmium Yellow" (a brilliant yellow pigment) and for electroplating in the industry. In the production of thin-film-solar cells,  $\text{CdCl}_2$  is used as an auxiliary material for electroplating of the CdS-buffer-layer.

### 4.3.4 Cadmium sulphide ( $\text{CdS}$ )

CdS is used in thin film solar cells as a junction partner in CIS<sup>6</sup> and Cd/Te solar cells. The CdS layer itself is not intended to absorb light but instead allows light to pass through to either the CIS or CdTe substrate. Therefore, the CdS layer must be thin (30 to 100 nanometre) in order to allow sufficient light to pass through. A purity of minimal 99.99% (4N) is required.

CdS is also used as pigment, that is valued for its good thermal stability in many polymers, for example in engineering plastics. By adding selenium in its formula it is possible to obtain colours ranging from a greenish yellow to red violet. Pigments do not necessarily need a purity as high as for solar cells.

## 4.4 Life cycle inventories

### 4.4.1 Cadmium, semi-conductor grade

This section is mainly based on the report of Ali et al. (2004), additional information was taken from Fthenakis & Kim (2005); Fthenakis (2004). The high purity cadmium metal (99.9999% or 6N) can be produced either by zone refining process techniques or vacuum distillation. Although both techniques have their distinct purification profiles, vacuum distillation is more appropriate since the yield is higher in less time (Ali et al. 2004).

Vacuum distillation reduces the atmospheric pressure above the cadmium metal in order to decrease the boiling point. The following setting is described at laboratory-scale (Ali et al. 2004):

- The system consists of two cylindrical crucibles. The evaporator and the condenser are made with high-density graphite (150 mm OD x 120 mm ID x 300 mm Ht / 200 mm Ht). The two cylinders are enclosed in a reactor of stainless steel, its vacuum is  $2.1 \cdot 10^{-3}$  Torr.
- The condensation temperature in the inner wall of the collector is maintained at around 270°C, i.e. below cadmium freezing point.
- The cadmium metal in the evaporator is heated up to a temperature of 450°C.
- On average, it takes 5 hours for the entire deposition of 2 kg cadmium. Three stages of purification are necessary to obtain a purity of 99.9999%. The reported yield varies within 70% and 86%, with a geometrical mean of 77%.

**Purification of cadmium:** For an industrial production. A larger reactor is assumed for a continuous production. Due to the better ability of control, the energy source is electricity. Fthenakis<sup>7</sup> reports an energy consumption from 12'200 kJ/kg to 15'800 kJ/kg. For this project, the geometric mean of 13'880 kJ/kg was used.

Three stages of purification are necessary to obtain a purity of 99.9999%. Each stage has an average yield of 77%. The total yield is 46% for a purification setting of 3 cycles. The loss of 54% is captured

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<sup>6</sup> CIS stands for Copper-Indium-Diselenid

<sup>7</sup> Personal communication of V. Fthenakis to N. Jungbluth, November 2006

in HEPA-Filters<sup>8</sup>, as described in Fthenakis (2004). The efficiency of HEPA filters in collecting particulates is reported with 99.97%, the rest is emitted to the air. It is assumed, that the filtrate can be used again for a new purification cycle.

**Infrastructure:** The infrastructure has been roughly assessed with the dataset of the organic chemical plant.

**Transports:** The transport has been calculated with the standard distances of ecoinvent (see Tab. 4.2).

Tab. 4.2 Standard distances for transports

| Typo of transport | Distance                   | Source:                    |
|-------------------|----------------------------|----------------------------|
| Railway           | 200 km in USA (production) | Frischknecht et al. (2004) |
| Lorry             | 100 km in USA (production) | Frischknecht et al. (2004) |

**Emissions and Waste:** As there are no other materials used in the processes, only the emissions of cadmium into the air has been taken into account:  $(1-99.97\%) * 2\% * 1 \text{ kg Cd} = 6 * 10^{-6} \text{ kg}$  per kilogram purified cadmium.

Tab. 4.3 Unit process raw data for cadmium, semi-conductor grade, semi-conductor grade

| Explanations                 | Name                                   | Location | Infrastructure-Process | Unit | cadmium, semiconductor-grade, at plant | uncertaintyType | StandardDeviation 95% | GeneralComment                                                       |
|------------------------------|----------------------------------------|----------|------------------------|------|----------------------------------------|-----------------|-----------------------|----------------------------------------------------------------------|
|                              | Location                               |          |                        |      | US                                     |                 |                       |                                                                      |
|                              | InfrastructureProcess                  |          |                        |      | 0                                      |                 |                       |                                                                      |
|                              | Unit                                   |          |                        |      | kg                                     |                 |                       |                                                                      |
| Technosphere                 | cadmium, primary, at plant             | GLO      | 0                      | kg   | 1.00E+0                                | 1               | 2.30                  | (5,4,2,1,5,5); calculated based on formula and loss of air emissions |
|                              | electricity, medium voltage, at grid   | US       | 0                      | kWh  | 3.86E+0                                | 1               | 2.30                  | (5,4,2,1,5,5); personal communication from Fthenakis                 |
|                              | transport, freight, rail               | RER      | 0                      | tkm  | 2.00E-1                                | 1               | 2.10                  | (4,5,na,na,na,na); standard distances in Europe and USA (each 200km) |
|                              | transport, lorry >16t, fleet average   | RER      | 0                      | tkm  | 2.00E-1                                | 1               | 2.10                  | (4,5,na,na,na,na); standard distances in Europe and USA (each 100km) |
|                              | chemical plant, organics               | RER      | 1                      | unit | 4.00E-10                               | 1               | 3.96                  | (5,4,2,1,5,5); assumption, based on annual production of 50'000 t    |
| air, high population density | Heat, waste                            |          |                        | MJ   | 1.39E+1                                | 1               | 2.30                  | (5,4,2,1,5,5); calculation based on used energy                      |
|                              | Cadmium                                |          |                        | kg   | 6.00E-6                                | 1               | 6.11                  | (5,4,2,1,5,5); literature and estimation                             |
| Outputs                      | cadmium, semiconductor-grade, at plant | US       | 0                      | kg   | 1.00E+0                                |                 |                       |                                                                      |

#### 4.4.2 Cadmium telluride, semi-conductor grade

Pinheiro (2006) reports different forms of CdTe used in close spaced sublimation (CSS)-techniques: single-crystal or polycrystalline wafers, powders, pellets or pieces, CdTe film deposited onto glass, molybdenum substrate (CdTe source-plate), sintered CdTe powder. Therefore, various techniques are used to produce CdTe in one or another form.

Guilinger (1999b) and Triboulet (2004) describes the crystal precipitation: High purity (5N to 6N) cadmium and tellurium are synthesized by melting the elements at a temperature of 1092°C in quartz tubing. The reaction is exothermic. The cadmium and tellurium must be of high purity because con-

<sup>8</sup> HEPA stand for *High Efficiency-Particulate Air*

taminates can cause an explosion. CdTe can only be produced in small batches of around 1.3 kg, because of the volatility of the chemical reaction.

Harman (1985) reports the preparation of CdTe for substrates: “Elements are sealed in evacuated, carbon-coated quartz tubes. The capsule is placed in a single furnace which is initially in a nearly horizontal position. A slight tilt in the furnace keeps the liquid and solid material at the small diameter end of the tube. The temperature of the furnace is raised to its final temperature of about 1110°C, at the conical tip end approximately 100°C lower. The temperature of the growth tube is held for about 16 to 20hrs, then the furnace is rotated about 90° to a vertical position. After the temperature is stabilized, the tube is lowered at a rate of about 1.2mm/hr until the solution has completely solidified.”

**Production and purification of cadmium telluride:** Since no production details about the use of resources are available, we assume the following production:

- Resources: The reaction is according “the stoichiometric quantities of high purity elements “ (Harman (1985)):  $1 \text{ Mol Te (} = 127.6 \text{ g)} + 1 \text{ Mol Cd (} 112.4\text{g)} \rightarrow 1 \text{ Mol CdTe (} 240 \text{ g)}$  According to Frischknecht et al. (2004), a yield of 95% was assumed.
- Energy: Fthenakis (see Footnote 7) report the needed energy for different processes and/or producers: Minimal 37'400 kJ/kg produced CdTe have to be used, the maximal value is 53'600 kJ/kg. In this project, the geometrical mean of 44'770 kJ/kg is used for an average production process. As the process needs a precise controlling, electricity is used as energy source.

**Infrastructure:** The infrastructure has been roughly assessed with the dataset of the organic chemical plant.

**Transports:** see Tab. 4.2 in Section 4.4.1

**Emission and Waste:** With a yield of 95% and the mentioned HEPA-filters, the emissions are:  $(1-99.97 \%) * 2 \% * 1 \text{ kg CdTe} = 6 * 10^{-6} \text{ kg of CdTe}$ . The emissions are accounted as “cadmium emission into air, high population density”.



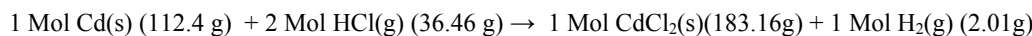
Tab. 4.4 Unit process raw data for cadmium telluride, semi-conductor grade

| Explanations                 | Name                                             | Location | Infrastructure-Process | Unit | cadmium telluride, semiconductor-grade, at plant | uncertaintyType | StandardDeviation | 95% | GeneralComment                                       |
|------------------------------|--------------------------------------------------|----------|------------------------|------|--------------------------------------------------|-----------------|-------------------|-----|------------------------------------------------------|
|                              | Location                                         |          |                        |      | US                                               |                 |                   |     |                                                      |
|                              | InfrastructureProcess                            |          |                        |      | 0                                                |                 |                   |     |                                                      |
|                              | Unit                                             |          |                        |      | kg                                               |                 |                   |     |                                                      |
|                              | electricity, medium voltage, at grid             | US       | 0                      | kWh  | 1.24E+1                                          | 1               | 1.63              |     | (5,3,1,3,3,5); personal communication (Fthenakis)    |
|                              | transport, freight, rail                         | RER      | 0                      | tkm  | 2.11E-1                                          | 1               | 2.10              |     | (4,5,na,na,na,na); standard distances for US (200km) |
|                              | transport, lorry >16t, fleet average             | RER      | 0                      | tkm  | 1.05E-1                                          | 1               | 2.10              |     | (4,5,na,na,na,na); standard distances for US (100km) |
|                              | chemical plant, organics                         | RER      | 1                      | unit | 4.00E-10                                         | 1               | 3.82              |     | (4,5,2,5,5,5); assumption                            |
|                              | cadmium, semiconductor-grade, at plant           | US       | 0                      | kg   | 4.93E-1                                          | 1               | 2.34              |     | (5,5,2,5,5,5); calculated based on formula           |
|                              | tellurium, semiconductor-grade, at plant         | GLO      | 0                      | kg   | 5.60E-1                                          | 1               | 2.34              |     | (5,5,2,5,5,5); calculated based on formula           |
| air, high population density | Heat, waste                                      |          |                        | MJ   | 7.58E-1                                          | 1               | 1.68              |     | (5,5,2,3,3,5); estimation                            |
|                              | Cadmium                                          |          |                        | kg   | 6.00E-6                                          | 1               | 5.42              |     | (5,5,2,3,3,5); estimation                            |
|                              | cadmium telluride, semiconductor-grade, at plant | US       | 0                      | kg   | 1.00E+0                                          |                 |                   |     |                                                      |

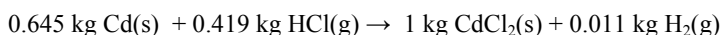
#### 4.4.3 Cadmium chloride, semi-conductor grade

Cadmium chloride can be produced either by reaction of molten cadmium and chlorine gas at 600°C or by dissolving cadmium metal (or the oxide) in hydrochloric acid (Schulte-Schrepping & Piscator 2002). As data from literature is not available, the process and material consumption has been roughly estimated.

**Production of cadmium chloride:** Anhydrous cadmium chloride can be prepared by the action of anhydrous chlorine or hydrogen chloride gas on heated cadmium metal. The following reaction has been assumed:



or with an efficiency of 95% for synthesis, see Frischknecht et al. (2004):



As no further information about the energy consumption are available, an estimation based on the enthalpy has been made: The heat capacity of cadmium is  $26.02 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$ , as the melting temperature of cadmium is about 320°C, we consider a difference of temperature of 300°C for heating the cadmium. The heat of fusion of cadmium is  $6'210 \text{ J}\cdot\text{mol}^{-1}$ , we estimate an overall efficiency of 50%.

Heat capacity:  $300 \text{ K} * 26.02 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1} = 7.8 \text{ kJ}$

Heat of Fusion:  $6.2 \text{ kJ}$

Sum:  $14 \text{ kJ} = 0.014 \text{ MJ} = 0.0039 \text{ kWh} \rightarrow 0.0078 \text{ kWh/mol}$  (incl. efficiency of 50%)

As no information about purification is available, presumably the environmental impacts are underestimated.

**Infrastructure:** The infrastructure has been roughly assessed with the available dataset of the organic chemical plant.

**Transports:** see Tab. 3.7 in Section 4.4.1

**Emission and Waste:** From the molten cadmium, some emissions are considered. We assume the same efficiency of production process as for the purification of cadmium (see section 4.3.1) Therefore,

we assume that from 0.645 kg Cadmium 77% are converted into cadmium chloride, the rest is captured in HEPA-filters with an efficiency of 99.97%:  $(1-99.97\%) * 2\% * 0.645 \text{ kg Cd} = 3.876 * 10^{-6} \text{ kg}$  per kilogram cadmium chloride. The generated hydrogen is also emitted into the air.

Tab. 4.5 Unit process raw data for cadmium chloride

| Explanations                 | Name                                            | Location | Infrastructure-Process | Unit | cadmium chloride, semiconductor-grade, at plant | uncertaintyType | StandardDeviation 95% | GeneralComment                                                       |
|------------------------------|-------------------------------------------------|----------|------------------------|------|-------------------------------------------------|-----------------|-----------------------|----------------------------------------------------------------------|
|                              | Location                                        |          |                        |      | US                                              |                 |                       |                                                                      |
|                              | InfrastructureProcess                           |          |                        |      | 0                                               |                 |                       |                                                                      |
|                              | Unit                                            |          |                        |      | kg                                              |                 |                       |                                                                      |
|                              | electricity, medium voltage, at grid            | US       | 0                      | kWh  | 4.26E-2                                         | 1               | 2.30                  | (5,4,2,1,5,5); assumption,                                           |
|                              | transport, freight, rail                        | RER      | 0                      | tkm  | 2.13E-1                                         | 1               | 2.10                  | (4,5,na,na,na,na); standard distances in Europe and USA (each 200km) |
|                              | transport, lorry >16t, fleet average            | RER      | 0                      | tkm  | 1.07E-1                                         | 1               | 2.10                  | (4,5,na,na,na,na); standard distances in Europe and USA (each 100km) |
|                              | chemical plant, organics                        | RER      | 1                      | unit | 4.00E-10                                        | 1               | 3.81                  | (4,5,2,1,5,5); assumption                                            |
|                              | cadmium, semiconductor-grade, at plant          | US       | 0                      | kg   | 6.46E-1                                         | 1               | 2.30                  | (5,4,2,1,5,5); calculated based on formula                           |
|                              | hydrochloric acid, 30% in H2O, at plant         | RER      | 0                      | kg   | 4.19E-1                                         | 1               | 2.30                  | (5,4,2,1,5,5); calculated based on formula                           |
| air, high population density | Heat, waste                                     |          |                        | MJ   | 1.53E-1                                         | 1               | 2.33                  | (5,5,2,1,5,5); estimation                                            |
|                              | Cadmium                                         |          |                        | kg   | 3.88E-6                                         | 1               | 6.16                  | (5,5,2,1,5,5); estimation                                            |
|                              | Hydrogen                                        |          |                        | kg   | 1.15E-2                                         | 1               | 2.55                  | (5,5,2,1,5,5); estimation                                            |
|                              | cadmium chloride, semiconductor-grade, at plant | US       | 0                      | kg   | 1.00E+0                                         |                 |                       |                                                                      |

#### 4.4.4 Cadmium sulphide, semi-conductor grade

This part is mainly based on the reports of Guilinger (1999a) and Faria & Chiola (1981). Guilinger (1999a) reported that in 1999 Asarco in the United States was the only producer of semi-conductor grade cadmium sulphide. It is assumed, that the production today also takes place in the US. Cadmium sulphide is being produced commercially by two methods:

- **Hydrometallurgical:**  
According to Guilinger (1999a), this aqueous technique of producing CdS involves a number of handling steps and is a complex procedure. According to Faria & Chiola (1981), is the normal procedure “dissolve a cadmium salt in water, preferably the chloride or sulphate, and then bubble hydrogen sulfide through the solution to precipitate essentially all of the cadmium in the form of cadmium sulfide.” The reported yield is quite high, anionic and cationic impurities co-precipitates with the cadmium sulfide. The following purification involves extensive washing and leaching, until a high purity of the output product can be reached.
- **Thermal:**  
This technique involves the fusing of cadmium and sulphide material. The production rate can be high, but size distribution and purity of the final product can be a problem. Therefore re-processing is necessary to achieve a consistent product. This process is generally used to produce lower grade CdS for use in pigments.  
According to Faria & Chiola (1981), the purification of thermal cadmium sulphide can be done with hydrogen sulfide: “luminescent grade cadmium sulphide is placed in a quartz tube reactor and a flow of nitrogen is maintained while the temperature is raised to about 500°C. Hydrogen sulfide is used to replace nitrogen and the temperature in the reactor is raised to about 1000°C. The temperature is held for 0.5h and then cooled down to 500°C under the hy-

drogen sulfide flow. The hydrogen sulfide is shut off and nitrogen is introduced while cooling the material to ambient temperature.”

**Production of cadmium sulphide:** Since no specific information about the production process is available, this part is mainly based on assumptions. It is assumed that the reaction is according to the stoichiometric quantities:  $1 \text{ Mol CdCl}_2 (= 183.3 \text{ g}) + 1 \text{ Mol H}_2\text{S} (34.1) \rightarrow 1 \text{ Mol CdS} (144.5 \text{ g}) + 1 \text{ Mol HCl} (112.4 \text{ g})$  According to Frischknecht et al. (2004), a yield of 95% was assumed. As no information about the energy consumption is available, the proxy-value of cadmium chloride (see section 4.4.3) is used instead (14 kJ per kg produced cadmium sulphide). As no information about a purification are available, presumably the environmental impacts are underestimated.

**Infrastructure:** The infrastructure has been roughly assessed with the available dataset of the organic chemical plant.

**Transports:** see Tab. 3.7 in Section 4.4.1

**Emission and Waste:** Guilinger (1999a) stated that “(The production) does generate a cadmium contaminated liquid waste stream that is environmentally difficult to handle and neutralize.” Therefore, a cadmium stream into the water is assumed. As a first estimation, the related value from zinc production has been chosen ( $2.60 \cdot 10^{-6}$  kg cadmium per kg cadmium sulphide). However, this value is fairly uncertain.

Tab. 4.6 Unit process raw data for cadmium sulphide

|                                      | Name                                                | Location              | Category | SubCategory         | InfrastructureProcess | Unit | cadmium sulphide, semiconductor-grade, at plant | UncertaintyType | StandardDeviation95% | GeneralComment                                      |
|--------------------------------------|-----------------------------------------------------|-----------------------|----------|---------------------|-----------------------|------|-------------------------------------------------|-----------------|----------------------|-----------------------------------------------------|
|                                      | Location                                            | InfrastructureProcess | Unit     |                     |                       |      | US 0 kg                                         |                 |                      |                                                     |
| product                              | cadmium sulphide, semiconductor-grade, at plant     | US                    | -        | -                   | 0                     | kg   | 1                                               |                 |                      |                                                     |
| technosphere                         | cadmium chloride, semiconductor-grade, at plant     | US                    | -        | -                   | 0                     | kg   | 1.34E+0                                         | 1               | 2.30                 | (5,4,2,1,5,5); calculated based on formula          |
|                                      | hydrogen sulphide, H <sub>2</sub> S, at plant       | RER                   | -        | -                   | 0                     | kg   | 2.48E-1                                         | 1               | 2.31                 | (5,4,2,5,5,5); calculated based on formula          |
|                                      | chemical plant, organics                            | RER                   | -        | -                   | 1                     | unit | 4.00E-10                                        | 1               | 3.82                 | (4,5,2,5,5,5); assumption                           |
|                                      | electricity, medium voltage, production US, at grid | US                    | -        | -                   | 0                     | kWh  | 3.89E-3                                         | 1               | 1.93                 | (4,5,5,1,4,5); assumption                           |
|                                      | transport, freight, rail                            | RER                   | -        | -                   | 0                     | tkm  | 3.17E-1                                         | 1               | 2.09                 | (4,5,na,na,na,na); standard distances in US (200km) |
|                                      | transport, lorry >16t, fleet average                | RER                   | -        | -                   | 0                     | tkm  | 1.58E-1                                         | 1               | 2.09                 | (4,5,na,na,na,na); standard distances in US (100km) |
| water, unspecified                   | Cadmium, ion                                        | -                     | water    | unspecified         | -                     | kg   | 2.60E-6                                         | 1               | 4.00                 | (5,5,2,1,5,5); estimation                           |
| emission water, unspecified          | Chloride                                            | -                     | water    | unspecified         | -                     | kg   | 5.16E-1                                         | 1               | 4.00                 | (5,5,2,1,5,5); estimation                           |
| emission air, low population density | Heat, waste                                         | -                     | air      | low population dens | -                     | MJ   | 1.40E-2                                         | 1               | 2.33                 | (5,5,2,1,5,5); estimation                           |

Kommentar [hja2931]: er-setzen

Tab. 4.7 EcoSpold Meta Information of cadmium purification and cadmium compounds

| ReferenceFunction           | 401 Name                       | cadmium, semiconductor-grade, at plant                                                                                                                                                 | cadmium telluride, semiconductor-grade, at plant                                                                                                                                                      | cadmium chloride, semiconductor-grade, at plant                                                                                                                                                             | cadmium sulphide, semiconductor-grade, at plant                                                                                                                                                        |
|-----------------------------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Geography                   | 662 Location                   | US                                                                                                                                                                                     | US                                                                                                                                                                                                    | US                                                                                                                                                                                                          | US                                                                                                                                                                                                     |
| ReferenceFunction           | 493 InfrastructureProcess      | 0                                                                                                                                                                                      | 0                                                                                                                                                                                                     | 0                                                                                                                                                                                                           | 0                                                                                                                                                                                                      |
| ReferenceFunction           | 403 Unit                       | kg                                                                                                                                                                                     | kg                                                                                                                                                                                                    | kg                                                                                                                                                                                                          | kg                                                                                                                                                                                                     |
| DataSetInformation          | 201 Type                       | 1                                                                                                                                                                                      | 1                                                                                                                                                                                                     | 1                                                                                                                                                                                                           | 1                                                                                                                                                                                                      |
|                             | 202 Version                    | 2.0                                                                                                                                                                                    | 2.0                                                                                                                                                                                                   | 2.0                                                                                                                                                                                                         | 2.0                                                                                                                                                                                                    |
|                             | 203 energyValues               | 0                                                                                                                                                                                      | 0                                                                                                                                                                                                     | 0                                                                                                                                                                                                           | 0                                                                                                                                                                                                      |
|                             | 205 LanguageCode               | en                                                                                                                                                                                     | en                                                                                                                                                                                                    | en                                                                                                                                                                                                          | en                                                                                                                                                                                                     |
|                             | 206 LocalLanguageCode          | de                                                                                                                                                                                     | de                                                                                                                                                                                                    | de                                                                                                                                                                                                          | de                                                                                                                                                                                                     |
| DataEntryBy                 | 302 Person                     | 44                                                                                                                                                                                     | 44                                                                                                                                                                                                    | 44                                                                                                                                                                                                          | 44                                                                                                                                                                                                     |
|                             | 304 QualityNetwork             | 1                                                                                                                                                                                      | 1                                                                                                                                                                                                     | 1                                                                                                                                                                                                           | 1                                                                                                                                                                                                      |
| ReferenceFunction           | 400 DataSetRelatesToProduct    | 1                                                                                                                                                                                      | 1                                                                                                                                                                                                     | 1                                                                                                                                                                                                           | 1                                                                                                                                                                                                      |
|                             | 402 IncludedProcesses          | The energy consumption and transports are considered for, emissions are estimated.                                                                                                     | The energy consumption and transports are considered for, emissions are estimated.                                                                                                                    | The energy consumption, emissions and transports are estimated.                                                                                                                                             | The energy consumption, emissions and transports are estimated.                                                                                                                                        |
|                             | 404 Amount                     | 1                                                                                                                                                                                      | 1                                                                                                                                                                                                     | 1                                                                                                                                                                                                           | 1                                                                                                                                                                                                      |
|                             | 490 LocalName                  | Cadmium, Halbleiter, ab Werk                                                                                                                                                           | Cadmiumtellurid, Halbleiter, ab Werk                                                                                                                                                                  | Cadmiumchlorid, Halbleiter, ab Werk                                                                                                                                                                         | Cadmiumsulfid, Halbleiter, ab Werk                                                                                                                                                                     |
|                             | 491 Synonyms                   |                                                                                                                                                                                        |                                                                                                                                                                                                       |                                                                                                                                                                                                             |                                                                                                                                                                                                        |
|                             | 492 GeneralComment             | This datasets describes the production of high-grade cadmium (99.9999%) from metallurgical cadmium. This purity of cadmium can be used in the production of various cadmium compounds. | This datasets describes the production of high-grade cadmium telluride (99.999%) from cadmium and tellurium. This purity of cadmium telluride can be used in the production of thin film solar cells. | This datasets describes the production of high-grade cadmium chloride (99.999%) from cadmium and hydrochloric acid. This purity of cadmium chloride can be used in the production of thin film solar cells. | This datasets describes the production of high-grade cadmium sulphide (99.99%) from cadmium and sulfuric acid. This purity of cadmium sulphide can be used in the production of thin film solar cells. |
|                             | 494 InfrastructureIncluded     | 1                                                                                                                                                                                      | 1                                                                                                                                                                                                     | 1                                                                                                                                                                                                           | 1                                                                                                                                                                                                      |
|                             | 495 Category                   | metals                                                                                                                                                                                 | metals                                                                                                                                                                                                | metals                                                                                                                                                                                                      | metals                                                                                                                                                                                                 |
|                             | 496 SubCategory                | extraction                                                                                                                                                                             | extraction                                                                                                                                                                                            | extraction                                                                                                                                                                                                  | extraction                                                                                                                                                                                             |
|                             | 497 LocalCategory              | Metalle                                                                                                                                                                                | Metalle                                                                                                                                                                                               | Metalle                                                                                                                                                                                                     | Metalle                                                                                                                                                                                                |
|                             | 498 LocalSubCategory           | Gewinnung                                                                                                                                                                              | Gewinnung                                                                                                                                                                                             | Gewinnung                                                                                                                                                                                                   | Gewinnung                                                                                                                                                                                              |
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|                             | 502 CASNumber                  | 000744-43-9                                                                                                                                                                            | 001306-25-8                                                                                                                                                                                           | 010108-64-2                                                                                                                                                                                                 | 001306-23-6                                                                                                                                                                                            |
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| Technology                  | 692 Text                       | Production of high-purity cadmium by vacuum distillation                                                                                                                               | Production of cadmium telluride from cadmium and tellurium                                                                                                                                            | Production of cadmium chloride by reaction of cadmium with hydrogen chloride                                                                                                                                | Production of cadmium sulphide by hydrometallurgical process                                                                                                                                           |
| Representativeness          | 722 Percent                    |                                                                                                                                                                                        |                                                                                                                                                                                                       |                                                                                                                                                                                                             |                                                                                                                                                                                                        |
|                             | 724 ProductionVolume           | unknown                                                                                                                                                                                | unknown                                                                                                                                                                                               | unknown                                                                                                                                                                                                     | unknown                                                                                                                                                                                                |
|                             | 725 SamplingProcedure          | literature and own estimations                                                                                                                                                         | literature and own estimations                                                                                                                                                                        | literature and own estimations                                                                                                                                                                              | literature and own estimations                                                                                                                                                                         |
|                             | 726 Extrapolations             | none                                                                                                                                                                                   | Emissions of cadmium into the air are extrapolated from cadmium purification                                                                                                                          | Cadmium emissions are extrapolated from cadmium purification.                                                                                                                                               | Energy consumption is extrapolated from CdCl due to lack of data and similarity of processes                                                                                                           |
|                             | 727 UncertaintyAdjustments     | none                                                                                                                                                                                   | none                                                                                                                                                                                                  | none                                                                                                                                                                                                        | none                                                                                                                                                                                                   |
| DataGeneratorAndPublication | 751 Person                     | 44                                                                                                                                                                                     | 44                                                                                                                                                                                                    | 44                                                                                                                                                                                                          | 44                                                                                                                                                                                                     |
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|                             | 757 ReferenceToPublishedSource | 10                                                                                                                                                                                     | 10                                                                                                                                                                                                    | 10                                                                                                                                                                                                          | 10                                                                                                                                                                                                     |
|                             | 758 Copyright                  | 1                                                                                                                                                                                      | 1                                                                                                                                                                                                     | 1                                                                                                                                                                                                           | 1                                                                                                                                                                                                      |
|                             | 759 AccessRestrictedTo         |                                                                                                                                                                                        |                                                                                                                                                                                                       |                                                                                                                                                                                                             |                                                                                                                                                                                                        |
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## Part XI

# Tantalum Powder

Data v2.1 (2009)

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Final report of updated Swiss National Life Cycle Inventory Database "ecoinvent", version 2.1



## Comments to version 2.1

This part and all data correspond to the part and data reported in version 2.0. No changes were made.

## Summary

The inventories of mining of tantalum containing ores and the production of capacitor grade tantalum powder are documented in this report. The inventory is based on the environmental reporting of the most important producer. The production of capacitor grade tantalum powder is based on theoretical models, similarity assumptions, and estimations. No recycling (secondary tantalum) is modelled. Hence, this dataset represents primary tantalum powder.

The production of tantalum metal, as it is used in the steel industry for alloying, follows a different process path. It includes the processing of  $K_7TaF_7$  with subsequent remelting in electron beam furnaces, which gives much higher burdens than the process described in this chapter. Therefore, the tantalum powder dataset described in this report is not recommended for use to represent tantalum metal manufacturing



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# 1 Introduction

The main uses of Tantalum are in the alloying of high temperature steels and as metal powder in capacitors used in the microelectronic industry. Today the biggest share of Tantalum goes into the capacitors used in mobile applications such as mobile phones, handheld and laptops. Due to its easy processing and favourable electric properties it is widely used. Although the price for Tantalum has risen drastically, discussed alternatives – alumina or niobium based capacitors – are in their infancy. The price boom has led to a proliferation of small scale mining activities mainly in the Republic Congo during the Civil War in 1998 – 2004 that led to uncontrolled exploitation (Behrendt et al., 2007).

Since the production of capacitor grade Tantalum is extensive and the material itself is used in microelectronics in comparatively large amounts, it plays a key role in the environmental assessment.

## 2 Material Characterisation

Tantalum is dark, dense, ductile, very hard, easily fabricated, and highly conductive of heat and electricity. The metal is renowned for its resistance to corrosion by acids; in fact, at temperatures below 150 °C tantalum is almost completely inert to attack by the normally aggressive aqua regia. It can be dissolved with hydrofluoric acid or acidic solutions containing the fluoride ion and sulphur trioxide, as well as with a solution of potassium hydroxide. Tantalum's high melting point of 3017°C (boiling point 5458°C) is exceeded only by tungsten and rhenium (Wikipedia Contributors, 2006.).

## 3 Reserves and Production

Tantalum is 54th in order of abundance of the elements in the Earth's crust (2.1 g/t). In many deposits, it occurs in association with the much more abundant niobium (24 g/t). The most important tantalum-containing minerals are tantalite, wodginite, microlite (the tantalum-rich end member of the pyrochlore series), and columbite (Tab. 3.1).

In the past, most of commercial tantalum came as a by-product of tin smelting especially in Thailand. Since these deposits are now largely depleted (Forrest, 2004) the tin production in this area has been decreased and is today of minor importance for tantalum production. Therefore in future the tantalum from tin slag will be of reduced importance.

**Tab. 3.1** Composition of important tantalum-containing minerals (Andersson et al., 1996).

| Mineral        | Composition                                                          | Ta <sub>2</sub> O <sub>5</sub> content, % | Nb <sub>2</sub> O <sub>5</sub> content, % |
|----------------|----------------------------------------------------------------------|-------------------------------------------|-------------------------------------------|
| Tantalite      | (Fe, Mn)Ta <sub>2</sub> O <sub>6</sub>                               | 42 – 84                                   | 2 – 40                                    |
| Microlite      | (Na, Ca)(Ta, Nb) <sub>2</sub> O <sub>6</sub> F                       | 60 – 70                                   | 5 – 10                                    |
| Columbite      | (Fe, Mn)(Nb, Ta) <sub>2</sub> O <sub>6</sub>                         | 1 – 40                                    | 40 – 75                                   |
| Wodginite      | (Ta, Nb, Sn, Mn, Fe, Mn) <sub>16</sub> O <sub>32</sub>               | 45 – 56                                   | 3 – 15                                    |
| Yttrotantalite | (Y, U, Ca)(Ta, Fe <sup>3+</sup> ) <sub>2</sub> O <sub>6</sub>        | 14 – 27                                   | 41 – 56                                   |
| Fergusonite    | (RE <sup>3+</sup> )(Nb, Ta)O <sub>4</sub> *                          | 4 – 43                                    | 14 – 46                                   |
| Strueverite    | (Ti, Ta, Nb, Fe) <sub>2</sub> O <sub>4</sub>                         | 7 – 13                                    | 9 – 14                                    |
| Tapiolite      | (Fe, Mn)(Nb, Ta) <sub>2</sub> O <sub>6</sub>                         | 40 – 85                                   | 8 – 15                                    |
| Euxenite       | (Y, Ca, Ce, U, Th)(Nb, Ta, Ti) <sub>2</sub> O <sub>6</sub>           | 1 – 6                                     | 22 – 30                                   |
| Samarskite     | (Fe, Ca, U, Y, Ce) <sub>2</sub> (Nb, Ta) <sub>2</sub> O <sub>6</sub> | 15 – 30                                   | 40 – 55                                   |

\* RE = rare earth element.

### 3.1 Primary Resources

The principal raw material with the highest potential for extracting tantalum is found in the pegmatitic deposits in Australia. The typical Ta<sub>2</sub>O<sub>5</sub> concentration in processible pegmatitic deposits suitable to processing, is less than ca. 0.1%. The uranium- and thorium- containing minerals yttrotantalite, strueverite, euxenite, and samarskite are sometimes relatively high radioactive, which limits their workability (Andersson et al., 1996). Today ores with contents of 200 ppm (0.02%) of Ta<sub>2</sub>O<sub>5</sub> are workable by means of open pit and bulk hydrometallurgical technology (Gille et al., 2006).

### **3.1.1 Ores**

Tantalum is mainly gained from Pegmatite as Tantalite. The tantalum production rates from tin slags (25%) and from secondary resources (20%) imply a global share of primary production from Tantalite ores of 55%. The main producer of tantalum is the company Sons of Gwalia (SGW) that mines today 50% of the known primary Tantalite reserves (Polyakov & Polyakova, 2003). The company has been producing around 55-60% of the world's tantalum from two mines in Western Australia, and in 2003 this figure reached almost 70% (Forrest, 2004). The reserves at Greenbushes and Wodgina mines together amount to 49'000 t Ta<sub>2</sub>O<sub>5</sub> at 240 ppm and an additional 47'000 t at 162 ppm (Sons of Gwalia, 2003).

The USGS data shows the high dynamic of the mine production during the last ten years. Today the production of ores is estimated to amount 1'500 t and 1'900 tantalum (contained in various intermediates and commodities) for the years 2004 and 2005 respectively (USGS, 2006).

### **3.1.2 Tantalum containing slags**

An important potential tantalum reserve lies in the so called Southeast Asian Tin Belt, stretching from China to Indonesia through Thailand and Malaysia, and consists of cassiterite (SnO<sub>2</sub>) that contains some tantalum and niobium. During tin smelting, the tantalum and niobium become concentrated in the slag (Andersson et al., 1996). This slag can be processed directly for tantalum and niobium recovery. Roskill Information Services (Gille et al., 2006) estimated that 25% of the 2002 worldwide demand in tantalum was covered by the recovery of tin slags from the Southeast Asian Tin Belt. Since many of these operations are closing down due to depletion and accompanying decreased profits in the tin market, this source today is of minor importance.

## **3.2 Secondary Resources**

Due to the high price of tantalum, the market of secondary resources remains attractive. In 2002 Roskill Information Services (Gille et al., 2006) estimated that 20% of the tantalum originates from tantalum bearing scrap.

## **3.3 Production**

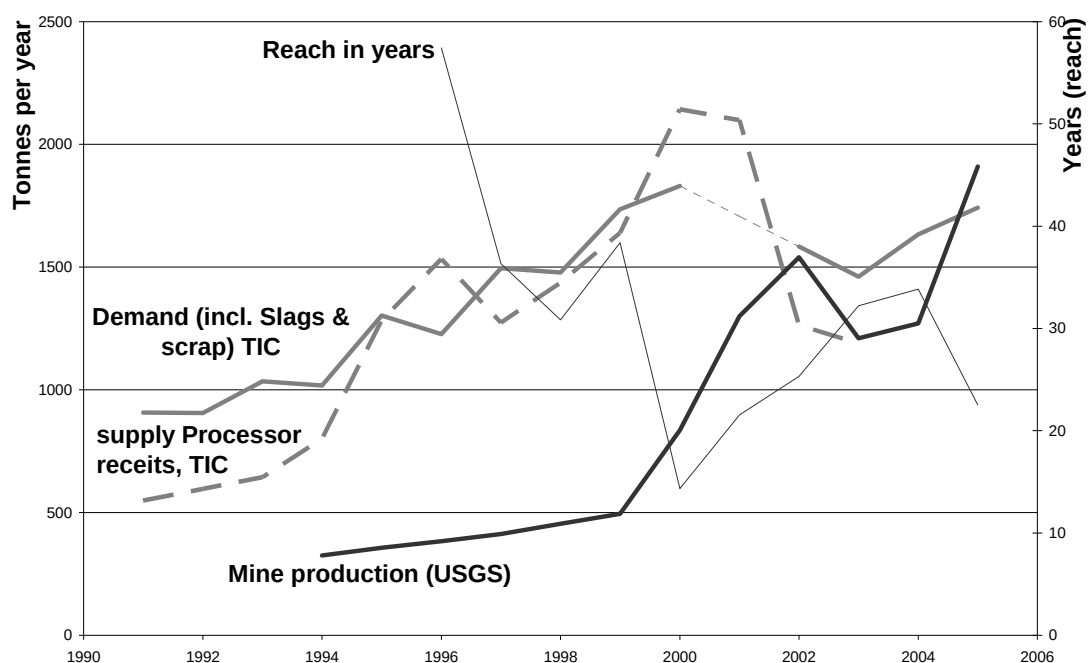
The Tantalum Niobium International Study Centre TIC used to keep track of the production and demand of worldwide tantalum market until 2001. Data gathered from various sources show the development in the last ten years (Tab. 3.2). The supply of primary tantalum is largely monopolised by two large producers: HC Stark/Bayer GP (Germany) and the Cabot Corp. (U.S.). These two companies account for 85% of the world production of tantalum powder, tantalum wire, and tantalum oxides (Polyakov & Polyakova (2003)).

### 3. Reserves and Production

**Tab. 3.2** Compilation of Production, supply and demand of tantalum. Data has been drawn from different sources indicated below.

| YEAR | Tantalum Supply,<br>Production from tantalite, slags & scrap | Tantalum Demand     | Mine production <sup>5</sup> | Reserve <sup>5</sup> | Reserve Base <sup>5</sup> | Reserve Reach <sup>6</sup> |
|------|--------------------------------------------------------------|---------------------|------------------------------|----------------------|---------------------------|----------------------------|
| Year | t Ta contained                                               | t Ta contained      | t Ta contained               | '000 t Ta cont.      | '000 t Ta cont.           | years                      |
| 1991 | 549 <sup>1</sup>                                             | 907 <sup>1</sup>    |                              |                      |                           |                            |
| 1992 | 596 <sup>1</sup>                                             | 905 <sup>1</sup>    |                              |                      |                           |                            |
| 1993 | 644 <sup>1</sup>                                             | 1035 <sup>1</sup>   |                              |                      |                           |                            |
| 1994 | 800 <sup>1</sup>                                             | 1017 <sup>1</sup>   | 325                          |                      |                           |                            |
| 1995 | 1281 <sup>1</sup>                                            | 1303 <sup>1</sup>   | 356                          |                      |                           |                            |
| 1996 | 1532 <sup>1</sup>                                            | 1227 <sup>1</sup>   | 383                          | 22                   | 35                        | 57                         |
| 1997 | 1274 <sup>1</sup>                                            | 1495 <sup>1</sup>   | 413                          | 15                   | 26                        | 36                         |
| 1998 | 1437 <sup>1</sup>                                            | 1478 <sup>1</sup>   | 454                          | 14                   | 24                        | 31                         |
| 1999 | 1639 <sup>1</sup>                                            | 1736 <sup>1</sup>   | 495                          | 19                   | 24                        | 38                         |
| 2000 | 2143 <sup>1</sup>                                            | 1766 <sup>1</sup>   | 836                          | 12                   | 36                        | 14                         |
| 2001 | 2026 <sup>2</sup>                                            | (1664) <sup>3</sup> | 1300                         | 28                   | 60                        | 22                         |
| 2002 | 1219 <sup>2</sup>                                            | (1563) <sup>3</sup> | 1540                         | 39                   | 110                       | 25                         |
| 2003 | 1147 <sup>2</sup>                                            | 1461 <sup>4</sup>   | 1210                         | 39                   | 110                       | 32                         |
| 2004 | n. a.                                                        | 1633 <sup>4</sup>   | 1270                         | 43                   | 150                       | 34                         |
| 2005 | n. a.                                                        | 1742 <sup>4</sup>   | 1910                         | 43                   | 150                       | 23                         |

<sup>1</sup> dme (2002); <sup>2</sup> Anonymous (2004), comprise production from slag and tantalite only; <sup>3</sup> Interpolation between 2000 and 2003; <sup>4</sup> Forrest (2004); <sup>5</sup> USGS (2006) and preceding; <sup>6</sup> Obtained dividing the number in column "Reserve" by the corresponding number in column "Mine production"



**Fig. 3.1** Compilation of data on production and demand of tantalum. Different sources are used (see Tab. 3.2).



## 4 Use

The Brussels-based Tantalum Niobium International Study Centre (TIC) estimates that in 2003 50% of the tantalum went into powder for capacitors, 15% into metal for chemical plants in corrosion resistant alloys, 10% into temperature resistant superalloys, 10% into cutting tools as TaC and 15% into speciality applications (Forrest, 2004). Most probably the special applications correspond to a great extent to the application in the military sector as liner material and in anti-armour warheads (Muller, 1996). TIC estimated that tantalum demand in 2003 reached 1'900 tonnes  $Ta_2O_5$  and increases at a rate of 7 to 8% to 2'200 tonnes in 2005 (Forrest, 2004).

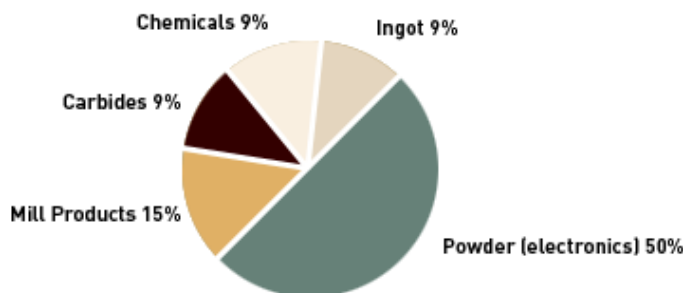


Fig. 4.1 Applications of tantalum in 2003 (Sons of Gwalia, 2006).

The major use for tantalum, as metal powder, is in the production of electronic components, mainly capacitors and some high-end audio-grade resistors. Tantalum electrolytic capacitors exploit the characteristic of tantalum to form a protective oxide surface layer, using tantalum foil as one plate of the capacitor, the oxide as the dielectric, and an electrolytic solution as the other plate (Wikipedia Contributors, 2006).

Today, the tantalum electrolytic capacitor industry requires approximately 68 percent of the world's tantalum production. Tantalum capacitors are characterized by a very high volumetric efficiency of capacitance and very high reliability under extreme conditions. Compared with aluminum or ceramic capacitors, tantalum capacitors show a distinctly lower temperature and frequency dependence of their electrical properties. The typical application area of application of tantalum capacitors is shown in Fig. 4.2 (H.C. Starck, 2006).

Beside electronics, tantalum is used as alloying element in metals and superalloys as well as carbide for hard metals (Fig. 5.1). However, these applications are not covered by this dataset and are not discussed further.

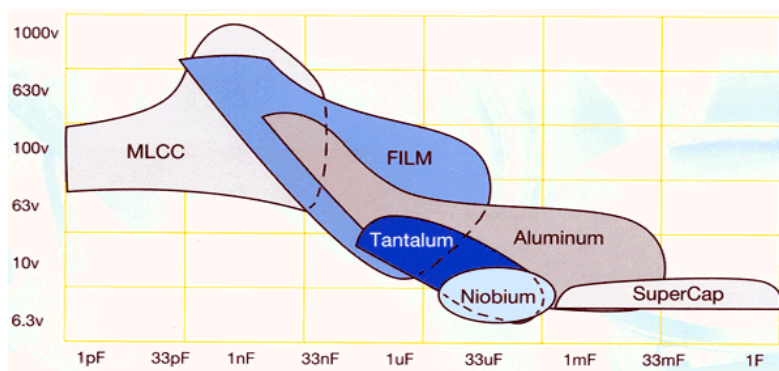


Fig. 4.2 Application of tantalum in capacitors (H.C. Starck, 2006).

## 5 System Characterisation

Depending on the intended product and company, a multitude of processes are applied to extract the tantalum and to manufacture products from it. Different technologies exist to produce metallic tantalum products:

- From its **heptafluoride** by means of Electrochemic reduction (KF/KCl,) or metallothermic reduction with potassium (Yoon et al., 2005).
- From its **pentoxide** by means of a reduction with gaseous magnesium.

Both processes are to be found in the schematic in Fig. 4. As feedstock, slag from tin production is used apart from ores. Low grade tin slags are either enriched with scrap to form synthetic ores that can be subjected to the liquid-liquid extraction with MIBK (methyl isobutyl ketone, 4-methyl pentan-2-one) or else pass, together with tantalum containing scrap from electronic scrap, the calcination route. Secondary tantalum raw materials are always extracted by hydrometallurgical processes via tantalum pentachloride or oxide (Fig. 5.1).

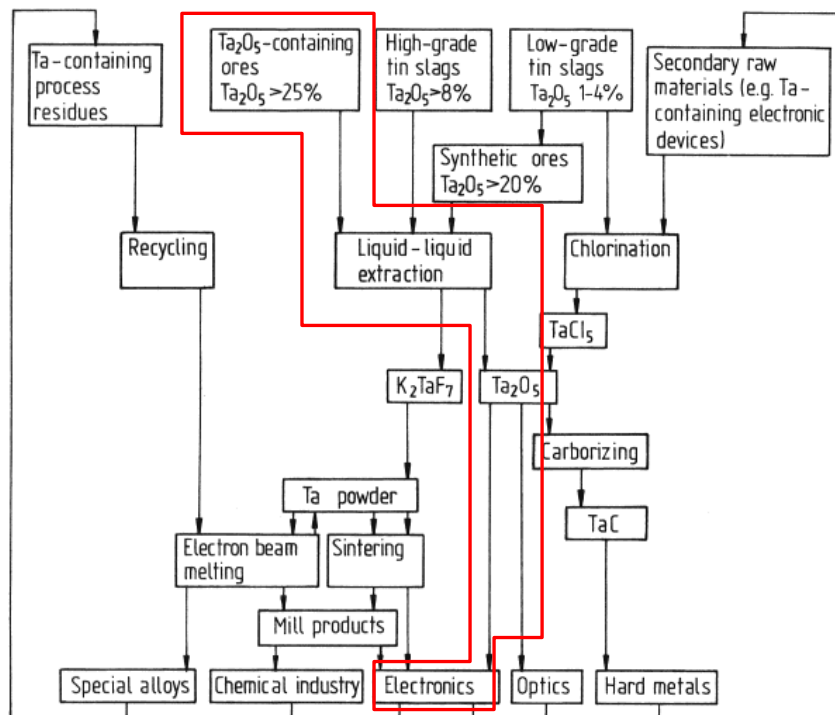


Fig. 5.1 General overview of the production of tantalum products from different feeds (Andersson et al., 1996) and selection for the inventoried system (red).

Tantalum powders or semi finished products such as wires and sheets for applications in alloys and steel industry are mainly processed by the reduction of K<sub>2</sub>TaF<sub>7</sub> with sodium. The powder obtained is then pressed to bars, that are formed to ingots by electron beam melting and shaped to the desired form. This process therefore differs much from the capacitor powder production. Further information on the respective process is given in Albrecht & Ingalls (1983) and is not further discussed here.

In this report no overview of all processes can be given, but only the technologies relevant to the production of capacitor grade tantalum powder is regarded. Three different processes are applied in capacitor grade tantalum powder production (H.C. Starck, 2006), namely:

- Electron beam melted process (via  $K_2TaF_7$ )
- Na-reduced process (via  $K_2TaF_7$ )
- Mg-reduced process (via  $Ta_2O_5$ )

For capacitor powders, e.g. from H.C. Starck, all these three different processes are applied in dependence of the desired application. To cover the needs for higher volumetric capacitance and lower rated voltage, the magnesium reduced process is employed (Fig. 5.2). Since the miniaturisation of capacitors that are used in microelectronics require ever higher specific electric capacitance, the magnesium reduced process via the tantalum oxide is considered relevant in the context of this inventory.

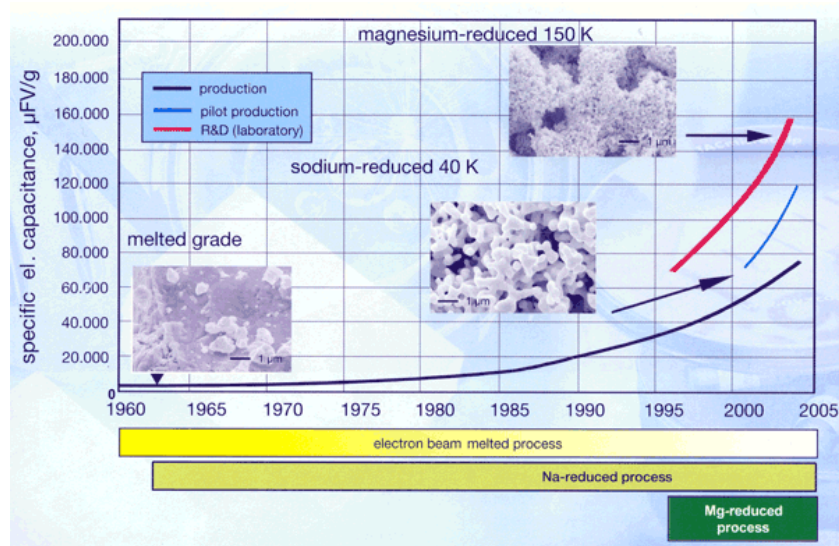


Fig. 5.2 Tantalum powder production processes according to the application fields (H.C. Starck, 2006).

### 5.1.1 Production of Tantalum powder in ecoinvent

This inventory describes the production of tantalum powder for the use in capacitors in electric applications. The tantalum powder is produced from virgin tantalite ore to tantalumpentoxide which is reduced with magnesium vapour to tantalum powder. No secondary production from consumer scrap and also no by-product tantalum from tin production is considered. This dataset is exclusively meant to be used in the context of electronic devices, since the production of tantalum for alloying purposes follows a different route (Fig. 5.1).

The metadata of the established inventory is given in (Tab. 5.1), their exchanges are shown in (Fig. 5.4). The unit process data for the individual steps are presented in separate data for tantalite mining (Tab. 5.3), extraction (Tab. 5.5) and reduction (Tab. 5.7).

Tab. 5.1 Ecoinvent meta information for the processes for the production of 1 kg tantalum powder for capacitors.

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>                  | <b>Tantalum, powder, capacitor-grade, at regional storage</b>                                                                                                                                                                                                                                                                                                                                                                            |
| Location                     | GLO                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Infrastructure Process       | 0                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Unit                         | kg                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Dataset Version              | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Included Processes           | Includes the mining of tantalite, its transport to the metallurgical plant, purification and reduction steps. Infrastructure is assessed on related metallurgical and mining operations.                                                                                                                                                                                                                                                 |
| Amount                       | 1                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Local Name                   | Tantal, Pulver, für Kondensatoren, ab Regionallager                                                                                                                                                                                                                                                                                                                                                                                      |
| Synonyms                     | tantalium                                                                                                                                                                                                                                                                                                                                                                                                                                |
| General Comment              | This dataset is functional for the use of tantalum as powder in capacitors of portable electronic devices. It is not valid for the use as an alloying element in steel. Only primary production from tantalite / pegmatite ore is considered. Tantalum as by-product from tin production is not considered.                                                                                                                              |
| Start Date                   | 2000                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| End Date                     | 2003                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Data valid for entire period | 1                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Geography Text               |                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Technology Text              | Open pit mining and beneficiation at Wodgina and Greenbushes in Australia, reduction from pentoxide at HC Starck, Germany. Relates to state of the art technology, where purified Ta <sub>2</sub> O <sub>5</sub> powder is reduced by magnesium vapour in a rotary kiln or a multiple hearth furnace. Production from secondary materials such as post consumer scrap is disregarded, as is the by product tantalum from tin production. |
| Representativeness           | 80%                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Production Volume            | 1045.2 t per year                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Sampling Procedure           | Literature                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Extrapolations               | see Geography and Technology                                                                                                                                                                                                                                                                                                                                                                                                             |
| Uncertainty Adjustments      | This dataset is established on for the use of tantalum in capacitors. Any other application is not covered by this dataset.                                                                                                                                                                                                                                                                                                              |

### 5.1.2 Mining of Tantalumpentoxide

Most of the tantalum is mined in the two operations Greenbushes and Wodgina in Western Australia. The resulting concentrates contain around 40% Ta<sub>2</sub>O<sub>5</sub>. An estimated worldwide 10% of Niobium supply originates from joint extraction with tantalite deposits (Polyakov & Polyakova (2003)). However, this coupled product is not modelled within this dataset.

The overall recovery of tantalum from ore ranges between 72-74% (73% is assumed in this study), upgrading (concentrating) the ore from 0.1% to 38% Ta<sub>2</sub>O<sub>5</sub> (Burt, 1996). Assuming an average ore concentration of 194 ppm (weighted mean of Greensbushes and Wodgina deposits, see Section 3.1 Primary Resources) 7067 t ore is required to produce 2.63 t concentrate containing 1 t Ta<sub>2</sub>O<sub>5</sub>.

The tantalite is currently mined in open pits – one underground pit in Greenbushes is suspended but may resume operation in the future. The dust emissions are approximated in both amount and particle

distribution approximated with specific values from copper mining from the CEPMEIP-Database<sup>1</sup>. The infrastructure has been approximated with the respective dataset for a generic non ferrous metal mine with 20'000 Mt ore throughput in 50 year lifetime. The use of conveyor belts has been assumed to be of the same specific value per tonne ore mined as in the copper production as approximation (see Tab. 5.2). Land use and occupation has been taken from Sons of Gwalia (2003). The remaining mass after separating the tantalite is the tailings that must be disposed of. The transport services reflect the transportation of tantalum oxide to the harbour by train (60 km) and further by ship to Germany (14'000 km) and train inland to the manufacturer (400 km).

**Tab. 5.2** Specific values for infrastructure employed for mining and of tantalite concentrate

| Specific Values                             | Unit                     | Value    | Source & Comment             |
|---------------------------------------------|--------------------------|----------|------------------------------|
| Land use                                    | m <sup>2</sup> / t ore   | 7.85E-02 | Sons of Gwalia (2003)        |
| Land occupation                             | m <sup>2</sup> a / t ore | 7.85E-02 | Sons of Gwalia (2003)        |
| Conveyor belts                              | m / t ore                | 7.57E-05 | Proxy with copper production |
| Dust emissions (Total solid particles, TSP) | g / t ore                | 101.72   | CEPMEIP <sup>1</sup>         |
| Dust emissions, PM10                        | g / t ore                | 50       | CEPMEIP <sup>1</sup>         |
| Dust emissions, PM2.5                       | g / t ore                | 5        | CEPMEIP <sup>1</sup>         |

**Tab. 5.3** Process data of mining and beneficiation per 1 t Ta<sub>2</sub>O<sub>5</sub> contained in the resulting tantalite concentrate.

| Mining and Beneficiation of Tantalite                   | 1 t Ta <sub>2</sub> O <sub>5</sub> contained in concentrate |          | Source & Comment      |
|---------------------------------------------------------|-------------------------------------------------------------|----------|-----------------------|
| Inputs                                                  |                                                             |          |                       |
| Ore                                                     | t                                                           | 7067     | Calculation           |
| Ta <sub>2</sub> O <sub>5</sub> in ore                   | kg                                                          | 1370     | Calculation           |
| Electricity                                             | kWh                                                         | 155466   | Sons of Gwalia (2003) |
| Water Consumption                                       | m <sup>3</sup>                                              | 5212     | Sons of Gwalia (2003) |
| Fuel Consumption                                        | MJ                                                          | 828720   | Sons of Gwalia (2003) |
| exploitation infrastrutrue, conveyor belts              | m                                                           | 0.535    | Approximation         |
| infrastructure of mining (open pit)                     | unit                                                        | 1.99E-04 | Mine                  |
| Land transformation                                     | m <sup>2</sup>                                              | 554      |                       |
| Land occupation                                         | m <sup>2</sup>                                              | 554      |                       |
| Concentrate transport by train                          | tkm                                                         | 1.21     |                       |
| Concentrate transport by ship                           | tkm                                                         | 36.8     |                       |
| Outputs                                                 |                                                             |          |                       |
| Tailings disposed                                       | t                                                           | 7064     | Calculation           |
| PM<2.5                                                  | kg                                                          | 35.3     |                       |
| PM2.5-10                                                | kg                                                          | 318      |                       |
| PM>10                                                   | kg                                                          | 365      |                       |
| Water effluents                                         | m <sup>3</sup>                                              | 5.21     |                       |
| Concentrate at 38% Ta <sub>2</sub> O <sub>5</sub>       | kg                                                          | 2632     |                       |
| Ta <sub>2</sub> O <sub>5</sub> contained in concentrate | kg                                                          | 1000     |                       |

<sup>1</sup> [http://www.air.sk/tno/cepmeip/em\\_factors.php](http://www.air.sk/tno/cepmeip/em_factors.php), accessed in 2003

### 5.1.3 Production of Ta<sub>2</sub>O<sub>5</sub>

As explained in the introduction of 5 *System Characterisation*, the adequate process for reducing the tantalite is assumed to make use of magnesium (see Fig. 5.2). HC Stark/Bayer patented a process with gaseous magnesium (Shekhter et al., 2001) is taken here as the method to produce tantalum powder. The process considered for the production of capacitor grade tantalum powder consists of the following steps (Gille et al., 2006, Fig. 5.3).

- Breakdown with HF
- Extraction with MIBK
- Selective extraction of Niobium
- Stripping of tantalum from MIBK
- Precipitation of oxide hydrates
- Calcination to Ta<sub>2</sub>O<sub>5</sub>
- Reduction of Ta<sub>2</sub>O<sub>5</sub> with magnesium vapour

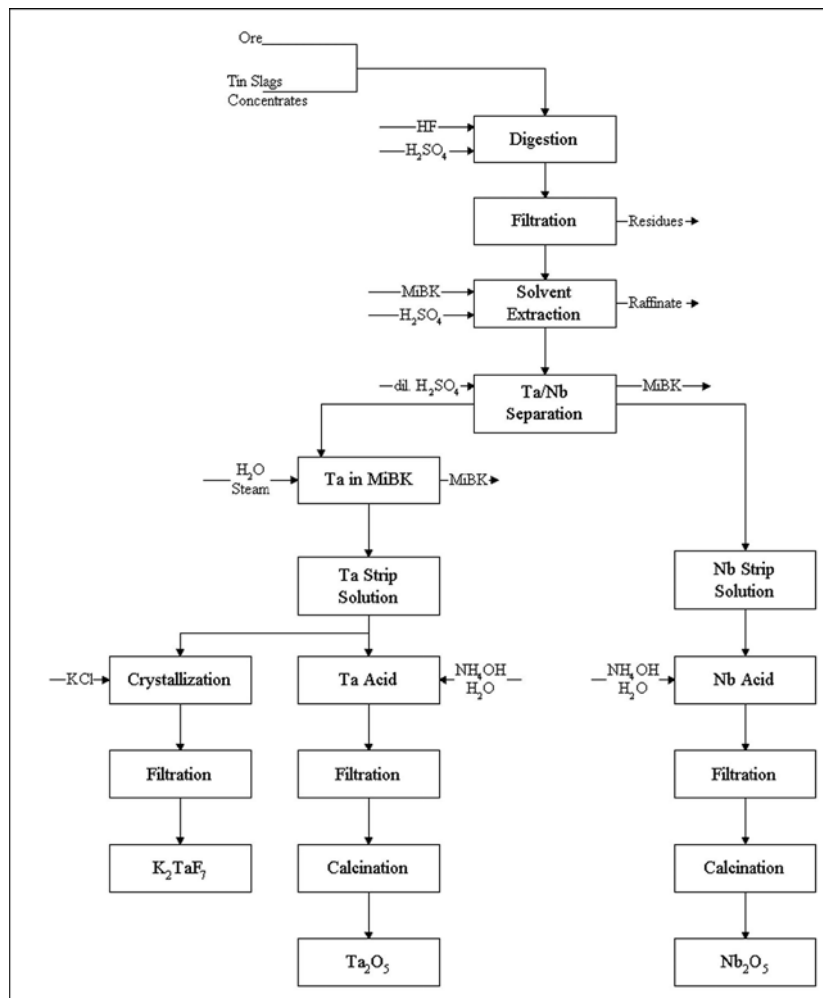
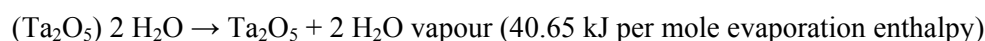


Fig. 5.3 Hydrometallurgical Processing of Niobium and Tantalum (Eckert et al., 2001)

The ore concentrate, up to 40% Ta<sub>2</sub>O<sub>5</sub> (in this study 38%, see 5.1.2 *Mining of Tantalumpentoxide*), is dissolved in concentrated hydrofluoric acid (Andersson et al., 1996). Petrova et al. (1997) report extraction efficiency for artificial tantalum concentrate (including TaC bearing scrap) of 80-85% in traditional industries in 1996 and 95% with optimised parameters.

Gupta (2003) gives a stoichiometry in the leaching process of 16 moles of HF per 2 moles H<sub>2</sub>TaO<sub>6</sub> in solution. Eckert (1996) indicates large amounts of HF (2 t), H<sub>2</sub>SO<sub>4</sub> (3 t) and NH<sub>3</sub> (700 kg) required in 1990 to produce one tonne pure Ta<sub>2</sub>O<sub>5</sub>. Although the process advanced since then regarding regenerating the acids, no quantitative indication on the specific net requirements of the process chemicals can be made. In this study therefore the values of Eckert (1996) (also cited in Polyakov & Polyakova, 2003) are used.

The following reaction was assumed for subsequent calcination (dehydration):



This results in a theoretical energy requirement of 0.23 MJ heat per kg oxide, assuming an efficiency of 80%. The electricity demand for pumping was disregarded, because no information on this was available. Ammonia recovery is assumed to amount to 99.9% of the used ammonia in a 28% NH<sub>3</sub> solution (Eckert, 1996), the rest being emitted into the air. Also the spent HF is recovered as diluted solution. No process information for recovery of the spent reactants is available. Both recovered solutions are therefore regarded here to be by-products and leave the system burden free without substitution for process chemical inputs.

Tab. 5.4 Assumptions Extraction of tantalum oxide form Tantalite concentrate

| Specific Values         | Unit | Value    | Remarks / Source                                                                  |
|-------------------------|------|----------|-----------------------------------------------------------------------------------|
| Extraction efficiency   | %    | 95%      | Assumption                                                                        |
| Infrastructure facility | unit | 4.00E-10 | approximated with module "chemical plant, organics": 4.2 ha, output of 50'000 t/a |

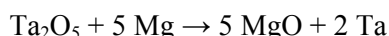
Tab. 5.5 Extraction of tantalum oxide form Tantalite concentrate

| Extraction                                |     | 1 kg Ta <sub>2</sub> O <sub>5</sub> | Remarks / Source                                                                                                                                                |
|-------------------------------------------|-----|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inputs</b>                             |     |                                     |                                                                                                                                                                 |
| Concentrate fed                           | kg  | 2.77                                | Tantalite obtained from mining and beneficiation, see Tab. 5.3                                                                                                  |
| HF                                        | kg  | 2                                   | Polyakov & Polyakova (2003)                                                                                                                                     |
| H <sub>2</sub> SO <sub>4</sub>            | kg  | 3                                   | Polyakov & Polyakova (2003)                                                                                                                                     |
| NH <sub>3</sub>                           | kg  | 0.7                                 | Polyakov & Polyakova (2003)                                                                                                                                     |
| Energy for Calcination                    | MJ  | 0.23                                | Evaporation of Ta <sub>2</sub> O <sub>5</sub> x 2 H <sub>2</sub> O with 40,651 kJ per mole evaporation enthalpy                                                 |
| Electricity for pumping                   | kWh | -                                   | Not available                                                                                                                                                   |
| sludge to disposal                        | kg  | 14                                  | solids, precipitated salts: CaF <sub>2</sub> , CaSO <sub>4</sub> , TiO <sub>2</sub> , Fe <sub>2</sub> O <sub>3</sub> . assumed to be sold (Eckert et al., 2001) |
| HF recovered                              | kg  | 2.00                                | Cut-off, considered as by-product                                                                                                                               |
| Ammonia recovered                         | kg  | 0.70                                | Cut-off, considered as by-product                                                                                                                               |
| <b>Outputs</b>                            |     |                                     |                                                                                                                                                                 |
| Ammonia emitted                           | kg  | 0.00070                             | 0.1 % of the ammonia employed.                                                                                                                                  |
| Ta <sub>2</sub> O <sub>5</sub> , purified | kg  | 1                                   | Contains 0.82 kg tantalum                                                                                                                                       |

### 5.1.4 Reduction to Tantalum powder

Only little details are known from this reductive step, therefore the reasoning is based on the respective patent. The reduction is effectuated by means of magnesium vapour that is brought into contact with  $\text{Ta}_2\text{O}_5$  powder in a continuous process, e.g. in a rotary kiln or a multiple hearth furnace. The heat required is assumed to be supplied by electricity. The resulting product contains impurities in the order of 0.2% to 0.25%, most of it being oxygen (Andersson et al., 1996). In the calculation for this inventory therefore a process efficiency of 100% was assumed. An overview of the specific values and assumptions used is given in Tab. 5.6.

The energy required is calculated upon the reaction stoichiometry:



Normalised to 1 kg Ta output an amount of 1.22 kg  $\text{Ta}_2\text{O}_5$  and 0.336 kg Mg is needed as input materials. Assuming that Magnesium is added 10% in excess (Shekhter et al. (2001)), 0.369 kg Magnesium required is calculated ( $= 0.336 \text{ kg} * 110\%$ ). The unreacted magnesium is supposed to be recovered. Hence no burdens are allocated to it.

The reducing agent magnesium passes two phase transitions from solid state to liquid (8.95 kJ / mol) and from liquid to gas (127.6 kJ / mol). The total theoretical energy requirement therefore amounts to 136.5 kJ / mol with a stoichiometry of 13.81 moles Mg per kg Ta.

The magnesium oxide is removed from the tantalum powder with diluted sulphuric acid and cleaned with ultra pure water. Since no quantitative information on this last cleaning step is available, 3 kg of sulphuric acid and 2 kg of ultra pure water per kg Ta product were assumed. Furthermore it is assumed that the resulting magnesium sulphate in solution can be processed further. Considered as by-product it leaves the system burden free without substituting raw materials.

**Tab. 5.6** Specific values and assumptions used for calculation in the reduction step.

| Reduction                  |          | 1 kg Ta powder | Remarks / sources                                                               |
|----------------------------|----------|----------------|---------------------------------------------------------------------------------|
| <b>Specific values</b>     |          |                |                                                                                 |
| Evaporation enthalpy of Mg | kJ / mol | 127.6          |                                                                                 |
| Enthalpy of fusion of Mg   | kJ / mol | 8.95           |                                                                                 |
| Magnesium stoichiometry    | %        | 110%           | $\text{Ta}_2\text{O}_5 + 5 \text{ Mg} \rightarrow 5 \text{ MgO} + 2 \text{ Ta}$ |
| Density Argon              | g/l      | 1.784          | norm pressure                                                                   |
| Density Oxygen             | g/l      | 1.429          | norm pressure                                                                   |
| reduction efficiency       | %        | 100%           |                                                                                 |



**Tab. 5.7** Process data for the reduction of 1 kg tantalumpentoxide with magnesium vapour.

| Reduction                                 |     | 1 kg Ta powder | Remarks / sources                                                                                                     |
|-------------------------------------------|-----|----------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Input</b>                              |     |                |                                                                                                                       |
| Ta <sub>2</sub> O <sub>5</sub> , purified | kg  | 1.22           | See. Tab. 5.5. Assuming reduction efficiency of 100%                                                                  |
| Magnesium                                 | kg  | 0.369          | with Mg 10% in excess to stoichiometry                                                                                |
| Electricity for reduction                 | kWh | 0.720          | Assuming 80% efficiency in heat production and a total theoretical energy of 136.5 kJ / mol Mg for evaporation        |
| Argon                                     | kg  | 0.0892         | Assuming 5 m3 Volume per 100 kg tantalum powder                                                                       |
| Oxygen                                    | kg  |                |                                                                                                                       |
| Dilute sulphuric acid                     | kg  | 3              | estimation                                                                                                            |
| ultrapure water                           | kg  | 2              | estimation                                                                                                            |
| <b>Output</b>                             |     |                |                                                                                                                       |
| Wastewater                                | kg  | 5.594          | Loaded with SO <sub>4</sub> , solved 0.557 kg MgO, 0.0369 kg solved unreacted Mg, by-product, not counted, estimation |
| Pure Tantalum Powder                      | kg  | 1              |                                                                                                                       |

### 5.1.5 Transport and Infrastructure

The concentrate is assumed to be transported by train to the nearest port (60 km), by ship to Germany (14'000 km), then by train to the facility (400 km). Based on the ecoinvent quality-guidelines, developed for Europe, the chemicals are assumed to be transported 600 km by train and 100 km by lorry also in this case. An overview over the resulting transport service is given in Tab. 5.8. In this study, the infrastructure for extraction and reduction of tantalum pentoxide is estimated based on the module "chemical plant, organics", that has a built area of about 4.2 ha and an average output of 50'000 t/a (Althaus et al., 2003).

**Tab. 5.8** Inventory and calculations for transport and infrastructure.

|                         | Unit | Value    | remarks                                                                                         |
|-------------------------|------|----------|-------------------------------------------------------------------------------------------------|
| Chemicals by train      | tkm  | 7.7      | 12.7 kg over 600 km, estimation using ecoinvent standard distance for Europe                    |
| Chemicals by lorry      | tkm  | 1.3      | 12.7 kg over 100 km, estimation using ecoinvent standard distance for Europe                    |
| Infrastructure facility | Unit | 4.00E-10 | approximated with module "chemical plant, organics": 4.2 ha total surface, output of 50'000 t/a |

## 5. System Characterisation

| General Flow information                    |    |              |  |                   | Representation in ecoinvent |                                                          |                        |                            |                 |          |                                                               |            |            | Uncertainty information |                |      |          |                  |
|---------------------------------------------|----|--------------|--|-------------------|-----------------------------|----------------------------------------------------------|------------------------|----------------------------|-----------------|----------|---------------------------------------------------------------|------------|------------|-------------------------|----------------|------|----------|------------------|
| Input                                       |    | Process Name |  | Output            |                             | Remarks                                                  | Category               | Sub-category               | Infra-structure | Location | Modul name in ecoinvent                                       | Mean value | Mean value | Unit                    | Source         | Type | StDv 95% | General Comment  |
| Ta2O5 in ore                                | ML | Mining       |  |                   |                             | 9.08 t ore at 194 ppm Tantalite, 73% extraction yield    | resource               | in ground                  |                 |          | Tantalum, 39.5% in tantalite, 7.7E-5% in crude ore, in ground | 1.76E+00   |            | kg                      | 0              | 1    | 1.13     | (2,2,2,1,1,4,12) |
| Electricity                                 | ML |              |  |                   |                             | Environmental Report 2003                                | electricity            | production mix             | No              | UCTE     | electricity, medium voltage, production UCTE, at grid         | 2.00E+02   |            | kWh                     | SGW (2003)     | 1    | 1.14     | (2,3,2,1,1,4,2)  |
| Water Consumption                           | ML |              |  |                   |                             | Environmental Report 2003                                | resource               | in water                   |                 |          | Water, river                                                  | 6.70E+00   |            | m3                      | SGW (2003)     | 1    | 1.13     | (2,2,1,1,1,4,12) |
| Fuel Consumption                            | ML |              |  |                   |                             | Environmental Report 2003                                | construction processes | machinery                  | No              | GLO      | diesel, burned in building machine                            | 1.07E+03   |            | MJ                      | SGW (2003)     | 1    | 1.13     | (2,2,2,1,1,4,4)  |
| exploitation infrastructure, conveyor belts | ML |              |  |                   |                             | same values as in copper exploitation (7.6E-5 m / t ore) | construction processes | machinery                  | Yes             | RER      | conveyor belt, at plant                                       | 6.87E-04   |            | m                       | Althaus (2003) | 1    | 3.36     | (5,5,1,1,3,5,9)  |
| infrastructure of mining (underground)      | ML |              |  |                   |                             | no underground mining                                    | metals                 | extraction                 | Yes             | GLO      | non-ferrous metal mine, underground                           | 0.00E+00   |            | unit                    | SGW (2003)     | 1    | 3.36     | (5,5,1,1,3,5,9)  |
| infrastructure of mining (surface)          | ML |              |  |                   |                             | env. 20'000'000 t during 50 years assumed                | metals                 | extraction                 | Yes             | GLO      | non-ferrous metal mine, surface                               | 2.56E-07   |            | unit                    | estimation     | 1    | 3.36     | (5,5,1,1,3,5,9)  |
| Land transformation from nature             | ML |              |  |                   |                             | meanvalue of 1999 through 2003 (7.8E-2 m2 / t ore)       | resource               | land                       |                 |          | Transformation, from unknown                                  | 7.13E-01   |            | m2                      | SGW (2003)     | 1    | 2.05     | (2,2,2,1,3,2,8)  |
| Land transformation to mine                 | ML |              |  |                   |                             | meanvalue of 1999 through 2003 (7.8E-2 m2 / t ore)       | resource               | land                       |                 |          | Transformation, to mineral extraction site                    | 7.13E-01   |            | m2                      | SGW (2003)     | 1    | 2.05     | (2,2,2,1,3,2,8)  |
| Land use for mining                         | ML |              |  |                   |                             | steady state assumed                                     | resource               | land                       |                 |          | Occupation, mineral extraction site                           | 7.13E-01   |            | m2a                     | estimation     | 1    | 1.57     | (2,2,2,1,3,2,7)  |
| Land transformation from mine               | ML |              |  |                   |                             | meanvalue of 1999 through 2003 (7.8E-2 m2 / t ore)       | resource               | land                       |                 |          | Transformation, from mineral extraction site                  | 7.13E-01   |            | m2                      | SGW (2003)     | 1    | 2.05     | (2,2,2,1,3,2,8)  |
| Land transformation to nature               | ML |              |  |                   |                             | meanvalue of 1999 through 2003 (7.8E-2 m2 / t ore)       | resource               | land                       |                 |          | Transformation, to unknown                                    | 7.13E-01   |            | m2                      | SGW (2003)     | 1    | 2.05     | (2,2,2,1,3,2,8)  |
| Concentrate transport train                 | ML |              |  |                   |                             | mine->port (AUS): 60 km / port (GER)->facility: 400 km   | transport systems      | train                      | No              | RER      | transport, freight, rail                                      | 1.56E-03   |            | tkm                     | estimation     | 1    | 2.02     | (2,1,1,1,1,4,5)  |
| Concentrate transport ship                  | ML |              |  |                   |                             | port (AUS)->port (GER): 14000 km                         | transport systems      | ship                       | No              | RER      | transport, barge                                              | 4.74E-02   |            | tkm                     | estimation     | 1    | 2.02     | (2,1,1,1,1,4,5)  |
|                                             | ML |              |  | Tailings disposed |                             |                                                          | waste management       | residual material landfill | No              | GLO      | disposal, non-sulfidic tailings, off-site                     | 9.08E+03   |            | kg                      |                | 1    | 1.13     | (2,2,2,1,1,4,6)  |
|                                             | ML |              |  | PM<2.5            |                             | 4.9% of 0.102 kg dust per tonne ore mined                | air                    | low population density     |                 |          | Particulates, < 2.5 um                                        | 4.54E-02   |            | kg                      | CEPMEIP        | 1    | 2.10     | (2,2,2,2,3,5,26) |
|                                             | ML |              |  | PM2.5-10          |                             | 44.2% of 0.102 kg dust per tonne ore mined               | air                    | low population density     |                 |          | Particulates, > 2.5 um, and < 10um                            | 4.09E-01   |            | kg                      | CEPMEIP        | 1    | 1.62     | (2,2,2,2,3,5,25) |
|                                             | ML |              |  | PM>10             |                             | 50.8% of 0.102 kg dust per tonne ore mined               | air                    | low population density     |                 |          | Particulates, > 10 um                                         | 4.70E-01   |            | kg                      | CEPMEIP        | 1    | 3.10     | (2,2,2,2,3,5,27) |
|                                             | ML |              |  | Water effluents   |                             | not accounted for                                        |                        |                            |                 |          |                                                               | 6.70E+00   |            | m3                      |                | 1    | 1.13     | (2,2,2,1,1,4,6)  |

Fig. 5.4 Flows for "tantalum, powder, capacitor-grade, at regional storage" and their representation in the ecoinvent database.

## 5. System Characterisation

| General Flow information  |    |              |           |                    |  | Representation in ecoinvent                                                                                                      |                                                                                                                                                    |                        |                 |          |                                                 |                                                        |            | Uncertainty information |                 |      |          |                  |                 |
|---------------------------|----|--------------|-----------|--------------------|--|----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-----------------|----------|-------------------------------------------------|--------------------------------------------------------|------------|-------------------------|-----------------|------|----------|------------------|-----------------|
| Input                     |    | Process Name |           | Output             |  | Remarks                                                                                                                          | Category                                                                                                                                           | Sub-category           | Infra-structure | Location | Modul name in ecoinvent                         | Mean value                                             | Mean value | Unit                    | Source          | Type | StDv 95% | General Comment  |                 |
|                           |    |              |           |                    |  |                                                                                                                                  |                                                                                                                                                    |                        |                 |          |                                                 |                                                        |            |                         |                 |      |          |                  |                 |
| HF                        | mL | Extraction   |           |                    |  |                                                                                                                                  | chemicals                                                                                                                                          | inorganics             | No              | GLO      | hydrogen fluoride, at plant                     | 2.44E+00                                               |            | kg                      | Polyakov (2003) | 1    | 1.16     | (2,2,3,1,1,4,4)  |                 |
| H2SO4                     | mL |              |           |                    |  |                                                                                                                                  | chemicals                                                                                                                                          | inorganics             | No              | RER      | sulphuric acid, liquid, at plant                | 3.66E+00                                               |            | kg                      | Polyakov (2003) | 1    | 1.16     | (2,2,3,1,1,4,4)  |                 |
| NH3                       | mL |              |           |                    |  |                                                                                                                                  | chemicals                                                                                                                                          | inorganics             | No              | RER      | ammonia, liquid, at regional storehouse         | 8.55E-01                                               |            | kg                      | Polyakov (2003) | 1    | 1.16     | (2,2,3,1,1,4,4)  |                 |
| Energy for Calcination    | mL |              |           |                    |  | Dehydration approximated with 40,6 kJ per mole evaporationenthalpy for water.                                                    | natural gas                                                                                                                                        | heating systems        | No              | RER      | heat, natural gas, at industrial furnace >100kW | 2.80E-01                                               |            | MJ                      |                 | 1    | 1.88     | (5,5,2,1,4,5,2)  |                 |
|                           |    |              | mL        | sludge to disposal |  | solids, precipitated salts: CaF2, CaSO4, TiO2, Fe2O3. assumed to be sold                                                         |                                                                                                                                                    |                        |                 |          |                                                 | 1.71E+01                                               |            | kg                      | Eckert (1996)   | 1    | 1.13     | (2,2,2,1,1,4,6)  |                 |
|                           |    |              | mL        | HF recovered (40%) |  | used as by-product                                                                                                               |                                                                                                                                                    |                        |                 |          |                                                 | 2.44E+00                                               |            | kg                      |                 | 1    |          |                  |                 |
|                           |    |              | mL        | Ammonia recovered  |  | used as by-product                                                                                                               |                                                                                                                                                    |                        |                 |          |                                                 | 8.54E-01                                               |            | kg                      |                 | 1    |          |                  |                 |
|                           |    |              | mL        | Ammonia emitted    |  |                                                                                                                                  | air                                                                                                                                                | low population density |                 |          | Ammonia                                         | 8.55E-04                                               |            | kg                      |                 | 1    | 1.52     | (2,2,2,1,1,4,16) |                 |
| Magnesium                 | mL |              | Reduction |                    |  |                                                                                                                                  | with Mg 10% in excess                                                                                                                              | metals                 | extraction      | No       | RER                                             | magnesium, at plant                                    | 3.69E-01   |                         | kg              |      | 1        | 1.38             | (4,5,1,1,1,5,4) |
| Electricity for reduction | mL |              |           |                    |  |                                                                                                                                  | Assuming 80% efficiency in heat production and a total of 136.5 MJ / mol Mg for evaporation. Heating-up considered by heat exchange from reaction. | electricity            | production mix  | No       | UCTE                                            | electricity, medium voltage, production UCTE, at grid  | 7.20E-01   |                         | kWh             | 0    | 1        | 1.38             | (4,5,1,1,1,5,2) |
| Argon                     | mL |              |           |                    |  | Assuming 5 m3 Volume per 100 kg Tantalum powder                                                                                  | chemicals                                                                                                                                          | inorganics             | No              | RER      | argon, liquid, at plant                         | 8.92E-02                                               |            | kg                      |                 | 1    | 1.38     | (4,5,1,1,1,5,4)  |                 |
| Dilute sulphuric acid     | mL |              |           |                    |  | estimations                                                                                                                      | chemicals                                                                                                                                          | inorganics             | No              | RER      | sulphuric acid, liquid, at plant                | 3.00E+00                                               |            | kg                      | estimation      | 1    | 1.38     | (4,5,1,1,1,5,4)  |                 |
| ultrapure water           | mL |              |           |                    |  | estimations                                                                                                                      | water supply                                                                                                                                       | production             | No              | GLO      | water, ultrapure, at plant                      | 2.00E+00                                               |            | kg                      | estimation      | 1    | 1.38     | (4,5,1,1,1,5,4)  |                 |
|                           |    | mL           |           | Wastewater         |  | Loaded with SO4 and Mg, byproduct, not counted approximated with module "chemical plant, organics": 4.2 ha. output of 50'000 t/a | chemicals                                                                                                                                          | organics               | Yes             | RER      | chemical plant, organics                        | 4.00E-10                                               |            | unit                    |                 | 1    | 3.55     | (5,5,2,1,4,5,9)  |                 |
| Infrastructure facility   | mL |              |           |                    |  | 12.4 kg over 600 km                                                                                                              | transport systems                                                                                                                                  | train                  | No              | RER      | transport, freight, rail                        | 7.45E+00                                               |            | tkm                     | estimation      | 1    | 2.02     | (2,1,1,1,1,4,5)  |                 |
| Chemicals by train        | mL |              |           |                    |  | 12.4 kg over 100 km                                                                                                              | transport systems                                                                                                                                  | road                   | No              | RER      | transport, lorry >16t, fleet average            | 1.24E+00                                               |            | tkm                     | estimation      | 1    | 2.02     | (2,1,1,1,1,4,5)  |                 |
| Chemicals by lorry        | mL |              |           |                    |  |                                                                                                                                  |                                                                                                                                                    | metals                 | extraction      | No       | GLO                                             | tantalum, powder, capacitor-grade, at regional storage | 1.00E+00   |                         | kg              |      |          |                  |                 |

Fig. 5.4 (cont.) Flows for "tantalum, powder, capacitor-grade, at regional storage" and their representation in the ecoinvent database

## 6 Data Quality Considerations

The production process is rather new and not much is known from public sources about quantitative process measures. Therefore the data has been established based on reasoning and rough assumptions. Direct emissions others than dust and ammonia have been omitted due to lack of data. This might restrain the quality of this inventory, if omitted emissions prove to be important in the overall assessment. However, no estimation of the relative importance of this omission can be performed on the basis of the current information. Infrastructure requirement has been approximated with the related technology of copper mining and chemical plants. The uncertainty has been calculated with the pedigree approach. The data describe the production process supplying the demands for the use in capacitors for electronic devices. This dataset should not be used when tantalum is a substantial part of the system modelled (e.g. comparing different capacitor materials). It has to be stressed that this dataset is solely valid for tantalum powder that is used in capacitors. For all other uses of tantalum – namely: tantalum based superalloys in high temperature machinery and tantalum carbides in hard metals – this dataset is not applicable.

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## Part XI

# Tin

Data v2.1 (2009)

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Classen M., Althaus H.-J., Blaser S., Tuchs Schmid M., Jungbluth N., Doka G., Faist Emmenegger M. and Scharnhorst W. (2009) Life Cycle Inventories of Metals. Final report ecoinvent data v2.1, No 10. EMPA Dübendorf, Swiss Centre for Life Cycle Inventories, Dübendorf, CH, Online-Version under: <http://www.ecoinvent.ch>.

Final report of updated Swiss National Life Cycle Inventory Database "ecoinvent", version 2.1

## Comments on version 2.1

This part and all data correspond to the part and data reported in version 2.0. No changes were made.

## Summary

The production of tin is inventoried in the modules described in this report. Data reported are mainly based on theoretical models, similarity assumptions and estimations. Secondary sources of tin (recycling) is not considered, thus the overall data quality is poor.

Within ecoinvent, these data are used for representation of materials of minor importance. If in an LCA study one of these materials is of importance, the data should not be used.

This part is the excerpt from “Part VIII Other metals” in Althaus et al. (2004) that has been reorganised for the actual version 2.0. No changes from the original dataset v 1.1 are reported, hence no adjustments have been accomplished for the update to version 2.0. Note that the flow names in the tables displaying unit processes have not been updated to reflect the name changes with version 2.0.





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# 1 Introduction

The information available for these inventories was rather sparse. Thus the data quality must be described as poor. One important shortcoming is the neglecting of tin recycling. Within ecoinvent, these data are used for representation of materials of minor importance. If in an LCA study one of these materials is of importance, the data should not be used.

## 2 Reserves and Resources Tin

The most important tin ore is cassiterite ( $\text{SnO}_2$ ), although small amounts of tin are recovered from sulphide minerals such as stannite ( $\text{Cu}_2\text{FeSnS}_4$ ). Although most tin is obtained from mining tin ores, cassiterite is also found in association with ores of tungsten, tantalum and lead and minor quantities of tin are recovered as by-products of mining these metals.

Tin occurs in both primary (typically acid igneous intrusive rocks) and secondary deposits such as alluvial and eluvial deposits. Only about 20 percent of the world's tin deposits occur as primary hard-rock veins or lodes. These ores contain predominantly high temperature minerals and commonly occur in close association with silicic granites. About 80 % of the world's tin deposits occur not as primary lodes, but as unconsolidated secondary or placer deposits in river beds and valleys or on the sea ground. The largest concentration of both onshore and offshore placers is in the extensive tin belt of South-East Asia, stretching from China in the north, through Thailand, Burma and Malaysia, to the islands of Indonesia in the south.

Compared to commercially viable deposits of copper, lead, zinc, nickel and bauxite, tin deposits are generally small. Further, tin is almost always found closely allied to the granite from which it originates.<sup>1</sup>

The tin production given in Tab. 2.1 is taken from the itri homepage<sup>2</sup> that refers to the bi-monthly 'Tin Monitor' published by CRU International Ltd.

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<sup>1</sup> <http://www.miningtrading.com/tin/index.html>,  
[http://www.tintechnology.com/library/detail/library\\_/Tin%20Mining.htm](http://www.tintechnology.com/library/detail/library_/Tin%20Mining.htm), access date 14.03.03

<sup>2</sup> <http://www.itri.co.uk/market.htm>, access date 14.03.03

Tab. 2.1 Western World Tin Production by Country (tonnes) <sup>2</sup>

| Country      | 1995        |                     | 1996        |                     |
|--------------|-------------|---------------------|-------------|---------------------|
|              | Refined Tin | Tin-in-Concentrates | Refined Tin | Tin-in-Concentrates |
| Malaysia     | 41,000      | 6,402               | 38,100      | 5,175               |
| Indonesia    | 44,200      | 46,058              | 48,900      | 51,024              |
| Thailand     | 8,200       | 1,784               | 11,000      | 1,299               |
| Brazil       | 17,000      | 19,500              | 17,600      | 19,200              |
| Bolivia      | 18,000      | 16,300              | 16,700      | 14,802              |
| Peru         | -           | 22,020              | 500         | 27,002              |
| Australia    | -           | 7,750               | -           | 8,496               |
| Portugal     | -           | 4,616               | -           | 4,625               |
| UK           | -           | 2,000               | -           | 2,000               |
| Africa       | -           | 3,200               | -           | 3,200               |
| Rest of West | 14,000      | 2,000               | 13,000      | 2,000               |
| TOTAL        | 142,400     | 131,630             | 145,800     | 138,823             |

Note: China and Russia are major tin producers but production data is not currently verifiable.

Tin is also gained from recycling. On one hand there is the recycling of "new scrap" that can be done easily. On the other hand there is the recovery of tin from tinplate which is becoming increasingly difficult, as the change to electrolytic methods of tinning is leading to very thin coatings which sometimes diffuse into the steel sheet. The recycling of this material will continue to be a technically and economically difficult task. In principle, two processes are used today for recovering tin from tinplate: the alkaline electrolytic method and the alkaline chemical method. Processing tin-containing alloys is easier than recovering tin from scrap tinplate. However, the tin content of many alloys has decreased over recent years, and cheaper materials have in some cases replaced alloys. Both these developments have tended to limit the potential for recovering secondary tin, and also explain why the amount of recovered tin in Europe has decreased from 15 800 t in 1980 to 13 300 t in 1990. (Graf (1997))

### 3 Characterisation of Tin

This information is based on Graf (1997).

Tin (Sn) exists in two crystalline modifications, the  $\alpha$ - and  $\beta$ -forms.  $\alpha$ -tin is also called grey tin while  $\beta$ -tin is referred to as white tin. Some physical properties of  $\alpha$ - and  $\beta$ -tin are listed in Tab. 3.1.

In the periodic table, tin lies on the boundary between metals and non-metals. The transformation of the  $\alpha$ - to the  $\beta$ -modification is accompanied by a complete change of lattice structure, affecting the physical, chemical, and mechanical properties. The transformation of  $\beta$ -tin (white tin) into  $\alpha$ -tin (grey tin) is of practical importance, as it involves a volume increase of 21 %. The transformation process requires a high energy of activation, and can be very strongly hindered. White  $\beta$ -tin can therefore exist for many years at  $-30\text{ }^{\circ}\text{C}$ . The presence of  $\alpha$ -tin seed crystals is important for the transformation process, and these are formed by repeated phase transitions. Foreign "elements" also affect the transformation temperature and rate. These can consist of impurities and deformations. Tin vapour consists of  $\text{Sn}_2$  molecules.

**Tab. 3.1 Physical properties of  $\alpha$ - and  $\beta$ -tin (Graf (1997))**

|                                                                       |                                          |
|-----------------------------------------------------------------------|------------------------------------------|
| Natural isotopes                                                      | 10                                       |
| Relative atomic mass                                                  | 118.69                                   |
| Transformation temperature $\alpha$ -Sn $\leftrightarrow$ $\beta$ -Sn | 286.2 K                                  |
| Enthalpy of transformation                                            | 1966 J/mol                               |
| Density of $\beta$ -tin (20 °C)                                       | 7.286 g/cm <sup>3</sup>                  |
| Density of $\alpha$ -tin                                              | 5.765 g/cm <sup>3</sup>                  |
| Density of liquid tin (240 °C)                                        | 6.992 g/cm <sup>3</sup>                  |
| Molar heat capacity of $\beta$ -tin (25 °C)                           | 27.0 J mol <sup>-1</sup> K <sup>-1</sup> |
| Molar heat capacity of liquid tin                                     | 28.5 J mol <sup>-1</sup> K <sup>-1</sup> |
| Melting point                                                         | 505.06 K                                 |
| Enthalpy of fusion                                                    | 7029 J/mol                               |
| Boiling point                                                         | 2876 K                                   |
| Enthalpy of vaporization                                              | 295'763 J/mol                            |
| Coefficient of thermal conductivity of $\beta$ -tin at 0 °C           | 0.63 W cm <sup>-1</sup> K <sup>-1</sup>  |
| Specific electrical resistivity of $\alpha$ -tin (0 °C)               | 5×10 <sup>-6</sup> Wm                    |
| Specific electrical resistivity of $\beta$ -tin (25 °C)               | 11.15×10 <sup>-6</sup> Wm                |

Tin is stable in dry air, but is considerably more rapidly oxidised at a relative humidity of 80 %. Bright metallic tin becomes dull within 100 d even in indoor atmospheres. Oxygen is rapidly and irreversibly chemisorbed, and the oxide layer formed grows at an exponentially increasing rate. Typical impurities present after metallurgical production (e.g., Sb, Tl, Bi, and Fe) promote oxidation. Treatment with carbonate or chromate solutions leads to passivation. For the use of tin in food packaging it is important that lactic, malic, citric, tartaric, and acetic acids do not react at all at normal temperatures or do so to a negligible extent, especially in the absence of atmospheric oxygen. This is also true for alcohols and hydrocarbons.

## 4 Use / Application of Tin

Important applications for Tin are:<sup>3</sup>

- Food packaging and culinary uses: Tinplate (sheet steel coated with a thin layer of tin) is the primary material for food canning and shares the beverage can market with aluminium.
- Electronics: The principal use of tin in electronics is in the use of solders for the joining of electronic components. But there are many other varied uses in the general electronics field.
- Tin chemicals: The biggest single use of tin chemicals is as PVC stabilisers, but besides this they are used in a very wide range of applications, for example as polymerisation catalysts in silicone resins and polyurethane foam manufacture, and in ceramic pigments.
- Plumbing solders: Plumbing solders containing lead are being replaced by safe tin-silver (usually 96.5 percent tin/3.5 percent silver) or tin-copper (97 to 99 percent tin/1 to 3 percent copper) alloys, particularly for drinking water systems.
- Engineering alloys: Tin is used as an alloy with other metals.
- Pewter and bronze: Pewter (an alloy containing about 92 percent tin with small amounts of other metals added for strength) has long been known for its decorative appearance and ease of working as a craft metal. Tin bronzes (alloys of copper with 5-20% tin) are renowned for their use in art castings and as bell metals.
- Dental amalgams: Dental amalgam contains about 13 percent tin by weight together with silver and mercury.
- Anti-corrosion and engineering coatings: Tin and tin-alloy coatings are widely used in the manufacture of bearings and in many kinds of machinery and fabricated parts, both for their anti-corrosion and lubricant properties.
- Wine capsules: A tin coating acted as a barrier layer in traditional lead-based capsules to prevent contact between the lead and the wine. Tin-lead capsules are now being replaced with other materials but the most prestigious and closest in feel to the traditional capsules, are made from almost pure tin.

Consumption of tin in the western world is around 200'000 tonnes per year. The dominant consumers are Europe (33%), USA (17%) and Japan (12%). The distribution of tin consumption is unusually weighted toward Europe where 46% of all beverage containers are tinplate in contrast to the USA where tinplate is not used for beverage containers. Consumption can be divided into four segments, tinplate (30%), solder (29%), alloys and coatings (25%) and chemicals (16%).<sup>4</sup>

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<sup>3</sup> <http://www.tintechnology.com/library/detail/library /Applications%20for%20Tin.htm>, access date 14.03.03

<sup>4</sup> <http://www.murchisonunited.com.au/operations/renison.asp#production>, access date 28.03.03

## 5 System Characterisation

This chapter is mainly based on Graf (1997).

### 5.1 Overview

Tin is mined, beneficiated, smelted and refined. Fig. 5.1 gives an overview of the processes.

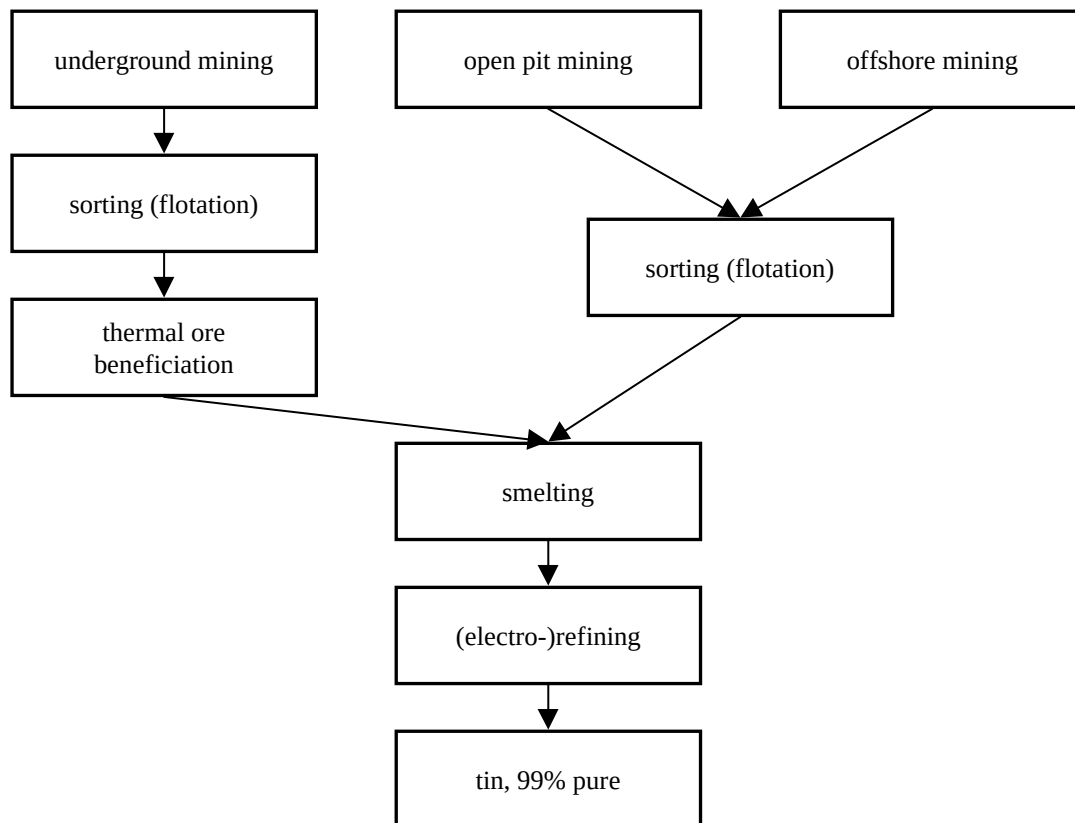


Fig. 5.1 Overview of tin production

### 5.2 Mining and beneficiation

No precise information is found on the amount of tin ore mined in the different types of mines. Based on descriptive information<sup>5</sup> it is therefore assumed that 20% of the tin ore is gained from underground mines and 40% each from open pit mines and from offshore deposits.

#### 5.2.1 Underground mining

Drilling and blasting is only needed in underground mining where depths can reach 1000 meters. Common technology for non-ferrous metal mining is applied. Tin ores are separated from the gangue using sorting methods based on density. The large variation in achievable tin concentration of about 6% to 60% in the ore is due to different sizes and intergrowth of the cassiterite with other minerals.

<sup>5</sup> <http://www.tintechnology.com/library/detail/library/Tin%20Mining.htm>, access date 14.03.03 and <http://www.murchisonunited.com.au/operations/renison.asp#production>, access date 28.03.03

Recoveries of around 75% of the tin are achieved by this beneficiation. Unacceptably high losses of material would occur if ore concentrates with higher tin contents were produced. Therefore, a pyrometallurgical "thermal ore beneficiation" stage is necessary prior to the actual reduction process. This is a volatilisation process that exploits the fact that the iron compounds and other slag formers have low vapour pressures at 1000 – 1500 °C, while SnO and especially SnS volatilise very readily. This technique can also be used to treat tin-containing slags. In the process where tin ore or slag is heated together with pyrite (FeS<sub>2</sub>) the SnO reacts to SnS and the FeS<sub>2</sub> to FeO. With careful process control, 90 – 95 % of the tin can be recovered as an oxidised product containing 40 – 60 % Sn.

Although this is at the lower end of the commercial viability we assume an average tin content of 0.1% in the crude ore that is mined underground and an average recovery of the tin minerals of 70%. Thus 1.43 tons of material has to be mined to gain 1 kg of tin from underground mines. With the assumed average density of 1.5 tons/m<sup>3</sup> crude ore this corresponds to 0.95 m<sup>3</sup> crude ore.

For a lack of specific information the energy consumption and the amount of explosives used for mining are calculated based on the values derived for platinum group metal (PGM) mining. The assumption underlying the calculation is that the respective consumption is proportional to the mass of crude ore extracted. The electricity needed for mechanical beneficiation is taken from Richter et al. (1996) and the water consumption is estimated. The in- and outputs for the thermal beneficiation step given in Tab. 5.2 are calculated with the values given in Tab. 5.1.

**Tab. 5.1 Values used for calculation of in- and output flows of the thermal beneficiation step.**

| <b>Values used for calculation of heat for thermal beneficiation</b>                           |          |              |
|------------------------------------------------------------------------------------------------|----------|--------------|
| Tin content after mechanical beneficiation [%]                                                 | 30       | assumption   |
| Thermal capacity of input [J/kg/K]                                                             | 235.91   | Value for Sn |
| Enthalpy of fusion [J/kg]                                                                      | 59221    | Value for Sn |
| Enthalpy of vaporisation [J/kg]                                                                | 2491894  | Value for Sn |
| Efficiency of kiln                                                                             | 0.25     | assumption   |
| Heating to [K]                                                                                 | 1250     | assumption   |
| Molar weight of Sn [kg/mol]                                                                    | 0.11869  |              |
| <b>Values used for calculation of SO<sub>2</sub> emissions to air of thermal beneficiation</b> |          |              |
| Part of SnO evaporating as SnS                                                                 | 0.9      | assumption   |
| Overall efficiency of abatement                                                                | 0.6      | assumption   |
| Molar weight of SO <sub>2</sub> [kg/mol]                                                       | 0.064058 |              |
| <b>Values used for calculation of FeS<sub>2</sub> input for thermal beneficiation</b>          |          |              |
| Molar weight of FeS <sub>2</sub> [kg/mol]                                                      | 0.119967 |              |
| Utilisation of S in FeS <sub>2</sub>                                                           | 0.7      | assumption   |



**Tab. 5.2 In- and outputs for underground mining and beneficiation of tin ore per kg of tin**

| <b>Underground mining of tin</b>                                    | <b>Flows per kg tin</b> | <b>Source</b>                            |
|---------------------------------------------------------------------|-------------------------|------------------------------------------|
| Crude ore [kg]                                                      | 1430                    | calculation based on ore composition     |
| Diesel, burned in building machines [MJ]                            | 98.96                   | calculation based on PGM                 |
| Electricity [kWh]                                                   | 64.66                   | calculation based on PGM                 |
| blasting [kg]                                                       | 0.66                    | calculation based on PGM                 |
| mine infrastructure [unit]                                          | 5.34*10 <sup>-14</sup>  | calculation based on PGM                 |
| <b>mechanical beneficiation of tin ore from underground mining</b>  |                         |                                          |
| Electricity [kWh]                                                   | 1.43                    | Richter et al. 1996                      |
| Water [m <sup>3</sup> ]                                             | 1.43                    | assumption                               |
| <b>thermal ore beneficiation of tin ore from underground mining</b> |                         |                                          |
| heat from heavy fuel oil furnace [MJ]                               | 37.95                   | calculation                              |
| Electricity [kWh]                                                   | 4                       | calculation based on Richter et al. 1996 |
| FeS <sub>2</sub> input [kg]                                         | 1.179                   | calculation based on assumptions         |
| SO <sub>2</sub> emissions to air [kg]                               | 0.9715                  | calculation based on assumptions         |

### 5.2.2 Open pit mining

For open pit mining it is assumed that no cap of the deposit exists. Water cannons operating at pressures of up to 1.5 MPa are used to separate the ore from the overburden. The mixture of water and heavy sand with about 70% tin content is then fed to the treatment plant.

The tin content in placer deposits lies in the range of 0.01 to 3%. We assume an average tin content of 0.1% in the crude ore that is mined and an average recovery of the tin minerals of 90%. Thus 1.11 tons of material has to be mined to gain 1 kg of tin from open pit mines. With the assumed average density of 1.5 tons/m<sup>3</sup> crude ore this corresponds to 0.74 m<sup>3</sup> crude ore. The thickness of the recoverable open deposit is assumed to be 5 meters. Thus 0.148 m<sup>2</sup> of land is transformed to mining area per kg of tin recovered. It is assumed that a mining site is depleted within 10 years and left for natural recultivation afterwards.

For a lack of specific information the energy consumption used for mining, the infrastructure needed therefore and the dust emissions generated by it are calculated based on the average values derived for iron ore and bauxite mining. The assumption underlying the calculation is that the respective consumption is proportional to the mass of crude ore extracted. No explosives are assumed to be used. The water consumption is estimated.

### 5.2.3 Offshore mining

Marine deposits are mined several kilometres offshore in depths of around 50 meters below sea level using bucket line dredges. Thus a similar mixture of water and heavy sand as in the open pit mining with water jets is won.

The tin content in placer deposits lies in the range of 0.01 to 3%. We assume an average tin content of 0.1% in the crude ore that is mined and an average recovery of the tin minerals of 90%. Thus 1.11 tons of material has to be mined to gain 1 kg of tin from offshore deposits. With the assumed average density of 1.5 tons/m<sup>3</sup> crude ore this corresponds to 0.74 m<sup>3</sup> crude ore. The thickness of the recoverable offshore deposit is assumed to be 5 meters. Thus 0.148 m<sup>2</sup> of benthos is transformed to mining area per kg of tin recovered. It is assumed that a mining site is depleted within 10 years and left for natural recultivation afterwards.

For a lack of specific information the energy consumption of open pit mining of tin ore is used assuming that the electricity used in open pit mining is replaced by diesel, burned in building machines with 5 times lower efficiency.

### 5.3 Production of crude tin

The beneficiated ores with the average tin content of 68% (20% with 50% Sn from underground mining, 80% with 70% Sn from open pit and offshore mining) are heated together with powdered coal in a reverbaratory furnace at 1200 °C. Thus a tin containing slag is formed. By a second melting in a sloping hearth the metallic tin runs off (97% purity) and a slag with about 8% Sn content is formed. This slag is recycled to the beneficiation step.

According to Richter et al. (1996), 15% to 20% carbon is added to the ore for smelting and 75 to 100 kg heating oil per tonne of ore is burned for heating. About 10-14% of the input ore mass is turned to slag that is used in a secondary smelting. This process uses 550 to 1000 kWh electricity (in an electric furnace) and 1 to 10 kg of electrodes per ton of slag. Beside that, 122 kg coal, 67 kg of gypsum and 39 kg of limestone are needed per ton of slag. 136 kg of secondary slag has to be landfilled per ton of primary slag input. These values are used to calculate the in- and outputs of the smelting of tin (see Tab. 5.3).

**Tab. 5.3 In- and outputs for smelting per kg pure tin (based on Richter et al. (1996))**

|                                                  |                          |                                     |                                    |                                     |
|--------------------------------------------------|--------------------------|-------------------------------------|------------------------------------|-------------------------------------|
| Sn concentration in Input                        | 0.68                     |                                     |                                    |                                     |
| Sn concentration in output                       | 1                        |                                     |                                    |                                     |
| Sn concentration in slag                         | 0.08                     |                                     |                                    |                                     |
| fraction of ore concentrate input to slag output | 0.12                     |                                     |                                    |                                     |
| fraction of Sn that can be extracted of slag     | 0.7                      |                                     |                                    |                                     |
| carbon content in coal and electrode             | 0.9                      |                                     |                                    |                                     |
| <b>Ore smelting</b>                              |                          | <b>per kg ore input</b>             | <b>per kg Sn input</b>             | <b>per kg Sn output</b>             |
| ore input                                        |                          | 1                                   | 1.47                               | 1.48 <sup>1</sup>                   |
| coal input [kg]                                  |                          | 0.175                               | 0.257                              | 0.259                               |
| heating oil [kg]                                 |                          | 87.5                                | 129                                | 130                                 |
| slag to slag smelting output [kg]                |                          | 0.12                                | 0.176                              | 0.178                               |
| carbon dioxide to air [kg]                       |                          | 0.578                               | 0.849                              | 0.855                               |
| <b>Slag smelting</b>                             | <b>per kg slag input</b> | <b>per kg ore input<sup>2</sup></b> | <b>per kg Sn input<sup>3</sup></b> | <b>per kg Sn output<sup>4</sup></b> |
| electrical power [kWh]                           | 0.775                    | 0.0930                              | 0.137                              | 0.138                               |
| electrodes input [kg]                            | 0.0055                   | 0.000660                            | 0.000971                           | 0.000977                            |
| coal input [kg]                                  | 0.122                    | 0.0146                              | 0.0215                             | 0.0217                              |
| gypsum input [kg]                                | 0.067                    | 0.00804                             | 0.0118                             | 0.0119                              |
| limestone input [kg]                             | 0.039                    | 0.00468                             | 0.00688                            | 0.00693                             |
| slag to landfill output [kg]                     |                          |                                     |                                    | 0.136 <sup>5</sup>                  |
| carbon dioxide to air [kg]                       | 0.421                    | 0.050                               | 0.074                              | 0.075                               |

1 calculated from value per input, tin fraction in slag, fraction of ore input to slag and fraction of extraction from slag

2 calculated from input per kg slag and slag to smelting per kg ore input

3 calculated from input per kg slag and slag to smelting per kg Sn input

4 calculated from input per kg slag and slag to smelting per kg Sn output

5 value given in Richter et al. (1996)

### 5.4 Refining

The crude tin obtained by the reduction process is insufficiently pure for most applications. Refining can be done pyrometallurgically or by electro-refining.

Electro-refining can only be operated at low current densities, leading to low process rates and inefficient utilisation of energy. Thus electro-refining is only worthwhile if the tin contains high concentrations of noble metals. It is therefore not considered in this study.

There are different types of pyrometallurgical refining depending on the impurities that have to be removed. All the processes have in common that liquid tin is treated in a way that the impurities form intermetallic compounds that can be removed. For removing arsenic it is even necessary to have a vacuum over the liquid metal. Graf (1997) gives an energy demand of 400 to 700 kWh per ton tin for heating and providing the vacuum. The average of this value is used as electricity input for lack of other information.

## **5.5 Emissions**

### **5.5.1 Air**

Beside the emissions from heat generation there are dust emissions from mining, SO<sub>2</sub> emissions from beneficiation and CO<sub>2</sub> emissions from the reduction of SnO with carbon.

### **5.5.2 Water**

No information about emissions to water is available. The beneficiation process being predominantly mechanical is not believed to be a relevant source of water emissions. Some emissions are likely to occur from waste gas treatment in the thermal beneficiation of the ore from underground mining.

## **5.6 Waste**

The overburden of mining is refilled in the mine and not treated as waste. Acid rock drainage (ARD) that is possibly occurring where sulfidic ores are present is thus not modelled for this waste fraction. The only waste considered as such is the secondary slag that is landfilled. No detailed information about the composition of this slag was found. From the composition of the crude ore it is assumed that the slag contains mainly Sn, SiO<sub>2</sub>, Fe, Ta<sub>2</sub>O<sub>5</sub>, TiO<sub>2</sub>, Al<sub>2</sub>O<sub>3</sub>, CaO, Nb<sub>2</sub>O<sub>5</sub> and As.

For a lack of better data the disposal of slag from unalloyed steel making in the electric arc furnace (EAF) is inventoried as a proxy.

## **5.7 Tin production inecoinvent**

The background information to the process "tin, at regional storage" are given in Tab. 5.4, flows are shown in Fig. 5.2.

## **5.8 Data quality**

Data uncertainty is derived using the Pedigree matrix as described in (Frischknecht et al. (2003)).

Most data for the tin production are assumptions based on qualitative information and theoretical models. Thus uncertainty is rather high. These data are of insufficient quality for LCA where tin or tin derived products (e.g. solders) play an important role.

Tab. 5.4 Meta information for the tin production processes

|                                       |                                                                                                                                                                                                                               |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>                           | <b>tin, at regional storage</b>                                                                                                                                                                                               |
| Location                              | RER                                                                                                                                                                                                                           |
| Infrastructure Process                | 0                                                                                                                                                                                                                             |
| Unit                                  | kg                                                                                                                                                                                                                            |
| Data Set Version                      | 2.0                                                                                                                                                                                                                           |
| Included Processes                    | The cradle to gate inventory of world-wide primary tin production includes mining, beneficiation, smelting and refining of tin ore and tin respectively. Also included are the transports from the major producers to Europe. |
| Amount                                | 1                                                                                                                                                                                                                             |
| Local Name                            | Zinn, ab Regionallager                                                                                                                                                                                                        |
| Synonyms                              |                                                                                                                                                                                                                               |
| General Comment to reference function | Most data are calculated based on assumptions and theoretical models. Their overall quality is poor.                                                                                                                          |
| CAS Number                            | 7440-31-5                                                                                                                                                                                                                     |
| Start Date                            | 1996                                                                                                                                                                                                                          |
| End Date                              | 2003                                                                                                                                                                                                                          |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                             |
| Other Period Text                     |                                                                                                                                                                                                                               |
| Geography text                        | Production data relate to the global average. Transports relate to the consumption in Europe                                                                                                                                  |
| Technology text                       | Data relate to an assumed average technology for mining, beneficiation, smelting and refining.                                                                                                                                |
| Representativeness [%]                |                                                                                                                                                                                                                               |
| Production Volume                     | > 280'000 tonnes per year (worldwide)                                                                                                                                                                                         |
| Sampling Procedure                    | based on literature and internet                                                                                                                                                                                              |
| Extrapolations                        | none                                                                                                                                                                                                                          |
| Uncertainty Adjustments               | none                                                                                                                                                                                                                          |

| General Flow Information                |                                                              |            | Representation in ecoinvent                                                                                                                                                                        |                        |                        |                 |          |                                                       |            |      | Uncertainty Informations |      |                        |                  |  |
|-----------------------------------------|--------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------------------------|-----------------|----------|-------------------------------------------------------|------------|------|--------------------------|------|------------------------|------------------|--|
| Input                                   | Process Name                                                 | Output     | Remarks                                                                                                                                                                                            | Category               | Sub-category           | Infra-structure | Location | Modul name in ecoinvent                               | Mean value | Unit | Source for mean value    | Type | Standard Deviation 95% | General Comment  |  |
| tin ore ore (resource)                  | Underground mining and beneficiation (20% of tin production) | waste heat | recovery of tin from ore = 70%; 20% of tin production; cassiterite = 0.127% in mined material, Sn = 79 % in cassiterite --> 1 kg Sn in ore corresponds to 790kg of ore and to 1t of mined material | resource               | in ground              |                 |          | Tin, 79% in cassiterite, 0.1% in crude ore, in ground | 2.86E+01   | kg   | calculation              | 1    | 1.415                  | (4,5,3,5,1,5,12) |  |
| land use (mine) [m2a]                   |                                                              |            | no land use inventoried for underground mining                                                                                                                                                     |                        |                        |                 |          |                                                       |            |      |                          |      |                        |                  |  |
| diesel, burned in building machine [MJ] |                                                              |            | recovery of tin from ore = 70%; 20% of tin production                                                                                                                                              | construction processes | machinery              | No              | GLO      | diesel, burned in building machine                    | 2.83E+01   | MJ   | calculation based on PGM | 1    | 2.382                  | (5,5,1,5,3,5,5)  |  |
| electricity                             |                                                              |            |                                                                                                                                                                                                    | electricity            | production mix         | No              | UCTE     | electricity, medium voltage, production UCTE, at grid | 1.85E+01   | kWh  | calculation based on PGM | 1    | 1.415                  | (3,5,1,5,3,5,2)  |  |
| blasting [kg explosive]                 |                                                              |            |                                                                                                                                                                                                    | construction processes | civil engineering      | No              | RER      | blasting                                              | 1.90E-01   | kg   | calculation based on PGM | 1    | 1.415                  | (3,5,1,5,3,5,1)  |  |
| infrastruct ure (mine) [unite]          |                                                              |            | input includes infrastructure needed for mining and beneficiation (buildings, access roads,...) (exclusive building machines)                                                                      | metals                 | extraction             | Yes             | GLO      | non ferrous metal mine, underground                   | 5.34E-14   | unit | calculation based on PGM | 1    | 3.161                  | (3,5,1,5,3,5,9)  |  |
|                                         |                                                              |            | from electricity use                                                                                                                                                                               | air                    | low population density |                 |          | Heat, w aste                                          | 6.65E+01   | MJ   | calculation              | 1    | 1.415                  | (3,5,1,5,3,5,13) |  |

Fig. 5.2 Flows for "tin, at regional storage" and its representation in the ecoinvent database

| General Flow Information |                                                                              |            | Representation in ecoinvent                                                                                                              |                        |                        |                 |          |                                                       |            |      | Uncertainty Informations                 |      |                        |                  |  |
|--------------------------|------------------------------------------------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------------------------|-----------------|----------|-------------------------------------------------------|------------|------|------------------------------------------|------|------------------------|------------------|--|
| Input                    | Process Name                                                                 | Output     | Remarks                                                                                                                                  | Category               | Sub-category           | Infra-structure | Location | Modul name in ecoinvent                               | Mean value | Unit | Source for mean value                    | Type | Standard Deviation 95% | General Comment  |  |
| electricity              | Mechanical beneficiation of tin from underground                             | waste heat | recovery of tin from ore = 70%; 20% of tin production                                                                                    | electricity            | production mix         | No              | UCTE     | electricity, medium voltage, production UCTE, at grid | 4.09E-01   | kWh  | Richter et al. 1996                      | 1    | 1.415                  | (3,5,1,5,3,5,2)  |  |
| water                    |                                                                              |            | 1m3 / ton crude ore assumed                                                                                                              | resource               | in water               |                 |          | Water, unspecified natural origin                     | 4.09E-01   | m3   | assumption                               | 1    | 1.69                   | (5,5,1,5,3,5,12) |  |
|                          |                                                                              |            | from electricity use                                                                                                                     | air                    | low population density |                 |          | Heat, waste                                           | 1.47E+00   | MJ   | calculation                              | 1    | 1.415                  | (3,5,1,5,3,5,13) |  |
| heat                     | Thermal beneficiation of tin from underground mining (20% of tin production) |            | recovery of tin from ore = 70%; 20% of tin production                                                                                    | oil                    | heating systems        | No              | RER      | heat, heavy fuel oil, at industrial furnace 1MW       | 1.08E+01   | MJ   | calculation                              | 1    | 1.415                  | (3,5,1,5,3,5,1)  |  |
| electricity              |                                                                              |            |                                                                                                                                          | electricity            | production mix         | No              | UCTE     | electricity, medium voltage, production UCTE, at grid | 1.14E+00   | kWh  | calculation based on Richter et al. 1996 | 1    | 1.415                  | (3,5,1,5,3,5,2)  |  |
| Pyrite                   |                                                                              |            | recovery of tin from ore = 70%; 20% of tin production; limestone as proxy for mining of pyrite --> acid rock drainage (ARD) is neglected | construction materials | additives              | No              | CH       | limestone, at mine                                    | 3.37E-01   | kg   | calculation based on assumptions         | 1    | 1.462                  | (4,5,1,5,3,5,4)  |  |
|                          |                                                                              |            | from electricity use                                                                                                                     | air                    | low population density |                 |          | Heat, waste                                           | 4.11E+00   | MJ   | calculation                              | 1    | 1.415                  | (3,5,1,5,3,5,13) |  |
|                          |                                                                              |            | recovery of tin from ore = 70%; 20% of tin production                                                                                    | air                    | low population density |                 |          | Sulfur dioxide                                        | 2.78E-01   | kg   | calculation based on assumptions         | 1    | 1.938                  | (5,5,1,5,3,5,31) |  |

Fig. 5.2 Flows for "tin, at regional storage" and its representation in the ecoinvent database (cont.)

| General Flow Information                |                                                                              |              | Representation in ecoinvent                                                                                                                                                                        |                        |                        |                 |          |                                                       |            | Uncertainty Informations |                                       |      |                        |                  |
|-----------------------------------------|------------------------------------------------------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------------------------|-----------------|----------|-------------------------------------------------------|------------|--------------------------|---------------------------------------|------|------------------------|------------------|
| Input                                   | Process Name                                                                 | Output       | Remarks                                                                                                                                                                                            | Category               | Sub-category           | Infra-structure | Location | Modul name in ecoinvent                               | Mean value | Unit                     | Source for mean value                 | Type | Standard Deviation 95% | General Comment  |
| tin ore ore (resource)                  | Mining and beneficiation of tin from open pit mining (40% of tin production) |              | recovery of tin from ore = 90%; 40% of tin production; cassiterite = 0.127% in mined material, Sn = 79 % in cassiterite --> 1 kg Sn in ore corresponds to 790kg of ore and to 1t of mined material | resource               | in ground              |                 |          | Tin, 79% in cassiterite, 0.1% in crude ore, in ground | 4.44E-01   | kg                       | calculation                           | 1    | 1.415                  | (4,5,3,5,1,5,12) |
| land use (mine) [m2a]                   |                                                                              |              | recovery of tin from ore = 90%; 40% of tin production                                                                                                                                              | resource               | land                   |                 |          | Occupation, mineral extraction site                   | 5.92E-01   | m2a                      | calculation based on assumptions      | 1    | 1.938                  | (5,5,1,5,3,5,7)  |
| land transformation (mine) [m2]         |                                                                              |              |                                                                                                                                                                                                    | resource               | land                   |                 |          | Transformation, from unknown                          | 5.92E-02   | m2                       | calculation based on assumptions      | 1    | 2.382                  | (5,5,1,5,3,5,8)  |
| land transformation (mine) [m2]         |                                                                              |              |                                                                                                                                                                                                    | resource               | land                   |                 |          | Transformation, to mineral extraction site            | 5.92E-02   | m2                       | calculation based on assumptions      | 1    | 2.382                  | (5,5,1,5,3,5,8)  |
| diesel, burned in building machine [MJ] |                                                                              |              |                                                                                                                                                                                                    | construction processes | machinery              | No              | GLO      | diesel, burned in building machine                    | 4.06E+00   | MJ                       | calculation based on Iron and bauxite | 1    | 2.382                  | (5,5,1,5,3,5,5)  |
| electricity                             |                                                                              |              |                                                                                                                                                                                                    | electricity            | production mix         | No              | UCTE     | electricity, medium voltage, production UCTE, at grid | 2.46E-01   | kWh                      | calculation based on Iron and bauxite | 1    | 1.69                   | (5,5,1,5,3,5,2)  |
| water                                   |                                                                              |              |                                                                                                                                                                                                    | resource               | in water               |                 |          | Water, unspecified natural origin                     | 6.36E-01   | m3                       | assumption                            | 1    | 1.69                   | (5,5,1,5,3,5,12) |
| infrastructure (mine) [unite]           |                                                                              |              | input includes infrastructure needed for mining and beneficiation (buildings, access roads, machines...) (exclusive building machines); iron mine as proxy                                         | metals                 | extraction             | Yes             | GLO      | mine, iron                                            | 9.85E-11   | unit                     | calculation based on Iron and bauxite | 1    | 3.376                  | (5,5,1,5,3,5,9)  |
|                                         |                                                                              | waste heat   | from electricity use                                                                                                                                                                               | air                    | low population density |                 |          | Heat, waste                                           | 8.86E-01   | MJ                       | calculation                           | 1    | 1.69                   | (5,5,1,5,3,5,13) |
|                                         |                                                                              | dust, coarse | recovery of tin from ore = 90%; 40% of tin production                                                                                                                                              | air                    | low population density |                 |          | Particulates, > 10 um                                 | 1.79E-01   | kg                       | calculation based on Iron and bauxite | 1    | 1.938                  | (5,5,1,5,3,5,25) |
|                                         |                                                                              | dust, medium |                                                                                                                                                                                                    | air                    | low population density |                 |          | Particulates, > 2.5 um, and < 10um                    | 1.61E-01   | kg                       | calculation based on Iron and bauxite | 1    | 2.382                  | (5,5,1,5,3,5,26) |
|                                         |                                                                              | dust, fine   |                                                                                                                                                                                                    | air                    | low population density |                 |          | Particulates, < 2.5 um                                | 1.79E-02   | kg                       | calculation based on Iron and bauxite | 1    | 3.376                  | (5,5,1,5,3,5,27) |

Fig. 5.2 Flows for "tin, at regional storage" and its representation in the ecoinvent database (cont.)

5. System Characterisation

| General Flow Information                |                                                                              |            | Representation in ecoinvent                                                                                                                                                                        |                        |                        |                 |          |                                                       |            |      | Uncertainty Informations                 |      |                        |                  |  |
|-----------------------------------------|------------------------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------------------------|-----------------|----------|-------------------------------------------------------|------------|------|------------------------------------------|------|------------------------|------------------|--|
| Input                                   | Process Name                                                                 | Output     | Remarks                                                                                                                                                                                            | Category               | Sub-category           | Infra-structure | Location | Modul name in ecoinvent                               | Mean value | Unit | Source for mean value                    | Type | Standard Deviation 95% | General Comment  |  |
| tin ore ore (resource)                  | Mining and beneficiation of tin from offshore mining (40% of tin production) |            | recovery of tin from ore = 90%; 40% of tin production; cassiterite = 0.127% in mined material, Sn = 79 % in cassiterite --> 1 kg Sn in ore corresponds to 790kg of ore and to 1t of mined material | resource               | in ground              |                 |          | Tin, 79% in cassiterite, 0.1% in crude ore, in ground | 4.44E-01   | kg   | calculation                              | 1    | 1.415                  | (4,5,3,5,1,5,12) |  |
| land use (mine) [m2a]                   |                                                                              |            | recovery of tin from ore = 90%; 40% of tin production                                                                                                                                              | resource               | land                   |                 |          | Occupation, mineral extraction site                   | 5.92E-01   | m2a  | calculation based on assumptions         | 1    | 1.938                  | (5,5,1,5,3,5,7)  |  |
| land transformation (mine) [m2]         |                                                                              |            |                                                                                                                                                                                                    | resource               | land                   |                 |          | Transformation, from sea and ocean                    | 5.92E-02   | m2   | calculation based on assumptions         | 1    | 2.382                  | (5,5,1,5,3,5,8)  |  |
| land transformation (mine) [m2]         |                                                                              |            |                                                                                                                                                                                                    | resource               | land                   |                 |          | Transformation, to industrial area, benthos           | 5.92E-02   | m2   | calculation based on assumptions         | 1    | 2.382                  | (5,5,1,5,3,5,8)  |  |
| diesel, burned in building machine [MJ] |                                                                              |            |                                                                                                                                                                                                    | construction processes | machinery              | No              | GLO      | diesel, burned in building machine                    | 8.49E+00   | MJ   | calculation based on Iron and bauxite    | 1    | 2.382                  | (5,5,1,5,3,5,5)  |  |
| coal                                    | Ore smelting                                                                 |            | 31.4 MJ/kg                                                                                                                                                                                         | hard coal              | fuels                  | No              | UCTE     | hard coal mix, at regional storage                    | 2.59E-01   | kg   | calculation based on Richter et al. 1996 | 1    | 1.415                  | (3,5,1,5,3,5,3)  |  |
| heat                                    |                                                                              |            | 41.2 MJ/kg                                                                                                                                                                                         | oil                    | heating systems        | No              | RER      | heat, heavy fuel oil, at industrial furnace 1MW       | 5.34E+00   | MJ   | calculation based on Richter et al. 1996 | 1    | 1.415                  | (3,5,1,5,3,5,1)  |  |
|                                         |                                                                              | waste heat | from burning of coal                                                                                                                                                                               | air                    | low population density |                 |          | Heat, waste                                           | 8.39E+00   | MJ   | calculation                              | 1    | 1.415                  | (3,5,1,5,3,5,13) |  |
|                                         |                                                                              | CO2        | from burning of coal                                                                                                                                                                               | air                    | low population density |                 |          | Carbon dioxide, fossil                                | 8.55E-01   | kg   | calculation based on Richter et al. 1997 | 1    | 1.415                  | (3,5,1,5,3,5,14) |  |

Fig. 5.2 Flows for "tin, at regional storage" and its representation in the ecoinvent database (cont.)



## 5. System Characterisation

| General Flow Information  |                         |        | Representation in ecoinvent  |                                                                    |                   |                            |                |                                                       |                                                                                  |                                                       | Uncertainty Informations                 |                                          |                               |                 |                  |                 |
|---------------------------|-------------------------|--------|------------------------------|--------------------------------------------------------------------|-------------------|----------------------------|----------------|-------------------------------------------------------|----------------------------------------------------------------------------------|-------------------------------------------------------|------------------------------------------|------------------------------------------|-------------------------------|-----------------|------------------|-----------------|
| Input                     | Process Name            | Output | Remarks                      | Category                                                           | Sub-category      | Infra-structure            | Location       | Modul name in ecoinvent                               | Mean value                                                                       | Unit                                                  | Source for mean value                    | Type                                     | Standard Deviation 95%        | General Comment |                  |                 |
| electricity               | Secondary slag smelting |        |                              | electricity                                                        | production mix    | No                         | UCTE           | electricity, medium voltage, production UCTE, at grid | 1.38E-01                                                                         | kWh                                                   | calculation based on Richter et al. 1996 | 1                                        | 1.415                         | (3,5,1,5,3,5,2) |                  |                 |
| electrodes                |                         |        | 31.4 MJ/kg                   | metals                                                             | extraction        | No                         | RER            | anode, aluminium electrolysis                         | 9.77E-04                                                                         | kg                                                    | calculation based on Richter et al. 1996 | 1                                        | 1.415                         | (3,5,1,5,3,5,3) |                  |                 |
| coal                      |                         |        | 31.4 MJ/kg                   | hard coal                                                          | fuels             | No                         | UCTE           | hard coal mix, at regional storage                    | 2.17E-02                                                                         | kg                                                    | calculation based on Richter et al. 1996 | 1                                        | 1.415                         | (3,5,1,5,3,5,3) |                  |                 |
| gypsum                    |                         |        |                              | construction materials                                             | others            | No                         | CH             | gypsum, mineral, at mine                              | 1.19E-02                                                                         | kg                                                    | calculation based on Richter et al. 1996 | 1                                        | 1.415                         | (3,5,1,5,3,5,3) |                  |                 |
| limestone                 |                         |        |                              | construction materials                                             | additives         | No                         | CH             | limestone, at mine                                    | 6.93E-03                                                                         | kg                                                    | calculation based on Richter et al. 1996 | 1                                        | 1.415                         | (3,5,1,5,3,5,3) |                  |                 |
|                           |                         |        | w aste heat                  | from electricity use and burning of coal and electrodes            | air               | low population density     |                |                                                       | Heat, w aste                                                                     | 1.21E+00                                              | MJ                                       | calculation                              | 1                             | 1.415           | (3,5,1,5,3,5,13) |                 |
|                           |                         |        | CO2                          | from burning of coal                                               | air               | low population density     |                |                                                       | Carbon dioxide, fossil                                                           | 7.47E-02                                              | kg                                       | calculation based on Richter et al. 1997 | 1                             | 1.415           | (3,5,1,5,3,5,24) |                 |
|                           |                         |        | w aste                       | dust from unalloyed electric steel production to landfill as proxy | w aste management | residual material landfill | No             | CH                                                    | disposal, dust, unalloyed EAF steel, 15.4% w ater, to residual material landfill | 1.36E-01                                              | kg                                       | Richter et al. 1996                      | 1                             | 1.415           | (3,5,1,5,3,5,6)  |                 |
| electricity               |                         |        | Refining                     |                                                                    |                   | electricity                | production mix | No                                                    | UCTE                                                                             | electricity, medium voltage, production UCTE, at grid | 5.50E-01                                 | kWh                                      | assumption based on Graf 1997 | 1               | 1.415            | (3,5,1,5,3,5,2) |
|                           |                         |        |                              |                                                                    | w aste heat       | from electricity use       | air            | low population density                                |                                                                                  |                                                       | Heat, w aste                             | 1.98E+00                                 | MJ                            | calculation     | 1                | 1.415           |
| Transport ocean freighter | Transport to Europe     |        | 10'000 km, 1.2 kg per kg tin | transport systems                                                  | ship              | No                         | OCE            | transport, transoceanic freight ship                  | 1.20E+01                                                                         | tkm                                                   | assumption                               | 1                                        | 2.382                         | (5,5,1,5,3,5,5) |                  |                 |
| Transport lorry           |                         |        | 500 km, 1.2 kg per kg tin    | transport systems                                                  | road              | No                         | RER            | transport, lorry 32t                                  | 6.00E-01                                                                         | tkm                                                   | assumption                               | 1                                        | 2.382                         | (5,5,1,5,3,5,5) |                  |                 |
|                           |                         |        | refined tin (99%)            |                                                                    | metals            | extraction                 | No             | RER                                                   | tin, at regional storage                                                         | 1.00E+00                                              | kg                                       |                                          |                               |                 |                  |                 |

Fig. 5.2 Flows for "tin, at regional storage" and its representation in the ecoinvent database (cont.)

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## Part VIII

# Magnesium and Magnesium Alloys

Data v2.1 (2009)

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Final report of updated Swiss National Life Cycle Inventory Database "ecoinvent", version 2.1

## Comments on version 2.1

This part and all data correspond to the part and data reported in version 2.0. No changes were made.

## Acknowledgement

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## Comments on this part

This part corresponds to the original part “Other Metals” in Althaus et al. (2004) that described the metals lead, zinc and magnesium and has been suspended for this version 2.0. The original content has been distributed over the new part VII on lead, zinc, cadmium and indium as well as this part that now exclusively is designated to the description of magnesium and its alloys AZ91 and AlMg3. Chapter “Cumulative Results and Interpretation” of the original report has been deleted for the current update.

## Summary

Magnesium, metal alloys containing magnesium, i.e. AZ91, AlMg3, and die castings are important construction materials, e.g. in the automobile sector or for electric equipment. The environmental impacts of magnesium production are investigated as a unit process over the life cycle from raw material extraction until the gate of the factory. The investigation is based on environmental data provided by a producer and on a literature research. High uncertainties still exist for the average energy and SF<sub>6</sub> use in magnesium industry.

## Keywords

AlMg3, Aluminium alloys, AZ91, Magnesium, Magnesium alloys, Magnesium die casting



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# 1 Introduction

The metal magnesium is getting more and more important as a construction material for different technical purposes. It is used as a lightweight material and as an addition to metal alloys such as MgAl3. This inventory for magnesium and related materials is meant to be used as a background unit process for the life cycle assessments of technical products such as e.g. equipment for electricity transmission or photovoltaic systems.

## 2 Reserves and Resources of Materials

Magnesium can be extracted from seawater, from minerals (dolomite, magnesite and brucite) or can be gained by recycling materials. It is the eighth most abundant element in the earth's crust. Seawater contains about 0.13wt% Magnesium (Hoy-Petersen et al. (1999)).

Norsk Hydro uses about 207 000 tonnes of dolomite with an approximate magnesium content of 26 330 tonnes and seawater to produce 106 000t of magnesium (Hydro Magnesium (1999b)).

## 3 Characterisation of the materials

Magnesium is a silvery white, light metal. It burns in air with an intense white flame. Pure magnesium has a high resistance to corrosion because its galvanic activity is low (Hoy-Petersen et al. (1999)).

## 4 Production and Use of Magnesium

The most important producer in Europe is Norsk Hydro with an annual production of 105'600 tonnes in 1998 while the total annual production was about 372'000 tonnes. Tab. 4.1 shows an overview of the production places of magnesium. It is important to know that the import of magnesium from the eastern world plays an important role for meeting the western worlds demand. But there are not many shipments between Europe and America. Various investment projects are going on to increase the production capacities due to the increasing demand. Due to the possibility to produce magnesium from seawater there is no shortcoming of resources for production.

Tab. 4.1 Magnesium production and consumption in 1996-97 (Aroule (1998))

| Countries and companies            | Production | Type of production process                              | Im-/ Exports |
|------------------------------------|------------|---------------------------------------------------------|--------------|
|                                    | 1'000t/a   |                                                         | 1'000t/a     |
| <b>Western world (8 producers)</b> | <b>254</b> | Mainly use of SF <sub>6</sub> for oxidation protection. | 52 Import    |
| Norsk Hydro                        | 106        | Electrolytic                                            |              |
| Others                             | 148        |                                                         |              |
| <b>Eastern world</b>               | <b>128</b> | Mainly use of SO <sub>2</sub> for oxidation protection. |              |
| CIS countries (3 producers)        | 48         |                                                         | -80 Export   |
| China (200 producers)              | 80         |                                                         |              |
| <b>Total</b>                       | <b>372</b> | 80% electrolytic, 20% thermal                           |              |

Magnesium is used for different technical purposes. The consumption of magnesium in the western world is summarized in Fig. 4.1 (Avedesian & Baker (1999)). Aluminium alloys and production of die-castings are the main purposes for consumption.

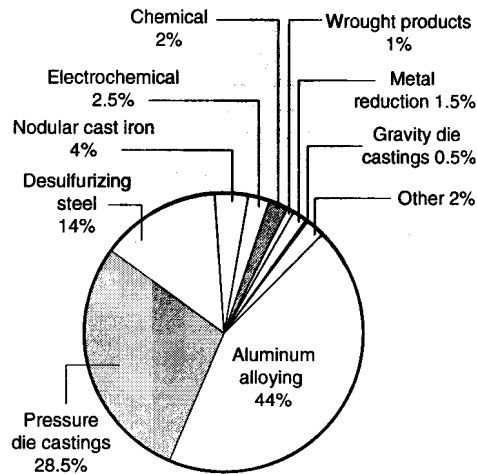


Fig. 4.1 Consumption of magnesium in Western world in 1997 (total 333'700 tonnes). Amount used for magnesium alloys (die castings) is shown shaded (Avedesian & Baker (1999)).

## 5 System Characterisation

The main production processes are electrolysis of magnesium chloride melts (made from resources like seawater and dolomite) and the metallothermic reduction of magnesium oxide with a share of 80% and 20%, respectively for the total production (Hoy-Petersen et al. (1999)). The different producers in the world use different processes. Fig. 5.1 shows a block diagram for the process used by Norsk Hydro that is investigated here. Magnesium is produced from seawater and dolomite. The production of  $\text{Mg}(\text{OH})_2$  includes calcination of dolomite, mixing with sea water and precipitation and filtration of  $\text{Mg}(\text{OH})_2$ . Thereafter  $\text{Mg}(\text{OH})_2$  filter cake is calcinated to  $\text{MgO}$ . Water free  $\text{MgCl}_2$  is prepared through pelletisation and chlorination of  $\text{MgO}$ .  $\text{MgCl}_2$  is electrolysed and ingots are produced in the foundry (Haagensen et al. (1996)).



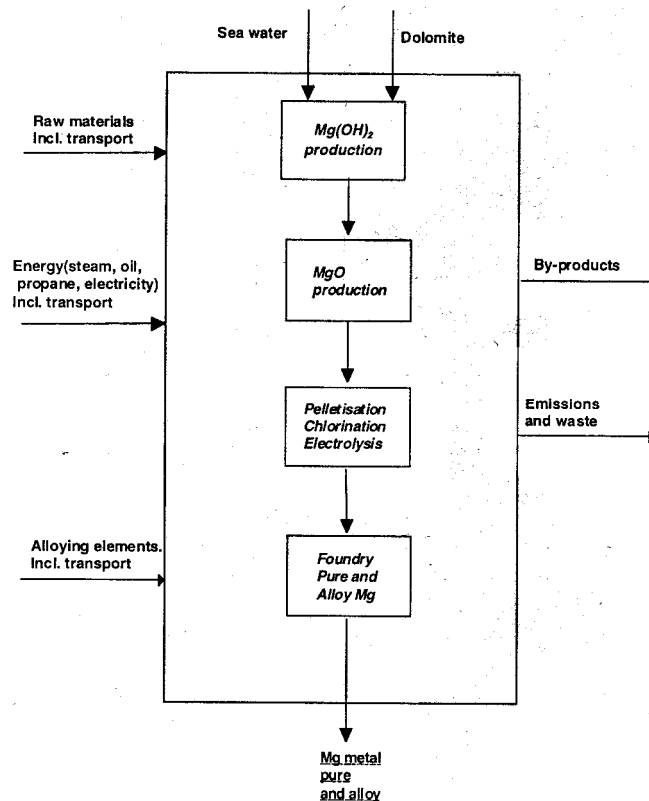


Fig. 5.1 Block diagram of the dolomite and seawater process for magnesium production in Porsgrunn, NO (Haagensen et al. (1996))

## 5.1 Life Cycle Inventories

The inventory is split up into the primary production of magnesium and the further production of magnesium alloys. Then the manufacturing of diecastings is investigated. The infrastructure for magnesium production is assumed roughly.

### 5.1.1 Magnesium

The life cycle inventory investigates the production of magnesium from seawater for a modern production plant based on data from a Norwegian producer. All processes from seawater uptake until the magnesium leaves the plant are included in the inventory.

Energy is used for the preparation of dolomite lime, making  $\text{MgCl}_2$  cell feed from seawater and acid generation. Tab. 5.1 shows an overview for the energy requirements of magnesium production investigated by different authors. Different methodologies are used for the reporting of the energy requirements. Large differences exist for the reported energy requirements for the production of one kilogram magnesium. But, it has to be considered that some of the data have been investigated in the seventies. Thus a higher energy efficiency (a lower specific energy consumption) can be assumed for today's production.

**Tab. 5.1 Energy requirement of magnesium production as investigated in different studies. It has to be noted that assumptions for the calculation of the total energy requirements vary among different studies**

| Products and processes                                                     | Different types of energy use | Further remarks about type of energy use and references, >year of first publication                                                                                                            |
|----------------------------------------------------------------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                            | MJ/kg                         |                                                                                                                                                                                                |
| Electrolysis of magnesium                                                  | 46.8<br>-50.4                 | Electricity use (Hoy-Petersen et al. (1999:568,>1977-87)).                                                                                                                                     |
| Electrolysis of magnesium                                                  | 36<br>-66.6                   | Electricity use (Celik et al. (1996)).                                                                                                                                                         |
| Silicothermic reduction of magnesium                                       | 61<br>-72                     | Electricity use (Celik et al. (1996)).                                                                                                                                                         |
| Metallothermic (Pidgeon process) reduction of magnesium                    | 108                           | Total energy requirement (Hoy-Petersen et al. (1999,>1980)).                                                                                                                                   |
| Metallothermic (Bolzano process) reduction of magnesium                    | 25<br>-26                     | Energy requirement of furnace (Wilson et al. (1995,>1980-81)).                                                                                                                                 |
| Recycling of magnesium                                                     | 10.8                          | Electricity use (Kammer et al. (2000))                                                                                                                                                         |
| Recycling of magnesium                                                     | 6.7                           | Total energy requirement for recycling (Wilson et al. (1995,>1974)).                                                                                                                           |
| Magnesium                                                                  | 267                           | Total energy requirement (Wilson et al. (1995,>1974)).                                                                                                                                         |
| Magnesium                                                                  | 325                           | Total energy requirement at Dow in 1978. Reduction of 15%-20% over ten years (Hoy-Petersen et al. (1999,>1980)).                                                                               |
| Magnesium                                                                  | 378                           | Total energy requirement, Split up: 5%, 23%, 62%, 10% for dolomite preparation, MgCl <sub>2</sub> cell feed from seawater, electrolysis, and acid regeneration, respectively (Barclay (1977)). |
| Magnesium                                                                  | 358                           | Total energy requirement, Split up: 6%, 45%, 45%, 4% for seawater treatment, evaporation, electrolysis, and acid recovery, respectively (Barclay (1977)).                                      |
| Magnesium                                                                  | 329<br>-487                   | Total energy requirement (Boustead & Hancock (1979:345,>1972-75)).                                                                                                                             |
| Magnesium                                                                  | 144                           | Total energy requirement <sup>1</sup> (Albright & Haagenzen (1997); Haagenzen et al. (1996)).                                                                                                  |
| Magnesium alloy AZ91                                                       | 151                           | Total energy requirement (Albright & Haagenzen (1997); Haagenzen et al. (1996)).                                                                                                               |
| Magnesium sheet manufacture                                                | 75                            | Total energy requirement for the production stage (Barclay (1977); Wilson et al. (1995)).                                                                                                      |
| Magnesium extrusions manufacture                                           | 83                            | Total energy requirement for the production stage (Barclay (1977); Wilson et al. (1995)).                                                                                                      |
| Magnesium die-casting production                                           | 21<br>-45                     | Total energy requirement for the production stage (Barclay (1977); Wilson et al. (1995)).                                                                                                      |
| Magnesium (from magnesite) alloy AZ91 cast part from diecaster             | 229 / 220 / 214               | Total energy requirement. 50% / 60% / 70% yield from die-caster. Including recycling of diecaster returns (Hydro Magnesium (1999a)).                                                           |
| Magnesium (from seawater and dolomite) alloy AZ91 cast part from diecaster | 205 / 182 / 171               | Total energy requirement. 40% / 60% / 80% yield from die-caster. Including recycling of diecaster returns (Hydro Magnesium (1999b)).                                                           |

<sup>1</sup> Johanna Öster Haagenzen (Norsk Hydro, email communication 10.8.2000): The total energy demand for the production of pure Mg, including hydro power electricity, fossil fuel and other sources such as steam. The system boundary is from cradle to gate for "all" raw materials (life cycle perspective). This means that production of hydropower electricity is included in the 144MJ/kg, as well as extraction, production and transport of fossil fuels and so forth.

The inventory for the production of magnesium is mainly based on the environmental reports of Norsk Hydro for the year 1998<sup>2</sup> and on an LCA for the production of magnesium and magnesium diecastings by Norsk Hydro. Tab. 5.2 shows the results of the cumulative life cycle inventory published by Norsk Hydro for magnesium and magnesium die-castings (Albright & Haagenzen (1997); Haagenzen et al. (1996); Hydro Magnesium (1999a); Hydro Magnesium (1999b); Norsk Hydro (1999)). There is no explanation for the origin of some of the emissions, e.g. dioxins.

Production data are proprietary information. Therefore, these reports present only aggregated figures of the life cycle inventory. Thus it was unavoidable to disaggregate some data with uncertain assumptions or use cumulative data in order to achieve an inventory table for this study.

**Tab. 5.2 Cumulative energy consumption, direct emissions and waste data for the production of 1kg magnesium and AZ91 part from die-casting at Norsk Hydro. All stages in the life cycle from cradle to gate are considered in the inventory data for the alloys (Albright & Haagenzen (1997); Hydro Magnesium (1999a); Hydro Magnesium (1999b))**

|                                 | Type of energy, emission or waste                 | Unit | LCI<br>Magnesium<br>kg | LCI AZ91,<br>60% yield,<br>Porsgrunn<br>kg | LCI AZ91,<br>60% yield,<br>Becancour<br>kg |
|---------------------------------|---------------------------------------------------|------|------------------------|--------------------------------------------|--------------------------------------------|
| <b>Energy use</b>               | Total energy consumption                          | MJ   | 144.0                  | 182.0                                      | 220.0                                      |
|                                 | Hydro power electricity                           | kWh  | 21.5                   | 27.2                                       | 30.0                                       |
|                                 | Energy from fossil fuels                          | MJ   | 49.1                   | 62.0                                       | 90.0                                       |
| <b>Emissions</b>                | Chlorinated hydrocarbons to air                   | kg   | 1.37E-05               | 1.24E-05                                   | -                                          |
|                                 | CO <sub>2</sub>                                   | kg   | -                      | 7.0                                        | 8.2                                        |
|                                 | Dioxins to air                                    | kg   | 2.33E-11               | 2.10E-10                                   | -                                          |
|                                 | Hydrogen chloride                                 | kg   | 8.0E-4 - 4.4E-3        | -                                          | -                                          |
|                                 | NO <sub>x</sub>                                   | kg   | -                      | 0.013                                      | 0.022                                      |
|                                 | Sulphur dioxide (SF <sub>6</sub> cast operations) | kg   | 2.42E-03               | 24.0                                       | 24.0                                       |
|                                 | Sulphur hexafluoride                              | kg   | ca. 0.0007             | 0.0018                                     | 0.0012                                     |
| <b>Characterisation results</b> | Global warming effect incl. SF <sub>6</sub>       | kg   | 19.0                   | 50.00                                      | 37.00                                      |
|                                 | Global warming effect excl. SF <sub>6</sub>       | kg   | 6.0                    | 8.0                                        | 9.0                                        |
|                                 | Acidification                                     | kg   | 0.02                   | 0.025                                      | -                                          |
|                                 | Winter smog                                       | kg   | 0.015                  | 0.017                                      | -                                          |
|                                 | Solid waste                                       | kg   | 0.5                    | 2.0                                        | 1.3                                        |

The newest figure for the cumulative energy use of total 144MJ/kg (Albright & Haagenzen (1997)) is used. This figure has to be disaggregated to the type of energy uses. It is assumed that about 60% of the energy is delivered by electricity produced in Norway with a conversion efficiency of 82% (according to the ratios shown in Tab. 5.2). Other energy uses are steam consumption, the use of gas and due to the transports. This is estimated with 40% or 49MJ cumulative energy demand and a conversion efficiency of 67%. Thus 39MJ/kg of natural gas are used. The uncertainty of the data for the energy use is high as the summary in Tab. 5.1 shows because a full disaggregating of the found data is not possible.

An important factor for the environmental impacts is the use of SF<sub>6</sub>. It is used to protect molten magnesium from oxidation in the cast house, in the production of ingots and at the diecaster. Sulphur dioxide has been known to fulfil this function for a long time, but SO<sub>2</sub> is toxic in higher concentrations and in contact with humidity it will enhance corrosion of steel equipment. The quantities used in different

<sup>2</sup> Environmental information on <http://www.magnesium.hydro.com/> (26.7.2000). The facility in Porsgrunn was closed down 1st of April 2002. A new activity has been established in a part of the old factory. This new remelting facility is called Hydro Magnesium Alloys. Process scrap from magnesium users and pure magnesium from China are used to produce magnesium alloys (Personal communication, Liv Kari Sivertsen, 14.5.02).

plants varies between 0.1kg and 11kg SF<sub>6</sub> per ton produced magnesium components. The use of SF<sub>6</sub> for die-casting is not included in this figure (Gjestland et al. (1996)). Norsk Hydro has reduced the emissions from formerly 4kg/t to 0.7kg/t in 1998.<sup>3</sup> Estimates for the worldwide average specific consumption vary between 1.3kg/t and 7kg/t (Maiss & Brenninkmeijer (1998:3084)). We assume an average emission figure of 2.9kg/t for the magnesium production with data for the Norsk Hydro plants (0.7kg/t, 34% production share) and other plants (4kg/t).

The inventory for other emissions (e.g. SO<sub>2</sub>, dioxin, chlorinated hydrocarbons to water) is based on the data published in the environmental reports of Norsk Hydro (Tab. 5.2, Norsk Hydro (1999)). Charcoal is used for the chlorination of magnesia in an amount of 0.45kg/t (Hoy-Petersen et al. (1999:562)). The import from Asia by ship has been considered for this material. The theoretical emission of 1.32kg CO<sub>2</sub>/kg magnesium has been calculated from the charcoal use which is used as a reduction agent. The unit process raw data and data quality indicators for magnesium production are shown in Tab. 5.3. Transport of raw materials have been estimated with standard distances.

Large uncertainties exist for data on energy use and SF<sub>6</sub> emissions during magnesium production. For the energy use and the SF<sub>6</sub> emission a literature review has been used to estimate the standard deviation. The highest uncertainties exist for the SF<sub>6</sub> emissions which might be as high as 11kg/t. Emission data are used for cumulative emissions. Thus they overestimate the direct emissions in an unknown range.

**Tab. 5.3 Unit process raw data and data quality indicators for the production of magnesium. References and assumptions for the inventory are presented in the text.**

|                | Name                                                                    | magnesium,<br>at plant | Uncertainty<br>Type | Standard<br>Deviation<br>95% | General Comment                                                                        |
|----------------|-------------------------------------------------------------------------|------------------------|---------------------|------------------------------|----------------------------------------------------------------------------------------|
|                | Location<br>Infrastructure Process<br>Unit                              | RER<br>0<br>kg         |                     |                              |                                                                                        |
| Tecnosphere    | charcoal, at plant                                                      | 4.50E-01               | 1                   | 1.26                         | (3,4,1,3,1,5); Literature                                                              |
|                | sulphur hexafluoride, liquid, at plant                                  | 2.90E-03               | 1                   | 1.22                         | (1,3,1,3,1,5); Estimation based on literature                                          |
|                | tap water, at user                                                      | 2.48E+02               | 1                   | 1.22                         | (1,3,1,3,1,5); Data from env. report                                                   |
|                | disposal, decarbonising waste, 30% water, to residual material landfill | 1.86E-01               | 1                   | 1.57                         | (1,3,1,3,4,5); Environmental report, type of waste not clear                           |
|                | disposal, separator sludge, 90% water, to hazardous waste incineration  | 5.74E-02               | 1                   | 1.57                         | (1,3,1,3,4,5); Data from environmental report, type of sludges not clear               |
|                | treatment, sewage, from residence, to wastewater treatment, class 2     | 2.48E-01               | 1                   | 1.31                         | (1,3,1,3,3,5); Data from env. report                                                   |
|                | transport, transoceanic freight ship                                    | 6.20E+00               | 1                   | 2.05                         | (4,na,na,1,na,na); Imports of Mg from China, imports of charcoal from overseas 10000km |
|                | transport, freight, rail                                                | 2.72E-01               | 1                   | 2.10                         | (4,5,na,na,na,na); Standard distance 600km                                             |
|                | electricity, medium voltage, production NORDEL, at grid                 | 2.15E+01               | 1                   | 1.25                         | desaggregated from cumulative energy                                                   |
|                | natural gas, burned in industrial furnace >100kW                        | 3.87E+01               | 1                   | 2.00                         | desaggregated from cumulative energy demand with assumption of 67% conversion          |
|                | transport, lorry >16t, fleet average                                    | 1.43E-01               | 1                   | 2.10                         | (4,5,na,na,na,na); Standard distance                                                   |
|                | magnesium plant                                                         | 2.00E-10               | 1                   | 3.10                         | (4,4,1,3,1,5); 50a of uses with a production of 100000t/a                              |
|                | limestone, milled, packed, at plant                                     | 1.95E+00               | 1                   | 1.31                         | (1,3,1,3,3,5); Estimation for dolomite use                                             |
|                | Magnesium, 0.13% in water                                               | 7.52E-01               | 1                   | 1.22                         | (1,3,1,3,3,3); Data from env. report                                                   |
|                | Carbon dioxide, biogenic                                                | 1.32E+00               | 1                   | 1.24                         | (4,1,3,1,1,3); Calculation from charcoal use, 2.93kg/kg                                |
| Resource use   | Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin                | 2.33E-11               | 1                   | 3.01                         | (1,3,1,3,1,3); Data from env. report                                                   |
|                | Heat, waste                                                             | 4.97E+01               | 1                   | 1.22                         | (4,1,1,1,1,3); Calculated from electricity                                             |
|                | Hydrogen chloride                                                       | 2.00E-03               | 1                   | 2.01                         | (1,3,1,3,1,3); Data from env. report                                                   |
|                | Sulfur dioxide                                                          | 2.42E-03               | 1                   | 2.01                         | (1,3,1,3,1,3); Data from env. report                                                   |
|                | Sulfur hexafluoride                                                     | 2.90E-03               | 1                   | 4.10                         | Estimation, large variations for different producers                                   |
|                | Chlorinated solvents, unspecified                                       | 2.48E-08               | 1                   | 2.01                         | (1,3,1,3,1,3); Data from env. report                                                   |
| Reference flow | magnesium, at plant                                                     | 1.00E+00               | -                   | -                            | -                                                                                      |

<sup>3</sup> Johanna Öster Haagenen (Norsk Hydro, email communication 8.8.2000): The total amount of SF<sub>6</sub> is depending on the total weight of magnesium handled. Norsk Hydro is in the process of changing production routines and SF<sub>6</sub> will be phased out. Hydro is also selling SO<sub>2</sub> diecast equipment and competence to diecasters and is offering service. Some diecasters have never used SF<sub>6</sub> and are still using SO<sub>2</sub>.

### 5.1.2 Magnesium Plant

The land use area and building size for a magnesium production plant has been estimated roughly with 1km<sup>2</sup> from a photo of the Norsk Hydro factory in Porsgrun.<sup>4</sup> Land transformation has been calculated according to standard methodology. The unit process raw data and data quality indicators for the magnesium production plant with an annual production capacity of 100 000t and a life time of 50 years are shown in Tab. 5.4. Data for the necessary equipment (likely also metals and bricks) were not available. They can be considered to be not important in comparison to the amount of produced magnesium.

**Tab. 5.4 Unit process raw data and data quality indicators for the magnesium production plant and data quality considerations.**

|                | Name                                       | magnesium plant  | Uncertainty Type | Standard Deviation 95% | General Comment                                  |
|----------------|--------------------------------------------|------------------|------------------|------------------------|--------------------------------------------------|
|                | Location<br>Infrastructure Process<br>Unit | RER<br>1<br>unit |                  |                        |                                                  |
| Tecnosphere    | Occupation, industrial area                | 5.00E+05         | 1                | 1.60                   | (4,4,1,3,1,5); Estimation from map               |
| Resource use   | Transformation, from unknown               | 5.00E+07         | 1                | 2.10                   | (4,4,1,3,1,5); Application of standard procedure |
|                | Transformation, to industrial area         | 1.00E+06         | 1                | 2.10                   | (4,4,1,3,1,5); Application of standard procedure |
| Reference flow | magnesium plant                            | 1.00E+06         | -                | -                      | -                                                |

### 5.1.3 Magnesium-Aluminium Alloys

#### AZ91D

The magnesium alloy AZ91D (also known as M11916 or DIN GD-MgAl9Zn1) is the most common one for the production of die-castings. It contains 9.1% aluminium and small amounts of copper, zinc and manganese. Environmental impacts of production are only known from a summarized inventory (Albright & Haagensen (1997)). The inventory data derived from this investigation is shown in Tab. 5.5. The energy use has been disaggregated in the same way as for magnesium. Inventory data for the infrastructure are not considered as it is assumed that the production will take place in the magnesium production plant. Specific data for the inert gas use were not available. It can be assumed that it is not much more than for the pure magnesium (which is already included in the inventory).

**Tab. 5.5 Unit process raw data and data quality indicators for the production of magnesium alloy AZ91. References and assumptions for the inventory are presented in the text.**

|                | Name                                                    | magnesium-alloy, AZ91, at plant | Uncertainty Type | Standard Deviation 95% | General Comment                         |
|----------------|---------------------------------------------------------|---------------------------------|------------------|------------------------|-----------------------------------------|
|                | Location<br>Infrastructure Process<br>Unit              | RER<br>0<br>kg                  |                  |                        |                                         |
| Tecnosphere    | magnesium, at plant                                     | 9.09E-01                        | 1                | 1.20                   | (2,2,1,1,1,5); Literature               |
|                | copper, at regional storage                             | 1.02E-03                        | 1                | 1.20                   | (2,2,1,1,1,5); Literature               |
|                | manganese, at regional storage                          | 3.05E-03                        | 1                | 1.20                   | (2,2,1,1,1,5); Literature               |
|                | zinc, primary, at regional storage                      | 7.11E-03                        | 1                | 1.20                   | (2,2,1,1,1,5); Literature               |
|                | Mg-silicon, at plant                                    | 2.03E-03                        | 1                | 1.20                   | (2,2,1,1,1,5); Literature               |
|                | transport, freight, rail                                | 2.11E-02                        | 1                | 2.10                   | (4,5,na,na,na,na); Standard distances   |
|                | electricity, medium voltage, production NORDEL, at grid | 1.51E+00                        | 1                | 1.30                   | (4,2,1,1,1,5); Literature disaggregated |
|                | transport, lorry >16t, fleet average                    | 1.06E-02                        | 1                | 2.10                   | (4,5,na,na,na,na); Standard distances   |
|                | aluminium, production mix. cast alloy, at plant         | 9.24E-02                        | 1                | 1.30                   | (4,2,1,1,1,5); Literature disaggregated |
| Emissions      | Heat, waste                                             | 5.43E+00                        | 1                | 1.30                   | (4,2,1,1,1,5); Literature disaggregated |
| Reference flow | magnesium-alloy, AZ91, at plant                         | 1.00E+00                        | -                | -                      | -                                       |

<sup>4</sup> Confirmed by Liv Kari Sivertsen, Norsk Hydro, NO, 14.5.02 in a personal communication.

## AlMg3

The production of AlMg3 (AA 5754 or EN AW-5754) is also investigated with the content of different metals used for the alloy. This aluminium alloy contains 3% of magnesium and little amounts of other metals (Lang (1999:9)).

The additional cumulative energy for producing the alloy is 7MJ/kg (conversion efficiency estimated with 82%) and it is assumed that it is delivered by the electricity mix of the aluminium industry. Transports are considered for the alloying metals but not for the main ingredients, as it is assumed that the alloy production is close to the magnesium or aluminium production plant. A material loss of 1.5% during production (assumed by Albright & Haagenen (1997 for magnesium alloys)) is considered for the inventory. It is assumed that the production takes place at the magnesium (or aluminium) production plant. Thus an additional infrastructure is not considered. The inventory is shown in Tab. 5.6. Information for further magnesium alloys can be found in the handbook of KAMMER *et al.* (2000)).

**Tab. 5.6 Unit process raw data and data quality indicators for the production of magnesium aluminium alloy AlMg3. References and assumptions for the inventory are presented in the text.**

|                | Name                                            | aluminium alloy, AlMg3, at plant | Uncertainty Type | Standard Deviation 95% | General Comment                            |
|----------------|-------------------------------------------------|----------------------------------|------------------|------------------------|--------------------------------------------|
|                | Location<br>Infrastructure Process Unit         | RER<br>0 kg                      |                  |                        |                                            |
| Technosphere   | magnesium, at plant                             | 3.05E-02                         | 1                | 1.20                   | (2,2,1,1,1,5); Literature                  |
|                | cast iron, at plant                             | 4.06E-03                         | 1                | 1.20                   | (2,2,1,1,1,5); Literature                  |
|                | chromium, at regional storage                   | 3.05E-03                         | 1                | 1.20                   | (2,2,1,1,1,5); Literature                  |
|                | copper, at regional storage                     | 1.02E-03                         | 1                | 1.20                   | (2,2,1,1,1,5); Literature                  |
|                | manganese, at regional storage                  | 5.08E-03                         | 1                | 1.20                   | (2,2,1,1,1,5); Literature                  |
|                | zinc, primary, at regional storage              | 2.03E-03                         | 1                | 1.20                   | (2,2,1,1,1,5); Literature                  |
|                | MG-silicon, at plant                            | 4.06E-03                         | 1                | 1.20                   | (2,2,1,1,1,5); Literature                  |
|                | transport, freight, rail                        | 3.86E-03                         | 1                | 2.10                   | (4,5,na,na,na,na); Standard distance 200km |
|                | electricity mix, aluminium industry             | 1.59E+00                         | 1                | 1.30                   | (4,2,1,1,1,5); Literature disaggregated    |
|                | transport, lorry >16t, fleet average            | 1.93E-03                         | 1                | 2.10                   | (4,5,na,na,na,na); Standard distance 100km |
|                | aluminium, production mix, cast alloy, at plant | 9.65E-01                         | 1                | 1.20                   | (2,2,1,1,1,5); Literature                  |
|                | Heat, waste                                     | 5.74E+00                         | 1                | 1.20                   | (2,2,1,1,1,5); Literature                  |
|                | aluminium alloy, AlMg3, at plant                | 1.00E+00                         | -                | -                      | -                                          |
| Emissions      |                                                 |                                  |                  |                        |                                            |
| Reference flow |                                                 |                                  |                  |                        |                                            |

### 5.1.4 Magnesium Die Castings

The diecasting process is used for the production of metal parts in a predefined form. SF<sub>6</sub> is also used during the die casting process in order to prevent molten magnesium from oxidation. Different authors have investigated the SF<sub>6</sub> use. GJESTLAND *et al.* (1996:39) reported that up to 1kg SF<sub>6</sub> per ton of Mg component might be used and emitted during the die casting process. In Germany, magnesium die casting producers used between 0.25 and 10 kg SF<sub>6</sub> per ton of magnesium or in some cases only SO<sub>2</sub>. The average use in 1995 and 1996 was 4.3 and 2.4 kg/t, respectively. Due to the decreasing amounts used, a prognosis assumed that until 2000 the average use will decrease to 1 kg/t (Schwarz & Leisewitz (1996:41)). For Swiss die casters the amount of SF<sub>6</sub> is reported to be 4kg SF<sub>6</sub>/t Mg-die castings. This is equal to an annual amount of 120kg (Klaus & Dinkel (1999:41)). A producer of die casting equipment<sup>5</sup> quantifies much smaller values for the usage of SF<sub>6</sub> and SO<sub>2</sub> with 0.012 to 0.018kg SF<sub>6</sub> and 0.049 to 0.075kg SO<sub>2</sub> per tonne of magnesium. These data could not be verified. For our inventory we assume based on the literature data 1kg SF<sub>6</sub>/t.

The cumulative energy use required to produce a finished die casting, including normal processing efficiencies, amounts to 21-45MJ/kg or 1.9-4.1kWh electricity/kg (Wilson *et al.* (1995:660)). It is as-

<sup>5</sup> RAUCH Fertigungstechnik, Austria, 22.8.2000, <http://www.rauch-ft.com/englisch/magnes/gas.html>.

sumed that only electricity is used (2.22kWh/kg). The inventory for die castings is shown in Tab. 5.7. Data for the infrastructure were not available.

**Tab. 5.7 Unit process raw data and data quality indicators for magnesium die-casting. References and assumptions for the inventory are presented in the text**

|              | Name                                                  | magnesium-<br>alloy, AZ91,<br>diecasting, at<br>plant | Uncertainty<br>Type | Standard<br>Deviation<br>95% | General Comment                            |
|--------------|-------------------------------------------------------|-------------------------------------------------------|---------------------|------------------------------|--------------------------------------------|
|              | Location<br>Infrastructure Process<br>Unit            | RER<br>0<br>kg                                        |                     |                              |                                            |
| Technosphere | magnesium-alloy, AZ91, at plant                       | 1.67E+00                                              | 1                   | 1.20                         | (2,2,1,1,1,5); Literature                  |
|              | sulphur hexafluoride, liquid, at plant                | 1.00E-03                                              | 1                   | 1.20                         | (2,2,1,1,1,5); Literature                  |
|              | transport, freight, rail                              | 3.35E-01                                              | 1                   | 2.10                         | (4,5,na,na,na,na); Standard distance       |
|              | electricity, medium voltage, production UCTE, at grid | 2.22E+00                                              | 1                   | 1.50                         | Literature estimation                      |
|              | transport, lorry >16t, fleet average                  | 1.67E-01                                              | 1                   | 2.10                         | (4,5,na,na,na,na); Standard distance 100km |
| Emissions    | Heat, waste                                           | 8.00E+00                                              | 1                   | 1.30                         | (4,2,1,1,1,5); Literature                  |
|              | Sulfur hexafluoride                                   | 1.00E-03                                              | 1                   | 1.60                         | (2,2,1,1,1,5); Literature                  |

### 5.1.5 Recycling of Magnesium

The possibilities for the recycling of magnesium are described by KAMMER *et al.* ( (2000:49ff)). To-day about 17% of the magnesium supply is produced by recycling of magnesium wastes. Mainly pure wastes with a known composition are used for the recycling. The recycled magnesium is used mainly during the production of aluminium, iron and steel as a desulphurisation or reduction agent.

The use of electricity for the recycling is much lower than for the production of primary magnesium. KAMMER *et al.* ( (2000)) estimated it to be only 3 kWh/kg in comparison to about 20 kWh/kg for the primary production. Different processes can be used depending on the type of input material. The processes need normally also a protecting gas like sulphur hexafluoride, but the tendency is to use SO<sub>2</sub> to reduce the greenhouse gas emission.

Due to substantial lack of data we refrain from presenting unit process raw data for the recycling of magnesium. One might use the figures for magnesium and reduce the energy use to one-tenth in a first assumption. Other impacts especially for the SF<sub>6</sub> emissions can be expected to be about the same.

## 5.2 Data Quality Considerations

The simplified approach with a pedigree matrix has been used for calculating most of the standard deviations. Large uncertainties exist for data on energy use and SF<sub>6</sub> emissions during magnesium production. For the energy use and the SF<sub>6</sub> emission a literature review has been used to estimate the standard deviation. The highest uncertainties exist for the SF<sub>6</sub> emissions which might be as high as 11 kg/t. Also the value for transport distances of materials is very uncertain.

## 5.3 Conclusions

The environmental impacts of magnesium production have been investigated as a unit process over the life cycle from raw material extraction until the gate of the factory. The investigation is based on environmental data provided by a producer and on a literature research. High uncertainties still exist for the average energy and SF<sub>6</sub> use in magnesium industry. The inventory can be used in life cycle assessment for complex products where magnesium or metal alloys are used.

## Abbreviations

### English

AZ91 Magnesium-aluminium (9.1%) alloy

AlMg3 Aluminium-magnesium (3%) alloy

## Appendices: EcoSpold Meta Information

| Type                 | ID  | Field name // ID           | 67                                                                                                                                              | 2994                                                                                             | 481                                                                                                                                                                | 491                                                                                                                                                                                                                                                                                                                               | 492                                                                                                                                                                |
|----------------------|-----|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ReferenceFunction    | 401 | Name                       | aluminium alloy, AlMg3, at plant                                                                                                                | magnesium plant                                                                                  | magnesium, at plant                                                                                                                                                | magnesium-alloy, AZ91, at plant                                                                                                                                                                                                                                                                                                   | magnesium-alloy, AZ91, diecasting, at plant                                                                                                                        |
| Geography            | 662 | Location                   | RER                                                                                                                                             | RER                                                                                              | RER                                                                                                                                                                | RER                                                                                                                                                                                                                                                                                                                               | RER                                                                                                                                                                |
| ReferenceFunction    | 493 | InfrastructureProcess      | 0                                                                                                                                               | 1                                                                                                | 0                                                                                                                                                                  | 0                                                                                                                                                                                                                                                                                                                                 | 0                                                                                                                                                                  |
| ReferenceFunction    | 403 | Unit                       | kg                                                                                                                                              | unit                                                                                             | kg                                                                                                                                                                 | kg                                                                                                                                                                                                                                                                                                                                | kg                                                                                                                                                                 |
| DataSetInformation   | 201 | Type                       | 1                                                                                                                                               | 1                                                                                                | 1                                                                                                                                                                  | 1                                                                                                                                                                                                                                                                                                                                 | 1                                                                                                                                                                  |
| DataSetInformation   | 202 | Version                    | 2.0                                                                                                                                             | 2.0                                                                                              | 2.0                                                                                                                                                                | 2.0                                                                                                                                                                                                                                                                                                                               | 2.0                                                                                                                                                                |
| DataSetInformation   | 203 | energyValues               | 0                                                                                                                                               | 0                                                                                                | 0                                                                                                                                                                  | 0                                                                                                                                                                                                                                                                                                                                 | 0                                                                                                                                                                  |
| DataSetInformation   | 205 | LanguageCode               | en                                                                                                                                              | en                                                                                               | en                                                                                                                                                                 | en                                                                                                                                                                                                                                                                                                                                | en                                                                                                                                                                 |
| DataSetInformation   | 206 | LocalLanguageCode          | de                                                                                                                                              | de                                                                                               | de                                                                                                                                                                 | de                                                                                                                                                                                                                                                                                                                                | de                                                                                                                                                                 |
| DataEntryBy          | 302 | Person                     | 41                                                                                                                                              | 41                                                                                               | 41                                                                                                                                                                 | 41                                                                                                                                                                                                                                                                                                                                | 41                                                                                                                                                                 |
| DataEntryBy          | 304 | QualityNetwork             | 1                                                                                                                                               | 1                                                                                                | 1                                                                                                                                                                  | 1                                                                                                                                                                                                                                                                                                                                 | 1                                                                                                                                                                  |
| ReferenceFunction    | 400 | DataSetRelatesToProduct    | 1                                                                                                                                               | 1                                                                                                | 1                                                                                                                                                                  | 1                                                                                                                                                                                                                                                                                                                                 | 1                                                                                                                                                                  |
| ReferenceFunction    | 402 | IncludedProcesses          | Production of alloy from primary metals, no extra infrastructure.                                                                               | Building infrastructure and land use, no machinery.                                              | Resource extraction until gate of the factory including infrastructure.                                                                                            | Production of alloy from primary metals, no extra infrastructure.                                                                                                                                                                                                                                                                 | Diecasting process, no extra infrastructure.                                                                                                                       |
| ReferenceFunction    | 404 | Amount                     | 1                                                                                                                                               | 1                                                                                                | 1                                                                                                                                                                  | 1                                                                                                                                                                                                                                                                                                                                 | 1                                                                                                                                                                  |
| ReferenceFunction    | 490 | LocalName                  | Aluminium-Legierung, AlMg3, ab Werk                                                                                                             | Magnesiumwerk                                                                                    | Magnesium, ab Werk                                                                                                                                                 | Magnesium-Legierung, AZ91, ab Werk                                                                                                                                                                                                                                                                                                | Magnesium-Legierung, AZ91, Druckguss, ab Werk                                                                                                                      |
| ReferenceFunction    | 491 | Synonyms                   | AA 5754//EN AW-5754                                                                                                                             |                                                                                                  |                                                                                                                                                                    | M11916//DIN GD-MgAl9Zn1                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                    |
|                      |     |                            | The production of AlMg3 is investigated with the content of different metals used for the alloy. This aluminium alloy contains 3% of magnesium. | Production plant for magnesium with an annual capacity of 100'000t and a life time of 50 years.  | Large uncertainties exist for data on energy use and SF6 emissions during magnesium production. This inventory describes an assumption for the average production. | The magnesium alloy AZ91D is the most common one for the production of die-castings. It contains 9.1% aluminium and small amounts of zinc and manganese. Environmental impacts of production are only known from a summarized inventory. It is assumed that magnesium for this alloy has been produced in a new production plant. | Large uncertainties exist for data on energy use and SF6 emissions during magnesium diecasting. This inventory describes an assumption for the average production. |
| ReferenceFunction    | 492 | GeneralComment             |                                                                                                                                                 |                                                                                                  |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                    |
| ReferenceFunction    | 494 | InfrastructureIncluded     | 1                                                                                                                                               | 1                                                                                                | 1                                                                                                                                                                  | 1                                                                                                                                                                                                                                                                                                                                 | 1                                                                                                                                                                  |
| ReferenceFunction    | 495 | Category                   | metals                                                                                                                                          | metals                                                                                           | metals                                                                                                                                                             | metals                                                                                                                                                                                                                                                                                                                            | metals                                                                                                                                                             |
| ReferenceFunction    | 496 | SubCategory                | extraction                                                                                                                                      | extraction                                                                                       | extraction                                                                                                                                                         | extraction                                                                                                                                                                                                                                                                                                                        | extraction                                                                                                                                                         |
| ReferenceFunction    | 497 | LocalCategory              | Metalle                                                                                                                                         | Metalle                                                                                          | Metalle                                                                                                                                                            | Metalle                                                                                                                                                                                                                                                                                                                           | Metalle                                                                                                                                                            |
| ReferenceFunction    | 498 | LocalSubCategory           | Gewinnung                                                                                                                                       | Gewinnung                                                                                        | Gewinnung                                                                                                                                                          | Gewinnung                                                                                                                                                                                                                                                                                                                         | Gewinnung                                                                                                                                                          |
| ReferenceFunction    | 499 | Formula                    |                                                                                                                                                 |                                                                                                  | Mg                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                    |
| ReferenceFunction    | 501 | StatisticalClassification  |                                                                                                                                                 |                                                                                                  |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                    |
| ReferenceFunction    | 502 | CASNumber                  |                                                                                                                                                 |                                                                                                  | 007439-95-4                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                    |
| TimePeriod           | 601 | StartDate                  | 1998                                                                                                                                            | 1998                                                                                             | 1998                                                                                                                                                               | 1998                                                                                                                                                                                                                                                                                                                              | 1998                                                                                                                                                               |
| TimePeriod           | 602 | EndDate                    | 1998                                                                                                                                            | 1998                                                                                             | 1998                                                                                                                                                               | 1998                                                                                                                                                                                                                                                                                                                              | 1998                                                                                                                                                               |
| TimePeriod           | 603 | DataValidForEntirePeriod   | 1                                                                                                                                               | 1                                                                                                | 1                                                                                                                                                                  | 1                                                                                                                                                                                                                                                                                                                                 | 1                                                                                                                                                                  |
| TimePeriod           | 611 | OtherPeriodText            | Environmental report                                                                                                                            | Environmental report                                                                             | Environmental report                                                                                                                                               | Environmental report                                                                                                                                                                                                                                                                                                              | Environmental report                                                                                                                                               |
| Geography            | 663 | Text                       | Data for 1 producer                                                                                                                             | Data for 1 producer                                                                              | Data from literature and for 1 producer                                                                                                                            | Data for 1 producer                                                                                                                                                                                                                                                                                                               | Data for 1 producer                                                                                                                                                |
|                      |     |                            | Production of aluminium-magnesium alloy from primary metals, no recycling of input materials.                                                   | Infrastructure (land use and buildings) for the production of magnesium not including equipment. | Production of magnesium from seawater and dolomite in a modern production plant based on data from a Norwegian producer.                                           | Production of magnesium alloy from primary metals, no recycling of input materials.                                                                                                                                                                                                                                               | Diecasting of AZ91, no recycling                                                                                                                                   |
| Technology           | 692 | Text                       |                                                                                                                                                 |                                                                                                  |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                    |
| Representativeness   | 722 | Percent                    |                                                                                                                                                 | 30                                                                                               | 30                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                    |
| Representativeness   | 724 | ProductionVolume           | ca. 146000t/a                                                                                                                                   | 105600 tonnes in 1998                                                                            | 372000 tonnes in 1998                                                                                                                                              | Not known                                                                                                                                                                                                                                                                                                                         | ca. 100000 t/a                                                                                                                                                     |
| Representativeness   | 725 | SamplingProcedure          | Environmental data for an important producer                                                                                                    | Estimation with map.                                                                             | Environmental data for an important producer                                                                                                                       | Environmental data from an important producer                                                                                                                                                                                                                                                                                     | Environmental data from an important producer                                                                                                                      |
| Representativeness   |     |                            | From 1 producer to average Europe                                                                                                               | From 1 producer to average Europe                                                                | From 1 producer to average Europe, energy uses disaggregated from cumulated figures. Type of electricity mix not known.                                            | From 1 producer to average Europe                                                                                                                                                                                                                                                                                                 | From 1 producer to average Europe                                                                                                                                  |
| Representativeness   | 726 | Extrapolations             |                                                                                                                                                 |                                                                                                  |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                    |
| Representativeness   | 727 | UncertaintyAdjustments     | none                                                                                                                                            | none                                                                                             | none                                                                                                                                                               | none                                                                                                                                                                                                                                                                                                                              | none                                                                                                                                                               |
| DataGeneratorAndPubl | 751 | Person                     | 41                                                                                                                                              | 41                                                                                               | 41                                                                                                                                                                 | 41                                                                                                                                                                                                                                                                                                                                | 41                                                                                                                                                                 |
| DataGeneratorAndPubl | 756 | DataPublishedIn            | 2                                                                                                                                               | 2                                                                                                | 2                                                                                                                                                                  | 2                                                                                                                                                                                                                                                                                                                                 | 2                                                                                                                                                                  |
| DataGeneratorAndPubl | 757 | ReferenceToPublishedSource | 10                                                                                                                                              | 10                                                                                               | 10                                                                                                                                                                 | 10                                                                                                                                                                                                                                                                                                                                | 10                                                                                                                                                                 |
| DataGeneratorAndPubl | 758 | Copyright                  | 1                                                                                                                                               | 1                                                                                                | 1                                                                                                                                                                  | 1                                                                                                                                                                                                                                                                                                                                 | 1                                                                                                                                                                  |
| DataGeneratorAndPubl | 759 | AccessRestrictedTo         | 0                                                                                                                                               | 0                                                                                                | 0                                                                                                                                                                  | 0                                                                                                                                                                                                                                                                                                                                 | 0                                                                                                                                                                  |
| DataGeneratorAndPubl | 760 | CompanyCode                |                                                                                                                                                 |                                                                                                  |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                    |
| DataGeneratorAndPubl | 761 | CountryCode                |                                                                                                                                                 |                                                                                                  |                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                    |
| DataGeneratorAndPubl | 762 | PageNumbers                | chapter magnesium                                                                                                                               | chapter magnesium                                                                                | chapter magnesium                                                                                                                                                  | chapter magnesium                                                                                                                                                                                                                                                                                                                 | chapter magnesium                                                                                                                                                  |



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## Part XIV

# **Non Ferrous Metal Winning Auxiliary Processes**

Data v2.1 (2009)

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Final report of updated Swiss National Life Cycle Inventory Database "ecoinvent", version 2.1

## Comments on version 2.1

This part corresponds to the original report in Althaus et al. (2004). Chapter “6 Cumulative Results and Interpretation” of the original report has been deleted for the current update.

Besides of these reported changes no alteration of the original text took place for version 2.0. Some of the reference flow names have been changed compared to the preceding versions. They have been updated with the actual names of version 2.0.

For version 2.1, a new model for emissions of sulfidic tailing disposal is introduced. The dataset “disposal sulfidic tailings, offsite” is changed and the report adapted accordingly.

## Summary

In this part, the modelling of disposal of sulphidic and non sulphidic tailings and overburdens and that of mine and smelter infrastructure is described. These modules are used in the production of different non ferrous metals.

The long term emissions from sulfidic tailings due to acid rock drainage are believed to be of major importance for the LCI of metals from sulphidic ores. For version 2.1 a dedicated tailings disposal model was taken up.



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# 1 Introduction

Several production processes of the non ferrous metal industry are similar to each other. Namely the winning of the metal bearing ore and the beneficiation processes depend mainly on throughput of minerals. Process specific differences are rather linked to the type of metal carrying ore than to the desired metals present in the ore. Most of the non ferrous metals originate from sulphidic ores which present different implication for processing than e.g. oxidic ores. Different metals, e.g. nickel and copper, often are coexistent in the sulphidic ore and are won jointly by the same metallurgical processes. Also infrastructure of mines, beneficiation facilities, metallurgical plants and refineries do not differ greatly from one metal winning process to another. Due to these similarities several auxiliary processes and infrastructure are not differentiated further in ecoinvent. The modelled disposal-processes for overburden and tailings are valid for all non-ferrous-metal industries. The only difference is made for sulphidic tailings, which undergo acid rock drainage – reactions with enhanced leaching and considerable acidification. In v2.1 of ecoinvent a dedicated model for sulfidic tailings was introduced.

## 2 System Characterisation

In this chapter several auxiliary processes of the non-ferrous metal industry are discussed:

- disposal processes,
- land use and
- infrastructure.

Fig. 2.1 gives an overview over these impacts discussed in this chapter and the modules in ecoinvent they are inventoried in. Basically three types of activities can be distinguished when inventorying mining operations.

**Operation:** Extraction of ore effects a direct type of impact. For each tonne concentrate excavated a certain amount of land is transformed to become an extraction site for a certain time. Additionally infrastructure like hauling trucks, bucket wheel excavator and conveyor belts are required. The demand in these resources (land) and processes (infrastructure) are inventoried directly within the respective mining module, they are direct proportional to the mass<sup>1</sup> of ore processed. Also the leaching of metal from ore is considered as extraction, the land use for leaching – which is considerable – therefore also is where appropriate inventoried within the mining module. In the case of froth flotation the ore is ground in mills. The mill heads and jaws are severely strained, the consumption in these parts is also inventoried as direct impact in terms of demand in austenitic manganese steel (2.1.2 Direct land use).

**Infrastructure:** The infrastructure is inventoried as separate modules, each for underground mine, open pit mine, beneficiation and smelter. These are pure infrastructure modules and comprise the building of the mining, beneficiation, etc. facilities, roads, docks etc inclusively the disposal. These modules do not refer to the direct impact in the mineral extraction site where the ore is mined, but to the facilities for ore-processing, maintenance buildings, staff accommodation, etc. The functional unit is infrastructure units per kilogram raw-material processed, thus the functional unit refers to the input (2.2 Infrastructure in the non-ferrous metal industry).

**Tailings / overburden:** The extraction of metal ore is actually a concentration process. First, the ore has to be separated from rock, be it to access the veins, be it to clean the extracted material from unwanted fractions. This unwanted material is referred to as overburden. In a second step, the metal bearing ore is concentrated further using different mechanical and/or chemical beneficiation methods. The unwanted mineral fraction is referred to as tailings. Tailings consist of ground mineral and often form a slurry after separation, which is disposed in large ponds lined with dams. Apart from the considerable land use, also emissions arise from disposing the material. It is assumed, that the disposed material is alien to the place where it is disposed (off-site) and therefore virtually all of the substances leached over time have to be regarded as emissions. The functional unit of these modules is the mass of tailings or overburden disposed (c.f. 2.3 Disposal of Overburden and Tailings).

**Rehabilitation of tailings:** In this study no rehabilitation of the tailings and overburden disposal site is considered. It is assumed, that the disposal site never will turn to its natural state and consequently no energy and materials are used for rehabilitation. This situation probably presents the norm in a global view.

**Kommentar [gd1]:** d.h. infinite land use?

<sup>1</sup> This assumption might be biased in the case of infrastructure like processing plants. The choice of the material's volume would probably more correct. Since the density is not exactly known in most cases and the difference in densities is expected to be small from ore to ore, the mass is chosen as functional unit.



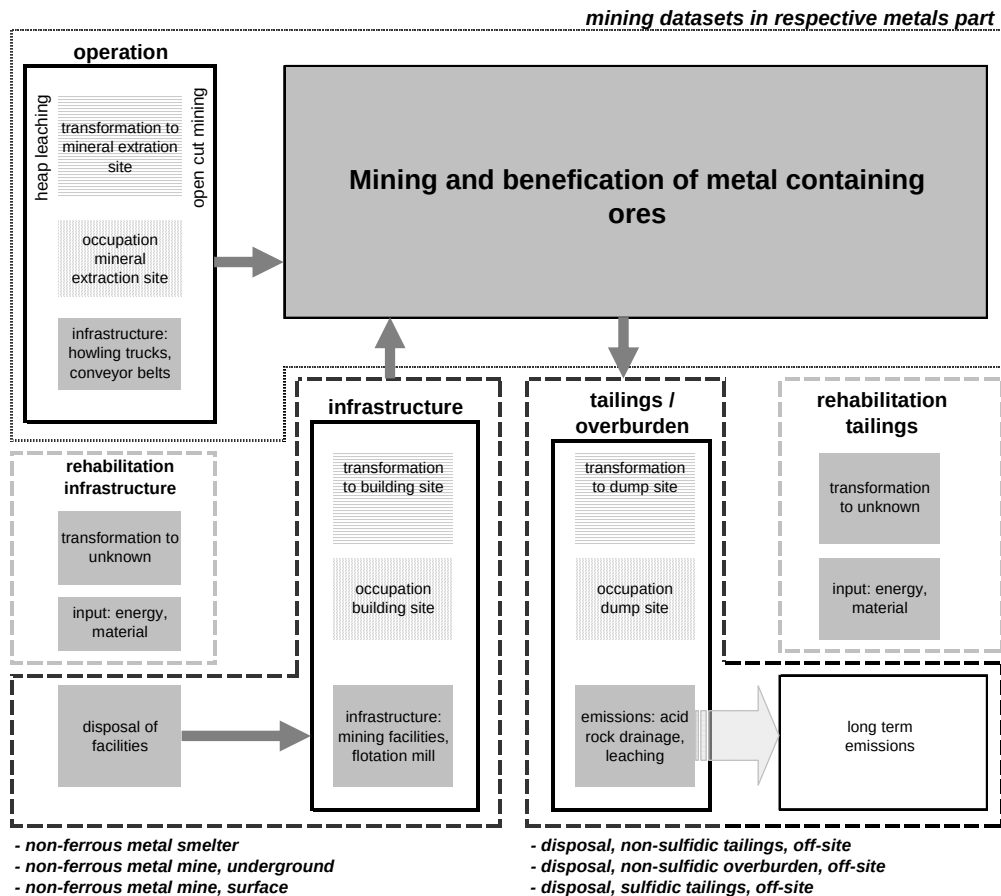


Fig. 2.1 Overview over infrastructure, land use and rehabilitation of the mineral extraction site. Rehabilitation of infrastructure (production site) and tailings are not considered in this study.

## 2.1 Land Transformation and Land Use in Mining and Beneficiation

The effects of mining and beneficiation to land resources can be differentiated into three types:

- Land used by facilities (roads, buildings, stationary equipment)
- Direct land use through exploration and beneficiation (excavation, eventually heap leaching)
- Land used for disposal purposes (stockpiles, tailings)

These three effects are treated in ecoinvent in different modules. As no detailed data for the working of different ore was available, the same values were used for all the modules dealing with non-ferrous metals.

### 2.1.1 Facilities

The land use of the facilities comprise roads, buildings, stationary equipment like screens, feeder, conveyor belts, docks, crusher, mills, cyclones, etc. A good dependency of the Area covered with the yearly production capacity of the operation is found in Martens et al. (2002). Results and assumptions are shown in Tab. 2.1. This land use is inventoried in the infrastructure modules

- “non-ferrous metal mine, surface”
- “non-ferrous metal mine, underground”
- “non-ferrous metal smelter”

**Tab. 2.1** Specific land use by facilities in open cut and underground mining. These figures comprise Beneficiation facilities. Data for land use is a function of the production capacity<sup>2</sup> as described in (Martens et al. (2002)).

| Infrastructure        | Unit                | Open cut   | Underground |
|-----------------------|---------------------|------------|-------------|
| Yield, capacity       | t/a                 | 400'000    | 400'000     |
| Lifetime              | a                   | 50         | 50          |
| Land use              | m <sup>2</sup>      | 122'011    | 184'830     |
| Total yield, lifetime | t                   | 20'000'000 | 20'000'000  |
| Specific land use     | m <sup>2</sup> /t   | 0.0061     | 0.0092      |
| land occupation       | a m <sup>2</sup> /t | 0.305      | 0.462       |

### 2.1.2 Direct land use

According to Martens et al. (2002) a total land transformation of 20 m<sup>2</sup> per thousand tonnes mined copper ore is effected. This figure of 0.02 m<sup>2</sup>/t raw ore is taken as proxy for all non ferrous metals except for aluminium, tin and magnesium. It includes the effects of tailing ponds (42% of the land transformation), leaching heaps (17%), stockpiles (13%), open cuts (10%), facilities (16%) and lowering of the ground through underground mining (2%), covering world-wide mining activities including open cut (80% of the ore) and underground (20%). In this study the lowering of the ground is assumed not to be ecologically effective. Where heap-leaching is effectuated, the value for land transformation adds up to 0.0059 m<sup>2</sup>/t ore, whereas no direct land use for underground mining is considered (Tab. 2.2).

**Tab. 2.2** Land use through direct exploration of the mine. A lifetime of 30 years was assumed. The lowering of the ground through underground mining is not regarded as ecologically relevant and thus is not inventoried.

| Specific land use | open pit              |                         |
|-------------------|-----------------------|-------------------------|
|                   | transformation        | occupation              |
| Unit              | m <sup>2</sup> /t ore | a m <sup>2</sup> /t ore |
| Leaching heaps    | 0.00340               | 0.102                   |
| Open cut          | 0.00250               | 0.075                   |
| Total             | 0.00590               | 0.177                   |

<sup>2</sup> Log-log dependency:  $\text{Area}_{\text{open cut}}[\text{m}^2] = 23.869 \text{ Yield}_{\text{open cut}}[\text{t/a}]^{0.662}$  and  $\text{Area}_{\text{underground}}[\text{m}^2] = 0.322 \text{ Yield}_{\text{underground}}[\text{t/a}]^{1.028}$

### 2.1.3 Land use for waste rock disposal

In mining large fractions of overburden has prior to extraction to be disposed on stockpiles. For open cut operations the ratio between ore and overburden ranges e. g. for nickel typically from 0.3 to 2 (Hilbrans & Hinrichs (1999)). Tailings consist of the unwanted part of the ground ore (gangue) which is separated and usually is disposed as slurry in large ponds.

According to reported values for tailing ponds in IPPC (2002), an average land use of 0.051 m<sup>2</sup>/t and 6.62 m<sup>2</sup>a/t tailings disposed is calculated (Tab. 2.3) assuming an average mine operation of 30 years. The land transformation is assumed to be from “unknown” to “disposal site” for the duration of 100 years after which it is rehabilitated in a natural way. Assuming that the whole disposal area for the tailings and overburdens has to be cleared prior to any disposal activities, i. e. at the very start of the mine operation, the duration of the whole land use is 130 years.

**Tab. 2.3 Assumptions and calculations regarding land use and eventual rehabilitation from tailings. Duration of mine operation phase is 30 years. The tailings are covered with 0.8 m of earth and a density of 0.8 kg / l. After 100 year the disposal site is assumed to be rehabilitated naturally.**

|                            | <b>Dimensio<br/>ns</b> | <b>Tailings</b> | <b>Lifetime<br/>Production</b> | <b>Land<br/>transformat<br/>ion</b> | <b>Land<br/>occupation</b> | <b>Coverage</b> | <b>Coverage</b> |
|----------------------------|------------------------|-----------------|--------------------------------|-------------------------------------|----------------------------|-----------------|-----------------|
|                            | [ha]                   | [t / a]         | [Mt]                           | [m <sup>2</sup> / t]                | [m <sup>2</sup> ·a / t]    | [t]             | [kg / t]        |
| Grapengerg                 | 35                     | 455'000         | 13'650'000                     | 0.026                               | 3.33                       | 2268            | 0.17            |
| Aitik                      | 1450                   | 17'700'000      | 531'000'000                    | 0.027                               | 3.55                       | 93960           | 0.18            |
| Pyhäsalmi                  | 100                    | 180'000         | 5'400'000                      | 0.19                                | 24.1                       | 6480            | 1.2             |
| Zinkgruvan                 | 66                     | 425'000         | 12'750'000                     | 0.052                               | 6.73                       | 4276.8          | 0.34            |
| Total<br>geometric<br>mean |                        |                 | 26'580'000                     | 0.051                               | 6.62                       | 8766            | 0.33            |

## 2.2 Infrastructure in the non-ferrous metal industry

For the inventory of the aluminium production a detailed estimation of the sintering and electrolysis has been made in this study. These figures together with the inventories of the blast furnace, electric arc and blast oxygen converter of the inventory of the iron production built the base for infrastructure estimation of the non-ferrous metal production.

Since it can be assumed that the environmental impact due to infrastructure is very small compared to the energetic intensive, high-throughput production processes in concern, a non-specific module for mining, beneficiation, smelter and refinery is set up. Tab. 2.4 gives an overview over the process-steps involved in the non-ferrous metal industry. Depending on the prevailing of a process step and raw material input the factor is calculated.

Tab. 2.4 Process-steps involved in the non-ferrous metal modules.

| metal / ore      | Raw material    | concentration | yield    | mining under-ground | mining surface | Beneficiation | sintering | Hydro-metallurgical | Metallo-thermal     | electric arc furnace | pyrometal lurgical blast | refinery |
|------------------|-----------------|---------------|----------|---------------------|----------------|---------------|-----------|---------------------|---------------------|----------------------|--------------------------|----------|
| Mn – ore         | Rock            | 35.7% (14.2%) | 69.2%    | 30%                 | 70%            | no            | -         | -                   | -                   | -                    | -                        | -        |
| Cr – ore         | Rock            | 25.5% (11.6%) | 79%      | 79%                 | 21%            |               |           |                     |                     |                      |                          |          |
| Pb/Zn – ore      | Rock            | 5 / 9%        | 68 / 85% | 100                 |                |               |           |                     |                     |                      |                          |          |
| Pb – metal       | Concentrate     | 53%           | 90%      |                     |                |               | yes       |                     |                     |                      | 100%                     | yes      |
| Zn – metal       | Concentrate     | 53%           | 90%      |                     |                |               | no        | 80%                 |                     |                      | 20%                      | no       |
| PGM -ore         | Rock            | 3.E-04%       | 90%      | 100%                | no             | yes           |           |                     |                     |                      |                          |          |
| PGM – metal      | Concentrate     | 5.1E-3%       |          |                     |                |               | yes       | yes                 |                     |                      | yes                      | yes      |
| Cu/Mo – ore      | Rock            | 0.99%         | 81%      | 30%                 | 70%            | yes           | yes       |                     |                     |                      |                          |          |
| Cu – primary     | Concentrate     | 29.7%         | 96.8%    |                     |                |               | Yes       | 10%                 |                     | yes                  | yes                      | no       |
| Cu – SX/EW       | Rock            | 0.99%         | 70%      | no                  | 100%           | no            | yes       | yes                 |                     |                      |                          | Yes      |
| Cu – second.     | Scrap / blister | 78%*          | 90*      |                     |                |               |           |                     |                     |                      | yes                      | Yes      |
| Cu – blister     | Concentrate     | 24.7%         | 97.8%    |                     |                |               | yes       |                     |                     |                      | yes                      |          |
| Ni – ore         | Rock            | 1.13% (0.76%) | 82%      | 100%                |                | yes           |           |                     |                     |                      |                          |          |
| Ni – metal       | concentrate     | 8.13%         | 97%      |                     |                |               | yes       |                     |                     |                      | yes, div.                | no       |
| Ni – ferronickel | Rock            | 1.98% (1.04%) | 57.5%    |                     |                | yes           |           |                     |                     | yes                  |                          | yes      |
| Cr – ferro-Cr    | Concentrate     | 30.7%         | 91.2%    |                     |                |               | no        |                     |                     | 100%                 |                          | no       |
| Mn – ferro-Mn    | Ore             | 42.4%         | 76.1%    |                     |                |               | yes       |                     |                     | 83%                  | 17%                      | No       |
| Cr – metal       | Diverse         | Diverse       | 91%      |                     |                |               |           | Electrolysis 25%    | Alumino-thermic 75% |                      |                          |          |
| Mn – metal       | Ferro-Mn / ore  | 74.5%         | 60%      |                     |                |               |           | 25%                 |                     |                      |                          | No       |

### 2.2.1 Process description

The following paragraphs discuss the main process steps in the non-ferrous metal industry.

**Mining:** Mining is differentiated into open-pit mining and underground mining. Depending on the hardness, the ore can be mined in some cases without explosives. Such soft ore is directly extracted with bucket loader, whereas hard material has to be blasted. Techniques in underground mining do not differ vastly from those in surface mining, especially if the layer mined is high enough for big vehicles. Usually vehicles used underground tend to be smaller with adaptation to low height and low off-gas emissions. Another difference between surface and underground mining is the need for air circulation and cleaning. Hence, beside a few specific features – like hoists in underground mining – no major differences exist.

**Beneficiation:** Most ores are subjected to a first concentration step after being mined. In some cases the only beneficiation possible is a mere classification by size and/or gravity, which yields only a little concentration effect. The beneficiation step of these ores is not considered in this study. In most other cases a beneficiation including a froth flotation is carried out: After mining, the ore is first ground. In a next step it is subjected to gravity concentration to separate the metallic particles from the metal bearing minerals. After this first concentration step, flotation is carried out to remove the gangue from the sulphidic minerals. For this lime and several organic chemicals are added. Tailings usually are led to tailing heaps or ponds. Flotation mills consist of several crushers and screens, ball and rod mills, belt feeder, pumps, filters, float cells, thickener and usually a magnet, dust collector and buildings – truck shop, crusher building, concentrator building and reagent building. In this setting it resembles the hydroxide plant in the aluminium production where the ore also is ground and cleaned from gangue. An example of a medium sized beneficiation plant (flotation mill) is shown in Fig. 2.2.

The beneficiation infrastructure in this study is not covered with a specific module, but approximated with the inventory of the aluminium hydroxide plant, which itself is a beneficiation (specifications: Medium sized plant. Output: 1'220'000 t/a, life time: 50 years). If the inventory of this module is related to its input rather than to its output, the demand has to be corrected by a factor of 1.40, i. e. the production of 1 kg aluminium hydroxide requires 1.4 kg bauxite (ore in the aluminium production). The beneficiation of 1 kg ore therefore requires  $8.40 \cdot 10^{-12}$  units of the module “aluminium hydroxide, plant”. IPPC (2001) reports a demand of 300 g steel due to abrasion of the mill jaws per tonne ground ore, the steel being inventoried as “chromium steel 18/8, at plant”.

**Metallurgy:** The process routes to produce different non-ferrous metals are very divers. As seen in Tab. 2.4 the metallurgical steps usually consist of a sintering / roasting step followed either by a pyrometallurgical or an electrolytic step. The pyrolytical step usually consists of a smelter followed by a converter. In some cases an additional refining step is necessary to yield the grade necessary for trade. The products usually leave the plant after re-melting, casting or preparation steps as powder, ingots, etc. An example of a metallurgical process is given in Fig. 2.4 (copper production).

**Refining:** Some metals like the platinum group metals are won from residues from electrolysis and other residues like dross or from scrap containing some metal values. These Refineries resemble chemical plants where metals subsequently are separated from each other using techniques as solvent—solvent extraction or precipitation/filtration.

Refineries are approximated with the module of “chemical plant, organics”.  $4 \cdot 10^{-10}$  units of this infrastructure module are afforded per kg chemicals. This value is taken also in this study for all those processes, where the refined metal is won by a cascade of purification processes. This is true for the Platinum group metals. For reasons of transparency, no specific module is made for the non-ferrous metal industry. If appropriate in a module, the refinery infrastructure is inventoried as “chemical plant, organics” and clearly is marked as proxy.



**Fig. 2.2** Picture of a flotation mill with crusher, concentrator, thickener and several flotation cells. The output of the mill is stated to be 6'000 – 10'000 t per day (around 2.5 Mt per year).

### **2.2.2 Non-ferrous metal mine, underground in Ecoinvent**

The infrastructure use for a mine is estimated in Tab. 2.5. It comprises land use for mining facilities, production of buildings and equipment including their disposal. No rehabilitation of the industrial site is considered. The values for the surface mine differ only in specific points from those for the underground mine. The data represent a rough estimate that is based on qualitative information like pictures and process descriptions of actual mines.

Land use and occupation is chosen as explained in 2.1.1 Facilities, it thus comprises also the beneficiation facilities.

### **2.2.3 Non-ferrous metal mine, surface in Ecoinvent**

The infrastructure use for a mine is estimated in Tab. 2.6. It comprises land use for mining facilities, production of buildings and equipment including their disposal. No rehabilitation of the industrial site is considered. The values for the surface mine differ only in specific points from those for the underground mine. The data represent a rough estimate that is based on qualitative information like pictures and process descriptions of actual mines.

Land use and occupation is chosen as explained in 2.1.1 Facilities, it thus comprises also the beneficiation facilities.

Tab. 2.5 Infrastructure mine, underground. Assumptions made and inventory chosen.

| Mine, underground                                                                                    | Demand      | Unit           | Material / Module          | Life time | Life time demand |
|------------------------------------------------------------------------------------------------------|-------------|----------------|----------------------------|-----------|------------------|
| Yearly production capacity                                                                           | 400'000'000 | kg             | Yearly production capacity | 50 a      | 400'000'000      |
| Area needed total [m <sup>2</sup> ]                                                                  | 184'830     | m <sup>2</sup> | Transformed area           | 50 a      | 184'830          |
| Administration building (50x50x10 m <sup>3</sup> )                                                   | 25'000      | m <sup>3</sup> | Building, multi-storey     | 50 a      | 25'000           |
| conveyor to storage [m]                                                                              | 300         | m              | conveyor belt, at plant    | 25 a      | 600              |
| Storage building (hall for 100'000 t = 50'000 m <sup>3</sup> ; average height 10m) [m <sup>2</sup> ] | 5'000       | m <sup>2</sup> | Building, hall             | 50 a      | 5'000            |
| 10 Hoists [kg]                                                                                       | 100'000     | kg             | Industrial machine, heavy  | 25 a      | 200'000          |
| 3 Truck shop (50x50)                                                                                 | 7'500       | m <sup>2</sup> | Building, hall             | 50 a      | 7'500            |
| 2 Warehouses (50x50x10 m <sup>3</sup> )                                                              | 50'000      | m <sup>3</sup> | Building, multi-storey     | 50 a      | 50'000           |
| Accommodation building (100x100x10 m <sup>3</sup> )                                                  | 100'000     | m <sup>3</sup> | Building, multi-storey     | 25 a      | 200'000          |
| Fix Equipment (cranes, screens)                                                                      | 25'000      | kg             | Industrial machine, heavy, | 25 a      | 50'000           |
| Mobile Equipment (vehicles)                                                                          | 120'000     | kg             | Industrial machine, heavy  | 10 a      | 600'000          |

Tab. 2.6 Infrastructure mine, surface. Assumptions made and inventory chosen.

| Mine, surface                                                                                        | Demand      | Unit           | Material / Module          | Life time | Life time demand |
|------------------------------------------------------------------------------------------------------|-------------|----------------|----------------------------|-----------|------------------|
| Yearly production capacity                                                                           | 400'000'000 | kg             | Yearly production capacity | 50 a      | 400'000'000      |
| Area needed total [m <sup>2</sup> ]                                                                  | 122'011     | m <sup>2</sup> | Transformed area           | 50 a      | 122'011          |
| Administration building (50x50x10 m <sup>3</sup> )                                                   | 25'000      | m <sup>3</sup> | Building, multi-storey     | 50 a      | 25'000           |
| conveyor to storage [m]                                                                              | 3'000       | m              | conveyor belt, at plant    | 25 a      | 6'000            |
| Storage building (hall for 100'000 t = 50'000 m <sup>3</sup> ; average height 10m) [m <sup>2</sup> ] | 5'000       | m <sup>2</sup> | Building, hall             | 50 a      | 5'000            |
| 4 Truck shop (50x50)                                                                                 | 10'000      | m <sup>2</sup> | Building, hall             | 50 a      | 10'000           |
| 2 Warehouse (50x50x10 m <sup>3</sup> )                                                               | 50'000      | m <sup>3</sup> | Building, multi-storey     | 50 a      | 50'000           |
| Accommodation (100x100x10 m <sup>3</sup> )                                                           | 100'000     | m <sup>3</sup> | Building, multi-storey     | 25 a      | 200'000          |
| Fix Equipment 5 x 5t (cranes, screens)                                                               | 25'000      | kg             | Industrial machine, heavy  | 25 a      | 50'000           |
| Mobile Equipment 12 x 10t (vehicles)                                                                 | 120'000     | kg             | Industrial machine, heavy  | 10 a      | 600'000          |

## 2.2.4 Non-ferrous metal, smelter in Ecoinvent

The infrastructure for the metallurgical step is inventoried as one unspecific module valid for a wide range of metal industries, including hydro- and pyrometallurgical processes. Its functional unit is [kg] ore concentrate in input. In this study the infrastructure of the metallurgical step is approximated roughly with:

1. sintering / roasting furnace (rotary kiln)
2. electrolysis plant (50% of the cases) and
3. blast furnace, followed by a electric arc converter (50% of the cases)

These steps build up the system of metallurgical facilities to process a certain amount of ore concentrate. This is shown in Fig. 2.3. With an average input of 6'775'000 t ore concentrate this module is required in the inventories with a factor  $1.48 \cdot 10^{-10}$  per kg concentrate processed.

Step 1) is inventoried using the infrastructure of the aluminium oxide plant, which itself represents a roasting (oxidising) step. Therefore the module “aluminium oxide, plant” (Medium sized plant. Output: 800'000 t/a, life time: 50 years) is chosen, which has to be scaled by factor 8.5 to produce the roasted concentrate required.

Step 2) represents the hydrometallurgical process route for which a fraction of 50% of the concentrate roasted is chosen, i. e. as an approximation it is assumed that one half of the metal is produced hydrometallurgically. The required infrastructure is approximated with the electrolysis of aluminium, “aluminium electrolysis, plant” (Medium sized plant. Output: 130'000 t/a, life time: 50 years. Site is shared with casting plant. Total land use is allocated 50% to electrolysis, 50% to casting). Also this module has to be scaled with a factor of 17.

Step 3) is also weighted with 50% and includes two steps, the smelting and a subsequent converting. Here too, existing infrastructure modules from the iron industry are chosen as proxy for the smelting furnace and the converter. In this case it is the module “blast furnace” (Assumptions for medium sized plant. Output: 1'500'000 t/a, life time: 50 years. Site is shared with BOF converter.) and “blast oxygen furnace converter” (Output: 1'500'000 t/a, life time: 50 a). The yield for the blast furnace is 91% (including sintering and smelting), yielding a factor of 1.22.

In most cases – as shown in Fig. 2.4 – the off-gas is captured and the sulphur dioxide from the roasting / smelting is used in the production of sulphidic acid. It is assumed that 0.7 kg  $H_2SO_4$  is produced from the off-gas of 1 kg concentrate roasted (Ayres et al. (2002)). Since most of the ores are of the sulphidic type, and also non-sulphidic ore smelters have elaborate abatement technologies for off-gas, additional 1.90 units “chemical plant, organics” (lifetime: 50 a; production: 50'000 t / a).



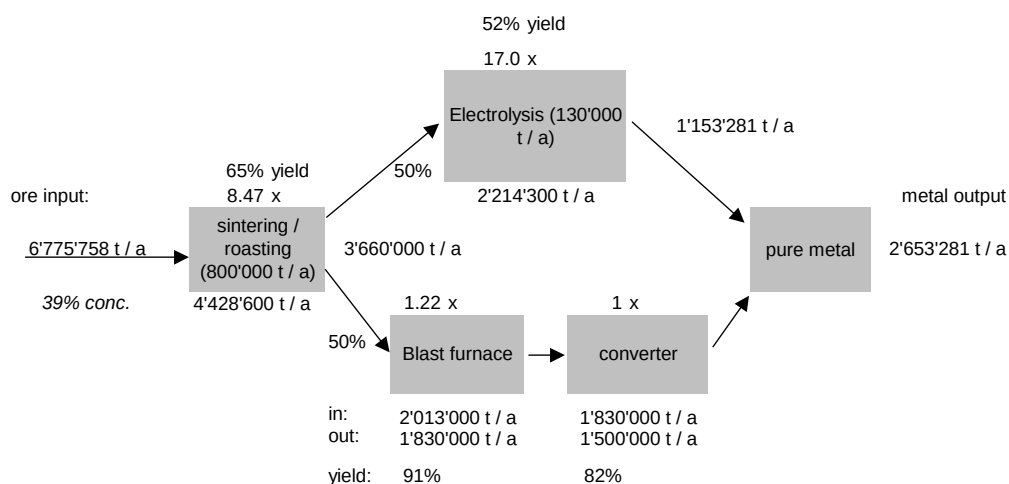


Fig. 2.3 Scheme of the infrastructure modules building up the metallurgical step. The modules are scaled to a specific throughput, which is set to be 1'500'000 t output per year at the converter. The modules for sintering/roasting, electrolysis and blast furnace are scaled to meet the requirements. The concentration in the ore is calculated to be 39%, which is true for many metals.

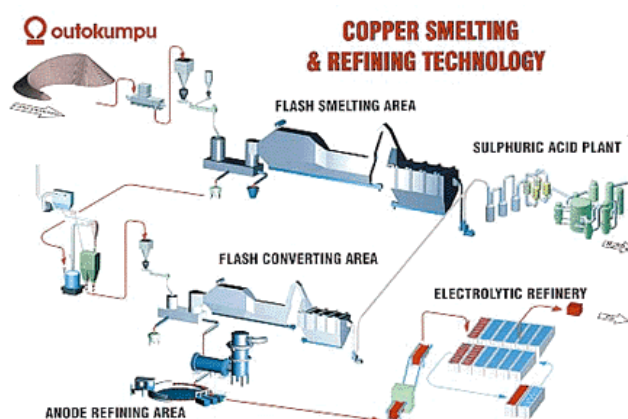


Fig. 2.4 Example of a pyrometallurgical process step in the copper industry<sup>3</sup>. Off-gas is captured in the sulphuric acid plant. The smelting furnace is followed by a flash converter, the metal leaves after a final refining step. This final refining is part of the metallurgical step and is not to be confused with the process denoted as refining in this study, which is more of a chemical plant than a smelter.

<sup>3</sup> Picture provided by Outokumpu Inc., <http://www.outokumpu.com>, accessed at 01.03.2003

Tab. 2.7 Ecoinvent meta information for non-ferrous-metal mines and smelters.

| Name                                  | non-ferrous metal mine, surface                                                                                            | non-ferrous metal mine, underground                                                                                        | non-ferrous metal smelter                                                                                                                                                                                                                                     |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location                              | GLO                                                                                                                        | GLO                                                                                                                        | GLO                                                                                                                                                                                                                                                           |
| Infrastructure Process                | 1                                                                                                                          | 1                                                                                                                          | 1                                                                                                                                                                                                                                                             |
| Unit                                  | unit                                                                                                                       | unit                                                                                                                       | unit                                                                                                                                                                                                                                                          |
| Data Set Version                      | 2.0                                                                                                                        | 2.0                                                                                                                        | 2.0                                                                                                                                                                                                                                                           |
| Included Processes                    | Includes the mine facilities. Does not include the land used for the actual mining operations                              | Includes the mine facilities. Does not include the land used for the actual mining operations                              | Includes facilities for roasting, electrolysis and for the blast furnace and converter processes.                                                                                                                                                             |
| Amount                                | 1                                                                                                                          | 1                                                                                                                          | 1                                                                                                                                                                                                                                                             |
| Local Name                            | Buntmetall Mine, Tagbau                                                                                                    | Buntmetall Mine, Untertagbau                                                                                               | Buntmetall Hüttenwerk                                                                                                                                                                                                                                         |
| Synonyms                              |                                                                                                                            |                                                                                                                            |                                                                                                                                                                                                                                                               |
| General Comment to reference function | The module is designed for unspecific use. It is based on rough estimations. Lifetime: 50a, output: 400'000 t ore per year | The module is designed for unspecific use. It is based on rough estimations. Lifetime: 50a, output: 400'000 t ore per year | The module is designed for unspecific use. It is based on inventories from aluminium and iron industry. Lifetime: 50a, input: 6'750'000 t ore concentrate (39%), output: 2'650'000 t metal per year                                                           |
| CAS Number                            |                                                                                                                            |                                                                                                                            |                                                                                                                                                                                                                                                               |
| Start Date                            | 2003                                                                                                                       | 2003                                                                                                                       | 2003                                                                                                                                                                                                                                                          |
| End Date                              | 2003                                                                                                                       | 2003                                                                                                                       | 2003                                                                                                                                                                                                                                                          |
| Data Valid For Entire Period          | 1                                                                                                                          | 1                                                                                                                          | 1                                                                                                                                                                                                                                                             |
| Other Period Text                     |                                                                                                                            |                                                                                                                            |                                                                                                                                                                                                                                                               |
| Geography text                        | No geographical context, inventory based on calculations                                                                   | No geographical context, inventory based on calculations                                                                   | No geographical context, inventory based on calculations                                                                                                                                                                                                      |
| Technology text                       | Assumed technology                                                                                                         | Assumed technology                                                                                                         | Infrastructure modules from aluminium and iron production are chosen as proxy: aluminium oxide, plant / aluminium hydroxide, plant / electrolysis, plant / blast furnace / blast furnace converter. Their outputs have been scaled to fit average operations. |
| Representativeness [%]                |                                                                                                                            |                                                                                                                            |                                                                                                                                                                                                                                                               |
| Production Volume                     | nA                                                                                                                         | nA                                                                                                                         | nA                                                                                                                                                                                                                                                            |
| Sampling Procedure                    | Literature                                                                                                                 | Literature                                                                                                                 | Literature                                                                                                                                                                                                                                                    |
| Extrapolations                        | see Geography and Technology                                                                                               | see Geography and Technology                                                                                               | see Geography and Technology                                                                                                                                                                                                                                  |
| Uncertainty Adjustments               | none                                                                                                                       | none                                                                                                                       | none                                                                                                                                                                                                                                                          |



## 2. System Characterisation

| General Flow Information                  |                                 |        | Representation in ecoinvent                   |                        |              |                 |          |                                                  |            | Uncertainty Informations |                       |      |                        |                                 |
|-------------------------------------------|---------------------------------|--------|-----------------------------------------------|------------------------|--------------|-----------------|----------|--------------------------------------------------|------------|--------------------------|-----------------------|------|------------------------|---------------------------------|
| Input                                     | Process Name                    | Output | Remarks                                       | Category               | Sub-category | Infra-structure | Location | Modul name in ecoinvent                          | Mean value | Unit                     | Source for mean value | Type | Standard Deviation 95% | General Comment                 |
| Land transformation, initial state        | non-ferrous metal mine, surface |        |                                               | resource               | land         |                 |          | Transformation, from unknown                     | 1.22E+05   | m2                       | rough estimate        | 1    | 5                      | (nA,nA,nA,nA,nA,nA,8); estimate |
| Land transformation, to mining facilities |                                 |        |                                               | resource               | land         |                 |          | Transformation, to industrial area               | 1.22E+05   | m2                       | rough estimate        | 1    | 2                      | (nA,nA,nA,nA,nA,nA,8)           |
| Land Occupation                           |                                 |        |                                               | resource               | land         |                 |          | Occupation, industrial area                      | 6.10E+06   | m2a                      | rough estimate        | 1    | 2                      | (nA,nA,nA,nA,nA,nA,8)           |
| Buildings                                 |                                 |        |                                               | construction processes | buildings    |                 |          | building, multi-storey                           | 2.75E+05   | m3                       | rough estimate        | 1    | 3                      | (nA,nA,nA,nA,nA,nA,9)           |
| Halls                                     |                                 |        |                                               | construction processes | buildings    |                 |          | building, hall                                   | 1.50E+04   | m2                       | rough estimate        | 1    | 3                      | (nA,nA,nA,nA,nA,nA,9)           |
| Machines                                  |                                 |        |                                               | construction processes | machinery    |                 |          | industrial machine, heavy, unspecified, at plant | 6.50E+05   | kg                       | rough estimate        | 1    | 3                      | (nA,nA,nA,nA,nA,nA,9)           |
| conveyor belts                            |                                 |        |                                               | construction processes | machinery    |                 |          | conveyor belt, at plant                          | 6.00E+03   | m                        | rough estimate        | 1    | 3                      | (nA,nA,nA,nA,nA,nA,9)           |
|                                           | open-pit mining facilities      |        | Lifetime: 50a, output: 400'000 t ore per year | metals                 | extraction   |                 |          | non-ferrous metal mine, surface                  | 1.00E+00   | unit                     |                       |      |                        |                                 |

Fig. 2.6 Flows for “non-ferrous metal mine, surface” and its representation in the ecoinvent database.

| General Flow Information   |                           |                       | Representation in ecoinvent                                                                 |           |              |                 |          |                                |            | Uncertainty Informations |                       |      |                        |                 |
|----------------------------|---------------------------|-----------------------|---------------------------------------------------------------------------------------------|-----------|--------------|-----------------|----------|--------------------------------|------------|--------------------------|-----------------------|------|------------------------|-----------------|
| Input                      | Process Name              | Output                | Remarks                                                                                     | Category  | Sub-category | Infra-structure | Location | Modul name in ecoinvent        | Mean value | Unit                     | Source for mean value | Type | Standard Deviation 95% | General Comment |
| Roasting / Sintering       | non-ferrous metal smelter | smelteries facilities |                                                                                             | metals    | extraction   |                 |          | aluminium oxide, plant         | 8.47E+00   | unit                     | rough estimate        | 1    | 5                      | estimate        |
| Electrolytical plant (50%) |                           |                       |                                                                                             | metals    | extraction   |                 |          | aluminium electrolysis, plant  | 1.70E+01   | unit                     | rough estimate        | 1    | 5                      | estimate        |
| Smelter (50%)              |                           |                       |                                                                                             | metals    | extraction   |                 |          | blast furnace                  | 1.22E+00   | unit                     | rough estimate        | 1    | 5                      | estimate        |
| Converter (50%)            |                           |                       |                                                                                             | metals    | extraction   |                 |          | blast oxygen furnace converter | 1.00E+00   | unit                     | rough estimate        | 1    | 5                      | estimate        |
| Off-gas treatment          |                           |                       |                                                                                             | chemicals | organics     |                 |          | chemical plant, organics       | 1.90E+00   | unit                     | rough estimate        | 1    | 5                      | estimate        |
|                            |                           |                       | Lifetime: 50a, input: 6'750'000 t ore concentrate (39%), output: 2'650'000 t metal per year | metals    | extraction   |                 |          | non-ferrous metal smelter      | 1.00E+00   | unit                     |                       |      |                        |                 |

Fig. 2.7 Flows for “non-ferrous metal smelter” and its representation in the ecoinvent database.

## 2.3 Disposal of Overburden and Tailings

This chapter deals with ecological burdens due to the disposal of sulphidic overburden and tailings from mining and Beneficiation. For modelling these deposits, three separate modules are generated:

- “disposal, sulfidic tailings, off-site”
- “disposal, non-sulfidic tailings, off-site”
- “disposal, non-sulfidic overburden, off-site”

Major impacts are emissions to water, land transformation and land use. Because of the acidic environment in the tailings, a considerable leaching occurs, whereas the overburden remains more or less inert. Overburden, which is refilled, is considered to have no ecological burden.

### 2.3.1 Process and infrastructure

To access an ore deposit, overburden has to be removed first. This material does not contain ores, therefore is not processed and is usually deposited as crushed rock or finely ground meal but chemically unchanged, either in the mining pit (backfill) or on piles. Overburden often is used as building material for the lining of the tailings ponds.

The mined ore then is separated from the gangue, i.e. from the unwanted minerals. This usually is done by grinding and flotation. The resulting slurry then is transported to so-called tailings stacks. This disposed material covers a sometimes considerable area, it often contains sulphur, ore-borne metals and different reagents from the beneficiation step. The sulphide minerals in the tailings are oxidised to sulphuric acid, which leads to a low pH and high degree of leaching of metals. Tailings stacks are about 20 m in thickness. Its borders are lined by dams and the excess water is led to a cascade of ponds for further settling of solids. In the tailings ponds the slurry consequently sediments and thickens. If the dam height is reached, the dams usually get enhanced, finally up to several tens of metres high (IPPC (2002)).

After the mines operation phase, these ponds are usually rehabilitated. Often the tailings pond is covered with earth, while the cleaning ponds stay to form natural surface waters.

### 2.3.2 Emissions

#### Water

The overburden is assumed to emit no relevant pollutants. The low pH in sulphidic tailings however is a distinctly different chemical environment than the overburden's original location. Also tailings are a finely ground rock meal, mixed with water and exposed to atmospheric oxygen. This leads to additional emissions from formerly inert minerals.

The tailings are often subject to acid rock drainage (ARD), where sulphur oxidises in the presence of metal, water and oxygen promoted by bacterial activity to form sulphuric acid. The reactions are complex and persistent. Mines worked by the Romans in Spain and the Vikings in Scandinavia continue to generate ARD today. This makes it an even bigger issue than cyanide leakage, which is chemically unstable and does not persist in the environment. Once exposed to oxygen ARD reactions on mineral surfaces will continue for thousands of years.

#### Air

No considerable emissions to air are reported. In arid climates wind-blown erosion can be a relevant emission pathway for tailings.

### 2.3.3 Disposal of Sulphidic Tailings in ecoinvent

Tailings are a slurry waste produced during metal ore refining (beneficiation). Tailings are usually poured into tailings impoundments, where water is removed and the slurry solids are stacked. Tailings can contain sulfides, which are biochemically oxidised to sulfuric acid. The acidic pH then promotes washout of many metals (Acid Rock Drainage, ARD). Large masses of tailings are produced per kilogram metal especially if ore grades are low. Tailings impoundments represent a considerable environmental burden in the primary metal process chain, because tailings contain large amounts of toxic metals and large masses of tailings are produced: the current annual worldwide production of all types of tailings is estimated to be several hundred million metric tonnes per year, which surpasses the mass naturally eroded from the earth's surface by rivers (Gardner & Sampat 1998). In the ecoinvent database v2.0 typically sulfidic tailings are produced in production of primary copper, nickel, zinc, lead, cobalt, molybdenum, and platinum group metals. Tailings from aluminium ore beneficiation (redmud) are heeded in the respective chapter. Tailings from uranium ore beneficiation is heeded in the nuclear process chain (ecoinvent report 6, part VII).

A dataset for sulfidic tailings was created in ecoinvent v1 in 2003. However, this dataset only included land use and land transformation aspects, but no emissions for a lack of reliable composition data and tailings deposit models. For ecoinvent v2 this data gap is filled with emissions data from a specifically designed tailings model. This complemented dataset should provide a first informed estimate of the full magnitude of sulfidic tailings burdens for non-ferrous metal products and encourage further research for more detailed models for the ecoinvent v3. In order to avoid overestimations, the modelling assumptions – within the ecoinvent framework – were chosen to be conservative. An extended report on the sulfidic tailings model is freely available online (Doka 2008).

#### Tailings landfill model

The tailings landfill model was based on the modelling framework applied in other ecoinvent landfill process models (ecoinvent report 13: Doka 2003). These models are based on a top-down approach. Literature data on measurements of deposited compositions and currently measured concentrations in landfill leachate are used to calculate the currently occurring emissions of the landfill. Heeding chief parameters like hydrology and pH development these emissions are then projected into the future. Thus these models incorporate physical and chemical conditions, while being rooted in the currently observed emission behaviour. More details on general modelling assumptions of these models can be taken from (Doka 2003). The resulting tailings model and its transfer coefficients are not applicable to other waste types.

A large database with literature data on tailings composition and measured tailings leachate concentrations mainly from copper, zinc, lead, nickel and molybdenum mining sites is assembled<sup>4</sup>. The geometric mean of established data is taken to represent a global average sulfidic tailings composition and leachate composition. The relation between leachate concentration and tailings composition quantifies how easily this particular element is transferred to the aqueous phase and thus how high its mobility is. To make comparisons with other landfill models, elemental transfer coefficients for the tailings landfill are calculated, although this is currently not necessary, as only one tailings material composition is considered here. Roughly 2600 individual data points are used to create the model of tailings emissions for 39 chemical elements.

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<sup>4</sup> Data was taken from ATSDR 1997; ATSDR 2006; Boulet & Larocque 1998; Dorronsoro et al. 2002; Downing & Gravel 2006; EPA 1994; Garcia-Meza et al. 2006; Hadjibiros et al. 2005; Hernandez et al. 2007; IAEA 2003; IPPC 2004; Koerth 2002; Laliberte & Tremblay 2002; Lazar 2002; Matheis et al. 1998; McLemore et al. 2006; MIMI 2004; Muwanga 1997; Nash 2000; OK Tedi 2002; Robinson et al. 2004; Thoms 2006; Tyler et al. 2004; Van Zyl 2002; Wels et al. 2000; Ye et al. 2002

Based on acid-base balances, it is assumed that no pH-induced changes in leachate concentrations occur. I.e. the *currently observed* emission behaviour is taken to represent the *average long-term* situation. This is a conservative approach, since the literature data includes data from impoundments that have not, or not yet developed an acidic pH. The long-term mobility of metals tends therefore to be underestimated.

Chemical weathering of tailings depends on climate conditions. In wet climates large volumes of downward leaching through the tailings body are possible. In arid climates leaching is slow or in reverse direction, because water evaporation transports soluble tailings materials to the surface. Surface erosion by wind is a relevant emission pathway in arid climates (cf. Fig. 2.8). In this study, however, only emissions via bottom leaching are considered.

While the landfill models in (Doka 2003) were based on Swiss climate, for the tailings model the climate and precipitation that is relevant to tailings sites is heeded. Based on the location of the world's largest producers of sulfidic tailings, a weighted average precipitation rate of 770 mm/a was calculated (Spiess 2002, FAO 2000). In a global average situation, tailings deposits are rarely vegetated due to the extreme type of ground. Evapotranspiration of precipitated water is therefore neglected. It is assumed that tailings deposits are ideally constructed in such a way that mechanical erosion by surface runoff is excluded. All precipitation water therefore infiltrates the landfill body. All major mining sites are outside permafrost regions, so that a normal water circulation is possible. The height of the tailings landfill is assumed to be 50 meters. With a tailings density of  $2200 \text{ kg/m}^3$  a mass of  $110'000 \text{ kg}$  tailings is deposited per square meter tailings landfill surface. Thus, an amount of 0.007 litres precipitation water per kilogram tailings infiltrates annually.



Fig. 2.8 Wind blown tailings in Clayton mine, Idaho, USA (Hammarstrom et al. 2002)



Fig. 2.9 Nickel Tailings in Sudbury Ontario, Canada (Burtynsky 1996)

A part of the infiltrated water flows preferentially through the tailings. Due to the fine grain size of tailings, the amount of preferentially flowing water is assumed to be only 5%, and the residence time of preferential water is assumed to be 18 weeks ( $=T_p$ ). The water content of stabilised tailings is assumed to be 30 w-%. It is assumed that preferentially flowing water has little contact with the tailings materials and carries no pollutants from the landfill<sup>5</sup>. Based on model formulas given in Doka

<sup>5</sup> If landfills dry out periodically, then preferential water might wash out evaporites and be significantly burdened. This effect is not heeded here.



(2003:III.30ff.) it follows that the amount of effective, i.e. polluted annual leachate volume is 0.0067 litres per kilogram tailings ( $=V_{eff}$ ).

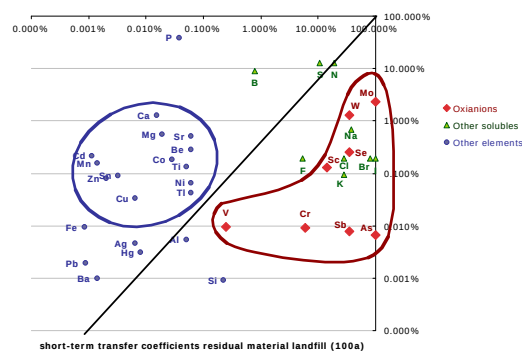
Transfer coefficients are calculated for two time horizons, 100 years (short-term) and 60'000 years (long-term). In Doka (2003) a long-term time frame of 60'000 years was chosen to represent an 'ecological planning horizon': by that time the Swiss midlands will probably be covered again with glaciers, that remodel the Swiss landscape. At that time the Swiss ecosphere will be redefined, creating new boundary conditions for environmental goals. The emissions over 60'000 years therefore represent a coarse mean expectation value of the burden inflicted on the ecosphere as we know it now and are concerned about. In other regions this ecological planning horizon might be dissimilar. Especially in tropical regions very long periods of climatic stability can be expected<sup>6</sup>. For simplicity a long-term time frame of 60'000 years for all tailings sites was chosen.

Due to lack of data, certain transfer coefficients are estimated by chemical similarity. Transfer coefficients for halogens are based on data from fluorine. Transfer coefficients for nitrogen are based on sulfur. Transfer coefficients for scandium are based on the arithmetic mean of antimony and selenium. Transfer coefficients for tungsten are based on the arithmetic mean of molybdenum and selenium. All these proxies are for completeness only and make up only very minor contributions to the LCIA results for this dataset (less than 0.1 percent).

Even after 60'000 years only a part of the deposited tailings are leached. With the modelled transfer coefficients roughly 8% of the toxic potential in tailings is emitted, and 92% are retained in the impoundment.

### Comparison with residual material landfill

The transfer coefficients of the tailings landfill can be compared with the transfer coefficients of a residual material landfill<sup>7</sup> from (Doka 2003), as shown in Fig. 2.10.



**Fig. 2.10 Comparison of short-term transfer coefficients for residual material landfill (x axis) and tailings deposit (y axis)**

<sup>6</sup> In such regions ecological planning horizons might be defined by very rare events like climate changes after meteorite impact (cf. Cretaceous-Tertiary impact winter 65 million years ago) or by very slow processes (glaciation after continental drift). For example, about 400 million years ago most regions that are tropical today were close to the south pole. However, currently tropical regions drift towards the equator and not towards the poles.

<sup>7</sup> The Swiss residual material landfill is for inert, inorganic waste materials sometimes solidified with cement.

Tab. 2.8 Average sulfidic tailings composition from literature survey

| Element | Content in tailings mg/kg (geomean) | Data points (n) | Geometric Standard Deviation |
|---------|-------------------------------------|-----------------|------------------------------|
| S       | 14600                               | 57              | 1670%                        |
| N       | 23.1                                | 3               | 11600%                       |
| P       | 493                                 | 85              | 384%                         |
| B       | 310                                 | 1               | n.a.                         |
| Cl      | 2                                   | 1               | n.a.                         |
| Br      | 0.229                               | 3               | 208%                         |
| F       | 826                                 | 30              | 376%                         |
| I       | n.a.                                | 0               | n.a.                         |
| Ag      | 14.4                                | 54              | 688%                         |
| As      | 284                                 | 87              | 919%                         |
| Ba      | 460                                 | 107             | 290%                         |
| Cd      | 7.16                                | 84              | 617%                         |
| Co      | 18.6                                | 62              | 404%                         |
| Cr      | 42.6                                | 114             | 234%                         |
| Cu      | 277                                 | 142             | 540%                         |
| Hg      | 0.882                               | 62              | 757%                         |
| Mn      | 875                                 | 109             | 426%                         |
| Mo      | 7.02                                | 61              | 371%                         |
| Ni      | 22.8                                | 96              | 487%                         |
| Pb      | 468                                 | 139             | 857%                         |
| Sb      | 79.2                                | 44              | 812%                         |
| Se      | 6.81                                | 13              | 661%                         |
| Sn      | 11.6                                | 35              | 402%                         |
| V       | 57                                  | 69              | 255%                         |
| Zn      | 791                                 | 140             | 762%                         |
| Be      | 1.45                                | 18              | 355%                         |
| Sc      | 6.13                                | 76              | 166%                         |
| Sr      | 112                                 | 92              | 302%                         |
| Ti      | 6.29                                | 52              | 11500%                       |
| Tl      | 2.6                                 | 10              | 736%                         |
| W       | 10.9                                | 19              | 414%                         |
| Si      | 154000                              | 24              | 306%                         |
| Fe      | 45400                               | 70              | 323%                         |
| Ca      | 11900                               | 63              | 423%                         |
| Al      | 27800                               | 60              | 262%                         |
| K       | 9600                                | 56              | 229%                         |
| Mg      | 7400                                | 55              | 455%                         |
| Na      | 1460                                | 42              | 507%                         |

Tab. 2.9 Modelled short- and long-term transfer coefficients for the sulfidic tailings deposit given as "kg emitted per kg deposited"

| Element    | Short-term transfer coefficients (100a) | Long-term transfer coefficients (60'000a) |
|------------|-----------------------------------------|-------------------------------------------|
| Sulfur     | 12.7%                                   | 100%                                      |
| Nitrogen   | 12.7%                                   | 100%                                      |
| Phosphor   | 39.4%                                   | 100%                                      |
| Boron      | 8.77%                                   | 100%                                      |
| Chlorine   | 0.195%                                  | 69%                                       |
| Bromium    | 0.195%                                  | 69%                                       |
| Fluorine   | 0.195%                                  | 69%                                       |
| Iodine     | 0.195%                                  | 69%                                       |
| Silver     | 0.00457%                                | 2.74%                                     |
| Arsenic    | 0.00666%                                | 3.92%                                     |
| Barium     | 0.000991%                               | 0.595%                                    |
| Cadmium    | 0.218%                                  | 100%                                      |
| Cobalt     | 0.185%                                  | 100%                                      |
| Chromium   | 0.00898%                                | 5.24%                                     |
| Copper     | 0.0335%                                 | 20.1%                                     |
| Mercury    | 0.00317%                                | 1.9%                                      |
| Manganese  | 0.159%                                  | 95.5%                                     |
| Molybdenum | 2.3%                                    | 100%                                      |
| Nickel     | 0.0653%                                 | 39.2%                                     |
| Lead       | 0.00195%                                | 1.17%                                     |
| Antimony   | 0.0077%                                 | 4.51%                                     |
| Selenium   | 0.255%                                  | 78.4%                                     |
| Tin        | 0.0923%                                 | 55.4%                                     |
| Vanadium   | 0.00937%                                | 5.47%                                     |
| Zinc       | 0.0795%                                 | 47.7%                                     |
| Beryllium  | 0.283%                                  | 100%                                      |
| Scandium   | 0.131%                                  | 41.4%                                     |
| Strontium  | 0.524%                                  | 100%                                      |
| Titanium   | 0.134%                                  | 80.3%                                     |
| Thallium   | 0.0426%                                 | 25.6%                                     |
| Tungsten   | 1.28%                                   | 89.2%                                     |
| Silicon    | 0.000913%                               | 0.548%                                    |
| Iron       | 0.00955%                                | 5.73%                                     |
| Calcium    | 1.27%                                   | 100%                                      |
| Aluminium  | 0.00546%                                | 3.28%                                     |
| Potassium  | 0.0941%                                 | 43.2%                                     |
| Magnesium  | 0.568%                                  | 100%                                      |
| Sodium     | 0.676%                                  | 98.3%                                     |

It is apparent that oxianions have reduced relative mobility in the tailings deposit (red area in Fig. 2.10). Oxianions are more mobile at high pH values. Since residual landfills have high pH values, but tailings deposit often have low pH values, the modelled behaviour is realistic. On the other hand there is a cluster of elements with higher mobility in tailings deposits (blue area in Fig. 2.10), that contains elements that are easier soluble at low pH values. While the uncertainties in these coarse models are high, it is an encouraging fact that they succeed in depicting *chemically reasonable mobility behaviour*, without any information on such mobility characteristics being entered into the models beforehand. The results of the model are reasonable and not overly dependent on the variable nature of the fundamental data. Otherwise such clustering of elemental types could not be distinctly observed. It is a sign that the sample size of the applied literature data is large enough to represent *typical* tailings.

### **Tailings impoundment land use**

Land use is based on a generic final height of a tailings impoundment of 50 meters. Thus, per kilogram tailings with a density of  $2200 \text{ kg/m}^3$  a land area of  $9.1 \cdot 10^{-6} \text{ m}^2$  is necessary. The land is assumed to be converted from unknown origin to dump site. After an operational phase of 30 years, spontaneous restoration of a consolidated state is assumed to be reached after one hundred years. The total occupation time is therefore 130 years and the land occupation per kilogram tailings  $1.18 \cdot 10^{-3} \text{ m}^2\text{a}$ .

### **Outlook**

Representation of sulfidic tailings disposal in the ecoinvent database is still very coarse: there is only one single dataset for *any* sulfidic tailings deposit. Much remains to be done, which was not possible due to time and money constraints: in the future the tailings disposal inventory should be differentiated according to the mined metal, the tailings site climate and other important conditions.

### **Data quality**

Uncertainty in this dataset is high, but not only due to a high aggregation level resulting from a worldwide average. The subject matter is inherently uncertain due to the stochastic processes in the earth's crust, leading to very diverse ore bodies, and thus to very variable data relating to tailings even for one single mining site. Hence, the level of uncertainty is high and likely to remain high, even if in the future more differentiated tailings models for single sites or regions are created. The sample sizes for almost all relevant (toxic) elements are in the range of 80 – 140 data points. Presuming the literature data is representative, these sample sizes are large enough to yield significant LCIA results for the observed data variability.

The uncertainty range for the emission data is derived according to the procedures applied in Doka (2003). In a worst case, all tailings material will be weathered or eroded within 60'000 years. This worst case is taken to represent the upper confidence value of a lognormal distribution. Together with the inventoried mean emission, the geometric standard deviation is defined.

Rehabilitation and tailings management as it is done in certain parts, can alter the ecological impacts greatly. Due to the lack of reliable data, no deliberate rehabilitation is assumed here.

Tab. 2.10 Inventoried exchanges for the disposal of sulfidic tailings.

| Name                         | Category | SubCategory        | Unit                                  | disposal, sulfidic tailings, off-site |
|------------------------------|----------|--------------------|---------------------------------------|---------------------------------------|
|                              |          |                    | Location<br>InfrastructureProces<br>s | GLO                                   |
|                              |          |                    | Unit                                  | 0                                     |
|                              |          |                    |                                       | kg                                    |
| Sulfate                      | water    | ground-            | kg                                    | 0.00559                               |
| Nitrate                      | water    | ground-            | kg                                    | 0.000013                              |
| Phosphate                    | water    | ground-            | kg                                    | 0.000595                              |
| Boron                        | water    | ground-            | kg                                    | 0.0000272                             |
| Chloride                     | water    | ground-            | kg                                    | 3.91E-09                              |
| Bromine                      | water    | ground-            | kg                                    | 4.47E-10                              |
| Fluoride                     | water    | ground-            | kg                                    | 0.00000161                            |
| Silver, ion                  | water    | ground-            | kg                                    | 6.59E-10                              |
| Arsenic, ion                 | water    | ground-            | kg                                    | 1.89E-08                              |
| Barium                       | water    | ground-            | kg                                    | 4.56E-09                              |
| Cadmium, ion                 | water    | ground-            | kg                                    | 1.56E-08                              |
| Cobalt                       | water    | ground-            | kg                                    | 3.44E-08                              |
| Chromium VI                  | water    | ground-            | kg                                    | 3.82E-09                              |
| Copper, ion                  | water    | ground-            | kg                                    | 9.29E-08                              |
| Mercury                      | water    | ground-            | kg                                    | 2.79E-11                              |
| Manganese                    | water    | ground-            | kg                                    | 0.00000139                            |
| Molybdenum                   | water    | ground-            | kg                                    | 0.000000162                           |
| Nickel, ion                  | water    | ground-            | kg                                    | 1.49E-08                              |
| Lead                         | water    | ground-            | kg                                    | 9.14E-09                              |
| Antimony                     | water    | ground-            | kg                                    | 6.1E-09                               |
| Selenium                     | water    | ground-            | kg                                    | 1.74E-08                              |
| Tin, ion                     | water    | ground-            | kg                                    | 1.07E-08                              |
| Vanadium, ion                | water    | ground-            | kg                                    | 5.35E-09                              |
| Zinc, ion                    | water    | ground-            | kg                                    | 0.000000629                           |
| Beryllium                    | water    | ground-            | kg                                    | 4.11E-09                              |
| Scandium                     | water    | ground-            | kg                                    | 8.05E-09                              |
| Strontium                    | water    | ground-            | kg                                    | 0.000000587                           |
| Titanium, ion                | water    | ground-            | kg                                    | 8.41E-09                              |
| Thallium                     | water    | ground-            | kg                                    | 1.11E-09                              |
| Tungsten                     | water    | ground-            | kg                                    | 0.00000014                            |
| Silicon                      | water    | ground-            | kg                                    | 0.00000141                            |
| Iron, ion                    | water    | ground-            | kg                                    | 0.00000434                            |
| Calcium, ion                 | water    | ground-            | kg                                    | 0.000151                              |
| Aluminum                     | water    | ground-            | kg                                    | 0.00000152                            |
| Potassium, ion               | water    | ground-            | kg                                    | 0.000000904                           |
| Magnesium                    | water    | ground-            | kg                                    | 0.000042                              |
| Sodium, ion                  | water    | ground-            | kg                                    | 0.00000985                            |
| Sulfate                      | water    | ground-, long-term | kg                                    | 0.0383                                |
| Nitrate                      | water    | ground-, long-term | kg                                    | 0.0000893                             |
| Phosphate                    | water    | ground-, long-term | kg                                    | 0.000916                              |
| Boron                        | water    | ground-, long-term | kg                                    | 0.000283                              |
| Chloride                     | water    | ground-, long-term | kg                                    | 0.00000138                            |
| Bromine                      | water    | ground-, long-term | kg                                    | 0.000000158                           |
| Fluoride                     | water    | ground-, long-term | kg                                    | 0.000569                              |
| Silver, ion                  | water    | ground-, long-term | kg                                    | 0.000000395                           |
| Arsenic, ion                 | water    | ground-, long-term | kg                                    | 0.0000111                             |
| Barium                       | water    | ground-, long-term | kg                                    | 0.00000273                            |
| Cadmium, ion                 | water    | ground-, long-term | kg                                    | 0.00000714                            |
| Cobalt                       | water    | ground-, long-term | kg                                    | 0.0000186                             |
| Chromium VI                  | water    | ground-, long-term | kg                                    | 0.00000223                            |
| Copper, ion                  | water    | ground-, long-term | kg                                    | 0.0000556                             |
| Mercury                      | water    | ground-, long-term | kg                                    | 1.67E-08                              |
| Manganese                    | water    | ground-, long-term | kg                                    | 0.000835                              |
| Molybdenum                   | water    | ground-, long-term | kg                                    | 0.00000686                            |
| Nickel, ion                  | water    | ground-, long-term | kg                                    | 0.00000893                            |
| Lead                         | water    | ground-, long-term | kg                                    | 0.00000547                            |
| Antimony                     | water    | ground-, long-term | kg                                    | 0.00000357                            |
| Selenium                     | water    | ground-, long-term | kg                                    | 0.00000532                            |
| Tin, ion                     | water    | ground-, long-term | kg                                    | 0.00000643                            |
| Vanadium, ion                | water    | ground-, long-term | kg                                    | 0.00000311                            |
| Zinc, ion                    | water    | ground-, long-term | kg                                    | 0.000377                              |
| Beryllium                    | water    | ground-, long-term | kg                                    | 0.00000145                            |
| Scandium                     | water    | ground-, long-term | kg                                    | 0.00000253                            |
| Strontium                    | water    | ground-, long-term | kg                                    | 0.000111                              |
| Titanium, ion                | water    | ground-, long-term | kg                                    | 0.00000504                            |
| Thallium                     | water    | ground-, long-term | kg                                    | 0.000000664                           |
| Tungsten                     | water    | ground-, long-term | kg                                    | 0.00000961                            |
| Silicon                      | water    | ground-, long-term | kg                                    | 0.000843                              |
| Iron, ion                    | water    | ground-, long-term | kg                                    | 0.0026                                |
| Calcium, ion                 | water    | ground-, long-term | kg                                    | 0.0117                                |
| Aluminum                     | water    | ground-, long-term | kg                                    | 0.000911                              |
| Potassium, ion               | water    | ground-, long-term | kg                                    | 0.00414                               |
| Magnesium                    | water    | ground-, long-term | kg                                    | 0.00735                               |
| Sodium, ion                  | water    | ground-, long-term | kg                                    | 0.00142                               |
| Transformation, from unknown | resource | land               | m2                                    | 0.00000909                            |
| Transformation, to dump site | resource | land               | m2                                    | 0.00000909                            |
| Occupation, dump site        | resource | land               | m2a                                   | 0.00118                               |

### 2.3.4 Disposal, non-sulphidic tailings, off-site in ecoinvent

The effects modelled in this module are

- land transformation
- land use

Tailings without considerable content of sulphur do not undergo ARD reactions. The pollution potential is therefore lower than in sulfidic tailings for the classic heavy metal pollutants. However also non-sulfidic tailings are mechanically compromised (ground into a dust), mixed with water and exposed to atmospheric oxygen. This can result in distinctly different behaviour compared to the original setting in the ore vein. No model for non-sulfidic tailings was created here, so no emissions to soil or water are considered, solely the land transformation and land use are modelled with similar figures as in 2.3.3.

The flux of this module and its representation in ecoinvent are presented in Fig. 2.12.

#### Data quality

Land use and occupation are calculated based on figures from only few sites. Rehabilitation and tailings management as it is done in certain parts, can alter the ecological impacts greatly. Due to the lack of reliable data, no rehabilitation is assumed here.

### 2.3.5 Disposal of overburden in ecoinvent

The effects modelled in this module are

- land transformation
- land use

Overburden is assumed to be piled on spherical heaps with a diameter of 20 m and a volume of 16'755 m<sup>3</sup>. With an in-situ density of 1.8 t / m<sup>3</sup> this is 0.0418 m<sup>2</sup> and 5.43 m<sup>2</sup>a per tonne overburden disposed. The land is assumed to be converted from unknown origin to dump site. Restoration of the natural state is assumed to be reached after one hundred years naturally, yielding an occupation of 130 years with the mine operation duration included.

No rehabilitation by human activities is assumed. Rehabilitation of coarse material often is done by re-vegetation. Per m<sup>2</sup> a mass of roughly 0.5 kg seeds would be necessary, which is 0.021 kg per tonne, transported over a distance of 100 km ( = 2.1 10<sup>-3</sup> tkm / t).

Overburden, which is refilled, is not inventoried at all.

The flux of this module and its representation in ecoinvent are presented in Fig. 2.13.

#### Data quality

Land use and occupation are calculated assuming a spherical heap. This can lead to overestimate the land occupation of bigger piles. Rehabilitation and tailings management as it is done in certain parts, can alter the ecological impacts greatly. Due to the lack of reliable data, no rehabilitation is assumed here.

Tab. 2.11 Ecoinvent meta information for disposal processes in PGM winning operations (to be completed).

| Name                                  | disposal, sulfidic tailings, off-site                                                                                                                                                                                                                                   | disposal, non-sulfidic overburden, off-site                                                                                                                              | disposal, non-sulfidic tailings, off-site   |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| Location                              | GLO                                                                                                                                                                                                                                                                     | GLO                                                                                                                                                                      | GLO                                         |
| Infrastructure Process                | 0                                                                                                                                                                                                                                                                       | 0                                                                                                                                                                        | 0                                           |
| Unit                                  | kg                                                                                                                                                                                                                                                                      | kg                                                                                                                                                                       | kg                                          |
| Data Set Version                      | 2.1                                                                                                                                                                                                                                                                     | 2.1                                                                                                                                                                      | 2.1                                         |
| Included Processes                    | Waste-specific short- and long-term emissions to ground water from tailings impoundment leachate. Land transformation and occupation.                                                                                                                                   |                                                                                                                                                                          |                                             |
| Amount                                | 1                                                                                                                                                                                                                                                                       | 1                                                                                                                                                                        | 1                                           |
| Local Name                            | Entsorgung, sulfidische Berge, neben Gelände deponiert                                                                                                                                                                                                                  | Entsorgung, nicht-sulfidischer Abraum, neben Gelände deponiert                                                                                                           |                                             |
| Synonyms                              |                                                                                                                                                                                                                                                                         |                                                                                                                                                                          |                                             |
| General Comment to reference function |                                                                                                                                                                                                                                                                         | The module is designed for unspecific use for disposal of sulphidic tailings in non-ferrous metal extraction. The module includes land use. No rehabilitation is assumed |                                             |
| CAS Number                            |                                                                                                                                                                                                                                                                         |                                                                                                                                                                          |                                             |
| Start Date                            | 2003                                                                                                                                                                                                                                                                    | 2003                                                                                                                                                                     | 2003                                        |
| End Date                              | 2003                                                                                                                                                                                                                                                                    | 2003                                                                                                                                                                     | 2003                                        |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                                                       | 1                                                                                                                                                                        | 1                                           |
| Other Period Text                     | Average waste composition from literature references. Transfer coefficients from prospective, tailings-specific model.                                                                                                                                                  |                                                                                                                                                                          |                                             |
| Geography text                        |                                                                                                                                                                                                                                                                         | No geographical context, inventory based on calculations                                                                                                                 | Land use of average global mining operation |
| Technology text                       | Mechanically stable tailings impoundment. Spontaneous renaturation after closure.                                                                                                                                                                                       | No rehabilitation is assumed                                                                                                                                             |                                             |
| Representativeness [%]                |                                                                                                                                                                                                                                                                         |                                                                                                                                                                          |                                             |
| Production Volume                     | nA                                                                                                                                                                                                                                                                      | nA                                                                                                                                                                       | nA                                          |
| Sampling Procedure                    | Tailings model based on observed leachate concentrations in literature. Extrapolated to 60'000 years heeding chemical characteristics. Initial waste composition from various literature sources.                                                                       | Literature                                                                                                                                                               | Literature                                  |
| Extrapolations                        | see Geography and Technology                                                                                                                                                                                                                                            | see Geography and Technology                                                                                                                                             |                                             |
| Uncertainty Adjustments               | Uncertainty of waste input composition data derived from large literature survey. Mean long-term emissions are the emissions during a generic ecological planning horizon (60'000a). Maximal long-term emissions are the complete emissions of all landfilled material. | none                                                                                                                                                                     | none                                        |

## 2. System Characterisation

| Expt<br>anal<br>on | InputGroup | OutputGroup | Name                                      | Locati<br>on | Category           | SubCat<br>egory | Infras<br>truct<br>urePr<br>ocess | Unit | disposal,<br>sulfidic tailings,<br>off-site | Uncertainty<br>Type                                                                         | Standard<br>Deviation<br>95% | GeneralComment |
|--------------------|------------|-------------|-------------------------------------------|--------------|--------------------|-----------------|-----------------------------------|------|---------------------------------------------|---------------------------------------------------------------------------------------------|------------------------------|----------------|
| 662<br>493<br>403  |            |             | Location<br>InfrastructureProcess<br>Unit |              |                    |                 |                                   |      | GLO<br>0<br>kg                              |                                                                                             |                              |                |
| 4                  |            | 4           | Sulfate                                   | water        | ground-            | kg              | 0.005589578                       | 1    | 3259%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Nitrate                                   | water        | ground-            | kg              | 1.30341E-05                       | 1    | 3436%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Phosphate                                 | water        | ground-            | kg              | 0.000594832                       | 1    | 1503%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Boron                                     | water        | ground-            | kg              | 2.72284E-05                       | 1    | 698%                                        | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Chloride                                  | water        | ground-            | kg              | 3.90518E-09                       | 1    | 1749%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Bromine                                   | water        | ground-            | kg              | 4.47031E-10                       | 1    | 816%                                        | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Fluoride                                  | water        | ground-            | kg              | 1.61371E-06                       | 1    | 2107%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Iodide                                    | water        | ground-            | kg              | 0                                 | 1    | 2345%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Silver, ion                               | water        | ground-            | kg              | 6.58755E-10                       | 1    | 7929%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Arsenic, ion                              | water        | ground-            | kg              | 1.89113E-08                       | 1    | 13021%                                      | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Barium                                    | water        | ground-            | kg              | 4.55655E-09                       | 1    | 2213%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Cadmium, ion                              | water        | ground-            | kg              | 1.56101E-08                       | 1    | 5100%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Cobalt                                    | water        | ground-            | kg              | 3.43619E-09                       | 1    | 2392%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Chromium VI                               | water        | ground-            | kg              | 3.82071E-09                       | 1    | 1353%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Copper, ion                               | water        | ground-            | kg              | 9.29308E-08                       | 1    | 4537%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Mercury                                   | water        | ground-            | kg              | 2.79266E-11                       | 1    | 9593%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Manganese                                 | water        | ground-            | kg              | 1.39292E-06                       | 1    | 2662%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Molybdenum                                | water        | ground-            | kg              | 1.61573E-07                       | 1    | 1677%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Nickel, ion                               | water        | ground-            | kg              | 1.49016E-08                       | 1    | 3605%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Lead                                      | water        | ground-            | kg              | 9.14304E-09                       | 1    | 1229%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Antimony                                  | water        | ground-            | kg              | 6.09724E-09                       | 1    | 10321%                                      | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Selenium                                  | water        | ground-            | kg              | 1.73658E-08                       | 1    | 5736%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Tin, ion                                  | water        | ground-            | kg              | 1.07277E-08                       | 1    | 2509%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Vanadium, ion                             | water        | ground-            | kg              | 5.34745E-09                       | 1    | 1509%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Zinc, ion                                 | water        | ground-            | kg              | 6.28537E-07                       | 1    | 8028%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Beryllium                                 | water        | ground-            | kg              | 4.10953E-09                       | 1    | 1842%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Scandium                                  | water        | ground-            | kg              | 8.05141E-09                       | 1    | 649%                                        | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Bromine                                   | water        | ground-            | kg              | 5.87202E-07                       | 1    | 1317%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Titanium, ion                             | water        | ground-            | kg              | 8.41026E-09                       | 1    | 10096%                                      | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Thallium                                  | water        | ground-            | kg              | 1.10784E-09                       | 1    | 7804%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Tungsten                                  | water        | ground-            | kg              | 1.39689E-07                       | 1    | 2153%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Silicon                                   | water        | ground-            | kg              | 1.4071E-06                        | 1    | 2391%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Iron, ion                                 | water        | ground-            | kg              | 4.33674E-06                       | 1    | 2126%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Calcium, ion                              | water        | ground-            | kg              | 0.000150632                       | 1    | 2238%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Aluminum                                  | water        | ground-            | kg              | 1.51915E-06                       | 1    | 1648%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Potassium, ion                            | water        | ground-            | kg              | 9.0406E-06                        | 1    | 1018%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Magnesium                                 | water        | ground-            | kg              | 4.20117E-05                       | 1    | 2716%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Sodium, ion                               | water        | ground-            | kg              | 9.84907E-06                       | 1    | 3298%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Sulfate                                   | water        | ground-, long-term | kg              | 0.038302333                       | 1    | 188%                                        | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Nitrate                                   | water        | ground-, long-term | kg              | 8.93154E-05                       | 1    | 189%                                        | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Phosphate                                 | water        | ground-, long-term | kg              | 0.000915639                       | 1    | 136%                                        | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Boron                                     | water        | ground-, long-term | kg              | 0.000283128                       | 1    | 207%                                        | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Chloride                                  | water        | ground-, long-term | kg              | 1.3761E-06                        | 1    | 472%                                        | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Bromine                                   | water        | ground-, long-term | kg              | 1.57524E-07                       | 1    | 472%                                        | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Fluoride                                  | water        | ground-, long-term | kg              | 0.000568637                       | 1    | 472%                                        | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Iodide                                    | water        | ground-, long-term | kg              | 0                                 | 1    | 472%                                        | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Silver, ion                               | water        | ground-, long-term | kg              | 3.94534E-07                       | 1    | 6318%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Arsenic, ion                              | water        | ground-, long-term | kg              | 1.11139E-05                       | 1    | 4529%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Barium                                    | water        | ground-, long-term | kg              | 2.73022E-06                       | 1    | 26947%                                      | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Cadmium, ion                              | water        | ground-, long-term | kg              | 7.14215E-06                       | 1    | 442%                                        | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Cobalt                                    | water        | ground-, long-term | kg              | 1.85558E-05                       | 1    | 455%                                        | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Chromium VI                               | water        | ground-, long-term | kg              | 2.22598E-06                       | 1    | 3473%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Copper, ion                               | water        | ground-, long-term | kg              | 5.55941E-05                       | 1    | 1103%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Mercury                                   | water        | ground-, long-term | kg              | 1.6729E-06                        | 1    | 8904%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Manganese                                 | water        | ground-, long-term | kg              | 0.000834545                       | 1    | 467%                                        | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Molybdenum                                | water        | ground-, long-term | kg              | 6.86218E-06                       | 1    | 282%                                        | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Nickel, ion                               | water        | ground-, long-term | kg              | 8.925E-06                         | 1    | 687%                                        | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Lead                                      | water        | ground-, long-term | kg              | 5.47094E-06                       | 1    | 14097%                                      | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Antimony                                  | water        | ground-, long-term | kg              | 3.56598E-06                       | 1    | 3982%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Selenium                                  | water        | ground-, long-term | kg              | 5.32268E-06                       | 1    | 439%                                        | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Tin, ion                                  | water        | ground-, long-term | kg              | 6.43109E-06                       | 1    | 566%                                        | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Vanadium, ion                             | water        | ground-, long-term | kg              | 3.11472E-06                       | 1    | 3340%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Zinc, ion                                 | water        | ground-, long-term | kg              | 0.000376615                       | 1    | 612%                                        | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Beryllium                                 | water        | ground-, long-term | kg              | 1.44861E-06                       | 1    | 423%                                        | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Scandium                                  | water        | ground-, long-term | kg              | 2.53012E-06                       | 1    | 607%                                        | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Bromine                                   | water        | ground-, long-term | kg              | 0.000111379                       | 1    | 378%                                        | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Titanium, ion                             | water        | ground-, long-term | kg              | 5.03971E-06                       | 1    | 488%                                        | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Thallium                                  | water        | ground-, long-term | kg              | 6.64224E-07                       | 1    | 919%                                        | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Tungsten                                  | water        | ground-, long-term | kg              | 9.61238E-06                       | 1    | 321%                                        | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Silicon                                   | water        | ground-, long-term | kg              | 0.000842706                       | 1    | 29169%                                      | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Iron, ion                                 | water        | ground-, long-term | kg              | 0.002598786                       | 1    | 3210%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Calcium, ion                              | water        | ground-, long-term | kg              | 0.01174368                        | 1    | 319%                                        | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Aluminum                                  | water        | ground-, long-term | kg              | 0.000910663                       | 1    | 5346%                                       | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Potassium, ion                            | water        | ground-, long-term | kg              | 0.004139383                       | 1    | 624%                                        | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Magnesium                                 | water        | ground-, long-term | kg              | 0.007353726                       | 1    | 373%                                        | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Sodium, ion                               | water        | ground-, long-term | kg              | 0.001425113                       | 1    | 361%                                        | uncertainty is calculated from uncertainties in waste composition and transfer coefficients |                              |                |
|                    |            | 4           | Transformation, from unknown              | resource     | land               | m2              | 9.09091E-06                       | 1    | 200%                                        | (3.3.1.1.1.5) & basic uncertainty of 2;                                                     |                              |                |
|                    |            | 4           | Transformation, to dump site              | resource     | land               | m2              | 9.09091E-06                       | 1    | 200%                                        | (3.3.1.1.1.5) & basic uncertainty of 2;                                                     |                              |                |
|                    |            | 4           | Occupation, dump site                     | resource     | land               | m2a             | 0.001181818                       | 1    | 158%                                        | (3.3.1.1.1.5) & basic uncertainty of 1.5;                                                   |                              |                |
| 0                  |            | 0           | Disposal, sulfidic tailings, off-site     | GLO          | waste              | manresidual m   | 0                                 | kg   | 1                                           |                                                                                             |                              |                |

Fig. 2.11 Flows for “disposal, sulphidic tailings, offsite” and its representation in the ecoinvent database.

## 2. System Characterisation

| General Flow Information     |                                             |                                             | Representation in ecoinvent       |                  |                            |                 |          |                                             |            | Uncertainty Informations |                       |      |                        |                 |
|------------------------------|---------------------------------------------|---------------------------------------------|-----------------------------------|------------------|----------------------------|-----------------|----------|---------------------------------------------|------------|--------------------------|-----------------------|------|------------------------|-----------------|
| Input                        | Process Name                                | Output                                      | Remarks                           | Category         | Sub-category               | Infra-structure | Location | Modul name in ecoinvent                     | Mean value | Unit                     | Source for mean value | Type | Standard Deviation 95% | General Comment |
| Land transformation, ground  | disposal, non-sulfidic overburden, off-site | disposal, non-sulfidic overburden, off-site |                                   | resource         | land                       |                 |          | Transformation, from unknown                | 5.10E-05   | m2                       | (IPPC, 2002)          | 1    | 2.06463                | (1,3,1,3,3,4,8) |
| Land transformation, tailing |                                             |                                             |                                   | resource         | land                       |                 |          | Transformation, to dump site                | 5.10E-05   | m2                       | (IPPC, 2002)          | 1    | 2.06463                | (1,3,1,3,3,4,8) |
| Land use for tailings        |                                             |                                             | average lifetime of 130 a assumed | resource         | land                       |                 |          | Occupation, dump site                       | 6.63E-03   | m2a                      | (IPPC, 2002)          | 1    | 1.58045                | (1,3,1,3,3,4,7) |
|                              |                                             |                                             |                                   | waste management | residual material landfill | No              | GLO      | disposal, non-sulfidic overburden, off-site | 1.00E+00   | kg                       |                       |      |                        |                 |

Fig. 2.12 Flows for “disposal, non-sulphidic tailings, offsite” and its representation in the ecoinvent database. The two tailings disposal modules are identical, except that from non sulphuric disposal no emissions are inventoried.

| General Flow Information        |                                           |                                           | Representation in ecoinvent       |                  |                            |                 |          |                                           |            | Uncertainty Informations |                       |      |                        |                 |
|---------------------------------|-------------------------------------------|-------------------------------------------|-----------------------------------|------------------|----------------------------|-----------------|----------|-------------------------------------------|------------|--------------------------|-----------------------|------|------------------------|-----------------|
| Input                           | Process Name                              | Output                                    | Remarks                           | Category         | Sub-category               | Infra-structure | Location | Modul name in ecoinvent                   | Mean value | Unit                     | Source for mean value | Type | Standard Deviation 95% | General Comment |
| Land transformation, ground     | disposal, non-sulfidic tailings, off-site | disposal, non-sulfidic tailings, off-site |                                   | resource         | land                       |                 |          | Transformation, from unknown              | 5.10E-05   | m2                       | (IPPC, 2002)          | 1    | 2.15521                | (4,4,1,3,3,5,8) |
| Land transformation, overburden |                                           |                                           |                                   | resource         | land                       |                 |          | Transformation, to dump site              | 5.10E-05   | m2                       | (IPPC, 2002)          | 1    | 2.15521                | (4,4,1,3,3,5,8) |
| Land use for overburden         |                                           |                                           | average lifetime of 130 a assumed | resource         | land                       |                 |          | Occupation, dump site                     | 6.63E-03   | m2a                      | (IPPC, 2002)          | 1    | 1.6872                 | (4,4,1,3,3,5,7) |
|                                 |                                           |                                           |                                   | waste management | residual material landfill | No              | GLO      | disposal, non-sulfidic tailings, off-site | 1.00E+00   | kg                       |                       |      |                        |                 |

Fig. 2.13 Flows for “disposal, non-sulphidic overburden, offsite” and its representation in the ecoinvent database.



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## Part XV

# Metal processing

Data v2.1 (2009)

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## **Summary and Comments on version 2.1**

This part contains information and LCI of different metal forming and coating processes and with welding. The modules include only the processes themselves and not the metals being formed, coated or welded.

The part corresponds to the original report in Althaus et al. (2004). The chapters “Cumulative Results and Interpretation” of the original report have been deleted for the current update.

Note that the flow names in the tables displaying unit processes have not been updated to reflect the name changes with version 2.0. This namely affects the designation of transportation service flows.

Version 2.1 remains unchanged compared to version 2.0 except for the meta information to the process “zinc coating, pieces, adjustment per um”.





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# 1 Introduction

This part deals with the forming and coating processes and with welding of different metals.

Metals, can be subjected to plastic deformation in the solid state by the application of external forces, i.e. it can be formed into new permanent shapes. Here, the term "forming" means the intentional and controlled change of the geometry of a work piece, usually from a simple geometry to a complex one with regard to the shape, size, accuracy and tolerances, appearance and properties. The mass, composition, and state of the material remain unchanged. If, however, the plastic limit is exceeded so that the change in geometry is not controlled, forming with defects occurs, and the desired product is not obtained. The phenomenon of plasticity results from the crystalline structure of metals. Their engineering and industrial use depends on forming technology (Kümmerling (1997)).

Coating of metals is done for different purposes. In the version ecoinvent data v 1.01 zinc coating and enamelling of steel and selective coating for solar collectors is addressed.

Many different welding techniques are industrially applied. Here two different techniques for welding steel and one for welding aluminium are assessed.

All the processing data have in common that they relate only to the actual processes. Thus e.g. coating of steel does not include the steel sheet to be coated. The same is true for forming and welding.

## 2 Steel forming

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Thanks goes to Dr. Christoph Zeltner, from Stahl Gerlafingen AG in Gerlafingen, Switzerland, for the information about the consumed energy of the hot and cold rolling process and the wire drawing process; for the help in understanding the forming processes and the guided tour through the steel forming plant in Gerlafingen.

### 2.1 General remarks

The output of raw steel production are semi-finished products categorised in cast ingots, slabs, billets and beam blanks (for further details on their properties see (IPPC (2001a), page 11)). At this point of the production steps, the steel is relatively weak due to large and uneven metal crystals or grains. Rolling the semi-finished products causes the coarse grain structure to recrystallise into a much finer grain structure, giving greater toughness, shock resistance and tensile strength CRC (1998). Therefore in this study, as a first processing step, the raw steel output (see Part II, chapter 4.9 of this report) is hot rolled to different flat or long products.

After this first step, further forming processes like cold rolling, wire drawing, tubes drawing or section bar rolling can be applied. All those processes are described in this chapter/section.

### 2.2 Hot rolling of steel and chromium steel

The term "hot rolling" or "hot forming" is used when heat is applied to the workpiece before forming it between electrically powered rollers. In general, the temperature for hot forming is higher than the recrystallization temperature (temperature ranging from 1050 to 1300 °C). If the temperature is too high, a coarse-grained structure can be formed, which is undesirable (Kümmerling (1997)) and (IPPC (2001a)).

In contrast to further processing steps like for example cold rolling, for the hot rolling process steel and stainless steel are treated in approximately the same way. Therefore in this study, only one module and process description for those two materials was made.

The advantages of hot forming include (Kümmerling (1997)):

- Great formability of the workpiece,
- Small forces required, as the yield stress of the workpiece is low,
- Large degree of deformation possible in one forming step, with consequent reduction in processing time,
- Beneficial effect on the structure, and hence on the properties of the workpiece,
- Little or no work hardening (if this is not desired).

The most significant disadvantages of hot forming are:

- Very high cost of heating the workpiece compared with the energy required for forming,
- Capital investment for the heating furnaces,

- Very severe tool wear due to the very hard and brittle scale that forms on the surface of the work-piece,
- Reduced working life of the tools due to the thermal load and increased wear.

Typical in-going slabs are 400 to 2500 mm wide and 40 to 500 mm thick and can be rolled to a thickness of 1.2 to 20 mm by the hot rolling process. Products obtained from hot rolling are usually classified in two basic types according to their shape: flat and long products. In (IPPC (2001a)) production volume of hot rolled products is given: "Total EU production in 1996 of hot rolled (HR) products was 127.8 million tonnes of which flats accounted for 79.2 million tonnes (ca. 62%). Germany is the largest manufacturer of the flats, with 22.6 million tonnes followed by France with 10.7 million tonnes, Belgium 9.9 million tonnes, Italy 9.7 million tonnes and UK 8.6 million tonnes. The vast majority of HR flat products is wide strip. The remaining 38 % of HR products are long products with about 48.5 million tonnes in 1996. The two major manufacturing countries are Italy with about 11.5 million tonnes and Germany with 10.3 million tonnes; followed by UK (7 million tonnes) and Spain (6.8 million tonnes). The largest part of the long products sector in tonnage terms is the production of wire rod which stands for roughly a third of the total production followed by reinforcing bars and merchant bars with an approximate share of one quarter of the production each." Fig. 2.1 shows an overview of products manufactured for various purposes and applications.

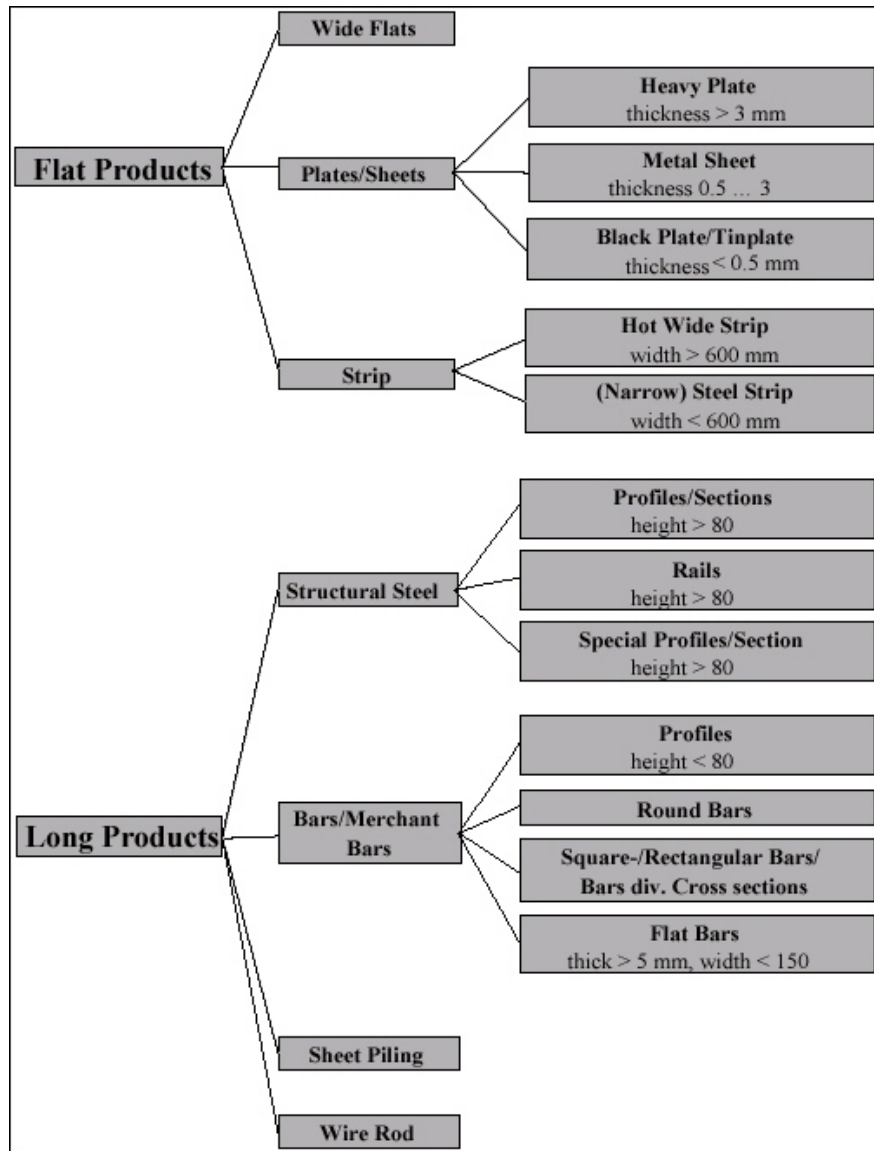


Fig. 2.1 Overview of Hot Rolled Products (IPPC (2001a)).

Hot rolling usually comprise the following process steps (IPPC (2001a)):

- Conditioning of the input (scarfing, grinding),
- Heating to rolling temperature,
- Descaling,
- Rolling (roughing including width reduction, rolling to final dimension and properties),
- Finishing (trimming, slitting, cutting).

A short description of every single process step is given below in this chapter.

The infrastructure of a hot rolling mill comprises buildings where production processes take place and for storage, surroundings and machinery for the single process steps (e.g.: furnace, rolls, cutting devices, etc.) Infrastructure is described in detail in chapter 1.1.

### 2.2.1 Conditioning of the input (scarfing, grinding)

In IPPC (2001a) the surface conditioning process steps are described as follow:

"Slabs, blooms and billets can have surface defects, like cracks, folds or seams. Surface preparation of rolling stock is necessary to ensure flawless surfaces of the rolled product. Surface defects of semifinished products, can also be removed by the following processes.

#### Scarfig

Scarfig removes surface defects of carbon steel grades by an oxy-fuel flame. The flame is used to rapidly melt and oxidise the steel surface while a separate supply of high pressure oxygen to the cutter propels the slag product from the surface. (...)

The scale from machine scarfig is washed from the steel surface by high pressure water and led to a water pit where the scale is removed by crane. The water is treated and recirculated. (...)

Dust emissions from scarfig operation are collected and, usually, abated by electrostatic precipitators (wet or dry).

The consumption of fuel (natural gas, liquefied petroleum or plant gases) and oxygen for scarfig depends on the dimensions of the rolling stock. Variations of 20 % in consumption and emissions are possible when the e.g. thickness of a slab is increased from 200 mm to 250 mm.

#### Grinding

For stainless steels and special steel grades, thermal removal of the surface defects is not possible. Therefore defects have to be eliminated by grinding. (...) Grinding machines are equipped with a roller table on which the work piece is supported and on which it can be moved. The grinding wheel is mounted on a carriage that permits it to be moved back and forth over the work piece. The unit is positioned in an enclosure, which collects the dust generated during grinding and ducts it to a bag filter before discharge to atmosphere."

### 2.2.2 Heating to rolling temperature

Reheating is done in different types of furnaces such as pit furnaces, pusher type furnaces, walking beam furnaces, rotary hearth furnaces (which are the most usual type of continuous reheating furnaces) or others. For details of these furnace types consult (IPPC (2001a)) and (Brandt & Weiler (1997)). These furnaces are fired directly, usually by oil, natural gas or plant gases such as COG<sup>1</sup> and BFG<sup>2</sup>. Depending on the heating media, different waste gas emissions will occur (mainly SO<sub>2</sub> and NO<sub>x</sub>). Integrated steel plants try to valorise as much as possible the gases arising within the works for internal use. Hence blast furnace gas, coke oven gas, BOF gas, natural gas and a lot of different combinations of these gases (so-called mixed gases) are used in continuous furnaces IPPC (2001a). No data was available about the detailed consumptions of those gases, therefore in this study the whole energy consumption for the reheating process is inventoried as natural gas. Emissions resulting from burning the natural gas in the furnace, are not inventoried separately, because they are already included in the heat module used "natural gas, burned in industrial furnace low-NO<sub>x</sub> >100kW".

### 2.2.3 Descaling

Scale, which is formed during heating, must be removed prior to rolling in order to avoid a contamination of the stock surface by scale impressed by the rolls (so-called 'rolled in matter').

---

<sup>1</sup> COG: Coke oven gas

<sup>2</sup> BFG: Blast furnace gas

A common method of descaling is breaking and spraying off the scale by means of high-pressure water. Water with pressures of 120 to 250 bar (exceptionally 600 bar) is applied via flat jet nozzles onto the material surface (IPPC (2001a)).

The main wastes of descaling are iron oxides, oil free and oily scale and waste water.

#### **2.2.4 Rolling (roughing including width reduction, rolling to final dimension and properties)**

Hot rolling mills are classified by the type of product that they produce and by their design features. For a more detailed insight in the different mill layouts and manufacturing sequences, see (IPPC (2001a)), where information for blooming/slabbing mills, hot strip mills, plate mills, bar and rod mills, structural/section mills, tube mills are reported. In this study no differentiation for the various mill types are made.

Edging is usually applied in the manufacture of strip and plate for width reduction. Roughing depicts the first bulk reduction done on the incoming hot roll stock for the production of strip, rod and sections. Generally, the roughing train consists of one or more horizontal stands for thickness reduction, including edgers (vertical stands) situated in front of the stand for width adjustment (IPPC (2001a)).

Energy is used in form of electricity and the water consumption depends on whether the water flow design is an open, a semiclosed or a closed circuit. Dust and fugitive oil are emitted to the air.

#### **2.2.5 Finishing (trimming, slitting, cutting)**

In general, the finishing mill consists of a number of rolling stands positioned after one another. The stands have staggered roll gaps so that the thickness reduction to the requested final thickness is done in one pass of the transfer bar. Crop shears arranged in front of the finishing stands serve to cut the ends of the rough strip. Hot strips with length up to 2 km and even more are wound up on so named down (sometimes up-) coilers at the end of the rolling mill by speeds up to 20 m/s and more. The finishing train of rod mills usually consists of rolling blocks with up to 10 sets of alternating horizontal and vertical roller pairs (IPPC (2001a)).

#### **2.2.6 Environmental issues of hot rolling**

The main environmental issues of hot rolling are emissions to air, especially NO<sub>x</sub> and SO<sub>x</sub>; the energy consumption of furnaces; (fugitive) dust emissions from product handling, rolling or mechanical surface treatment; oil- and solid-containing effluents and oil-containing wastes (IPPC (2001a)).

Contaminated cooling and process water arising from processes of hot and cold forming of steel is treated in internal waste water treatment plants. The reference contained no data about the composition of the untreated wastewater but reported details about the pollutants composition of the purified water, when leaving the treatment plant (Tab. 2.1).

The purifying process generates 5 kg sludge per cubic meter of treated water, which is assumed to be disposed in a residual material landfill type with cement solidification. The composition of the arising sludge is not known and is therefore calculated / estimated by G. Doka (see also Doka (2003)). As no special heavy metal precipitating agents are used, the unknown sludge composition is derived proportionally to the pollutant profile in the purified wastewater (Tab. 2.1). The water content of the sludge is assumed to be 20%.

The values for the amount and composition of the treated wastewater are taken as proxy for the other steel forming processes.



**Tab. 2.1 Composition of purified waste water and of the arising sludge**

|                          | Amount in purified waste water | Amount in waste water treatment sludge |
|--------------------------|--------------------------------|----------------------------------------|
|                          | kg/m <sup>3</sup>              | kg/kg                                  |
| TSS                      | 2.65E-02                       | -                                      |
| COD                      | 6.88E-03                       | -                                      |
| unspecified hydrocarbons | 3.15E-04                       | splitted in H and C                    |
| Hydrogen                 | -                              | 1.10E-03                               |
| Carbon                   | -                              | 9.92E-03                               |
| Copper                   | 4.84E-05                       | 1.69E-03                               |
| Zinc                     | 3.74E-05                       | 1.31E-03                               |
| Cadmium                  | 2.50E-05                       | 8.75E-04                               |
| Aluminium                | 2.37E-04                       | 8.28E-03                               |
| Lead                     | 5.00E-05                       | 1.75E-03                               |
| Chromium                 | 9.00E-05                       | 3.15E-03                               |
| Chromium 6+              | 5.00E-06                       | -                                      |
| Manganese                | 1.02E-04                       | 3.57E-03                               |
| Iron                     | 7.75E-04                       | 2.71E-02                               |
| Nickel                   | 1.41E-04                       | 4.95E-03                               |
| Mercury                  | 5.00E-06                       | 1.75E-04                               |
| Chlorine                 | 2.24E-04                       | 7.82E-03                               |
| Oxygen                   | n.a.                           | 7.28E-01                               |

### 2.2.7 Hot rolling in ecoinvent

Mean values are generally calculated as the geometric mean of the range given in IPPC (2001a). Where the range is given from zero to a maximal value, and the geometric mean can't therefore be calculated, the arithmetic mean is taken. The background information to the process "hot rolling, steel" is given in Tab. 2.2, flows are shown in Fig. 2.2.

#### Data quality

Data is taken almost exclusively from IPPC (2001a), some estimations are also made. The values can vary considerably depending on the degree of deformation.

Uncertainty is calculated assuming a lognormal distribution and the high value given in IPPC (2001a) representing the 95 % geometric standard deviation. Where no range is given in IPPC (2001a), the data quality is estimated by mean of the pedigree matrix, according to (Frischknecht et al. (2003)).

Tab. 2.2 Ecoinvent meta information for "hot rolling, steel"

|                                       |                                                                                                                                                                                                                      |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name                                  | hot rolling, steel                                                                                                                                                                                                   |
| Location                              | RER                                                                                                                                                                                                                  |
| Infrastructure Process                | 0                                                                                                                                                                                                                    |
| Unit                                  | kg                                                                                                                                                                                                                   |
| Data Set Version                      | 2.0                                                                                                                                                                                                                  |
| Included Processes                    | Includes the process steps scarfing, grinding heating, de-scaling, rolling and finishing. Semi-closed water circuit with water treatment plant is also included. Does not include the material being rolled          |
| Amount                                | 1                                                                                                                                                                                                                    |
| Local Name                            | Warm walzen, Stahl                                                                                                                                                                                                   |
| Synonyms                              | Hot forming                                                                                                                                                                                                          |
| General Comment to reference function | To achieve greater toughness, shock resistance and tensile strength, the raw steel production outputs cast ingots, slabs, billets and beam blanks are hot rolled to long or flat products or semi-finished products. |
| Start Date                            | 1997                                                                                                                                                                                                                 |
| End Date                              | 2002                                                                                                                                                                                                                 |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                    |
| Other Period Text                     |                                                                                                                                                                                                                      |
| Geography text                        | Data-set is representative for European Union                                                                                                                                                                        |
| Technology text                       | Average technique for EU. Assumption: heating 100% with natural gas. Furnaces of about 10 MW are approximated by furnace "> 100 kW"                                                                                  |
| Representativeness [%]                |                                                                                                                                                                                                                      |
| Production Volume                     | 127.8 million tonnes in 1996                                                                                                                                                                                         |
| Sampling Procedure                    | Literature                                                                                                                                                                                                           |
| Extrapolations                        | none                                                                                                                                                                                                                 |
| Uncertainty Adjustments               | The fact that no data were available about the percentage of companies represented by the study cited is considered in the uncertainties.                                                                            |

## 2. Steel forming

| General Flow Information |                              |                         | Representation in ecoinvent                                                                                                                                                                                                                                                                                                                                                |                   |                            |                 |          |                                                                                  |            | Uncertainty Informations |                       |      |                    |                                                           |
|--------------------------|------------------------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------------------------|-----------------|----------|----------------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------------------------------------|
| Input                    | Process Name                 | Output                  | Remarks                                                                                                                                                                                                                                                                                                                                                                    | Category          | Sub-category               | Infra-structure | Location | Modul name in ecoinvent                                                          | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                                           |
| Propane                  | hot rolling, steel, scarfing |                         | Variations of 20% in consumption and emissions are possible when the e.g. thickness of a slab is increased from 200 to 250 mm.                                                                                                                                                                                                                                             | oil               | fuels                      | No              | RER      | propane/ butane, at refinery                                                     | 5.30E-04   | kg                       | IPCC 2001             | 1    | 1.22               | (1,5,1,1,1,3,1)                                           |
| Oxygen                   |                              |                         |                                                                                                                                                                                                                                                                                                                                                                            | chemicals         | inorganics                 | No              | RER      | oxygen, liquid, at plant                                                         | 7.15E-03   | kg                       | IPCC 2001             | 1    | 1.22               | (1,5,1,1,1,3,1)                                           |
|                          |                              | Dust <2.5               | Dust emitted in cleaned waste gas. Corresponding geometric mean value of the concentration is 0.024 g/m3. The air flow therefore is 0.373 m3/kg dust. It is assumed that particle size of the emitted dust is below 2.5 micrometers, because the source of the dust is a combustion process.                                                                               | air               | unspecified                |                 |          | Particulates, < 2.5 um                                                           | 8.94E-06   | kg                       | IPCC 2001             | 1    | 8.94               | calculated from max value assuming lognormal distribution |
|                          |                              | Chromium dust to air    | Values in source for oxides, metal content calculated                                                                                                                                                                                                                                                                                                                      | air               | unspecified                |                 |          | Chromium                                                                         | 2.00E-08   | kg                       | IPCC 2001             | 1    | 4.47               |                                                           |
|                          |                              | Nickel dust to air      | Values in source for oxides, metal content calculated                                                                                                                                                                                                                                                                                                                      | air               | unspecified                |                 |          | Nickel                                                                           | 2.00E-08   | kg                       | IPCC 2001             | 1    | 2.24               |                                                           |
|                          |                              | Manganese dust to air   | Values in source for oxides, metal content calculated                                                                                                                                                                                                                                                                                                                      | air               | unspecified                |                 |          | Manganese                                                                        | 8.94E-08   | kg                       | IPCC 2001             | 1    | 2                  |                                                           |
|                          |                              | Lead dust to air        | Values in source for oxides, metal content calculated                                                                                                                                                                                                                                                                                                                      | air               | unspecified                |                 |          | Lead                                                                             | 1.77E-08   | kg                       | IPCC 2001             | 1    | 1.16               |                                                           |
|                          |                              | Copper dust to air      | Values in source for oxides, metal content calculated                                                                                                                                                                                                                                                                                                                      | air               | unspecified                |                 |          | Copper                                                                           | 7.18E-08   | kg                       | IPCC 2001             | 1    | 1.07               |                                                           |
|                          |                              | Iron dust to air        | calculated from mass balance (Dust - Sum(Cr-,Ni-,Mn-,Pb-,Cu-Oxides))                                                                                                                                                                                                                                                                                                       | air               | unspecified                |                 |          | Iron                                                                             | 6.06E-06   | kg                       | IPCC 2001             | 1    | 8.9                |                                                           |
|                          |                              | Filter Dust to landfill | From 4.4% to 58.7 % of dust from air cleaning devices are landfilled. Given data represent few manufacturers, therefore 20 % of total dust from air cleaning device is assumed to be landfilled. The rest is assumed to be recycled and thus not inventoried. The proxy chosen to represent the dust stands for dust disposal from steelmaking, thus being quite accurate. | w aste management | residual material landfill | No              | CH       | disposal, dust, unalloyed EAF steel, 15.4% w ater, to residual material landfill | 4.42E-04   | kg                       | IPCC 2001             | 1    | 1.47               |                                                           |
|                          |                              | NOx                     | Assuming the air flow calculated by mean of the dust emissions.                                                                                                                                                                                                                                                                                                            | air               | unspecified                |                 |          | Nitrogen oxides                                                                  | 1.31E-05   | kg                       | IPCC 2001             | 1    | 2.05               | (1,5,1,1,1,3,23)                                          |
|                          |                              | CO                      | Assuming the air flow calculated by mean of the dust emissions.                                                                                                                                                                                                                                                                                                            | air               | unspecified                |                 |          | Carbon monoxide, fossil                                                          | 2.24E-05   | kg                       | IPCC 2001             | 1    | 1.22               | (1,5,1,1,1,3,24)                                          |
|                          |                              | Scale                   | >99% recycled --> not inventoried                                                                                                                                                                                                                                                                                                                                          |                   |                            |                 |          |                                                                                  | n.a.       |                          | IPCC 2001             |      |                    |                                                           |

Fig. 2.2 Flows for "hot rolling, steel" and their representation in the ecoinvent database

## 2. Steel forming

| General Flow Information |                              |                           | Representation in ecoinvent                                                                                                                                                                                        |                  |                            |                 |          |                                                                                 |            | Uncertainty Informations |                       |      |                    |                                                           |
|--------------------------|------------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------------|-----------------|----------|---------------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------------------------------------|
| Input                    | Process Name                 | Output                    | Remarks                                                                                                                                                                                                            | Category         | Sub-category               | Infra-structure | Location | Modul name in ecoinvent                                                         | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                                           |
| Natural Gas              | hot rolling, steel, grinding | m Dust 2.5-10             | It was assumed, that the same air flow as for scarfing is arising. As the source of the dust is mechanical abrasion, it was estimated, that half of the dust is between 2.5 and 10 micrometers                     | air              | unspecified                |                 |          | Particulates, > 2.5 um, and < 10um                                              | 5.11E-06   | kg                       | IPCC 2001             | 1    | 1.83               | calculated from max value assuming lognormal distribution |
|                          |                              | m Dust >10                | It was assumed, that the same air flow as for scarfing is arising. As the source of the dust is mechanical abrasion, it was estimated, that half of the dust is bigger than 10 micrometers                         | air              | unspecified                |                 |          | Particulates, > 10 um                                                           | 5.11E-06   | kg                       | IPCC 2001             | 1    | 1.83               |                                                           |
|                          |                              | m Chromium (in Dust)      |                                                                                                                                                                                                                    | air              | unspecified                |                 |          | Chromium                                                                        | 3.54E-07   | kg                       | IPCC 2001             | 1    | 5.77               |                                                           |
|                          |                              | m Nickel (in Dust)        |                                                                                                                                                                                                                    | air              | unspecified                |                 |          | Nickel                                                                          | 2.24E-07   | kg                       | IPCC 2001             | 1    | 1.83               |                                                           |
|                          |                              | m Manganese (in Dust)     |                                                                                                                                                                                                                    | air              | unspecified                |                 |          | Manganese                                                                       | 1.29E-07   | kg                       | IPCC 2001             | 1    | 1.58               |                                                           |
|                          |                              | m Swarf                   | No data available about amount of swarf output from grinding. Swarf is assumed to be completely recycled --> not inventoried                                                                                       |                  |                            |                 |          |                                                                                 | n.a.       |                          | IPCC 2001             |      |                    |                                                           |
|                          |                              | m Scale (dry material)    | >99% recycled --> not inventoried                                                                                                                                                                                  |                  |                            |                 |          |                                                                                 | 2.65E-03   | kg                       | IPCC 2001             | 1    | 13.2               | calculated from max value assuming lognormal distribution |
|                          | hot rolling, steel, furnace  | m Dust                    | In continuous furnaces blast furnace gas, coke oven gas, BOF gas, natural gas and different combinations of these gases (mixed gases) are used. In this study the whole consumption is inventoried as natural gas. | natural gas      | heating systems            | No              | RER      | natural gas, burned in industrial furnace >100kW                                | 1.56E+00   | MJ                       | IPCC 2001             | 1    | 1.41               |                                                           |
|                          |                              | m NOx from low-NOx burner | Data for the furnace is given in (IPCC 2001) but not used in this study, because the flows are already inventoried in the utilised EI-Module for natural gas in industrial furnace >100kW.                         |                  |                            |                 |          |                                                                                 | 3.16E-06   | kg                       | IPCC 2001             | 1    | 3.16               |                                                           |
|                          |                              | m SO2                     |                                                                                                                                                                                                                    |                  |                            |                 |          |                                                                                 | 2.74E-04   | kg                       | IPCC 2001             | 1    | 1.83               |                                                           |
|                          |                              | m CO                      |                                                                                                                                                                                                                    |                  |                            |                 |          |                                                                                 | 1.34E-05   | kg                       | IPCC 2001             | 1    | 44.7               |                                                           |
|                          |                              | m Hydrocarbon             |                                                                                                                                                                                                                    |                  |                            |                 |          |                                                                                 | 6.52E-05   | kg                       | IPCC 2001             | 1    | 13                 |                                                           |
|                          |                              | m Furnace Scale           | Only 10.3% of total scale is landfilled and thus inventoried. More detailed values for single types of Mills are available in (IPCC 2001, p.65). Deposition of flue gas cleaning device as proxy                   | waste management | residual material landfill | No              | CH       | disposal, average incineration residue, 0% water, to residual material landfill | 2.50E-06   | kg                       | IPCC 2001             | 1    | 2                  |                                                           |
|                          |                              |                           |                                                                                                                                                                                                                    |                  |                            |                 |          |                                                                                 | 2.23E-04   | kg                       | IPCC 2001             | 1    | 21.7               |                                                           |

Fig. 2.2 Flows for "hot rolling, steel" and their representation in the ecoinvent database (cont.)

## 2. Steel forming

| General Flow Information   |                                 |                                    | Representation in ecoinvent                                                                                                                                |                   |                               |                 |          |                                                                                  |            | Uncertainty Informations |                       |      |                    |                                                           |
|----------------------------|---------------------------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------|-----------------|----------|----------------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------------------------------------|
| Input                      | Process Name                    | Output                             | Remarks                                                                                                                                                    | Category          | Sub-category                  | Infra-structure | Location | Modul name in ecoinvent                                                          | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                                           |
| Oil, Grease, Hydraulic Oil | hot rolling, steel, descaling   | Oil Free Scale                     | Only 3.6% of total scale is landfilled and thus is inventoried. Main component of scale is assumed to be soot (elemental carbon).                          | w aste management | residual material landfill    | No              | CH       | disposal, basic oxygen furnace w astes, 0% w ater, to residual material landfill | 4.83E-04   | kg                       | IPCC 2001             | 1    | 4.47               | calculated from max value assuming lognormal distribution |
|                            |                                 | Oily Scale                         | Only 5.6% of total scale is landfilled and thus is inventoried. Oily w aste are aproximated used mineral oil                                               | w aste management | hazardous w aste incineration | No              | CH       | disposal, used mineral oil, 10% w ater, to hazardous w aste incineration         | 2.13E-04   | kg                       | IPCC 2001             | 1    | 9.49               |                                                           |
|                            | hot rolling, steel, hot rolling | Particulates (oxides, dust) <2.5   | Value calculated basing on the value given for the oil and grease emissions to air (as hydrocarbons) and the fact that they correspond to 7% of the input. | chemicals         | organics                      | No              | RER      | lubricating oil, at plant                                                        | 4.04E-03   | kg                       | IPCC 2001             | 1    | 2.83               |                                                           |
|                            |                                 | Particulates (oxides, dust) 2.5-10 | Particulates originating from mill stands; inventoried as smaller than 2.5 micrometers because of high temperatures.                                       | air               | unspecified                   |                 |          | Particulates, < 2.5 um                                                           | 8.94E-06   | kg                       | IPCC 2001             | 1    | 4.47               | (1,5,1,1,1,3,26)                                          |
|                            |                                 | Particulates (oxides, dust) >10    | Particulates originating from coil handling stands; low er temperature produce coarser particles                                                           | air               | unspecified                   |                 |          | Particulates, > 2.5 um, and < 10um                                               | 2.24E-05   | kg                       | IPCC 2001             | 1    | 2.05               |                                                           |
|                            |                                 | Filter dust                        | Particulates originating from coil handling stands; low er temperature produce coarser particles                                                           | air               | unspecified                   |                 |          | Particulates, > 10 um                                                            | 2.24E-05   | kg                       | IPCC 2001             | 1    | 1.56               |                                                           |
|                            |                                 | Fugitive Oil Emissions             | Collected from rolling stands                                                                                                                              | w aste management | residual material landfill    | No              | CH       | disposal, dust, unalloyed EAF steel, 15.4% w ater, to residual material landfill | 8.72E-04   | kg                       | IPCC 2001             | 1    | 8.72               | calculated from max value assuming lognormal distribution |
|                            |                                 | Metallic By-Products               | No values are given to make a distinction between aromatic- und aliphatic hydrocarbons. Inventoried as NMVOC.                                              | air               | unspecified                   |                 |          | NMVOC, non-methane volatile organic compounds, unspecified origin                | 2.83E-04   | kg                       | IPCC 2001             | 1    | 2.83               |                                                           |
|                            |                                 |                                    | Particulates originating from coil handling stands; low er temperature produce coarser particles                                                           |                   |                               |                 |          |                                                                                  | 1.02E-01   | kg                       | IPCC 2001             | 1    | 1.46               |                                                           |

Fig. 2.2 Flows for "hot rolling, steel" and their representation in the ecoinvent database (cont.)

## 2. Steel forming

| General Flow Information                       |                                                 |                | Representation in ecoinvent                                                                                                                                                                                                                                                                                                                   |                        |              |                 |          |                                   |            | Uncertainty Informations |                       |      |                    |                                                           |
|------------------------------------------------|-------------------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|--------------|-----------------|----------|-----------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------------------------------------|
| Input                                          | Process Name                                    | Output         | Remarks                                                                                                                                                                                                                                                                                                                                       | Category               | Sub-category | Infra-structure | Location | Modul name in ecoinvent           | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                                           |
| Waste process water                            | hot rolling, steel, Waste water treatment plant | m <sup>3</sup> | Infrastructure for waste water treatment plant is comprised in the infrastructure module for the whole rolling mill plant. Value for energy use is comprised in the broad range of uncertainty the energy data for the rolling mill have. Water is containing suspended solids, oil, etc. but no exact data about composition were available. | resource               | in water     |                 |          | Water, unspecified natural origin | 5.50E-03   | m <sup>3</sup>           | IPCC 2001             | 1    | 2                  | calculated from max value assuming lognormal distribution |
| Flocculating agents (Ca(OH) <sub>2</sub> ;95%) |                                                 | m <sup>3</sup> | Assumption: same amount per cubic meter as for waste water from cold steel rolling                                                                                                                                                                                                                                                            | construction materials | binder       | No              | CH       | lime, hydrated, loose, at plant   | 4.97E-05   | kg                       | IPCC 2001             | 1    | 1.2                |                                                           |
| Polyelectrolytes                               |                                                 | m <sup>3</sup> | Assumption: same amount per cubic meter as for waste water from cold steel rolling. Personal communication by G. Doka --> inorganic chemicals as proxy                                                                                                                                                                                        | chemicals              | inorganics   | No              | GLO      | chemicals inorganic, at plant     | 1.45E-08   | kg                       | IPCC 2001             | 1    | 2                  |                                                           |
|                                                |                                                 | m <sup>3</sup> | Treated effluent is assumed to be discharged in surface water --> not inventoried                                                                                                                                                                                                                                                             |                        |              |                 |          |                                   | 5.50E-03   | m <sup>3</sup>           | IPCC 2001             | 1    | 2                  |                                                           |
|                                                |                                                 | m <sup>3</sup> |                                                                                                                                                                                                                                                                                                                                               | water                  | unspecified  |                 |          | Suspended solids, unspecified     | 1.46E-04   | kg                       | IPCC 2001             | 1    | 7.3                |                                                           |
|                                                |                                                 | m <sup>3</sup> |                                                                                                                                                                                                                                                                                                                                               | water                  | unspecified  |                 |          | COD, Chemical Oxygen Demand       | 3.78E-05   | kg                       | IPCC 2001             | 1    | 1.72               |                                                           |
|                                                |                                                 | m <sup>3</sup> |                                                                                                                                                                                                                                                                                                                                               | water                  | unspecified  |                 |          | BOD5, Biological Oxygen Demand    | 3.80E-05   | kg                       | Calculated from COD   | 1    | 3.44               |                                                           |
|                                                |                                                 | m <sup>3</sup> |                                                                                                                                                                                                                                                                                                                                               | water                  | unspecified  |                 |          | DOC, Dissolved Organic Carbon     | 1.20E-05   | kg                       | Calculated from COD   | 1    | 3.44               |                                                           |
|                                                |                                                 | m <sup>3</sup> |                                                                                                                                                                                                                                                                                                                                               | water                  | unspecified  |                 |          | TOC, Total Organic Carbon         | 1.20E-05   | kg                       | Calculated from COD   | 1    | 3.44               |                                                           |
|                                                |                                                 | m <sup>3</sup> |                                                                                                                                                                                                                                                                                                                                               | water                  | unspecified  |                 |          | Hydrocarbons, unspecified         | 1.73E-06   | kg                       | IPCC 2001             | 1    | 1.73               |                                                           |

Fig. 2.2 Flows for "hot rolling, steel" and their representation in the ecoinvent database (cont.)

## 2. Steel forming

| General Flow Information                        |              |                        | Representation in ecoinvent                                                                                                                                           |                   |                            |                 |          |                                                                               |            | Uncertainty Informations |                       |      |                    |                                                           |
|-------------------------------------------------|--------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------------------------|-----------------|----------|-------------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------------------------------------|
| Input                                           | Process Name | Output                 | Remarks                                                                                                                                                               | Category          | Sub-category               | Infra-structure | Location | Modul name in ecoinvent                                                       | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                                           |
| hot rolling, steel, Waste water treatment plant | m            | Copper (in DW)         |                                                                                                                                                                       | w ater            | unspecified                |                 |          | Copper, ion                                                                   | 2.66E-07   | kg                       | IPCC 2001             | 1    | 5.37               | calculated from max value assuming lognormal distribution |
|                                                 | m            | Zinc (in DW)           |                                                                                                                                                                       | w ater            | unspecified                |                 |          | Zinc, ion                                                                     | 2.06E-07   | kg                       | IPCC 2001             | 1    | 9.35               |                                                           |
|                                                 | m            | Cadmium (in DW)        |                                                                                                                                                                       | w ater            | unspecified                |                 |          | Cadmium, ion                                                                  | 1.38E-07   | kg                       | IPCC 2001             | 1    | 2                  |                                                           |
|                                                 | m            | Aluminum (in DW)       |                                                                                                                                                                       | w ater            | unspecified                |                 |          | Aluminum                                                                      | 1.30E-07   | kg                       | IPCC 2001             | 1    | 5.92               |                                                           |
|                                                 | m            | Lead (in DW)           |                                                                                                                                                                       | w ater            | unspecified                |                 |          | Lead                                                                          | 2.75E-07   | kg                       | IPCC 2001             | 1    | 2                  |                                                           |
|                                                 | m            | Chromium (in DW)       |                                                                                                                                                                       | w ater            | unspecified                |                 |          | Chromium, ion                                                                 | 4.95E-07   | kg                       | IPCC 2001             | 1    | 2                  |                                                           |
|                                                 | m            | Chromium6+ (in DW)     |                                                                                                                                                                       | w ater            | unspecified                |                 |          | Chromium VI                                                                   | 2.75E-08   | kg                       | IPCC 2001             | 1    | 2                  |                                                           |
|                                                 | m            | Manganese (in DW)      |                                                                                                                                                                       | w ater            | unspecified                |                 |          | Manganese                                                                     | 5.61E-07   | kg                       | IPCC 2001             | 1    | 2.55               |                                                           |
|                                                 | m            | Iron (in DW)           |                                                                                                                                                                       | w ater            | unspecified                |                 |          | Iron, ion                                                                     | 4.26E-06   | kg                       | IPCC 2001             | 1    | 2.58               |                                                           |
|                                                 | m            | Nickel (in DW)         |                                                                                                                                                                       | w ater            | unspecified                |                 |          | Nickel, ion                                                                   | 7.78E-07   | kg                       | IPCC 2001             | 1    | 14.1               |                                                           |
|                                                 | m            | Mercury (in DW)        |                                                                                                                                                                       | w ater            | unspecified                |                 |          | Mercury                                                                       | 2.75E-08   | kg                       | IPCC 2001             | 1    | 2                  |                                                           |
|                                                 | m            | Free Chlorine (in DW)  |                                                                                                                                                                       | w ater            | unspecified                |                 |          | Chloride                                                                      | 1.23E-06   | kg                       | IPCC 2001             | 1    | 2.24               | (2,5,1,1,1,3,6)                                           |
|                                                 | m            | Water Treatment Sludge | Assumption: same amount and composition per cubic meter as for waste water from cold steel rolling. Only a part of the sludge goes to landfill -- > 59.4% inventoried | w aste management | residual material landfill | No              | CH       | disposal, sludge from steel rolling, 20% water, to residual material landfill | 1.63E-02   | kg                       | IPCC 2001             | 1    | 1.22               |                                                           |

Fig. 2.2 Flows for "hot rolling, steel" and their representation in the ecoinvent database (cont.)

## 2. Steel forming

| General Flow Information |                               |             | Representation in ecoinvent                                                                                                                                                                                                                                                                                                                                        |                   |                              |                 |          |                                                                           |            | Uncertainty Informations |                       |      |                    |                             |
|--------------------------|-------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------------------|-----------------|----------|---------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------|
| Input                    | Process Name                  | Output      | Remarks                                                                                                                                                                                                                                                                                                                                                            | Category          | Sub-category                 | Infra-structure | Location | Modul name in ecoinvent                                                   | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment             |
| Primary steel            | hot rolling, steel, Overall   | Heat w aste | Only amount that is going to leave the system as scrap to recycling. Assumption basing on the values for cold sheet rolling taking into account a better formability do to the high temperature                                                                                                                                                                    | metals            | extraction                   | No              | RER      | steel, converter, unalloyed, at plant                                     | 5.00E-02   | kg                       | IPCC 2001             | 1    | 1.31               | (4,5,1,1,1,3,3)             |
| Electricity              |                               |             | In the reference data was available only for the rolling process in the narrow sense (70-140 kWh/t product), but additional electricity (estimated 40 kWh) is used for the process steps grinding, descaling and packing.                                                                                                                                          | electricity       | production mix               | No              | UCTE     | electricity, medium voltage, production UCTE, at grid                     | 1.40E-01   | kWh                      | IPCC 2001             | 1    | 1.14               | (3,3,1,1,1,3,2)             |
| Rolling shop             |                               |             |                                                                                                                                                                                                                                                                                                                                                                    | air               | unspecified                  |                 |          | Heat, w aste                                                              | 5.04E-01   | MJ                       | IPCC 2001             | 1    | 1.09               | (2,1,1,1,1,3,1)             |
| Lorry 32t                |                               |             |                                                                                                                                                                                                                                                                                                                                                                    | metals            | processing                   | Yes             | RER      | rolling mill                                                              | 1.62E-09   | unit                     | IPCC 2001             | 1    | 3.04               | (3,4,1,1,1,4,9)             |
|                          | hot rolling, steel, Packaging | Waste       | Includes the transports of all materials not produced at the plant and the transport to disposal of the w aste. 32t-lorry was taken as a proxy for all this transports, therefore transports to the municipal incineration are overestimated by one third (not relevant for the overall process). Distance to hazardous w aste disposal is inventoried with 100 km | transport systems | road                         | No              | RER      | transport, lorry 32t                                                      | 1.71E-02   | tkm                      | Estimation            | 1    | 2.09               | (4,5,n.a.,n.a.,n.a.,n.a.,5) |
| Paper; cardboard         |                               |             |                                                                                                                                                                                                                                                                                                                                                                    | paper & cardboard | cardboard & corrugated board | No              | RER      | packaging, corrugated board, mixed fibre, single w all, at plant          | 4.72E-12   | kg                       | EAA 2000              | 1    | 1.38               | (4,5,1,1,3,3,4)             |
| Plastics                 |                               |             |                                                                                                                                                                                                                                                                                                                                                                    | plastics          | processing                   | No              | RER      | packaging film, LDPE, at plant                                            | 1.89E-11   | kg                       | EAA 2000              | 1    | 1.38               | (4,5,1,1,3,3,4)             |
| Steel                    |                               |             |                                                                                                                                                                                                                                                                                                                                                                    | metals            | extraction                   | No              | RER      | steel, converter, unalloyed, at plant                                     | 2.36E-11   | kg                       | EAA 2000              | 1    | 1.38               | (4,5,1,1,3,3,4)             |
| Steel coils              |                               |             |                                                                                                                                                                                                                                                                                                                                                                    | metals            | processing                   | No              | RER      | sheet rolling, steel                                                      | 2.36E-11   | kg                       | EAA 2000              | 1    | 1.38               | (4,5,1,1,3,3,4)             |
| Wood                     |                               |             |                                                                                                                                                                                                                                                                                                                                                                    | w ooden materials | extraction                   | No              | RER      | saw n timber, softw ood, raw , air dried, u=20%, at plant                 | 1.62E-17   | m3                       | EAA 2000              | 1    | 1.38               | (4,5,1,1,3,3,4)             |
|                          |                               |             | Packing material is disposed as mixed w aste with municipal w aste as proxy                                                                                                                                                                                                                                                                                        | w aste management | municipal incineration       | No              | CH       | disposal, municipal solid w aste, 22.9% w ater, to municipal incineration | 7.08E-11   | kg                       | Estimation            | 1    | 1.31               | (2,5,1,1,3,3,6)             |
|                          |                               |             | recycled --> not inventoried                                                                                                                                                                                                                                                                                                                                       |                   |                              |                 |          |                                                                           | 5.00E-02   | kg                       | Estimation            | 1    |                    |                             |
|                          |                               |             |                                                                                                                                                                                                                                                                                                                                                                    | metals            | processing                   | No              | RER      | hot rolling, steel                                                        | 1.00E+00   | kg                       |                       |      |                    |                             |

Fig. 2.2 Flows for "hot rolling, steel" and their representation in the ecoinvent database (cont.)



## 2.3 Sheet rolling of steel

For many applications, the products of hot rolling are unsatisfactory, e.g., with respect to cross section, surface quality, dimensional accuracy, and general finish, so that cold rolling is necessary. Cold rolled products are mainly strips and sheets with high quality surface finish and precise metallurgical properties for use in high specification products.

In cold rolling, the properties of hot rolled strip products, e.g. thickness (typical final thickness from 0.1 to 3 mm), mechanical and technological characteristics, are changed by compression between rollers without previous heating of the input. However, its temperature can increase during the forming process, as the energy expended in forming is almost completely converted into heat. Frictional energy at the contact surfaces of the workpiece, if established, is also converted to heat. In rapid adiabatic processes the temperature can increase by  $>100\text{ }^{\circ}\text{C}$  (IPPC (2001a); Wiegels (1997)).

Contrarily to hot rolling, the process route of cold rolling is different for steel (carbon-or (low-)alloy steel) and for chromium steel (high alloy- or stainless steel). Therefore in this study those processes are described in two different chapters (see chapter 1.1 for chromium steel sheet).

The advantages of cold forming include (Wiegels (1997)):

For the workpiece:

- Good dimensional accuracy and close tolerances
- Good surface properties
- Very small dimensions (sheet thickness ca.  $10\text{ }\mu\text{m}$ , wire diameter ca.  $10\text{ }\mu\text{m}$ )
- Work hardening (if desired)

For the tool:

- Reduced wear (effective lubricants can be used, the thermal load is low, and scaling does not occur)

Generally:

- No energy is required for heating the workpiece

Cold rolled sheets and plates production (wide strip) was 39.6 millions tonnes in 1996. The main producing countries were Germany with 10.6 million tonnes followed by France with 6.3 million tonnes, Italy 4.3 million tonnes, UK 4.0 million tonnes and Belgium 3.9 million tonnes. Cold rolled narrow strip, obtained from cold rolling narrow hot strip or from slitting and cold rolling hot rolled sheet, amounted to about 8.3 millions tonnes in 1994 (2.7 million tonnes of cold rolled and 5.5 million tonnes of slit strip). In 1994, Germany produced about 35 % of the slit strip, with 1.9 million tonnes, followed by Italy and France each with a production of 0.9 million tonnes. (IPPC (2001a)).

The cold rolling mills for steel usually comprise the following process steps (IPPC (2001a)):

- Continuous pickling line,
- Cold rolling,
- Annealing,
- Temper mills,
- Inspection and finishing lines,
- Packaging lines for coils or sheets,

- Roll shop.

A short description of every single process step is given below.

The infrastructure of cold rolling mill for carbon steel is described in chapter 1.1. It comprises buildings where production processes take place and for storage, surroundings and machinery for the single process steps (e.g.: furnace, rolls, cutting devices, etc.).

### **2.3.1 Pickling**

The entire surface of the hot rolled coil is covered with a thin layer of scale containing oxides, which must be removed prior to cold rolling. Normal steels are usually pickled in sulfuric acid at 95–100 °C, or hydrochloric acid at 60–70 °C in three or four fully enclosed pickling baths. Before the actual pickling process, there is usually a stretcher–bend–leveling device which also causes mechanical descaling. This preliminary descaling is very important, as there must be a certain percentage of free iron surface in the pickling bath, so that pickling can start. Once the strip or sheet is pickled, it must be thoroughly rinsed with de-mineralised (or equal quality) water and subsequently dried. Oiling is done either with rolling oil or anticorrosive oil. The used acid can be regenerated, in the case of sulfuric acid by precipitating iron sulfate as its heptahydrate in a crystallizer. Hydrochloric acid can be regenerated by decomposing the iron chloride to hydrochloric acid and iron oxide, by a spray process at 450 °C, or in a fluidized bed at 850 °C (IPPC (2001a); Wiegels (1997))

### **2.3.2 Cold rolling**

The following description is taken from (IPPC (2001a); Wiegels (1997)). After pickling, the descaled hot-rolled strip is cold rolled passing through sets of rolls on reversing mills or on continuous in-line mills. Many types of stands are used in the production of cold-rolled strip, e.g., two-high, four-high, or larger.

Low carbon steels which are less hard, are generally rolled in multi-stand tandem mills due to a higher capacity of these mills. They consist of a series of four or six-high stands. The strip enters the first stand and undergoes an initial thickness reduction, further reductions are achieved in each subsequent stand until the final gauge is attained. For low carbon strip generally an emulsion of 0.5 to 4 % (10 - 20 % for direct application) of oil in water is necessary for lubrication, cooling of the strip, work and back-up rolls, and for removal of iron-particles. In the rolling process the wide strip is mainly deformed in a longitudinal direction, with transverse deformation being practically negligible. For optimum strip surface cleanliness it is important to avoid contamination of the tandem emulsion by the hydraulic oil, by grease or by the cooling water (used to cool the emulsion).

In reversing stands, one roll pass can be immediately followed by another by reversing the direction of the rollers. Reversing stands with many rolls exert a high pressure on the strip because the diameters of the working rolls are small.

Flat cold-rolled strip achieve thickness variations of only a few thousandths of a millimetre and a high-quality. Cold rolling causes work-hardening of the steel. This is reduced by an annealing stage.

### **2.3.3 Annealing**

The following chapter is taken from (IPPC (2001a))

The basic stages of the annealing process comprise:

- Heating to annealing temperature (above 650 °C),
- Holding at annealing temperature,

- Cooling.

This annealing process may be carried out in batch furnaces or continuous furnaces. The annealing cycle has a significant influence on the mechanical properties and, consequently, on the formability of the steel strip. The main parameter controlling the annealing cycle is the temperature profile.

Continuous annealing and batch annealing are considered to be complementary processes and not fully interchangeable. Production related issues are integral to the selection of the annealing technique.

### **Batch Annealing**

Prior to the annealing process the strip may be cleaned (degreased) to produce a cleaner surface. The most frequently used cleaning agents are phosphates, alkaline silicates, caustic soda and soda ash. The steel sheet is unwound from the coil and passed through cleaning tanks, which may be stirred to improve the cleaning effect. Subsequently, the steel is rinsed with water and recoiled. The degreasing solution can be regenerated and recycled.

The cold rolled coiled strip is stacked in a hood furnace for annealing. The combustion chamber (space between heating and protecting hood) is heated by oil or gas burners. Heat passes through the protective hood into where the steel coils are stacked. A circulating fan provides as uniform temperature distribution as possible. The atmosphere in conventional plants is usually HNX gas (a nitrogen-hydrogen mixture with a hydrogen content close to the flammability limit). The atmosphere can also be 100 % H<sub>2</sub> as in parts of the working cycle of high convection furnaces.

The heat passes into the coils through their outer edges, so that these areas are always hotter than the inner windings, especially during heating up. The heat treatment causes the organic residues of the emulsion to partly burn off; partly a distillation process takes place. Products of the reactions taking place are CO/CO<sub>2</sub>, H<sub>2</sub>, FeO<sub>x</sub> and CH<sub>4</sub>. The strip is heated to re-crystallisation temperature, and annealed at about 700 °C resulting in complete re-crystallisation of the cold rolled steel. For cooling the coils, the heating hood is removed. The cooling effect can be enhanced by spraying water on the protecting hood; by covering it with a cooling bell and blowing air on it or by using a cooling by-pass system which cools the protective atmosphere under the protective hood. The time necessary for annealing depends on the annealing temperature and the weight of the charge and can take 2 to 7 days. Annealing in this type of furnace is very slow and also the cooling rate is not suitable for certain steel grades. Therefore an annealing process in a 100 % hydrogen atmosphere was developed yielding in shorter annealing times and more rapid cooling.

### **Continuous Annealing**

For continuous annealing the coils are welded together on the entry side of the plant and subjected to the following process steps:

- Alkaline/electrolytic cleaning of the strips.
- Heating and holding at the required annealing temperature.
- Cooling (slow jet cooling, high gas jet cooling (using cooling rates up to 150 °C/s, hot water quench (HOWAQ), overaging, roll cooling, final cooling, misting jet cooling).

The continuous annealing is carried out by passing the steel strip through a multi-zone heating furnace with heating chamber, annealing chamber, cooling zone, tempering zone and a second cooling zone. The steel is heated to a range between 650 °C and 830 °C and then cooled by gas jets, gas-water sprays, contact rolls or water quenching, depending on the desired metallurgical properties. These furnaces are usually fired by gas (direct or indirect) or electrically heated. The steel can be protected by an inert gas atmosphere or a reducing gas atmosphere in parts of the furnace. Continuous annealing gives a shorter processing time (about 10 minutes), less intermediate storage, uniform mechanical properties and a superior surface cleanliness. Continuous annealing is more appropriate for the production of higher strength steels. The layout of the furnace facilitates energy recovery from the exhaust gas.

### 2.3.4 Tempering

After annealing the surface finish and mechanical properties of the steel are modified in line with the customer requirements. This is done by temper rolling which consists of subjecting the strip to a light rolling pass with a thickness reduction between 0.3 and 2 %. Before tempering the strip temperature must be less than 50 °C. Temper rolling is carried out in the temper mill which comprises typically one or two four-high stands, although two-high or six-high stands are also possible (IPPC (2001a)).

### 2.3.5 Inspecting and finishing

Finishing comprises slitting into coils of different width strip and transversal shearing to obtain sheet. Additional steps such as straightening, sampling, oiling and marking are also completed.

The finishing process comprises the following operations (IPPC (2001a)):

- Dimensional control (width, thickness and length),
- Inspection for surface defects and their removal,
- Sampling to determine the mechanical and technological properties, the strip roughness structure and the content of residue of particles from roll wear and carbon on the strip surface (statistical sampling),
- Trimming the coils to exact width,
- Straightening the strips to optimum flatness,
- Oiling the strips with electrostatic oiling machines or with oiling machines equipped with sprays or rolls oiling machines (anticorrosive oils or prelubes),
- Marking the finished products with coil number, production date, etc.
- Welding of different smaller coils to bigger coils.

### 2.3.6 Packaging of coils or sheets

The material, now in its final form, is usually packed ready for despatch to the customer. The packaging is designed to ensure that no damage occurs either during storage or transport to the end user. Control measures also have to be taken to prevent damage from mechanical and climatic sources. In addition the lifting and transport equipment in the production plant should also be designed and operated to minimize damage to the steel. After finishing, packing of the individual items ordered by the customers is done. The packing materials used includes: steel bands, paper, plastic, wood, seaworthy packing, special packing (IPPC (2001a)).

### 2.3.7 Maintenance of the rolls

The activities generally consist of the dismantling of the chocks, the grinding of the rolls, the texturing of the work rolls and the reassembling of the rolls. To satisfy the required strip tolerances and surface standards the rolls must be regularly reground. During this grinding operation the rolls are usually cooled and lubricated with a grinding emulsion. The spent emulsion can be recirculated and filtered, but partial replacement is required periodically. In addition, used oil and grinding sludge are produced which require disposal (IPPC (2001a)).

### **2.3.8 Environmental issues of sheet rolling of steel**

The main environmental issues of cold rolling are: acidic wastes and waste water; degreaser fume, acidic and oil mist emissions to air; oil-containing wastes and waste water (see chapter 2.2.6, Tab. 2.1); dust, e.g. from descaling and decoiling; NO<sub>x</sub> from mixed acid pickling and combustion gases from furnace firing. As for acid emissions to air from cold rolling, these may arise from pickling and acid regeneration processes. Emissions differ, depending on the pickling process used – basically the acid used. Only little data were available for dust emissions from steel handling and descaling operations (IPPC (2001a)).

### **2.3.9 Steel sheet rolling in ecoinvent**

Mean values are generally calculated as the geometric mean of the range given in IPPC (2001a). Where the range is given from zero to a maximal value, and the geometric mean can't therefore be calculated, the arithmetic mean is taken. The background information to the process "sheet rolling, steel" is given in Tab. 2.3, flows are shown in Fig. 2.3.

As no information on the shares of the different technologies for pickling, milling and annealing is available it is assumed that the respective shares are 50% each.

Water emissions with the treated wastewater are given in chapter 2.2.6, Tab. 2.1.

#### **Data quality**

Data is taken almost exclusively taken from IPPC (2001a), some estimations are also made. The values can vary considerably depending on the degree of deformation.

Uncertainty is calculated assuming a lognormal distribution and the high value given in IPPC (2001a) representing the 95% geometric standard deviation. Where no range is given in IPPC (2001a), the data quality is estimated by mean of the pedigree matrix, according to (Frischknecht et al. (2003)).

Tab. 2.3 Ecoinvent meta information for "sheet rolling, steel"

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>                           | <b>sheet rolling, steel</b>                                                                                                                                                                                                                                                                                                                                                                                                              |
| Location                              | RER                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Infrastructure Process                | 0                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Unit                                  | kg                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Data Set Version                      | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Included Processes                    | Includes the process steps continuous pickling line, cold rolling, annealing, tempering, inspecting and finishing, packing coils or sheets, roll maintenance. Does not include the material being rolled                                                                                                                                                                                                                                 |
| Amount                                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Local Name                            | Blech walzen, Stahl                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Synonyms                              | cold rolling                                                                                                                                                                                                                                                                                                                                                                                                                             |
| General Comment to reference function | This process is to be used only for un- and low-alloyed steel. For many applications, the products of hot rolling are unsatisfactory, e.g., with respect to cross section, surface quality, dimensional accuracy, and general finish, so that cold rolling is necessary. Cold rolled products are mainly strips and sheets with high quality surface finish and precise metallurgical properties for use in high specification products. |
| Start Date                            | 1997                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| End Date                              | 2002                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Other Period Text                     |                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Geography text                        | Data-set is representative for European Union                                                                                                                                                                                                                                                                                                                                                                                            |
| Technology text                       | Average technique for EU, for un- and low-alloyed steel. A 50-50 mix is inventoried where different technologies are applied for the same process step (e.g. pickling)                                                                                                                                                                                                                                                                   |
| Representativeness [%]                |                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Production Volume                     | Little less than 50 million tonnes in 1996                                                                                                                                                                                                                                                                                                                                                                                               |
| Sampling Procedure                    | Literature                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Extrapolations                        | none                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Uncertainty Adjustments               | The fact that no data were available about the percentage of companies represented by the study cited is considered in the uncertainties.                                                                                                                                                                                                                                                                                                |

## 2. Steel forming

| General Flow Information            |                                                                      |                                            | Representation in ecoinvent                                                                                                                                                                                                                                                       |                  |                              |                 |          |                                                                        |            | Uncertainty Informations |                       |      |                    |                                                           |
|-------------------------------------|----------------------------------------------------------------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------------------------|-----------------|----------|------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------------------------------------|
| Input                               | Process Name                                                         | Output                                     | Remarks                                                                                                                                                                                                                                                                           | Category         | Sub-category                 | Infra-structure | Location | Modul name in ecoinvent                                                | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                                           |
| primary steel                       | sheet rolling, steel, HCl-pickling (Assumption 50% of sheet rolling) |                                            | Only amount that is going to leave the system as waste to recycling.                                                                                                                                                                                                              | metals           | extraction                   | No              | RER      | steel, converter, unalloyed, at plant                                  | 8.55E-02   | kg                       | IPCC 2001             | 1    | 1.31               | (4,5,1,1,1,3,3)                                           |
| HCl (33%) with regeneration         |                                                                      |                                            |                                                                                                                                                                                                                                                                                   | chemicals        | inorganics                   | No              | RER      | hydrochloric acid, 30% in H2O, at plant                                | 3.97E-04   | kg                       | IPCC 2001             | 1    | 1.13               | calculated from max value assuming lognormal distribution |
| Anticorrosive oils                  |                                                                      |                                            | Lubricating oil as proxy                                                                                                                                                                                                                                                          | chemicals        | organics                     | No              | RER      | lubricating oil, at plant                                              | 7.07E-05   | kg                       | IPCC 2001             | 1    | 1.41               |                                                           |
| Steam for bath heating              |                                                                      |                                            |                                                                                                                                                                                                                                                                                   | chemicals        | organics                     | No              | RER      | heat, unspecific, in chemical plant                                    | 2.29E-02   | MJ                       | IPCC 2001             | 1    | 1.53               |                                                           |
| Electrical energy                   |                                                                      |                                            |                                                                                                                                                                                                                                                                                   | electricity      | production mix               | No              | UCTE     | electricity, medium voltage, production UCTE, at grid                  | 4.81E-03   | kWh                      | IPCC 2001             | 1    | 2.31               |                                                           |
| Caloric energy (natural gas or LPG) |                                                                      |                                            |                                                                                                                                                                                                                                                                                   | natural gas      | heating systems              | No              | RER      | natural gas, burned in industrial furnace >100kW                       | 3.16E-02   | MJ                       | IPCC 2001             | 1    | 1.58               |                                                           |
| Industrial & demineralised water    |                                                                      |                                            |                                                                                                                                                                                                                                                                                   | water supply     | production                   | No              | CH       | water, deionised, at plant                                             | 2.54E-02   | kg                       | IPCC 2001             | 1    | 2.55               |                                                           |
|                                     |                                                                      | Dust to air                                | Concentration multiplied with the amount of exhaust gas from pickling tanks. Particle size from chemical process: assumption 2.5-10 µm.                                                                                                                                           | air              | unspecified                  |                 |          | Particulates, > 2.5 µm, and < 10µm                                     | 5.00E-07   | kg                       | IPCC 2001             | 1    | 2                  | (1,5,1,1,1,3,23)                                          |
|                                     |                                                                      | HCl to air                                 |                                                                                                                                                                                                                                                                                   | air              | unspecified                  |                 |          | Hydrogen chloride                                                      | 1.29E-07   | kg                       | IPCC 2001             | 1    | 2.05               |                                                           |
|                                     |                                                                      | Sludge cake                                | The sludge can be approximated by Fe(OH)3. The amount of Fe in Fe(OH)3 (=76.66%) is inventoried. About two thirds are recycled --> 35% landfilled is inventoried                                                                                                                  | soil             | unspecified                  |                 |          | Iron                                                                   | 3.05E-05   | kg                       | IPCC 2001             | 1    | 5.28               | calculated from max value assuming lognormal distribution |
|                                     |                                                                      | Oil contained waste                        | No value is given in reference. As the oil input is about the same for HCl and H2SO4, it is assumed, that also the output is about the same. Only 22.8% which are landfilled are inventoried, the rest is recycled. Oil containing wastes are approximated with used mineral oil. | waste management | hazardous waste incineration | No              | CH       | disposal, used mineral oil, 10% waste, to hazardous waste incineration | 5.55E-06   | kg                       | IPCC 2001             | 1    | 1.22               | (1,5,1,1,1,3,6)                                           |
|                                     |                                                                      | Scale dust (collected by dedusting system) | Reference contains neither composition nor amount of collected dust. No information is available about the share that can be recycled.                                                                                                                                            |                  |                              |                 |          |                                                                        | n.a.       |                          | IPCC 2001             |      |                    |                                                           |
|                                     |                                                                      | Strip head / tail scrap                    | recycled --> not inventoried                                                                                                                                                                                                                                                      |                  |                              |                 |          |                                                                        | 1.50E-02   | kg                       | IPCC 2001             | 1    |                    |                                                           |

Fig. 2.3 Flows for "sheet rolling, steel" and their representation in the ecoinvent database

## 2. Steel forming

| General Flow Information |                                                                           |                                                                                                                                 | Representation in ecoinvent                                                                                                                                      |                  |                              |                 |          |                                                                        |            | Uncertainty Informations |                       |      |                    |                                                           |
|--------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------------------------|-----------------|----------|------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------------------------------------|
| Input                    | Process Name                                                              | Output                                                                                                                          | Remarks                                                                                                                                                          | Category         | Sub-category                 | Infra-structure | Location | Modul name in ecoinvent                                                | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                                           |
| H2SO4 (96%)              | sheet rolling, steel, H2SO4-pickling<br>(Assumption 50% of sheet rolling) | H2<br>SO4<br>SO2<br>Oil contained waste<br>Sludge cake<br>Scale dust (collected by dedusting system)<br>Strip head / tail scrap |                                                                                                                                                                  | chemicals        | inorganics                   | No              | RER      | sulphuric acid, liquid, at plant                                       | 4.18E-03   | kg                       | IPCC 2001             | 1    | 1.2                | calculated from max value assuming lognormal distribution |
| Anticorrosive oils       |                                                                           |                                                                                                                                 | Lubricating oil as proxy                                                                                                                                         | chemicals        | organics                     | No              | RER      | lubricating oil, at plant                                              | 1.06E-04   | kg                       | IPCC 2001             | 1    | 1.41               |                                                           |
| Steam for bath heating   |                                                                           |                                                                                                                                 |                                                                                                                                                                  | chemicals        | organics                     | No              | RER      | heat, unspecific, in chemical plant                                    | 3.54E-02   | MJ                       | IPCC 2001             | 1    | 1.41               |                                                           |
| Electrical energy        |                                                                           |                                                                                                                                 |                                                                                                                                                                  | electricity      | production mix               | No              | UCTE     | electricity, medium voltage, production UCTE, at grid                  | 4.39E-03   | kWh                      | IPCC 2001             | 1    | 1.58               |                                                           |
|                          |                                                                           |                                                                                                                                 |                                                                                                                                                                  | air              | unspecified                  |                 |          | Hydrogen                                                               | 7.27E-10   | kg                       | IPCC 2001             | 1    | 1.41               |                                                           |
|                          |                                                                           |                                                                                                                                 |                                                                                                                                                                  | air              | unspecified                  |                 |          | Sulfate                                                                | 3.46E-08   | kg                       | IPCC 2001             | 1    | 1.41               |                                                           |
|                          |                                                                           |                                                                                                                                 |                                                                                                                                                                  | air              | unspecified                  |                 |          | Sulfur dioxide                                                         | 3.16E-07   | kg                       | IPCC 2001             | 1    | 1.58               |                                                           |
|                          |                                                                           |                                                                                                                                 | Only 22.8% which are landfilled are inventoried, the rest is recycled. Oily wastes are approximated by used mineral oil                                          | waste management | hazardous waste incineration | No              | CH       | disposal, used mineral oil, 10% water, to hazardous waste incineration | 5.55E-06   | kg                       | IPCC 2001             | 1    | 1.41               | calculated from max value assuming lognormal distribution |
|                          |                                                                           |                                                                                                                                 | The sludge can be approximated by Fe(OH)3. The amount of Fe in Fe(OH)3 (=76.66%) is inventoried. About two thirds are recycled --> 35% landfilled is inventoried | soil             | unspecified                  |                 |          | Iron                                                                   | 1.47E-05   | kg                       | IPCC 2001             | 1    | 1.83               |                                                           |
|                          |                                                                           |                                                                                                                                 | Reference contains neither composition nor amount of collected dust. No information is available about the share that can be recycled.                           |                  |                              |                 |          |                                                                        | n.a.       |                          | IPCC 2001             |      |                    |                                                           |
|                          |                                                                           |                                                                                                                                 | recycled --> not inventoried                                                                                                                                     |                  |                              |                 |          |                                                                        | 2.04E-02   | kg                       | IPCC 2001             | 1    | 1.1                | calculated from max value assuming lognormal distribution |

Fig. 2.3 Flows for "sheet rolling, steel" and their representation in the ecoinvent database (cont.)



## 2. Steel forming

| General Flow Information            |                                                                     |        | Representation in ecoinvent                                                                                             |                  |                              |                 |          |                                                                                |            | Uncertainty Informations |                       |      |                    |                                                           |
|-------------------------------------|---------------------------------------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------|------------------|------------------------------|-----------------|----------|--------------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------------------------------------|
| Input                               | Process Name                                                        | Output | Remarks                                                                                                                 | Category         | Sub-category                 | Infra-structure | Location | Module name in ecoinvent                                                       | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                                           |
| Industrial & demineralised water    | sheet rolling, steel, Tandem Mill (Assumption 50% of sheet rolling) | mL     |                                                                                                                         | water supply     | production                   | No              | CH       | water, deionised, at plant                                                     | 1.18E-02   | kg                       | IPCC 2001             | 1    | 1.69               | calculated from max value assuming lognormal              |
| Rolling oil                         |                                                                     | mL     |                                                                                                                         | chemicals        | organics                     | No              | RER      | lubricating oil, at plant                                                      | 3.87E-04   | kg                       | IPCC 2001             | 1    | 2.58               |                                                           |
| Other oils (hydraulic, Morgoil...)  |                                                                     | mL     | no quantitative information                                                                                             |                  |                              |                 |          |                                                                                | n.a.       |                          | IPCC 2001             |      |                    |                                                           |
| Steam                               |                                                                     | mL     |                                                                                                                         | chemicals        | organics                     | No              | RER      | heat, unspecific, in chemical plant                                            | 8.66E-03   | MJ                       | IPCC 2001             | 1    | 1.73               | calculated from max value assuming lognormal distribution |
| Electrical energy                   |                                                                     | mL     |                                                                                                                         | electricity      | production mix               | No              | UCTE     | electricity, medium voltage, production UCTE, at grid                          | 3.40E-02   | kWh                      | IPCC 2001             | 1    | 1.22               |                                                           |
| Caloric energy (natural gas or LPG) |                                                                     | mL     |                                                                                                                         | natural gas      | heating systems              | No              | RER      | natural gas, burned in industrial furnace >100kW                               | 3.00E-03   | MJ                       | IPCC 2001             | 1    | 6                  |                                                           |
|                                     |                                                                     | mL     | Size distribution: mechanical process, no burning process involved.                                                     | air              | unspecified                  |                 |          | Particulates, > 2.5 um, and < 10um                                             | 2.40E-05   | kg                       | IPCC 2001             | 1    | 2.05               | (1,5,1,1,1,3,26)                                          |
|                                     |                                                                     | mL     | Size distribution: mechanical process, no burning process involved.                                                     | air              | unspecified                  |                 |          | Particulates, > 10 um                                                          | 2.40E-05   | kg                       | IPCC 2001             | 1    | 1.56               | (1,5,1,1,1,3,25)                                          |
|                                     |                                                                     | mL     |                                                                                                                         | air              | unspecified                  |                 |          | NM VOC, non-methane volatile organic compounds, unspecified origin             | 3.50E-06   | kg                       | IPCC 2001             | 1    | 2.05               | (1,5,1,1,1,3,23)                                          |
|                                     |                                                                     | mL     |                                                                                                                         | air              | unspecified                  |                 |          | NM VOC, non-methane volatile organic compounds, unspecified origin             | 9.85E-06   | kg                       | IPCC 2001             | 1    | 2.05               | (1,5,1,1,1,3,23)                                          |
|                                     |                                                                     | mL     | Only 22.8% which are landfilled are inventoried, the rest is recycled. Only wastes are approximated by used mineral oil | waste management | hazardous waste incineration | No              | CH       | disposal, used mineral oil, 10% water, to hazardous waste incineration         | 2.28E-08   | kg                       | IPCC 2001             | 1    | 1.22               | (1,5,1,1,1,3,6)                                           |
|                                     |                                                                     | mL     | recycled --> not inventoried                                                                                            |                  |                              |                 |          |                                                                                | 4.06E+00   | kg                       | IPCC 2001             | 1    | 1.62               | calculated from max value assuming lognormal distribution |
|                                     |                                                                     | mL     | Most of it is recycled --> 12.4% landfilled is inventoried                                                              | waste management | residual material landfill   | No              | CH       | disposal, basic oxygen furnace wastes, 0% water, to residual material landfill | 7.20E-05   | kg                       | IPCC 2001             | 1    | 1.29               |                                                           |
|                                     |                                                                     | mL     | recycled --> not inventoried                                                                                            |                  |                              |                 |          |                                                                                | 8.50E-03   | kg                       | IPCC 2001             | 1    |                    |                                                           |
|                                     |                                                                     | mL     | Dust 2.5-10 to air                                                                                                      |                  |                              |                 |          |                                                                                |            |                          |                       |      |                    |                                                           |
|                                     |                                                                     | mL     | Dust >10 to air                                                                                                         |                  |                              |                 |          |                                                                                |            |                          |                       |      |                    |                                                           |
|                                     |                                                                     | mL     | Hydrocarbons to air                                                                                                     |                  |                              |                 |          |                                                                                |            |                          |                       |      |                    |                                                           |
|                                     |                                                                     | mL     | Oil to air                                                                                                              |                  |                              |                 |          |                                                                                |            |                          |                       |      |                    |                                                           |
|                                     |                                                                     | mL     | Oil contaminated waste (disposed)                                                                                       |                  |                              |                 |          |                                                                                |            |                          |                       |      |                    |                                                           |
|                                     |                                                                     | mL     | Emulsion (internally recycled)                                                                                          |                  |                              |                 |          |                                                                                |            |                          |                       |      |                    |                                                           |
|                                     |                                                                     | mL     | sludge cake                                                                                                             |                  |                              |                 |          |                                                                                |            |                          |                       |      |                    |                                                           |
|                                     |                                                                     | mL     | Scrap, off-cuts, etc.                                                                                                   |                  |                              |                 |          |                                                                                |            |                          |                       |      |                    |                                                           |

Fig. 2.3 Flows for "sheet rolling, steel" and their representation in the ecoinvent database (cont.)

## 2. Steel forming

| General Flow Information            |                                                                           |        | Representation in ecoinvent                                |                   |                            |                 |          |                                                                                  |            | Uncertainty Informations |                       |      |                    |                                                           |
|-------------------------------------|---------------------------------------------------------------------------|--------|------------------------------------------------------------|-------------------|----------------------------|-----------------|----------|----------------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------------------------------------|
| Input                               | Process Name                                                              | Output | Remarks                                                    | Category          | Sub-category               | Infra-structure | Location | Modul name in ecoinvent                                                          | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                                           |
| Industrial & demineralised w ater   | sheet rolling, steel, Reversing Mill<br>(Assumption 50% of sheet rolling) | mL     |                                                            | w ater supply     | production                 | No              | CH       | w ater, deionised, at plant                                                      | 1.73E-02   | kg                       | IPCC 2001             | 1    | 1.73               | calculated from max value assuming lognormal distribution |
| Rolling oil                         |                                                                           |        |                                                            | chemicals         | organics                   | No              | RER      | lubricating oil, at plant                                                        | 5.24E-05   | kg                       | IPCC 2001             | 1    | 1.05               |                                                           |
| Electrical energy                   |                                                                           |        |                                                            | electricity       | production mix             | No              | UCTE     | electricity, medium voltage, production UCTE, at grid                            | 3.37E-02   | kWh                      | IPCC 2001             | 1    | 1.01               |                                                           |
| Caloric energy (natural gas or LPG) |                                                                           |        |                                                            | natural gas       | heating systems            | No              | RER      | natural gas, burned in industrial furnace >100kW                                 | 1.17E-05   | MJ                       | IPCC 2001             | 1    | 1.02               |                                                           |
|                                     |                                                                           |        |                                                            | air               | unspecified                |                 |          | NM VOC, non-methane volatile organic compounds, unspecified origin               | 4.61E-06   | kg                       | IPCC 2001             | 1    | 1.1                | (1,5,1,1,1,3,6)                                           |
|                                     |                                                                           |        |                                                            | air               | unspecified                |                 |          | NM VOC, non-methane volatile organic compounds, unspecified origin               | 5.40E-06   | kg                       | IPCC 2001             | 1    | 1.22               |                                                           |
|                                     |                                                                           |        | Most of it is recycled --> 12.4% landfilled is inventoried | w aste management | residual material landfill | No              | CH       | disposal, basic oxygen furnace w astes, 0% w ater, to residual material landfill | 1.21E-04   | kg                       | IPCC 2001             | 1    | 1.03               | calculated from max value assuming lognormal distribution |
|                                     |                                                                           |        | recycled --> not inventoried                               |                   |                            |                 |          |                                                                                  | 4.37E-03   | m3                       | IPCC 2001             | 1    | 1.03               |                                                           |
|                                     |                                                                           |        | recycled --> not inventoried                               |                   |                            |                 |          |                                                                                  | 1.00E-02   | kg                       | IPCC 2001             |      |                    |                                                           |
|                                     |                                                                           |        |                                                            |                   |                            |                 |          |                                                                                  |            |                          |                       |      |                    |                                                           |

Fig. 2.3 Flows for "sheet rolling, steel" and their representation in the ecoinvent database (cont.)

## 2. Steel forming

| General Flow Information            |                                                                            |                                 | Representation in ecoinvent                                                                                    |                  |                              |                 |          |                                                                        |            | Uncertainty Informations |                       |      |                    |                                                           |
|-------------------------------------|----------------------------------------------------------------------------|---------------------------------|----------------------------------------------------------------------------------------------------------------|------------------|------------------------------|-----------------|----------|------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------------------------------------|
| Input                               | Process Name                                                               | Output                          | Remarks                                                                                                        | Category         | Sub-category                 | Infra-structure | Location | Modul name in ecoinvent                                                | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                                           |
| Electrical energy                   | sheet rolling, steel, Batch annealing<br>(Assumption 50% of sheet rolling) |                                 |                                                                                                                | electricity      | production mix               | No              | UCTE     | electricity, medium voltage, production UCTE, at grid                  | 1.18E-02   | kWh                      | IPCC 2001             | 1    | 1.41               | calculated from max value assuming lognormal distribution |
| Caloric energy (natural gas or LPG) |                                                                            |                                 |                                                                                                                | natural gas      | heating systems              | No              | RER      | natural gas, burned in industrial furnace >100kW                       | 3.41E-01   | MJ                       | IPCC 2001             | 1    | 1.1                |                                                           |
| Inert gas (H2,N2)                   |                                                                            |                                 | no quantitative information                                                                                    |                  |                              |                 |          |                                                                        | n.a.       |                          | IPCC 2001             |      |                    |                                                           |
|                                     |                                                                            | Oil containing waste (disposed) | Oily wastes are approximated by used mineral oil. Only 22.8% landfilled are inventoried, the rest is recycled. | waste management | hazardous waste incineration | No              | CH       | disposal, used mineral oil, 10% water, to hazardous waste incineration | 2.00E-05   | kg                       | IPCC 2001             | 1    | 1.22               | (1,5,1,1,1,3,6)                                           |
|                                     |                                                                            | SO2 - reversing mill            | Arithmetic mean value, because the lower value of the range is zero.                                           | air              | unspecified                  |                 |          | Sulfur dioxide                                                         | 5.00E-07   | kg                       | IPCC 2001             | 1    | 2                  | calculated from max value assuming lognormal distribution |
|                                     |                                                                            | SO2 - continuous mills          | Arithmetic mean value, because the lower value of the range is zero.                                           | air              | unspecified                  |                 |          | Sulfur dioxide                                                         | 1.03E-05   | kg                       | IPCC 2001             | 1    | 2                  |                                                           |
|                                     |                                                                            | NOx - reversing mill            |                                                                                                                | air              | unspecified                  |                 |          | Nitrogen oxides                                                        | 4.50E-05   | kg                       | IPCC 2001             | 1    | 10                 |                                                           |
|                                     |                                                                            | NOx - continuous mills          |                                                                                                                | air              | unspecified                  |                 |          | Nitrogen oxides                                                        | 5.59E-06   | kg                       | IPCC 2001             | 1    | 22.4               |                                                           |

Fig. 2.3 Flows for "sheet rolling, steel" and their representation in the ecoinvent database (cont.)

## 2. Steel forming

| General Flow Information                                    |                                                                              |                                      | Representation in ecoinvent                                                                                                   |                   |                               |                 |          |                                                                          |            | Uncertainty Informations |                       |      |                    |                                                           |
|-------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------|-----------------|----------|--------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------------------------------------|
| Input                                                       | Process Name                                                                 | Output                               | Remarks                                                                                                                       | Category          | Sub-category                  | Infra-structure | Location | Modul name in ecoinvent                                                  | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                                           |
| Chemicals for alkaline and electrolytic pre-cleaning        | sheet rolling, steel, Continuous annealing (Assumption 50% of sheet rolling) |                                      | no quantitative information                                                                                                   |                   |                               |                 |          |                                                                          | n.a.       |                          | IPCC 2001             |      |                    |                                                           |
| Electrical energy                                           |                                                                              |                                      |                                                                                                                               | electricity       | production mix                | No              | UCTE     | electricity, medium voltage, production UCTE, at grid                    | 2.82E-02   | kWh                      | IPCC 2001             | 1    | 1.18               | calculated from max value assuming lognormal distribution |
| Caloric energy (natural gas or LPG)                         |                                                                              |                                      |                                                                                                                               | natural gas       | heating systems               | No              | RER      | natural gas, burned in industrial furnace >100kW                         | 5.36E-01   | MJ                       | IPCC 2001             | 1    | 1.38               |                                                           |
| Steam                                                       |                                                                              |                                      |                                                                                                                               | chemicals         | organics                      | No              | RER      | heat, unspecific, in chemical plant                                      | 5.95E-02   | MJ                       | IPCC 2001             | 1    | 1.22               | (1,5,1,1,1,3,1)                                           |
| Inert gas (H2,N2)                                           |                                                                              |                                      | no quantitative information                                                                                                   |                   |                               |                 |          |                                                                          | n.a.       |                          | IPCC 2001             |      |                    |                                                           |
| Wet temper rolling agent (for rough pass, electrical sheet) |                                                                              |                                      | no quantitative information                                                                                                   |                   |                               |                 |          |                                                                          | n.a.       |                          | IPCC 2001             |      |                    |                                                           |
| Anticorrosive oils                                          |                                                                              |                                      | no quantitative information                                                                                                   |                   |                               |                 |          |                                                                          | n.a.       |                          | IPCC 2001             |      |                    |                                                           |
|                                                             |                                                                              | SO2 - reversing mill                 | Arithmetic mean value, because the low er value of the range is zero.                                                         | air               | unspecified                   |                 |          | Sulfur dioxide                                                           | 5.00E-07   | kg                       | IPCC 2001             | 1    | 2                  | calculated from max value assuming lognormal distribution |
|                                                             |                                                                              | SO2 - continuous mills               | Arithmetic mean value, because the low er value of the range is zero.                                                         | air               | unspecified                   |                 |          | Sulfur dioxide                                                           | 1.03E-05   | kg                       | IPCC 2001             | 1    | 2                  |                                                           |
|                                                             |                                                                              | NOx - reversing mill                 |                                                                                                                               | air               | unspecified                   |                 |          | Nitrogen oxides                                                          | 4.50E-05   | kg                       | IPCC 2001             | 1    | 10                 |                                                           |
|                                                             |                                                                              | NOx - continuous mills               |                                                                                                                               | air               | unspecified                   |                 |          | Nitrogen oxides                                                          | 5.59E-06   | kg                       | IPCC 2001             | 1    | 22.4               |                                                           |
|                                                             |                                                                              | Waste heat (steam, can be recovered) | recycled --> not inventoried                                                                                                  |                   |                               |                 |          |                                                                          | 2.69E-02   | MJ                       | IPCC 2001             | 1    | 1.45               |                                                           |
|                                                             |                                                                              | Sludge cake (disposed)               | Containing oil. Oily w astes are approxiamted by used mineral oil. Most of it is recycled --> 12.4% landfilled is inventoried | w aste management | hazardous w aste incineration | No              | CH       | disposal, used mineral oil, 10% w ater, to hazardous w aste incineration | 5.70E-06   | kg                       | IPCC 2001             | 1    | 5.11               |                                                           |
|                                                             |                                                                              | Scale dust                           | no quantitative information                                                                                                   |                   |                               |                 |          |                                                                          | n.a.       |                          | IPCC 2001             |      |                    |                                                           |
|                                                             |                                                                              | Oil sludge from pre-cleaning         | no quantitative information                                                                                                   |                   |                               |                 |          |                                                                          | n.a.       |                          | IPCC 2001             |      |                    |                                                           |

Fig. 2.3 Flows for "sheet rolling, steel" and their representation in the ecoinvent database (cont.)

## 2. Steel forming

| General Flow Information         |                                                                |                           | Representation in ecoinvent                                                                                                                                   |                   |                              |                 |          |                                                                        |            | Uncertainty Informations |                       |      |                    |                                                           |
|----------------------------------|----------------------------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------------------|-----------------|----------|------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------------------------------------|
| Input                            | Process Name                                                   | Output                    | Remarks                                                                                                                                                       | Category          | Sub-category                 | Infra-structure | Location | Module name in ecoinvent                                               | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                                           |
| Industrial & demineralised water | sheet rolling, steel, Tempering                                | Oil contaminated water    |                                                                                                                                                               | water supply      | production                   | No              | CH       | water, deionised, at plant                                             | 2.82E-03   | kg                       | IPCC 2001             | 1    | 1.41               | calculated from max value assuming lognormal distribution |
| Electrical energy                |                                                                |                           |                                                                                                                                                               | electricity       | production mix               | No              | UCTE     | electricity, medium voltage, production UCTE, at grid                  | 1.52E-02   | kWh                      | IPCC 2001             | 1    | 2.74               |                                                           |
| Tempering fluid concentrate      |                                                                |                           | 5% solutions of wet temper rolling agents, nitrite based. Lubricating oil as proxy                                                                            | chemicals         | organics                     | No              | RER      | lubricating oil, at plant                                              | 5.74E-04   | kg                       | IPCC 2001             | 1    | 5.22               |                                                           |
| Anticorrosive oils               |                                                                |                           | Lubricating oil as proxy                                                                                                                                      | chemicals         | organics                     | No              | RER      | lubricating oil, at plant                                              | 4.47E-05   | kg                       | IPCC 2001             | 1    | 1.12               |                                                           |
|                                  |                                                                |                           | Oily wastes are approximated by used mineral oil. Only 22.8% landfilled are inventoried, the rest is recycled.                                                | waste management  | hazardous waste incineration | No              | CH       | disposal, used mineral oil, 10% water, to hazardous waste incineration | 1.32E-05   | kg                       | IPCC 2001             | 1    | 1.12               |                                                           |
|                                  | sheet rolling, steel, Finishing (cutting, inspection, packing) | Tempering / cooling fluid | recycled --> not inventoried                                                                                                                                  |                   |                              |                 |          |                                                                        | n.a.       |                          | IPCC 2001             |      |                    | calculated from max value assuming lognormal distribution |
|                                  |                                                                | Sludge cake               | recycled --> not inventoried                                                                                                                                  |                   |                              |                 |          |                                                                        | 3.51E-04   | kg                       | IPCC 2001             | 1    | 1.41               |                                                           |
|                                  |                                                                | Leakage oil               |                                                                                                                                                               | soil              | unspecified                  |                 |          | Oils, unspecified                                                      | 3.85E-05   | kg                       | IPCC 2001             | 1    | 1.12               |                                                           |
| Anticorrosive oils               |                                                                |                           | Lubricating oil as proxy                                                                                                                                      | chemicals         | organics                     | No              | RER      | lubricating oil, at plant                                              | 2.83E-04   | kg                       | IPCC 2001             | 1    | 1.41               |                                                           |
| Singing ink                      |                                                                |                           | Assumed as irrelevant --> input not inventoried, inventoried for transport and in a waste mix for waste disposal                                              |                   |                              |                 |          |                                                                        | 6.32E-06   | kg                       | IPCC 2001             | 1    | 12.6               |                                                           |
| Film (clear)                     |                                                                |                           |                                                                                                                                                               | plastics          | processing                   | No              | RER      | packaging film, LDPE, at plant                                         | 8.49E-05   | kg                       | IPCC 2001             | 1    | 1.06               |                                                           |
| Paper (blue)                     |                                                                |                           | No description available in reference --> assumption kraft paper as proxy                                                                                     | paper & cardboard | packaging papers             | No              | RER      | kraft paper, unbleached, at plant                                      | 2.68E-05   | kg                       | IPCC 2001             | 1    | 2.98               |                                                           |
| VCI Paper                        |                                                                |                           | No description available in reference --> assumption kraft paper as proxy                                                                                     | paper & cardboard | packaging papers             | No              | RER      | kraft paper, bleached, at plant                                        | 1.60E-06   | kg                       | IPCC 2001             | 1    | 1.06               |                                                           |
| FLEX-Stripes                     |                                                                |                           | Reference contained no further description of the product. Steel sheet was chosen as proxy. Volume was calculated with an assumed sheet-thickness of 0.55 mm. | metals            | extraction                   | No              | RER      | steel, electric, un- and low-alloyed, at plant                         | 2.36E-04   | kg                       | IPCC 2001             | 1    | 1.1                |                                                           |
| FLEX-Stripes                     |                                                                |                           |                                                                                                                                                               | metals            | processing                   | No              | RER      | sheet rolling, steel                                                   | 2.36E-04   | kg                       | IPCC 2001             | 1    | 1.1                |                                                           |
| Wood                             |                                                                |                           |                                                                                                                                                               | wooden materials  | extraction                   | No              | RER      | sawn timber, softwood, raw, air dried, u=20%, at plant                 | 1.85E-06   | m <sup>3</sup>           | IPCC 2001             | 1    | 1.05               |                                                           |
| Electrical energy                |                                                                |                           |                                                                                                                                                               | electricity       | production mix               | No              | UCTE     | electricity, medium voltage, production UCTE, at grid                  | 7.86E-03   | kWh                      | IPCC 2001             | 1    | 1.41               |                                                           |
|                                  |                                                                | Oil contaminated waste    | Only 22.8% landfilled are inventoried, the rest is recycled. Oil containing waste is approximated by used mineral oil                                         | waste management  | hazardous waste incineration | No              | CH       | disposal, used mineral oil, 10% water, to hazardous waste incineration | 2.93E-05   | kg                       | IPCC 2001             | 1    | 1.07               |                                                           |
|                                  |                                                                | Scrap                     | recycled --> not inventoried                                                                                                                                  |                   |                              |                 |          |                                                                        | 3.16E-02   | kg                       | IPCC 2001             | 1    | 1.26               |                                                           |

Fig. 2.3 Flows for "sheet rolling, steel" and their representation in the ecoinvent database (cont.)

## 2. Steel forming

| General Flow Information  |                                        |        | Representation in ecoinvent                                                                                                                    |                   |                               |                 |          |                                                                                |            | Uncertainty Informations |                       |      |                    |                                                           |
|---------------------------|----------------------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------|-----------------|----------|--------------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------------------------------------|
| Input                     | Process Name                           | Output | Remarks                                                                                                                                        | Category          | Sub-category                  | Infra-structure | Location | Modul name in ecoinvent                                                        | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                                           |
| Grinding emulsion (fresh) | sheet rolling, steel, roll maintenance | mL     | Lubricating oil as proxy                                                                                                                       | chemicals         | organics                      | No              | RER      | lubricating oil, at plant                                                      | 5.66E-06   | kg                       | IPCC 2001             | 1    | 14.1               | calculated from max value assuming lognormal distribution |
| Shot grain (SBT)          |                                        | mL     | One of the most important abrasive materials is taken as proxy                                                                                 | chemicals         | inorganics                    | No              | RER      | silicon carbide, at plant                                                      | 4.47E-05   | kg                       | IPCC 2001             | 1    | 1.12               |                                                           |
| Soluble oil concentrate   |                                        | mL     | Lubricating oil as proxy                                                                                                                       | chemicals         | organics                      | No              | RER      | lubricating oil, at plant                                                      | 5.48E-06   | kg                       | IPCC 2001             | 1    | 1.1                |                                                           |
| Chromium salt (PRETEX)    |                                        | mL     | Na-dichromate as proxy                                                                                                                         | chemicals         | inorganics                    | No              | RER      | sodium dichromate, at plant                                                    | 1.50E-04   | kg                       | IPCC 2001             | 1    | 1.07               |                                                           |
|                           |                                        | mL     | Size distribution: mechanical process w without burning process                                                                                | air               | unspecified                   |                 |          | Particulates, > 2.5 um, and < 10um                                             | 2.09E-08   | kg                       | IPCC 2001             | 1    | 2.05               | (1,5,1,1,1,3,26)                                          |
|                           |                                        | mL     | Size distribution: mechanical process w without burning process                                                                                | air               | unspecified                   |                 |          | Particulates, > 10 um                                                          | 2.09E-08   | kg                       | IPCC 2001             | 1    | 1.56               | (1,5,1,1,1,3,25)                                          |
|                           |                                        | mL     |                                                                                                                                                | air               | unspecified                   |                 |          | Chromium                                                                       | 4.18E-09   | kg                       | IPCC 2001             | 1    | 5.06               | (1,5,1,1,1,3,22)                                          |
|                           |                                        | mL     | Water treatment sludge as proxy                                                                                                                | w aste management | residual material landfill    | No              | CH       | disposal, sludge from steel rolling, 20% w ater, to residual material landfill | 1.41E-04   | kg                       | IPCC 2001             | 1    | 1.41               | calculated from max value assuming lognormal distribution |
|                           |                                        | mL     | inventoried in infrastructure module --> not separately inventoried                                                                            |                   |                               |                 |          |                                                                                | 9.49E-07   | kg                       | IPCC 2001             | 1    | 1.05               |                                                           |
|                           |                                        | mL     | Only 22.8% landfilled are inventoried, the rest is recycled. Containing oil. Oily w astes are approximated by used mineral oil                 | w aste management | hazardous w aste incineration | No              | CH       | disposal, used mineral oil, 10% w ater, to hazardous w aste incineration       | 1.21E-05   | kg                       | IPCC 2001             | 1    | 1.51               |                                                           |
|                           |                                        | mL     | Only 22.8% landfilled are inventoried, the rest is recycled. Oily w astes are approximated by used mineral oil                                 | w aste management | hazardous w aste incineration | No              | CH       | disposal, used mineral oil, 10% w ater, to hazardous w aste incineration       | 3.40E-05   | kg                       | IPCC 2001             | 1    | 1.15               |                                                           |
|                           |                                        | mL     | SBT (Shot Blast Texturing): This is a mechanical application of the texture onto the roll using grit blasting. Therefore the size distribution | air               | unspecified                   |                 |          | Particulates, > 2.5 um, and < 10um                                             | 2.24E-05   | kg                       | IPCC 2001             | 1    | 1.12               |                                                           |
|                           |                                        | mL     | SBT (Shot Blast Texturing): This is a mechanical application of the texture onto the roll using grit blasting. Therefore the size distribution | air               | unspecified                   |                 |          | Particulates, > 10 um                                                          | 2.24E-05   | kg                       | IPCC 2001             | 1    | 1.12               |                                                           |

Fig. 2.3 Flows for "sheet rolling, steel" and their representation in the ecoinvent database (cont.)

## 2. Steel forming

| General Flow Information          |                                                 |        | Representation in ecoinvent       |                                                                                                                                                                |                        |                               |          |                         |                                                                               | Uncertainty Informations |                       |            |                    |                 |                                                           |                                                           |
|-----------------------------------|-------------------------------------------------|--------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-------------------------------|----------|-------------------------|-------------------------------------------------------------------------------|--------------------------|-----------------------|------------|--------------------|-----------------|-----------------------------------------------------------|-----------------------------------------------------------|
| Input                             | Process Name                                    | Output | Remarks                           | Category                                                                                                                                                       | Sub-category           | Infra-structure               | Location | Modul name in ecoinvent | Mean value                                                                    | Unit                     | Source for mean value | Type       | Standard Deviation | General Comment |                                                           |                                                           |
| Demineralized Water               | sheet rolling, steel, hydr fluids               | mL     | Disposed waste: rejects           |                                                                                                                                                                | w ater supply          | production                    | No       | CH                      | w ater, deionised, at plant                                                   | 7.73E-02                 | kg                    | IPCC 2001  | 1                  | 1.55            | calculated from max value assuming lognormal distribution |                                                           |
| Grease, oil, hydraulic oil        |                                                 |        |                                   | Lubricating oil as proxy                                                                                                                                       | chemicals              | organics                      | No       | RER                     | lubricating oil, at plant                                                     | 1.64E-03                 | kg                    | IPCC 2001  | 1                  | 1.1             |                                                           |                                                           |
| Rolling oil                       |                                                 |        |                                   | Lubricating oil as proxy                                                                                                                                       | chemicals              | organics                      | No       | RER                     | lubricating oil, at plant                                                     | 9.95E-04                 | kg                    | IPCC 2001  | 1                  | 2.21            |                                                           |                                                           |
| Hydraulic & lubrication oils      |                                                 |        |                                   | Lubricating oil as proxy                                                                                                                                       | chemicals              | organics                      | No       | RER                     | lubricating oil, at plant                                                     | 4.24E-04                 | kg                    | IPCC 2001  | 1                  | 2.12            |                                                           |                                                           |
| Electrical energy                 |                                                 |        |                                   |                                                                                                                                                                | electricity            | production mix                | No       | UCTE                    | electricity, medium voltage, production UCTE, at grid                         | 8.33E-02                 | kWh                   | IPCC 2001  | 1                  | 1.5             |                                                           |                                                           |
| Steam                             |                                                 |        |                                   | Steam used as energy source                                                                                                                                    | chemicals              | organics                      | No       | RER                     | heat, unspecific, in chemical plant                                           | 4.47E-02                 | MJ                    | IPCC 2001  | 1                  | 1.12            |                                                           |                                                           |
|                                   | sheet rolling, steel, water and water treatment | mL     | Disposed waste: Used oil + w ater | Rejects made mainly of steel. Only 34% landfilled are inventoried, the rest is recycled.                                                                       | w aste management      | municipal incineration        | No       | CH                      | disposal, steel in car shredder residue, 0% w ater, to municipal incineration | 7.91E-03                 | kg                    | IPCC 2001  | 1                  | 2               | (1,5,1,1,1,3,12)                                          |                                                           |
|                                   |                                                 |        |                                   | Only 22.8% landfilled are inventoried, the rest is recycled. Oily w astes are approximated by used mineral oil                                                 | w aste management      | hazardous w aste incineration | No       | CH                      | disposal, used mineral oil, 10% w ater, to hazardous w aste incineration      | 7.63E-04                 | kg                    | IPCC 2001  | 1                  | 1.05            |                                                           |                                                           |
| Cooling w ater                    |                                                 |        |                                   | Sum of the different process steps                                                                                                                             | resource               | in w ater                     |          |                         | Water, cooling, unspecified natural origin                                    | 2.70E-02                 | m3                    | assumption | 1                  | 1.22            |                                                           |                                                           |
|                                   |                                                 |        |                                   | One part of the incomin cooling and process w ater leaves the system as not contaminated evaporated w ater or included in discarded w aste --> not inventoried |                        |                               |          |                         |                                                                               | 2.12E-02                 | m3                    | assumption | 1                  |                 |                                                           |                                                           |
| Flocculating agents (Ca(OH)2;95%) |                                                 |        |                                   |                                                                                                                                                                | construction materials | binder                        | No       | CH                      | lime, hydrated, loose, at plant                                               | 5.42E-05                 | kg                    | assumption | 1                  | 1.2             |                                                           | calculated from max value assuming lognormal distribution |
| Polyelectrolytes                  |                                                 |        |                                   | Personal communication by G. Doka --> inorganic chemicals as proxy                                                                                             | chemicals              | inorganics                    | No       | GLO                     | chemicals inorganic, at plant                                                 | 1.58E-08                 | kg                    | assumption | 1                  | 2               |                                                           |                                                           |

Fig. 2.3 Flows for "sheet rolling, steel" and their representation in the ecoinvent database (cont.)

## 2. Steel forming

| General Flow Information                        |              |                        | Representation in ecoinvent                                                                                                                                                                                                                                                                                                                                                                              |                  |                            |                 |          |                                                                               |            | Uncertainty Informations |                       |      |                    |                  |
|-------------------------------------------------|--------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------------|-----------------|----------|-------------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|------------------|
| Input                                           | Process Name | Output                 | Remarks                                                                                                                                                                                                                                                                                                                                                                                                  | Category         | Sub-category               | Infra-structure | Location | Modul name in ecoinvent                                                       | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment  |
| sheet rolling, steel, water and water treatment | mL           | Treated waste water    | One part of the incoming cooling and process water leaves the system as treated effluent; it is assumed to be discharged in surface water --> not inventoried. Infrastructure for waste water treatment plant is comprised in the infrastructure module for the whole rolling mill plant. Value for energy use is comprised in the broad range of uncertainty the energy data for the rolling mill have. |                  |                            |                 |          |                                                                               | 6.00E-03   | m <sup>3</sup>           | assumption            | 1    |                    |                  |
|                                                 |              | TSS                    |                                                                                                                                                                                                                                                                                                                                                                                                          | water            | unspecified                |                 |          | Suspended solids, unspecified                                                 | 1.59E-04   | kg                       | assumption            | 1    | 1.57               | (2,5,1,1,1,3,32) |
|                                                 |              | COD                    |                                                                                                                                                                                                                                                                                                                                                                                                          | water            | unspecified                |                 |          | COD, Chemical Oxygen Demand                                                   | 4.13E-05   | kg                       | assumption            | 1    | 1.57               | (2,5,1,1,1,3,32) |
|                                                 |              | BOD                    |                                                                                                                                                                                                                                                                                                                                                                                                          | water            | unspecified                |                 |          | BOD5, Biological Oxygen Demand                                                | 4.13E-05   | kg                       | assumption            | 1    | 2.79               | (5,5,5,5,5,5,32) |
|                                                 |              | DOC                    |                                                                                                                                                                                                                                                                                                                                                                                                          | water            | unspecified                |                 |          | DOC, Dissolved Organic Carbon                                                 | 1.33E-05   | kg                       | assumption            | 1    | 2.79               | (5,5,5,5,5,5,32) |
|                                                 |              | TOC                    |                                                                                                                                                                                                                                                                                                                                                                                                          | water            | unspecified                |                 |          | TOC, Total Organic Carbon                                                     | 1.33E-05   | kg                       | assumption            | 1    | 2.79               | (5,5,5,5,5,5,32) |
|                                                 |              | Hydrocarbons           |                                                                                                                                                                                                                                                                                                                                                                                                          | water            | unspecified                |                 |          | Hydrocarbons, unspecified                                                     | 1.89E-06   | kg                       | assumption            | 1    | 3.05               | (2,5,1,1,1,3,34) |
|                                                 |              | Cu                     |                                                                                                                                                                                                                                                                                                                                                                                                          | water            | unspecified                |                 |          | Copper, ion                                                                   | 2.90E-07   | kg                       | assumption            | 1    | 1.86               | (2,5,1,1,1,3,37) |
|                                                 |              | Zn                     |                                                                                                                                                                                                                                                                                                                                                                                                          | water            | unspecified                |                 |          | Zinc, ion                                                                     | 2.24E-07   | kg                       | assumption            | 1    | 1.86               | (2,5,1,1,1,3,37) |
|                                                 |              | Cd                     |                                                                                                                                                                                                                                                                                                                                                                                                          | water            | unspecified                |                 |          | Cadmium, ion                                                                  | 1.50E-07   | kg                       | assumption            | 1    | 1.86               | (2,5,1,1,1,3,37) |
|                                                 |              | Al                     |                                                                                                                                                                                                                                                                                                                                                                                                          | water            | unspecified                |                 |          | Aluminum                                                                      | 1.42E-06   | kg                       | assumption            | 1    | 1.86               | (2,5,1,1,1,3,37) |
|                                                 |              | Pb                     |                                                                                                                                                                                                                                                                                                                                                                                                          | water            | unspecified                |                 |          | Lead                                                                          | 3.00E-07   | kg                       | assumption            | 1    | 1.86               | (2,5,1,1,1,3,37) |
|                                                 |              | Cr                     |                                                                                                                                                                                                                                                                                                                                                                                                          | water            | unspecified                |                 |          | Chromium, ion                                                                 | 5.40E-07   | kg                       | assumption            | 1    | 1.86               | (2,5,1,1,1,3,37) |
|                                                 |              | Cr6+                   |                                                                                                                                                                                                                                                                                                                                                                                                          | water            | unspecified                |                 |          | Chromium VI                                                                   | 3.00E-08   | kg                       | assumption            | 1    | 1.86               | (2,5,1,1,1,3,37) |
|                                                 |              | Mn                     |                                                                                                                                                                                                                                                                                                                                                                                                          | water            | unspecified                |                 |          | Manganese                                                                     | 6.12E-07   | kg                       | assumption            | 1    | 1.86               | (2,5,1,1,1,3,37) |
|                                                 |              | Fe                     |                                                                                                                                                                                                                                                                                                                                                                                                          | water            | unspecified                |                 |          | Iron, ion                                                                     | 4.65E-06   | kg                       | assumption            | 1    | 1.86               | (2,5,1,1,1,3,37) |
|                                                 |              | Ni                     |                                                                                                                                                                                                                                                                                                                                                                                                          | water            | unspecified                |                 |          | Nickel, ion                                                                   | 8.49E-07   | kg                       | assumption            | 1    | 1.86               | (2,5,1,1,1,3,37) |
|                                                 |              | Hg                     |                                                                                                                                                                                                                                                                                                                                                                                                          | water            | unspecified                |                 |          | Mercury                                                                       | 3.00E-08   | kg                       | assumption            | 1    | 1.86               | (2,5,1,1,1,3,37) |
|                                                 |              | Free Chlorine          |                                                                                                                                                                                                                                                                                                                                                                                                          | water            | unspecified                |                 |          | Chloride                                                                      | 1.34E-06   | kg                       | assumption            | 1    | 1.57               | (2,5,1,1,1,3,33) |
|                                                 |              | Water treatment sludge | Only a part of the sludge goes to landfill --> 66.5% inventoried                                                                                                                                                                                                                                                                                                                                         | waste management | residual material landfill | No              | CH       | disposal, sludge from steel rolling, 20% water, to residual material landfill | 2.00E-02   | kg                       | assumption            | 1    | 1.57               | (2,5,1,1,1,3,43) |

Fig. 2.3 Flows for "sheet rolling, steel" and their representation in the ecoinvent database (cont.)



## 2. Steel forming

| General Flow Information |                               |                    | Representation in ecoinvent                                                                                                                                                                                                                                                                                                                                      |                   |                        |                 |          |                                                                         |            | Uncertainty Informations |                       |      |                    |                                |
|--------------------------|-------------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------------|-----------------|----------|-------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|--------------------------------|
| Input                    | Process Name                  | Output             | Remarks                                                                                                                                                                                                                                                                                                                                                          | Category          | Sub-category           | Infra-structure | Location | Modul name in ecoinvent                                                 | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                |
| Rolling Shop (CR)        | sheet rolling, steel, Overall |                    |                                                                                                                                                                                                                                                                                                                                                                  | metals            | processing             | Yes             | RER      | rolling mill                                                            | 1.62E-09   | unit                     | Estimation            | 1    | 3.04               | (3,4,1,1,1,4,9)                |
| Lorry 32t                |                               |                    | Includes the transports of all materials not produced at the plant and the transport to disposal of the waste. 32t-lorry was taken as a proxy for all this transports, therefore transports to the municipal incineration are overestimated by one third (not relevant for the overall process). Distance to hazardous waste disposal is inventoried with 100 km | transport systems | road                   | No              | RER      | transport, lorry 32t                                                    | 2.98E-02   | tkm                      | Estimation            | 1    | 2.09               | (4,5,n.a.,n.a.,n.a.,n.a.,5)    |
|                          |                               | Waste-mix          | As a proxy, packing materials are mixed and disposed as municipal waste                                                                                                                                                                                                                                                                                          | waste management  | municipal incineration | No              | CH       | disposal, municipal solid waste, 22.9% water, to municipal incineration | 3.58E-04   | kg                       | Estimation            | 1    | 1.22               | (2,5,1,1,1,3,6)                |
|                          |                               | Waste Heat         |                                                                                                                                                                                                                                                                                                                                                                  | air               | unspecified            |                 |          | Heat, waste                                                             | 8.04E-01   | MJ                       | IPCC 2001             | 1    | 1.56               | (n.a.,5,n.a.,n.a.,n.a.,n.a.,7) |
|                          |                               | Cold rolling sheet |                                                                                                                                                                                                                                                                                                                                                                  | metals            | processing             | No              | RER      | sheet rolling, steel                                                    | 1.00E+00   | kg                       |                       |      |                    |                                |

Fig. 2.3 Flows for "sheet rolling, steel" and their representation in the ecoinvent database (cont.)

## 2.4 Sheet rolling of chromium steel

The process route for high alloy steel (stainless steel) requires an initial annealing step prior to pickling. This is necessary because, after the hot rolling process, the strip cools slowly, and carbon tends to precipitate at the grain boundaries, owing to its reduced solubility at lower temperature. Annealing causes the carbon to go back into solution. Rapid cooling of the strip then prevents the carbon from precipitation by diffusion. Further more, several additional annealing and/or pickling steps during rolling may be necessary.

Due to the different material characteristics, the field of application of chromium steel is very different from the one of steel. The production volume of chromium steel is much smaller (estimated 5 %) than the one of carbon and low alloyed steel<sup>3</sup>.

The cold rolling mills for chromium steel usually comprise the following process steps (based on IPPC (2001a)):

- Annealing (I) and pickling (I) of chromium steel,
- Cold rolling,
- Annealing (II) and pickling (II) of chromium steel,
- Tempering,
- Inspecting and finishing,
- Packing coils or sheets,
- Roll maintenance.

A short description of the process steps which differs from the steel sheet rolling (annealing and pickling I & II and cold rolling) is given below. For the other process steps see chapter 2.3.

The infrastructure of cold rolling mill for stainless steel is described in chapter 1.1. It comprises buildings where production processes take place and for storage, surroundings and machinery for the single process steps (e.g.: furnace, rolls, cutting devices, etc.).

### 2.4.1 Annealing (I) and pickling (I) of chromium steel

The process of first annealing and pickling of chromium steel is described in (IPPC (2001a)).

Due to the presence of an oxide scale and chromium depleted layer formed during the hot rolling the steel must also be descaled / pickled prior to cold rolling. In addition, the greater hardness of the steel compared with carbon steels necessitates an initial annealing (I). The annealing process has three components, heating to annealing temperature, temperature equalisation and cooling. Both continuous and batch annealing processes may be used.

Ferritic grades are generally annealed as tightly wound coils in a batch heat treatment facility. Heating may be achieved via gas firing or electrical heating. An inert protective atmosphere (nitrogen/hydrogen) is required for metallurgical reasons. Ferritic steels are generally annealed at steel temperatures of up to 800 °C.

For continuous annealing of austenitic steels the steel coil is unwound and passed through one or more heat treatment furnaces. An oxidizing atmosphere is needed to form a scale, which is rich in oxygen to allow a better pickling. The waste gases are exhausted via flues under natural or forced draft. Strip temperatures typically up to 1100 °C are necessary for annealing austenitic steels. The steel strip then

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<sup>3</sup> Personal communication on 25. February 03 from Dr. Ch. Zeltner, Stahl Gerlafingen, Swiss Steel Ltd.

passes through the cooling section and may be cooled using gas jets, air, water sprays or water quenching. Continuous annealing is generally combined with a descaling/pickling plant to form a continuous annealing and pickling line.

After annealing the steel is descaled to produce a steel surface suitable for cold rolling. Mechanical descaling such as shot blasting or scale breaking may be used prior to chemical pickling to remove the heavy scale burden. However, careful control of such mechanical processes is necessary to minimise damage to the strip surface, which could affect the final product quality. Final scale removal including the removal of the chromium depleted layer beneath the scale is generally done by pickling in a mixture of nitric and hydrofluoric acids at temperatures of up to 70 °C.

The chemical reactions that take place during the pickling of stainless steels are complex and involve the removal of the adherent scale layer by the combined action of the acids used. For further details consult directly the source (IPPC (2001a)).

Pickling line design is usually the same as for pickling of low alloy steels. Except, that on exit from the last pickling tank the steel strip is rinsed with water to remove traces of acid, dried, but no application of oil is necessary because of the corrosion resistance of the steel.

### **2.4.2 Cold rolling**

The cold rolling mill is described in (IPPC (2001a)) as follows:

After the initial annealing and pickling the steel is generally rolled to the required thickness on reversing cluster mills for a number of passes until the desired dimensions are achieved or until work hardening necessitates further annealing. Reversing stands with many rolls exert a high pressure on the strip because the diameters of the working rolls are small. These stands are therefore mainly used for steels with work-hardening properties, e.g., stainless steels and electric sheet steels.

The process of cold reduction generates heat, part of which is transferred to the rolling oil which is sprayed onto the steel and rolls for cooling as well as lubrication. This heat is subsequently dissipated through heat exchangers to cooling water systems.

Mineral oils are generally used as rolling oil and close control of the oil cleanliness is necessary for optimum performance. This may be achieved using oil filtration circuits, which incorporate media element filters or powder pre-coat filter systems. Although the use of mineral oils predominates, emulsion cooling systems similar to those used for carbon steels may be employed. In such cases additional precautions must be taken to control oil cleanliness in order to avoid strip marking. Collection hoods are fitted to the mill and operate under continuous extraction to remove oil mist that is generated.

### **2.4.3 Annealing (II) and pickling (II) of chromium steel**

Annealing and pickling are described in (IPPC (2001a); Wiegels (1997)).

Prior to the final annealing process, degreasing of the strip may be required to remove any contaminants from the steel surface. This can be achieved using an alkali based cleaning systems, which may be incorporated in the entry section of continuous annealing lines. The steel coil is unwound and passed through a series of tanks containing the cleaning solutions. Agitation of the solution is usually incorporated to enhance the cleaning. Afterwards the strip is rinsed with water. De-mineralised water can be used for the final rinsing step.

The process route for final annealing and pickling is dictated by the surface finish required. Examples of this are EN standard finish '2R' which requires a bright annealing treatment and surface finish '2B' which requires annealing and pickling.

Bright annealing is generally conducted in an electrically heated or gas fired furnace with a protective inert atmosphere of nitrogen and/or hydrogen. Both batch annealing and continuous annealing may be used. Degreasing of the steel surface is required to prevent surface discolouration.

Material with a '2B' surface finish is generally processed on continuous annealing and pickling lines. The furnace design and operation is generally similar to that for the annealing of hot band material described above. The steel is unwound and passed through one or more furnaces, which are usually gas fired. Annealing takes place in an oxygen rich atmosphere. This is necessary to ensure that the scale generated in the furnace is of a composition that is readily removable by the downstream chemical pickling processes. Stainless steels are to an increasing extent being annealed in bright annealing plants, as this process enables the bright surface produced by cold rolling, and typical of special steels, to be retained.

Unlike the descaling of hot band material described above, mechanical descaling techniques cannot be employed to assist in the removal of this scale due to the surface damage which would result for the final cold rolled strip. Therefore only chemical pickling is applied. This is done in the same manner as described above for hot band pickling. Scale of high alloy steel, due to the presence of oxides of alloying elements, cannot always be removed in just a single acid descaling step. Additional treatment or pre-treatment processes may be necessary. These may include scale removal by an initial electrolytic descaling process (using e.g. sodium sulfate) installed prior to the mixed acid section.

#### **2.4.4 Environmental issues of sheet rolling of chromium steel**

Unlike sulfuric acid or hydrochloric acid pickling solutions, regeneration is not possible, so that the spent acids must be rendered harmless and disposed off (Wiegels (1997)). For other environmental issues, see chapter 2.3.8.

#### **2.4.5 Chromium steel sheet rolling in ecoinvent**

Mean values are generally calculated as the geometric mean of the range given in IPPC (2001a). Where the range is given from zero to a maximal value, and the geometric mean can't therefore be calculated, the arithmetic mean is taken. The background information to the process "sheet rolling, chromium steel" is given in Tab. 2.4, flows are shown in Fig. 2.4.

Water emissions with the treated wastewater are given in chapter 2.2.6, Tab. 2.1.

#### **Data quality**

Data is almost exclusively taken from IPPC (2001a), some estimations are also made. The values can vary considerably depending on the degree of deformation.

Uncertainty is calculated assuming a lognormal distribution and the high value given in IPPC (2001a) representing the 95 % geometric standard deviation. Where no range is given in IPPC (2001a), the data quality is estimated by mean of the pedigree matrix, according to (Frischknecht et al. (2003)).

Tab. 2.4 Ecoinvent meta information for "sheet rolling, chromium steel"

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>                           | <b>sheet rolling, chromium steel</b>                                                                                                                                                                                                                                                                                                                                                                                                         |
| Location                              | RER                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Infrastructure Process                | 0                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Unit                                  | kg                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Data Set Version                      | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Included Processes                    | Includes the process steps annealing (I) and pickling (I), cold rolling, annealing (II) and pickling (II), tempering, inspecting and finishing, packing coils or sheets, roll maintenance. Does not include the material being rolled.                                                                                                                                                                                                       |
| Amount                                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Local Name                            | Blech walzen, Chromstahl                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Synonyms                              |                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| General Comment to reference function | This process is to be used only for high alloyed (chromium) steel. For many applications, the products of hot rolling are unsatisfactory, e.g., with respect to cross section, surface quality, dimensional accuracy, and general finish, so that cold rolling is necessary. Cold rolled products are mainly strips and sheets with high quality surface finish and precise metallurgical properties for use in high specification products. |
| Start Date                            | 1997                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| End Date                              | 2002                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Other Period Text                     |                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Geography text                        | Data-set is representative for European Union                                                                                                                                                                                                                                                                                                                                                                                                |
| Technology text                       | Average technique for EU, for high-alloyed or chromium steel.                                                                                                                                                                                                                                                                                                                                                                                |
| Representativeness [%]                |                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Production Volume                     | Less than 2.5 million tonnes per year (Zeltner --> 5%Stahl)                                                                                                                                                                                                                                                                                                                                                                                  |
| Sampling Procedure                    | Literature                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Extrapolations                        | none                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Uncertainty Adjustments               | The fact that no data were available about the percentage of companies represented by the study cited is considered in the uncertainties.                                                                                                                                                                                                                                                                                                    |

## 2. Steel forming

| General Flow Information            |                                                       |                                                    | Representation in ecoinvent                                                                                                                  |              |                 |                 |          |                                                       |            | Uncertainty Informations |                       |      |                    |                                                           |
|-------------------------------------|-------------------------------------------------------|----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|-----------------|----------|-------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------------------------------------|
| Input                               | Process Name                                          | Output                                             | Remarks                                                                                                                                      | Category     | Sub-category    | Infra-structure | Location | Modul name in ecoinvent                               | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                                           |
| Primary chromium steel              | sheet rolling, chromium steel, Pickling and annealing |                                                    | Only part of the steel that leaves the system as scrap to recycling is inventoried                                                           | metals       | extraction      | No              | RER      | chromium steel 18/8, at plant                         | 5.16E-02   | kg                       | IPCC 2001             | 1    | 1.31               | (4,5,1,1,1,3,3)                                           |
| Industrial & demineralised water    |                                                       |                                                    |                                                                                                                                              | water supply | production      | No              | CH       | water, deionised, at plant                            | 9.06E-02   | kg                       | IPCC 2001             | 1    | 6.06               | calculated from max value assuming lognormal distribution |
| Shot                                |                                                       |                                                    | One of the most important abrasive materials is taken as proxy                                                                               | chemicals    | inorganics      | No              | RER      | silicon carbide, at plant                             | 2.12E-03   | kg                       | IPCC 2001             | 1    | 1.41               |                                                           |
| Nitric acid (HNO3 70%)              |                                                       |                                                    | 1.4 kg HNO3 (50%) is used as proxy for 1 kg HNO3 (70%)                                                                                       | chemicals    | inorganics      | No              | RER      | nitric acid, 50% in H2O, at plant                     | 7.67E-03   | kg                       | IPCC 2001             | 1    | 1.83               |                                                           |
| Hydrofluoric acid (HF 70%)          |                                                       |                                                    |                                                                                                                                              | chemicals    | inorganics      | No              | GLO      | hydrogen fluoride, at plant                           | 4.33E-03   | kg                       | IPCC 2001             | 1    | 1.73               |                                                           |
| Hydrogen peroxide (H2O2 35%)        |                                                       |                                                    | 0.7 kg of hydrogen peroxide (50%) is used as proxy for 1 kg H2O2 (35%)                                                                       | chemicals    | inorganics      | No              | RER      | hydrogen peroxide, 50% in H2O, at plant               | 3.83E-03   | kg                       | IPCC 2001             | 1    | 1.83               |                                                           |
| Compressed air                      |                                                       |                                                    | Electrical energy needed to compress the air is assumed to be included in the overall process electricity demand. The air is not inventoried |              |                 |                 |          |                                                       | n.a.       |                          | IPCC 2001             |      |                    |                                                           |
| Electrical energy                   |                                                       |                                                    |                                                                                                                                              | electricity  | production mix  | No              | UCTE     | electricity, medium voltage, production UCTE, at grid | 1.56E-01   | kWh                      | IPCC 2001             | 1    | 1.25               | calculated from max value assuming lognormal distribution |
| Caloric energy (natural gas or LPG) |                                                       |                                                    |                                                                                                                                              | natural gas  | heating systems | No              | RER      | natural gas, burned in industrial furnace >100kW      | 2.37E-03   | MJ                       | IPCC 2001             | 1    | 1.39               |                                                           |
| Steam (acid heating)                |                                                       |                                                    |                                                                                                                                              | chemicals    | organics        | No              | RER      | heat, unspecified, in chemical plant                  | 1.73E-04   | MJ                       | IPCC 2001             | 1    | 1.73               |                                                           |
|                                     |                                                       | Metal oxide and/or scale particles, shot fragments | no quantitative information                                                                                                                  |              |                 |                 |          |                                                       | n.a.       |                          | IPCC 2001             |      |                    |                                                           |
|                                     |                                                       | Dust2.5-10                                         | Particle size distribution from mechanical descaling is assumed to be 50-50 smaller and bigger than 10um                                     | air          | unspecified     |                 |          | Particulates, > 2.5 um, and < 10um                    | 7.07E-06   | kg                       | IPCC 2001             | 1    | 1.41               | calculated from max value assuming lognormal distribution |
|                                     |                                                       | Dust>10                                            |                                                                                                                                              | air          | unspecified     |                 |          | Particulates, > 10 um                                 | 7.07E-06   | kg                       | IPCC 2001             | 1    | 1.41               |                                                           |
|                                     |                                                       | HF                                                 |                                                                                                                                              | air          | unspecified     |                 |          | Hydrogen fluoride                                     | 8.25E-07   | kg                       | IPCC 2001             | 1    | 4.12               |                                                           |
|                                     |                                                       | NOx                                                |                                                                                                                                              | air          | unspecified     |                 |          | Nitrogen oxides                                       | 1.10E-04   | kg                       | IPCC 2001             | 1    | 36.5               |                                                           |
|                                     |                                                       | SO2                                                |                                                                                                                                              | air          | unspecified     |                 |          | Sulfur dioxide                                        | 1.00E-06   | kg                       | IPCC 2001             | 1    | 1.22               |                                                           |

Fig. 2.4 Flows for "sheet rolling, chromium steel" and their representation in the ecoinvent database

## 2. Steel forming

| General Flow Information                                                                                                                                      |                                                          |                                                  | Representation in ecoinvent                                                                                                                                                                         |                  |                              |                 |          |                                                                                 |            | Uncertainty Informations |                       |      |                    |                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------------------------|-----------------|----------|---------------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------------------------------------|
| Input                                                                                                                                                         | Process Name                                             | Output                                           | Remarks                                                                                                                                                                                             | Category         | Sub-category                 | Infra-structure | Location | Modul name in ecoinvent                                                         | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                                           |
| <div>Oil (mineral) <math>\text{m}^3</math></div> <div>Oil filtration media <math>\text{m}^3</math></div> <div>Electrical energy <math>\text{MWh}</math></div> | sheet rolling, chromium steel, Pickling and annealing    | $\text{m}^3$ Sludge cake                         | The sludge can be approximated by $\text{Fe}(\text{OH})_3$ . The amount of Fe in $\text{Fe}(\text{OH})_3$ (=76.66%) is inventoried. About two thirds are recycled --> 35% landfilled is inventoried | soil             | unspecified                  |                 |          | Iron                                                                            | 5.48E-03   | kg                       | IPCC 2001             | 1    | 2.74               | calculated from max value assuming lognormal distribution |
|                                                                                                                                                               |                                                          | $\text{m}^3$ Spent pickle bath / mixed acid      | no quantitative information. It is assumed to be recycled --> not inventoried                                                                                                                       |                  |                              |                 |          |                                                                                 | n.a.       |                          | IPCC 2001             |      |                    |                                                           |
|                                                                                                                                                               |                                                          | $\text{m}^3$ SO2 from furnace                    |                                                                                                                                                                                                     | air              | unspecified                  |                 |          | Sulfur dioxide                                                                  | 1.50E-05   | kg                       | IPCC 2001             | 1    | 2                  | calculated from max value assuming lognormal distribution |
|                                                                                                                                                               |                                                          | $\text{m}^3$ NOx from furnace                    |                                                                                                                                                                                                     | air              | unspecified                  |                 |          | Nitrogen oxides                                                                 | 1.22E-05   | kg                       | IPCC 2001             | 1    | 122                |                                                           |
|                                                                                                                                                               | sheet rolling, chromium steel, High alloy reversing mill | $\text{m}^3$ Oil                                 | Mean value of two mean values                                                                                                                                                                       | air              | unspecified                  |                 |          | NM VOC, non-methane volatile organic compounds, unspecified origin              | 7.61E-05   | kg                       | IPCC 2001             | 1    | 1.2                |                                                           |
|                                                                                                                                                               |                                                          | $\text{m}^3$ Oil contaminated waste filter media | Assumption 50% filterdust. Only 12.4% landfilled are inventoried. The proxy chosen to represent the dust stands for dust disposal from steelmaking, thus being quite accurate.                      | waste management | residual material landfill   | No              | CH       | disposal, dust, unalloyed EAF steel, 15.4% water, to residual material landfill | 1.39E-04   | kg                       | IPCC 2001             | 1    | 1.25               |                                                           |
|                                                                                                                                                               |                                                          | $\text{m}^3$ Oil contaminated waste filter media | Assumption 50% oil contaminated waste. Only the 12.4% being landfilled are inventoried                                                                                                              | waste management | hazardous waste incineration | No              | CH       | disposal, used mineral oil, 10% water, to hazardous waste incineration          | 1.39E-04   | kg                       | IPCC 2001             | 1    | 1.25               |                                                           |
|                                                                                                                                                               |                                                          | $\text{m}^3$ Scrap, off-cuts, etc.               | Recycled --> not inventoried                                                                                                                                                                        |                  |                              |                 |          |                                                                                 | 2.00E-02   | kg                       | IPCC 2001             | 1    |                    |                                                           |
|                                                                                                                                                               |                                                          |                                                  |                                                                                                                                                                                                     |                  |                              |                 |          |                                                                                 |            |                          |                       |      |                    |                                                           |
|                                                                                                                                                               |                                                          |                                                  |                                                                                                                                                                                                     |                  |                              |                 |          |                                                                                 |            |                          |                       |      |                    |                                                           |
|                                                                                                                                                               |                                                          |                                                  |                                                                                                                                                                                                     |                  |                              |                 |          |                                                                                 |            |                          |                       |      |                    |                                                           |
|                                                                                                                                                               |                                                          |                                                  |                                                                                                                                                                                                     |                  |                              |                 |          |                                                                                 |            |                          |                       |      |                    |                                                           |

Fig. 2.4 Flows for "sheet rolling, chromium steel" and their representation in the ecoinvent database (cont.)

## 2. Steel forming

| General Flow Information         |                                                                         |        | Representation in ecoinvent                                                                                                                                   |                   |                              |                 |          |                                                                        |            | Uncertainty Informations |                       |      |                    |                                                           |
|----------------------------------|-------------------------------------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------------------|-----------------|----------|------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------------------------------------|
| Input                            | Process Name                                                            | Output | Remarks                                                                                                                                                       | Category          | Sub-category                 | Infra-structure | Location | Modul name in ecoinvent                                                | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                                           |
| Industrial & demineralised water | sheet rolling, chromium steel, Tempering                                | m      |                                                                                                                                                               | water supply      | production                   | No              | CH       | water, deionised, at plant                                             | 2.82E-03   | kg                       | IPCC 2001             | 1    | 1.41               | calculated from max value assuming lognormal distribution |
| Electrical energy                |                                                                         |        |                                                                                                                                                               | electricity       | production mix               | No              | UCTE     | electricity, medium voltage, production UCTE, at grid                  | 1.52E-02   | kWh                      | IPCC 2001             | 1    | 2.74               |                                                           |
| Tempering fluid concentrate      |                                                                         |        | Good recycling rate --> assumed as not relevant                                                                                                               |                   |                              |                 |          |                                                                        | 5.74E-04   | kg                       | IPCC 2001             | 1    | 5.22               |                                                           |
| Anticorrosive oils               |                                                                         |        | Lubricating oil as proxy                                                                                                                                      | chemicals         | organics                     | No              | RER      | lubricating oil, at plant                                              | 4.47E-05   | kg                       | IPCC 2001             | 1    | 1.12               |                                                           |
|                                  | sheet rolling, chromium steel, Finishing (cutting, inspection, packing) | m      | Oil containing wastes are approximated by used mineral oil. Only the 22.8% being landfilled are inventoried                                                   | waste management  | hazardous waste incineration | No              | CH       | disposal, used mineral oil, 10% water, to hazardous waste incineration | 1.32E-05   | kg                       | IPCC 2001             | 1    | 1.12               | calculated from max value assuming lognormal distribution |
|                                  |                                                                         |        | Recycled --> not inventoried                                                                                                                                  |                   |                              |                 |          |                                                                        | n.a.       |                          | IPCC 2001             |      |                    |                                                           |
|                                  |                                                                         |        | Recycled --> not inventoried                                                                                                                                  |                   |                              |                 |          |                                                                        | 2.83E-03   | kg                       | IPCC 2001             | 1    | 1.41               |                                                           |
|                                  |                                                                         |        |                                                                                                                                                               | soil              | industrial                   |                 |          | Oils, unspecified                                                      | 3.85E-05   | kg                       | IPCC 2001             | 1    | 1.12               |                                                           |
|                                  |                                                                         |        | Lubricating oil as proxy                                                                                                                                      | chemicals         | organics                     | No              | RER      | lubricating oil, at plant                                              | 2.83E-04   | kg                       | IPCC 2001             | 1    | 1.41               |                                                           |
|                                  |                                                                         |        | Assumed as irrelevant --> input not inventoried, inventoried for transport and in a waste mix for waste disposal                                              |                   |                              |                 |          |                                                                        | 6.32E-06   | kg                       | IPCC 2001             | 1    | 12.6               |                                                           |
|                                  |                                                                         |        |                                                                                                                                                               | plastics          | processing                   | No              | RER      | packaging film, LDPE, at plant                                         | 8.49E-05   | kg                       | IPCC 2001             | 1    | 1.06               |                                                           |
|                                  |                                                                         |        | No description available in reference --> assumption kraft paper as proxy                                                                                     | paper & cardboard | packaging papers             | No              | RER      | kraft paper, unbleached, at plant                                      | 2.68E-05   | kg                       | IPCC 2001             | 1    | 2.98               |                                                           |
|                                  |                                                                         |        | No description available in reference --> assumption kraft paper as proxy                                                                                     | paper & cardboard | packaging papers             | No              | RER      | kraft paper, bleached, at plant                                        | 1.60E-06   | kg                       | IPCC 2001             | 1    | 1.06               |                                                           |
|                                  |                                                                         |        | Reference contained no further description of the product. Steel sheet was chosen as proxy. Volume was calculated with an assumed sheet-thickness of 0.55 mm. | metals            | extraction                   | No              | RER      | steel, electric, un- and low-alloyed, at plant                         | 2.36E-04   | kg                       | IPCC 2001             | 1    | 1.1                |                                                           |
|                                  |                                                                         |        |                                                                                                                                                               | metals            | processing                   | No              | RER      | sheet rolling, steel                                                   | 2.36E-04   | kg                       | IPCC 2001             | 1    | 1.1                |                                                           |
|                                  |                                                                         |        |                                                                                                                                                               | wooden materials  | extraction                   | No              | RER      | sawn timber, softwood, raw, air dried, u=20%, at plant                 | 1.85E-06   | m3                       | IPCC 2001             | 1    | 1.05               |                                                           |
|                                  |                                                                         |        |                                                                                                                                                               | electricity       | production mix               | No              | UCTE     | electricity, medium voltage, production UCTE, at grid                  | 7.86E-03   | kWh                      | IPCC 2001             | 1    | 1.41               |                                                           |
|                                  |                                                                         |        | Oil containing wastes are approximated by used mineral oil. Only the 22.8% being landfilled are inventoried                                                   | waste management  | hazardous waste incineration | No              | CH       | disposal, used mineral oil, 10% water, to hazardous waste incineration | 2.93E-05   | kg                       | IPCC 2001             | 1    | 1.07               |                                                           |
|                                  |                                                                         |        | recycled --> not inventoried                                                                                                                                  |                   |                              |                 |          |                                                                        | 3.16E-02   | kg                       | IPCC 2001             | 1    | 1.26               |                                                           |
|                                  |                                                                         |        | Oil contaminated water                                                                                                                                        |                   |                              |                 |          |                                                                        |            |                          |                       |      |                    |                                                           |
|                                  |                                                                         |        | Tempering / cooling fluid                                                                                                                                     |                   |                              |                 |          |                                                                        |            |                          |                       |      |                    |                                                           |
|                                  |                                                                         |        | Sludge cake                                                                                                                                                   |                   |                              |                 |          |                                                                        |            |                          |                       |      |                    |                                                           |
|                                  |                                                                         |        | Leakage oil                                                                                                                                                   |                   |                              |                 |          |                                                                        |            |                          |                       |      |                    |                                                           |
| Anticorrosive oils               | sheet rolling, chromium steel, Finishing (cutting, inspection, packing) | m      |                                                                                                                                                               |                   |                              |                 |          |                                                                        |            |                          |                       |      |                    | calculated from max value assuming lognormal distribution |
| Singing ink                      |                                                                         |        |                                                                                                                                                               |                   |                              |                 |          |                                                                        |            |                          |                       |      |                    |                                                           |
| Film (clear)                     |                                                                         |        |                                                                                                                                                               |                   |                              |                 |          |                                                                        |            |                          |                       |      |                    |                                                           |
| Paper (blue)                     |                                                                         |        |                                                                                                                                                               |                   |                              |                 |          |                                                                        |            |                          |                       |      |                    |                                                           |
| VCI Paper                        |                                                                         |        |                                                                                                                                                               |                   |                              |                 |          |                                                                        |            |                          |                       |      |                    |                                                           |
| FLEX-Stripes                     |                                                                         |        |                                                                                                                                                               |                   |                              |                 |          |                                                                        |            |                          |                       |      |                    |                                                           |
| FLEX-Stripes                     |                                                                         |        |                                                                                                                                                               |                   |                              |                 |          |                                                                        |            |                          |                       |      |                    |                                                           |
| Wood                             |                                                                         |        |                                                                                                                                                               |                   |                              |                 |          |                                                                        |            |                          |                       |      |                    |                                                           |
| Electrical energy                |                                                                         |        |                                                                                                                                                               |                   |                              |                 |          |                                                                        |            |                          |                       |      |                    |                                                           |
|                                  |                                                                         |        | Oil contaminated waste                                                                                                                                        |                   |                              |                 |          |                                                                        |            |                          |                       |      |                    |                                                           |
|                                  |                                                                         |        | Scrap                                                                                                                                                         |                   |                              |                 |          |                                                                        |            |                          |                       |      |                    |                                                           |

Fig. 2.4 Flows for "sheet rolling, chromium steel" and their representation in the ecoinvent database (cont.)



## 2. Steel forming

| General Flow Information  |                                                 |                         | Representation in ecoinvent                                                                                                                    |                   |                               |                 |          |                                                                                |            | Uncertainty Informations |                       |      |                    |                                                           |
|---------------------------|-------------------------------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------|-----------------|----------|--------------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------------------------------------|
| Input                     | Process Name                                    | Output                  | Remarks                                                                                                                                        | Category          | Sub-category                  | Infra-structure | Location | Modul name in ecoinvent                                                        | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                                           |
| Grinding emulsion (fresh) | sheet rolling, chromium steel, roll maintenance |                         | Lubricating oil as proxy                                                                                                                       | chemicals         | organics                      | No              | RER      | lubricating oil, at plant                                                      | 5.66E-06   | kg                       | IPCC 2001             | 1    | 14.1               | calculated from max value assuming lognormal distribution |
| Shot grain (SBT)          |                                                 |                         | One of the most important abrasive materials is taken as proxy                                                                                 | chemicals         | inorganics                    | No              | RER      | silicon carbide, at plant                                                      | 4.47E-05   | kg                       | IPCC 2001             | 1    | 1.12               |                                                           |
| Soluble oil concentrate   |                                                 |                         | Lubricating oil as proxy                                                                                                                       | chemicals         | organics                      | No              | RER      | lubricating oil, at plant                                                      | 5.48E-06   | kg                       | IPCC 2001             | 1    | 1.1                |                                                           |
| Chromium salt (PRETEX)    |                                                 |                         | Inorganic salt containing chromium as proxy                                                                                                    | paintings         | production                    | No              | RER      | wood preservative, inorganic salt, containing Cr, at plant                     | 1.50E-04   | kg                       | IPCC 2001             | 1    | 1.07               |                                                           |
|                           |                                                 | Dust2.5-10 to air       | Size distribution: mechanical process without burning process                                                                                  | air               | unspecified                   |                 |          | Particulates, > 2.5 um, and < 10um                                             | 2.09E-08   | kg                       | IPCC 2001             | 1    | 2.05               | (1,5,1,1,1,3,26)                                          |
|                           |                                                 | Dust>10                 | Size distribution: mechanical process without burning process                                                                                  | air               | unspecified                   |                 |          | Particulates, > 10 um                                                          | 2.09E-08   | kg                       | IPCC 2001             | 1    | 1.56               | (1,5,1,1,1,3,25)                                          |
|                           |                                                 | Chromium                |                                                                                                                                                | air               | unspecified                   |                 |          | Chromium                                                                       | 4.18E-09   | kg                       | IPCC 2001             | 1    | 5.06               | (1,5,1,1,1,3,22)                                          |
|                           |                                                 | Grinding Sludge         | Water treatment sludge as proxy                                                                                                                | w aste management | residual material landfill    | No              | CH       | disposal, sludge from steel rolling, 20% w ater, to residual material landfill | 1.41E-04   | kg                       | IPCC 2001             | 1    | 1.41               | calculated from max value assuming lognormal distribution |
|                           |                                                 | Grinding wheel          | inventoried in infrastructure module                                                                                                           |                   |                               |                 |          |                                                                                | 9.49E-07   | kg                       | IPCC 2001             | 1    | 1.05               |                                                           |
|                           |                                                 | Waste emulsion          | Oil containig w astes are approximated by used mineral oil. Only the 22.8% being landfilled are inventoried                                    | w aste management | hazardous w aste incineration | No              | CH       | disposal, used mineral oil, 10% w ater, to hazardous w aste incineration       | 1.21E-05   | kg                       | IPCC 2001             | 1    | 1.51               |                                                           |
|                           |                                                 | Oil contaminated w aste | Oil containig w astes are approximated by used mineral oil. Only the 22.8% being landfilled are inventoried                                    | w aste management | hazardous w aste incineration | No              | CH       | disposal, used mineral oil, 10% w ater, to hazardous w aste incineration       | 3.40E-05   | kg                       | IPCC 2001             | 1    | 1.15               |                                                           |
|                           |                                                 | SBT-dust2.5-10          | SBT (Shot Blast Texturing): This is a mechanical application of the texture onto the roll using grit blasting. Therefore the size distribution | air               | unspecified                   |                 |          | Particulates, > 2.5 um, and < 10um                                             | 2.24E-05   | kg                       | IPCC 2001             | 1    | 1.12               |                                                           |
|                           |                                                 | SBT-dust>10             | SBT (Shot Blast Texturing): This is a mechanical application of the texture onto the roll using grit blasting. Therefore the size distribution | air               | unspecified                   |                 |          | Particulates, > 10 um                                                          | 2.24E-05   | kg                       | IPCC 2001             | 1    | 1.12               |                                                           |

Fig. 2.4 Flows for "sheet rolling, chromium steel" and their representation in the ecoinvent database (cont.)

## 2. Steel forming

| General Flow Information     |                                                                           |                                  | Representation in ecoinvent                                                                                                                                                                                                                                                                                                                                      |                   |                              |                 |          |                                                                              |            | Uncertainty Informations |                       |      |                    |                                                           |
|------------------------------|---------------------------------------------------------------------------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------------------|-----------------|----------|------------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------------------------------------|
| Input                        | Process Name                                                              | Output                           | Remarks                                                                                                                                                                                                                                                                                                                                                          | Category          | Sub-category                 | Infra-structure | Location | Module name in ecoinvent                                                     | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                                           |
| Demineralized Water          | sheet rolling, chromium steel, Hydraulic fluids and lubricants management |                                  |                                                                                                                                                                                                                                                                                                                                                                  | water supply      | production                   | No              | CH       | water, deionised, at plant                                                   | 7.73E-02   | kg                       | IPCC 2001             | 1    | 1.55               | calculated from max value assuming lognormal distribution |
| Grease, oil, hydraulic oil   |                                                                           |                                  | Lubricating oil as proxy                                                                                                                                                                                                                                                                                                                                         | chemicals         | organics                     | No              | RER      | lubricating oil, at plant                                                    | 1.64E-03   | kg                       | IPCC 2001             | 1    | 1.1                |                                                           |
| Rolling oil                  |                                                                           |                                  | Lubricating oil as proxy                                                                                                                                                                                                                                                                                                                                         | chemicals         | organics                     | No              | RER      | lubricating oil, at plant                                                    | 9.95E-04   | kg                       | IPCC 2001             | 1    | 2.21               |                                                           |
| Hydraulic & lubrication oils |                                                                           |                                  | Lubricating oil as proxy                                                                                                                                                                                                                                                                                                                                         | chemicals         | organics                     | No              | RER      | lubricating oil, at plant                                                    | 4.24E-04   | kg                       | IPCC 2001             | 1    | 2.12               |                                                           |
| Electrical energy            |                                                                           |                                  |                                                                                                                                                                                                                                                                                                                                                                  | electricity       | production mix               | No              | UCTE     | electricity, medium voltage, production UCTE, at grid                        | 8.33E-02   | kWh                      | IPCC 2001             | 1    | 1.5                |                                                           |
| Steam                        |                                                                           |                                  | Steam used as energy source                                                                                                                                                                                                                                                                                                                                      | chemicals         | organics                     | No              | RER      | heat, unspecific, in chemical plant                                          | 4.47E-02   | MJ                       | IPCC 2001             | 1    | 1.12               |                                                           |
|                              | sheet rolling, chromium steel, Overall process                            | Disposed waste: rejects          | Rejects made mainly of steel. Only the 34% being landfilled are inventoried, the rest is recycled.                                                                                                                                                                                                                                                               | waste management  | municipal incineration       | No              | CH       | disposal, steel in car shredder residue, 0% water, to municipal incineration | 7.91E-03   | kg                       | IPCC 2001             | 1    | 2                  | (1,5,1,1,1,3,12)                                          |
|                              |                                                                           | Disposed waste: Used oil + water | Oil containing wastes are approximated by used mineral oil. Only the 22.8% being landfilled are inventoried                                                                                                                                                                                                                                                      | waste management  | hazardous waste incineration | No              | CH       | disposal, used mineral oil, 10% water, to hazardous waste incineration       | 7.63E-04   | kg                       | IPCC 2001             | 1    | 1.05               |                                                           |
| Cooling water                |                                                                           |                                  | Sum of the different process steps                                                                                                                                                                                                                                                                                                                               | resource          | in water                     |                 |          | Water, cooling, unspecified natural origin                                   | 5.32E-02   | m3                       | IPCC 2001             | 1    | 1.22               |                                                           |
|                              |                                                                           | Cooling and process water        | One part of the incoming cooling and process water leaves the system as not contaminated evaporated water or included in discarded waste --> not inventoried                                                                                                                                                                                                     |                   |                              |                 |          |                                                                              | 4.80E-02   | m3                       | IPCC 2001             | 1    | 1.22               |                                                           |
| Rolling Shop (CR)            |                                                                           |                                  |                                                                                                                                                                                                                                                                                                                                                                  | metals            | processing                   | Yes             | RER      | rolling mill                                                                 | 1.62E-09   | unit                     | Estimation            | 1    | 3.04               |                                                           |
| Lorry 32t                    |                                                                           |                                  | Includes the transports of all materials not produced at the plant and the transport to disposal of the waste. 32t-lorry was taken as a proxy for all this transports, therefore transports to the municipal incineration are overestimated by one third (not relevant for the overall process). Distance to hazardous waste disposal is inventoried with 100 km | transport systems | road                         | No              | RER      | transport, lorry 32t                                                         | 2.54E-02   | tkm                      | Estimation            | 1    | 1.05               |                                                           |
|                              |                                                                           | Waste-mix                        |                                                                                                                                                                                                                                                                                                                                                                  | waste management  | municipal incineration       | No              | CH       | disposal, municipal solid waste, 22.9% water, to municipal incineration      | 3.58E-04   | kg                       | EAA 2000              | 1    | 1.22               |                                                           |
|                              |                                                                           | Heat Waste                       |                                                                                                                                                                                                                                                                                                                                                                  | air               | unspecified                  |                 |          | Heat, waste                                                                  | 1.64E+00   | MJ                       | IPCC 2001             | 1    | 1.56               |                                                           |
|                              |                                                                           | Cold rolling sheet               |                                                                                                                                                                                                                                                                                                                                                                  | metals            | processing                   | No              | RER      | sheet rolling, chromium steel                                                | 1.00E+00   | kg                       |                       | 1    | 1.22               |                                                           |
|                              |                                                                           |                                  |                                                                                                                                                                                                                                                                                                                                                                  |                   |                              |                 |          |                                                                              |            |                          |                       |      |                    |                                                           |

Fig. 2.4 Flows for "sheet rolling, chromium steel" and their representation in the ecoinvent database (cont.)

## 2. Steel forming

| General Flow Information                       |                                                            |                        | Representation in ecoinvent                                                                                                                                                                                                                                                                                                                                                                             |                        |                            |                 |          |                                                                               |            | Uncertainty Informations |                       |      |                    |                                                           |
|------------------------------------------------|------------------------------------------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|----------------------------|-----------------|----------|-------------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------------------------------------|
| Input                                          | Process Name                                               | Output                 | Remarks                                                                                                                                                                                                                                                                                                                                                                                                 | Category               | Sub-category               | Infra-structure | Location | Modul name in ecoinvent                                                       | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                                           |
| Flocculating agents (Ca(OH) <sub>2</sub> :95%) | sheet rolling, chromium steel, Waste water treatment plant | Treated waste water    |                                                                                                                                                                                                                                                                                                                                                                                                         | construction materials | binder                     | No              | CH       | lime, hydrated, loose, at plant                                               | 4.74E-05   | kg                       | IPCC 2001             | 1    | 1.2                | calculated from max value assuming lognormal distribution |
| Polyelectrolytes                               |                                                            |                        | Personal communication by G. Doka --> inorganic chemicals as proxy                                                                                                                                                                                                                                                                                                                                      | chemicals              | inorganics                 | No              | GLO      | chemicals inorganic, at plant                                                 | 1.38E-08   | kg                       | IPCC 2001             | 1    | 2                  |                                                           |
|                                                |                                                            |                        | One part of the incomin cooling and process water leaves the system as treated effluent; it is assumed to be discharged in surface water --> not inventoried. Infrastructure for waste water treatment plant is comprised in the infrastructure module for the whole rolling mill plant. Value for energy use is comprised in the broad range of uncertainty the energy data for the rolling mill have. |                        |                            |                 |          |                                                                               | 5.25E-03   | m <sup>3</sup>           | IPCC 2001             |      |                    |                                                           |
|                                                |                                                            |                        |                                                                                                                                                                                                                                                                                                                                                                                                         | water                  | unspecified                |                 |          | Suspended solids, unspecified                                                 | 1.39E-04   | kg                       | IPCC 2001             | 1    | 1.57               | (2,5,1,1,1,3,32)                                          |
|                                                |                                                            |                        |                                                                                                                                                                                                                                                                                                                                                                                                         | water                  | unspecified                |                 |          | COD, Chemical Oxygen Demand                                                   | 3.61E-05   | kg                       | IPCC 2001             | 1    | 1.57               | (2,5,1,1,1,3,32)                                          |
|                                                |                                                            |                        |                                                                                                                                                                                                                                                                                                                                                                                                         | water                  | unspecified                |                 |          | BOD5, Biological Oxygen Demand                                                | 1.65E-06   | kg                       | IPCC 2001             | 1    | 1.2                | (5,5,5,5,5,5,32)                                          |
|                                                |                                                            |                        |                                                                                                                                                                                                                                                                                                                                                                                                         | water                  | unspecified                |                 |          | DOC, Dissolved Organic Carbon                                                 | 2.54E-07   | kg                       | IPCC 2001             | 1    | 2                  | (5,5,5,5,5,5,32)                                          |
|                                                |                                                            |                        |                                                                                                                                                                                                                                                                                                                                                                                                         | water                  | unspecified                |                 |          | TOC, Total Organic Carbon                                                     | 1.96E-07   | kg                       | IPCC 2001             | 1    | 2.79               | (5,5,5,5,5,5,32)                                          |
|                                                |                                                            |                        |                                                                                                                                                                                                                                                                                                                                                                                                         | water                  | unspecified                |                 |          | Hydrocarbons, unspecified                                                     | 1.65E-06   | kg                       | IPCC 2001             | 1    | 3.05               | (2,5,1,1,1,3,34)                                          |
|                                                |                                                            |                        |                                                                                                                                                                                                                                                                                                                                                                                                         | water                  | unspecified                |                 |          | Copper, ion                                                                   | 2.54E-07   | kg                       | IPCC 2001             | 1    | 1.86               | (2,5,1,1,1,3,37)                                          |
|                                                |                                                            |                        |                                                                                                                                                                                                                                                                                                                                                                                                         | water                  | unspecified                |                 |          | Zinc, ion                                                                     | 1.96E-07   | kg                       | IPCC 2001             | 1    | 1.86               | (2,5,1,1,1,3,37)                                          |
|                                                |                                                            |                        |                                                                                                                                                                                                                                                                                                                                                                                                         | water                  | unspecified                |                 |          | Cadmium, ion                                                                  | 1.31E-07   | kg                       | IPCC 2001             | 1    | 1.86               | (2,5,1,1,1,3,37)                                          |
|                                                |                                                            |                        |                                                                                                                                                                                                                                                                                                                                                                                                         | water                  | unspecified                |                 |          | Aluminum                                                                      | 1.24E-06   | kg                       | IPCC 2001             | 1    | 1.86               | (2,5,1,1,1,3,37)                                          |
|                                                |                                                            |                        |                                                                                                                                                                                                                                                                                                                                                                                                         | water                  | unspecified                |                 |          | Lead                                                                          | 2.63E-07   | kg                       | IPCC 2001             | 1    | 1.86               | (2,5,1,1,1,3,37)                                          |
|                                                |                                                            |                        |                                                                                                                                                                                                                                                                                                                                                                                                         | water                  | unspecified                |                 |          | Chromium, ion                                                                 | 4.73E-07   | kg                       | IPCC 2001             | 1    | 1.86               | (2,5,1,1,1,3,37)                                          |
|                                                |                                                            |                        |                                                                                                                                                                                                                                                                                                                                                                                                         | water                  | unspecified                |                 |          | Chromium VI                                                                   | 2.63E-08   | kg                       | IPCC 2001             | 1    | 1.86               | (2,5,1,1,1,3,37)                                          |
|                                                |                                                            |                        |                                                                                                                                                                                                                                                                                                                                                                                                         | water                  | unspecified                |                 |          | Manganese                                                                     | 5.35E-07   | kg                       | IPCC 2001             | 1    | 1.86               | (2,5,1,1,1,3,37)                                          |
|                                                |                                                            |                        |                                                                                                                                                                                                                                                                                                                                                                                                         | water                  | unspecified                |                 |          | Iron, ion                                                                     | 4.07E-06   | kg                       | IPCC 2001             | 1    | 1.86               | (2,5,1,1,1,3,37)                                          |
|                                                |                                                            |                        |                                                                                                                                                                                                                                                                                                                                                                                                         | water                  | unspecified                |                 |          | Nickel, ion                                                                   | 7.42E-07   | kg                       | IPCC 2001             | 1    | 1.86               | (2,5,1,1,1,3,37)                                          |
|                                                |                                                            |                        |                                                                                                                                                                                                                                                                                                                                                                                                         | water                  | unspecified                |                 |          | Mercury                                                                       | 2.63E-08   | kg                       | IPCC 2001             | 1    | 1.86               | (2,5,1,1,1,3,37)                                          |
|                                                |                                                            |                        |                                                                                                                                                                                                                                                                                                                                                                                                         | water                  | unspecified                |                 |          | Chloride                                                                      | 1.17E-06   | kg                       | IPCC 2001             | 1    | 1.57               | (2,5,1,1,1,3,33)                                          |
|                                                |                                                            | Water treatment sludge | Assumption: same amount per cubic meter as for waste water from cold steel rolling. Only a part of the sludge goes to landfill --> 66.5% inventoried                                                                                                                                                                                                                                                    | waste management       | residual material landfill | No              | CH       | disposal, sludge from steel rolling, 20% water, to residual material landfill | 1.75E-02   | kg                       | IPCC 2001             | 1    | 1.57               | (2,5,1,1,1,3,43)                                          |

Fig. 2.4 Flows for "sheet rolling, chromium steel" and their representation in the ecoinvent database (cont.)

## 2.5 Wire drawing of steel and chromium steel

Wire drawing is a process in which wire rods/wires are reduced in size by drawing them through cone-shaped openings of a smaller cross section, so called dies. The input usually is wire rod of diameters ranging from 5.5 to 16 mm obtained from hot rolling mills in form of coils.

The European Union has the world largest wire drawing industry, followed by Japan and North America. It produces about 6 million tonnes of wire per year. Including the various wire products, such as barbed wire, grill, fencing, netting, nails etc, the production of the sector reaches more than 7 million tonnes per year. The largest producer of steel wire is Germany with 32 % (about 1.09 million tonnes) of EU wire production, followed by Italy (approx. 22 %, 1.2 million tonnes), UK, Benelux (mainly Belgium), France and Spain.

A typical wire drawing process line comprises the following steps (IPPC (2001a)):

- Pre-treatment of the wire rod (mechanical descaling, pickling).
- Dry or wet drawing (usually several drafts with decreasing die sizes).
- In some cases heat treatment (continuous-/discontinuous annealing, patenting, oil hardening).
- Finishing.

The final diameter size of dry drawn wire is between one and two millimetres, wet drawn wire has an even smaller diameter. A short description of every single process step is given below.

Wire is manufactured in different grades of steel: low carbon steel with carbon content of up to 0.25 %, high carbon steel with a carbon content of over 0.25 %, stainless and other alloy steel. Non-alloy steel wire can be uncoated or coated with zinc, copper, brass, tin, nickel, chrome, plastic or varnish. Wire is sent as coils to further processing, like coating and manufacturing of finished products (e.g. cable, mesh, barbed wire, wire fencing, grill, springs, nails).

The infrastructure of steel and chromium steel wire drawing is described in 1.1. It comprises buildings where production processes take place and for storage, surroundings and machinery for the single process steps (e.g.: furnace, rolls, cutting devices, etc.).

### 2.5.1 Pre-treatment of the wire rod

Pre-treatment of wire rod is described in (IPPC (2001a)).

"Air-cooling after rolling produces an iron oxide-layer (scale) on the wire rod surface. This layer is very hard, brittle and not deformable and must therefore be removed before any further processing can take place. Two techniques for descaling wire rods are applied: mechanical descaling and chemical pickling.

#### **Mechanical descaling of wire rod**

In the most common mechanical descaling method applied: wire bending, the wire is bent to cause the brittle scale to peel off. Other descaling techniques, like sanding, brushing or shot blasting, are used as a finishing step after wire bending to remove loose scale or are used as stand-alone techniques. Wire bending, generates a relatively coarse scale, which generates only negligible amounts of dust.

#### **Chemical descaling (pickling) of wire rod**

In chemical descaling the mill scale is removed by dissolving in acid. Acids used are:

- Sulphuric or hydrochloric acid for low carbon wire.
- Hydrochloric acid for high carbon wire.

Pickling is mostly done in a batch process. Each roll of wire rod is immersed in the acid bath. The acid slowly dissolves the oxide layer by transforming it into iron chlorides or sulphates. When pickling in HCl an  $H_2$ -inhibitor is used to suppress the reaction:  $2\text{HCl} + \text{Fe} \rightarrow \text{FeCl}_2 + H_2$  and reduce the unwanted loss of iron. After pickling the wire rod coils are rinsed in water. This is usually done in cascade rinsing (e.g. rinsing 3 times) to yield maximum rinsing efficiency and minimum water consumption.

### **Application of Soap (Lubricant) Carrier**

In some cases, soap carrier is applied to enhance the adhesion of the lubricant to the wire. A wide range of soap carriers is available. Selection is made on economical basis and on required properties in downstream processes. Traditional soap carriers include lime, borax and Zn-phosphate. Modern soap carriers are typically mixtures of soluble salts, e.g. sodium and potassium sulphate, chloride, borax, phosphate or silicate. They are adapted to a particular soap and a particular drawing situation. Soap carrier is applied before drawing by dipping the wire in a watery soap carrier solution. This can be done in a batch process, usually associated with chemical descaling, or continuous in association with mechanical descaling.

Consumption and emissions associated with the application of soap carrier vary in dependence of the type of soap carrier, thickness applied and the wire rod diameter. No quantitative data is available at the moment.

If  $\text{Zn-PO}_4$  coating is used, a thick  $\text{FePO}_4$  sludge or cake is formed containing traces of zinc, soluble phosphates and chlorate or nitrite coming from the  $\text{Zn-PO}_4$  bath. Also spent soap carrier bath may arise as liquid waste. If drying is done in gas - or fuel fired ovens, the usual combustion products are emitted."

### **2.5.2 Dry drawing of wire**

Dry drawing is typically used to draw wire rod ( $> 5.5\text{ mm}$ ) to a product diameter of 1 - 2 mm. The diameter of wire is reduced by passing it through a series of dies with diminishing diameter. Before entering the dies, the wire is passed through a dry lubricant. In most cases, soap-based lubricants are used with the soap formulation depending on choice of fats from which the soap is produced, choice of fillers or choice of additives. In exceptional cases (e.g. special steels, wire with special metal coatings) other lubricants like pastes or oils can be used. The drawing operation heats both (wire and drawing die) through friction of the wire. Cooling is done indirectly, by cooling the capstans in contact with the wire with water (IPPC (2001a)).

### **2.5.3 Wet drawing of wire**

Wet drawing is typically used to draw wire with an intermediate product diameter of 1 - 2 mm to final diameter. The wire is also passed through a series of dies with diminishing diameter, but wire, dies and capstans are immersed in a lubricant liquid that provides lubrication and cooling. Typically, soap or oil emulsions (for some applications plain oil) are used. The heat resulting from drawing operation is taken up by the lubricant, which is cooled indirectly with water (IPPC (2001a)).

### **2.5.4 Heat treatment of wire**

Heat treatment of wire has different goals. Therefore different types of heat treatment are applied depending on the type of steel (low carbon / high carbon / stainless) and the final use (desired ductility and strength). Heat treatment also removes soap and lubricant residues thermally. The following heat treatment types are applied (IPPC (2001a)):

- Batch annealing,
- Continuous (In-Line) annealing,
- Patenting,
- Oil hardening and tempering,
- Stress-relieving.

A considerable percentage of the output of the wire industry does not need any heat-treatment. The heavy deformation of the metal crystal structure caused by drawing increases the hardness and the strength of the wire in the axial direction (IPPC (2001a)).

### **Batch annealing**

Drawing profoundly deforms the shape of the metal crystals of the wire. Annealing is one of the different methods to re-obtain a suited crystal shape. Batch annealing, typically used for low carbon steel wire to obtain a very soft and ductile end product, is done in bell or pot furnaces, in chambers filled with a either neutral or reducing protective gas (nitrogen, hydrogen, nitrogen/hydrogen mixtures and partly oxidized natural gas) and typically heated by gas or fuel. The warming up from room to peak temperature (roughly 700 °C) takes several hours; also cooling down again takes several hours. In order to keep an over-pressure in the "pots" or "bells", a fraction of the protective gas is continuously purged. In some cases, the wire is oiled immediately after annealing (IPPC (2001a)).

### **Continuous (In-Line) annealing**

Continuous annealing or strand-annealing has a similar goal as batch annealing: restore a suited crystal texture to the steel in the wire after drawing. But the desired crystal shape and metal properties are different from those that are desired when batch-annealing is applied. Continuous annealing is a typical heat treatment method for low carbon products. Strand-annealing is a fast continuous process. The wire is heated up to the recrystallisation temperature (500 - 700 °C) commonly by passing through a molten lead bath, is kept at this temperature for a few seconds, and is cooled down again by quenching in a water bath. This can be followed by in-line pickling with warm or cold HCl, other acids or electrolytes, to remove oxides which also partly dissolves the potentially dragged lead. Pickling is followed by a rinsing cascade (IPPC (2001a)).

For stainless steel, the heat-treatment is done under a protective gas atmosphere as if no protective atmosphere would be used, the stainless steel wire would be oxidized; removal of these oxides would require pickling with special acids. The heating can be done by electricity (resistance heating, inductive heating) or by combustion (IPPC (2001a)).

### **Patenting**

Patenting is a heat-treatment method typically used for high-carbon and alloyed steel products to create a special crystal structure, which allows easy further deformation. In contrast to annealing where iron and iron/carbon compounds tend to be separated, patenting yields a structure where the carbon is homogeneously distributed in the iron. Patenting is done by heating the wire to 850 - 1000 °C, then cooling fast to 450 - 600 °C and keeping the wire for a while at this temperature, and finally quenching in water (IPPC (2001a)).

### **Oil hardening and tempering**

Oil hardening and tempering creates a special crystal structure in the steel with a high percentage of martensite, resulting in increased hardness and wear resistance combined with good toughness. First, the wire is heated to 850 – 1000 °C, followed by rapid cooling. Oil-hardening is always followed by a tempering or stress-relieving step, in order to remove the stresses caused by the extremely fast cooling. This is done by heating the wire again to 300 – 500 °C. Typically a normal oven is used, with electrical heating or direct heating with a typical combustion gas, but also inductive heating can be used (IPPC (2001a)).

### **Stress-relieving**

The goal of stress-relieving is to remove internal stresses in the wire caused by a previous processing steps, without changing the shape and structure of the steel crystal. The internal stresses can be caused by deformation (mechanical stresses) or by fast cooling (thermal stresses). Stress-relieving as a stand-alone heat-treatment step is a typical process step for the production of PC-strand (wire for prestressed concrete). Stress-relieving can be done at various temperatures (200 - 500 °C), dependent on the desired characteristics of the final product. Typically a normal oven is used, with electrical heating or direct heating with a typical combustion gas, but also inductive heating can be used. After stress-relieving, the wire is cooled relatively slowly in air or water (IPPC (2001a)).

## **2.5.5 Environmental issues of wire drawing of steel**

The main environmental aspects of wire drawing are: air emissions from pickling, acidic wastes and waste water; fugitive soap dust (dry drawing), spent lubricant and effluents (wet drawing), combustion gas from furnaces and emissions and lead-containing wastes from lead baths (IPPC (2001a)).

## **2.5.6 Wire drawing of steel and chromium steel in ecoinvent**

For mechanical wire rod preparation, only data about wire bending were available. No data were available about the application of soap carrier, for batch annealing, for oil hardening and tempering, annealing of stainless steel wire and stress relieving. For the heat treatment available data were inventoried only with 25 %, because 75 % of drawn steel underlies no further tempering.

The output is wire drawing for 1 kg of product. If coated wire is needed, one has to apply an additional module, e.g. "zinc coating, coils" or "zinc coating, pieces".

Mean values are generally calculated as the geometric mean of the range given in IPPC (2001a). Where the range is given from zero to a maximal value, and the geometric mean can't therefore be calculated, the arithmetic mean is taken. The background information to the process "wire drawing, steel" is given in Tab. 2.5, flows are shown in Fig. 2.5.

Wastewater is estimated based on the wastewater from hot and cold rolling.

### **Data quality**

Data is taken almost exclusively taken from IPPC (2001a), some estimations are also made. The values can vary considerably depending on the degree of deformation.

Uncertainty is calculated assuming a lognormal distribution and the high value given in IPPC (2001a) representing the 95 % geometric standard deviation. Where no range is given in IPPC (2001a), the data quality is estimated by mean of the pedigree matrix, according to (Frischknecht et al. (2003)).

Tab. 2.5 Ecoinvent meta information for "wire drawing, steel"

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name                                  | wire drawing, steel                                                                                                                                                                                                                                                                                                                                                                                                   |
| Location                              | RER                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Infrastructure Process                | 0                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Unit                                  | kg                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Data Set Version                      | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Included Processes                    | Includes the process steps pre-treatment of the wire rod (mechanical descaling, pickling), dry or wet drawing (usually several drafts with decreasing die sizes), in some cases heat treatment (continuous-/discontinuous annealing, patenting, oil hardening) and Finishing. Does not include coating and the material being rolled                                                                                  |
| Amount                                | 1                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Local Name                            | Draht ziehen, Stahl                                                                                                                                                                                                                                                                                                                                                                                                   |
| Synonyms                              |                                                                                                                                                                                                                                                                                                                                                                                                                       |
| General Comment to reference function | Wire drawing is a process in which wire rods/wires are reduced in diameter by drawing them through cone-shaped openings of a smaller cross section, so called dies. The input usually is wire rod of diameters ranging from 5.5 to 16 mm obtained from hot rolling mills in form of coils. The final diameter size of dry drawn wire is between one and two millimetres, wet drawn wire has an even smaller diameter. |
| Start Date                            | 1997                                                                                                                                                                                                                                                                                                                                                                                                                  |
| End Date                              | 2002                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Other Period Text                     |                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Geography text                        | Data-set is representative for European Union                                                                                                                                                                                                                                                                                                                                                                         |
| Technology text                       | Average technique for EU. The processes of steel and stainless steel aren't fundamentally different, thus this module covers both materials                                                                                                                                                                                                                                                                           |
| Representativeness [%]                |                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Production Volume                     | 6 to 7 million tonnes in 1996                                                                                                                                                                                                                                                                                                                                                                                         |
| Sampling Procedure                    | Literature                                                                                                                                                                                                                                                                                                                                                                                                            |
| Extrapolations                        | none                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Uncertainty Adjustments               | The fact that no data were available about the percentage of companies represented by the study cited is considered in the uncertainties.                                                                                                                                                                                                                                                                             |



## 2. Steel forming

| General Flow Information |                           |                                                      | Representation in ecoinvent                                                                                                       |                  |                            |                 |          |                                                                                |            | Uncertainty Informations |                       |      |                    |                                                           |
|--------------------------|---------------------------|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------------|-----------------|----------|--------------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------------------------------------|
| Input                    | Process Name              | Output                                               | Remarks                                                                                                                           | Category         | Sub-category               | Infra-structure | Location | Modul name in ecoinvent                                                        | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                                           |
| primary steel            | wire drawing, steel, Wire | Scale (FeO and Fe3O4)                                | Only part of the steel that leaves the system as scrap to recycling is inventoried. Assumed to be 50% of scrap from sheet rolling | metals           | extraction                 | No              | RER      | steel, converter, unalloyed, at plant                                          | 4.28E-02   | kg                       | Estimation            | 1    | 1.31               | (4,5,1,1,1,3,3)                                           |
|                          |                           |                                                      | Coarse scale, amount of dust is negligible. BOF waste that is used as proxy contains more CaO than is expected for the scale.     | waste management | residual material landfill | No              | CH       | disposal, basic oxygen furnace wastes, 0% water, to residual material landfill | 3.16E-03   | kg                       | IPCC 2001             | 1    | 1.58               | calculated from max value assuming lognormal distribution |
| HCL (32%)                |                           |                                                      | Value consider 50% of plants with, and 50% of plants without acid recovery                                                        | chemicals        | inorganics                 | No              | RER      | hydrochloric acid, 30% in H2O, at plant                                        | 1.21E-02   | kg                       | IPCC 2001             | 1    | 1.29               |                                                           |
| H2SO4 (98%)              |                           |                                                      | Value consider 50% of plants with, and 50% of plants without acid recovery                                                        | chemicals        | inorganics                 | No              | RER      | sulphuric acid, liquid, at plant                                               | 8.84E-03   | kg                       | IPCC 2001             | 1    | 1.41               |                                                           |
| H2 inhibitor             |                           |                                                      |                                                                                                                                   | chemicals        | organics                   | No              | GLO      | chemicals organic, at plant                                                    | 3.16E-05   | kg                       | IPCC 2001             | 1    | 3.16               |                                                           |
|                          |                           | HCL pickling (acid vapour, droplets) to air          | Assumption: the same percentage of the input is evaporated as it is in the pickling process of cold sheet rolling                 | air              | unspecified                |                 |          | Hydrogen chloride                                                              | 3.93E-06   | kg                       | IPCC 2001             | 1    | 1.62               | (1,5,1,1,3,3,31)                                          |
|                          |                           |                                                      |                                                                                                                                   | air              | unspecified                |                 |          | Hydrogen                                                                       | 1.54E-09   | kg                       | IPCC 2001             | 1    | 1.62               | (1,5,1,1,3,3,31)                                          |
|                          |                           |                                                      |                                                                                                                                   | air              | unspecified                |                 |          | Sulfate                                                                        | 7.32E-08   | kg                       | IPCC 2001             | 1    | 1.31               | (1,5,1,1,3,3,15)                                          |
|                          |                           |                                                      |                                                                                                                                   | air              | unspecified                |                 |          | Sulfur dioxide                                                                 | 6.68E-07   | kg                       | IPCC 2001             | 1    | 1.31               | (1,5,1,1,3,3,15)                                          |
|                          |                           |                                                      |                                                                                                                                   |                  |                            |                 |          |                                                                                |            |                          |                       |      |                    |                                                           |
|                          |                           | SO4 from H2SO4 pickling (aerosols, droplets of acid) | Value consider 50% of plants with, and 50% of plants without acid recovery                                                        | water            | unspecified                |                 |          | Chloride                                                                       | 9.98E-06   | kg                       | IPCC 2001             | 1    | 1.37               | calculated from max value assuming lognormal distribution |
|                          |                           |                                                      | Value consider 50% of plants with, and 50% of plants without acid recovery                                                        | water            | unspecified                |                 |          | Sulfate                                                                        | 7.35E-06   | kg                       | IPCC 2001             | 1    | 1.5                |                                                           |
|                          |                           |                                                      | Negligible if correct H2-inhibitor is used                                                                                        |                  |                            |                 |          |                                                                                | 4.46E-07   | kg                       | IPCC 2001             | 1    | 1.41               |                                                           |
|                          |                           |                                                      | Value consider 50% of plants with, and 50% of plants without acid recovery                                                        | water            | unspecified                |                 |          | Iron, ion                                                                      | 9.34E-06   | kg                       | IPCC 2001             | 1    | 1.57               |                                                           |
|                          |                           | SO2 from H2SO4 pickling (aerosols, droplets of acid) |                                                                                                                                   |                  |                            |                 |          |                                                                                |            |                          |                       |      |                    |                                                           |
|                          |                           | Cl- from spent HCL                                   |                                                                                                                                   |                  |                            |                 |          |                                                                                |            |                          |                       |      |                    |                                                           |
|                          |                           | SO4-- from spent H2SO4                               |                                                                                                                                   |                  |                            |                 |          |                                                                                |            |                          |                       |      |                    |                                                           |
|                          |                           | H+ from spent acid                                   |                                                                                                                                   |                  |                            |                 |          |                                                                                |            |                          |                       |      |                    |                                                           |
|                          |                           | Fe++ from spent acid                                 |                                                                                                                                   |                  |                            |                 |          |                                                                                |            |                          |                       |      |                    |                                                           |

Fig. 2.5 Flows for "wire drawing, steel" and their representation in the ecoinvent database

## 2. Steel forming

| General Flow Information                                       |                                  |                                                                                   | Representation in ecoinvent                                                                                      |                   |                               |                 |          |                                                                          |            | Uncertainty Informations |                       |      |                    |                                                           |
|----------------------------------------------------------------|----------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------|-----------------|----------|--------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------------------------------------|
| Input                                                          | Process Name                     | Output                                                                            | Remarks                                                                                                          | Category          | Sub-category                  | Infra-structure | Location | Modul name in ecoinvent                                                  | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                                           |
| Dry draw ing lubricant (preformulated alkali soap + additives) | wire drawing, steel, Dry drawing | Spent dry draw ing lubricant                                                      | Assumption: lubricant is composed 50-50 of alkali soap and lubricating oil                                       | w ashing agents   | tensides                      | No              | RER      | soap, at plant                                                           | 1.00E-03   | kg                       | IPCC 2001             | 1    | 2                  | calculated from max value assuming lognormal distribution |
| Dry draw ing lubricant (lubricating oil)                       |                                  |                                                                                   |                                                                                                                  | chemicals         | organics                      | No              | RER      | lubricating oil, at plant                                                | 1.00E-03   | kg                       | IPCC 2001             | 1    | 2                  |                                                           |
|                                                                |                                  |                                                                                   | Oil contaminated w aste is approximated by used mineral oil                                                      | w aste management | hazardous w aste incineration | No              | CH       | disposal, used mineral oil, 10% w ater, to hazardous w aste incineration | 2.00E-03   | kg                       | IPCC 2001             | 1    | 2                  |                                                           |
|                                                                |                                  |                                                                                   |                                                                                                                  |                   |                               |                 |          |                                                                          | n.a.       |                          | IPCC 2001             |      |                    |                                                           |
|                                                                | wire drawing, steel, Wet drawing | Spent Wet Draw ing Emulsion (including sludge removed by filtration/decant ation) |                                                                                                                  |                   |                               |                 |          |                                                                          | negligible |                          | IPCC 2001             |      |                    | calculated from max value assuming lognormal distribution |
| Preformulated concentrated soap emulsion                       |                                  |                                                                                   |                                                                                                                  | w ashing agents   | tensides                      | No              | RER      | soap, at plant                                                           | 1.58E-03   | kg                       | IPCC 2001             | 1    | 3.16               |                                                           |
| Preformulated concentrated oil emulsion                        |                                  |                                                                                   |                                                                                                                  | chemicals         | organics                      | No              | RER      | lubricating oil, at plant                                                | 1.58E-03   | kg                       | IPCC 2001             | 1    | 3.16               |                                                           |
|                                                                |                                  |                                                                                   | Oil contaminated w aste (containing COD (20-80 g/l) and Metals (0.2-1 g/l)) is approximated by used mineral oil. | w aste management | hazardous w aste incineration | No              | CH       | disposal, used mineral oil, 10% w ater, to hazardous w aste incineration | 6.08E-02   | kg                       | IPCC 2001             | 1    | 3.54               |                                                           |
|                                                                |                                  | Broken dies                                                                       |                                                                                                                  |                   |                               |                 |          |                                                                          | negligible |                          | IPCC 2001             |      |                    |                                                           |

Fig. 2.5 Flows for "wire drawing, steel" and their representation in the ecoinvent database (cont.)

## 2. Steel forming

| General Flow Information                |                                           |                                              | Representation in ecoinvent                                                                                                                                                                                                                                                                           |                   |                            |                 |          |                                                                                 |            | Uncertainty Informations |                       |      |                    |                                                           |
|-----------------------------------------|-------------------------------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------------------------|-----------------|----------|---------------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------------------------------------|
| Input                                   | Process Name                              | Output                                       | Remarks                                                                                                                                                                                                                                                                                               | Category          | Sub-category               | Infra-structure | Location | Modul name in ecoinvent                                                         | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                                           |
| Lead                                    | wire drawing, steel, Continuous annealing | Lead w aste (lead oxides + spent bath cover) | Only an estimated 25% of the output of the wire industry is annealed.                                                                                                                                                                                                                                 | metals            | extraction                 | No              | RER      | lead, at regional storage                                                       | 9.68E-04   | kg                       | IPCC 2001             | 1    | 3.87               | calculated from max value assuming lognormal distribution |
| Energy for heating the lead bath - gas  |                                           | SS in quench water overflow                  | energy consumption calculated as energy needed for heating up the lead bath from ambient to melting temperature Overall efficiency of 25% assumed. Assumption: 50% of plants using gas and 50% of plants using fuel.                                                                                  | natural gas       | heating systems            | No              | RER      | natural gas, burned in industrial furnace >100kW                                | 7.03E-05   | MJ                       | IPCC 2001             | 1    | 1.38               | (4,5,1,1,3,3,1)                                           |
| Energy for heating the lead bath - fuel |                                           | Pb in quench water overflow                  |                                                                                                                                                                                                                                                                                                       | oil               | heating systems            | No              | RER      | light fuel oil, burned in industrial furnace 1MW, non-modulating                | 7.03E-05   | MJ                       | IPCC 2001             | 1    | 1.38               | (4,5,1,1,3,3,1)                                           |
|                                         |                                           | Pb to air                                    | Only an estimated 25% of the output of the wire industry is annealed--> only 25% inventoried as dust from EAF process. The proxy chosen to represent the waste probably overestimates some heavy metals (e.g. Cr).                                                                                    | w aste management | residual material landfill | No              | CH       | disposal, dust, unalloyed EAF steel, 15.4% water, to residual material landfill | 9.68E-04   | kg                       | IPCC 2001             | 1    | 3.87               | calculated from max value assuming lognormal distribution |
|                                         |                                           | Dust to air                                  | inventoried in the water treatment plant                                                                                                                                                                                                                                                              |                   |                            |                 |          |                                                                                 | n.a.       |                          | IPCC 2001             |      |                    |                                                           |
|                                         |                                           | TOC to air                                   | inventoried in the water treatment plant                                                                                                                                                                                                                                                              |                   |                            |                 |          |                                                                                 | 1.77E-06   | kg                       | IPCC 2001             | 1    | 3.16               |                                                           |
|                                         |                                           | Heating emissions                            | Reference contained no data about the air flow of waste air. Thus the amount of exhaust gas from HCL-pickling tanks from cold sheet rolling is assumed. Only an estimated 25% of the output of the wire industry is annealed.                                                                         | air               | unspecified                |                 |          | Lead                                                                            | 7.91E-09   | kg                       | IPCC 2001             | 1    | 5.11               | (2,5,1,1,3,3,22)                                          |
|                                         |                                           |                                              | Reference contained no data about the air flow of waste air. Thus the amount of exhaust gas from HCL-pickling tanks from cold sheet rolling is assumed. Only an estimated 25% of the output of the wire industry is annealed. Assumption: Dust emissions from lead bath are between 2.5 and 10 um.    | air               | unspecified                |                 |          | Particulates, > 2.5 um, and < 10um                                              | 1.37E-07   | kg                       | IPCC 2001             | 1    | 1.62               | (2,5,1,1,3,3,25)                                          |
|                                         |                                           |                                              | Reference contained no data about the air flow of waste air. Thus the amount of exhaust gas from HCL-pickling tanks from cold sheet rolling is assumed. Only an estimated 25% of the output of the wire industry is annealed. NMVOC used as proxy with the assumption that C-content of NMVOC is 80%. | air               | unspecified                |                 |          | NMVOC, non-methane volatile organic compounds, unspecified origin               | 2.21E-07   | kg                       | IPCC 2001             | 1    | 2.1                | (2,5,1,1,3,3,23)                                          |
|                                         |                                           |                                              | Emissions arising from fuel and gas consumption are included in the inventoried burner modules                                                                                                                                                                                                        |                   |                            |                 |          |                                                                                 | n.a.       |                          | IPCC 2001             |      |                    |                                                           |

Fig. 2.5 Flows for "wire drawing, steel" and their representation in the ecoinvent database (cont.)

## 2. Steel forming

| General Flow Information                    |                                             |                                                                                            | Representation in ecoinvent                                                                                                                                                                       |             |                 |                 |          |                                                  |            | Uncertainty Informations |                       |      |                    |                                                           |
|---------------------------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------|-----------------|----------|--------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------------------------------------|
| Input                                       | Process Name                                | Output                                                                                     | Remarks                                                                                                                                                                                           | Category    | Sub-category    | Infra-structure | Location | Modul name in ecoinvent                          | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                                           |
| HCL (32% HCL)                               | in-line HCL pickling - continuous annealing | Total Cl-<br>Total H+<br>Fe<br>HCl (emissions from scrubber)<br>Lubricant burn-off<br>Dust | Only an estimated 25% of the output of the wire industry is annealed.                                                                                                                             | chemicals   | inorganics      | No              | RER      | hydrochloric acid, 30% in H2O, at plant          | 7.91E-03   | kg                       | IPCC 2001             | 1    | 3.16               | calculated from max value assuming lognormal distribution |
| Energy for heating the HCl-bath (gas, fuel) |                                             |                                                                                            | Often the bath is heated by residual heat in the wire, therefore, as reference contained no data -> not inventoried                                                                               |             |                 |                 |          |                                                  | n.a.       |                          | IPCC 2001             |      |                    |                                                           |
|                                             |                                             |                                                                                            | Only an estimated 25% of the output of the wire industry is annealed.                                                                                                                             | water       | unspecified     |                 |          | Chloride                                         | 1.14E+00   | kg                       | IPCC 2001             | 1    | 1.35               | calculated from max value assuming lognormal distribution |
|                                             |                                             |                                                                                            | Only an estimated 25% of the output of the wire industry is annealed.                                                                                                                             |             |                 |                 |          |                                                  | 3.32E-05   | kg                       | IPCC 2001             | 1    | 1.35               |                                                           |
|                                             |                                             |                                                                                            | Only an estimated 25% of the output of the wire industry is annealed.                                                                                                                             | water       | unspecified     |                 |          | Iron, ion                                        | 4.84E-04   | kg                       | IPCC 2001             | 1    | 1.44               |                                                           |
|                                             |                                             |                                                                                            | negligible                                                                                                                                                                                        |             |                 |                 |          |                                                  | 5.93E-09   | kg                       | IPCC 2001             | 1    | 2                  |                                                           |
|                                             |                                             |                                                                                            | no data available                                                                                                                                                                                 |             |                 |                 |          |                                                  | n.a.       |                          | IPCC 2001             |      |                    |                                                           |
|                                             | wire drawing, steel, Patenting              |                                                                                            | Particle size from chemical process: assumption 2.5-10 um. No data available --> not inventoried                                                                                                  |             |                 |                 |          |                                                  | n.a.       |                          | IPCC 2001             |      |                    |                                                           |
| Lead                                        |                                             |                                                                                            | Patenting is done typically for high alloyed steel. Amount of chromium steel wire is about 10% of total steel wires. --> This step is inventoried for only 10% of steel wire.                     | metals      | extraction      | No              | RER      | lead, at regional storage                        | 3.16E-04   | kg                       | IPCC 2001             | 1    | 3.16               | calculated from max value assuming lognormal distribution |
| Energy                                      |                                             |                                                                                            | Reference contained no data about the energy consumption. As proxy it was calculated the energy needed for heating up the steel wire and the lead bath from ambient to the requested temperature. | natural gas | heating systems | No              | RER      | natural gas, burned in industrial furnace >100kW | 4.13E-01   | MJ                       | IPCC 2001             | 2    | 1.38               | (4,5,1,1,3,3,1)                                           |

Fig. 2.5 Flows for "wire drawing, steel" and their representation in the ecoinvent database (cont.)

## 2. Steel forming

| General Flow Information |                                |                                               | Representation in ecoinvent                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                  |                            |                 |          |                                                                                 |            | Uncertainty Informations |                       |      |                    |                                                           |
|--------------------------|--------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------------|-----------------|----------|---------------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------------------------------------|
| Input                    | Process Name                   | Output                                        | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Category         | Sub-category               | Infra-structure | Location | Modul name in ecoinvent                                                         | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                                           |
|                          | wire drawing, steel, Patenting | m Lead waste (lead oxides + spent bath cover) | Patenting is done typically for high alloyed steel. Amount of chromium steel wire is about 10% of total steel wires --> only 10% inventoried as dust from EAF process. The proxy chosen to represent the waste probably overestimates some heavy metals (e.g. Cr)                                                                                                                                                                                                                                                             | waste management | residual material landfill | No              | CH       | disposal, dust, unalloyed EAF steel, 15.4% water, to residual material landfill | 3.16E-04   | kg                       | IPCC 2001             | 1    | 3.16               | calculated from max value assuming lognormal distribution |
|                          |                                | m SS                                          | inventoried in the water treatment plant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                  |                            |                 |          |                                                                                 | n.a.       |                          | IPCC 2001             |      |                    |                                                           |
|                          |                                | m Pb                                          | inventoried in the water treatment plant                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                  |                            |                 |          |                                                                                 | 7.07E-07   | kg                       | IPCC 2001             | 1    | 3.16               |                                                           |
|                          |                                | m Pb to air                                   | Reference contained no data about the air flow of waste air. As proxy, the amount of exhaust gas from HCL-pickling tanks from cold sheet rolling is taken. Patenting is done typically for high alloyed steel. Amount of chromium steel wire is about 10% of total steel wires --> only 10% inventoried                                                                                                                                                                                                                       | air              | unspecified                |                 |          | Lead                                                                            | 1.41E-09   | kg                       | IPCC 2001             | 1    | 5.11               | (2,5,1,1,3,3,22)                                          |
|                          |                                | m Dust to air                                 | Reference contained no data about the air flow of waste air. As proxy, the amount of exhaust gas from HCL-pickling tanks from cold sheet rolling is taken. Assumption: Dust emissions from lead bath are bigger than 10 um. Amount of chromium steel wire is about 10% of total steel wires --> only 10% inventoried                                                                                                                                                                                                          | air              | unspecified                |                 |          | Particulates, > 10 um                                                           | 5.48E-08   | kg                       | IPCC 2001             | 1    | 1.62               | (2,5,1,1,3,3,25)                                          |
|                          |                                | m CO from furnace                             | Reference contained no data about the air flow of waste air. As proxy, the amount of exhaust gas from annealing from cold sheet rolling is taken. Patenting is done typically for high alloyed steel. Amount of chromium steel wire is about 10% of total steel wires --> only 10% inventoried. As gas is burned with O2-shortage, this CO-emission is added to the emissions inventoried in the module for burning the natural gas. Amount of chromium steel wire is about 10% of total steel wires --> only 10% inventoried | air              | unspecified                |                 |          | Carbon monoxide, fossil                                                         | 5.48E-06   | kg                       | IPCC 2001             | 1    | 5.11               | (2,5,1,1,3,3,17)                                          |
|                          |                                | m NOx from furnace                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                  |                            |                 |          |                                                                                 | negligible |                          | IPCC 2001             |      |                    |                                                           |
|                          |                                | m SO2 from furnace                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                  |                            |                 |          |                                                                                 | negligible |                          | IPCC 2001             |      |                    |                                                           |

Fig. 2.5 Flows for "wire drawing, steel" and their representation in the ecoinvent database (cont.)

## 2. Steel forming

| General Flow Information          |                                                  |                                                                                                                                                                                                                                                                                                                                                  | Representation in ecoinvent                                                                                                                                                                                                                                                                                                                                                                              |                        |                                                          |                  |                            |                                 |            |                                                                               | Uncertainty Informations |      |                    |                  |      |                  |
|-----------------------------------|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|----------------------------------------------------------|------------------|----------------------------|---------------------------------|------------|-------------------------------------------------------------------------------|--------------------------|------|--------------------|------------------|------|------------------|
| Input                             | Process Name                                     | Output                                                                                                                                                                                                                                                                                                                                           | Remarks                                                                                                                                                                                                                                                                                                                                                                                                  | Category               | Sub-category                                             | Infra-structure  | Location                   | Modul name in ecoinvent         | Mean value | Unit                                                                          | Source for mean value    | Type | Standard Deviation | General Comment  |      |                  |
| Flocculating agents (Ca(OH)2;95%) | wire drawing, steel, Waste water treatment plant | <div><div>Treated waste water</div><div>TSS</div><div>COD</div><div>BOD</div><div>DOC</div><div>TOC</div><div>Hydrocarbons</div><div>Cu</div><div>Zn</div><div>Cd</div><div>Al</div><div>Pb</div><div>Cr</div><div>Cr6+</div><div>Mn</div><div>Fe</div><div>Ni</div><div>Hg</div><div>Free Chlorine</div><div>Water treatment sludge</div></div> | Assumption: waste water treatment is comparable with the WWT of hot and cold rolling                                                                                                                                                                                                                                                                                                                     | construction materials | binder                                                   | No               | CH                         | lime, hydrated, loose, at plant | 6.32E-05   | kg                                                                            | IPCC 2001                | 1    | 1.31               | (2,5,1,1,3,3,4)  |      |                  |
| Polyelectrolytes                  |                                                  |                                                                                                                                                                                                                                                                                                                                                  | inorganic chemicals as proxy (personal communication with G. Doka). Assumption: waste water treatment is comparable with the WWT of hot and cold rolling                                                                                                                                                                                                                                                 | chemicals              | inorganics                                               | No               | GLO                        | chemicals inorganic, at plant   | 1.84E-08   | kg                                                                            | IPCC 2001                | 1    | 1.31               | (2,5,1,1,3,3,4)  |      |                  |
|                                   |                                                  |                                                                                                                                                                                                                                                                                                                                                  | One part of the incoming cooling and process water leaves the system as treated effluent; it is assumed to be discharged in surface water --> not inventoried. Infrastructure for waste water treatment plant is comprised in the infrastructure module for the whole rolling mill plant. Value for energy use is comprised in the broad range of uncertainty the energy data for the rolling mill have. |                        |                                                          |                  |                            |                                 | 7.00E-03   | m3                                                                            | IPCC 2001                |      |                    |                  |      |                  |
|                                   |                                                  |                                                                                                                                                                                                                                                                                                                                                  | Assumption: waste water treatment is comparable with the WWT of hot and cold rolling                                                                                                                                                                                                                                                                                                                     | water                  | unspecified                                              |                  |                            | Suspended solids, unspecified   | 1.86E-04   | kg                                                                            | IPCC 2001                | 1    | 1.57               | (2,5,1,1,1,3,32) |      |                  |
|                                   |                                                  |                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                          | water                  | unspecified                                              |                  |                            | COD, Chemical Oxygen Demand     | 4.81E-05   | kg                                                                            | IPCC 2001                | 1    | 1.57               | (2,5,1,1,1,3,32) |      |                  |
|                                   |                                                  |                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                          | water                  | unspecified                                              |                  |                            | BOD5, Biological Oxygen Demand  | 4.81E-05   | kg                                                                            | IPCC 2001                | 1    | 2.79               | (5,5,5,5,5,5,32) |      |                  |
|                                   |                                                  |                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                          | water                  | unspecified                                              |                  |                            | DOC, Dissolved Organic Carbon   | 1.53E-05   | kg                                                                            | IPCC 2001                | 1    | 2.79               | (5,5,5,5,5,5,32) |      |                  |
|                                   |                                                  |                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                          | water                  | unspecified                                              |                  |                            | TOC, Total Organic Carbon       | 1.53E-05   | kg                                                                            | IPCC 2001                | 1    | 2.79               | (5,5,5,5,5,5,32) |      |                  |
|                                   |                                                  |                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                          | water                  | unspecified                                              |                  |                            | Hydrocarbons, unspecified       | 2.20E-06   | kg                                                                            | IPCC 2001                | 1    | 3.05               | (2,5,1,1,1,3,34) |      |                  |
|                                   |                                                  |                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                          | water                  | unspecified                                              |                  |                            | Copper, ion                     | 3.39E-07   | kg                                                                            | IPCC 2001                | 1    | 1.86               | (2,5,1,1,1,3,37) |      |                  |
|                                   |                                                  |                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                          | water                  | unspecified                                              |                  |                            | Zinc, ion                       | 2.62E-07   | kg                                                                            | IPCC 2001                | 1    | 1.86               | (2,5,1,1,1,3,37) |      |                  |
|                                   |                                                  |                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                          | water                  | unspecified                                              |                  |                            | Cadmium, ion                    | 1.75E-07   | kg                                                                            | IPCC 2001                | 1    | 1.86               | (2,5,1,1,1,3,37) |      |                  |
|                                   |                                                  |                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                          | water                  | unspecified                                              |                  |                            | Aluminum                        | 1.66E-06   | kg                                                                            | IPCC 2001                | 1    | 1.86               | (2,5,1,1,1,3,37) |      |                  |
|                                   |                                                  |                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                          | water                  | unspecified                                              |                  |                            | Lead                            | 3.50E-07   | kg                                                                            | IPCC 2001                | 1    | 1.86               | (2,5,1,1,1,3,37) |      |                  |
|                                   |                                                  |                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                          | water                  | unspecified                                              |                  |                            | Chromium, ion                   | 6.30E-07   | kg                                                                            | IPCC 2001                | 1    | 1.86               | (2,5,1,1,1,3,37) |      |                  |
|                                   |                                                  |                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                          | water                  | unspecified                                              |                  |                            | Chromium VI                     | 3.50E-08   | kg                                                                            | IPCC 2001                | 1    | 1.86               | (2,5,1,1,1,3,37) |      |                  |
|                                   |                                                  |                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                          | water                  | unspecified                                              |                  |                            | Manganese                       | 7.14E-07   | kg                                                                            | IPCC 2001                | 1    | 1.86               | (2,5,1,1,1,3,37) |      |                  |
|                                   |                                                  |                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                          | water                  | unspecified                                              |                  |                            | Iron, ion                       | 5.42E-06   | kg                                                                            | IPCC 2001                | 1    | 1.86               | (2,5,1,1,1,3,37) |      |                  |
|                                   |                                                  |                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                          | water                  | unspecified                                              |                  |                            | Nickel, ion                     | 9.90E-07   | kg                                                                            | IPCC 2001                | 1    | 1.86               | (2,5,1,1,1,3,37) |      |                  |
|                                   |                                                  |                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                          | water                  | unspecified                                              |                  |                            | Mercury                         | 3.50E-08   | kg                                                                            | IPCC 2001                | 1    | 1.86               | (2,5,1,1,1,3,37) |      |                  |
|                                   |                                                  |                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                          | water                  | unspecified                                              |                  |                            | Chloride                        | 1.57E-06   | kg                                                                            | IPCC 2001                | 1    | 1.57               | (2,5,1,1,1,3,33) |      |                  |
|                                   |                                                  |                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                          | Water treatment sludge | Same composition as for cold rolling of steel is assumed | waste management | residual material landfill | No                              | CH         | disposal, sludge from steel rolling, 20% water, to residual material landfill | 3.50E-02                 | kg   | IPCC 2001          | 1                | 1.57 | (2,5,1,1,1,3,43) |

**Fig. 2.5** Flows for "wire drawing, steel" and their representation in the ecoinvent database (cont.)

## 2. Steel forming

| General Flow Information |                                      |                                       | Representation in ecoinvent                                                                                                                                                                                                                                                   |                   |                              |                |          |                                                                         |            | Uncertainty Informations |                                        |      |                    |                                |
|--------------------------|--------------------------------------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------------------|----------------|----------|-------------------------------------------------------------------------|------------|--------------------------|----------------------------------------|------|--------------------|--------------------------------|
| Input                    | Process Name                         | Output                                | Remarks                                                                                                                                                                                                                                                                       | Category          | Sub-category                 | Infrastructure | Location | Module name in ecoinvent                                                | Mean value | Unit                     | Source for mean value                  | Type | Standard Deviation | General Comment                |
| Water                    | wire drawing, steel, Overall process | water out                             | Water used for different purposes: rinsing, cooling, dilution, quenching water. Sum of the different process steps, mainly consisting of cooling water, which amount has huge variations. Assumption: wastewater treatment is comparable with the WWT of hot and cold rolling | resource          | in water                     |                |          | Water, cooling, unspecified natural origin                              | 3.50E-02   | m <sup>3</sup>           | IPCC 2001                              | 1    | 1.38               | (4,5,1,1,3,3,12)               |
|                          |                                      |                                       | One part of the incoming cooling and process water leaves the system as not contaminated evaporated water or included in discarded waste --> not inventoried                                                                                                                  |                   |                              |                |          |                                                                         | 2.80E-02   | m <sup>3</sup>           | IPCC 2001                              | 1    | 1.38               | (4,5,1,1,3,3,12)               |
| Electrical energy        |                                      |                                       | No data available in IPCC 2001; therefore values from Stahl Gerlafingen were taken                                                                                                                                                                                            | electricity       | production mix               | No             | UCTE     | electricity, medium voltage, production UCTE, at grid                   | 1.41E-01   | kWh                      | Personal communication Dr. Ch. Zeltner | 1    | 1.24               | (3,5,1,3,1,3,2)                |
| Rolling Shop (CR)        |                                      |                                       |                                                                                                                                                                                                                                                                               | metals            | processing                   | Yes            | RER      | rolling mill                                                            | 1.62E-09   | unit                     | Estimation                             | 1    | 3.04               | (3,4,1,1,1,4,9)                |
| Lorry 32t                | wire drawing, steel, Packaging       |                                       |                                                                                                                                                                                                                                                                               | transport systems | road                         | No             | RER      | transport, lorry 32t                                                    | 3.16E-02   | tkm                      | Estimation                             | 1    | 2.09               | (4,5,n.a.,n.a.,n.a.,n.a.,5)    |
| Paper; cardboard         |                                      |                                       |                                                                                                                                                                                                                                                                               | paper & cardboard | cardboard & corrugated board | No             | RER      | packaging, corrugated board, mixed fibre, single wall, at plant         | 4.72E-12   | kg                       | EAA 2000                               | 1    | 1.38               | (4,5,1,1,3,3,4)                |
| Plastics                 |                                      |                                       |                                                                                                                                                                                                                                                                               | plastics          | processing                   | No             | RER      | packaging film, LDPE, at plant                                          | 1.89E-11   | kg                       | EAA 2000                               | 1    | 1.38               | (4,5,1,1,3,3,4)                |
| Steel                    |                                      |                                       |                                                                                                                                                                                                                                                                               | metals            | extraction                   | No             | RER      | steel, converter, unalloyed, at plant                                   | 2.36E-11   | kg                       | EAA 2000                               | 1    | 1.38               | (4,5,1,1,3,3,4)                |
| Steel coils              | wire drawing, steel, Waste treatment | Waste                                 |                                                                                                                                                                                                                                                                               | metals            | processing                   | No             | RER      | sheet rolling, steel                                                    | 2.36E-11   | kg                       | EAA 2000                               | 1    | 1.38               | (4,5,1,1,3,3,4)                |
| Wood                     |                                      |                                       |                                                                                                                                                                                                                                                                               | wooden materials  | extraction                   | No             | RER      | sawn timber, softwood, raw, air dried, u=20%, at plant                  | 1.62E-17   | m <sup>3</sup>           | EAA 2000                               | 1    | 1.38               | (4,5,1,1,3,3,4)                |
|                          |                                      |                                       |                                                                                                                                                                                                                                                                               | waste management  | municipal incineration       | No             | CH       | disposal, municipal solid waste, 22.9% water, to municipal incineration | 7.08E-11   | kg                       | EAA 2000                               | 1    | 1.22               | (2,5,1,1,1,3,6)                |
|                          |                                      |                                       |                                                                                                                                                                                                                                                                               | air               | unspecified                  |                |          | Heat, waste                                                             | 5.09E-01   | MJ                       | IPCC 2001                              | 1    | 1.56               | (n.a.,5,n.a.,n.a.,n.a.,n.a.,7) |
|                          |                                      | Drawing wire, steel or chromium steel |                                                                                                                                                                                                                                                                               | metals            | processing                   | No             | RER      | wire drawing, steel                                                     | 1.00E+00   | kg                       |                                        |      |                    |                                |

Fig. 2.5 Flows for "wire drawing, steel" and their representation in the ecoinvent database (cont.)

## 2.6 Pipe drawing of steel and chromium steel

The production process and volume of drawn pipes (or seamless tubes) is described in (IPPC (2001a)).

Steel tubes cover three product categories, which differ considerably in manufacturing process, raw materials, and investment requirements. These categories are classified as:

- Seamless tubes.
- Welded steel tubes of circular cross-section over 406.4 mm in outside diameter.
- Welded steel tubes of circular cross-section up to and including 406.4 mm in outside diameter together with welded steel tubes of non-circular cross-section of any perimeter.

In steel tube manufacture, the EU, which produced 11.8 million tonnes in 1996, (20.9 % of total world production) is the largest producer followed by Japan and the U.S. The European steel tube industry has a highly concentrated structure. Five countries – Germany (3.2 million tonnes), Italy (3.2 million tonnes), France (1.4 million tonnes), United Kingdom (1.3 million tonnes) and Spain (0.9 million tonnes) - account for roughly 90 % of total EU production. About 26.4 % of the tubes produced are seamless tubes, of the remaining 73.6 % welded tubes about 18.9 % are of large diameter. The major part of the total steel tube sector is small diameter tubes with a share of 54.7 %.

Nevertheless this type of pipes represent only about one quarter of total production, in this study the focus was laid on the manufacturing of seamless pipes. The reason is that, if LCI data is needed about welded tubes, they can be gained easily, combining different ecoinvent modules. In fact welded tubes are produced from cold rolled or hot rolled steel strip; hot rolled wide strip; or thick plate. In principal the production sequence comprises bending of the strip or plate to an open tube and closing the gap by welding. Therefore ecoinvent modules of steel, of steel sheet production and welding are needed.

Seamless tube production consists basically of the following manufacturing steps:

- Heating of the input.
- Piercing (press-piercing or oblique rolling).
- Elongation/Stretching.
- Final Rolling.
- Heat Treatment.

The most important types of plants used today are mandrel mills, plug mills, push bench mills and pilger mills. Raw material for the production of unalloyed and medium-alloy steel tubes are continuously cast round billets or blooms, the raw material for high-chrome steel tubes are mainly round bars, rolled from cast billets.

The input is heated to rolling temperature in rotary hearth furnaces fired with natural gas or oil. Then a hollow billet (ingot) is produced by piercing. The most important hole-piercing process uses inclined rollers. Longitudinal rolling, transverse rolling, or presses are used for the elongating stage. The most important one is the continuous tube rolling process, as it is one of the most efficient and productive processes.

Tubes for undemanding applications can be shipped immediately after rolling. For more exacting quality requirements, especially good toughness properties, separate heat treatment is usually carried out after rolling. This can include normalising and tempering.

The infrastructure of steel and chromium steel pipes drawing is described in 1.1. It comprises buildings where production processes take place and for storage, surroundings and machinery for the single process steps (e.g.: furnace, rolls, cutting devices, etc.).



### **2.6.1 Pipe drawing of steel and chromium steel in ecoinvent**

No data was available about the production process of seamless pipes. As an approximation it was decided to compare the tube production process with one of the forming processes for steel and chromium steel, described above, for which data was available. The wire drawing process was evaluated being the one that describes the pipes drawing process the best. As the geometry of the pipe is more complex as the geometry of the wire, pipe drawing is inventoried with 110% of wire drawing.

The background information to the process "drawing of pipes, steel" is given in Tab. 2.6, flows are shown in Fig. 2.6.

#### **Data quality**

Uncertainty is estimated by mean of the pedigree matrix, according to (Frischknecht et al. (2003)).

## **2.7 Section bar rolling of steel**

The production process of section bar rolling is not described in details in the studied literature and no data about material and energy flows were available. An extract of the description found in (IPPC (2001a)) is reported below.

As most other types of mills, section rolling mills are divided into roughing, intermediate and finishing stands, preceded by a descaling machine installed between the reheating furnace and the roughing mill. On leaving the finishing stand or train, the bars are divided to cooling bed length or to commercial length by saws. After cooling on the cooling bed, the bars are straightened on mechanical or hydraulic straightening machines and cut to length by saws as required.

As conclusion from this description and the lack of data, it was decided to combine other ecoinvent modules to reach an approximation of the relevant flows of the section bar rolling process.

The infrastructure of steel section bar rolling is described in 1.1. It comprises buildings where production processes take place and for storage, surroundings and machinery fore the single process steps (e.g.: furnace, rolls, cutting devices, etc.).

### **2.7.1 Section bar rolling of steel in ecoinvent**

Due to the lack of data, the section bar rolling process is approximated by inventorying 10% of the wire drawing process.

The background information to the process "section bar rolling, steel" is given in Tab. 2.6, flows are shown in Fig. 2.7.

#### **Data quality**

Uncertainty is estimated by mean of the pedigree matrix, according to (Frischknecht et al. (2003)).

Tab. 2.6 Ecoinvent meta information for "drawing of pipes, steel" and "section bar rolling, steel"

| Name                                  | drawing of pipes, steel                                                                                                                                                                                             | section bar rolling, steel                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location                              | RER                                                                                                                                                                                                                 | RER                                                                                                                                                                                                                                                                                                                                                                  |
| Infrastructure Process                | 0                                                                                                                                                                                                                   | 0                                                                                                                                                                                                                                                                                                                                                                    |
| Unit                                  | kg                                                                                                                                                                                                                  | kg                                                                                                                                                                                                                                                                                                                                                                   |
| Data Set Version                      | 2.0                                                                                                                                                                                                                 | 2.0                                                                                                                                                                                                                                                                                                                                                                  |
| Included Processes                    | The module describes the manufacturing of seamless tubes. It is not representative for welded tubes. It includes 110% of the wire drawing process. Does not include the material being rolled                       | The module describes the rolling process of section bar. It includes 50% of the wire drawing process. Does not include the material being rolled                                                                                                                                                                                                                     |
| Amount                                | 1                                                                                                                                                                                                                   | 1                                                                                                                                                                                                                                                                                                                                                                    |
| Local Name                            | Rohr ziehen, Stahl                                                                                                                                                                                                  | Profil walzen, Stahl                                                                                                                                                                                                                                                                                                                                                 |
| Synonyms                              | drawing of tubes                                                                                                                                                                                                    | profile                                                                                                                                                                                                                                                                                                                                                              |
| General Comment to reference function | Seamless tube production consists basically of the following manufacturing steps: Heating of the input, Piercing (press-piercing or oblique rolling), Elongation/Stretching, Final Rolling and Heat Treatment.      | As most other types of mills, section rolling mills are divided into roughing, intermediate and finishing stands, preceded by a descaling machine installed between the reheating furnace and the roughing mill. After cooling on the cooling bed, the bars are straightened on mechanical or hydraulic straightening machines and cut to length by saws as required |
| Start Date                            | 1997                                                                                                                                                                                                                | 1997                                                                                                                                                                                                                                                                                                                                                                 |
| End Date                              | 2002                                                                                                                                                                                                                | 2002                                                                                                                                                                                                                                                                                                                                                                 |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                   | 1                                                                                                                                                                                                                                                                                                                                                                    |
| Other Period Text                     |                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                      |
| Geography text                        | Data-set is representative for European Union                                                                                                                                                                       | Data-set is representative for European Union                                                                                                                                                                                                                                                                                                                        |
| Technology text                       | Average technique for EU. Because of the lack of data about the production process, it was approximated with the production process of steel wire drawing.                                                          | Average technique for EU. Because of the lack of data about the production process, it was approximated with the production process of wire drawing.                                                                                                                                                                                                                 |
| Representativeness [%]                |                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                      |
| Production Volume                     | 3.3 million tonnes in 1996                                                                                                                                                                                          | No data available                                                                                                                                                                                                                                                                                                                                                    |
| Sampling Procedure                    | Literature and estimation                                                                                                                                                                                           | Literature and estimation                                                                                                                                                                                                                                                                                                                                            |
| Extrapolations                        | see technology                                                                                                                                                                                                      | see technology                                                                                                                                                                                                                                                                                                                                                       |
| Uncertainty Adjustments               | The extrapolation made for the description of the technology process and the fact that no data were available about the percentage of companies represented by the study cited are considered in the uncertainties. | The extrapolation made for the description of the technology process and the fact that no data were available about the percentage of companies represented by the study cited are considered in the uncertainties.                                                                                                                                                  |

| General Flow Information |                         |              | Representation in ecoinvent |          |              |                 |          |                          |            | Uncertainty Informations |                       |      |                    |                 |
|--------------------------|-------------------------|--------------|-----------------------------|----------|--------------|-----------------|----------|--------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------|
| Input                    | Process Name            | Output       | Remarks                     | Category | Sub-category | Infra-structure | Location | Modul name in ecoinvent  | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment |
| Tube                     | drawing of pipes, steel | Tube drawing |                             | metals   | processing   | No              | RER      | wire drawing, steel      | 1.10E+00   | kg                       | Estimation            | 1    | 1.38               | (4,5,1,1,3,3,4) |
|                          |                         |              |                             | metals   | processing   | No              | RER      | draw ing of pipes, steel | 1.00E+00   | kg                       |                       |      |                    |                 |

Fig. 2.6 Flows for "drawing of pipes, steel" and their representation in the ecoinvent database

| General Flow Information |                            |                       | Representation in ecoinvent |          |              |                 |          |                            |            | Uncertainty Informations |                       |      |                    |                 |
|--------------------------|----------------------------|-----------------------|-----------------------------|----------|--------------|-----------------|----------|----------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------|
| Input                    | Process Name               | Output                | Remarks                     | Category | Sub-category | Infra-structure | Location | Modul name in ecoinvent    | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment |
| Section bar extruding    | section bar rolling, steel | Section bar extruding |                             | metals   | processing   | No              | RER      | wire draw ing, steel       | 5.00E-01   | kg                       | Estimation            | 1    | 1.38               | (4,5,1,1,3,3,4) |
|                          |                            |                       |                             | metals   | processing   | No              | RER      | section bar rolling, steel | 1.00E+00   | kg                       |                       |      |                    |                 |

Fig. 2.7 Flows for "section bar rolling, steel" and their representation in the ecoinvent database

## **2.8 Infrastructure of the rolling mill**

### **2.8.1 Land use and buildings of the rolling mill**

Data on land use of rolling mills are based on the map of the steel producing and forming plant of Stahl Gerlafingen, an enterprise of the Swiss Steel Ltd.

The total area of the plant was read from a map with the scale of 1:6000. The buildings were estimated covering 25 % of the total area of the plant, 75 % are therefore surroundings, which are further distinguished in 5 % green area and 95 % built up or asphalted surface.

The area of the buildings belonging to the steel manufacturing part of the plant could be read from the map<sup>4</sup>, they were estimated to be occupying 20 % of the total build-up surface of the plant. Therefore the total area, the surroundings and the built-up area belonging to the rolling mill amount to 80 % of the total area of the plant. Only the surface of the administration buildings was divided into equal parts (4000 m<sup>2</sup>) for the steel manufacturing and the steel rolling process. The total area of the rolling mill is 310'000 square meters, all other values were calculated basing on this value. The buildings were estimated to be 10 m high and to have a lifetime of 50 years.

### **2.8.2 Machinery and disposal of the machinery of the rolling mill**

Absolutely no data about the amount of machinery was available, therefore it was necessary to make a rough estimation of the volume, respectively the volume of the main components of the machinery. It is assumed that the machinery is basically composed of steel on a foundation of concrete. The volume of the machinery was estimated to be 0.5 % of the buildings volume of the rolling process. The concrete fundament is assumed to have the same mass as the machines. Density of the steel is 7850 kg/m<sup>3</sup>, that of the concrete 2300 kg/m<sup>3</sup>. The lifetime of the machinery was estimated to be 25 years.

The yearly output of the visited plant amounts to 616000 tonnes of formed steel.

The module for concrete disposal from building includes recycling and disposal and it is assumed that 100 % of the steel is recycled.

### **2.8.3 Infrastructure of the rolling mill in ecoinvent**

The background information to the process "rolling mill" is given in Tab. 2.7, flows are shown in Fig. 2.8.

#### **Data quality**

Uncertainty is estimated by mean of the pedigree matrix, according to (Frischknecht et al. (2003)).

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<sup>4</sup> Personal communication on 25. February 03 from Dr. Ch. Zeltner, Stahl Gerlafingen, Swiss Steel Ltd.

Tab. 2.7 Ecoinvent meta information for the "rolling mill"

|                                       |                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name                                  | rolling mill                                                                                                                                                                                                                                                                                                                                 |
| Location                              | RER                                                                                                                                                                                                                                                                                                                                          |
| Infrastructure Process                | 1                                                                                                                                                                                                                                                                                                                                            |
| Unit                                  | unit                                                                                                                                                                                                                                                                                                                                         |
| Data Set Version                      | 1                                                                                                                                                                                                                                                                                                                                            |
| Included Processes                    | Includes land occupation data for and land transformation data to buildings, asphalted and green surroundings; land transformation data from unknown; the buildings; company internal roads, parking lots etc.; the main material of which the machinery is composed and their disposal.                                                     |
| Amount                                | 1                                                                                                                                                                                                                                                                                                                                            |
| Local Name                            | Walzwerk                                                                                                                                                                                                                                                                                                                                     |
| Synonyms                              |                                                                                                                                                                                                                                                                                                                                              |
| General Comment to reference function | The values for plant area were taken from a map received during the visit to a electro-steel manufacturing and forming plant in Gerlafingen, Switzerland. The output was 616000 tonnes of formed steel in 2002. The Lifetime is assumed to be 50 years for infrastructure like buildings and routes, and 25 years for machinery and furnaces |
| Start Date                            | 2002                                                                                                                                                                                                                                                                                                                                         |
| End Date                              | 2002                                                                                                                                                                                                                                                                                                                                         |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                                                                                                                            |
| Other Period Text                     |                                                                                                                                                                                                                                                                                                                                              |
| Geography text                        | The visited plant is in the range of size typical for European union (about a quarter of the maximal size). Data-set is representative for European Union                                                                                                                                                                                    |
| Technology text                       | The visited plant produces hot rolled merchant bars, wire rod, concrete reinforcing bars, mats and cold rolled topar-S, topar-Rc and topar-Rs                                                                                                                                                                                                |
| Representativeness [%]                |                                                                                                                                                                                                                                                                                                                                              |
| Production Volume                     | No data available                                                                                                                                                                                                                                                                                                                            |
| Sampling Procedure                    | plant map and estimation                                                                                                                                                                                                                                                                                                                     |
| Extrapolations                        | see technology and geography                                                                                                                                                                                                                                                                                                                 |
| Uncertainty Adjustments               | The extrapolation made for the description of the technology process, the geographical extrapolations, the estimations for machinery weight and the fact that no data were available about the percentage of companies represented by the study cited are considered in the uncertainties.                                                   |

## 2. Steel forming

| General Flow Information                  |              |              | Representation in ecoinvent                                                       |                        |                     |                 |          |                                                                 |            | Uncertainty Informations |                              |      |                    |                             |
|-------------------------------------------|--------------|--------------|-----------------------------------------------------------------------------------|------------------------|---------------------|-----------------|----------|-----------------------------------------------------------------|------------|--------------------------|------------------------------|------|--------------------|-----------------------------|
| Input                                     | Process Name | Output       | Remarks                                                                           | Category               | Sub-category        | Infra-structure | Location | Modul name in ecoinvent                                         | Mean value | Unit                     | Source for mean value        | Type | Standard Deviation | General Comment             |
| Land occupation-buildings                 | rolling mill |              |                                                                                   | resource               | land                |                 |          | Occupation, industrial area, built up                           | 3.88E+03   | m <sup>2</sup> a         | Vist to one site, estimation | 1    | 1.62               | (4,5,1,3,1,n.a.,7)          |
| Land occupation-surroundings              |              |              | The value for land use of the buildings and the green surroundings was subtracted | resource               | land                |                 |          | Occupation, industrial area                                     | 1.10E+04   | m <sup>2</sup> a         | Vist to one site, estimation | 1    | 1.62               | (4,5,1,3,1,n.a.,7)          |
| Land occupation-surroundings-green        |              |              |                                                                                   | resource               | land                |                 |          | Occupation, industrial area, vegetation                         | 5.81E+02   | m <sup>2</sup> a         | Vist to one site, estimation | 1    | 1.62               | (4,5,1,3,1,n.a.,7)          |
| Land transformation from                  |              |              |                                                                                   | resource               | land                |                 |          | Transformation, from unknown                                    | 3.10E+02   | m <sup>2</sup>           | Vist to one site, estimation | 1    | 2.1                | (4,5,1,3,1,n.a.,8)          |
| Land transformation to buildings          |              |              |                                                                                   | resource               | land                |                 |          | Transformation, to industrial area, built up                    | 7.75E+01   | m <sup>2</sup>           | Vist to one site, estimation | 1    | 2.1                | (4,5,1,3,1,n.a.,8)          |
| Land transformation to surroundings       |              |              |                                                                                   | resource               | land                |                 |          | Transformation, to industrial area                              | 2.21E+02   | m <sup>2</sup>           | Vist to one site, estimation | 1    | 2.1                | (4,5,1,3,1,n.a.,8)          |
| Land transformation to surroundings-green |              |              |                                                                                   | resource               | land                |                 |          | Transformation, to industrial area, vegetation                  | 1.16E+01   | m <sup>2</sup>           | Vist to one site, estimation | 1    | 2.1                | (4,5,1,3,1,n.a.,8)          |
| Building-rolling process                  |              |              |                                                                                   | construction processes | buildings           | Yes             | CH       | building, hall, steel construction                              | 7.35E+01   | m <sup>2</sup>           | Vist to one site, estimation | 1    | 3.09               | (4,5,1,3,1,n.a.,9)          |
| Building-administration                   |              |              |                                                                                   | construction processes | buildings           | Yes             | RER      | building, multi-storey                                          | 4.00E+01   | m <sup>3</sup>           | Vist to one site, estimation | 1    | 3.09               | (4,5,1,3,1,n.a.,9)          |
| Fundament of the machinery                |              |              |                                                                                   | construction materials | concrete            | No              | CH       | concrete, normal, at plant                                      | 2.51E+01   | m <sup>3</sup>           | Vist to one site, estimation | 1    | 1.3                | (4,5,1,3,1,n.a.,4)          |
| Machinery                                 |              |              |                                                                                   | metals                 | extraction          | No              | RER      | steel, electric, un- and low -alloyed, at plant                 | 5.77E+04   | kg                       | Vist to one site, estimation | 1    | 1.3                | (4,5,1,3,1,n.a.,4)          |
| Lorry 32t                                 |              |              |                                                                                   | transport systems      | road                | No              | RER      | transport, lorry 32t                                            | 3.75E+04   | tkm                      | Vist to one site, estimation | 1    | 2.09               | (4,5,n.a.,n.a.,n.a.,n.a.,5) |
|                                           |              | Concrete     |                                                                                   | waste management       | building demolition | No              | CH       | disposal, building, concrete, not reinforced, to final disposal | 5.77E+04   | kg                       | Vist to one site, estimation | 1    | 1.3                | (4,5,1,3,1,n.a.,6)          |
|                                           |              | Steel        | recycled --> not inventoried                                                      |                        |                     |                 |          |                                                                 | 5.77E+04   | kg                       | Vist to one site, estimation | 1    | 1.3                | (4,5,1,3,1,n.a.,6)          |
|                                           |              | Rolling mill |                                                                                   | metals                 | processing          | Yes             | RER      | rolling mill                                                    | 1.00E+00   | unit                     |                              |      |                    |                             |

Fig. 2.8 Flows for "rolling mill" and their representation in the ecoinvent database

## 3 Aluminium forming

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 Review: Roland Hischier

As for steel, the outputs of the aluminium production are slabs with relatively coarse shape, which can be further formed to sheets, foils or section bars.

### 3.1 Sheet rolling of aluminium

The proceedings of rolling aluminium from an initial rough form to a well defined sheet is similar to the processes taking place in steel forming, therefore it is not described in every detail.

An overview of the different process steps taking place, when Aluminium is rolled to sheet form, is given in Fig. 3.1.

The basic input material for sheet fabrication is primary or secondary aluminium ingot of 500 to 700 mm thickness and a weight up to 25 tonnes. After heating the aluminium slab to around 500 °C, it is hot rolled, typically to 2-8 mm thickness, followed by cold rolling to the final gauge, 0.2 to 6 mm (EAA (2000)).

As the process steps are similar to those of steel forming, also the infrastructure needed to roll aluminium slabs into sheet form, is assumed to be similar. It is balanced the same amount of rolling mill per volume unit of rolled material, as it is for the forming processes of steel.

#### 3.1.1 Environmental issues of sheet rolling of aluminium

In the data referred in the considered literature (EAA (2000)), the main air and water emissions, are arising from the combustion of energy carriers. Emissions to land are mainly hazardous and oily wastes.

#### 3.1.2 Sheet rolling of aluminium in ecoinvent

The inventory data reported were obtained from an industry-wide survey carried out in 1998. For the sheet rolling industry segment, the survey coverage was 71% related to the total production output (EAA (2000)).

The basic input material, primary aluminium is not balanced. The scrap losses of aluminium are mainly internally recycled, this recycling process is not balanced. A small amount of scrap material is not recycled, but is lost in waste. To obtain an even mass balance for 1 kilogram of output product, it is necessary to balance the 12 grams lost in waste as input of primary aluminium (Fig. 3.1).

The background information to the process "sheet rolling, aluminium" is given in Tab. 3.1, flows are shown in Fig. 3.1

#### Data quality

A standard precision of 5% percent of the value is given in EAA (2000)(), to obtain a more detailed view of the data quality, the uncertainty is estimated by completing the pedigree matrix, according to (Frischknecht et al. (2003)).

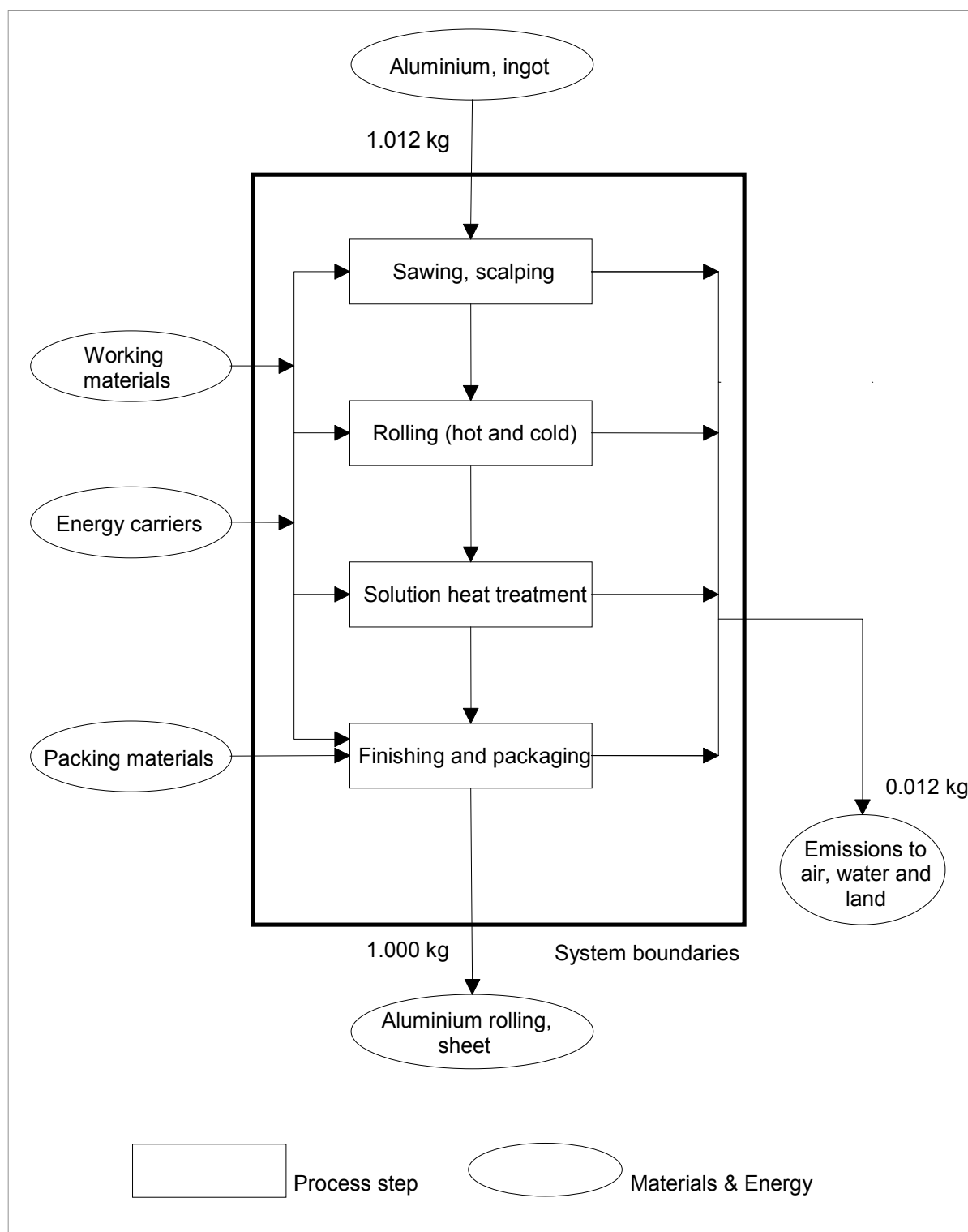


Fig. 3.1 Aluminium sheet rolling process; values are aluminium flows (modified from EAA (2000))



Tab. 3.1 Ecoinvent meta information for "sheet rolling, aluminium"

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>                           | <b>sheet rolling, aluminium</b>                                                                                                                                                                                                                                                                                                                                                                                                       |
| Location                              | RER                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Infrastructure Process                | 0                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Unit                                  | kg                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Data Set Version                      | 1                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Included Processes                    | All the process steps, which can be attributed to semi-fabrication (sawing, scalping, hot rolling, cold rolling, solution heat treatment, finishing and packaging), are included. Does not include the material being rolled, only the amount of scrap lost in waste is balanced as primary aluminium input. Includes the transport of the materials to the plant, but does not include the transport of the product to the customer. |
| Amount                                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Local Name                            | Blech walzen, Aluminium                                                                                                                                                                                                                                                                                                                                                                                                               |
| Synonyms                              | profile rolling//section bar rolling                                                                                                                                                                                                                                                                                                                                                                                                  |
| General Comment to reference function | Aluminium ingots of 500-700 mm thickness and up to 25 tonnes are rolled, first hot, then cold, to a final sheet thickness ranging from 0.2 to 6 mm. The module can be applied also for section bar rolling of aluminium.                                                                                                                                                                                                              |
| Start Date                            | 2000                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| End Date                              | 2002                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Other Period Text                     | Data refers to a survey made in 1998                                                                                                                                                                                                                                                                                                                                                                                                  |
| Geography text                        | Dataset is representative for European Union                                                                                                                                                                                                                                                                                                                                                                                          |
| Technology text                       | Average technique for European Union. The infrastructure was assumed to be the same as for rolling steel.                                                                                                                                                                                                                                                                                                                             |
| Representativeness [%]                | 71                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Production Volume                     | No data available                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Sampling Procedure                    | Literature, estimation for transport and infrastructure                                                                                                                                                                                                                                                                                                                                                                               |
| Extrapolations                        | see technology and sampling procedure                                                                                                                                                                                                                                                                                                                                                                                                 |
| Uncertainty Adjustments               | Precision of the data set is referred to be of 5 % by the reference, nevertheless uncertainty was estimated by filling out the uncertainty scoring                                                                                                                                                                                                                                                                                    |

### 3. Aluminium forming

| General Flow Information |                                   |              | Representation in ecoinvent                                                                                                                           |                   |                 |                 |          |                                                                    |            | Uncertainty Informations |                       |      |                    |                             |
|--------------------------|-----------------------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-----------------|-----------------|----------|--------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------|
| Input                    | Process Name                      | Output       | Remarks                                                                                                                                               | Category          | Sub-category    | Infra-structure | Location | Modul name in ecoinvent                                            | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment             |
| Primary Aluminium        | sheet rolling, aluminium, rolling |              | Rolling aluminium casuses scrap losses wich cannot be recovered, therefore 0.012 kg of primary aluminium per per kg of material rolled is inventoried | metals            | extraction      | No              | RER      | aluminium, primary, at plant                                       | 1.20E-02   | kg                       | EAA2000               | 1    | 1.07               | (1,2,1,1,1,3,3)             |
| Rolling Oil              |                                   |              |                                                                                                                                                       | chemicals         | organics        | No              | RER      | lubricating oil, at plant                                          | 3.80E-03   | kg                       | EAA2000               | 1    | 1.07               | (1,2,1,1,1,3,11)            |
| Water                    |                                   |              |                                                                                                                                                       | resource          | in water        |                 |          | Water, cooling, unspecified natural origin                         | 6.00E-03   | m3                       | EAA2000               | 1    | 1.07               | (1,2,1,1,1,3,12)            |
| Fuel Oil                 |                                   |              |                                                                                                                                                       | oil               | heating systems | No              | RER      | light fuel oil, burned in industrial furnace 1MW, non-modulating   | 1.45E-02   | MJ                       | EAA2000               | 1    | 1.07               | (1,2,1,1,1,3,1)             |
| Gas                      |                                   |              |                                                                                                                                                       | natural gas       | heating systems | No              | RER      | natural gas, burned in industrial furnace low - NOx >100kW         | 2.11E+00   | MJ                       | EAA2000               | 1    | 1.07               | (1,2,1,1,1,3,1)             |
| Electricity              |                                   |              |                                                                                                                                                       | electricity       | production mix  | No              | UCTE     | electricity, medium voltage, production UCTE, at grid              | 5.47E-01   | kWh                      | EAA2000               | 1    | 1.07               | (1,2,1,1,1,3,2)             |
| Rolling mill             |                                   |              | It is assumed, that the same volume of material is rolled per year. Density of steel is 7850 kg/m3 and for aluminium 2700 kg/m3                       | metals            | processing      | Yes             | RER      | rolling mill                                                       | 4.72E-09   | unit                     | EAA2000               | 1    | 3.36               | (4,5,1,3,4,5,9)             |
| Lorry 32 t               |                                   |              |                                                                                                                                                       | transport systems | road            | No              | RER      | transport, lorry 32t                                               | 5.07E-03   | tkm                      | EAA2000               | 1    | 2.09               | (4,5,n.a.,n.a.,n.a.,n.a.,5) |
|                          |                                   | VOC to air   |                                                                                                                                                       | air               | unspecified     |                 |          | NM VOC, non-methane volatile organic compounds, unspecified origin | 4.40E-04   | kg                       | EAA2000               | 1    | 2                  | (1,2,1,1,1,3,23)            |
|                          |                                   | Water        |                                                                                                                                                       |                   |                 |                 |          |                                                                    | 6.00E-03   | m3                       | EAA2000               | 1    | 1.45               | (4,5,2,1,3,5,4)             |
|                          |                                   | COD in Water |                                                                                                                                                       | water             | unspecified     |                 |          | COD, Chemical Oxygen Demand                                        | 7.90E-05   | kg                       | EAA2000               | 1    | 1.51               | (1,2,1,1,1,3,32)            |
|                          |                                   | BOD          |                                                                                                                                                       | water             | unspecified     |                 |          | BOD5, Biological Oxygen Demand                                     | 7.90E-05   | kg                       | EAA2000               | 1    | 2.79               | (5,5,5,5,5,32)              |
|                          |                                   | DOC          |                                                                                                                                                       | water             | unspecified     |                 |          | DOC, Dissolved Organic Carbon                                      | 2.60E-05   | kg                       | EAA2000               | 1    | 2.79               | (5,5,5,5,5,32)              |
|                          |                                   | TOC          |                                                                                                                                                       | water             | unspecified     |                 |          | TOC, Total Organic Carbon                                          | 2.60E-05   | kg                       | EAA2000               | 1    | 2.79               | (5,5,5,5,5,32)              |

Fig. 3.2 Flows for "sheet rolling, aluminium" and their representation in the ecoinvent database

### 3. Aluminium forming

| General Flow Information                                                                                                                   |                                     |                          | Representation in ecoinvent                                                               |                   |                              |                 |          |                                                                         |            | Uncertainty Informations |                       |      |                    |                  |
|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|--------------------------|-------------------------------------------------------------------------------------------|-------------------|------------------------------|-----------------|----------|-------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|------------------|
| Input                                                                                                                                      | Process Name                        | Output                   | Remarks                                                                                   | Category          | Sub-category                 | Infra-structure | Location | Module name in ecoinvent                                                | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment  |
| <div> <div>Paper; cardboard-alsh</div> <div>Plastics-alsh</div> <div>Steel-alsh</div> <div>Steel-sh-alsh</div> <div>Wood-alsh</div> </div> | sheet rolling, aluminium, rolling   | Hazardous waste          | contains 15% aluminium                                                                    | waste management  | underground deposit          | No              | DE       | disposal, hazardous waste, 0% water, to underground deposit             | 4.80E-03   | kg                       | EAA2000               | 1    | 1.07               | (1,2,1,1,1,3,6)  |
|                                                                                                                                            |                                     | Oil                      | Oil is reprocessed/burnt; here it is assumed to be incinerated; contains 5% aluminium     | waste management  | hazardous waste incineration | No              | CH       | disposal, used mineral oil, 10% water, to hazardous waste incineration  | 2.30E-03   | kg                       | EAA2000               | 1    | 1.07               | (1,2,1,1,1,3,6)  |
|                                                                                                                                            |                                     | Solid waste, unspecified | Municipal waste as proxy; contains 33% aluminium                                          | waste management  | municipal incineration       | No              | CH       | disposal, municipal solid waste, 22.9% water, to municipal incineration | 6.50E-03   | kg                       | EAA2000               | 1    | 1.07               | (1,2,1,1,1,3,6)  |
|                                                                                                                                            |                                     | Scrap                    | aluminium losses during the rolling process that are not included in the different wastes | waste management  | municipal incineration       | No              | CH       | disposal, aluminium, 0% water, to municipal incineration                | 9.02E-03   | kg                       | EAA2000               | 1    | 1.09               | (2,2,1,1,1,3,6)  |
|                                                                                                                                            |                                     | Heat waste               |                                                                                           | air               | unspecified                  |                 |          | Heat, waste                                                             | 1.97E+00   | MJ                       | EAA2000               | 1    | 1.07               | (1,2,1,1,1,3,13) |
|                                                                                                                                            | sheet rolling, aluminium, packaging |                          |                                                                                           | paper & cardboard | cardboard & corrugated board | No              | RER      | packaging, corrugated board, mixed fibre, single wall, at plant         | 1.00E-04   | kg                       | EAA2000               | 1    | 1.07               | (1,2,1,1,1,3,4)  |
|                                                                                                                                            |                                     |                          |                                                                                           | plastics          | processing                   | No              | RER      | packaging film, LDPE, at plant                                          | 4.00E-04   | kg                       | EAA2000               | 1    | 1.07               | (1,2,1,1,1,3,4)  |
|                                                                                                                                            |                                     |                          |                                                                                           | metals            | extraction                   | No              | RER      | steel, converter, unalloyed, at plant                                   | 5.00E-04   | kg                       | EAA2000               | 1    | 1.07               | (1,2,1,1,1,3,4)  |
|                                                                                                                                            |                                     |                          |                                                                                           | metals            | processing                   | No              | RER      | sheet rolling, steel                                                    | 5.00E-04   | kg                       | EAA2000               | 1    | 1.07               | (1,2,1,1,1,3,4)  |
|                                                                                                                                            |                                     |                          |                                                                                           | wooden materials  | extraction                   | No              | RER      | sawn timber, softwood, raw, air dried, u=20%, at plant                  | 1.85E-07   | m <sup>3</sup>           | EAA2000               | 1    | 1.07               | (1,2,1,1,1,3,4)  |
|                                                                                                                                            |                                     | Sheet rolling, Aluminium |                                                                                           | metals            | processing                   | No              | RER      | sheet rolling, aluminium                                                | 1.00E+00   | kg                       |                       |      |                    |                  |

Fig. 3.2 Flows for "sheet rolling, aluminium" and their representation in the ecoinvent database (cont.)

## 3.2 Section bar extrusion of aluminium

"Extrusion is the process of forcing heated aluminium through a die under extreme force to produce the desired shape. Extrusions can be made in literally thousands of complex shapes. Billets are pre-heated and forced by a ram through one or more dies to achieve the desired cross-section. The product is long in relation to its cross-sectional dimensions and has a cross section other than that of rod and bar and pipe and tube"<sup>5</sup>. The starting material for aluminium extrusion production is an extrusion billet (or extrusion ingot), coming from either primary or secondary aluminium production.

An overview of the different process steps taking place, when Aluminium is extruded to a profile, is given in Fig. 3.3.

The same amount of rolling mill infrastructure balanced per volume unit of treated material for the steel forming processes, is inventoried for the section bar extrusion of aluminium.

### 3.2.1 Environmental issues of section bar extruding of aluminium

In the data referred in the considered literature (EAA (2000)), the main air and water emissions, are arising from the combustion of energy carriers. Emissions to land are mainly hazardous and oily wastes.

### 3.2.2 Section bar extruding of aluminium in ecoinvent

The inventory data reported were obtained from an industry-wide survey carried out in 1998. For the section bar extruding industry segment, the survey coverage was 30 % related to the total production output (EAA (2000)).

The basic input material, primary aluminium is not balanced. The scrap losses of aluminium are mainly internally recycled, this recycling process is not balanced. A small amount of scrap material is not recycled, but is lost in waste. To obtain an even mass balance for 1 kilogram of output product, it is necessary to balance the 13 grams lost in waste as input of primary aluminium (Fig. 3.3).

The background information to the process "section bar extrusion, aluminium" is given in Tab. 3.2, flows are shown in Fig. 3.2.

#### Data quality

A standard precision of 5% percent of the value is given in EAA (2000)(), to obtain a more detailed view of the data quality, the uncertainty is estimated by completing the pedigree matrix, according to (Frischknecht et al. (2003)).

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<sup>5</sup> [http://www.energysolutionscenter.org/HeatTreat/MetalsAdvisor/aluminum/process\\_descriptions/extrusion.htm](http://www.energysolutionscenter.org/HeatTreat/MetalsAdvisor/aluminum/process_descriptions/extrusion.htm), accessed 24.03.03

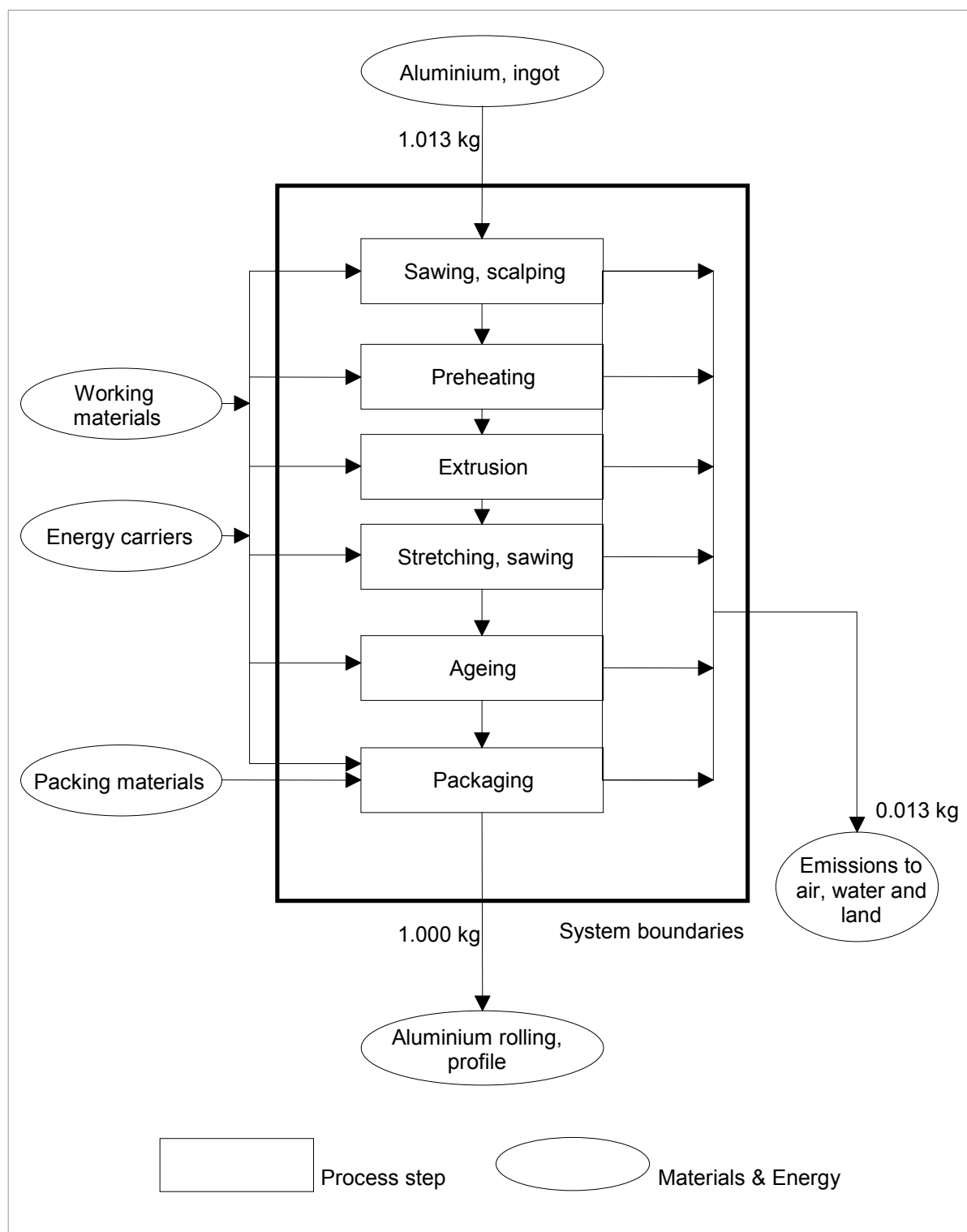


Fig. 3.3 Aluminium profile extrusion process; values are aluminium flows (modified from EAA (2000))

Tab. 3.2 Ecoinvent meta information for "section bar extrusion, aluminium"

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>                           | <b>section bar extrusion, aluminium</b>                                                                                                                                                                                                                                                                                                                                                                                     |
| Location                              | RER                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Infrastructure Process                | 0                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Unit                                  | kg                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Data Set Version                      | 1                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Included Processes                    | All the process steps, which can be attributed to semi-fabrication (sawing, scalping, preheating, extrusion, stretching, sawing, ageing and packaging), are included. Does not include the material being extruded; only the amount of scrap lost in waste is balanced as primary aluminium input. Includes the transport of the materials to the plant, but does not include the transport of the product to the customer. |
| Amount                                | 1                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Local Name                            | Profil extrudieren, Aluminium                                                                                                                                                                                                                                                                                                                                                                                               |
| Synonyms                              | profile extrusion                                                                                                                                                                                                                                                                                                                                                                                                           |
| General Comment to reference function | Heated aluminium is forced through one or more dies under extreme force to produce the desired shape. A great number of different shapes, and therefore possible applications, can be achieved by extrusion of aluminium.                                                                                                                                                                                                   |
| Start Date                            | 2000                                                                                                                                                                                                                                                                                                                                                                                                                        |
| End Date                              | 2002                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Other Period Text                     | Data refers to a survey made in 1998                                                                                                                                                                                                                                                                                                                                                                                        |
| Geography text                        | Data-set is representative for European Union                                                                                                                                                                                                                                                                                                                                                                               |
| Technology text                       | Average technique for European Union. The infrastructure was assumed to be the same as for rolling steel.                                                                                                                                                                                                                                                                                                                   |
| Representativeness [%]                | 30                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Production Volume                     | No data available                                                                                                                                                                                                                                                                                                                                                                                                           |
| Sampling Procedure                    | Literature, estimation for transport and infrastructure                                                                                                                                                                                                                                                                                                                                                                     |
| Extrapolations                        | see technology and sampling procedure                                                                                                                                                                                                                                                                                                                                                                                       |
| Uncertainty Adjustments               | Precision of the data set is referred to be of 5 % by the reference, nevertheless uncertainty was estimated by filling out the uncertainty scoring                                                                                                                                                                                                                                                                          |

### 3. Aluminium forming

| General Flow Information |                                             |                           | Representation in ecoinvent                                                                                                     |                   |                               |                 |          |                                                                                 |            | Uncertainty Informations |                       |      |                    |                             |
|--------------------------|---------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------|-----------------|----------|---------------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------|
| Input                    | Process Name                                | Output                    | Remarks                                                                                                                         | Category          | Sub-category                  | Infra-structure | Location | Module name in ecoinvent                                                        | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment             |
| Primary Aluminium        | section bar extrusion, aluminium, exxtusion |                           |                                                                                                                                 | metals            | extraction                    | No              | RER      | aluminium, primary, at plant                                                    | 1.30E-02   | kg                       | EAA2000               | 1    | 1.09               | (1,3,1,1,1,3,3)             |
| Caustic soda             |                                             |                           |                                                                                                                                 | chemicals         | inorganics                    | No              | RER      | soda, pow der, at plant                                                         | 2.80E-02   | kg                       | EAA2000               | 1    | 1.09               | (1,3,1,1,1,3,4)             |
| Water                    |                                             |                           |                                                                                                                                 | resource          | in w ater                     |                 |          | Water, cooling, unspecified natural origin                                      | 1.70E-02   | m 3                      | EAA2000               | 1    | 1.09               | (1,3,1,1,1,3,12)            |
| Fuel Oil                 |                                             |                           |                                                                                                                                 | oil               | heating systems               | No              | RER      | light fuel oil, burned in industrial furnace 1MW, non-modulating                | 2.77E-02   | M J                      | EAA2000               | 1    | 1.09               | (1,3,1,1,1,3,1)             |
| Gas                      |                                             |                           |                                                                                                                                 | natural gas       | heating systems               | No              | RER      | natural gas, burned in industrial furnace low - NOx >100kW                      | 3.00E+00   | M J                      | EAA2000               | 1    | 1.09               | (1,3,1,1,1,3,1)             |
| Electricity              |                                             |                           |                                                                                                                                 | electricity       | production mix                | No              | UCTE     | electricity, medium voltage, production UCTE, at grid                           | 1.14E+00   | kWh                      | EAA2000               | 1    | 1.09               | (1,3,1,1,1,3,2)             |
| Rolling mill             |                                             |                           | It is assumed, that the same volume of material is rolled per year. Density of steel is 7850 kg/m3 and for aluminium 2700 kg/m3 | metals            | processing                    | Yes             | RER      | rolling mill                                                                    | 4.72E-09   | unit                     | EAA2000               | 1    | 3.36               | (4,5,1,3,4,5,9)             |
| Lorry 32t                |                                             |                           |                                                                                                                                 | transport systems | road                          | No              | RER      | transport, lorry 32t                                                            | 2.25E-02   | tkm                      | EAA2000               | 1    | 2.09               | (4,5,n.a.,n.a.,n.a.,n.a.,5) |
|                          |                                             | Hazardous waste           | contains 15% aluminium                                                                                                          | w aste management | underground deposit           | No              | DE       | disposal, hazardous w aste, 0% w ater, to underground deposit                   | 1.60E-03   | kg                       | EAA2000               | 1    | 1.09               | (1,3,1,1,1,3,6)             |
|                          |                                             | Oil                       | Oil is reprocessed/burnt; here it is assumed to be incinerated. contains 5% aluminium                                           | w aste management | hazardous w aste incineration | No              | CH       | disposal, used mineral oil, 10% w ater, to hazardous w aste incineration        | 1.70E-03   | kg                       | EAA2000               | 1    | 1.09               | (1,3,1,1,1,3,6)             |
|                          |                                             | Spent bath/sludge         | 70% of filter dust w ith 0% w ater, as proxy for sludge w ith 30% w ater. Contains 15% aluminium                                | w aste management | residual material landfill    | No              | CH       | disposal, filter dust Al electrolysis, 0% w ater, to residual material landfill | 2.03E-02   | kg                       | EAA2000               | 1    | 1.09               | (1,3,1,1,1,3,6)             |
|                          |                                             | Solid w aste, unspecified | Municipal w aste as proxy. contains 33% aluminium                                                                               | w aste management | municipal incineration        | No              | CH       | disposal, municipal solid w aste, 22.9% w ater, to municipal incineration       | 2.23E-02   | kg                       | EAA2000               | 1    | 1.09               | (1,3,1,1,1,3,6)             |
|                          |                                             | Scrap                     | Losses during the rolling process that are not included in the different w astes                                                | w aste management | municipal incineration        | No              | CH       | disposal, aluminium, 0% w ater, to municipal incineration                       | 9.66E-04   | kg                       | EAA2000               | 1    | 1.09               | (2,2,1,1,1,3,6)             |
|                          |                                             | Heat w aste               |                                                                                                                                 | air               | unspecified                   |                 |          | Heat, w aste                                                                    | 4.11E+00   | M J                      | EAA2000               | 1    | 1.09               | (1,3,1,1,1,3,13)            |

Fig. 3.4 Flows for "section bar extrusion, aluminium" and their representation in the ecoinvent database

### 3. Aluminium forming

| General Flow Information |                                             |                              | Representation in ecoinvent |                   |                              |                 |          |                                                                  |            | Uncertainty Informations |                       |      |                    |                 |
|--------------------------|---------------------------------------------|------------------------------|-----------------------------|-------------------|------------------------------|-----------------|----------|------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------|
| Input                    | Process Name                                | Output                       | Remarks                     | Category          | Sub-category                 | Infra-structure | Location | Modul name in ecoinvent                                          | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment |
| Paper, cardboard-alpr    | section bar extrusion, aluminium, packaging | Profile extruding, Aluminium |                             | paper & cardboard | cardboard & corrugated board | No              | RER      | packaging, corrugated board, mixed fibre, single w all, at plant | 3.00E-03   | kg                       | EAA2000               | 1    | 1.09               | (1,3,1,1,1,3,4) |
| Plastic-alpr             |                                             |                              |                             | plastics          | processing                   | No              | RER      | packaging film, LDPE, at plant                                   | 2.10E-03   | kg                       | EAA2000               | 1    | 1.09               | (1,3,1,1,1,3,4) |
| Steel-alpr               |                                             |                              |                             | metals            | extraction                   | No              | RER      | steel, converter, unalloyed, at plant                            | 9.00E-04   | kg                       | EAA2000               | 1    | 1.09               | (1,3,1,1,1,3,4) |
| Steel-sh-alpr            |                                             |                              |                             | metals            | processing                   | No              | RER      | sheet rolling, steel                                             | 9.00E-04   | kg                       | EAA2000               | 1    | 1.09               | (1,3,1,1,1,3,4) |
| Wood-alpr                |                                             |                              |                             | w ooden materials | extraction                   | No              | RER      | saw n timber, softw ood, raw , air dried, u=20%, at plant        | 5.19E-05   | m 3                      | EAA2000               | 1    | 1.09               | (1,3,1,1,1,3,4) |
|                          |                                             |                              |                             | metals            | processing                   | No              | RER      | section bar extrusion, aluminium                                 | 1.00E+00   | kg                       |                       |      |                    |                 |

**Fig. 3.4** Flows for "section bar extrusion, aluminium" and their representation in the ecoinvent database (cont.)



## 4 Copper and copper alloy forming

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Review: Roland Hischier

Copper and copper alloys are melted continuously or in batches and cast for the production of shapes suited for the further fabrication step. The cast products are precursors for a variety of materials such as sheets, strips, sections, bars, rods, wires and tubes (IPPC (2001b)).

### 4.1 Sheet rolling of copper

Slabs/cakes of copper or copper alloy are the start material for the fabrication of sheets and strips. The material is preheated in gas or oil fired furnaces, hot and cold rolled and then sent to the finishing operation, which includes re-rolling, cutting to length and width. Surface milling, annealing, pickling, washing and drying are required as intermediate steps to produce high quality strips and sheets.

Hot rolling is usually done with a dual rolling mill equipped with benches up to 200 m and a final coiling device. The cooling water for the rolls has small amounts of lubricant added for improving the attachment to the steel rolls. The vapour generated is vented and the vent gas is de-misted prior to release to atmosphere.

Further cold rolling operations are then performed. Cold rolling results in a hardened metal. In most cases the coil is annealed prior to cold rolling. Annealing is done under reducing conditions to avoid oxidation. A protection gas “exogas” or nitrogen/hydrogen mixtures are used. Exogas is produced at site from natural gas in a special reactor, which is indirectly fired. N<sub>2</sub> and H<sub>2</sub> are purchased and stored at site in special tanks. The N<sub>2</sub>/H<sub>2</sub> protection gas mixtures are produced from the storage tanks by mixing the components in the ratio required. For annealing before cold rolling bell type furnaces are used with electric heating or indirectly fired by natural gas or fuel oil. Tower type furnaces are applied for intermediate annealing of pre-rolled coils.

The sheet thickness is further reduced by stepwise cold rolling operations on different reversing mills. For cold rolling different types of mills called Duo, Quarto, Sexto, Sendzimir (12 rolls) are used. Rolling mill designs as single-stand which are combined to an in-line multi-stand rolling mill are also applied. The combination of the different types applied depends on the thickness of the sheet on the coils and on the finishing status the coil has already been reached which.

During cold rolling an emulsion or oil is used for roll protection. Therefore, the roll stands are vented and the ventilation gases are cleaned by mechanical filters, wet electrostatic precipitators or scrubbing. The emulsion and the oil is cleaned from metal and cracked oil particles are removed by paper or textile band filters.

This description given in (IPPC (2001b)) shows, that the principle for sheet rolling of copper is almost the same of the one for sheet rolling of steel, chromium steel or aluminium. An approximative overview of the process is given in

As no data were available, the infrastructure it is assumed to be the same as for steel rolling. The same amount of infrastructure per volume unit is balanced.

#### 4.1.1 Environmental issues of sheet rolling of copper

The main environmental issues of the refined copper industry are air and water pollution. The facilities generally have their own wastewater treatment facilities and wastewater recycling is usually practised. Most potential wastes are reused. Some local aspects, like noise, are relevant to the industry. Due to the hazardous nature of some solid and liquid waste streams, there is also a significant risk for soil

contamination unless they are stored and handled correctly. Fugitive emissions can be highly significant (IPPC (2001b)).

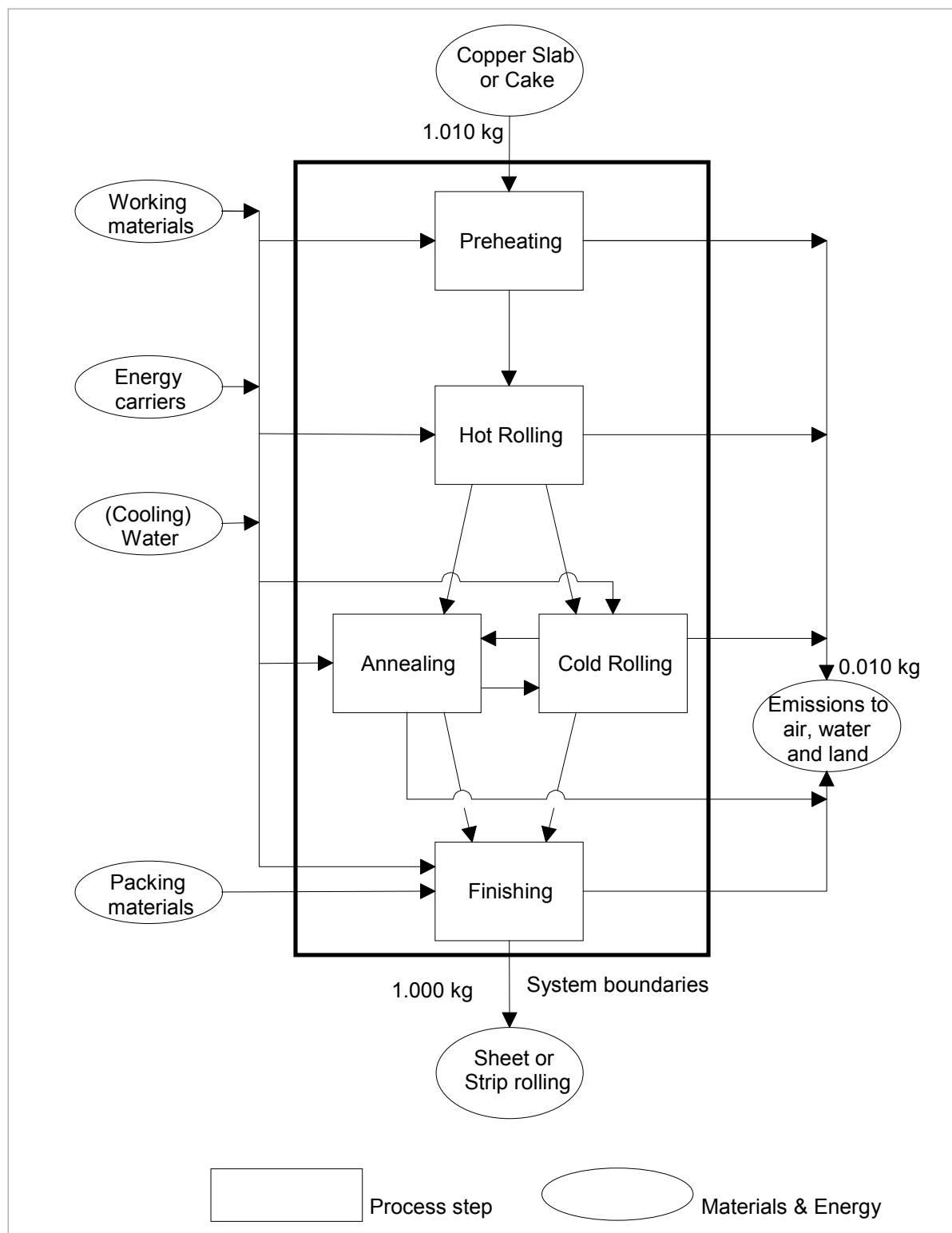


Fig. 4.1 Copper sheet rolling process; values are copper flows

### 4.1.2 Sheet rolling of copper in ecoinvent

As no data were available in literature for the process of copper sheet rolling, the values for the different in- and outputs are estimated by the values of aluminium sheet production and the hot rolling process of steel.

The background information to the process "sheet rolling, copper" is given in Tab. 4.1, flows are shown in Fig. 4.2.

#### Data quality

Data quality is estimated by filling in the pedigree matrix, according to (Frischknecht et al. (2003)), taking into account the different more or less rough estimations and assumptions by approximating the copper sheet rolling with the values of the aluminium sheet rolling process.

**Tab. 4.1 Ecoinvent meta information for "sheet rolling, copper"**

| Name                                  | sheet rolling, copper                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location                              | RER                                                                                                                                                                                                                                                                                                                                          |
| Infrastructure Process                | 0                                                                                                                                                                                                                                                                                                                                            |
| Unit                                  | kg                                                                                                                                                                                                                                                                                                                                           |
| Data Set Version                      | 1                                                                                                                                                                                                                                                                                                                                            |
| Included Processes                    | Includes the process-steps: preheating, hot rolling, cold rolling, annealing and finishing. Does not include the material being rolled, only the amount of scrap lost in waste is balanced as primary copper input. Includes the transport of the materials to the plant, but does not include the transport of the product to the customer. |
| Amount                                | 1                                                                                                                                                                                                                                                                                                                                            |
| Local Name                            | Blech walzen, Kupfer                                                                                                                                                                                                                                                                                                                         |
| Synonyms                              |                                                                                                                                                                                                                                                                                                                                              |
| General Comment to reference function | The process of copper sheet rolling is similar to the process of aluminium sheet rolling. Also the final outputs are similar in shape, even if the material properties and thus the applications are very different                                                                                                                          |
| Start Date                            | 2000                                                                                                                                                                                                                                                                                                                                         |
| End Date                              | 2002                                                                                                                                                                                                                                                                                                                                         |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                                                                                                                            |
| Other Period Text                     | Data on which assumptions are basing are from 1998                                                                                                                                                                                                                                                                                           |
| Geography text                        | Data-set is representative for European Union                                                                                                                                                                                                                                                                                                |
| Technology text                       | Technique describes the average of European Union. Data are estimated basing on the data of aluminium sheet rolling, set in relation by mean of the volume of material worked.                                                                                                                                                               |
| Representativeness [%]                |                                                                                                                                                                                                                                                                                                                                              |
| Production Volume                     | No data available                                                                                                                                                                                                                                                                                                                            |
| Sampling Procedure                    | Estimation basing on Aluminium data                                                                                                                                                                                                                                                                                                          |
| Extrapolations                        | see technology                                                                                                                                                                                                                                                                                                                               |
| Uncertainty Adjustments               | The fact that no data were available about the percentage of companies represented by the study cited and the assumptions made, are considered in the uncertainties.                                                                                                                                                                         |

#### 4. Copper and copper alloy forming

| General Flow Information |                                |            | Representation in ecoinvent                                                                                                                                                                                                                  |                   |                 |                 |          |                                                                    |            | Uncertainty Informations |                       |      |                    |                             |  |
|--------------------------|--------------------------------|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-----------------|-----------------|----------|--------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------|--|
| Input                    | Process Name                   | Output     | Remarks                                                                                                                                                                                                                                      | Category          | Sub-category    | Infra-structure | Location | Modul name in ecoinvent                                            | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment             |  |
| Primary Copper           | sheet rolling, copper, rolling |            | It is assumed, that about the same volume of primary material is lost due to scratch losses as it is for aluminium sheet production                                                                                                          | metals            | extraction      | No              | RER      | copper, at regional storage                                        | 3.96E-02   | kg                       | Estimation            | 1    | 1.45               | (4,5,2,1,3,5,3)             |  |
| Rolling oil              |                                |            | Basing on values for hot rolling of steel and aluminium sheet rolling                                                                                                                                                                        | chemicals         | organics        | No              | RER      | lubricating oil, at plant                                          | 4.00E-03   | kg                       | Estimation            | 1    | 1.45               | (4,5,2,1,3,5,11)            |  |
| Water                    |                                |            | Basing on values for hot rolling of steel and aluminium sheet rolling considering the temperatures reached in the process                                                                                                                    | resource          | in w ater       |                 |          | Water, cooling, unspecified natural origin                         | 6.00E-03   | m 3                      | Estimation            | 1    | 1.45               | (4,5,2,1,3,5,12)            |  |
| Fuel Oil                 |                                |            | Basing on values for hot rolling of steel and aluminium sheet rolling. As no data w ere available about the fraction of oil and gas burners used, the amount of thermal energy is distributed by one half for fuel oil and natural gas each. | oil               | heating systems | No              | RER      | light fuel oil, burned in industrial furnace 1MW, non-modulating   | 8.00E-01   | M J                      | Estimation            | 1    | 1.45               | (4,5,2,1,3,5,1)             |  |
| Gas                      |                                |            |                                                                                                                                                                                                                                              | natural gas       | heating systems | No              | RER      | natural gas, burned in industrial furnace low - NOx >100kW         | 8.00E-01   | M J                      | Estimation            | 1    | 1.45               | (4,5,2,1,3,5,1)             |  |
| Electricity              |                                |            | Basing mainly on the values of aluminium sheet rolling                                                                                                                                                                                       | electricity       | production mix  | No              | UCTE     | electricity, medium voltage, production UCTE, at grid              | 1.81E+00   | kWh                      | Estimation            | 1    | 1.45               | (4,5,2,1,3,5,2)             |  |
| Rolling Mill-Cu          |                                |            | It is assumed, that the same volume of material is rolled per year. Density of steel is 7850 kg/m3 and for copper 8920 kg/m3                                                                                                                 | metals            | processing      | Yes             | RER      | rolling mill                                                       | 1.43E-09   | unit                     | Estimation            | 1    | 3.18               | (4,5,2,1,3,5,9)             |  |
| Lorry 32 t               |                                |            | The packing and working material is assumed to be transported over 300 km in Europe before using it in the rolling mill                                                                                                                      | transport systems | road            | No              | RER      | transport, lorry 32t                                               | 1.41E-02   | tkm                      | Estimation            | 1    | 2.09               | (4,5,n.a.,n.a.,n.a.,n.a.,5) |  |
|                          |                                | VOC to air | 10% of rolling oil assumed (based on values of steel and aluminum rolling)                                                                                                                                                                   | air               | unspecified     |                 |          | NM VOC, non-methane volatile organic compounds, unspecified origin | 4.00E-04   | kg                       | Estimation            | 1    | 2.19               | (4,5,2,1,3,5,23)            |  |
|                          |                                | Water      | Same value as for aluminium rolling assumed                                                                                                                                                                                                  |                   |                 |                 |          |                                                                    | 6.00E-03   | m 3                      | Estimation            | 1    | 1.45               | (4,5,2,1,3,5,4)             |  |
|                          |                                | COD        | Same value as for aluminium rolling assumed                                                                                                                                                                                                  | w ater            | unspecified     |                 |          | COD, Chemical Oxygen Demand                                        | 7.90E-05   | kg                       | Estimation            | 1    | 1.73               | (4,5,2,1,3,5,32)            |  |
|                          |                                | BOD        | Same value as for aluminium rolling assumed                                                                                                                                                                                                  | w ater            | unspecified     |                 |          | BOD5, Biological Oxygen Demand                                     | 7.90E-05   | kg                       | Estimation            | 1    | 2.79               | (5,5,5,5,5,32)              |  |
|                          |                                | DOC        | Same value as for aluminium rolling assumed                                                                                                                                                                                                  | w ater            | unspecified     |                 |          | DOC, Dissolved Organic Carbon                                      | 2.06E-05   | kg                       | Estimation            | 1    | 2.79               | (5,5,5,5,5,32)              |  |
|                          |                                | TOC        | Same value as for aluminium rolling assumed                                                                                                                                                                                                  | w ater            | unspecified     |                 |          | TOC, Total Organic Carbon                                          | 2.06E-05   | kg                       | Estimation            | 1    | 2.79               | (5,5,5,5,5,32)              |  |

Fig. 4.2 Flows for "sheet rolling, copper" and their representation in the ecoinvent database

4. Copper and copper alloy forming

| General Flow Information                                                                                             |                                  |                          | Representation in ecoinvent                                                                                                                                                             |                   |                              |                 |          |                                                                         |            | Uncertainty Informations |                       |      |                    |                  |
|----------------------------------------------------------------------------------------------------------------------|----------------------------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------------------|-----------------|----------|-------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|------------------|
| Input                                                                                                                | Process Name                     | Output                   | Remarks                                                                                                                                                                                 | Category          | Sub-category                 | Infra-structure | Location | Module name in ecoinvent                                                | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment  |
| <div> <div>Paper; cardboard</div> <div>Plastics</div> <div>Steel</div> <div>Steel-sheet</div> <div>Wood</div> </div> | sheet rolling, copper, rolling   | Hazardous waste          | Contains 15% copper                                                                                                                                                                     | waste management  | underground deposit          | No              | DE       | disposal, hazardous waste, 0% water, to underground deposit             | 1.55E-02   | kg                       | Estimation            | 1    | 1.45               | (4,5,2,1,3,5,6)  |
|                                                                                                                      |                                  | Oil                      | Oil is reprocessed/burnt; here it is assumed to be incinerated; Contains 5% copper                                                                                                      | waste management  | hazardous waste incineration | No              | CH       | disposal, used mineral oil, 10% water, to hazardous waste incineration  | 2.40E-03   | kg                       | Estimation            | 1    | 1.45               | (4,5,2,1,3,5,6)  |
|                                                                                                                      |                                  | Solid waste, unspecified | Municipal waste as proxy; Contains 33% copper                                                                                                                                           | waste management  | municipal incineration       | No              | CH       | disposal, municipal solid waste, 22.9% water, to municipal incineration | 2.32E-02   | kg                       | Estimation            | 1    | 1.45               | (4,5,2,1,3,5,6)  |
|                                                                                                                      |                                  | Scrap                    | Losses during the rolling process that are not included in the different wastes. Recycled                                                                                               |                   |                              |                 |          |                                                                         | 2.96E-02   | kg                       | Estimation            | 1    | 1.09               | (2,2,1,1,1,3,6)  |
|                                                                                                                      |                                  | Heat waste               |                                                                                                                                                                                         | air               | unspecified                  |                 |          | Heat, waste                                                             | 6.51E+00   | MJ                       | Estimation            | 1    | 1.45               | (4,5,2,1,3,5,13) |
|                                                                                                                      | sheet rolling, copper, packaging |                          | The same amount of packing material as for one cubic meter of aluminium sheet is inventoried. Densities are 2700 kg/m <sup>3</sup> for aluminium and 8920 kg/m <sup>3</sup> for copper. | paper & cardboard | cardboard & corrugated board | No              | RER      | packaging, corrugated board, mixed fibre, single wall, at plant         | 3.30E-04   | kg                       | Estimation            | 1    | 1.45               | (4,5,2,1,3,5,4)  |
|                                                                                                                      |                                  |                          |                                                                                                                                                                                         | plastics          | processing                   | No              | RER      | packaging film, LDPE, at plant                                          | 1.32E-03   | kg                       | Estimation            | 1    | 1.45               | (4,5,2,1,3,5,4)  |
|                                                                                                                      |                                  |                          |                                                                                                                                                                                         | metals            | extraction                   | No              | RER      | steel, converter, unalloyed, at plant                                   | 1.65E-03   | kg                       | Estimation            | 1    | 1.45               | (4,5,2,1,3,5,4)  |
|                                                                                                                      |                                  |                          |                                                                                                                                                                                         | metals            | processing                   | No              | RER      | sheet rolling, steel                                                    | 1.65E-03   | kg                       | Estimation            | 1    | 1.45               | (4,5,2,1,3,5,4)  |
|                                                                                                                      |                                  |                          |                                                                                                                                                                                         | wooden materials  | extraction                   | No              | RER      | sawn timber, softwood, raw, air dried, u=20%, at plant                  | 6.12E-07   | m <sup>3</sup>           | Estimation            | 1    | 1.45               | (4,5,2,1,3,5,4)  |
|                                                                                                                      |                                  | Copper Sheet             |                                                                                                                                                                                         | metals            | processing                   | No              | RER      | sheet rolling, copper                                                   | 1.00E+00   | kg                       |                       |      |                    |                  |

Fig. 4.2 Flows for "sheet rolling, copper" and their representation in the ecoinvent database (cont.)

## 4.2 Wire drawing of copper

Wire production consists in a first step of wire rod production, which can be compared and approximated by the process of sheet rolling with a smaller degree of deformation. In a second working step, the wire rod is drawn through a series of dies and reduced to its final cross section. The description of this process can well be approximated by the description of steel wire drawing, described in chapter 2.5.

As no data were available, the infrastructure it is assumed to be the same (per volume unit) as for steel wire drawing.

### 4.2.1 Environmental issues of wire drawing of copper

The environmental issues of copper wire drawing, are the same as for the copper sheet production (see chapter 4.1.1).

### 4.2.2 Wire drawing of copper in ecoinvent

No data were available in literature about the production process of copper wire drawing. As the production process can be divided in two main steps, the assumptions and approximations for the balanced values are made separately for these two steps.

The first step, the rolling of the wire rod, is similar to the rolling of the copper sheet. The second is similar in the technique applied, but not in the values resulting, to the drawing of steel wire, because of the greater formability of copper. Values balanced were estimated basing on aluminium and steel forming processes. The background information to the process "wire drawing, copper" is given in Tab. 4.2, flows are shown in Fig. 4.3.

#### Data quality

Data quality is estimated by filling in the pedigree matrix, according to (Frischknecht et al. (2003)), taking into account the different more or less rough estimations and assumptions by approximating the copper wire drawing with the adapted values of the aluminium sheet rolling process.

Tab. 4.2 Ecoinvent meta information for "wire drawing, copper"

|                                       |                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>                           | <b>wire drawing, copper</b>                                                                                                                                                                                                                                                                                                            |
| Location                              | RER                                                                                                                                                                                                                                                                                                                                    |
| Infrastructure Process                | 0                                                                                                                                                                                                                                                                                                                                      |
| Unit                                  | kg                                                                                                                                                                                                                                                                                                                                     |
| Data Set Version                      | 1                                                                                                                                                                                                                                                                                                                                      |
| Included Processes                    | Includes the production of wire rod and the further drawing of this to wire. Does not include the material being rolled or drawn; only the amount of scrap lost in waste is balanced as primary copper input. Includes the transport of the materials to the plant, but does not include the transport of the product to the customer. |
| Amount                                | 1                                                                                                                                                                                                                                                                                                                                      |
| Local Name                            | Draht ziehen, Kupfer                                                                                                                                                                                                                                                                                                                   |
| Synonyms                              |                                                                                                                                                                                                                                                                                                                                        |
| General Comment to reference function | Wire rod production is comparable to sheet rolling leading to another final shape. Further drawing leads to wires with cross sections ranging from 1.6 to 3.5 mm and higher.                                                                                                                                                           |
| Start Date                            | 2000                                                                                                                                                                                                                                                                                                                                   |
| End Date                              | 2002                                                                                                                                                                                                                                                                                                                                   |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                                                                                                                      |
| Other Period Text                     | Data on which assumptions are basing are from 1998                                                                                                                                                                                                                                                                                     |
| Geography text                        | Data-set is representative for European Union                                                                                                                                                                                                                                                                                          |
| Technology text                       | Technique describes the average of European Union. Data are estimated basing on the data of aluminium sheet rolling, set in relation by mean of the volume of material worked.                                                                                                                                                         |
| Representativeness [%]                |                                                                                                                                                                                                                                                                                                                                        |
| Production Volume                     | No data available                                                                                                                                                                                                                                                                                                                      |
| Sampling Procedure                    | Estimation basing on Aluminium data                                                                                                                                                                                                                                                                                                    |
| Extrapolations                        | see technology                                                                                                                                                                                                                                                                                                                         |
| Uncertainty Adjustments               | The fact that no data were available about the percentage of companies represented by the study cited and the assumptions made, are considered in the uncertainties.                                                                                                                                                                   |

#### 4. Copper and copper alloy forming

| General Flow Information |                               |            | Representation in ecoinvent                                                                                                                                                                                                  |                   |                 |                 |          |                                                                    |            | Uncertainty Informations |                       |      |                    |                             |
|--------------------------|-------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-----------------|-----------------|----------|--------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------|
| Input                    | Process Name                  | Output     | Remarks                                                                                                                                                                                                                      | Category          | Sub-category    | Infra-structure | Location | Modul name in ecoinvent                                            | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment             |
| Primary Copper           | wire drawing, copper, rolling |            | It is assumed, that for each step, the wire rod rolling and the wire drawing, about 50% of primary material is lost due to scratch losses as it is for copper sheet production. Thus in the same amount is inventoried.      | metals            | extraction      | No              | RER      | copper, at regional storage                                        | 3.96E-02   | kg                       | Estimation            | 1    | 0.08               | (4,5,2,1,3,5,3)             |
| Rolling oil              |                               |            | The wire rod production step requires less rolling oil than sheet rolling (factor 0.8 assumed), due to the smaller degree of deformation. For the wire drawing step the same amount as for sheet rolling is assumed.         | chemicals         | organics        | No              | RER      | lubricating oil, at plant                                          | 7.20E-03   | kg                       | Estimation            | 1    | 0.01               | (4,5,2,1,3,5,11)            |
| Water                    |                               |            | The rod production step needs the same amount of cooling water as sheet rolling, as the material is heated to the same temperature. The wire drawing step requires less water (factor 0.8 assumed) because of friction heat. | resource          | in water        |                 |          | Water, cooling, unspecified natural origin                         | 1.08E-02   | m3                       | Estimation            | 1    | 0.01               | (4,5,2,1,3,5,12)            |
| Fuel Oil                 |                               |            | Basing on the amount of material to be heated. As no data were available about the fraction of oil and gas burners used, the amount of thermal energy is distributed by one half for fuel oil and natural gas each.          | oil               | heating systems | No              | RER      | light fuel oil, burned in industrial furnace 1MW, non-modulating   | 8.00E-01   | MJ                       | Estimation            | 1    | 0.83               | (4,5,2,1,3,5,1)             |
| Gas                      |                               |            |                                                                                                                                                                                                                              | natural gas       | heating systems | No              | RER      | natural gas, burned in industrial furnace low-NOx >100kW           | 8.00E-01   | MJ                       | Estimation            | 1    | 0.83               | (4,5,2,1,3,5,1)             |
| Electricity              |                               |            | Wire rod rolling requires less, wire drawing more electrical energy for the mechanical deformation step, than sheet rolling                                                                                                  | electricity       | production mix  | No              | UCTE     | electricity, medium voltage, production UCTE, at grid              | 3.61E+00   | kWh                      | Estimation            | 1    | 3.61               | (4,5,2,1,3,5,2)             |
| Rolling Mill-Cu          |                               |            | It is assumed, that the same volume of material is rolled per year. Density of steel is 7850 kg/m3 and for copper 8920 kg/m3                                                                                                 | metals            | processing      | Yes             | RER      | rolling mill                                                       | 1.43E-09   | unit                     | Estimation            | 1    | 0                  | (4,5,2,1,3,5,9)             |
| Lorry 32 t               |                               |            | The packing and working material is assumed to be transported over 300 km in Europe before using it in the rolling mill                                                                                                      | transport systems | road            | No              | RER      | transport, lorry 32t                                               | 1.50E-02   | tkm                      | Estimation            | 1    | 0.03               | (4,5,n.a.,n.a.,n.a.,n.a.,5) |
|                          |                               | VOC to air | 10% of rolling oil assumed (based on values of steel and aluminum rolling)                                                                                                                                                   | air               | unspecified     |                 |          | NM VOC, non-methane volatile organic compounds, unspecified origin | 7.20E-04   | kg                       | Estimation            | 1    | 0                  | (4,5,2,1,3,5,23)            |

Fig. 4.3 Flows for "wire drawing, copper" and their representation in the ecoinvent database



#### 4. Copper and copper alloy forming

| General Flow Information                                                                                             |                                 |                            | Representation in ecoinvent                                                                                                                                                                     |                   |                              |                |          |                                                                         |            | Uncertainty Informations |                       |      |                    |                  |
|----------------------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------------------|----------------|----------|-------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|------------------|
| Input                                                                                                                | Process Name                    | Output                     | Remarks                                                                                                                                                                                         | Category          | Sub-category                 | Infrastructure | Location | Module name in ecoinvent                                                | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment  |
| <div> <div>Paper; cardboard</div> <div>Plastics</div> <div>Steel</div> <div>Steel-sheet</div> <div>Wood</div> </div> | wire drawing, copper, rolling   | m Water                    | Same value as for aluminium rolling assumed                                                                                                                                                     |                   |                              |                |          |                                                                         | 1.08E-02   | m <sup>3</sup>           | Estimation            | 1    | 0.01               | (4,5,2,1,3,5,4)  |
|                                                                                                                      |                                 | m COD                      | Same value as for aluminium rolling assumed                                                                                                                                                     | water             | unspecified                  |                |          | COD, Chemical Oxygen Demand                                             | 7.90E-05   | kg                       | Estimation            | 1    | 0                  | (4,5,2,1,3,5,32) |
|                                                                                                                      |                                 | m BOD                      | Same value as for aluminium rolling assumed                                                                                                                                                     | water             | unspecified                  |                |          | BOD5, Biological Oxygen Demand                                          | 7.90E-05   | kg                       | Estimation            | 1    | 2.79               | (5,5,5,5,5,32)   |
|                                                                                                                      |                                 | m DOC                      | Same value as for aluminium rolling assumed                                                                                                                                                     | water             | unspecified                  |                |          | DOC, Dissolved Organic Carbon                                           | 2.06E-05   | kg                       | Estimation            | 1    | 2.79               | (5,5,5,5,5,32)   |
|                                                                                                                      |                                 | m TOC                      | Same value as for aluminium rolling assumed                                                                                                                                                     | water             | unspecified                  |                |          | TOC, Total Organic Carbon                                               | 2.06E-05   | kg                       | Estimation            | 1    | 2.79               | (5,5,5,5,5,32)   |
|                                                                                                                      |                                 | m Hazardous waste          | Contains 15% copper                                                                                                                                                                             | waste management  | underground deposit          | No             | DE       | disposal, hazardous waste, 0% water, to underground deposit             | 1.55E-02   | kg                       | Estimation            | 1    | 0.03               | (4,5,2,1,3,5,6)  |
|                                                                                                                      |                                 | m Oil                      | Oil is reprocessed/burnt; here it is assumed to be incinerated; Contains 5% copper                                                                                                              | waste management  | hazardous waste incineration | No             | CH       | disposal, used mineral oil, 10% water, to hazardous waste incineration  | 4.32E-03   | kg                       | Estimation            | 1    | 0                  | (4,5,2,1,3,5,6)  |
|                                                                                                                      |                                 | m Solid waste, unspecified | Municipal waste as proxy; Contains 33% copper                                                                                                                                                   | waste management  | municipal incineration       | No             | CH       | disposal, municipal solid waste, 22.9% water, to municipal incineration | 2.32E-02   | kg                       | Estimation            | 1    | 0.04               | (4,5,2,1,3,5,6)  |
|                                                                                                                      |                                 | m Scrap                    | Losses during the rolling process that are not included in the different wastes. Recycled                                                                                                       |                   |                              |                |          |                                                                         | 2.95E-02   | kg                       | Estimation            | 1    | 0.06               | (2,2,1,1,1,3,6)  |
|                                                                                                                      |                                 | m Heat waste               |                                                                                                                                                                                                 | air               | unspecified                  |                |          | Heat, waste                                                             | 1.30E+01   | MJ                       | Estimation            | 1    | 13                 | (4,5,2,1,3,5,13) |
|                                                                                                                      | wire drawing, copper, packaging | m                          | The same amount of packing material as for one cubic meter of aluminium sheet product is inventoried. Densities are 2700 kg/m <sup>3</sup> for aluminium and 8920 kg/m <sup>3</sup> for copper. | paper & cardboard | cardboard & corrugated board | No             | RER      | packaging, corrugated board, mixed fibre, single wall, at plant         | 3.30E-04   | kg                       | Estimation            | 1    | 0                  | (4,5,2,1,3,5,4)  |
|                                                                                                                      |                                 | m                          |                                                                                                                                                                                                 | plastics          | processing                   | No             | RER      | packaging film, LDPE, at plant                                          | 1.32E-03   | kg                       | Estimation            | 1    | 0                  | (4,5,2,1,3,5,4)  |
|                                                                                                                      |                                 | m                          |                                                                                                                                                                                                 | metals            | extraction                   | No             | RER      | steel, converter, unalloyed, at plant                                   | 1.65E-03   | kg                       | Estimation            | 1    | 0                  | (4,5,2,1,3,5,4)  |
|                                                                                                                      |                                 | m                          |                                                                                                                                                                                                 | metals            | processing                   | No             | RER      | sheet rolling, steel                                                    | 1.65E-03   | kg                       | Estimation            | 1    | 0                  | (4,5,2,1,3,5,4)  |
|                                                                                                                      |                                 | m                          |                                                                                                                                                                                                 | wooden materials  | extraction                   | No             | RER      | sawn timber, softwood, raw, air dried, u=20%, at plant                  | 6.12E-07   | m <sup>3</sup>           | Estimation            | 1    | 0                  | (4,5,2,1,3,5,4)  |
|                                                                                                                      |                                 | m Copper wire              |                                                                                                                                                                                                 | metals            | processing                   | No             | RER      | wire drawing, copper                                                    | 1.00E+00   | kg                       |                       |      |                    |                  |

Fig. 4.3 Flows for "wire drawing, copper" and their representation in the ecoinvent database (cont.)

## **4.3 Contouring of brass and bronze**

### **4.3.1 Fundamentals of forming of copper alloys**

“Billets of copper or copper alloy are preheated and extruded and drawn into tubes with a variety of diameters and thickness in a number of “passes”. The drawing machines use a variety of oils and soaps as lubricant that contaminates the tubes and off cuts. The products are normally annealed and degreased before transport and the off cuts are de-oiled in a furnace or other degreasing process before being returned to the furnace for melting.

Copper tubes may be also produced utilising an extrusion press with piercer in which billet sections are extruded to tube shell pieces; the tube shell pieces are then rolled in a breakdown roller (Pilger mill) and finally drawn to size in drawing blocks. The oil utilised (in small quantities) for the drawing operations is neutralised in in-line degreasing/pickling systems often connected with the annealing section.

Products are annealed in a variety of furnaces under reducing conditions using as protection gas “exogas” or hydrogen/nitrogen mixtures.

Wire, tube, strip and some other materials are pickled to produce a bright finish before packing or, in the case of copper wire, to remove oxide from the surface prior to wire drawing. Sulphuric acid solutions are used and sometimes a mixture of dilute sulphuric and nitric acids are used. In the latter case nitric fumes can be emitted if higher strength acids are used.

Surface pickling of sheets and strips is usually performed with sulphuric acid (8 - 10%) but for some special alloys, mixtures of sulphuric and nitric acids are used in continuous lines or automatic systems. Gas cleaning is used to remove nitric acid fumes. The acid is changed from time to time in order to ensure that surface quality is not affected.

Spent acid is sent for treatment and recovery of metal in internal or external treatment plants. The pickling systems are vented for worker protection. The products are rinsed and the rinse water is sent for treatment, sludges are recycled if possible. For surface degreasing of rolled products, detergents containing water are used. The spent water is cleaned by ultra filtration.” (IPPC (2001b))

### **4.3.2 Environmental issues of forming of copper alloys**

Beside the energy consumption of the process, the emissions of metals and hydrocarbons to water from degreasing and pickling are an environmental issue. IPPC (2001b) gives the values in Tab. 4.3 for the annual loads discharged to water from a copper semis production plant. Hydrocarbon emissions are not measured. Emissions of “protection gas” can also be a problem:

“Typical CO production in a shaft furnace used for wire-rod or semis production is 2000 to 11000 grams per tonne of copper. In some installations after-burning is used to remove hydrocarbons from the gases when scrap coated with organic matter is processed. CO is also destroyed at the same time and the emissions are reported to be ~ 45 grams per tonne of copper.” (IPPC (2001b))

### **4.3.3 Contouring of brass and bronze in ecoinvent**

The processes are theoretically modelled according to the information in IPPC (2001b). Most of the values for the flows in these processes are estimated from similar processes since very few quantitative data is available.

The background information to the process "anodising, aluminium sheet" is given in Tab. 4.4, flows are shown in Fig. 4.4 and Fig. 4.5.

### Data quality

The uncertainty is estimated by completing the pedigree matrix, according to (Frischknecht et al. (2003)). A lognormal distribution is assumed. The different more or less rough estimations and assumptions are taken into account. Overall data quality is poor.

**Tab. 4.3 Annual loads discharged to water from a copper semis production plant (IPPC (2001b))**

| Emission | kg/a | kg/m3 effluent |
|----------|------|----------------|
| Cu       | 11   | 3.14E-04       |
| Ni       | 3    | 8.57E-05       |
| Zn       | 25   | 7.14E-04       |
| Pb       | 1    | 2.86E-05       |
| Cr       | 1    | 2.86E-05       |
| As       | 0.01 | 2.86E-07       |
| Cd       | 0.01 | 2.86E-07       |
| Hg       | 0.01 | 2.86E-07       |
| Sn       | 1    | 2.86E-05       |

Effluent discharge: 35000 m3/a.

**Tab. 4.4 Ecoinvent meta information for contouring of brass and bronze**

| Name                                  | contour, brass                                                                                             | contour, bronze                                                                                            |
|---------------------------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Location                              | RER                                                                                                        | RER                                                                                                        |
| Infrastructure Process                | 0                                                                                                          | 0                                                                                                          |
| Unit                                  | kg                                                                                                         | kg                                                                                                         |
| Data Set Version                      | 2.0                                                                                                        | 2.0                                                                                                        |
| Included Processes                    | Heating, rolling, degreasing / pickling and waste water treatment.                                         | Heating, rolling, degreasing / pickling and waste water treatment.                                         |
| Amount                                | 1                                                                                                          | 1                                                                                                          |
| Local Name                            | Profilieren, Messing                                                                                       | Profilieren, Bronze                                                                                        |
| Synonyms                              | rolling brass                                                                                              | rolling bronze                                                                                             |
| General Comment to reference function | Most of the data used stem from processes similar to contouring brass. Overall data quality is poor.       | Most of the data used stem from processes similar to contouring bronze. Overall data quality is poor.      |
| Start Date                            | 1997                                                                                                       | 1997                                                                                                       |
| End Date                              | 2002                                                                                                       | 2002                                                                                                       |
| Data Valid For Entire Period          | 1                                                                                                          | 1                                                                                                          |
| Other Period Text                     |                                                                                                            |                                                                                                            |
| Geography text                        |                                                                                                            |                                                                                                            |
| Technology text                       | Technique describes the average of European Union. Data are estimated basing on data of similar processes. | Technique describes the average of European Union. Data are estimated basing on data of similar processes. |
| Representativeness [%]                |                                                                                                            |                                                                                                            |
| Production Volume                     | Unknown                                                                                                    | Unknown                                                                                                    |
| Sampling Procedure                    | Literature                                                                                                 | Literature                                                                                                 |
| Extrapolations                        | None                                                                                                       | None                                                                                                       |
| Uncertainty Adjustments               | None                                                                                                       | None                                                                                                       |

4. Copper and copper alloy forming

| General flow information |                        |             |                                                                                                      | Representation in ecoinvent |                 |                  |           |                                                                    |            |      | Uncertainty       |      |          |                        |
|--------------------------|------------------------|-------------|------------------------------------------------------------------------------------------------------|-----------------------------|-----------------|------------------|-----------|--------------------------------------------------------------------|------------|------|-------------------|------|----------|------------------------|
| Input                    | Process Name           | Output      | Remarks                                                                                              | Cate gory                   | Sub category    | Infra struc ture | Loca tion | Modul name in ecoinvent                                            | Mean value | Unit | Source mean value | Type | StDv 95% | General Comment        |
| brass                    | contour, brass, part 1 |             | Same as for copper sheet rolling                                                                     | metals                      | extraction      | No               | CH        | brass, at plant                                                    | 3.96E-02   | kg   | Estimate          | 1    | 2.39     | (4,5,5,5,5,4)          |
| rolling oil              |                        |             |                                                                                                      | chemicals                   | organics        | No               | RER       | lubricating oil, at plant                                          | 4.00E-03   | kg   | Estimate          | 1    | 2.39     | (4,5,5,5,5,4)          |
| sulphuric acid           |                        |             | for degreasing / pickling. Amount as HCl for degreasing / pickling of steel sheet for powder coating | chemicals                   | inorganics      | No               | RER       | sulphuric acid, liquid, at plant                                   | 3.40E-01   | kg   | Estimate          | 1    | 2.39     | (4,5,5,5,5,4)          |
| water                    |                        |             | Same as for copper sheet rolling                                                                     | water supply                | production      | No               | RER       | tap water, at user                                                 | 6.00E+00   | kg   | Estimate          | 1    | 2.39     | (4,5,5,5,5,4)          |
| electricity general      |                        |             |                                                                                                      | electricity                 | production mix  | No               | UCTE      | electricity, medium voltage, production UCTE, at grid              | 1.81E+00   | kWh  | Estimate          | 1    | 2.56     | (5,5,5,5,5,2)          |
| heat for curing          |                        |             |                                                                                                      | natural gas                 | heating systems | No               | RER       | heat, natural gas, at industrial furnace low-NOx >100kW            | 1.60E+00   | M J  | Estimate          | 1    | 2.39     | (4,5,5,5,5,1)          |
| Transport                |                        |             |                                                                                                      | transport systems           | road            | No               | RER       | transport, lorry 32t                                               | 3.84 E-02  | tkm  | Estimate          | 1    | 2.09     | (4,5,n.a.,n.a.,n.a.,5) |
| Transport                |                        |             |                                                                                                      | transport systems           | train           | No               | RER       | transport, freight, rail                                           | 2.30E-01   | tkm  | Estimate          | 1    | 2.09     | (4,5,n.a.,n.a.,n.a.,5) |
| Infrastructure           |                        |             |                                                                                                      | metals                      | processing      | Yes              | RER       | rolling mill                                                       | 1.43E-09   | unit | Estimate          | 1    | 4.25     | (5,5,5,5,5,9)          |
|                          |                        | Waste heat  |                                                                                                      | air                         | unspecified     |                  |           | Heat, waste                                                        | 6.52E+00   | M J  | calculation       | 1    | 2.56     | (5,5,5,5,5,13)         |
|                          |                        | CO to air   |                                                                                                      | air                         | unspecified     |                  |           | NM VOC, non-methane volatile organic compounds, unspecified origin | 4.50E-05   | kg   | IPPC 2001         | 1    | 2.61     | (4,5,5,5,5,31)         |
|                          |                        | waste water | Same value as for hot rolling of steel assumed                                                       |                             |                 |                  |           |                                                                    | 5.50E-03   | m3   | Estimate          |      |          |                        |
|                          |                        | COD         |                                                                                                      | water                       | unspecified     |                  |           | COD, Chemical Oxygen Demand                                        | 7.90E-05   | kg   | Estimate          | 1    | 2.79     | (5,5,5,5,5,32)         |
|                          |                        | BOD         |                                                                                                      | water                       | unspecified     |                  |           | BOD5, Biological Oxygen Demand                                     | 7.90E-05   | kg   | Estimate          | 1    | 2.79     | (5,5,5,5,5,32)         |
|                          |                        | DOC         |                                                                                                      | water                       | unspecified     |                  |           | DOC, Dissolved Organic Carbon                                      | 2.06E-05   | kg   | Estimate          | 1    | 2.79     | (5,5,5,5,5,32)         |
|                          |                        | TOC         |                                                                                                      | water                       | unspecified     |                  |           | TOC, Total Organic Carbon                                          | 2.06E-05   | kg   | Estimate          | 1    | 2.79     | (5,5,5,5,5,32)         |

Fig. 4.4 Flows for "contour, brass" and their representation in the ecoinvent database

4. Copper and copper alloy forming

| General flow information |              |                                |                                                                                                 | Representation in ecoinvent |                            |                |          |                                                                                |            |      | Uncertainty       |      |          |                 |
|--------------------------|--------------|--------------------------------|-------------------------------------------------------------------------------------------------|-----------------------------|----------------------------|----------------|----------|--------------------------------------------------------------------------------|------------|------|-------------------|------|----------|-----------------|
| Input                    | Process Name | Output                         | Remarks                                                                                         | Category                    | Sub category               | Infrastructure | Location | Modul name in ecoinvent                                                        | Mean value | Unit | Source mean value | Type | StDv 95% | General Comment |
| contour, brass, part 2   |              | Cu                             |                                                                                                 | water                       | unspecified                |                |          | Copper, ion                                                                    | 1.73E-06   | kg   | IPPC 2001         | 1    | 6.45     | (5,5,5,5,5,35)  |
|                          |              | Ni                             |                                                                                                 | water                       | unspecified                |                |          | Nickel, ion                                                                    | 4.71E-07   | kg   | IPPC 2001         | 1    | 6.45     | (5,5,5,5,5,35)  |
|                          |              | Zn                             |                                                                                                 | water                       | unspecified                |                |          | Zinc, ion                                                                      | 3.93E-06   | kg   | IPPC 2001         | 1    | 6.45     | (5,5,5,5,5,35)  |
|                          |              | Pb                             |                                                                                                 | water                       | unspecified                |                |          | Lead                                                                           | 1.57E-07   | kg   | IPPC 2001         | 1    | 6.45     | (5,5,5,5,5,35)  |
|                          |              | Cr                             | Relation Cr / Cr VI as for waste water from pickling steel                                      | water                       | unspecified                |                |          | Chromium, ion                                                                  | 1.51E-07   | kg   | IPPC 2001         | 1    | 6.45     | (5,5,5,5,5,35)  |
|                          |              | Cr VI                          |                                                                                                 | water                       | unspecified                |                |          | Chromium VI                                                                    | 6.29E-09   | kg   | IPPC 2001         | 1    | 6.45     | (5,5,5,5,5,35)  |
|                          |              | As                             |                                                                                                 | water                       | unspecified                |                |          | Arsenic, ion                                                                   | 1.57E-09   | kg   | IPPC 2001         | 1    | 6.45     | (5,5,5,5,5,35)  |
|                          |              | Cd                             |                                                                                                 | water                       | unspecified                |                |          | Cadmium, ion                                                                   | 1.57E-09   | kg   | IPPC 2001         | 1    | 6.45     | (5,5,5,5,5,35)  |
|                          |              | Hg                             |                                                                                                 | water                       | unspecified                |                |          | Mercury                                                                        | 1.57E-09   | kg   | IPPC 2001         | 1    | 6.45     | (5,5,5,5,5,35)  |
|                          |              | Sn                             |                                                                                                 | water                       | unspecified                |                |          | Tin, ion                                                                       | 1.57E-07   | kg   | IPPC 2001         | 1    | 6.45     | (5,5,5,5,5,35)  |
|                          |              | Waste water treatment sludges  | Same value as for hot rolling of steel assumed                                                  | waste management            | residual material landfill | No             | CH       | disposal, sludge from steel rolling, 20%water, to residual material landfill   | 1.63E-02   | kg   | IPPC 2001         | 1    | 2.56     | (5,5,5,5,5,6)   |
|                          |              | Waste from filters and sludges | Zn rich filter dust from EAF steel used as proxy; Amount estimated as for hot rolling of steel. | waste management            | residual material landfill | No             | CH       | disposal, dust, unalloyed EAF steel, 15.4%water, to residual material landfill | 4.42E-04   | kg   | Estimate          | 1    | 2.56     | (5,5,5,5,5,6)   |
|                          |              | contouring brass               |                                                                                                 | metals                      | processing                 | No             | RER      | contour, brass                                                                 | 1.00E+00   | kg   |                   |      |          |                 |

Fig. 4.4 Flows for "contour, brass" and their representation in the ecoinvent database (cont.)

4. Copper and copper alloy forming

| General flow information |                         |             |                                                                                                      | Representation in ecoinvent |                 |                |          |                                                                    |            |      | Uncertainty       |      |          |                        |
|--------------------------|-------------------------|-------------|------------------------------------------------------------------------------------------------------|-----------------------------|-----------------|----------------|----------|--------------------------------------------------------------------|------------|------|-------------------|------|----------|------------------------|
| Input                    | Process Name            | Output      | Remarks                                                                                              | Category                    | Sub category    | Infrastructure | Location | Modul name in ecoinvent                                            | Mean value | Unit | Source mean value | Type | StDv 95% | General Comment        |
| bronze                   | contour, bronze, part 1 |             | Same as for copper sheet rolling                                                                     | metals                      | extraction      | No             | CH       | bronze, at plant                                                   | 3.96E-02   | kg   | calculation       | 1    | 2.39     | (4,5,5,5,5,4)          |
| rolling oil              |                         |             |                                                                                                      | chemicals                   | organics        | No             | RER      | lubricating oil, at plant                                          | 4.00E-03   | kg   | Estimate          | 1    | 2.39     | (4,5,5,5,5,4)          |
| sulphuric acid           |                         |             | for degreasing / pickling. Amount as HCl for degreasing / pickling of steel sheet for powder coating | chemicals                   | inorganics      | No             | RER      | sulphuric acid, liquid, at plant                                   | 3.40E-01   | kg   | Estimate          | 1    | 2.39     | (4,5,5,5,5,4)          |
| water                    |                         |             | Same as for copper sheet rolling                                                                     | water supply                | production      | No             | RER      | tap water, at user                                                 | 6.00E+00   | kg   | Estimate          | 1    | 2.39     | (4,5,5,5,5,4)          |
| electricity general      |                         |             |                                                                                                      | electricity                 | production mix  | No             | UCTE     | electricity, medium voltage, production UCTE, at grid              | 1.81E+00   | kWh  | Estimate          | 1    | 2.56     | (5,5,5,5,5,2)          |
| heat for curing          |                         |             |                                                                                                      | natural gas                 | heating systems | No             | RER      | heat, natural gas, at industrial furnace low-NOx >100kW            | 1.60E+00   | MJ   | Estimate          | 1    | 2.39     | (4,5,5,5,5,1)          |
| Transport                |                         |             |                                                                                                      | transport systems           | road            | No             | RER      | transport, lorry 32t                                               | 3.84E-02   | tkm  | Estimate          | 1    | 2.09     | (4,5,n.a.,n.a.,n.a.,5) |
| Transport                |                         |             |                                                                                                      | transport systems           | train           | No             | RER      | transport, freight, rail                                           | 2.30E-01   | tkm  | Estimate          | 1    | 2.09     | (4,5,n.a.,n.a.,n.a.,5) |
| Infrastructure           |                         |             |                                                                                                      | metals                      | processing      | Yes            | RER      | rolling mill                                                       | 1.43E-09   | unit | Estimate          | 1    | 4.25     | (5,5,5,5,5,9)          |
|                          |                         | Waste heat  |                                                                                                      | air                         | unspecified     |                |          | Heat, waste                                                        | 6.52E+00   | MJ   | calculation       | 1    | 2.56     | (5,5,5,5,5,13)         |
|                          |                         | CO to air   |                                                                                                      | air                         | unspecified     |                |          | NM VOC, non-methane volatile organic compounds, unspecified origin | 4.50E-05   | kg   | IPPC 2001         | 1    | 2.61     | (4,5,5,5,5,31)         |
|                          |                         | waste water |                                                                                                      |                             |                 |                |          |                                                                    | 5.50E-03   | m3   | Estimate          |      |          |                        |
|                          |                         | COD         | Same value as for hot rolling of steel assumed                                                       | water                       | unspecified     |                |          | COD, Chemical Oxygen Demand                                        | 7.90E-05   | kg   | Estimate          | 1    | 2.79     | (5,5,5,5,5,32)         |
|                          |                         | BOD         |                                                                                                      | water                       | unspecified     |                |          | BOD5, Biological Oxygen Demand                                     | 7.90E-05   | kg   | Estimate          | 1    | 2.79     | (5,5,5,5,5,32)         |
|                          |                         | DOC         |                                                                                                      | water                       | unspecified     |                |          | DOC, Dissolved Organic Carbon                                      | 2.06E-05   | kg   | Estimate          | 1    | 2.79     | (5,5,5,5,5,32)         |
|                          |                         | TOC         |                                                                                                      | water                       | unspecified     |                |          | TOC, Total Organic Carbon                                          | 2.06E-05   | kg   | Estimate          | 1    | 2.79     | (5,5,5,5,5,32)         |

Fig. 4.5 Flows for "contour, bronze" and their representation in the ecoinvent database

4. Copper and copper alloy forming

| General flow information |              |                                |                                                                                                 | Representation in ecoinvent |                            |                |          |                                                                                |            |      | Uncertainty       |      |          |                 |
|--------------------------|--------------|--------------------------------|-------------------------------------------------------------------------------------------------|-----------------------------|----------------------------|----------------|----------|--------------------------------------------------------------------------------|------------|------|-------------------|------|----------|-----------------|
| Input                    | Process Name | Output                         | Remarks                                                                                         | Category                    | Sub category               | Infrastructure | Location | Modul name in ecoinvent                                                        | Mean value | Unit | Source mean value | Type | StDv 95% | General Comment |
| contour, bronze, part 2  | ML           | Cu                             |                                                                                                 | water                       | unspecified                |                |          | Copper, ion                                                                    | 0.00E+00   | kg   | IPPC 2001         | 1    | 6.45     | (5,5,5,5,5,35)  |
|                          | ML           | Ni                             |                                                                                                 | water                       | unspecified                |                |          | Nickel, ion                                                                    | 0.00E+00   | kg   | IPPC 2001         | 1    | 6.45     | (5,5,5,5,5,35)  |
|                          | ML           | Zn                             |                                                                                                 | water                       | unspecified                |                |          | Zinc, ion                                                                      | 0.00E+00   | kg   | IPPC 2001         | 1    | 6.45     | (5,5,5,5,5,35)  |
|                          | ML           | Pb                             |                                                                                                 | water                       | unspecified                |                |          | Lead                                                                           | 0.00E+00   | kg   | IPPC 2001         | 1    | 6.45     | (5,5,5,5,5,35)  |
|                          | ML           | Cr                             | Relation Cr / Cr VI as for waste water from pickling steel                                      | water                       | unspecified                |                |          | Chromium, ion                                                                  | 0.00E+00   | kg   | IPPC 2001         | 1    | 6.45     | (5,5,5,5,5,35)  |
|                          | ML           | Cr VI                          |                                                                                                 | water                       | unspecified                |                |          | Chromium VI                                                                    | 0.00E+00   | kg   | IPPC 2001         | 1    | 6.45     | (5,5,5,5,5,35)  |
|                          | ML           | As                             |                                                                                                 | water                       | unspecified                |                |          | Arsenic, ion                                                                   | 0.00E+00   | kg   | IPPC 2001         | 1    | 6.45     | (5,5,5,5,5,35)  |
|                          | ML           | Cd                             |                                                                                                 | water                       | unspecified                |                |          | Cadmium, ion                                                                   | 0.00E+00   | kg   | IPPC 2001         | 1    | 6.45     | (5,5,5,5,5,35)  |
|                          | ML           | Hg                             |                                                                                                 | water                       | unspecified                |                |          | Mercury                                                                        | 0.00E+00   | kg   | IPPC 2001         | 1    | 6.45     | (5,5,5,5,5,35)  |
|                          | ML           | Sn                             |                                                                                                 | water                       | unspecified                |                |          | Tin, ion                                                                       | 0.00E+00   | kg   | IPPC 2001         | 1    | 6.45     | (5,5,5,5,5,35)  |
|                          | ML           | Waste water treatment sludges  | Same value as for hot rolling of steel assumed                                                  | waste management            | residual material landfill | No             | CH       | disposal, sludge from steel rolling, 20%water, to residual material landfill   | 1.63E-02   | kg   | IPPC 2001         | 1    | 2.56     | (5,5,5,5,5,6)   |
|                          | ML           | Waste from filters and sludges | Zn rich filter dust from EAF steel used as proxy; Amount estimated as for hot rolling of steel. | waste management            | residual material landfill | No             | CH       | disposal, dust, unalloyed EAF steel, 15.4%water, to residual material landfill | 4.42E-04   | kg   | Estimate          | 1    | 2.56     | (5,5,5,5,5,6)   |
|                          | ML           | contouring brass               |                                                                                                 | metals                      | processing                 | No             | RER      | contour, bronze                                                                | 1.00E+00   | kg   |                   |      |          |                 |

Fig. 4.5 Flows for "contour, bronze" and their representation in the ecoinvent database (cont.)

## 5 Zinc coating of steel

Authors: Silvio Blaser, Hans.Jörg Althaus, EMPA Dübendorf  
Review: Roland Hischier

The hot-dip galvanising of steel involves the application of a zinc coating onto the surface of steel by a process wherein the steel is immersed into molten zinc to apply the coating. The zinc reacts with the steel to form the galvanised coatings. The time the steel is immersed in the zinc along with post-galvanising treatment controls the coating thickness, appearance and other characteristics. Hot dip galvanised coatings are applied to steel to improve the anti-corrosion performance of the steel to ensure that it lasts as long as possible with a minimum of maintenance or to provide a decorative appearance. Since zinc has to be heated to a temperature of approximately 455 °C or higher for the galvanising process to be implemented correctly, the process is referred to as a “hot dip” process.

There are two common practices for applying zinc in hot-dip galvanising. One involves the application of zinc onto steel sheet as it is passed as a continuous ribbon of steel through a bath of molten zinc at high speeds. Hence, the term “continuous” hot-dip galvanising.

The other process involves the application of a zinc coating to the surface of steel parts after the parts have been fabricated. This process is not a continuous process in that the parts are immersed as a discrete “batch” into the zinc bath. Hence, the names “batch”, “after fabrication” and “general” galvanising are applied to this process (extracted from Pankert (1997) and internet<sup>6,7</sup>).

### 5.1 Zinc coating of steel coils

The production process is described in (IPPC (2001a)) and two URL-addresses<sup>6,8</sup>.

As mentioned above, the continuous galvanising process is used to describe the application of a zinc coating onto the surface of steel sheet. In the hot dip coating process, the steel is continuously passed through the molten metal, at speeds up to 200 meters per minute. The sheet thickness might be as thin as 0.25 mm or less to as thick as 6.3 mm. In the molten-metal bath, an alloying reaction between the two metals takes place, leading to a good bond between coating and substrate. As the alloy layer is hard and brittle, steel is immersed for a very brief time, in order to keep the alloy layer very thin to accommodate subsequent forming into the final shape. As the strip emerges from the molten bath, it drags out excess metal, much like when an object is pulled rapidly from a container of water. Through the use of an air-wiping process, a controlled amount (thickness) of coating usually expressed as weight of coating per unit area is allowed to remain on the strip surface.

Metals suitable for the use in hot dip coating are those, which have a melting point low enough to avoid any thermal changes in the steel product, like for example aluminium, lead, tin, and zinc. In this study only the case of zinc was analysed.

In general continuous coating lines for sheet comprise the following steps:

- An entry-end welder to join the trailing edge of one coil to the leading edge of the succeeding coil,
- Surface cleaning by means of chemical and/or thermal treatment,
- Heat treatment; annealing furnace to heat the steel to high temperatures to impart the desired mechanical properties (strength and formability) to the steel sheet,

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<sup>6</sup> <http://www.galvinform.com/GalvInfoNote%20PDF's/GalvInfo9.pdf>, accessed 03.03.03

<sup>7</sup> <http://www.corp.indgalv.com.au/technical/pdf/zfile02.pdf>, accessed 03.03.03

<sup>8</sup> <http://www.galvinform.com/GalvInfoNote%20PDF's/ginote2.pdf>, accessed 03.03.03



- Immersion in a bath of molten metal (zinc for this study),
- Finishing treatment (cooling section, temper mill, tension leveller, surface treatment section and a recoiler).

Continuous wire galvanising plants involve the following steps: pickling, fluxing, galvanising, finishing. Wire is hot dip coated with zinc (galvanised) or tin. Zinc coating is mainly applied as corrosion protection. Tin gives the wire a shine appearance and also provides a solder adhesive layer. As the flows for the two processes are similar, in this study only the continuous galvanising lines for sheets and trips were inventoried. A more detailed description of every single step is given below.

The infrastructure of continuous hot dip zinc coating comprises buildings where galvanising processes take place and for storage, surroundings and machinery fore the single process steps. The assumed lifetime for the infrastructure is 50 years. As no information is available, the infrastructure is approximated by the infrastructure for rolling.

The production of continuous hot dip coating lines in EU 15 was 15 Mt in 1997. The vast majority of coated products, as shown in Fig. 5.1, were galvanised steel.

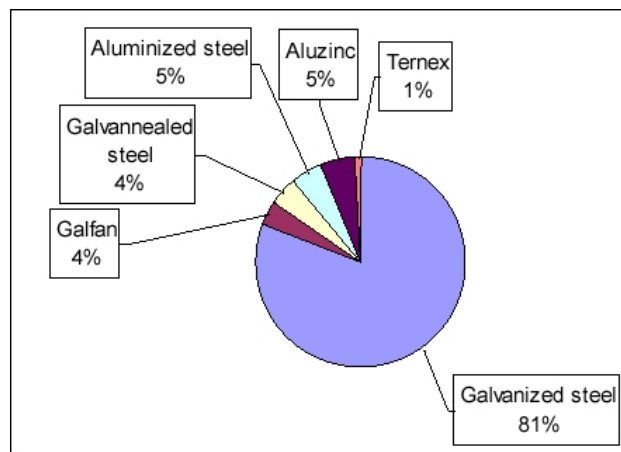


Fig. 5.1 Production share for different hot dip coatings IPPC (2001a)

### 5.1.1 Entry section

As one coil is being processed through the coating bath, another is being welded to the trailing edge of the first coil. Thus, the process is truly “continuous”, the coating line may indeed operate continuously for days without interruption. No data is available about the welding process therefore it is neglected.

Hot dip galvanising of cold rolled strip is more common than galvanising hot rolled strip. The processing is similar, except that for hot rolled products additional descaling (pickling, to remove hot mill scale) is necessary. Pickling of unannealed cold rolled coil is done for activation. Both processes are done in hydrochloric acid, followed by rinsing. The technology is the same as the pickling process described in the cold rolling section 2.3.1.

It is essential that the surface of the steel coil is free of impurities such as grease, oil or abraded iron fines to ensure the entire surface is exposed to the coating product and strong adherence is achieved. Although degreasing is not necessary when the strip is subsequently heat treated, it is often applied in any case. The cleaning process involves several or all of the following steps:

- Alkaline degreasing via immersion or by spraying; maybe coupled with brushing.
- Alkaline degreasing by an electrolytic system; maybe coupled with brushing.

- Rinsing with water with intermediate brushing between rinsing tanks.
- Drying.

Description is taken from (IPPC (2001a)) and the internet <sup>9</sup>.

### 5.1.2 Heat treatment

From the cleaning section, the strip passes into the heating (annealing) furnace to soften the full-hard strip, to impart the desired strength and formability to the steel and to allow the steel to reach the required temperature before dipping. In the annealing furnace, the strip is maintained under a reducing gas atmosphere to remove any vestiges of oxides on the steel surface. The gas atmosphere is composed of hydrogen and nitrogen. This oxide-reduction step, iron oxide being converted to iron by reaction with the hydrogen, is very important to obtain complete wetting of the steel surface during the short time that it is immersed into the coating bath.

The following types of furnaces are used:

- Sendzimir Furnace; this type of furnace is no longer used in modern coating lines, but some older plants might still operate this horizontal continuous furnace,
- Direct flame furnace (D.F.F; non oxidizing type); this type of furnace cleans the coil surface and does not require any prior degreasing of the metal, but air emissions are generated as the remaining oil from the surface is burnt,
- Radiant Tube Furnace (R.T.F.); this furnace is a variation with indirect heating. The combustion gases circulate through radiant tubes and do not get in contact with the strip. The annealing furnace is combined with an alkaline degreasing equipment.

Fuels used are desulphured coke oven gas and (inventoried in this study) natural gas. Description is taken from (IPPC (2001a)) and the internet <sup>10</sup>.

### 5.1.3 Hot dipping (galvanising)

The exit end of the furnace is connected directly to the molten coating bath metal to prevent the heated steel strip from re-oxidising (in the air) prior to reacting with the molten coating metal. The galvanising bath consists of one or more tanks, usually made of ceramic material. These pots, which can be either fixed or mobile, contain the molten metal at a temperature of 440 – 490 °C, through which the strip is passed. The bath contains zinc and any other required additive (i.e. antimony, lead or aluminium). The bath contains a sufficient amount of molten zinc in order to prevent the wide fluctuations in the operating temperature that may appear under maximum operating conditions. Most pots are nowadays heated by electrical induction systems. However, natural gas as fuel for heating the pots is an alternative.

As the steel passes through the molten zinc bath the surface is coated, to some extent with layers of different iron-zinc alloys, but due to the high speed of the steel strip (max. 180 - 200 m/min) and the short exposure time, the coating (between 20 and 45 µm thick (Frischknecht et al. (1996))) mainly consist of zinc.

Temperature control of the bath is essential, as a high temperature will increase the rate of oxidation at the surface of the molten bath, resulting in increased ash generation. A low bath temperature will increase the viscosity of the molten coating metal, thus inhibiting the formation of a thin coating layer.

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<sup>9</sup> <http://www.galvinform.com/GalvInfoNote%20PDF's/ginote2.pdf>, accessed 03.03.03

<sup>10</sup> <http://www.galvinform.com/GalvInfoNote%20PDF's/ginote2.pdf>, accessed 03.03.03

The bath temperature is therefore optimised to meet the individual quality requirements of the final coated product.

In the coating bath, the strip passes around a submerged roll and then exits the bath in a vertical direction. At the exit point, a set of air or nitrogen knives (high pressure air jets) wipe off excess molten metal leaving behind a closely controlled thickness of molten metal. The system is usually controlled by an automatic gauge, which measures the thickness of the coating using X-ray measurement technology or a similar technique. Description is taken from (IPPC (2001a)) and the internet<sup>10</sup>.

#### **5.1.4 Cooling, post treatment and finishing**

The coating is then gradually cooled by air coolers to allow the metal to freeze on the steel surface. Freezing of the coating, or solidification, has to be accomplished before the strip contacts another roll to avoid transferring the coating onto the roll. Thus, coating-process lines usually have a high tower above the coating bath.

Following the coating, the steel strips are usually subjected to post-treatment in order to prevent surface damages and flaws, like white rust caused by water condensation in case of insufficient air access. Voluminous corrosion products as zinc hydroxides (white rust) affect further processing, such as e.g. painting. The formation of condensate during transportation and storage is to be generally prevented.

The resistance against white rust is considerably increased by the application of post treatment, like:

- Oiling; applying a wet film coat with thickness between 0.25 - 3 g/m<sup>2</sup>/side,
- Passivation; a treatment with solvents containing chromic acids, applied by spraying or applicator-rolls, with coat thickness of a few nanometer only,
- Phosphating; applying a phosphate coat of approx. 1 - 1.8 g/m<sup>2</sup>.

To give the steel a special surface appearance, smoothness etc. as required by the customers or to meet width tolerances, the following finishing operations may be applied:

- Mini or No Spangle Treatment; the strip is either cooled more rapidly or steam sprayed with a chemical additive (usually phosphate based) or sprayed with zinc powder,
- Mat Mill; the strip is skin-passed to obtain a mat finish surface.
- For certain applications, an edge cutting must be done in order to obtain the width tolerance needed and to avoid little dog bone defects.

The continuous strip is sheared at the weld that was made at the entry end of the line to remove the weld as well as to maintain coil-to-coil identity. Description is taken from (IPPC (2001a)) and the internet<sup>11</sup>.

#### **5.1.5 Environmental issues of continuous zinc coating process of steel**

Main environmental issues concerning this sub-sector are acidic air emissions, wastes and wastewater; air emissions and energy consumption of furnaces, Zinc-containing residues, oil- and chrome-containing wastewater (IPPC (2001a)).

As wastewater is treated in the factory, post treatment wastewater composition and ancillary products for the water treatment as given in IPPC (2001a) are used to calculate the expenditures and the water emissions of the water treatment process.

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<sup>11</sup> <http://www.galvinfo.com/GalvInfoNote%20PDF's/GalvInfo9.pdf>, accessed 03.03.03

### 5.1.6 Continuous hot dip zinc coating in ecoinvent

In (IPPC (2001a)) most values are given in ranges, the mean values are generally calculated as the geometric mean of the given range. When the lower value of the range is zero, then the arithmetic mean of the range is taken (what correspond to half of the maximal value). Values taken from (Frischknecht et al. (1996)) are already mean values and do not have any information about minima and maxima or distribution.

The background information to the process "zinc coating, coils" is given in Tab. 5.1, flows are shown in Fig. 5.2.

#### Data quality

For values represented by the geometric mean, uncertainty is calculated assuming a lognormal distribution and the high value given in (IPPC (2001a)) representing the 95% geometric standard deviation. For the arithmetic means of ranges given in (IPPC (2001a)) and the values from (Frischknecht et al. (1996)), the uncertainty is estimated by completing the pedigree matrix, according to (Frischknecht et al. (2003)).

Tab. 5.1 Ecoinvent meta information for "zinc coating, coils"

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>                           | <b>zinc coating, coils</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Location                              | RER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Infrastructure Process                | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Unit                                  | m <sup>2</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Data Set Version                      | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Included Processes                    | Includes the process steps surface cleaning (by means of chemical and/or thermal treatment), heat treatment, immersion in a bath of molten zinc and finishing treatment. Also includes zinc input and transportation to coiling plant. Does not include the steel strip or sheet being coated.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Amount                                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Local Name                            | Bandverzinkung                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Synonyms                              | hot dip coating//galvanizing//sendzimir                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| General Comment to reference function | This module is applicable, if continuous zinc coating (galvanising) of hot or cold rolled strip- or wire-coils has to be inventoried. Data given per tonne of coated product are transformed by dividing them by the mean surface area of 64 m <sup>2</sup> /t (corresponding to ca. 4.2 mm thickness). The zinc coating layer is between 20 to 45 µm thick. The ecoinvent module "zinc coating, pieces, adjustment per um" can be used for adjusting for thinner or thicker zinc layers of coated coils, but it 1.53 µm of it have to be inventoried per um layer thickness adjustment. This is because the zinc consumption per um is higher for coil coating than it is for piece coating. For the chemicals and the zinc, 100 km of 32t lorry transport were inventoried. The softened industrial water is assumed to be produced at the coating plant. Continuous wire coating can be approximated with this module. |
| Start Date                            | 1996                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| End Date                              | 2002                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Other Period Text                     | Reference from 1996 refers partly to data from 1993.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Geography text                        | Data-set is representative for European Union                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Technology text                       | Average technique for EU.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Representativeness [%]                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Production Volume                     | 12 million tonnes in 1997                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Sampling Procedure                    | Literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Extrapolations                        | see technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Uncertainty Adjustments               | Some values are from single or very few plants. For other values no data were available about the percentage of companies represented by the study cited. This was considered in the uncertainties.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

## 5. Zinc coating of steel

| General Flow Information |                                 |                                                    | Representation in ecoinvent                                                                                                                                                                                                                                                        |                  |                              |                 |          |                                                                        |            | Uncertainty Informations |                          |      |                    |                                                           |
|--------------------------|---------------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|------------------------------|-----------------|----------|------------------------------------------------------------------------|------------|--------------------------|--------------------------|------|--------------------|-----------------------------------------------------------|
| Input                    | Process Name                    | Output                                             | Remarks                                                                                                                                                                                                                                                                            | Category         | Sub-category                 | Infra-structure | Location | Modul name in ecoinvent                                                | Mean value | Unit                     | Source for mean value    | Type | Standard Deviation | General Comment                                           |
| HCl-in                   | zinc coating, coils, Pickling   | HCl to air                                         |                                                                                                                                                                                                                                                                                    | chemicals        | inorganics                   | No              | RER      | hydrochloric acid, 30% in H <sub>2</sub> O, at plant                   | 3.40E-01   | kg                       | Frischknecht et al. 1996 | 1    | 1.11               | (1,n.A.,3,1,1,n.A.,4)                                     |
| Anticorrosive oils       |                                 | Sludge cake                                        | Assumption: same amount per m2 product surface as for HCl-pickling in cold rolling sheet. Lubricating oil as proxy                                                                                                                                                                 | chemicals        | organics                     | No              | RER      | lubricating oil, at plant                                              | 1.77E-02   | kg                       | IPPC 2001                | 1    | 1.41               | see text                                                  |
|                          |                                 | Oil contained waste                                | Assumption: same amount per m2 product surface as for HCl-pickling in cold rolling sheet. The sludge can be aproximated by Fe(OH) <sub>3</sub> . The amount of Fe in Fe(OH) <sub>3</sub> (=76.66%) is inventoried. About two thirds are recycled --> 35% landfilled is inventoried | air              | unspecified                  |                 |          | Hydrogen chloride                                                      | 5.05E-02   | kg                       | Frischknecht et al. 1996 | 1    | 2                  | (2,n.A.,2,1,1,n.A.,23)                                    |
|                          |                                 | Strip head / tail scrap                            | Assumption: same amount per m2 product surface as for HCl-pickling in cold rolling sheet. Only 22.8% which are landfilled are inventoried, the rest is recycled. Oil containig wastes are approximated with used mineral oil.                                                      | soil             | unspecified                  |                 |          | Iron                                                                   | 9.94E-03   | kg                       | IPPC 2001                | 1    | 5.28               | calculated from max value assuming lognormal distribution |
|                          |                                 |                                                    | Assumption: same amount per kg product as for HCl-pickling in cold rolling sheet. Recycled --> not inventoried                                                                                                                                                                     | waste management | hazardous waste incineration | No              | CH       | disposal, used mineral oil, 10% water, to hazardous waste incineration | 1.39E-06   | kg                       | IPPC 2001                | 1    | 1.56               | (1,5,1,1,1,3,41)                                          |
| Detergents               | zinc coating, coils, Degreasing | NaOH as proxy                                      |                                                                                                                                                                                                                                                                                    |                  |                              |                 |          |                                                                        | 4.69E-01   | kg                       | IPPC 2001                | 1    |                    |                                                           |
|                          |                                 | NaOH as proxy                                      |                                                                                                                                                                                                                                                                                    | chemicals        | inorganics                   | No              | RER      | sodium hydroxide, 50% in H <sub>2</sub> O, production mix, at plant    | 3.52E-03   | kg                       | IPPC 2001                | 1    | 2                  | calculated from max value assuming lognormal distribution |
|                          |                                 | Na as proxy (amount refers to Na in NaOH)          |                                                                                                                                                                                                                                                                                    | air              | unspecified                  |                 |          | Sodium                                                                 | 4.49E-07   | kg                       | IPPC 2001                | 1    | 2.05               | (1,5,1,1,1,3,23)                                          |
|                          |                                 | Na as proxy (amount refers to Na in NaOH)          |                                                                                                                                                                                                                                                                                    | water            | unspecified                  |                 |          | Sodium, ion                                                            | 4.04E-03   | kg                       | IPPC 2001                | 1    | 1.56               | (1,5,1,1,1,3,33)                                          |
|                          |                                 | Oil                                                | Assumption: detergents not contained in the oily sludge. Oil containing wastes are approximated by used mineral oil disposal                                                                                                                                                       | waste management | hazardous waste incineration | No              | CH       | disposal, used mineral oil, 10% water, to hazardous waste incineration | 7.03E-04   | kg                       | IPPC 2001                | 1    | 2                  | calculated from max value assuming lognormal distribution |
|                          |                                 | Oily sludge                                        | Oil containing wastes are approximated by used mineral oil disposal                                                                                                                                                                                                                | waste management | hazardous waste incineration | No              | CH       | disposal, used mineral oil, 10% water, to hazardous waste incineration | 2.81E-03   | kg                       | IPPC 2001                | 1    | 1.56               | (1,5,1,1,1,3,41)                                          |
|                          |                                 | Fe <sub>2</sub> O <sub>3</sub> , FeSO <sub>4</sub> | Reference contained no data                                                                                                                                                                                                                                                        |                  |                              |                 |          |                                                                        | n.a.       |                          | IPPC 2001                |      |                    |                                                           |

**Fig. 5.2** Flows for "zinc coating, coils" and their representation in the ecoinvent database

5. Zinc coating of steel

| General Flow Information                          |                                                  |                                    | Representation in ecoinvent                                                                                                                                                                                                                                                                                                                           |           |              |                 |          |                                         |            | Uncertainty Informations |                          |      |                    |                                                           |
|---------------------------------------------------|--------------------------------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------|-----------------|----------|-----------------------------------------|------------|--------------------------|--------------------------|------|--------------------|-----------------------------------------------------------|
| Input                                             | Process Name                                     | Output                             | Remarks                                                                                                                                                                                                                                                                                                                                               | Category  | Sub-category | Infra-structure | Location | Modul name in ecoinvent                 | Mean value | Unit                     | Source for mean value    | Type | Standard Deviation | General Comment                                           |
| Hydrogen                                          | zinc coating, coils, Heat treatment              | CO-RTFurnace                       |                                                                                                                                                                                                                                                                                                                                                       | chemicals | inorganics   | No              | RER      | hydrogen, liquid, at plant              | 1.92E-03   | kg                       | IPPC 2001                | 1    | 1.83               | calculated from max value assuming lognormal distribution |
| Nitrogen                                          |                                                  |                                    |                                                                                                                                                                                                                                                                                                                                                       | chemicals | inorganics   | No              | RER      | nitrogen, liquid, at plant              | 6.30E-01   | kg                       | IPPC 2001                | 1    | 2.48               |                                                           |
| Ammonia                                           |                                                  |                                    |                                                                                                                                                                                                                                                                                                                                                       | chemicals | inorganics   | No              | RER      | ammonia, liquid, at regional storehouse | 3.06E-03   | kg                       | Frischknecht et al. 1996 | 1    | 3.58               |                                                           |
|                                                   | zinc coating, coils, treatment                   | NOx-RTFurnace                      | included in the inventoried module for natural gas burning (overall process)                                                                                                                                                                                                                                                                          |           |              |                 |          |                                         | 8.30E-05   | kg                       | IPPC 2001                | 1    | 4.39               |                                                           |
|                                                   |                                                  |                                    | included in the inventoried module for natural gas burning (overall process)                                                                                                                                                                                                                                                                          |           |              |                 |          |                                         | 1.19E-03   | kg                       | IPPC 2001                | 1    | 1.38               |                                                           |
|                                                   |                                                  |                                    |                                                                                                                                                                                                                                                                                                                                                       | metals    | extraction   | No              | RER      | zinc for coating, at regional storage   | 6.79E-01   | kg                       | IPPC 2001                | 1    | 1.74               |                                                           |
| Zinc                                              | zinc coating, coils, treatment                   | Plate scrap Zn containing material | recycled --> not inventoried                                                                                                                                                                                                                                                                                                                          |           |              |                 |          |                                         | 1.15E-01   | kg                       | IPPC 2001                | 1    | 4.9                | (1,5,1,1,1,3,4)                                           |
|                                                   |                                                  |                                    | recycled --> not inventoried                                                                                                                                                                                                                                                                                                                          |           |              |                 |          |                                         | 3.52E-02   | kg                       | IPPC 2001                | 1    | 2                  |                                                           |
|                                                   |                                                  |                                    | Reference contained no data. Assumption: Same amount per m2 product surface as for cold rolling chromium steel sheet. Lubricating oil as proxy                                                                                                                                                                                                        | chemicals | organics     | No              | RER      | lubricating oil, at plant               | 3.54E-02   | kg                       | IPPC 2001                | 1    | 1.22               |                                                           |
| Oil-in                                            | zinc coating, coils, treatment plant             | Oil-out                            | Nowadays, the oily protection is mainly applied by electrostatic oily machines, which produce negligible emission levels                                                                                                                                                                                                                              |           |              |                 |          |                                         | negligible |                          | IPPC 2001                |      |                    |                                                           |
|                                                   |                                                  |                                    | Reference contained no data                                                                                                                                                                                                                                                                                                                           |           |              |                 |          |                                         | n.a.       |                          | IPPC 2001                |      |                    |                                                           |
|                                                   |                                                  |                                    | Reference contained no data                                                                                                                                                                                                                                                                                                                           |           |              |                 |          |                                         | n.a.       |                          | IPPC 2001                |      |                    |                                                           |
| Corrosion inhibitors, partly Zn-based             | zinc coating, coils, Waste water treatment plant | Waste water                        | Reference contained no data. Assumption: Same amount per kg product as for steel forming process. Aluminium sulphate as proxy.                                                                                                                                                                                                                        | chemicals | inorganics   | No              | RER      | aluminium sulphate, powder, at plant    | 6.07E-05   | kg                       | IPPC 2001                | 1    | 1.22               | (2,5,1,1,1,3,4)                                           |
| Biocides: sodium hypochlorite, sodium bromide     |                                                  |                                    | One part of the incoming cooling and process water leaves the system as treated effluent; it is assumed to be discharged in surface water --> not inventoried. Infrastructure for waste water treatment plant is comprised in the infrastructure module. Value for energy use is comprised in the range of uncertainty of inventoried overall energy. |           |              |                 |          |                                         | 6.72E-03   | m3                       | IPPC 2001                |      |                    |                                                           |
| Flocculating agents: Aluminium sulphate, polymers |                                                  |                                    |                                                                                                                                                                                                                                                                                                                                                       |           |              |                 |          |                                         |            |                          |                          |      |                    |                                                           |

Fig. 5.2 Flows for "zinc coating, coils" and their representation in the ecoinvent database (cont.)

5. Zinc coating of steel

| General Flow Information                         |              |                                       | Representation in ecoinvent                                                                                                                                            |                   |                            |                 |          |                                                                                |            | Uncertainty Informations |                          |      |                    |                                                           |
|--------------------------------------------------|--------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------------------------|-----------------|----------|--------------------------------------------------------------------------------|------------|--------------------------|--------------------------|------|--------------------|-----------------------------------------------------------|
| Input                                            | Process Name | Output                                | Remarks                                                                                                                                                                | Category          | Sub-category               | Infra-structure | Location | Modul name in ecoinvent                                                        | Mean value | Unit                     | Source for mean value    | Type | Standard Deviation | General Comment                                           |
| zinc coating, coils, Waste water treatment plant | m            | TSS                                   |                                                                                                                                                                        | w ater            | unspecified                |                 |          | Suspended solids, unspecified                                                  | 5.33E-06   | kg                       | IPPC 2001                | 1    | 8.53               | calculated from max value                                 |
|                                                  | m            | Iron                                  |                                                                                                                                                                        | w ater            | unspecified                |                 |          | Iron, ion                                                                      | 1.15E-05   | kg                       | IPPC 2001                | 1    | 8.17               | assuming                                                  |
|                                                  | m            | Nickel                                |                                                                                                                                                                        | w ater            | unspecified                |                 |          | Nickel, ion                                                                    | 1.98E-07   | kg                       | IPPC 2001                | 1    | 3.16               | lognormal distribution                                    |
|                                                  | m            | Zinc                                  |                                                                                                                                                                        | w ater            | unspecified                |                 |          | Zinc, ion                                                                      | 1.56E-06   | kg                       | IPPC 2001                | 1    | 20                 | (1,5,1,1,1,3,35)                                          |
|                                                  | m            | Lead                                  |                                                                                                                                                                        | w ater            | unspecified                |                 |          | Lead                                                                           | 1.31E-07   | kg                       | IPPC 2001                | 1    | 5.06               | calculated from max value                                 |
|                                                  | m            | Chromium                              |                                                                                                                                                                        | w ater            | unspecified                |                 |          | Chromium, ion                                                                  | 1.44E-06   | kg                       | IPPC 2001                | 1    | 2                  | assuming lognormal                                        |
|                                                  | m            | Cr6+                                  |                                                                                                                                                                        | w ater            | unspecified                |                 |          | Chromium VI                                                                    | 6.72E-08   | kg                       | IPPC 2001                | 1    | 2                  | Twice the uncertainty of COD                              |
|                                                  | m            | COD                                   |                                                                                                                                                                        | w ater            | unspecified                |                 |          | COD, Chemical Oxygen Demand                                                    | 4.28E-04   | kg                       | IPPC 2001                | 1    | 5.48               | calculated from max value assuming lognormal distribution |
|                                                  | m            | BOD                                   |                                                                                                                                                                        | w ater            | unspecified                |                 |          | BOD5, Biological Oxygen Demand                                                 | 4.28E-04   | kg                       | calculated from COD      | 1    | 11                 |                                                           |
|                                                  | m            | DOC                                   |                                                                                                                                                                        | w ater            | unspecified                |                 |          | DOC, Dissolved Organic Carbon                                                  | 1.99E-05   | kg                       | calculated from TOC      | 1    | 13.4               |                                                           |
|                                                  | m            | TOC                                   |                                                                                                                                                                        | w ater            | unspecified                |                 |          | TOC, Total Organic Carbon                                                      | 1.99E-05   | kg                       | IPPC 2001                | 1    | 6.69               | calculated from max value assuming lognormal distribution |
|                                                  | m            | Hydrocarbons                          |                                                                                                                                                                        | w ater            | unspecified                |                 |          | Hydrocarbons, unspecified                                                      | 1.16E-06   | kg                       | IPPC 2001                | 1    | 3.05               | (1,5,1,1,1,3,34)                                          |
|                                                  | m            | Oil and fat                           |                                                                                                                                                                        | w ater            | unspecified                |                 |          | Oils, unspecified                                                              | 6.94E-06   | kg                       | IPPC 2001                | 1    | 6.08               |                                                           |
|                                                  | m            | Phosphorus                            |                                                                                                                                                                        | w ater            | unspecified                |                 |          | Phosphorus                                                                     | 1.17E-07   | kg                       | IPPC 2001                | 1    | 9.35               | calculated from max value assuming lognormal distribution |
|                                                  | m            | Water treatment sludge (dry material) | Waste w ater treatment sludge from steel forming processes is inventoried as proxy. 1.2 times the amount is inventoried to account for the w ater content of the proxy | w aste management | residual material landfill | No              | CH       | disposal, sludge from steel rolling, 20% w ater, to residual material landfill | 7.26E-03   | kg                       | IPPC 2001                | 1    | 3.87               |                                                           |
|                                                  | m            | Chemical treatment sludges            | Waste w ater treatment sludge from steel forming processes is inventoried as proxy. 1.2 times the amount is inventoried to account for the w ater content of the proxy | w aste management | residual material landfill | No              | CH       | disposal, sludge from steel rolling, 20% w ater, to residual material landfill | 2.25E-03   | kg                       | IPPC 2001                | 1    | 5.06               | (1,5,1,1,1,3,35)                                          |
|                                                  | m            | Chloride                              |                                                                                                                                                                        | w ater            | unspecified                |                 |          | Chloride                                                                       | 3.13E-05   | kg                       | Frischknecht et al. 1996 | 1    | 1.56               | (1,5,1,1,1,3,33)                                          |

Fig. 5.2 Flows for "zinc coating, coils" and their representation in the ecoinvent database (cont.)



5. Zinc coating of steel

| General Flow Information   |                                      |        | Representation in ecoinvent                                                                                           |                   |                 |                 |          |                                                       |            | Uncertainty Informations |                        |      |                    |                                                           |
|----------------------------|--------------------------------------|--------|-----------------------------------------------------------------------------------------------------------------------|-------------------|-----------------|-----------------|----------|-------------------------------------------------------|------------|--------------------------|------------------------|------|--------------------|-----------------------------------------------------------|
| Input                      | Process Name                         | Output | Remarks                                                                                                               | Category          | Sub-category    | Infra-structure | Location | Modul name in ecoinvent                               | Mean value | Unit                     | Source for mean value  | Type | Standard Deviation | General Comment                                           |
| Demineralised Water        | zinc coating, coils, Overall process | Water  |                                                                                                                       | w ater supply     | production      | No              | CH       | w ater, deionised, at plant                           | 4.03E+00   | kg                       | IPPC 2001              | 1    | 1.55               |                                                           |
| Process and cooling w ater |                                      |        |                                                                                                                       | resource          | in w ater       | Yes             | CH       | Water, unspecified natural origin                     | 1.18E-02   | m 3                      | IPPC 2001              | 1    | 1.22               | (1,5,1,1,1,3,12)                                          |
|                            |                                      |        | Process and cooling w ater evaporated (uncontaminated --> not inventoried) or already included in the w aste disposal |                   |                 |                 |          |                                                       | 9.16E-03   | m 3                      | IPPC 2001              | 1    | 1.22               | (1,5,1,1,1,3,12)                                          |
| Natural gas                |                                      |        |                                                                                                                       | natural gas       | heating systems | No              | RER      | natural gas, burned in industrial furnace >100kW      | 1.84E+01   | M J                      | IPPC 2001              | 1    | 1.47               | calculated from max value assuming lognormal distribution |
| Electricity                |                                      |        |                                                                                                                       | electricity       | production mix  | No              | UCTE     | electricity, medium voltage, production UCTE, at grid | 3.41E-01   | kWh                      | IPPC 2001              | 1    | 1.78               |                                                           |
| Hot w ater                 |                                      |        |                                                                                                                       | chemicals         | organics        | No              | RER      | heat, unspecific, in chemical plant                   | 4.64E-01   | M J                      | IPPC 2001              | 1    | 1.48               |                                                           |
| Soda                       |                                      |        |                                                                                                                       | chemicals         | inorganics      | No              | RER      | soda, pow der, at plant                               | 1.67E-03   | kg                       | Frisknecht et al. 1996 | 1    | 1.11               | (1,n.A.,3,1,1,n.A.,12)                                    |
| Transport, Lorry 32t       |                                      |        |                                                                                                                       | transport systems | road            | No              | RER      | transport, lorry 32t                                  | 1.52E-01   | tkm                      | Frisknecht et al. 1996 | 1    | 2.09               | (4,5,n.a.,n.a.,n.a.,n.a.,5)                               |
|                            |                                      |        | recycled                                                                                                              |                   |                 |                 |          |                                                       | n.a        |                          | IPPC 2001              |      |                    |                                                           |
|                            |                                      |        | Particle size from mechanical process: assumption 50 % >10 um.                                                        | air               | unspecified     |                 |          | Particulates, > 10 um                                 | 5.70E-03   | kg                       | Frisknecht et al. 1996 | 1    | 1.57               | (1,5,2,1,1,3,25)                                          |
|                            |                                      |        | Particle size from chemical process: assumption 2.5-10 um. In addition 50 % of the mechanical abrasion                | air               | unspecified     |                 |          | Particulates, > 2.5 um, and < 10um                    | 1.71E-02   | kg                       | Frisknecht et al. 1996 | 1    | 2.05               | (1,5,2,1,1,3,26)                                          |
|                            |                                      |        |                                                                                                                       | air               | unspecified     |                 |          | Ammonia                                               | 1.09E-01   | kg                       | Frisknecht et al. 1996 | 1    | 2.05               | (1,5,2,1,1,3,23)                                          |
|                            |                                      |        |                                                                                                                       | air               | unspecified     |                 |          | Arsenic                                               | 3.71E-09   | kg                       | Frisknecht et al. 1996 | 1    | 1.57               | (1,5,2,1,1,3,31)                                          |
|                            |                                      |        |                                                                                                                       | air               | unspecified     |                 |          | Cadmium                                               | 2.56E-08   | kg                       | Frisknecht et al. 1996 | 1    | 1.57               | (1,5,2,1,1,3,31)                                          |
|                            |                                      |        |                                                                                                                       | air               | unspecified     |                 |          | Lead                                                  | 6.63E-08   | kg                       | Frisknecht et al. 1996 | 1    | 1.57               | (1,5,2,1,1,3,31)                                          |
|                            |                                      |        |                                                                                                                       | air               | unspecified     |                 |          | Zinc                                                  | 1.31E-05   | kg                       | Frisknecht et al. 1996 | 1    | 1.57               | (1,5,2,1,1,3,31)                                          |
|                            |                                      |        |                                                                                                                       | air               | unspecified     |                 |          | Iron                                                  | 2.73E-05   | kg                       | Frisknecht et al. 1996 | 1    | 1.57               | (1,5,2,1,1,3,31)                                          |
|                            |                                      |        | 10% of rolling mill for steel forming per kg output as rough approximation                                            | metals            | processing      | Yes             | RER      | rolling mill                                          | 2.54E-12   | unit                     | estimation             | 1    | 3.07               | (4,4,1,3,1,4,9)                                           |
|                            |                                      |        |                                                                                                                       | air               | unspecified     |                 |          | Heat, w aste                                          | 1.23E+00   | M J                      | IPPC 2001              | 1    | 1.22               | (1,5,1,1,1,3,13)                                          |
|                            |                                      |        |                                                                                                                       | metals            | processing      | No              | RER      | zinc coating, coils                                   | 1.00E+00   | m 2                      |                        |      |                    |                                                           |
| Infrastructure             |                                      |        |                                                                                                                       |                   |                 |                 |          |                                                       |            |                          |                        |      |                    |                                                           |

Fig. 5.2 Flows for "zinc coating, coils" and their representation in the ecoinvent database (cont.)

## 5.2 Zinc coating of pieces

The production process is described in (IPPC (2001a)) and on the internet<sup>12</sup>.

In a sense, the general or “batch” process is the same as the continuous process in that the objective is to apply a continuous coating of corrosion-resistant zinc onto the surface of steel. However, the practices to achieve this end result are very different.

As stated above, the second galvanising process involves the application of zinc onto a “fabricated” shape. That means, the steel is shaped into the final product, and then dipped into molten zinc to apply a zinc coating. The items to be coated in batch galvanising plants are steel fabrications, such as nails, screws and other very small items; lattice grates, construction parts, structural components, light poles and many more. Tubes are sometimes also galvanised in conventional batch coating plants. These items are coated either one at a time or, in the case of small parts, as a number of parts contained in a “basket”. Hence, the terms “batch” or “after fabrication” are applied to this process.

Galvanising, usually comprises the following process steps:

- Degreasing – physically cleaning and degreasing steel parts,
- Pickling – removing scale, rust, and oxide from steel parts,
- Fluxing – applying a flux coating to reduce surface oxide and allow for the growth of a zinc-iron intermetallic layer necessary for proper galvanising,
- Galvanising – dipping parts in the kettle containing molten zinc and lead.
- Finishing – excessiv zinc removal, cooling, inspection.

A galvanising plant essentially consists of a series of treatment or process baths. The steel is moved between tanks and dipped into the baths by overhead cranes. A more detailed description of every single step is given below.

The infrastructure of batch hot dip zinc coating, comprises buildings where galvanising processes take place and for storage, surroundings and machinery fore the single process steps. The mean lifetime for the infrastructure is 50 years. As no information is available, the infrastructure is approximated by the infrastructure for rolling.

In 1997 the tonnage of galvanised steel was about 5 million. The largest share was produced in Germany with 1.4 million tonnes and 185 galvanising plants (in 1997). Second largest producer was Italy with 0.8 million tonnes (74 plants), followed by UK and Ireland with 0.7 million tonnes (88 plants) and France 0.7 million tonnes (69 plants). The zinc consumption of the EU galvanising industry (excluding Greece and Luxembourg) in 1997 was 381188 t. Galvanised steel is used in construction, transport, agriculture, power transmission and other applications where good corrosion protection and long life are essential.

### 5.2.1 Degreasing of steel

During the degreasing step organic contaminants such as coolants and lubricants are removed from the fabricated steel in a hot alkali solution. These surface contaminants need to be removed prior to pickling so that the surface can be “wetted” by the pickling solution. The steel is dipped in baths containing surfactants, which remove oil and grease from the metal surface by emulsifying. The resulting unstable emulsions float on the surface of the bath and can be removed by gravitational separators, skimmers, micro- or ultrafiltration etc.

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<sup>12</sup> <http://www.oit.doe.gov/nice3/factsheets/ferrotech.pdf>, accessed 03.03.03

Concentration, bath temperature and immersion time of the work-pieces determine the efficiency of the degreasing baths. The normal temperature range for indirectly heated degreasing baths is 30 – 70 °C, although in some cases hot degreasing is applied at a temperature of about 85 °C. The baths consists of a sodium-hydroxide solution (1 - 10 %) plus other alkaline reagents, such as soda, sodium silicate, condensed alkaline phosphates and borax, and specific surfactants, emulsifying agents and dispersion agents. After degreasing, rinsing is necessary to prevent carry-over of degreasing agents which would shorten the life of pickling baths and reduce the re-usability of the bath.

An alternative method is acidic degreasing; but it is not considered in this study.

Description is taken from (IPPC (2001a)) and the internet<sup>13</sup>.

### 5.2.2 Pickling of steel

Pickling involves the immersion of parts into an acid solution (typically either heated sulfuric acid or ambient temperature hydrochloric acid) to remove surface scale or rust (both oxides of iron), casting skin, roll skin or grid. Therefore a galvanising plant usually comprises a series of pickling baths with different acid concentrations ranging from 2 - 16 %, normally 12 – 16 % when freshly prepared. To prevent excessive pickling of steel items, especially in pickling high tensile steels, and to protect the steel pickling vats, pickling inhibitors (e.g. hexamethylenetetramine) are added to the bath.

During operation the iron content of the pickling bath increases, while the amount of free acid decreases, making it necessary to top up the bath occasionally by adding fresh acid. Iron-(II)-chloride has a limited solubility in HCl. When this maximum is reached, pickling becomes impossible, but usually the pickling bath has to be replaced even sooner at lower FeCl<sub>2</sub> concentrations. Discarding concentrations of 170 g FeCl<sub>2</sub>/l (=75 g Fe<sup>2+</sup> /l) and 100 to 120 g Fe/l have been reported. Raising the temperature of the pickling bath can enable its use at relatively high concentrations of FeCl<sub>2</sub>, e.g. 175 to 200 g/l with the bath at 35 °C, with some increase in emissions.

Pickling in plants with open pre-treatment is usually done at ambient air temperature; plants with an enclosed pre-treatment sometimes operate with higher acid temperatures. Gaseous emissions of the pickling agent can arise from the pickling baths, depending on concentration and temperature of the bath, and from the pickled items. Hydrogen vesicles formed during the pickling process can also contain acid droplets.

Sometimes it is necessary to clean the suspension devices of zinc coatings, to remove faulty coatings from steel fabrications or to de-zinc fabrications whose coatings have to be renewed. This is commonly done by dipping in diluted pickle acid. This process step is called stripping. When pickling and stripping are carried out in the same treatment vat, pickle liquors are created which contain iron- and zinc chloride.

Rinsing is a very important step in the galvanising process as it prolongs the life of subsequent treatment baths, reduces the generation of waste and increases the re-usability of by-products. After degreasing and pickling the fabricated steel is therefore rinsed/dipped in water baths, which are sometimes heated. Water from rinsing can be used to prepare fresh pickling or degreasing baths as a way of recycling water and minimising aqueous blowdown. Description is taken from (IPPC (2001a)) and the internet<sup>14</sup>

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<sup>13</sup> <http://www.galvininfo.com/GalvInfoNote%20PDF's/GalvInfo9.pdf>, accessed 03.03.03

<sup>14</sup> <http://www.galvininfo.com/GalvInfoNote%20PDF's/GalvInfo9.pdf>, accessed 03.03.03

### 5.2.3 Fluxing of steel

The purpose of fluxing is to enable liquid zinc to wet the surface of the steel, a necessary prerequisite for the galvanising reaction, and with ammonium chloride-containing fluxes, to provide additional pickling (cleaning of the surface) during hot dipping. At temperatures of above 200 °C the ammonium chloride in the flux decomposes into  $\text{NH}_3$  and  $\text{HCl}$ , which results in an additional pickling effect. Fluxing can be either “dry” or “wet”.

In dry fluxing, the steel is immersed in a fluxing bath, usually an aqueous solution of zinc chloride and ammonium chloride, typically maintained at 40 – 80 °C. Cold fluxing is possible but reduces the potential for air-drying of the work after removal from the flux bath.

Typical bath characteristics are:

- $\text{ZnCl}_2$  150 - 300 g/l
- $\text{NH}_4\text{Cl}$  150 - 300 g/l
- Density: 1.15 – 1.30 g/ml
- Dissolved iron: < 2 g/l

pH of flux baths may be in the range of 1 to 5.0 (normally adjusted to approximately 4.5).

Ammonium chloride in a typical good flux often accounts for 40 – 60 % of the total flux salt. It provides quick drying and better removal of iron oxides from the surfaces of the items, but also causes more fume, ash and dross formation during the coating process. Zinc chloride prevents oxidation of the work-piece surfaces. This is particularly important when the drying time is long.

The iron content of the flux bath is extremely important for process control, economy and environment. To reduce the environmental impact of ammonium chloride during dipping, some galvanisers have changed to 'smoke-reduced' fluxes in which ammonium chloride has partly or completely been substituted by potassium chloride.

After withdrawing the work-pieces from the flux bath, some of the water from the adhering fluxing fluid evaporates. Further drying is sometimes achieved in dedicated drying. Drying of the work helps reduce splashing and ejection of metal from the zinc bath as the work is dipped, a benefit which is increased if the work retains heat after leaving the dryer, i.e. if preheat is applied.

A small number of galvanising works, especially those with exceptionally demanding fluxing requirements for intricate parts, operate an alternative process, called wet fluxing. In this process the fluxing agents flow as a layer of molten salt on the surface of the galvanising bath. Steel parts to be galvanised are passed through the flux layer into the zinc bath. Then the molten salt layer is drawn back from the surface by means of a rake to allow the steel parts to be withdrawn from the galvanising bath without further contact with the flux. Description is taken from (IPPC (2001a)) and the internet<sup>15</sup>

### 5.2.4 Galvanising of steel

The fluxed steel parts are slowly lowered into a bath of molten zinc. The steel reacts with the zinc to form a coating consisting of a series of zinc-iron alloy layers topped by a layer of pure zinc when the parts are withdrawn from the bath. The batch process requires that the part be immersed for much longer times than for the continuous process, typically measured in minutes (up to 30 minutes for the heaviest structural parts), not seconds. There are two reasons for needing longer immersion times. One is to allow the part to reach the bath temperature, because immersion of a relatively cold thick-walled large pipe, for example, means that as the steel is first immersed, it freezes a skin of zinc onto its surface. For the coating to bond metallurgically to the steel, the pipe has to be heated to “remelt” the zinc. Then, additional time is required to develop the iron/zinc alloy bond zone. For most steels, a longer

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<sup>15</sup> <http://www.galvinform.com/GalvInfoNote%20PDF's/GalvInfo9.pdf>, accessed 03.03.03

immersion time provides a thicker alloy layer (in Frischknecht et al. (1996), values of 65 µm for both-sided and of 130 µm for one-sided coating are given), a thicker alloy bond layer is often desired to provide a longer life to the final product, that is, a longer life before the onset of rust. Like the zinc itself, the alloy layer is galvanically protective to the steel part; thus, a thicker alloy layer means longer life. Yes, the alloy layer is hard and brittle, but since the part is already fabricated, there is no need for additional forming.

The molten zinc has a temperature of 440 – 475 °C. Kettle dimensions vary greatly; typical dimensions are 7 m long by 1.4 m wide by 2.6 m deep. The kettle is normally heated externally, commonly by gas or oil-fired burners.

The zinc bath usually also contains very small amounts of other metals, which either are impurities of the zinc input or are added as alloying elements. A typical bath composition is: Zinc 98.9 wt-%, lead 1.0 wt-%, iron 0.03 wt-%, aluminium 0.002 wt-%, cadmium 0.02 wt-%, traces of other metals (e.g. tin, copper). Aluminium and lead are added because of their influence on the thickness and the appearance of the coating (lead causes the formation of the typical, attractive large spangled surface, which through the years was “the way to identify galvanised coatings”). Lead helps to wet the steel before galvanising and the zinc to flow from the surface after galvanising. In some instances today, bismuth is being substituted for lead to achieve free drainage of the excess zinc. Lead can also be used to protect the kettle. The thickness of the kettle's steel walls is measured on a regular basis to prevent breaking. A small number of plants carry out 'high temperature galvanising' using refractory lined vessels, which allow the zinc bath to be operated at higher temperatures, usually about 530 °C. Description is taken from (IPPC (2001a)) and the internet<sup>16</sup>

### **5.2.5 Finishing**

The steel parts are withdrawn from the zinc bath; excessive zinc is removed by wiping or, in some cases, by rattling. They are then cooled and inspected. Small surface imperfections are repaired and the parts are removed from the jigs and made ready for dispatch. After hot dip coating some steel products are quenched in water to give them special properties. As protection against white rust, the products may be covered with oil emulsions or may be chromated. Other details of variations of this process step for pipes and small components are also reported in (IPPC (2001a)).

### **5.2.6 Environmental issues of the batch process for zinc coating of steel**

The main environmental issues for batch galvanising are emissions to air (HCl from pickling, and dust and gaseous compounds from the kettle); spent process solutions (degreasing solutions, pickling baths and flux baths), oily wastes (e.g. from cleaning of degreasing baths) and zinc-containing residues (filter dust, zinc ash, hard zinc), (IPPC (2001a)).

As wastewater is treated in the factory, post treatment wastewater composition and ancillary products for the water treatment as given in IPPC (2001a) are used to calculate the expenditures and the water emissions of the water treatment process.

### **5.2.7 Hot dip zinc coating (batch process) in ecoinvent**

In (IPPC (2001a)) most values are given in ranges, the mean values are generally calculated as the geometric mean of the given range. When the lower value of the range is zero, then the arithmetic mean of the range is taken (what correspond to half of the maximal value). Values taken from

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<sup>16</sup> <http://www.galvinfo.com/GalvInfoNote%20PDF's/GalvInfo9.pdf>, accessed 03.03.03

(Frischknecht et al. (1996)) are already mean values and do not have any information about minima and maxima or distribution.

The background information to the process "zinc coating, pieces" is given in Tab. 5.2, flows are shown in Fig. 5.3

### Data quality

For values represented by the geometric mean, uncertainty is calculated assuming a lognormal distribution and the high value given in (IPPC (2001a)) representing the 95% geometric standard deviation. For the arithmetic means of ranges given in (IPPC (2001a)) and the values from (Frischknecht et al. (1996)), the uncertainty is estimated by completing the pedigree matrix, according to (Frischknecht et al. (2003)).

**Tab. 5.2** Ecoinvent meta information for "zinc coating, pieces"

| Name                                  | zinc coating, pieces                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location                              | RER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Infrastructure Process                | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Unit                                  | m2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Data Set Version                      | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Included Processes                    | Includes the process steps degreasing, pickling, fluxing, galvanising (melt zinc coating) and finishing. Does not include the fabricates being coated.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Amount                                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Local Name                            | Stückverzinkung                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Synonyms                              | hot dip coating//galvanising                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| General Comment to reference function | This module is applicable, if batch zinc coating of semi-finished steel products, that's shape doesn't change anymore, has to be balanced. Data given per tonne of coated product had to be transformed dividing it by the mean surface area of 60 m2/t. For the chemicals and the zinc, 100 km of 32t lorry transport were balanced. The softened industrial water is assumed to be produced at the coating plant. The coating layer is 65um thick, if the surface is coated on both sides and 130 µm thick, if coating is done on one side only. Due to the fact, that data are given per square meter and not per kg, the energy inventoried for heavy pieces is to low and for light pieces to high. If the data would have been given per kg, the error would have been the contrary. |
| Start Date                            | 1996                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| End Date                              | 2002                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Other Period Text                     | Reference from 1996 refers partly to data from 1993.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Geography text                        | Data-set is representative for European Union                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Technology text                       | Average technique for EU.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Representativeness [%]                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Production Volume                     | 5 million tonnes in 1997                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Sampling Procedure                    | Literature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Extrapolations                        | see technology                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Uncertainty Adjustments               | The fact that no data were available about the percentage of companies represented by the study cited is considered in the uncertainties.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

5. Zinc coating of steel

| General Flow Information |                                  |        | Representation in ecoinvent                                                                                                                                                       |                   |                               |                 |          |                                                                          |            | Uncertainty Informations |                       |      |                    |                                                           |
|--------------------------|----------------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------|-----------------|----------|--------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------------------------------------|
| Input                    | Process Name                     | Output | Remarks                                                                                                                                                                           | Category          | Sub-category                  | Infra-structure | Location | Modul name in ecoinvent                                                  | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                                           |
| Degreasing agent         | zinc coating, pieces, Degreasing | mL     | solution of 10 % needed, --> 1/5 of 50% solution is inventoried                                                                                                                   | chemicals         | inorganics                    | No              | RER      | sodium hydroxide, 50% in H2O, production mix, at plant                   | 6.33E-03   | kg                       | IPPC 2001             | 1    | 2                  | calculated from max value assuming lognormal distribution |
| Water_                   |                                  |        |                                                                                                                                                                                   | resource          | in w ater                     |                 |          | Water, unspecified natural origin                                        | 1.67E-04   | m 3                      | IPPC 2001             | 1    | 2                  |                                                           |
| Oil_                     |                                  |        | Assumption: each energy carrier is used from the same number of plants                                                                                                            | oil               | heating systems               | No              | RER      | heat, light fuel oil, at industrial furnace 1MW                          | 3.35E-01   | M J                      | IPPC 2001             | 1    | 2                  |                                                           |
| Natural gas              |                                  |        |                                                                                                                                                                                   | natural gas       | heating systems               | No              | RER      | natural gas, burned in industrial furnace >100kW                         | 3.35E-01   | M J                      | IPPC 2001             | 1    | 2                  |                                                           |
| Electricity              |                                  |        |                                                                                                                                                                                   | electricity       | production mix                | No              | UCTE     | electricity, medium voltage, production UCTE, at grid                    | 9.29E-02   | kWh                      | IPPC 2001             | 1    | 2                  |                                                           |
| w aste heat              |                                  |        |                                                                                                                                                                                   | chemicals         | organics                      | No              | RER      | heat, unspecific, in chemical plant                                      | 3.35E-01   | M J                      | IPPC 2001             | 1    | 2                  |                                                           |
|                          | zinc coating, pieces, Pickling   | mL     | Including discarded degreasing bath. Recycled --> not inventoried                                                                                                                 |                   |                               |                 |          |                                                                          | 4.50E-02   | kg                       | IPPC 2001             | 1    | 2                  | (1,5,1,1,1,3,41)                                          |
|                          |                                  |        | Liquor and sludge                                                                                                                                                                 |                   |                               |                 |          |                                                                          |            |                          |                       |      |                    |                                                           |
|                          |                                  | mL     | disposed                                                                                                                                                                          | w aste management | hazardous w aste incineration | No              | CH       | disposal, used mineral oil, 10% w ater, to hazardous w aste incineration | 2.67E-03   | kg                       | IPPC 2001             | 1    | 1.56               |                                                           |
|                          |                                  |        |                                                                                                                                                                                   | chemicals         | inorganics                    | No              | RER      | hydrochloric acid, 30% in H2O, at plant                                  | 3.20E-01   | kg                       | IPPC 2001             | 1    | 2.09               |                                                           |
| Hydrochloric acid (30%)  |                                  | mL     | hexamethylenetetramine                                                                                                                                                            | chemicals         | organics                      | No              | GLO      | chemicals organic, at plant                                              | 1.67E-03   | kg                       | IPPC 2001             | 1    | 2                  |                                                           |
| Inhibitor                |                                  |        |                                                                                                                                                                                   | resource          | in w ater                     |                 |          | Water, unspecified natural origin                                        | 2.92E-04   | m 3                      | IPPC 2001             | 1    | 2                  |                                                           |
| Water_                   |                                  | mL     |                                                                                                                                                                                   | electricity       | production mix                | No              | UCTE     | electricity, medium voltage, production UCTE, at grid                    | 1.04E-01   | kWh                      | IPPC 2001             | 1    | 4                  | calculated from max value assuming lognormal distribution |
| Electricity              |                                  |        | Assumption: same amount of w aste gas per kg of product as for HCl-pickling in cold sheet rolling of steel                                                                        | air               | unspecified                   |                 |          | Hydrogen chloride                                                        | 1.18E-06   | kg                       | IPPC 2001             | 1    | 7.07               |                                                           |
|                          |                                  | mL     | Assumption: same amount of w aste gas per kg of product as for HCl-pickling in cold sheet rolling of steel. Assumption: particle from chemical process are betw een 2.5 and 10 um | air               | unspecified                   |                 |          | Particulates, > 2.5 um, and < 10um                                       | 5.00E-05   | kg                       | IPPC 2001             | 1    | 2                  |                                                           |
|                          |                                  |        | assumed as recycled --> not inventoried                                                                                                                                           |                   |                               |                 |          |                                                                          | 3.33E-01   | l                        | IPPC 2001             | 1    | 2                  |                                                           |
|                          |                                  | mL     | Discarded pickle liquor                                                                                                                                                           |                   |                               |                 |          |                                                                          | n.a.       |                          | IPPC 2001             |      |                    |                                                           |

Fig. 5.3 Flows for "zinc coating, pieces" and their representation in the ecoinvent database

5. Zinc coating of steel

| General Flow Information                                                                                                                           |                                |                         | Representation in ecoinvent                                                                                                                                                     |           |              |                 |          |                                                      |            | Uncertainty Informations |                       |      |                    |                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------|-----------------|----------|------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------------------------------------|
| Input                                                                                                                                              | Process Name                   | Output                  | Remarks                                                                                                                                                                         | Category  | Sub-category | Infra-structure | Location | Modul name in ecoinvent                              | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                                           |
| <div>Hydrochloric acid (30%)</div> <div>Water</div> <div>Flux agent (ZnCl<sub>2</sub> x NH<sub>4</sub>Cl)</div> <div>Water</div> <div>Energy</div> | zinc coating, pieces, Pickling | Hydrogen chloride       | Assumption: same amount of waste gas per kg of product as for HCl-pickling in cold sheet rolling of steel                                                                       | air       | unspecified  |                 |          | Hydrogen chloride                                    | 1.18E-06   | kg                       | IPPC 2001             | 1    | 7.07               | calculated from max value assuming lognormal distribution |
|                                                                                                                                                    |                                | Dust                    | Assumption: same amount of waste gas per kg of product as for HCl-pickling in cold sheet rolling of steel. Assumption: particle from chemical process are between 2.5 and 10 um | air       | unspecified  |                 |          | Particulates, > 2.5 um, and < 10um                   | 5.00E-05   | kg                       | IPPC 2001             | 1    | 2                  |                                                           |
|                                                                                                                                                    |                                | Waste acid and sludge   | assumed as recycled --> not inventoried                                                                                                                                         |           |              |                 |          |                                                      | 3.33E-01   | l                        | IPPC 2001             | 1    | 2                  |                                                           |
|                                                                                                                                                    |                                | Discarded pickle liquor | Reference contained no data about composition and amount of discarded pickling liquor.                                                                                          |           |              |                 |          |                                                      | n.a.       |                          | IPPC 2001             |      |                    |                                                           |
|                                                                                                                                                    | zinc coating, pieces,          | Spent strip liquor      |                                                                                                                                                                                 | chemicals | inorganics   | No              | RER      | hydrochloric acid, 30% in H <sub>2</sub> O, at plant | 5.00E-02   | kg                       | IPPC 2001             | 1    | 2                  | calculated from max value assuming lognormal distribution |
|                                                                                                                                                    |                                |                         |                                                                                                                                                                                 | resource  | in water     |                 |          | Water, unspecified natural origin                    | 5.83E-05   | m <sup>3</sup>           | IPPC 2001             | 1    | 2                  |                                                           |
|                                                                                                                                                    | zinc coating, pieces, Fluxing  | Spent flux              | assumed as recycled --> not inventoried                                                                                                                                         |           |              |                 |          |                                                      | 2.24E-02   | kg                       | IPPC 2001             | 1    | 1.12               |                                                           |
|                                                                                                                                                    |                                |                         | Inorganic chemicals as proxy                                                                                                                                                    | chemicals | inorganics   | No              | GLO      | chemicals inorganic, at plant                        | 2.50E-02   | kg                       | IPPC 2001             | 1    | 2                  | calculated from max value assuming lognormal distribution |
|                                                                                                                                                    |                                |                         |                                                                                                                                                                                 | resource  | in water     |                 |          | Water, unspecified natural origin                    | 1.67E-04   | m <sup>3</sup>           | IPPC 2001             | 1    | 2                  |                                                           |
|                                                                                                                                                    |                                |                         | Energy for heating up the flux bath is in some case supplied by internally recovered heat                                                                                       |           |              |                 |          |                                                      | n.a.       |                          | IPPC 2001             |      |                    | calculated from max value assuming lognormal distribution |
|                                                                                                                                                    |                                |                         | assumed as recycled --> not inventoried                                                                                                                                         |           |              |                 |          |                                                      | 4.08E-02   | kg                       | IPPC 2001             | 1    | 2.45               |                                                           |
|                                                                                                                                                    |                                |                         |                                                                                                                                                                                 |           |              |                 |          |                                                      | n.a.       |                          | IPPC 2001             |      |                    |                                                           |

Fig. 5.3 Flows for "zinc coating, pieces" and their representation in the ecoinvent database (cont.)



5. Zinc coating of steel

| General Flow Information |                               |                                    | Representation in ecoinvent                                                                                                                                                                                                                            |                  |                            |                 |          |                                                                                 |            | Uncertainty Informations |                       |      |                    |                                                           |
|--------------------------|-------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------------|-----------------|----------|---------------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------------------------------------|
| Input                    | Process Name                  | Output                             | Remarks                                                                                                                                                                                                                                                | Category         | Sub-category               | Infra-structure | Location | Modul name in ecoinvent                                                         | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment                                           |
| Water                    | zinc coating, pieces, Rinsing |                                    | Inventoried for the overall process                                                                                                                                                                                                                    | resource         | in water                   |                 |          | Water, unspecified natural origin                                               | 1.67E-04   | m 3                      | IPPC 2001             | 1    | 2                  | calculated from max value assuming lognormal distribution |
| Zinc                     |                               |                                    | Values from the two references are almost the same                                                                                                                                                                                                     | metals           | extraction                 | No              | RER      | zinc for coating, at regional storage                                           | 1.05E+00   | kg                       | IPPC 2001             | 1    | 3.16               |                                                           |
| Recovered zinc           |                               |                                    | recycling --> not inventoried                                                                                                                                                                                                                          |                  |                            |                 |          |                                                                                 | 1.25E-01   | kg                       | IPPC 2001             | 1    | 2                  |                                                           |
| Oil                      |                               |                                    | Assumption: 49% of the plants is using fuel as energy carrier for heating up the zinc bath                                                                                                                                                             | oil              | heating systems            | No              | RER      | heat, light fuel oil, at industrial furnace 1MW                                 | 1.25E+01   | MJ                       | IPPC 2001             | 1    | 2.36               |                                                           |
| Natural gas              |                               |                                    | Assumption: 49% of the plants is using natural gas as energy carrier for heating up the zinc bath                                                                                                                                                      | natural gas      | heating systems            | No              | RER      | natural gas, burned in industrial furnace >100kW                                | 1.25E+01   | MJ                       | IPPC 2001             | 1    | 2.36               |                                                           |
| Electricity              |                               |                                    | Assumption: 2% of the plants is using electricity as energy carrier for heating up the zinc bath                                                                                                                                                       | electricity      | production mix             | No              | UCTE     | electricity, medium voltage, production UCTE, at grid                           | 1.41E-01   | kWh                      | IPPC 2001             | 1    | 2.36               |                                                           |
|                          |                               | Extracted gas from kettle          |                                                                                                                                                                                                                                                        |                  |                            |                 |          |                                                                                 | 7.07E+01   | m 3                      | IPPC 2001             | 1    | 2.83               | calculated from max value assuming lognormal distribution |
|                          |                               | Dust>10                            |                                                                                                                                                                                                                                                        | air              | unspecified                |                 |          | Particulates, > 10 um                                                           | 2.58E-03   | kg                       | IPPC 2001             | 1    | 3.87               |                                                           |
|                          |                               | Zinc                               |                                                                                                                                                                                                                                                        | air              | unspecified                |                 |          | Zinc                                                                            | 4.47E-04   | kg                       | IPPC 2001             | 1    | 3.16               |                                                           |
|                          |                               | Hydrogen chloride                  |                                                                                                                                                                                                                                                        | air              | unspecified                |                 |          | Hydrogen chloride                                                               | 1.00E-04   | kg                       | IPPC 2001             | 1    | 1.41               |                                                           |
|                          |                               | Lead                               |                                                                                                                                                                                                                                                        |                  |                            |                 |          |                                                                                 | negligible |                          | IPPC 2001             |      |                    |                                                           |
|                          |                               | Ash                                | Floating on the surface of the galvanizing bath, ash is skimmed off before removing the dipped work-pieces. Zinc content is 40 to 90 % and makes it valuable for recycling either directly in the plant by secondary zinc industry --> not inventoried |                  |                            |                 |          |                                                                                 | 1.67E-01   | kg                       | IPPC 2001             | 1    | 2.5                | calculated from max value assuming lognormal distribution |
|                          |                               | Dross                              | Returned to the secondary zinc industry for recovery of the zinc content or to the zinc chemicals industry for the manufacture of zinc oxide --> not inventoried                                                                                       |                  |                            |                 |          |                                                                                 | 2.04E-01   | kg                       | IPPC 2001             | 1    | 2.45               |                                                           |
|                          |                               | Combustion gas: (NOx, CO/CO2, SO2) | Inventoried in the fuel combustion modules                                                                                                                                                                                                             |                  |                            |                 |          |                                                                                 | 2.12E+01   | m 3                      | IPPC 2001             | 1    | 2.55               |                                                           |
|                          |                               | Filter dust                        | Dust from low -alloyed EAF-steel as proxy                                                                                                                                                                                                              | waste management | residual material landfill | No              | CH       | disposal, dust, unalloyed EAF steel, 15.4% water, to residual material landfill | 2.45E-01   | kg                       | IPPC 2001             | 1    | 2.45               |                                                           |

Fig. 5.3 Flows for "zinc coating, pieces" and their representation in the ecoinvent database (cont.)

5. Zinc coating of steel

| General Flow Information |              |                                                   | Representation in ecoinvent                                                                                                                                                                                                                                                                                                                                                                                                                          |                  |                            |                |          |                                                                               |            | Uncertainty Informations |                       |      |                    |                  |  |
|--------------------------|--------------|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------------------------|----------------|----------|-------------------------------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|------------------|--|
| Input                    | Process Name | Output                                            | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                              | Category         | Sub-category               | Infrastructure | Location | Module name in ecoinvent                                                      | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment  |  |
| Flocculating agents      | mL           | zinc coating, pieces, Waste water treatment plant | Reference contained no data. Assumption: Same amount per kg product as for coil coating. Aluminium sulphate as proxy.                                                                                                                                                                                                                                                                                                                                | chemicals        | inorganics                 | No             | RER      | aluminium sulphate, powder, at plant                                          | 1.51E-05   | kg                       | IPPC 2001             | 1    | 1.22               | (2,5,1,1,1,3,4)  |  |
|                          |              |                                                   | One part of the incoming cooling and process water leaves the system as treated effluent; it is assumed to be discharged in surface water --> not inventoried. Infrastructure for waste water treatment plant is comprised in the infrastructure module. Value for energy use is comprised in the range of uncertainty of inventoried overall energy. Assumption: ratio between treated and untreated water is the same as for coil coating process. |                  |                            |                |          |                                                                               | 1.67E-03   | m3                       | IPPC 2001             | 1    | 1.22               | (2,5,1,1,1,3,4)  |  |
|                          |              |                                                   | Assumption: same amount per m3 treated water, as for coil coating process                                                                                                                                                                                                                                                                                                                                                                            | water            | unspecified                |                |          | Suspended solids, unspecified                                                 | 3.74E-06   | kg                       | IPPC 2001             | 1    | 1.57               | (2,5,1,1,1,3,32) |  |
|                          |              |                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      | water            | unspecified                |                |          | Iron, ion                                                                     | 4.10E-07   | kg                       | IPPC 2001             | 1    | 5.06               | (2,5,1,1,1,3,35) |  |
|                          |              |                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      | water            | unspecified                |                |          | Nickel, ion                                                                   | 1.67E-08   | kg                       | IPPC 2001             | 1    | 5.06               | (2,5,1,1,1,3,35) |  |
|                          |              |                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      | water            | unspecified                |                |          | Zinc, ion                                                                     | 2.63E-07   | kg                       | IPPC 2001             | 1    | 5.06               | (2,5,1,1,1,3,35) |  |
|                          |              |                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      | water            | unspecified                |                |          | Lead                                                                          | 2.51E-08   | kg                       | IPPC 2001             | 1    | 5.06               | (2,5,1,1,1,3,35) |  |
|                          |              |                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      | water            | unspecified                |                |          | Chromium, ion                                                                 | 1.10E-07   | kg                       | IPPC 2001             | 1    | 5.06               | (2,5,1,1,1,3,35) |  |
|                          |              |                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      | water            | unspecified                |                |          | Chromium VI                                                                   | 1.67E-08   | kg                       | IPPC 2001             | 1    | 5.06               | (2,5,1,1,1,3,35) |  |
|                          |              |                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      | water            | unspecified                |                |          | COD, Chemical Oxygen Demand                                                   | 2.20E-04   | kg                       | IPPC 2001             | 1    | 1.57               | (2,5,1,1,1,3,32) |  |
|                          |              |                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      | water            | unspecified                |                |          | BOD5, Biological Oxygen Demand                                                | 2.20E-04   | kg                       | IPPC 2001             | 1    | 2.5                | (5,5,5,5,5,32)   |  |
|                          |              |                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      | water            | unspecified                |                |          | DOC, Dissolved Organic Carbon                                                 | 4.95E-06   | kg                       | IPPC 2001             | 1    | 2.45               | (5,5,5,5,5,32)   |  |
|                          |              |                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      | water            | unspecified                |                |          | TOC, Total Organic Carbon                                                     | 4.95E-06   | kg                       | IPPC 2001             | 1    | 1.57               | (2,5,1,1,1,3,32) |  |
|                          |              |                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      | water            | unspecified                |                |          | Hydrocarbons, unspecified                                                     | 1.98E-06   | kg                       | IPPC 2001             | 1    | 3.05               | (2,5,1,1,1,3,34) |  |
|                          |              |                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      | water            | unspecified                |                |          | Oils, unspecified                                                             | 1.73E-06   | kg                       | IPPC 2001             | 1    | 1.57               | (2,5,1,1,1,3,32) |  |
|                          |              |                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      | water            | unspecified                |                |          | Phosphorus                                                                    | 1.50E-06   | kg                       | IPPC 2001             | 1    | 1.57               | (2,5,1,1,1,3,33) |  |
|                          |              |                                                   | Waste water treatment sludge from steel forming processes is inventoried as proxy. Assumption: same amount per m3 treated water, as for coil coating process.                                                                                                                                                                                                                                                                                        | waste management | residual material landfill | No             | CH       | disposal, sludge from steel rolling, 20% water, to residual material landfill | 1.51E-03   | kg                       | IPPC 2001             | 1    | 1.22               | (2,5,1,1,1,3,6)  |  |

Fig. 5.3 Flows for "zinc coating, pieces" and their representation in the ecoinvent database (cont.)

5. Zinc coating of steel

| General Flow Information |                                       |                   | Representation in ecoinvent                                                                                                                                                                                         |                   |              |                 |          |                                         |            | Uncertainty Informations |                        |      |                    |                             |
|--------------------------|---------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--------------|-----------------|----------|-----------------------------------------|------------|--------------------------|------------------------|------|--------------------|-----------------------------|
| Input                    | Process Name                          | Output            | Remarks                                                                                                                                                                                                             | Category          | Sub-category | Infra-structure | Location | Modul name in ecoinvent                 | Mean value | Unit                     | Source for mean value  | Type | Standard Deviation | General Comment             |
| Lorry 32t                | zinc coating, pieces, Overall process | Water             |                                                                                                                                                                                                                     | transport systems | road         | No              | RER      | transport, lorry 32t                    | 1.34E-01   | tkm                      | Frisknecht et al. 1996 | 1    | 2.09               | (4,5,n.a.,n.a.,n.a.,n.a.,5) |
| Soda                     |                                       |                   |                                                                                                                                                                                                                     | chemicals         | inorganics   | No              | RER      | soda, powder, at plant                  | 1.67E-03   | kg                       | Frisknecht et al. 1996 | 1    | 2.05               | (1,5,2,1,1,3,5)             |
| Ammonia                  |                                       |                   |                                                                                                                                                                                                                     | chemicals         | inorganics   | No              | RER      | ammonia, liquid, at regional storehouse | 5.10E-04   | kg                       | Frisknecht et al. 1996 | 1    | 1.22               | (1,5,2,1,1,3,4)             |
| Water (softened)         |                                       |                   |                                                                                                                                                                                                                     | water supply      | production   | No              | CH       | water, deionised, at plant              | 3.33E+00   | kg                       | Frisknecht et al. 1996 | 1    | 1.23               | (2,5,2,1,1,3,12)            |
|                          | zinc coating, pieces, Overall process | Heat waste        | Process and cooling water evaporated (uncontaminated --> not inventoried) or already included in the waste disposal. Assumption: ratio between treated and untreated water is the same as for coil coating process. |                   |              |                 |          |                                         | 2.51E-03   | m3                       | Frisknecht et al. 1996 | 1    | 1.23               | (2,5,2,1,1,3,12)            |
|                          |                                       |                   | 10% of rolling mill for steel forming per kg output as rough approximation                                                                                                                                          | metals            | processing   | Yes             | RER      | rolling mill                            | 2.71E-12   | unit                     | estimation             | 1    | 3.15               | (4,5,1,3,3,4,9)             |
|                          |                                       |                   |                                                                                                                                                                                                                     | air               | unspecified  |                 |          | Heat, waste                             | 1.26E-01   | MJ                       | IPPC 2001              | 1    | 1.22               | (1,5,1,1,1,3,13)            |
|                          |                                       |                   | Particle size from mechanical process: assumption 50 % >10 um.                                                                                                                                                      | air               | unspecified  |                 |          | Particulates, > 10 um                   | 2.58E-05   | kg                       | Frisknecht et al. 1996 | 1    | 1.57               | (1,5,2,1,1,3,25)            |
|                          |                                       |                   | Particle size from chemical process: assumption 2.5-10 um. In addition 50 % of the mechanical abrasion                                                                                                              | air               | unspecified  |                 |          | Particulates, > 2.5 um, and < 10um      | 7.73E-05   | kg                       | Frisknecht et al. 1996 | 1    | 2.05               | (1,5,2,1,1,3,26)            |
|                          |                                       |                   |                                                                                                                                                                                                                     | air               | unspecified  |                 |          | Hydrogen chloride                       | 2.04E-04   | kg                       | Frisknecht et al. 1996 | 1    | 2.05               | (1,5,2,1,1,3,23)            |
|                          |                                       |                   |                                                                                                                                                                                                                     | air               | unspecified  |                 |          | Ammonia                                 | 7.27E-04   | kg                       | Frisknecht et al. 1996 | 1    | 2.05               | (1,5,2,1,1,3,23)            |
|                          |                                       |                   |                                                                                                                                                                                                                     | air               | unspecified  |                 |          | Arsenic                                 | 3.16E-09   | kg                       | Frisknecht et al. 1996 | 1    | 1.57               | (1,5,2,1,1,3,31)            |
|                          |                                       |                   |                                                                                                                                                                                                                     | air               | unspecified  |                 |          | Cadmium                                 | 2.18E-08   | kg                       | Frisknecht et al. 1996 | 1    | 1.57               | (1,5,2,1,1,3,31)            |
|                          |                                       |                   |                                                                                                                                                                                                                     | air               | unspecified  |                 |          | Lead                                    | 5.67E-08   | kg                       | Frisknecht et al. 1996 | 1    | 1.57               | (1,5,2,1,1,3,31)            |
|                          |                                       |                   |                                                                                                                                                                                                                     | air               | unspecified  |                 |          | Zinc                                    | 1.12E-05   | kg                       | Frisknecht et al. 1996 | 1    | 1.57               | (1,5,2,1,1,3,31)            |
|                          |                                       |                   |                                                                                                                                                                                                                     | air               | unspecified  |                 |          | Iron                                    | 2.33E-05   | kg                       | Frisknecht et al. 1996 | 1    | 1.57               | (1,5,2,1,1,3,31)            |
|                          |                                       |                   |                                                                                                                                                                                                                     | water             | unspecified  |                 |          | Chloride                                | 3.13E-05   | kg                       | Frisknecht et al. 1996 | 1    | 1.57               | (1,5,2,1,1,3,31)            |
|                          |                                       |                   |                                                                                                                                                                                                                     | water             | unspecified  |                 |          | Zinc, ion                               | 3.13E-05   | kg                       | Frisknecht et al. 1996 | 1    | 1.57               | (1,5,2,1,1,3,31)            |
|                          |                                       |                   |                                                                                                                                                                                                                     | water             | unspecified  |                 |          | Iron, ion                               | 9.38E-05   | kg                       | Frisknecht et al. 1996 | 1    | 1.57               | (1,5,2,1,1,3,31)            |
| Infrastructure           |                                       | coating of pieces |                                                                                                                                                                                                                     | metals            | processing   | No              | RER      | zinc coating, pieces                    | 1.00E+00   | m2                       |                        |      |                    |                             |

Fig. 5.3 Flows for "zinc coating, pieces" and their representation in the ecoinvent database (cont.)

### **5.3 Zinc coating of pieces, correction of layer thickness per $\mu\text{m}$**

If zinc layer thickness other than the double-sided 65  $\mu\text{m}$  or the one-sided 130  $\mu\text{m}$  are requested, the amount of zinc needed and the arising heavy metals emissions (evaporating from zinc) have to be adapted proportionally to the  $\mu\text{m}$ -change of layer thickness. The remaining flows depend basically on the coated piece surface area and have therefore not to be changed.

#### **5.3.1 Zinc coating of pieces, correction of layer thickness per $\mu\text{m}$ in ecoinvent**

Per  $\mu\text{m}$  of additional or subtracted layer thickness, the module has to be inventoried one time. This module can be applied also for changing the layer thickness when continuous galvanising of coils is done. In this case the module has to be inventoried 1.53 times per  $\mu\text{m}$  of changed thickness layer, because the zinc consumption per  $\mu\text{m}$  is higher. Data are taken from (Frischknecht et al. (1996)) where no information about minima and maxima or distribution are given.

The background information to the process "zinc coating, pieces, adjustment per  $\mu\text{m}$ " is given in Tab. 5.3, flows are shown in Fig. 5.4.

#### **Data quality**

The uncertainty is estimated by completing the pedigree matrix, according to (Frischknecht et al. (2003)). A lognormal distribution is assumed.

Tab. 5.3 Ecoinvent meta information for "zinc coating, pieces, adjustment per um"

|                                       |                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>                           | <b>zinc coating, pieces, adjustment per um</b>                                                                                                                                                                                                                                                                                                                                   |
| Location                              | RER                                                                                                                                                                                                                                                                                                                                                                              |
| Infrastructure Process                | 0                                                                                                                                                                                                                                                                                                                                                                                |
| Unit                                  | m2                                                                                                                                                                                                                                                                                                                                                                               |
| Data Set Version                      | 1                                                                                                                                                                                                                                                                                                                                                                                |
| Included Processes                    | Includes the process steps galvanising (melt zinc coating) and finishing. Does not include the fabricates being coated.                                                                                                                                                                                                                                                          |
| Amount                                | 1                                                                                                                                                                                                                                                                                                                                                                                |
| Local Name                            | Stückverzinkung, Korrektur pro um                                                                                                                                                                                                                                                                                                                                                |
| Synonyms                              | hot dip coating/galvanising                                                                                                                                                                                                                                                                                                                                                      |
| General Comment to reference function | This module is applicable if the coating layer is thicker than the one given for "zinc coating, pieces", or for "zinc coating, coils". Applying it for the latter, the module has to be multiplied by the factor 1.53 for correct use. The Dataset is not applicable for thinner coating layers. Data given per square meter of product surface corrected by one $\mu\text{m}$ . |
| Start Date                            | 1996                                                                                                                                                                                                                                                                                                                                                                             |
| End Date                              | 2002                                                                                                                                                                                                                                                                                                                                                                             |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                                                                                                                                                                |
| Other Period Text                     | Reference from 1996 refers partly to data from 1993.                                                                                                                                                                                                                                                                                                                             |
| Geography text                        | Data-set is representative for European Union                                                                                                                                                                                                                                                                                                                                    |
| Technology text                       | Average technique for EU.                                                                                                                                                                                                                                                                                                                                                        |
| Representativeness [%]                |                                                                                                                                                                                                                                                                                                                                                                                  |
| Production Volume                     | No data available                                                                                                                                                                                                                                                                                                                                                                |
| Sampling Procedure                    | Literature                                                                                                                                                                                                                                                                                                                                                                       |
| Extrapolations                        | none                                                                                                                                                                                                                                                                                                                                                                             |
| Uncertainty Adjustments               | The fact that no data were available about the percentage of companies represented by the study cited is considered in the uncertainties.                                                                                                                                                                                                                                        |

5. Zinc coating of steel

| General Flow Information |                                                    |                | Representation in ecoinvent |                   |              |                 |          |                                                    |            | Uncertainty Informations |                        |      |                    |                             |
|--------------------------|----------------------------------------------------|----------------|-----------------------------|-------------------|--------------|-----------------|----------|----------------------------------------------------|------------|--------------------------|------------------------|------|--------------------|-----------------------------|
| Input                    | Process Name                                       | Output         | Remarks                     | Category          | Sub-category | Infra-structure | Location | Modul name in ecoinvent                            | Mean value | Unit                     | Source for mean value  | Type | Standard Deviation | General Comment             |
| Lorry                    | zinc coating, pieces, adjustment per $\mu\text{m}$ |                |                             | transport systems | road         | No              | RER      | transport, lorry 32t                               | 1.50E-03   | tkm                      | Frisknecht et al. 1996 | 1    | 2.09               | (4,5,n.a.,n.a.,n.a.,n.a.,5) |
| Zinc                     |                                                    |                |                             | metals            | extraction   | No              | RER      | zinc for coating, at regional storage              | 1.54E-02   | kg                       | Frisknecht et al. 1996 | 1    | 1.22               | (1,5,2,1,1,3,12)            |
|                          |                                                    | As             |                             | air               | unspecified  |                 |          | Arsenic                                            | 4.86E-11   | kg                       | Frisknecht et al. 1996 | 1    | 1.57               | (1,5,2,1,1,3,31)            |
|                          |                                                    | Cd             |                             | air               | unspecified  |                 |          | Cadmium                                            | 3.35E-10   | kg                       | Frisknecht et al. 1996 | 1    | 1.57               | (1,5,2,1,1,3,31)            |
|                          |                                                    | Pb             |                             | air               | unspecified  |                 |          | Lead                                               | 8.72E-10   | kg                       | Frisknecht et al. 1996 | 1    | 1.57               | (1,5,2,1,1,3,31)            |
|                          |                                                    | Zn             |                             | air               | unspecified  |                 |          | Zinc                                               | 1.72E-07   | kg                       | Frisknecht et al. 1996 | 1    | 1.57               | (1,5,2,1,1,3,31)            |
|                          |                                                    | Coated product |                             | metals            | processing   | No              | RER      | zinc coating, pieces, adjustment per $\mu\text{m}$ | 1.00E+00   | m <sup>2</sup>           |                        |      |                    |                             |

Fig. 5.4 Flows for "zinc coating, pieces, adjustment per  $\mu\text{m}$ " and their representation in the ecoinvent database

## 6 Welding

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Review: Roland Hischier

### 6.1 Definition and fundamentals of welding

Welding is defined (DIN 1910, Part 1: 07.81 and ISO 857: 1990) as the bonding of several parts made of the same or different materials to form a single unit, where, in the weld region, either the temperature is raised to the melting point, the parts are pressed together, or both. Weld filler materials are also used as necessary. The welded joint can be separated only by destroying it. This definition applies to the welding of both metals and thermoplastic materials (Brandt & Weiler (1997)).

Weldability (see DIN 8528, Part 1: 06.73): A component fabricated by welding is weldable if, after fabrication, it fulfils all the functions required of it at its point of use for the duration of its expected lifetime. It is not weldable if it does not fulfil these functions as a result of any defects arising from a welding process. One of the most important preconditions for a good welded joint is that impurities on the surface of the workpiece such as thick scale, rust, oil, and paint should be removed as completely as possible. It is certainly possible to weld over a protective coating, but this can affect the integrity of the weld (Brandt & Weiler (1997)).

The methods of welding metals can be divided into two groups:

In fusion welding of metals, the junction area is melted to form a pool of molten metal, which consists of parent metal, either alone or with added filler metal (minus metal lost in the process). The zone surrounding the molten pool is to a greater or lesser extent affected by changes to its structure and properties, depending on the degree and duration of heating and the rate of cooling. This depends on the heat flux, which in turn is very dependent on the welding method.

In compression welding of metals, the workpieces are pressed together at the welding region, and, in most processes, are also heated. The heat is applied by various means, e.g., electrically (resistance or induction heating), or by flames (Brandt & Weiler (1997)).

This study concentrates on the fusion welding, differing between gas welding of steel, arc welding of steel and arc welding of aluminium. Fig. 6.1 shows the different processes and sub-processes of the fusion welding.

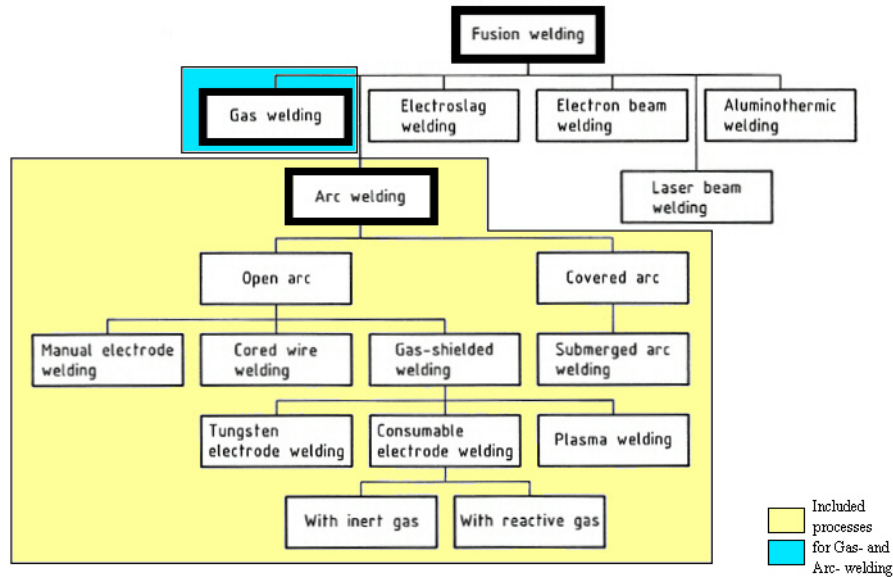


Fig. 6.1 Fusion welding processes (Brandt & Weiler (1997)).

### 6.1.1 Environmental issues of the welding process

Environmental issues in the welding process are mainly discussed in literature from a point of view of health and safety for the welder. Incorrect operation of a welding process can lead to hazards to personnel, especially from electric current, atmospheric pollutants (smoke, vapours, and gases), radiation, and noise. Precaution against fire and explosion is also very important. Fire can be produced directly by flames, or can begin slowly due to unnoticed molten material impelled over long distances, often only becoming serious after some hours. Fuel gases (e.g. acetylene, propane, etc.) and oxygen, constitute the main source of danger for explosions.

The concentrations of atmospheric pollutants such as airborne particulates (fume), vapours, and gases must not exceed the legal limits, i.e., the MAK ("Maximale Arbeitsplatzkonzentrationen"; maximal concentration on the workplace) values. Those limits are only valuable to guarantee human health when workers are exposed to the harming gases during the workprocess. They are not setting limits and they don't tell anything about the environmental pollution itself, because they can be maintained simply by diluting the concentration by mixing the polluted air with fresh air.

The airborne particulates contain lead, chromium, cadmium, nickel, mercury, copper, and zinc. They are formed as metal vapour during the processes of welding and soldering, and usually become oxidised and suspended in the air as fine dust. The harmful gases formed can include nitrogen oxides (NO, NO<sub>2</sub>) from flames, ozone and sometimes phosgene from the intense light emitted by electric arcs, and carbon monoxide when welding with a carbon dioxide gas shield.

## 6.2 Manual Gas Welding of Steel

In manual gas welding (or autogenic welding; descriptions taken from Brandt & Weiler (1997)) the heat required is produced by burning a mixture of acetylene and oxygen. The flame impinges directly on the welding area, and the filler metal is fed by hand.

In manual gas welding, the rate of heat flow to the workpiece is low compared with other welding methods, and the process is comparatively slow. For this reason, and because of the high thermal conductivity of metal, the structure of the neighbouring parent metal is considerably affected. In spite of its disadvantages, ca. 10% of all welding is by the autogenic method.



The autogenic torch is simple, lightweight, and cheap. It is much used for repairs and for simple jobs in the workshop. The gas supply consists simply of oxygen and acetylene bottles with tubing to convey these gases to the welding area. The main uses of autogenic torches are for welding pipelines in chemical plant construction, general industrial plants, shipbuilding, and heating plants, where its value depends more on the manual skill of the welder than on high melting rates.

Acetylene is used exclusively as the fuel gas in gas welding, as of all the hydrocarbons it has the highest heat of formation per unit mass, and gives the highest combustion temperature (Fig. 6.2), combustion rate, and hence heat flux and heat transfer from the flame to the workpiece. This is especially useful for the rapid heating of small surfaces. To prevent scaling and decarburization of the workpiece, the acetylene flame is adjusted to a neutral state, i.e., with no excess of either oxygen or fuel gas (volume ratio of oxygen to acetylene ca. 1.05). The rest of the oxygen required by the stoichiometry is provided by the surrounding air.

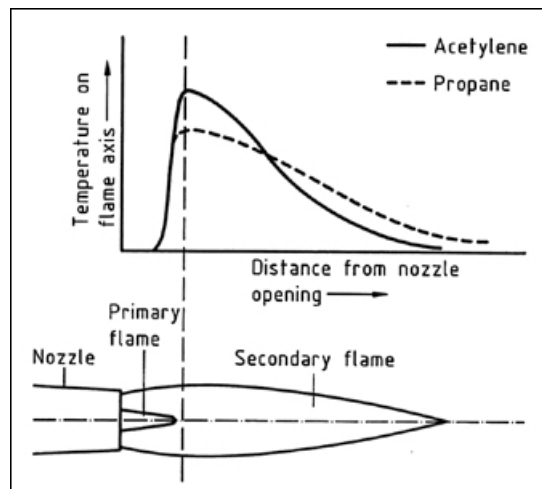


Fig. 6.2 Temperature profile along the flame axis when burning acetylene and propane Brandt & Weiler (1997).

As autogenic welding is done manually with no or very little infrastructure- and material demand for the machinery, those inputs are neglected in this study.

### 6.2.1 The manual gas welding process of steel in ecoinvent

For a lack of information on acetylene and oxygen consumption and on emissions the following assumptions are made:

The amount of acetylene used is estimated assuming that the energy needed for gas welding is five times the energy needed for automated arc welding (MAG). The heating value of acetylene is 51.2 MJ/kg (Bach et al. (1999)).

The volume ratio of oxygen / acetylene = 1.05. Thus the mol ratio of oxygen / acetylene also = 1.05. With the molar weight of  $O_2 = 32$  g/mol and of  $C_2H_2 = 26$  g/mol the mass ratio of oxygen / acetylene is calculated to be  $32 / (26 \cdot 1.05) = 1.17$ .

The metal, the nitrogen oxide and the ozone emissions are taken from the MAG welding process.

The CO and  $CO_2$  emissions are calculated to reflect the formation by the acetylene burning. Because of the neutral to slightly reducing conditions chosen for welding, the CO /  $CO_2$  ratio is rather high. It is assumed to be 1:100 (mol/mol). Thus per kg (= 38.46 mol) of acetylene burned 3.35 kg  $CO_2$  (= 76.16 mol) and 0.021 kg CO (= 0.7616 mol) are formed.

Furthermore it is assumed that 1% of the acetylene is emitted to air by lighting the torch and changing the bottles.

The transports of the steel feed and gas is inventoried according to the quality guidelines (see Frischknecht et al. (2003))

The background information to the process "welding, gas, steel" is given in Tab. 6.1, flows are shown in Fig. 6.3.

### Data quality

The uncertainty is estimated by completing the pedigree matrix, according to (Frischknecht et al. (2003)). A lognormal distribution is assumed.

**Tab. 6.1** Ecoinvent meta information for "welding, gas, steel"

| <b>Name</b>                           | <b>welding, gas, steel</b>                                                                                                                                 |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location                              | RER                                                                                                                                                        |
| Infrastructure Process                | 0                                                                                                                                                          |
| Unit                                  | m                                                                                                                                                          |
| Data Set Version                      | 2.0                                                                                                                                                        |
| Included Processes                    | Includes the acetylene welding of non alloyed steel and the transports of the filler rod and the protective gas to the place of use.                       |
| Amount                                | 1                                                                                                                                                          |
| Local Name                            | Schweissen, Autogen, Stahl                                                                                                                                 |
| Synonyms                              |                                                                                                                                                            |
| General Comment to reference function | Based partly on rather rough estimates and on input data of poor quality (acetylene). Not to be used if welding is of importance in the system considered. |
| Start Date                            | 1997                                                                                                                                                       |
| End Date                              | 2002                                                                                                                                                       |
| Data Valid For Entire Period          | 1                                                                                                                                                          |
| Other Period Text                     |                                                                                                                                                            |
| Geography text                        | Data-set is representative for Europe                                                                                                                      |
| Technology text                       | Acetylene welding with a wire consumption of 0.0536 kg/m                                                                                                   |
| Representativeness [%]                |                                                                                                                                                            |
| Production Volume                     | No data available                                                                                                                                          |
| Sampling Procedure                    | Literature                                                                                                                                                 |
| Extrapolations                        | none                                                                                                                                                       |
| Uncertainty Adjustments               | none                                                                                                                                                       |

## 6. Welding

| General Flow Information |                     |               | Representation in ecoinvent                                        |                        |                |                 |          |                                                       |            | Uncertainty Informations |                                                            |      |                    |                             |
|--------------------------|---------------------|---------------|--------------------------------------------------------------------|------------------------|----------------|-----------------|----------|-------------------------------------------------------|------------|--------------------------|------------------------------------------------------------|------|--------------------|-----------------------------|
| Input                    | Process Name        | Output        | Remarks                                                            | Category               | Sub-category   | Infra-structure | Location | Modul name in ecoinvent                               | Mean value | Unit                     | Source for mean value                                      | Type | Standard Deviation | General Comment             |
| Steel                    | welding, arc, steel |               |                                                                    | metals                 | extraction     | No              | RER      | steel, low -alloyed, at plant                         | 5.36E-02   | kg                       | calculated                                                 | 1    | 1.38               | (4,5,1,1,1,5,4)             |
| Argon                    |                     |               |                                                                    | chemicals              | inorganics     | No              | RER      | argon, liquid, at plant                               | 2.37E-02   | kg                       | calculated                                                 | 1    | 1.38               | (4,5,1,1,1,5,4)             |
| Carbon dioxide           |                     |               |                                                                    | chemicals              | inorganics     | No              | RER      | carbon dioxide liquid, at plant                       | 4.11E-03   | kg                       | calculated                                                 | 1    | 1.38               | (4,5,1,1,1,5,4)             |
| Oxygen                   |                     |               |                                                                    | chemicals              | inorganics     | No              | RER      | oxygen, liquid, at plant                              | 9.15E-04   | kg                       | calculated                                                 | 1    | 1.38               | (4,5,1,1,1,5,4)             |
| Electricity              |                     |               |                                                                    | electricity            | production mix | No              | UCTE     | electricity, medium voltage, production UCTE, at grid | 2.75E-02   | kWh                      | calculated from personal communication, Tenconi SA, Airolo | 1    | 1.32               | (3,5,1,3,1,5,2)             |
| transport rail           |                     |               | steel 600 km, gases 100 km. Gases twice the weight for the bottles | transport systems      | train          | No              | RER      | transport, freight, rail                              | 6.66E-02   | tkm                      | estimated                                                  | 1    | 2.09               | (4,5,n.A.,n.A.,n.A.,n.A.,5) |
| transport lorry          |                     |               | steel and gases 50 km. Gases twice the weight for the bottles      | transport systems      | road           | No              | RER      | transport, lorry 32t                                  | 5.55E-03   | tkm                      | estimated                                                  | 1    | 2.09               | (4,5,n.A.,n.A.,n.A.,n.A.,5) |
| Infrastructure           |                     |               | power transistor                                                   | construction processes | buildings      | No              | RER      | electronics for control units                         | 1.00E-05   | kg                       | estimated                                                  | 1    | 3.59               | (5,5,5,3,3,5,9)             |
|                          |                     | Cr (II, III)  |                                                                    | air                    | unspecified    |                 |          | Chromium                                              | 2.81E-05   | kg                       | Künniger 1998                                              | 1    | 5.42               | (1,5,5,3,3,5,22)            |
|                          |                     | Cr (VI)       |                                                                    | air                    | unspecified    |                 |          | Chromium VI                                           | 3.00E-08   | kg                       | Künniger 1998                                              | 1    | 5.42               | (1,5,5,3,3,5,22)            |
|                          |                     | Cu            |                                                                    | air                    | unspecified    |                 |          | Copper                                                | 2.10E-07   | kg                       | Künniger 1998                                              | 1    | 1.92               | (1,5,5,3,3,5,31)            |
|                          |                     | Fe            |                                                                    | air                    | unspecified    |                 |          | Iron                                                  | 8.74E-05   | kg                       | Künniger 1998                                              | 1    | 1.92               | (1,5,5,3,3,5,31)            |
|                          |                     | Mn            |                                                                    | air                    | unspecified    |                 |          | Manganese                                             | 1.13E-05   | kg                       | Künniger 1998                                              | 1    | 5.42               | (1,5,5,3,3,5,22)            |
|                          |                     | Mo            |                                                                    | air                    | unspecified    |                 |          | Molybdenum                                            | 2.16E-06   | kg                       | Künniger 1998                                              | 1    | 5.42               | (1,5,5,3,3,5,22)            |
|                          |                     | Ni            |                                                                    | air                    | unspecified    |                 |          | Nickel                                                | 1.13E-05   | kg                       | Künniger 1998                                              | 1    | 5.42               | (1,5,5,3,3,5,22)            |
|                          |                     | Zn            |                                                                    | air                    | unspecified    |                 |          | Zinc                                                  | 4.20E-07   | kg                       | Künniger 1998                                              | 1    | 5.42               | (1,5,5,3,3,5,22)            |
|                          |                     | partikel <2.5 | sum of oxides of the metal emissions and the fluoride above        | air                    | unspecified    |                 |          | Particulates, < 2.5 um                                | 1.41E-04   | kg                       | calculated                                                 | 1    | 3.41               | (4,5,5,3,3,5,20)            |
|                          |                     | CO            |                                                                    | air                    | unspecified    |                 |          | Carbon monoxide, fossil                               | 6.67E-05   | kg                       | Künniger 1998                                              | 1    | 5.42               | (1,5,5,3,3,5,17)            |
|                          |                     | NOx           |                                                                    | air                    | unspecified    |                 |          | Nitrogen oxides                                       | 2.00E-06   | kg                       | Künniger 1998                                              | 1    | 1.92               | (1,5,5,3,3,5,16)            |
|                          |                     | O3            |                                                                    | air                    | unspecified    |                 |          | Ozone                                                 | 9.06E-06   | kg                       | Künniger 1998                                              | 1    | 1.92               | (1,5,5,3,3,5,31)            |
|                          |                     | Waste Heat    |                                                                    | air                    | unspecified    |                 |          | Heat, waste                                           | 9.90E-02   | MJ                       | calculated                                                 | 1    | 1.32               | (3,5,1,3,1,5,13)            |
|                          |                     | MAG welding   |                                                                    | metals                 | processing     | No              | RER      | welding, arc, steel                                   | 1.00E+00   | m                        |                                                            |      |                    |                             |

Fig. 6.3 Flows for "welding, gas, steel" and their representation in the ecoinvent database

### 6.3 Arc welding of steel

The following description of the welding processes is taken from Brandt & Weiler (1997). A good overview of the arc welding process in German language is given in Nies (1992).

In arc welding the heat source is the electric arc between the welding electrode and the workpiece. The arc may be exposed, but is then nearly always surrounded by a gas shield (see below), or it may be covered by a layer of powder (submerged arc welding). In both cases, access of air (oxygen, nitrogen, and water vapour) is restricted.

Protective gases are either inert (almost exclusively argon), reactive (carbon dioxide), or mixed gases with a high proportion of inert gas, e.g., argon, and 1–9 vol.% oxygen or 18 vol.% carbon dioxide. These not only have a protective effect, but in some cases a metallurgical effect also.

Welding powders used in the submerged arc process (DIN 8557, Part 1: 04.81 and EN 756: 1992) stabilise the arc and reduce the rate of heat loss and also have a protective effect. Metallurgical effects include deoxidisation, prevention or replacement of burn-off losses, and sometimes addition of alloying elements. The powders consist mainly of oxides of silicon, manganese, calcium, barium, magnesium, titanium, sodium, potassium, and iron, and also some fluorides, e.g., of calcium and sodium/aluminium. The many types of powder can be classified according to their principal components (manganese silicate type, calcium silicate type, basic aluminate type, etc.).

The electrical energy is fed to the arc from controllable and variable sources of electric current, e.g., transformers, rectifiers, or converters. Arc welding processes operate at a higher heat flow rate than gas welding, so that higher welding and melting rates can be achieved and the heat-affected zones are less extensive.

There are several different types of arc welding processes (see also Fig. 6.1).

#### Manual Arc Welding

In manual arc welding with coated electrodes, the arc is formed between the melting end of the electrode and the workpiece, causing the coating to melt and partially vaporise. The weld pool and the molten droplets during transfer to the workpiece are protected from the surrounding air by the gases produced from the coating.

The coating also performs a number of other functions:

- It facilitates metallurgical reactions that clean the weld pool
- It stabilises the arc
- It compensates for the loss of alloying elements from the cored wire and the parent metal
- It can introduce seed nuclei into the weld pool and hence grain refine the weld metal
- It can control the basicity or acidity of the melt
- It can alloy with the material of the welded joint and hence modify its chemical composition

Cored wire is usually used to achieve a composition that matches that of the alloy being welded as closely as possible. The special requirements of manual electrodes are met with the aid of the coating used. The coatings on manual welding electrodes have compositions similar to those of submerged arc welding powders, and usually contain silicon, manganese, calcium, titanium, magnesium, aluminum, potassium, and sodium, as well as gas formers such as cellulose for certain applications.

Two important types of steel electrode are:

- The rutile (titanium dioxide) type with a high proportion of titanium dioxide in the coating, with which simple manual welding tasks can be quickly carried out

- The basic lime (calcium carbonate) type with a high proportion of calcium carbonate in the coating, which can produce welds with the best quality characteristics, especially toughness

High-performance electrodes have coatings containing a high proportion of metal in powder form. These give a higher melting rate than normal electrodes for the same electricity consumption.

### **Welding with Cored Wire**

Cored wire is a tubular metal electrode filled with material similar to that used for the coating on normal electrodes. Its particular advantage is that it can be fed from a long coil, and can be welded continuously. This gives a faster average welding speed than with manual electrode welding. During welding, the cored wire slides through a burner close to the welding area. Shielding gas can also be injected. The use of cored wires containing gas-producing substances can enable welding to be carried out without a gas shield under difficult conditions, e.g., in wind.

### **Tungsten Inert Gas (TIG) Welding**

In these processes, the arc is formed between the tip of an infusible tungsten electrode and the workpiece. The function of the electrode is only to conduct electricity and produce the arc. The filler metal is fed to the weld pool from one side, either by hand in rod form or by an automatic system as wire. Argon is usually used as the protective gas, occasionally helium or argon–helium mixture. The tungsten electrode is alloyed with up to 4% thorium dioxide to reduce the work of electron emission at the tip of the electrode and to increase the maximum current.

Welds of the highest quality can be produced by TIG welding. It is best to use direct current with the workpiece as the positive pole when welding high-alloy (chromium–nickel) steels, nickel-based alloys, titanium, zirconium, tantalum, and their alloys. Alternating current is used when welding metals that form high-melting oxides, e.g., aluminium and its alloys, as continuous reversal of the current direction continuously removes the oxide film from the weld pool.

Pulsed current arc welding is a variation of the TIG process. This is very useful for welding thin sheet metal, as it minimises the extent of the heat-affected zone and hence the distortion of the sheet. The weld pool is so small that dripping of molten metal does not occur, even when welding in difficult situations. For high-quality TIG welding, transistorised pulsed current welding robots are available that can be very precisely controlled by a variety of computer programs.

### **Plasma Welding**

With the plasma arc, the gas rate is usually much higher than with the TIG arc. Also, the temperature of the arc is increased by a factor of 2–4 and the energy density by a factor of ca. 10. This is achieved by greatly constricting a TIG arc together with the protective gas stream surrounding it with the aid of a water-cooled copper nozzle, forcing the entire gas stream to pass through the arc.

In plasma welding, the plasma jet is always surrounded by a protective gas layer. As the plasma jet usually penetrates into the sheet, the result is a key-lock effect similar to that produced by autogenic rightward welding. There is an upper limit to the weldable sheet thickness, but the welding speed is significantly higher than in other arc welding processes. The amount of heat applied per unit length of weld is small.

A very wide range of power outputs can be used in plasma welding, and the process can therefore be used for sheet thicknesses in the approximate range 0.01–10 mm. So-called microplasma welding is a low-power process using an "untransferred" arc, i.e., between the tungsten electrode and the nozzle wall. In the medium to higher power range, the plasma welding arc is "transferred," i.e., it is between the tungsten electrode and the workpiece. The usual plasma gases are argon and hydrogen.

Plasma welding is also used for resurfacing metal, which can be plated over large areas to a thickness of ca. 2–8 mm. It is mainly used in the construction of chemical equipment. There are two variations: plasma welding of powder coatings and plasma welding of wire.

In plasma welding of powder coatings, the powder is fed from a holding vessel to the plasma stream, where it melts. In the hot wire process, two wires are fed in parallel to the welding zone. Electric current from an auxiliary source is passed through the wires, heating them (because of their ohmic resistance) to a temperature just below their melting point close to the welding zone. The plasma burner oscillates and can therefore apply metal to a wide fusion zone.

### **Metal Inert Gas (MIG) and Metal Active Gas (MAG) Welding**

In MIG and MAG welding, high-power processes, the arc is produced between a wire electrode, fed from a coil, and the workpiece. This method of supplying the electric current gives a high electrode current density (ca. 100 A/mm<sup>2</sup>) and hence a high melting rate.

The wire electrode is fed to the electric arc by a motor whose speed can be varied according to the wire diameter and electric current. The current to the welding wire is fed through a tubular contact in the "welding pistol" close to the arc. Only direct current is used in the MIG and MAG process, the electrode usually being the positive pole. The metal is then transferred as small droplets and penetration is deep. If the electrode is negative, larger drops of metal are transferred and there is less penetration.

The difference between MIG and MAG lies in the gas used:

- The metal inert gas (MIG) process uses an inactive, i.e., noble gas shield
- In the metal active gas (MAG) process the protective gas is active, e.g., carbon dioxide or mixtures or argon with carbon dioxide and/or oxygen is used

The MIG process, with argon (or sometimes helium) as the protective gas, can be used to weld workpieces of thickness >3 mm made of non-ferrous metals and alloy steels. With pulsed direct current of variable magnitude and pulse frequency, the transfer of droplets can be controlled. In this pulsed arc or pulsed MIG process, each pulse of current causes a droplet of metal that has been melted in the background current phase to be thrown off. The droplet size and frequency can be influenced by the type of pulse used. The pulsed MIG process is very suitable for welding in difficult situations.

The MAG process is mainly used for welding plain carbon and low-alloy steels. As the protective gas has an oxidising effect, the filler metal must contain a high proportion of deoxidising components (e.g., manganese, silicon, aluminium). The metallurgical reaction that takes place gives increased nucleation and hence grain-refined weld metal.

### **Sources of Electric Current**

Power transistors are increasingly used for controlling the electric current. They can be used as digital switches or analog regulators. As their power losses are higher, analog regulators are used mainly for lower-power equipment. As switches (high frequency), they can be in the primary or secondary circuit, i.e., the transistors can be between the mains supply and the transformer or the transformer can be between the mains and the transistors. With the first arrangement the equipment is lightweight, but very sensitive to voltage peaks. Equipment of the second type is heavier.

### **Submerged Arc Welding**

In submerged arc welding, the arc produced between the automatically fed filler wire and the workpiece is covered by powder, and is located in a cavity formed within powder that has become molten. The powder flows in the direction of welding, preceding the welding zone of the metal sheet. Only a

small amount of powder is consumed by the welding process, and the unused powder is sucked away and reused. The powder blanket essentially performs the same function as the coating on manual electrodes. The welding speed is high, and in equipment with several filler wires the melting capacity is also high. The process can be used only in situations where the sheet metal is not steeply sloped.

### **Narrow Gap Welding**

Heavy section workpieces can be joined economically and with minimum effect on the parent metal by the so-called narrow gap welding process. The best of the various methods for filling a vertical narrow gap with weld metal is the MIG/MAG process. In this gas-shielded process, the filler wire and protective gas are cooled in a sword-shaped sheathed tube and fed to the welding point in the narrow gap. There are two systems:

- Two wires, which have been subjected to plastic deformation to give a predetermined internal stress, are guided into the two corners of the welding gap, giving a defect-free root position
- In the second system, a wire preformed to a waved shape so that it alternately fills each corner of the region to be welded inside the narrow gap performs this function.

Narrow gap welding can also be carried out by the submerged technique. While gas-shielded narrow gap welding can be used for gap widths of ca. 10 mm, the gap must be wider for submerged welding.

#### **6.3.1 Automated arc welding of steel in ecoinvent**

The module provides values per meter weld seam by MAG-welding with a mixture of 83% Ar, 13% CO<sub>2</sub> and 4% O<sub>2</sub> as protective gas. The gas demand is 16 l/min. The low alloyed steel wire is 1.2 mm in diameter and 6 m/min is used. The welding speed is 1 m/min<sup>17</sup>. Thus the consumption of wire per meter of weld seam is 0.0536 kg.

The power consumption is calculated based on the value given in Künniger & Richter (1997) and the assumption that the energy input corresponds linearly to the amount of wire used. The emissions (except for the particulate emission that is calculated as the sum of all the metal emission) are given in Künniger & Richter (1998).

The transports of the steel feed and gas is inventoried according to the quality guidelines (see Frischknecht et al. (2003))

For the source of power (infrastructure) the unspecific electronic device is used as proxy. A value of 1E-5 kg/m is assumed (this value corresponds to the life time production of 10 km of weld seam per kg power transistor).

The background information to the process "welding, arc, steel" is given in Tab. 6.2, flows are shown in Fig. 6.4.

### **Data quality**

The uncertainty is estimated by completing the pedigree matrix, according to (Frischknecht et al. (2003)). A lognormal distribution is assumed.

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<sup>17</sup> personal communication, G. Willhelm, EMPA Dübendorf, 08.05.03

Tab. 6.2 Ecoinvent meta information for "welding, arc, steel"

|                                       |                                                                                                                                                                                              |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>                           | <b>welding, arc, steel</b>                                                                                                                                                                   |
| Location                              | RER                                                                                                                                                                                          |
| Infrastructure Process                | 0                                                                                                                                                                                            |
| Unit                                  | m                                                                                                                                                                                            |
| Data Set Version                      | 2.0                                                                                                                                                                                          |
| Included Processes                    | Includes the MAG-welding of non alloyed steel and the transports of the filler rod and the protective gas to the place of use.                                                               |
| Amount                                | 1                                                                                                                                                                                            |
| Local Name                            | Schweissen, Lichtbogen, Stahl                                                                                                                                                                |
| Synonyms                              | MAG/MIG                                                                                                                                                                                      |
| General Comment to reference function | Different arc welding technologies can have very different emissions. Only to be used for the technology describes or as a rough proxy if welding is not important in the system considered. |
| Start Date                            | 1997                                                                                                                                                                                         |
| End Date                              | 2002                                                                                                                                                                                         |
| Data Valid For Entire Period          | 1                                                                                                                                                                                            |
| Other Period Text                     |                                                                                                                                                                                              |
| Geography text                        | Data-set is representative for Europe                                                                                                                                                        |
| Technology text                       | MAG-welding of unalloyed steel with 83% Ar, 13% CO <sub>2</sub> and 4% O <sub>2</sub> protective gas and a wire consumption of 0.0536 kg/m.                                                  |
| Representativeness [%]                |                                                                                                                                                                                              |
| Production Volume                     | No data available                                                                                                                                                                            |
| Sampling Procedure                    | Literature                                                                                                                                                                                   |
| Extrapolations                        | none                                                                                                                                                                                         |
| Uncertainty Adjustments               | none                                                                                                                                                                                         |



## 6. Welding

| General Flow Information |                     |                 | Representation in ecoinvent                                                           |                   |              |                 |          |                                   |            | Uncertainty Informations |                        |      |                    |                             |  |
|--------------------------|---------------------|-----------------|---------------------------------------------------------------------------------------|-------------------|--------------|-----------------|----------|-----------------------------------|------------|--------------------------|------------------------|------|--------------------|-----------------------------|--|
| Input                    | Process Name        | Output          | Remarks                                                                               | Category          | Sub-category | Infra-structure | Location | Modul name in ecoinvent           | Mean value | Unit                     | Source for mean value  | Type | Standard Deviation | General Comment             |  |
| Steel                    | welding, gas, steel |                 |                                                                                       | metals            | extraction   | No              | RER      | steel, low -alloyed, at plant     | 5.36E-02   | kg                       | calculated             | 1    | 1.38               | (4,5,1,1,1,5,4)             |  |
| acetylen                 |                     |                 |                                                                                       | chemicals         | organics     | No              | CH       | acetylene, at regional storehouse | 9.76E-03   | kg                       | estimated / calculated | 1    | 1.68               | (4,5,1,1,4,5,1)             |  |
| oxygen                   |                     |                 |                                                                                       | chemicals         | inorganics   | No              | RER      | oxygen, liquid, at plant          | 1.14E-02   | kg                       | calculated             | 1    | 1.68               | (4,5,1,1,4,5,4)             |  |
| transport rail           |                     |                 | steel 600 km, acetylene 600 km, oxygen 100 km. Gases twice the weight for the bottles | transport systems | train        | No              | RER      | transport, freight, rail          | 4.62E-02   | tkm                      | estimated              | 1    | 2.09               | (4,5,n.A.,n.A.,n.A.,n.A.,5) |  |
| transport lorry          |                     |                 | steel, acetylene and oxygen 50 km. Gases twice the weight for the bottles             | transport systems | road         | No              | RER      | transport, lorry 32t              | 4.80E-03   | tkm                      | estimated              | 1    | 2.09               | (4,5,n.A.,n.A.,n.A.,n.A.,5) |  |
| Infrastructure           |                     |                 | neglected                                                                             |                   |              |                 |          |                                   | 0.00E+00   |                          |                        |      |                    |                             |  |
|                          |                     | Cr (II, III)    |                                                                                       | air               | unspecified  |                 |          | Chromium                          | 2.81E-05   | kg                       | Künniger 1998          | 1    | 5.42               | (1,5,5,3,3,5,22)            |  |
|                          |                     | Cr (VI)         |                                                                                       | air               | unspecified  |                 |          | Chromium VI                       | 3.00E-08   | kg                       | Künniger 1998          | 1    | 5.42               | (1,5,5,3,3,5,22)            |  |
|                          |                     | Cu              |                                                                                       | air               | unspecified  |                 |          | Copper                            | 2.10E-07   | kg                       | Künniger 1998          | 1    | 1.92               | (1,5,5,3,3,5,31)            |  |
|                          |                     | Fe              |                                                                                       | air               | unspecified  |                 |          | Iron                              | 8.74E-05   | kg                       | Künniger 1998          | 1    | 1.92               | (1,5,5,3,3,5,31)            |  |
|                          |                     | Mn              |                                                                                       | air               | unspecified  |                 |          | Manganese                         | 1.13E-05   | kg                       | Künniger 1998          | 1    | 5.42               | (1,5,5,3,3,5,22)            |  |
|                          |                     | Mo              |                                                                                       | air               | unspecified  |                 |          | Molybdenum                        | 2.16E-06   | kg                       | Künniger 1998          | 1    | 5.42               | (1,5,5,3,3,5,22)            |  |
|                          |                     | Ni              |                                                                                       | air               | unspecified  |                 |          | Nickel                            | 1.13E-05   | kg                       | Künniger 1998          | 1    | 5.42               | (1,5,5,3,3,5,22)            |  |
|                          |                     | Zn              |                                                                                       | air               | unspecified  |                 |          | Zinc                              | 4.20E-07   | kg                       | Künniger 1998          | 1    | 5.42               | (1,5,5,3,3,5,22)            |  |
|                          |                     | partikel <2.5   |                                                                                       | air               | unspecified  |                 |          | Particulates, < 2.5 um            | 1.41E-04   | kg                       | calculated             | 1    | 3.41               | (4,5,5,3,3,5,20)            |  |
|                          |                     | CO2             | sum of oxides of the metal emissions and the fluoride above                           | air               | unspecified  |                 |          | Carbon dioxide, fossil            | 3.24E-02   | kg                       | calculated             | 1    | 1.73               | (4,5,5,3,3,5,14)            |  |
|                          |                     | CO              |                                                                                       | air               | unspecified  |                 |          | Carbon monoxide, fossil           | 2.03E-04   | kg                       | calculated             | 1    | 5.47               | (4,5,5,3,3,5,17)            |  |
|                          |                     | NOx             |                                                                                       | air               | unspecified  |                 |          | Nitrogen oxides                   | 2.00E-06   | kg                       | Künniger 1998          | 1    | 1.92               | (1,5,5,3,3,5,16)            |  |
|                          |                     | O3              |                                                                                       | air               | unspecified  |                 |          | Ozone                             | 9.06E-06   | kg                       | Künniger 1998          | 1    | 1.92               | (1,5,5,3,3,5,31)            |  |
|                          |                     | acetylen to air |                                                                                       | air               | unspecified  |                 |          | Ethyne                            | 9.76E-05   | kg                       | estimated              | 1    | 3.22               | (5,5,5,5,5,23)              |  |
|                          |                     | waste heat      |                                                                                       | air               | unspecified  |                 |          | Heat, waste                       | 5.00E-01   | MJ                       | calculated             | 1    | 1.68               | (4,5,1,1,4,5,13)            |  |
|                          |                     | Gas welding     |                                                                                       | metals            | processing   | No              | RER      | welding, gas, steel               | 1.00E+00   | m                        |                        |      |                    |                             |  |

Fig. 6.4 Flows for "welding, arc, steel" and their representation in the ecoinvent database

## 6.4 Arc welding of aluminium

This chapter is mainly based on Brandt & Weiler (1997):

The main problem in welding of aluminium is dealing with the film of aluminium oxide scale which adheres extremely strongly and has a very high melting point. As the thermal conductivity of aluminium metal is also very high, there must be an ample heat supply so the weld pool does not solidify too rapidly or contain nonmetallic particles or gas-filled pores.

The simplest process, welding with manual electrodes, is widely used and is completely satisfactory for simple joints. Qualitatively good welds can be produced by the TIG and MIG processes. The preferred protective gas is helium. The alternating current removes the layer of aluminium oxide. If the weld is carefully prepared and the auxiliary materials have been correctly chosen (surface cleanliness), mechanically sound low-porosity welded joints are obtained. Recent investigations have indicated that high-power processes such as submerged arc welding are also suitable.

Owing to the low melting point of aluminium, the low viscosity of the molten metal, and the small difference in density between the metal and its oxide, manual welding of aluminium and its alloys needs highly skilled welders.

### 6.4.1 Arc welding of aluminium in ecoinvent

Because no quantitative information on aluminium arc welding could be obtained, the following assumptions are made:

The volume of filler rod used is estimated to be the same as for MAG welding of steel, i.e.  $6.79 \times 10^{-6} \text{ m}^3$ . Thus 0.0183 kg/m weld seam of filler rod is used. The filler rod is assumed to be composed of 90% Al and 10% Si.

Energy consumption is calculated from the current and the voltage commonly used for aluminium MIG welding<sup>18</sup>. With 150 A and 15V a power of 2250 W are obtained. Assuming a welding speed of 0.5 m/min the energy consumption is calculated to 75 Wh/m weld seam.

Helium is assumed to be used as protective gas. A flow of 10 l/min is assumed. Thus 20 l/m weld seam helium is consumed.

The total particulate matter emissions are assumed to be as high as for MAG welding of steel. The composition however is different. Zn, Al and Si are thought to be dominant. We assume 60% Al, 30% Zn and 10% Si.

The background information to the process "welding, arc, aluminium" is given in Tab. 6.3, flows are shown in Fig. 6.5.

### Data quality

The uncertainty is estimated by completing the pedigree matrix, according to (Frischknecht et al. (2003)). A lognormal distribution is assumed. The data are based on rough estimations and of poor overall quality.

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<sup>18</sup> <http://www.brazing.com/techguide/procedures/aluminium.asp>, accessed 17.03.03

Tab. 6.3 Ecoinvent meta information for "welding, arc, aluminium"

|                                       |                                                                                                                        |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Name</b>                           | <b>welding, arc, aluminium</b>                                                                                         |
| Location                              | RER                                                                                                                    |
| Infrastructure Process                | 0                                                                                                                      |
| Unit                                  | m                                                                                                                      |
| Data Set Version                      | 1                                                                                                                      |
| Included Processes                    | Includes the arc welding of aluminium and the transports of the filler rod and the protective gas to the place of use. |
| Amount                                | 1                                                                                                                      |
| Local Name                            | Schweissen, Lichtbogen, Aluminium                                                                                      |
| Synonyms                              | MAG                                                                                                                    |
| General Comment to reference function | Based on rather rough estimates. Not to be used if welding is of importance in the system considered.                  |
| Start Date                            | 1997                                                                                                                   |
| End Date                              | 2002                                                                                                                   |
| Data Valid For Entire Period          | 1                                                                                                                      |
| Other Period Text                     |                                                                                                                        |
| Geography text                        | Data-set is representative for Europe                                                                                  |
| Technology text                       | MIG Welding of aluminium with helium as protective gas.                                                                |
| Representativeness [%]                |                                                                                                                        |
| Production Volume                     | No data available                                                                                                      |
| Sampling Procedure                    | Literature                                                                                                             |
| Extrapolations                        | none                                                                                                                   |
| Uncertainty Adjustments               | none                                                                                                                   |

## 6. Welding

| General Flow Information |                         |                   | Representation in ecoinvent |                        |                |                 |          |                                                       |            | Uncertainty Informations |                       |      |                    |                             |  |
|--------------------------|-------------------------|-------------------|-----------------------------|------------------------|----------------|-----------------|----------|-------------------------------------------------------|------------|--------------------------|-----------------------|------|--------------------|-----------------------------|--|
| Input                    | Process Name            | Output            | Remarks                     | Category               | Sub-category   | Infra-structure | Location | Modul name in ecoinvent                               | Mean value | Unit                     | Source for mean value | Type | Standard Deviation | General Comment             |  |
| Aluminium                | welding, arc, aluminium |                   |                             | metals                 | extraction     | No              | RER      | aluminium, production mix, at plant                   | 1.65E-02   | kg                       | estimate              | 1    | 1.93               | (5,5,3,3,5,3)               |  |
| Silizium                 |                         |                   |                             | metals                 | extraction     | No              | RER      | aluminium, production mix, at plant                   | 1.83E-03   | kg                       | estimate              | 1    | 1.93               | (5,5,3,3,5,3)               |  |
| Helium                   |                         |                   |                             | chemicals              | inorganics     | No              | RER      | argon, crude, liquid, at plant                        | 3.57E-03   | kg                       | estimate              | 1    | 1.93               | (5,5,3,3,5,3)               |  |
| Electricity              |                         |                   |                             | electricity            | production mix | No              | UCTE     | electricity, medium voltage, production UCTE, at grid | 7.50E-02   | kWh                      | estimate              | 1    | 1.93               | (5,5,3,3,5,2)               |  |
| transport rail           |                         |                   |                             | transport systems      | train          | No              | RER      | transport, freight, rail                              | 1.53E-02   | tkm                      | estimate              | 1    | 2.09               | (4,5,n.A.,n.A.,n.A.,n.A.,5) |  |
| transport lorry          |                         |                   |                             | transport systems      | road           | No              | RER      | transport, lorry 32t                                  | 1.27E-03   | tkm                      | estimate              | 1    | 2.09               | (4,5,n.A.,n.A.,n.A.,n.A.,5) |  |
| Infrastructure           |                         |                   |                             | construction processes | buildings      | No              | RER      | electronics for control units                         | 1.00E-05   | kg                       | estimate              | 1    | 3.59               | (5,5,3,3,5,9)               |  |
|                          |                         | He                |                             | air                    | unspecified    |                 |          | Helium                                                | 3.57E-06   | kg                       | estimate              | 1    | 2.16               | (5,5,3,3,5,31)              |  |
|                          |                         | Al                |                             | air                    | unspecified    |                 |          | Aluminum                                              | 8.45E-05   | kg                       | estimate              | 1    | 2.16               | (5,5,3,3,5,31)              |  |
|                          |                         | Zn                |                             | air                    | unspecified    |                 |          | Zinc                                                  | 4.23E-05   | kg                       | estimate              | 1    | 2.16               | (5,5,3,3,5,31)              |  |
|                          |                         | Si                |                             | air                    | unspecified    |                 |          | Silicon                                               | 1.41E-05   | kg                       | estimate              | 1    | 2.16               | (5,5,3,3,5,31)              |  |
|                          |                         | Particulates      |                             | air                    | unspecified    |                 |          | Particulates, < 2.5 um                                | 1.41E-04   | kg                       | estimate              | 1    | 3.59               | (5,5,3,3,5,27)              |  |
|                          |                         | CO                |                             | air                    | unspecified    |                 |          | Carbon monoxide, fossil                               | 6.67E-05   | kg                       | estimate              | 1    | 2.16               | (5,5,3,3,5,31)              |  |
|                          |                         | NOx               |                             | air                    | unspecified    |                 |          | Nitrogen oxides                                       | 2.00E-06   | kg                       | estimate              | 1    | 2.16               | (5,5,3,3,5,31)              |  |
|                          |                         | O3                |                             | air                    | unspecified    |                 |          | Ozone                                                 | 9.06E-06   | kg                       | estimate              | 1    | 2.16               | (5,5,3,3,5,31)              |  |
|                          |                         | Waste Heat        |                             | air                    | unspecified    |                 |          | Heat, waste                                           | 2.70E-01   | MJ                       | estimate              | 1    | 1.93               | (5,5,3,3,5,13)              |  |
|                          |                         | Welding Aluminium |                             | metals                 | processing     | No              | RER      | w elding, arc, aluminium                              | 1.00E+00   | m                        |                       |      |                    |                             |  |

Fig. 6.5 Flows for "welding, arc, aluminium" and their representation in the ecoinvent database

## 7 Enamelling

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Review: Mischa Classen, Hans-Jörg Althaus, Rolf Frischknecht

### Acknowledgement

We thank Gabriela Lowitz for the information provided in a personal communication.

### Summary

Enamel is a glass of a particular chemical composition and physical nature determined for the surface protection of metal. The major interest for the enamelling process in this database is the application for hot water boilers. The inventory has been elaborated for the European situation. The basic inventory is split up into the major process stages degreasing, corroding, cleaning and firing as well as an inventory for the materials used for enamelling. The life cycle inventory is based on the data investigated in a study for the new single layer enamelling process. General assumptions are used for transport distances. The inventory of enamelling can be considered as a basis for the assessment of this process in connection with technical applications.

### 7.1 Introduction

Enamel is a glass of a particular chemical composition and physical nature determined for the surface protection of metal. This results in a composite material of glass and metal, which combines positive qualities of both materials. The process of firing this glass on metal is called enamelling and thus the object, which has had its surface protected and improved, is referred to as enamelled (Wendel-Email (2002)). Some of the early vitreous enamelled relics date back to the 13th Century BC (IVE (2002)).

The word enamel comes from the High German word “smelzan” and later from the Old French “es-mail”. In American English it is referred to as Porcelain Enamel. It should not be confused with paint, which is sometimes called ‘enamel’. Paints cannot be enamel. They do not have the hardness, heat resistance and colour stability that is only available with real vitreous enamel (IVE (2002)).

### 7.2 Resources of Enamel

First of all an enamel frit is melted from natural and synthetic materials in furnaces. Silica sand (40%-49%), boric anhydride (15%-17.5%) and sodium monoxide (15%-17%) are the major ingredients of enamel frits (Lowitz (1998)). After that, it is ground to become enamel slurry or powder enamel. Depending on the nature and size of the object to be enamelled, the metal pieces get a wet or a dry application of enamel. This is done by dipping, spraying, pouring, flooding or powdering. Afterwards the objects are fired at temperatures of between 550°C and 1,000°C.

### 7.3 Characterisation of the Product

The smooth glass-like surface is hard; it is scratch, chemical and fire resistant. It is easy to clean and hygienic (IVE (2002)).

### 7.4 Application of Enamels

One can recognise enamel as the material used to produce the now highly collectable advertising signs produced during the early 20th Century. The “Hovis” and “Virol” signs were part of the everyday street scene. Cook stoves will almost certainly have a vitreous enamelled oven and the higher quality

cookers will use it on the outer parts. A cast iron or steel bath may also have been vitreous enamelled. Less obvious are the storage silos on farms, usually blue or green (IVE (2002)).

Enamel is become more used as a protective coating for hot water tanks, like solar boilers and water heaters, since the availability of appropriate powder application equipment for such products (Ditmer (2002)).

Due to its mechanical, thermal, optical and chemical properties, enamel offers a good surface protection for any enamelable metal. Not only can metals be protected against corrosion by being enamelled, but they can also be made more interesting by the use of colour (Wendel-Email (2002)). Vitreous enamel can be applied to most metals. For jewellery and decorative items it is often applied to gold, silver, copper and bronze. For the more common uses, it is applied to steel or cast iron. There are some specialised uses on stainless steel and aluminium (IVE (2002)).

## 7.5 System Characterisation

Fig. 7.1 shows a typical layout of a modern powder enamelling plant for hot water tanks. This plant has a production capacity of circa 100 -120 products / hour and consists of the following equipment :

- Shot blaster
- Powder enamel system
- Automatic transfer device
- Enamelling furnace
- Cooling tunnel

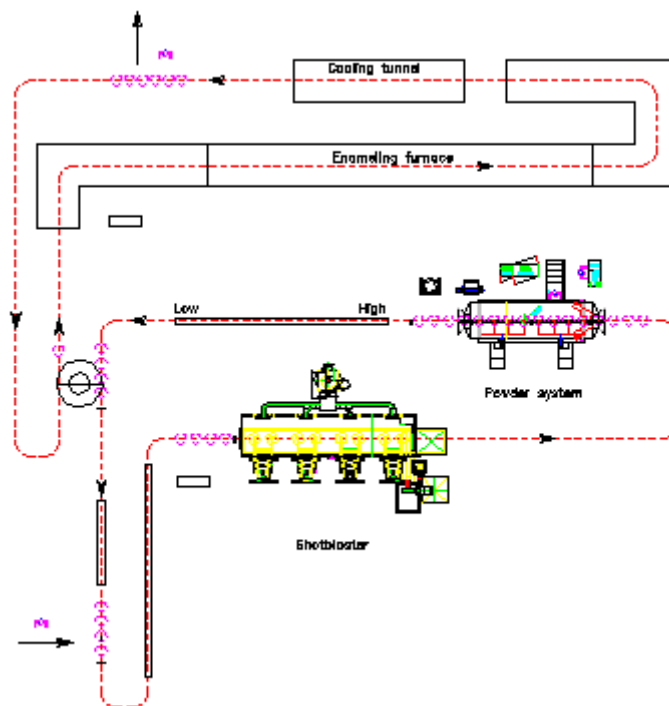


Fig. 7.1 Layout of a powder enamelling plant for hot water tanks (Ditmer (2002)).

The boilers are loaded onto a conveyor near the testing station after the welding department and then they are transported to the shot blaster. They remain stationary in front of the sliding doors until this machine is ready for operation. The doors will open automatically at the end of each cycle and the boilers will move to the blasting positions where they are clamped into a centring device. During the

first stage the boilers will be centred and rotated, while a rigid multi nozzle lance enters the product to blast the interior. Remaining shot medium will be washed away by means of compressed air during the next stage. At the end of this cycle boilers will be moved to the automatic powder application system, which also comprises of two stages (Ditmer (2002)).

At the first stage the boilers will be centred while the interior is sprayed with powder enamel by means of extended automatic guns mounted on a PLC controlled reciprocator. At the second stage, the spraying operation will be repeated in order to obtain the required enamel layer thickness. An automatic transfer unit picks the boilers one-by-one from the pre-treatment / application conveyor and places them carefully on the furnace conveyor (Ditmer (2002)). Fig. 7.2 shows a picture of an enamelling plant for hot water tanks.

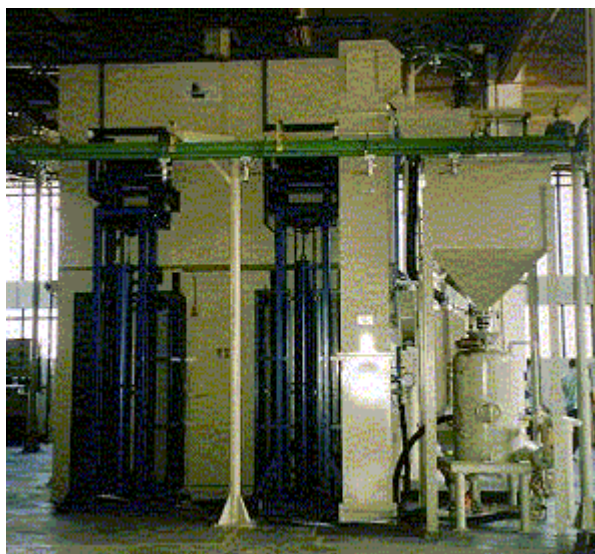


Fig. 7.2 Picture of an enamelling plant for hot water tanks (Ditmer (2002)).

## 7.6 Life Cycle Inventory of Enamelling

The environmental impacts of enamelling processes have been investigated by Lowitz (1998), right side of Tab. 7.1). Three different processes were investigated in this Ph.d. thesis per  $\text{m}^2$  of enamelled steel sheet:

- At the time of investigation the major process used two stages with a basic and an upper enamel coating which is fired in two production stages. The enamel use is together  $710\text{g}/\text{m}^2$ .
- An improved process uses also two layers which are fired in one process. The enamel use is together  $390\text{g}/\text{m}^2$ .
- A new production process uses only one enamel layer and one firing process. This powder coating technology is used in the meantime by many manufacturers. The enamel use is  $600\text{g}/\text{m}^2$ .

The life cycle inventory is based on the data provided by Lowitz (1998) for the new single layer enamelling process. According to an actual statement of the author this is the most important process in the mean time. They have been investigated from 1993-1998 in Germany. Further information about the type of steel sheet were not available.

Tab. 7.1 shows the unit process raw data and the data investigated by Lowitz (1998). The major interest for the enamelling process in this database is the application for hot water boilers. An important share of enamelled boilers is imported to Switzerland from different countries. Thus the inventory has been elaborated for the European situation.

The basic inventory is split up into the major process stages degreasing, corroding, nickel-plating, cleaning and firing as well as an inventory for the materials used for enamelling. According to the information provided by an expert<sup>19</sup> is the process of nickel-plating not used for the enamelling of boilers (which are the major product of interest). It has thus not been considered for the inventory.

Gas, oil and electricity are used as energy carriers by different manufacturers, but the share of these energy carriers in today plants was not known. Lowitz (1998) assumed that all energy is delivered by electricity which was at this time the dominating energy used. The energy for the pre-treatment processes was not split up to the different stages. For some special chemicals there are no background data available. Their use has been assessed with the generic data for organic or inorganic chemicals. Transports of materials are calculated with the standard transport distances. Data for some process specific emissions are also shown in the study of Lowitz (1998). They are considered for the unit process raw data.

Data for the infrastructure and land use of such facilities were not available. They have been roughly assumed with data for a metal coating plant. More Information about threshold limits for effluents and air emissions in Germany can be found in dev (1999). But, it was not possible to relate these information to the functional unit investigated in this inventory. Further background information for the enamelling process is given by Petzold (Petzold & Pöschmann (1992)).

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<sup>19</sup> Personal communication with E. Voß, Wendel-Email, 14.6.2002. Corroding or sandblasting are used as pre-treatment processes.



Tab. 7.1 Unit process raw data and data quality indicators for the enamelling process (Lowitz (1998))

|                           | Name                                                                   | Location | Infrastruct | Unit    | enamelling | Uncertain<br>Standard<br>Deviation<br>95% | GeneralComment                                                                         | degreasing | corroding | nickel-plating<br>(not<br>considered) | cleaning | enamel<br>materials | firing  |
|---------------------------|------------------------------------------------------------------------|----------|-------------|---------|------------|-------------------------------------------|----------------------------------------------------------------------------------------|------------|-----------|---------------------------------------|----------|---------------------|---------|
|                           | Location                                                               |          |             |         | RER        |                                           |                                                                                        | DE         | DE        | DE                                    | DE       | DE                  | DE      |
|                           | InfrastructureProcess<br>Unit                                          |          |             |         | 0<br>m2    |                                           |                                                                                        | 0<br>m2    | 0<br>m2   | 0<br>m2                               | 0<br>m2  | 0<br>m2             | 0<br>m2 |
| output                    | enamelling                                                             |          | RER         | 0 m2    | 1.00E+0    |                                           |                                                                                        |            |           |                                       |          |                     |         |
| technosphere              | electricity, medium voltage, production UCTE, at grid                  | UCTE     | 0 kWh       | 1.44E+1 | 1          | 1.35                                      | (3,4,2,3,3,5); Other energy carriers possible. Some manufacturers use also gas or oil. | 2.57E+0    | 2.57E+0   | 2.57E+0                               | 2.57E+0  | 1.17E+0             | 5.56E+0 |
|                           | aluminium oxide, at plant                                              | RER      | 0 kg        | 4.60E-2 | 1          | 1.26                                      | (3,4,2,3,1,5); Literature                                                              |            |           |                                       |          | 4.60E-2             |         |
|                           | Borax, anhydrous, powder, at plant                                     | RER      | 0 kg        | 2.89E-1 | 1          | 1.26                                      | (3,4,2,3,1,5); Literature                                                              |            |           |                                       |          | 2.89E-1             |         |
|                           | boric acid, anhydrous, powder, at plant                                | RER      | 0 kg        | -       | 1          | 1.26                                      | (3,4,2,3,1,5); Literature                                                              |            |           | 1.00E-2                               |          |                     |         |
|                           | chemicals inorganic, at plant                                          | GLO      | 0 kg        | 4.02E-2 | 1          | 1.35                                      | (3,4,2,3,3,5); Literature, special chemicals (corroding agent, nickel sulphate)        |            | 2.20E-2   | 3.00E-2                               |          | 1.82E-2             |         |
|                           | chemicals organic, at plant                                            | GLO      | 0 kg        | 1.80E-2 | 1          | 1.35                                      | (3,4,2,3,3,5); Literature, Organic adsorbers                                           |            |           |                                       |          | 1.80E-2             |         |
|                           | limestone, milled, packed, at plant                                    | CH       | 0 kg        | 3.43E-2 | 1          | 1.26                                      | (3,4,2,3,1,5); Literature                                                              |            |           |                                       |          | 3.43E-2             |         |
|                           | potassium nitrate, as K2O, at regional storehouse                      | RER      | 0 kg        | 2.70E-2 | 1          | 1.26                                      | (3,4,2,3,1,5); Literature                                                              |            |           |                                       |          | 2.70E-2             |         |
|                           | silica sand, at plant                                                  | CH       | 0 kg        | 3.05E-1 | 1          | 1.26                                      | (3,4,2,3,1,5); Literature                                                              |            |           |                                       |          | 3.05E-1             |         |
|                           | soap, at plant                                                         | RER      | 0 kg        | 5.00E-3 | 1          | 1.26                                      | (3,4,2,3,1,5); Literature, Cleaning agents                                             | 5.00E-3    |           |                                       |          |                     |         |
|                           | sulphuric acid, liquid, at plant                                       | RER      | 0 kg        | 3.00E-2 | 1          | 1.26                                      | (3,4,2,3,1,5); Literature                                                              |            | 3.00E-2   |                                       |          |                     |         |
|                           | metal coating plant                                                    | RER      | 1 unit      | 3.33E-7 | 1          | 3.95                                      | (5,5,1,3,5,na); Rough estimation                                                       |            |           |                                       |          |                     |         |
|                           | tap water, at user                                                     | RER      | 0 kg        | 7.30E+1 | 1          | 1.26                                      | (3,4,2,3,1,5); Literature                                                              | 2.00E+0    | 1.10E+1   | 2.00E+0                               | 6.00E+1  | 6.00E-4             |         |
|                           | transport, lorry 32t                                                   | RER      | 0 tkm       | 7.95E-2 | 1          | 2.09                                      | (4,5,na,na,na,na); Standard distance 100km                                             | 5.00E-4    | 5.20E-3   | 4.00E-3                               | -        | 7.38E-2             | -       |
|                           | transport, freight, rail                                               | RER      | 0 tkm       | 4.77E-1 | 1          | 2.09                                      | (4,5,na,na,na,na); Standard distance 600km                                             | 3.00E-3    | 3.12E-2   | 2.40E-2                               | -        | 4.43E-1             | -       |
|                           | treatment, sewage, from residence, to wastewater treatment, class 2    | CH       | 0 m3        | 7.30E-2 | 1          | 1.60                                      | (3,4,2,3,4,5); Specific emissions not known                                            | 2.00E-3    | 1.10E-2   | 2.00E-3                               | 6.00E-2  | 6.00E-7             | -       |
|                           | disposal, separator sludge, 90% water, to hazardous waste incineration | CH       | 0 kg        | 9.80E-2 | 1          | 1.26                                      | (3,4,2,3,1,5); Literature, Corroding bast                                              |            | 5.00E-2   |                                       |          | 1.50E-2             | 3.30E-2 |
| emission air, unspecified | Heat, waste                                                            | -        | - MJ        | 1.44E+1 | 1          | 1.26                                      | (3,4,2,3,1,5); Literature                                                              | 2.57E+0    | 2.57E+0   | 2.57E+0                               | 2.57E+0  | 1.17E+0             | 5.56E+0 |
|                           | Particulates, > 2.5 um, and < 10um                                     | -        | - kg        | 1.20E-2 | 1          | 2.11                                      | (4,4,2,3,1,5); Calculation, emission from combustion.                                  |            |           |                                       |          |                     | 1.20E-2 |
|                           | Sulfur dioxide                                                         | -        | - kg        | 2.00E-3 | 1          | 1.63                                      | (4,4,2,3,1,5); Literature                                                              |            | 2.00E-3   |                                       |          |                     |         |
|                           | Iron                                                                   | -        | - kg        | -       | 2          | 1.63                                      | (4,4,2,3,1,5); Calculation (not considered)                                            |            |           | 1.60E-2                               |          |                     |         |
|                           | Nickel                                                                 | -        | - kg        | -       | 3          | 1.63                                      | (4,4,2,3,1,5); Calculation (not considered)                                            |            |           | 3.00E-3                               |          |                     |         |

## 7.7 Data Quality Considerations

Tab. 7.1 shows the data quality indicators for the inventory of the enamelling process. The simplified approach with a pedigree matrix has been used for calculating the standard deviation. The inventory is based on an estimate for the process at a time when it was new and not in common use. Nowadays it became the standard procedure. Large uncertainties exist for data for transport distances of materials. Other types of energy use are also possible for this process. Thus in general the data quality is not very reliable.

## 7.8 Conclusions

The inventory of enamelling can be considered as a basis for the assessment of this process in connection with technical applications. It is not meant to be used for the direct comparison of this process with alternative processes. For that, detailed information about product properties and type of application would be necessary.

## Appendices: EcoSpold Meta Information

|                    |     |                            |                                                                                                                                                                                                                                                                                                                                                    |
|--------------------|-----|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ReferenceFunction  | 401 | Name                       | enamelling                                                                                                                                                                                                                                                                                                                                         |
| Geography          | 662 | Location                   | RER                                                                                                                                                                                                                                                                                                                                                |
| ReferenceFunction  | 493 | InfrastructureProcess      | 0                                                                                                                                                                                                                                                                                                                                                  |
| ReferenceFunction  | 403 | Unit                       | m2                                                                                                                                                                                                                                                                                                                                                 |
| DataSetInformation | 201 | Type                       | 1                                                                                                                                                                                                                                                                                                                                                  |
|                    | 202 | Version                    | 1.0                                                                                                                                                                                                                                                                                                                                                |
|                    | 203 | energyValues               | 0                                                                                                                                                                                                                                                                                                                                                  |
|                    | 205 | LanguageCode               | en                                                                                                                                                                                                                                                                                                                                                 |
|                    | 206 | LocalLanguageCode          | de                                                                                                                                                                                                                                                                                                                                                 |
| DataEntryBy        | 302 | Person                     | 41                                                                                                                                                                                                                                                                                                                                                 |
|                    | 304 | QualityNetwork             | 1                                                                                                                                                                                                                                                                                                                                                  |
| ReferenceFunction  | 400 | DataSetRelatesToProduct    | 1                                                                                                                                                                                                                                                                                                                                                  |
|                    | 402 | IncludedProcesses          | The basic inventory is split up into the major process stages degreasing, corroding, cleaning and firing as well as an inventory for the materials used for enamelling. All stages of enamelling are summed up not including the metal sheets treated. Data for infrastructure and land are estimated roughly with data for a metal coating plant. |
|                    | 404 | Amount                     | 1                                                                                                                                                                                                                                                                                                                                                  |
|                    | 490 | LocalName                  | Emaillieren                                                                                                                                                                                                                                                                                                                                        |
|                    | 491 | Synonyms                   | porcelain enamel                                                                                                                                                                                                                                                                                                                                   |
|                    | 492 | GeneralComment             | Enamel is a glass of a particular chemical composition and physical nature determined for the surface protection of metal.                                                                                                                                                                                                                         |
|                    | 494 | InfrastructureIncluded     | 0                                                                                                                                                                                                                                                                                                                                                  |
|                    | 495 | Category                   | metals                                                                                                                                                                                                                                                                                                                                             |
|                    | 496 | SubCategory                | processing                                                                                                                                                                                                                                                                                                                                         |
|                    | 497 | LocalCategory              | Metalle                                                                                                                                                                                                                                                                                                                                            |
|                    | 498 | LocalSubCategory           | Verarbeitung                                                                                                                                                                                                                                                                                                                                       |
|                    | 499 | Formula                    |                                                                                                                                                                                                                                                                                                                                                    |
|                    | 501 | StatisticalClassification  |                                                                                                                                                                                                                                                                                                                                                    |
|                    | 502 | CASNumber                  |                                                                                                                                                                                                                                                                                                                                                    |
| TimePeriod         | 601 | StartDate                  | 1993                                                                                                                                                                                                                                                                                                                                               |
|                    | 602 | EndDate                    | 1998                                                                                                                                                                                                                                                                                                                                               |
|                    | 603 | DataValidForEntirePeriod   | 1                                                                                                                                                                                                                                                                                                                                                  |
|                    | 611 | OtherPeriodText            | Time of data collection.                                                                                                                                                                                                                                                                                                                           |
| Geography          | 663 | Text                       | The inventory has been elaborated for the European situation based on data investigated in Germany.                                                                                                                                                                                                                                                |
| Technology         | 692 | Text                       | Single layer enamelling process. Powder coating technology. 600g enamel per m2 of steel metal sheet. Specifications of steel not known.                                                                                                                                                                                                            |
| Representativeness | 722 | Percent                    |                                                                                                                                                                                                                                                                                                                                                    |
|                    | 724 | ProductionVolume           | Not known.                                                                                                                                                                                                                                                                                                                                         |
|                    | 725 | SamplingProcedure          | Literature review, data provided by producers.                                                                                                                                                                                                                                                                                                     |
|                    | 726 | Extrapolations             | Study for Germany used as a basis for the assessment of the European average.                                                                                                                                                                                                                                                                      |
|                    | 727 | UncertaintyAdjustments     | none                                                                                                                                                                                                                                                                                                                                               |
| DataGenerator      | 751 | Person                     | 41                                                                                                                                                                                                                                                                                                                                                 |
| AndPublication     | 756 | DataPublishedIn            | 2                                                                                                                                                                                                                                                                                                                                                  |
|                    | 757 | ReferenceToPublishedSource | 10                                                                                                                                                                                                                                                                                                                                                 |
|                    | 758 | Copyright                  | 1                                                                                                                                                                                                                                                                                                                                                  |
|                    | 759 | AccessRestrictedTo         | 0                                                                                                                                                                                                                                                                                                                                                  |
|                    | 760 | CompanyCode                |                                                                                                                                                                                                                                                                                                                                                    |
|                    | 761 | CountryCode                |                                                                                                                                                                                                                                                                                                                                                    |
|                    | 762 | PageNumbers                | Enamelling                                                                                                                                                                                                                                                                                                                                         |

## 8 Selective Coatings on Metal Sheets for Solar Absorbers

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Review: Hans-Jörg Althaus, Rolf Frischknecht

### Acknowledgement

The investigation of different selective coatings for solar absorbers took place in a difficult market. The last years have shown a high competition between different coating technologies and manufacturers. Environmental arguments played an important role in this competition and thus different manufacturers were sceptical in providing data or information for our background study. We thank Davorin Pavic und Thorsten Heinze, Interpane Solar Beschichtungs Gesellschaft, V. Handke, IZT, Berlin, Michael Grupp, Alanod Aluminium-Veredlung GmbH & Co, Dr. K. Raghunathan, IPTE, Kai Bohn, ChromeCoat A/S, Mr. Rossy and Mr. Roulet, Energie Solaire SA, CH, Billy Byrom, Aetsolar, US, Alternat Energy Technologies LLC, US and other persons of different manufacturers for the information provided in a personal communication. Thanks are also due to Paul Gantenbein, SPF Rapperswil for his comments on the manuscript.

### Summary

In order to reduce the energy loss through heat emission, solar absorbers have a spectrally, selective surface coating on the metal sheet. This chapter investigates unit process data for selective coatings made by vacuum deposition and wet processes. Information provided by different manufacturers are the basis for the gate to gate inventories per m<sup>2</sup> coated material. The inventory for the sputter or black chrome coatings on copper and stainless steel and the aluminium oxide-based layer pigmented with colloidal nickel process is based on a reliable environmental report or data provided by producers with figures for all known relevant energy and material uses and emissions. Data for the physical vapour deposition process and the Black Majic on copper are only rough estimations with known data gaps for the use of materials. The inventories of metal coating processes can be considered as a reliable basis for the assessment of this process in relation to the production of the whole solar absorbers.

A direct comparison of different selective coatings based on these data is not possible. Some environmental aspects could not be quantified for some processes. System boundaries differ for the different processes investigated. A comparison makes only sense for the amount of heat delivered by different solar absorbers including the production and dismantling of these appliances and not for the coating process alone. Thus the product properties (e.g. absorption and emittance, life time, further materials, etc.) of the solar collector have to be taken into account if a comparison for solar absorbers with different coatings is made based on the data investigated here. The analysis of solar systems shows that coating has a minor impacts in comparison to the other components of the system.

### 8.1 Introduction

Solar absorbers are usually black, as dark surfaces demonstrate a particularly high degree of solar light absorption. The level of solar absorption ( $\alpha_{\text{sol}}$ ) indicates the amount of solar radiation being absorbed that means not being reflected. As the absorber warms up to a temperature higher than the ambient temperature, it gives off a great part of the accumulated solar energy in form of long-wave heat radiation. The ratio of absorbed energy to emitted heat is indicated by the degree of emission. The loss of heat energy in the long wave length is increased by heat radiation (Solarserver (2002)).

In order to reduce energy loss through heat emission, efficient absorbers have a selective surface coating. This coating enables the conversion of a high proportion of the solar radiation into heat. The coat-

ing influences not only the solar absorption of the surface (efficiency of collecting the radiation), but also the emission (how easily the surface emits the longer wavelength infrared radiation). Commonly quoted parameters of performance for solar selective surfaces are solar absorption ( $\alpha_{\text{sol}}$ ) and thermal emittance ( $\epsilon_{\text{th}}$ ). The usual coatings show a solar absorption of over 0.9.

Galvanic selective coatings include black chrome on copper and stainless steel, black nickel, and aluminium oxide with nickel. Another type of wet process is the Black Majic coating. Vacuum deposited coatings include sputter coatings on copper, physical vapour deposition (PVD) coatings on aluminium and titanium-nitride-oxide layer. Solar paints which can be mechanically applied to the absorbers (with either brushes or sprays), are less or not at all selective, as they have a high level of emittance.

The last years show a great shift in the share of different coating technologies. With a high competition on the market, environmental arguments played an important role in the marketing strategies of some companies. Thus there was some sceptics in providing data for a report like this.

This chapter investigates different selective coating procedures for metal sheets. Only the process, but not the substrate coated are included in the inventory. The aim is to investigate background data for the environmental impacts of the coating processes. Nevertheless, as the end consumer buys a solar collector and not a coating a direct comparison of the environmental impacts per square metre doesn't make much sense. But these data are used to investigate the environmental impacts of different solar collectors including all necessary materials and processes (see Jungbluth (2003)).

## 8.2 Resources used for Selective Coatings

Only small amounts of materials (mainly metals used as substrate) and chemicals (mainly acids) are directly used for the coating process. For the final product, the solar absorber, materials like steel or copper are used in much higher quantities (Jungbluth (2003)).

## 8.3 Characterisation of Metal Coatings

Tabor (1955) introduced the use of spectrally selective surfaces for solar collectors, and several types of selective surfaces have been developed since. Several coating methods to prepare selective surfaces are used in the manufacture of solar absorber. A good overview for the processes used by different producers is given by (Remmers (1998)) and (Lampert (n.d.)). Tab. 8.1 shows the main characteristics of different processes.

Tab. 8.1 Characterisation of processes to make selective coatings

| General                          | Types                                  | Substrates                         | Coating and process materials                                                |
|----------------------------------|----------------------------------------|------------------------------------|------------------------------------------------------------------------------|
| oxidation                        | natural                                | copper, iron                       | metal oxides                                                                 |
| painting                         | spray coating, mechanical              | aluminium, galvanized steel        | black paint                                                                  |
| dry process or vacuum deposition | physical vapour deposition, sputtering | copper, aluminium                  | chromium, tin, argon, nitrogen, oxygen, titanium-nitrite-oxide, silica oxide |
| wet process                      | electroplated (galvanic) or sol-gel    | aluminium, copper, stainless steel | aluminium oxide, black chrome, tin and nickel.                               |

Oxide coatings were the first type of coatings used in solar absorbers. Metals used in early solar absorbers, such as copper and iron underwent natural oxidation, and thus reach a desirable absorption. However, as the oxidation processes occur naturally, they are difficult to control, which results in a change in the thermal emittance of the material and eventual degradation of the layer with a subsequent change of the energy efficiency of the absorber.

For low temperature heat collection, or not for year round use, the black paint is in practice a cost-effective option. Selective paint on polished aluminium or galvanic steel reaches a solar absorption of  $\alpha=0.85\sim0.91$  and a thermal emittance of  $\varepsilon=0.23\sim0.40$ . However, at high temperatures, the paint is likely to melt or burn off the surface, releasing volatile organic compounds to the environment. This type of coating is not widespread in the European market.

Electroplated coatings are widely used in the solar absorber industry. These coatings are applied to the absorber plate metal in a wet process. Black Chrome is plated from chromic acid bath solutions. The black chrome and black nickel coating exhibit solar absorption and thermal emittance values ranging from  $\alpha=0.92\sim0.98$  and  $\varepsilon=0.06\sim0.25$ , respectively, and is thermally stable at 325°C on nickel-coated copper and stainless steel substrates. There are also anodised electroplating coloured selective absorption coatings on a aluminium surface ( $\alpha=0.85\sim0.91$ ,  $\varepsilon=0.23\sim0.40$ ). Coatings on stainless steel have also a potential market in solar thermal applications where glazing is not required and in higher temperature systems including parabolic concentrators. Another type of wet process is the sol-gel technique used for Black Majic coatings.

Vacuum deposition processes are a newer technology for flat plate absorbers with an increasing share of producers. Physical vapour deposition (PVD) depicts a group of processes where at least one of the reactants, not in a gas phase, is evaporated in a vacuum chamber by either heat, sputtering (atomisation), or arc-evaporation. The processes are used also for evacuated collectors and as the receivers in high temperature solar thermal systems. A process of vacuum based selective coating of black chrome uses a high power electron beam gun to evaporate the chromium oxide, which gets deposited on pre-heated copper substrate. PVD coatings an aluminium exhibits solar absorption and thermal emittance values ranging from  $\alpha=0.92\sim0.96$  and  $\varepsilon=0.03\sim0.07$ , respectively (Alanod (2002)).

Further information about these processes can be found in different subchapters of chapter 8.6 and in the literature (Duffie & Beckmann (1991); Haefer (1987)).

## 8.4 Application of Selective Coatings

The selective coatings investigated for this dataset are used for solar absorbers. In the last years there was a shift in the market share of products sold. The total market for selective coatings in the EU is estimated to be 1.2 Mio. m<sup>2</sup> per year (not including vacuum tube absorbers and painted absorbers). The market share for black chrome coatings is decreasing in the last years. This is mainly due to the lower prices for PVD coatings. Today about 35% of the coatings on the market are made by black chrome coatings.<sup>20</sup> And about 20% of solar absorbers sold in Germany are electroplated.<sup>21</sup>

## 8.5 System Characterisation

Different types of selective coatings for solar absorbers are investigated here as far as data or at least some information on environmental aspects were available. There are other selective coating processes for solar absorbers which could not be investigated in this study due to the lack of data. A short description of these coating processes is provided in the next chapter. Readers interested in details are referred to the literature.

## 8.6 Life Cycle Inventories for Selective Coatings

Existing life cycle assessments for solar absorbers showed that the impact of the selective coating as such and differences between different processes are not very important for the total environmental

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<sup>20</sup> Personal communication, Werner Ledderboge, Gibo-Solar, and Kai Bohn, ChromeCoat A/S, 7.2002.

<sup>21</sup> Personal communication V. Handke, IZT, Berlin, 5.2002.

impacts caused by the absorber (Althaus (1999); Streicher et al. (2002)). More important is the choice and the amount of the substrate materials like copper and aluminium used for the construction and the overall energy efficiency of the solar absorber. Nevertheless there is a big discussion about the environmental impacts of different coating technologies. Main concerns of the discussion were i.e. the energy use for production and the emissions of chrome and acids with the effluents. Thus a detailed investigation of data for the coatings might help to better understand the influence of the coating for the total environmental impacts due to the production of a solar absorber.

Most of the data for the life cycle inventories have been derived from questionnaires sent to different manufacturers. The report (IPPC (2002)) does not contain many helpful information specific enough for the processes investigated here.

The unit process raw data are investigated gate-to-gate and refer to the coating of one m<sup>2</sup> metal sheet. The metal sheet itself (substrate) is not included in the inventory. Production processes outside of Europe are assumed with an European electricity mix. For production processes with only single manufacturers the country specific electricity mix is assumed. Standard transport distances have been used for the delivery of materials if not stated otherwise.

### **8.6.1 Vacuum Deposition Processes**

#### **Sputter coating on copper sheet substrates with chromium and tin oxide**

The life cycle inventory for metal coatings on copper made by a sputtering process is based on data provided by a German manufacturer<sup>22</sup> for its production in 2001. An environmental report of this manufacturer for 2000 (Interpane (2000)) was available for comparison. The production process has been improved further on. In 2002 the waste heat of the new production plant is used to replace fuel oil for room heating. The company and the environmental report is certified according to EU eco-audit EMAS and DIN/ISO 14001. Tab. 8.2 shows the unit process raw data and the data quality indicators for this coating technology. The product is sold under the brand name Sunselect. Data for 2001 are shown for the total annual production and per m<sup>2</sup>. Data for the previous year are only shown per m<sup>2</sup>.

The data provided by Interpane could directly be used for the life cycle inventory. The total production in 2001 was 440000 m<sup>2</sup>. The copper sheet has a thickness of 0,2 mm and a weight of ca. 1,8 kg/m<sup>2</sup>. Interpane is applying a sputtering process (cathode evaporation method) for the coating of absorbers. Argon, nitrogen and oxygen are used as process gasses for the process. The target materials chromium and tin are used with an efficiency of 50%. Water is mainly used for cooling and thus it does not have to be treated afterwards. Materials are mainly bought from German producers. The amount of copper sheet substrate coated is not included in the inventory for the coating process, but the amount of copper wastes is included as a demand. These relative clean wastes are recycled.

The emissions stated in the environmental report are not considered here because they are derived from literature values for diesel and electricity use. Instead the burning of the fuels is considered with ecoinvent background data. The transport distances for recycled wastes are not included in the inventory. The used copper sheets (substrate) are not cleaned or pre-treated in a special way by the supplier. The quality is the same as the ones used by other coating manufacturers.

(Frischknecht et al. (1996)) has also estimated data for this process without specifying the origin. Lampert (n.d.) gives some information for a black nickel coating made by magnetron sputtering in Sweden which is also shown in Tab. 8.2.

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<sup>22</sup> Information provided in a questionnaire by Davorin Pavic, Interpane Solar Beschichtungs Gesellschaft, 6.2002.

Tab. 8.2 Unit process raw data and data quality indicators for selective coatings made by sputtering

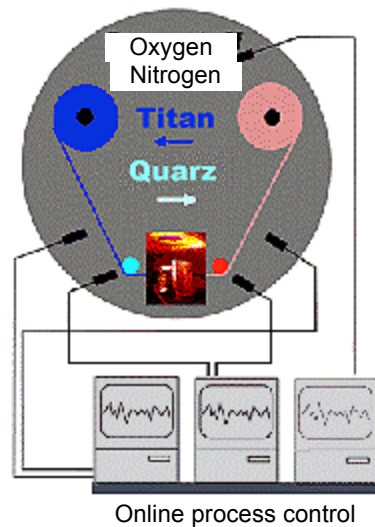
|                                       | Name                                                                    | Location | Infrastructure | Unit | selective coating, copper sheet, sputtering | Uncertainty Typ<br>Standard Deviation 95% | GeneralComment                                                     | Interpane 2001 | Interpane 2000 | Sputtering, Frischknecht et al .1996 | black nickel sputtering |
|---------------------------------------|-------------------------------------------------------------------------|----------|----------------|------|---------------------------------------------|-------------------------------------------|--------------------------------------------------------------------|----------------|----------------|--------------------------------------|-------------------------|
|                                       |                                                                         |          |                |      |                                             |                                           |                                                                    | DE<br>0<br>m2  | DE<br>0<br>m2  | CH<br>0<br>m2                        | SE<br>0<br>m2           |
| output                                | selective coating, copper sheet, sputtering                             | DE       | 0              | m2   | 1.00E+0                                     |                                           |                                                                    | 4.40E+5        | 1.00E+0        |                                      |                         |
| technosphere                          | electricity, medium voltage, at grid                                    | DE       | 0              | kWh  | 3.47E+0                                     | 1 1.07                                    | (1,2,1,1,1,3); Questionnaire, total electricity use in the factory | 1.53E+6        | 3.91E+0        | 1.67E+0                              | 2.00E+0                 |
|                                       | light fuel oil, at regional storage                                     | RER      | 0              | kg   | 6.20E-3                                     | 1 1.07                                    | (1,2,1,1,1,3); Questionnaire, including heating                    | 2.73E+3        | 1.31E-2        |                                      |                         |
|                                       | diesel, burned in building machine                                      | GLO      | 0              | MJ   | 2.25E-1                                     | 1 1.07                                    | (1,2,1,1,1,3); Questionnaire, use for vehicles in plant operation  | 9.92E+4        | 4.32E-1        |                                      |                         |
|                                       | lubricating oil, at plant                                               | RER      | 0              | kg   | 7.73E-4                                     | 1 1.07                                    | (1,2,1,1,1,3); Questionnaire                                       | 3.40E+2        | 7.41E-4        |                                      | 2.80E-5                 |
|                                       | chromium, at regional storage                                           | RER      | 0              | kg   | 7.40E-3                                     | 1 1.07                                    | (1,2,1,1,1,3); Questionnaire                                       | 3.26E+3        | 9.50E-3        |                                      |                         |
|                                       | copper, at regional storage                                             | RER      | 0              | kg   | 8.18E-2                                     | 1 1.07                                    | (1,2,1,1,1,3); Copper production wastes                            | 3.60E+4        | 1.59E-1        |                                      |                         |
|                                       | argon, liquid, at plant                                                 | RER      | 0              | kg   | 3.28E-3                                     | 1 1.07                                    | (1,2,1,1,1,3); Questionnaire (plus helium)                         | 1.44E+3        | 3.11E-3        |                                      | 1.19E-3                 |
|                                       | nitrogen, liquid, at plant                                              | RER      | 0              | kg   | 1.82E-3                                     | 1 1.07                                    | (1,2,1,1,1,3); Questionnaire                                       | 8.00E+2        | 4.90E-4        |                                      |                         |
|                                       | oxygen, liquid, at plant                                                | RER      | 0              | kg   | 9.09E-4                                     | 1 1.07                                    | (1,2,1,1,1,3); Questionnaire                                       | 4.00E+2        | 1.74E-3        |                                      |                         |
|                                       | corrugated board, mixed fibre, single wall, at plant                    | RER      | 0              | kg   | 1.77E-2                                     | 1 1.07                                    | (1,2,1,1,1,3); Questionnaire                                       | 7.77E+3        | 2.18E-2        |                                      |                         |
|                                       | polyethylene, LDPE, granulate, at plant                                 | RER      | 0              | kg   | 5.18E-4                                     | 1 1.07                                    | (1,2,1,1,1,3); Plastic packagings                                  | 2.28E+2        | 9.59E-3        |                                      |                         |
|                                       | kraft paper, bleached, at plant                                         | RER      | 0              | kg   | 5.55E-4                                     | 1 1.10                                    | (1,2,1,1,1,3); Questionnaire                                       | 2.44E+2        | 7.41E-4        |                                      |                         |
|                                       | tin, at regional storage                                                | RER      | 0              | kg   | 8.62E-3                                     | 1 1.07                                    | (1,2,1,1,1,3); Questionnaire                                       | 3.79E+3        | 9.50E-3        |                                      |                         |
|                                       | tap water, at user                                                      | RER      | 0              | kg   | 2.95E+0                                     | 1 1.07                                    | (1,2,1,1,1,3); Questionnaire                                       | 1.30E+6        | 3.76E+0        |                                      |                         |
|                                       | metal coating plant                                                     | RER      | 1              | unit | 3.33E-7                                     | 1 3.00                                    | (1,2,1,1,1,3); Questionnaire                                       | 1.47E-1        | 1.00E-5        |                                      |                         |
|                                       | transport, lorry 32t                                                    | RER      | 0              | tkm  | 1.23E-2                                     | 1 2.09                                    | (4,5,na,na,na,na); Standard distance 100km                         | 5.43E+3        | 2.16E-2        |                                      | 1.22E-4                 |
|                                       | transport, freight, rail                                                | RER      | 0              | tkm  | 2.09E-1                                     | 1 2.09                                    | (4,5,na,na,na,na); Standard distance 600km                         | 9.21E+4        | 3.89E-1        |                                      | 7.30E-4                 |
|                                       | treatment, sewage, from residence, to wastewater treatment, class 2     | CH       | 0              | m3   | 2.27E-4                                     | 1 1.07                                    | (1,2,1,1,1,3); Questionnaire                                       | 1.00E+2        | 1.13E-3        |                                      |                         |
|                                       | disposal, used mineral oil, 10% water, to hazardous waste incineration  | CH       | 0              | kg   | 7.73E-4                                     | 1 1.07                                    | (1,2,1,1,1,3); Questionnaire                                       | 3.40E+2        | 7.32E-4        |                                      |                         |
|                                       | disposal, municipal solid waste, 22.9% water, to municipal incineration | CH       | 0              | kg   | 3.41E-3                                     | 1 1.07                                    | (1,2,1,1,1,3); Questionnaire                                       | 1.50E+3        | 1.09E-2        |                                      |                         |
| emission air, high population density | Heat, waste                                                             | -        | -              | MJ   | 1.25E+1                                     | 1 1.07                                    | (1,2,1,1,1,3); Calculation                                         | 5.49E+6        | 1.41E+1        | 6.00E+0                              | 7.20E+0                 |
|                                       | Carbon dioxide, fossil                                                  | -        | -              | kg   |                                             |                                           | Not considered                                                     | 9.10E+5        | 2.36E+0        |                                      |                         |
|                                       | Carbon monoxide, fossil                                                 | -        | -              | kg   |                                             |                                           | Not considered                                                     | 1.72E+2        | 5.30E-4        |                                      |                         |
|                                       | Nitrogen oxides                                                         | -        | -              | kg   |                                             |                                           | Not considered                                                     | 2.00E+3        | 5.35E-3        |                                      |                         |
|                                       | Sulfur dioxide                                                          | -        | -              | kg   |                                             |                                           | Not considered                                                     | 3.96E+3        | 1.02E-2        |                                      |                         |
|                                       | Particulates, > 10 um                                                   | -        | -              | kg   |                                             |                                           | Not considered                                                     | 1.36E+3        | 3.50E-3        |                                      |                         |
|                                       | Methane, fossil                                                         | -        | -              | kg   |                                             |                                           | Not considered                                                     | 2.13E+3        | 5.49E-3        |                                      |                         |
|                                       | NMVOOC, non-methane volatile organic compounds, unspecified origin      | -        | -              | kg   |                                             |                                           | Not considered                                                     | 1.85E+2        | 6.19E-4        |                                      |                         |
| resource, in ground                   | Molybdenum, xx% in ???, yy% in crude ore, in ground                     | -        | -              | kg   |                                             |                                           | Not considered                                                     |                | -              | 3.00E-6                              |                         |

### PVD coatings on copper sheet substrates with titanium-nitrite-oxide

The life cycle inventory for metal coatings on copper made by a physical vapour deposition (PVD) process is based on information found in the internet for a German producer (Tinox (2002)). Tab. 8.3 shows the unit process raw data and data quality indicators for this coating process. Copper is coated by TINOX in a process with a titanium-nitrite-oxide ( $\text{TiN}_x$ ,  $\text{TiO}_y$  and a small amount of  $\text{TiC}$ ) layer in a vacuum chamber. Fig. 8.1 shows a schematic production process in which 200m<sup>2</sup> copper sheets substrate are coated per batch. An electron beam gun is used for the evaporation process. The layer is coated with  $\text{SiO}_2$  for protection and antireflection (Lampert (n.d.)).

Quantitative data for the use of nitrogen and oxygen in the process were not available. The same figures as for the sputtering process are used. An eventually necessary pre-treatment of the copper sheet outside of the factory and losses of coated material are not included in the data.





**Fig. 8.1** TINOX batch reactive evaporation process. An electron beam gun is installed in a chamber. Oxygen and nitrogen are steadily added to the process that is controlled online (Tinox (2002))

**Tab. 8.3** Unit process raw data and data quality indicators for selective coatings made by physical vapour deposition

|                                       | Name                                                        | Location | Infrastructure | Unit | selective coating, copper sheet, physical vapour deposition | UncertaintyT<br>StandardDev<br>ation95% | GeneralComment                                                      |
|---------------------------------------|-------------------------------------------------------------|----------|----------------|------|-------------------------------------------------------------|-----------------------------------------|---------------------------------------------------------------------|
|                                       | Location<br>InfrastructureProcess<br>Unit                   |          |                |      | DE<br>0<br>m2                                               |                                         |                                                                     |
| output                                | selective coating, copper sheet, physical vapour deposition | DE       | 0              | m2   | 1.00E+0                                                     |                                         |                                                                     |
| technosphere                          | electricity, medium voltage, at grid                        | DE       | 0              | kWh  | 1.20E+0                                                     | 1 1.09                                  | (2,2,1,3,1,3); Public information. Electricity use for the process. |
|                                       | lubricating oil, at plant                                   | RER      | 0              | kg   | 1.68E-5                                                     | 1 1.09                                  | (2,2,1,3,1,3); Public information.                                  |
|                                       | silica sand, at plant                                       | CH       | 0              | kg   | 8.90E-4                                                     | 1 1.09                                  | (2,2,1,3,1,3); Public information.                                  |
|                                       | titanium dioxide, production mix, at plant                  | RER      | 0              | kg   | 1.00E-3                                                     | 1 1.09                                  | (2,2,1,3,1,3); Titanium nitrite oxide                               |
|                                       | nitrogen, liquid, at plant                                  | RER      | 0              | kg   | 1.82E-3                                                     | 1 1.62                                  | (5,na,1,3,3,5); Estimation                                          |
|                                       | oxygen, liquid, at plant                                    | RER      | 0              | kg   | 9.09E-4                                                     | 1 1.62                                  | (5,na,1,3,3,5); Estimation                                          |
|                                       | metal coating plant                                         | RER      | 1              | unit | 3.33E-7                                                     | 1 3.90                                  | (5,na,1,1,5,na); Estimation                                         |
|                                       | transport, lorry 32t                                        | RER      | 0              | tkm  | 4.63E-4                                                     | 1 2.09                                  | (4,5,na,na,na,na); Standard distance 100km                          |
|                                       | transport, freight, rail                                    | RER      | 0              | tkm  | 2.78E-3                                                     | 1 2.09                                  | (4,5,na,na,na,na); Standard distance 600km                          |
| emission air, high population density | Heat, waste                                                 | -        | -              | MJ   | 4.32E+0                                                     | 1 1.09                                  | (2,2,1,3,1,3); Public information.                                  |

### PVD coatings on aluminium sheet substrates

(Alanod (2002)) uses a selective PVD coating for aluminium sheet substrate. During the hot rolling process an additional unwanted oxide film forms, which is incorporated into the rolled surface during the cold rolling process, thereby leading to irregularities in the surface. In order to avoid these irregularities to the texture or the chemical composition of the aluminium, Alanod degreases the surface and then it is electro-chemically brightened. Here the top layer of the strip is removed, so that the pure metal now forms the strip surface. As a further result of this process, micro roughness will be evened out.

This pre-treated aluminium surface is then oxidised in an acid anodising bath with the help of electrical energy. During this process a very dense but very thin oxide film develops, the so-called barrier layer. The thickness of this layer depends on the level of the direct voltage employed during the anodising process. The barrier layer, which consists mainly from aluminium oxide, is a good electrical insulator.

The multi-layer selective coating is deposited on the anodised aluminium substrate. Three optically active layers are applied continuously in an "air-to-air" thin film process.

The last production step incorporates the sealing of the pores, whose diameter is in the range of 15 nm. To enable this, the strip is passed through a bath of almost boiling water, which is responsible for the formation of so-called boehmite crystals thereby closing the pores. The oxide layer on the top are optimised for a good solar absorption ( $\alpha$ ). After the final drying process, the anodised surface is mechanically as well as chemically protected. The anodised aluminium coil is now ready for use.<sup>23</sup>

Quantitative data for energy and resource uses and emissions of this process were not available.

## 8.6.2 Wet Processes

### Aluminium oxide-based layer pigmented with colloidal nickel on aluminium sheet substrates

Selective coatings with an aluminium oxide-based layer pigmented with colloidal nickel are used by one company in Germany. The coatings are made in Slovakia by anodisation in phosphoric acid followed by electrolysis in a nickel pigmentsing bath. The inventory is based on previous published data (Frischknecht et al. (1996)). These data have been provided by the manufacturer in 1993. The production process has been improved in the meantime, but new data were not available (Thermosolar (2002)).<sup>24</sup> Tab. 8.4 shows the unit process raw data and the data quality indicators for this selective coating process. Losses of coated materials are not included in the inventory.

**Tab. 8.4 Unit process raw data and data quality indicators for aluminium oxide-based layer pigmented with colloidal nickel.**

|                                       | Name                                                                 | Location | Infrastructure | Unit | selective coating, aluminium sheet, nickel pigmented aluminium oxide | Uncertainty | Standard Deviation | General Comment                                               |
|---------------------------------------|----------------------------------------------------------------------|----------|----------------|------|----------------------------------------------------------------------|-------------|--------------------|---------------------------------------------------------------|
|                                       | Location Infrastructure Process Unit                                 |          |                |      | SK 0 m2                                                              |             |                    |                                                               |
| product                               | selective coating, aluminium sheet, nickel pigmented aluminium oxide | SK       | 0              | m2   | 1.00E+0                                                              |             |                    |                                                               |
| technosphere                          | tap water, at user                                                   | RER      | 0              | kg   | 5.20E+2                                                              | 1           | 1.14               | (2,2,3,1,1,3); Questionnaire                                  |
|                                       | electricity, medium voltage, at grid                                 | SK       | 0              | kWh  | 1.81E+0                                                              | 1           | 1.14               | (2,2,3,1,1,3); Questionnaire, electricity use for the process |
|                                       | light fuel oil, burned in industrial furnace 1MW, non-modulating     | RER      | 0              | MJ   | 4.10E+1                                                              | 1           | 1.14               | (2,2,3,1,1,3); Questionnaire                                  |
|                                       | sodium hydroxide, 50% in H2O, production mix, at plant               | RER      | 0              | kg   | 1.64E-1                                                              | 1           | 1.14               | (2,2,3,1,1,3); Questionnaire                                  |
|                                       | nickel, 99.5%, at plant                                              | GLO      | 0              | kg   | 2.18E-2                                                              | 1           | 1.14               | (2,2,3,1,1,3); Questionnaire                                  |
|                                       | phosphoric acid, fertiliser grade, 70% in H2O, at plant              | GLO      | 0              | kg   | 2.69E-1                                                              | 1           | 1.14               | (2,2,3,1,1,3); Questionnaire                                  |
|                                       | nitric acid, 50% in H2O, at plant                                    | RER      | 0              | kg   | 1.06E-1                                                              | 1           | 1.14               | (2,2,3,1,1,3); Questionnaire                                  |
|                                       | boric acid, anhydrous, powder, at plant                              | RER      | 0              | kg   | 1.80E-2                                                              | 1           | 1.14               | (2,2,3,1,1,3); Questionnaire                                  |
|                                       | transport, lorry 32t                                                 | RER      | 0              | tkm  | 5.79E-2                                                              | 1           | 2.09               | (4,5,na,na,na,na); Standard distance 100km                    |
|                                       | transport, freight, rail                                             | RER      | 0              | tkm  | 3.47E-1                                                              | 1           | 2.09               | (4,5,na,na,na,na); Standard distance 600km                    |
|                                       | metal coating plant                                                  | RER      | 1              | unit | 3.33E-7                                                              | 1           | 1.10               | (2,2,3,1,1,3); Estimation                                     |
| emission air, high population density | Heat, waste                                                          | -        | -              | MJ   | 6.50E+0                                                              | 1           | 1.14               | (2,2,3,1,1,3); Questionnaire                                  |
| emission water, river                 | Aluminum                                                             | -        | -              | kg   | 1.60E-3                                                              | 1           | 5.02               | (2,2,3,1,1,3); Questionnaire                                  |
|                                       | Nickel, ion                                                          | -        | -              | kg   | 5.00E-4                                                              | 1           | 5.02               | (2,2,3,1,1,3); Questionnaire                                  |
|                                       | Suspended solids, unspecified                                        | -        | -              | kg   | 1.60E-2                                                              | 1           | 1.53               | (2,2,3,1,1,3); Questionnaire                                  |

### Black chrome coatings on copper sheet substrates

First a nickel layer is deposited on the metal sheet for the black chrome coating of copper substrates. Then chrome in the oxidized form is used for the second layer. The company (mti (2002)) uses chromium metal and chromium oxide (Cr<sub>2</sub>O<sub>3</sub>) for the selective black chrome. This type of chrome (Cr<sup>3+</sup>) is less poisonous than the well known Cr<sup>6+</sup>.

Black chrome coatings with nickel underlay are produced by ChromeCoat A/S in DK. Several years of field experience showed that this type of coating is a durable selective surface for solar absorbers. The chemical solution is used mainly in a closed loop. Waste products are not directly emitted to the envi-

<sup>23</sup> Personal communication Michael Grupp, Alanod Aluminium-Veredlung GmbH & Co. KG, 6.2002.

<sup>24</sup> Personal communication with Mr. Novak, Thermosolar AG, SK, 7.2002.

ronment. Rinsing water is treated by ion exchangers and returned back to the line. The polluted water from the regeneration of the ion exchangers passes a vacuum evaporator that separates water from metals, acids, etc.. The remaining waste is treated in a chemical treatment plant. Some data for this process have been provided in a questionnaire (Tab. 8.6), but this does not include the exact composition of the wastes.<sup>25</sup>

Data for the continuous process of black chrome coatings at MTI, US were available in a product brochure and from a questionnaire.<sup>26</sup> MTI Solar electrodeposits black chromium on nickel plated copper in a continuous plating process. This coating is marketed under the trade name KROSOL. The major process stages are:

1. Alkaline cleaning of copper to remove any oils on the surface and wash
2. Neutralizing with the acid to remove any oxidation and wash
3. Plating nickel on copper and wash
4. Plating black chrome on the nickel plated copper and wash
5. Wash and dry

The company MTI uses about 2.3kWh of electricity per m<sup>2</sup> of coated substrate including all pre-processes like cleaning. A major environmental concern for this process was the waste water from galvanic process. About 50% of the production costs at MTI are necessary for the treatment process for the effluents (Remmers (1998)). Tab. 8.6 shows the quantitative data for air emissions and the use of acids for this factory. Tab. 8.5 shows the concentration of water pollutants in the effluents from the MTI black chrome coating process. The bold figures have been used to assess the environmental impacts due to the treatment of these effluents (Doka (2003)). One square metre of coated copper sheet has a weight of 1.225kg. MTI has coated 8 Mio. m<sup>2</sup> in 30 years of production.

**Tab. 8.5 Concentration of water pollutants in the effluents from the MTI black chrome coating process**

| Water emissions        | Unit | Amount to public treatment plant |
|------------------------|------|----------------------------------|
| BOD                    | mg/l | <2                               |
| COD                    | mg/l | <b>2</b>                         |
| Total Petroleum HCs    | mg/l | 0.1                              |
| Total Suspended Solids | mg/l | 3.7                              |
| Chromium               | mg/l | <b>0.35</b>                      |
| Nickel                 | mg/l | <b>0.3</b>                       |
| Copper                 | mg/l | <b>0.49</b>                      |
| Silver                 | mg/l | <0.01 ( <b>0.005</b> )           |
| Cadmium                | mg/l | <0.01 ( <b>0.005</b> )           |
| Lead                   | mg/l | <0.1 ( <b>0.05</b> )             |
| Zinc                   | mg/l | <b>0.21</b>                      |

Streicher (2002) investigated an energy use of 12.4kWh/m<sup>2</sup> for the producer Gibo-Solar, but this production plant has been closed.<sup>20</sup> The electricity use due to the chrome coating process has been estimated by Hofstetter *et al.* (1992) with 0.6kWh/m<sup>2</sup>. This value is not reliable because it was estimated based on laboratory experiments for normal black chrome processes made in the 1980's.

<sup>25</sup> Information provided in marketing leaflets and in a questionnaire by ChromeCoat, Kai Bohn, 7.2002.

<sup>26</sup> Personal communication with Dr. Kidambi Raghunathan, IPTE, Inc., US, 9.2002 and MTI Solar Selective Absorber Strip in Rolls and Modular Assemblies, MTI Materials Technology Inc., US.

The inventory for black chrome coatings on copper is included in the database calculated as an average for the information provided by different producers. Tab. 8.6 shows the unit process raw data for the coating process. The inventory is calculated with an European electricity mix. Losses of coated materials are not included in the inventory. Also transports from the US to Europe are so far not included in the inventory.

**Tab. 8.6 Unit process raw data and data quality indicators for selective coatings made by black chrome coating of copper coils. Inventory data from different production plants**

|                                       | Name                                                                       | Location | Infrastructure | Unit | selective coating, copper sheet, black chrome | Uncertainty T Standard Deviation 95%                                  | GeneralComment | MTI     | Chrome Coat | Gibo Solar  |
|---------------------------------------|----------------------------------------------------------------------------|----------|----------------|------|-----------------------------------------------|-----------------------------------------------------------------------|----------------|---------|-------------|-------------|
|                                       | Location InfrastructureProcess Unit                                        |          |                |      | RER 0 m2                                      |                                                                       |                | US 0 m2 | DK 0 m2     | DE, PL 0 m2 |
| output                                | selective coating, copper sheet, black chrome                              | RER      | 0              | m2   | 1.00E+0                                       |                                                                       |                | n.a.    | 1.00E+0     | 1.00E+0     |
| technosphere                          | electricity, medium voltage, production UCTE, at grid                      | UCTE     | 0              | kWh  | 2.66E+0                                       | 1 1.09 (2,2,1,1,1,3); Questionnaires, electricity use in the factory  |                | 2.31E+0 | 3.00E+0     | 1.24E+1     |
|                                       | natural gas, burned in industrial furnace >100kW                           | RER      | 0              | MJ   | 5.45E+0                                       | 1 1.09 (2,2,1,1,1,3); Questionnaire                                   |                | 1.09E+1 | -           | n.a.        |
|                                       | chemicals inorganic, at plant                                              | GLO      | 0              | kg   | 6.85E-4                                       | 1 1.09 (2,2,1,1,1,3); Questionnaire, proprietary mix plating solution |                | 1.37E-3 | -           | n.a.        |
|                                       | boric acid, anhydrous, powder, at plant                                    | RER      | 0              | kg   | 1.88E-4                                       | 1 1.09 (2,2,1,1,1,3); Questionnaires                                  |                | 3.75E-4 | -           | n.a.        |
|                                       | chromium oxide, flakes, at plant                                           | RER      | 0              | kg   | 9.72E-3                                       | 1 1.09 (2,2,1,1,1,3); Questionnaires                                  |                | 1.80E-3 | 1.76E-2     | n.a.        |
|                                       | fluosilicic acid, 22% in H <sub>2</sub> O, at plant                        | RER      | 0              | kg   | 3.12E-4                                       | 1 1.09 (2,2,1,1,1,3); Questionnaires                                  |                | 6.24E-4 | -           | n.a.        |
|                                       | nickel, 99.5%, at plant                                                    | GLO      | 0              | kg   | 2.02E-2                                       | 1 1.09 (2,2,1,1,1,3); Questionnaires                                  |                | 3.76E-4 | 4.00E-2     | n.a.        |
|                                       | sulphuric acid, liquid, at plant                                           | RER      | 0              | kg   | 3.68E-3                                       | 1 1.09 (2,2,1,1,1,3); Questionnaires                                  |                | 7.35E-3 | -           | n.a.        |
|                                       | soap, at plant                                                             | RER      | 0              | kg   | 2.04E-3                                       | 1 1.09 (2,2,1,1,1,3); Questionnaires                                  |                | 4.08E-3 | -           | n.a.        |
|                                       | tap water, at user                                                         | RER      | 0              | kg   | 1.09E+1                                       | 1 1.09 (2,2,1,1,1,3); Questionnaires                                  |                | 2.12E+1 | 5.88E-1     | n.a.        |
|                                       | metal coating plant                                                        | RER      | 1              | unit | 3.33E-7                                       | 1 3.90 (5,na,1,1,5,na); Estimation                                    |                |         |             | n.a.        |
|                                       | transport, lorry 32t                                                       | RER      | 0              | tkm  | 3.68E-3                                       | 1 2.09 (4,5,na,na,na,na); Standard distance 100km                     |                | 1.60E-3 | 5.76E-3     | -           |
|                                       | transport, freight, rail                                                   | RER      | 0              | tkm  | 2.21E-2                                       | 1 1.10 (4,5,na,na,na,na); Standard distance 600km                     |                | 9.58E-3 | 3.46E-2     | -           |
|                                       | treatment, black chrome coating effluent, to wastewater treatment, class 2 | CH       | 0              | m3   | 8.49E-3                                       | 1 1.09 (2,2,1,1,1,3); Questionnaires                                  |                | 1.70E-2 | -           | n.a.        |
|                                       | disposal, hazardous waste, 0% water, to underground deposit                | DE       | 0              | kg   | 2.00E-1                                       | 1 1.09 (2,2,1,1,1,3); Water is evaporated or recycled                 |                | -       | 4.00E-1     | n.a.        |
| emission air, high population density | Heat, waste                                                                | -        | -              | MJ   | 9.56E+0                                       | 1 1.09 (2,2,1,1,1,3); Calculation                                     |                | 8.32E+0 | 1.08E+1     | 4.45E+1     |
|                                       | Chromium                                                                   | -        | -              | kg   | 1.50E-6                                       | 1 5.01 (2,2,1,1,1,3); Questionnaires                                  |                | 1.50E-6 | n.a.        | n.a.        |
|                                       | Nickel                                                                     | -        | -              | kg   | 3.00E-7                                       | 1 5.01 (2,2,1,1,1,3); Questionnaires                                  |                | 3.00E-7 | n.a.        | n.a.        |

### Black chrome coatings on stainless steel sheet substrates

Live cycle inventory data were available for the deposition of black chrome on stainless steel in a galvanic batch process (Energie Solaire (2002)).<sup>27</sup> The main process stages used for the process at this Swiss production plant are:

- 1: Cleaning with electrolytic wash bath at ambient temperature (20°C)
- 2: Activation
- 3: Black chrome coating at ambient temperature (20°C)
- 4: Recuperation baths
- 5: Reduction bath

The detailed inventory for black chrome coating on stainless steel with the above mentioned process stages is included in the database. Tab. 8.7 shows the unit process raw data for the black chrome coating process. Rinsing water is recycled through a water treatment plant. There are no process specific air and water emissions. A part of the used water is evaporated in the process. Effluents with a chrome (III) content of 1.35% are recycled. Losses of coated materials are not included in the inventory. In tests this type of coating resists without any alteration to humidity, UV, high temperature – up to 530° – as well as acid rain and salted mist.

<sup>27</sup> Data and information provided in personal communication by Energie Solaire, CH.

**Tab. 8.7 Unit process raw data and data quality indicators for selective coatings made by black chrome coating of stainless steel**

|                                       | Name                                                   | Location | Infrastructure | Unit | selective coating, stainless steel sheet, black chrome CH 0 m2 | Uncertainty | Standard Deviation | GeneralComment                                                  |
|---------------------------------------|--------------------------------------------------------|----------|----------------|------|----------------------------------------------------------------|-------------|--------------------|-----------------------------------------------------------------|
|                                       | Location<br>InfrastructureProcess<br>Unit              |          |                |      |                                                                |             |                    |                                                                 |
| output                                | selective coating, stainless steel sheet, black chrome | CH       | 0              | m2   | 1.00E+0                                                        |             |                    |                                                                 |
| technosphere                          | electricity, medium voltage, at grid                   | CH       | 0              | kWh  | 1.35E+0                                                        | 1           | 1.09               | (2,2,1,1,1,3); Questionnaire, electricity use for the process   |
|                                       | chromium oxide, flakes, at plant                       | RER      | 0              | kg   | 3.50E-2                                                        | 1           | 1.09               | (2,2,1,1,1,3); Questionnaire, coating, wastes are recycled      |
|                                       | hydrochloric acid, 30% in H2O, at plant                | RER      | 0              | kg   | 5.00E-2                                                        | 1           | 1.09               | (2,2,1,1,1,3); Questionnaire, activation                        |
|                                       | sodium hydroxide, 50% in H2O, production mix, at plant | RER      | 0              | kg   | 9.50E-2                                                        | 1           | 1.09               | (2,2,1,1,1,3); Questionnaire, wash bath                         |
|                                       | sodium sulphate, powder, at plant                      | RER      | 0              | kg   | 3.00E-2                                                        | 1           | 1.22               | (2,2,1,1,3,3); Questionnaire, sodium bisulfite, reduction of Cr |
|                                       | sulphuric acid, liquid, at plant                       | RER      | 0              | kg   | 5.00E-3                                                        | 1           | 1.09               | (2,2,1,1,1,3); Questionnaire, Ph correction                     |
|                                       | tap water, at user                                     | RER      | 0              | kg   | 4.00E-1                                                        | 1           | 1.09               | (2,2,1,1,1,3); Questionnaire, water is evaporated.              |
|                                       | metal coating plant                                    | RER      | 1              | unit | 3.33E-7                                                        | 1           | 3.90               | (5,na,1,1,5,na); Estimation                                     |
|                                       | transport, lorry 32t                                   | RER      | 0              | tkm  | 2.15E-2                                                        | 1           | 2.09               | (4,5,na,na,na,na); Standard distance 100km                      |
| emission air, high population density | transport, freight, rail                               | RER      | 0              | tkm  | 1.29E-1                                                        | 1           | 1.10               | (4,5,na,na,na,na); Standard distance 600km                      |
|                                       | Heat, waste                                            | -        | -              | MJ   | 4.86E+0                                                        | 1           | 1.09               | (2,2,1,1,1,3); Calculation                                      |

### Black Majic coating on copper sheet substrates

Thermafin Manufacturing<sup>28</sup> in the USA uses a single-phase nanocrystalline technology for solar selective coatings in a sol-gel based wet process (Alternate Energy Technologies (2002)). The brand name for this process is “Black Majic”. The following list contains a description of the process stages used for the process:

1. obtain top-grade copper with microfinish
2. pre-clean copper prior to coating
3. activate copper surface with nickel
4. coat with Black Majic, these are crystals raised on the surface with a tin-nickel alloy plating solution
5. cure final product with a protective glass layer

Tab. 8.8 shows the properties of the tin-nickel plating solution. The alloy contains about 65% tin and 35% nickel. The amount of chemicals has been roughly estimated with the concentrations shown in the table and 50% of the amount of water used for the process.

<sup>28</sup> Personal communication with Billy Byrom, Alternate Energy Technologies LLC, 7.2002.

Tab. 8.8 Properties of tin-nickel alloy plating solution.<sup>28</sup>

|                                                            | <i>g/L</i>    | <i>Molarity</i> |
|------------------------------------------------------------|---------------|-----------------|
| Stannous chloride, $\text{SnCl}_2$                         | 49            | 0.26            |
| (or $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$ )            | (58)          | 0.26            |
| Nickel chloride, $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$ | 300           | 1.26            |
| Ammonium bifluoride, $\text{NH}_4\text{HF}_2$              | 56            | 1.0             |
| Ammonium hydroxide, $\text{NH}_4\text{OH}$                 | to pH 2.0–2.5 |                 |
| By analysis:                                               |               |                 |
| Stannous tin, $\text{Sn}^{++}$                             | 26–37.5       | 0.22–0.32       |
| Nickel, Ni                                                 | 60–83         | 1.0–1.4         |
| Total fluorine*                                            | 34–45         | 1.8–2.4         |
| Temperature, °C                                            | 65–71         |                 |
| Cathode current density, A/m <sup>2</sup>                  | 100–300       |                 |
| Anode current density, A/m <sup>2</sup>                    | to 500        |                 |

\*Total fluorine should be at least as high as total tin:  $\text{Sn(II)} + \text{Sn(IV)}$ ; see text.

The crystals grow in a tin-nickel alloy plating solution. Hydrogen fluoride gas is used in the process. Alcohol is used as a solvent during the Black Majic coating (Remmers (1998)). The coating is fully encapsulated with a permanent protective glass layer made out of silica oxide. Some data for this process were available from a questionnaire. Tab. 8.9 shows the unit process raw data and data quality indicators for the Black Majic coating process. The electricity use has been assessed roughly by the company. The European mix is used instead of the mix for US, which has not been investigated for the background data. No information was available about process specific air and water emissions. Water emissions are estimated to be half of the amount of chemicals used for the process. Losses of coated materials are not included in the inventory.

**Tab. 8.9 Unit process raw data and data quality indicators for selective coatings made by Black Majic coating of copper.**

|                                       | Name                                                                            | Location | Infrastructure | Unit | selective coating, copper sheet, black majic | Uncertainty | Standard Deviation | GeneralComment                                                                   | Aetsolar     |
|---------------------------------------|---------------------------------------------------------------------------------|----------|----------------|------|----------------------------------------------|-------------|--------------------|----------------------------------------------------------------------------------|--------------|
|                                       | Location<br>InfrastructureProcess<br>Unit                                       |          |                |      | US<br>0<br>m2                                |             |                    |                                                                                  | US<br>0<br>a |
| output                                | selective coating, copper sheet, black majic                                    | US       | 0              | m2   | 1.00E+0                                      |             |                    |                                                                                  | 8.26E+4      |
| technosphere                          | electricity, medium voltage, at grid                                            | US       | 0              | kWh  | 6.31E-1                                      | 1           | 1.22               | (4,2,1,1,1,3); Questionnaire, 1000kWh/week                                       | 5.21E+4      |
|                                       | fatty alcohol, petrochemical, at plant                                          | RER      | 0              | kg   | 6.05E-3                                      | 1           | 5.26               | (5,2,1,1,1,3); Rough estimation, basic uncertainty = 5                           | 5.00E+2      |
|                                       | ammonia, liquid, at regional storehouse                                         | RER      | 0              | kg   | 1.69E-2                                      | 1           | 5.26               | (5,2,1,1,1,3); Rough estimation, basic uncertainty = 5                           | 1.40E+3      |
|                                       | hydrochloric acid, 30% in H2O, at plant                                         | RER      | 0              | kg   | 1.02E-1                                      | 1           | 5.26               | (5,2,1,1,1,3); Rough estimation, basic uncertainty = 5                           | 8.40E+3      |
|                                       | hydrogen fluoride, at plant                                                     | GLO      | 0              | kg   | 1.36E-2                                      | 1           | 5.26               | (5,2,1,1,1,3); Rough estimation, basic uncertainty = 5                           | 1.12E+3      |
|                                       | nickel, 99.5%, at plant                                                         | GLO      | 0              | kg   | 2.37E-2                                      | 1           | 5.26               | (5,2,1,1,1,3); Rough estimation, basic uncertainty = 5                           | 1.96E+3      |
|                                       | tin, at regional storage                                                        | RER      | 0              | kg   | 1.02E-2                                      | 1           | 5.26               | (5,2,1,1,1,3); Rough estimation, basic uncertainty = 5                           | 8.40E+2      |
|                                       | tap water, at user                                                              | RER      | 0              | kg   | 6.78E-1                                      | 1           | 1.09               | (2,2,1,1,1,3); Questionnaire                                                     | 5.60E+4      |
|                                       | metal coating plant                                                             | RER      | 1              | unit | 3.33E-7                                      | 1           | 3.68               | (3,na,1,1,5,na); Estimation                                                      |              |
|                                       | transport, lorry >16t, fleet average                                            | RER      | 0              | tkm  | 1.72E-2                                      | 1           | 2.09               | (4,5,na,na,na,na); Standard distance 100km                                       | 1.42E+3      |
|                                       | transport, freight, rail                                                        | RER      | 0              | tkm  | 1.03E-1                                      | 1           | 1.10               | (4,5,na,na,na,na); Standard distance 600km                                       | 8.53E+3      |
|                                       | disposal, hazardous waste, 25% water, to hazardous waste incineration           | CH       | 0              | kg   | 1.72E-1                                      | 1           | 1.09               | (2,2,1,1,1,3); Nickel sludge from waste water, estimation                        | 1.42E+4      |
|                                       | treatment, sewage, unpolluted, from residence, to wastewater treatment, class 2 | CH       | 0              | m3   | 6.90E-4                                      | 1           | 1.09               | (2,2,1,1,1,3); Questionnaire, estimation that emitted substances are not reduced | 5.70E+1      |
| emission air, high population density | Heat, waste                                                                     | -        | -              | MJ   | 2.27E+0                                      | 1           | 1.09               | (2,2,1,1,1,3); Calculation                                                       | 1.88E+5      |
| emission water, river                 | Ammonium, ion                                                                   | -        | -              | kg   | 1.69E-5                                      | 1           | 1.51               | (5,na,1,1,1,3); Estimation 0.1%of use as emission to river                       | 1.40E+0      |
|                                       | Chloride                                                                        | -        | -              | kg   | 1.02E-4                                      | 1           | 1.51               | (5,na,1,1,1,3); Estimation 0.1%of use as emission to river                       | 8.40E+0      |
|                                       | Fluoride                                                                        | -        | -              | kg   | 1.36E-5                                      | 1           | 1.51               | (5,na,1,1,1,3); Estimation 0.1%of use as emission to river                       | 1.12E+0      |
|                                       | Nickel, ion                                                                     | -        | -              | kg   | 2.37E-5                                      | 1           | 1.51               | (5,na,1,1,1,3); Estimation 0.1%of use as emission to river                       | 1.96E+0      |
|                                       | Tin, ion                                                                        | -        | -              | kg   | 1.02E-5                                      | 1           | 1.51               | (5,na,1,1,1,3); Estimation 0.1%of use as emission to river                       | 8.40E-1      |

### 8.6.3 Coating Plant

The unit process raw data for the manufacturing plant in Tab. 8.10 have been assessed with data provided by different producers. Production data are rounded in the table due to secrecy reasons. The data for the coating plant have been normalized for a production capacity of 100000m<sup>2</sup>/a. The life time of the plant is estimated to be 30 years. The share of different production activities at (Interpane (2000)) for the total land use has been confirmed in a personal communication. The averaged inventory for the infrastructure is used for all coating processes. Data for necessary machinery were not available.

**Tab. 8.10 Unit process raw data and data quality indicators for selective coatings manufacturing plant.**

|                   | Name                                           | Location | Category | SubCategory | Infrastructure | Unit | metal coating plant | Uncertainty | Standard Deviation | GeneralComment                 | coating plant | coating plant | coating plant | coating plant | coating plant |
|-------------------|------------------------------------------------|----------|----------|-------------|----------------|------|---------------------|-------------|--------------------|--------------------------------|---------------|---------------|---------------|---------------|---------------|
|                   | Location<br>InfrastructureProcess<br>Unit      |          |          |             |                |      | RER                 |             |                    |                                | DE            | CH            | DK            | US            | US            |
| output            | metal coating plant                            | RER      | -        | -           | 1              | unit | 1.00E+0             |             |                    |                                | 1             | 1             | 1             | 1             | 1             |
| technosphere      | building, hall                                 | CH       | -        | -           | 1              | m2   | 5.24E+2             | 1           | 3.02               | (3,3,1,1,1,na); Questionnaires | 1.00E+3       | 1.80E+2       | 5.00E+1       | 2.45E+3       | 1.40E+3       |
| resource, land    | Occupation, industrial area, built up          | -        | resource | land        | -              | m2a  | 3.39E+4             | 1           | 1.52               | (3,3,1,1,1,na); 50a occupation | 1.11E+5       | 1.80E+4       | 1.05E+4       | 1.47E+5       | 4.20E+4       |
|                   | Occupation, industrial area, vegetation        | -        | resource | land        | -              | m2a  | 1.48E+4             | 1           | 1.52               | (3,3,1,1,1,na); 50a occupation | 4.80E+4       | 1.26E+4       | 9.00E+3       | 7.35E+4       | -             |
|                   | Transformation, from unknown                   | -        | resource | land        | -              | m2   | 1.13E+3             | 1           | 2.02               | (3,3,1,1,1,na); Calculation    | 3.70E+3       | 6.00E+2       | 3.50E+2       | 4.90E+3       | 1.40E+3       |
|                   | Transformation, to industrial area, vegetation | -        | resource | land        | -              | m2   | 4.92E+2             | 1           | 2.02               | (3,3,1,1,1,na); Questionnaires | 1.60E+3       | 4.20E+2       | 3.00E+2       | 2.45E+3       | -             |
|                   | Transformation, to industrial area, built up   | -        | resource | land        | -              | m2   | 6.38E+2             | 1           | 2.02               | (3,3,1,1,1,na); Questionnaires | 2.10E+3       | 1.80E+2       | 5.00E+1       | 2.45E+3       | 1.40E+3       |
| Annual production |                                                |          |          |             |                | m2   | 1E+6                |             |                    |                                | 4E+5          | 1E+4          | 2E+5          | 3E+5          | 8E+4          |

## 8.7 Data Quality Considerations

The simplified approach with a pedigree matrix has been used for calculating most of the standard deviation. Information for the aluminium oxide-based layer pigmented with colloidal nickel has been

provided by a manufacturer some years ago and is thus out of date. The inventory for the sputtering process is based on an externally verified environmental report with figures for all known energy uses and major emissions. Data for the PVD process are only a rough estimation with known data gaps for the use of materials. Data for the black chrome coatings on stainless steel and copper are reliable based on unverified information directly provided by the manufacturer. The data quality of the Black Majic coating process on copper is not very reliable. Only few data were available. Thus the unit process data have been derived by using some assumptions.

It has to be noted that the system boundaries for the investigation of different processes differ as the losses of coated materials, additional energy uses e.g. for cars, etc. are not included for all coating processes in the same depth. Thus a direct comparison of coating processes with these inventories is not possible. Also transports from the coating plant to the final customer, which might be important, have not been included in the assessment.

The overall data quality is sufficient for the aim of this section, the provision of background data for the life cycle inventory of solar absorbers. The results per square metre are not meant to be used for a direct comparison of coated absorbers. For that purpose the inclusion of all data for the absorber and the consideration of product properties is necessary (especially solar absorbance and thermal emittance, further materials, life time, etc.).

## 8.8 Cumulative Results and Interpretation

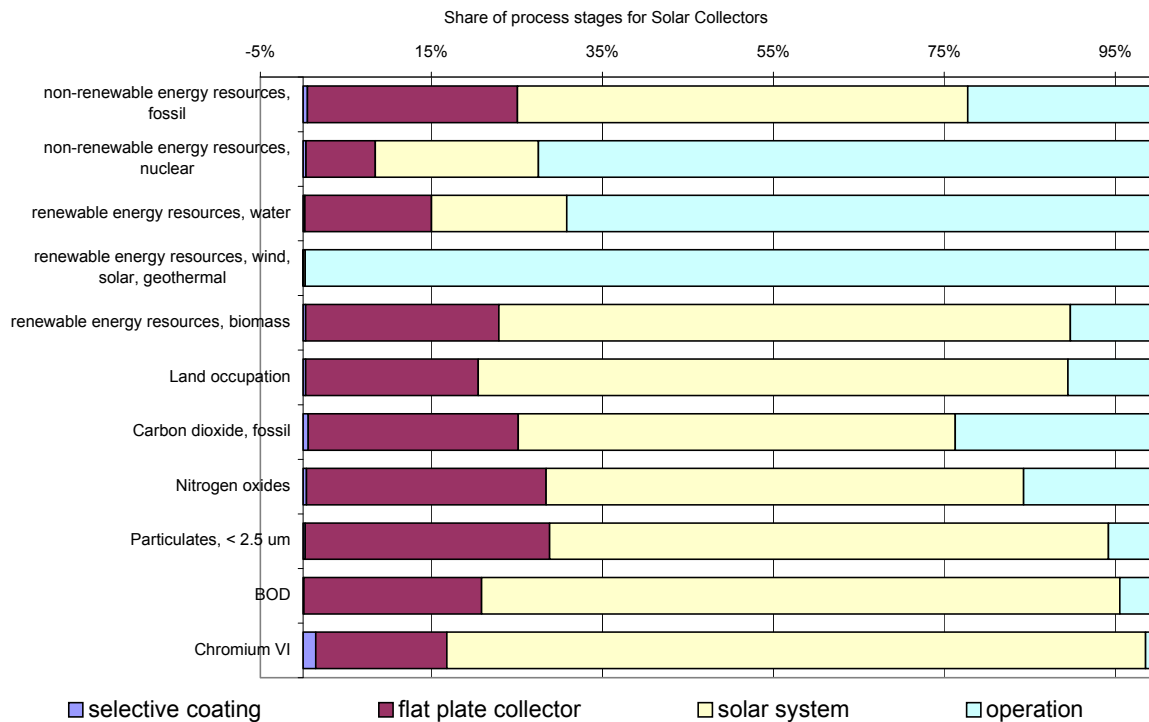
Tab. 8.11 shows selected LCI results and the cumulative energy demand for selective coatings on metal sheets. Instead of comparing different products, we just show minimum, maximum and mean values for the products investigated here. There might be quite large differences among the different coatings. But there is no one possibility that is the worst or the best in all aspects. Thus a very detailed comparison would be necessary.

**Tab. 8.11 Minimum, maximum and mean LCI results and the cumulative energy demand for selective coatings on metal sheets**

|                            |                          |                                                     |       | selective<br>coating | selective<br>coating | selective<br>coating |
|----------------------------|--------------------------|-----------------------------------------------------|-------|----------------------|----------------------|----------------------|
| Name                       |                          |                                                     |       | MIN                  | MEAN                 | MAX                  |
| Location                   |                          |                                                     | Unit  | m2                   | m2                   | m2                   |
| Infrastructure             |                          |                                                     |       | 0                    | 0                    | 0                    |
| <b>LCIA results</b>        |                          |                                                     |       |                      |                      |                      |
|                            | cumulative energy demand | non-renewable energy resources, fossil              | MJ-Eq | 7.1                  | 27.1                 | 72.8                 |
|                            | cumulative energy demand | non-renewable energy resources, nuclear             | MJ-Eq | 4.70                 | 11.03                | 16.30                |
|                            | cumulative energy demand | renewable energy resources, water                   | MJ-Eq | 4.2E-1               | 2.3E+0               | 3.0E+0               |
|                            | cumulative energy demand | renewable energy resources, wind, solar, geothermal | MJ-Eq | 8.6E-2               | 3.3E-1               | 9.6E-1               |
|                            | cumulative energy demand | renewable energy resources, biomass                 | MJ-Eq | 2.2E-1               | 3.8E-1               | 6.1E-1               |
| <b>LCI results</b>         |                          |                                                     |       |                      |                      |                      |
| resource                   | Land occupation          | total                                               | m2a   | 5.9E-2               | 1.3E-1               | 2.3E-1               |
| air                        | Carbon dioxide, fossil   | total                                               | kg    | 5.8E-1               | 2.2E+0               | 5.5E+0               |
| air                        | NM VOC                   | total                                               | kg    | 2.4E-4               | 1.0E-3               | 2.4E-3               |
| air                        | Nitrogen oxides          | total                                               | kg    | 1.1E-3               | 5.3E-3               | 1.1E-2               |
| air                        | Sulphur dioxide          | total                                               | kg    | 7.9E-4               | 2.3E-2               | 4.9E-2               |
| air                        | Particulates, < 2.5 um   | total                                               | kg    | 1.4E-4               | 9.6E-4               | 1.7E-3               |
| water                      | BOD                      | total                                               | kg    | 3.2E-4               | 3.7E-3               | 1.5E-2               |
| soil                       | Cadmium                  | total                                               | kg    | 2.5E-10              | 2.1E-9               | 8.7E-9               |
| <b>Further LCI results</b> |                          |                                                     |       |                      |                      |                      |
| water                      | Chromium VI              | total                                               | kg    | 1.6E-6               | 1.5E-4               | 6.4E-4               |



The following Fig. 8.2 shows the share of the selective coating in comparison to other components of the solar system (Jungbluth (2003)). It is quite clear that the impacts from the coating can be neglected in comparison to the production of the collector itself (especially the metal sheet), other components of the system (hot water storage) and in comparison to the operation of the system. Thus the discussion for the environmental improvement of solar collector systems should rather focus on the selective coating, but more on the amount and type of other bulk materials used.



**Fig. 8.2** Share of selective coating in comparison to other aspects for a flat plate collector with a black chrome coating (Jungbluth (2003))

## 8.9 Conclusions

The inventory of metal coating processes can be considered as a basis for the assessment of this process in connection with the production of solar absorbers. A direct comparison of different selective coatings based on these data is not advisable. Some environmental aspects could not be quantified for some processes. System boundaries differ for the different processes investigated. A comparison makes only sense for the amount of heat delivered by different solar absorbers including the production and dismantling of these appliances and not for the coating process alone. Thus the product properties (e.g. absorption and emittance, life time, further materials, etc.) of the solar collector have to be taken into account if a comparison for solar absorbers with different coatings is made based on the data investigated here.

## Abbreviations

|                           |                            |
|---------------------------|----------------------------|
| $\alpha_{\text{sol}}$     | solar absorption           |
| $\varepsilon_{\text{th}}$ | thermal emittance          |
| PVD                       | Physical Vapour Deposition |

## Appendices: EcoSpold Meta Information

| ReferenceFunction  | Name                      | selective coating, copper sheet, sputtering                                                                                               | selective coating, copper sheet, physical vapour deposition                                                                                                                                 | selective coating, aluminium sheet, nickel pigmented aluminium oxide                                                                     | selective coating, copper sheet, black chrome                                                                                                                                                                                                           | selective coating, stainless steel sheet, black chrome                                                                                                                                                                                  | selective coating, copper sheet, black majic                                                          | metal coating plant                                                                                                                  |
|--------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Geography          | Location                  | DE                                                                                                                                        | DE                                                                                                                                                                                          | SK                                                                                                                                       | RER                                                                                                                                                                                                                                                     | CH                                                                                                                                                                                                                                      | US                                                                                                    | RER                                                                                                                                  |
| ReferenceFunction  | InfrastructureProcess     | 0                                                                                                                                         | 0                                                                                                                                                                                           | 0                                                                                                                                        | 0                                                                                                                                                                                                                                                       | 0                                                                                                                                                                                                                                       | 0                                                                                                     | 1                                                                                                                                    |
| ReferenceFunction  | Unit                      | m2                                                                                                                                        | m2                                                                                                                                                                                          | m2                                                                                                                                       | m2                                                                                                                                                                                                                                                      | m2                                                                                                                                                                                                                                      | m2                                                                                                    | unit                                                                                                                                 |
| DataSetInformation | Type                      | 1                                                                                                                                         | 1                                                                                                                                                                                           | 1                                                                                                                                        | 1                                                                                                                                                                                                                                                       | 1                                                                                                                                                                                                                                       | 1                                                                                                     | 1                                                                                                                                    |
|                    | Version                   | 1.0                                                                                                                                       | 1.0                                                                                                                                                                                         | 1.0                                                                                                                                      | 1.0                                                                                                                                                                                                                                                     | 1.0                                                                                                                                                                                                                                     | 1.0                                                                                                   | 1.0                                                                                                                                  |
|                    | energyValues              | 0                                                                                                                                         | 0                                                                                                                                                                                           | 0                                                                                                                                        | 0                                                                                                                                                                                                                                                       | 0                                                                                                                                                                                                                                       | 0                                                                                                     | 0                                                                                                                                    |
|                    | LanguageCode              | en                                                                                                                                        | en                                                                                                                                                                                          | en                                                                                                                                       | en                                                                                                                                                                                                                                                      | en                                                                                                                                                                                                                                      | en                                                                                                    | en                                                                                                                                   |
|                    | LocalLanguageCode         | de                                                                                                                                        | de                                                                                                                                                                                          | de                                                                                                                                       | de                                                                                                                                                                                                                                                      | de                                                                                                                                                                                                                                      | de                                                                                                    | de                                                                                                                                   |
|                    | Person                    | 41                                                                                                                                        | 41                                                                                                                                                                                          | 41                                                                                                                                       | 41                                                                                                                                                                                                                                                      | 41                                                                                                                                                                                                                                      | 41                                                                                                    | 41                                                                                                                                   |
|                    | QualityNetwork            | 1                                                                                                                                         | 1                                                                                                                                                                                           | 1                                                                                                                                        | 1                                                                                                                                                                                                                                                       | 1                                                                                                                                                                                                                                       | 1                                                                                                     | 1                                                                                                                                    |
| ReferenceFunction  | DataSetRelatesToPr        | 1                                                                                                                                         | 1                                                                                                                                                                                           | 1                                                                                                                                        | 1                                                                                                                                                                                                                                                       | 1                                                                                                                                                                                                                                       | 1                                                                                                     | 1                                                                                                                                    |
|                    | IncludedProcesses         | Gate to gate inventory for the process including energy use in the plant, infrastructure and packaging, relevant materials and emissions. | Gate to gate inventory for the process including infrastructure, excluding process specific emissions. Not including eventual pre-treatment of copper sheets by supplier, packages, wastes. | Gate to gate inventory for the process including infrastructure and relevant materials and emissions. Not including wastes and packages. | Gate to gate inventory for the process including infrastructure, wastes, effluents. Not including packages.                                                                                                                                             | Gate to gate inventory for the process including infrastructure. No information for waste and packages.                                                                                                                                 | Gate to gate inventory for the process including infrastructure, wastes. No information for packages. | Basic assumption for land use and buildings, no machinery data.                                                                      |
|                    | Amount                    | 1                                                                                                                                         | 1                                                                                                                                                                                           | 1                                                                                                                                        | 1                                                                                                                                                                                                                                                       | 1                                                                                                                                                                                                                                       | 1                                                                                                     | 1                                                                                                                                    |
|                    | LocalName                 | Selektive Beschichtung, Kupferblech, Sputtern                                                                                             | Selektive Beschichtung, Kupferblech, Vakuumverfahren                                                                                                                                        | Selektive Beschichtung, Aluminiumblech, Nickelpigmentiertes Aluminiumoxid                                                                | Selektive Beschichtung, Kupferblech, Schwarzchrom                                                                                                                                                                                                       | Selektive Beschichtung, rostfreies Stahlblech, Schwarzchrom                                                                                                                                                                             | Selektive Beschichtung, Kupferblech, Schwarzkristall                                                  | Metall-Beschichtungswerk                                                                                                             |
|                    | Synonyms                  | PVD // physical vapour deposition // sunselect                                                                                            | TINOX                                                                                                                                                                                       |                                                                                                                                          | KROSOL                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                         |                                                                                                       |                                                                                                                                      |
|                    | GeneralComment            | Inventory for selective coatings of solar absorbers.                                                                                      | Inventory for selective coatings of solar absorbers.                                                                                                                                        | Inventory for selective coatings of solar absorbers ( $\alpha=0.85-0.91$ , $\epsilon=0.23-0.40$ ).                                       | Inventory for selective coatings of solar absorbers. For black chrome and black nickel the solar absorbance is $\alpha=0.91-0.98$ and thermal emittance is $\epsilon=0.08-0.15$ .                                                                       | Inventory for selective coatings of solar absorbers. For black chrome and black nickel the solar absorbance is $\alpha=0.91-0.98$ and thermal emittance is $\epsilon=0.08-0.15$ .                                                       | Inventory for selective coatings of solar absorbers.                                                  | Inventory of plant for making selective coatings on metal sheets. Production capacity 100000 m2 of coatings per year. Life time 30a. |
|                    | InfrastructureInclude     | 1                                                                                                                                         | 1                                                                                                                                                                                           | 1                                                                                                                                        | 1                                                                                                                                                                                                                                                       | 1                                                                                                                                                                                                                                       | 1                                                                                                     | 1                                                                                                                                    |
|                    | Category                  | metals                                                                                                                                    | metals                                                                                                                                                                                      | metals                                                                                                                                   | metals                                                                                                                                                                                                                                                  | metals                                                                                                                                                                                                                                  | metals                                                                                                | metals                                                                                                                               |
|                    | SubCategory               | refinement                                                                                                                                | refinement                                                                                                                                                                                  | refinement                                                                                                                               | refinement                                                                                                                                                                                                                                              | refinement                                                                                                                                                                                                                              | refinement                                                                                            | refinement                                                                                                                           |
|                    | LocalCategory             | Metalle                                                                                                                                   | Metalle                                                                                                                                                                                     | Metalle                                                                                                                                  | Metalle                                                                                                                                                                                                                                                 | Metalle                                                                                                                                                                                                                                 | Metalle                                                                                               | Metalle                                                                                                                              |
|                    | LocalSubCategory          | Veredelung                                                                                                                                | Veredelung                                                                                                                                                                                  | Veredelung                                                                                                                               | Veredelung                                                                                                                                                                                                                                              | Veredelung                                                                                                                                                                                                                              | Veredelung                                                                                            | Veredelung                                                                                                                           |
|                    | Formula                   |                                                                                                                                           |                                                                                                                                                                                             |                                                                                                                                          |                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                         |                                                                                                       |                                                                                                                                      |
|                    | StatisticalClassification |                                                                                                                                           |                                                                                                                                                                                             |                                                                                                                                          |                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                         |                                                                                                       |                                                                                                                                      |
|                    | CASNumber                 |                                                                                                                                           |                                                                                                                                                                                             |                                                                                                                                          |                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                         |                                                                                                       |                                                                                                                                      |
| TimePeriod         | StartDate                 | 2001                                                                                                                                      | 1997                                                                                                                                                                                        | 1993                                                                                                                                     | 2000                                                                                                                                                                                                                                                    | 2001                                                                                                                                                                                                                                    | 2001                                                                                                  | 2000                                                                                                                                 |
|                    | EndDate                   | 2001                                                                                                                                      | 2000                                                                                                                                                                                        | 1993                                                                                                                                     | 2001                                                                                                                                                                                                                                                    | 2001                                                                                                                                                                                                                                    | 2001                                                                                                  | 2001                                                                                                                                 |
|                    | DataValidForEntireP       | 1                                                                                                                                         | 1                                                                                                                                                                                           | 1                                                                                                                                        | 1                                                                                                                                                                                                                                                       | 1                                                                                                                                                                                                                                       | 1                                                                                                     | 1                                                                                                                                    |
| TimePeriod         | OtherPeriodText           | Environmental report.                                                                                                                     | Publication date.                                                                                                                                                                           | Date of information.                                                                                                                     | Data for the production in these years.                                                                                                                                                                                                                 | Information provided for this year.                                                                                                                                                                                                     | Date of information.                                                                                  | Environmental report and personal communication.                                                                                     |
| Geography          | Text                      | Production plant in DE. No data for SE.                                                                                                   | Production plant in DE.                                                                                                                                                                     | Production plant in SK.                                                                                                                  | Production plants data for US, DK. No data for Gibo-Solar PL (DE).                                                                                                                                                                                      | Production plant data for CH.                                                                                                                                                                                                           | Production plant data for US.                                                                         | Production plants in DE, DK, US and CH.                                                                                              |
| Technology         | Text                      | Sputtering of chromium and tin                                                                                                            | TINOX (titanium-nitride-oxide) PVD process.                                                                                                                                                 | Highly selective, aluminium oxide-based layer pigmented with colloidal nickel                                                            | Black chrome coating with nickel underlay. Alkaline cleaning of copper to remove any oils on the surface. Neutralizing with the acid to remove any oxidation. Plating nickel on copper. Plating black chrome on the nickel plated copper. Wash and dry. | Black chrome coating. The main process stages used for the process are: Cleaning with electrolytic wash bath at ambient temperature (20°C), Activation, Black coating at ambient temperature (20°C), Recuperation baths, Reduction bath | Black Majic crystal coating with nickel underlay.                                                     | Sputtering, black majic and black chrome coating.                                                                                    |
| Representativeness | Percent                   | 70                                                                                                                                        |                                                                                                                                                                                             |                                                                                                                                          | 75                                                                                                                                                                                                                                                      | 100                                                                                                                                                                                                                                     | 100                                                                                                   |                                                                                                                                      |
|                    | ProductionVolume          | 229000m2 in 2000. 460000m2 in 2001                                                                                                        | Data for the only known company.                                                                                                                                                            | One Company on the market.                                                                                                               | 8Mio m2 in about 30a by one company. Annual production about 500000 m2.                                                                                                                                                                                 | Data for one company.                                                                                                                                                                                                                   | Data for 1 company. Annual production about 100000 m2.                                                | About 700000 m2 in 2000.                                                                                                             |
|                    | SamplingProcedure         | Environmental report and personal communication.                                                                                          | Publication.                                                                                                                                                                                | Personal communication.                                                                                                                  | Personal communication and public information                                                                                                                                                                                                           | Personal communication.                                                                                                                                                                                                                 | Personal communication and public information                                                         | Environmental report and personal communication.                                                                                     |

## 9 Soft solders (lead-free) and soldering

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### Abstract

This chapter contains the life cycle inventory information about the production of lead-free soft solders. The inventory is meant to be used as a background dataset for technical processes using soft solder. The product investigated is Sn97Cu3 that is made from tin (97%) and copper (3%). It is lead-free. The investigation is based on data provided by a producer and information from literature. Figures for the energy use and for wastes have been reported for a production plant. The use of other materials and process emissions to air have been reported in literature. General assumptions are used for transport distances. Furthermore data for the production plant were available. Energy uses of different soldering processes are discussed. But, due to the large variations possible a generic inventory has not been elaborated.

### 9.1 Introduction

This chapter contains the life cycle inventory information about the production and application of lead-free soft solders. The inventory is meant to be used as a background dataset for technical processes using soft solder. The lead-free product investigated is Sn97Cu3 that is made from tin (97%) and copper (3%). It is used i.e. for soldering of copper plates for the production of solar collectors.

### 9.2 Reserves and resources of product

Solders can be produced from a wide range of different metals. Important ingredients are tin, lead, copper, silver, cadmium, zinc and aluminium. Other metals and materials used in small proportions are i.e. phosphorus, arsenic, bismuth and iron (DIN (1997)).

### 9.3 Characterisation and use of soft solder

Relevant solders are mainly alloys of tin and lead with small amounts of other metals. They are used for a wide variety of engineering joining and repairing tasks, ranging from plumbing uses to the mass production of radiators/heat exchangers and electronic printed wiring boards (HMIP (1994)). In the context of this study soft solders used for the construction of solar absorbers are the main interest. One product that can be used for this application is Sn97Cu3 that is made from tin (97%) and copper (3%).<sup>29</sup> It does not contain lead.

### 9.4 System characterisation

The inventory is split up into the two stages of solder production and use during soldering. Data for solder production in Switzerland could not be found. Thus an European production scenario is assumed. The soldering itself is described in this report. Quantitative data from different case studies are

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<sup>29</sup> Personal communication and information provided by Frank Schröer, Felder GmbH Löttechnik.

shown. But the soldering could not be investigated as a generic inventory due to the large variation of possible processes.

#### **9.4.1 Production of solder**

Her Majesty's Inspectorate of Pollution a regulatory board in Great Britain (HMIP (1994)) provides a description for the production process of solders:

*An important production route begins with the recovery and refining of scrap tin and lead materials. The process and its releases are outlined on the upper part of block flow diagram (Fig. 9.1); raw materials are any suitable scrap tin or lead bearing items, alloys and drosses. Frequently dross recovery services are provided to end users by the solder manufacturer.*

*Metallic scrap and highly metallic drosses are melted in kettles which are indirectly heated by oil, gas or electricity. Material with high dross or flux content is melted in sloping hearth reverberatory furnaces or rotary furnaces. These are directly heated using oil or gas. At the temperatures involved, organic type fluxes are vaporised or burnt off and appropriate gas treatment is necessary. During melting, oxide and residual fluxes float to the surface of the metal and are removed by drossing into drums or special containers using rakes, perforated ladles or, on large installations, machines.*

*The molten alloy is best refined, if necessary, by pyrometallurgical techniques although electrolytic refining is an effective alternative. In pyrometallurgical refining, the temperature is first lowered to precipitate high melting point impurities present in some quantity, such as iron and copper, which collect on the surface and are removed. Additions of chemicals at specific temperatures are then used to remove the remaining impurities. For example, sulphur is added to remove residual copper as the sulphide and ammonium chloride to eliminate zinc as zinc chloride. These form drosses which are sold for recovery of the metal. Sodium carbonate is added at the conclusion of the drossing to take up any excess chemicals.*

*Electrolytic refining uses anodes of crude solder and thin sheets of refined solder as the cathodes in a cell in which the electrolyte of hydrofluorosilicic acid or hydrofluoroboric acid is circulated. Tin and lead are deposited on the cathode whilst the impurities remain on the anode as a slime. Periodically the cathodes are removed, washed with water and melted. The impurity slime is brushed from the anode, washed and filtered. All metallic process waste is normally sold for recovery of the metal. Refined solder is cast into master alloy ingots by pumping or pouring into moulds followed by an appropriate air or water cooling method.*

*Solders are manufactured to national, international and company standard specifications. There is an end user driven move to lead-free or low lead alloys where this is technically acceptable, e.g. in gas plumbing uses. The annual usage ratio of bought-in to recovered materials varies markedly between different producers from over 90% to under 50%.*

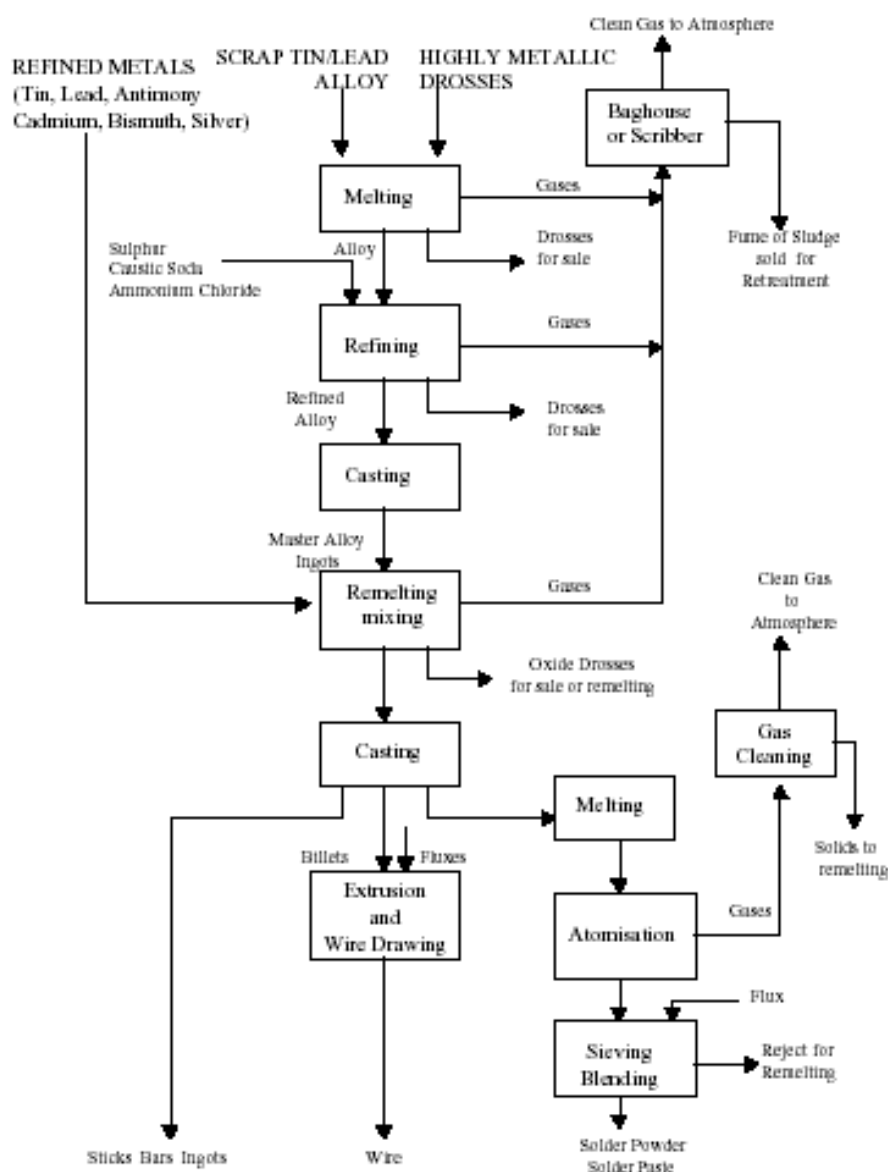


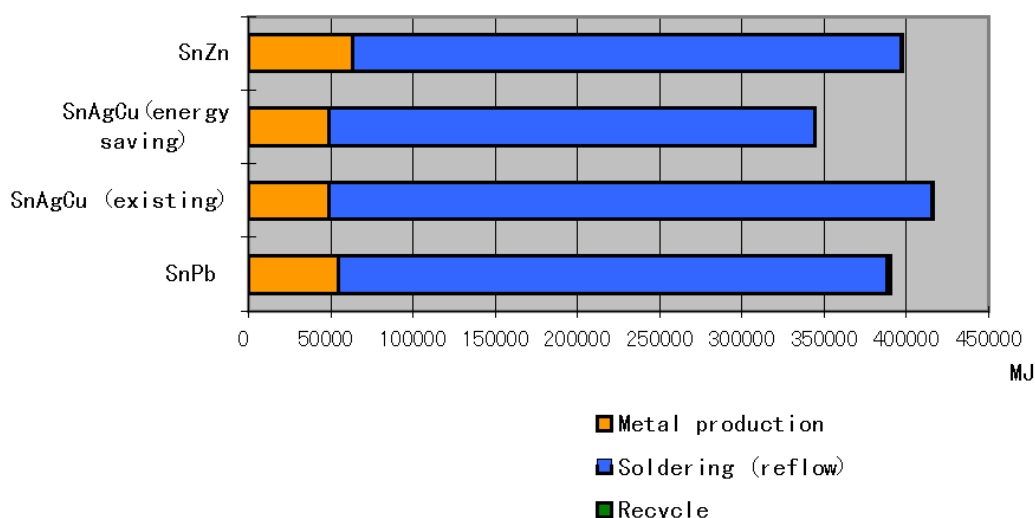
Fig. 9.1 Typical flow sheet for solder manufacturing (HMIP (1994))

### 9.4.2 Soldering

For the use of solders the solder itself and the materials to be fixed have to be heated to a temperature higher than the melting point of the solder. The source of heat depends on the type of application.

## 9.5 Life cycle inventories

A review of existing studies for the life cycle of solder products shows that the production of metals and the use of solder are the most important stages in the life cycle. The direct energy use for the production of the solder is not so important. Fig. 9.2 shows an investigation of different types of solders used in the production of printed wiring boards (PWB) (Hamano et al. (2001)).



**Fig. 9.2** The difference of total energy consumption of PWB. Use of 400kg solder in one reflow line (Hamano et al. (2001))

The life cycle inventory is investigated for the production of lead-free solders. The soldering itself could not be investigated with quantitative data because it depends on the type of process used which might vary considerable between different producers and for different products.

### 9.5.1 Production of lead-free soft solder

The life cycle inventory for the production of soft solder (Sn97Cu3) is based on the information provided by a manufacturer in Germany (see footnote 29) and some additional information from literature. These basic information are shown on the right side of Tab. 7.1.

Seegerberg (1995) has estimated the electricity use for the production of solder (solder mixing, casting and atomisation) to be 0.325kWh/kg of solder. Oelichmann (1997) has used this inventory and extended it for the use of some more auxiliary materials. He calculated an electricity use of about 0.5kWh/kg. Data for a new process for the atomisation of solder powders are provided by a company (Atomising Systems Ltd (2002)). This process uses about 0.17kWh electricity per kg metal. Other estimates give the energy use for solder production to be 2 and 5 MJ/kg.<sup>30</sup> The Felder GmbH produces about 2000t of this type of solder (Sn97Cu3) per year (which makes only a small part (1%) of its total production). About 500MWh (0.25kWh/kg) of electricity and 1666GJ (0.83MJ/kg) of natural gas are used for the production of this specific product in one year. These figures are used in the inventory.

The new process of (Atomising Systems Ltd (2002)) uses about 1.4l of inert gas per kg metal. HMIP (1994) shows data for maximum releases to the air during the production of solders (Tab. 9.1). Data for air emissions related to a specific amount of product were not available. Process specific emissions have been estimated with the maximum releases (Tab. 9.1) and the flow rate of inert gas (0.62m<sup>3</sup> nitrogen per kg or 0.8kg/kg, investigated by Oelichmann (1997)). The emissions of ammonia, hydrogen chloride, NMVOC, and the main metals copper and tin are calculated with the flow rate of nitrogen for the specific product investigated here.

<sup>30</sup> Personal communication Markus Stutz, Motorola, 17.4.2002 and Email by Otmar Deubzer, IZM, Berlin, 31.5.2002.

**Tab. 9.1 Maximum releases of the process to the air HMIP (1994)**

| Substance                                                                          | Abated release concentration mg/m <sup>3</sup> |
|------------------------------------------------------------------------------------|------------------------------------------------|
| Total particulate matter                                                           | 10                                             |
| Lead and its compounds                                                             | 2                                              |
| Antimony, copper, tin and their compounds taken together                           | 5                                              |
| Arsenic and nickel and their compounds taken together                              | 1                                              |
| Chlorine                                                                           | 10                                             |
| Chlorides (as HCl)                                                                 | 30                                             |
| Ammonia                                                                            | 15                                             |
| Volatile organic compounds (measured as total carbon excluding particulate matter) | 50                                             |

Tab. 7.1 shows all unit process raw data for the production of soft solders. The metal use has been calculated from the content in the product and a loss as metal ashes of 1%. The metal content of the solder is 90%, the rest is flux. Transport distances have been used according to the general guidelines. The producer provided also data for the use of water for cooling. Metal ashes (10g/kg) and packages (0.01m<sup>3</sup>/kg) which are recycled according to the information of a producer are not considered as disposed wastes for the inventory. Data for other wastes have been reported by the producer.

### 9.5.2 Solder production plant

The inventory for the land use of a solder production plant has been elaborated according to the general guidelines. Data for the land use of one production plant (3300m<sup>2</sup> of buildings and a total area of 4500m<sup>2</sup>) have been reported by the producer. The production of 2000t is equal to 1% share of the total production. Thus the total production of this plant is estimated to be 200'000t/a. Tab. 7.1 shows the unit process raw data for a generic solder production plant with a production capacity of 100000 tonnes per year and a life time of 50 years.

### 9.5.3 Soldering

Seegerberg (1995) has estimated the electricity use for the use of solder in screen printing to be 5.58kWh/kg of solder. Hamano et al. (2001) have investigated the use of solder during printing wiring board (PWB) production (Tab. 9.2). They reported a much higher specific energy use between 739 and 918MJ/kg which makes this process to the most important one in the life cycle of solder products. Oelichmann (1997) reported an electricity use of about 200kWh per kg of used solder for the reflow soldering process.

**Tab. 9.2 The energy data of one reflow line Hamano et al. (2001). Type of energy use not specified**

| MJ/kg                  | Soldering (reflow) |
|------------------------|--------------------|
| SnPb                   | 835                |
| SnAgCu (existing)      | 918                |
| SnAgCu (energy saving) | 739                |
| SnZn                   | 835                |

Soft solder is used for the construction of solar collectors. Here a copper sheet and copper tubes are fixed by solder. The two components can be heated up together in an oven, by an electric heater or they can be soldered by using heat from propane or hydrogen combustion. The energy use might vary considerable between different companies due to different technologies and the standard of energy saving technologies.<sup>29</sup>

Fig. 9.3 shows the manufacturing of a solar collector with copper pipes (Soltop (2002)).



**Fig. 9.3 Soldering of solar collectors**

Generic unit process raw data for the use of soft solders could not be investigated. The energy use, the type of energy and the equipment used show great variations, which makes it impossible to provide a generic inventory. Instead it is recommended to investigate the energy use for melting separately for each process using soft solders with plant specific data.



Tab. 9.3 Unit process raw data and data quality indicators for the production and use of soft solders and the solder production plant. References are presented in the text

|                           | Name                                                                    | Location | Infrastructure | Unit | soft solder, Sn97Cu3, at plant | solder production plant | Uncertainty Standard deviation | GeneralComment                                                                   | soft solder | atomisation                 | solder mixing and casting | atomisation and mixing | solder  | soldering |
|---------------------------|-------------------------------------------------------------------------|----------|----------------|------|--------------------------------|-------------------------|--------------------------------|----------------------------------------------------------------------------------|-------------|-----------------------------|---------------------------|------------------------|---------|-----------|
|                           | Location InfrastructureProcess Unit                                     |          |                |      | RER 0 kg                       | RER 1 unit              |                                |                                                                                  | DE 0 a      | UK 0 kg                     | SE 0 kg                   | SE 0 kg                | SE 0 kg | SE 0 kg   |
| output                    | soft solder, Sn97Cu3, at plant                                          | RER      | 0              | kg   | 1.00E+0                        | 0                       |                                |                                                                                  | 2.00E+6     | 1.00E+0                     | 1.00E+0                   | 1.00E+0                | 1.00E+0 | 1.00E+0   |
|                           | solder production plant                                                 | RER      | 1              | unit | 0                              | 1.00E+0                 |                                |                                                                                  | -           | -                           | -                         | -                      | -       | -         |
| technosphere              | natural gas, burned in industrial furnace >100kW                        | RER      | 0              | MJ   | 8.28E-1                        | -                       | 1                              | 1.10 (2,3,1,3,1,3); Questionnaire                                                | 1.66E+6     | -                           | -                         | -                      | -       | -         |
|                           | electricity, medium voltage, production UCTE, at grid                   | UCTE     | 0              | kWh  | 2.50E-1                        | -                       | 1                              | 1.10 (2,3,1,3,1,3); Questionnaire                                                | 5.00E+5     | 1.67E-1                     | 3.60E-1                   | 1.52E-1                | 3.25E-1 | 5.58E+0   |
|                           | chemicals organic, at plant                                             | GLO      | 0              | kg   | 1.11E-1                        | -                       | 1                              | 1.14 (3,3,2,3,1,3); Assumption for additives based on literature                 | -           | -                           | -                         | 1.11E-1                | -       | -         |
|                           | nitrogen, liquid, at plant                                              | RER      | 0              | kg   | 7.74E-1                        | -                       | 1                              | 1.14 (3,3,2,3,1,3); Literature, Inert gas                                        | -           | 1.74E-3                     | -                         | 7.74E-1                | -       | -         |
|                           | copper, at regional storage                                             | RER      | 0              | kg   | 2.73E-2                        | -                       | 1                              | 1.10 (2,3,1,3,1,3); 1% losses                                                    | 5.45E+4     | -                           | 4.98E-3                   | -                      | -       | -         |
|                           | tin, at regional storage                                                | RER      | 0              | kg   | 8.82E-1                        | -                       | 1                              | 1.10 (2,3,1,3,1,3); 1% losses                                                    | 1.76E+6     | -                           | 9.00E-1                   | -                      | -       | -         |
|                           | tap water, at user                                                      | RER      | 0              | kg   | 7.50E-3                        | -                       | 1                              | 1.10 (2,3,1,3,1,3); Produced waste water                                         | 1.50E+4     | -                           | -                         | -                      | -       | -         |
|                           | transport, lorry 32t                                                    | RER      | 0              | tkm  | 1.41E-1                        | -                       | 1                              | 2.09 (4,5,na,na,na,na); Standard distance 100km, 50km                            | -           | -                           | -                         | -                      | -       | -         |
|                           | transport, freight, rail                                                | RER      | 0              | tkm  | 3.26E-1                        | -                       | 1                              | 2.09 (4,5,na,na,na,na); Standard distance 100, 200, 600km                        | -           | -                           | -                         | -                      | -       | -         |
|                           | solder production plant                                                 | RER      | 1              | unit | 2.00E-10                       | -                       | 1                              | 3.01 (2,3,1,3,1,3); Calculation for plant with 100'000t/a and a life time of 50a | -           | -                           | -                         | -                      | -       | -         |
|                           | disposal, municipal solid waste, 22.9% water, to municipal incineration | CH       | 0              | kg   | 2.50E-2                        | -                       | 1                              | 1.10 (2,3,1,3,1,3); Unspecified wastes                                           | 5.00E+4     | -                           | -                         | -                      | -       | -         |
|                           | disposal, separator sludge, 90% water, to hazardous waste incineration  | CH       | 0              | kg   | 7.50E-3                        | -                       | 1                              | 1.10 (2,3,1,3,1,3); Waste water from production                                  | 1.50E+4     | -                           | -                         | -                      | -       | -         |
| emission air, unspecified | Heat, waste                                                             | -        | -              | MJ   | 9.00E-1                        | -                       | 1                              | 1.10 (2,3,1,3,1,3); Calculation                                                  | -           | -                           | -                         | -                      | -       | -         |
|                           | Ammonia                                                                 | -        | -              | kg   | 9.25E-6                        | -                       | 1                              | 2.33 (5,3,1,3,5,5); Maximum estimation in inert gas flow                         | -           | -                           | -                         | -                      | -       | -         |
|                           | Copper                                                                  | -        | -              | kg   | 9.29E-8                        | -                       | 1                              | 6.10 (5,3,1,3,5,5); Estimation (3% of 5mg/m3 in flow rate of inert gas)          | -           | -                           | -                         | -                      | -       | -         |
|                           | Hydrogen chloride                                                       | -        | -              | kg   | 1.86E-5                        | -                       | 1                              | 2.51 (5,3,1,3,5,5); Maximum estimation in inert gas flow                         | -           | -                           | -                         | -                      | -       | -         |
|                           | NMVOOC, non-methane volatile organic compounds, unspecified origin      | -        | -              | kg   | 3.08E-5                        | -                       | 1                              | 2.51 (5,3,1,3,5,5); Maximum estimation in inert gas flow                         | -           | -                           | -                         | -                      | -       | -         |
|                           | Particulates, > 10 um                                                   | -        | -              | kg   | 6.17E-6                        | -                       | 1                              | 2.94 (5,3,1,3,5,5); Maximum estimation in inert gas flow                         | -           | -                           | -                         | -                      | -       | -         |
|                           | Tin                                                                     | -        | -              | kg   | 3.00E-6                        | -                       | 1                              | 6.10 (5,3,1,3,5,5); Estimation (97% of 5mg/m3 in flow rate of inert gas)         | -           | -                           | -                         | -                      | -       | -         |
| resource, land            | Occupation, industrial area, built up                                   | -        | -              | m2a  | -                              | 8.25E+4                 | 1                              | 1.51 (2,3,1,3,1,3); Standard 50a occupation                                      | 1.65E+3     | -                           | -                         | -                      | -       | -         |
|                           | Occupation, industrial area, vegetation                                 | -        | -              | m2a  | -                              | 3.00E+4                 | 1                              | 1.51 (2,3,1,3,1,3); Standard 50a occupation                                      | 6.00E+2     | -                           | -                         | -                      | -       | -         |
|                           | Transformation, from unknown                                            | -        | -              | m2   | -                              | 2.25E+3                 | 1                              | 2.01 (2,3,1,3,1,3); Questionnaire                                                | 4.50E+1     | -                           | -                         | -                      | -       | -         |
|                           | Transformation, to industrial area, built up                            | -        | -              | m2   | -                              | 1.65E+3                 | 1                              | 2.01 (2,3,1,3,1,3); Questionnaire                                                | 3.30E+1     | -                           | -                         | -                      | -       | -         |
|                           | Transformation, to industrial area, vegetation                          | -        | -              | m2   | -                              | 6.00E+2                 | 1                              | 2.01 (2,3,1,3,1,3); Questionnaire                                                | 1.20E+1     | -                           | -                         | -                      | -       | -         |
| technosphere              | building, hall                                                          | CH       | 1              | m2   | -                              | 6.00E+2                 | 1                              | 3.01 (2,3,1,3,1,3); Questionnaire                                                | -           | -                           | -                         | -                      | -       | -         |
| source                    |                                                                         |          |                |      |                                |                         |                                |                                                                                  | Felder      | Atomising Systems Ltd. 2002 | Oelichmann 1997           | Seegerberg 1995        |         |           |

## 9.6 Data quality considerations

Tab. 7.1 shows the data quality indicators for the inventory of production and use of soft solders. The simplified approach with a pedigree matrix has been used for calculating the standard deviation. Large uncertainties exist for data on emissions and for the transport distances of materials.

## 9.7 Conclusions

The inventory for the production of soft solders can be considered as a reliable basis for the assessment of this material in connection with technical applications. It is not meant to be used for the comparison of different solders. For that, detailed information about product properties and type of application would be necessary.

## Abbreviations

DIN (1997) is a valuable source for the description of abbreviations in the context of soldering materials.

## Appendices: EcoSpold meta information

|                    |                                |                                                                                                                                                                                                                                                                                            |                                                                                                        |
|--------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| ReferenceFunction  | 401 Name                       | soft solder, Sn97Cu3, at plant                                                                                                                                                                                                                                                             | solder production plant                                                                                |
| Geography          | 662 Location                   | RER                                                                                                                                                                                                                                                                                        | RER                                                                                                    |
| ReferenceFunction  | 493 InfrastructureProcess      | 0                                                                                                                                                                                                                                                                                          | 1                                                                                                      |
| ReferenceFunction  | 403 Unit                       | kg                                                                                                                                                                                                                                                                                         | unit                                                                                                   |
| DataSetInformation | 201 Type                       | 1                                                                                                                                                                                                                                                                                          | 1                                                                                                      |
|                    | 202 Version                    | 1.0                                                                                                                                                                                                                                                                                        | 1.0                                                                                                    |
|                    | 203 energyValues               | 0                                                                                                                                                                                                                                                                                          | 0                                                                                                      |
|                    | 205 LanguageCode               | en                                                                                                                                                                                                                                                                                         | en                                                                                                     |
|                    | 206 LocalLanguageCode          | de                                                                                                                                                                                                                                                                                         | de                                                                                                     |
| DataEntryBy        | 302 Person                     | 41                                                                                                                                                                                                                                                                                         | 41                                                                                                     |
|                    | 304 QualityNetwork             | 1                                                                                                                                                                                                                                                                                          | 1                                                                                                      |
| ReferenceFunction  | 400 DataSetRelatesToProduct    | 1                                                                                                                                                                                                                                                                                          | 1                                                                                                      |
|                    | 402 IncludedProcesses          | Gate to gate including solder mixing and casting, atomisation and paste mixing, including estimate for emissions and infrastructure.                                                                                                                                                       | Infrastructure of a solder production plant including land use and buildings, not including equipment. |
|                    | 404 Amount                     | 1                                                                                                                                                                                                                                                                                          | 1                                                                                                      |
|                    | 490 LocalName                  | Weichlot, Sn97Cu3, ab Werk                                                                                                                                                                                                                                                                 | Produktionsanlage Lötmaterialien                                                                       |
|                    | 491 Synonyms                   |                                                                                                                                                                                                                                                                                            |                                                                                                        |
|                    | 492 GeneralComment             | The inventory is meant to be used as a background dataset for technical processes using soft solder. The lead-free product investigated is Sn97Cu3 that is made from tin (97%) and copper (3%). The investigation is based on data provided by a producer and information from literature. | Inventory for a production plant with an annual capacity of 100'000t and a life time of 50a.           |
|                    | 494 InfrastructureIncluded     |                                                                                                                                                                                                                                                                                            |                                                                                                        |
|                    | 495 Category                   | metals                                                                                                                                                                                                                                                                                     | metals                                                                                                 |
|                    | 496 SubCategory                | extraction                                                                                                                                                                                                                                                                                 | refinement                                                                                             |
|                    | 497 LocalCategory              | Metalle                                                                                                                                                                                                                                                                                    | Metalle                                                                                                |
|                    | 498 LocalSubCategory           | Gewinnung                                                                                                                                                                                                                                                                                  | Veredelung                                                                                             |
|                    | 499 Formula                    |                                                                                                                                                                                                                                                                                            |                                                                                                        |
|                    | 501 StatisticalClassification  |                                                                                                                                                                                                                                                                                            |                                                                                                        |
|                    | 502 CASNumber                  |                                                                                                                                                                                                                                                                                            |                                                                                                        |
| TimePeriod         | 601 StartDate                  | 1995                                                                                                                                                                                                                                                                                       | 2000                                                                                                   |
|                    | 602 EndDate                    | 2000                                                                                                                                                                                                                                                                                       | 2000                                                                                                   |
|                    | 603 DataValidForEntirePeriod   | 1                                                                                                                                                                                                                                                                                          | 1                                                                                                      |
|                    | 611 OtherPeriodText            | Time of publications. Most data for 2000.                                                                                                                                                                                                                                                  | Data for 2000                                                                                          |
| Geography          | 663 Text                       | Producer in Germany and literature data for Europe                                                                                                                                                                                                                                         | Producer in Germany.                                                                                   |
| Technology         | 692 Text                       | Production of lead-free soft solder from metals tin and copper.                                                                                                                                                                                                                            | Infrastructure building and land use.                                                                  |
| Representativeness | 722 Percent                    |                                                                                                                                                                                                                                                                                            |                                                                                                        |
|                    | 724 ProductionVolume           | 2000t for this product.                                                                                                                                                                                                                                                                    | 100000 tonnes per year.                                                                                |
|                    | 725 SamplingProcedure          | Questionnaire and literature                                                                                                                                                                                                                                                               | Questionnaire                                                                                          |
|                    | 726 Extrapolations             | From one producer to European average.                                                                                                                                                                                                                                                     | From one producer to European average.                                                                 |
|                    | 727 UncertaintyAdjustments     | none                                                                                                                                                                                                                                                                                       | none                                                                                                   |
| DataGenerator      | 751 Person                     | 41                                                                                                                                                                                                                                                                                         | 41                                                                                                     |
| AndPublication     | 756 DataPublishedIn            | 2                                                                                                                                                                                                                                                                                          | 2                                                                                                      |
|                    | 757 ReferenceToPublishedSource | 10                                                                                                                                                                                                                                                                                         | 10                                                                                                     |
|                    | 758 Copyright                  | 1                                                                                                                                                                                                                                                                                          | 1                                                                                                      |
|                    | 759 AccessRestrictedTo         | 0                                                                                                                                                                                                                                                                                          | 0                                                                                                      |
|                    | 760 CompanyCode                |                                                                                                                                                                                                                                                                                            |                                                                                                        |
|                    | 761 CountryCode                |                                                                                                                                                                                                                                                                                            |                                                                                                        |
|                    | 762 PageNumbers                | Soft Solders                                                                                                                                                                                                                                                                               | Soft Solders                                                                                           |

## 10 Powder coating

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### 10.1 Definition and fundamentals of powder coating

In powder coating the coating material is applied to the workpiece in the form of dry thermoplastic or thermosetting powder. The powder particles are heated and melt to form a sealed film. The thermoplastic powders melt and fuse on heating whereas the thermosetting powders forms chemical cross-links. Two main application processes are used: electrostatic spraying and fluidized-bed coating. (Meyer (1997))

The cleaning of the workpiece before coating can either be done mechanically (e.g. steam / high pressure water blasting) or chemically. Chemical degreasing includes solvent degreasing, alkaline degreasing, pickling, phosphating and chromating. The last two methods are mainly applied to enhance corrosion resistance either of the workpiece between degreasing and coating or after coating.

#### 10.1.1 Environmental issues of powder coating

The cleaning of the workpiece is likely to be the most important environmental issue of powder coating. Depending on the method used this step emits VOC to air or metals and hydrocarbons to water.

According to Meyer (1997), powder coating is the coating method that produces the least environmental pollution because the occurrence of paint slurry is avoided, the separation of dust – air mixtures is simple, and the coating powders are solvent-free and generally cure by polyaddition.

Dust emissions from powder coating are generally below 5 mg coating powder per cubic meter of exhaust air. Emissions from the driers (basically organics) are generally also extremely low.

Coating powders without hazardous components (e.g., lead pigments) do generally not have toxic or fibrogenic effects. The fraction of respirable dust that can penetrate the lungs is  $\ll 10\%$  (w/w).

### 10.2 Coating powder

Thermosetting coating powders are produced exclusively by the extruder process. Most commonly used are powders on the basis of epoxy-polyester (ca. 60%), polyester-trisglycidyl isocyanurate (ca. 20%), epoxy (ca. 15%) and polyester-polyurethane (ca. 5%). (Percentage of European production in 1998 (Meyer (1997))).

Inorganic pigments (e.g., titanium dioxide, iron oxides, and chromium oxides) are best suited for coating powders because of their thermal stability, their inert behaviour towards other constituents of the powder, their mechanical stability to withstand the extrusion process and because they don't affect the binder viscosity too much. Lead pigments are sometimes still used for exterior-grade powder coatings. In Europe the use of cadmium pigments is declining.

A 1:1 mixture of epoxy and polyester resin is chosen for modelling the powder inecoinvent. Powders of this composition are mainly used for exterior architectural applications. The pigment inventoried is  $\text{TiO}_2$ . Thus the coating powder inecoinvent stands for a white powder for applications e.g. in the building sector.

The background information to the process "coating powder, at plant" is given in Tab. 10.1, flows are shown in Fig. 10.1.

## Data quality

The uncertainty is estimated by completing the pedigree matrix, according to (Frischknecht et al. (2003)). A lognormal distribution is assumed.

**Tab. 10.1** Ecoinvent meta information for "coating powder, at plant"

| Name                                  | coating powder, at plant                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location                              | RER                                                                                                                                                                                                                                                                                                                                                                                     |
| Infrastructure Process                | 0                                                                                                                                                                                                                                                                                                                                                                                       |
| Unit                                  | kg                                                                                                                                                                                                                                                                                                                                                                                      |
| Data Set Version                      | 2.0                                                                                                                                                                                                                                                                                                                                                                                     |
| Included Processes                    | Powder fabrication and packing in buckets. The functional unit refers to 1 kg of powder. For transports the buckets (0.0133 kg/kg powder) have to be taken into account separately                                                                                                                                                                                                      |
| Amount                                | 1                                                                                                                                                                                                                                                                                                                                                                                       |
| Local Name                            | Pulverlack, für Pulverbeschichten                                                                                                                                                                                                                                                                                                                                                       |
| Synonyms                              |                                                                                                                                                                                                                                                                                                                                                                                         |
| General Comment to reference function | A 1:1 mixture of epoxy and polyester resin is assumed. Powders of this composition are mainly used for exterior architectural applications. The pigment inventoried is TiO <sub>2</sub> . Other pigments (and thus other colours) could have very different ecoprofiles. Since many different powders are available on the market, the representation of a specific powder can be poor. |
| Start Date                            | 1996                                                                                                                                                                                                                                                                                                                                                                                    |
| End Date                              | 2003                                                                                                                                                                                                                                                                                                                                                                                    |
| Data Valid For Entire Period          | 1                                                                                                                                                                                                                                                                                                                                                                                       |
| Other Period Text                     |                                                                                                                                                                                                                                                                                                                                                                                         |
| Geography text                        | Swiss waste treatment assumed                                                                                                                                                                                                                                                                                                                                                           |
| Technology text                       | Extrusion and grinding. This technology is commonly used with thermosetting powders.                                                                                                                                                                                                                                                                                                    |
| Representativeness [%]                |                                                                                                                                                                                                                                                                                                                                                                                         |
| Production Volume                     | Unknown                                                                                                                                                                                                                                                                                                                                                                                 |
| Sampling Procedure                    | Literature                                                                                                                                                                                                                                                                                                                                                                              |
| Extrapolations                        | None                                                                                                                                                                                                                                                                                                                                                                                    |
| Uncertainty Adjustments               | None                                                                                                                                                                                                                                                                                                                                                                                    |

| General flow information   |              |                          |                                             | Representation in ecoinvent |                              |                |          |                                                                             |            |      |                                         | Uncertainty |          |                        |  |
|----------------------------|--------------|--------------------------|---------------------------------------------|-----------------------------|------------------------------|----------------|----------|-----------------------------------------------------------------------------|------------|------|-----------------------------------------|-------------|----------|------------------------|--|
| Input                      | Process Name | Output                   | Remarks                                     | Category                    | Sub category                 | Infrastructure | Location | Module name in ecoinvent                                                    | Mean value | Unit | Source mean value                       | Type        | StdV 95% | General Comment        |  |
| Epoxy resin                | m            | coating powder, at plant |                                             | plastics                    | monomers                     | No             | RER      | epoxy resin, liquid, at plant                                               | 2.93E-01   | kg   | assumption based on Richter et al. 1996 | 1           | 1.24     | (3,3,1,1,1,5,4)        |  |
| Polyester resin            | m            |                          |                                             | paintings                   | production                   | No             | RER      | polyester resin, unsaturated, at plant                                      | 2.93E-01   | kg   | assumption based on Richter et al. 1996 | 1           | 1.24     | (3,3,1,1,1,5,4)        |  |
| pigment                    | m            |                          | TiO2 as pigment                             | chemicals                   | inorganics                   | No             | RER      | titanium dioxide, production mix, at plant                                  | 3.54E-01   | kg   | assumption based on Richter et al. 1996 | 1           | 1.24     | (3,3,1,1,1,5,4)        |  |
| Additives                  | m            |                          | binder, defrothing agent, stabilizers       | chemicals                   | organics                     | No             | GLO      | chemicals organic, at plant                                                 | 6.97E-02   | kg   | assumption based on Richter et al. 1996 | 1           | 1.24     | (3,3,1,1,1,5,4)        |  |
| extrusion                  | m            |                          |                                             | plastics                    | processing                   | No             | RER      | extrusion, plastic pipes                                                    | 1.01E+00   | kg   | calculation                             | 1           | 1.33     | (3,3,1,1,3,5,2)        |  |
| electricity                | m            |                          | grinding and general consumption            | electricity                 | production mix               | No             | UCTE     | electricity, medium voltage, production UCTE, at grid                       | 1.83E+00   | kWh  | assumption based on extrusion process   | 1           | 1.33     | (3,3,1,1,3,5,2)        |  |
| plastic bucket, material   | m            |                          | assumption: buckets of 15 kg at 0.2 kg tara | plastics                    | polymers                     | No             | RER      | polyethylene, LDPE, granulate, at plant                                     | 1.33E-02   | kg   | assumption                              | 1           | 1.33     | (3,3,1,1,3,5,3)        |  |
| plastic bucket, production | m            |                          |                                             | plastics                    | processing                   | No             | RER      | extrusion, plastic pipes                                                    | 1.33E-02   | kg   | calculation                             | 1           | 1.33     | (3,3,1,1,3,5,2)        |  |
| Transport                  | m            |                          | Raw and packaging materials 100 km          | transport systems           | road                         | No             | RER      | transport, lorry 32t                                                        | 1.02E-01   | tkm  | assumption                              | 1           | 2.09     | (4,5,n.a.,n.a.,n.a.,5) |  |
| Transport                  | m            |                          | Raw and packaging materials 600 km          | transport systems           | train                        | No             | RER      | transport, freight, rail                                                    | 6.14E-01   | tkm  | assumption                              | 1           | 2.09     | (4,5,n.a.,n.a.,n.a.,5) |  |
| Infrastructure             | m            |                          |                                             | chemicals                   | organics                     | Yes            | RER      | chemical plant, organics                                                    | 4.00E-10   | unit | assumption                              | 1           | 3.00     | (5,5,5,5,5,10)         |  |
|                            | m            | waste heat               |                                             | air                         | unspecified                  |                |          | Heat, waste                                                                 | 6.59E+00   | MJ   |                                         | 1           | 1.33     | (3,3,1,1,3,5,13)       |  |
|                            | m            | NM VOC                   |                                             | air                         | unspecified                  |                |          | NM VOC, non-methane volatile organic compounds, unspecified origin          | 1.00E-04   | kg   | assumption                              | 1           | 3.22     | (5,5,5,5,5,23)         |  |
|                            | m            | production waste         |                                             | waste management            | hazardous waste incineration | No             | CH       | disposal, emulsion paint remains, 0% water, to hazardous waste incineration | 1.00E-02   | kg   | assumption                              | 1           | 2.56     | (5,5,5,5,5,6)          |  |
|                            | m            | coating powder, packed   |                                             | metals                      | processing                   | No             | RER      | coating powder, at plant                                                    | 1.00E+00   | kg   |                                         |             |          |                        |  |

Fig. 10.1 Flows for "coating powder, at plant" and their representation in the ecoinvent database

### 10.3 Powder coating of steel

The 2 mm thick steel sheet is cleaned by alkaline degreasing and pickling. The data for these two processes from zinc coating of steel are used. The sheet is coated with a layer of 80 µm and cured at 220 °C. The heat energy consumption for the curing is calculated using the energy consumption reported by a Swiss producer<sup>31</sup>. Since the data in this source is given for steel and aluminium together, the curing temperature and the thermal capacity of the steel and aluminium sheet are used to estimate the share of the energy demand for each of these materials.

The dataset includes neither the steel sheet nor its transport to the coating plant.

The background information to the process "powder coating, steel" is given in Tab. 10.2, flows are shown in Fig. 10.2.

#### Data quality

The uncertainty is estimated by completing the pedigree matrix, according to (Frischknecht et al. (2003)). A lognormal distribution is assumed.

Tab. 10.2 Ecoinvent meta information for "powder coating, steel"

| Name                                  | powder coating, steel                                                                                                                                                           |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location                              | RER                                                                                                                                                                             |
| Infrastructure Process                | 0                                                                                                                                                                               |
| Unit                                  | m <sup>2</sup>                                                                                                                                                                  |
| Data Set Version                      | 2.0                                                                                                                                                                             |
| Included Processes                    | Phosphating of the steel sheet, powder coating and heat curing. No transports of the steel sheet to the coating plant or back are inventoried.                                  |
| Amount                                | 1                                                                                                                                                                               |
| Local Name                            | Pulverbeschichten, Stahl                                                                                                                                                        |
| Synonyms                              |                                                                                                                                                                                 |
| General Comment to reference function | The powder inventoried is mainly used in exterior architectural applications. The coating thickness is 80 µm. The heat consumption is calculated for a sheet of 2 mm thickness. |
| Start Date                            | 1996                                                                                                                                                                            |
| End Date                              | 2003                                                                                                                                                                            |
| Data Valid For Entire Period          | 1                                                                                                                                                                               |
| Other Period Text                     |                                                                                                                                                                                 |
| Geography text                        | Swiss waste treatment assumed                                                                                                                                                   |
| Technology text                       | Phosphating, coating and heat curing. Treatment of the waste water is assumed.                                                                                                  |
| Representativeness [%]                |                                                                                                                                                                                 |
| Production Volume                     | Unknown                                                                                                                                                                         |
| Sampling Procedure                    | Literature                                                                                                                                                                      |
| Extrapolations                        | None                                                                                                                                                                            |
| Uncertainty Adjustments               | None                                                                                                                                                                            |

<sup>31</sup> [http://www.schweizer-metallbau.ch/framedef.cfm?Navistring=06&aktuell\\_content=06](http://www.schweizer-metallbau.ch/framedef.cfm?Navistring=06&aktuell_content=06), accessed 9.2.2004

| General flow information                     |                               |                                    |                                                       | Representation in ecoinvent |                              |                |          |                                                                      |            |      |                                                     | Uncertainty information |          |                        |  |
|----------------------------------------------|-------------------------------|------------------------------------|-------------------------------------------------------|-----------------------------|------------------------------|----------------|----------|----------------------------------------------------------------------|------------|------|-----------------------------------------------------|-------------------------|----------|------------------------|--|
| Input                                        | Process Name                  | Output                             | Remarks                                               | Category                    | Sub category                 | Infrastructure | Location | Modul name in ecoinvent                                              | Mean value | Unit | Source mean value                                   | Type                    | StDv 95% | General Comment        |  |
| coating powder                               | powder coating, steel, part 1 |                                    | coating of 80 um, density 1200 kg/m3, 0.05%loss       | metals                      | processing                   | No             | RER      | coating powder, at plant                                             | 9.65E-02   | kg   | calculation                                         | 1                       | 2.39     | (4,5,5,5,5,4)          |  |
| HCl                                          |                               |                                    | for degreasing / pickling                             | chemicals                   | inorganics                   | No             | RER      | hydrochloric acid, 30%in H2O, at plant                               | 3.40E-01   | kg   | IPPC 2001                                           | 1                       | 124      | (3,3,1,1,5,4)          |  |
| Anticorrosive oils                           |                               |                                    |                                                       | chemicals                   | organics                     | No             | RER      | lubricating oil, at plant                                            | 1.72E-02   | kg   | IPPC 2001                                           | 1                       | 124      | (3,3,1,1,5,4)          |  |
| sodium hydroxide                             |                               |                                    |                                                       | chemicals                   | inorganics                   | No             | RER      | sodium hydroxide, 50%in H2O, production mix, at plant                | 3.42E-03   | kg   | IPPC 2001                                           | 1                       | 124      | (3,3,1,1,5,4)          |  |
| Flocculating agent for waste water treatment |                               |                                    |                                                       | chemicals                   | inorganics                   | No             | RER      | aluminium sulphate, powder, at plant                                 | 5.90E-05   | kg   | IPPC 2001                                           | 1                       | 124      | (3,3,1,1,5,4)          |  |
| water                                        |                               |                                    |                                                       | water supply                | production                   | No             | RER      | water, decarbonised, at plant                                        | 6.54E-06   | kg   | IPPC 2001                                           | 1                       | 124      | (3,3,1,1,5,4)          |  |
| electricity                                  |                               |                                    |                                                       | electricity                 | production mix               | No             | UCTE     | electricity, medium voltage, production UCTE, at grid                | 3.00E-01   | kWh  | Richter et al. 1996                                 | 1                       | 124      | (3,3,1,1,5,2)          |  |
| electricity general                          |                               |                                    |                                                       | electricity                 | production mix               | No             | UCTE     | electricity, medium voltage, production UCTE, at grid                | 1.00E+00   | kWh  | assumption                                          | 1                       | 2.56     | (5,5,5,5,5,2)          |  |
| heat for curing                              |                               |                                    |                                                       | natural gas                 | heating systems              | No             | RER      | heat, natural gas, at industrial furnace low-NOx >100kW              | 3.44E+01   | MJ   | calculation from data from Schweizer Metallbau 2001 | 1                       | 194      | (4,5,5,4,5,1)          |  |
| Transport                                    |                               |                                    | Powder (incl. 0.0133 kg bucket) and chemicals: 100 km | transport systems           | road                         | No             | RER      | transport, lorry 32t                                                 | 1.26E-01   | tkm  | assumption                                          | 1                       | 2.09     | (4,5,n.a.,n.a.,n.a.,5) |  |
| Transport                                    |                               |                                    | Powder (incl. 0.0133 kg bucket) and chemicals: 600 km | transport systems           | train                        | No             | RER      | transport, freight, rail                                             | 7.58E-01   | tkm  | assumption                                          | 1                       | 2.09     | (4,5,n.a.,n.a.,n.a.,5) |  |
| Infrastructure                               |                               |                                    |                                                       | chemicals                   | organics                     | Yes            | RER      | chemical plant, organics                                             | 4.00E-10   | unit | assumption                                          | 1                       | 4.25     | (5,5,5,5,5,9)          |  |
|                                              |                               | Waste heat                         |                                                       | air                         | unspecified                  |                |          | Heat, waste                                                          | 4.68E+00   | MJ   | calculation                                         | 1                       | 2.56     | (5,5,5,5,5,13)         |  |
|                                              |                               | NM VOC to air                      | 0.5%of powder input                                   | air                         | unspecified                  |                |          | NM VOC, non-methane volatile organic compounds, unspecified origin   | 4.82E-04   | kg   | assumption                                          | 1                       | 3.22     | (5,5,5,5,5,23)         |  |
|                                              |                               | HCl to air                         |                                                       | air                         | unspecified                  |                |          | Hydrogen chloride                                                    | 5.05E-02   | kg   | Frischknecht et al. 1996                            | 1                       | 2.00     | (2,n.A.,2,1,1,n.A.,23) |  |
|                                              |                               | NaOH,(PO4)2-air                    | Na as proxy (amount refers to Na in NaOH)             | air                         | unspecified                  |                |          | Sodium                                                               | 4.37E-07   | kg   | IPPC 2001                                           | 1                       | 2.05     | (1,5,1,1,3,23)         |  |
|                                              |                               | NaOH,(PO4)2-water                  | Na as proxy (amount refers to Na in NaOH)             | water                       | unspecified                  |                |          | Sodium, ion                                                          | 3.93E-03   | kg   | IPPC 2001                                           | 1                       | 1.56     | (1,5,1,1,3,33)         |  |
|                                              |                               | Waste (from degreasing / pickling) |                                                       | waste management            | hazardous waste incineration | No             | CH       | disposal, hazardous waste, 25%water, to hazardous waste incineration | 3.10E-02   | kg   | IPPC 2001                                           | 1                       | 124      | (3,3,1,1,5,6)          |  |

Fig. 10.2 Flows for "powder coating, steel" and their representation in the ecoinvent database



| General flow information      |              |                                       |                                                                                                                                                                                                                                                                                                                            | Representation in ecoinvent |                            |                |          |                                                                              |            |      | Uncertainty information  |      |          |                                                           |
|-------------------------------|--------------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------------------|----------------|----------|------------------------------------------------------------------------------|------------|------|--------------------------|------|----------|-----------------------------------------------------------|
| Input                         | Process Name | Output                                | Remarks                                                                                                                                                                                                                                                                                                                    | Category                    | Sub category               | Infrastructure | Location | Module name in ecoinvent                                                     | Mean value | Unit | Source mean value        | Type | StdV 95% | General Comment                                           |
| powder coating, steel, part 2 | m            | Waste water                           | Treated waste water. Total emissions are estimated based on the degreasing / pickling process in the zinc coating process of steel. Infrastructure for waste water treatment plant is comprised in the infrastructure module. Value for energy use is comprised in the range of uncertainty of inventoried overall energy. |                             |                            |                |          |                                                                              | 6.54E-03   | m3   | IPPC 2001                |      |          |                                                           |
|                               | m            | TSS                                   |                                                                                                                                                                                                                                                                                                                            | water                       | unspecified                |                |          | Suspended solids, unspecified                                                | 5.19E-06   | kg   | IPPC 2001                | 1    | 8.53     | calculated from max value assuming lognormal distribution |
|                               | m            | Iron                                  |                                                                                                                                                                                                                                                                                                                            | water                       | unspecified                |                |          | Iron, ion                                                                    | 1.12E-05   | kg   | IPPC 2001                | 1    | 8.17     |                                                           |
|                               | m            | Nickel                                |                                                                                                                                                                                                                                                                                                                            | water                       | unspecified                |                |          | Nickel, ion                                                                  | 1.92E-07   | kg   | IPPC 2001                | 1    | 3.16     |                                                           |
|                               | m            | Zinc                                  |                                                                                                                                                                                                                                                                                                                            | water                       | unspecified                |                |          | Zinc, ion                                                                    | 1.52E-06   | kg   | IPPC 2001                | 1    | 20.02    |                                                           |
|                               | m            | Lead                                  |                                                                                                                                                                                                                                                                                                                            | water                       | unspecified                |                |          | Lead                                                                         | 1.28E-07   | kg   | IPPC 2001                | 1    | 124      | (3,3,1,1,5,6)                                             |
|                               | m            | Chromium                              |                                                                                                                                                                                                                                                                                                                            | water                       | unspecified                |                |          | Chromium, ion                                                                | 1.41E-06   | kg   | IPPC 2001                | 1    | 2.00     | calculated from max value assuming lognormal distribution |
|                               | m            | Cr6+                                  |                                                                                                                                                                                                                                                                                                                            | water                       | unspecified                |                |          | Chromium VI                                                                  | 6.54E-08   | kg   | IPPC 2001                | 1    | 2.00     |                                                           |
|                               | m            | COD                                   |                                                                                                                                                                                                                                                                                                                            | water                       | unspecified                |                |          | COD, Chemical Oxygen Demand                                                  | 4.16E-04   | kg   | IPPC 2001                | 1    | 5.48     | Twice the uncertainty of COD                              |
|                               | m            | BOD                                   |                                                                                                                                                                                                                                                                                                                            | water                       | unspecified                |                |          | BOD5, Biological Oxygen Demand                                               | 4.16E-04   | kg   | calculated from COD      | 1    | 10.95    |                                                           |
|                               | m            | DOC                                   |                                                                                                                                                                                                                                                                                                                            | water                       | unspecified                |                |          | DOC, Dissolved Organic Carbon                                                | 1.93E-05   | kg   | calculated from TOC      | 1    | 13.38    | Twice the uncertainty of TOC                              |
|                               | m            | TOC                                   |                                                                                                                                                                                                                                                                                                                            | water                       | unspecified                |                |          | TOC, Total Organic Carbon                                                    | 1.93E-05   | kg   | IPPC 2001                | 1    | 6.69     | calculated from max value assuming lognormal distribution |
|                               | m            | Hydrocarbons                          |                                                                                                                                                                                                                                                                                                                            | water                       | unspecified                |                |          | Hydrocarbons, unspecified                                                    | 1.12E-06   | kg   | IPPC 2001                | 1    | 124      | (3,3,1,1,5,6)                                             |
|                               | m            | Oil and fat                           |                                                                                                                                                                                                                                                                                                                            | water                       | unspecified                |                |          | Oils, unspecified                                                            | 6.75E-06   | kg   | IPPC 2001                | 1    | 6.08     | calculated from max value assuming lognormal distribution |
|                               | m            | Phosphorus                            |                                                                                                                                                                                                                                                                                                                            | water                       | unspecified                |                |          | Phosphorus                                                                   | 1.14E-07   | kg   | IPPC 2001                | 1    | 9.35     |                                                           |
|                               | m            | Chloride                              |                                                                                                                                                                                                                                                                                                                            | water                       | unspecified                |                |          | Chloride                                                                     | 3.13E-05   | kg   | Frischknecht et al. 1996 | 1    | 124      | (3,3,1,1,5,6)                                             |
|                               | m            | Water treatment sludge (dry material) | Waste water treatment sludge from steel forming processes is inventoried as proxy. 1.2 times the amount is inventoried to account for the water content of the proxy                                                                                                                                                       | waste management            | residual material landfill | No             | CH       | disposal, sludge from steel rolling, 20%water, to residual material landfill | 7.06E-03   | kg   | IPPC 2001                | 1    | 3.87     | calculated from max value assuming lognormal distribution |
|                               | m            | Chemical treatment sludges            |                                                                                                                                                                                                                                                                                                                            | waste management            | residual material landfill | No             | CH       | disposal, sludge from steel rolling, 20%water, to residual material landfill | 2.19E-03   | kg   | IPPC 2001                | 1    | 124      | (3,3,1,1,5,6)                                             |
|                               | m            | Waste (bucket including paint rests)  |                                                                                                                                                                                                                                                                                                                            | waste management            | municipal incineration     | No             | CH       | disposal, polyethylene, 0.4% water, to municipal incineration                | 2.61E-03   | kg   | calculation              | 1    | 2.56     | (5,5,5,5,5,6)                                             |
|                               | m            | Powder coating, steel sheet           | refers to a 2 mm thick sheet                                                                                                                                                                                                                                                                                               | metals                      | processing                 | No             | RER      | powder coating, steel                                                        | 1.00E+00   | m2   |                          |      |          |                                                           |

Fig. 10.2 Flows for "powder coating, steel" and their representation in the ecoinvent database (cont.)

## 10.4 Powder coating of aluminium

The 2 mm thick aluminium sheet is cleaned by alkaline degreasing and chromatising. Some of the data for these processes from powder coating of steel sheets are used to account for the Cr and hydrocarbon emissions to water. Other emissions are neglected because for them the pickling of steel seemed not suitable as proxy. The sheet is coated with a layer of 80 µm and cured at 220 °C. The heat energy consumption for the curing is calculated using the energy consumption reported by a Swiss producer<sup>32</sup>. Since the data in this source is given for steel and aluminium together, the curing temperature and the thermal capacity of the steel and aluminium sheet are used to estimate the share of the energy demand for each of these materials.

The dataset includes neither the aluminium sheet nor its transport to the coating plant.

The background information to the process "powder coating, aluminium" is given in Tab. 10.3, flows are shown in Fig. 10.3.

### Data quality

The uncertainty is estimated by completing the pedigree matrix, according to (Frischknecht et al. (2003)). A lognormal distribution is assumed. The data are based on rough estimations and of poor overall quality.

**Tab. 10.3** Ecoinvent meta information for "powder coating, aluminium sheet"

| Name                                  | powder coating, aluminium sheet                                                                                                                                                 |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location                              | RER                                                                                                                                                                             |
| Infrastructure Process                | 0                                                                                                                                                                               |
| Unit                                  | m <sup>2</sup>                                                                                                                                                                  |
| Data Set Version                      | 2.0                                                                                                                                                                             |
| Included Processes                    | Chromatising of the aluminium sheet, powder coating and heat curing. No transports of the aluminium sheet to the coating plant or back are inventoried.                         |
| Amount                                | 1                                                                                                                                                                               |
| Local Name                            | Pulverbeschichten, Aluminiumblech                                                                                                                                               |
| Synonyms                              |                                                                                                                                                                                 |
| General Comment to reference function | The powder inventoried is mainly used in exterior architectural applications. The coating thickness is 80 µm. The heat consumption is calculated for a sheet of 2 mm thickness. |
| Start Date                            | 1996                                                                                                                                                                            |
| End Date                              | 2003                                                                                                                                                                            |
| Data Valid For Entire Period          | 1                                                                                                                                                                               |
| Other Period Text                     |                                                                                                                                                                                 |
| Geography text                        | Swiss waste treatment assumed                                                                                                                                                   |
| Technology text                       | Chomatizing, coating and heat curing. Treatment of the waste water is assumed.                                                                                                  |
| Representativeness [%]                |                                                                                                                                                                                 |
| Production Volume                     | Unknown                                                                                                                                                                         |
| Sampling Procedure                    | Literature                                                                                                                                                                      |
| Extrapolations                        | None                                                                                                                                                                            |
| Uncertainty Adjustments               | None                                                                                                                                                                            |

<sup>32</sup> [http://www.schweizer-metallbau.ch/framedef.cfm?Navistring=06&aktuell\\_content=06](http://www.schweizer-metallbau.ch/framedef.cfm?Navistring=06&aktuell_content=06), accessed 9.2.2004

| General flow information |                                         |            |                                                       | Representation in ecoinvent |                 |                |          |                                                                    |            |      |                                                     | Uncertainty information |          |                        |  |
|--------------------------|-----------------------------------------|------------|-------------------------------------------------------|-----------------------------|-----------------|----------------|----------|--------------------------------------------------------------------|------------|------|-----------------------------------------------------|-------------------------|----------|------------------------|--|
| Input                    | Process Name                            | Output     | Remarks                                               | Category                    | Sub category    | Infrastructure | Location | Modul name in ecoinvent                                            | Mean value | Unit | Source mean value                                   | Type                    | StDv 95% | General Comment        |  |
| coating powder           | powder coating, aluminium sheet, part 1 |            | coating of 80 um, density 1200 kg/m3, 0.05%loss       | metals                      | processing      | No             | RER      | coating powder, at plant                                           | 9.65E-02   | kg   | calculation                                         | 1                       | 2.39     | (4,5,5,5,5,4)          |  |
| HCl                      |                                         |            | for degreasing / chromatising                         | chemicals                   | inorganics      | No             | RER      | hydrochloric acid, 30%in H2O, at plant                             | 2.80E-02   | kg   | Richter et al. 1996                                 | 1                       | 124      | (3,3,1,1,1,5,4)        |  |
| chromic acid             |                                         |            |                                                       | chemicals                   | inorganics      | No             | RER      | chromium oxide, flakes, at plant                                   | 7.00E-03   | kg   | Richter et al. 1996                                 | 1                       | 124      | (3,3,1,1,1,5,4)        |  |
| lime                     |                                         |            |                                                       | construction materials      | binder          | No             | CH       | lime, hydrated, packed, at plant                                   | 1.90E-02   | kg   | Richter et al. 1996                                 | 1                       | 124      | (3,3,1,1,1,5,4)        |  |
| sodium hydroxide         |                                         |            |                                                       | chemicals                   | inorganics      | No             | RER      | sodium hydroxide, 50%in H2O, production mix, at plant              | 1.60E-02   | kg   | Richter et al. 1996                                 | 1                       | 124      | (3,3,1,1,1,5,4)        |  |
| chemicals organic        |                                         |            |                                                       | chemicals                   | organics        | No             | GLO      | chemicals organic, at plant                                        | 9.00E-03   | kg   | Richter et al. 1996                                 | 1                       | 124      | (3,3,1,1,1,5,4)        |  |
| water                    |                                         |            |                                                       | water supply                | production      | No             | RER      | water, decarbonised, at plant                                      | 6.54E-06   | kg   | IPPC 2001                                           | 1                       | 124      | (3,3,1,1,1,5,4)        |  |
| electricity              |                                         |            |                                                       | electricity                 | production mix  | No             | UCTE     | electricity, medium voltage, production UCTE, at grid              | 3.00E-01   | kWh  | Richter et al. 1996                                 | 1                       | 124      | (3,3,1,1,1,5,2)        |  |
| electricity general      |                                         |            |                                                       | electricity                 | production mix  | No             | UCTE     | electricity, medium voltage, production UCTE, at grid              | 1.00E+00   | kWh  | assumption                                          | 1                       | 2.56     | (5,5,5,5,5,2)          |  |
| heat for curing          |                                         |            |                                                       | natural gas                 | heating systems | No             | RER      | heat, natural gas, at industrial furnace low-NOx >100kW            | 2.42E+01   | MJ   | calculation from data from Schweizer Metallbau 2001 | 1                       | 1.94     | (4,5,5,5,4,5,1)        |  |
| Transport                |                                         |            | Powder (incl. 0.0133 kg bucket) and chemicals: 100 km | transport systems           | road            | No             | RER      | transport, lorry 32t                                               | 2.70E-02   | tkm  | assumption                                          | 1                       | 2.09     | (4,5,n.a.,n.a.,n.a.,5) |  |
| Transport                |                                         |            | Powder (incl. 0.0133 kg bucket) and chemicals: 600 km | transport systems           | train           | No             | RER      | transport, freight, rail                                           | 1.62E-01   | tkm  | assumption                                          | 1                       | 2.09     | (4,5,n.a.,n.a.,n.a.,5) |  |
| Infrastructure           |                                         |            |                                                       | chemicals                   | organics        | Yes            | RER      | chemical plant, organics                                           | 4.00E-10   | unit | assumption                                          | 1                       | 4.25     | (5,5,5,5,5,9)          |  |
|                          |                                         | Waste heat |                                                       | air                         | unspecified     |                |          | Heat, waste                                                        | 4.68E+00   | MJ   | calculation                                         | 1                       | 2.56     | (5,5,5,5,5,13)         |  |
|                          |                                         | NM VOC     | 0.5%of powder input                                   | air                         | unspecified     |                |          | NM VOC, non-methane volatile organic compounds, unspecified origin | 4.82E-04   | kg   | assumption                                          | 1                       | 3.22     | (5,5,5,5,5,23)         |  |

Fig. 10.3 Flows for "powder coating, aluminium sheet" and their representation in the ecoinvent database

| General flow information                |              |                                       |                                                                                                                                                                                                                                                                                                                                                                                                        | Representation in ecoinvent |                              |                |          |                                                                              |            |      | Uncertainty information |      |          |                                                           |
|-----------------------------------------|--------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|------------------------------|----------------|----------|------------------------------------------------------------------------------|------------|------|-------------------------|------|----------|-----------------------------------------------------------|
| Input                                   | Process Name | Output                                | Remarks                                                                                                                                                                                                                                                                                                                                                                                                | Category                    | Sub category                 | Infrastructure | Location | Module name in ecoinvent                                                     | Mean value | Unit | Source mean value       | Type | StdV 95% | General Comment                                           |
| powder coating, aluminium sheet, part 2 | m            | Waste water                           | Treated waste water. Total emissions are estimated based on the degreasing / pickling process in the zinc coating process of steel. Infrastructure for waste water treatment plant is comprised in the infrastructure module. Value for energy use is comprised in the range of uncertainty of inventoried overall energy. Chromium and organic emissions taken from cleaning of steel sheet as proxy. |                             |                              |                |          |                                                                              | 6.54E-03   | m3   | IPPC 2001               |      |          |                                                           |
|                                         | m            | Chromium                              |                                                                                                                                                                                                                                                                                                                                                                                                        | water                       | unspecified                  |                |          | Chromium, ion                                                                | 1.41E-06   | kg   | IPPC 2001               | 1    | 2.00     | calculated from max value assuming lognormal distribution |
|                                         | m            | Cr6+                                  |                                                                                                                                                                                                                                                                                                                                                                                                        | water                       | unspecified                  |                |          | Chromium VI                                                                  | 6.54E-08   | kg   | IPPC 2001               | 1    | 2.00     | calculated from max value assuming lognormal distribution |
|                                         | m            | COD                                   |                                                                                                                                                                                                                                                                                                                                                                                                        | water                       | unspecified                  |                |          | COD, Chemical Oxygen Demand                                                  | 4.16E-04   | kg   | IPPC 2001               | 1    | 5.48     | calculated from max value assuming lognormal distribution |
|                                         | m            | BOD                                   |                                                                                                                                                                                                                                                                                                                                                                                                        | water                       | unspecified                  |                |          | BOD5, Biological Oxygen Demand                                               | 4.16E-04   | kg   | calculated from COD     | 1    | 10.95    | Twice the uncertainty of COD                              |
|                                         | m            | DOC                                   |                                                                                                                                                                                                                                                                                                                                                                                                        | water                       | unspecified                  |                |          | DOC, Dissolved Organic Carbon                                                | 1.93E-05   | kg   | calculated from TOC     | 1    | 13.38    | Twice the uncertainty of TOC                              |
|                                         | m            | TOC                                   |                                                                                                                                                                                                                                                                                                                                                                                                        | water                       | unspecified                  |                |          | TOC, Total Organic Carbon                                                    | 1.93E-05   | kg   | IPPC 2001               | 1    | 6.69     | calculated from max value assuming lognormal distribution |
|                                         | m            | Hydrocarbons                          |                                                                                                                                                                                                                                                                                                                                                                                                        | water                       | unspecified                  |                |          | Hydrocarbons, unspecified                                                    | 1.12E-06   | kg   | IPPC 2001               | 1    | 1.24     | (3,3,1,1,5,6)                                             |
|                                         | m            | Oil and fat                           |                                                                                                                                                                                                                                                                                                                                                                                                        | water                       | unspecified                  |                |          | Oils, unspecified                                                            | 6.75E-06   | kg   | IPPC 2001               | 1    | 6.08     | calculated from max value assuming lognormal distribution |
|                                         | m            | Water treatment sludge (dry material) | Waste water treatment sludge from steel forming processes is inventoried as proxy. 12 times the amount is inventoried to account for the water content of the proxy.                                                                                                                                                                                                                                   | waste management            | residual material landfill   | No             | CH       | disposal, sludge from steel rolling, 20%water, to residual material landfill | 7.06E-03   | kg   | IPPC 2001               | 1    | 3.87     | calculated from max value assuming lognormal distribution |
|                                         | m            | Chemical treatment sludges            |                                                                                                                                                                                                                                                                                                                                                                                                        | waste management            | residual material landfill   | No             | CH       | disposal, sludge from steel rolling, 20%water, to residual material landfill | 2.19E-03   | kg   | IPPC 2001               | 1    | 1.24     | (3,3,1,1,5,6)                                             |
|                                         | m            | Waste (from chromatising)             |                                                                                                                                                                                                                                                                                                                                                                                                        | waste management            | hazardous waste incineration | No             | CH       | disposal, hazardous waste, 25%water, to hazardous waste incineration         | 1.27E-01   | kg   | Richter et al. 1996     | 1    | 1.24     | (3,3,1,1,5,6)                                             |
|                                         | m            | Waste (bucket including paint rests)  |                                                                                                                                                                                                                                                                                                                                                                                                        | waste management            | municipal incineration       | No             | CH       | disposal, polyethylene, 0.4% water, to municipal incineration                | 1.29E-03   | kg   | calculation             | 1    | 2.56     | (5,5,5,5,5,6)                                             |
|                                         | m            | Powder coating, aluminium sheet       | refers to a 2 mm thick sheet                                                                                                                                                                                                                                                                                                                                                                           | metals                      | processing                   | No             | RER      | powder coating, aluminium sheet                                              | 1.00E+00   | m2   |                         |      |          |                                                           |

Fig. 10.3 Flows for "powder coating, aluminium sheet" and their representation in the ecoinvent database (cont.)

# 11 Anodising of aluminium

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## 11.1 Definition and fundamentals of anodising

This chapter is mainly based on Lyle & Granger (1997).

Anodising is an electro-chemical process converting the aluminium surface to aluminium oxide. Two principal types of anodised coatings are recognized: barrier and porous.

The barrier coating is formed by aluminium ions from the base metal combined with the oxygen of water in the electrolyte. If the oxide is appreciably soluble in the electrolyte a relatively thick, porous oxide layer can develop.

The most common electrolytes for conventional thick porous anodised coatings are sulphuric acid solutions. These coatings are decorative or protective and range in thickness from 4 to 30 µm. Chromic acid anodised coatings are thinner and less abrasion resistant than the sulphuric acid coatings, but they are more flexible and more resistant to deformation and heating without crazing. Oxalic acid anodised coatings are somewhat harder than the sulphuric acid coatings.

Anodic coatings are coloured by certain alloying elements. Aluminium – silicon alloys provide durable grey coatings widely used as exterior architectural finishes. Chromium and copper in solid solution yield light gold finishes, and manganese imparts tan to brown shades. Porous anodic coatings can be impregnated with organic and inorganic dyes and pigments during anodising by incorporating the dyes and pigments in the electrolyte or during a post-anodising treatment.

Many anodic coatings are porous and subsequently are sealed to improve resistance to staining, resistance to corrosion, and durability of the colours. Sealing is accomplished by immersion in hot water.

### 11.1.1 Environmental issues of anodising

The use of solvents and acids for the cleaning, pickling and anodising of the aluminium is likely to be the most important environmental issue of anodising. These steps emit VOC to air and / or metals and hydrocarbons to water.

## 11.2 Anodising in ecoinvent

The anodising of aluminium involves the following steps:

- Mechanical pre-treatment (sanding, brushing, polishing, abrasive blasting) is made to eliminate defects and irregularities. This step is not always done and therefore inventoried only for 50% of the anodised sheets.
- Chemical pre-treatment (degreasing, pickling) is made to degrease and deoxidise the surface and to eliminate heavy metals from the surface.
- Electrolytic oxidation builds up the oxide layer.
- Post-treatment (sealing) hardens and consolidates the oxide layer.

Process data used in ecoinvent is based on Richter et al. (1996), who report data from 1995 for two anodising plants in Switzerland. Data correspond to an oxide layer thickness of 20 µm.

The background information to the process "anodising, aluminium sheet" is given in Tab. 11.1, flows are shown in Fig. 11.1.

### Data quality

The uncertainty is estimated by completing the pedigree matrix, according to (Frischknecht et al. (2003)). A lognormal distribution is assumed.

**Tab. 11.1** Ecoinvent meta information for "anodising, aluminium sheet"

| Name                                  | anodising, aluminium sheet                                                                                                                                                           |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location                              | RER                                                                                                                                                                                  |
| Infrastructure Process                | 0                                                                                                                                                                                    |
| Unit                                  | m <sup>2</sup>                                                                                                                                                                       |
| Data Set Version                      | 2.0                                                                                                                                                                                  |
| Included Processes                    | Mechanical surface treatment (50% of the workpieces), degreasing, pickling, anodising (20 µm layer thickness) and sealing. Waste water treatment is included. Dying is not included. |
| Amount                                | 1                                                                                                                                                                                    |
| Local Name                            | Eloxieren, Aluminiumblech                                                                                                                                                            |
| Synonyms                              | Anodisieren                                                                                                                                                                          |
| General Comment to reference function | Colourless anodising is inventoried. The coating thickness is 20 µm.                                                                                                                 |
| Start Date                            | 1996                                                                                                                                                                                 |
| End Date                              | 2003                                                                                                                                                                                 |
| Data Valid For Entire Period          | 1                                                                                                                                                                                    |
| Other Period Text                     |                                                                                                                                                                                      |
| Geography text                        | Swiss waste treatment assumed                                                                                                                                                        |
| Technology text                       | Porous anodising                                                                                                                                                                     |
| Representativeness [%]                |                                                                                                                                                                                      |
| Production Volume                     | Unknown                                                                                                                                                                              |
| Sampling Procedure                    | Literature / survey                                                                                                                                                                  |
| Extrapolations                        | None                                                                                                                                                                                 |
| Uncertainty Adjustments               | None                                                                                                                                                                                 |

| General flow information |                                                  |                 |                                                                               | Representation in ecoinvent |                              |                |          |                                                                      |            |      | Uncertainty information |      |          |                  |
|--------------------------|--------------------------------------------------|-----------------|-------------------------------------------------------------------------------|-----------------------------|------------------------------|----------------|----------|----------------------------------------------------------------------|------------|------|-------------------------|------|----------|------------------|
| Input                    | Process Name                                     | Output          | Remarks                                                                       | Category                    | Sub category                 | Infrastructure | Location | Modul name in ecoinvent                                              | Mean value | Unit | Source mean value       | Type | StdV 95% | General Comment  |
| electricity              | mechanical surface treatment (50% of workpieces) |                 |                                                                               | electricity                 | production mix               | No             | UCTE     | electricity, medium voltage, production UCTE, at grid                | 4.00E-02   | kWh  | Richter et al. 1996     | 1    | 1.08     | (1,3,2,3,1,1,2)  |
| lubricating oil          |                                                  |                 |                                                                               | chemicals                   | organics                     | No             | RER      | lubricating oil, at plant                                            | 2.75E-02   | kg   | Richter et al. 1996     | 1    | 1.08     | (1,3,2,3,1,1,4)  |
| saw dust                 |                                                  |                 | 80g, u=20% assumed<br>--> 66.7 g atro; @<br>188.6 kg/m3<br>(bulk) --> 3.53E-4 | wood energy                 | fuels                        | No             | RER      | wood chips, mixed, from industry, u=40% at plant                     | 1.77E-04   | m3   | Richter et al. 1996     | 1    | 1.08     | (1,3,2,3,1,1,4)  |
|                          |                                                  | dust            | size assumed by author                                                        | air                         | unspecified                  |                |          | Particulates, > 2.5 um, and < 10um                                   | 1.35E-05   | kg   | Richter et al. 1996     | 1    | 2.01     | (1,3,2,3,1,1,26) |
|                          |                                                  | saw dust waste  |                                                                               | waste management            | municipal incineration       | No             | CH       | disposal, wood untreated, 20%water, to municipal incineration        | 4.00E-02   | kg   | Richter et al. 1996     | 1    | 1.08     | (1,3,2,3,1,1,6)  |
|                          | degreasing (100% of workpieces)                  | oil waste       |                                                                               | waste management            | municipal incineration       | No             | CH       | disposal, bitumen sheet, 15% water, to municipal incineration        | 2.75E-02   | kg   | Richter et al. 1996     | 1    | 1.08     | (1,3,2,3,1,1,6)  |
|                          |                                                  | aluminium waste |                                                                               | waste management            | municipal incineration       | No             | CH       | disposal, aluminium, 0% water, to municipal incineration             | 1.50E-02   | kg   | Richter et al. 1996     | 1    | 1.08     | (1,3,2,3,1,1,6)  |
| heat                     |                                                  |                 |                                                                               | oil                         | heating systems              | No             | RER      | heat, light fuel oil, at industrial furnace 1M W                     | 1.96E+00   | MJ   | Richter et al. 1996     | 1    | 1.08     | (1,3,2,3,1,1,1)  |
| organic solvent          |                                                  |                 |                                                                               | chemicals                   | organics                     | No             | GLO      | chemicals organic, at plant                                          | 6.00E-03   | kg   | Richter et al. 1996     | 1    | 1.08     | (1,3,2,3,1,1,4)  |
| tenside                  |                                                  |                 |                                                                               | washing agents              | tensides                     | No             | RER      | ethoxylated alcohols, unspecified, at plant                          | 6.00E-03   | kg   | Richter et al. 1996     | 1    | 1.08     | (1,3,2,3,1,1,4)  |
|                          |                                                  | emulsion waste  |                                                                               | waste management            | hazardous waste incineration | No             | CH       | disposal, hazardous waste, 25%water, to hazardous waste incineration | 1.25E-03   | kg   | Richter et al. 1996     | 1    | 1.08     | (1,3,2,3,1,1,6)  |

Fig. 11.1 Flows for "anodising, aluminium sheet" and their representation in the ecoinvent database

| General flow information |              |                                   |        | Representation in ecoinvent                                                                                     |              |                 |                        |              |                                                       |               | Uncertainty information |                      |      |             |                 |
|--------------------------|--------------|-----------------------------------|--------|-----------------------------------------------------------------------------------------------------------------|--------------|-----------------|------------------------|--------------|-------------------------------------------------------|---------------|-------------------------|----------------------|------|-------------|-----------------|
| Input                    | Process Name |                                   | Output | Remarks                                                                                                         | Cate<br>gory | Sub<br>category | Infra<br>struc<br>ture | Loca<br>tion | Modul name in<br>ecoinvent                            | Mean<br>value | Unit                    | Source mean<br>value | Type | StDv<br>95% | General Comment |
| heat                     | m            | pickling<br>(100% of              |        |                                                                                                                 | oil          | heating systems | No                     | RER          | heat, light fuel oil, at industrial furnace 1M W      | 9.80E-01      | M J                     | Richter et al. 1996  | 1    | 1.08        | (1,3,2,3,1,1,1) |
| sodium hydroxide         | m            |                                   |        |                                                                                                                 | chemicals    | inorganics      | No                     | RER          | sodium hydroxide, 50%in H2O, production mix, at plant | 4.80E-02      | kg                      | Richter et al. 1996  | 1    | 1.08        | (1,3,2,3,1,1,4) |
| organic solvent          | m            |                                   |        |                                                                                                                 | chemicals    | organics        | No                     | GLO          | chemicals organic, at plant                           | 1.00E-02      | kg                      | Richter et al. 1996  | 1    | 1.08        | (1,3,2,3,1,1,4) |
| nitric acid              | m            |                                   |        |                                                                                                                 | chemicals    | inorganics      | No                     | RER          | nitric acid, 50%in H2O, at plant                      | 3.00E-02      | kg                      | Richter et al. 1996  | 1    | 1.08        | (1,3,2,3,1,1,4) |
| electricity              | m            | anodising (100%<br>of workpieces) |        | power consumption is proportional to the oxyde layer thickness. Data given here correspond to a layer of 20 um. | electricity  | production mix  | No                     | UCTE         | electricity, medium voltage, production UCTE, at grid | 4.80E+00      | kWh                     | Richter et al. 1996  | 1    | 1.08        | (1,3,2,3,1,1,2) |
| sulphuric acid           | m            |                                   |        |                                                                                                                 | chemicals    | inorganics      | No                     | RER          | sulphuric acid, liquid, at plant                      | 1.46E-01      | kg                      | Richter et al. 1996  | 1    | 1.08        | (1,3,2,3,1,1,4) |
| aluminium sulphate       | m            |                                   |        |                                                                                                                 | chemicals    | inorganics      | No                     | RER          | aluminium sulphate, powder, at plant                  | 1.40E-02      | kg                      | Richter et al. 1996  | 1    | 1.08        | (1,3,2,3,1,1,4) |
| heat                     | m            |                                   |        |                                                                                                                 | oil          | heating systems | No                     | RER          | heat, light fuel oil, at industrial furnace 1M W      | 4.90E+00      | M J                     | Richter et al. 1996  | 1    | 1.08        | (1,3,2,3,1,1,1) |
| organic additive         | m            |                                   |        |                                                                                                                 | chemicals    | organics        | No                     | GLO          | chemicals organic, at plant                           | 1.00E-02      | kg                      | Richter et al. 1996  | 1    | 1.08        | (1,3,2,3,1,1,4) |

Fig. 11.1 Flows for "anodising, aluminium sheet" and their representation in the ecoinvent database (cont.)



## 11. Anodising of aluminium

| General flow information                     |                                                          |                          |                                                                                | Representation in ecoinvent |                      |                |          |                                                                 |            |      | Uncertainty information                  |      |          |                        |
|----------------------------------------------|----------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------|-----------------------------|----------------------|----------------|----------|-----------------------------------------------------------------|------------|------|------------------------------------------|------|----------|------------------------|
| Input                                        | Process Name                                             | Output                   | Remarks                                                                        | Category                    | Sub category         | Infrastructure | Location | Modul name in ecoinvent                                         | Mean value | Unit | Source mean value                        | Type | StdV 95% | General Comment        |
| sodium chloride                              | anodising, aluminium sheet; overall (100% of workpieces) |                          |                                                                                | chemicals                   | inorganics           | No             | RER      | sodium chlorate, powder, at plant                               | 1.10E-02   | kg   | Richter et al. 1996                      | 1    | 1.08     | (1,3,2,3,1,1,4)        |
| sodium hydroxide                             |                                                          |                          |                                                                                | chemicals                   | inorganics           | No             | RER      | sodium hydroxide, 50%in H2O, diaphragm cell, at plant           | 7.38E-02   | kg   | Richter et al. 1996                      | 1    | 1.08     | (1,3,2,3,1,1,4)        |
| Flocculating agent for waste water treatment |                                                          |                          |                                                                                | chemicals                   | inorganics           | No             | RER      | aluminium sulphate, powder, at plant                            | 6.00E-05   | kg   | Richter et al. 1996                      | 1    | 1.08     | (1,3,2,3,1,1,4)        |
| plastic (unspec.)                            |                                                          |                          |                                                                                | plastics                    | polymers             | No             | RER      | polyethylene, LDPE, granulate, at plant                         | 1.00E-03   | kg   | Richter et al. 1996                      | 1    | 1.08     | (1,3,2,3,1,1,4)        |
| aluminium                                    |                                                          |                          |                                                                                | metals                      | extraction           | No             | RER      | aluminium, primary, at plant                                    | 1.88E-03   | kg   | Richter et al. 1996                      | 1    | 1.08     | (1,3,2,3,1,1,4)        |
| white spirit                                 |                                                          |                          |                                                                                | paintings                   | production           | No             | RER      | white spirit, at plant                                          | 1.25E-04   | kg   | Richter et al. 1996                      | 1    | 1.08     | (1,3,2,3,1,1,4)        |
| water                                        |                                                          |                          |                                                                                | water supply                | production           | No             | RER      | water, decarbonised, at plant                                   | 7.50E+02   | kg   | Richter et al. 1996                      | 1    | 1.08     | (1,3,2,3,1,1,12)       |
| Transport                                    |                                                          |                          | Lubricating oil, chemicals, plastic and aluminium: 100 km                      | transport systems           | road                 | No             | RER      | transport, lorry 32t                                            | 6.17E-02   | tkm  | assumption                               | 1    | 2.09     | (4,5,n.a.,n.a.,n.a.,5) |
| Transport                                    |                                                          |                          | Lubricating oil, chemicals, plastic and aluminium: 600 km                      | transport systems           | train                | No             | RER      | transport, freight, rail                                        | 3.70E-01   | tkm  | assumption                               | 1    | 2.09     | (4,5,n.a.,n.a.,n.a.,5) |
| Infrastructure                               |                                                          |                          |                                                                                | chemicals                   | organics             | Yes            | RER      | chemical plant, organics                                        | 4.00E-10   | unit | assumption                               | 1    | 4.25     | (5,5,5,5,5,9)          |
|                                              |                                                          | Waste heat               |                                                                                | air                         | unspecified          |                |          | Heat, waste                                                     | 1.74E+01   | MJ   | calculation                              | 1    | 2.56     | (5,5,5,5,5,13)         |
|                                              |                                                          | Waste water to treatment | inventoried as unpolluted water since water emissions are directly inventoried | waste management            | wastewater treatment | No             | CH       | treatment, sewage, unpolluted, to wastewater treatment, class 3 | 7.50E-01   | m3   | calculation based on Richter et al. 1996 | 1    | 1.08     | (1,3,2,3,1,1,12)       |
|                                              |                                                          | Cl- to water             |                                                                                | water                       | unspecified          |                |          | Chloride                                                        | 6.66E-03   | kg   | Richter et al. 1996                      | 1    | 1.51     | (1,3,2,3,1,1,33)       |
|                                              |                                                          | Al to water              |                                                                                | water                       | unspecified          |                |          | Aluminum                                                        | 1.50E-03   | kg   | Richter et al. 1996                      | 1    | 5.01     | (1,3,2,3,1,1,35)       |
|                                              |                                                          | Pb to water              |                                                                                | water                       | unspecified          |                |          | Lead                                                            | 2.00E-05   | kg   | Richter et al. 1996                      | 1    | 5.01     | (1,3,2,3,1,1,35)       |
|                                              |                                                          | Cd to water              |                                                                                | water                       | unspecified          |                |          | Cadmium, ion                                                    | 7.50E-07   | kg   | Richter et al. 1996                      | 1    | 5.01     | (1,3,2,3,1,1,35)       |
|                                              |                                                          | Fe to water              |                                                                                | water                       | unspecified          |                |          | Iron, ion                                                       | 2.25E-05   | kg   | Richter et al. 1996                      | 1    | 5.01     | (1,3,2,3,1,1,35)       |
|                                              |                                                          | Cu to water              |                                                                                | water                       | unspecified          |                |          | Copper, ion                                                     | 1.50E-05   | kg   | Richter et al. 1996                      | 1    | 5.01     | (1,3,2,3,1,1,35)       |
|                                              |                                                          | Mn to water              |                                                                                | water                       | unspecified          |                |          | Manganese                                                       | 1.28E-04   | kg   | Richter et al. 1996                      | 1    | 5.01     | (1,3,2,3,1,1,35)       |
|                                              |                                                          | Na to water              |                                                                                | water                       | unspecified          |                |          | Sodium, ion                                                     | 4.33E-02   | kg   | Richter et al. 1996                      | 1    | 5.01     | (1,3,2,3,1,1,35)       |
|                                              |                                                          | Ni to water              |                                                                                | water                       | unspecified          |                |          | Nickel, ion                                                     | 2.18E-04   | kg   | Richter et al. 1996                      | 1    | 5.01     | (1,3,2,3,1,1,35)       |
|                                              |                                                          | Zn to water              |                                                                                | water                       | unspecified          |                |          | Zinc, ion                                                       | 4.50E-05   | kg   | Richter et al. 1996                      | 1    | 5.01     | (1,3,2,3,1,1,35)       |
|                                              |                                                          | Sn to water              |                                                                                | water                       | unspecified          |                |          | Tin, ion                                                        | 1.25E-04   | kg   | Richter et al. 1996                      | 1    | 5.01     | (1,3,2,3,1,1,35)       |
|                                              |                                                          | sulphate to water        |                                                                                | water                       | unspecified          |                |          | Sulfate                                                         | 1.13E-01   | kg   | Richter et al. 1996                      | 1    | 1.51     | (1,3,2,3,1,1,33)       |
|                                              |                                                          | anodising (20 um)        |                                                                                | metals                      | processing           | No             | RER      | anodising, aluminium sheet                                      | 1.00E+00   | m2   | Richter et al. 1996                      |      |          |                        |

Fig. 11.1 Flows for "anodising, aluminium sheet" and their representation in the ecoinvent database (cont.)

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## Part XVI

# Antimony

Data v2.1 (2009)

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Final report of updated Swiss National Life Cycle Inventory Database "ecoinvent", version 2.1

## **Comments on this part**

The production of antimony is inventoried in the modules described in this report. Data reported are mainly based on literature and theoretical models.





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# 1 Introduction

Antimony and its sulphide have been known and used for several thousand years. It occurs natively and is found in over 100 mineral species. The predominant ore mineral of Antimony is its sulphide, stibnite ( $\text{Sb}_2\text{S}_3$ ). Antimony is mainly mined from complex sulfidic ores with iron, copper, lead, mercury or silver.

## 2 Reserves and Resources of Antimony

Antimony, like copper, lead and zinc, is a markedly chalcophilic (sulphur loving) element. Its estimated abundance in the Earth's crust is 0.2 to 0.5 ppm. Antimony occurs natively and is found in over 100 mineral species. The predominant ore mineral of Antimony is its sulphide, stibnite ( $\text{Sb}_2\text{S}_3$ ). Antimony is mainly mined from complex sulfidic ores with iron, copper, lead, mercury or silver (USGS 2007). China produces more than 80% of the world antimony, mainly from stibnite. The largest mine in China is Xikuangshan mine in Hunan Province.

Secondary antimony stems mainly from the recycling of lead acid batteries in which antimony is used as an alloying element (ca 2% w/w). In the mid 1990s, about 30% of the European and 22% of the United States antimony production originated from secondary antimony from this source (Ullmann 2006). Antimony is almost exclusively recycled at lead smelters and consequently used in lead-antimony alloys, mainly lead acid batteries.

The world-wide secondary antimony production share today is considerably lower than it was ten years ago, mainly for three reasons:

1. the antimony content in lead acid batteries decreased in the last decades due to the introduction of low-maintenance batteries in the 1970s. Thus the main source for secondary material decreased and is still decreasing.
2. the world antimony production increased from ca. 90'000 t/a in the mid 1990s to over 130'000 t/a in 2005 and 2006.
3. Europe and the U.S. already had a high consumption of lead acid batteries before 1970 when the main producers of antimony (China, Russia & Bolivia) still had a very low car (and therefore lead acid battery) density. Thus, countries with high potential for secondary production like Europe and the U.S. are insignificant contributors to the global antimony supply (less than 3%). The secondary production share in significant contributing countries like China, Russia or Bolivia is low. Thus, world average secondary production share is a lot lower than the European or the U.S. share.

As a consequence of 1) and 2), recycled antimony dropped from 22% in the mid 1990s to only 15% of the total U.S. market supply in 1999 (USGS 2007). Today, the share of recycled antimony on total global production is estimated to be less than 5%.

### 3 Characterisation of the material

The element antimony (Sb, 7440-36-0) is a blue-white metalloid. In its elemental form it is a silvery white, brittle, fusible, crystalline solid that exhibits poor electrical and heat conductivity properties and vaporizes at low temperatures. A metalloid, antimony resembles a metal in its appearance and in many of its physical properties, but does not chemically react as a metal. It is also attacked by oxidizing acids and halogens. Antimony and some of its alloys expand on cooling (negative thermal expansion). Some physical properties are listed in Tab. 3.1.

Tab. 3.1 Some physical properties of antimony.

| Physical properties of antimony <sup>1</sup>                   |         |
|----------------------------------------------------------------|---------|
| Melting point °C                                               | 630.9   |
| Latent heat of fusion, kJ/mol                                  | 19.87   |
| Boiling point °C                                               | 1587    |
| Latent heat of vaporisation, kJ/mol                            | 77.14   |
| Vapour pressure, Pa @ 630.9 °C                                 | 2.49E-9 |
| Density, g/cm <sup>3</sup> @ 26 °C                             | 6.684   |
| Mean specific heat, J g <sup>-1</sup> K <sup>-1</sup> 0–100 °C | 0.21    |

Antimony is comparable in its toxicological behaviour to arsenic and bismuth. Poisoning with antimony and its compounds can result from acute and chronic exposure, especially from exposure to airborne particles in the workplace.

Commercial grades of antimony trioxide (the principal commercial form of processed antimony) contain 99.2 to 99.8 % trioxide with iron, lead and arsenic as main impurities. Commercial antimony metal is usually at least 99% antimony with no more than 0.3% arsenic impurity. The largest amount on the market is Chinese No. 2 grade, which has minimum 99.65% antimony. Antimony metal prepared for semiconductor use in electronics is 99.999 plus pure. (USGS 2007)

Antimony sulphide ore is marketed in at least three forms: “lump sulphide ore”, “clean sulphide concentrate” and “Chinese concentrate” (USGS 2007). In the 1990s, much of the high grade metal and the trioxide was produced in the consuming countries from imported concentrates, metal or crude oxides. However, a trend from concentrate towards metal and oxide imports was observed during the last decades (USGS 2007). The U.S. geological survey estimates the U.S. imports of ores and crude oxides in 1999 to approximately 7’700 tons. On the other hand, 39’000 tons of refined metal and oxides were imported in the same year, more than 20’000 tons of which from China. Since the end of the 1990s, China closed down several antimony production sites, impede illegal (or informal) operations and set up a policy of no more selling ore concentrates but refined products. Thus, it can be expected that today, most of the antimony trioxide and metal production takes place in China.

<sup>1</sup> <http://www.3rd1000.com/elements/Antimony.htm>; accessed 23.09.2008

## 4 Use / Application of the Material

Flame retardants were estimated to account for 60% of global antimony consumption in 2000. The remaining 40% were equally shared between Alloys, mainly lead alloys primarily used in lead acid batteries, and other uses. The principal commercial form of processed antimony is the trioxide, which is mainly used in flame retardants for plastics (USGS 2007). Antimony compounds in the form of Oxides, Sulfides, Sodium Antimonate, and Antimony Trichloride are also used in the making of ceramic enamels, glass, paints, and pottery. Also, Antimony Sulfide is one of the ingredients of safety matches.

Antimony is increasingly being used in the semiconductor industry in the production of diodes, infrared detectors, and Hall-Effect devices. As an alloy, this metalloid greatly increases Lead's hardness and mechanical strength. The most important use of metallic antimony is as a hardener in lead for storage batteries. Uses include:

- Batteries
- Antifriction Alloys
- Type Metal
- Small Arms and Tracer Ammunition
- Cable Sheathing
- Matches
- Medicines
- Plumbing
- Soldering - Some "Lead-Free" Solders Contain 5% Sb
- Main and Big-End Bearings in Internal Combustion Engines (as Alloy)
- Used in the past to treat Schistosomiasis; today Praziquantel is universally used
- Used in Linotype Printing Machines

## 5 System Characterisation

This chapter describes the mining of sulphidic antimony deposits and the related extraction of antimony.

Mining processes are similar for all sulfidic ores. Open pit mining and underground mining are both applied for antimony. No information was found on the share of the two types of mining. The ores are concentrated to up to 60% Sb content by froth flotation. Low grade ores (1.5 to 25% Sb) are roasted to produce antimony oxide. Medium grade ores are smelted in blast furnaces to produce crude metal. High grade ores (>45% Sb) are liquitated, i.e. heated in reducing atmosphere to melt the sulphide and separate it from the gangue. The blast furnace process is the most important process for the extraction of antimony.

Since not enough information was found on antimony ore composition, mining and beneficiation, the mining and beneficiation of molybdenite from coupled production with copper is used as proxy. This seems reasonable since molybdenum occurs in about the same concentration as antimony together with copper and since the beneficiation processes are basically the same.

The metallurgical processes in the Antimony production are comparable to those in lead and zinc production. Since no quantitative data was available, the production of Antimony is approximated with the smelting process of lead.

Tab. 5.1 gives the meta information to the processes while Fig. 5.1 and Fig. 5.2 show the LCI flow-sheets for the datasets.

Tab. 5.1 Meta information for the mining of antimony bearing ore and the smelting of antimony.

| Name                                  | stibnite ore, 70% stibnite, at mine                                                                                                                                | antimony, at refinery                                                                                                        |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Location                              | CN                                                                                                                                                                 | CN                                                                                                                           |
| Infrastructure Process                | 0                                                                                                                                                                  | 0                                                                                                                            |
| Unit                                  | kg                                                                                                                                                                 | kg                                                                                                                           |
| Data Set Version                      | 2.0                                                                                                                                                                | 2.0                                                                                                                          |
| Included Processes                    | The module includes the mining and the beneficiation step with the mining infrastructure and disposal of overburden and tailings (see part "Auxiliary Processes"). | Smelting of concentrate and disposal of wastes.                                                                              |
| Amount                                | 1                                                                                                                                                                  | 1                                                                                                                            |
| Local Name                            | Stibnitzerz, 70% Stibnit, ab Mine                                                                                                                                  | Antimon, ab Raffinerie                                                                                                       |
| Synonyms                              |                                                                                                                                                                    |                                                                                                                              |
| General Comment to reference function | The model represents mining and beneficiation of Stibnite in China. China produces more than 80% of the global Antimony.                                           | The model represents Antimony production in China. China produces more than 80% of the global Antimony.                      |
| CAS Number                            |                                                                                                                                                                    | 7440-36-0                                                                                                                    |
| Start Date                            | 1994                                                                                                                                                               | 1994                                                                                                                         |
| End Date                              | 2003                                                                                                                                                               | 2003                                                                                                                         |
| Data Valid For Entire Period          | 1                                                                                                                                                                  | 1                                                                                                                            |
| Other Period Text                     |                                                                                                                                                                    |                                                                                                                              |
| Geography text                        | The model represents mining and beneficiation of Antimony in China. China produces more than 80% of the global Antimony.                                           | The model represents Antimony production in China. China produces more than 80% of the global Antimony.                      |
| Technology text                       | This module is approximated with values from molybdenum mining and beneficiation (coupled with copper mining)                                                      | The data represent a mixture of blast furnace, rotary kiln and electrowinning process. It is approximated from lead smelting |
| Representativeness [%]                |                                                                                                                                                                    |                                                                                                                              |
| Production Volume                     |                                                                                                                                                                    |                                                                                                                              |
| Sampling Procedure                    | literature                                                                                                                                                         | literature                                                                                                                   |
| Extrapolations                        | see Geography and Technology                                                                                                                                       | see Geography and Technology                                                                                                 |
| Uncertainty Adjustments               | none                                                                                                                                                               | none                                                                                                                         |

## 5. System Characterisation

|    | B                           | F                                                    | G        | J                          | K    | L                                      | M               | N                         | R                |
|----|-----------------------------|------------------------------------------------------|----------|----------------------------|------|----------------------------------------|-----------------|---------------------------|------------------|
|    | Explanations                | Name                                                 | Location | Infrastructure-<br>Process | Unit | stibnite ore, 70%<br>stibnite, at mine | uncertaintyType | StandardDeviation<br>n95% | GeneralComment   |
| 3  |                             | Location                                             |          |                            |      | CN                                     |                 |                           |                  |
| 4  |                             | InfrastructureProcess                                |          |                            |      | 0                                      |                 |                           |                  |
| 5  |                             | Unit                                                 |          |                            |      | kg                                     |                 |                           |                  |
| 6  |                             |                                                      |          |                            |      |                                        |                 |                           |                  |
| 7  | Technosphere                | electricity, medium voltage, at grid                 | CN       | 0                          | kWh  | 3.98E+0                                | 1               | 1.14                      | (2,3,2,1,1,4,2)  |
| 8  |                             | limestone, milled, packed, at plant                  | CH       | 0                          | kg   | 1.98E-1                                | 1               | 1.04                      | reported values  |
| 9  |                             | electricity, hydropower, at run-of-river power plant | RER      | 0                          | kWh  | 0                                      | 1               | 1.14                      | (2,3,2,1,1,4,2)  |
| 10 |                             | blasting                                             | RER      | 0                          | kg   | 1.57E-1                                | 1               | 1.13                      | (2,2,2,1,1,4,4)  |
| 11 |                             | diesel, burned in building machine                   | GLO      | 0                          | MJ   | 7.94E-3                                | 1               | 1.13                      | (2,2,2,1,1,4,4)  |
| 12 |                             | chromium steel 18/8, at plant                        | RER      | 0                          | kg   | 1.17E-1                                | 1               | 1.20                      | reported values  |
| 13 |                             | chemicals inorganic, at plant                        | GLO      | 0                          | kg   | 1.21E-1                                | 1               | 2.00                      | reported values  |
| 14 |                             | sodium cyanide, at plant                             | RER      | 0                          | kg   | 5.49E-3                                | 1               | 2.00                      | reported values  |
| 15 |                             | chemicals organic, at plant                          | GLO      | 0                          | kg   | 3.54E-2                                | 1               | 2.00                      | reported values  |
| 16 |                             | conveyor belt, at plant                              | RER      | 1                          | m    | 1.19E-5                                | 1               | 3.36                      | (5,5,1,1,3,5,9)  |
| 17 |                             | aluminium hydroxide, plant                           | RER      | 1                          | unit | 1.32E-9                                | 1               | 3.07                      | (3,2,2,1,3,4,9)  |
| 18 |                             | non-ferrous metal mine, surface                      | GLO      | 1                          | unit | 1.73E-9                                | 1               | 3.36                      | (5,5,1,1,3,5,9)  |
| 19 |                             | non-ferrous metal mine, underground                  | GLO      | 1                          | unit | 7.53E-10                               | 1               | 3.36                      | (5,5,1,1,3,5,9)  |
| 20 |                             | disposal, sulfidic tailings, off-site                | GLO      | 0                          | kg   | 2.70E+2                                | 1               | 1.13                      | (2,2,2,1,1,4,6)  |
| 34 | resource, in water          | Water, river                                         |          |                            | m3   | 2.32E-1                                | 1               | 1.13                      | (2,2,1,1,1,4,12) |
| 35 | resource, land              | Occupation, mineral extraction site                  |          |                            | m2a  | 8.18E-3                                | 1               | 1.51                      | (2,2,2,1,1,2,7)  |
| 36 |                             | Transformation, to mineral extraction site           |          |                            | m2   | 2.73E-4                                | 1               | 2.01                      | (2,2,2,1,1,2,8)  |
| 37 |                             | Transformation, from unknown                         |          |                            | m2   | 2.73E-4                                | 1               | 2.01                      | (2,2,2,1,1,2,8)  |
| 38 | resource, in ground         | Stibnite, in ground                                  |          |                            | kg   | 7.00E-1                                | 1               | 1.13                      | (2,2,2,1,1,4,12) |
| 40 | air, low population density | Antimony                                             |          |                            | kg   | 1.44E-8                                | 1               | 1.59                      | (3,2,1,1,3,4,31) |
| 41 |                             | Arsenic                                              |          |                            | kg   | 1.09E-7                                | 1               | 1.59                      | (3,2,1,1,3,4,31) |
| 42 |                             | Cadmium                                              |          |                            | kg   | 7.94E-9                                | 1               | 1.59                      | (3,2,1,1,3,4,31) |
| 43 |                             | Carbon dioxide, fossil                               |          |                            | kg   | 8.70E-2                                | 1               | 2.30                      | (5,3,6,5,5,24)   |
| 44 |                             | Chromium                                             |          |                            | kg   | 7.24E-6                                | 1               | 1.59                      | (3,2,1,1,3,4,31) |
| 45 |                             | Copper                                               |          |                            | kg   | 3.61E-6                                | 1               | 1.59                      | (3,2,1,1,3,4,31) |
| 46 |                             | Heat, waste                                          |          |                            | MJ   | 1.43E+1                                | 1               | 1.14                      | (2,3,2,1,1,4,13) |
| 47 |                             | Lead                                                 |          |                            | kg   | 1.01E-6                                | 1               | 1.59                      | (3,2,1,1,3,4,31) |
| 48 |                             | Manganese                                            |          |                            | kg   | 6.89E-5                                | 1               | 1.59                      | (3,2,1,1,3,4,31) |
| 49 |                             | Mercury                                              |          |                            | kg   | 3.61E-9                                | 1               | 1.59                      | (3,2,1,1,3,4,31) |
| 50 |                             | Nickel                                               |          |                            | kg   | 5.78E-6                                | 1               | 1.59                      | (3,2,1,1,3,4,31) |
| 51 |                             | Particulates, < 2.5 um                               |          |                            | kg   | 3.67E-2                                | 1               | 2.69                      | (3,2,1,1,5,4,23) |
| 52 |                             | Particulates, > 10 um                                |          |                            | kg   | 3.55E-3                                | 1               | 2.69                      | (3,2,1,1,5,4,19) |
| 53 |                             | Particulates, > 2.5 um, and < 10um                   |          |                            | kg   | 3.19E-2                                | 1               | 3.69                      | (3,2,1,1,5,4,20) |
| 54 |                             | Selenium                                             |          |                            | kg   | 3.61E-9                                | 1               | 1.59                      | (3,2,1,1,3,4,31) |
| 55 |                             | Zinc                                                 |          |                            | kg   | 1.37E-5                                | 1               | 1.59                      | (3,2,1,1,3,4,31) |
| 58 |                             | Fluorine                                             |          |                            | kg   | 6.89E-5                                | 1               | 1.59                      | (3,2,1,1,3,4,31) |
| 59 |                             | Cobalt                                               |          |                            | kg   | 1.44E-6                                | 1               | 1.59                      | (3,2,1,1,3,4,31) |
| 60 |                             | Carbon disulfide                                     |          |                            | kg   | 1.57E-2                                | 1               | 2.31                      | (4,2,2,5,4,23)   |
| 61 |                             | Boron                                                |          |                            | kg   | 7.24E-7                                | 1               | 1.59                      | (3,2,1,1,3,4,31) |
| 62 |                             | Beryllium                                            |          |                            | kg   | 1.88E-7                                | 1               | 1.59                      | (3,2,1,1,3,4,31) |
| 63 | water, river                | Arsenic, ion                                         |          |                            | kg   | 6.83E-7                                | 1               | 10.00                     | reported values  |
| 64 |                             | Cadmium, ion                                         |          |                            | kg   | 7.36E-8                                | 1               | 10.00                     | reported values  |
| 65 |                             | Chromium, ion                                        |          |                            | kg   | 1.27E-7                                | 1               | 10.00                     | reported values  |
| 66 |                             | Copper, ion                                          |          |                            | kg   | 1.83E-6                                | 1               | 10.00                     | reported values  |
| 67 |                             | Lead                                                 |          |                            | kg   | 6.48E-7                                | 1               | 10.00                     | reported values  |
| 68 |                             | Mercury                                              |          |                            | kg   | 8.76E-9                                | 1               | 10.00                     | reported values  |
| 69 |                             | Nickel, ion                                          |          |                            | kg   | 5.65E-6                                | 1               | 10.00                     | reported values  |
| 70 |                             | Zinc, ion                                            |          |                            | kg   | 1.76E-5                                | 1               | 10.00                     | reported values  |
| 71 |                             | Aluminum                                             |          |                            | kg   | 2.00E-5                                | 1               | 10.00                     | reported values  |
| 72 |                             | BOD5, Biological Oxygen Demand                       |          |                            | kg   | 2.42E-3                                | 1               | 10.00                     | reported values  |
| 73 |                             | Calcium, ion                                         |          |                            | kg   | 1.59E-1                                | 1               | 10.00                     | reported values  |
| 74 |                             | Cobalt                                               |          |                            | kg   | 1.81E-7                                | 1               | 10.00                     | reported values  |
| 75 |                             | COD, Chemical Oxygen Demand                          |          |                            | kg   | 2.42E-3                                | 1               | 10.00                     | reported values  |
| 76 |                             | Cyanide                                              |          |                            | kg   | 6.07E-4                                | 1               | 1.84                      | (4,2,2,3,4,4,33) |
| 77 |                             | DOC, Dissolved Organic Carbon                        |          |                            | kg   | 9.46E-4                                | 1               | 10.00                     | reported values  |
| 78 |                             | Iron, ion                                            |          |                            | kg   | 6.72E-5                                | 1               | 10.00                     | reported values  |
| 79 |                             | Manganese                                            |          |                            | kg   | 5.72E-6                                | 1               | 10.00                     | reported values  |
| 80 |                             | Nitrogen, organic bound                              |          |                            | kg   | 5.28E-3                                | 1               | 10.00                     | reported values  |
| 81 |                             | Dissolved solids                                     |          |                            | kg   | 1.20E-3                                | 1               | 10.00                     | reported values  |
| 82 |                             | Sulfate                                              |          |                            | kg   | 5.47E-1                                | 1               | 10.00                     | reported values  |
| 83 |                             | TOC, Total Organic Carbon                            |          |                            | kg   | 9.46E-4                                | 1               | 10.00                     | reported values  |
| 84 | Outputs                     | stibnite ore, 70% stibnite, at mine                  | CN       | 0                          | kg   | 1.00E+0                                |                 |                           |                  |

Fig. 5.1 Flows for " stibnite ore, 70% stibnite, at mine " (CN) in the ecoinvent database



## 5. System Characterisation

|    | B                           | F                                                                    | G        | J                      | K    | S                     | T               | U                    | Y                                           |
|----|-----------------------------|----------------------------------------------------------------------|----------|------------------------|------|-----------------------|-----------------|----------------------|---------------------------------------------|
|    | Explanations                | Name                                                                 | Location | Infrastructure-Process | Unit | antimony, at refinery | uncertaintyType | StandardDeviation95% | GeneralComment                              |
| 3  |                             | Location                                                             |          |                        |      | CN                    |                 |                      |                                             |
| 4  |                             | InfrastructureProcess                                                |          |                        |      | 0                     |                 |                      |                                             |
| 5  |                             | Unit                                                                 |          |                        |      | kg                    |                 |                      |                                             |
| 6  |                             |                                                                      |          |                        |      |                       |                 |                      |                                             |
| 7  | Technosphere                | electricity, medium voltage, at grid                                 | CN       | 0                      | kWh  | 1.53E-1               | 1               | 1.33                 | (3,3,2,1,3,5,2)                             |
| 8  |                             | limestone, milled, packed, at plant                                  | CH       | 0                      | kg   | 6.47E-1               | 1               | 1.25                 | (3,3,2,1,3,3,4)                             |
| 21 |                             | disposal, lead smelter slag, 0% water, to residual material landfill | GLO      | 0                      | kg   | 4.10E-1               | 1               | 1.25                 | (3,3,2,1,3,3,6)                             |
| 22 |                             | hard coal, burned in industrial furnace 1-10MW                       | RER      | 0                      | MJ   | 6.74E+0               | 1               | 1.25                 | (3,3,2,1,3,3,2)                             |
| 23 |                             | heavy fuel oil, burned in industrial furnace 1MW, non-modulating     | RER      | 0                      | MJ   | 2.75E-1               | 1               | 1.25                 | (3,3,2,1,3,3,1)                             |
| 24 |                             | iron ore, 65% Fe, at beneficiation                                   | GLO      | 0                      | kg   | 6.15E-2               | 1               | 1.25                 | (3,3,2,1,3,3,2)                             |
| 25 |                             | stibnite ore, 70% stibnite, at mine                                  | CN       | 0                      | kg   | 1.79E+0               | 1               | 1.33                 | (3,3,2,1,3,5,4)                             |
| 26 |                             | natural gas, burned in industrial furnace >100kW                     | RER      | 0                      | MJ   | 5.58E-1               | 1               | 1.25                 | (3,3,2,1,3,3,1)                             |
| 27 |                             | nitrogen, liquid, at plant                                           | RER      | 0                      | kg   | 2.00E-2               | 1               | 1.25                 | (3,3,2,1,3,3,2)                             |
| 28 |                             | oxygen, liquid, at plant                                             | RER      | 0                      | kg   | 2.03E-1               | 1               | 1.25                 | (3,3,2,1,3,3,4)                             |
| 29 |                             | silica sand, at plant                                                | DE       | 0                      | kg   | 6.08E-1               | 1               | 1.25                 | (3,3,2,1,3,3,4)                             |
| 30 |                             | transport, freight, rail                                             | RER      | 0                      | tkm  | 5.28E+0               | 1               | 1.31                 | (4,3,1,1,3,3,2)                             |
| 31 |                             | transport, lorry >16t, fleet average                                 | RER      | 0                      | tkm  | 9.31E-1               | 1               | 1.31                 | (4,3,1,1,3,3,2)                             |
| 32 |                             | resource correction, PbZn, silver, negative                          | GLO      | 0                      | kg   | 0                     | 1               | 1.00                 | calculation                                 |
| 33 |                             | resource correction, PbZn, lead, positive                            | GLO      | 0                      | kg   | 0                     | 1               | 1.00                 | calculation                                 |
| 39 | air, unspecified            | Carbon dioxide, fossil                                               |          |                        | kg   | 2.85E-1               | 1               | 1.58                 | (4,2,2,3,4,4,14)                            |
| 40 | air, low population density | Antimony                                                             |          |                        | kg   | 2.19E-6               | 1               | 5.04                 | (3,4,2,1,1,4,22)                            |
| 41 |                             | Arsenic                                                              |          |                        | kg   | 3.92E-6               | 1               | 5.04                 | (3,4,2,1,1,4,22)                            |
| 42 |                             | Cadmium                                                              |          |                        | kg   | 3.15E-6               | 1               | 1.55                 | (3,4,2,1,1,4,16)                            |
| 47 |                             | Lead                                                                 |          |                        | kg   | 3.02E-4               | 1               | 5.04                 | (3,4,2,1,1,4,22)                            |
| 49 |                             | Mercury                                                              |          |                        | kg   | 8.04E-8               | 1               | 5.04                 | (3,4,2,1,1,4,22)                            |
| 51 |                             | Particulates, < 2.5 um                                               |          |                        | kg   | 1.60E-4               | 1               | 3.04                 | (3,4,2,1,1,4,27)                            |
| 52 |                             | Particulates, > 10 um                                                |          |                        | kg   | 4.57E-5               | 1               | 1.46                 | (3,4,2,1,1,4,29)                            |
| 53 |                             | Particulates, > 2.5 um, and < 10um                                   |          |                        | kg   | 1.14E-4               | 1               | 1.28                 | (3,4,2,1,1,4,28)                            |
| 56 |                             | Sulfur dioxide                                                       |          |                        | kg   | 3.19E-2               | 1               | 1.19                 | (3,4,2,1,1,4,15)                            |
| 57 |                             | Dioxins, measured as 2,3,7,8-tetrachlorodibenzo-p-dioxin             |          |                        | kg   | 4.84E-11              | 1               | 10.00                | reported                                    |
| 63 | water, river                | Arsenic, ion                                                         |          |                        | kg   | 9.67E-7               | 1               | 5.07                 | (1,3,1,3,3,4,35); values from Zn-production |
| 64 |                             | Cadmium, ion                                                         |          |                        | kg   | 2.51E-6               | 1               | 5.07                 | (1,3,1,3,3,4,35); values from Zn-production |
| 66 |                             | Copper, ion                                                          |          |                        | kg   | 4.84E-6               | 1               | 5.07                 | (1,3,1,3,3,4,35); values from Zn-production |
| 67 |                             | Lead                                                                 |          |                        | kg   | 1.74E-5               | 1               | 5.07                 | (1,3,1,3,3,4,35); values from Zn-production |
| 68 |                             | Mercury                                                              |          |                        | kg   | 4.16E-8               | 1               | 5.07                 | (1,3,1,3,3,4,35); values from Zn-production |
| 72 |                             | BOD5, Biological Oxygen Demand                                       |          |                        | kg   | 3.76E-4               | 1               | 1.58                 | (1,3,1,3,3,4,32); values from Zn-production |
| 75 |                             | COD, Chemical Oxygen Demand                                          |          |                        | kg   | 5.64E-4               | 1               | 1.58                 | (1,3,1,3,3,4,32); values from Zn-production |
| 77 |                             | DOC, Dissolved Organic Carbon                                        |          |                        | kg   | 2.20E-4               | 1               | 1.58                 | (1,3,1,3,3,4,32); values from Zn-production |
| 83 |                             | TOC, Total Organic Carbon                                            |          |                        | kg   | 2.20E-4               | 1               | 1.58                 | (1,3,1,3,3,4,32); values from Zn-production |
| 85 |                             | antimony, at refinery                                                | CN       | 0                      | kg   | 1.00E+0               | 1               | 1.58                 | (1,3,1,3,3,4,32); values from Zn-production |

Fig. 5.2 Flows for "antimony, at refinery" in the ecoinvent database

### 5.1.1 Data Quality Considerations

Not much is known from public sources about quantitative process measures. Therefore the data has been established based on theoretical reasoning and rough assumptions, extrapolating the data from molybdenum mining and lead refining. This dataset approximates metallic antimony and is not to be used when antimony is used as flame retardant (as an oxide) or when it is a substantial part of the system modelled.

## 6 Literature

- Reuter et al. (2004) Reuter M. A., Boin U., Rem P., Yang Y., Fraunholz N. and Schaik A. v. (2004) The Optimization of Recycling: Integrating the Resource, Technological, and Life Cycles. In: *JOM*, 56, pp. 33-37, Online-Version under: <http://www.ingentaconnect.com/content/tms/jom/2004/00000056/00000008/art0007>
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